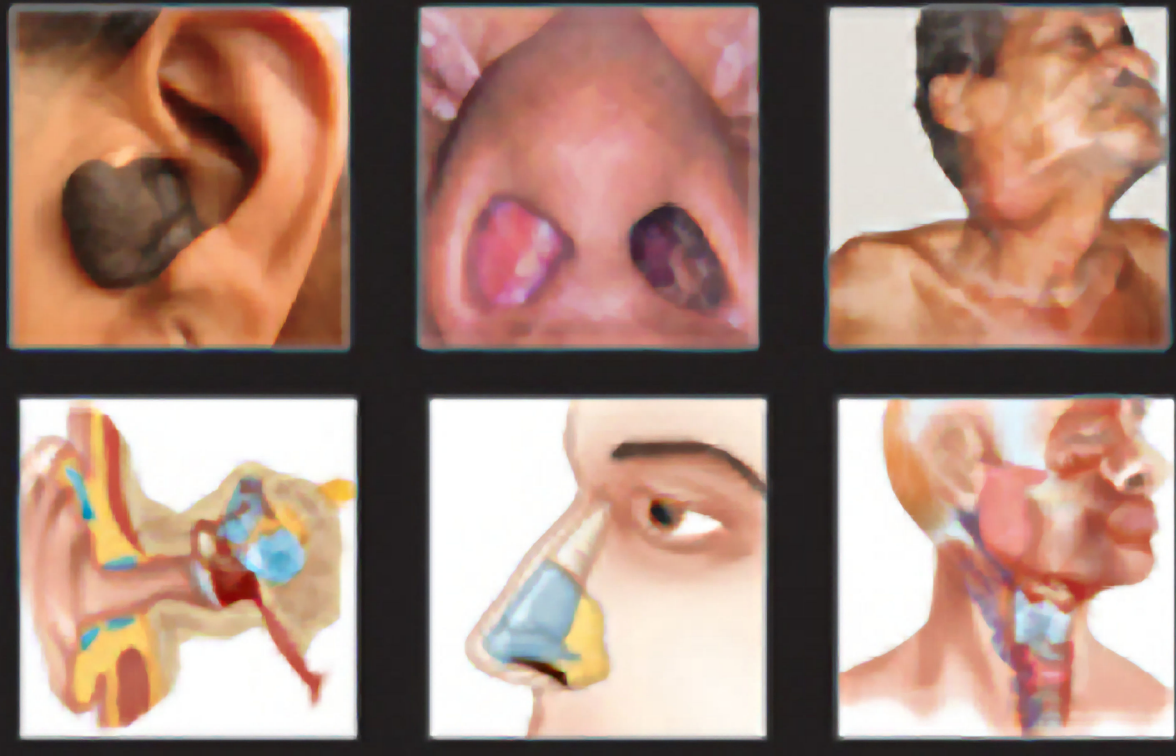
EAR, NOSE & THROAT

Mohan Bansal

DISEASES OF EAR, NOSE & THROAT



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DISEASES OF
EAR, NOSE AND THROAT

DISEASES OF
EAR, NOSE AND THROAT
Head and Neck Surgery

Mohan Bansal MS PhD FICS FACS
Honorary Professor, Otorhinolaryngology
Faculty of Medical Sciences
Charotar University of Science and Technology (CHARUSAT)
Changa, Anand, Gujarat, India
Consultant, Ear, Nose, Throat, Head and Neck Surgeon
Anand, Gujarat

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Jaypee Brothers Medical Publishers (P) Ltd

Headquarters

Jaypee Brothers Medical Publishers (P) Ltd
4838/24, Ansari Road, Daryaganj
New Delhi 110002, India
Phone: +91-11-43574357
Fax: +91-11-43574314
Email: jaypee@jaypeebrothers.com

Overseas Offices

J.P. Medical Ltd
83, Victoria Street, London
SW1H 0HW (UK)
Phone: +44-2031708910
Fax: +02-03-0086180
Email: info@jpmedpub.com

Jaypee-Highlights Medical Publishers Inc
City of Knowledge, Bld. 237, Clayton
Panama City, Panama
Phone: +507-317-0496
Fax: +507-301-0499
Email: cservice@jphmedical.com

Jaypee Brothers Medical Publishers (P) Ltd
17/1-B Babar Road, Block-B, Shaymali
Mohammadpur, Dhaka-1207
Bangladesh
Mobile: +08801912003485
Email: jaypeedhaka@gmail.com

Jaypee Brothers Medical Publishers (P) Ltd
Shorakhute, Kathmandu
Nepal
Phone: +00977-9841528578
Email: jaypee.nepal@gmail.com

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DEDICATED TO

Almighty Lord, my parents, teachers, family, patients and students

Shri Ramakrishna Paramhansa

He indeed is blessed, in whom all the qualities of head and heart are fully developed and evenly balanced. He acquits himself admirably well in whatever position he may be placed. He is full of guileless faith and love for God, and yet his dealings with others leave nothing to be desired. When he is engaged in worldly affairs, he is a thorough man of business. In the assembly of the learned, he establishes his claims as a man of superior learning, and in debates, he shows wonderful powers of reasoning. To his parents, he is obedient and affectionate; to his relations and friends, he is loving and sweet; to his neighbors, he is kind and sympathetic and always ready to do goods; to his wife, he is the god of love. Such a man is indeed perfect.

Holy Mother Sri Sarada Devi

If you want peace, do not find fault with others. Rather see your own faults. Learn to make the world your own. No one is stranger, my child; the whole world is your own.

Swami Vivekananda

We are responsible for what we are, and whatever we wish ourselves to be, we have the power to make ourselves. If what we are now has been the result of our own past actions, it certainly follows that whatever we wish to be in future can be produced by our present actions. Man is man, so long as he is struggling to rise above nature, and this nature is both internal and external.

PREFACE

As long as I live, I learn.

• **Bhagwan Shri Ramakrishna Dev** •

Diseases of Ear, Nose and Throat, which represents Otorhinolaryngology Head and Neck Surgery in all of its diversity, is created to fill the need of contemporary definitive book. The reader will find boxes, tables, flow charts, line diagrams and photographs, which serve to enhance learning. The book is comprehensive and of broader scope and is designed for students, residents and practitioners alike. It offers a balanced presentation of content and emphasizes the practical features of clinical diagnosis and patient management. The students will like the simplicity, directness and clarity. Each chapter includes clear, compelling, and up-to-date discussions and expertly executed and generously sized art. The brevity, conciseness, readable format and easy accessibility of key information will facilitate efficient use in any practice setting. Each page is carefully laid out to place related text, figures, and tables near one another to minimize the need for page turning. To provide an overview, each chapter begins with the list of its content and ends with Further Reading section. Each chapter has Clinical Highlights section for the quick revision of the students. This section has been especially prepared for answering frequently asked MCQs, short-answer questions and oral/viva questions. The Appendix contains Top 101 Clinical Secrets and Problem-Oriented Cases which will be of immense use and interest to the readers.

I would like to acknowledge my parents, late Shri Ramchandra and Smt Kalawati Devi Bansal, for enabling me to survive comfortably during my seemingly endless years of education. My family has unswervingly endorsed the time required for this mission, so heartfelt love and thanks go to my wife, Sushma, as well as our children Tejal and Mohit and his wife Astha. My loyal assistant for the last 10 years, Tejal Patel, has provided amounts of all-round care to cover for my time. I wish to thank my professor friends who spared their valuable time in reviewing the chapters.

The process of learning is truly life-long. Creating this text allows me to continue to become invigorated and inspired by otolaryngology. I hope that my quest to document significant and up-to-date information has been successful. My sincere hope is that readers, everywhere, will benefit from this book. I invite readers and educators to send their suggestions so that I can include them in the next edition. The structure, content, and production values of this book will be shaped by its relationship with educators and readers.

Mohan Bansal
(mohanbansal@yahoo.com)

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For this book *Diseases of Ear, Nose and Throat*, I have enjoyed the opportunity of collaborating with a group of dedicated and talented professionals. I would like to recognize and thank the members of the book team, who indeed worked hard, to bring this book to you. Shri Jitendar P Vij (Chairman and Managing Director), Jaypee Brothers Medical Publishers, illuminated the path for this book with his creative ideas and dedication. The insights and skills of Dr Richa Saxena (Editor-in-Chief) helped in polishing this book to best meet the needs of students and faculty alike. Mr Ankit Vij (Managing Director), the young and dynamic leader, took personal interest and laid out each page of the book to achieve the best possible placement of text, figures, and other elements. The suggestions from Mr Saket Budhiraja (Director-Sales and Marketing) were very practical and meaningful. Mr Tarun Duneja (Director- Publishing) demonstrated his untiring expertise during each step of the production process. I would like to thank Ms Sunita Katla (Publishing Manager) for her efforts towards the finalisation of the book. I would also like to thank Mr KK Raman (Production Manager), Ms Samina Khan (PA to Director-Publishing), Mr Amit Rai (Editor), Mr Ashutosh Srivastava (Assistant Editor) and Mr Kapil Dev Sharma (DTP Operator) for their work with efficiency. Ms Seema Dogra's (Cover Designer) and Mr Sumit Kumar's (Graphic Designer) artistic ability, organizational skills, attention to detail and understanding of illustration preferences greatly enhance the visual appeal and style of figures. They are consummate professionals whose efforts I truly appreciate. Tejal Patel, my assistant, shepherded the manuscript and electronic files. Sushma coordinated the development of many supplements that support this text. Dr Rimpal Chauhan, Chandani, Priti, Falguni, Rina, Rashmi, Tejal, Bimal and Hansika, my students, have collaborated on the illustrations for this book. The PG seminars, Journal Club meetings and case discussion at PSMC, Karamsad, Anand, Gujarat, are very enriching. So I am thankful to Prof Ravi Tiwari, Prof Girish Mishra, Prof Yojana Sharma, Dr Hiren Soni, Dr Siddharth Shah, Dr Nimesh Patel and PG students for their valuable and meaningful discussions. I feel immense pleasure to express my heartfelt emotions to my PhD guide Prof Vikas Sinha (Prof, ENT, and Dean, MP Shah Medical College, Jamnagar) and Prof Nitin Nagarkar (Govt Medical College, Chandigarh) and faculties of BJMC, Ahmedabad, Prof R Vishwakarma, Prof Bela J Prajapati, Dr Neena H Bhalodiya, Dr BK Kedia, Dr Kalpesh Patel, and Dr Divang Gupta, Dr Shaun and Dr Shashank for their kind cooperation and friendly help.

Under the GSE program of Rotary Foundation, I visited some of the best medical centers in the USA including the Mayo Hospital with my friend Prof Ranjan Aiyar. I appreciate his whole-hearted support. I am happy to express my thanks to my friend Prof Mohan Jagade with whom I received the Garnett Passé and Rodney William Memorial Foundation, International Educational Scholarship for attending the 16th World Congress of ORL, Head and Neck Surgery, in Australia.

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- Arun Agarwal, Maulana Azad Medical College, New Delhi
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- Anirban Biswas, Kolkata, West Bengal
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- Shelly Chadha, Maulana Azad Medical College, New Delhi
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- Jyoti Dabholkar, Seth GSMC & KEM Hospital, Mumbai, Maharashtra
- Sudip Kumar Das, Institute of Postgraduate Medical Education and Research Medical College, Kolkata, West Bengal
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1 Anatomy and Physiology of Ear

Look at the anvil of a blacksmith – how it is hammered and beaten; yet it moves not from its place. Let men learn patience and endurance from it.

—Sri Ramakrishna Dev

Points of Focus

- **TEMPORAL BONE**

ANATOMY OF EXTERNAL EAR

- **AURICLE:** Incisura Terminalis, Endaural Incision, Frost Bite, Sebaceous Cysts, Grafts
- **EXTERNAL AUDITORY CANAL (EAC):** Fissures of Santorini, Foramen of Huschke
- **TYMPANIC MEMBRANE:** Pars Tensa, Pars Flaccida

MIDDLE EAR ANATOMY

- **PARTS OF MIDDLE EAR:** Epi, Meso, and Hypotympanum
- **BOUNDARIES OF MIDDLE EAR:** Tegmental, Jugular, Carotid, Mastoid, Labyrinthine, and Membranous Walls
- **OSSICLES:** Malleus, Incus, Stapes
- **INTRATYMPANIC MUSCLES:** Tensor Tympani, Stapedius
- **INTRATYMPANIC NERVES:** Tympanic Plexus, Tympanic Branch (Jacobson) of Glossopharyngeal, Chorda Tympani Nerve
- **MIDDLE EAR MUCOSA AND COMPARTMENTS:** Prussak's Space, Anterior and Posterior Attic Compartments, Inferior Incudal Space, Anterior and Posterior Pouches of Von Troltsch
- **MASTOID ANTRUM:** Macewen's Triangle
- **TYPES OF MASTOID:** Cellular, Diploeic, and Acellular
  - ✓ Mastoid Air Cells: Zygomatic, Tegmen, Perisinus, Retrofacial, Perilabyrinthine, Peritubal, Tip, Marginal, and Squamous Cells
- **KORNER'S SEPTUM**
- **BLOOD SUPPLY AND LYMPHATIC DRAINAGE OF EAR**

ANATOMY OF INTERNAL EAR

- **BONY LABYRINTH**
  - ✓ Vestibule: Oval window, Spherical recess, Elliptical Recess, Mike's Dot, Vestibular Crest and Cochlear Recess, Aqueduct of Vestibule.
  - ✓ Semicircular Canals: Superior, Lateral, Posterior, and Crus Commune
  - ✓ Cochlea: Modiolus, Osseous Spiral Lamina, Rosenthal's Canal, Scala Vestibuli, Scala Tympani, Promontory, Helicotrema, Round Window, Aqueduct of Cochlea
- **MEMBRANOUS LABYRINTH**
  - ✓ Cochlear Duct: Basilar membrane, Reissner's Membrane, Stria Vascularis
  - ✓ Utricle and Saccule
  - ✓ Semicircular Ducts
  - ✓ Endolymphatic Duct and Sac
- **INNER EAR FLUIDS:** Perilymph and Endolymph
- **ORGAN OF CORTI:** Tunnel Of Corti, Inner and Outer Hair Cells, Deiter and Hensen's Cells, and Tectorial Membrane
- **VESTIBULAR RECEPTORS**
  - ✓ Cristae: Cupula (Type 1 and 2 cells)
  - ✓ Maculae: Striola and Otolithic Membrane
- **BLOOD SUPPLY OF LABYRINTH**
- **INTERNAL AUDITORY CANAL:** Contents and Auditory Nerve

DEVELOPMENT OF EAR

CENTRAL CONNECTIONS (NEURAL PATHWAYS)

- **AUDITORY NEURAL PATHWAYS:** Eighth Nerve, Cochlear Nuclei, Olivary Complex (Superior), Lateral Lemniscus, Inferior Colliculus, Medial Geniculate Body and Auditory Cortex
- **CENTRAL VESTIBULAR CONNECTIONS:** Vestibular Nerve and Vestibular Nuclei; Functions of Vestibular

Contd...

Contd...

Nuclei: Vestibuloocular Reflexes, Vestibulospinal Tract, Vestibulocerebellar Tract, Autonomic Symptoms, Motion Awareness

PHYSIOLOGY OF HEARING

- **CONDUCTION OF SOUND**
  - ✓ Transformer Action of Middle Ear: Hydraulic Action of Tympanic Membrane, Curved Membrane Effect, Lever Action of the Ossicles
  - ✓ Phase Differential between Oval and Round Window: Acoustic Separation of Two Windows
  - ✓ Natural Resonance of External and Middle Ear
- **TRANSDUCTION OF MECHANICAL ENERGY TO ELECTRICAL IMPULSES**
  - ✓ Round Window Reflex
  - ✓ Transduction: Traveling Wave Theory of von Bekesy, Tonotopic Gradient in Cochlea
  - ✓ Functions of Hair Cells
  - ✓ Electrical Potentials: Endocochlear Potential, Cochlear Microphonics, Summating Potential, and Compound Action Potential
- **SEMICIRCULAR CANALS FUNCTIONS**
  - ✓ Nystagmus: Flow of Endolymph, Rotating Chair Test
- **UTRICLE AND SACCULE FUNCTIONS**
- **MAINTENANCE OF BODY EQUILIBRIUM:** Sensory component, Motor component
  - ✓ Push and Pull System, Pathophysiology, and Compensation
- **CLINICAL HIGHLIGHTS**

TEMPORAL BONE

The temporal bone has an interesting multifaceted anatomy. The important structures present and their complicated anatomic interrelations make the temporal bone surgery a challenge.

- *Relations*: It articulates with five cranial bones: parietal, sphenoid, occipital, zygomatic and mandible. This pyramidal shaped bone forms part of the base and lateral side of skull (Fig. 1). The petrous part separates middle cranial fossa from the posterior cranial fossa.
- *Contents*: It houses the hearing and vestibular organs. The important structures which pass through it include internal carotid artery, internal jugular vein and facial nerve. So the temporal bone houses following structures:
  - Bony portion of external ear
  - Middle ear containing malleus, incus and stapes
  - Internal ear containing peripheral portions of auditory and vestibular system
  - Fallopian canal containing facial nerve
  - Osseous canal for the internal carotid artery
  - Bony covering for the sigmoid sinus and the jugular bulb
- *Parts*: The four portions of temporal bones are referred as separate bones and include
  - Squamous
  - Petrous
  - Tympanic
  - Mastoid

Ear

For the sake of description ear is divided into three parts (Fig. 2):

1. External ear
2. Middle ear
3. Internal ear

ANATOMY OF EXTERNAL EAR

The external ear is divided into auricle (pinna) and external acoustic or auditory canal (EAC). The tympanic membrane separates external ear from the middle ear.

AURICLE

The auricle is made up of (except its lobule) a framework of a single piece of yellow elastic cartilage (Fig. 3), which is covered

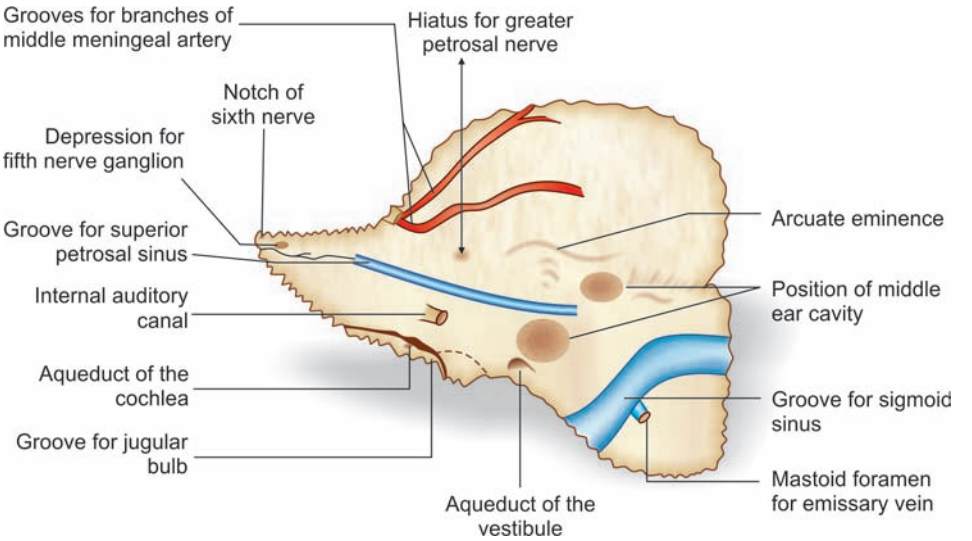


Fig. 1: Intracranial view of petrous and squamous parts of temporal bone

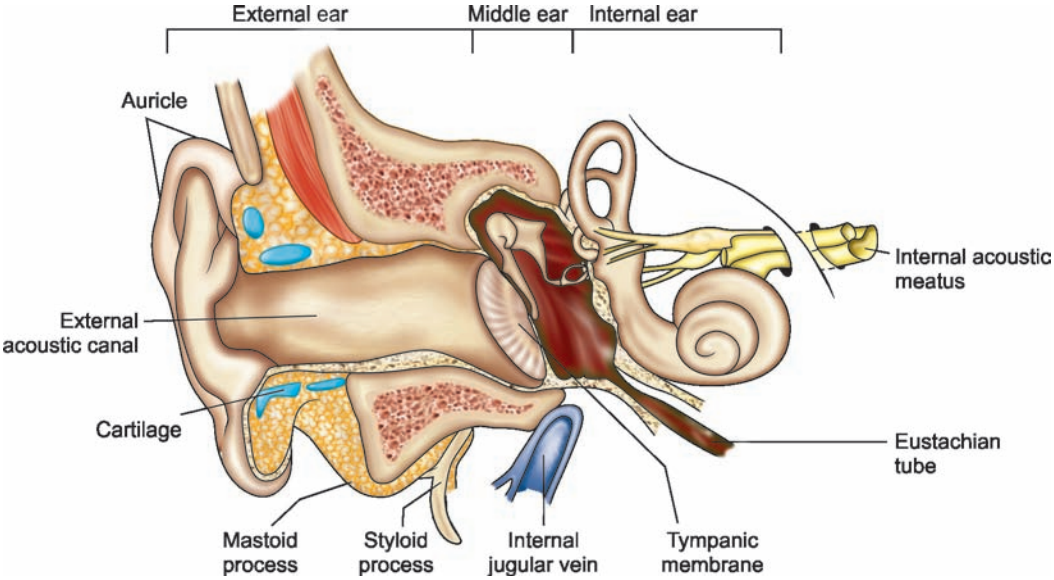


Fig. 2: Three parts of the ear: external, middle and internal

with skin. The skin is adherent to the perichondrium on its lateral surface while it is comparatively loose on the medial surface. Epithelium is squamous keratinizing. Sebaceous glands and hair follicles are found in the subcutaneous tissue. Adipose tissue is present only in the lobule. There are various elevations and depressions, which can be seen on the lateral surface of pinna (Fig. 4).

- *Incisura Terminalis*: This area is devoid of cartilage and lies between the tragus and crus of the helix.

Endaural incision: It is made in incisura terminalis for the surgery of EAC and middle ear. It does not cut through the auricular cartilage.

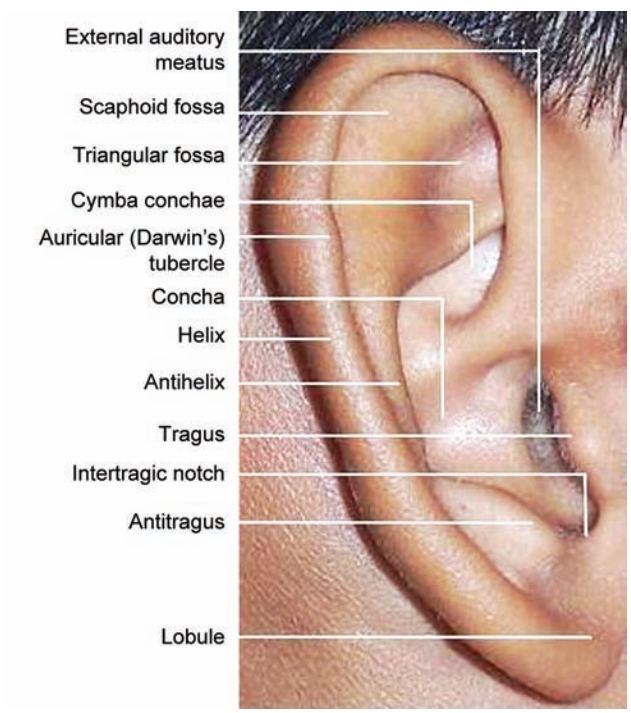


Fig. 3: External features of auricle

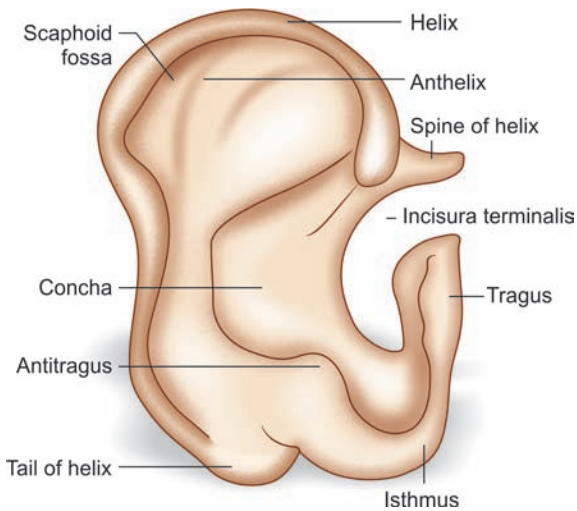
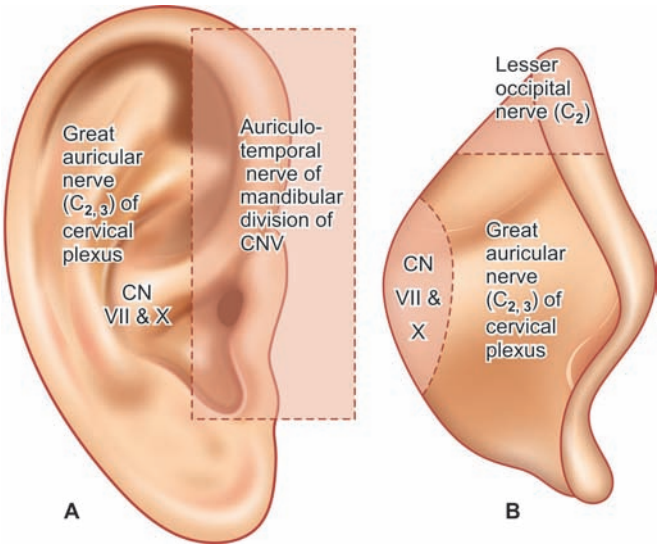


Fig. 4: Auricle cartilage: external features

Frost bite: The outer surface of pinna is more prone to frost bite because the skin is adherent to the underlying perichondrium. There is no subcutaneous tissue.
Sebaceous cysts: They are more common on medial surface of pinna.

- *Grafts in rhinoplasty*: The conchal cartilage is frequently used to correct depressed nasal bridge. The composite grafts of the skin and cartilage can be used for repair of defects of ala of nose.
- *Grafts in tympanoplasty*: Tragal and conchal cartilage and perichondrium and fat from lobule are often used during tympanoplasty operations.

- *Nerve Supply* (Figs 5A and B): (See otalgia in Chapter Otologic Symptoms and Examination)
  1. *Auriculotemporal nerve (CN V_3)*: It is a branch of mandibular division of trigeminal nerve and supplies anterosuperior part of lateral surface of pinna including tragus and crus of helix.



Figs 5A and B: Nerve supply of right pinna. (A) Lateral surface; (B) Medial surface

2. *CN VII (facial nerve)*: It innervates the skin of lateral concha and antihelix, lobule and mastoid.
3. *CN X (vagus nerve)*: Its auricular branch (Arnold's nerve) supplies to concha and postauricular skin.
4. *Greater auricular nerve* ($C_{2,3}$): This nerve of cervical plexus supplies most of the medial surface of auricle and posterior part of lateral surface and the postauricular region.
5. *Lesser occipital nerve (C_2):* This nerve of cervical plexus supplies upper part of medial surface of auricle and postauricular region.

EXTERNAL AUDITORY CANAL

- *Dimensions*: External auditory canal (EAC) measures about 24 mm and extends from the concha to the tympanic membrane. Its anterior wall is 6 mm longer than the posterior wall. EAC is usually divided into 2 parts: (1) cartilaginous and (2) bony. Its outer one-third (8 mm) is cartilaginous and its inner two-third (16 mm) is bony.
- *Direction*: EAC is 'S' shaped and not straight. Its outer one-third cartilaginous part is directed upwards, backwards and medially while it's inner two-third bony part is directed downwards, forwards and medially.

For examining the tympanic membrane, the pinna is pulled upwards, backwards and laterally, which brings the two parts of EAC in alignment.

- *Cartilaginous EAC:* It is a continuation of the cartilage that forms the framework of the pinna.

Fissures of Santorini: Transverse slits in the floor of cartilaginous EAC called "fissures of Santorini" provide passages for infections and neoplasms to and from the surrounding soft tissue (especially parotid gland). The parotid and mastoid infections can manifest in the EAC.

- *Skin Glands*: The skin of the cartilaginous canal (Fig. 6) is thick and contains ceruminous and pilosebaceous glands that secrete wax. The hydrophobic, slightly acidic (pH 6.0–6.5) cerumen is formed in this part of EAC.

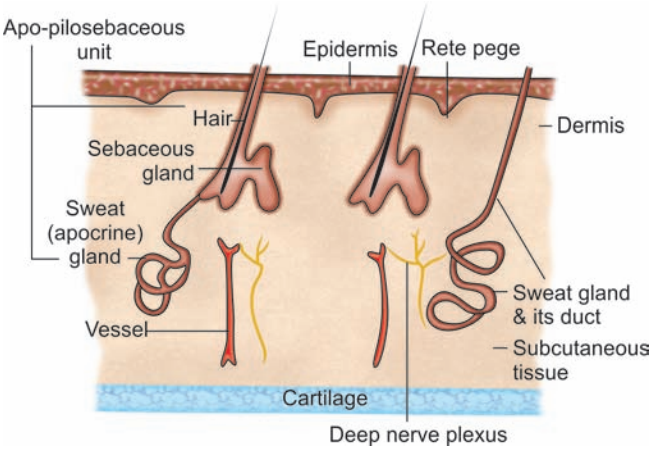


Fig. 6: Skin of cartilaginous external auditory canal

Hair follicles are present only in the outer cartilaginous canal and therefore furuncles (staphylococcal infection of hair follicles) are seen only in the cartilaginous EAC.

- Bony EAC: It is mainly formed by the tympanic portion of temporal bone but roof is formed by the squamous part of the temporal bone (Fig. 7). In the anterosuperior region, squamous part articulates with tympanic bone (tympanosquamous suture). Inferiorly and medially squamous part joins with the lateral superior portion of the petrous bone (petrosquamous suture). Skin of the bony EAC is thin and continuous over the tympanic membrane skin is devoid of subcutaneous layer, hair follicles and ceruminous glands.
  - Isthmus: Approximately 6 mm lateral to tympanic membrane, bony EAC has a narrowing called the isthmus.

Foreign body impacted medial to bony isthmus of EAC are difficult to remove.

  - *Recess*: Anteroinferior part of the deep bony meatus, medial to the isthmus has a recess, which is called the anterior recess.

The anterior recess of bony EAC acts as a cesspool for discharge and debris.

  - *Foramen of Huschke*: In children and occasionally in adults, anteroinferior bony EAC may have a deficiency that is called foramen of Huschke.

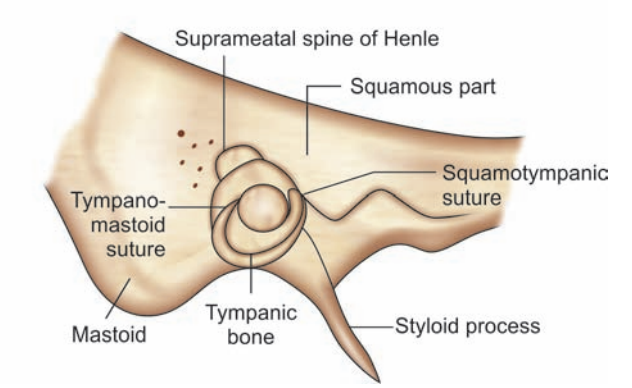


Fig. 7: Lateral view of temporal bone showing endomeatal spines and sutures

Foramen of Huschke permits spread of infections to and from EAC and parotid.

- *Relations of Bony EAC*
  - Superior: Middle cranial fossa
  - Inferior: Parotid gland
  - Posterior: Mastoid antrum and air cells and the facial nerve
  - Anterior: Temporomandibular joint (TMJ)
  - Medial: Tympanic membrane
  - Lateral: Cartilaginous EAC

Acute mastoiditis causes sagging of posterosuperior part of deeper bony EAC because it is related with the mastoid antrum.

- *Epithelial Migration*: The skin of EAC has a unique self-cleansing mechanism. This migratory process continues from the medial to lateral side. The sloughed epithelium is extruded out as a component of cerumen.
- *Nerve Supply* (Fig. 8)**:** (See otalgia in Chapter Otologic Symptoms and Examination)
  - *Auriculotemporal nerve (CN V_3):* It is a branch of mandibular division of trigeminal nerve and supplies anterosuperior wall of external auditory canal.
  - *CN X (vagus nerve)*: Its auricular branch (Arnold's nerve) supplies to inferoposterior external auditory canal.
  - *CN VII (facial nerve)*: It innervates the skin of the mastoid and posterior external auditory canal.

- *Hitzelberger's sign*: The hypoesthesia of posterior meatal wall occurs due to the pressure on facial nerve (sensory fibers are affected early) in patients with acoustic neuroma.
- *Vasovagal reflex*: While cleaning the EAC, patient may develop coughing, bradycardia, syncope and even cardiac arrest. They can occur because of Arnold's branch of vagus nerve.
- *Appetite*: Because of vagal innervation, instilling spirit in EAC before meal can stimulate appetite.
- *Ramsay Hunt syndrome*: Vesicles of herpes zoster oticus occur on mastoid and posterior meatal wall which indicate that this part of external ear has facial nerve innervation.

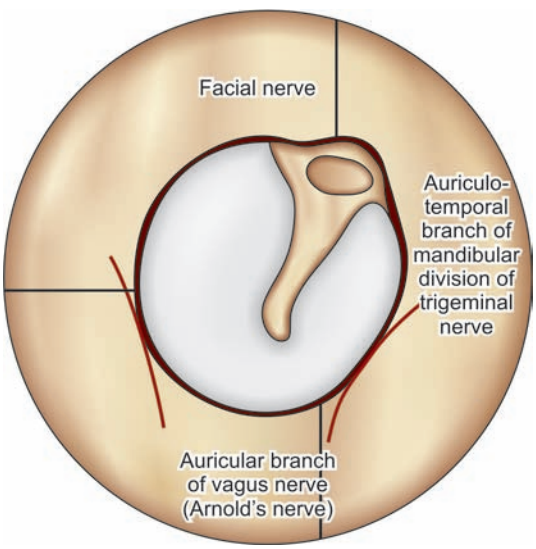


Fig. 8: Nerve supply of EAC

TYMPANIC MEMBRANE (FIG. 9)

- *Dimensions*: Its dimensions are: 9–10 mm height and 8–9 mm width. It is 0.1 mm thick.
- *Position*: Tympanic membrane (TM) is a partition wall between the EAC and the middle ear. It is positioned obliquely. It forms angle of 55° with deep EAC. Its posterosuperior part is more lateral than its anteroinferior part.
- *Parts*: Tympanic membrane consists of two parts: (1) pars tensa and (2) pars flaccida.
  - *Pars tensa*: It forms most of tympanic membrane (TM).
    - Annulus tympanicus: TM is thickened in the periphery and forms a fibrocartilaginous ring called the annulus tympanicus that fits in the tympanic sulcus.
    - Umbo: The central part of TM near the tip of malleus is tended inwards and is called the umbo.
    - Cone of light: A bright cone of light radiating from the tip of malleus to the periphery in the anteroinferior quadrant is usually seen during otoscopy.
  - *Pars flaccida (Shrapnell's membrane)*: It is situated above the lateral process of malleus between the notch of Rivinus and the anterior and posterior malleal folds. It is not as tense as pars tensa and may appear little pinkish.
- *Structure:* Tympanic membrane consists of the following three layers (Fig. 10):

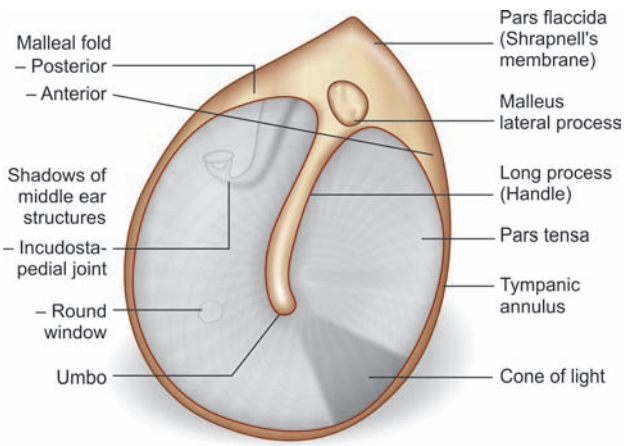


Fig. 9: Tympanic membrane showing attic, malleus handle, umbo, cone of light and structures of middle ear seen through it on otoscopy

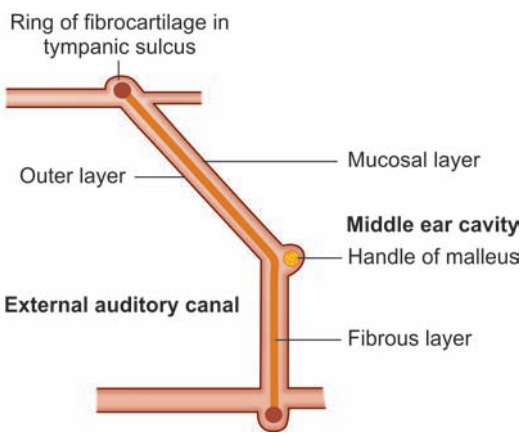


Fig. 10: Three layers of tympanic membrane

a. *Outer epithelial layer*: It is continuous with the EAC skin.
b. *Middle fibrous layer*: It encloses the handle of malleus and consists of three types of fibers: radial, circular and parabolic. In comparison to pars tensa, this layer is very thin in pars flaccida and not organized into various fibers.
c. *Inner mucosal layer:* It is continuous with the middle ear mucosa.

- *Otoscopy:* Normal tympanic membrane is shiny and pearly-gray in color. Its lateral surface is concave, which is more marked at the tip of malleus (umbo). Attic area lies above the lateral process of malleus and is slightly pinkish. Its transparency varies from person to person. Some middle ear structures can usually be seen through the membrane such as incudostapedial joint.
- *Mobility (Seigalization):* A normal tympanic membrane is mobile, which can be tested with pneumatic otoscope or Siegel's speculum.
- *Nerve Supply:* (See otalgia in Chapter of Otologic Symptoms and Examination)
  - *Auriculotemporal nerve (CN V_3):* It is a branch of mandibular division of trigeminal nerve and supplies anterior half of lateral surface of TM.
  - *CN X (vagus nerve):* Its auricular branch (Arnold's nerve) supplies to posterior half of lateral surface of TM.
  - *CN IX (glossopharyngeal nerve)*: Its tympanic branch (Jacobson's nerve) supplies to medial surface of tympanic membrane.

MIDDLE EAR ANATOMY

The middle ear cleft (Fig. 11), which is lined by mucous membrane and filled with air, consists of the middle ear, eustachian tube, aditus ad antrum, mastoid antrum and mastoid air cells. Middle ear is a 1 to 2 cm^3 air filled cavity that houses ossicles, stapedius and tensor tympani muscles and chorda tympani nerve and tympanic plexus.

Relations of Middle Ear Cleft

- Roof: Tegmen plate separates it from middle cranial fossa and its contents like meninges and temporal lobe of cerebrum.
- Floor: Jugular bulb
- Medial: Labyrinth. Lateral semicircular canal lie posterosuperior to facial nerve.
- Posterior: Sigmoid venous sinus
- Anterior: Petrous part of internal carotid artery lying in carotid canal.

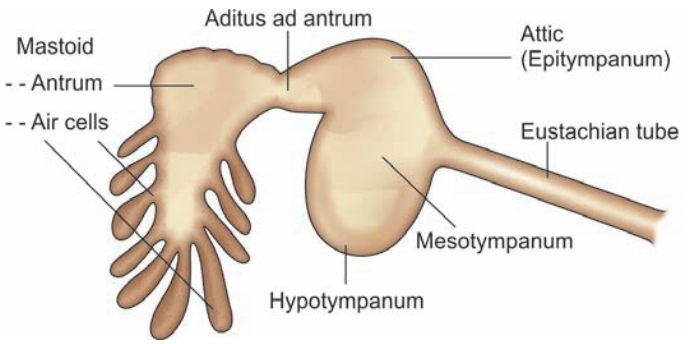


Fig. 11: Parts of middle ear cleft

- Posteromedial: Posteromedial to mastoid air cells is situated cerebellum in the posterior cranial fossa.
  - Cranial nerves:
    a. CN V and CN VI: They lie close to the apex of the petrous pyramid.
    b. CN VII: The horizontal tympanic part is situated in the medial wall of middle ear, while vertical mastoid part runs between the middle ear and mastoid air cells system.

PARTS OF MIDDLE EAR (TYMPANUM)

The dimensions of middle ear are shown in Figure 12. The tympanum (Fig. 13) is traditionally divided into three parts—mesotympanum, epitympanum and hypotympanum.

1. *Mesotympanum*: This is the portion of middle ear that lies at the level of pars tensa.
2. *Epitympanum (attic):* This is the portion of middle ear that lies above the level of pars tensa and medial to Shrapnell's membrane and the bony lateral attic wall.
3. *Hypotympanum*: This is the portion of middle ear that lies below the level of pars tensa.
   - *Protympanum*: The portion of middle ear around the eustachian tube opening is termed as protympanum.

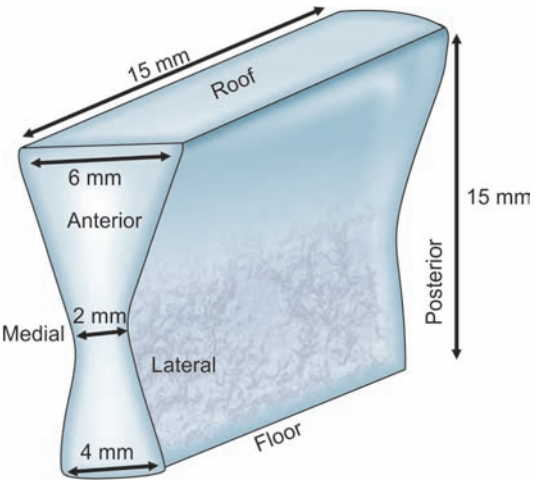


Fig. 12: Dimensions of tympanum

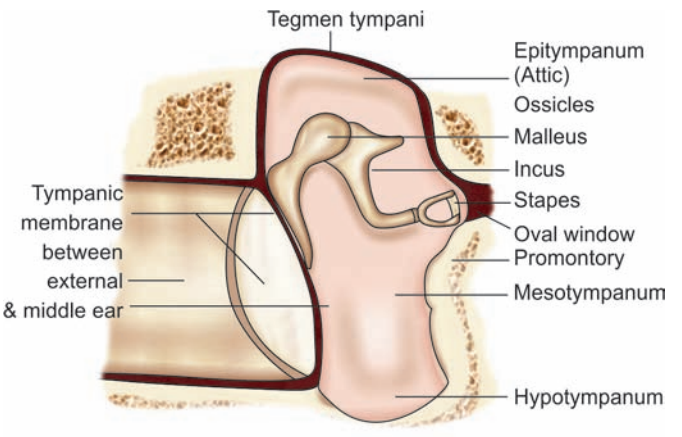


Fig. 13: Parts of middle ear seen on coronal section

BOUNDARIES OF MIDDLE EAR (FIG. 14)

Middle ear has six boundaries: roof, floor, and medial, lateral, anterior and posterior walls.

1. *Roof (Tegmental wall):* It is formed by tegmen tympani (a thin plate of bone), which extends posteriorly to form the roof of the aditus and antrum (tegmen antri). Tegmen tympani separates middle ear from the middle cranial fossa.
2. *Floor (Jugular wall):* The floor, a thin plate of bone, separates tympanic cavity from the jugular bulb.

The floor of middle ear may be congenitally dehiscent. In such cases, jugular bulb projects into the middle ear and is at greater risk of injury during surgery because it is just covered by middle ear mucosa.

3. *Anterior (Carotid wall)*: The anterior wall, a thin plate of bone, which separates the middle ear cavity from internal carotid artery, has following features:
   a. *Eustachian tube*: It connects the middle ear with naso-pharynx. It aerates and drains the middle ear. See chapter Disorders of Eustachian Tube.

Malfunctioning of eustachian tube is common cause of ear infections especially in children.

   b. *Canal of tensor tympani muscle*: It is situated in the roof of eustachian tube.
   c. Canal for chorda tympani nerve.
   d. Attachment of anterior malleolar ligament.
4. *Posterior (Mastoid wall)*: It lies close to the mastoid air cells and presents following structures:
   a. *Pyramid*: It is a bony projection through the summit of which appears the tendon of the stapedius muscle that is inserted to the neck of stapes.
   b. *Aditus ad antrum*: It is an opening through which mastoid antrum opens into the attic. It lies above the pyramid. Its relations are following:
      i. Medial: Bony prominence of the horizontal semicircular canal.
      ii. Lateral: Fossa incudis, to which is attached the short process of incus.
      iii. Inferior: Fallopian canal for facial nerve.
   c. *Facial nerve*: The vertical mastoid part of the fallopian canal for facial nerve runs in the posterior wall just behind the pyramid.

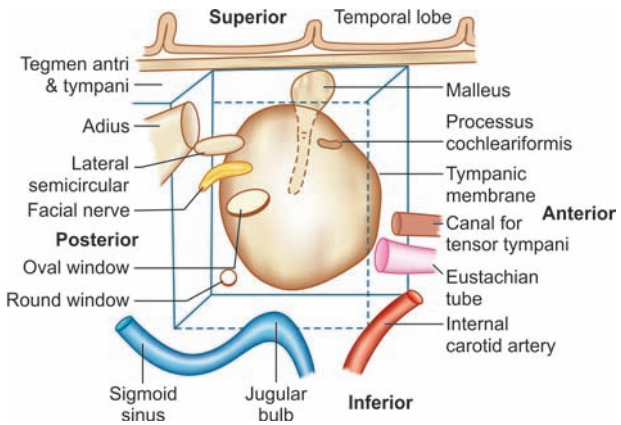


Fig. 14: Six boundaries of tympanum. Medial wall is seen through the tympanic membrane

   d. *Facial (Suprapyramidal) recess* (Fig. 15): This recess is a depression in the posterior wall lateral to the pyramid. Its boundaries are following:
      i. Medial: Vertical part of CN VII.
      ii. Lateral: Chorda tympani (branch of 7th CN) and tympanic annulus.
      iii. Superior: Fossa incudis, in which lies short process of incus.
   e. *Sinus (Infrapyramidal) tympani*: This deep recess lies medial to the pyramid. It is bounded by the subiculum below and the ponticulus above.

In the intact canal wall mastoidectomy, middle ear is approached (posterior tympanotomy or facial recess approach) through the facial recess without disturbing posterior meatal wall (Fig. 16).

5. *Medial (Labyrinthine wall) (Figs 17 and 18)*: It is formed by the lateral wall of labyrinth. It presents following structures:
   a. Promontory: It is a bony bulge which is due to the basal coil of cochlea.
   b. Oval window (Fenestra vestibuli): The footplate of stapes is placed in this window.

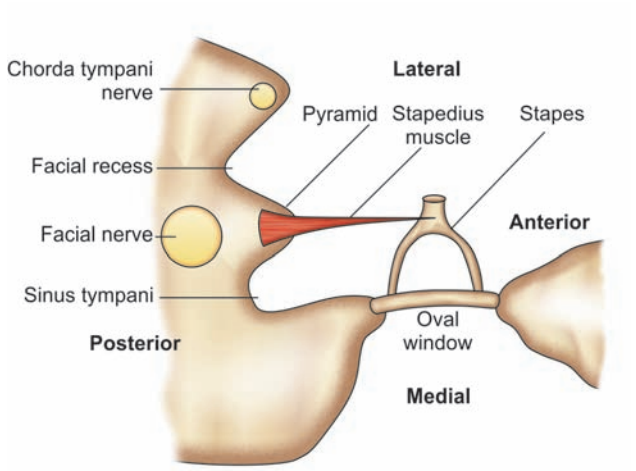


Fig. 15: Facial recess and sinus tympani relations with facial nerve and pyramidal eminence

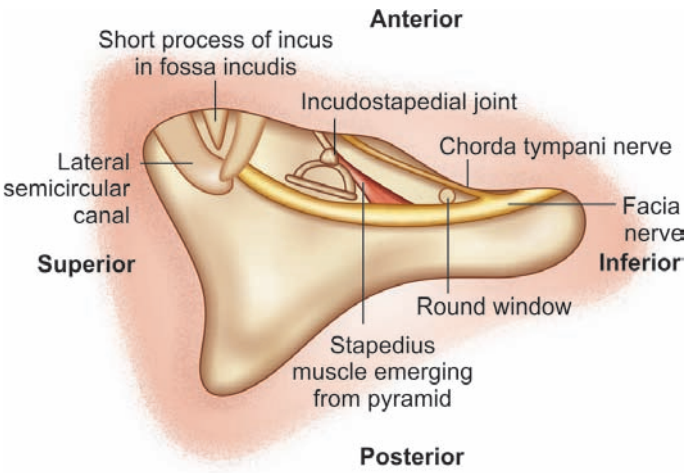


Fig. 16: Posterior tympanotomy. Structures of middle ear seen through the opening of facial recess

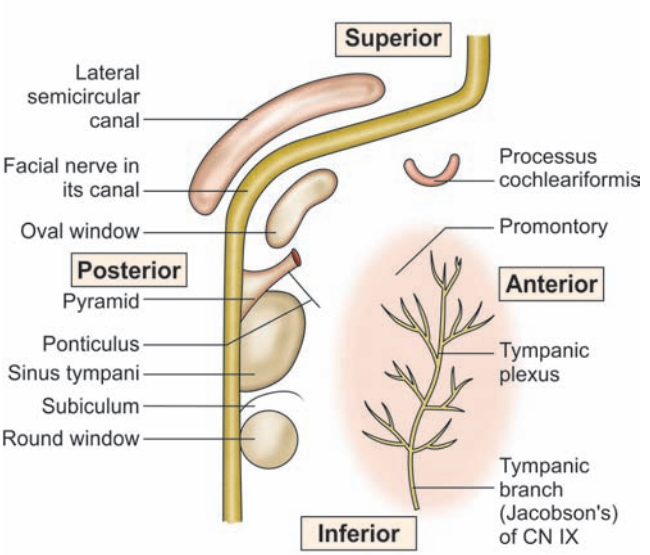


Fig. 17: Medial wall of middle ear

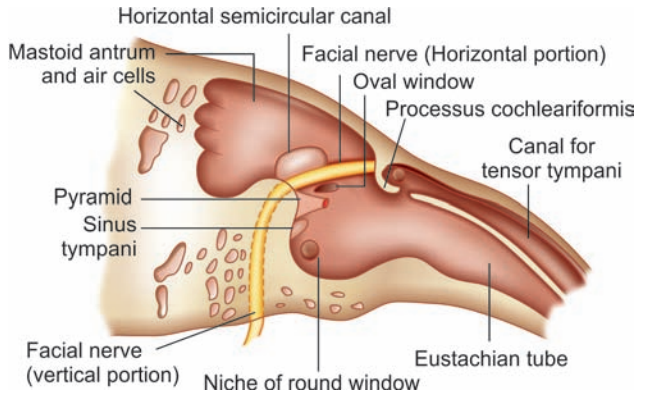


Fig. 18: Medial wall of middle ear cleft

c. Round window (Fenestra cochleae): It is covered by the secondary tympanic membrane.
d. Horizontal tympanic part of fallopian canal for facial nerve: It lies above the oval window.

The tympanic segment of facial nerve canal may be congenitally dehiscent and the exposed facial nerve becomes vulnerable to injuries or infection.

e. Lateral semicircular canal: It lies above the fallopian canal, facial nerve.
f. Processus cochleariformis: It is a hook-like projection, which lies anterior to the oval window. The tendon of tensor tympani takes a turn on this process and then is inserted on the neck of malleus.

Processus cochleariformis is an important surgical landmark for the level of the genu of the facial nerve.

6. *Lateral (Membranous wall) (Fig. 19)*:
   a. Tympanic membrane: Lateral wall is formed mainly by the tympanic membrane. Some structures of the middle ear (such as long process of incus, incudostapedial joint, round window and eustachian tube) can be seen through the normal semitransparent tympanic membrane.

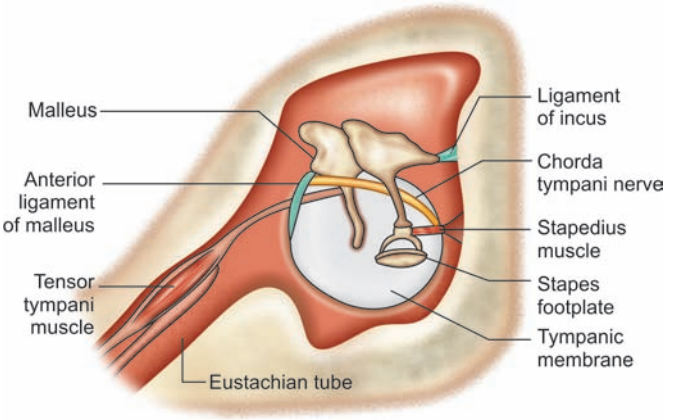


Fig. 19: Right tympanic membrane, ossicles and eustachian tube seen from medial side

b. Scutum: An upper part of epitympanum is formed by outer bony attic wall called scutum.

OSSICLES

The ossicles (Fig. 20) conduct sound energy from the tympanic membrane to the oval window. There are three middle ear ossicles—malleus, incus and stapes.

1. *Malleus (hammer):* It consists of a head, neck, handle (manubrium), a lateral and an anterior process. It is the largest ossicle and measures 8 mm in length.
   a. Head and neck: They lie in the attic.
   b. Manubrium: It is embedded in the fibrous layer of the tympanic membrane.
   c. Lateral process: It appears as a knob-like projection on the outer surface of the tympanic membrane and provides attachments to the anterior and posterior malleal folds.
2. *Incus (anvil):* It consists of following parts:
   a. Body and short process: They lie in the attic.
   b. Long process: It hangs vertically and forms incudostapedial joint with the head of stapes.
3. *Stapes (stirrup):* This smallest bone of body measures about 3.5 mm. It consists of head, neck, anterior and posterior

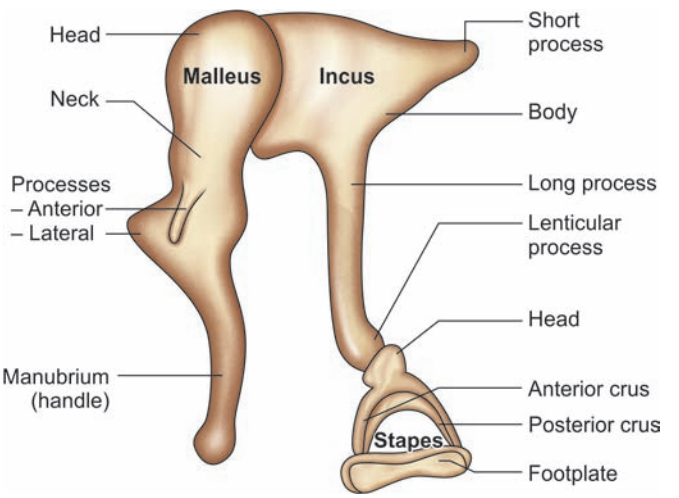


Fig. 20: Middle ear ossicles

crura and footplate. The footplate is positioned in the oval window by annular ligament.

INTRATYMPANIC MUSCLES

There are two middle ear muscles: tensor tympani and the stapedius.

1. *Tensor tympani*: It runs above the eustachian tube. Its tendon turns round the processus cochleariformis and passes laterally. It tenses the tympanic membrane.
   a. Origin: Bony tunnel above the osseous part of eustachian tube.
   b. Insertion: Just below the neck of malleus.
   c. Nerve supply: It develops from the 1st branchial arch and is supplied by a branch of mandibular division of trigeminal nerve (CN V_3).
2. *Stapedius*: On contraction it dampens the loud sounds and prevents noise trauma to the inner ear.
   a. Origin: Conical cavity and canal within pyramid.
   b. Insertion: It inserts to the neck stapes.
   c. Nerve supply: It is developed from the second branchial arch and is supplied by a branch of CN VII (nerve to stapedius of facial nerve).

- *Functions:* Acoustic reflex protects ear from loud sounds.
  a. Dampening of middle ear mechanics: Loud sounds (80 dB and above) cause contraction of stapedius that limits stapes movement.
  b. Gain control mechanism: Acoustic reflex keep cochlear input more constant and expand dynamic range.
  c. Reduction in self generated noise: Stapedius muscle contracts with chewing and vocalization.

INTRATYMPANIC NERVES (FIG. 21)

- *Tympanic plexus (Nerve supply of middle ear):* The tympanic nerve plexus, which lies on the promontory, supplies to the medial surface of the tympanic membrane, tympanic cavity, mastoid air cells and the bony eustachian tube. It is formed by following nerves:
  - Tympanic branch (Jacobson) of glossopharyngeal: It carries secretomotor fibers to the parotid gland. The pathway of secretomotor fibers to the parotid gland consists of inferior salivary nucleus > CN IX > Jacobson's tympanic branch > Tympanic plexus > Lesser petrosal nerve > Otic ganglion > Auriculotemporal nerve > Parotid gland.

Section of Jacobson's nerve is carried out in cases of Frey's syndrome.

  - Sympathetic fibers: Caroticotympanic nerves come from the sympathetic plexus, which is present round the internal carotid artery.
- *Chorda tympani nerve*: This branch of the facial nerve enters the middle ear through posterior canaliculus. It runs on the medial surface of the tympanic membrane. It lies between the malleus and long process of incus, above the insertion of tensor tympani. It carries gustatory fibers from the anterior two-third of tongue and parasympathetic secretomotor fibers to the submaxillary and sublingual salivary glands.

MIDDLE EAR MUCOSA

Middle ear mucosa wraps ossicles, muscles, ligaments and nerves like peritoneum wraps various viscera in the abdomen. These mucosal folds divide the middle ear into various compartments. So, all the middle ear structures lie outside the mucous membrane. Mucous membrane of the nasopharynx is continuous with that of the middle ear cleft.

Middle ear cavity is lined by ciliated columnar epithelium in its anterior and inferior part and mucosa changes to cuboidal type in the posterior part. Attic and mastoid air cells are lined by flat, nonciliated epithelium. Eustachian tube is lined by ciliated pseudostratified columnar epithelium with several mucous glands in the submucosa.

COMPARTMENTS AND FOLDS OF MIDDLE EAR (FIGS 22 AND 23)

Ossicles and their mucosal folds separate mesotympanum from epitympanum (attic).

1. *Compartments of Epitympanum*
   a. Prussak's space: Its boundaries, which limit spread of infection to other compartments, are following:
      i. Lateral: Membrana flaccida (Shrapnell's membrane)

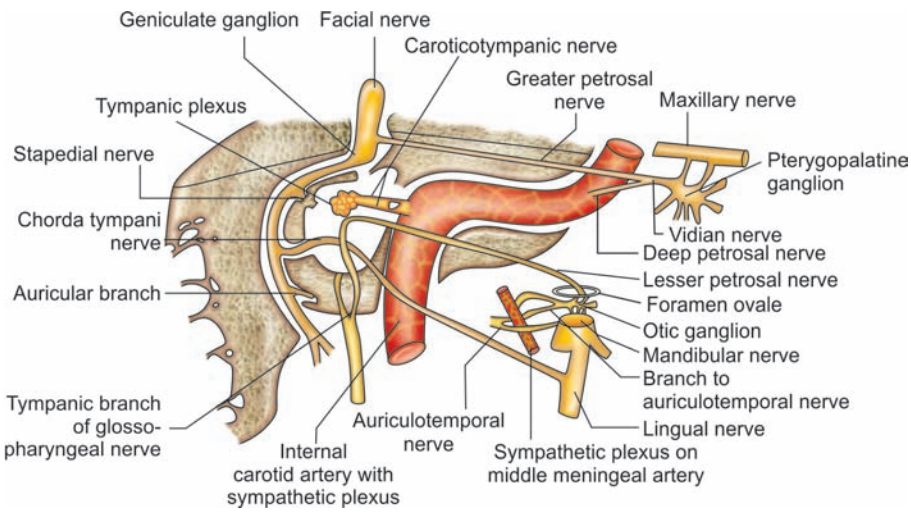


Fig. 21: Nerves in relation with the middle ear. Note secretomotor pathway of salivary, lacrimal and nasal glands

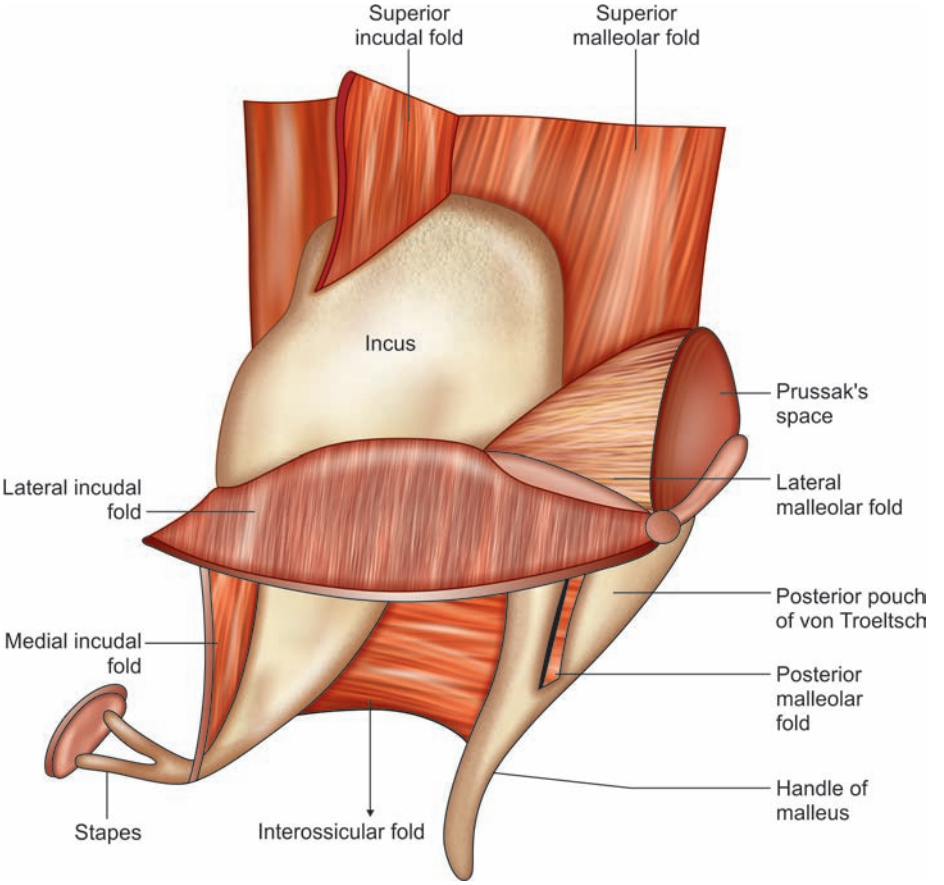


Fig. 22: Posterosuperior and lateral view of right tympanic cavity showing compartments and folds of middle ear (after Proctor)

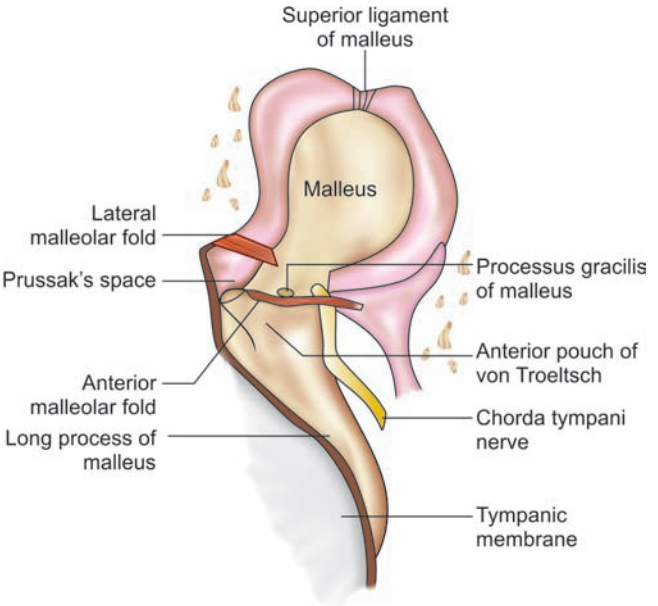


Fig. 23: Prussak's space and anterior pouch of von Troeltsch

ii. Medial: Neck of malleus
iii. Floor: Lateral process of malleus
iv. Roof: Fibers of lateral malleolar ligament arising from neck of malleus and inserting along the rim of notch of Rivinus

b. *Attic compartments*: Transversely placed superior malleolar fold divides attic into two compartments—smaller anterior and larger posterior. The space between the lateral malleolar fold and lateral incudal fold provides communication with Prussak's space.

i. Anterior attic compartment
ii. Posterior attic compartment: Superior incudal fold divides this space into following two divisions: medial and lateral spaces.

2. *Compartments of Mesotympanum*: In the upper part of mesotympanum there are following three compartments.

a. Inferior incudal space: Its boundaries are following
i. Superior: Lateral incudal fold
ii. Medial: Medial incudal fold
iii. Lateral: Posterior malleolar fold extending from neck of malleus to posterosuperior margin of tympanic sulcus.
iv. Anterior: Interossicular fold that lies between long process of incus and upper two-third of handle of malleus.

b. Anterior pouch of von Troeltsch: It lies between the following boundaries:
i. Medial: Anterior malleolar fold extending from neck of malleus to anterosuperior margin of tympanic sulcus
ii. Lateral: Portion of the tympanic membrane anterior to handle of malleus

c. Posterior pouch of von Troeltsch: It is situated between the following boundaries:
i. Medial: Posterior malleolar fold extending from neck of malleus to posterosuperior margin of tympanic sulcus.

ii. Lateral: Portion of the tympanic membrane posterior to handle of malleus.

MASTOID ANTRUM

This air-containing space (9 mm height, 14 mm width and 7 mm depth) is situated in the upper part of mastoid. Its boundaries are following:

- *Roof*: It is formed by the tegmen antri, which separates mastoid antrum from the middle cranial fossa.
- *Lateral wall:* It is formed by a 1.5 cm thick plate of squamous part of temporal bone which is marked on the lateral surface of mastoid by suprameatal (Macewen's) triangle (Fig. 24). It is covered by postaural skin.
  - Boundaries of Macewen's triangle
    - Linea temporalis (temporal line): A ridge of bone extending posteriorly from the zygomatic process (marking the lower margin of temporalis muscle and approximating the floor of middle cranial fossa)
    - EAC: Posterosuperior margin of EAC.
    - Tangent: A tangent to the posterior margin of EAC.
- *Medial wall*: It is formed by the petrous bone and related to the
  - Posterior semicircular canal
  - Endolymphatic sac
  - Dura of posterior cranial fossa
- *Anterior*: Anteriorly mastoid antrum communicates with the attic through the aditus ad antrum. Medial to lateral relations are following:
  - Facial nerve canal
  - Aditus ad antrum and facial recess lie between tympanum and mastoid antrum (see posterior wall of middle ear in the section of boundaries of middle ear)
  - Deep bony external auditory canal (EAC)
- *Posterior wall:* It is formed by mastoid bone and communicates with mastoid air cells.
  - Sigmoid sinus curves downwards.
- *Floor:* It is formed by mastoid bone and communicates with mastoid air cells. Other deeper relations from medial to lateral sides are
  - Jugular bulb medial to facial canal.
  - Digastric ridge which gives origin of posterior belly of digastric muscle.
  - Origin of sternocleidomastoid muscle.

TYPES OF MASTOID (FIG. 25)

The mastoid consists of "honeycomb" air cells, which lie underneath the bony cortex. Depending on its development, three types of mastoid are described: cellular, diploeic and acellular.

a. *Cellular (Well-pneumatized):* Mastoid cells are well developed with thin intervening septa.
b. *Diploeic*: Mainly there are marrow spaces with few air cells.
c. *Acellular (Sclerotic):* There are neither cells nor marrow spaces.

Mastoid Air Cells (Figs 26 to 28)

Mastoid antrum, which is present in all types of mastoids, is the most constant mastoid air cell. In sclerotic mastoid, antrum is usually small and sigmoid sinus may be anteriorly positioned. In cases of mastoiditis, abscesses may form in these air cells and result in various types of intra and extra cranial complications (See chapter Complications of Suppurative Otitis Media).

The mastoid air cells are traditionally divided into several groups, which include:

a. Zygomatic cells: In the root of zygoma.
b. Tegmen cells: In the tegmen tympani.
c. Perisinus cells: Present over the sinus plate.
d. Retrofacial cells: Present round the fallopian canal of facial nerve.
e. Perilabyrinthine cells: They are located above, below and behind the labyrinth. The cells, which are present in the arch of superior semicircular canal, may communicate with the petrous apex.
f. Peritubal: They are present around the eustachian tube. These and the hypotympanic cells communicate with the petrous apex.
g. Tip cells: These large cells lie in the tip of mastoid medial and lateral to the digastric ridge.
h. Marginal cells: These cells, which lie behind the sinus plate, may extend into the occipital bone.
i. Squamous cells: They lie in the squamous part of temporal bone.

KORNER'S SEPTUM

Mastoid develops from the squamous and petrous parts of temporal bone. In some cases petrosquamosal suture persists as

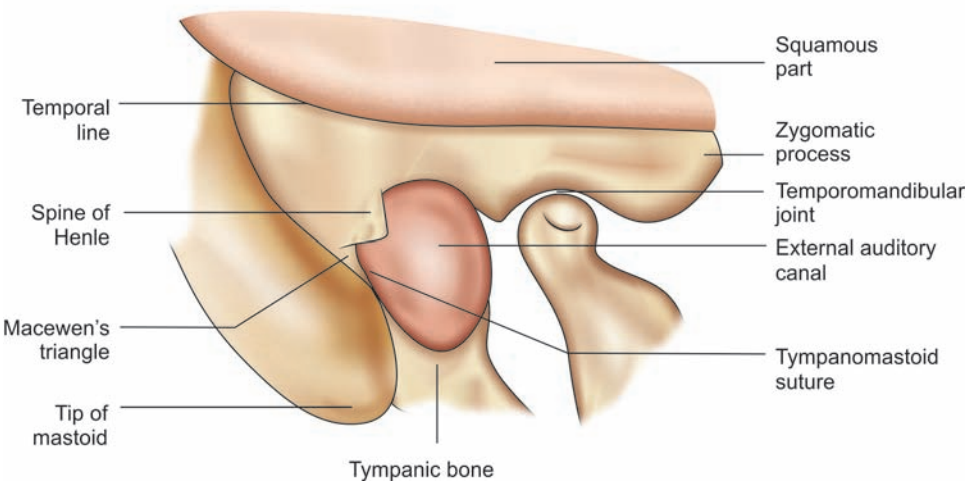


Fig. 24: Macewen's triangle. Surface landmark for mastoid antrum

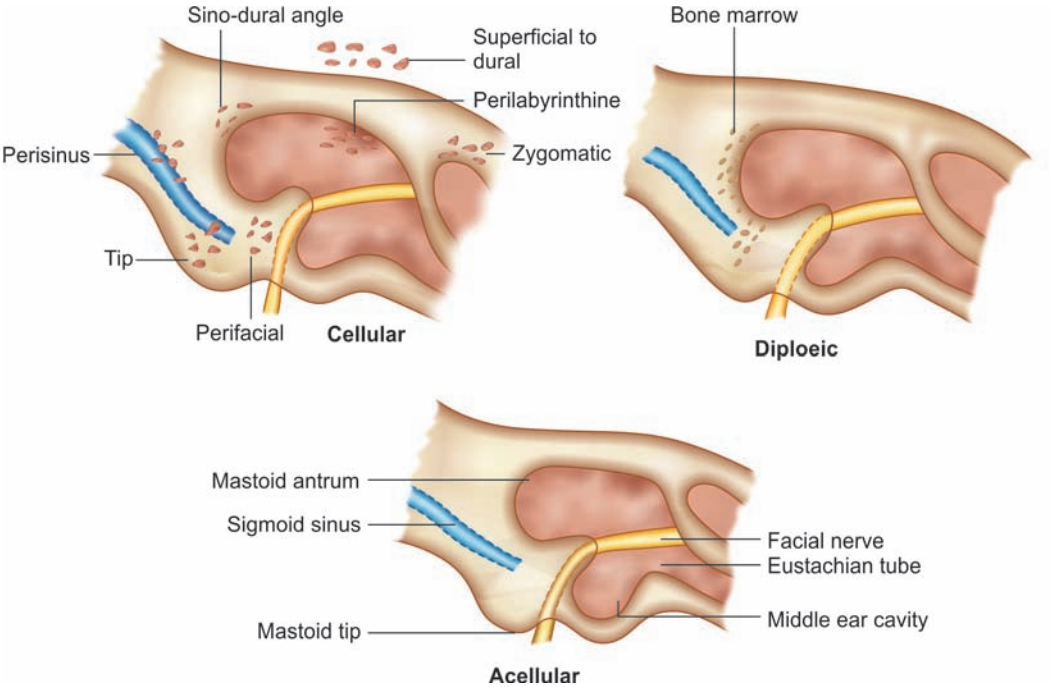


Fig. 25: Three types of mastoid: Cellular, diploeic and acellular

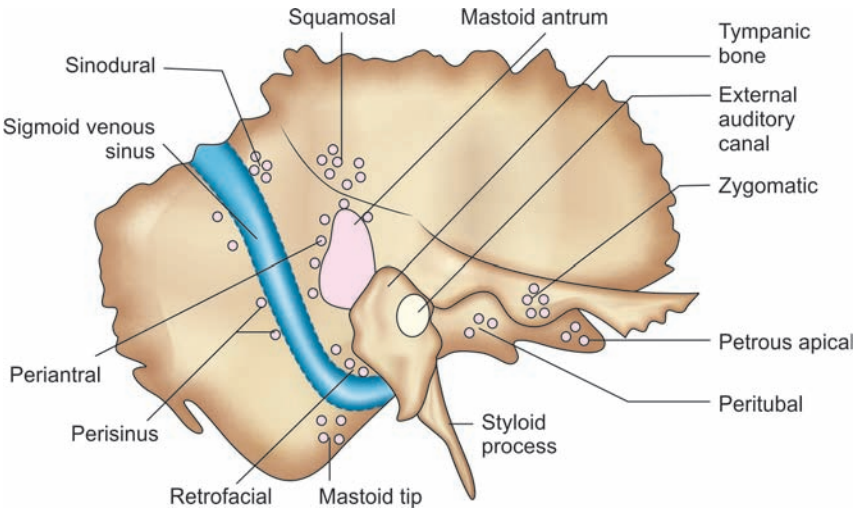


Fig. 26: Air cells of temporal bone

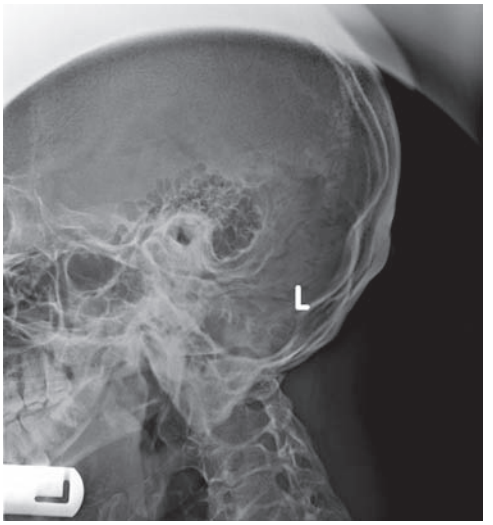


Fig. 27: X-ray mastoid left showing normal pneumatization
Source: Dr Jayesh Patel, Consultant Radiologist, Anand, Gujarat

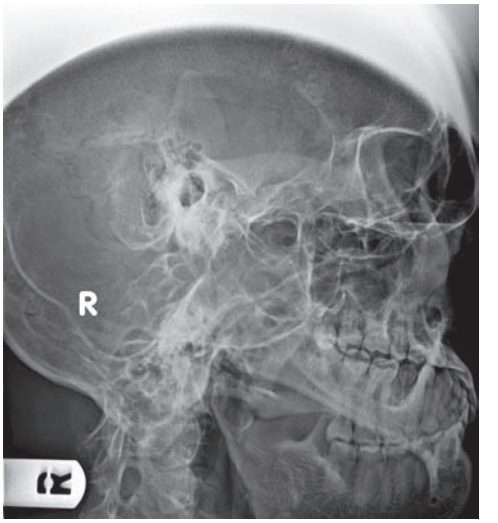


Fig. 28: X-ray mastoid right showing partial loss of pneumatization
Source: Dr Jayesh Patel, Consultant Radiologist, Anand, Gujarat

a bony plate called Korner's septum, which separates superficial squamosal cells from the deep petrosal cells. During the mastoid surgery, Korner's septum causes difficulty in locating the antrum and the deeper cells.

If not recognized, Korner's septum leads to incomplete removal of disease during mastoidectomy. Mastoid antrum can be entered into only after the removal of Korner's septum.

BLOOD SUPPLY

Arterial Supply

Following branches of external and internal carotid arteries supply blood to middle ear:

1. *External Carotid Artery*
   a. Maxillary artery
      i. *Anterior tympanic artery: Major contributor*
      ii. Middle meningeal artery
         - Petrosal branch
         - Superior tympanic artery: It traverses along the canal for tensor tympanic muscle.
      iii. Artery of pterygoid canal: Branch that runs along eustachian tube.
   b. Posterior auricular artery
      i. *Stylomastoid artery: Major contributor*
   c. Ascending pharyngeal artery
      i. Tympanic branch
2. *Internal Carotid Artery*: petrous part
   a. Caroticotympanic branches.

Venous Drainage

Veins from the middle ear cleft drain into pterygoid venous plexus, superior petrosal sinus and sigmoid sinus.

LYMPHATIC DRAINAGE OF EAR

The lymphatics of middle ear drain into retropharyngeal and parotid nodes. Eustachian tube lymphatics drain into retropharyngeal group of lymph nodes (Table 1). Internal ear does not have any lymphatics.

ANATOMY OF INTERNAL EAR

The internal ear (labyrinth), which has organs of both hearing and balance, is divided into bony and membranous labyrinth. The membranous labyrinth is filled with endolymph. Perilymph is filled in the space present between membranous and bony labyrinths.

BONY LABYRINTH

Bony labyrinth (Fig. 29) consists of three parts: vestibule, semicircular canals and cochlea. The lateral wall of labyrinth is medial

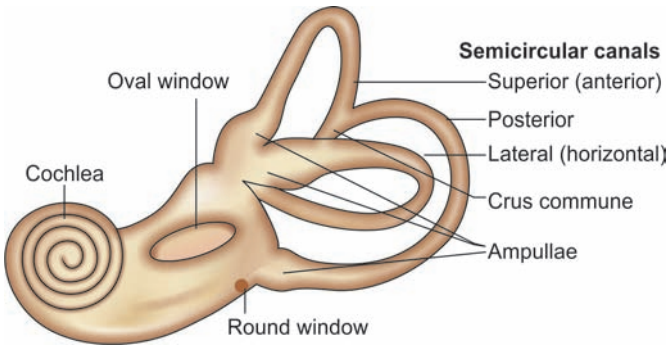


Fig. 29: Bony labyrinth of left side. External features seen from lateral side

wall of middle ear. The medial wall of labyrinth is the lateral limit of internal auditory canal (IAC).

a. Vestibule: This central chamber of the labyrinth (5 mm) has following structures:

1. *Lateral wall*: It has oval window.
   i. Oval window (Fenestra vestibuli): It lies in the lateral wall and closed by footplate of stapes surrounded by annular ligament.
2. *Medial wall (Fig. 30)*: It shows following structures:
   i. Spherical recess: It is situated anteriorly and lodges the saccule. Perforations of maculae cribrosa media provides passage for fibers of inferior vestibular nerve.
   ii. Elliptical recess: It is situated posteriorly and lodges the utricle. The perforations of maculae cribrosa superior (Mike's dot) provide passage to nerve fibers that supply to utricle and ampulla of superior and lateral semicircular canals (SCC).
   iii. Vestibular crest and cochlear recess: The spherical and elliptical recesses are separated from each other by vestibular crest. Inferiorly vestibular crest splits to enclose cochlear recess for cochlear nerve fibers.
   iv. Opening of aqueduct of vestibule: It is present below the elliptical recess. Through this passes the endolymphatic duct.

Large vestibular aqueduct syndrome: An enlarged vestibular aqueduct is associated with sensorineural hearing loss, Pendred's syndrome and anatomic defects of cochlear modiolus.

3. *Posterosuperior region:*
   i. Five openings of semicircular canals: They are present in the posterosuperior part of vestibule.
4. *Anterior*: Cochlea opens into the anterior region of vestibule.

TABLE 1 Lymphatic drainage of ear

Nodes	*Region*
Preauricular and parotid nodes	Auricle: Concha, tragus, fossa triangularis Cartilaginous external auditory canal
Infra-auricular nodes	Auricle: Lobule and antitragus
Postauricular, deep cervical and spinal accessory nodes	Auricle: Helix and antihelix
Retropharyngeal nodes draining into upper deep cervical nodes	Middle ear and eustachian tube

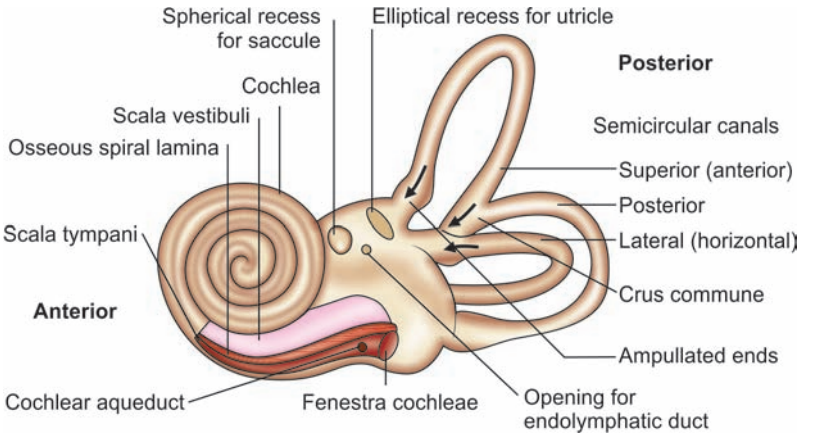


Fig. 30: Medial wall of left bony labyrinth seen from lateral side after the removal of its lateral wall

b. *Semicircular Canals (Fig. 31)*: There are three SCCs: lateral (horizontal), posterior and superior (anterior). Each canal occupies 2/3rd of a circle and has a diameter of 0.8 mm. They lie in planes at right angles to one another. Each canal has two ends: ampullated and nonampullated. All the three ampullated ends and nonampullated end of lateral SCC open independently and directly into the vestibule.

1. *Superior SCC*: It is 15–20 mm long and situated transverse to the axis of petrous part of temporal bone. Its antero-lateral end is ampullated and opens in the superolateral part of vestibule.
2. *Lateral SCC*: It is 12–15 mm long and projects as a rounded bulge into the middle ear, aditus and antrum. It makes an angle of 30° with the horizontal plane. Its anterior end is ampullated and opens into the upper part of vestibule. The posterior nonampullated end opens into the lower part of vestibule below the orifice of crus commune.
3. *Posterior SCC*: It is 18–22 mm long and situated parallel and close to the posterior surface of petrous part of temporal bone. Its lower end is ampullated and opens into the lower part of vestibule. Its upper limb joins the crus commune.
   - *Crus commune*: The nonampullated ends of posterior and superior canals join and form a crus commune (4 mm length), which then opens into the medial part of vestibule. So, the three SCCs open into the vestibule by five openings.

c. *Cochlea (Figs 32 and 33)*: The bony cochlea, which is a coiled tube, looks like snail. Cochlear canal makes 2.5–2.75 turns round a central pyramid of bone called modiolus. The cochlear tube is 30 mm long. It is 5 mm from base to apex and 9 mm around its base.

1. *Modiolus*: The base of modiolus, which is directed towards internal acoustic meatus, transmits vessels and nerves to the cochlea. The apex lies medial to tensor tympani muscle.
2. *Osseous spiral lamina*: A thin plate of bone called osseous spiral lamina, winds spirally around the modiolus like the thread of a screw. This bony lamina gives attachment to the basilar membrane and divides the bony cochlear tube into three compartments: scala vestibuli, scala tympani and scala media (membranous cochlea).
3. *Rosenthal's canal:* The spiral ganglions are situated in Rosenthal's canal, which runs along the osseous spiral lamina.
4. *Scala vestibuli*: This upper most channel is continuous with vestibule and closed at oval window by the stapes foot plate.
5. *Scala tympani*: This lowermost channel is closed by secondary tympanic membrane of round window (RW).

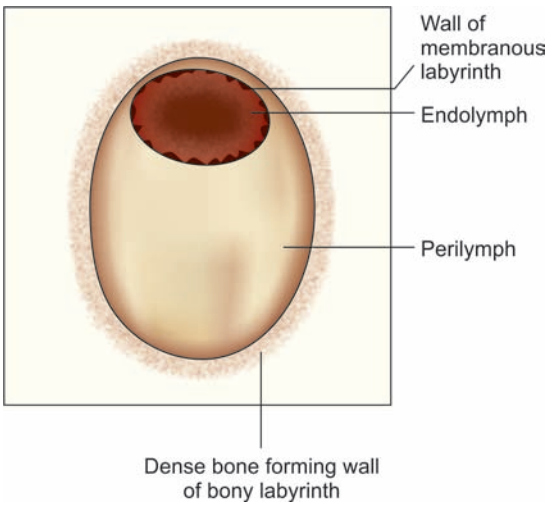


Fig. 31: Cut section of semicircular canal

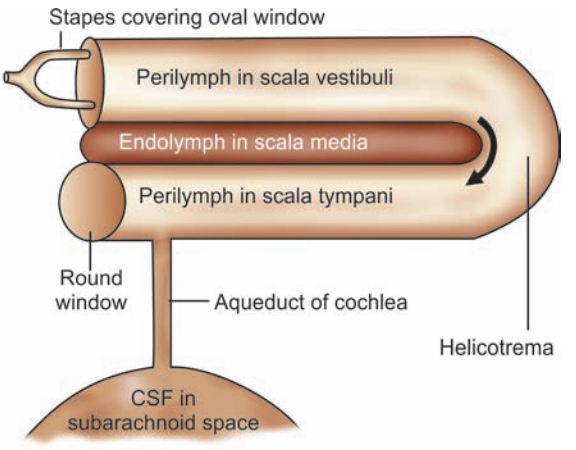


Fig. 32: Cochlea: Peri- and endolymphatic systems relations with cerebrospinal fluid (CSF)

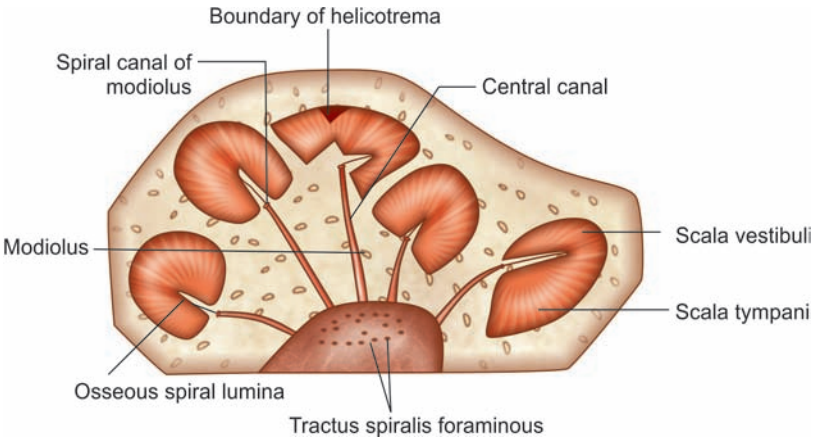


Fig. 33: Cut section of bony cochlea

6. Promontory: The promontory, a bony bulge in the medial wall of middle ear, represents the basal coil of cochlea.
7. Helicotrema: The scala vestibuli and scala tympani, which communicate with each other at the apex of cochlea through an opening called helicotrema, are filled with perilymph.
8. *Round window (fenestra cochlea):* On the lateral wall of internal ear (medial wall of middle ear), scala vestibuli is closed by the stapes footplate, while the scala tympani is closed by secondary TM of RW.
9. *Aqueduct of cochlea*: The scala tympani is connected with the subarachnoid space through the aqueduct of cochlea. It is thought to regulate perilymph and pressure in bony labyrinth.

MEMBRANOUS LABYRINTH

Membranous labyrinth (Fig. 34) consists of cochlear duct, utricle, saccule, three semicircular ducts and endolymphatic duct and sac.

- *Cochlear Duct (Membranous Cochlea or Scala Media) (Fig. 35)* This blind coiled tube, which appears triangular on cross-section, is connected to the saccule through ductus reunions. It is bounded by the following three walls:
  a. Basilar membrane: It supports the organ of Corti. Its length increases as it proceeds from the basal coil to the apical coil. So, the higher frequencies of sound are heard at the basal coil while lower tones at the apical coil. The inner thin area is called zona arcuata while outer thick area is called zona pectinata.

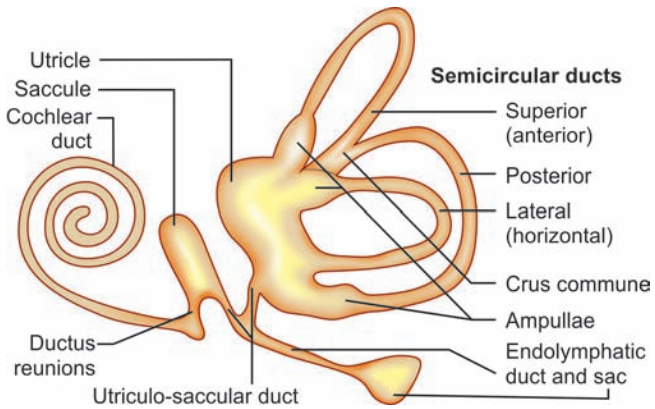


Fig. 34: Membranous labyrinth of left side: External features

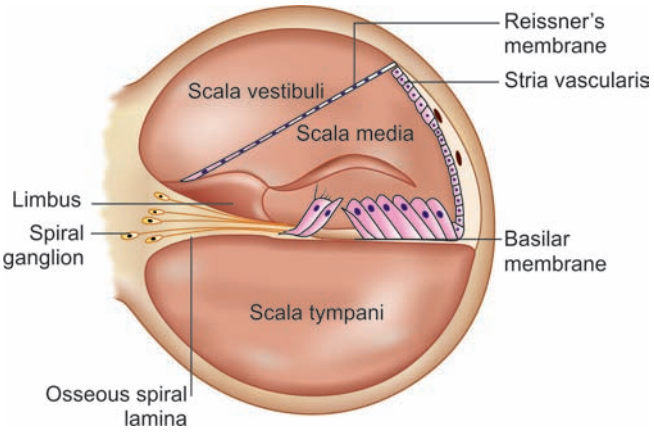


Fig. 35: Structure of cochlear canal after its cut section

   b. Reissner's membrane: It separates scala media from the scala vestibuli.
   c. Stria vascularis: It contains vascular epithelium and secrets endolymph.
- *Utricle:* The utricle, which is oblong and irregular, has anteriorly upward slope at an approximate angle of 30°. It lies in the posterior part of bony vestibule and receives the five openings of the three semicircular ducts. The utricle (4.33 mm^2) is bigger than saccule (2.4 mm^2) and lies superior to saccule. The utricle is connected with the saccule through utriculosaccular duct. Its sensory epithelium, which is called macula, is concerned with linear acceleration and deceleration.
- *Saccule:* The saccule lies anterior to the utricle opposite the stapes footplate in the bony vestibule. Its sensory epithelium, macula responds to linear acceleration and deceleration. The saccule is connected to the cochlea through the thin reunion duct.

The distended saccule in Meniere's disease can be surgically decompressed by perforating the footplate because it lies against the stapes footplate.

- *Semicircular Ducts*: The three semicircular ducts, which open in the utricle, correspond exactly to the three bony canals. The ampullated end contains a thickened ridge of neuroepithelium, which is called crista ampullaris.

- *Endolymphatic Duct and Sac*: The ducts from utricle and saccule unite and form utriculosaccular duct, which continues as endolymphatic duct that passes through the vestibular aqueduct. The terminal part of the endolymphatic duct is dilated and forms endolymphatic sac that is situated between the two layers of dura on the posterior surface of the petrous bone. Endolymphatic sac consists of both an intraosseous and an extraosseous portion. The endolymphatic duct and sac are thought to be involved in the reabsorption and regulation of endolymph.

- Endolymphatic sac is thought to regulate pressure of membranous labyrinth.
- Endolymphatic sac is decompressed, drained or shunt in Meniere's disease.

INNER EAR FLUIDS

Perilymph fills the space between bony and membranous labyrinth while endolymph fills the entire membranous labyrinth (Table 2).

Perilymph

It resembles extracellular fluid and is rich in sodium ions. The aqueduct of cochlea provides communication between scala tympani and subarachnoid space. Perilymph percolates through the arachnoid type connective tissue present in the aqueduct of cochlea.

- *Source*: There are two theories:
  1. Filtrate of blood serum from the capillaries of spiral ligament.
  2. CSF reaching labyrinth via aqueduct of cochlea.

Endolymph

It resembles intracellular fluid and is rich in potassium ions. Protein and glucose contents are less than in perilymph.

- *Source*: They are believed to be following:
  1. Stria vascularis.
  2. Dark cells of utricle and ampullated ends of semicircular ducts.
- *Absorption*: There are following two opinions regarding the absorption of endolymph:
  1. Endolymphatic sac: *The longitudinal flow theory* believes that from cochlear duct endolymph reaches saccule, utricle and endolymphatic duct and is then absorbed by endolymphatic sac.
  2. Stria vascularis: *The radial flow theory* believes that endolymph is secreted as well as absorbed by the stria vascularis.

TABLE 2 Differences between the composition of endolymph, perilymph and CSF**

	Endolymph	*Perilymph*	*CSF*
Na^+ (mEq/L)	3	150	152
K^+ (mEq/L)	150	3–5	4
Cl^- (mEq/L)	130	125	
Protein (mg/dL)	126	200–400	20–50
Glucose (mg/dL)	10–40	85	70

**Values vary from the site of collection such as cochlea, saccule and endolymphatic sac in cases of endolymph and scala tympani and vestibuli in cases of perilymph.

ORGAN OF CORTI

This (Fig. 36) sensory organ of the hearing, is situated on the basilar membrane. It is spread like a ribbon along the entire length of basilar membrane. It consists of following important components:

1. *Tunnel of Corti*: This tunnel, which is situated between the inner and outer rods, contains a fluid called cortilymph. The functions of the rods and cortilymph are yet not clear.
2. *Hair Cells*: These important receptor cells of hearing transduce sound energy into electrical energy. There are two types of hair cells—inner and outer. At low magnification stereocilia (evaginations of membrane on the apical surface) appears as hairs. The stereocilia have mechanically activated ion channels which are opened by the sound stimuli. With the advancement of age there is generalized reduction in the number of hair cells. Differences between inner and outer hair cells are given in Table 3.
   a. Inner hair cells: Inner hair cells (IHCs) form a single row and are richly supplied by afferent cochlear fibers. These flask-shaped cells are very important in the transmission of auditory impulses. Their nerve fibers are mainly afferent.
   b. Outer hair cells: Outer hair cells (OHCs) are arranged in three or four rows and mainly receive efferent innervation from the olivary complex. These cylindrical cells modulate the function of inner hair cells. Their nerve fibers are mainly efferent.
   Nerve supply: Ninety five percent of afferent fibers of spiral ganglion of cochlear nerve supply the IHCs. The OHCs get only 5% of the cochlear nerve fibers. Efferent fibers, which are mainly for the OHCs, come from the superior olivary complex through the olivocochlear bundle. Hair cells are innervated by dendrites of bipolar cells of spiral ganglion. Each cochlea sends auditory information to both sides of brain.
   c. Supporting Cells: Deiter's cells, which are situated between the outer hair cells, provide support to OHC. Cells of Hensen are situated outside the Dieter's cells.
   d. Tectorial Membrane: The tectorial membrane, which overlies the organ of Corti, consists of gelatinous matrix and delicate fibers. The shearing force between the hair cells and tectorial membrane stimulate the hair cells.

VESTIBULAR RECEPTORS

Peripheral vestibular receptors are of two types: cristae and maculae.

1. *Cristae (Fig. 37)*: They lie in the ampullated ends of the three semicircular ducts and respond to angular acceleration. On a crest-like mound of connective tissue lie the sensory epithelial hair cells, which are covered by cupula. In the crista of lateral SCC, the polarization is towards the utricle whereas in the cristae of superior and posterior canals, polarization is away from the utricle.
   a. Cupula: The cilia of epithelial hair cells project into cupula

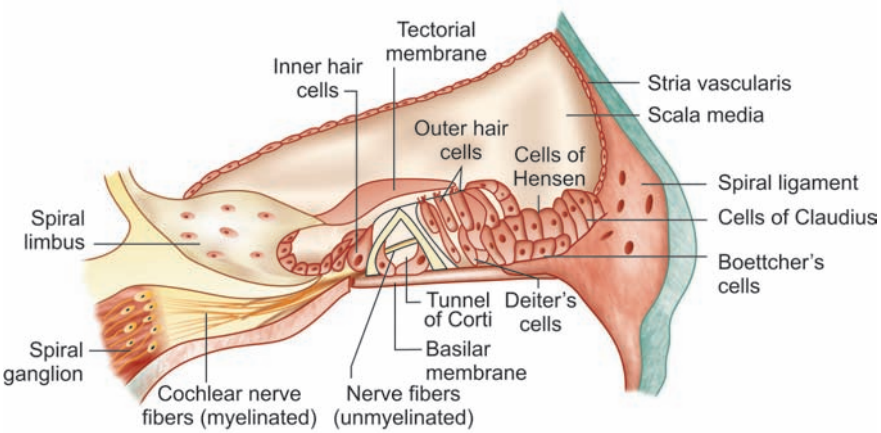


Fig. 36: Structure of organ of Corti

TABLE 3 Difference between inner hair cells (IHCs) and outer hair cells (OHCs)

	Inner hair cells	*Outer hair cells*
Cells numbers	3500	12000
Rows	One	Three or four
Shape	Flask	Cylindrical
Nerve supply	Mainly afferent fibers	Mainly efferent fibers
Development	Early	Late
Function	Transmit auditory stimuli	Modulate function of inner hair cells
Ototoxicity	More resistant	More sensitive and easily damaged
High intensity noise	More resistant	More sensitive and easily damaged
Generation of otoacoustic emissions	No	Yes

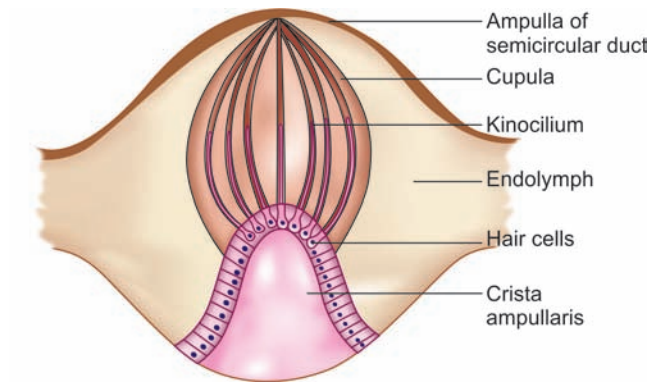


Fig. 37: Crista, hair cells and cupula. Cut section of ampulla of semicircular duct

that consists of a gelatinous mass (complex carbohydrates or glycoproteins and proteoglycans arranged in filamentous network), which extends from the surface of crista to the ceiling of the ampulla. The cupula, which is thought to be secreted by the supporting cells, forms a water tight partition. With the movements of endolymph, cupula can be displaced to any one side like a swing door. The gelatinous mass of cupula, which consists of polysaccharide, contains canals into which project the cilia of sensory hair cells. The altered cupula mechanics may result in clinical manifestations of peripheral vestibular disorders such as vascular, viral or bacterial and vestibular neuronitis. The mechanism governing caloric nystagmus under earth gravity and zero gravity in space is not clear. It seems that a direct thermal effect on the SCC afferents play only a small role.

b. *Sensory epithelial hair cells (Fig. 38)*: The sensory hair cells are of two types: type 1 and type 2. From the upper surface of each cell projects a kinocilium and multiple stereocilia. The kinocilium, which is thicker than stereocilia, is located on the edge of the cell. Sensory cells are surrounded by supporting cells which have microvilli on their upper surface. Hair cells of both types may have contact with the same nerve calyce.

   i. Type 1 cells: These cells are found only in birds and mammals. They are flask-shaped and correspond to the IHC of organ of Corti. Each cell has a single large cup-like nerve terminal that surrounds the base.

   ii. Type 2 cells: They are cylindrical and have multiple nerve terminals at the base. They resemble OHC of organ of Corti.

2. *Maculae (Fig. 39)*: They lie in otolith organs (utricle and saccule). Macula of the utricle is situated in its floor in a horizontal plane in the dilated superior portion of the utricle. Macula of saccule is situated in its medial wall in a vertical plane. The macula utriculi (approximately 33,000 hair cells) are larger than saccular macula (approximately 18,000 hair cells). The striola, which is a narrow curved line in center, divides the macula into two areas. They appreciate position of head in

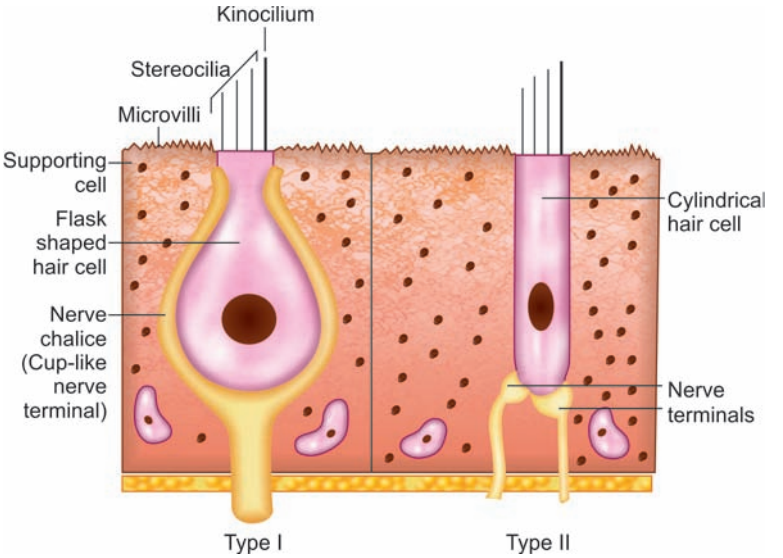


Fig. 38: Hair cells of vestibular organs

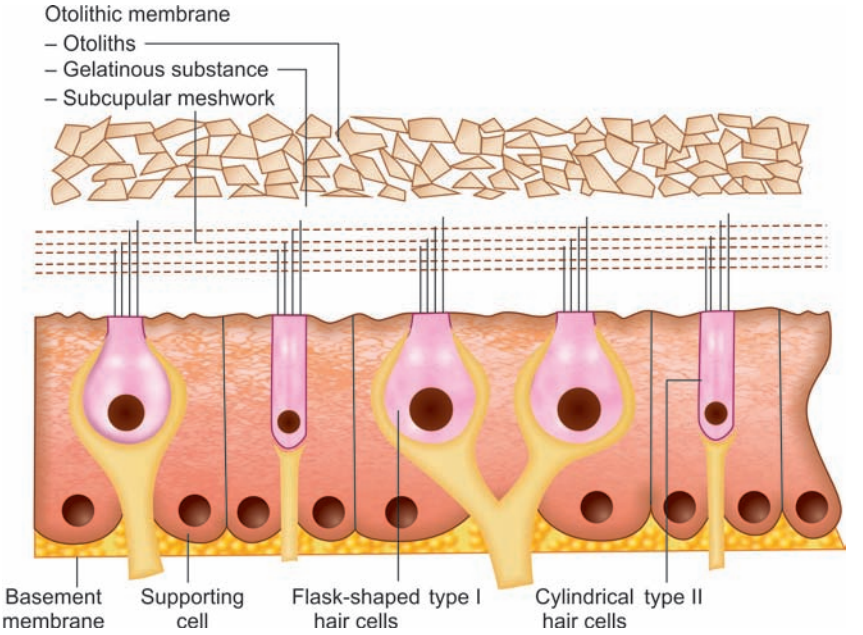


Fig. 39: Macula of otolith organs: utricle and saccule

response to gravity and linear acceleration. A macula consists mainly of two parts: a sensory neuroepithelium and an otolith membrane.

Similar to presbyacusis, degenerative changes occur in macular hair cells and otoconia with ageing.

a. *Sensory neuroepithelium:* It is made up of type1 and type 2 cells, which are similar to the hair cells of the crista. Type I cells are in higher concentration in the area of striola and change orientation (mirror-shaped) along the line of striola with opposite polarity. The kinocilia face striola in the utricular macula, whereas in saccule, they face away from the striola. The polarity and curvilinear shape of striola offer CNS wide range of neural information of angles in all the three dimensions for optimal perception and compensatory correction. During tilt, translational head movements and positioning, visual stimuli combined with receptors of neck muscles, joint and ligaments play an important part.

b. *Otolithic membrane:* The otoconial membrane consists of a gelatinous mass, a subgelatinous space and the crystals of calcium carbonate called otoliths (otoconia or statoconia). The otoconia, which are multitude of small cylindrical and hexagonally shaped bodies with pointed ends, consists of an organic protein matrix together with crystallized calcium carbonate. The otoconia (3–19 μm long) lie on the top of the gelatinous mass. The cilia of hair cells project into the gelatinous layer. The linear, gravitational and head tilt movements result into the displacement of otolithic membrane, which stimulate the hair cells lying in different planes.

BLOOD SUPPLY OF LABYRINTH

- *Internal Auditory (Labyrinthine) Artery*: Labyrinth is supplied by internal auditory artery which is a branch of anterior inferior cerebellar artery that arises from basilar artery. The labyrinthine artery may directly arise from the basilar artery.
  - *Branches*: Internal auditory artery divides into two following branches:
    - *i. Anterior vestibular artery:* It supplies to utricle and lateral and superior SCC.
    - *ii. Common cochlear artery*: It further divides into two following branches:
      1. Main cochlear artery: It supplies to cochlea (80%)
      2. Vestibulocochlear artery: It again divides into two following branches:
         a. Posterior vestibular artery: It supplies to saccule and posterior SCC.
         b. Cochlear branch: It supplies to cochlea (20%).

The cochlea does not have any collateral arterial circulation.

- *Venous Drainage*: It is through internal auditory vein, vein of cochlear aqueduct and vein of vestibular aqueduct. These veins drain into the inferior petrosal and sigmoid sinuses.

INTERNAL AUDITORY CANAL

Internal auditory canal (IAC) is about 1 cm long and passes into petrous part of temporal bone in a lateral direction. It is lined by dura. At its lateral end (fundus) IAC is closed by a vertical cribriform plate of bone that separates it from labyrinth (Fig. 40). A transverse crest divides this plate into smaller upper and larger lower parts. Upper part is further divided into anterior and posterior quadrant by a vertical crest called Bill's bar.

Contents

1. Vestibulocochlear nerve.
2. Facial nerve including nervous intermedius.
3. Internal auditory artery and vein.

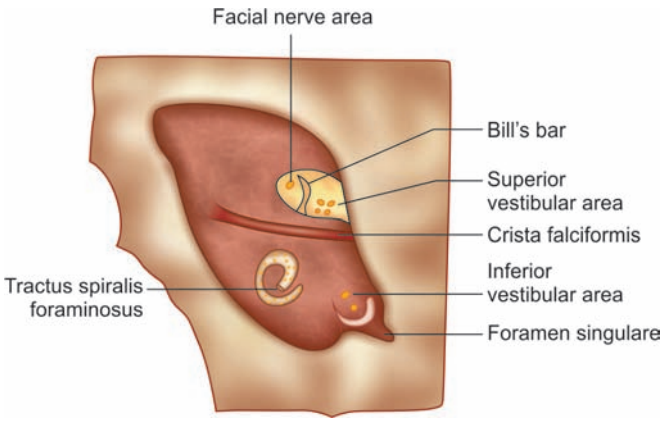


Fig. 40: Fundus of right internal auditory canal (IAC) as seen from IAC

For further details of IAC and cerebellopontine (CP) angle see chapter Tumors of Ear and Cerebellopontine Angle.

Vestibulocochlear (Auditory) Nerve

In the internal auditory canal the vestibular and cochlear nerves merge and form vestibulocochlear nerve (CN VIII). A small branch of anterior inferior cerebellar artery (AICA), which can be used as a landmark during vestibular schwannoma surgery, runs between the CN VII and CN VIII on the brainstem.

DEVELOPMENT OF EAR

The embryologic source and the time of development of external and middle ears are independent of the inner ear development. Therefore malformed and non-functional inner ear can have normal external and middle ears and vice versa (Table 4).

Auricle

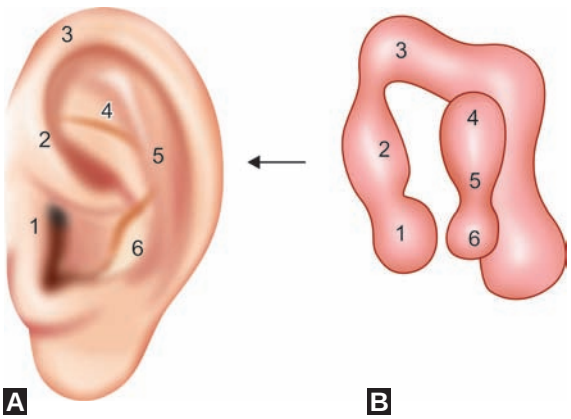
In the sixth week of embryonic life, six tubercles (Hillocks of His) appear around the first branchial cleft (Figs 41A and B). They progressively grow and coalesce and form the auricle. Tragus develops from the tubercle, which arise from the first branchial arch. The remaining pinna develops from the rest of the five tubercles of second arch. By the 20th week, pinna attains adult shape. Initially, pinna is located low on the side of the neck but later on it moves to a more lateral and cranial position.

Preauricular sinus or cyst: It is commonly seen between the tragus and crus of helix. It is the result of the faulty fusion between the first and the second arch tubercles.

TABLE 4 Development of ear (Timings shown in the week of gestation)**

Development	*Beginning*	*Completion*
Vestibule	3	20
Cochlea	3	20
Middle ear	3	30
Auricle	6	20
EAC	6	28

**Gulya AJ. Developmental anatomy of ear. Glasscock and Shambaugh. Surgery of Ear. WB Saunders (1990)



Figs 41A and B: Development of pinna (A) from six hillocks of His (B) around the first branchial cleft (1 from first and 2–6 from second branchial arch)

External Auditory Canal

External auditory canal (EAC) develops from the first branchial cleft (Fig. 42). External ear canal gets fully formed by the 28th week. In the 16th embryonic week, cells proliferate from the bottom of ectodermal cleft and form a meatal plug.

Atresia of canal: The recanalization of meatal plug, which begins from the deeper part near the tympanic membrane and progresses outwards, forms the epithelial lining of the bony meatus. This is the reason why deeper meatus is sometimes developed while there is atresia of canal in the outer part.

Tympanic Membrane

It develops from all the three germinal layers.

1. *Ectoderm*: Outer epithelial layer is formed by the ectoderm.
2. *Mesoderm:* The middle fibrous layer develops from the mesoderm.
3. *Endoderm*: Inner mucosal layer is formed by the endoderm.

Middle Ear (Fig. 42)

- *Endoderm of Tubotympanic Recess*: The eustachian tube, tympanic cavity, attic, antrum and mastoid air cells are derived from the endoderm of tubotympanic recess which arises from the first and partly from the second pharyngeal pouches.
- *First Branchial Arch*: Malleus and incus develops from mesoderm of the first arch.
- *Second Branchial Arch***:** The stapes superstructures develop from the second arch.
- *Otic Capsule*: The stapes footplate and annular ligament are derived from the otic capsule. See chapter for the details of Branchial Apparatus.

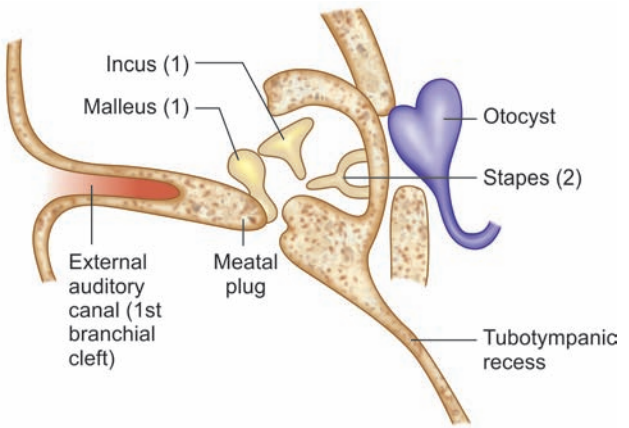


Fig. 42: Development of external and middle ears. 1 (Meckel's cartilage) and 2 (Reichert's cartilage) branchial arches

Inner Ear

Development of the inner ear, which begins in third week of fetal life, is complete by the 16th week (Fig. 43).

- *Auditory Placode*: The auditory placode, which is thickened ectoderm of hind brain, gets invaginated and forms auditory vesicle (otocyst).
- *Auditory Vesicle*: The auditory vesicle differentiates into endolymphatic duct and sac, utricle, semicircular ducts, saccule and cochlea.
  - Development of pars superior (semicircular canals and utricle) takes place earlier than pars inferior (saccule and cochlea). The pars superior is phylogenetically older part of labyrinth.

The cochlea develops by 20 weeks of gestation and the fetus can hear in the womb of the mother. The great Indian epic of Mahabharata, which was written thousands of years ago, mentions that Abhimanyu son of great warrior Arjun while in his mother's womb heard conversation (regarding the art of battle ground) of his mother and father.

CENTRAL CONNECTIONS (NEURAL PATHWAYS)

AUDITORY NEURAL PATHWAYS

The auditory pathway (Fig. 44) from peripheral to center consists of **e**ighth nerve, **c**ochlear nuclei, **o**livary complex (superior), **l**ateral lemniscus, **i**nferior colliculus, **m**edial geniculate body and **a**uditory cortex (*ECOLIMA* mnemonic) (Table 5).

- *Hair Cells*: Ninety five percent of afferent fibers of spiral ganglion of cochlear nerve supply the IHCs. The OHCs get only 5% of the cochlear nerve fibers. Efferent fibers, which are mainly for the OHCs, come from the superior olivary complex through the olivocochlear bundle. Hair cells are innervated by dendrites of bipolar cells of spiral ganglion. Each cochlea sends auditory information to both sides of brain.

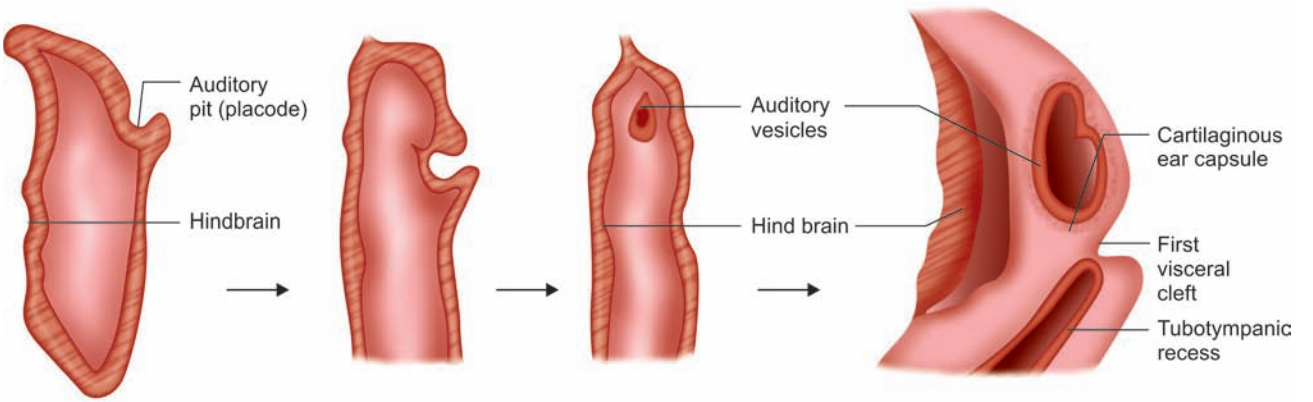


Fig. 43: Development of labyrinth

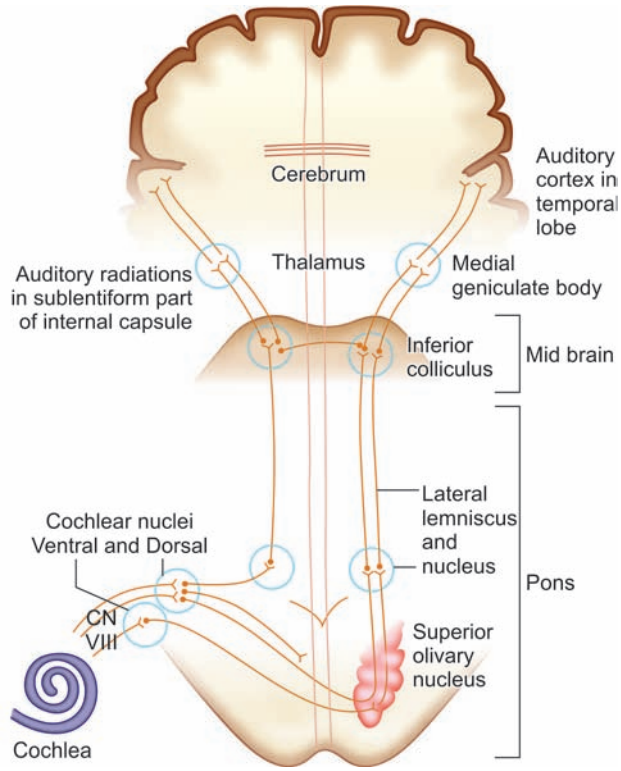


Fig. 44: Central auditory pathways

TABLE 5 Ascending auditory pathways, from below upwards

First order neurons	*Bipolar neurons of spiral ganglion in cochlear nerve*
Second order neurons	Dorsal and ventral cochlear nuclei
Third order neurons	Superior olivary complex in pons. From here fibers travel in lateral lemniscus in pons
Fourth order neurons	Inferior colliculus in midbrain
Fifth order neurons	Medial geniculate body in thalamus. From here fibers go to auditory cortex in temporal lobe of the cerebrum through the auditory radiations in sublentiform part of internal capsule

- *Cochlear Nerve:* Axons of these bipolar cells form the cochlear nerve, which ends in the both the dorsal and ventral ipsilateral cochlear nuclei.
- *Brainstem:*
  - *Cochlear nuclei:* The cochlear nuclei send neural information to both sides of the brain.
  - *Superior olivary nucleus, lateral lemniscus and inferior colliculus*: From the cochlear nuclei, some of the axons go directly to inferior colliculus (both ipsilateral and contralateral) while other goes via superior olivary nucleus and lateral lemniscus (both ipsilateral and contralateral). So, the auditory fibers travel via both ipsilateral as well as contralateral routes and have multiple decussation points.
- *Thalamus*: From the inferior colliculus axons go to the medial geniculate body of metathalamus via inferior brachium.
- *Cerebrum*: From the medial geniculate body axons go to the primary auditory cortex of temporal lobe of the cerebrum via the sublentiform part of internal capsule. Each side of ear is represented in both the cerebral hemispheres. The area of hearing is situated in the superior temporal gyrus (Brodmann's area 41).

CENTRAL VESTIBULAR CONNECTIONS (FIG. 45)

Vestibular Nerve

The Scarpa's ganglion, which lies in the lateral part of the internal acoustic meatus, contains bipolar cells. The peripheral processes of these bipolar cells innervate the sensory epithelium of the labyrinth. The central processes aggregate and form the vestibular nerve. A significant feature of vestibular neurons is their high frequency of resting discharge with an average of 90/sec. The majority of vestibular nerve fibers terminate in vestibular nuclei but some go directly to the cerebellum.

Numbers of both vestibular hair cells and nerve cells in Scarpa's ganglion are found to be reduced in the older people (18,000 in young adults, decreasing to around 12,000 at the age of 80 years).

- *Branches:* The vestibular nerve has two branches superior and inferior.
  - *Superior vestibular nerve:* It supplies the cristae of superior and lateral SCC, macula of utricle and the anterosuperior portion of the macula of the saccule.
  - *Inferior vestibular nerve*: It innervates the crista of posterior SCC and main portion of the macula of the saccule.

Vestibular Nuclei

They are four in number: superior, inferior (descending), medial and lateral. They receive afferents not only through vestibular nerve but also from cerebellum, reticular systems, spinal cord and contralateral vestibular nuclei (Table 6).

Functions of Efferents from Vestibular Nuclei

The information received from the labyrinths, eyes and proprioceptive systems is integrated in CNS. The efferents from vestibular nuclei perform following functions:

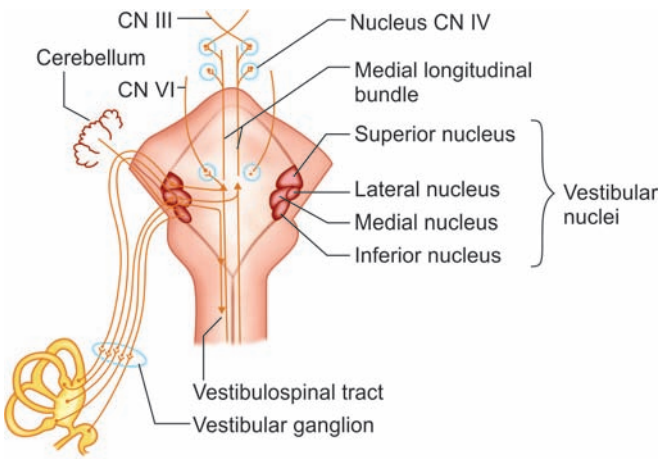


Fig. 45: Vestibular pathways

TABLE 6 Afferent and efferent connections of vestibular nuclei

Afferents	*Efferents*
1. Peripheral vestibular receptors (semicircular canals, utricle and saccule)	1. Nuclei of CN III, IV, VI via medial longitudinal bundle
2. Cerebellum	2. Cerebellum (vestibulocerebellar fibers)
3. Spinal cord	3. Vestibulospinal tract
4. Vestibular nuclei of the opposite side	4. Vestibular nuclei of the opposite side
5. Reticular formation	5. Autonomic nervous system
	6. Cerebral cortex (temporal lobe) through thalamus

1. *Vestibuloocular Reflexes:* The medial longitudinal bundle is the pathway for vestibuloocular reflexes and explains the genesis of nystagmus. It helps in stabilizing the gaze so that image is fixed on the fovea of retina during the head movement.
2. *Equilibrium*:
   a. *Vestibulospinal tract*: It coordinates the movements of head, neck and body in the maintenance of balance.
   b. *Vestibulocerebellar tract*: It coordinates input information to maintain the body balance.
3. *Autonomic Symptoms*: Autonomic nervous system explains nausea, vomiting, palpitation, sweating and pallor seen in vestibular disorders such as Meniere's disease.
4. *Motion Awareness:* The temporal lobe is responsible for subjective awareness of motion.

PHYSIOLOGY OF HEARING

The pinna collects sound signal* from the environment, which passes through EAC and vibrates the tympanic membrane (Fig. 46). Vibrations of the tympanic membrane are transmitted to the stapes footplate through the chain of ossicles.** Vibrations of stapes footplate result in the pressure changes in the labyrinthine fluids*** that make movement of the basilar membrane and thus stimulate the hair cells of the organ of Corti. The IHC of cochlea act as transducers and convert the mechanical energy into electrical impulses which travel along the auditory nerve.

Components of Hearing Physiology

The physiology of hearing is broadly divided into three divisions:

1. Conduction of mechanical sound energy (external and middle ear conductive apparatus).
2. Transduction of mechanical sound energy into electrical impulses (cochlear sensory system).
3. Conduction of electrical impulses to brain (CN VIII, brainstem, thalamus and temporal lobe neural pathways).

CONDUCTION OF SOUND

Pinna

Pinna serves following functions because of its shape and location. It increases sound pressure by 6 dB (2 times)

1. *Collection*: Gather sound arriving from an arc of 135°
2. *Localization:* Determine the origin of sound
3. *Concentration:* Horn-shaped concha acts like a megaphone and concentrate the sound at the entrance of EAC.

External Auditory Canal

Along with pinna it can increase sound pressure at the tympanic membrane by 15–22 dB at 4000 Hz.

Impedance Matching Mechanism (Transformer Action) of Middle Ear (Figs 47 and 48)

When the air-conducted sound travels to the cochlear fluids most of the sound energy is reflected away.**** Middle ear compensates for this loss of sound energy.

Middle ear converts sound of greater amplitude, but lesser force, to that of lesser amplitude and greater force. This function of the middle ear is called impedance matching mechanism or the transformer action. The following are the different functions of various structures of the conducting mechanism of the hearing:

- *Hydraulic Action of Tympanic Membrane*: The area of tympanic membrane is much larger than the stapes footplate. Therefore, tympanic membrane provides large hydraulic ratio between the tympanic membrane and stapes footplate. The effective vibratory area of tympanic membrane is about two-third. The effective areal ratio between tympanic membrane and stapes footplate is about 17:1. This mechanical advantage is provided by the tympanic membrane.
- *Curved Membrane Effect:* Movements of tympanic membrane are more at the periphery than the center, which provide some leverage.
- *Lever Action of the Ossicles*: Ossicular chain conducts sound from tympanic membrane to the oval window. Lever action of the ossicles (Handle of malleus is 1.3 times longer than long process of the incus) provides a mechanical advantage of 1.3.

The product of areal ratio (hydraulic action of tympanic membrane) and lever action of ossicles is 22.1 (17 x 1.3). It offers a 25 dB increase in sound energy arriving to cochlea.

Phase Differential Between Oval and Round Window (Figs 46 and 48)

Both oval and round windows provide free movement of cochlear fluids in scala vestibuli and scala tympani respectively. Sound waves do not reach the oval and round windows simultaneously. The preferential pathway to oval window receives sound vibrations first and round window acts as a relief window. When the

* Waves of compression and rarefaction that is capable of producing sound. Speed of sound in the air at 20° C at sea level is 344 m (1120 ft).
** Sound travels faster in liquids and solids than in the air.
***When sound energy passes from air to liquid medium, most of its energy is reflected back because of the impedance offered by the liquid.
**** A person under water cannot hear any sound made in the air because 99.9% of the sound energy is reflected away from the surface of water because of the impedance offered by water.

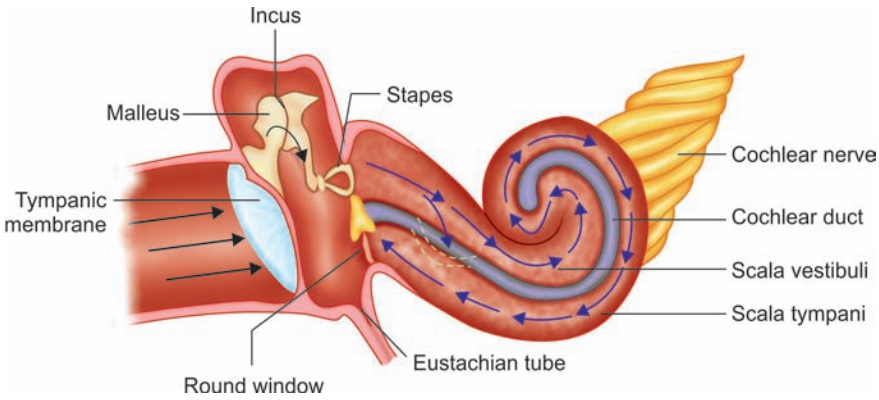


Fig. 46: Physiology of hearing. Arrows show sound waves

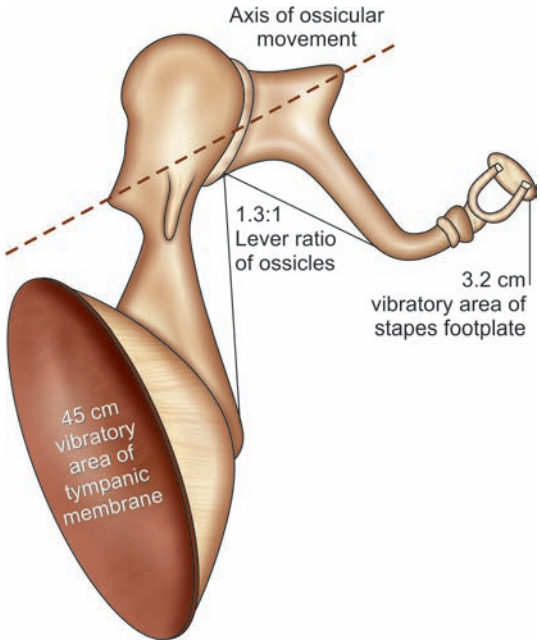


Fig. 47: Transformer function of middle ear. Hydraulic effect of tympanic membrane (17:1) and lever action of ossicles (1.3:1)

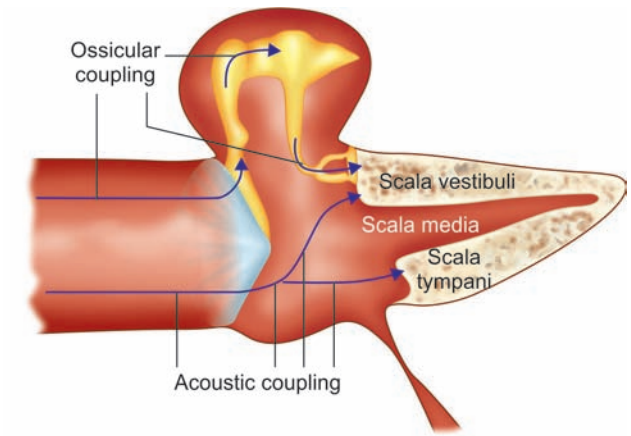


Fig. 48: Ossicular and acoustic coupling pathways. Acoustic coupling is caused by the middle ear pressure that results from EAC sound pressure and motion of eardrum. Ossicular coupling (coupled motion of tympanic membrane and ossicles including stapes footplate) is 60 dB more than acoustic coupling. Oval and round windows are spatially separated

oval window is receiving wave of compression, the round window is at the phase of rarefaction.

If only one window is functioning as in otosclerosis there will be no movement of cochlear fluids.

- *Acoustic Separation of Two Windows*: The sound should not reach both oval and round windows simultaneously. An intact tympanic membrane with the help of intact ossicular chain provides preferential pathway to oval window. The presence of air in the middle ear delays the pathway to round window. If the sound waves strike both the windows simultaneously, they would cancel each other's effect and there will not be any movement of the perilymph. This acoustic separation of two windows is provided by the tympanic membrane and a cushion of air in the middle ear around the round window.
- *Aeration:* Patent eustachian tube provides aeration to the middle ear.

Natural Resonance of External and Middle Ear

Natural resonances of the external and middle ear allow certain frequencies of sound to pass more easily to the inner ear. The greatest sensitivity of the sound transmission is between 500 and 3000 Hz (speech frequencies).

Following are the natural resonances:

- External auditory canal (EAC): 3000–4000 Hz
- Tympanic membrane: 800–1600 Hz.
- Ossicular chain: 500–2000 Hz
- Middle ear: 800 Hz

The noise induced hearing loss usually occurs between 3 and 6 kHz with a peak at 4 kHz because that is the resonant frequency of EAC.

TRANSDUCTION OF MECHANICAL ENERGY TO ELECTRICAL IMPULSES

Round Window Reflex

The round window membrane moves in response to the movement of footplate of stapes. When stapes is pressed pressure is exerted to scala vestibuli perilymph which is transferred to scala

media and than to scala tympani. The pressure is ultimately transferred to round window which bulges into middle ear.

Organ of Corti (Fig. 49)

Pressure in scala media causes downward movement of basilar membrane. Along with the basilar membrane, organ of Corti move up and down with sound stimulus. This causes a shearing action between tectorial membrane and the reticular lamina and results in bending of stereocilia.

Transduction

Transduction is the conversion of mechanical energy to electrical energy. Movements of the stapes footplate are transmitted to the cochlear fluids, which move the basilar and tectorial membranes differentially and sets up shearing force that bends the stereocilia. Movement of stereocilia opens and closes ion channels and produces receptor potential in the IHCs. This cochlear microphonics triggers the nerve impulse by releasing neurotransmitters onto afferent nerve fibers (Fig. 50).

1. *Traveling Wave Theory of von Bekesy*: In response to sinusoidal sound, the movement of the basilar membrane appears as traveling wave, which moves from the base to the apex. Depending on the frequency, a particular segment of the basilar membrane achieves maximum amplitude. Each wave is weak at the onset but becomes stronger as it reaches its natural resonant frequency.
2. *Tonotopic Gradient in Cochlea*: Tonotopic map of basilar membrane determines the site of largest peak of the wave. Higher frequencies are represented in the basal turn and the progressively lower tones towards the apex of the cochlea (Fig. 51). High frequency waves travel a short distance and die. Low frequency waves travel a long distance and die.

Functions of Hair Cells

1. *Inner Hair Cells:* They are believed to be the classic auditory receptor cells which signal the brain about the presence of specific sound.
2. *Outer Hair Cells*: They have been shown to shorten and lengthen when stimulated by sound. A protein called prestin provides OHCs their ability to contract. They are thought to have following functions:

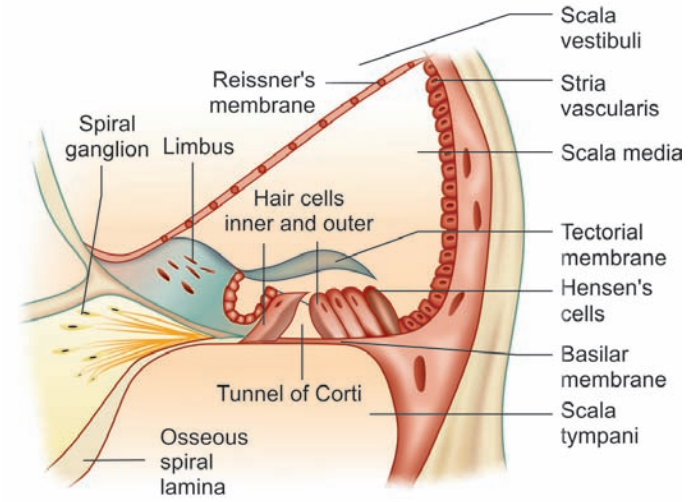


Fig. 49: Organ of Corti

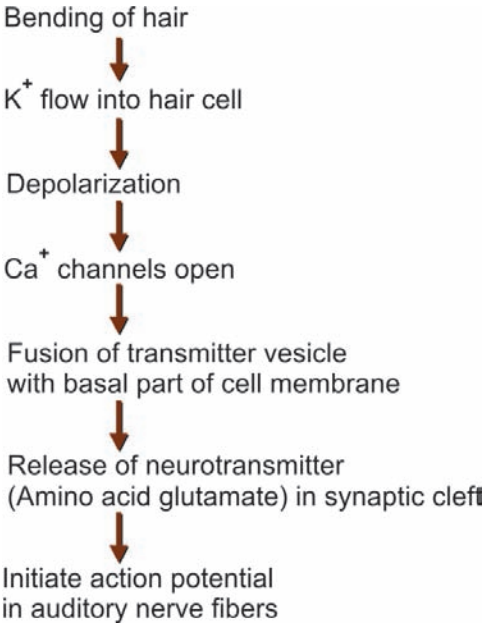


Fig. 50: Flow chart showing major steps of transduction

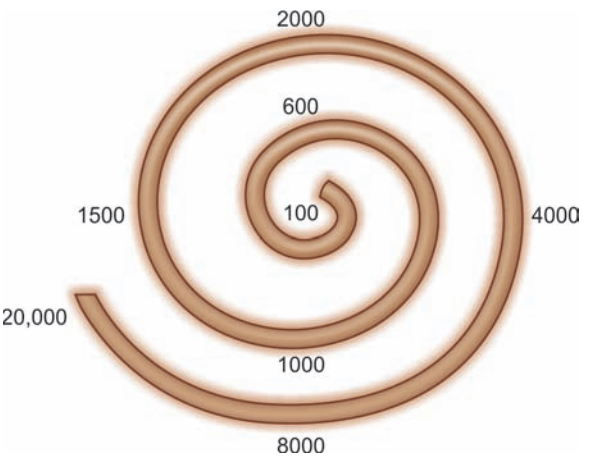


Fig. 51: Tonotopic gradient in cochlea. Higher frequencies are represented in the basal turn and the progressively lower tones towards the apex of the cochlea

a. *Amplification:* OHCs amplify effect of sound stimuli to their adjacent IHCs.
b. *Sharpening:* OHCs sharpen the frequency response of adjacent IHCs.
c. *Inhibitory:* Efferent stimulation of OHCs may be responsible for decreasing the responsiveness of cochlea.
d. *Cochlear microphonics*: OHCs are responsible for cochlear microphonic effect of electrocochleography.
e. *Otoacoustic emissions*: OHCs produce otoacoustic emissions that can be recorded and used to screen newborns for hearing loss. See Chapters Hearing Evaluation and Hearing Impairment in Infants and Young Children.

Cochlear hair cells in birds regenerate after noise-induced or ototoxic loss but its significance in humans is yet to be elucidated.

Electrical Potentials

Endocochlear potential, cochlear microphonics (CM) and summating potential (SP) are from cochlea while the compound

action potential (AP) is from the cochlear nerve fibers. Both CM and SP are receptor potentials similar to other sensory end-organs.

- *Endocochlear Potential*: This resting potential of +80 mV direct current (DC) is recorded from scala media. This energy source for cochlear transduction is generated from stria vascularis by Na^+/K^+ -ATPase pump. Endolymph has high K^+ concentration. It acts as a battery and helps in driving the current through the hair cells when they move after exposure to any sound stimulus.
- *Cochlear Microphonics:* Cochlear microphonics (CM) is an alternating current (AC) potential. Basilar membrane moves in response to sound stimulus. Changes occur in electrical resistance at the tips of OHC. Flow of K^+ through the outer hair cells produces voltage fluctuations and called CM.

Cochlear microphonics is absent in the part of cochlea where the outer hair cells are damaged.

- *Summating Potential*: Summating potential (SP) is a DC potential, which may be either negative or positive. It is produced by hair cells. It follows the "envelop" of stimulating sound and is superimposed on cochlear nerve action potential. This is a rectified derivative of sound signal. Probably it arises from IHCs with a small contribution from OHCs.

Summating potential of cochlea helps in the diagnosis of Ménière's diseases.

- *Compound (Auditory Nerve) Action Potential:* It is the neural discharge of auditory nerve. It follows all or none phenomena so has all or none response to auditory nerve fibers. Each nerve fiber has optimum stimulus frequency for which the threshold is lowest. Amplitude increases while latency decreases with intensity over 40–50 dB range. The following features differentiate it from CM and SP:
  a. No gradation
  b. Latency
  c. Propagation
  d. Post-response refractory period

MEDIAL GENICULATE BODY AND TEMPORAL LOBE AUDITORY CORTEX

They are organized into isofrequency layers arranged tonotopically from low frequency to high frequency. Most cells respond to binaural stimulation. Their main function appears to be sound localization. Neurons can either summate excitatory responses from both ears or excitatory response from one ear and inhibitory response from other.

PHYSIOLOGY OF VESTIBULAR SYSTEM

Vestibular system is traditionally divided into two parts: peripheral and central.

- *Peripheral Vestibular System*: It consists of semicircular ducts (dynamic labyrinth), utricle and saccule (static labyrinth) and vestibular nerve. Each vestibular receptor is precisely oriented to detect head movement in a specific direction or plane (Table 7). All receptors are tonically active.
- *Central Vestibular System:* It includes vestibular nuclei and tracts that integrate vestibular impulses with other systems to maintain body balance.

TABLE 7 Vestibular receptors and direction or plane of head movement

Vestibular Receptors	*Direction or Plane of Head Movement*
Horizontal semicircular canal (SCC)	Horizontal head turning (Angular acceleration)
Superior and posterior semicircular canal	Pitching the front to back and side to side
Otolith organs (Utricle and saccule)	Linear head movement (vertical and horizontal), tilting and gravity

SEMICIRCULAR CANALS FUNCTIONS

Semicircular canals respond to angular acceleration and deceleration. The three canals, which lie in three different planes, are situated at right angles to each other (Fig. 52). Any change in position of head can be detected. The one that lies at right angle to the axis of rotation is most stimulated. For example: The horizontal canal responds maximum to rotation on the vertical axis.

Nystagmus

The stimulation of SCCs produces nystagmus. The direction of nystagmus depends on the plane of the canal being stimulated. The nystagmus is horizontal from horizontal (lateral) canal; rotatory from the superior (anterior) canal; and vertical from the posterior canal.

- *Flow of Endolymph*: The flow of endolymph displaces the cupula and stimulates the epithelial hair cells (Fig. 53) of crista in the ampulla of the SSC. The flow of endolymph towards the ampulla or utricle is called ampullopetal or utriculopetal. The flow of endolymph away from ampulla or utricle is called ampullofugal or utriculofugal. The quick component of horizontal nystagmus is always opposite to the direction of flow of endolymph in the horizontal SCC. In lateral SCC ampullopetal displacement of stereocilia increases (stimulatory) the firing rate whereas ampullofugal displacement decreases (inhibitory) the firing rate. The opposite happens in posterior and superior canals.
- *Rotating Chair Test*: In the rotating chair test, when patient is rotating to the right and then abruptly stopped, the endolymph continues to move to the right due to inertia. Here endolymph movement in the lateral SCC will be

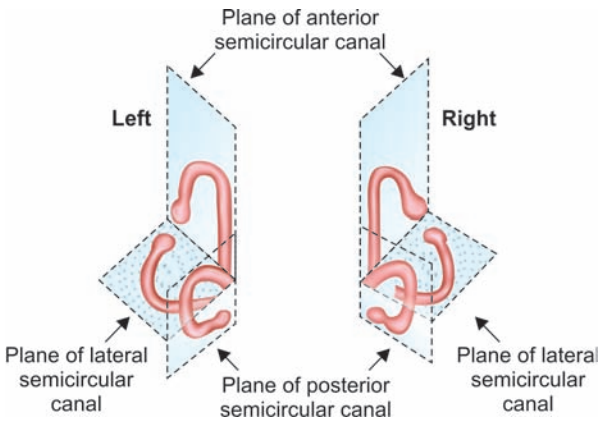


Fig. 52: Planes of three semicircular canals as seen from posterior side

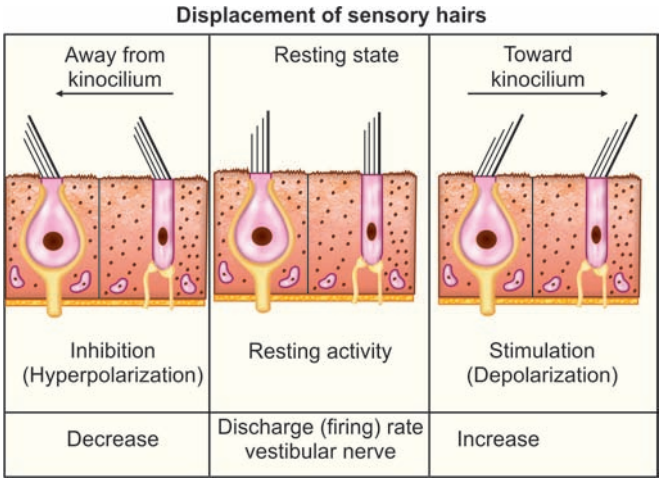


Fig. 53: Vestibular hair cells. Displacement of stereocilia toward kinocilium leads to depolarization (stimulation) and increases the vestibular nerve discharge rate

ampullopetal for left canal and the horizontal nystagmus will be directed to the left.

UTRICLE AND SACCULE FUNCTIONS

They respond to the linear acceleration and deceleration or gravitational pull during the head tilts. The sensory hair cells of the macula lie in different planes. Macula of the utricle is situated in its floor in a horizontal plane in the dilated superior portion of the utricle. Macula of saccule is situated in its medial wall in a vertical plane. During the head tilts hair cells are stimulated by displacement of otolithic membrane. The functions of saccule and utricle are similar but the saccule is also seen to respond to sound vibrations.

- Saccular macula responds to the tilting of head. If the head is tilted to left side, left saccular macula is stimulated while right saccular macula will remain static.
- Utricular macula responds to forward and backward movement of head.

Striola

It is a narrow curved line in center that divides the macula into two areas. Type I cells are in higher concentration in the area of striola and change orientation (mirror-shaped) along the line of striola with opposite polarity. The kinocilia face striola in the utricular macula, whereas in saccule, they face away from the striola. The polarity and curvilinear shape of striola offer CNS wide range of neural information of angles in all the three dimensions for optimal perception and compensatory correction. The bending of utricular hair cells away from striola causes depolarization (stimulation) whereas bending of saccular hair cells towards striola causes hyperpolarization (inhibition). During tilt, translational head movements and positioning, visual stimuli combined with receptors of neck muscles, joint and ligaments play an important part.

MAINTENANCE OF BODY EQUILIBRIUM

Sensory Component

The vestibular system records changes in the head position, linear or angular acceleration and deceleration and gravitational effects. The central nervous system receives information not only from the vestibular system but also from other sensory systems, which include visual, auditory and somatosensory (muscles, joints, tendons, skin). All this information is integrated and utilized in the regulation of equilibrium and body posture. Cerebellum, which is connected to vestibular receptors further helps in coordinating muscular movements, which vary in their rate, range, force and duration.

Motor Component

The standing and walking need not only sensory integration (from vestibular, somatosensory and visual systems) but also motor commands, which are fine-tuned through the frontal cerebral lobes, cerebellum and basal ganglia. Disorder of any of these systems can lead to dizziness.

Push and Pull System

The balance system, which includes vestibular, visual and somatosensory organs, can be compared with a two-sided push and pull system. In a neutral position, push and pull of one side is equal to that of the other side. If one side is pulling more than the other, the body balance is disturbed. During the turning or tilting, a temporary change in the push and pull force of one system is taken care of by the appropriate reflexes and motor outputs to the eyes (vestibuloocular reflex), neck (vestibulocervical reflex) and trunk and limbs (vestibulospinal reflex), which maintains new position of head and body. If any component of push and pull system of one side is diseased than it results in vertigo and ataxia.

- *Example*: Turning the head to the right direction produces an increase in the resting spontaneous outflow of action potential in the nerve coming from right horizontal SCC. Simultaneously there occurs decrease activity in left vestibular nerve. The CNS compares the input coming from each vestibule. There is no sense of movement when input is equal. The CNS interprets asymmetric input not only as a head rotation but also generates compensatory eye movements and postural adjustment.

Pathophysiology: Imbalance due to SCC-mediated vestibuloocular and vestibulospinal pathways results in abnormal sense of rotation, while imbalance due to otolith-mediated vestibuloocular and vestibulospinal pathways may manifest as vertical diplopia, abnormal sense of upright, sense of tilting and a tendency to lean or fall to affected side.
Compensation : With the help of other sensory, visual and kinesthetic inputs, the CNS is capable of rebalancing itself, that is, compensation occurs after an injury to peripheral vestibular system.

- *Unilateral vestibular loss*: Slowly progressive lesions of vestibular system do not cause spontaneous vertigo and vertigo of acute unilateral vestibular loss improves over the ensuing days.
- *Bilateral vestibular loss*: Bilateral vestibular loss (usually due to ototoxicity) results in inefficient vestibuloocular and vestibulospinal systems. It produces a permanent imbalance (especially in the dark) and oscillopsia (fixed objects appear to jump with any head movement). But as long as there is symmetry in vestibular input, even in cases of bilateral loss, there is no vertigo.

Diseases and Evaluation of Vestibular System

For the evaluation of nystagmus, dizzy patient and diseases of vestibular system see following chapters in this book:

1. Evaluation of Vertigo
2. Peripheral Vestibular Disorders
3. Central Vestibular Disorders

Clinical Highlights

1. **Pinna** has to be pulled upwards, backwards and laterally to see the tympanic membrane in adults. It attains 90–95% of adult size by 5–6 years of life.
2. **Auricular cartilage**: It gives shape to pinna and is absent in the lobule. Dehiscences may be seen in outer cartilaginous ear canal.
3. **Foramen of Huschke**: It lies in the vicinity of the "fissure of Santorini".
4. **External auditory canal**: It is smaller than Eustachian tube and is 24 mm in length. Outer one-third (8 mm) is cartilaginous and inner two-third (16 mm) is bony. Bony EAC does not contain ceruminous glands or hair follicles. The pH is acidic in normal healthy ear canals.
5. **Tympanic membrane**: It develops from all the three germinal layers: ectoderm (outer epithelial layer), mesoderm (middle fibrous layer) and endoderm (inner mucosal layer).
   a. Red tympanic membrane may be normal in a crying child.
   b. Retracted tympanic membrane shows prominent lateral process of malleus and foreshortened handle of malleus.
   c. Bulging tympanic membrane loses all landmarks.
6. **Dimensions of middle ear**: Vertical and anteroposterior dimensions are 15 mm each. Transverse dimension is 2 mm in mesotympanum (narrowest between the umbo and promontory), 6 mm in epitympanum and 4 mm in hypotympanum.
7. **Tegmen tympani**: Plate of bone separating attic of middle ear from middle cranial fossa.
8. **Ossicles**: Malleus and incus are derived from Meckel's cartilage of the first branchial (pharyngeal) arch. Stapes superstructure and styloid process develop from Reichert's cartilage of second branchial arch. Stapes footplate and annular ligament are derived from the otic capsule.
9. **Stapes**: It is smallest bone of body and weighs about 2.5 mg**.** Stapes footplate covers oval window. Area of stapes footplate is 3.2 sq mm.
10. **Stapedius muscle**: It is supplied by CN VII. Its paralysis causes hyperacusis (phonophobia).
11. **Prussak's space**: It is bounded by pars flaccida (laterally), neck of malleus (medially), lateral process of malleus (inferiorly) and lateral malleal ligament (superiorly). Posteriorly, it opens into epitympanum.
12. **Von Troeltsch anterior pouch**: It is situated between the pars tensa and anterior malleolar fold.
13. **Mastoid tip**: Mastoid tip is absent at the time of birth but mastoid antrum is present. Mastoid tip does not develop till 2 years. Therefore postaural incision for mastoid exploration in children needs modification to avoid injury to the facial nerve.
14. **Macewen's triangle**: This surgical landmark on the lateral surface of the temporal bone acts as a guide to the mastoid antrum. The cribriform area of Macewen's triangle is bounded by temporal line and posterosuperior margin of bony meatus.
15. **Mastoid antrum**: In an adult, it lies 12–15 mm deep to suprameatal triangle. But at the time of birth it lies just 2 mm deep to suprameatal triangle. The thickness of the bone increases up to puberty at the rate of 1 mm per year.
16. **Korner's septum**: This bony plate, which is sometimes seen during mastoid surgery, separates superficial squamous cells from the deeper petrosal air cells. Mastoid antrum lies medial to it.
17. **Citelli's angle (Sinudural angle)**: It lies between the sigmoid sinus and middle fossa dura mater.
18. **Bill's island**: This thin plate of bone left on sigmoid sinus during mastoidectomy helps in retracting the sigmoid sinus. It should not be confused with Bill's bar, which lies in the fundus of internal auditory canal.
19. **Solid angle**: This area of bony labyrinth lies between the three semicircular canals.
20. **Trautmann's triangle**: This area is bounded by the bony labyrinth anteriorly, sigmoid sinus posteriorly and the dura or superior petrosal sinus superiorly.
21. **Donaldson's line**: This line is a surgical landmark for endolymphatic sac. It passes through horizontal SCC bisecting the posterior SCC. The endolymphatic sac that appears as thickening of the posterior cranial fossa dura is situated inferior to Donaldson's line.
22. **Utricle**: Utricle is the 1st part of membranous labyrinth to appear during the intrauterine life.
23. **Semicircular canals**: They connect with the utricle via 5 (five) openings**.** Semicircular canals of two sides are paired synergistically (horizontal canals of both sides; and one side posterior with opposite side superior).
24. **Ear development**: The middle ear, malleus, incus, stapes, labyrinth and the cochlea are fully developed by birth. Hair cells in the vestibular and cochlear end organs are derived from ectoderm. The parotid glands also develop from ectoderm.
25. **Otic capsule (Bony labyrinth)**: It ossifies from 14 centers. Ossification, which starts at 16th week of intrauterine life, ends by 20–21st week of gestation.
26. **Cochlea**: The coiled tube of this snail like structure makes 2.5–2.75 turns and measures 32 mm in length.
27. **Modiolus**: This central bony axis of cochlea measures 5 mm in length.
28. **Endolymph**: It is present in membranous labyrinth and is produced by stria vascularis in scala media and dark cells of the vestibular labyrinth. It is absorbed by endolymphatic sac.
29. **Perilymph**: Scala tympani and vestibuli are filled with perilymph.
30. **Endolymphatic sac**: The endolymphatic sac that appears as thickening of the posterior cranial fossa dura is situated posterior to the posterior semicircular canal below the Donaldson's line.

Contd...

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31. **Vestibular function tests**: Some of the common tests include galvanic stimulation, fistula test and cold caloric test.
32. **Area ratio (hydraulic ratio)**: Area of adult tympanic membrane is 90 mm^2, of which 45–55 mm^2 is functional. Area of stapes footplate is 3.2 mm^2. Area ratio (hydraulic ratio) is 14:1 to 17:1.
33. **Lever ratio**: It is 1.3:1 between the handle of malleus and the long process of incus.
34. **Axis of ossicular rotation**: It passes through anterior process of malleus and short process of incus.
35. **Loss of sound during ear transmission**: Average loss of sound during ear transmission is 20–30 dB.
36. **Outer hair cells**
37. **Cochlear microphonics**: This electrical activity occurs in the inner ear in the vicinity of hair cells in response to sound stimulation.

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2 Anatomy and Physiology of Nose and Paranasal Sinuses

He is truly a man to whom money is only a servant; but, on the other hand, those who do not know how to make a proper use of it, hardly deserve to be called men.

Sri Ramakrishna Dev

Points of Focus

- **INTRODUCTION**

ANATOMY OF NOSE

- **EXTERNAL NOSE**
  - ✓ Osteocartilaginous Framework of Nose
  - ✓ Nasal Musculature Nasal Skin
  - ✓ Development
- **INTERNAL NOSE**
  - ✓ Vestibule of Nose
  - ✓ Nasal Septum
  - ✓ Middle Meatus and Osteomeatal Complex
    - – Concha Bullosa
    - – Bulla Ethmoidalis
    - – Hiatus Semilunaris
    - – Uncinate Process
    - – Ethmoidal Infundibulum
    - – Agger Nasi
  - ✓ Linings of Internal Nose
  - ✓ Blood Supply of Nose
  - ✓ Little's Area or Kiesselbach's Plexus
  - ✓ Submucosal Vascular Plexus
  - ✓ Nerve Supply of Nasal Cavity
  - ✓ Lymphatic Drainage

ANATOMY OF PARANASAL SINUSES

- ✓ Maxillary Sinus (Antrum of Highmore)
- ✓ Frontal Sinus
- ✓ Ethmoidal Sinuses
  - – Agger Nasi Cells
  - – Grand (Basal) Lamella
  - – Haller Cells
  - – Onodi Cells
- ✓ Sphenoid Sinus
- ✓ Mucous Membrane of Paranasal Sinuses
- ✓ Mucus Drainage of Sinuses
- ✓ Lymphatic Drainage
- ✓ Blood Supply
- ✓ Nerve Supply
- ✓ Development of Paranasal Sinuses

PHYSIOLOGY OF NOSE

- **RESPIRATION**
  - ✓ Nasal Cycle
- **AIR-CONDITIONING OF INSPIRED AIR**
  - ✓ Filtration and Purification
  - ✓ Temperature Control
  - ✓ Humidification
- **PROTECTION OF AIRWAY**
  - ✓ Mucociliary Mechanism
  - ✓ Factors Affecting Ciliary Beating
  - ✓ Kartagener's Syndrome
- **VOCAL RESONANCE**
- **NASAL REFLEXES**
- **OLFACTION**
  - ✓ Olfactory Pathways
  - ✓ Vomeronasal Organ (VNO) of Jacobson

PHYSIOLOGY OF PARANASAL SINUSES

- **FUNCTIONS**
- **VENTILATION OF SINUSES**
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

The nasal cavity and paranasal air sinuses are lined by mucosa, which serve to warm, moisten and filter inspired air. Each side nasal cavity begins at the external nares and posteriorly opens through the choanae into the nasopharynx. The two nasal cavities are separated from each other by midline nasal septum and bounded laterally by three shelf-like bones, the conchae (turbinates). All the four paranasal air sinuses drain into meatuses between the conchae on the lateral nasal wall and are named after the bones that contain them (maxillary, ethmoid, frontal and sphenoid). Figure 1 is the coronal section of head showing the nasal and oral cavities, sinuses and orbits and related structures. The ophthalmic (CN V_1) and maxillary (CN V_2) divisions of the trigeminal nerve (CN V) provide the sensory innervation of nose and paranasal sinuses. The olfactory nerves (CN 1) carry smell information from the upper part of the nasal cavity. The glands of mucosa are supplied by the postganglionic fibers from the pterygopalatine ganglion, which receive preganglionic fibers from the greater petrosal branch of facial nerve (CN VII). CT anatomy is described in chapter Diagnostic Imaging

ANATOMY OF NOSE

EXTERNAL NOSE

The external nose has important cosmetic value and enhances personality and beauty of an individual. Figure 2 shows a crooked nose with "C" shaped deformity and soft tissue anatomic landmarks (reference points). This pyramidal structure is made up of osteocartilaginous framework, which is covered by muscles and skin.

Osteocartilaginous Framework of Nose

The upper one-third of external nose is bony (nasal bones) and forms bridge (root) of the nose while lower two-third is cartilaginous and forms dorsum of the nose (Figs 3 to 5).

- *Bony Part*: The two nasal bones meet in the midline and rest on the nasal process of the frontal bone. They are held between the frontal processes of the maxillae.
- *Cartilaginous Part:* It is made up of mainly upper and lower lateral cartilages and septal cartilages. The various cartilages are connected with one another and with the adjoining bones by perichondrium and periosteum.
  - *Upper lateral cartilages*: They are attached to the under surface of the nasal bones above and extend up to the lower lateral cartilages below. Both side cartilages fuse with each other and with the upper border of the septal cartilage and form dorsal surface of the nose. The lower free margin, which can be seen intranasally, forms limen vestibuli or nasal valve.
  - *Lower lateral cartilages*: This U-shaped alar cartilage has two crura lateral and medial. The lateral crus, which overlaps lower margin of upper lateral cartilage, forms the ala while medial crus lies in columella.
  - *Lesser alar (or sesamoid) cartilages*: They may be two or more in number and lie above and lateral to alar cartilages. Most of the lower free margin of ala consists of fibrofatty tissue and not the alar cartilage.
  - *Septal cartilage*: The anterosuperior border of septal cartilage, which supports the dorsum of cartilaginous part of the nose, extends from under surface of nasal bones to the nasal tip. It is discussed in detail in other section of this chapter.

Due to loss of septal cartilage support, cases of septal abscess and excessive removal of septal cartilage (submucosal resection of septum) result in supra tip depression deformity.

Nasal Musculature

The facial muscles, which bring about movements of the nose, include procerus, nasalis (transverse and alar parts), levator labii superioris alaeque nasi (muscle with the longest name) and depressor septi.

Nasal Skin

The skin, which covers nasal bones and upper lateral cartilages, is thin and freely mobile. But skin covering the alar cartilages is tick and adherent and contains many sebaceous glands.

Hypertrophy of sebaceous glands of external nose skin results in a lobulated tumor called rhinophyma (see chapter Diseases of External Nose).

Dangerous Area of Face (Danger Triangle Area)
This triangular area, venous drainage of which goes intracranially, extends from nasion to angles of mouth and includes external nose and upper lip. The inferior ophthalmic vein, which receives angular vein, drains into cavernous sinus. The infection of this area has the potential to cause cavernous sinus thrombosis.

Development

Above the roof of stomodeum, the mesenchymal frontonasal process grows downward and merges with the maxillary processes, which arise from 1st branchial arch. The ectodermal thickening of olfactory placode invaginates as a pit between the frontonasal process and lateral nasal process. The frontonasal process forms median nasal process and upper lip philtrum. The lateral nasal process forms the lower lateral cartilage and lobule of the lateral portion of nose. The olfactory placode invaginates internally to rest high in the nasal cavity and forms olfactory epithelium.

Congenital cleft lip deformity occurs due to the failure of the fusion of medial frontonasal process and lateral maxillary process.

INTERNAL NOSE

The nasal septum divides the internal nose into two halves right and left nasal cavities. The nasal cavities communicate with the exterior through anterior nares (nostrils) and with the nasopharynx through posterior nasal choanae. The anterior and inferior skin-lined portion of internal nose is called vestibule and posterior mucosa-lined portion makes nasal cavity proper. Each nasal cavity has four boundaries lateral and medial walls, roof and floor.

- Lateral Wall of Nasal Cavity
  - *Bones*: The lateral wall is formed by following bones:
    - Nasal bone

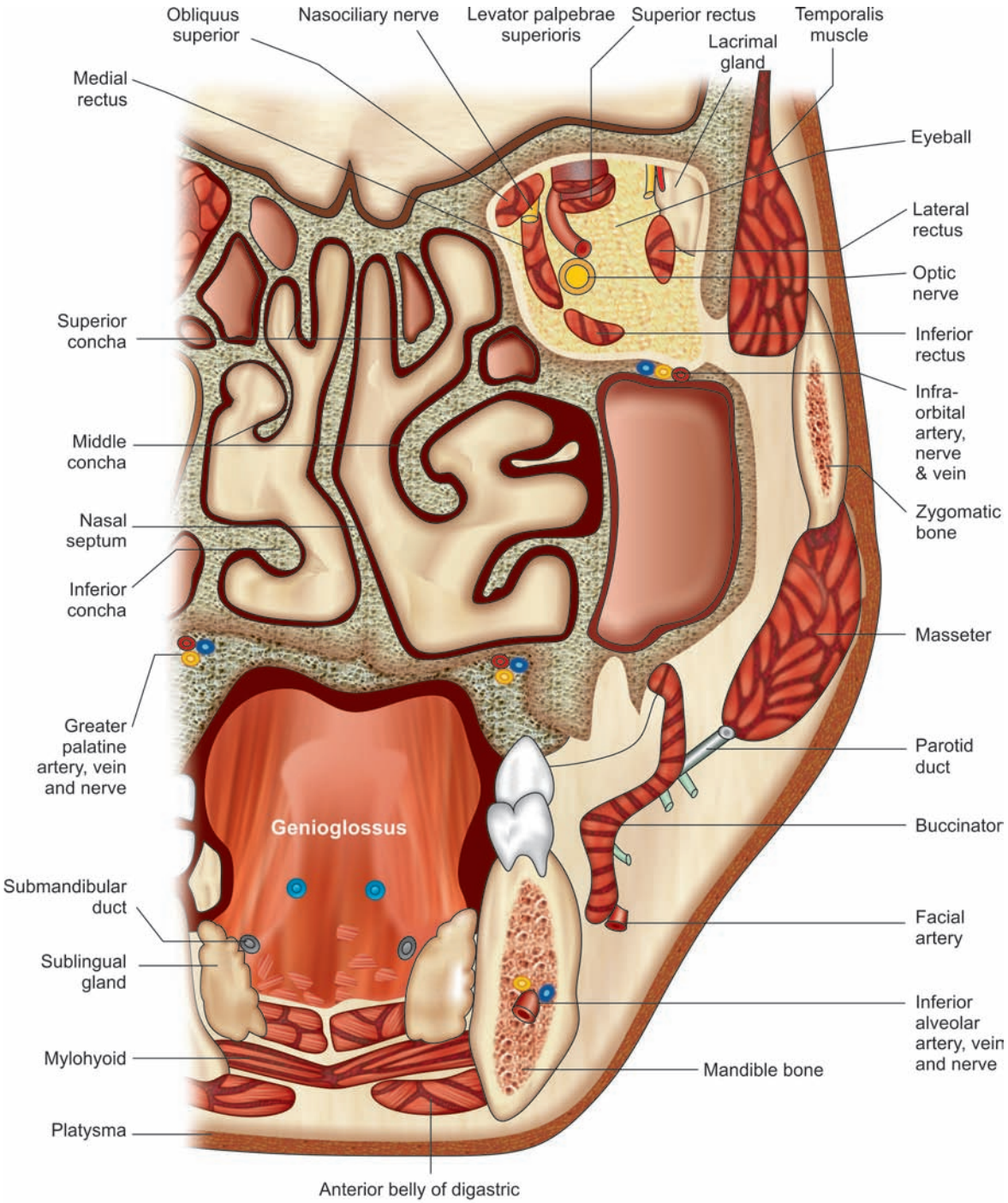


Fig. 1: Coronal section of head showing nose, sinuses, oral cavity and orbits

- Maxilla: Frontal process and medial surface maxilla and medial wall of maxillary sinus
- Lacrimal bone
- Inferior turbinate
- Ethmoid: Lateral mass of ethmoidal bone
- Palatine bone: Perpendicular plate
- Sphenoid: Medial pterygoid plate

▸ *Turbinates*: Three scroll-like bony projections, inferior, middle and superior turbinates (conchae) are seen over the lateral wall. Sometimes, a fourth turbinate, concha suprema is also seen (Figs 6 and 7).

- *Inferior turbinate*: This is the largest turbinate and is a separate bone.
- *Middle turbinate*: This is the part of ethmoidal bone and is described in other section of this chapter.
- *Superior turbinate*: This is the smallest turbinate and a part of ethmoidal bone and may get pneumatized by one or more ethmoidal air cells. It is situated posterior and superior to the middle turbinate and so not usually visible on anterior rhinoscopy examination.
- *Supreme turbinate*: It may be seen lying above the superior turbinate in some cases.

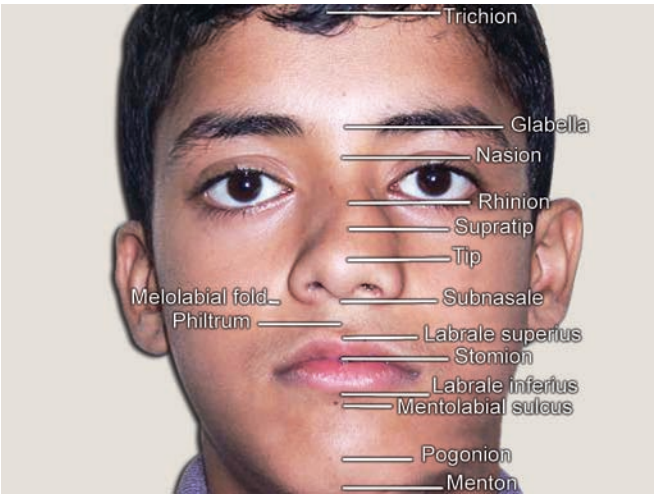


Fig. 2: Crooked nose ("C" shaped) Note soft tissue anatomic landmarks (reference points)

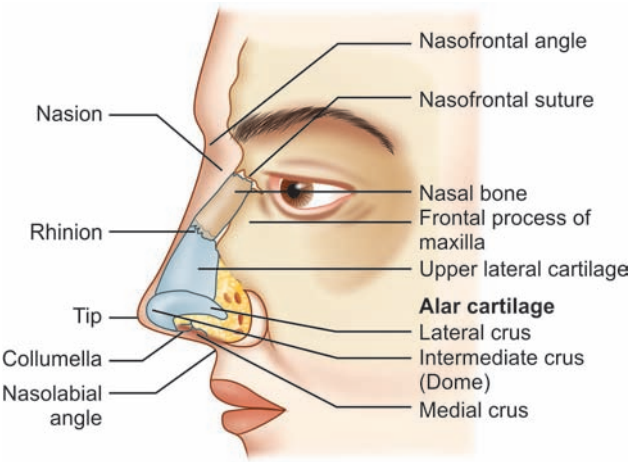


Fig. 3: External nose structure—lateral view

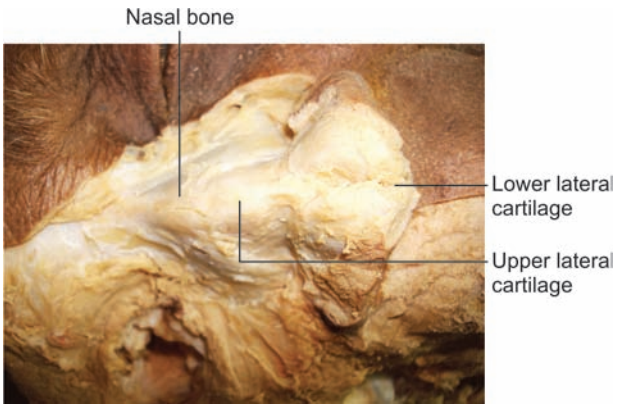


Fig. 4: Cadaveric dissection of external nose showing nasal bones and upper and lower lateral cartilages

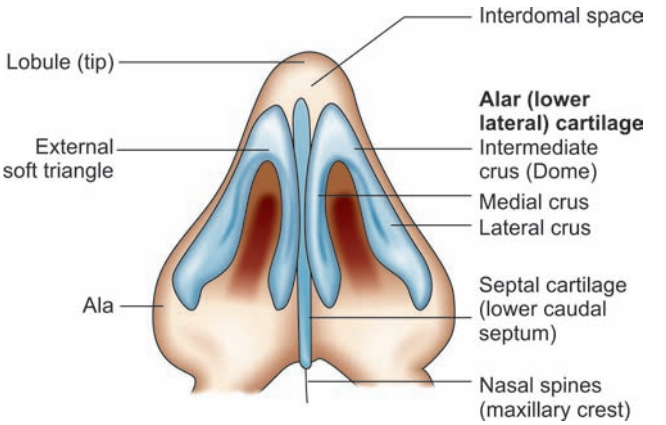


Fig. 5: External nose structure basal view

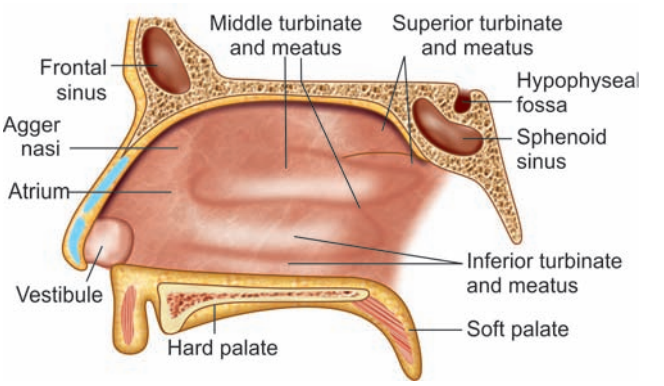


Fig. 6: Lateral wall of nasal cavity showing turbinates and meatuses

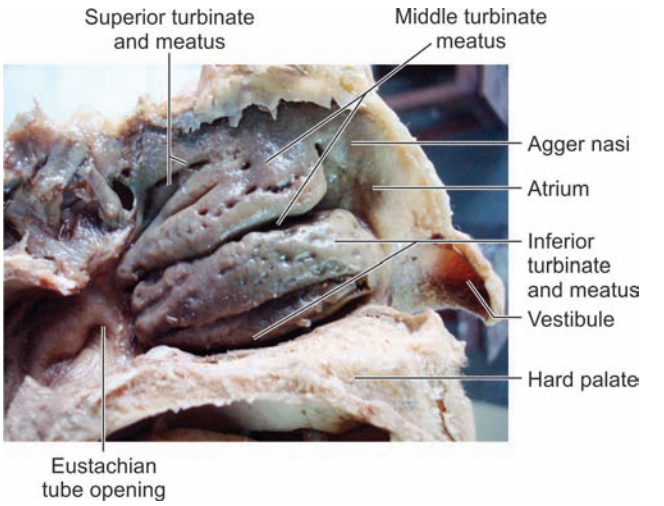


Fig. 7: Lateral wall of nasal cavity of cadaver showing turbinates and meatuses

- *Meatuses*: The three corresponding spaces present below and lateral to each turbinate are inferior, middle and superior meatus. (Figs 8 to 10)
  - *Inferior meatus*: It is present along the whole length of the lateral wall. Nasolacrimal duct opens in the anterior part of inferior meatus. The opening of nasolacrimal duct is guarded by Hasner's mucosal valve.
  - *Middle meatus*: It lies below the middle turbinate and is described in other section of this chapter.
  - *Superior meatus*: It is present only in the posterior third of lateral wall of nose. Posterior ethmoidal sinuses open in this space.
- *Sphenoethmoidal Recess*: It lies above the superior turbinate. The sphenoid sinus opens into this recess

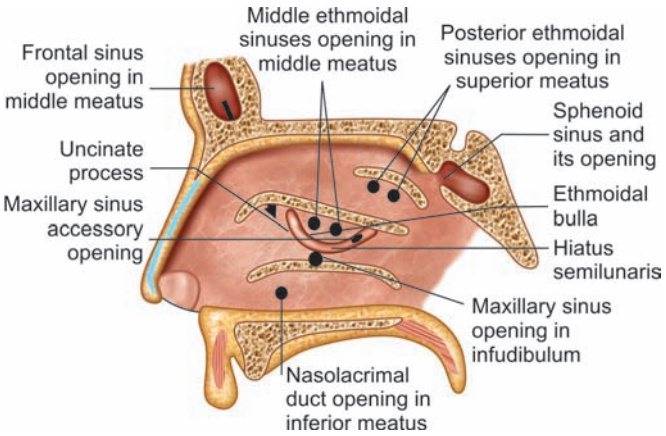


Fig. 8: Openings of paranasal sinuses. Lateral wall of nose after removal of turbinates

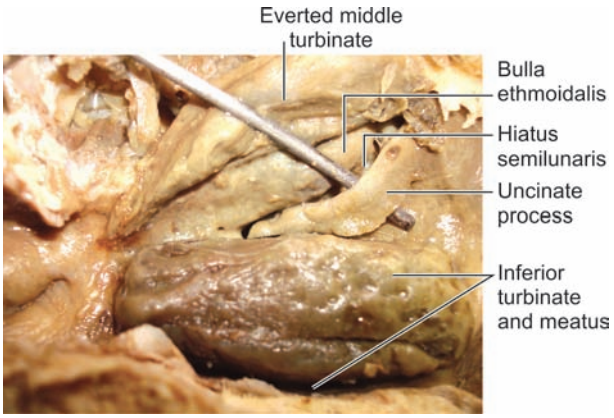


Fig. 9: Cadaveric dissection of osteomeatal complex. Middle turbinate is reflected upward. Probe lies in the ethmoidal infundibulum and is coming out through the lower attachment of uncinate process

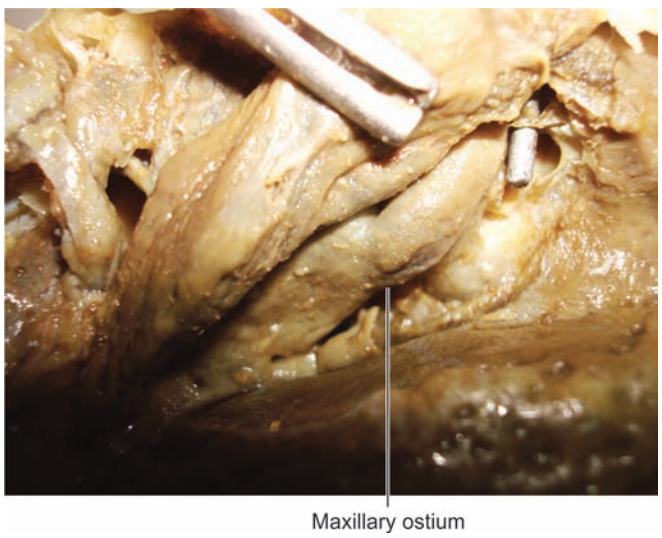


Fig. 10: Cadaveric dissection of osteomeatal complex. Middle turbinate is reflected upward. Uncinate process is removed and the ethmoidal infundibulum is opened. The probe is showing the frontal sinus opening. Note the maxillary ostium

medial to the superior turbinate about 1 cm above the upper margin of posterior choana close to the posterior border of septum.

- *Atrium*: This shallow depression lies in front of the middle meatus and above the vestibule.

- Medial Wall: It is formed by the nasal septum, which has been described in other section of this chapter.
- Roof: It has three parts: anterior sloping part (nasal bones), middle horizontal part (cribriform plate of ethmoid through which olfactory nerves pass) and posterior sloping part (body of sphenoid bone).
- Floor: Floor of the nose makes roof (hard palate) of the oral cavity. It is made up of two bones: palatine process of the maxilla (anterior three-fourth) and horizontal plate of the palatine bone (posterior one-fourth).

Vestibule of Nose

This anteroinferior portion of nasal cavity is lined by skin, which contains sebaceous glands, hair follicles and the hair (vibrissae). For the internal and external nasal valves see chapter Nasal Septum.

- *Limen Nasi (nasal valve):* This area is the greatest constriction of respiratory tract. Its boundaries include
  - *Floor*: Floor of the nose.
  - *Superior and lateral:* The caudal margin of upper lateral cartilage.
  - *Medial:* Columella and lower part of the nasal septum up to mucocutaneous junction.

Injudicious resection of lateral nasal cartilage during rhinoplasty can produce collapse (obstruction) of nose during inspiration.

Nasal Septum

Nasal septum (Figs 11 and 12) can be divided into three parts: septum proper, membranous septum and columellar septum. The columellar and membranous parts can be moved from side to side.

1. *Columellar Septum*: It is covered on either side by skin. The columella contains medial crura of lower lateral cartilages, which are joined together with fibrous tissue.
2. *Membranous Septum*: It lies between the columella and the caudal border of septal cartilage and consists of only double layer of skin. There is no bony or cartilaginous support in membranous septum.
3. *Septum Proper*: It is covered with mucous membrane and consists of osteocartilaginous framework. The principal constituents of septum proper are the perpendicular plate of ethmoid, the vomer and a large quadrilateral septal cartilage, which is wedged between vomer and ethmoid plate. Other bones, which make very small contributions, include crest of nasal bones, nasal spine of frontal bone, rostrum of sphenoid, crests of palatine and maxilla and the anterior nasal spine of maxilla.

- Septal Cartilage: This large quadrilateral septal cartilage is wedged between vomer and ethmoid plate.
  - *Inferior margin*: It lies in a groove of vomer and rests anteriorly on anterior nasal spine. It may get dislocated from anterior nasal spine (caudal septal deviation) or vomerine groove (septal spur). This occurs due to trauma. These deformities compromise the nasal airway.
  - *Superior margin*: Septal cartilage fuses with the upper lateral cartilages of external nose. Therefore septal deviation may be associated with deviation of cartilaginous part of external nose.

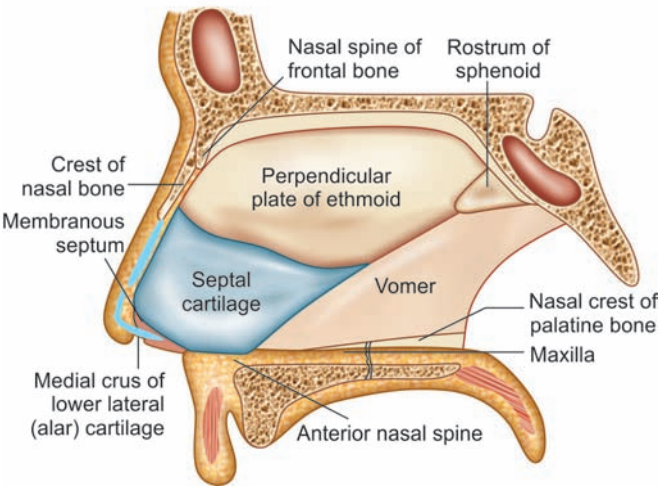


Fig. 11: Structure of the nasal septum lateral view. Bony and cartilaginous part seen after removing the mucosa. Posterior triangular segment of cartilaginous septum generally overlaps the bony septum

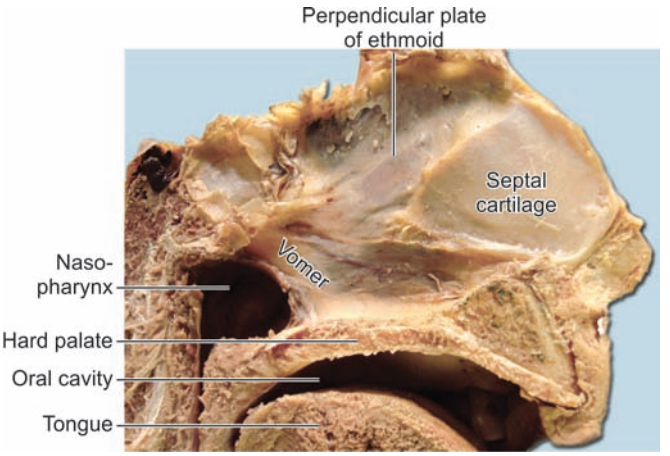


Fig. 12: Nasal septum of a cadaver

Septal cartilage also provides support to the tip and dorsum of cartilaginous part of external nose. Septal cartilage destruction, which can be caused due to septal abscess, injuries, tuberculosis or excessive removal during septal surgery, results in depression of lower part of dorsum of nose and drooping of the nasal tip.

Middle Meatus and Osteomeatal Complex (Figs 9, 10 and 13)

Middle meatus space lies below the middle turbinate. It is present in the posterior half of the lateral wall. The osteomeatal complex includes uncinate process, maxillary ostium, middle turbinate, bulla ethmoidalis and ethmoid infundibulum. It is important as ostium of frontal, maxillary and anterior ethmoidal sinuses are present in this area.

Any mucosal swelling and congenital anomaly of osteomeatal unit can cause obstruction, stasis and repeated infections of the upstream sinuses. Functional endoscopic sinus surgery (FESS) puts stress on this complex, which need normal restoration to enhance sinus drainage.

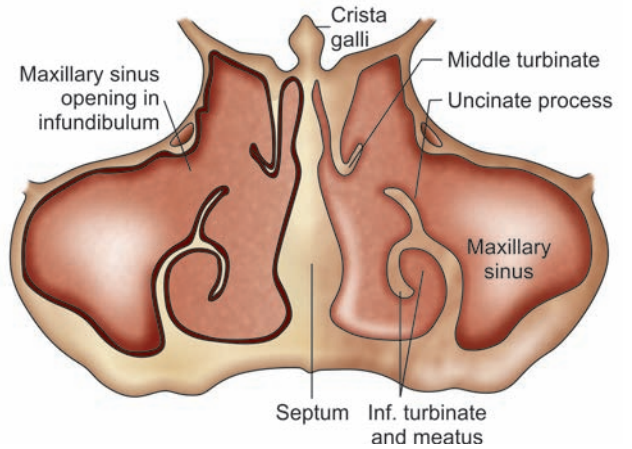


Fig. 13: Osteomeatal complex. Coronal section of nose and paranasal sinuses at the level of osteomeatal complex. The uncinate process is in sagittal plane bounding the infundibulum in which opens the ostium of maxillary sinus

- Middle Turbinate: This lower ethmoidal turbinate is attached to the lateral wall through the ground basal lamella. Its anterior one-third lies in sagittal plane and is attached to the lateral edge of cribriform plate, which forms the floor of anterior cranial fossa. Its middle one-third lies in frontal plane and is attached to lamina papyracea, which forms medial wall of orbit. Its posterior one-third, which forms roof of middle meatus, runs horizontally and is attached to lamina papyracea and medial wall of maxillary sinus.

Fracture of middle turbinate can damage cribriform plate and cause SF rhinorrhea and anosmia.

  - *Paradoxical*: In some cases lateral surface of middle turbinate is convex. It can cause narrowing of the middle meatus. It can affect ventilation and mucociliary clearance in osteomeatal unit.

Concha Bullosa: Concha bullosa is pneumatization of the middle turbinate. It is present in 30% of the population and is usually asymptomatic. It drains into the frontal recess directly or through agger nasi cells. The enlarged middle turbinate can affect ventilation and mucociliary clearance in osteomeatal unit. The obstruction to the drainage system of concha bullosa results in symptoms. It requires endoscopic sinus surgery (removal of medial wall of concha or entire concha bullosa).

- Bulla Ethmoidalis: Middle ethmoidal air cells form this rounded bulge and they open on or above it. As the bulla ethmoidalis lies anterior to the ground lamella of middle turbinate, these air cells are considered part of anterior group of ethmoidal cells. If not pneumatized, bulla ethmoidalis may remain like a solid bony prominence. It may extend superiorly to base of skull and posteriorly to ground lamella of middle turbinate.
  - *Lateral Sinus of Grunwald*: Sometimes there are spaces above or behind the bulla ethmoidalis, which are called suprabullar or retrobullar recesses respectively and together form lateral sinus of Grunwald. It is bounded superiorly by the base of skull, laterally by lamina papyracea, inferiorly by the bulla, and medially opens (through hiatus semilunaris superior) in to the middle meatus bounded by the middle turbinate.

- Hiatus Semilunaris: This two-dimensional gap lies between the posterosuperior free border of uncinate process and front of bulla. It opens laterally into a three-dimensional funnel-shaped space called ethmoidal infundibulum.
- Uncinate Process: This small thin lamina of ethmoid bone forms floor and medial wall of the ethmoidal infundibulum. This sickle shape structure runs from anterosuperior to posteroinferior direction.
  - *Two borders*: The free posterosuperior border is sharp and runs parallel to anterior surface of bulla ethmoidalis. The anteroinferior border is attached to the lateral wall of nose.
  - *Two ends*: Posteroinferior end is attached to inferior turbinate and divides the lower membranous part of middle meatus into anterior and posterior fontanelle, which are devoid of bone. If perforated they open into the maxillary sinus. The anterosuperior end may be inserted laterally on the lamina papyracea, upwards into the base of skull or medially into the middle turbinate and accounts for the variations in the drainage of frontal sinus.
- Ethmoidal Infundibulum: Frontal sinus opens into the anterosuperior part of infundibulum just posterior to posterior wall (opening) of agger nasi cells (curved ridge running downwards and forwards above the atrium). Anterior ethmoidal air cells open into the infundibulum. Maxillary sinus, which may have accessory openings, opens into posterior part of the infundibulum.
  - *Boundaries*:
    - *Medial:* Uncinate process and frontal process of maxilla and sometimes lacrimal bone.
    - *Lateral*: Lamina papyracea.
- Agger Nasi: It is an elevation that lies just anterior to the attachment of middle turbinate and if pneumatized contains agger nasi cells, which communicate with frontal recess.

Large agger nasi cells can constrict the frontal recess and can impair the frontal sinus drainage.

Linings of Internal Nose

- *Skin of Nasal Vestibule*: Vestibule is lined by skin (stratified squamous epithelium), which contains hair, hair follicles and sebaceous glands.
- *Olfactory Epithelium*: The olfactory epithelium that is paler in color lines the olfactory region, which includes roof of nasal cavity and area above superior concha.
- *Respiratory Mucosa*: The respiratory mucous membrane, which covers the lower two-third of the nasal cavity, shows variable thickness. It is thickest over nasal conchae especially at their ends. It is thick over the nasal septum and thin in the meatuses and floor of the nose. This respiratory mucous membrane is pseudostratified ciliated columnar epithelium and contain plenty of goblet cells. It is highly vascular and contains erectile tissue. The submucosal layer contains both racemose and tubular glands that secrete serous and mucous secretions for the surface mucous blanket. The deeper specialized vascular plexus (erectile tissue) consists of arterioles, capillaries, vascular sinusoids, venous plexuses and venules.

Blood Supply of Nose

Branches of both external and internal carotid arteries supply the nose (Box 1). Sphenopalatine and greater palatine branches of internal maxillary artery (branch of external carotid artery) supply posterior inferior part of internal nose. Branches of facial artery (branch of external carotid artery) supply anterior inferior nasal cavity. Anterior and posterior ethmoidal branches of ophthalmic artery (branch of internal carotid artery) supply superior part of nasal cavity (Figs 14 and 15). For the retrocolumellar vein see chapter Diseases of External Nose and Epistaxis.

Little's area or Kiesselbach's plexus in the anteroinferior part of nasal septum just above the vestibule is the vascular area, where anterior ethmoidal, sphenopalatine, greater palatine and septal branch of superior labial arteries and their corresponding veins form an anastomosis. This is the most common site for epistaxis and bleeding polyp (fibroangioma) of septum.

Submucosal Vascular Plexus

The submucosal venous plexus of respiratory mucosa of nose, which consists of arterioles, capillaries, vascular sinusoids, venous plexuses and venules, resembles erectile tissue of genitalia. This deeper specialized vascular plexus is present in the erectile tissue of inferior turbinate and adjacent septum and posterior part of middle turbinate. It is under the control of autonomic nervous system. Stimulation of sympathetic vasomotor nerves causes vasoconstriction (noradrenaline secretion), while parasympathetic stimulation causes vasodilation (acetylcholine secretion) and watery nasal discharge (secretomotor). Stronger mechanical or chemical stimulation of internal nose may cause apnea and bradycardia.

Nerve Supply of Nasal Cavity (Figs 16 and 17)

- *Olfactory Nerves*: These central filaments of the olfactory cells, which are arranged into 12–20 nerves, carry sense of smell from the olfactory region of nose. They pass through the cribriform plate and end in the olfactory bulb.

Olfactory nerves carry meningeal sheaths of dura, arachnoid and pia mater and because of this injury to these nerves can result in CSF rhinorrhea and meningitis.

Box 1: Blood Supply of Nose

1. Internal Carotid System

- Ophthalmic Artery
- Anterior ethmoidal artery
- Posterior ethmoidal artery

2. External Carotid System

1. *Maxillary Artery*
   - Branches of sphenopalatine artery: Nasopalatine, posterior nasal septal branches and posterior lateral nasal branches.
   - Greater palatine artery
   - Nasal branch of anterior superior dental artery, which is a branch of infraorbital artery.
2. *Facial Artery*
   - Superior labial artery

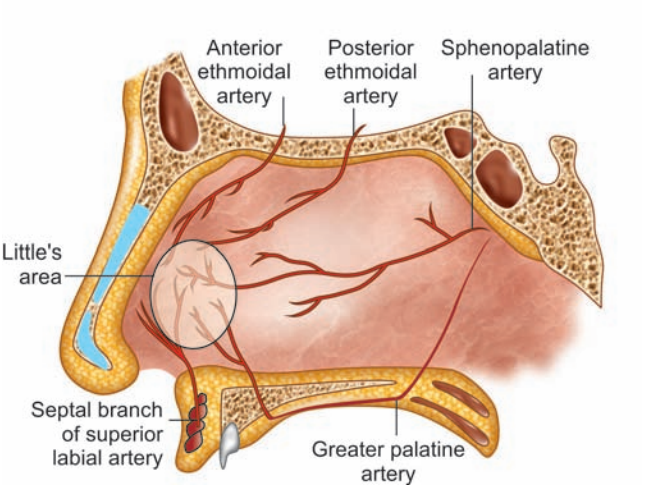


Fig. 14: Blood supply of nasal septum

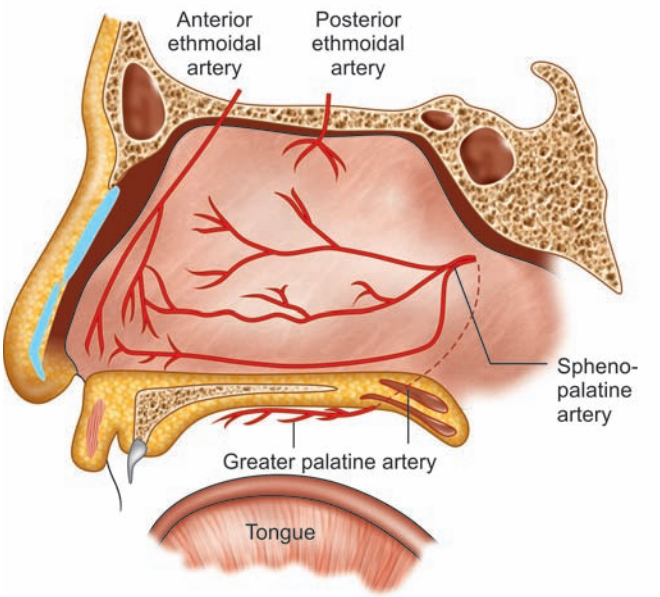


Fig. 15: Blood supply of lateral wall of nose

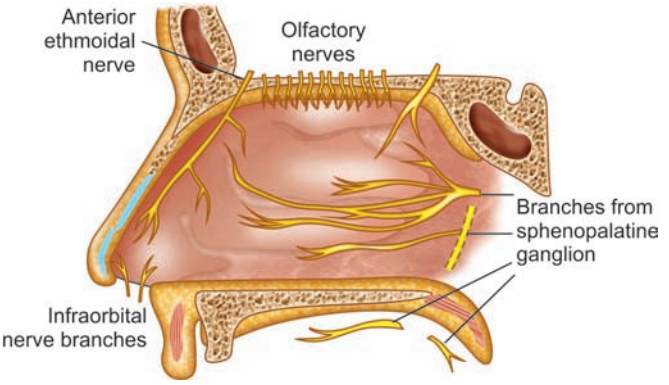


Fig. 16: Nerve supply of lateral wall of nose

- *Trigeminal Nerve*: The ophthalmic division (anterior ethmoidal nerve) and maxillary division (sphenopalatine and infraorbital branches) carry common sensation (such as touch and irritation from acrid odors). Both these divisions supply to the skin of external nose.
  - Branches of infraorbital nerve supply vestibule of nose.
  - The posterior two-third of nasal cavity (septum and lateral wall) is supplied by branches of sphenopalatine ganglion, which pass through the sphenopalatine foramen situated near the posterior end of middle turbinate, where it can be blocked by placing a pledget of cotton of 4% xylocaine.
  - Anterosuperior part of the nasal cavity (lateral wall and septum) is supplied by anterior ethmoidal nerve, which can be blocked by placing the cotton pledget impregnated in 4% xylocaine high up inside of nasal bones.

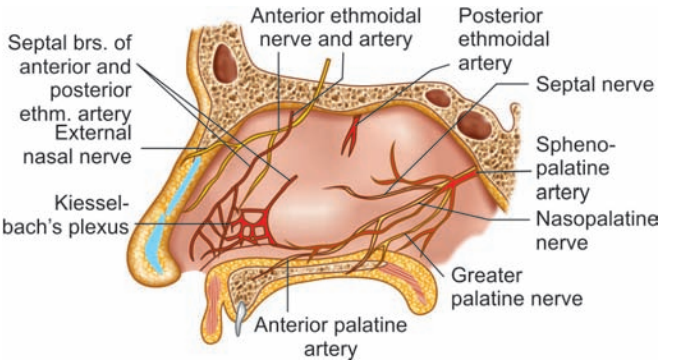


Fig. 17: Artery (red color) and nerve (yellow color) supply of nasal septum

- *Autonomic Nervous System*
  - *Parasympathetic*: Parasympathetic nerve fibers are secretomotor and supply the nasal glands. The greater superficial petrosal nerve, which carry preganglionic fibers from the parasympathetic nucleus situated in brain stem, joins deep petrosal nerve (postganglionic vasomotor sympathetic fibers) and forms nerve to pterygoid canal (vidian nerve), which reaches to the sphenopalatine ganglion where only parasympathetic fibers relay. The branches of sphenopalatine ganglion pass through sphenopalatine foramen and supply nasal cavity. Parasympathetic fibers, which supply to nasal blood vessels, cause vasodilation.
  - *Sympathetic*: The preganglionic sympathetic nerve fibers come from upper two thoracic segments of spinal cord (origin from hypothalamus) and relay in superior cervical ganglion. The postganglionic fibers form a plexus around internal carotid artery, from which deep petrosal nerve arises that joins the preganglionic parasympathetic fibers of greater petrosal nerve to form the nerve to pterygoid canal (vidian nerve). These postganglionic sympathetic vasomotor fibers reach the nasal cavity without relaying in the sphenopalatine ganglion and cause vasoconstriction of nasal vessels (decongestion of nasal cavity).

Some surgeons are of opinion that excessive rhinorrhea (vasomotor and allergic rhinitis) can be controlled by section of the vidian nerve (vidian neurectomy).

- *Facial Nerve*: It supplies to the muscles of the external nose.

Lymphatic Drainage

- *Submandibular Lymph Nodes*: The external nose and anterior part of nasal cavity drain into submandibular lymph nodes.
- *Retropharyngeal Nodes*: Posterior part of nasal cavity drains into upper jugular nodes and retropharyngeal nodes, which drain to upper jugular nodes.
- *Perineural Intracranial Spread*: The perineural intracranial spread of cancer is possible through the lymphatics of the upper nasal cavity, which communicate with subarachnoid space along the olfactory nerves.

ANATOMY OF PARANASAL SINUSES

On each side there are four paranasal air sinuses in four cranial bones: frontal, maxilla, ethmoid and sphenoid. They are divided into two groups:

1. *Anterior Group*: The sinuses, which open anterior to basal lamella of middle turbinate in the middle meatus, form anterior group of paranasal sinuses. They are maxillary, frontal and anterior ethmoid sinuses.
2. *Posterior Group*: The sinuses, which open posterior and superior to basal lamella of middle turbinate, form posterior group of paranasal sinuses. They are posterior ethmoid and sphenoid sinuses. The posterior ethmoidal sinuses open in the superior meatus and sphenoid sinuses open in sphenoethmoidal recess.

Maxillary Sinus (Antrum of Highmore) (Fig. 18)

It is the first to develop in human fetus. This largest paranasal sinus (15 ml capacity in adult). It occupies the body of maxilla and is pyramidal shape. The base faces towards lateral wall of nose and apex is directed laterally into the zygomatic process.

- *Boundaries*
  - *Anterior wall*: The anterior facial surface of maxilla (canine fossa) is related to cheek.
  - *Posterior wall*: It is in relation with the infratemporal and pterygopalatine fossa.
  - *Medial wall*: It is thin and membranous at places and faces middle and inferior meatuses.

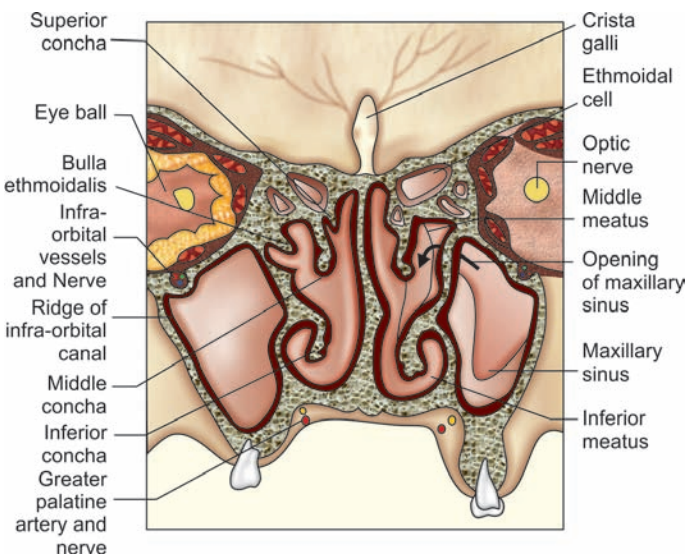


Fig. 18: Coronal section of nose and paranasal sinuses seen from behind

  - *Floor:* It is situated about 1 cm below the level of floor of nose in adults. Until 3 years of age, sinus floor is 4–5 mm above the nasal floor. It is formed by alveolar process of the maxilla. The roots of all the molars, second premolar and sometimes first premolar (depending on the extent of pneumatization and age of the person), are situated in the floor of maxillary sinuses. These teeth roots are separated from the sinus mucosa by a thin lamina of bone, which may be dehiscent. The chances of oroantral fistulae are high after the extraction of these teeth, the infection of which can result in maxillary sinusitis.
  - *Roof*: The roof of the maxillary sinus is the floor of the orbit and is traversed by infraorbital nerve and vessels.
- *Ostium of Maxillary Sinus*: It is situated higher in medial wall and opens in the posterior part of ethmoidal infundibulum.
- *Accessory Ostium*: In 30% of population, an accessory ostium, which may be quite large, is seen behind and in front of the natural main ostium. The maxillary sinus does not drain through accessory ostium and is bypassed by the mucus blanket.

During endoscopic sinus surgery, accessory maxillary ostium is joined with the natural maxillary ostium to prevent recirculation of the mucopus into the maxillary sinus.

Frontal Sinus

The frontal sinus is situated above and deep to the supraorbital margin. It lies between the inner and outer tables of the lower part of frontal bone. The shape and size of this loculated sinus vary (very large to absent). The bilateral frontal sinuses are often asymmetric. The intervening bony septum, which is thin and often obliquely placed, may be deficient in some cases. A very large sinus may extend into the roof of the orbit.

- *Relations*:
  - *Anterior wall* of the sinus is related to the forehead skin.
  - *Floor* is in relation with orbit.
  - *Posterior wall* relations are meninges and frontal lobe of brain.
- *Ostium* of frontal sinus is situated in its floor and opens into the frontal recess, which depending upon the attachment of uncinate process opens either in the infundibulum or medial to the uncinate process into the middle meatus.

Ethmoidal Sinuses (Fig. 19)

Ethmoidal sinuses are thin walled air cavities in lateral masses of ethmoid bone. They vary in number (3–18) and lie between upper third of lateral nasal wall and the medial wall of orbit. Clinically they are divided into two groups: anterior and posterior. Anterior ethmoid group opens into the middle meatus. Posterior ethmoid group opens into the superior meatus and some in sphenoethmoidal recess.

- Boundaries
  - *Roof*: It is closed by the frontal bone**,** which forms the floor of anterior cranial fossa.
  - *Lateral wall*: Lamina papyracea separates it laterally from the orbit.

Lamina papyracea is paper thin and can be easily damaged during intranasal surgery and destroyed by ethmoidal infections. Optic nerve, which is at risk during posterior ethmoid surgery, is in close relationship with posterior group of ethmoidal air cells.

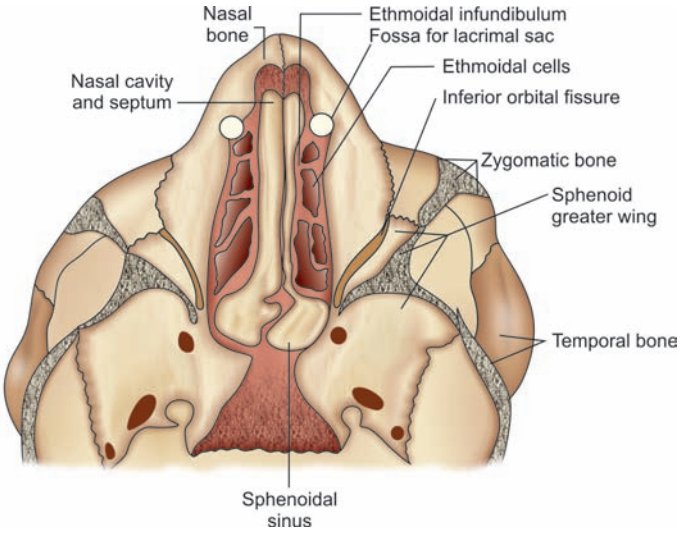


Fig. 19: Axial section of nose and paranasal sinuses in the upper part showing ethmoidal and sphenoidal sinuses

- *Agger nasi cells*: These are most anterior of anterior ethmoid cells and lie in close proximity of frontal recess. They are situated at the agger ridge, which is present just anterior to anterosuperior attachment of middle turbinate.
- *Grand (Basal) lamella*: This bony insertion of middle turbinate into the skull base and lateral nasal wall separates anterior from posterior ethmoid cells. Grand lamella can be divided into three parts. Anterior one-third inserts into lamina cribrosa, middle one-third (oblique anterosuperior to posteroinferior course) into lamina papyracea and posterior one-third horizontal part inserts into lateral nasal wall.

- Big agger nasi cells can obstruct drainage of frontal sinus and their removal provides better view of nasofrontal duct during endoscopic sinus surgery.
- Haller cells: These ethmoid cells extend into the roof of maxillary sinus in the region of maxillary sinus ostium. These cells may remain asymptomatic or affect maxillary sinus ventilation and drainage resulting in recurrent or chronic maxillary sinusitis. They are present in 10% of population.
- Onodi cells: These are posterior ethmoid cells and extend either laterally or superiorly along the sphenoid sinus. The optic nerve can lie within them. Onodi cells must be recognized during the endoscopic sinus surgery on posterior ethmoid to avoid optic nerve injury.

Sphenoid Sinus (Figs 20 and 21)

The two sphenoid sinuses, one on each side are rarely symmetrical. They occupy body of sphenoid bone and are separated by a thin bony septum, which is usually obliquely situated and may even be deficient. The ostium, which is situated in the upper part of anterior wall, drains into sphenoethmoidal recess.

The anterior wall of sphenoid sinus is 7 cm away from nasal sill at 30° angle.

- *Relations*: The relations are important during the endoscopic sinus surgery and trans-sphenoidal hypophysectomy. The extent and relations of sphenoid sinus depend upon the degree of pneumatization, which may extend into the wings of sphenoid, pterygoid plates and clivus.
  - *Anterior part*: The superior relations are olfactory tract, optic chiasma and frontal lobe. The lateral wall relations are optic nerve, internal carotid artery and maxillary nerve. These structures may be dehiscent in the lateral wall of sinus.
  - *Posterior part*: Roof of sinus is floor of sella turcica (pituitary gland fossa). Lateral wall is related to cavernous sinus, which contains internal carotid artery and CN III, IV, V and ophthalmic and maxillary divisions CN V (trigeminal).

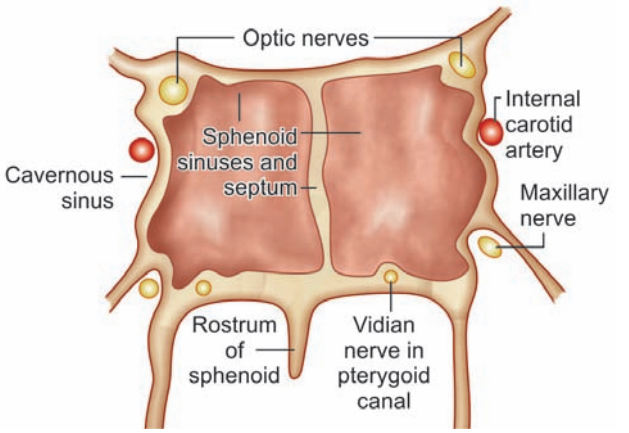


Fig. 20: Sphenoidal sinuses coronal section showing important structures situated in relation with sphenoid sinus walls

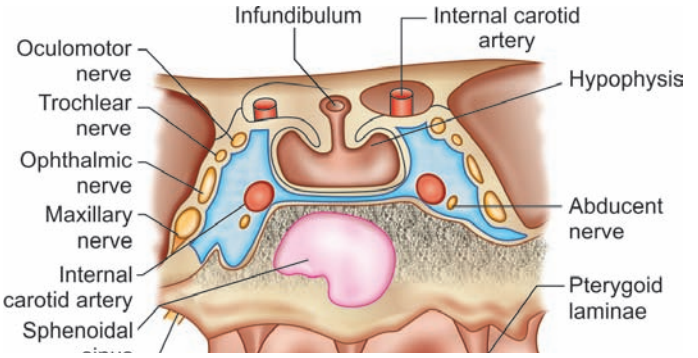


Fig. 21: Relations of sphenoid sinus with cavernous sinus and pituitary fossa and gland

Mucous Membrane of Paranasal Sinuses

The mucous membrane of paranasal sinuses is thinner and less vascular and is continuous with that of the nasal cavity through the sinuses' openings. It is ciliated pseudostratified columnar epithelium with numerous mucoserous glands and goblet cells. They produce mucus blanket which has two layers: inner thin serous layer or sol phase and outer viscous mucus layer or gel phase. Cilia, which help in drainage of mucus, are more marked near the ostia.

Mucus Drainage of Sinuses

Mucus secretions of paranasal sinuses travel to their ostium in a spiral manner. The cilia propel mucus through the ostium into the respective meatuses.

The mucus from anterior groups of sinuses passes along the lateral pharyngeal gutter. Hypertrophy of lateral pharyngeal lymphoid occurs in infection of anterior sinuses. The mucus from posterior group of sinuses spreads over the posterior pharyngeal wall.

- *Maxillary Sinus*: In the maxillary sinus, secretion transports (stellate pattern) begin from floor and along all the four walls (anterior, posterior, medial and lateral) and roof and converge at the natural ostium.
- *Frontal Sinus*: Frontal sinus secretions flow in both directions along medial aspect of ostium. They flow out of sinus along floor and inferior parts of anterior and posterior walls. From the medial aspect of ostium, secretions flow superiorly and then laterally along the roof of frontal sinus.

The plain CT scan without contrast is the first line of screening study of the nose and paranasal sinuses. The plain PNS X-rays do not offer adequate views of osteomeatal complex, sphenoid and ethmoid sinuses due to the overlapping of structures.

Lymphatic Drainage

The lymphatic of sinuses form a capillary network in mucosa. They drain into upper deep cervical nodes lateral retropharyngeal, nodes either directly or through group of lymph nodes.

Blood Supply

The paranasal sinuses are supplied by the branches of both external carotid artery (facial artery and sphenopalatine branch of maxillary artery) and internal carotid artery (anterior and posterior ethmoidal branches of ophthalmic artery) (Table 1). Sphenopalatine artery, which divides into two main branches, enters the nasal cavity through sphenopalatine foramen (posterior to middle turbinate).

The septal branch of sphenopalatine artery passes across the inferior aspect of anterior surface of sphenoid sinus and can be damaged during ESS sphenoidectomy.

Nerve Supply

Paranasal sinuses are mainly supplied by branches of CN V (trigeminal nerve) (Table 2).

- The supraorbital, supratrochlear and anterior and posterior ethmoidal nerves are branches of ophthalmic division of (CN V_1).
- The greater palatine, posterolateral nasal and superior nasal branches of the infraorbital nerve (branches of maxillary division of CN V) supply maxillary sinus.
- The sphenopalatine nerve, which also carries parasympathetic secretomotor fibers, is the branch of maxillary division of CN V.

Development of Paranasal Sinuses

The paranasal sinuses, which develop as out-pouchings from the mucous membrane of nose, continue to grow during childhood and early adult life (Table 2). At the time of birth, only the maxillary and ethmoidal sinuses are present. Radiologically, maxillary sinus is distinguished at 4–5 months, ethmoid at 1 year and sphenoid at 4 years and frontal at 6 years of age.

At birth, both frontal as well as sphenoid sinuses are absent and therefore not clinically significant in young children.

PHYSIOLOGY OF NOSE

The functions of the nose include (1) respiration, (2) air conditioning of inspired air, (3) protection of lower airway, (4) vocal resonance, (5) nasal reflex and (6) olfaction.

RESPIRATION

Humans are natural nose breathers. A newborn with bilateral choanal atresia may asphyxiate to death if immediate airway management is not done. Mouth breathing is learned later on in life. The nose allows breathing during eating.

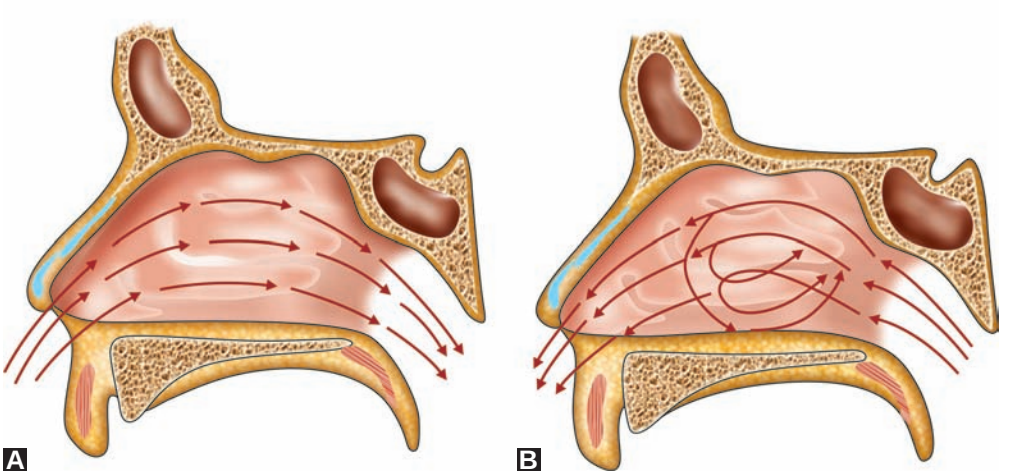
Inspiratory and expiratory air currents during quite respiration pass between the turbinates and nasal septum (Figs 22A and B). Little air passes below (i.e. inferior meatus) and above

TABLE 1 Blood and nerve supplies of paranasal sinuses

Sinuses	*Arteries*	*Nerves*
Frontal	Supraorbital, supratrochlear	Supraorbital, supratrochlear
Maxillary	Maxillary (main) and facial	Maxillary
Anterior ethmoidal	Anterior ethmoidal	Anterior ethmoidal
Posterior ethmoidal	Posterior ethmoid and sphenopalatine	Posterior ethmoid and sphenopalatine
Sphenoidal	Posterior ethmoid and sphenopalatine	Posterior ethmoid and sphenopalatine

TABLE 2 Development and growth of paranasal sinuses

Sinus	*At birth*	*Adult size*	*Growth*	*Radiological appearance (Age)*
Maxillary	Present	15 years	Biphasic growth: Birth–3 years, 7–12 year	4–5 months
Ethmoid	Present	12 years	Size increases up to 12 years	1 year
Frontal	Absent	13–18 years	Invades frontal bone (2–4 yrs), size increases until teens	6 years
Sphenoid	Absent	12–15 years	Reaches sella turcica (7 yrs), dorsum sellae (late teens), basisphenoid (adult)	4 years



Figs 22A and B: Nasal airflow. (A) Inspiratory air flow. 50% of the total airflow passes through middle part of nose. 35% air passes through inferior meatus and only 15% through olfactory region; (B) Expiration. Limen nasi converts the expiratory air current into eddies under the cover of inferior and middle turbinates and ventilates sinuses

the level of turbinates (i.e. olfactory region). Sniffing helps weak odorous substances reaching olfactory area.

During expiration, friction offered at nasal valve converts air currents into eddies under cover of inferior and middle turbinates and ventilates the sinuses through their openings in the nose. Anterior end of inferior turbinate increases and decreases in size and thus regulates inflow of air.

Nasal Cycle

The alternate opening and closing of each side of nose is called nasal cycle. It varies every few hours and is characteristic of an individual. Kayser first described it in 1895. Nasal cycle, which is observed in up to 80% of normal subjects, has also been studied with the help of rhinomanometry. There occur rhythmic cyclical congestion and decongestion of nasal mucosa (perhaps under the control of autonomic innervation), which control the airflow through nasal cavity. Most of the people are not conscious about this alteration in airflow because the total resistance of nasal airflow remains constant.

AIR-CONDITIONING OF INSPIRED AIR

Nose acts as an "air-conditioner" and filters, purifies, moistens and warms the inspired air for lungs. In 24 hours we breathe 500 cubic feet of air. Air is efficiently filtered, humidified, adjusted to proper temperature and cleared of all the dust, bacteria and viruses by the nose.

- *Filtration and Purification*: Nasal vibrissae filter coarse particles (up to 3 μm). Finer particles of 0.5–0.3 μm (such as dust, pollen and bacteria) adhere to the mucus (electrostatic attraction with the nasal mucus blanket), which is secreted by mucous glands and spread all over the nasal mucous membrane. But particles < 0.5 μm pass through the nose into lower airways.
- *Temperature Control*: The turbinates double the surface area of nasal mucosa. The large surface of nasal mucosa is structurally adapted (highly vascular with cavernous venous spaces or sinusoids) to regulate temperature of the inspired air. The mucous membrane of middle and inferior turbinates and adjacent part of the septum controls the blood flow that regulates the size of turbinates. This "radiator" mechanism warms up the inspired cold air (which may be < 0°C) to near body temperature (37°C). Hot air is cooled to the body temperature.
- *Humidification*: The serous glands of nasal mucous membrane regulate the relative humidity (75–100%) of the inspired air, which is dry in winter and humid in summer. Approximately 1 liter of water is evaporated from nasal mucosa in 24 hours. During expiration, nose removes water (maintaining hydration) and heat (preventing hypothermia) from expired air.

Moisture facilitates the function of the ciliary epithelium. Fifty percent relative humidity (dry air) can stop ciliary function in 8–10 minutes, which predisposes to infections of the respiratory tract and affects gas exchange. Nasal obstruction affects gaseous exchange in the lungs, which results in pO_2 fall and pCO_2 rise that causes apnea spells during sleep.

PROTECTION OF AIRWAY

- *Enzymes and Immunoglobulins*: Enzymes and immunoglobulins are present in nasal secretions. Muramidase (lysozyme) kills bacteria and viruses. Immunoglobulins (IgA and IgE) and interferon provide immunity against upper respiratory tract infections.
- *Sneezing*: Foreign particles, irritating nasal mucosa are expelled by this protective reflex. Copious flow of nasal secretions (due to irritation) washes noxious substance out.

Mucociliary Mechanism

Nasal mucosa is rich in mucous and serous secretory glands (600–700 ml of nasal secretions in 24 hours), which form a mucus blanket spread over the mucosa. Mucus blanket consists of two layers superficial mucus layer and deep serous layer and floats (5–10 mm/minute) on the cilia. Cilia beat constantly (10–20 times per second at room temperature) like a "conveyer belt" towards the nasopharynx. The complete sheet of mucus blanket reaches into the pharynx in

10–20 minutes. This viscous mucus blanket entraps bacteria, viruses and dust particles from the inspired air and carries them into pharynx and gets swallowed into stomach and digested.

- *Ciliary Biphasic Beats*: The ciliary biphasic beats have two strokes rapid effective stroke (extended cilia reach superficial mucus layer) and slow recovery stroke, in which cilia bend (reach deep thin serous layer) and travel slowly in the reverse direction. The effective strokes move the mucus blanket in one direction.
- *Factors Affecting Ciliary Beating*: The ciliary movements can be affected by:
  - Environmental pollution
  - Humidity
  - Drying
  - Drugs such as adrenaline
  - Excessive heat and cold
  - Hypertonic and hypotonic solutions
  - Smoking (nicotine)
  - Infections: viral and bacterial
  - Parasympathetic system
  - Airborne external irritants: SO_2 and CO_2
  - The cilia and lysozyme act best at nasal secretion pH 7, the alteration of which (due to infections and certain nasal drops) can affect the functions of cilia and lysozyme.

Kartagener's syndrome: In this immotile cilia syndrome cilia are defective and cannot beat effectively and lead to stagnation of mucus. There is absence of dynein arm on the peripheral ciliary microtubules. Patient presents with triad of
1. Chronic rhinosinusitis (mucus accumulation in nose),
2. Bronchiectasis and
3. Situs inversus

VOCAL RESONANCE

Nose acts as a resonating chamber for the speech. For phonating nasal consonants (M/N/NG), sound passes via nasopharynx and nose. When either nose or nasopharynx is blocked (rhinolalia clausa), speech becomes denasal and M/N/Ng are pronounced as B/D/G respectively. Reverse happens in velopharyngeal insufficiency (rhinolalia aperta).

NASAL REFLEXES

- *Sneezing*: Irritation of nasal mucosa causes sneezing. If a finger is pressed under the columella it may abort sneezing.
- *Cardiopulmonary Responses*: Strong nasal stimuli result in profound cardiopulmonary responses such as breathing cessation and bradycardia. It is relevant to sleep apnea syndrome.
- *Appetite*: Good smell of food results in reflex secretion of saliva and gastric juice.
- *Nasobronchial and Nasopulmonary Reflexes*: They affect pulmonary functions.

Pulmonary resistance: Nasal obstruction increases pulmonary resistance, which can be reversed by treating nasal obstruction.
a. Nasal packing lowers pO_2, which becomes normal after removal of pack.
b. Chronic nasal obstruction (tonsil and adenoid hypertrophy) in children causes pulmonary hypertension and cor pulmonale, which can be reversed after their surgical treatment.

OLFACTION

Olfactory system is highly developed in animals and is important for their communication and survival and gives warnings of the environmental dangers. In humans it is comparatively less developed but plays an important role in enjoying delicious and sumptuous food.

Vapors of ammonia cause irritation and stimulate fibers of the trigeminal nerve (not olfactory). The olfaction is affected in patients of nose block and food tastes bland and unpalatable.

Olfactory Pathways

Olfactory system is an important constituent of limbic system.

- *Olfactory Receptor Cells*: Olfactory epithelium in the olfactory region of nose contains millions of olfactory receptor cells, peripheral processes of which reach the mucosal surface and expand into a ventricle that have several cilia and receive odorous substances.
- *Olfactory Nerves*: Central processes of the olfactory cells make olfactory nerves.
- *Olfactory Bulb*: Olfactory nerves pass through the cribriform plate of ethmoid and end in the mitral cells of the olfactory bulb.
- *Olfactory Tract*: Axons of mitral cells traverse in olfactory tract.
- *Cerebrum*: Olfactory tract carries smell to the prepiriform cortex and the amygdaloid nucleus.

Primary olfactory receptor cells are able to regenerate entirely while other special sensory primary neurons cannot.

Vomeronasal Organ of Jacobson

This accessory olfactory tissue is present in 1–3 mm tubule with an oval orifice. It is situated 1 cm behind the caudal end of septum and 3 mm above the nasal floor. The color of mucosa in this area is pale yellowish. Though electrovomerogram has been recorded in response to odorants yet its function in humans remain uncertain.

PHYSIOLOGY OF PARANASAL SINUSES

FUNCTIONS

The functions of the paranasal sinuses, which are not well proved, include:

- Air-conditioning of the inspired air (humidification and warming)

- Keep the nasal chambers moist
- Resonance to voice
- Protect the delicate structures in the orbit (eye) and the cranium (brain)
- Lighten the skull bones
- Rapid growth of face
- Absorption of shock to the face and skull
- Increasing the area of olfactory membrane
- Regulation of intranasal pressure.

VENTILATION OF SINUSES

Ventilation of paranasal sinuses takes place through their ostia. It is paradoxical and reverse to lungs. Sinuses get emptied of air during inspiration and filled with air during expiration. Inspiration causes negative pressure (–6 mm to –200 mm of H_2O) in the nose. Expiration causes positive pressure in the nose and sets up eddies that ventilate the sinuses.

Clinical Highlights

1. **Nasal septum**: The major constituent is quadrangular septal cartilage. Other includes perpendicular plate of ethmoid, vomer, sphenoid rostrum, frontal nasal spine, anterior nasal spine, crests of maxilla, palatine and nasal bones.
2. **Vomeronasal organ:** This vestigial structure, which was said to be related with olfaction, can sometimes be seen as a pit on the anteroinferior part of nasal septum.
3. **Nasolacrimal duct:** Direction of nasolacrimal duct is downwards, backwards and laterally. It courses anterior to maxillary ostium and opens into inferior meatus.
4. **Uncinate process**: This process of ethmoidal bone forms medial wall of ethmoidal infundibulum, which lies in the lateral wall of nose.
5. **Hiatus semilunaris**: This two-dimensional door like space lies between bulla ethmoidalis and uncinate process.
6. **Ethmoidal infundibulum**: It is situated in lateral wall of nose and communicates with nasal cavity through hiatus semilunaris. Opening of the maxillary sinus is situated near its floor.
7. **Posterior fontanelle:** It is situated in the lateral wall of middle meatus.
8. **Sphenopalatine ganglion**: It innervates nose, palate and lacrimal gland (associated with lacrimation). It is anesthetized by injecting xylocaine just above and behind the posterior end of middle turbinate and into the greater palatine foramen and canal.
9. **Sphenoid sinus:** In an adult the distance between its opening and anterior nasal spine is about 7 cm. Its cavity may show indentations of internal carotid artery, optic nerve, vidian nerve and maxillary division of trigeminal nerve. Sphenoid sinus drains into sphenoethmoidal recess.
10. **Sinuses draining anterior to basal lamella**: They are maxillary sinus, frontal sinus and anterior ethmoidal sinuses including agger nasi cells.
11. **Schneiderian membrane (mucosa):** This is the pseudostratified ciliated columnar respiratory mucosa of nose.
12. **Respiratory cilia:** The **length** is about 5–7 micron. **Rate of ciliary beat** is about 700 to 1000 beats per minute (10–20 times per second) at room temperature. Ratio of effective phase and recovery phase of ciliary beat is 1:3. **Nasal cilia** are approximately 0.3 micron thick.
13. **Mucous blanket:** The pH of **"mucous blanket"** of nose is 7.
14. **Nasal temperature:** Normal nasal temperature is 32° C. All nasal ciliary activity ceases at 7–10^0 C.
15. **Main current of nasal airflow during inspiration:** It is through the middle part of the nasal cavity (medial to middle turbinate and the middle meatus) in a parabolic curve.

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3 Anatomy and Physiology of Oral Cavity, Pharynx, and Esophagus

That knowledge which purifies the mind alone is true knowledge, all else is only a negation of knowledge. He alone is the true teacher who is illumined by the light of true knowledge

—Sri Ramakrishna Dev

Points of Focus

ORAL CAVITY
- ✓ Subsites of Cancer in Oral Cavity
- ✓ Lips
- ✓ Alveoli and Gingivae
- ✓ Floor of Mouth
- ✓ Buccal Mucosa
- ✓ Retromolar Trigone
- ✓ Hard Palate
- ✓ Tongue
- ✓ Prefacial and Post-facial Lymph Nodes

SALIVARY GLANDS
- ✓ Development, Structure, Acini, Functions of Saliva
- ✓ Parotid Gland: Surgical Landmarks of Facial Nerve
- ✓ Submandibular Salivary Glands
- ✓ Sublingual Salivary Glands
- ✓ Minor Salivary Glands

PHARYNX
- **PHARYNGEAL WALL:** Mucous Membrane, Pharyngobasilar Fascia, Muscular Coat, Buccopharyngeal Fascia
  - ✓ Pharyngeal Spaces
  - ✓ Killian's Dehiscence
- **WALDEYER'S RING**
  - ✓ Functions of Tonsils and Adenoids

NASOPHARYNX
- ✓ Extension, Epithelium, Communications, Boundaries
- ✓ Nasopharyngeal Bursa: Tornwald's Disease
- ✓ Rathke's Pouch: Craniopharyngioma
- ✓ Fossa of Rosenmuller
- ✓ Sinus of Morgagni
- ✓ Passavant's Ridge
- ✓ Functions of Nasopharynx
- **ADENOIDS**
  - ✓ Differences between Palatine Tonsils and Adenoids

OROPHARYNX
- ✓ Communications, Boundaries, Oropharyngeal Isthmus, Soft Palate, Base of Tongue, Lingual Tonsils, Valleculae, Faucial Tonsil, Anterior and Posterior Pillar
- ✓ Functions of Oropharynx
- **PALATINE (FAUCIAL) TONSILS**
  - ✓ Site and Extension, Surfaces and Poles, Tonsillar Crypts, Capsule, Bed of the Tonsil
  - ✓ Styloid Process

LARYNGOPHARYNX
- ✓ Pyriform Sinus (Fossa)
- ✓ Postcricoid Region
- ✓ Posterior Pharyngeal Wall
- ✓ Functions
- ✓ Hypopharyngeal Diverticulum

ESOPHAGUS
- ✓ Constrictions
- ✓ Structure
- ✓ Esophageal Sphincters: Upper and Lower

PHYSIOLOGY OF SWALLOWING
- ✓ Gastroesophageal Reflux

EMBRYOLOGY
- ✓ Pharyngeal Apparatus
- ✓ Neural, Muscular, and Skeletal Elements Derived from Pharyngeal Arches and Myotomes
- ✓ Skeletal Muscles Innervated by Cranial Nerves
- ✓ First Arch Syndrome: Treacher Collins Syndrome, Pierre

Contd...

Robin Sequence
- ✓ Structures Derived from Pharyngeal Pouches and Grooves
- ✓ DiGeorge Sequence
- ✓ Ectopic Parathyroid or Thymus
- ✓ Branchial Fistula
- ✓ Branchial Cyst
- ✓ Tongue
- ✓ Face: Cleft Lip
- ✓ Palate: Cleft Palate

➠ **CLINICAL HIGHLIGHTS**

ORAL CAVITY

It extends from the lips to the junction of hard and soft palate and circumvallate papillae (Figs 1 and 2). The subsites of oral cavity categorized by the American Joint Committee on Cancer Staging (AJCC) are shown in Box 1.

- Hard palate is included in the oral cavity whereas soft palate is the part of oropharynx. The anterior two-third of tongue is the part of oral cavity while the posterior one-third of tongue is included in oropharynx.

Lips

Lips form the anterior boundary of the oral cavity.

- *Parts*: The two surfaces of lip, skin and mucosal, become continuous with one another round its red margin, which is called vermilion.
- *Vermilion border*: The site along which the two lips meet with one another is marked by a line that indicates transition from a dry vermilion to a moist vermilion.
  - *The dry vermilion* with a pattern of wrinkles has clear cut boundary line between it and the skin proper, patterned with Cupid's bow in upper lip and central fullness in the lower lip.
  - *The smooth wet vermilion* merges without obvious surface change with mucosa lining of lip. Mucosal surface has mucous glands and supporting tissue within the muscle and mucosa.
- *Epithelium*: The lip is covered with non-keratinized stratified squamous epithelium which is transparent and contains no hairs, sebaceous glands, or pigments. Therefore it appears red. On the vermilion border the mucosa is closely applied to the orbicularis muscle and the distance between the epithelium and muscle is just 2 mm.

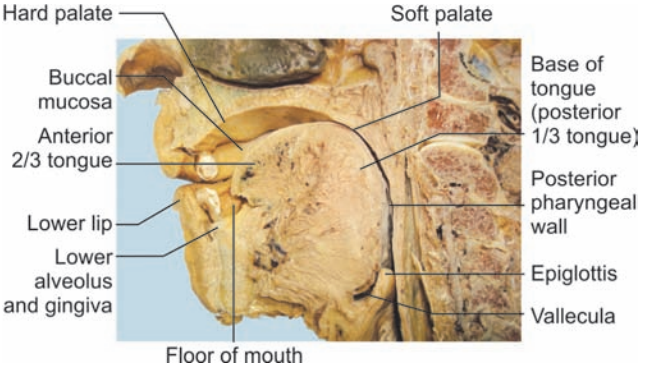


Fig. 1: Sagittal section head showing subsites of oral cavity and oropharynx

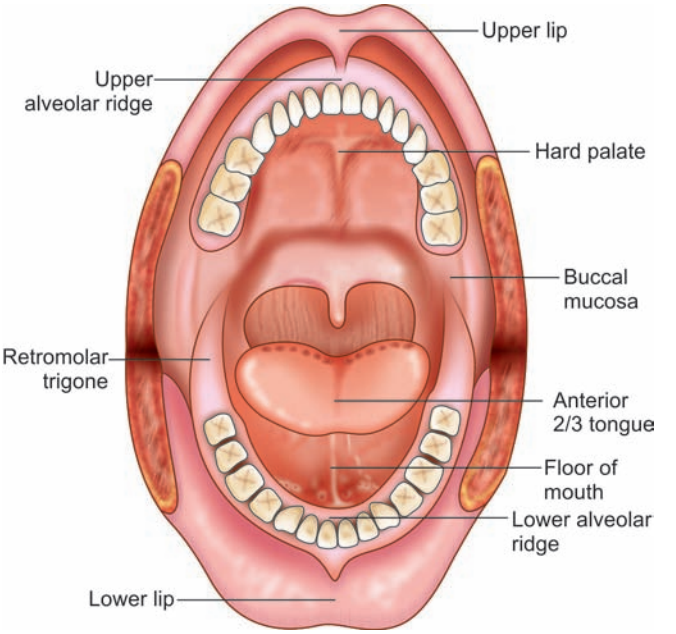


Fig. 2: Oral cavity overview

- An ulcerative squamous cell carcinoma lesion fixes the skin early to the deep substance of lip because of the 2 mm distance between the epithelium and muscle.
- Commissure of mouth: As far as possible, it should be spared during the surgery because it is very delicate and difficult to reconstruct.

- *Blood supply*: Small submental arteries and inferior and superior labial arteries which are branches of facial artery supply to lips.
- *Lymphatic drainage*:
  - *Lower lip*: Medial portion of lower lip drains into submental (level I) and lateral portion to submandibular nodes (level I).
  - *Upper lip*: It drains into preauricular, infraparotid and submandibular nodes.

Alveoli and Gingivae

Gums (gingivae) surround the teeth and cover the upper and lower alveolar ridges.

- *Lymphatic drainage*:
  - *Upper alveolus*:
    - Buccal aspect of mucosa drains to submandibular nodes.
    - Lingual aspect drains either directly to upper deep cervical or through the lateral retropharyngeal nodes.

Box 1: Subsites of cancer in oral cavity

1. **Lip (Vermilion surface)**
   a. Upper lip
   b. Lower lip
   c. Commissures
2. **Buccal mucosa (includes mucosa of cheek and inner surface of lips up to line of contact of opposing lip): Mucosal surfaces of**
   a. Upper and lower lips
   b. Cheeks
   c. Retromolar area
   d. Upper and lower buccoalveolar sulci
3. **Anterior two-third of tongue (anterior to circumvallate papillae)**
   a. Dorsal and ventral surfaces
   b. Lateral borders
   c. Tip
4. **Hard palate**
5. **Lower alveolar ridge (alveolus and gingiva)**
6. **Upper alveolar ridge (alveolus and gingiva)**
7. **Floor of mouth**
8. **Retromolar trigone**

Source: AJCC 1997

- *Lower alveolus*: The central part of both buccal and lingual surfaces of mucosa drains to submental whereas lateral parts drain to submandibular nodes.

Floor of Mouth (Fig. 3)

Floor of mouth is a crescent-shaped area that lies between the gingivae and undersurface of tongue. There are few minor salivary glands in the floor of mouth.

- *Examination*:
  - Anterior portion of the floor (frenulum, sublingual papillae with openings of submandibular ducts) is examined when patient raises the tip of tongue towards the hard palate.
  - Lateral portion of the U-shaped floor of mouth is examined by displacing the tongue in medial direction with the help of a tongue depressor.

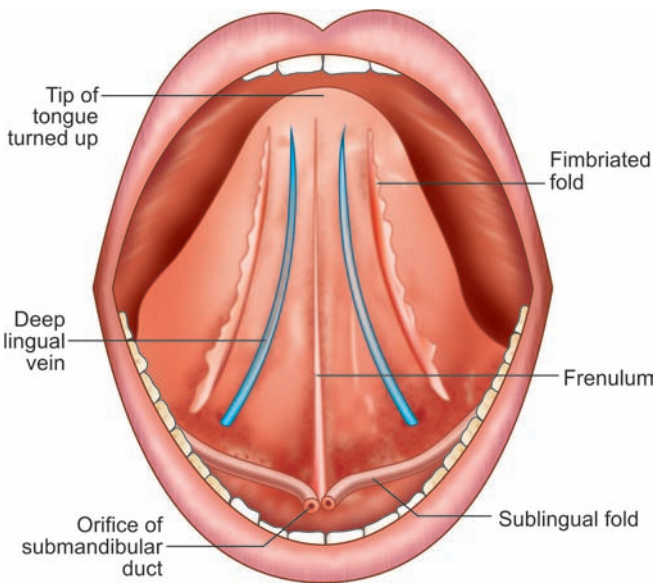


Fig. 3: Inferior surface of tongue and floor of mouth

- *Lymphatic Drainage*:
  - Anterior portion of floor of mouth drains into submental and submandibular nodes (level I). Lymphatics from this area also cross the midline.
  - Posterior portion drains into upper deep cervical nodes (level II and III).

Buccal Mucosa

It is covered with non-keratinizing stratified squamous epithelium. Buccal mucosa lines the inner surface of cheeks and lips and extends between the superior and inferior gingivobuccal sulcus. It extends posteriorly up to pterygomandibular raphe and anteriorly to the meeting line of lips. It covers parotid duct, minor salivary glands, buccinator muscle and upper and lower alveoli.

- *Lymphatic Drainage*: Buccal mucosa drains into submental and submandibular nodes and from there to the deep cervical lymph nodes.

Carcinoma of buccal mucosa is the second most common cancer of oral cavity after the tongue.

Retromolar Trigone

Retromolar trigone is a triangular area of mucosa that covers anterior border of the ascending ramus of mandible. It lies behind the third molar, where the pterygomandibular raphe is attached to the mandible. Its base lies posterior to the last molar while its apex is adjacent to the maxillary tuberosity. The pterygomandibular raphe provides origin to superior constrictor and buccinator muscles.

Hard Palate

Hard palate forms the roof of the oral cavity and floor of the nasal cavity. It contains high number of minor salivary glands.

In hard palate, minor salivary glands tumors such as adenoid cystic, mucoepidermoid and adenocarcinoma are more common than squamous cell carcinoma. Most of the squamous cell carcinoma of upper alveolus and hard palate arises from maxillary antrum.

- *Lymphatic drainage*:
  - Anterior part of palate drains to submandibular nodes.
  - Posterior part drains either directly to the upper deep cervical or via lateral retropharyngeal nodes.

Tongue

Anterior two-third of tongue is the content of the oral cavity (Fig. 4). Posterior one-third (base of tongue) is situated behind the circumvallate papillae and forms anterior wall oropharynx.

- *Parts*: Oral tongue can be divided into the tip, lateral borders and dorsum and ventral surfaces of the tongue.

- The interlacing tongue muscle fibers and constant movement of tongue (speech and chewing) facilitate spread and wide dissemination of tongue cancer.
- The palpation of tongue tumors and excision of cancer lesion with 2 cm safe margin are of paramount importance.

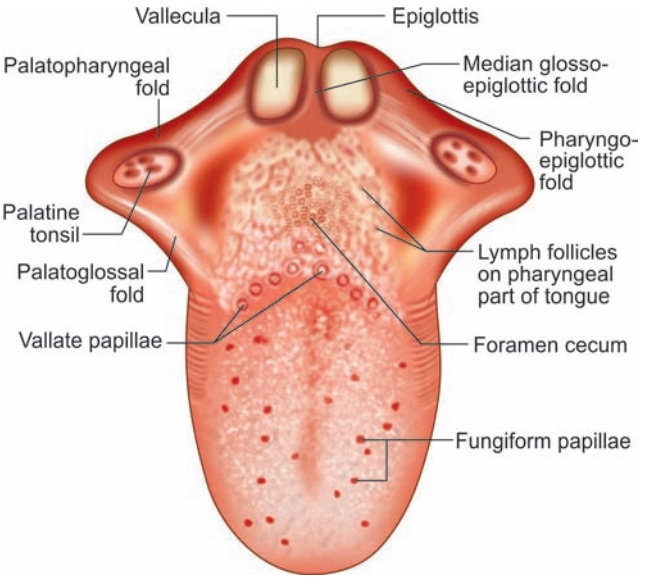


Fig. 4: Dorsum of the tongue and oropharynx

- *Lymphatic Drainage*:
  - Tip of tongue drains to submental and deep jugular cervical chain.
  - Sublingual surface drains to the submandibular nodes.
  - The rest of the anterior two-third of tongue drains to the deep jugular chain of lymph nodes. These level III nodes are situated between digastric and omohyoid muscles.
  - Central portion and base drain into deep cervical nodes of both sides.

Lymphatics from anterior portion of mouth can sometimes drain directly to lower jugular chain (levels III and IV).

The patients with cancer of lateral border of anterior two-third can have enlargement only of jugulodigastric and jugulomohyoid lymph nodes.

Pre-Facial and Post-Facial Lymph Nodes

These lymph nodes are present near the anteroinferior angle of masseter muscle where facial artery crosses the inferior margin of mandible. Prefacial and post-facial lymph nodes are often affected in cancer of mouth patients and should be removed along with the radical or modified radical neck dissection.

Taste Buds

They are present in tongue and palate and innervated by facial (anterior two-third tongue), glossopharyngeal (posterior one-third tongue) and vagus nerve (vallecula and epiglottis). Figure 5 shows neural pathways of taste sensation.

SALIVARY GLANDS

Salivary glands (Fig. 6) are categorized into two groups – major and minor:

1. Major: Major salivary glands are parotid and submandibular glands.
2. Minor: Among the minor salivary glands the largest is sublingual gland.

Development

The major salivary glands develop from ectoderm. The development begins during the 6th week of gestation as solid, ridge like ingrowths of oral epithelium, which continues to develop into tubules. These tubules later become the ductal system of the salivary glands.

Structure

Both serous and mucous cells are arranged into acini. They are drained by a series of ducts. An intercalated duct drains into a striated duct, which further opens into an excretory duct. Contractile myoepithelial cells, which surround both the acini and intercalated ducts, help in draining saliva through the ductal system.

- *Types of acini:* The parotid gland is composed of mainly serous acini while the sublingual and minor salivary glands are composed of predominantly mucinous acini. The submandibular gland consists of a mixture of serous and mucinous acini.

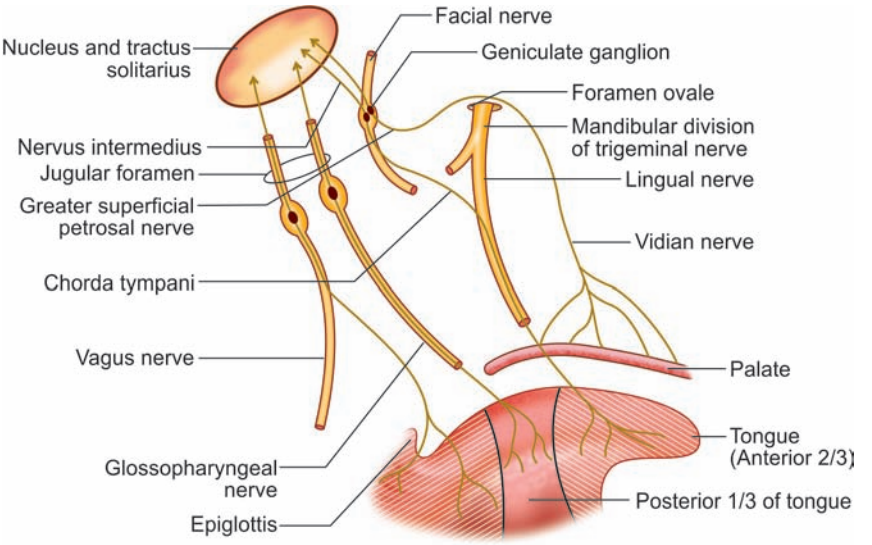


Fig. 5: Neural pathways of taste sensation

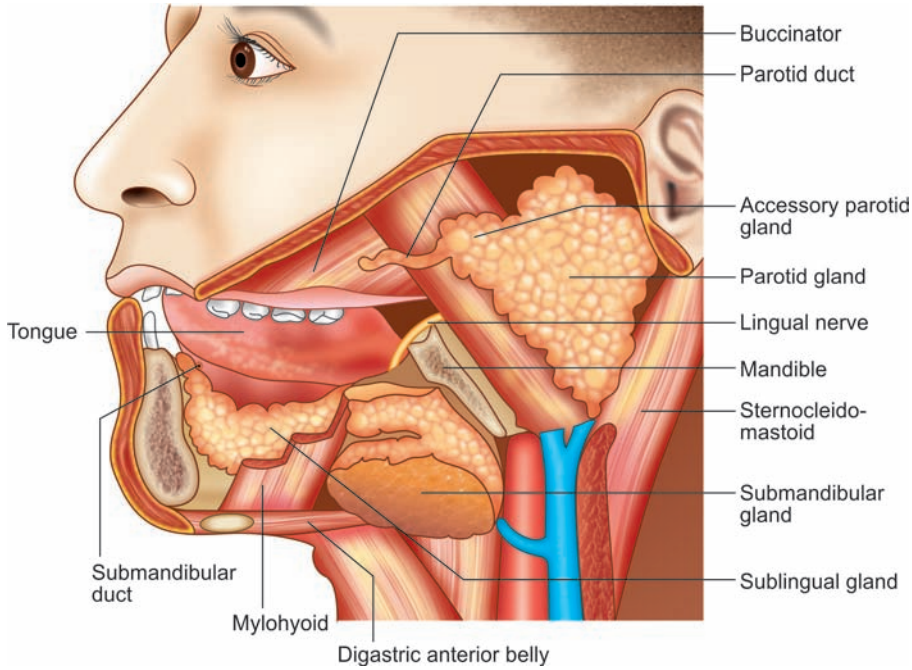


Fig. 6: Salivary glands (parotid, submandibular and sublingual) and their relations

Functions of Saliva

Mucin, a lubricating agent, which is secreted mainly by sublingual and minor salivary glands, is essential for speech, mastication and swallowing. Seromucous substance protects against proteolytic and hydrolytic enzymes produced in plaque, chemical agents and desiccation.

Parotid Gland (Fig. 7)

It is situated in the surrounding region of external auditory canal. This recess like region is bounded by ramus of mandible, external auditory canal and mastoid process. Its upper part extends just below zygoma and its lower part (tail of parotid) to the sternomastoid muscle. It extends forward over the masseter. Its capsule is derived from the investing layer of deep cervical fascia. Several lymph nodes are situated superficial (6–8 lymph nodes) to the gland and within the gland (10–12 lymph nodes). The intraparotid nodes are largely in the superficial lobe whereas deep lobe has very few nodes.

Facial nerve arbitrarily divides the gland into superficial (80%) and deep (20%) lobes. An accessory lobe superior to the parotid duct is present in some cases.

Its excretory Stensen's duct emerges anteriorly and runs over the masseter muscle and turns medially over the anterior margin of masseter, perforates the buccinator muscle and subsequently opens into the vestibule of mouth opposite the crown of upper second molar tooth.

The deep retromandibular lobe lies in the parapharyngeal space anterior to the carotid sheath (containing internal carotid artery, internal jugular vein and vagus nerve) and styloid process and posterior to infratemporal fossa. Medially superior constrictor muscle separates deep lobe from the tonsil and oropharynx. Medial to the facial nerve, following structures run through the deep parotid lobe:

1. External carotid artery and its terminal branches maxillary and superficial temporal arteries.
2. Maxillary and superficial temporal veins and their union retromandibular vein along with its anterior and posterior divisions.

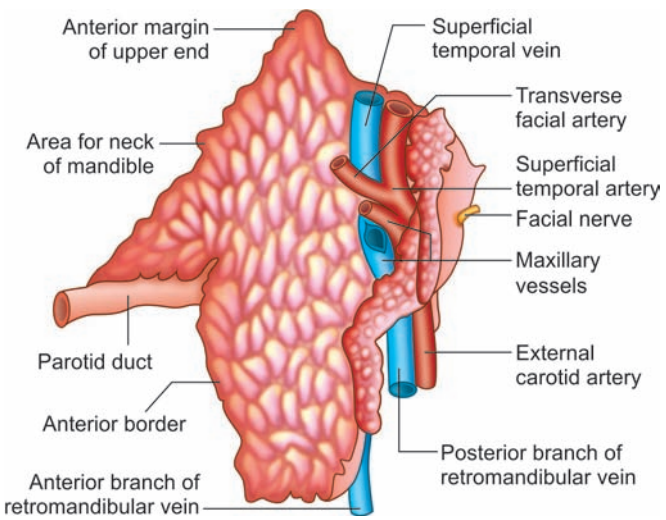


Fig. 7: Anteromedial surface of parotid gland

Surgical Landmarks of Facial Nerve during Parotid Gland Surgery

Facial nerve emerges from the stylomastoid foramen and enters the parotid through its posteromedial surface. Immediately after entering the parotid, the facial nerve divides into its two terminal divisions: upper temporozygomatic (giving temporal and zygomatic branches) and lower cervicofacial (giving mandibular and cervical branches). Buccal branch can arise from either upper or lower divisions. It is at great risk during the parotid surgery. Following landmarks help in identifying the facial nerve in the parotid gland:

- *Tragal cartilage pointer*: The tragal cartilage ends on a point. It is a sharp triangular extension of tragal cartilage, which seems to point towards facial nerve. Facial verve lies 1–1.5 cm medial and inferior to tragal point surrounded by a small aggregation of fat and overlain by a small vessel.

- *Tympanomastoid suture*: Facial nerve lies 6–8 mm deep to the suture.
- *Styloid process*: Facial nerve lies on the posterolateral aspect of the styloid process near its base.
- *Posterior belly of digastric*: Follow the posterior belly of digastric up to 5 mm below the bony meatal edge. The facial nerve lies between the mastoid and the posterosuperior part of the posterior belly of digastric muscle. The facial nerve passes downwards, forwards and laterally immediately above the upper border of digastric posterior belly.
- *Mastoid process*: Follow the anterior border of mastoid process up to the vaginal process of tympani bone. Facial nerve bisects the tympanomastoid angle at the tympanomastoid suture.
- *Peripheral branches*: The following branches may be followed proximally:
  - Temporal: It bisects a line drawn from tragus to lateral canthus of eye.
  - Buccal: It runs 1 cm above and parallel to Stensen's duct over the masseter.
  - Ramus mandibularis: It travels superficial to the facial vessels 2 cm below inferior border of mandible and 1 cm anterior to angle of mandible.

Parotid tumors may extend into the lateral parapharyngeal space by passing either medial or lateral to stylomandibular ligament, which is derived from the investing layer of deep cervical fascia and also separate parotid from submandibular gland.

Submandibular Salivary Glands

These paired salivary glands lie below the lateral part of body of mandible and fills the major portion of the submandibular triangle.

- *Lobes*: They have two lobes 1) larger superficial, and 2) smaller deep. Both the lobes are continuous around the posterior margin of mylohyoid muscle. The superficial lobe lies on the mylohyoid muscle and is bounded anteriorly by the anterior belly of digastric. The deep lobe (Fig. 8) lies on the hyoglossus muscle and terminate in an excretory duct.
- *Facial vessels*: The facial vein runs over the surface of the gland whereas facial artery has complex relation and may be embedded in the gland tissue.
- *Nerves*: The deep lobe of the gland lies on the hyoglossus muscle and closely related to the lingual nerve, submandibular parasympathetic ganglion and hypoglossal nerve. Marginal mandibular branch of facial nerve which runs near the lower border of mandible, lies superficial to capsule and facial vessels. It is at risk during the excision of submandibular gland.
- *Capsule*: The investing layer of deep cervical fascia splits and forms the capsule of the gland.
- *Duct:* Its excretory Wharton's duct emerges from the deep lobe and runs over the hyoglossus muscle between the lingual nerve (superiorly) and hypoglossal nerve (inferiorly). The duct drains into the floor of mouth at the sublingual papilla lateral to the tongue frenulum.
- *Lymph nodes*: Several lymph nodes are situated near the gland and within the superficial lobe of gland. The extra glandular nodes are present in front of and behind the facial artery at the edge of mandible close to the ramus mandibularis of the facial nerve.

The common disorders of submandibular salivary glands include sialadenitis and sialothiasis (stone formation).

Sublingual Salivary Glands

These paired minor salivary glands are present in the anterior part of floor of mouth. They are situated between the mucosa,

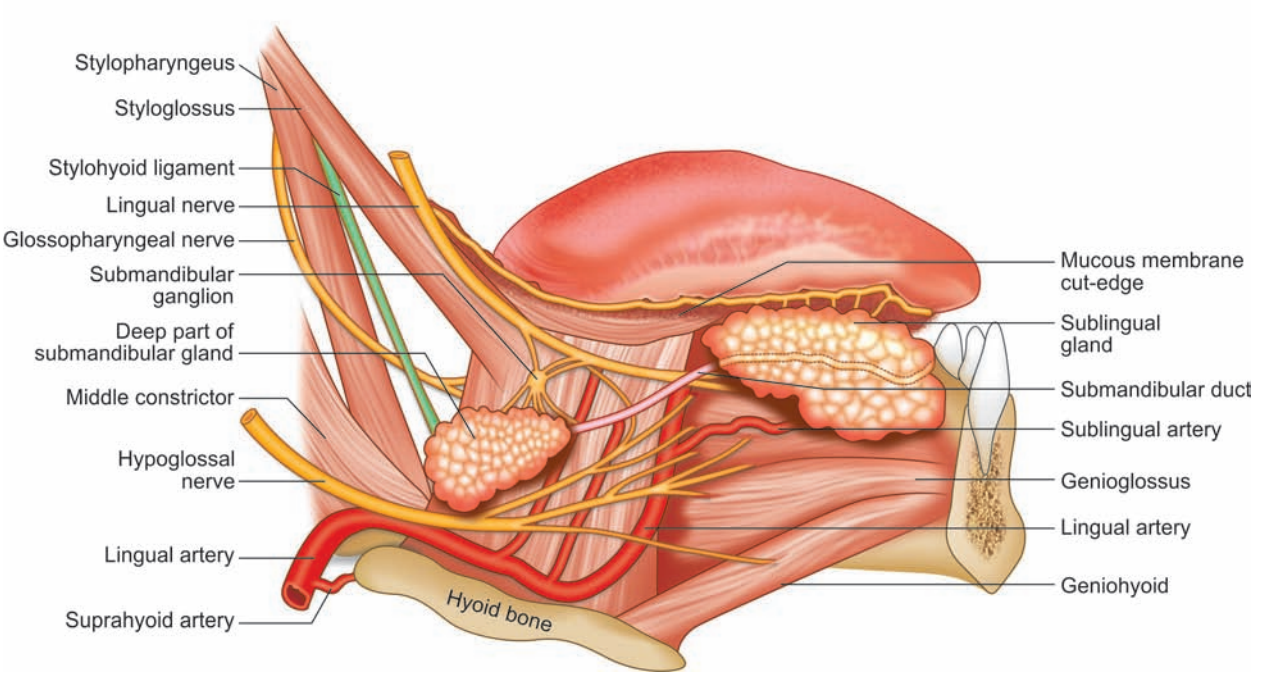


Fig. 8: Relations of the deeper part of submandibular gland after removal of mylohyoid muscle

mylohyoid muscle and anterior part of body of mandible. They have multiple excretory ducts which open either directly into the floor of mouth or drain into the Wharton's duct of submandibular gland.

The common disorder of sublingual salivary gland is ranula. It is a mucous extravasation cyst that takes an appearance of frog's belly (ranula).

Minor Salivary Glands

There are about 450 minor salivary glands in oral cavity situated in the mucosa of lips, cheeks, palate (250 on hard palate, 100 on soft palate and 10 on uvula), floor of mouth and retromolar region. They are present though in small numbers in sinuses, oropharynx, larynx and trachea.

The highest number (250) of minor salivary glands is situated in hard palate. The palatal glands are found in the posterolateral part especially gingivae of molar teeth.

They have mucous secreting acini and contribute to 10% of total salivary secretion.

The common disorders of minor salivary glands include mucous retention cyst of lower lip and hard palate tumors. The latter are usually (90%) malignant.

PHARYNX

Pharynx is a 12–14 cm long conical tube which forms upper part of the air and food passages. It extends from basiocciput and basisphenoid to the lower border of cricoid cartilage. Inferiorly, it becomes continuous with the esophagus through pharyngoesophageal junction (1.5 cm width), which is the narrowest part of gastrointestinal tract apart from the appendix. The width of pharynx base, which faces towards skull base, is 3.5 cm. Pharynx is traditionally divided into three parts: nasopharynx, oropharynx and hypopharynx (Fig. 9).

Pharyngeal Wall (Fig. 10)

From within outwards it consists of four layers mucous membrane, pharyngeal aponeurosis (pharyngobasilar fascia), muscular coat and buccopharyngeal fascia.

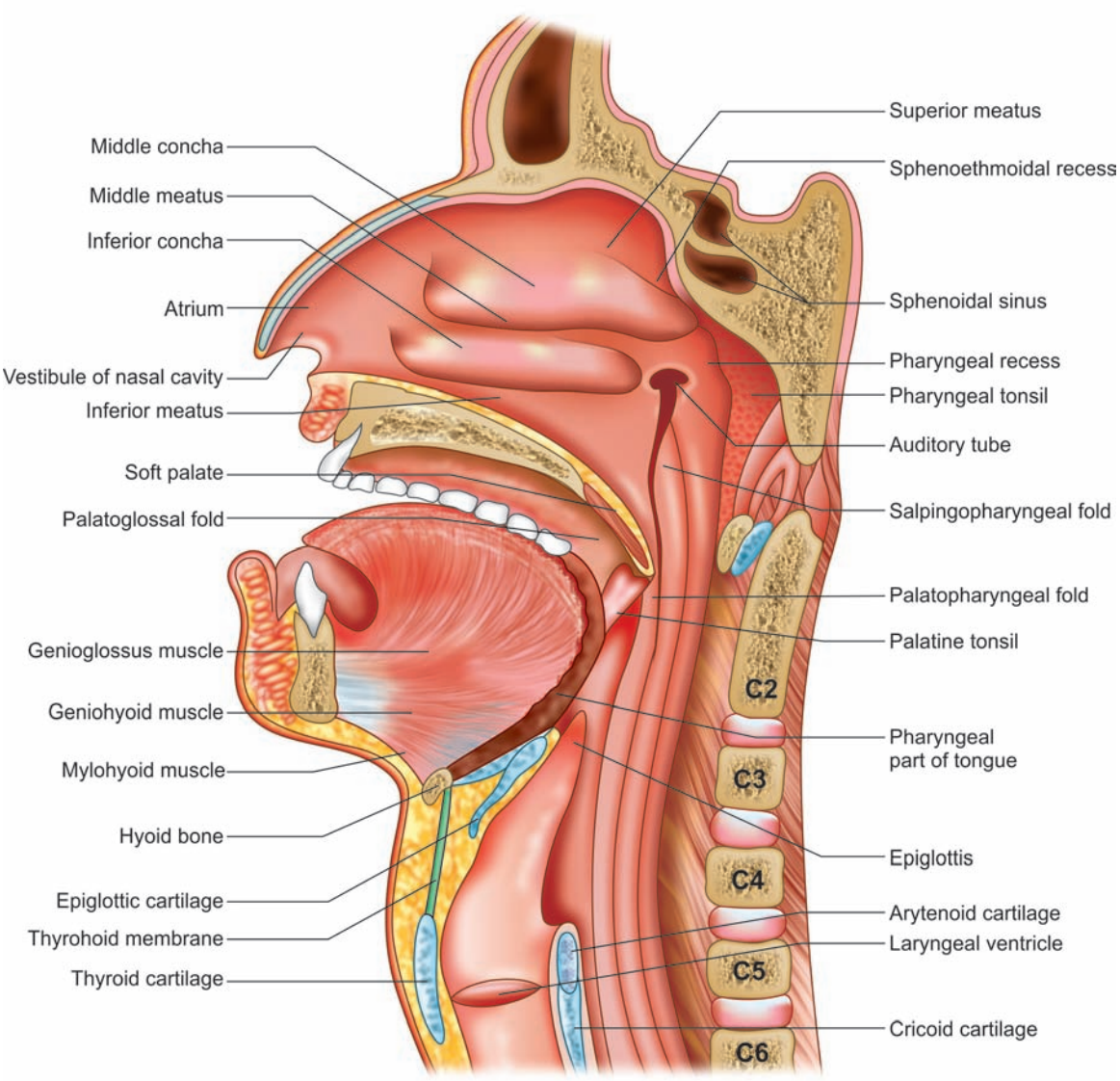


Fig. 9: Paramedian section of nose, mouth, pharynx and larynx

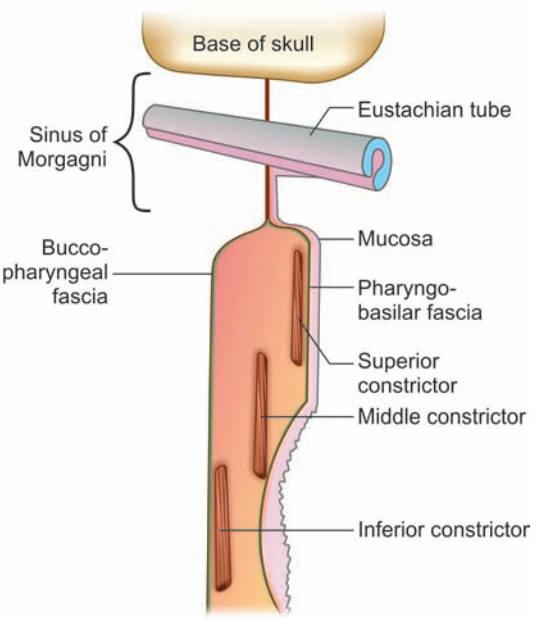


Fig. 10: Pharyngeal wall. Coronal section

- *Mucous membrane*: The pharyngeal mucosa, which is ciliated columnar in the nasopharynx and stratified squamous elsewhere, is continuous with mucous membrane of Eustachian tubes, nasal cavities, mouth, larynx and esophagus. There are numerous mucous glands.
- *Pharyngeal aponeurosis (pharyngobasilar fascia):* This fibrous layer lines the muscular coat and fills up the gap left in the muscular layer especially near skull base. It is thick between the superior border of superior constrictor muscle and base of skull and becomes thin and indistinct inferiorly.
- *Muscular coat*: It consists of two layers of muscles—external and internal
  - *External layer*: It consists of 3 horizontal circular muscles—superior, middle and inferior constrictor muscles (Fig. 11).
  - *Internal layer*: It consists of 3 vertical muscles stylopharyngeus, salpingopharyngeus and palatopharyngeus muscles.
- *Buccopharyngeal fascia*: This fascial layer covers outer surface of the constrictor muscles. In the upper part, it extends anteriorly and covers the buccinator muscle. Above the superior constrictor, it merges with pharyngeal aponeurosis.

Pharyngeal Spaces

The pharyngeal spaces where abscesses can form include retropharyngeal and parapharyngeal spaces (See chapter Deep Space Neck Infections).

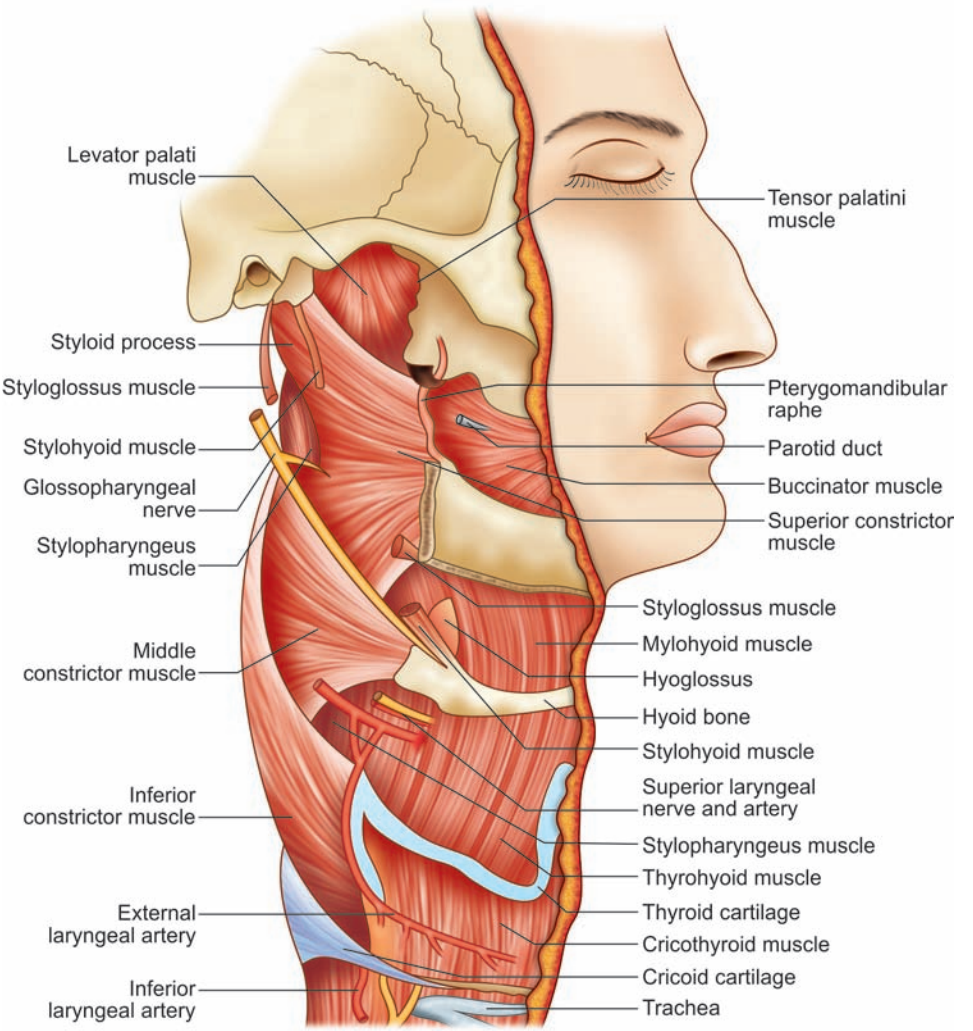


Fig. 11: Lateral view of pharynx showing constrictor muscles with their attachments and relations

Killian's Dehiscence

This potential gap lies between the two parts of inferior constrictor muscle: oblique fibers of thyropharyngeus and transverse fibers of cricopharyngeus. This weak area is common site for following:

- Perforation: Perforation during esophagoscopy. This gap's also called "gateway of tears."
- Herniation: Herniation of pharyngeal mucosa in cases of Zenker's diverticulum (pharyngeal pouch).

WALDEYER'S RING (FIG. 12)

Mucosa associated lymphoid tissue (MALT) in subepithelial layer of pharynx is aggregated at the entrance of aerodigestive tract and are collectively called Waldeyer's ring. Waldeyer's ring consists of:

1. Adenoids or Nasopharyngeal tonsil
2. Tubal tonsils (Fossa of Rosenmuller)
3. Lateral pharyngeal bands
4. Palatine tonsils
5. Nodules (Posterior pharyngeal wall).
6. Lingual tonsils

Functions of Tonsils and Adenoids

- *Immunology and host defenses*: T-lymphocytes in parafollicular region provide cell-mediated immunity against various viruses, bacteria and fungi. Once the pathogens enter into these lymphoid aggregations they are dealt by IgM and IgG antibodies, which are produced by plasma cells.
- *Sentinels at the portal of aerodigestive tract*: Oropharyngeal subepithelial masses of lymphoid tissues, which form the part of Waldeyer's ring, act as protective sentinels against harmful intruders into the air and food passages. The crypts in tonsils increase the surface area for contact with foreign substances.
- *Antibody production especially secretory IgA*: B-lymphocytes in the germinal centers of these lymphoid follicles produce antibodies IgA.

NASOPHARYNX

Nasopharynx is the uppermost part of the pharynx and lies behind the nasal cavities (Fig. 13).

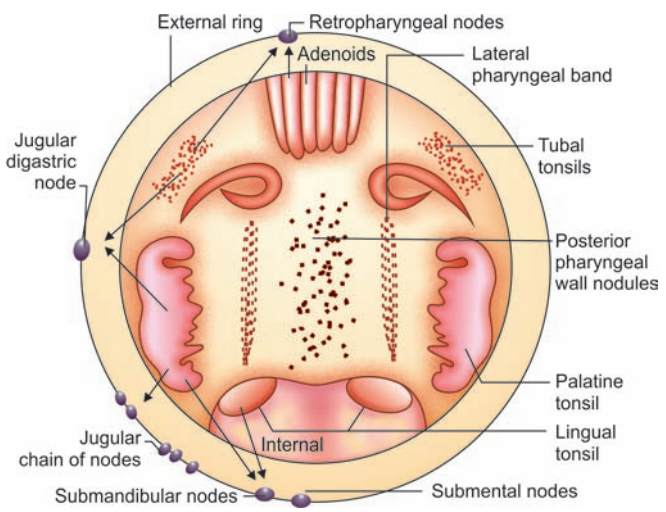


Fig. 12: Waldeyer's ring

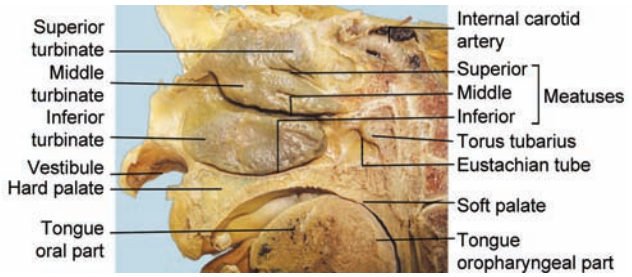


Fig. 13: Sagittal section of head showing lateral wall of nasal cavity and nasopharynx

- *Extension*: It extends from the base of skull to the soft palate (level of hard palate and atlas).
- *Epithelium*: Functionally, nasopharynx is the posterior extension of nasal cavity and is lined with pseudostratified ciliated columnar epithelium.
- *Communications*: It communicates with nasal cavity, middle ear cavity and oropharynx.

Boundaries

- *Roof and the posterior wall*: They imperceptibly merge with each other. Roof is formed by basisphenoid and basiocciput. Posterior wall is formed by the arch of the atlas vertebra that articulates with dense of axis and covered by prevertebral muscles and fascia.
  - *Nasopharyngeal tonsil (adenoids):* A subepithelial collection of lymphoid tissue, which is present at the junction of roof and posterior wall of nasopharynx, throws the overlying mucous membrane into radiating folds. They increase in size up to the age of 6 years and after that gradually atrophies. (See the chapter Adenotonsillectomy).
  - *Nasopharyngeal bursa*: This epithelial lined median recess, which is seen within the adenoid mass, extends from pharyngeal mucosa to periosteum of basiocciput. It represents the embryonic site of attachment of notochord to the pharyngeal endoderm.

Tornwald's disease: This is an abscess that develops in nasopharyngeal bursa. Infected bursa usually presents with persistent postnasal discharge or crusting.

  - *Rathke's pouch*: This dimple above the adenoids (junction of nasal septum and roof of nasopharynx) is the reminiscent of the buccal mucosal invagination, which forms the anterior lobe of pituitary.

A craniopharyngioma may arise from Rathke's pouch.

- *Floor*: Anteriorly soft palate; posteriorly communicates with the oropharynx through nasopharyngeal isthmus. It forms the roof of oropharynx.
- *Anterior wall*: It communicates with nasal cavities through posterior nasal apertures (choanae), which are separated from each other by the posterior border of the nasal septum. Posterior ends of nasal turbinates and meatuses can be seen through posterior choanae.
- *Lateral wall*. Structures seen on lateral wall are following:
  - *Pharyngeal opening of Eustachian tube*: It is situated 1.25 cm behind the posterior end of inferior turbinate.
  - *Torus tubarius*: It is an elevation, which is raised by the cartilage of Eustachian tube and bounds the Eustachian tube above and behind.

- *Tubal tonsil*: This collection of subepithelial lymphoid tissue is continuous with adenoid tissue and forms a part of the Waldeyer's ring. It is situated at the tubal elevation.

Enlarged or infected tubal tonsils can affect Eustachian tube.

- *Fossa of Rosenmuller*: It is a recess that is situated above and behind the tubal elevation.

Fossa of Rosenmuller is a common site for the nasopharyngeal carcinoma.

- *Salpingopharyngeal fold*: It is raised by the salpingopharyngeus muscle and extends from the lower end of the torus tubarius to the lateral pharyngeal wall.

Sinus of Morgagni

This is a space which lies between the base of the skull and upper free border of superior constrictor muscle. Following structures pass through this space:

1. Eustachian tube
2. Levator veli palatini muscle, and
3. Ascending palatine artery, which is a branch of the facial artery.

Passavant's Ridge

This mucosal ridge, which encircles the posterior and lateral walls of nasopharyngeal isthmus, is raised by fibers of palatopharyngeus muscle. When soft palate contracts during deglutition or speech, it makes firm contact with Passavant's ridge and cut off nasopharynx from the oropharynx.

Lymphatic Drainage

Lymphatic of the nasopharynx and the structures present on its boundaries drain into:

- Retropharyngeal and parapharyngeal lymph nodes.
- Upper deep cervical nodes either directly or indirectly through retropharyngeal and parapharyngeal lymph nodes.
- Spinal accessory chain of nodes: They are present in the posterior triangle of the neck.
- Contralateral cervical lymph nodes: Nasopharyngeal lymphatic may cross midline.

Functions of Nasopharynx

- *Airway*: A conduit for air to its way to the larynx and trachea.
- *Middle ear ventilation*: Eustachian tube ventilates the middle ear and equalizes air pressure on both sides of tympanic membrane. Dysfunctions of Eustachian tube can affect middle ear.
- *Resonance*: It is a part of resonating chamber for voice production. Nasopharyngeal obstruction and velopharyngeal incompetence produce changes in voice.
- *Drainage*: It is a drainage channel for nasal and nasopharyngeal mucus secretions.
- *Nasopharyngeal isthmus*: It cuts off nasopharynx from oropharynx during swallowing, vomiting, gagging and speech.

ADENOIDS

These nasopharyngeal tonsils, which are covered with ciliated columnar epithelium, are situated at the junction of the roof and posterior wall of the nasopharynx. Adenoids are composed of vertical ridges of lymphoid tissue, which are separated by deep clefts. Adenoids have no crypts and capsule whereas palatine tonsils have both capsules and crypts (Table 1). Adenoid tissue shows physiological enlargement during the first 6 years of life and then tends to regress and almost completely disappears by 20 years of age.

Blood Supply

Following branches of external carotid artery supply to adenoids:

1. Ascending palatine branch of facial artery.
2. Ascending pharyngeal artery.
3. Pharyngeal branch of the third part of maxillary artery.

Lymphatic Drainage

Adenoids drain into upper deep jugular nodes directly or through retropharyngeal and parapharyngeal nodes.

OROPHARYNX

This middle part of pharynx lies behind the oral cavity (Figs 14 and 15). It extends from the plane of hard palate above to the plane of hyoid bone below.

Communications

- *Anterior*: It communicates with oral cavity through oropharyngeal isthmus.
- *Superior*: It communicates with nasopharynx through nasopharyngeal isthmus.
- *Inferior*: It communicates with laryngopharynx.

Boundaries

1. *Posterior wall*: Retropharyngeal space that lies opposite the axis and upper part of the third cervical vertebra.
2. *Anterior wall*: The upper part of anterior wall communicates with oral cavity through oropharyngeal isthmus. The structures present in the lower part are: base of tongue, lingual tonsils and valleculae.
   a *Oropharyngeal isthmus*: It is bounded by following structures:
      i. *Above:* Soft palate
      ii. *Inferior*: Dorsal surface of tongue
      iii. *Lateral:* On either side, by palatoglossal arch (anterior tonsillar pillar)

TABLE 1 Differences between palatine tonsils and adenoids

	Adenoids	*Palatine Tonsils*
Number	Single	One on each side
Site	Nasopharynx	Tonsillar fossa in oropharynx
Crypts or Furrows	Only furrows	Only crypts
Capsule	Absent	Present
Epithelium	Ciliated columnar	Squamous stratified
In adults after 20 years of age	Absent	Present

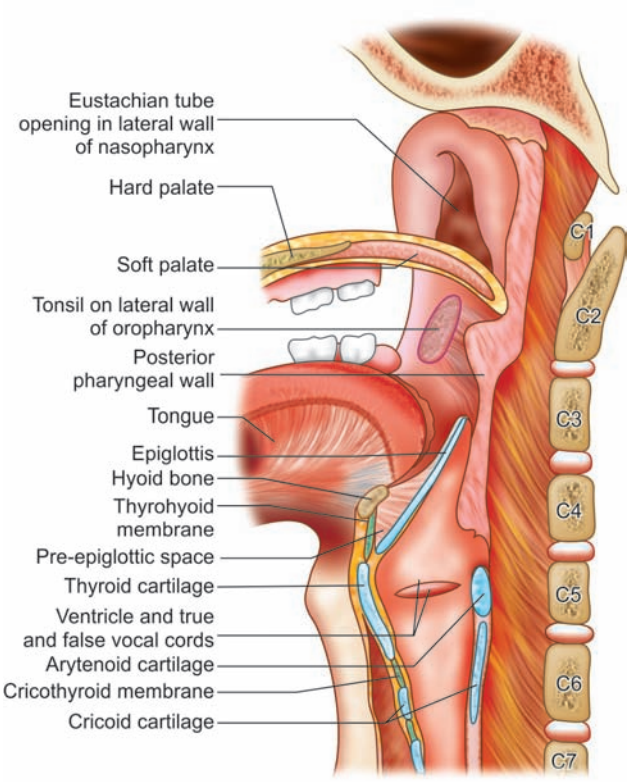


Fig. 14: Structure of pharynx sagittal section

b. *Base of tongue*: It lies posterior to circumvallate papillae and insertion of palatoglossal muscle.

c. *Lingual tonsils*: They are situated in the base of tongue.

Infection or compensatory enlargement of lingual tonsils may occur after tonsillectomy.

d. *Valleculae*: These cup-shaped spaces, one on each side, lie between the base of tongue and anterior surface of epiglottis. The median glossoepiglottic fold separates the two valleculae. Laterally they are bounded by the pharyngoepiglottic fold that is the upper limit of pyriform sinus of laryngopharynx.

Valleculae are common sites for retention cysts.

3. *Lateral wall*: Both anterior and posterior tonsillar pillars emerge from the soft palate and enclose tonsillar fossa, in which is situated the palatine tonsil. So the structures present in the lateral wall include:
   a. Palatine or faucial tonsil.
   b. Anterior pillar or palatoglossal arch: The palatoglossus muscle is present in this fold.
   c. Posterior pillar or palatopharyngeal arch: The palatopharyngeus muscle lies in this fold.
4. *Superior*: Anteriorly soft palate makes the roof of oropharynx. Posteriorly it communicates with nasopharynx through nasopharyngeal isthmus at the plane of hard palate and atlas vertebra.
5. *Inferior*: It communicates with laryngopharynx at the plane of upper border of epiglottis and the pharyngoepiglottic folds and third cervical vertebra.

Lymphatic Drainage (Fig. 16)

Oropharynx drains in the following lymph nodes:

- Upper jugular chain particularly the jugulodigastric (tonsillar) node.
- Retropharyngeal and parapharyngeal nodes: Soft palate, lateral and posterior pharyngeal walls and the base of tongue.
- Posterior cervical group.

Lymphatics of base of tongue drain bilaterally.

Functions of Oropharynx

- A common conduit for the passage of both air and food.
- Oropharyngeal phase of deglutition.
- Vocal tract for certain speech sounds.
- Taste: The base of tongue, soft palate, anterior pillars and posterior pharyngeal wall contain taste buds.
- Local defense and immunity.

PALATINE (FAUCIAL) TONSILS (FIGS 17 AND 18)

Site and Extension

Palatine tonsil is an ovoid mass of lymphoid tissue. It is situated in the lateral wall of oropharynx (tonsillar fossa) between the anterior (mucosal fold of palatoglossal muscle) and posterior

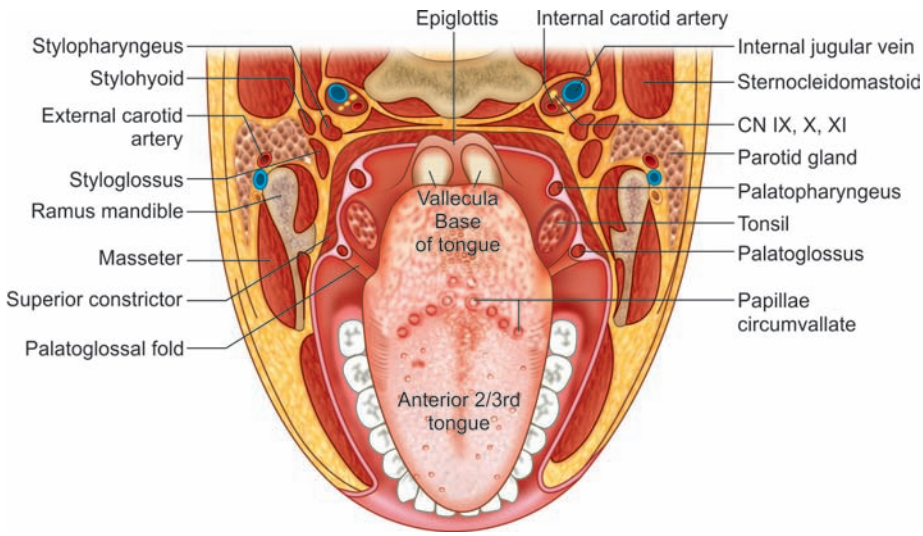


Fig. 15: Structures of oropharynx and their relations in axial section

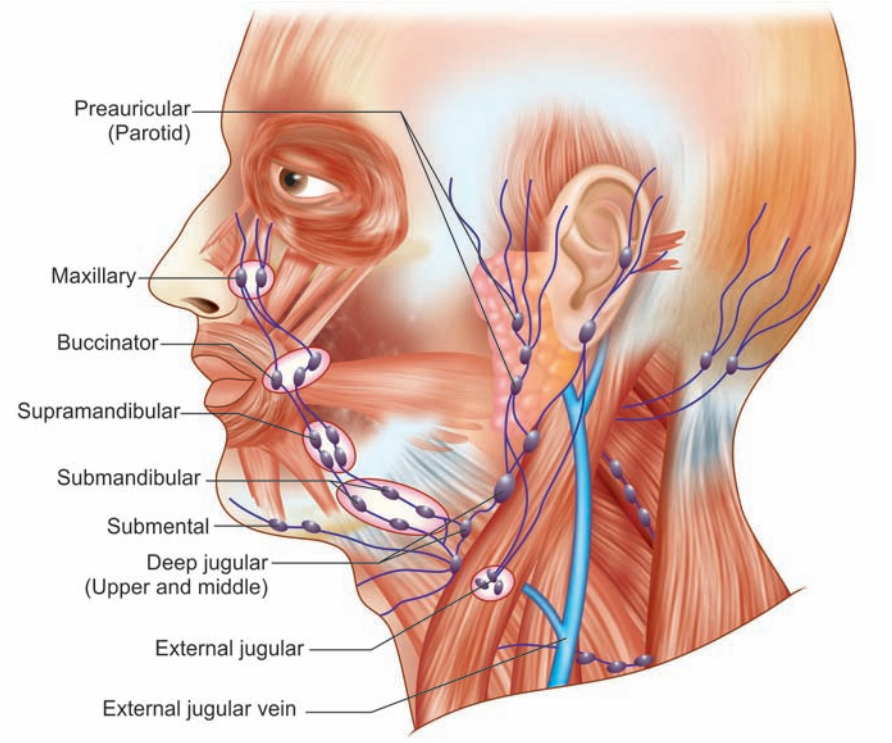


Fig. 16: Cervical lymph node chains involved in oropharyngeal tumors

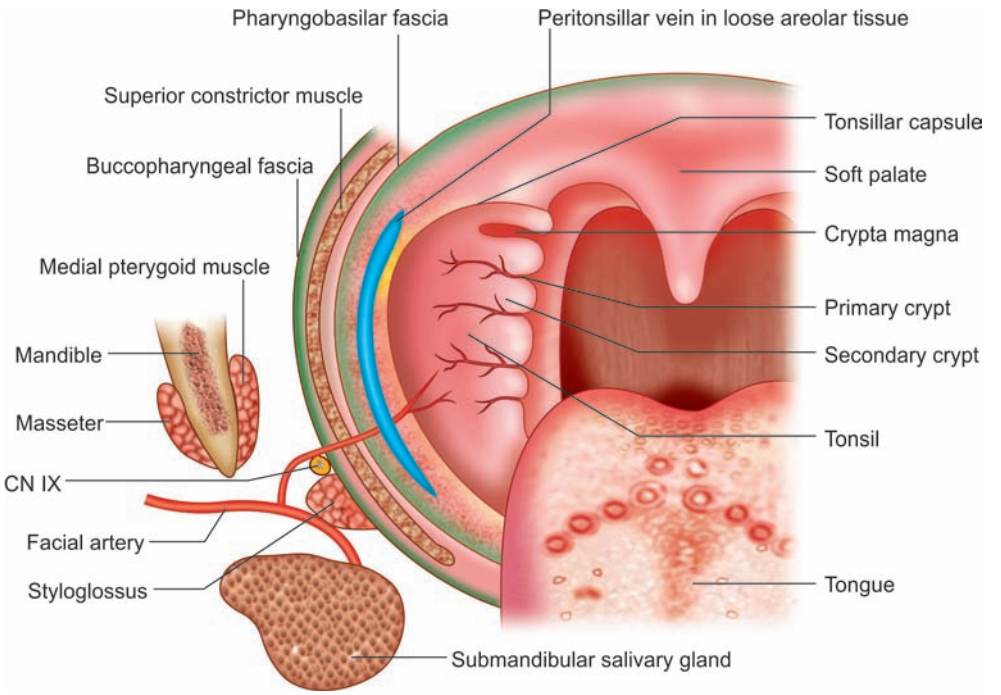


Fig. 17: Bed of tonsil

tonsillar pillars (mucosal fold of palatopharyngeal muscle). It extends superiorly into the soft palate, inferiorly into the tongue base and anteriorly into palatoglossal arch (formed by palatoglossal muscle).

Surfaces and Poles

A tonsil has two surfaces medial and lateral and two poles upper and lower. Surface epithelium of tonsil is continuous with the epithelium of oropharynx. Tonsillar crypts are tube like invaginations from the surface epithelium. Infections of tonsil can involve any of these components.

- *Supratonsillar fossa*: The medial surface of tonsillar upper pole has a semi-lunar fold, which extends between anterior and posterior pillars. This pole encloses a potential space called supratonsillar fossa.
- *Anterior tonsillar space*: At the lower pole, which is attached to the tongue, a triangular fold of mucous membrane extends from anterior pillar to the anteroinferior part of tonsil and encloses a space called anterior tonsillar space.

The tonsillolingual sulcus, which separates the tonsil from tongue, is a common site for carcinoma.

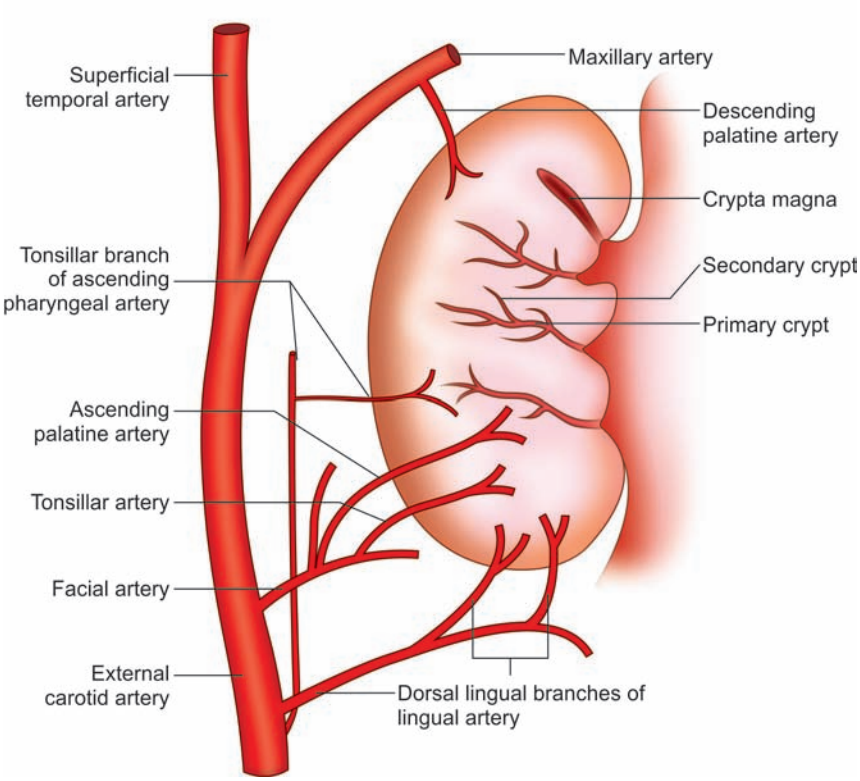


Fig. 18: Blood supply and crypts of tonsil

Tonsillar Crypts

The nonkeratinizing stratified squamous epithelium on medial surface of tonsil dips into the tonsillar mass and forms crypts (12–15). Openings of crypts can be seen on the medial surface of the tonsil.

- *Crypta magna or intratonsillar cleft*: It is situated near the upper part of tonsil. It is very large and deep and represents the ventral part of second pharyngeal pouch.
- *Secondary crypts*: They arise from the main crypts within the substance of tonsil.
- *Content*: Crypts may be filled with cheesy material, which consists of epithelial cells, bacteria and food debris and can be expressed out with pressure over the anterior tonsillar pillar.

Tonsils may be larger in childhood and usually regress in size near puberty. If they become the seat of disease, they may be removed.

Capsule

Lateral surface of tonsil is covered by a well-defined fibrous capsule, which is separated from the bed of tonsil by loose areolar tissue that allows easy dissection in this plane during tonsillectomy. In this same plane occurs the peritonsillar abscess. Some fibers of palatoglossus and palatopharyngeus muscles are attached to tonsillar capsule.

Bed of the Tonsil (Fig. 17)

The superior constrictor, glossopharyngeal nerve and stylo-glossus muscle form the bed of tonsil. Lateral to the superior constrictor muscles lies facial artery, submandibular salivary gland, posterior belly of digastric muscle, medial pterygoid muscle and the angle of mandible.

Styloid process: The styloid process when enlarged may be palpated intraorally in the lower part of tonsillar fossa. The glossopharyngeal nerve and styloid process can be approached through the tonsil bed after tonsillectomy.

Blood Supply (Fig. 18)

- *Arterial supply*: The main artery of tonsil is tonsillar branch of facial artery, which arises from external carotid artery. The other vessels, which supply the tonsil include:
  - Ascending pharyngeal artery from external carotid.
  - Ascending palatine, a branch of facial artery.
  - Dorsal lingual branch of lingual artery.
  - Descending palatine branch of maxillary artery.

The ascending pharyngeal, facial, lingual and maxillary arteries are all branches of external carotid artery that may need to be ligated in cases of refractory bleeding after tonsillectomy.

- *Venous drainage*: Veins from the tonsils drain into paraton-sillar veins, which are present on lateral surface of tonsil and drain into the common facial vein and pharyngeal venous plexus.

Lymphatic Drainage

Palatine tonsils drain into the jugulodigastric (tonsillar) nodes of upper deep cervical group, which are situated below the angle of mandible.

Nerve Supply

The sensory nerve supply to tonsil is from:

- Lesser palatine branches from sphenopalatine ganglion of maxillary division of trigeminal nerve (CN V_2).
- Glossopharyngeal nerve (CN IX).

LARYNGOPHARYNX

Hypopharynx, which is the lowest part of the pharynx, lies behind and partly lateral to the larynx (Figs 19 and 20). It extends superiorly from the plane of hyoid bone to the lower border of cricoid cartilage. Posteriorly, it lies on the 3rd, 4th, 5th, 6th cervical vertebrae.

Communications

- *Superior*: It is continuous with oropharynx at the level of hyoid bone.
- *Inferior*: It becomes continuous with esophagus at the level of lower border of cricoid cartilage and 6th cervical vertebra.
- *Anterior*: It communicates with larynx through the laryngeal inlet, which is bounded by the epiglottis, aryepiglottic folds and arytenoids.

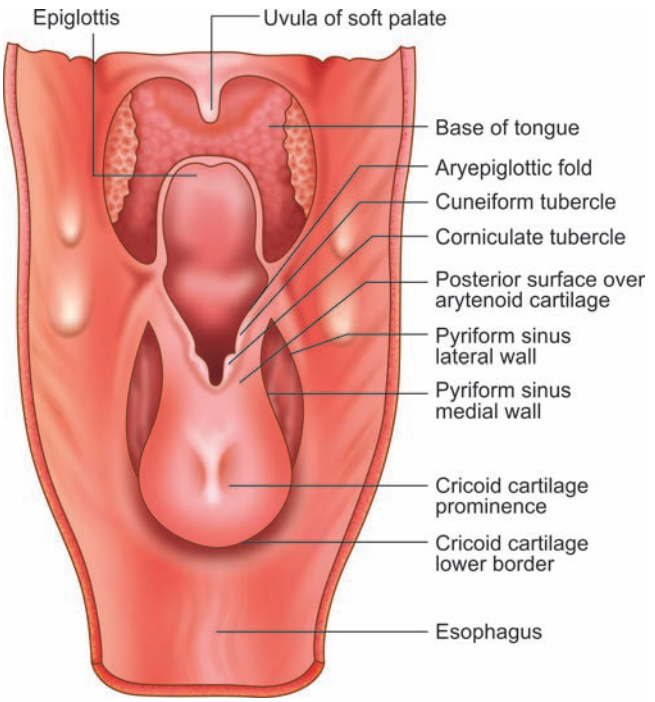


Fig. 19: Structures of hypopharynx. Posterior view of laryngopharynx

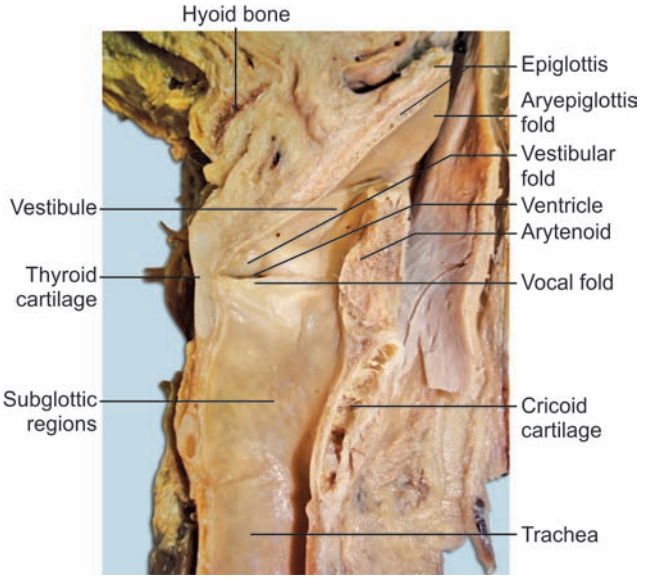


Fig. 20: Sagittal section of larynx and laryngopharynx

Hypopharynx Subsites

It is subdivided into three regions—pyriform sinus, postcricoid region and posterior pharyngeal wall.

- *Pyriform sinus (Fossa):* Each pyriform fossa, which lies on either side of the larynx, forms the lateral channel for food. Foreign bodies may lodge in the pyriform fossa.
  Boundaries
  - Lateral: Thyrohyoid membrane and the thyroid cartilage.
  - Medial: Aryepiglottic fold, posterolateral surface of arytenoid and cricoid cartilages.
  - Superior: Pharyngoepiglottic fold separates it from vallecula.
  - Inferior: It opens into the esophagus at the level of lower border of cricoid cartilage.

Internal laryngeal nerve, which runs submucosally in the lateral wall of pyriform fossa and causes referred earache in cases of carcinoma pyriform fossa, is accessed here for local anesthesia.

- *Postcricoid region*: This anterior wall of laryngopharynx (pharyngoesophageal junction) extends between the level of arytenoids and lower border of cricoid lamina.

Postcricoid region is a common site for carcinoma, which usually develops from Plummer-Vinson syndrome especially in females.

- *Posterior pharyngeal wall*: It extends from the superior level of hyoid bone to the level of inferior border of cricoid cartilage and between the apices of pyriform fosse.

Lymphatic Drainage (Fig. 21)

- *Pyriform sinus*: It drains into the upper jugular chain through the thyrohyoid membrane.

Pyriform fosse have rich lymphatic network and carcinoma of this region has high frequency of nodal metastases.

- *Posterior pharyngeal wall*: Lateral pharyngeal or parapharyngeal nodes and then to deep cervical lymph nodes.
- *Postcricoid region*: Parapharyngeal nodes and nodes of supraclavicular and paratracheal chain.

Functions

- Common pathway for air and food.
- Provides a vocal tract for resonance of certain speech sounds.
- Helps in deglutition.

Hypopharyngeal Diverticulum

In coordination between contraction of pharyngeal muscles and relaxation of cricopharyngeal sphincter at the upper end of esophagus can lead to hypopharyngeal diverticulum. The cricopharyngeal sphincter fails to relax when pharyngeal muscles are contracting.

ESOPHAGUS

Esophagus is a 25 cm long muscular tube that begins from the lower end of hypopharynx at the level of lower border of cricoid cartilage (C6) to the cardiac end of stomach (T11). Cervical

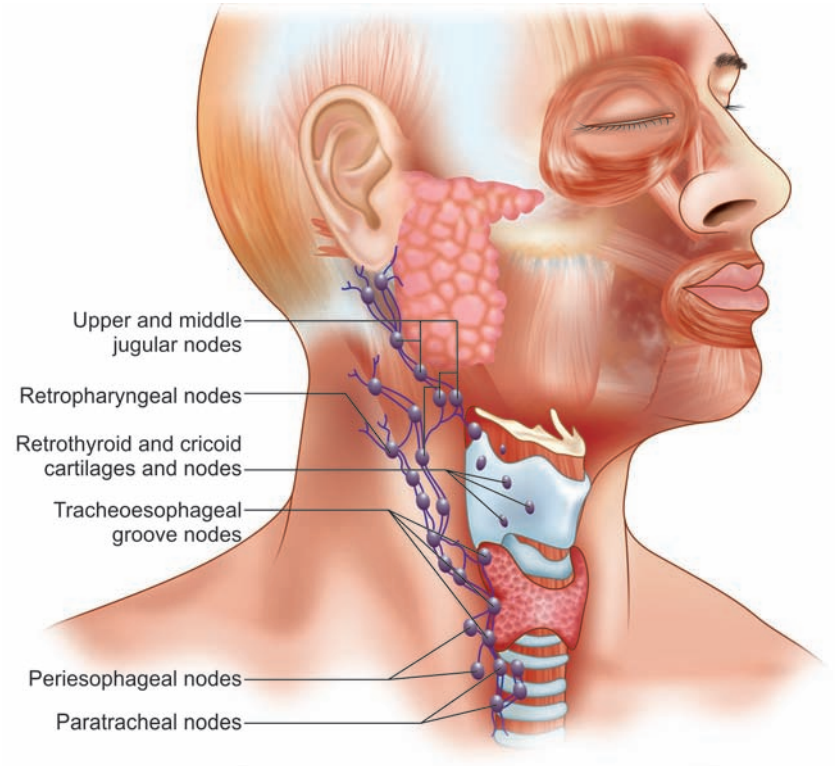


Fig. 21: Lymph node chains draining hypopharynx. Upper and middle jugular nodes are commonly affected but retropharyngeal, paratracheal, paraesophageal, and parapharyngeal space nodes can also be involved

esophagus inclines to the left from its origin to thoracic inlet. Thoracic esophagus also inclines to the left from T7 to the esophageal opening in the diaphragm (Fig. 22).

Constrictions

Esophagus shows four normal constrictions, which can be seen in esophagogram as well as during esophagoscopy. These constrictions, where foreign bodies can be held up, are at the following levels:

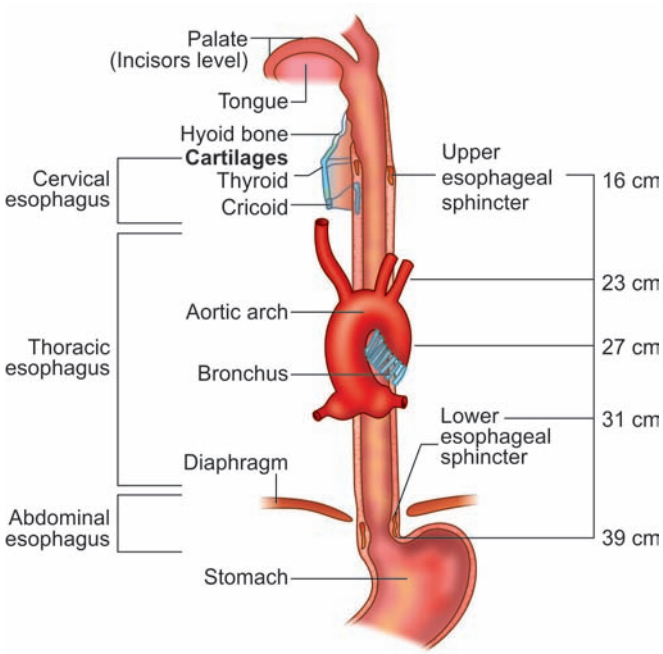


Fig. 22: Esophagus and its relations

- Pharyngoesophageal junction (C6)-15 cm from the upper incisors.
- Arch of aorta (T4)-22 cm from upper incisors.
- Left main bronchus (T5)-25 cm from upper incisors.
- Diaphragm (T10)-40 cm from upper incisors.

Structure

The esophageal wall consists of following four layers from within outwards:

- *Mucosa*: It is stratified squamous epithelium.
- *Submucosa.*
- *Muscular layer*: It has inner circular and outer longitudinal fibers. Circular fibers at the lower end get thickened and form cardiac sphincter. The muscle fibers are striated in upper third, smooth in the lower third and both striated and smooth in the middle third of esophagus.
- *Fibrous layer*: It loosely covers the esophagus.

Nerve Supply

- *Parasympathetic:* It is mediated by branches of vagus nerve (CN X) that has synaptic connections to myenteric (Auerbach's) plexus. Meissner's submucosal plexus is sparse in esophagus.
- *Sympathetic*: Sympathetic trunk.

Lymphatic Drainage

- *Cervical esophagus*: Deep cervical group of lymph nodes.
- *Thoracic esophagus*: Posterior mediastinal lymph nodes.
- *Abdominal esophagus*: Gastric lymph nodes.

Esophageal Sphincters

Esophageal manometry shows two high-pressure zones, which form the physiological sphincters of esophagus. The middle

portion of esophagus shows active peristalsis that is weaker in the upper part and becomes gradually stronger towards the lower part of esophagus.

- *Upper esophageal sphincter*: Upper esophageal sphincter (UES) is about 3–5 cm long and functions during the act of swallowing. It begins from the upper border of esophagus. This functional unit correlates anatomically with junction of the thyropharyngeus and cricopharyngeus, which are powerful striated muscles. UES is closed at rest and protects respiratory passage from regurgitation of esophageal contents. It does not allow air to enter into the esophagus except during swallowing.
- *Lower esophageal sphincter*: Lower esophageal sphincter (LES) is also about 3–5 cm long. This is relatively high pressure zone prevents gastric contents from refluxing into the esophagus. LES has a pressure of 10–25 mm Hg. It is situated at lower portion of esophagus just above the esophagogastric junction. It is more subtle and created by the asymmetrical arrangement of smooth muscle fibers in the distal esophageal wall. It opens in response to primary peristalsis and vomiting and allows air to escape from the stomach.
  - *Factors affecting LES tone*: They include the following: food, gastric distension, gastrointestinal hormones, drugs and smoking.

PHYSIOLOGY OF SWALLOWING

The process of swallowing is usually divided into three phases: oral (buccal), pharyngeal and esophageal. The swallowing center in medulla integrates functions of cranial nerves V, VII, IX, X and XII. Various mechanical and neuromuscular conditions can disrupt this process.

- *Oral phase*: The food is chewed, lubricated and converted into a bolus, which is then propelled into the pharynx. The tongue is elevated against the palate and food is propelled into oropharynx. It is a voluntary phase.
- *Pharyngeal phase*: Once the bolus of food comes into oropharynx, a series of reflex actions occurs, which do not allow the food to go into nasopharynx, oral cavity and larynx and facilitate carrying the food past oropharynx and laryngopharynx into the esophagus.
  - *Nasopharynx*: Closure of nasopharynx, which occurs when soft palate contracts against the Passavant's ridge, cuts of the nasopharynx from oropharynx.
  - *Oropharynx*: Closure of oropharyngeal isthmus, which occurs due to the sphincteric action of palatoglossal muscles when tongue contracts against palate, prevents the entry of food back into oral cavity.
  - *Larynx*: Closure of laryngeal inlet by contraction of aryepiglottic folds and closure of false and true cords prevent aspiration of food material into the larynx. There occurs temporary cessation of respiration and rising of larynx under the base of tongue. The function of epiglottis is not clear but is seen deflecting backwards while food passes into the pyriform fossa.
  - *Hypopharynx*: When pharyngeal muscles contract cricopharyngeus muscles relax and food passes from pharynx into the esophagus. These actions are well timed and synchronous.
- *Esophageal phase*: Once the food enters into the esophagus, cricopharyngeal sphincter closes and primary peristalsis of esophagus takes the food bolus down into the stomach. Gastroesophageal sphincter (GES) relaxes well before the arrival of peristaltic wave. Bolus of food is taken into the stomach with the peristaltic waves and then gastroesophageal sphincter closes.

Gastroesophageal Reflux

Reflux of food (gastric content) into esophagus is prevented with the following:

- Functioning gastroesophageal sphincter.
- Negative intrathoracic pressure and slight positive intra-abdominal pressure.
- Pinch-cock effect of diaphragm.
- Mucosal folds.
- Esophago-gastric angle.

Heartburn (Pyrosis) refers to the feeling of substernal burning and is highly specific of gastroesophageal reflux disease (GERD).

EMBRYOLOGY

Pharyngeal Apparatus (Fig. 23)

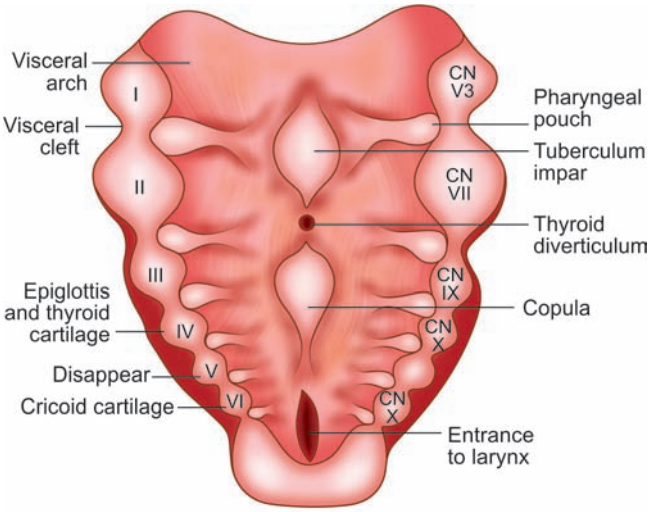
It consists of pharyngeal arches (1, 2, 3, 4 and 6), internal pouches (1, 2, 3 and 4) and external grooves (1, 2, 3 and 4). Figure 23 and 24 summarize their anatomic associations in fetus and adults.

Pharyngeal Arches (Fig. 24)

Table 2 summarizes the relationships among the nerves and adult structures derived from the pharyngeal arches and myotomes.

First Arch Syndrome: It occurs from abnormal development of I pharyngeal arch. The faulty migration of neural crest cells causes facial anomalies. The two well-described syndromes are following:

1. Treacher Collins Syndrome: It consists of mandibular and zygomatic hypoplasia and patients have down-slanted palpebral fissure, coloboma and malformed ear.
2. Pierre Robin Sequence: It is a triad of poor mandibular growth, cleft palate and posteriorly placed tongue.



Visceral arches and cleft, and pharyngeal pouch (I, II, III, IV, V, VI)

Fig. 23: Branchial apparatus showing branchial arches, pharyngeal pouches, and visceral clefts

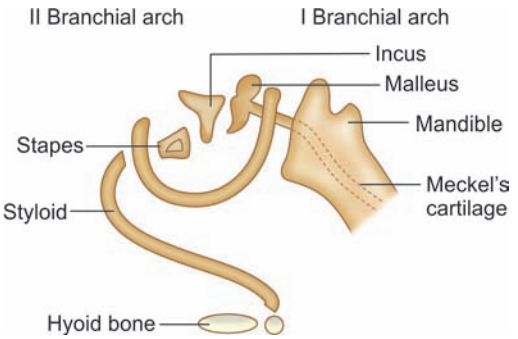


Fig. 24: Derivatives of branchial arches

Pharyngeal Pouches and Grooves

Table 3 summarizes the adult structures derived from the pharyngeal pouches and grooves.

DiGeorge Sequence: It consists of involvement of neural crest cells and the failure of III and IV pharyngeal pouches to differentiate into the parathyroids and thymus. The patient presents with following problems:

1. Immunological problems.
2. Hypocalcemia.
3. Cardiovascular defects such as persistent truncus arteriosus.
4. Abnormal ears.
5. Micrognathia.

- Ectopic parathyroid or thymus: They are generally found along the lateral aspect of neck and important during neck surgery. They result from abnormal migration of parathyroids and thymus from III and IV pharyngeal pouches to their adult positions.

- Branchial Fistula: This pharyngeal fistula occurs when II pharyngeal pouch and groove persist. It is generally found along the anterior border of sternocleidomastoid muscle.
- Branchial Cyst: This pharyngeal cyst forms when pharyngeal grooves, which normally obliterates, persist. It is usually located in the upper neck near the angle of mandible.

Thyroid Gland

Thyroid gland develops from midline thyroid diverticulum that forms in the floor of foregut and migrates caudally to its adult position. The thyroglossal duct, which is later obliterated, connects the thyroid diverticulum to the foregut that is indicated by foramen cecum in adults.

- Thyroglossal cyst or fistula: They occur in midline of neck when parts of thyroglossal duct persist. The cyst is generally seen near the hyoid bone but may also be found at the base of tongue (lingual cyst).
- Ectopic thyroid: Ectopic thyroid tissue may be found in the midline of neck along the course of thyroglossal duct.

TABLE 2 The neural, muscular and skeletal elements derived from pharyngeal arches and myotomes; and skeletal muscles innervated by cranial nerves

Arches and Myotomes	*Nerve*	*Muscles*	*Skeletal*
I arch	Mandibular division of trigeminal nerve (CN V3)	Mastication (Masseter, temporalis, medial and lateral pterygoids), mylohyoid, anterior belly of digastric and tensor tympani and palati	Mandibular process-mandible, maxillary process-maxilla (Meckel's cartilage), malleus, incus and sphenomandibular ligament
II arch	Facial nerve (CN VII)	Muscles of facial expression, posterior belly of digastric, stylohyoid, stapedius	Hyoid (lesser horn and upper body), styloid process, stapes and stylohyoid ligament
III arch	Glossopharyngeal nerve (CN IX)	Stylopharyngeus	Hyoid (greater horn and lower body)
IV arch	Vagus (CN X): External laryngeal nerve branch of superior laryngeal nerve	Cricothyroid and inferior constrictor	Thyroid cartilage
	Vagus (CN X): Pharyngeal branches to pharyngeal plexus	All the muscles of palate (except tensor palati) and pharynx (except stylopharyngeus and inferior constrictor)**	
V arch	Lost		
VI arch	Vagus nerve (CN X): Recurrent laryngeal nerve	All the intrinsic muscles of larynx (except cricothyroid) and inferior constrictor	Cricoid, arytenoid, corniculate, and cuneiform cartilages
Myotome	Accessory nerve (CN XI)	Trapezius and sternocleidomastoid	Skull and scapula
Occipital myotome	Hypoglossal nerve (CN X)	All the muscles of tongue except palatoglossus	
Preoptic myotome	Oculomotor (CN III)	All the extraocular muscles except lateral rectus and superior oblique	
	Trochlear (CN IV)	Superior oblique	
	Abducens (CN VI)	Lateral rectus	

** The development of pharyngeal and palatine muscles (innervated by vagus nerve) are controversial.

TABLE 3 Adult structures derived from the pharyngeal pouches and grooves

Pouches and Grooves	*Adult Structures*
I Pouch	Epithelial lining of eustachian tube and middle ear cavity
II Pouch	Epithelial lining of crypts of palatine tonsil
III Pouch	Inferior parathyroid gland and thymus
IV Pouch	Superior parathyroid gland and ultimobranchial body**
I Groove	Epithelial lining of external auditory canal
Rest of the Grooves	Obliterated

** Neural crest cells migrate into ultimobranchial body and form parafollicular cells of thyroid gland.

Tongue

- *Mucosa*:
  - Anterior two-third of the tongue is associated with I pharyngeal arch and innervated by lingual branch of mandibular division of trigeminal nerve (CN V_3).
  - Posterior one-third of tongue is associated with III pharyngeal arch and is innervated by glossopharyngeal nerve (CN IX).
- *Muscles*: Intrinsic and extrinsic muscles of tongue develop from occipital somites. Myoblasts from occipital somites migrate into the region of tongue.

Face

The face develops from following prominences:

1. Frontonasal prominence
2. A pair of maxillary prominences
3. A pair of mandibular prominences

Cleft Lip: It occurs when the maxillary prominence fails to fuse with medial nasal prominence.

Palate

1. *Intermaxillary segment and primary palate*: The two medial nasal prominences fuse together at the midline and develop into the following structures that lie anterior to incisive foramen:
   a. Philtrum of upper lip
   b. Four incisor teeth
   c. Primary palate
2. *Secondary palate*: It develops from two palatine shelves (outgrowths of maxillary prominences) that fuse together in midline posterior to incisive foramen.

Cleft Palate: It occurs when two palatine shelves fail to fuse together in midline or the primary palate.

Clinical Highlights

1. **Philtrum:** Vertical midline sulcus of upper lip extending from nasal columella to vermilion border.
2. **Waldeyer's ring:** This ring of lymphoid tissue is situated in both nasopharynx and oropharynx.
3. **Adenoids:** They are situated in the roof and posterior wall of nasopharynx and do not have a capsule on external surface. They do not have crypts. They are present at birth and usually disappear by puberty.
4. **Second pharyngeal pouch:** Palatine tonsil develops in the second pharyngeal pouch.
5. **Main blood supply to tonsil:** It is from the tonsillar branch of facial artery.
6. **Passavant's ridge:** It represents superior interdigitation of the superior constrictor and palatopharyngeal muscles.
7. **Killian-Jamieson's space:** It lies between cricopharyngeus and circular fibers of the esophagus.
8. **Killian dehiscence:** It is found between thyropharyngeus and cricopharyngeus.
9. **Structures passing between superior and middle constrictors:** Glossopharyngeal nerve and stylopharyngeus muscle.
10. **Structure passing between middle and inferior constrictor muscles:** Superior laryngeal artery and vein and internal laryngeal branch of superior laryngeal nerve.
11. **Rouviere's node:** This most superior node of the lateral group of retropharyngeal lymph nodes is common site of lymphatic metastasis from the nasopharynx.
12. **Structures passing between upper border of superior constrictor muscle and base of skull:** They are levator veli palatini, Eustachian tube, and ascending palatine artery.
13. **Lamier Hackemann's space:** It lies between circular and longitudinal fibers of the esophagus.

FURTHER READING

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4 Anatomy and Physiology of Larynx and Tracheobronchial Tree

All religions are true. God can be reached by different religions. Many rivers flow by many ways but they fall into the sea. There all are one. Every man should follow his own religion. A Christian should follow Christianity, and a Mohammedan Mohammedanism. For Hindu, the ancient path, the path of Aryan Rishis, is the best.

—Sri Ramakrishna Dev

Points of Focus

ANATOMY OF LARYNX

- **CARTILAGES:** Hyaline and Elastic; Thyroid, Cricoid, Epiglottis, Arytenoid, Corniculate (Santorini) and Cuneiform (Wrisberg)
- **JOINTS: CRICOARYTENOID AND CRICOTHYROID**
- **MEMBRANES AND LIGAMENTS**
  - ✓ Extrinsic: Thyrohyoid Membrane, Hyoepiglottic Ligament, Cricotracheal Membrane
  - ✓ Intrinsic: Cricovocal Membrane and Quadrangular Membrane
- **CAVITY OF THE LARYNX:** Inlet of larynx, Vestibule, Ventricle, Saccule, Subglottic region, Vestibular folds, Vocal folds, and Glottis
- **MUCOUS MEMBRANE OF THE LARYNX**
- **LYMPHATIC DRAINAGE**
- **SPACES OF THE LARYNX:** Pre-epiglottic (space of Boyer), Paraglottic space, and Reinke's space
- **FUNCTIONAL DIVISIONS OF VOCAL FOLDS**
- **PHASE DIFFERENCE**
- **MUSCLES OF LARYNX**
  - ✓ Intrinsic muscles: Vocal Cords (Abductors, Adductors, Tensors and Relaxers); Laryngeal Inlet (Openers and Closers)
  - ✓ Extrinsic Muscles
- **NERVE SUPPLY OF LARYNX:** Recurrent laryngeal nerve and Superior laryngeal nerve
- **DEVELOPMENT**
  - ✓ Infant Larynx
  - ✓ Growth of Larynx

FUNCTIONS OF LARYNX

- **PROTECTION OF LOWER AIRWAYS**
- **PHONATION AND SPEECH**
- **RESPIRATION**
- **FIXATION OF CHEST**

ANATOMY OF TRACHEOBRONCHIAL TREE

- **TRACHEA AND BRONCHI**
  - ✓ Eparterial Bronchus
- **TRACHEAL CARTILAGES**
- **MUCOSA**
- **BRONCHOPULMONARY SEGMENTS**
- **CLINICAL HIGHLIGHTS**

ANATOMY OF LARYNX

The larynx lies in the middle and anterior part of neck opposite the third to sixth cervical vertebrae. During swallowing and phonation larynx moves in vertical as well as anteroposterior direction. The passive side to side movement of larynx produces a grating sensation called laryngeal crepitus.

CARTILAGES (FIGS 1, 2 AND 3)

Larynx has 3 unpaired (thyroid, cricoid and epiglottis) and 3 paired cartilages (arytenoid, corniculate and cuneiform).

- *Types*:
- *Hyaline*: Thyroid, cricoid and most of the arytenoid cartilage (except its tip) are hyaline cartilages, which undergo

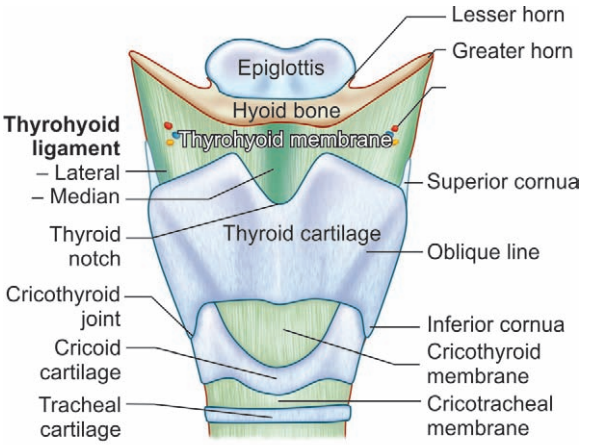

Fig. 1: Laryngeal framework—anterior view

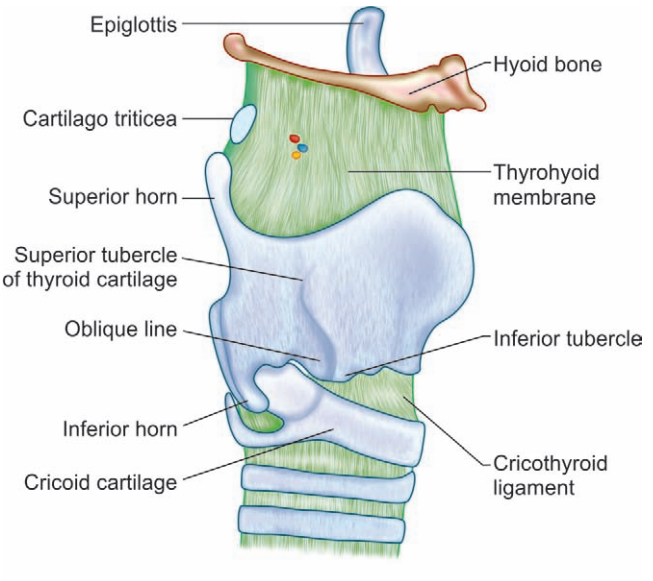

Fig. 2: Lateral view of larynx showing cartilages and ligaments

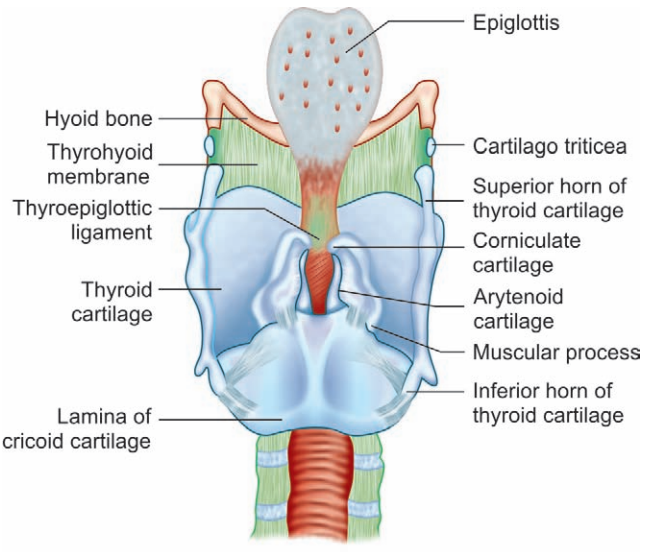

Fig. 3: Posterior view of larynx showing cartilages and ligaments

ossification. The ossification, which begins first in thyroid at the age of 25 years and later in cricoid and arytenoids, is complete by 65 years.

2. *Elastic*: The epiglottis, corniculate, cuneiform and tip of arytenoids are fibroelastic in nature. They do not ossify. Other example of elastic cartilage is auricular cartilage.

- *Thyroid*: This largest laryngeal cartilage has two alae, which meet anteriorly in midline and form an angle (Adam's apple) that is 90° in males and 120° in females. Vocal cords are attached in the middle of thyroid angle. Any airway obstruction above the vocal cord due to tumor or foreign body can be quickly, easily and effectively bypassed by piercing the cricothyroid membrane (cricothyrotomy).
- *Cricoid*: This ring shaped cartilage has narrow anterior arch and expanded posterior lamina, over which articulate arytenoids.
- *Epiglottis*: This leaf-like, yellow, elastic cartilage forms anterior wall of laryngeal inlet. Petiole, a stalk-like process of epiglottis attaches to the thyroid angle.
  - *Parts*: The anterior surface of epiglottis is attached to body of hyoid bone by hyoepiglottic ligament that divides epiglottis into two parts suprahyoid and infrahyoid.
  - *Pre-epiglottic space*: Anterior surface of infrahyoid epiglottis is separated from thyrohyoid membrane and thyroid cartilage by fat filled pre-epiglottic space, which may be invaded by carcinoma of supraglottic larynx or the base of tongue.
- *Arytenoid cartilages*: The pyramidal shape arytenoid cartilage has the following parts:
  - *Base*: It articulates with cricoid cartilage.
  - *Muscular process*: This lateral process provides attachment to intrinsic laryngeal muscles.
  - *Vocal process*: This anterior process provides attachment to vocal ligament of vocal cord.
  - *Apex:* Superiorly it supports the corniculate cartilage in aryepiglottic fold.
- *Corniculate cartilage (of Santorini):* This articulates with the apex of arytenoids cartilage.
- *Cuneiform cartilage (of Wrisberg)*: This rod shaped cartilage is situated in front of corniculate cartilage in the aryepiglottic fold.

JOINTS

- *Cricoarytenoid joint*: This synovial joint is formed between the base of arytenoid and a facet on the upper border of cricoid lamina. Two types of movements are possible at this joint; rotatory and gliding. The rotatory movement occurs at a vertical axis and abducts or adducts the vocal cord. Arytenoids glide laterally and medially and help in closing or opening the posterior part of glottis.
- *Cricothyroid joint*: This synovial joint is formed between the inferior cornua of thyroid cartilage and a facet on the cricoid cartilage.

MEMBRANES AND LIGAMENTS (FIGS 4 TO 6)

Extrinsic

- *Thyrohyoid membrane*: This membrane, which is pierced by neurovascular bundle of superior laryngeal vessels and internal laryngeal nerve, connects thyroid cartilage to hyoid bone.
- *Hyoepiglottic ligament*: It connects the epiglottic cartilage to the body of hyoid bone.

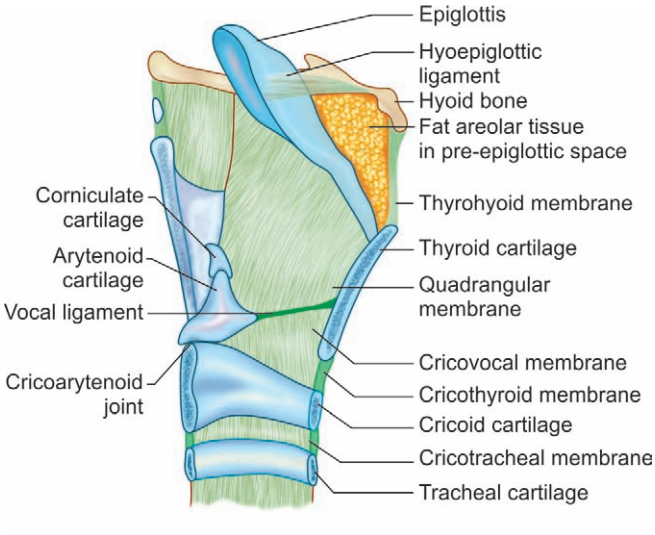

Fig. 4: Sagittal section of larynx

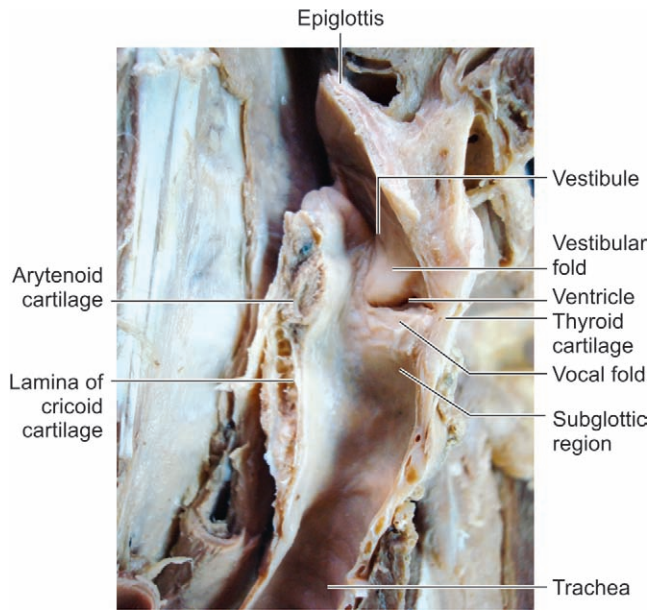

Fig. 5: Sagittal section of cadaveric larynx

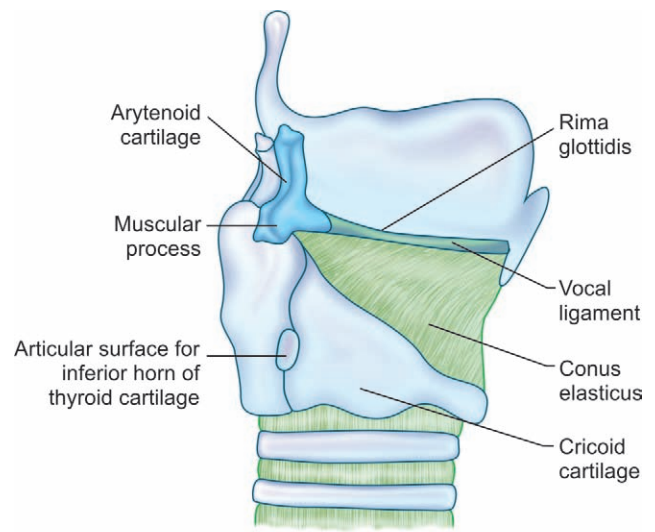

Fig. 6: Lateral view of larynx after removing right lamina of thyroid cartilages. See the laryngeal fibroelastic membrane and its attachment

c. *Cricotracheal membrane*: It connects cricoid cartilage to the first tracheal ring.

Intrinsic

- *Cricovocal membrane*: This triangular fibroelastic membrane has free upper border (vocal ligament), which stretches between middle of thyroid angle to the vocal process of arytenoids. The lower border is attached to the arch of cricoid cartilage.
  - *Conus elasticus*: The two sides of cricovocal membranes form conus elasticus. Subglottic foreign bodies sometimes get impacted in the region of conus elasticus.
  - *Cricothyroid membrane*: The anterior part of conus elasticus is thick and forms cricothyroid membrane, which connects thyroid cartilage to cricoid cartilage.

Any airway obstruction above the vocal cord due to tumor or foreign body can be quickly, easily and effectively bypassed by piercing the cricothyroid membrane (cricothyrotomy). Subglottic foreign bodies sometimes get impacted in the region of conus elasticus.

- *Quadrangular membrane*: This is not well defined. It stretches between the epiglottis and arytenoids cartilages. Its free lower border forms the vestibular ligament, which lies in the vestibular fold (false cord). Its upper border lies in aryepiglottic fold.

CAVITY OF THE LARYNX (FIGS 7 AND 8)

Superiorly laryngeal cavity communicates with laryngeal pharynx through the laryngeal inlet. Larynx ends at the lower border of cricoid cartilage (level of lower border of VI cervical vertebra) and becomes continuous with the trachea. The vestibular and vocal folds divide laryngeal cavity into three parts vestibule, ventricle and subglottic region.

- *Inlet of larynx*: During swallowing the aryepiglottic folds, tubercle of epiglottis and arytenoids approximate and close the laryngeal inlet completely. This oblique entrance has following boundaries:

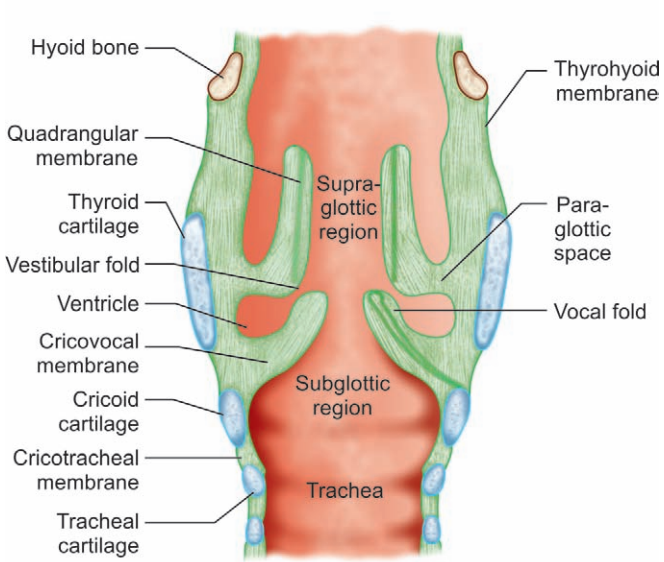

Fig. 7: Coronal section of larynx

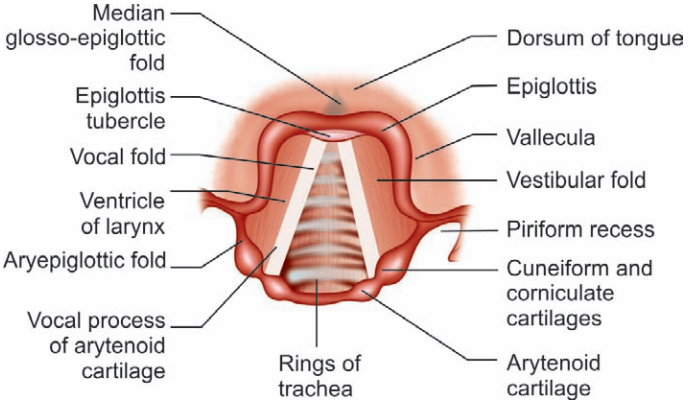

Fig. 8: Superior view of the inside of the larynx as seen during laryngoscopic examination during moderate respiration. Some of the subsites of oropharynx and laryngopharynx can also be noticed

- *Anterior:* Free margin of epiglottis.
- *Lateral 2 sides*: Aryepiglottic folds.
- *Posterior*: Interarytenoid fold.

- *Vestibule*: This region lies between the laryngeal inlet and vestibular folds. It has following boundaries:
  - *Anterior wall*: Posterior surface of epiglottis.
  - *Lateral sides*: Aryepiglottic folds .
  - *Posterior wall*: Arytenoids.
- *Ventricle (Sinus of larynx):* This deep elliptical space lies between vestibular and vocal folds.
- *Saccule*: This diverticulum of mucous membrane starts from the anterior part of ventricle and extends superiorly between vestibular folds and thyroid lamina. The secretions of mucous glands in the saccule provide lubrication for vocal cords.

- Laryngocele: This is an abnormally enlarged and distended saccule contains air.
- Retention cyst: The obstruction of duct of mucous gland in saccule can result in retention cyst.

- *Subglottic region (Infraglottic larynx):* The subglottic region extends from below the vocal cords to lower border of cricoid cartilage.
- *Vestibular folds (False cords):* These folds of mucous membrane one on each side contain vestibular ligament, fibers of thyroarytenoid muscle and mucous glands. They are situated anteroposteriorly across the laryngeal cavity.
- *Vocal folds (Vocal cords):* These pearly-white sharp bands one on each side extend between middle of thyroid angle and vocal processes of arytenoids. The vocal cord consists of a vocal ligament (upper edge of cricovocal membrane), which is covered with mucous membrane that has scanty subepithelial connective tissue.

Most of the laryngeal foreign bodies are seen in supraglottic region lying above the vocal cords.

- *Glottis (Rima glottidis):* This narrowest part of an adult laryngeal cavity lies between vocal cords and arytenoids of the two sides. Size and shape of glottis varies with the activities of the vocal cords. Anterior two-third of glottis is formed by membranous vocal cords while posterior one-third by vocal processes of arytenoids. Anteroposterior length of glottis is more in men (24 mm) than in women (16 mm).

Rima glottidis is the narrowest part of larynx in adults whereas in infants the narrowest part of larynx is subglottic region.

MUCOUS MEMBRANE OF THE LARYNX

Epithelium of the mucous membrane is ciliated columnar except over the vocal cords and upper part of the epiglottis and aryepiglottic folds where it is stratified squamous. It lines the larynx loosely except over the posterior surface of epiglottis, true vocal cords and corniculate and cuneiform cartilages.

Mucous glands are abundant on the epiglottis, posterior part of the aryepiglottic folds and in the saccule. Vocal folds do not have mucous glands.

LYMPHATIC DRAINAGE

- *Supraglottic*: Lymphatics drain into upper deep cervical nodes through the thyrohyoid membrane.
- *Infraglottic*: Lymphatics drain into prelaryngeal (Delphian node) and pretracheal nodes (through cricothyroid membrane) and then to lower deep cervical and mediastinal nodes. Some lymphatics pierce cricotracheal membrane and drain into lower deep cervical nodes.
- *Glottic*: Lymphatics in vocal cords are very scanty, hence glottic carcinoma rarely shows lymphatic metastases.

SPACES OF THE LARYNX

- *Pre-epiglottic space of Boyer*: Anterior surface of infrahyoid epiglottis is separated from thyrohyoid membrane and thyroid cartilage by fat filled pre-epiglottic space. So it has following boundaries:
  - *Anterior*: Upper part of thyroid cartilage and thyrohyoid membrane.
  - *Superior*: Hyoepiglottic ligament.
  - *Posterior*: Infrahyoid epiglottis.
  - *Inferior*: Thyroepiglottic ligament.
  - *Communication*: Laterally it is continuous with paraglottic space.

Pre-epiglottic space can be invaded by carcinoma of supraglottic larynx and the base of tongue.

- *Paraglottic space*: It communicates with pre-epiglottic space and is bounded by
  - *Anterolateral*: Thyroid cartilage and cricothyroid membrane.
  - *Inferomedial*: Conus elasticus.
  - *Medial*: Ventricle and quadrangular membrane.
  - *Posterior*: Anterior mucosa of pyriform fossa.

Growths invading paraglottic space destroy cricothyroid membrane and present in the neck. Ventricle tumors invade paraglottic space and then spread transglottically. Vocal cord tumors involving thyroarytenoid muscle invade paraglottic space and then subglottic and extralaryngeal region. Lateral supraglottic tumors can travel to subglottic region through the paraglottic space along the inner surface of thyroid. Pyriform fossa tumor can come into endolarynx and fix vocal folds through the posterior part of paraglottic space.

- *Reinke's space*: This potential space has scanty subepithelial connective tissues and lies under the epithelium of vocal cords. It is bounded by–
  - *Above and below*: Arcuate lines.
  - *Anterior*: Anterior commissure.
  - *Posterior*: Vocal process of arytenoids.

Reinke's edema: Edema of Reinke's space results in fusiform swelling of the membranous vocal cords.

FUNCTIONAL DIVISIONS OF VOCAL FOLDS

The three functional divisions (Fig. 9) of vocal folds, the stiffness of which are affected by voice disorders and laryngeal lesions, are cover, transition and body.

- *Cover*: Epithelium and superficial layer of lamina propria.
- *Transition*: Vocal ligament consists of intermediate and deep layers of lamina propria. Dense collagen fibers increase the stiffness and elastin fibers decrease the stiffness.

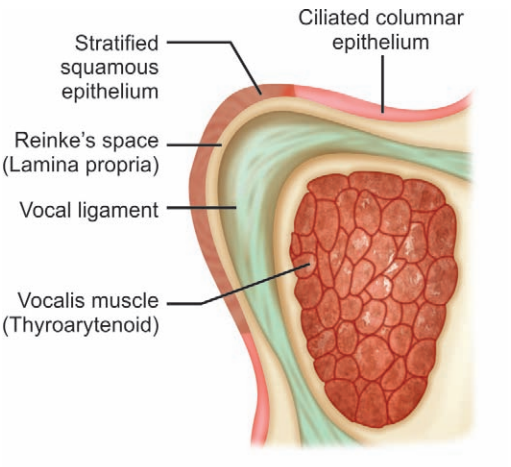

Fig. 9: Structure of vocal fold. Cross-section

- *Body*: It is the least pliable division and contains thyroarytenoid muscle.

Mass and stiffness determine the various glottis shapes during the typical cycle at modal pitch. During high-pitch phonation vocal folds are stretched (increased stiffness of cover) while during low-pitch they are contracted (lax cover).

PHASE DIFFERENCE

The two vocal fold edges are superior and inferior. The folds are less pliable at the anterior and posterior attachments. There occur two types of phase differences during the phonation (vibrations of the closed vocal folds) vertical and horizontal:

- *Vertical phase difference*: Inferior vocal fold edges separate before superior edges and return back in same order.
- *Horizontal phase difference*: The anterior part separates before posterior part and close in same order.

MUSCLES OF LARYNX

There are two types of laryngeal muscles intrinsic (connecting laryngeal cartilages to each other) and extrinsic (connecting larynx to the surrounding structures). Table 1 describes in brief the intrinsic laryngeal muscles. (Figs 10 to 13)

- *Intrinsic muscles* (Figs 5 to 7): They are further divided into two: 1) muscles acting on vocal cords and 2) muscles acting on laryngeal inlet.
  - *Vocal cords*
    - *Abductors:* Posterior cricoarytenoid
    - *Adductors*:
      - Lateral cricoarytenoid
      - Interarytenoid (transverse and oblique arytenoids)
      - Thyroarytenoid (external part)
    - *Tensors*: Cricothyroid
    - *Relaxers*

TABLE 1 Origin insertion and actions of intrinsic laryngeal muscles

Muscle	*Origin*	*Insertion*	*Direction of fibers*	*Action*
Posterior cricoarytenoid	Posterior surface of cricoid lamina	Front of arytenoid muscular process	Upwards and outwards	Abducts vocal folds
Lateral cricoarytenoid	Upper border of cricoid arch	Front of arytenoid muscular process	Upwards and backwards	Adducts vocal folds
Transverse interarytenoid	Back of one arytenoid and its muscular process	Back of another arytenoid and its muscular process	Transverse	Adducts vocal folds
Thyroarytenoid – External part	Lower half of thyroid angle	Anterolateral arytenoid surface	Backwards, upwards and outwards	Adducts vocal folds
Cricothyroid	Anterolateral surface of cricoid arch	Lower border and inferior cornua of thyroid	Backwards, upwards and outwards	Lengthens vocal folds (tensor)
Thyroarytenoid – Internal part (Vocalis)	Lower half of thyroid angle	Lateral surface arytenoid vocal process	Backwards, upwards and outwards	Shortens vocal folds & thicknes
Thyroepiglottic (Part of Thyroarytenoid)	Lower half of thyroid angle	Margin of epiglottic	Backwards, upwards and outwards	Opens laryngeal inlet
Oblique Interarytenoid	Back of arytenoid muscular process	Apex of opposite arytenoid	Upward towards opposite side arytenoid	Closes laryngeal inlet
Aryepiglottic (Prolongation of Oblique Interarytenoid)	Back of arytenoid muscular process	Margin of epiglottic	Upward towards opposite side aryepiglottic fold	Closes laryngeal inlet

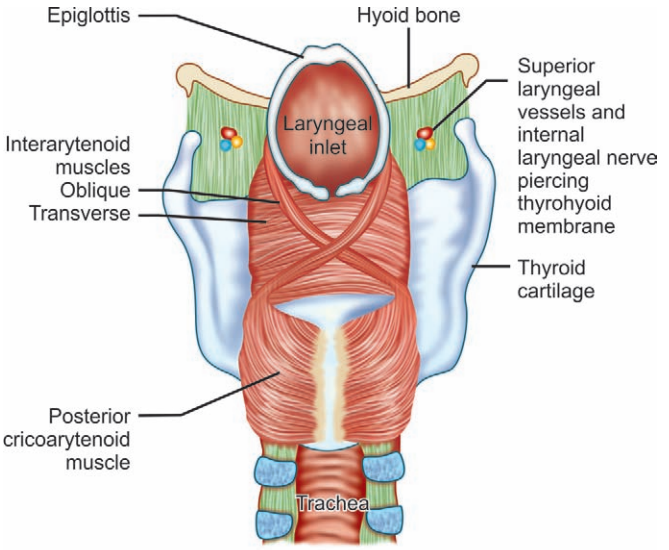

Fig. 10: Intrinsic muscles of larynx as seen on its posterior view

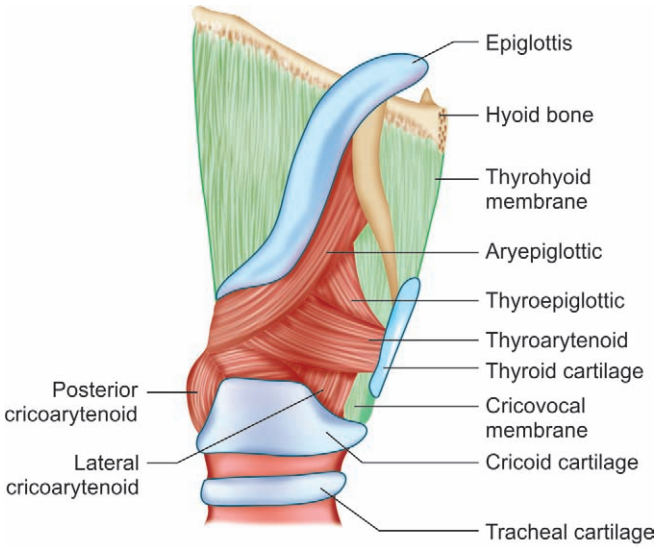

Fig. 11: Intrinsic muscles of larynx as seen on its lateral view

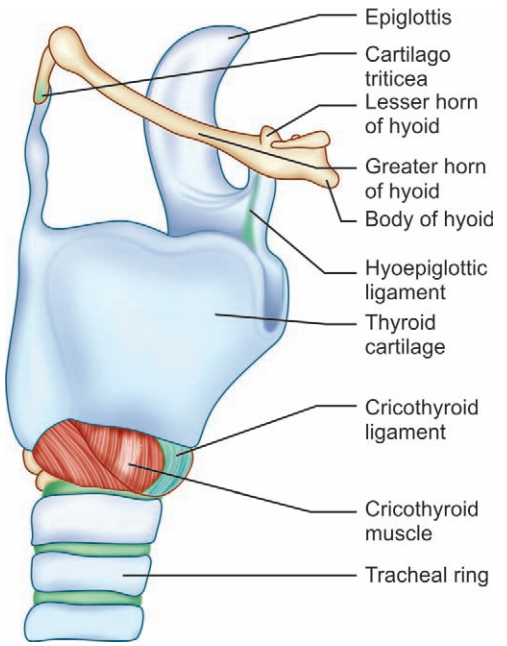

Fig. 12: Lateral view of larynx showing cricothyroid muscle, cartilages and ligaments

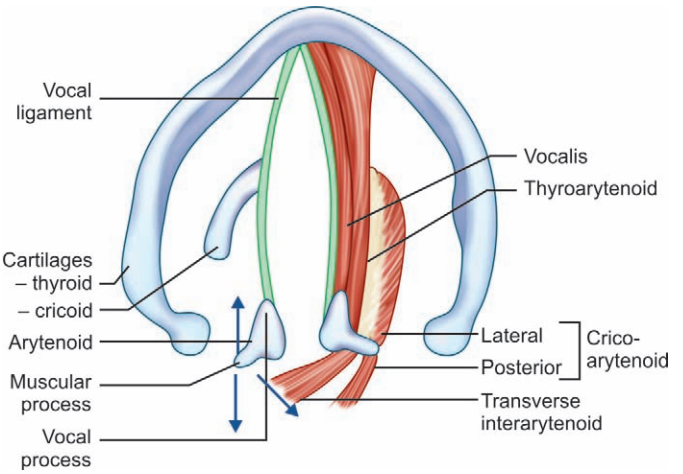

Fig. 13: Intrinsic muscles of larynx and their actions. Arrows indicate direction of movement of arytenoid's processes

  - Vocalis
  - Thyroarytenoid internal part
- *Laryngeal inlet*
  - *Openers*: Thyroepiglottic (part of thyroarytenoid)
  - *Closers*:
    - Interarytenoid (oblique part)
    - Aryepiglottic (posterior oblique part of inter-arytenoid)

- *Extrinsic muscles*: They are divided into two elevators and depressors of larynx.
  - *Elevators*.
    - *Primary elevators*: They are attached to the thyroid cartilage and include vertical pharyngeal muscles (stylopharyngeus, salpingopharyngeus and palatopharyngeus) and an infrahyoid muscle that is thyrohyoid.
    - *Secondary elevators*: They are attached to the hyoid bone and include suprahyoid muscles, which are mylohyoid (main), digastrics, stylohyoid and geniohyoid.
  - *Depressors*: They include infrahyoid strap muscles, which are sternohyoid, sternothyroid and omohyoid.

NERVE SUPPLY OF LARYNX

Larynx is innervated by the vagus nerve (Fig. 14). The motor neurons arise from the nucleus ambiguus of the medulla. The vagus nerve exits through the jugular foramen and then travels within the carotid sheath in the neck.

- *Motor supply*: All the intrinsic muscles of larynx are supplied by the recurrent laryngeal nerve except the cricothyroid muscle, which is supplied by the external laryngeal nerve, a branch of superior laryngeal nerve. Both recurrent and superior laryngeal nerves are the branches of vagus nerve (CN X), which carry the fibers of cranial part of accessory nerve (CN XI).
- *Sensory supply*: Larynx above the vocal cords is supplied by internal laryngeal nerve (branch of superior laryngeal). Larynx below the vocal cords is supplied by the internal branch of recurrent laryngeal nerve.
- *Recurrent laryngeal nerve*: The site of origin and course of right nerve is different from the left.
  - *Left recurrent laryngeal nerve*: It has intrathoracic course

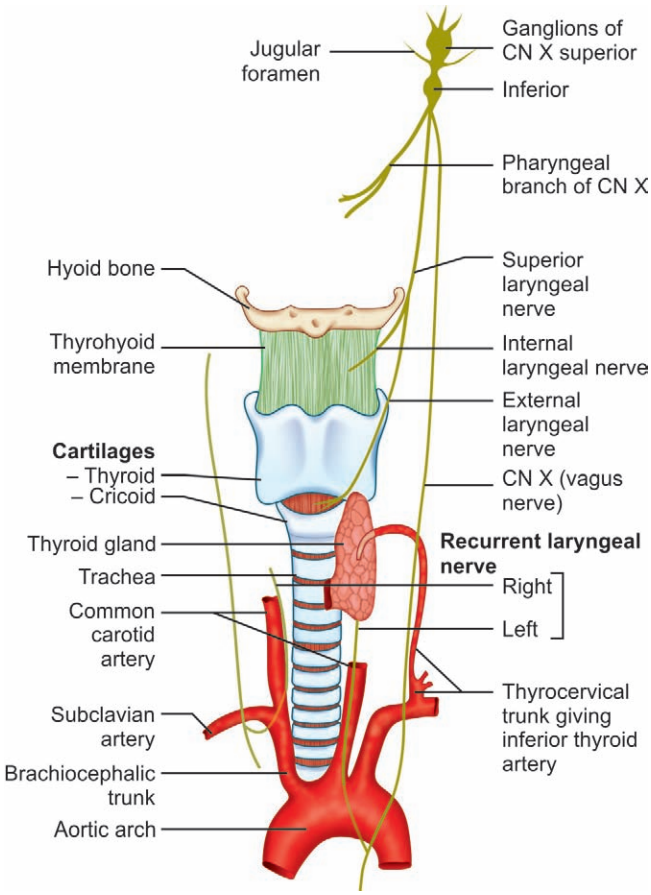

Fig. 14: Nerves supplying the larynx and their relations especially with arteries

and arises from the vagus in the mediastinum at the level of arch of aorta. It loops around aortic arch and ascends into the neck in the tracheoesophageal gutter.

- *Right recurrent laryngeal nerve*: It arises from the vagus at the level of subclavian artery and loops around subclavian artery and then ascends in the tracheoesophageal gutter. It does not have intrathoracic course.

- *Superior laryngeal nerve*: After arising from inferior ganglion of the vagus it descends behind the internal carotid artery, which lies in the carotid sheath. At the level of greater cornua of hyoid it divides into external and internal branches.
  - *External laryngeal nerve*: It travels in relation with the superior thyroid artery and supplies cricothyroid muscle.
  - *Internal laryngeal nerve*: It pierces the thyrohyoid membrane along with the superior laryngeal vessels and supplies sensory innervation to the mucosa of supraglottic larynx and hypopharynx.

The cricothyroid is the only intrinsic laryngeal muscle that is innervated by superior laryngeal nerve and not by the recurrent laryngeal nerve.

DEVELOPMENT

The hypobranchial eminence appears in the floor of primitive pharynx between the 2nd, 3rd and 4th branchial arches (Table 2). During the 3rd week of gestation, a median tracheobronchial groove appears in the floor of primitive foregut which deepens. The lateral septae (laryngotracheal septum) grow and fuse and separates the esophagus from trachea. The upper end of embryonic trachea forms the laryngeal inlet and lower end elongates and divides into two lateral lung buds, which grow out and form primitive pleural cavity. The hypobranchial eminence forms primitive epiglottis. Two swellings lateral to laryngeal fissure form arytenoids and aryepiglottic folds. The enveloping splanchnic mesoderm of buds forms connective tissue of lungs and bronchi. Further divisions of lung buds form the lung lobes and bronchial tree including bronchopulmonary segments down to alveolar sacs.

TABLE 2 Embryological development of larynx

Structure	*Source*
Laryngeal mucosa	Endoderm of cephalic part of foregut
Laryngeal cartilages	Mesenchyme
Epiglottis	Hypobranchial eminence
Upper part of thyroid cartilage	4th branchial arch
Lower part of thyroid cartilage, cricoid, corniculate, and cuneiform cartilages	6th branchial arch
Intrinsic muscles of larynx	6th branchial arch

Infant Larynx

In comparison to the adult larynx, infant larynx has following significant differences:

- *Position*: Infant larynx is situated higher in the neck. Vocal cords lie at C3/C4 level and during swallowing go up to C1/C2 level. The epiglottis and soft palate forms a nasopharyngeal channel for breathing during suckling. Infant can simultaneously swallow and breathe. The milk passes over dorsum of tongue and sides of epiglottis while breathing occurs through nasopharyngeal channel formed by epiglottis and soft palate. In adults vocal cords lie at C5 level.
- *Cartilages*: Laryngeal cartilages in infants are soft and collapse easily.
  - *Epiglottis*: It is omega shaped.
  - *Arytenoids*: They are relatively large and cover significant posterior part of glottis.
  - *Thyroid*: It is flat.
  - *Cricoid*: The diameter of cricoid is smaller than glottis.
- *Cricothyroid and thyrohyoid spaces*: They are very narrow. Hyoid bone overlaps thyroid and thyroid overlaps cricoid.

In infants, cricothyroid and thyrohyoid spaces are not appreciable landmarks especially during tracheostomy.

- *Size*: The larynx of an infant is smaller and has a narrower lumen.
- *Shape:* It is conical and funnel-shaped.
- *Submucosal tissue*: It is thick and loose and becomes easily edematous in response to trauma or inflammation.

The subglottis is the narrowest portion of larynx in children and has the complete cartilaginous ring (cricoid). So edema in this region occurs at the expense of lumen.

Growth of Larynx

Larynx has two phases of spurts in growth:

- *First spurt*: It occurs at 3 years of life and larynx grows in width and length.

First spurt of laryngeal growth in width and length at 3 years of life obviates the need for any airway surgery in certain congenital anomaly.

- *Second spurt*: It occurs during adolescence.
  - *Thyroid angle*: It develops.
  - *Position*: Larynx gradually descends.
  - *Vocal cords*: They descend to C5 level and increase in length especially in males and lead to voice changes (Table 3).

FUNCTIONS OF LARYNX

The four main functions of larynx are respiration, protection of lower respiratory tract, phonation and fixation of the chest.

PROTECTION OF LOWER AIRWAYS

Larynx protects the lower respiratory tract in following ways:

- *Sphincteric closure of laryngeal opening*: During swallowing food entry into air passage is prevented by the closure of three successive sphincters, so no food or vomitus can enter the larynx. The sphincters are following:
  - Laryngeal inlet
  - False cords
  - True cords
- *Cessation of respiration*: When food comes in contact with the oropharynx, reflex generated by afferent fibers of ninth nerve ceases the respiration temporarily.
- *Cough reflex*: Coughing dislodges and expels any foreign particle that comes in contact with respiratory mucosa. Larynx acts as a watchdog of lungs and immediately starts "barking" at the entry of any foreign body.

PHONATION AND SPEECH

- *Aerodynamic myoelastic theory of voice production*: Like a wind instrument, larynx produces voice.
- *Speech*: There are three phases in the production of speech: pulmonary, laryngeal and supraglottis/oral.
  - *Pulmonary phase*: It creates energy flow with inflation of lungs and expulsion of air. It provides a column of air to the larynx. The subglottic air pressure is generated by the exhaled air from the lungs with the help of contraction of thoracic and abdominal muscles.
  - *Laryngeal phase*: Vocal folds vibrate to create sound that is then modified in the next phase. The air pressure opens the adducted cords and small puffs of air are released. Vocal fold vibrations are not the result of laryngeal muscles. Vocal folds are adducted and pressure of moving air causes vibrations of the elastic vocal folds.
  - *Supraglottic/oral phase*: The laryngeal sound that is modified in the supraglottic/oral phase is considered a unique individual sound. Words of the sentences are formed by the muscles of pharynx, tongue and lips and teeth. The vibration of the vocal cords produces sound, which is amplified by mouth, pharynx, nose and chest. The modulator action of lips, tongue, palate, pharynx and teeth converts the sound into speech.

The air pressure produced by the lungs controls the intensity of sound. The frequency of vocal cord vibrations controls the pitch of sound. Different frequencies are produced with changes in length, breadth, elasticity and extension of vocal folds.

TABLE 3 Length of vocal cords in male and female children and adults

	Females		*Males*	
	Children	Adults	Children	Adults
Length of vocal cords in mm	6	15–19	8	17–23

RESPIRATION

The adduction of vocal cords during expiration and abduction of vocal cords during inspiration regulate the flow of air into the lungs.

FIXATION OF CHEST

Closed larynx helps in the fixation of chest wall, which facilitates the action of various thoracic and abdominal muscles. This function plays an important role during digging, pulling and climbing, coughing, vomiting, defecation, micturition and childbirth.

ANATOMY OF TRACHEOBRONCHIAL TREE

TRACHEA AND BRONCHI

Trachea is a dynamic organ. It expands and contracts longitudinally in response to the demands of respiration, swallowing and gravity. Trachea is about 10–12 cm long and extends from lower border of cricoid cartilage of larynx to the carina (level of V thoracic vertebra), where trachea divides into right and left main bronchi. Trachea has two parts—cervical and thoracic. As the bronchi divide, they become progressively smaller. The tracheobronchial tree conducts air from the upper respiratory tract to pulmonary alveoli.

Right main bronchus is wider shorter, and more vertical than left bronchus and hence more common site of foreign bodies.

- *Eparterial bronchus*: Eparterial means above the artery. It is the right upper lobe bronchus that passes over the right pulmonary artery. The left upper lobe bronchus passes under the left pulmonary artery.

A deep bronchoscopic biopsy (5 mm cup forceps) from eparterial right upper lobe divider (spur)/secondary carina can involve right pulmonary artery and results in disastrous bleeding.

TRACHEAL CARTILAGES

Trachea is supported by "U" shaped hyaline cartilages (16–20 in number) which have membranous interconnections. Posterior dehiscent portion of tracheal cartilages are closed by fibrous

tissue. The fibrous membranous part contains smooth and trachealis muscle fibers. As the bronchi divide, the cartilages become progressively smaller and less complete, until the alveoli are formed. Alveoli do not have any cartilage.

MUCOSA

Ciliated respiratory epithelium lines the tracheobronchial tree. This specialized membrane not only conditions the inspired air (including warming and humidifying) but also traps and expels small foreign bodies (1–5 mm particles), which are propelled back up to the pharynx and then swallowed. Larger foreign particles may trigger cough reflex.

BRONCHOPULMONARY SEGMENTS (FIGS 15 TO 17)

The bronchoscopist regards the lungs as being divided according to internal bronchial distribution, rather than by external fissures. The bronchoscopist needs an extended internal anatomic nomenclature that correlates closely with the endobronchial system and the locations description can be reliably described to another bronchoscopist. As s/he progresses to the more peripheral bronchopulmonary tree an "a" represents anterior segment and "b" posterior segments. A lesion in a sub-sub-subsegmental (fifth-order) bronchus in the left lung could be designated as LB1b1b.

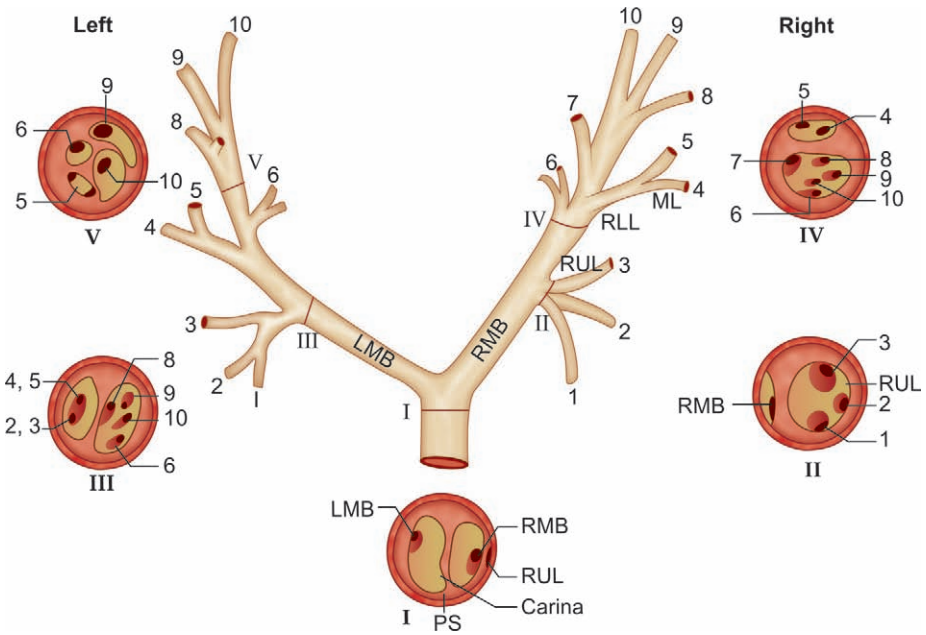

Fig. 15: Tracheobronchial tree and bronchopulmonary segments as seen through the bronchoscope. Right: RMB, right main bronchus; PS, posterior spur of carina; RUL, right upper lobe bronchus; RLL, right lower lobe bronchus; ML, middle lobe bronchus; 1, apical; 2, posterior; 3, anterior; 4, lateral of middle lobe; 5, medial of middle lobe; 6, apical of inferior lobe; 7, medial basal; 8, anterior basal; 9, lateral basal; and 10, posterior basal. Left: LMB, left main bronchus; 1, apical; 2, posterior; 3, anterior; 4, superior lingual; 5, inferior lingual; 6, apical of lower lobe; 8, anterior basal; 9, lateral basal; and 10, posterior basal

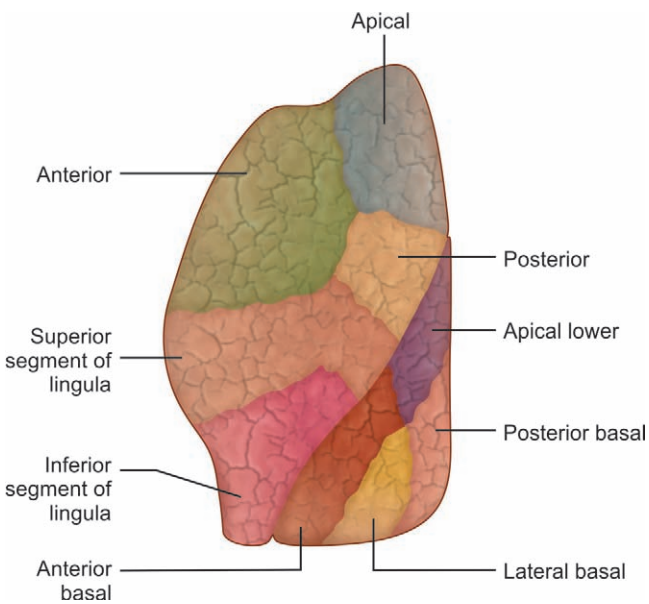

Fig.16 : Lateral view of left lung showing bronchopulmonary segments

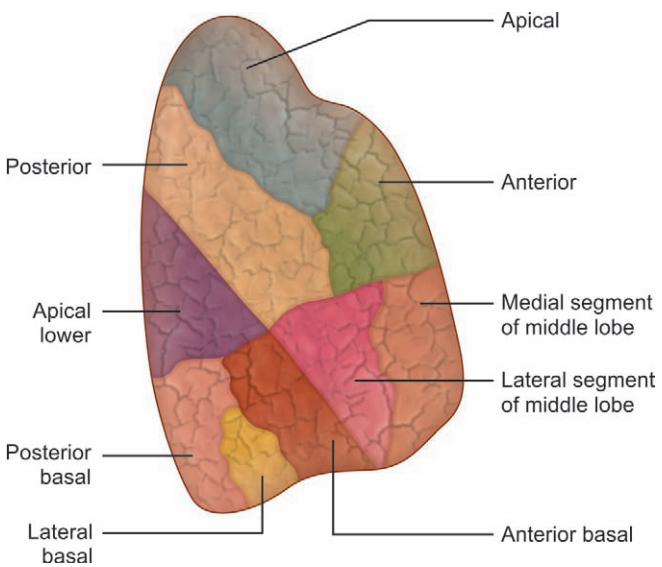

Fig. 17: Lateral view of right lung showing bronchopulmonary segments

Generation and order of bronchopulmonary segments are as follows:

- Trachea (zero generation).
- Main stem bronchi (first generation): Left and right (right lung accounts for 55–60% of total lung parenchyma).
- Lobar bronchi (second generation): Right upper, middle and lower lobes and left upper and its lingular part and lower lobes.
- Lobar segmental bronchi (Third generation): 10 segmental lobules (bronchopulmonary segments) on right side and 9 on left side (Table 4)

TABLE 4 Nomenclature of segmental lobules of pulmonary lobes

Lobes	*Segmental lobules*	*Number*
Right Bronchus (RB) and Lung		
Right upper lobe (RUL)	Apical	RB1 of RUL
	Posterior	RB2 of RUL
	Anterior	RB3 of RUL
Right middle lobe (RML)	Lateral	RB4 of RML
	Medial	RB5 of RML
Right lower lobe (RLL)	Superior (apical)	RB6 of RLL
	Medial basal	RB7 of RLL
	Anterior basal	RB8 of RLL
	Lateral basal	RB9 of RLL
	Posterior basal	RB10 of RLL
Left Bronchus (LB) and Lung		
Left upper lobe (LUL)	Apical/Posterior	LB1/2 of LUL
	Anterior	LB3 of LUL
Lingula of LUL	Superior	LB4 of lingula LUL
	Inferior	LB5 of lingula LUL
Left lower lobe (LLL)	Superior (apical)	LB6 of LLL
	Medial basal	LB7 of LLL
	Anterior basal	LB8 of LLL
	Lateral basal	LB9 of LLL
	Posterior basal	LB10 of LLL

Contd...

- Lobular bronchioles (first airway segment) of less than 0.7 mm diameter are 10th to 14th generation bronchi.

- The right main bronchus is shorter, wider and nearly vertical than the left. Therefore inhaled foreign bodies are more common in right main bronchus than the left.
- Aspiration is common into the posterior bronchus of the right upper lobe (Fig. 18).

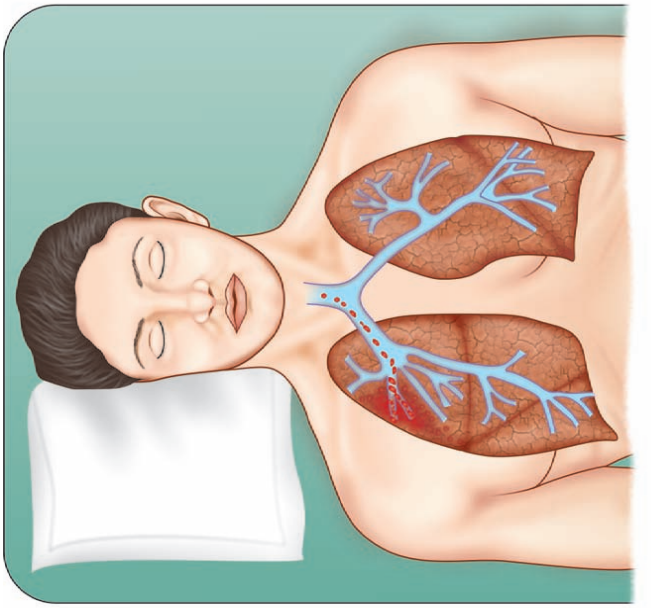

Fig. 18: Aspiration into the posterior bronchus of the right upper lobe in a patient in coma lying in right lateral position

Clinical Highlights

1. **Dimension of larynx**: Anteroposterior depth of male glottis is 24 mm.
2. **Ventricle of Morgagni**: It is the laryngeal ventricle that is situated in between true and false cords of larynx.
3. **Elastic cartilages**: The auricular, epiglottis, corniculate, cuneiform cartilages and apices of the arytenoid cartilages are elastic and do not undergo calcification.
4. **Hyaline cartilages**: The thyroid, cricoid and greater part of arytenoid cartilages are hyaline and undergo calcification.
5. **Wrisberg's cartilage**: This cuneiform fibroelastic cartilage is situated in aryepiglottic fold and does not undergo calcification.
6. **Cricoid cartilage**: It develops from VI branchial arch.
7. **Thyroid angle**: In male, it is 90° while in female it is 120°.
8. **Broyle's ligament**: This is a small ligament which connects both vocal cords at the anterior commissure to the thyroid cartilage.
9. **Epithelium of vocal cords**: It is nonkeratinizing stratified squamous.
10. **Hidden areas of the larynx**: The infrahyoid epiglottis, anterior commissure, subglottis, ventricle and apex of pyriform fossa are not properly visible with indirect laryngoscopy. The direct laryngoscopy examination is required to rule out malignancies of these areas.

11. **Laryngeal spaces**: The pre-epiglottic and paraglottic spaces may be invaded by carcinoma arising in the laryngeal mucosa. Edema of Reinke's space causes polypoid degeneration of vocal cords.
12. **Reinke' space**: This potential space with scanty subepithelial connective tissues lies under the epithelium of vocal cords and superficial to its elastic layer. It is limited above and below by the arcuate lines, anteriorly by anterior commissure and posteriorly by vocal process of arytenoids. Edema of Reinke's space results in fusiform swelling of the membranous vocal cords.
13. **Delphian node**: Prelaryngeal lymph node in the region of the thyroid isthmus.
14. **Motor nerve supply of larynx**: The recurrent laryngeal nerve supplies all the intrinsic muscles of the larynx except the cricothyroid (tenses the vocal cord), which is innervated by superior laryngeal nerve. Both these nerves are branches of vagus nerve. Therefore paralysis of recurrent laryngeal nerve does not affect cricothyroid.
15. **Sensory nerve supply of larynx**: The recurrent laryngeal nerve innervates laryngeal mucosa below the vocal fold whereas superior laryngeal nerve innervates laryngeal mucosa above the vocal fold.
16. **Galen's anastomosis**: It is the anastomosis between the branches of superior and recurrent laryngeal nerves (branches of vagus nerve).
17. **Non recurrent-recurrent laryngeal nerve (Inferior laryngeal nerve)**: It is often associated with anomalous retro-esophageal right subclavian artery.
18. **Adductors of vocal cord**: They are lateral cricoarytenoid, thyroarytenoid, and interarytenoid muscles.
19. **Tensor of vocal cord**: There is only one tensor muscle of the vocal cord and that is cricothyroid.
20. **Posterior cricoarytenoid muscle**: This is the only abductor muscle of the larynx. It is supplied by recurrent laryngeal nerve.
21. **Glottic chink**: In cadaveric position of the vocal cords vocal chink is 14 mm. In full abduction it is about 19 mm. During the whisper, the position of vocal cord is paramedian.
22. **Vocal cord palsy**: It is commonly caused by lesions of recurrent laryngeal nerve**.** Because of the intrathoracic course of left recurrent laryngeal nerve left vocal cord palsy is twice more common than right vocal cord palsy.
23. **Tracheobronchial tree and larynx**: Rate of topical absorption is highest.

FURTHER READING

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5 Anatomy of Neck

We should give everyone his due. What is not edible for a man, give to a cow; what is not edible for a cow, give to a dog; what is not edible for a dog; throw into a lake for fishes to eat. But never waste.

—Mother Sarada Devi.

Points of Focus

- **SURFACE ANATOMY**
  - ✓ Mastoid, Mandible, Hyoid, Thyroid and Cricoid Cartilages, Trachea, Thyroid gland, Sternocleidomastoid and Trapezius muscles, Carotid Arteries, Internal jugular Vein, Chassaignac's Triangle
- **TRIANGLES OF NECK**
  - ✓ Anterior Triangle: Submental, Submandibular, Carotid and Muscular
  - ✓ Posterior Triangle: Occipital and Supraclavicular (Subclavian)
- **CERVICAL FASCIA**
  - ✓ Superficial Cervical Fascia
  - ✓ Deep Cervical Fascia: Superficial (Investing) Layer, Middle (Visceral Pretracheal) Layer and Deep (Prevertebral) Layer and Carotid Sheath.
- **LYMPH NODES OF HEAD AND NECK**
  - ✓ Submental, Submandibular, Parotid, Post Auricular, Occipital, Facial, Malar, Superficial Group of Lateral Cervical, Deep Group, Internal Jugular Chain, Spinal Accessory Chain, Transverse Cervical Chain, Scalene Nodes, Anterior Cornical Nodes, Anterior Jugular Chain, Prelaryngeal, Pretracheal, Paratracheal Nodes
  - ✓ Classification
  - ✓ Levels of Cervical Lymph Nodes
  - ✓ Anterior Compartment Nodes
  - ✓ Ho's Triangle in Supraclavicular Zone or Fossa
- **NECK DISSECTION**
  - ✓ Classification of Neck Dissection
  - ✓ Radical Neck Dissection (RND)
  - ✓ Structures Removed, Structures not Removed, Incisions, Contraindications
  - ✓ Modified Neck Dissection
  - ✓ Selective Neck Dissection
  - ✓ Extended Neck Dissection
- **THYROID GLAND**
  - ✓ Blood Supply and Related Laryngeal Nerves
  - ✓ Superior Thyroid Artery, External Laryngeal Nerve and Joll's Triangle
  - ✓ Inferior Thyroid Artery, Recurrent Laryngeal Nerve and Beahr's Triangle
  - ✓ Venous Drainage
  - ✓ Lymphatic Drainage
- **PARATHYROID GLANDS**
- **DEVELOPMENT**
  - ✓ Thyroid Gland
  - ✓ Thyroglossal Cyst or Fistula
  - ✓ Ectopic Thyroid
- **CLINICAL HIGHLIGHTS**

This chapter will outline the clinically important areas of neck anatomy. The floor of mouth forms the upper limit of neck anteriorly while posteriorly upper border is base of skull. The lower boundaries include first rib and first thoracic vertebra.

SURFACE ANATOMY

The following landmarks of surface anatomy are important for neck examination as well as for planning the surgical incision (Fig. 1).

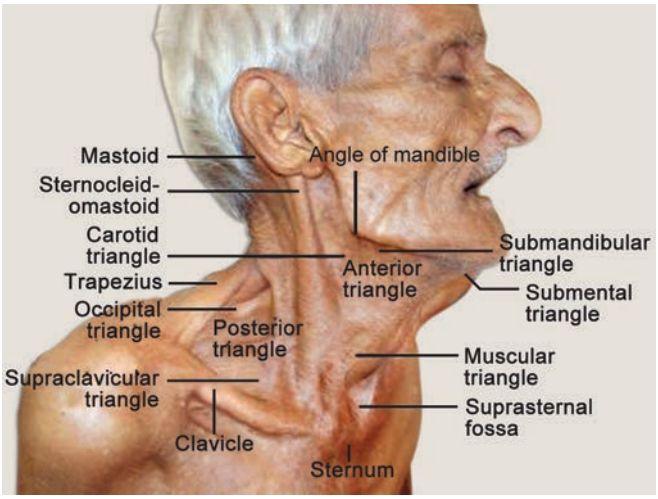

Fig. 1: Surface anatomy of neck. Anterior and posterior neck triangles separated by sternocleidomastoid muscle

- *Mastoid*: It provides insertion to sternocleidomastoid muscle. The transverse process of atlas can be palpated anterior and inferior to mastoid. Parotid gland is present between mastoid and angle and ramus of mandible.
- *Mandible*: The submandibular gland can be palpated just below the ramus of mandible.
  - The marginal mandibular branch of facial nerve crosses the lower border of mandible superficial to facial vessels just anterior to masseter muscle.
- *Hyoid*: The swellings of thyroglossal cyst and subhyoid bursitis are seen close to hyoid and move up on protruding out the tongue.
  - The marginal mandibular branch of facial nerve does not course below the greater cornu of hyoid bone. So the incision below the greater cornu will not damage this nerve.
- *Thyroid cartilage*: Adam's apple is more prominent in males than females. The widening of thyroid lamina occurs in cases of carcinoma of larynx.
- *Cricoid cartilage*: It lies at the level of sixth cervical vertebra and forms the lower limit of larynx and laryngopharynx. The cricothyroid membrane situated between thyroid and cricoid cartilages is the site of emergency tracheostomy.
- *Trachea*: The cervical trachea can be shifted to either side due to thyroid and superior mediastinal tumors.
- *Thyroid gland*: The thyroid isthmus can be palpated below the cricoid cartilage over second to fourth tracheal rings. The normal thyroid lobes lie deep to sternocleidomastoid muscle and are not palpable.
- *Sternocleidomastoid muscle*: Sternocleidomastoid (SCM) divides the anterolateral quadrangular part of neck into anterior and posterior triangles.
  - Origin: Two heads: (1) Thick tendinous sternal head arises from anterior and lateral surface of manubrium; (2) Muscular clavicular head arises from medial third of clavicle.
  - Insertion: Lateral aspect of mastoid and lateral half of superior nuchal line.
  - Relations: The internal jugular vein (IJV) lies deep between the clavicular and sternal heads of the SCM. The carotid sheath and deep jugular chain of lymph nodes lie deep to SCM. The tilting of the head to side relaxes SCM and facilitates palpation of deep cervical lymph nodes (levels II, III and IV) that accounts for 80% of all neck nodes.
    - Nerves: The accessory nerve and branches of cervical plexus (lesser occipital, great auricular and transverse cervical nerves) emerge from the junction of upper and middle third of the posterior border of SCM. The accessory nerve courses across the roof of posterior triangle and enters the anterior border of trapezius at the junction of its middle and lower third.
  - Actions: Ipsilateral flexion and opposite side rotation of head and anterior flexion when both side muscle act together.
  - Nerve supply: Spinal part of accessory nerve (motor) and anterior rami of $C_{2,3,4}$ (sensory).
  - Blood supply: Branches of occipital and superior thyroid arteries (branches of external carotid artery).
- *Trapezius muscle*: This large muscle makes the posterior boundary of posterior triangle and elevates the shoulder.
  - Nerve supply: Spinal part of accessory nerve (motor) and anterior rami of $C_{2,3,4}$ (sensory). Surgical damage to accessory nerve in posterior triangle is not uncommon and leads to malrotation of scapula and drooping of shoulder.
- *Carotid arteries*: The common carotid artery (CCA) is best palpated at the level of greater cornu of hyoid bone in the carotid triangle where it bifurcates into internal (ICA) and external carotid arteries (ECA). CCA runs medial and deep to IJV, while vagus nerve runs between the two in carotid sheath. CCA and ICA usually do not give any branches in the neck. IJV runs anterolateral to ICA. ECA terminates deep in the parotid gland near the neck of mandible and divides into superficial temporal and maxillary arteries. ECA is anterior and superficial to ICA and is identified for ligation in the carotid triangle by recognizing its branches.
- *Internal jugular vein:* It extends from lobule of the ear to the medial end of clavicle deep to SCM in carotid sheath. This thin-walled capitance vein emerges in neck from jugular foramen and is continuation of sigmoid sinus. It joins the subclavian vein and forms the brachiocephalic vein. Its tributaries in neck include facial, lingual, pharyngeal and superior and middle thyroid veins.
- *Chassaignac's triangle:* It is formed by longus colli, scalenus anterior and subclavian artery. Its apex faces towards tubercle of sixth cervical vertebra (Chassaignac's tubercle). The thoracic duct on left side joins the intersection of IJV and subclavian vein in this triangle.

TRIANGLES OF NECK

Sternocleidomastoid divides the anterolateral quadrangular part of neck into anterior and posterior triangles.

The digastric and omohyoid muscles divide these triangles into further sub-triangles (Fig. 2).

- *Anterior triangle*: It is bounded by the SCM, inferior border of mandible and cervical midline. The anterior and posterior bellies of digastric and superior belly of omohyoid divide the anterior triangle into four triangles:

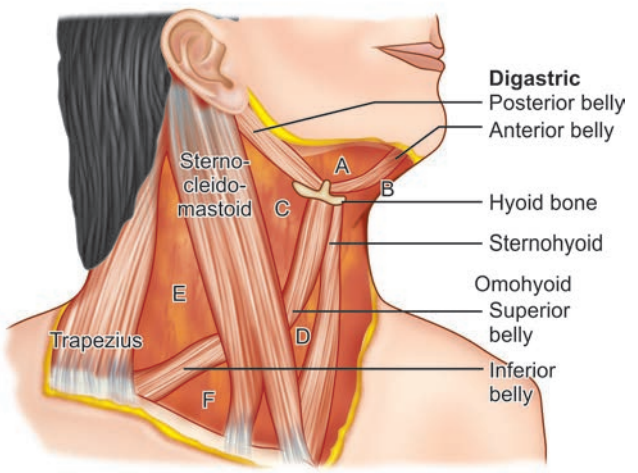

Fig. 2: Triangles of neck. Submandibular (A), submental (B), carotid (C), muscular (D), occipital (E) and supraclavicular (F)

- Submental: It is bounded by the two sides of anterior bellies of digastric muscle and hyoid bone and contains lymph nodes.
- Submandibular: It is bounded by the lower border of mandible and anterior and posterior bellies of digastric. Contents include submandibular salivary gland, lymph nodes, facial vessels and marginal branch of facial nerve.
- Carotid: It is bounded by the SCM, posterior belly of digastric and superior belly of omohyoid. Contents include carotid sheath and bifurcation of CCA and lymph nodes. Carotid body tumor occurs here.
- Muscular: This subhyoid triangle is bounded by SCM, superior belly of omohyoid and cervical midline. The contents include infrahyoid strap muscles and structures which lie deep to them in central compartment like thyroid and parathyroids, larynx and trachea, laryngopharynx and cervical esophagus and carotid sheath.

- *Posterior triangle*: It is bounded by the SCM, clavicle and anterior border of trapezius. The inferior belly of omohyoid divides the posterior triangle into two triangles:
  - Occipital: This lateral neck triangle is bounded by SCM, trapezius and inferior belly of omohyoid. Its contents include accessory nerve, lymph nodes, fibrofatty tissue and nerves of cervical plexus.
  - Supraclavicular (subclavian): It is bounded by SCM, inferior belly of omohyoid and clavicle. The contents include fibrofatty tissue, brachial plexus, subclavian vessels, thyrocervical trunk, Sibson's subpleural fascia and cervical pleura.

CERVICAL FASCIA

The neck has both superficial as well as deep fascia.

Superficial Cervical Fascia

The thin cervicocephalic superficial fascia lies immediately beneath the skin and invests platysma and muscles of facial expression. It extends from epicranium of scalp down to the thorax and axilla. Subplatysmal flap protects the skin blood supply as the vessels penetrate the superficial fascia. The space between superficial fascia and investing layer of deep cervical fascia contains following structures:

1. Adipose tissue
2. Sensory nerves
3. Vessels: External and anterior jugular veins
4. Superficial lymphatics

Deep Cervical Fascia

The deep cervical fascia has three layers—superficial investing layer, middle visceral (pretracheal) layer and deep prevertebral layer (Figs 3 and 4). At certain places these three layers coalesce and cannot be separated from each other. The neck spaces (submandibular peritonsillar, parapharyngeal, retropharyngeal, pretracheal and prevertebral) and their infections are described in Chapter 54: Deep Neck Infections.

1. *Superficial (investing) layer*: It surrounds the neck and splits and encloses two muscles (SCM and trapezius) and two glands (submandibular and parotid) and forms two spaces (suprasternal space of Burns and supraclavicular).
   a. Attachments
      i. Posterior: Ligamentum nuchae and cervical spines.
      ii. Anterior: Hyoid bone.
      iii. Superior: External occipital protuberance, superior nuchal lines, mastoid tip.
      iv. Inferior: Acromion of scapula, clavicle and sternum.
   b. Two laminas: At the mandible and parotid investing layer splits into two laminas. The internal lamina covers medial surface of pterygoid muscle up to the skull base tympanic bone, styloid process and forms stylomandibular ligament. The external lamina covers masseter and parotid and attaches to zygomatic arch.

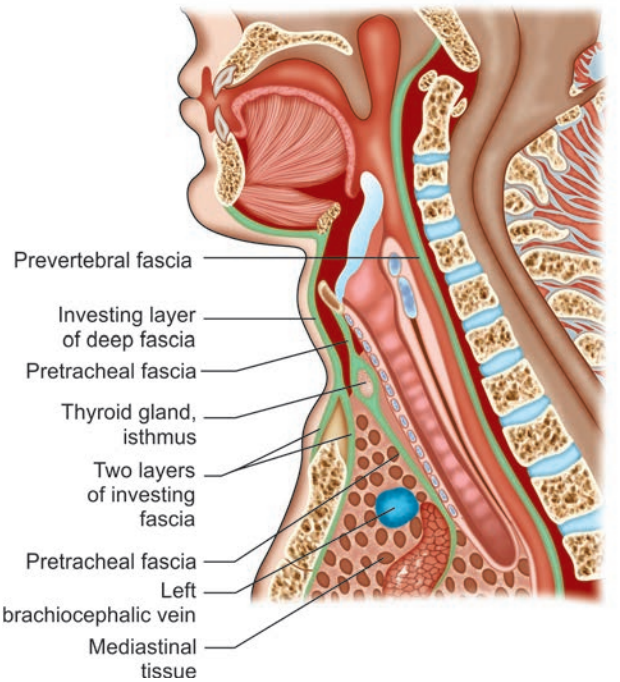

Fig. 3: Sagittal section of neck showing three layers of deep cervical fascia

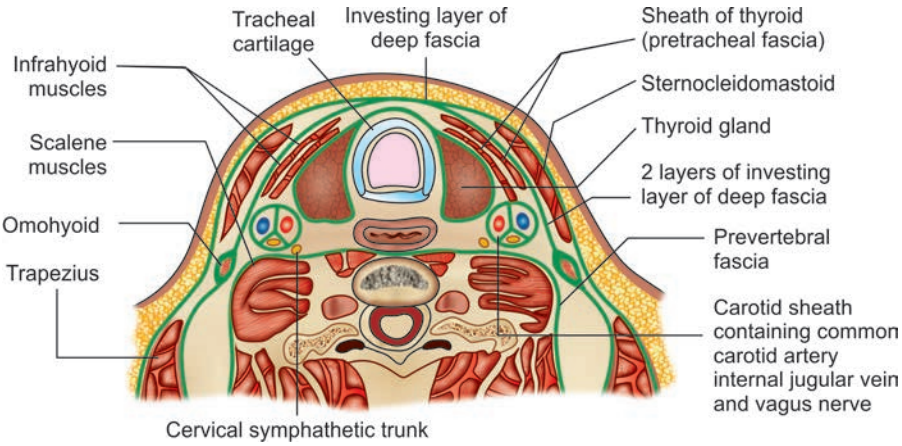

Fig. 4: Transverse section of lower part of neck showing three layers of deep cervical fascia

2. *Middle (visceral or pretracheal) layer*: In the upper part it envelops the pharyngeal constrictor and buccinator muscles and forms buccopharyngeal fascia. It is separated into two divisions—muscular and visceral.
   a. Attachments:
      i. Superiorly it is attached to hyoid bone and thyroid cartilage.
      ii. Posteriorly it fuses with the alar division of prevertebral fascia at T2 level and forms anterior wall of retropharyngeal space.
   b. Muscular division: It surrounds the strap muscles (sternohyoid, sternothyroid and thyrohyoid) and inserts inferiorly into the clavicle and sternum.
   c. Visceral division: It passes deep to the strap muscles and envelops pharynx, larynx, trachea, thyroid, parathyroid and esophagus. It extends inferiorly into the superior mediastinum and joins pericardium.
3. *Deep (prevertebral) layer*: Posteriorly it is attached to ligamentum nuchae and cervical spines. At the transverse processes of cervical vertebra it divides into two divisions: anterior (alar) and posterior (prevertebral).
   a. Prevertebral division: It extends from skull base down the length of vertebral column up to coccyx. Laterally it extends as axillary sheath. It forms the floor of posterior triangle and facilitates gliding movement of pharynx during the swallowing. It splits and encloses muscles (vertebral, deep posterior triangle and scalene), vessels (vertebral and subclavian) and nerves (phrenic and brachial plexus).
   b. Alar division: It lies between the visceral layer of middle layer and prevertebral division of deep layer of deep cervical fascia. Posteriorly visceral layer of middle layer fuses with the alar division of prevertebral fascia at T2 level and forms anterior wall of retropharyngeal space.
4. *Carotid sheath*: All the three layers contribute in the formation of carotid sheath which is present deep to SCM and contains IJV, common and internal carotid arteries, vagus nerve and about 80% lymph nodes of the neck. The cervical sympathetic chains lie deep to carotid sheath.

LYMPH NODES OF HEAD AND NECK

For the broad classification and the drainage of different groups of lymph nodes see Table 1.

- Submental nodes, two to eight in number, lie on the mylohyoid muscle in the submental triangle, which is situated between right and left anterior bellies of digastric muscles and the hyoid bone.
- Submandibular nodes, which are in relation to submandibular gland and facial artery, lie in submandibular (digastric) triangle, which is situated between anterior and posterior bellies of digastric muscle and bounded superiorly by the lower border of mandible and an imaginary line drawn between the angle of mandible and mastoid.
- Parotid nodes, which lie in relation to the parotid salivary gland, are extraglandular (preauricular and infraauricular) and intraglandular.
- Post auricular nodes (mastoid nodes) lie behind the auricle.
- Occipital nodes, at the apex of the posterior triangle, are situated both superficial and deep into splenius capitus muscle.
- Facial nodes, which lie along facial vessels, are grouped according to their location.
- Malar nodes lie near outer canthus.
- Superficial group of lateral cervical nodes lies along external jugular vein.
- Deep group, which consists of three chains: internal jugular, spinal accessory and transverse, lie deep to sternocleidomastoid muscle and in the posterior triangle.
- Internal jugular chain, which is further divided into upper (jugulodigastric node), middle and lower groups, lie anterior, lateral and posterior to internal jugular vein and extends from the digastric muscle to the subclavian vein.
- Spinal accessory chain lies along the spinal accessory nerve and the upper nodes of this chain coalesce with upper jugular nodes.
- Transverse cervical chain (supraclavicular nodes), in the lower part of the posterior triangle, lies horizontally, along the transverse cervical vessels. The posterior cervical triangle lies between posterior border of sternocleidomastoid, anterior border of trapezius and the clavicle below.
  - Scalene nodes: These are medial group of supraclavicular nodes.
- Anterior cervical nodes, which lie between the two carotids and below the level of hyoid bone, consist of two chains: anterior jugular chain and juxtavisceral chain.

TABLE 1 Classification of lymph nodes of head and neck

- Nodes of Upper Horizontal Chain
  - Submental nodes: Afferents: from chin, lower lip (only middle part), lower anterior gums, anterior floor of mouth and tip of tongue; Efferents: to submandibular nodes and internal chain.
  - Submandibular nodes: *Afferents* from lower lip (only lateral part), upper lip, cheek, nasal vestibule and anterior part of nasal cavity, gums and teeth (except lower anterior gums and teeth), medial canthus, soft palate, anterior pillar, anterior part of tongue, submandibular and sublingual salivary glands and floor of mouth (except central anterior part); *Efferents* to internal jugular chain.
  - Parotid nodes**:** *Afferents* from scalp (anterior to pinna), pinna, external auditory canal, face, buccal mucosa; *Efferents* to internal jugular and external jugular chain.
  - Post auricular nodes (mastoid nodes): *Afferents*: from scalp (posterior to pinna), posterior surface of pinna and skin of mastoid; *Efferents* to infra-auricular nodes and into internal jugular chain.
  - Occipital nodes: *Afferents* from scalp, skin of upper neck; *Efferents* to upper accessory chain of nodes.
  - Facial nodes (midmandibular, buccinator, infraorbital and malar): *Afferents* from upper and lower lids, nose, lips and cheek; *Efferents* to submandibular nodes.
- Lateral Cervical Nodes (superficial and deep): The superficial group drains into following deep internal jugular and transverse cervical nodes.
  - Internal jugular chain
    - ◊ Upper deep cervical group (jugulodigastric node): *Afferents* from oral cavity, oropharynx, nasopharynx, hypopharynx, larynx and parotid.
    - ◊ Middle deep cervical group: *Afferents* from oral cavity, oropharynx, hypopharynx, larynx, and thyroid.
    - ◊ Lower deep cervical group: *Afferents* from larynx, thyroid and cervical esophagus.
  - Spinal accessory chain. *Afferents* from scalp, skin of the neck, nasopharynx, occipital and postauricular nodes; *Efferents* to transverse cervical chain.
  - Transverse cervical chain (supraclavicular nodes). *Afferents* from accessory chain.
    - ◊ Medial supraclavicular (scalene nodes): *Afferents* from breast, lung, stomach, colon, ovary and testis.
- Anterior Cervical Nodes
  - Anterior jugular chain: *Afferents* from skin of anterior neck.
  - Juxtavisceral chain
    - ◊ Prelaryngeal node (Delphian node): *Afferents* from subglottic larynx and pyriform sinuses.
    - ◊ Pretracheal nodes: *Afferents* from thyroid gland the trachea; *Efferents* to paratracheal, lower internal jugular and anterior mediastinal nodes.
    - ◊ Paratracheal nodes (recurrent laryngeal nerve chain): *Afferents* from thyroid lobes, subglottic larynx, trachea and cervical esophagus.
- Lymph Nodes Not Clinically Palpable
  - Retropharyngeal nodes: Lateral (Rouviere) and medial groups: *Afferent* from nasal cavity, paranasal sinuses, hard and soft palate, nasopharynx, and posterior wall of pharynx; *Efferents* to upper internal jugular group.
  - Sublingual nodes. *Afferents* from anterior part of the floor of mouth and ventral surface of tongue; *Efferents* to submandibular or upper jugular nodes.

- Anterior jugular chain lies along anterior jugular vein.
- Juxtavisceral chain consists of prelaryngeal, pretracheal, and paratracheal nodes.
  - i. Prelaryngeal node (Delphian node) lies on cricothyroid membrane.
  - ii. Pretracheal nodes lie in front of the trachea, deep to pretracheal fascia.
  - iii. Paratracheal nodes (recurrent nerve chain) lie along recurrent laryngeal nerve.

- Lymph nodes not palpable clinically. They are retropharyngeal and sublingual.
  - a. Retropharyngeal nodes, behind the pharynx, are divided into lateral and medial groups.
    - i. Lateral group lies close to the base of skull at the level of atlas. Rouviere is the most superior node of this group
    - ii. Medial group lies near the midline but at a lower level.
  - b. Sublingual nodes lie deep along the lingual vessels.

The deep cervical jugular groups receive in addition to their direct areas of drainage all the efferents from the pericraniocervical ring, efferents from the superficial cervical nodes and efferents from other paravisceral deep nodes (such as retropharyngeal, infrahyoid, prelaryngeal, pretracheal, paratracheal and subclavian). All the lymph from the head and neck finally drains into ipsilateral lower deep cervical group (terminal group). Efferents from the latter converge and form (right and left) jugular lymph trunks that descends on its vein to its termination at the jugulosubclavian venous junction (thoracic duct on left side).

Levels of Cervical Lymph Nodes (Figs 5 and 6)

The cervical lymph nodes are divided into different levels (Table 2), which is very important in cases of head and neck malignancies. It is well accepted and recognized throughout the world. On the bases of these levels various types of neck dissections are described (Table 3).

- Anterior compartment nodes lie between the medial borders of SCM muscles (or carotid sheaths), hyoid bone above and suprasternal notch below.
- Ho's triangle in supraclavicular zone or fossa lies between the medial and lateral ends of upper border of clavicle at the point where neck meets the shoulder. Lymph nodes of this triangle are important in carcinoma of the nasopharynx. Metastasis in these nodes, which include lower part of levels IV and V, indicates an advanced stage of malignancy nasopharynx.

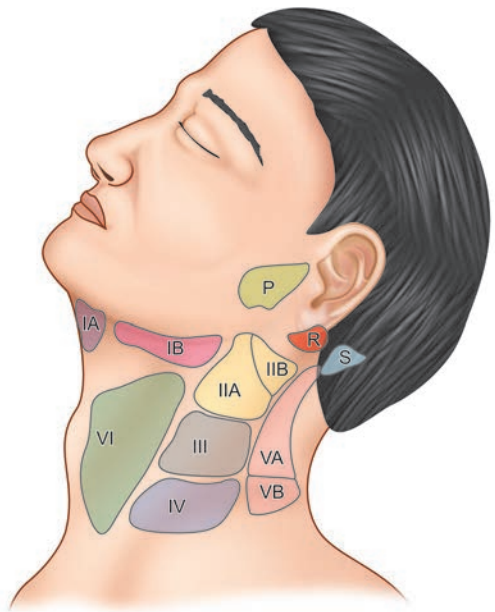

Fig. 5: Neck nodes regions.P, parotid preauricular; R, retroauricular, and S, suboccipital regions

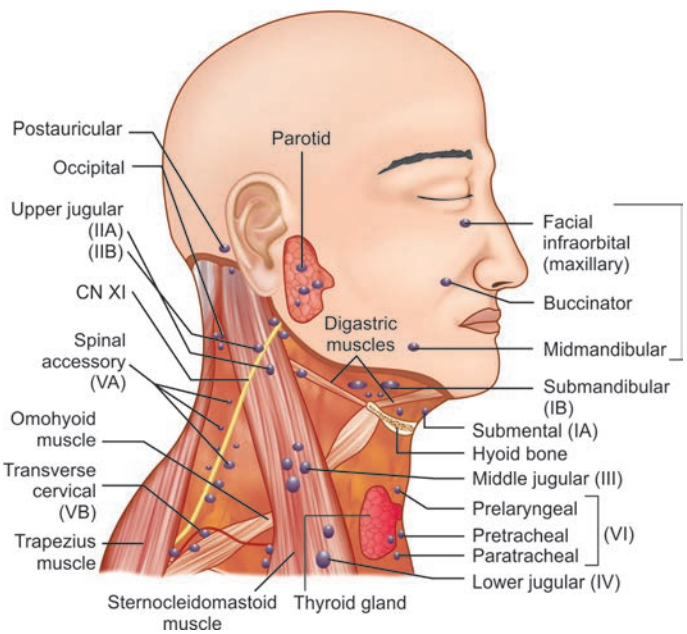

Fig. 6: Lymph nodes of head and neck. Neck nodes regions (levels) are mentioned in ( )

TABLE 2 Levels of cervical lymph nodes

1. Level I: Submental nodes and submandibular nodes
   a. Level IA: Submental nodes
   b. Level IB: Submandibular nodes
2. Level II: Upper jugular (between skull base and hyoid bone).
   a. Level IIA: Inferomedial to accessory nerve
   b. Level IIB: Superolateral to accessory nerve
3. Level III: Middle jugular (between hyoid bone and upper border of cricoid cartilage).
4. Level IV: Lower jugular (between upper border of cricoid cartilage and clavicle).
5. Level V: Posterior triangle group: subdivided into upper, middle and lower, corresponding to planes of levels II, III and IV.
   a. Level VA: Spinal accessory nodes above the level of cricoid cartilage
   b. Level VB: Transverse cervical chains and supra-clavicular nodes below the level of cricoid cartilage.
6. Level VI: Anterior compartment nodes: Prelaryngeal (Precricoid or Delphian), perithyroidal, pretracheal, paratracheal).
7. Level VII: Superior mediastinum nodes.
   - Lymph nodes not included:
     a P: Parotid-preauricular
     b R: Retroauricular (mastoid)
     c S: Suboccipital
     d. Others: Retropharyngeal and facial

TABLE 3 Classification of neck dissection

- *Radical neck dissection (RND):* Removes I, II, III, IV and V level of lymph nodes, sternocleidomastoid muscle, internal jugular vein, spinal accessory nerve, submandibular salivary gland, tail of the parotid and omohyoid muscle.
- Modified radical neck dissection
  - Type I: Preserves accessory cranial nerve (CN XI)
  - Type II: Preserves CN XI and internal jugular vein (IJV)
  - Type III: Preserves CN XI, IJV and sternocleidomastoid muscle (SCM)
- *Selective neck dissection:* Preserves CN XI, IJV and SCM.
  - Supra omohyoid (or anterolateral): Removes level I, II and III lymph nodes (cancer of oral cavity).
  - Lateral: Removes level II, III and IV lymph nodes (cancer of pharynx, hypopharynx and larynx)
  - Posterolateral: Removes level II, III, IV, V and suboccipital lymph nodes (cancer or melanoma of posterior scalp or posterior upper neck)
  - Anterior compartment: Removes level VI lymph nodes (cancer thyroid, subglottic, cervical trachea, hypopharynx)
- *Extended neck dissection*: Extended RND may include additional lymph node groups (retropharyngeal, parotid or level VI nodes) and nonlymphatic structures (external carotid artery, hypoglossal nerve, parotid gland, mastoid tip).

NECK DISSECTION

This procedure is used in the surgical management of cancer of the aerodigestive tract. It removes metastases to cervical lymph nodes and their surrounding fibrofatty tissues. See Table 3 for different types of neck dissection, their meanings and indications.

Radical Neck Dissection

- *Structures removed:* In radical neck dissection (RND), all lymph nodes (level of I to V) and structures closely related to them and present in the area between mandible and mastoid above, clavicle below, sternocleidomastoid, hyoid bone and contralateral anterior belly of digastric medially, and trapezius posteriorly, are removed. They include:
  - Lymph nodes: Level of I to V lymph nodes present in submental, submandibular, and posterior triangle and deep cervical internal jugular chain
  - Muscles: Sternocleidomastoid muscle and omohyoid muscle
  - Internal jugular vein
  - Spinal accessory nerve
  - Salivary glands: Submandibular salivary gland and tail of the parotid.
- *Structures not removed*: In RND, the lymph nodes regions, which are not approached and encountered are: postauricular, suboccipital, parotid (except those in the tail), facial, retropharyngeal and paratracheal. Following important structures, which though are encountered during the surgery, must be saved. Due care must be taken neither to injure nor to remove them as that can lead to morbidity and even mortality:
  - Arteries: Common carotid, internal carotid and external carotid arteries.
  - Nerves: Brachial plexus, phrenic nerve, vagus nerve, cervical sympathetic chain, marginal mandibular branch of facial, lingual and hypoglossal nerves.
- *Incisions:* Though RND alone can be performed, usually it is combined with the removal of primary tumor. So depending on the nature of primary growth, any one of the following common incisions may be employed:
  - Schobinger
  - Mc Fee
  - Hockey-stick
  - Extensions from Gluck-Sorenson's incision, used for laryngectomy with neck dissection.
- *Contraindications*: They include:
  - Untreatable primary cancer
  - Distant metastases
  - Neck nodes fixed to deeper lying important structures
  - Major systemic illnesses.

Modified Neck Dissection

Modified neck dissection is similar to radical neck dissection but with certain modifications, which include preservation of one or more of the following structures:

- Spinal accessory nerve
- Internal jugular vein
- Sternocleidomastoid muscle.

Selective Neck Dissection

In selective neck dissection, in addition to the three non-lymphatic structures (spinal accessory, sternocleidomastoid muscle and internal jugular vein) certain levels of cervical lymph nodes are not removed. There are different types of selective neck dissections (Table 3).

Extended Neck Dissection

In addition to the structures removed in RND, extended neck dissection includes other lymph node groups (retropharyngeal, parotid or level VI nodes) and non-lymphatic structures (external carotid artery, hypoglossal nerve, parotid gland, mastoid tip).

THYROID GLAND (FIG. 7)

It is made up of two lateral lobes and central isthmus. Lobes extend from thyroid cartilage to sixth tracheal ring. Isthmus overlies second to fourth tracheal rings. Sometimes a pyramidal lobe projects up from the isthmus on left side. Pretracheal layer of deep cervical fascia envelops the thyroid gland and trachea. Anterior relations include strap muscles, anterior jugular veins and SCM. Posterior relations are laryngopharynx, larynx, trachea and esophagus. The carotid sheath, which contains CCA, IJV and vagus nerve, lies lateral to the thyroid lobe. Cervical sympathetic trunk and thyrocervical trunk giving inferior thyroid artery lie posterior to carotid sheath.

Blood Supply and Related Laryngeal Nerves

- *Superior thyroid artery and external laryngeal nerve*: The external laryngeal nerve (ELN), a branch of superior laryngeal nerve supplying the cricothyroid muscle lies deep to the upper pole in sternothyrolaryngeal (Joll's) triangle. The superior thyroid vessels running down the upper pole are very close to ELN, which can be damaged if superior thyroid vessels are ligated too high. ELN injury will paralyze the cricothyroid muscle and limits the patient's vocal range noticed particularly by the singers.
  - Joll's triangle: Boundaries of Joll's triangle include:
    i. Lateral: Upper pole of thyroid gland and superior thyroid vessels.
    ii. Superior: Insertion of sternothyroid muscle.

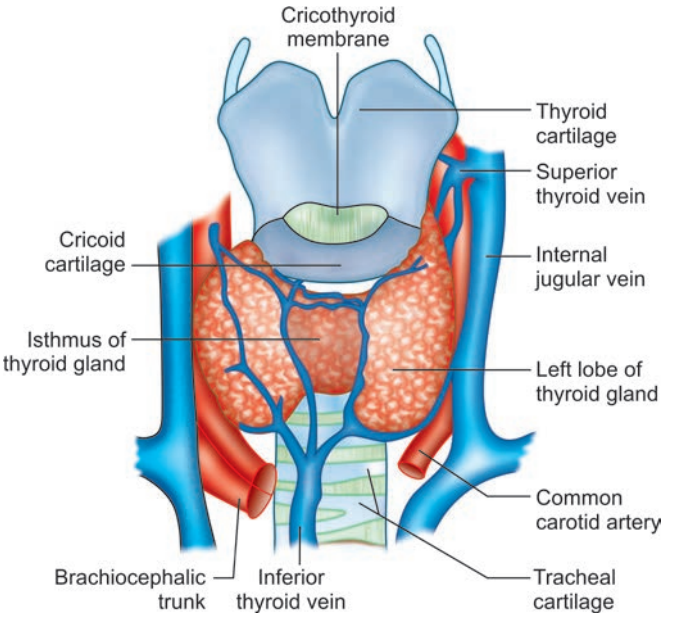

Fig. 7: Anterior surface of thyroid gland and its relations with carotid arteries and internal jugular vein

iii. Medial: Cervical midline.
iv. Floor: Cricothyroid muscle.
v. Roof: Strap muscles.

- *Inferior thyroid artery and recurrent laryngeal nerve*: The recurrent laryngeal nerve (RLN) can be identified in the lower part of tracheoesophageal groove where it forms a boundary of the Beahr's triangle. RLN can be anterior, posterior or pass through the branches of inferior thyroid artery, which is traditionally ligated away from the thyroid lobe. The injury to RLN results in vocal cord palsy and hoarseness of voice. At the level of second or third tracheal ring in the region of Berry's ligament, RLN is intimately close to or pierce posterior surface of thyroid lobe. Berry's ligament is posterior condensation of pretracheal layer of deep cervical fascia where thyroid gland is tethered to trachea. The inferior thyroid artery arises from the thyrocervical trunk and pierces the prevertebral fascia medial to carotid sheath and enters into the posterior part of thyroid lobe.
  - Beahr's triangle: The boundaries of Beahr's triangle include:
    i. Medial: RLN in the lower part of tracheoesophageal groove. Occasionally RLN lie lateral to tracheoesophageal groove.
    ii. Lateral: Common carotid artery
    iii. Superior: Inferior thyroid artery
- *Thyroidea ima artery*: When present it may arise from aortic arch or innominate artery.

Venous Drainage

- *Superior and middle thyroid veins*: They drain into internal jugular vein.
- *Inferior thyroid veins*: They are multiple and drain into brachiocephalic vein.

Lymphatic Drainage

The lymphatic drainage of thyroid gland can be grouped into two categories—major and minor:

- *Major*: Major lymphatic drainage goes to;
  - Middle deep cervical (jugular) node level III
  - Lower deep cervical (jugular) node level IV
  - Posterior triangle nodes level V
- *Minor*: Some of the lymphatics drain into:
  - Pre- and paratracheal nodes level VI
  - Superior mediastinal nodes level VII

PARATHYROID GLANDS (FIG. 8)

- *Superior parathyroids*: They lie near the junction of medial and posterior surfaces of upper third of thyroid lobe at the level of cricoid cartilage superior to inferior thyroid artery. They are situated in or near the thyroid capsule. Their position is more consistent than the inferior parathyroids.
- *Inferior parathyroids*: They are posteriorly placed on the lower poles inferior to inferior thyroid artery.

DEVELOPMENT

The neck skin is derived from cervical dermatomes (second to sixth cervical segment) and muscles (SCM, trapezius and strap) from cervical myotomes. The cervical viscera and structures innervated by cranial nerves develop from branchial apparatus, which has been described in Chapter 3: Anatomy and Physiology of Oral Cavity, Pharynx and Esophagus.

Thyroid Gland

Thyroid gland develops from midline thyroid diverticulum that forms in the floor of foregut and migrates caudally to its adult position. The thyroglossal duct, which is later obliterated, connects the thyroid diverticulum to the foregut that is indicated by foramen cecum in adults.

- Thyroglossal cyst or fistula: They occur in midline of neck when parts of thyroglossal duct persist. The cyst is generally seen near the hyoid bone but may also be found at the base of tongue (lingual cyst).
- Ectopic thyroid: Ectopic thyroid tissue may be found in the midline of neck along the course of thyroglossal duct.

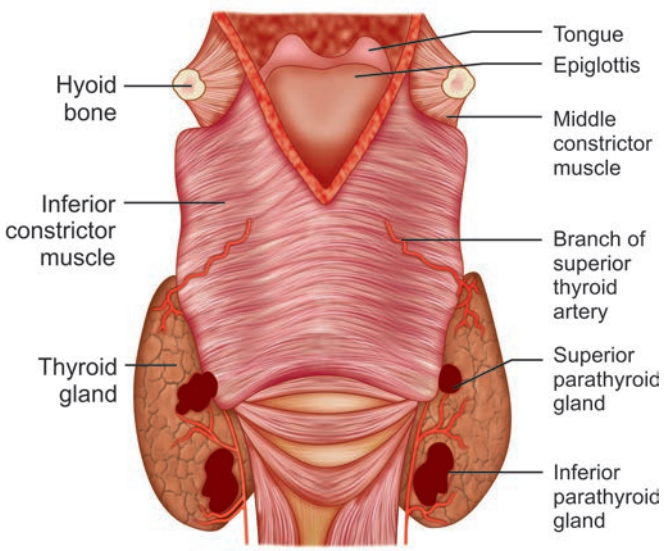

Fig. 8: Posterior surface of thyroid and parathyroid glands and laryngopharynx and cervical esophagus

Clinical Highlights

1. **Selective neck dissection:** Removal of only selective groups/levels of cervical lymph nodes.
2. **Nerves preserved in radical neck dissection:** They include vagus, hypoglossal and phrenic.
3. **Elective neck dissection:** In medullary carcinoma of thyroid, elective neck dissection is appropriate.

FURTHER READING

1. Robbins KT. Classification of neck dissection, current concepts and future consideration. Otolaryngol Clin North Am. 1998;31:639-55.
2. Shah JP. Patterns of lymph node metastases from squamous carcinomas of the upper aerodigestive tract. Am J Surg. 1990;60:405-9.

6 Bacteria and Antibiotics

You see many stars in the sky at night, but not when the sun rises. Can you therefore say that there are no stars in the heavens during the day? O man, because you cannot find God in the days of your ignorance, say not that there is no God.

—Sri Ramakrishna Dev

Points of Focus

BACTERIA

- ✓ Reservoir, Transmission, Diseases Antibiotics
- ✓ Gram-Positive Bacteria: Cocci, Rods, Aerobic or Facultative Anaerobic Rods, Anaerobic Rods
- ✓ Gram-Negative Bacteria: Cocci, Rods, Helical (and Microaerophilic) Facultative Anaerobic Rods, Anaerobic Straight to Helical Rods
- ✓ Normal Microflora

- **STAPHYLOCOCCI:** *Staphylococcus aureus*, MRSA
- **STREPTOCOCCI:** Group A β-Hemolytic Streptococci, *Streptococcus pneumoniae*, Oral streptococci, Anginosus-Milleri Group
- ***CORYNEBACTERIUM DIPHTHERIAE***
- **NEISSERIA SPECIES:** Gonococcus, Meningococcus
- ***MOREXELLA CATARRHALIS***
- ***HAEMOPHILUS INFLUENZAE***
- ***BORDETELLA PERTUSSIS***
- ***PSEUDOMONAS AERUGINOSA***
- **ENTEROBACTERIACEAE:** *Escherichia coli*, Klebsiella sp
- **ANAEROBES:** *Bacteroides fragilis*, *Fusobacterium necrophorum*
- **MICROAEROPHILIC BACTERIA:** Actinomyces
- **Mycobacteria:** *Mycobacterium tuberculosis*, Atypical Mycobacterium
- ***MYCOPLASMA PNEUMONIAE***
- **CHLAMYDIAE**
- **SPIROCHAETES:** Borrelia, *Treponema pallidum*

ANTIBIOTICS

- ✓ ENT Infections, Bacteria, Antibiotics
- ✓ Antimicrobial Resistance
- ✓ Drug Hypersensitivity Reactions
- ✓ Prerequisites of Antimicrobial Therapy
- ✓ Surgical Prophylaxis
- ✓ Sensitivity Testing: Stokes and Breakpoint Methods, PCR
- ✓ Classification

- **INHIBITORS OF BACTERIAL CELL WALL SYNTHESIS (BETA-LACTAM ANTIBIOTICS)**
  - ✓ Penicillins: Natural Penicillins, Penicillinase-Resistant Penicillins, Aminopenicillins, Carbapenems, Monobactams, Beta-lactamase Inhibitors
  - ✓ Cephalosporins: First to Fourth Generations
  - ✓ Glycopeptides: Vancomycin and Teicoplanin
  - ✓ Inhibitors of Nucleic Acid Synthesis
  - ✓ Quinolone Antibiotics
  - ✓ Sulphonamides and Trimethoprim
  - ✓ Rifamycins
  - ✓ Metronidazole
- **INHIBITORS OF BACTERIAL PROTEIN SYNTHESIS (RIBOSOMAL)**
  - ✓ Tetracyclines
  - ✓ Chloramphenicol
  - ✓ Macrolides (e.g. Erythromycin and azithromycin)
  - ✓ Fusidic Acid
  - ✓ Aminoglycosides
  - ✓ Streptogramins
  - ✓ Oxazolidinones (e.g. Linezolid)
- **ANTITUBERCULAR DRUGS**
- **NONSPECIFIC ANTISEPTICS**
  - ✓ Adverse Effects of Antimicrobials
- **CLINICAL HIGHLIGHTS**

BACTERIA

Bacteria are classified on the basis of Gram stain, shape and results of biochemical reactions. The normal microflora and infections of ear, nose and throat will be discussed in brief in this chapter. Tables 1 and 2 show Gram-positive and Gram-negative, rods and cocci and aerobic and anaerobic bacteria with their reservoir and transmission, diseases caused by them and the antibiotics to which they are sensitive.

Normal Microflora

The normal microflora of nasopharynx consists of Gram-positive and Gram–negative aerobic and anaerobic bacteria (Table 3). The nasopharynx can be colonized with potentially pathogenic bacteria without ill effect in majority of people. There seems to be a fine balance between underlying host risk factors. In bacteriotherapy, nasal oral streptococci are administered to prevent or reduce recurrent episodes of otitis media. The use of broad spectrum antibiotics can change the microflora of nasopharynx.

STAPHYLOCOCCI

They are Gram-positive and catalase-positive cocci that produce typical grape-like clusters on Gram stain. They are found on skin and mucous membrane. The coagulase test separates the genus into two groups. The main pathogenic species is *Staphylococcus aureus*, which coagulate plasma. Rests of the 23 species, which are coagulase-negative, can survive on inanimate objects such as implants, intravascular lines and prosthetic valves. Though thought to be nonpathogenic they are major cause for prosthesis failure.

Staphylococcus Aureus

This coagulase-positive *Staphylococcus* typically produces golden colonies on blood agar. Ten to forty percent of populations carry *S. aureus* in their anterior nares and perineum (especially in hospital inpatients). It is the most common cause for wound and skin sepsis. It can have direct invasion as well as spread through blood. It is the commonest organism found in blood cultures of patients with septicemia. The majority of

TABLE 1 Gram-positive bacteria with their reservoir and transmission, diseases caused by them and anti-biotics to which they are sensitive

Gram-positive bacteria	*Reservoir*	*Transmission*	*Diseases*	*Antibiotics*
Cocci				
Staphylococcus aureus	Nose, skin	Hands, sneezing, wounds, foods	Furuncles, carbuncles, impetigo, surgical infection pneumonia, toxic shock syndrome, endocarditis, food poisoning	MSSA: Beta-lactams; MRSA: Vancomycin, Fusidic acid
Streptococcus pyogenes	Throat, skin	Respiratory droplets, direct contact	Suppurative: Tonsillo-pharyngitis, scarlet fever, pyoderma / impetigo; Nonsuppurative: Rheumatic fever, glomerulonephritis	Beta-lactams, macrolides (erythromycin)
Streptococcus pneumoniae	Upper respiratory tract	Respiratory droplets (not highly communicable)	Sinusitis, otitis media, pneumonia, meningitis	Penicillin G, vancomycin +-rifampin (resistance cases)
Streptococcus viridans	Oropharynx		Dental caries, endocarditis	Penicillin G with aminoglycoside (endocarditis)
Rods				
Aerobic or Facultative Anaerobic Rods				
Corynebacterium diphtheriae	Oropharynx	Respiratory droplets	Diphtheria	Erythromycin and antitoxin
Nocardia asteroides	Soil, dust	Airborne, traumatic implantation	Subcutaneous abscess with granules (mycetoma), bronchopulmonary nocardiosis	Sulfonamides in high doses, trimethoprim + sulfamethoxazole
Mycobacterium tuberculosis	Lungs	Respiratory droplets, droplet nuclei	Tuberculosis	Multiple drug therapy: Isoniazid, rifampin, pyrazinamide, ethambutol, streptomycin
Gram-positive bacteria	*Reservoir*	*Transmission*	*Diseases*	*Antibiotics*
Nontuberculous Mycobacteria	Surface water, soil, cigarettes	Noncontagious	Mycobacterial lymphadenitis	Surgical removal
Mycobacterium leprae	Mucosa, skin, nerves	Nasal discharge	Leprosy	Multiple drug therapy: Dapsone and rifampin, clofazimine (for lepromatous)

Contd...

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Anaerobic Rods				
Clostridium tetani	Soil	Puncture wounds, trauma	Tetanus	Hyperimmune human globulin, metronidazole, penicillin
Clostridium difficile	Antibiotic associated diarrhea			Metronidazole, vancomycin
Clostridium botulinum causing Botulism and *Clostridium perfringens* causing gas gangrene and food poisoning are also spore formers				
Actinomyces israelii	Gingival crevices, female genital tract	Endogenous	Actinomycosis: Abscess with sulfur granules	Ampicillin or penicillin G and surgical drainage
Lactobacillus				
Non-Gram Staining				
Mycoplasma pneumoniae	Respiratory tract	Respiratory droplets, close contact	Pneumonia, pharyngitis	Macrolides e.g. erythromycin, azithromycin, clarithromycin

Abbreviations: MSSA, methicillin-sensitive Staphylococcus aureus; MRSA, methicillin-resistant Staphylococcus aureus

TABLE 2 Gram-negative bacteria with their reservoir and transmission, diseases caused by them and antibiotics to which they are sensitive

Gram-negative bacteria	*Reservoir*	*Transmission*	*Diseases*	*Antibiotics*
Cocci				
Neisseria meningitidis	Nasopharynx	Respiratory droplets	Meningitis, meningococcemia	Penicillin G, ceftriaxone
Neisseria gonorrhoeae	Genital tract	Sexual contact, birth	Gonorrhea	Ceftriaxone
Morexella catarrhalis is morphologically similar to Neisseria				
Rods				
Pseudomonas aeruginosa	Ubiquitous in water	Water aerosols, raw vegetables, flowers	Compromising condition, opportunistic infections e.g. cellulitis, pneumonia, septicemia	Antipseudomonal penicillin, third generation cephalosporins
Legionella pneumophila	Rivers, streams, A.C. water cooling tanks	Aerosols; no human transmission	Atypical pneumonia	Quinolones, macrolides, and rifampin in immunocompromised patients
Bordetella pertussis	Human (vaccinated)	Respiratory droplets	Whooping cough (Pertussis)	Erythromycin
Gram-negative bacteria	*Reservoir*	*Transmission*	*Diseases*	*Antibiotics*
Helical (and Microaerophilic)				
Campylobacter jejuni	Intestinal tracts of humans and domestic animals	Fecal-oral, primarily from poultry	Ubiquitous infectious diarrhea	Erythromycin, quinolones (Penicillin-resistant)
Helicobacter pylori	Humans	Fecal-oral, oral-oral	Gastritis, duodenal ulcers	Amoxicillin + clarithromycin + omeprazole
Facultative Anaerobic Rods				
Enterobacteriaceae				
Escherichia coli	Colon, vagina, urethra		Urinary tract infections, diarrhea, septicemia and meningitis	Quinolones
Klebsiella pneumoniae	Colon, upper respiratory tract	From own flora	Pneumonia, septicemia, urinary tract infection	As per sensitivity report
Proteus mirabilis / vulgaris	Colon, water, soil		Urinary tract infection, septicemia	As per susceptibility
Pasteurellaceae				
Haemophilus influenzae	Nasopharynx	Respiratory droplets, shared toys	Epiglottitis, sinusitis, otitis media, pneumonia, meningitis	Cefotaxime, ceftriaxone

Contd...

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Anaerobic Straight to Helical Rods				
Spirochetes				
Treponema pallidum	Genital tract	Sexually, across placenta	Syphilis	Benzathine penicillin
Borrelia burgdorferi	Nymphs, ticks	Nymphs, ticks	Lyme disease	Primary: Doxycycline, amoxicillin, azithromycin, clarithromycin; Secondary: Ceftriaxone
Chlamydiaceae				
Chlamydia trachomatis	Genital tract, eyes	Sexual contact, hand-to-eye	Lymphogranuloma venereum, trachoma	Doxycycline, azithromycin
Chlamydia psittaci	Parrots, turkeys	Dried secretions, feces	Psittacosis (Atypical pneumonia)	Doxycycline

TABLE 3 Nasopharyngeal microflora

Normal microflora	Oral streptococci, *Neisseria sp.*, *Haemophilus sp.*, *Candida albicans*, *Corynebacteria sp.* (diphtheroids), anaerobes
Potentially pathogenic bacteria	Pyogenic streptococci (Group A, C and G), *M. catarrhalis*, *S. pneumoniae*, *H. influenzae*, and *N. meningitidis*
Inhibiting adherence of pathogenic bacteria	Oral streptococci

S. aureus isolates are resistant to penicillin due to the production of penicillinase.

- Methicillin-resistant *S. aureus*: The isolation of epidemic methicillin-resistant *S. aureus* (MRSA) strains, which are resistant to beta-lactam antibiotics, has been increasing.

STREPTOCOCCI

They are Gram-positive catalase-negative cocci which produce chains in broth culture. They are separated into several groups (Table 1) using following tests:

1. Hemolysis on blood agar: Beta (clear), alpha (green) and gamma (nonhemolytic).
2. Lancefield grouping antisera (A-O): The main groups are A, B, C, D and G.
3. Biochemical tests.
4. Growth characteristics.
5. Ribosomal RNA analysis.

Beta-hemolytic Streptococci (Pyogenic streptococci) cause complete hemolysis of blood on blood agar. Group A streptococci (*Streptococcus pyogenes*) is the most important beta-hemolytic streptococci. Group C and G occasionally produce diseases similar to group A. Group B *streptococcus* (*Streptococcus agalactiae*) cause neonatal sepsis and invasive disease in diabetics, alcoholics and immunocompromised patients. Streptococci are generally sensitive to all the beta-lactam and macrolide groups of antibiotics.

Group A β-Hemolytic Streptococci (*Streptococcus pyogenes*)

Streptococcus pyogenes is the most important beta-hemolytic streptococci. They cause suppurative diseases of tonsillitis, skin sepsis (cellulitis), severe necrotizing fasciitis, septicemia, septic arthritis and streptococcal toxic shock syndromes. The nonsuppurative diseases caused by *Streptococcus pyogenes* include glomerulonephritis and rheumatic fever.

Streptococcus Pneumoniae (*Pneumococcus*)

Pneumococci produce incomplete or alpha hemolysis on blood agar with draughtsman-like colonies that are bile soluble and sensitive to optochin. They are the commonest bacteria in all age groups of otitis media, sinusitis and pneumonia. They can cause meningitis via hematogenous spread. Though usually sensitive to penicillin, they have started showing resistance to it.

- *Conjugate pneumococcal vaccines* (7, 9 and 11 valent): Reduce invasive disease, lobal pneumonia and acute otitis media.

Oral Streptococci (*Streptococcus viridans*)

Oral streptococci produce incomplete or alpha hemolysis on blood agar but are not sensitive to optochin. They are not groupable. They occasionally cause tooth decay and dental abscesses but commonly cause native valve endocarditis. Surprisingly they have been shown to reduce the incidence of recurrent otitis media in otitis-prone children.

Anginosus-Milleri Group (*Streptococcus anginosus* or *Streptococcus milleri*)

Streptococcus intermedius, *Streptococcus anginosus* and *Streptococcus constellatus* are included in this genus. They are one of the commonest microorganisms in deep pyogenic lesions, liver, lung and brain abscesses.

CORYNEBACTERIUM DIPHTHERIAE

Corynebacteria, which usually do not cause any disease, are Gram-positive pleomorphic bacilli which look like Chinese letters. The strains of *Corynebacterium diphtheriae* that are infected with lysogenic bacteriophage carrying "tox gene", produce exotoxin (diphtheria toxin). The diphtheria toxin inhibits cellular protein synthesis and results in pharyngeal damage and myocardial and neurological toxicity. Diphtheria is described in detail in Chapter Pharyngitis and Adenotonsillar Diseases.

NEISSERIA SPECIES

These Gram-negative diplococci appear in pairs. The two important members, which are identified on the basis of biochemical reactions and monoclonal antisera, are mentioned here.

1. *Neisseria gonorrhoeae* (*Gonococcus*): Gonococci produce sexually transmitted disease (STD) gonorrhea and cause cervicitis, urethritis and pelvic inflammation and occasionally pharyngitis.
2. *Neisseria meningitidis* (*Meningococcus*): Meningococci produce meningitis, septicemia and rarely septic arthritis.
   a. Conjugate serogroup C meningococcal vaccine: In young teenagers and children it reduces incidence of invasive serogroup C disease.

MOREXELLA CATARRHALIS

This Gram-negative diplococcus is morphologically similar to Neisseria. This is the third commonest cause of otitis media. It also causes acute sinusitis and infective exacerbation of chronic obstructive pulmonary disease (COPD). They produce beta-lactamase and are resistant to penicillins including ampicillin.

HAEMOPHILUS INFLUENZAE

Haemophilia are Gram-negative pleomorphic coccobacilli diplococci. *Haemophilus influenzae* is the important pathogen of this genus and identified by its poor growth on blood agar.

1. *Noncapsulated strains*: They are one of the commonest causes of otitis media and sinusitis.
2. *Capsular type b Strains*: They are associated with severe invasive disease such as meningitis, arthritis and epiglottitis.
   - Conjugated *H. influenzae* type b (Hib) capsular polysaccharide vaccine: A dramatic reduction in incidence of Hib invasive disease has been noted.

BORDETELLA PERTUSSIS

These small Gram-negative *coccobacilli* produce pertussis (whooping cough). Pertussis toxin mediates the paroxysmal cough and induces lymphocytosis. Pertussis is common in infants and children. The reservoir of infection is adult carriers and subclinical cases.

- *Vaccine*: The immunity from vaccine (primarily IgA) lasts for 5–10 years and wanes after childhood. Newborns have little immunity. Vaccinated humans of >10 years of age serve as reservoir. Immunity to actual pertussis is life long.

About 12–20% of afebrile adults with cough >2 weeks have pertussis.

PSEUDOMONAS AERUGINOSA

Pseudomonas is thin Gram-negative bacilli, which survive well in moist warm environments. They are usually resistant to most antibiotics and disinfectant. *Pseudomonas aeruginosa* is the commonest pseudomonad that produces opportunistic infections. Because of its pigments or pyocyanins production it can be recognized in culture and clinical samples. In chronic otitis externa *P. aeruginosa* colonizes in damaged ear canal. It can cause invasive disease in immunosuppressed and poorly controlled diabetic patients.

- *Ecology*: *P. aeruginosa* is ubiquitous water and soil microorganism that grows to very high titers overnight in standing water whether distilled or tap. Sources of infections include raw vegetables, respirators, sink drains, humidifiers and flowers. It can get on skin from fecal organism.

Pseudomonas aeruginosa: Susceptibilities are important because drug resistance is very common.

ENTEROBACTERIACEAE

These aerobic Gram-negative bacilli grow under aerobic and anaerobic conditions. They become part of nasopharyngeal flora in ICU ventilated patients and can produce lower respiratory tract infection.

- *Escherichia coli and Klebsiella spp* are commonly found in urinary and abdominal sepsis but occasionally produce infection in upper respiratory tract (URT) and middle ear.
- *Klebsiella rhinoscleromatis*: It can produce chronic granulomatous infection of URT.

ANAEROBES

Only a few of anaerobic genera produce infection. They are more likely to be isolated in large amount of pus than the swab.

1. Bacteroides fragilis: This Gram-negative bacillus is the commonest anaerobe found in polymicrobial infections.
2. Fusobacterium necrophorum: This fusiform Gram-negative bacillus occasionally produces necrobacillosis, which is a rapidly progressive invasive disease.

MICROAEROPHILIC BACTERIA

Microaerophilic bacteria can tolerate with very low concentration of oxygen.

1. Actinomyces: These Gram-negative bacilli can penetrate deep tissues after mucocutaneous injury and produce actinomycosis, which is a polymicrobial disease. Clumps of Gram-negative bacilli can be seen on biopsy. In pus they look like sulfur granules. *Actinomyces israelii*, *Actinomyces meyeri* and *Actinomyces odontolyticus*, the species usually isolated are generally sensitive to penicillin.

MYCOBACTERIA

These acid- and alcohol-fast bacilli (AFB) are seen on Ziehl-Neelsen stain. They grow slowly in aerobic conditions and are not isolated on routine agar media. The 25 Mycobacteria spp are usually split into two groups *Mycobacterium tuberculosis* and *Atypical mycobacteria*.

1. *Mycobacterium tuberculosis*: It grows slowly on the egg-containing Lowenstein-Jensen media. The visible colonies take 4–6 weeks to appear. The use of liquid media with automated detection of growth and DNA probes can decrease this isolation time.
2. Atypical mycobacterium: *Mycobacterium avium intracellulare*, *Mycobacterium kansasii*, *Mycobacterium scrofulaceum* and *Mycobacterium malmoense* are responsible for cervical lymphadenopathy in children. Disseminated *Mycobacterium avium intracellulare* can be an opportunistic infection in HIV patients.

MYCOPLASMA PNEUMONIAE

Mycoplasmas do not have cell wall and diagnosis is based on serological tests. *Mycoplasma pneumoniae* is the important pathogen of this group and common cause of epidemics of pneumonia in children and young adults. Occasionally it can cause otitis media. The macrolides, tetracyclines and fluoroquinolones are often effective.

CHLAMYDIAE

These tiny Gram-negative intracellular bacteria form inclusion bodies in the cells and replicate intracellularly. The macrolides, tetracyclines and fluoroquinolones are often effective.

1. *Species*: The following three species produce disease:
   - *i. Chlamydophilia psittaci*: Severe lower respiratory tract infection (LRTI).
   - *ii. Chlamydophilia trachomatis*: Ocular and genital infection.
   - *iii. Chlamydophilia pneumoniae*: Upper and lower RTI.
2. *Diagnosis*: It is based on following tests:
   - i. Rising titers of complement-fixing antibodies to whole cell antigen.
   - ii. Species specific antibody by microimmunofluorescence
   - iii. Species specific DNA by polymerase chain reaction (PCR).

SPIROCHAETES

These irregular curved motile bacteria are not isolated on routine culture.

- Borrelia species
- *Borrelia vincenti*: It is seen in acute necrotizing gingivitis and Vincent's angina, accompanied by anaerobic fusobacteria.
- *Borrelia burgdorferi*: It causes Lyme disease.
- *Treponema pallidum*: It causes STD syphilis.

ANTIBIOTICS

The use of antibiotics can disturb the delicate bacterial ecology of the patient and allow proliferation of resistant bacteria. It is becoming vital to realize the importance of appropriate use of antibiotics. The pace of development of antibiotic resistant is faster than the development of new antibiotics. Ototoxic antimicrobial agents are also discussed in Chapter 13: Sensorineural Hearing Loss.

The most common ear, nose and throat infections, bacteria responsible for them and antibiotics used to treat them are shown in Table 4.

Antimicrobial Resistance

The number of multiresistant Gram-negative bacteria is becoming an increasing problem in hospitals. Pneumococci are getting resistant to penicillin. The methicillin-resistant *Staphylococcus aureus* (MRSA) are getting resistant to beta-lactam agents (such as flucloxacillin), quinolones and macrolide antibiotics. Beta-lactamase can break down the beta-lactam bond of cephalosporin. Multidrug-resistant TB isolates are resulting in failures of AKT. Antimicrobial resistance can be acquired by the organisms by either (1) genetic mutation, selection of resistant clone or (2) acquisition of genetic material required to produce resistance in the form of plasmids, transposons, or naked DNA. The following mechanism underlies microbial resistance to antibiotics:

1. Ciprofloxacin: Reduced topoisomerase sensitivity to inhibitors. Changes in the structure of topoisomerase IV that inhibit nucleic acid synthesis.
2. Sulfonamides: Increased synthesis of p-aminobenzoic acid (PABA).

TABLE 4 Most common ear, nose and throat infections, bacteria responsible for them and antibiotics used to treat them

Infection disease	*Bacteria*	*Antibiotics first-line*	*Antibiotics second-line*
Croup with secondary bacterial infection	*S. aureus*, *H. influenzae*	Ampicillin + Sulbactam	2nd or 3rd generation cephalosporins
Epiglottitis	*H. influenzae*	Ampicillin + Sulbactam	2nd or 3rd generation cephalosporins
Pharyngitis	*S. pyogenes*, Mycoplasma, Chlamydia, Gonococci	Penicillin	Macrolides, 1st generation cephalosporins
Tonsillitis	*S. pyogenes*, mixed anaerobes	Penicillin	Clindamycin, amoxicillin + clavulanic acid, dicloxacillin and cephalexin
Deep cervical abscess	Mixed anaerobes, Pseudomonas	Clindamycin + quinolone (for pseudomonas)	
Sinusitis	Streptococcus sp., *H. influenzae*	Amoxicillin, trimethoprim + sulfamethoxazole	2nd or 3rd generation cephalosporins, quinolones, macrolides
Acute otitis media	Streptococcus sp., *H. influenzae*	Amoxicillin, macrolides	Cefaclor
Chronic otitis media	Pseudomonas, Proteus sp., *S. aureus*, mixed anaerobes	Amoxicillin, macrolides, ciprofloxacin	
Mastoiditis	Streptococcus sp., *H. influenzae*	Cephalosporins, macrolides	
Acute otitis externa	*S. aureus*, Pseudomonas, Proteus sp.	Ofloxacin + steroid ear drops	

3. Beta-lactams: Formation of drug-inactivating hydrolases.
4. Penicillins: Changes in penicillin-binding proteins (PBPs).
5. Extended-spectrum penicillins: Structural changes in porins.
6. Macrolides: Formation of methyltransferases that change receptors structure.
7. Isoniazid: A decrease in the expression of cat k gene that encodes catalase.
8. Rifampicin: Mutations of the gene that codes for DNA-dependent RNA polymerase.

Drug Hypersensitivity Reactions

1. IgE-mediated: It causes rapid onset anaphylaxis, angioedema and laryngospasm.
2. IgM and IgG antibodies fixed to cells: They cause vasculitis, neutropenia and positive Coomb's test.
3. Immune complex formation: It leads to vasculitis, serum sickness and interstitial nephritis.
4. T-cell mediated: It results into urticarial and maculopapular rashes and Stevens-Johnson syndrome.

Prerequisites of Antimicrobial Therapy

For the effective treatment of any infection, the treating physician should have the following knowledge of the antibiotic, causative organisms, course of disease and the patient:

1. *Microorganism*: The likely causative organism and its antimicrobial susceptibility.
   - Gram staining/culture and sensitivity test: for antibiotic specificity.
   - Local bacterial resistance pattern.
2. *Antimicrobial agents*: Its pharmacokinetics, spectrum of action, side effects, drug interactions. Lipid soluble chloramphenicol is well absorbed orally and crosses blood-brain barrier. Antibiotics excreted in urine are more effective in UTI.
3. *Route of administration and doses*: They depend upon type and severity of infection.
4. *Disease and site of infection*: The natural course of disease and site of infection. Antibiotics may be given for proven or suspected microorganism or for prophylaxis. Not all the antibiotics cross the blood-brain barrier. Lower pH in an abscess reduces the efficacy of aminoglycosides and macrolides.
5. *Patients*:
   - Age of the patient.
   - History of drug allergies and drugs taking.
   - Concurrent medical conditions: Renal and hepatic functions, immune status and pregnancy.

- *Minimum inhibitory concentration*: Minimum inhibitory concentration (MIC) is the lowest concentration of antimicrobial agent at which growth of the microorganism is completely inhibited.
- *Combination of antibiotics*: It can lead to additive (1+1=2), synergistic (1+1=>2) or antagonistic (1+1=<2) effects. Combination of penicillins and aminoglycosides is an example of synergy for enterococci and *P. aeruginosa*.

More than one antibiotic are preferred in the following conditions:
1. Polymicrobial infections
2. Broad-spectrum coverage of an unknown infection
3. For increasing bactericidal activity
4. To prevent emergence of resistant strains, e.g. tuberculosis.

Surgical Prophylaxis

Antibiotics should be given immediately before or during the surgery. The benefits of antibiotics from extending the duration of prophylaxis beyond wound closure are not proved. Extended prophylaxis can disturb patient natural flora and develop resistance.

Sensitivity Testing

1. *Stokes method*: Known concentrations of the antibiotic are impregnated into a disc. It is released into the agar onto which test and control microorganisms are inoculated. After an overnight inoculation the zone of inhibition of growth is compared with known test microorganism.
2. *Breakpoint method*: It is used to test multiple microorganisms which are inoculated onto different agar plates that contain high and low concentrations of a range of antibiotics.
3. *Polymerase chain reaction:* Polymerase chain reaction (PCR) is used for more slow growing microorganisms such as *Mycobacterium tuberculosis*. It detects antibiotic resistance gene.

Reasons of failed antibiotic treatment may be following:
1. Incorrect diagnosis and presumptive microorganism
2. Abscess requiring surgical drainage
3. Foreign body
4. Resistant microorganism
5. Patient noncompliance
6. Incorrect dose, duration and route of administration
7. Adverse reactions

Classification

Bactericidal drugs are preferred in severely immunocompromised patients. Bacteriostatic antibiotics depend greatly on active host defense mechanism. On the basis of cellular target sites, the main antibiotics can be categorized into following groups:

1. Inhibitors of bacterial cell wall synthesis (Beta-lactam antibiotics)
2. Inhibitors of nucleic acid synthesis
3. Inhibitors of bacterial protein synthesis.

INHIBITORS OF BACTERIAL CELL WALL SYNTHESIS (BETA-LACTAM ANTIBIOTICS)

They contain beta-lactam ring that mimics the cross-link of peptidoglycan "backbone" of cell wall. Inhibition of this cross-link weakens the cell wall and results in cell death by lysis.

Penicillins

These are bactericidal beta-lactams. Penicillin V and aminopenicillins are administered orally but remainders are given parenterally.

- Classification and spectrum: They are categorized into following groups on the basis of their bacterial activity:
  - *Natural penicillins* (e.g. Penicillin V): Gram-positive bacteria excluding staphylococci but including streptococci and pneumococcal meningitis; spirochetes, clostridia and corynebacteria and some Gram-negative cocci including meningococci, most of anaerobes

(excluding Bacteroides fragilis) and penicillin-sensitive gonococci. The majority of S. aureus isolates are resistant to penicillin due to the production of penicillinase.

- *Penicillinase-resistant penicillins* (e.g. Flucloxacillin): Additional activity against penicillin-resistant but MRSA. Other antistaphylococcal penicillins are methicillin, oxacillin and dicloxacillin.
- *Aminopenicillins* (e.g. Ampicillin and amoxicillin): Additional activity against Gram-negative bacilli including *Haemophilus influenzae* (upper and lower respiratory tract infections), *E.coli*, Proteus spp, *S. pyogenes*, most *S. pneumoniae*, and anaerobes and many Enterobacteriaceae.
  - *Expanded spectrum penicillins*: Increased activity against enterobacteriaceae and pseudomonads.
- *Carbapenems* (e.g. Imipenem and meropenem): Broad spectrum against Gram-positive and Gram-negative bacteria including beta-lactamase producing Gram-negative bacteria and anaerobes including *Bacteroides fragilis*.
- *Monobactams* (e.g. Aztreonam): Only active against Gram-negative aerobic bacteria including pseudomonas.
- *Beta-lactamase inhibitors* (e.g. Clavulanic acid): Increased activity against anaerobes and enterobacteriaceae. Clavulanic acid, sulbactam and tazobactam inhibit beta-lactamase and are used in combination with penicillins and cephalosporins.

- *Side effects*: Anaphylaxis, angioneurotic edema, urticaria, Stevens-Johnson syndrome, dermatitis, morbilliform eruptions, allergic vasculitis, reversible neutropenia (on prolonged high dose), epilepsy (in cases of renal failure) and diarrhea including *C. difficile* infection (with ampicillin and amoxicillin).

- The most common bacteria in acute otitis media and sinusitis are *S. pneumoniae*, *H. influenzae*, and *Morexella catarrhalis*. These diseases do not respond to amoxicillin if *H. influenzae* are nontypable, or *H. influenzae* and *M. catarrhalis* are beta-lactamase producers.
- Augmented penicillins with beta-lactamase inhibitors that are active against *H. influenzae*, *Morexella catarrhalis*, *S. aureus*, *S. pneumoniae* and most anaerobes include amoxicillin/clavulanate and ampicillin/sulbactam.
- Antipseudomonal penicillins are ticarcillin, carbenicillin and piperacillin.

Cephalosporins

They are closely related to penicillins and not active against enterococci. They are grouped into following four generations:

1. *First-generation cephalosporins*: They are administered orally and most active against Gram-positive cocci such as MSSA, streptococci and most pneumococci. They are active against *E. coli*, *Klebsiella pneumoniae*, and some proteus sp. They do not cross blood-brain barrier. They are used in surgical prophylaxis. Examples include cephalexin, cefazolin and cephradine.
2. *Second-generation cephalosporins*: They are active against Gram-positive cocci, *H. influenzae* and *Morexella catarrhalis*. They have increased Gram-negative coverage including some anaerobes. Most do not enter CNS. Both oral and intravenous (IV) formulations are available. Example includes cefotetan (*Bacteroides fragilis*) and cefaclor (*H. influenzae* and *M. catarrhalis*).
3. *Third-generation cephalosporins*: Wider-spectrum includes Gram-positive and Gram-negative cocci and many Gram-negative rods. These are injectable agents. They have increased activity against Gram-negative bacteria including *H. influenzae* and *M. catarrhalis*. Ceftazidime is highly active against *Pseudomona aeruginosa*. Cefotaxime and ceftriaxone cross blood-brain barrier and can treat meningitis. Cefoperazone does not enter CNS. Cefixime can be given orally. Ceftizoxime is active against *B. fragilis*.
4. *Fourth-generation cephalosporins*: They have increased activity against Gram-negative bacteria and some pseudomonads. Only fourth-generation cephalosporins are active against *B. fragilis*. Cefepime has one of the broadest antibiotic coverage and is typically used for nosocomially acquired mixed infections.

Microorganisms not covered by cephalosporins are: MRSA, Atypicals (Chlamydia, Mycoplasma), Listeria monocytogenes and enterococci (MALE).

- Side effects: Low incidence of anaphylaxis but cross-reactivity is present in about 10% of penicillin allergic patients and rashes can occur. Higher risk of *C. difficile* with third-generation cephalosporins. Phlebitis (IV injections), pain (IM injections).

Mild itchy rashes are common with penicillins and cephalosporins. A penicillin anaphylaxis patient should not be given cephalosporins. However, cephalosporins can be prescribed in patients with penicillin rashes.

Glycopeptides (e.g. Vancomycin and Teicoplanin)

They are not absorbed orally and do not penetrate blood-brain barrier. Glycopeptides are mainly (80%) excreted in urine.

- *Spectrum*: They only have Gram-positive activity. Resistance is uncommon but increasing especially in enterococci. It can be used in patients who are allergic to penicillin. They are very effective in resistant Gram-positive infections such as MRSA. Orally, vancomycin is a second-line of antibiotic in cases of *C. difficile* diarrhea.
- *Red-man syndrome*: Rapid infusion of vancomycin release histamine and cause renal toxicity especially with concomitant use of aminoglycosides.
- *Ototoxicity*: Tinnitus and high-frequency hearing loss may be antecedents to profound hearing loss. Similar to aminoglycosides cochlear hair cell degeneration is irreversible. The ototoxicity is related to a cumulative dose rather than the daily dose. Incidence is high in patients with advanced age, renal impairment and concomitant aminoglycoside therapy.

Vancomycin is the most commonly used antibiotic for methicillin-resistant strains of *S. aureus.*

INHIBITORS OF NUCLEIC ACID SYNTHESIS

Quinolone Antibiotics

They interfere with DNA gyrase that is required for supercoiling of bacterial DNA during transcription and replication. These bactericidal agents are administered orally and have wide tissue distribution except CSF.

- *Spectrum:* Gram-positive and Gram–negative bacteria including *H. influenzae*, *M. catarrhalis*, *N. gonorrhoeae*, *Salmonella typhi* (typhoid) and even pseudomonads. Newer quinolones are active against anaerobic bacteria and streptococci.
- *Side effects*: Epileptic seizures, increasing theophylline levels and photosensitivity (newer quinolones).
- *Resistance:* Because of the abundant use of quinolones in common respiratory and pharyngeal infections, resistant strains of pneumococci, streptococci and staphylococci have started appearing. It is suggested that they should primarily be used for pseudomonal infections.

The oral antibiotics that are active against pseudomonal infections are fluoroquinolones, which include ciprofloxacin, ofloxacin and levofloxacin.

Sulphonamides and Trimethoprim (Co-trimoxazole)

These bacteriostatic agents interfere with folic acid synthesis in bacteria and have good oral absorption and tissue penetration. It is unaffected by beta-lactamase and is appropriate for the treatment of chronic bronchitis, sinusitis and otitis media.

- *Spectrum*: Gram-positive and Gram–negative bacteria, *Pneumocystis carinii* and Nocardia spp. Trimethoprim is used in urinary and respiratory tract infection. The combination of sulfonamides and trimethoprim (co-trimoxazole) is very effective against *H. influenzae*, *M. catarrhalis* and some streptococci. It is not effective against anaerobes.
- *Side effects*: Allergic skin rashes, photosensitivity, Stevens-Johnson syndrome and aplastic anemia.

Sulphonamides are well known cause of fixed drug eruptions, photosensitivity rashes and Stevens-Johnson syndrome.

Rifamycins

They inhibit bacterial DNA-dependent RNA polymerase and transcription to mRNA. They are absorbed orally and have good tissue, CSF and intracellular penetration.

- *Spectrum*: Gram-positive bacteria and mycobacteria. It is used for prophylaxis against meningococcal and *H. influenzae type b* infection. It is used in combination with other drugs for mycobacterial and severe staphylococcal infections.
- *Side effects*: Hepatitis and occasionally hemolytic anemia.

Rifampicin is used for the prophylaxis and treatment of carrier of nasopharyngeal *H. influenzae* and meningococci infections.

Metronidazole

It produces toxic metabolites which damage bacterial DNA. Metronidazole is absorbed orally and penetrates tissue and crosses blood-brain barrier.

- *Spectrum*: Treatment and prophylaxis of anaerobic bacteria (including *Bacteroides fragilis* and *C. difficile*) and protozoa such as giardiasis and amoebiasis. Metronidazole is the antibiotic of choice for the treatment of pseudomembranous enterocolitis due to *C. difficile*.
- *Side effects*: Dizziness and metallic taste; Peripheral neuropathy and encephalopathy with long term use.
- *Caution*: Alcohol must be avoided in patients taking metronidazole.

C. difficile pseudomembranous enterocolitis: Metronidazole is the antibiotic of choice for the treatment of pseudomembranous enterocolitis due to *C. difficile*. Oral, vancomycin is a second-line of antibiotic in cases of *C. difficile* diarrhea.

INHIBITORS OF BACTERIAL PROTEIN SYNTHESIS (RIBOSOMAL)

Tetracyclines

Tetracyclines bind to the 30S ribosomal subunit and inhibit binding of tRNA to 30S ribosome. These bacteriostatic agents are absorbed orally and well distributed in tissues but do not penetrate CSF. Half life varies with formulations. Doxycycline can be given once a day.

- *Spectrum*: Broad spectrum antibiotic. Gram-positive bacteria including *S. aureus* and pneumococci; some of the Gram-negative bacteria such as *H. influenzae* and gonococci; atypical bacteria such as Chlamydia, mycoplasmas, reickettsiae and spirochetes; some anaerobic and protozoal activity. They are not usually used as first-line of antibiotic except in rickettsial infections.

Indications of tetracyclines: Back up to penicillin G in syphilis, prophylaxis in chronic bronchitis and for acne. Minocycline appears in high concentration in saliva and is used in meningococcal carriers.

- *Side effects*: Gastrointestinal intolerance (commonest), photosensitivity reaction and teeth enamel staining (in children).
  - *Teratogenic*: They may cause fetal growth inhibition and congenital limb abnormalities.
- *Caution*: In renal and liver failure; childhood (under the age of 10) and pregnancy. Tetracyclines are concentrated in developing bones and teeth.

Ampicillin, penicillin V, sulfonamides and tetracyclines, when given for > 10–14 days, interact with contraceptives and reduce their effectiveness. Other methods of contraception should be advised to fertile females for at least one cycle after the discontinuation of the antibiotic.

Chloramphenicol

This bacteriostatic agent binds to the 50S ribosome. Chloramphenicol is absorbed orally and distributed widely in tissues. It crosses blood-brain barrier. Currently it is used only topically for eye infections.

- *Spectrum*: Almost all Gram-positive cocci and Gram-negative bacteria and many anaerobic bacteria, mycoplasmas, rickettsiae, and spirochetes; bactericidal against *H. influenzae*, and Neisseria sp. Pseudomonas are resistant.

- *Side effects*: Grey baby syndrome in neonates, reversible dose-dependent bone marrow suppression and irreversible aplastic anemia (1 in 40,000).

Macrolides (Erythromycin and Azithromycin)

Macrolides inhibit the binding of tRNA to 50S ribosome in bacteria. These bacteriostatic agents are absorbed orally. Clarithromycin has longer half-life and azithromycin is concentrated intracellularly. Macrolides form alternative therapy for respiratory infection, chlamydial and campylobacter infections.

- *Erythromycin*: Spectrum: Aerobic and anaerobic Gram-positive cocci (except enterococci) and some Gram-negative anaerobes; Bordetella sp., mycoplasmas, rickettsiae, *C. diphtheriae*, Campylobacter, *T. pallidum*, *Borrelia burgdorferi* and Legionella sp.
- *Azithromycin*: It is more active against *Haemophilus influenzae* and *M. catarrhalis* (causing otitis media and sinusitis), Chlamydia, Gonococcus, and *Mycobacterium avium-intracellulare*.
- *Clarithromycin*: It has activity against *M. avium complex* (MAC) and *H. pylori*.
- Side effects: Nausea, vomiting, cramps, reversible jaundice, pseudomembranous colitis due to *Clostridium difficile* overgrowth and thrombophlebitis (in intravenous use).

Azithromycin is safe in pregnancy and does not inhibit drug metabolism. Clarithromycin causes reversible hearing loss at high dose and teratogenic in animals.

- Clindamycin is similar to macrolides but has better activity against anaerobes especially Bacteroides sp. It has no activity against mycoplasma.
- Lincomycin has similar spectrum to clindamycin but is poorly absorbed.

The antibiotics active against anaerobes include beta-lactams, clindamycin and metronidazole.

Fusidic Acid

It inhibits bacterial translocation. Resistance emerges rapidly unless a second antibiotic is given. It is absorbed well orally. Fusidic acid has good tissue penetration including brain abscess but cannot reach the CSF.

- *Spectrum*: *S. aureus*. It can be used in severe staphylococcal infections (bone and joint) in combination with another anti-staphylococcal antibiotic.
- *Side effects*: Cholestasis and jaundice especially with intravenous formulations.

Aminoglycosides

They bind to both subunits of ribosome, resulting in misreading of genetic code. Aminoglycosides are not absorbed orally and have reduced activity at a low pH. See Chapter 13: Sensorineural Hearing Loss for more details.

- *Spectrum*: They are given for severe Gram-negative and staphylococcal infections especially hospital acquired infection including *E. coli* and Serratia, Klebsiella, Proteus and Enterobacter sp. In combination with beta-lactam antibiotics they are particularly effective against Pseudomonas sp. Streptomycin is used for mycobacterial infections and brucellosis. Anaerobic infections are resistant to aminoglycoside.
- *Side effects*: Ototoxicity, nephrotoxicity and neuromuscular paralysis.
- *Caution*: Because of their narrow therapeutic window, therapeutic monitoring is mandatory. Doses must be adjusted according to the age, renal function and weight.

Once-daily gentamicin: Despite its short elimination half-life, gentamicin can be given once daily at high dose in the severe infections of aerobic Gram-negative rods. This regimen offers similar effectiveness and does not increase the risk of ototoxicity and development of resistance rather it results in less nephrotoxicity.

Streptogramins

A combination of compounds binds to the bacterial 50S ribosome. The use is limited to the treatment of resistant Gram-positive infections.

- *Spectrum*: Gram-positive activity including some multiresistant Enterococci, Mycoplasma sp., Chlamydia sp. and Legionella sp.
- *Side effects*: Thrombophlebitis.

Oxazolidinones (Linezolid)

These new class of antibiotics act on bacterial ribosome and are well absorbed orally.

- *Spectrum*: Bacteriostatic Gram-positive activity especially against resistant staphylococci such as MRSA and vancomycin-resistant enterococci (VRE).
- *Side effects*: Thrombocytopenia with prolonged therapy.

Antibiotics never used in pregnancy: Aminoglycosides, erythromycin estolate, clarithromycin, quinolones and tetracyclines.

ANTITUBERCULAR DRUGS

- *M. tuberculosis*: Combination drug therapy prevents emergence of resistance and provides possibly synergistic effects against *M. tuberculosis*.
  - *Drugs of primary regimens* (2–4 drugs): INH, rifampin, ethambutol and pyrazinamide. In cases of highly resistant organisms other drugs may be required.
  - *Back up drugs*: Aminoglycosides (streptomycin, amikacin, kanamycin), quinolones, capreomycin (marked hearing loss) and cycloserine (neurotoxic).
  - *Prophylaxis*: INH or rifampin. Both in multidrug resistance.
- *M. avium complex* (MAC):
  - *Treatment*: Clarithromycin + ethambutol + rifabutin.
  - *Prophylaxis*: Azithromycin (1 × week) or clarithromycin (daily).
- Common side effects:
  - *Isoniazid*: Hepatitis (age-dependent), peripheral neuritis (use vitamin B_6).
  - *Rifampicin*: Proteinuria, hepatitis, flu-like syndrome, thrombocytopenia, red-orange metabolites.

- *Ethambutol:* Dose-dependent retrobulbar neuritis results in decreased visual acuity and red-green discrimination.
- *Pyrazinamide*: Polyarthralgia, myalgia, hepatitis, rash, hyperuricemia, phototoxicity, increased synthesis of porphyrin.
- *Streptomycin*: Ototoxic (vertigo and hearing loss) and nephrotoxic.

NONSPECIFIC ANTISEPTICS

Following are some of the nonspecific antiseptics used in ENT practice:

1. Sulzberger's Powder: It consists of 2 gm iodine and boric acid powder to make 100 gm. It is used in the after care of tympanomastoid surgery and keeps the cavity clean and dry. It is blown into mastoid cavity.
2. 2% acetic acid solution.
3. 3% boric acid in 70% alcohol.
4. Aqueous merthiolate (thimerosal).
5. 25% M-cresyl acetate (Cresylate).
6. Povidone-iodine.
7. 2% Gentian violet in 95% alcohol.

Thimerosal and Cresylate are not used in patients with tympanic membrane perforation.

Adverse Effects of Antimicrobials

1. Anaphylaxis with antimicrobial agents is an uncommon event that occurs in about 1 in 100,000 treatments.
2. Metabolic bone marrow depression can occur with zidovudine, ganciclovir and sulphonamides.
3. Reversible granulocytopenia can occur with sulphonamides and 2–3 weeks course of high doses of penicillins and cephalosporins.
4. Hepatocellular damage (hepatitis) is most often seen with rifampicin and isoniazid (5% cases).
5. Granulomatous hepatitis though rare can occur with high-dose ampicillin and flucloxacillin and prolonged quinine therapy.
6. With antibiotics diarrhea can be caused by the alteration of colonic flora and over growth of *Clostridium difficile*.
7. The most commonly implicated antibiotics that can cause pseudomembranous colitis of *C. difficile* are clindamycin, cephalosporins, ampicillin and quinolones.
8. Nephrotoxicity and ototoxicity can occur with aminoglycosides and amphotericin.

Clinical Highlights

1. **Mitochondria** are found only in eukaryotic organisms. Both bacteria and viruses do not have mitochondria.
2. ***M. tuberculosis* and *M. avium intracellulare complex*** (MAI or MAC) are major problems in HIV-infected and AIDS patients.
3. **Major encapsulated organisms** include *S. pneumoniae*, *K. pneumoniae*, *H. influenzae*, *P. aeruginosa*, *N. meningitidis* and *Cryptococcus neoformans* (yeast).
4. **Staphylococci** are differentiated from streptococci by catalase test with hydrogen peroxide.
5. **In cases of sore throat, Gram staining** can identify Gram-positive cocci (alpha hemolytic streptococci) and Gram-negative cocci (Neisseria) which are normally present in throat. Gram staining cannot differentiate pathogens from non-pathogens.
6. **The most useful diagnosis of Lyme disease** is detection of IgM/IgG antibodies to the spirochete.
7. **In the screening for bacterial meningitis** a latex particle agglutination test, searches for polysaccharide in CSF.
8. **Capsule, cell wall and cytoplasmic membrane** are present in both Gram-positive and Gram-negative organisms. The outer membrane is present only in Gram-negative organisms.
9. **Peptidoglycan** of the cell wall is the net structure that offers major protection from osmotic damage and confers cell shape.
10. **The anti-phagocytic bacterial structures** include bacterial capsule, pili of *gonococci*, M-protein of *Strep. pyogenes*, and the A proteins of *S. aureus*.
11. **Lipid A** of outer membrane is most responsible for triggering Gram-negative shock.
12. **The most important virulence factor for *S. pneumoniae*** is the capsule. Without capsule pneumococci can not cause pneumonia or meningitis.
13. **M-protein (surface protein) antigen** is most useful in identifying the strains of Group A streptococci (Strep. pyogenes) that may induce glomerulonephritis. The large number of M-protein types and nonimmunogenic capsule of hyaluronic acid results in repeated Strep. infections.
14. **Cord virulence factor (trehalose dimycolate)** causes *Mycobacterium tuberculosis* to clump together and grow in a serpentine-like fashion.
15. **Facultative anaerobes** respire in the presence of oxygen and ferments in the absence of oxygen.
16. **Bacterial pneumonia** in persons with alcohol intoxication is usually caused by *Strep. pneumoniae*.
17. **The antimicrobials** that are eliminated by kidney and need major dose reductions to avoid nephrotoxicity include amphotericin B, gentamycin, imipenem-cilastatin and vancomycin.
18. **In patients with a deficiency of glucose-6-phosphate dehydrogenase,** the antimicrobials which are contraindicated include sulfonamides, TMP-SMX, isoniazid, primaquine, quinine and nitrofurantoin.
19. **Ceftriaxone** is the drug of choice in infections with beta-lactamase producing *H. influenzae*, *M. catarrhalis* and *N. gonorrhoeae*.
20. **Streptomycin** causes miscoding in bacterial protein synthesis.
21. **Tetracyclines** prevent binding of aminoacyl-tRNA.
22. **Chloramphenicol** inhibits peptidyltransferase.

23. **Beta-lactams** inhibit cross-linking of peptidoglycan chains in bacterial cell-wall synthesis.
24. **Drugs that can cross blood-brain barrier** can also appear in the milk of lactating mother and may affect the infant.
25. **In case of metronidazole hypersensitivity, vancomycin** is the most appropriate antibiotic to treat pseudomembranous colitis.
26. **Penicillin G** still remains the drug of choice in the treatment of syphilis caused by *T. pallidum*.

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7 Fungi and Viruses

It is in the mind alone that one feels pure or impure. A Man first makes his own mind guilty and then sees another's fault. Can you injure anybody by enumerating his faults? You only injure yourself.

—Holy Mother Sarada Devi

Points of Focus

FUNGI
- ✓ Otomycosis
- ✓ Rhinosporidiosis
- ✓ Invasive Fungal Sinusitis
- ✓ Noninvasive Fungal Sinusitis
- ✓ Allergic Fungal Sinusitis
- ✓ Candidiasis

ANTIFUNGAL THERAPY
- ✓ Polyenes: Amphotericin B, Nystatin
- ✓ Azoles: Imidazoles (Clotrimazole, Ketoconazole), Triazole (Fluconazole, Itraconazole, Voriconazole)
- ✓ Echinocandins (Capsofungin)
- ✓ Flucytosine

VIRUSES
- ✓ Diagnosis: Tissue culture, Immunoassay, PCR
- ✓ Coronavirus
- ✓ Adenovirus
- ✓ Rhinovirus
- ✓ Influenza
- ✓ Mumps
- ✓ Respiratory Syncytial Virus
- ✓ Parainfluenza Virus
- ✓ Herpes Simplex Virus: Primary Oral Disease, Reactivation Factors, Recurrent Labial Herpes, Bell's Palsy
- ✓ Epstein-Barr Virus
- ✓ Human Papilloma Virus

ANTIVIRALS
- ✓ Interferon (IFN)
- ✓ Nucleoside Analogue (Acyclovir, Valacyclovir, Famciclovir, and Ganciclovir)
- ✓ Influenza Neuraminidase Inhibitors (Oseltamivir, Zanamivir)
- ✓ Amantadine, Rimantadine
- ✓ Ribavirin
- ✓ Antiretroviral Therapy

PANDEMIC INFLUENZA A H1N1
- ✓ Epidemiology: Causative Organism, Transmission, Incubation Period, Communicability
- ✓ Clinical Features, Risk Factors, Complications, Emergency Warning Signs
- ✓ Investigations
- ✓ Treatment
- ✓ Infrastructure/Manpower/Material Support
- ✓ Standard Operating Procedures
- ✓ Antivirals: Oseltamivir, Zanamivir, Peramivir IV
- ✓ Oseltamivir-Resistant H1N1 Virus
- ✓ Supportive Therapy
- ✓ Immunomodulating Drugs
- ✓ Antibiotics
- ✓ Discharge Policy
- ✓ Chemoprophylaxis
- ✓ Non-Pharmaceutical Interventions
- ✓ Laboratory Tests
- ✓ H1N1 Flu Vaccination: Indications and Contraindications
- ✓ Measures for Preventing Spread of Swine Flu

CLINICAL HIGHLIGHTS

FUNGI

The two most common ear, nose and throat fungal infections are otomycosis and oropharyngeal candidiasis. In this section we will see the fungi and their common ear, nose and throat disorders (Table 1).

Mycotic Diseases of Otorhinolaryngology (Table 1)

1. *Otomycosis*: This superficial, diffuse, fungal infection of external ear canal is described in chapter Diseases of External Ear. In temperate regions *Candida* is more common while in tropical and subtropical regions *Aspergillus niger* is most common. It is common in warm and humid environment and in adult patients with seborrheic dermatitis, psoriasis and prolonged use of antibiotic steroid drops. It is less common in children.
2. *Rhinosporidiosis*: This chronic granulomatous fungal infection that involves nasal mucosa and conjunctiva is described in chapter Nasal Manifestation of Systemic Diseases. It is very common in southern India and Sri Lanka. Rhinosporidium seeberi, in tissue forms large, thick-walled sporangium-like structures containing many endospores.
3. *Invasive fungal sinusitis*: All the three varieties (acute fulminant, chronic invasive and chronic granulomatous) are described in chapter Nasal Manifestation of Systemic Diseases. Histopathology confirms the diagnosis. Rhizopus arrhizus, Absidia, and Rhizomucor are the commonest causes of acute fulminant sinusitis. Alternaria, Aspergillus, Bipolaris, Curvularia and Exserohilum are important causes of chronic invasive sinusitis.
4. *Noninvasive fungal sinusitis*: This chronic noninvasive fungal infection (sinus ball or sinus mycetoma) is described in chapter Nasal Manifestation of Systemic Diseases. It may become invasive on the development of immunosuppressor. *Aspergillus fumigatus* is the most common cause.
5. *Allergic fungal sinusitis*: This common cause of chronic rhinosinusitis is described in chapter Nasal Manifestation of Systemic Diseases.
6. *Candidiasis*: Pharyngeal candidiasis is an extension of oral candidiasis. Candida opportunistic pathogen can cause acute or chronic deep seated infection of mucosa in debilitated or immunocompromised patients. *Candida albicans* is the most common cause.

ANTIFUNGAL THERAPY

Many antifungal agents are toxic and used only topically. The invasive moulds need systemic antifungal therapy combined with surgery. Sensitivity testing of fungi is difficult and usually does not correlate with clinical response. The major systemic and topical antifungals are discussed below:

1. *Polyenes:* They bind to the sterols of fungal cell membranes and result in leakage of cell content.
   a. *Amphotericin B*: It is active against all fungi and *Leishmania sp*. It is usually given parenterally though topical applications have also been tried. Cerebrospinal fluid (CSF) penetration is poor. Amphotericin B into liposomes is better effective and less toxic
      i. Side effects: Fever, rigors, headache, hyperkalemia and renal toxicity.

Intravenous amphetamine B causes infusion related hypotension (via histamine release), fever and chills. They may be attenuated by prior administration of nonsteroidal anti-inflammatory drugs (NSAIDs), antihistamine and steroids.

   b. *Nystatin:* It is used topically for yeast infection such as otomycosis and candidiasis.
2. *Azoles***:** This largest group of antifungal agents inhibits cytochrome P450 and cell wall formation. They are active against *Candida sp*., dermatophytes and some *Aspergillus sp*.
3. *Imidazoles:*
   a. *Clotrimazole*: Topical clotrimazole is powerful against oral, vaginal and skin candidiasis and dermatophytes infections.
   b. *Ketoconazole:* It is more toxic than triazoles and causes hepatitis in 1:15,000 patients.

TABLE 1 Common ear, nose and throat fungi with their reservoir, diseases and antifungal treatment

Fungi	*ENT diseases*	*Treatment*	*Reservoir*
Aspergillus	Otomycosis, chronic invasive sinusitis, allergic fungal sinusitis (AFS)	AFS: Removal of polyp and allergic mucin + oral corticosteroids + antiallergic therapy	Air, soil, dust, decomposing plant
A. niger	Otomycosis	Topical nystatin, clotrimazole, econazole cream	
A. flavus, *A. fumigatus*	Otomycosis, acute fulminant sinusitis, noninvasive fungal sinusitis	Noninvasive fungal sinusitis: Removal of fungus ball	
Candida, e.g. *C. albicans, C. parapsilosis, C. tropicalis*	Otomycosis, candidiasis	Candidiasis: Nystatin (100,000 units/ml) or amphotericin B (100 mg/ml) oral suspension, clotrimazole troches, miconazole oral gel, oral fluconazole	Mouth and upper respiratory tract
Rhinosporidium seeberi	Rhinosporidiosis	Surgical excision	Stagnant pools of fresh water
Rhizopus arrhizus, Absidia, Rhizomucor	Acute fulminant sinusitis	Surgical excision + amphotericin B, itraconazole, voriconazole, posaconazole	Ubiquitous in environment
Alternaria, Bipolaris, Curvularia and Exserohilum	Chronic invasive sinusitis (CIS), allergic fungal sinusitis	CIS: Extensive surgical debridement + itraconazole, voriconazole, posaconazole	Worldwide distribution

4. *Triazoles*:
   a. *Fluconazole*: It covers *Candida sp* and *Cryptococcus* and has no activity against invasive moulds.
   b. *Itraconazole and voriconazole*: They have better activity against invasive fungi including *Aspergillus* and *Fusarium*.
5. *Echinocandins* (e.g. Capsofungin): These new antifungal inhibit cell-wall glucan synthesis. They are rapidly fungicidal against azole-resistant *Candida* and *Aspergillus*. They do cover other moulds.
6. *Flucytosine*: It disrupts fungal protein and DNA synthesis. It is used in severely active *Candida sp* and *Cryptococcus sp* only and also used in severe infections like meningitis. Dose is reduced in presence of renal failure. It is used in combination therapy to reduce the development of resistance.
   a. *Side effects*: Bone marrow toxicity and hepatotoxicity.
7. *Allylamines* (e.g. Terbinafine): They inhibit ergosterol synthesis and used in extensive skin and nail dermatophytes infection.
8. *Griseofulvin*: It treats fungal nail infection.

VIRUSES

Respiratory tract viral infections are the commonest diseases of humans worldwide. They account for 30% of all childhood deaths in developing world due to insufficient vaccine coverage, younger age of infection and concurrent malnutrition. Most preschool children have six to eight viral respiratory infections per year. Many viruses have seasonal activity. The outbreaks of parainfluenza types 1 and 2 occur in autumn and type 3 in spring. Respiratory syncytial virus outbreaks happen in winter. Viruses infect cells by binding host cell receptors. This section will cover the viruses that infect and are involved in the pathogenesis of neoplasm of ear, nose and throat regions (Table 2). HIV/AIDS is covered in separate chapter.

Diagnosis

Viral pathogens can be diagnosed with the help of following techniques:

1. *Tissue culture*: Nasopharyngeal secretions are preferred for inoculation in patients with respiratory tract infections.
2. *Immunoassay*:
   a. *IgM*: The specific immunoglobulin antibody can be estimated.
   b. *Rapid detection*: Immunofluorescent-labelled antibody against viral antigens.

- Polymerase chain reaction: Polymerase chain reaction (PCR) detects viral nucleic acid and has greater sensitivity.

Special Features of Viruses (Table 2)

1. *Coronavirus*: It is the largest human RNA virus and causes up to 10% cases of common cold especially in winter.
2. *Adenovirus*: It was originally isolated from human adenoid tissue and most children are asymptomatically infected. It causes outbreaks of upper and lower respiratory tract infection in small communities especially in winter. The pharyngoconjunctival fever occurs in swimming pool outbreaks.
3. *Rhinovirus*: This small virus is restricted to upper respiratory tract. Bradykinin release causes symptoms of rhinitis. Because of the over 100 serotypes, repeated infections occur and vaccination is not practical.
4. *Influenza*: Outbreaks begin suddenly, peaking within 4 weeks and last between 8–10 weeks. Pandemics of influenza are unpredictable. The pandemic of 1918 killed 20–40 million people. Pneumonia is usually due to secondary *Staphylococcus aureus* superinfection. Annual immunization is recommended for elderly, diabetics, immunosuppressed and patients with chronic liver, cardiac, respiratory and renal diseases. H1N1 influenza is described in other section of this chapter.
5. *Mumps*: The complications of this highly contagious virus include encephalitis and aseptic meningitis. Vaccination: Live attenuated trivalent vaccine mumps-measles-rubella (MMR).
6. *Respiratory syncytial virus*: This contagious virus causes winter outbreaks in infants. Epidemics last for months. Respiratory syncytial virus (RSV) is responsible for about 1 million childhood death every year in developing countries. Coryzal symptoms rapidly progress to respiratory distress. It affects mostly infants from 6 weeks to 6 months. Ribavirin aerosol inhalation is administered in severe pediatric infections.
7. *Parainfluenza virus*: Parainfluenza virus (PIV) causes nearly a third of all respiratory tract infections. PIV 4 is less common.
   a. *PIV 1 and 2*: Autumn croup outbreaks every 1-2 years.
   b. *PIV 3*: Endemic infantile bronchiolitis and pneumonia.
8. *Herpes simplex virus*:
   a. Primary oral disease presents with sore throat, pharyngitis, fever and painful vesicles on oropharynx and oral cavity, which persists for several days. After primary infection HSV tracks along sensory nerves and remains dormant in sensory ganglia. The diagnosis is usually clinical. Diagnostic tests include electron microscopy, antibody tests and tissue culture. Antivirals should be used immediately with the appearance of the symptoms. Gloves should be used if contact is likely. Preventive intermittent therapy is recommended in patients who have more than six to twelve episodes per year.
   b. Reactivation factors: They include trauma, stress, menstruation, fever, extremes of temperature, and ultraviolet light.
   c. Recurrent labial herpes: Initial itching, burning, tingling and pain last between 6 hours to 2 days. Vesicles involve lips and perioral skin and may extend into nostrils. Multiple recurrences at the same sight are common and may become secondarily infected with *S. aureus*. Complete healing usually takes 2 weeks.
   d. Bell's palsy: HSV-1 genome is detected in endoneurial fluid in patients with Bell's palsy.
9. *Epstein-Barr virus*: Epstein-Barr virus (EBV) infects B-lymphocytes and epithelial cells of both oral and genital tracts. Primary infection is usually subclinical but makes the person life-long carrier.
10. *Human papillomavirus*: Human papillomavirus (HPV) has predilection for human epithelial cells and causes papillomas. There are 60 types of HPV. HPV types 6 and 11 are associated with benign papillomas and HPV-18 with carcinomas.
    a. *Type 6 and 11*: They are acquired from infected maternal genital tract and may result in juvenile-onset recurrent

TABLE 2 Viruses that cause ear, nose and throat disorders with their structure, transmission, diseases and antiviral agents

Virus	*Structure and types*	*Transmission*	*Diseases*	*Antiviral agents*
Coronavirus	RNA virus; strains 229E, OC4	Aerosol	Common cold, pneumonia, acute otitis media (AOM)	
Adenovirus	Double stranded DNA virus; Types 1,2,3,5,7	Aerosol	Pharyngitis, exudative tonsillitis, pneumonia, pharyngoconjuntival fever, AOM, keratoconjunctivitis	Ribavirin in childhood pneumonia
Rhinovirus	Nonenveloped RNA virus	Aerosol, contact (hands, utensils)	Common cold, pneumonia, AOM	
Influenza Incubation 1-3 days	RNA orthomyxovirus; Subtypes A,B,C; Avian H5N1, H1N1 (swine flu)	Droplets;	Flu, bronchiolitis, pneumonia, acute sensorineural hearing loss (SNHL) anosmia	Zanamivir, oseltamivir, amantadine, rimantadine
Mumps Incubation 2-3 weeks	RNA virus	Respiratory droplet, close contact	Mumps (parotitis), orchitis, pancreatitis, acute SNHL	
Respiratory syncytial virus; Incubation 2-8 weeks	RNA virus; Serotypes A, B	Contact with respiratory secretions	Croup, bronchiolitis, bronchitis, AOM, anosmia	Ribavirin aerosol inhalation
Parainfluenza virus	Types 1,2,3,4	Person to person	Common cold, croup, bronchiolitis, pneumonia, anosmia	
Herpes simplex virus (HSV); Incubation 2-14 days	Double stranded DNA virus; Types 1, 2	Direct contact with mucosal secretions; oral (HSV-1), genitourinary (HSV-2)	Pharyngitis, gingivostomatitis, recurrent labial herpes, Bell's palsy, acute SNHL, anosmia	Famciclovir, valacyclovir, acyclovir,
Epstein-Barr virus	DNA virus		Infectious mononucleosis, hairy oral leukoplakia, nasopharyngeal carcinoma, Burkitt's lymphoma, B-cell lymphoma	Acyclovir
Human papilloma virus	Double stranded DNA papovavirus	Perinatal (types 6, 11)	Papillomas and carcinoma of skin, genitalia, larynx, recurrent respiratory papillomatosis	

respiratory papillomatosis. They may cause inverted papilloma and invasive cancer.

b. *Types 13 and 32*: They are confined to mouth and may cause Heck's tumor.

c. *Types 16 and 18*: They may cause laryngeal carcinoma.

ANTIVIRALS

The antiviral agents have a broad range of activity but theoretically. The antivirals with clinically proven indications are described below:

1. *Interferon*: This naturally occurring therapeutic agent is less effective and side effects limit its use.
   a. *Indications*: In conjunction with lamivudine and ribavirin alpha interferon (IFN) is used in chronic hepatitis B and C.
   b. *Side effects*: Flu-like symptoms and hematological complications.
2. *Nucleoside analogue (Acyclovir, valacyclovir, famciclovir and ganciclovir)*: Acyclovir and foscarnet inhibit herpes DNA polymerase. Acyclovir is widely distributed including CSF. High doses are required for varicella zoster virus (VZV), HSV in immunocompromised and HSV encephalitis.
   a. *Indications*: HSV and herpes zoster virus (HZV) infections, recurrent HSV-1 and 2 infections; prophylaxis in bone marrow and organ transplant patients; Ganciclovir is used in cytomegalovirus (esophagitis and retinitis) in immunocompromised patients.
3. *Influenza neuraminidase inhibitors*: They block the virus release, by inhibiting neuraminidase that prevents the clumping of viruses and their binding to the surface of cells which have been already infected.
   a. *Oseltamivir*: It is effective, safe and well tolerated and is the first line of antiviral in influenza.
   b. *Zanamivir*: It is administered by powder inhalation within 48 hours of symptoms for "at risk adults" only.
4. *Amantadine*: It is used for both prophylaxis and treatment of influenza A. Amantadine inhibits influenza A virus "uncoating." Its use is limited by its CNS side effects.
   a. *Rimantadine*: Amantadine and rimantadine influenza antivirals block membrane ion channels and inhibit induction of RNA synthesis.
5. *Ribavirin*: This synthetic analogue of guanosine has dose-limiting toxic effects. It is administered by aerosol inhalation. Ribavirin inhibits inosine-5-monophosphate dehydrogenase.
   a. *Indications*: Pediatric cases of severe RSV infection and hepatitis C.
6. *Antiretroviral therapy:* It is discussed in HIV infection.

- The viruses implicated in sudden sensorineural hearing loss include rubella, mumps, measles, herpes zoster (HZV), cytomegalovirus (CMV) and influenza.
- The viruses linked to postviral olfactory disorders include RSV, herpes, influenza and parainfluenza 3.
- Acute otitis media: Rhinovirus and RSV are most commonly detected. Adenovirus and Coronavirus are found less frequently.
- RSV can cause a rapid deterioration in children.
- EBV and HPV are associated with certain benign, premalignant and malignant conditions of head and neck.
- Acyclovir group of antivirals are highly effective while others have a limited role.
- Acyclovir, famciclovir or valacyclovir reduces the recurrent episodes of herpes infection by up to 80%.
- Vaccination is highly effective in controlling mumps, measles, rubella and influenza viruses but is ineffective against rhinovirus, RSV, EBV and HSV.

PANDEMIC INFLUENZA A H1N1 (SWINE FLU)

Influenza like illness caused by Influenza A (H1N1) (a reassorted influenza virus), was first reported from Mexico on 18th March, 2009. It rapidly spreaded to neighboring United States and Canada. Subsequently the disease spread to all the continents. As per WHO assessment the overall severity of Influenza pandemic is moderate implying that most people recovered from infection without the need for hospitalization or the medical care.

India reported its first case on 13th May, 2009. Majority of those who died had some underlying diseases and reported late to the identified health care facility. On 3rd August 2009, India registered its first swine flu death when a 14-year old girl became victim to this disease.

Epidemiology

- *Causative organism*: Genetic sequencing shows a new subtype of influenza A (H1N1) virus with segments from four influenza viruses:
  - North American Swine
  - North American Avian
  - Human Influenza
  - Eurasian Swine
- *Transmission*: The transmission is by droplet infection and fomites. H1N1 flu is contagious and similar to regular seasonal influenza viruses spread from person-to-person through coughing or sneezing by infected persons. People may become infected by touching a surface or object—with flu viruses on it and then touching their mouth, nose, or eyes. It is now causing wide spread human to human transmission.
- *Incubation period*: 1–7 days.
- *Communicability:* From 1 day before to 7 days after the onset of symptoms. If illness persists for more than 7 days, chances of communicability may persist till resolution of illness. Children or people with weakened immune systems may spread the virus for a longer period. The secondary attack rate ranges from 22–33%.

Clinical Features

Illness ranges from mild to severe. All types of flu can cause fever, coughing and/or sore throat, runny or stuffy nose, headaches and/or body aches, chills and fatigue.
  - The symptoms of H1N1 flu are similar to seasonal flu, but may be more severe and include additional symptoms such as vomiting and diarrhea.
  - This acute febrile illness presents with systemic symptoms ranging from mild fatigue to respiratory failure and death. The symptoms include fever (100°–104° F), sore throat, cough, rhinitis, myalgia, headache, photophobia, fatigue and dyspnea.
- *Course*: The vast majority of H1N1 flu patients recovered without any medical treatment but hospitalizations and deaths have occurred.
- *Risk factors*: About 70% of hospitalized H1N1 flu patients had one or more medical conditions, which include pregnancy, diabetes, heart disease, asthma and kidney disease. These medical conditions place patients in the "high risk" category. Adults of 65 years and older flu patients are prioritized for antiviral treatment to limit risk of complication. Associated risk conditions (COPD, diabetes, etc.) have higher risk of complications from any influenza.
- *Complications*: There is insufficient information to date about clinical complications of the current pandemic influenza A (H1N1) virus infection. Clinicians should expect complications to be similar to seasonal influenza: sinusitis, otitis media, croup, pneumonia, bronchiolitis, status asthmaticus, myocarditis, pericarditis, myositis, rhabdomyolysis, encephalitis, seizures, toxic shock syndrome and secondary bacterial pneumonia with or without sepsis. Individuals at extremes of age and with preexisting medical conditions are at higher risk of complications and exacerbation of the underlying conditions. The complications may be divided into two groups pulmonary and extrapulmonary.
  - *Pulmonary*: Viral pneumonia, secondary bacterial pneumonia with exacerbation of COPD and bronchial asthma.
  - *Extrapulmonary*: Myositis, rhabdomyolysis, myocarditis, encephalitis and transverse myelitis.
- *Emergency warning signs*: The sign and symptoms shown in Table 3 indicate that patients need emergency medical care.

Investigations

Routine investigations required for evaluation and management of a patient include hematological, biochemical, radiological and microbiological tests as necessary.

TABLE 3 Emergency warning signs

Children	*Adults*
• Dyspnea: Fast breathing or difficulty in breathing	• Dyspnea: Difficulty in breathing or shortness of breath
• Cyanosis: Bluish or gray skin color	• Pain or pressure in the chest or abdomen
• Dehydration: Not drinking enough fluids	• Sudden dizziness
• Vomiting: Severe or persistent	• Confusion
• Drowsy: Not waking up or not interacting	• Severe or persistent vomiting
• Irritable: Child does not want to be held	• Recurrence of flu-like symptoms with fever and worse cough
• Recurrence of flu-like symptoms with fever and worse cough	

Diagnosis

Confirmation of pandemic influenza A (H1N1) infection is through:

- Real time (RT) PCR, or
- Isolation of the virus in culture, or
- Four-fold rise in virus specific neutralizing antibodies

For confirmation of diagnosis, clinical specimens such as nasopharyngeal swab, throat swab, nasal swab, wash or aspirate and tracheal aspirate (for intubated patients) are to be obtained. The sample should be collected by a trained physician/microbiologist preferably before administration of the antiviral drug. Keep specimens at 4°C in viral transport media until transported for testing. The samples should be transported to designated laboratories within 24 hours. If they cannot be transported then it needs to be stored at -70°C. Paired blood samples at an interval of 14 days for serological testing should also be collected.

Treatment

- *Guiding principles of treatment* include:
  - Early implementation of infection control precautions to minimize nosocomial/household spread of disease.
  - Prompt treatment to prevent severe illness and death.
  - Early identification and follow-up of persons at risk.
- *Infrastructure/manpower/material support*
  - *Isolation facilities*: If dedicated isolation room is not available then patients can be cohorted in a well ventilated isolation ward with beds kept one meter apart.
  - *Manpower*: Dedicated doctors, nurses and paramedical workers.
  - *Equipment:* Portable X-ray machine, ventilators, large oxygen cylinders, pulse oxymeter .
  - *Supplies*: Adequate quantities of personal protection equipments (PPE), disinfectants and medications (oseltamivir, antibiotics and other medicines).
- *Standard operating procedures*: Reinforce standard infection control precautions:
  - All those entering the room must use high efficiency masks, gowns, goggles, gloves, cap and shoe cover.
  - Restrict number of visitors and provide them with PPE.
  - Provide antiviral prophylaxis to health care perso-nnel managing the case and ask them to monitor their own health twice a day.
  - Dispose waste properly by placing it in sealed impermeable bags labeled as Bio-Hazard.
- *Antivirals* (Table 4): There are four influenza antiviral drugs approved for use in the United States (oseltamivir, zanamivir, amantadine and rimantadine). Among these, oseltamivir (TAMIFLU®) and zanamivir (RELENZA®) are effective in treating the 2009 H1N1 flu. A fifth drug, Peramivir IV, has been authorized under an Emergency Use Authorization (EUA) to treat certain patients with suspected or confirmed 2009 H1N1 influenza virus infection. Amantadine and rimantadine are not effective against H1N1 flu.
  - The antiviral drugs can make illness milder and make patients feel better faster. They may also prevent serious influenza complications. The sooner a sick person is treated with an antiviral, the greater effect it will have on the patient's recovery.
  - When used to prevent the flu, antiviral drugs are about 70–90% effective.

Oseltamivir and zanamivir are approved for the treatment of H1N1 influenza and their side effects include nausea, vomiting, vertigo and bronchitis.

TABLE 4 General information on oseltamivir (TAMIFLU®), zanamivir (RELENZA®) and Peramivir IV

Oseltamivir	*Zanamivir*	*Peramivir IV*
• Approved to both treat and prevent influenza A and B virus infection • Effective in people 1 year of age and older • Available in capsule form and in liquid form that can be measured for proper dosage. For children who cannot swallow capsules, especially children less than 1 year old, the liquid form is preferred. If the child cannot swallow capsule then capsule may be opened and mixed with a sweet thick liquid • Most common side effects are nausea and vomiting. These are not usually severe and usually happen in the first 2 days of treatment	• Approved to treat and, in some cases, prevent influenza A and B virus infection • Effective in people 7 years and older and can prevent influenza A and B virus infection in people 5 years and older • Packaged in a medicine disk called a Rotadisk® and is inhaled by mouth using a delivery device called a Diskhaler® • Risk of increased difficulty breathing, especially among people with a history of underlying breathing problems. If patients have increased difficulty breathing, they should stop Relenza and get medical attention right away. • Children should NOT take if they have ever had a severe allergic reaction to it or to lactose • Not recommended for people with chronic lung disease such as asthma or chronic obstructive pulmonary disease (COPD)	• An experimental drug used to treat specifically following patients who are admitted to a hospital: – Not responding to either oral or inhaled antiviral therapy, or – Drug delivery by a route other than IV (e.g. enteral oseltamivir or inhaled zanamivir) is not expected to be dependable or is not feasible, or – The clinician judges IV therapy is appropriate due to other circumstances. • Peramivir IV is given intravenous, once a day for 5 days or more depending on the patients' condition. • Possible side-effects include diarrhea, nausea, vomiting and a decrease in white blood cell count

- *Oseltamivir medication*: Oseltamivir is the recommended drug both for prophylaxis and treatment. In the current phase, a suspect case would be provided oseltamivir.
  - *Dose for adults by weight:* It is also available as syrup (12 mg/ml). If needed dose and duration can be modified as per clinical condition.
  - - <15 kg: 30 mg BD for 5 days
  - - 15–23 kg: 45 mg BD for 5 days
  - - 24–40 kg: 60 mg BD for 5 days
  - - >40kg: 75 mg BD for 5 days
- *Dose for infants by age*:
  - < 3 months: 12 mg BD for 5 days
  - 3–5 months: 20 mg BD for 5 days
  - 6–11 months: 25 mg BD for 5 days
- *Adverse reactions*: Oseltamivir is generally well tolerated. Gastrointestinal side effects (transient nausea, vomiting) may increase with increasing doses, particularly above 300 mg/day. Occasionally, it may cause bronchitis, insomnia and vertigo. Less commonly angina, pseudomembranous colitis and peritonsillar abscess have also been reported. There have been rare reports of anaphylaxis and skin rashes. In children, most frequently reported side effect is vomiting. Infrequently, abdominal pain, epistaxis, bronchitis, otitis media, dermatitis and conjunctivitis have also been observed. There is no recommendation for dose reduction in patients with hepatic disease. Though rare reporting of fatal neuropsychiatiric illness in children and adolescents has been linked to oseltamivir, there is no scientific evidence for a causal relationship.
- *Oseltamivir-resistant H1N1 virus:* There have been isolated reports of small clusters of patients with an Oseltamivir-resistant H1N1 virus in late 2009. Reports have been sporadic and infrequent, with no conclusive evidence that Oseltamivir-resistant H1N1 viruses are circulating within communities or worldwide. The original reported clusters occurred in a single ward in a hospital, and both involved patients whose immune systems were severely compromised or suppressed. The person-to-person transmission of an oseltamivir-resistant virus has not been conclusively demonstrated. According to WHO, no evidence suggests that oseltamivir-resistant viruses are causing a different or more severe form of H1N1 illness to date.
- *Supportive therapy*: It includes IV fluids, parenteral nutrition, oxygen therapy ventilatory support, antibiotics for secondary infection and vasopressors for shock.
  - *Symptomatic*: Paracetamol or ibuprofen is prescribed for fever, myalgia and headache. For URI symptoms, short course of topical decongestants, saline nasal drops, throat lozenges and steam inhalation may be beneficial.
  - Patient is advised to drink plenty of fluids.
  - Smokers should avoid smoking.
  - Salicylate (aspirin) is contrindicated in any influenza patient due to its potential to cause Reye's syndrome.
- *Monitoring*: The suspected cases would be constantly monitored for clinical radiological evidence of lower respiratory tract infection and for hypoxia (respiratory rate, oxygen saturation, level of consciousness).
  - Patients with signs of tachypnea, dyspnea, respiratory distress and oxygen saturation less than 90% should be supplemented with oxygen therapy. Types of oxygen devices depend on the severity of hypoxic conditions which can be started from oxygen cannula, simple mask, partial re-breathing mask (mask with reservoir bag) and non rebreathing mask. In children, oxygen hood or head boxes can be used.
  - Patients with severe pneumonia and acute respiratory failure (SpO_2 < 90% and PaO_2 <60 mm Hg with oxygen therapy) must be supported with mechanical ventilation. Invasive mechanical ventilation is preferred choice. Non invasive ventilation is an option when mechanical ventilation is not available. To reduce spread of infectious aerosols, use of high-efficiency particulate air (HEPA) filters on expiratory ports of the ventilator circuit/high flow oxygen masks is recommended.
  - Maintain airway, breathing and circulation (ABC);
  - Maintain hydration, electrolyte balance and nutrition.

If the laboratory reports are negative, the patient would be discharged after giving full course of oseltamivir. Even if the test results are negative, all cases with strong epidemiological criteria need to be followed up.

- Immunomodulating drugs have not been found to be beneficial in treatment of ARDS or sepsis associated multi-organ failure. High dose corticosteroids in particular have no evidence of benefit and there is potential for harm. Low dose corticosteroids (Hydrocortisone 200-400 mg/day) may be useful in persisting septic shock (SBP < 90).
- *Antibiotics*: Suspected cases without pneumonia do not require antibiotic therapy. Antibacterial agents should be administered, if required, as per locally accepted clinical practice guidelines. Patient on mechanical ventilation should be administered antibiotics prophylactically to prevent hospital associated infections.
- *Discharge policy*: If the laboratory reports are negative, the patient would be discharged after giving full course of oseltamivir. Even if the test results are negative, all cases with strong epidemiological criteria need to be followed up.

It has been observed that some of the patients even though asymptomatic, continue to test positive for influenza A H1N1. A treated and recovered patient, even though testing positive, has very little possibility of infecting others. In view of the above, the following recommendations are made:

- Patients who responded to treatment after 2–3 days and become totally asymptomatic should be discharged after five days of treatment. There is no need for a repeat test.
- Patients who continue to have symptoms of fever, sore throat, etc. even on the fifth day should continue treatment for 5 more days. If the patient becomes asymptomatic during the course of treatment there is no need to test further.
- For patients who continue to be symptomatic even after 10 days of treatment or those cases with respiratory distress and in whom secondary infection is taken care of, and if patients continue to shed virus, then resistance of the patients to antiviral would be tested. The dose of antiviral may be adjusted on case-to-case basis.
- The family of patients discharged earlier should be educated on personal hygiene and infection control measures at home; children should not attend school during this period.
- *Chemoprophylaxis*: Oseltamivir is the drug of choice.

Prophylaxis should be provided till 10 days after last exposure (maximum period of 6 weeks).

- *Health care workers at high risk*: The treating physicians and other paramedical personnel at the isolation facility would be put on chemoprophylaxis. All health care personnel coming in contact with suspected, probable or confirmed cases.
- *Contacts*: Chemoprophylaxis is advised for those contacts with high-risk(with underlying systemic diseases; extremes of age (<5 years and>65 years). All close contacts of suspected, probable and confirmed cases. Close contacts include household /social contacts, family members, workplace or school contacts, fellow travelers, etc.
- *Clusters*: In phase 5, if the clusters are reported for the first time, and given that those exposed are known and can be traced easily, then family, social and community contacts should be given chemoprophylaxis.
- *Mass chemoprophylaxis*: The strategy of containment by geographic approach by giving oseltamivir to every individual in a prescribed geographic limit of 5 km from the epicenter (The village/city where the cluster is reported) would be applied:
  - If the virus is lethal and causing severe morbidity and high mortality.
  - Though affecting humans, is not efficiently transmitting in our population.
  - If the cluster is limited by natural geographic boundaries.

This strategic decision would be taken by the RRT in consultation with State Health Department/MOHFW, Government of India.

- *Doses of oseltamivir*:

  They are similar to therapeutic doses.
- *Non-pharmaceutical interventions*: Close contacts of suspected, probable and confirmed cases should be advised to remain at home (voluntary home quarantine) for at least 7 days after the last contact with the case. Monitoring of fever should be done for at least 7 days. Prompt testing and hospitalization must be done when symptoms are reported. All suspected cases, clusters of ILI/SARI cases need to be notified to the State Health Authorities and the Ministry of Health and Family Welfare, Govt. of India (Director, EMR and NICD).
- *Laboratory tests*: The samples are to be tested in BSL-3 or BSL 2+ laboratory with BSL-3 precautions. The apex laboratories are:
  - National Institute of Communicable Diseases, 22, Sham Nath Marg, Delhi [Tel. Nos. Influenza Monitoring Cell: 011-23921401; Director: 011-23913148])
  - National Institute of Virology, 20-A, Dr. Ambedkar Road, Pune-411001 [Tel.No. 020-26124386)]
  - There is a network of other laboratories that can test for Influenza A H1N1. This network is being expanded to include private laboratories. The updated list is available on the website of Ministry of Health and Family Welfare (*Source*: www.mohfw.nic.in).
- *Vaccination*: H1N1 vaccine is available in the following forms.
- Live Attenuated Influenza Vaccine (LAIV) nasal spray for H1N1 flu.
- 15 μg pre-filled syringes licensed for use in children age 4 and older.
- Multi-dose vials for persons 6 months of age and older.

The indications, contraindication, and doses of H1N1 "Flu Shot" and H1N1 LAIV "Nasal Spray Vaccine" are shown in Table 5.

- *Prevention of spread from contacts*: Table 6 enumerates everyday steps to protect people from the spread of swine flu.

TABLE 5 H1N1 flu vaccination. Indications and contraindications for injection and nasal spray

Age and conditions	*Injection*	*Nasal spray*
0–6 months	Contraindication	Contraindication
6 months to 9 years	2 doses	Contraindication in < 2 years
10–49 years	1 dose	Only in healthy individuals
> 49 years/Pregnant women	1 dose	Contraindication
Arthritis, Asthma, HIV/AIDS	1 dose	Contraindication
< 6 years with recurrent wheezing	1 dose	Contraindication
Aspirin therapy	1 dose	Contraindication
Guillain Barre syndrome within 6 months of getting flu vaccine	Contraindication	Contraindication
Allergy to eggs or nasal spray vaccine components	Contraindication	Contraindication

TABLE 6 Measures for preventing spread of swine flu

Advice for general population

- Cover nose and mouth with a tissue during coughing or sneezing and throw the tissue in the trash
- Wash hands often with soap and water, especially after coughing or sneezing. Alcohol-based hand cleaners are also effective
- Avoid touching eyes, nose or mouth
- Avoid close contact with sick people
- Sick patients should stay home until at least 24 hours after they no longer have a fever without the use of paracetamol
- Follow public health advice regarding school closures, avoiding crowds and other social distancing measures

Contd...

Contd...

Advice for H1N1 flu patients

- Stay at home, follow doctor's orders and watch for warning signs. Remain at home for 7 days after the symptoms begin or until symptom-free for 24 hours, whichever is longer
- Avoid close contact with others, especially pregnant women, young children and infants and people with chronic medical conditions (such as asthma, diabetes, or heart disease)
- Wear a facemask when sharing common spaces with other household members.
- Get plenty of rest
- Drink clear fluids such as water, broth, sports drinks, or electrolyte beverages made for infants to prevent becoming dehydrated
- Cover coughs and sneezes
- Clean hands with soap and water or an alcohol-based hand rub after using tissues, coughing, or sneezing into your hands

Advice for people taking care of H1N1 flu patients

- Avoid being face-to-face with the sick person. When holding a small child, place his/her chin on your shoulder
- Everyone in the household should clean his/her hands often, using soap and water or an alcohol-based hand rub
- The patients should cover coughs and clean their hands with soap and water or an alcohol-based hand rub often, especially after coughing and/or sneezing
- Be aware about the special care that might be needed in pregnancy, diabetes, heart disease, asthma, or emphysema, etc.
- Talk to doctor about taking oseltamivir or zanamivir to prevent getting the flu.
- Wear a facemask, when close contact is unavoidable
- Monitor yourself and household members for flu symptoms and contact a health care provider if symptoms occur
- Get medical care right away if the patient exhibits emergency warning signs

Clinical Highlights

1. **Acyclovir:** The major mechanism of HSV resistance to acyclovir is the loss of ability to produce viral thymidine kinase, which leads to monophosphorylation of acyclovir that is a prerequisite to inhibit DNA polymerases in HSV.
2. **Fluconazole:** It is the most suitable antifungal in an immunocompromised patient for the prophylaxis of *Cryptococcus neoformans*. It is effective orally, penetrates CSF and less toxic.
3. **Nystatin:** It interacts with ergosterol and forms artificial membrane (pores). Azole antifungals inhibit ergosterol synthesis.

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8 Human Immunodeficiency Virus Infection

Nothing else is necessary but these—love, sincerity, and patience. What is life but growth i.e. expansion, i.e. love. All love is life, all selfishness is death, and this is true here or hereafter. It is life to do good, it is death not to do good to others. Ninety percent of human brutes you see are dead are ghosts—or none lives, my boys, but he who loves.

—Swami Vivekananda

Points of Focus

- **HIV/AIDS**
  - ✓ Transmission
  - ✓ Human immunodeficiency virus (HIV)
  - ✓ Immunology
  - ✓ Course of Disease
  - ✓ AIDS Indicator Diseases
  - ✓ Diagnosis
  - ✓ Classification
  - ✓ CDC AIDS Case Definitions
  - ✓ Treatment
- **CERVICAL ADENOPATHY**
- **NEOPLASMS**
  - ✓ Kaposi's Sarcoma
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  - ✓ Lymphoepithelial Cysts of Parotid Gland
- **NOSE AND SINUSES**
- **NASOPHARYNX**
- **EAR**
  - ✓ External Ear
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- **ORAL CAVITY**
  - ✓ Candidiasis
  - ✓ Hairy Leukoplakia
  - ✓ Herpes Simplex Stomatitis
  - ✓ Gingivitis and Periodontal Disease
  - ✓ ANUG
  - ✓ Aphthous Ulcers
- **OCCUPATIONAL EXPOSURE**
  - ✓ Management of Needle Stick Injury
  - ✓ Prevention
  - ✓ Care of Endoscopes
- **CLINICAL HIGHLIGHTS**

HIV/AIDS

In 1986, the first case of acquired immunodeficiency syndrome (AIDS) was seen in India and surveillance of human immunodeficiency virus (HIV) infection/AIDS was started. National AIDS Control Organization (NACO) reports that Manipur, Maharashtra, Nagaland, Punjab and Daman & Diu (descending order) have high seropositivity rates per thousand persons screened for HIV. Among the cases of AIDS reported to NACO, maximum number of AIDS cases (in descending order) were in Tamil Nadu, followed by Maharashtra, Gujarat and Andhra Pradesh. About 85% were due to sexual relationship while nearly 3% were due to blood and blood products and about 2.5% due to perinatal transmission.

Transmission

The primary modes of HIV infection transmission and high-risk groups are:

a. *Sexual contact*: Homosexual (especially receptive anal intercourse), heterosexual (promiscuous individuals, prostitutes and truck drivers) and bisexual.
b. *Skin piercing devices*: Non-sterile needles, syringes and other skin piercing devices, and needle sharing in intravenous drug abuse.
c. *Transfusion of blood and its products*: Common in hemophiliac, thalassemic and dialysis patients.
d. *Mother to child*: Prenatal (transplacental), perinatal (during birth) and postnatal. The chances of vertical HIV

transmission from an infected mother to her infant depend on the stage of maternal disease, type of delivery, birth order and concurrent sexually transmitted disease (STD) such as syphilis. They are approximately 25%.

e. *Health care workers*: Contact with blood and body fluids such as amniotic, pleural, peritoneal or pericardial fluids. Risk from specimens of urine, stool, saliva, sputum, tears, sweat and vomitus is negligible.

Human Immunodeficiency Virus

HIV is composed of nucleocapsid core, structural proteins and a membrane envelope. It is a retrovirus on the Lentivirus subfamily. The nucleocapsid core contains two identical genomic RNA strands of about 10,000 nucleotides. HIV infection has a long incubation time and slow progression of disease.

- *Types*: HIV infection of human beings are of two types:
  a. *HIV Type1*: Most common and very pathogenic.
  b. *HIV Type2*: Less common and less pathogenic.

Immunology

- *CD4 count*: HIV infects CD4 T-lymphocytes and macrophages, which have CD4 surface marker and are associated with helper inducer function of the immune system. Once CD4 lymphocytes count falls below 500 cells/mm^3 (normal 600–1500 cell/mm^3), the myriad of AIDS defining pathological conditions start appearing. Death usually occurs within 2–3 years, once CD4 cell count falls below 200 cell/mm^3.
- *Immunity*: The slow but progressive impairment of both humoral and cell-mediated immunities makes the patient susceptible to opportunistic infections and neoplasms, culminating in AIDS. The impairment of macrophage function results in impaired chemotaxis and phagocytosis with increased susceptibility to candidiasis and toxoplasmosis. Decreased immunoglobulin production due to lack of T-cell stimulation of B cells makes the person vulnerable to encapsulated microorganisms such as *Streptococcus pneumoniae*.

Course of Disease

The HIV infection runs through the following stages:

I. *Initial viremia*: HIV infection begins with initial viremia, which produces mild fever, headache, body aches, macular skin rash and lymph nodes enlargement. This clinical picture, which resembles infectious mononucleosis, subsides in 1–2 weeks. The HIV is then taken up by lymphatic system (lymph nodes, tonsils and adenoids). This initial viremia lasts for a few weeks and then no virus is detected in plasma. Antibody test becomes positive in 2–4 months of infection.

II. *Latent period*: This asymptomatic period lasts for a long period of about 10 years, HIV is replicating in the lymphoid tissue. CD4 T-helper cell count and function deteriorates.

III. *Advanced disease*: AIDS picture starts appearing after several years, when CD4 T-cell count falls significantly and patient becomes susceptible to opportunistic infections. Death usually occurs within 2 years of the appearance of clinical signs and symptoms of AIDS.

AIDS Indicator Diseases

1. The most common index disease (> 60%) for the diagnosis of AIDS is *Pneumocystis carinii* pneumonia. CD4 count is a serum marker, which is used to measure the progression of AIDS.
2. The reactivation of the latent protozoan infection *Toxoplasma gondii* (consuming raw infected meat) is the most common cause of focal encephalitis, which shows hypodense ring enhancing lesion on CT scan.
3. The most common viral opportunistic pathogen is cytomegalovirus (CMV), which can cause chorioretinitis, pneumonia, esophagitis, colitis, encephalitis and hepatitis.
4. The common skin manifestations, which may also be present in head and neck region, are: candidiasis, tinea versicolor, molluscum contagiosum and *Staphylococcus aureus* infection.
5. ENT manifestations of AIDS are caused by opportunistic viruses, bacteria, fungi and protozoa and due to activation of the neoplastic processes. More than 50% of the patients with AIDS present with involvement of head and neck region.

Diagnosis

Anti-HIV antibodies, detected by enzyme-linked immunosorbent assay (ELISA) and Western blot, occur within 3 months of HIV infection. CD4 count and HIV viral load accurately ascertain the HIV infection and patient's immune status. The important tests, which are done to diagnose and treat HIV infection and its progression, are:

1. *ELISA test*: Screening test with high sensitivity of > 99.9%.
2. *Western blot test*: Confirmation test.
3. *CD4 cell count*: Normal count 600–1500/mm^3. CD4 lymphocyte percentage is more reliable than CD4 count. Falling counts indicate progression of disease. Count < 200/mm^3 and < 20% indicates high risk of AIDS progression.
4. *PCR*: Quantitative test to measure viral load, which correlates with progression of disease.

- Other tests which may also be carried out, include:
  - *Complete blood count*: Anemia, leukopenia especially lymphopenia and thrombocytopenia.
  - *Beta 2-Microglobulin level*: It is a prognostic test, which indicates macrophage monocyte stimulation. The levels rise at seroconversion and continue to rise with progression of disease.
  - *P-24 antigen*: The presence of P-24 core protein of HIV indicates active HIV replication. This is an earliest test, which becomes positive even before the seroconversion.

Classification

- *Three Clinical Categories*: They are:
  1. *Asymptomatic HIV infection (Initial viremia)*: Persistent generalized lymphadenopathy, or acute HIV infection, which produces mild fever, headache, body aches, macular skin rash and lymph nodes enlargement (clinical picture similar to infectious mononucleosis) for 1–2 weeks.
  2. *Symptomatic conditions*: Oral thrush, hairy leukoplakia and fungal sinusitis (See other sections in this chapter).
  3. *AIDS*: Some of the AIDS indicator diseases are mentioned above.
- *CD4 Count*: The disease can be categorized further as per the CD4 counts:
  1. 500 or more
  2. 200 to 499
  3. Less than 200

CDC AIDS Case Definitions

The Center for Disease Control and Prevention (CDC) classifies adults and adolescents AIDS cases into 3 groups for surveillance:

I. *Definitive AIDS diagnoses* (with or without laboratory evidence of HIV infection)
   a. Coccidiomycosis. Disseminated at a site other than or in addition to lungs or cervical or hilar nodes.
   a. Candidiasis: Esophagus, trachea, bronchi, or lungs.
   b. Cryptococcosis: Extrapulmonary.
   c. Cryptosporidiosis with persistent diarrhea > 1 month.
   d. Cytomegalovirus infection of an organ other than liver, spleen, or lymph nodes.
   e. Herpes simplex virus infection: Persistent mucocutaneous ulcer > 1 month; bronchitis, pneumonitis, or esophagitis of any duration.
   f. Kaposi sarcoma in < 60 years of age.
   g. Lymphoma brain primary in < 60 years of age.
   h. *Mycobacterium avium* complex or *Mycobacterium kansasii* disease. Disseminated at a site other than or in addition to lungs, skin, or cervical or hilar nodes.
   i. *Pneumocystis jiroveci* pneumonia.
   j. Progressive multifocal leukoencephalopathy.
   k. Toxoplasmosis of brain.

II. *Definitive AIDS diagnoses* (with laboratory evidence of HIV infection)
   a. Coccidiomycosis. Disseminated at a site other than or in addition to lungs or cervical or hilar nodes.
   b. HIV encephalopathy.
   c. Histoplasmosis. Disseminated at a site other than or in addition to lungs or cervical or hilar nodes.
   d. Isoporiasis with persistent diarrhea > 1 month.
   e. Kaposi sarcoma at any age.
   f. Lymphoma brain primary at any age.
   g. Other non-Hodgkin lymphoma of B cell or unknown immunologic phenotype.
   h. Any mycobacterial disease other than *Mycobacterium tuberculosis*. Disseminated at a site other than or in addition to lungs, skin, or cervical or hilar nodes.
   i. Extrapulmonary *Mycobacterium tuberculosis.*
   j. Recurrent *Salmonella* (nontyphoid) septicemia.
   k. HIV wasting syndrome.
   l. CD4 < 200 cells/mcL or < 14%.
   m. Pulmonary tuberculosis.
   n. Recurrent pneumonia.
   o. Invasive cervical cancer.

III. *Presumptive AIDS diagnoses* (with laboratory evidence of HIV infection)
   a. Esophageal candidiasis (recent onset of retrosternal pain on swallowing) and oral candidiasis.
   b. CMV retinitis: A characteristic lesion seen on serial ophthalmoscopy.
   c. Mycobacteriosis. Species of Acid-fast bacilli not identified by culture in stool or normally sterile body fluids or tissue from a site other than lungs, skin, or cervical or hilar nodes.
   d. Kaposi sarcoma. Red or violaceous plaque-like lesion on skin or mucous membrane.
   e. *P. jiroveci* pneumonia. * Dyspnea on exertion or dry cough of recent onset (3 months); and * X-ray chest showing diffuse bilateral interstitial infiltrates or gallium scan showing diffuse bilateral pulmonary disease; and * < 70 mm Hg arterial oxygen partial pressure or < 80% respiratory diffusing capacity or increased alveolar-arterial oxygen tension gradient; and * No finding of bacterial pneumonia.
   f. Toxoplasmosis of brain: * Recent onset of focal brain abnormality or reduced level of consciousness; and * Brain imaging showing mass lesion enhanced by contrast; and * Toxoplasmosis antibody or response to anti toxoplasmosis therapy.
   g. Recurrent pneumonia: * > 1 episode in a year; and * Clinical or radiological diagnosis of acute pneumonia.
   h. Pulmonary tuberculosis: * Apical or miliary infiltrates; and * Clinical or radiological response to AKT.

Treatment

It consists of four elements: treatment for opportunistic infections and malignancies, antiretroviral treatment, hematopoietic stimulating factors and prevention of opportunistic infections.

- *Treatment for opportunistic infections and malignancies*: Patients who have good response to antiretroviral treatment, do not need maintenance treatment of opportunistic infections such as—
  - *P. jiroveci*: Trimethoprim-sulfamethoxazole; trimethoprim with dapsone; pentamidine; primaquine and clindamycin; atovaquone, or trimetrexate.
  - *M. avium*: Clarithromycin with ethambutol.
  - *Toxoplasmosis*: Pyrimethamine with sulfadiazine and folinic acid.
  - *Lymphoma*: Chemotherapy, radiation and dexamethasone for CNS lesions.
  - *Cryptococcal meningitis*: Amphotericin B followed by fluconazole.
  - *Cytomegalovirus*: Valganciclovir, ganciclovir, foscarnet.
  - *Candidiasis* (esophageal or recurrent vaginal): Fluconazole.
  - *Herpes simplex and zoster*: Acyclovir, famciclovir, valacyclovir, foscarnet.
  - *Kaposi sarcoma*: Limited cutaneous lesions (observation, intralesional vinblastin); Extensive cutaneous lesions (chemotherapy, radiation); and visceral lesion (combination chemotherapy).
- *Antiretroviral treatment* (Table 1) Highly active antiretroviral therapy (HAART) has been reported to threefold decrease in mortality and a sixfold decrease in opportunistic infections. It consists of nucleoside reverse transcriptase inhibitors (NRTI), nonnucleoside reverse transcriptase inhibitors (NNRTI) and protease inhibitors (PI). Nucleoside analogues includes zidovudine (AZT), didanosine (ddI), zalcitabine (ddC) and stavudine (d4T). Reverse transcriptase inhibitor is lamivudine (3TC). Protease inhibitors are saquinovir, ritonavir and indinavir.
  - *Indications*: CD4 cell count < 350 cells/mcL (< 200 cells/mcL in cases of higher risk for toxicity); Symptomatic HIV disease; Very high viral loads > 100,000/mcL.

TABLE 1 Antiretrovirals

Nucleoside reverse transcriptase inhibitors (NRTI)	Zidovudine, lamivudine, tenofovir, emtricitabine, stavudine, didanosine, abacavir, apricitabine, elvucitabine, racivir
Non nucleoside reverse transcriptase inhibitors (NNRTI)	Nevirapine, efavirenz, etravirine, rilpivirine
Protease inhibitors (PI)	Ritonavir, atazanavir, lopinavir, darunavir, indinavir, saquinavir, nelfinavir, amprenavir, fos-amprenavir, tipranvavir
Integrase inhibitors	Raltegravir, elvitegravir
Receptor blockers	TNX 355
Co-receptor blockers	Maraviroc (R5), vicriviroc (R5), PRO 140 (R4), AMD 070 (R4)
Fusion inhibitors	Enfuvirtide
Maturation inhibitor	Bevirimat

- *Goal*: Viral load < 50–75 copies/mcL.
- *Monitoring*: Watch for toxicity and adverse reactions; CD4 cell count and viral load; Drug resistance.

- *Hematopoietic Stimulating Factors*:
  - *Epoetin alfa (erythropoietin)*: In cases of anemia it avoids blood transfusions.
  - *Human G-CSF (filgrastim) and granulocyte-macrophage colony-stimulating factor GM-CSF (sargramostim):* They increase neutrophil count of HIV patients.
- *Prevention of opportunistic infections*: The decision to begin prophylactic therapy of opportunistic infections depends on following factors:
  - CD4 count.
  - Evidence of severe immune suppression such as oral candidiasis.
  - Past history of infection.

CERVICAL ADENOPATHY

Baseline palpable adenopathy is common in HIV infection.

- *Indications for biopsy*: The conclusive fine needle aspiration cytology (FNAC) can avoid open biopsy. The indications of cervical lymph node biopsy include:
  - Single big node
  - Progressive enlargement of adenopathy
  - Marked constitutional symptoms
  - Cytopenia and raised erythrocyte sedimentation rate (ESR)
- *Causes*: Cervical lymphadenopathy can be due to:
  - Infection: Typical and atypical tuberculosis, histoplasmosis, toxoplasmosis and cat-scratch disease.
  - Malignancy: Lymphoma, carcinoma and Kaposi sarcoma.

NEOPLASMS

Four cancers are currently included in CDC classification of AIDS: Kaposi sarcoma, non-Hodgkin lymphoma, primary lymphoma of brain and invasive cervical carcinoma.

1. *Most common neoplasms*: They are–
   - Kaposi's sarcoma
   - Non-Hodgkin's lymphoma
   - Lymphoepithelial cysts of parotid gland
2. *Uncommon neoplasms*: They include–
   - Hodgkin's Lymphoma
   - Cutaneous neoplasms: Squamous cell carcinoma

Kaposi's Sarcoma

Kaposi's sarcoma is the most common malignant lesion of AIDS. It can occur at any stage of HIV infection

- *Clinical features*: This multicentric non-invasive neoplasm respects the fascial planes and involves skin, mucosa and viscera.
  - *Lesions*: Lesions appear as nonblanching, purplish (in light skinned people) or brownish (in dark skinned people) papules or nodules. These slightly raised lesions may vary in size (millimeters to centimeters).
  - *Site*: In oral cavity the lesions are most often palatal papules. The exophytic lesions of tongue and gingivae may also be seen. The lesions can involve posterior pharyngeal wall and other sites such as eyelids, conjunctiva, pinna, toe webs, face, neck and upper trunk.
- *Differential diagnoses*: It should be differentiated from angioma or pyogenic granuloma.
- *Diagnosis*: The lesion is diagnosed by biopsy that shows excessive proliferation of spindle cells of vascular origin, endothelial cells, extravasation of red blood cells and hemosidrin laden macrophages.
- *Treatment*:
  - *Systemic chemotherapy*: Rapidly progressive dermatologic or visceral disease needs liposomally encapsulated doxorubicin intravenously every 3 weeks.
  - *Local:* Milder forms do not need any specific treatment and usually respond to antiretroviral therapy. Other forms of local management include:
    - Localized radiation
    - Intralesional vinblastin
    - Cryotherapy
- *Prognosis*: The ultimate survival is usually determined by infections and not by Kaposi's sarcoma.

Non-Hodgkin's Lymphoma

It is the second most common AIDS-associated malignancy. Non-Hodgkin's lymphoma (NHL) is an aggressive late stage disease and occurs in 10–30% of AIDS patients.

1. *Types*: Both nodal, extranodal (about 90%) and CNS lymphomas (about 40%) may be seen when CD4 counts fall below 200/mm^3.
2. *Clinical features*: It presents with fever, night sweat, weight loss and non-tender rapidly enlarging neck mass.
3. *Treatment*: Chemotherapy and radiotherapy show poor control.

Lymphoepithelial Cysts of Parotid Gland

Lymphoepithelial cysts are bilateral and multiloculated and contained within parotid fascia. They usually involve the tail of parotid. Other parotid lesions include Kaposi's sarcoma, non-Hodgkin's lymphoma and parotitis.

- *Clinical features*: Asymptomatic bilateral parotid swellings are not uncommon in HIV infections. Reactive lymphocytes infiltrate parotid tissue. The salivary flow and amylase are normal.
- *Treatment:* Parotidectomy is advocated if needle aspiration

fails. Intralesional tetracycline sclerosis and low dose of radiation have also been tried.
- *Prognosis*: In children, it indicates strong anti-HIV response and longer survival.

NOSE AND SINUSES

Acute and chronic sinus problems are common.
- *Pathogens*: Both aerobic and anaerobic infections are common.
  - The unusual pathogens seen are: *Legionella, Alternaria, Cryptococcus, Candida* and *Acanthamoeba.*
  - Fungal sinusitis can be caused by *Aspergillus* and *Mucormycosis*, which is rapidly invasive and extends intracranially.
- *Treatment:* In persistent and recurrent cases of sinusitis, in addition to antibiotics, long course of guaifenesin and topical steroids help in resolving the disease.
  - Surgical treatment is considered in invasive fungal infections.
- *Indications of antral puncture sinus lavage*: It should be done if patient does not respond to medications and when local and systemic complications arise.

NASOPHARYNX

- Kaposi's sarcoma and non-Hodgkin's lymphoma can involve nasopharynx.
- Serous and acute otitis media occur due to adenoid hypertrophy.

EAR

- *External ear*: Kaposi's sarcoma of pinna and seborrheic dermatitis, which may get secondarily infected, are common.
  - Aural polyps due to *Pneumocystis carinii* are also reported.
- *Middle ear*: Serous and acute otitis media occur due to adenoid hypertrophy.
  - *Most common pathogens* (decreasing frequency): *Streptococcus pneumoniae, Haemophilus influenzae* and *Morexella catarrhalis*. *Staphylococcus*, *Pseudomonas* and *Pneumocystis carinii* are seen with greater frequencies.
- *Sensorineural hearing loss*: Usually due to cytomegalovirus affecting inner ear or CN VIII.
- *Facial palsy (Ramsay Hunt syndrome or Herpes zoster oticus)*: It has been reported in up to 16% of AIDS patients. Reactivation of latent varicella zoster virus can occur.
  - *Clinical features*: It presents with vesicular eruptions of ear and face, facial palsy, ear pain, tinnitus, vertigo and sensorineural hearing loss.

ORAL CAVITY

Oral candidiasis or hairy leukoplakia is highly suggestive of HIV infection in patients who have no other obvious cause of immunodeficiency. Kaposi's sarcoma lesions may ulcerate and become secondarily infected. Aphthous ulcers and xerostomia can also occur.

1. *Candidiasis*: Oral candidiasis is the most common oral manifestation of AIDS. It presents with tender, white and pseudomembranous lesions. HIV candidiasis patients have high rate of progression to AIDS. Candida infection of esophagus causes severe odynophagia.
   - *Types*: (1) The pseudomembranous lesions are removable white plaques. (2) Erythematous lesions are red friable plaques.
   - *Treatment*: Patients who do not respond to clotrimazole (10 mg troche 4 or 5 times a day) can be treated with fluconazole (50–100 mg orally once a day for 3–7 days).
     - *Angular cheilitis*: Fissures at the sides of the mouth are usually due to Candida. They are treated topically with ketoconazole cream (2%) twice a day.
     - *Candidal esophagitis*: Fluconazole 200 mg OD for 10–14 days. The patients, who do not respond, should be evaluated for other causes of esophagitis such as herpes simplex and CMV.
2. *Hairy leukoplakia*: This oral cavity lesion is seen almost exclusively in HIV patients. About half of the HIV infected patients with hairy leukoplakia develops AIDS in 16 months (Fig. 1).
   - *Pathogens*: It is caused by Epstein-Barr virus and papillomavirus.
   - *Clinical feature*: It manifests as white vertically corrugated patches on the lateral border of tongue. The lesion may be flat or slightly raised. It is not usually troubling to patients and sometimes regresses spontaneously. The vertical parallel lines have fine or thick hairy projections.
3. *Herpes simplex stomatitis*: Vesicular and extremely painful lesions may coalesce and form large ulcers of several centimeters. They are quite common.
   - Angular cheilitis can also be seen.
4. *Gingivitis and periodontitis*: Gingivitis and periodontitis presents as tender bleeding gums and erythema at the gum-line.
   - *Treatment*: It usually responds to professional dental cleaning and chlorhexidine rinses.
5. *Acute necrotizing ulcerative gingivitis (ANUG) and necrotizing stomatitis*: They indicate severe AIDS and present as red and swollen gingival lesions that undergo necrosis and destruction of periodontal soft tissue.
   - *Treatment*: Metronidazole 400 mg TDS for 5 days and referral to oral surgeons.
6. *Aphthous ulcers*: The treatment of these painful ulcers include:
   - Fluocinonide topical application six times a day.
   - Dexamethasone swishes (0.5 mg in 5 ml elixir TDS).
   - Lidocaine 10% spray.

Fig. 1: Hairy leukoplakia lateral margin of anterior two-third tongue. Note the slightly raised lesion with hairy projections

- Thalidomide 50 mg orally daily and increasing to 100–200 mg daily. This teratogenic analgesic should be prescribed only to refractory patients who are at zero risk of procreation. The most common side effects include sedation and peripheral neuropathy.

OCCUPATIONAL EXPOSURE

Health care providers such as doctors (especially surgeons), nurses and laboratory staff while handling the blood, body fluids and other secretions may contract the disease. Each and every sample handled should be considered as potentially infected. The risk is usually due to:

1. Needle-stick injury: Hollow injection needle is more dangerous than solid suture needle. The risk of contracting HIV infection is 1:250.
2. Cuts with contaminated knife or other sharp instruments.
3. Exposure of open wound including area of dermatitis or even normal skin to infected blood or body fluid. Gloves and gown/coat are protective.
4. Large mucous membrane exposure by splatter of blood and amniotic fluid.

Management of Needle Stick Injury

- Immediately wash the area thoroughly with water without soap and apply an antiseptic.
- Though the seroconversion takes time, ELISA test is done at the earliest to establish negative baseline for worker's compensation. The test is repeated at 6 weeks, 3 months and 6 months.
- Zidovudine therapy, 200 mg every 4 hours for 6 weeks, at the earliest has shown to decrease the rate of seroconversion after needle stick injury. The side effects of drug should be kept in mind.

Prevention

- Wear gloves. Wash hands before and after the contact.
- Handle the blood, body fluid and laboratory specimens as potentially infectious.
- Immediately place used syringes in impermeable container; never recap or manipulate needle.
- Wear gowns, protective glasses and mask when splatter with blood or body fluids is expected, such as bronchoscopy, tracheostomy and oral surgery.
- Linen soiled with blood and body fluid and secretions must be handled as potentially infectious.
- Wear mask for the opportunistic respiratory infections such as tuberculosis.

Care of Endoscopes

After the use, endoscopes should be wiped with alcohol and then immersed in 2% glutaraldehyde for 10 minutes. The flexible scopes then must be rinsed in water and air-dried.

Clinical Highlights

1. **Risk factors**: A single hollow bore needle prick has 1:200 chances of seroconversion.
2. **Human immunodeficiency virus (HIV):** Reverse transcriptase of the virus produces DNA from viral RNA.
3. **Immunity**: HIV affects the cell mediated immunity.
4. **CD4 count**: Opportunistic infections and malignancies arise when CD4 count falls below 200/mm^3.
5. **Kaposi's sarcoma**: It is the most common malignancy seen in AIDS patients.
6. **Oral manifestations of HIV**: They include oral candidiasis, hairy leukoplakia and recurrent aphthous ulcers.
7. **Side effects of indinavir (protease inhibitor)**: They include breast hypertrophy, central adiposity, hyperlipidemia, insulin resistance and nephrolithiasis.
8. **Ritonavir**: It inhibits HIV protease.
9. **Highly active antiretroviral therapy**: Highly active antiretroviral therapy (HAART) in HIV infection is associated with following:
   a. Reduced incidence of opportunistic infections.
   b. Decrease in viral mRNA copies/mL of blood.
   c. Increase in CD4 count.
   d. Decrease in rate of emergence of drug resistance.

FURTHER READING

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9 History and Examination

The education which does not help the common mass of people to equip themselves for the struggle for life, which does not bring out strength of character, a spirit of philanthropy, and the courage of a lion—is it worth the name? Real education is that which enables one to stand on his own legs.

—Swami Vivekananda

Points of Focus

- **OTORHINOLARYNGOLOGY**
- **HISTORY TAKING**
  - ✓ General Scheme of Case Taking
  - ✓ Common Complaints of Ear, Nose and Throat
- **PHYSICAL EXAMINATION**
  - ✓ Features of General Survey Examination
  - ✓ Local Examination
  - ✓ General Examination
- **GENERAL SET-UP**
  - ✓ Bull's Eye Lamp
  - ✓ Head Mirror
- **SWELLINGS AND ULCERS**
  - ✓ History
  - ✓ Examination of Swelling
  - ✓ Examination of Ulcers
- **EXAMINATION OF CRANIAL NERVES**
- **HEADACHE**
  - ✓ Definition
  - ✓ Pain Sensitive Structures for Headache
  - ✓ International Classification of Headache
  - ✓ Etiology
  - ✓ ENT Causes
  - ✓ History and Examination
  - ✓ Causes of Headache and their Features
  - ✓ Tension-type Headache
  - ✓ Depression Headache
  - ✓ Cluster Headache
  - ✓ Post-Traumatic Headache
  - ✓ Cough Headache
  - ✓ Giant Cell (Temporal or Cranial) Arteritis
- **FACIAL PAIN**
  - ✓ Trigeminal Neuralgia (Tic Douloureux)
  - ✓ Atypical Facial Pain
  - ✓ Glossopharyngeal Neuralgia
  - ✓ Post-Herpetic Neuralgia
- **TEMPOROMANDIBULAR DISORDERS**
  - ✓ Common Causes
  - ✓ Clinical Features
  - ✓ Myogenic Disorders
  - ✓ Meniscal Displacement
  - ✓ Imaging Studies: Orthopantomogram (OPG), CT scan, MRI
  - ✓ Treatment: Restricted Mouth Opening and Soft Diet, Analgesics, Muscle Relaxants, Anxiolytics, Antidepressants, Narcotics, Splint Therapy, Surgery
- **CLINICAL HIGHLIGHTS**

OTORHINOLARYNGOLOGY

Otorhinolaryngology (Oto: Ear; Rhino: Nose; Laryngo: Larynx-Throat), which is also called as otolaryngology, is the specialty that deals with diseases of ear, nose, throat, head and neck region. Originally, the specialty was commonly called as EENT (eyes, ear, nose and throat) and included ophthalmology as

well. But the two disciplines split many years ago because of the explosion of medical knowledge. In recent time the specialty is usually called as "Otolaryngology–head and neck surgery". The doctor who deals with the specialty is called otolaryngologist/Ear, neck and throat, head and neck surgeon.

Otolaryngology is both medical as well as surgical specialty. Approximately, only 10% of ENT patients require surgical interference. The otolaryngologist cares for all the age groups of patients such as children, adults and old.

The subdivisions, which exist within the specialty, are: otology, laryngology, rhinology and bronchoesophagology. The other subspecialties recognized in recent times because of information explosion are: pediatric otolaryngology, otolaryngologic allergy, rhinology and sinus surgery, facial plastic and reconstructive surgery and head and neck (oncology) surgery. Otology includes not only ear and temporal bone but also neurotology and skull base surgery. The recent developments have made some ENT surgeons to specialize in neurolaryngology, microvascular surgery, chemosensation (taste and smell disorders), audiology and speech disorders.

Evaluation of the patient with disease of ENT requires patience and practice because the examiner will have to deal with the narrow and darker cavities of the ear, nose, pharynx and larynx, which have relatively complex and complicated anatomy and physiology.

HISTORY TAKING

On the basis of adequate history, many conditions can be accurately diagnosed. Box 1 shows the general scheme of case taking.

1. *Personal particulars:* The name, age, sex, religion, social status, occupation and residential address of the patient are noted, as many a time they are quite relevant and associated with certain diseases.
2. *Chief complaints:* The complaints side of affection and their duration are recorded in a chronological order of their appearance. Table 1 shows the list of common complaints of ENT patients. It is very important to ask patient, "What made you to consult doctor?" The details of ear, nose and throat complaints are covered in the sections of ear, nose and paranasal sinuses, oral cavity and salivary glands, pharynx and esophagus, larynx and trachea and bronchus tree and neck.
3. *History of present illness:* It includes the details of all the complaints mentioned in the chief complaints that begin with the appearance of first symptom and extend up to the time of consultation. It usually consists of the mode of onset (sudden/gradual), preceding events causing onset, course of symptoms (progressive/constant/fluctuant; and continuous/intermittent), factors aggravating/relieving, other accompanying complaints, and the treatment taken. In cases of unilateral disorder note the side and site of affliction. If both the sides of ear and/or nose are affected then it should be mentioned which is worst affected. Many a time negative answers are equally important in arriving at a diagnosis. Inquiries are made of any systemic diseases, which patient might be suffering from such as diabetes, hypertension, coronary artery disease, liver or kidney disease, tuberculosis, HIV/AIDS and a bleeding disorder.
4. *History of past illness:* It includes history of similar complaints in the past and the treatment taken and history of surgeries, accidents, radiations and complications. All the diseases suffered by the patient in past whether seemingly relevant/irrelevant should be recorded in a chronological order.
5. *Drug history:* It records all the drugs which patient was/is taking such as steroids, chemotherapy, insulin, antihypertensive, diuretics, monoamine oxidase (MAO) inhibitors, contraceptive and hormone replacement therapy.
6. *History of allergy:* It must not be missed as the consequences can be life-threatening. Patients can be allergic to certain drugs, diets, pollens, fungi, animals and dusts.
7. *Personal history:* It inquires about the patient's occupation, personal habits (smoking, alcohol and chewing of *Paan*, *Sopari* and tobacco), food habits (vegetarian/nonvegetarian, regular/irregular, spicy food), lifestyle (exercise or sedentary), and marital status. In women menstrual history and number of pregnancies and miscarriages should be recorded.
8. *Family history:* It is important as many ENT diseases run in families and have genetic basis such as otospongiosis, certain types of sensorineural hearing loss, malignancies and autoimmune disorders. Infectious diseases such as STD, tuberculosis, mumps, pediculosis, scabies and diphtheria can affect other family members.
9. *History of immunization:* It should be asked in children.

Box 1: General scheme of case taking

- *History taking*
  - Personal particulars
  - Chief complaints with duration
  - History of present illness
  - Past history
  - Drug history
  - History of allergy
  - Personal history
  - Family history
  - History of immunization
- *Physical examination*
  - General survey
  - Local examination
  - General examination
- *Provisional clinical diagnosis*
- *Differential diagnosis*
- *Investigations*
  - Laboratory
  - Radiological
  - Biopsy
- *Final diagnosis*
- *Treatment*
  - Medical
  - Surgical
- *Progress and follow-up*
- *Termination*

PHYSICAL EXAMINATION

It consists of general survey, local examination and general examination.

TABLE 1 Common complaints of ear, nose and throat

Ear	*Nose/Sinuses*	*Oral cavity*	*Throat*	*Face*	*Neck*
Hearing loss	Blocking	Ulcers	Pain	Pain	Pain
Discharge	Sneezing	Swelling	Odynophagia	Cleft lip	Swelling
Pain	Disturb smell	Pain	Stridor	Swelling	Nodes
Noises	Discharge	Disturb taste/	Voice change	Injury	Goiter
Dizziness/Vertigo	Nose bleed	Salivation	Nasal voice	Parotid and	Injury
Itching	Post-nasal drip	Trismus	Dysphagia	Submandibular	Cyst
Deformities	Headache	Ankyloglossia	Snoring	lesions	Fistula
Swelling	Facial pain	Cleft palate	Cough	Jaw lesions	Sinus
Facial palsy	Swelling	Injury/FB (foreign	sputum		
Injury/Foreign	Deformities	body)	Injury/FB		
body(FB)	Injury/FB	Halitosis			

General: Fever, headache, migraine, vomiting, convulsions, loss of weight and anorexia.
Personal: Chewing of *Paan, Sopari* or Tobacco, smoking, alcohol, swimming, exposure to dust and noise

Systemic: Anemia, asthma, bleeding tendencies, allergy, epilepsy, jaundice, tuberculosis, kidney trouble, hernia, acidity/reflux gastroesophageal reflux disease (GERD), peptic ulcer, diabetes, hypertension, chest pain, heart attack, eye problems and pregnancy.

General Survey

The general survey (Table 2) includes general assessment of illness, mental state, intelligence, build, nutrition, attitude, decubitus (patient's position in bed), skin color and eruptions and vital parameters pulse, blood pressure, respiration and temperature.

Cyanosis is not detectable in patients with severe anemia.

Local Examination

It includes inspection and palpation of the affected region. Percussion and auscultation are important in a few ENT conditions. Examination of the draining lymph nodes is the essential component of local examination. Detailed description of ENT examination is covered in the sections of ear, nose and paranasal sinuses, oral cavity and salivary glands, pharynx and esophagus, larynx, trachea and bronchus and neck. The general set up of otorhinolaryngological head and neck examination will be discussed in this chapter. ENT head and neck examination also includes eyes and cranial nerves.

Following a specific order facilitates complete examination without missing any region. It is always advisable to proceed from outside to inside. For example begin from external nose to nasal cavity. The comprehensive examination of ear, nose, throat, head and neck include following components.

1. *Ears:* For detail, see chapter Otological Symptoms and Examination.
2. *Nose*: For detail, see chapter Nasal Symptoms and Examination.
3. *Oral cavity*: For detail, see chapter Oral Symptoms and Examination.
4. *Pharynx*: For the details of oropharynx, nasopharynx, and laryngopharynx see chapter Pharyngeal Symptoms and Examination.
5. *Larynx*: For detail, see chapter Laryngeal Symptoms and Examination.
6. *Head*: The examination of head includes face and scalp.
7. *Neck*: Note movements, neck veins, lymph nodes, carotid pulsation and thyroid glands. For detail see chapter Cervical Symptoms and Examination.
8. *Systemic examination*: It should include respiratory and neurologic examination as indicated.

General Examination

It is especially required in chronic or systemic diseases, indoor patients and before surgery.

It consists of examination of all the systems such as gastrointestinal, respiratory, cardiovascular and neurological.

GENERAL SET-UP

Bull's Eye Lamp

The lamp house can be tilted and rotated and raised or lowered on the floor stand according to the need. It is a powerful source of light and provides an illumination of about 200 candle power without any image of the filament. The patient is seated erect on a stool or in a chair opposite the examiner. She is examined in a semi-dark room. The head of the patient leans slightly forward towards the examiner (Fig. 1). The child feels comfortable in mother lap but should be held properly (Fig. 2). Bull's eye lamp is placed on the left (some prefer right side) of the patient at the level of his shoulder (Fig. 3). The examiner adjusts a head mirror on his/her right eye (some prefer left eye), which reflects light from the Bull's eye lamp onto the area of interest (Figs 4A and B).

Head Mirror

It is a concave mirror with a central hole of 19 mm diameter. The mirror has a focal length of about 25 cm and a diameter of 89 mm (3-1/2 inch). It provides not only good illumination but also permits freedom to use both hands for other activities.

The perfect and comfortable use of Bull's lamp and head mirror needs some coordination, which requires practice and patience. However, some doctors prefer headlights. Both the electric bulb and fiber optic headlight of various types are available. Chapter Instruments discusses some of the routinely used OPD instruments that are used for examining ENT patients.

SWELLINGS AND ULCERS (FIG. 5)

History

The history taking (Table 3) should enquire following information from the patient:

1. *Duration*: A small painless lump or ulcer may be present for a long time before noticed (accidentally) by the patient or relatives.

TABLE 2 Features of general survey examination

Illness	Severity: Mild, moderate, severe or emergency
Intelligence	Can doctor rely on patient's history?
Level of consciousness Glasgow coma scale	I: Fully oriented to time and place. Obey commands II: Disoriented to time and place. Localizing to pain III: Utter occasional words. Flexing to pain IV: No words. Extending to pain V: No response
Build and nutrition	Cachexia is usually due to malignancy
Attitude	Neck retraction and rigidity in meningitis
Gait: Pattern of walking	Ataxic gait in cerebellar lesions
Faces	Adenoid faces (adenoid hypertrophy), Risus Sardonicus (Tetanus), mask face (Parkinsonism) and moon face (Cushing's syndrome)
Decubitus	Curled upon one side away from light in cerebral irritation
Pallor of skin, nail-bed, conjunctiva	Anemia, massive bleeding, shock, intense emotions
Cyanosis	Peripheral (nail-bed) and central (tongue)
Jaundice	Look sclera, nail-bed, ear lobule, and nasal tip
Skin	Look for macules (nevi and erythema), papules (wart and granulomas), vesicles (filled with fluids), pustules (filled with pus) and wheals (urticaria)
Nails	Koilonychia, clubbing
Pulse	Rate (fast/slow), rhythm (regular/irregular), tension and force, volume (pulse pressure), character and condition of arterial wall
Respiration	Tachypnea (fever, shock, hypoxia, metabolic acidosis, tetany, hysteria), slow and deep respiration (cerebral compression), Cheyne-Stokes irregular respiration (overventilation alternating with apnea)
Temperature	Continuous, remittent (fluctuating), or intermittent—daily (quotidian), alternate days (tertian), every two days (quartan)
Blood pressure	Hypotension, hypertension

Fig. 1: Patient's position. Patient seated erect and head leans slightly forward

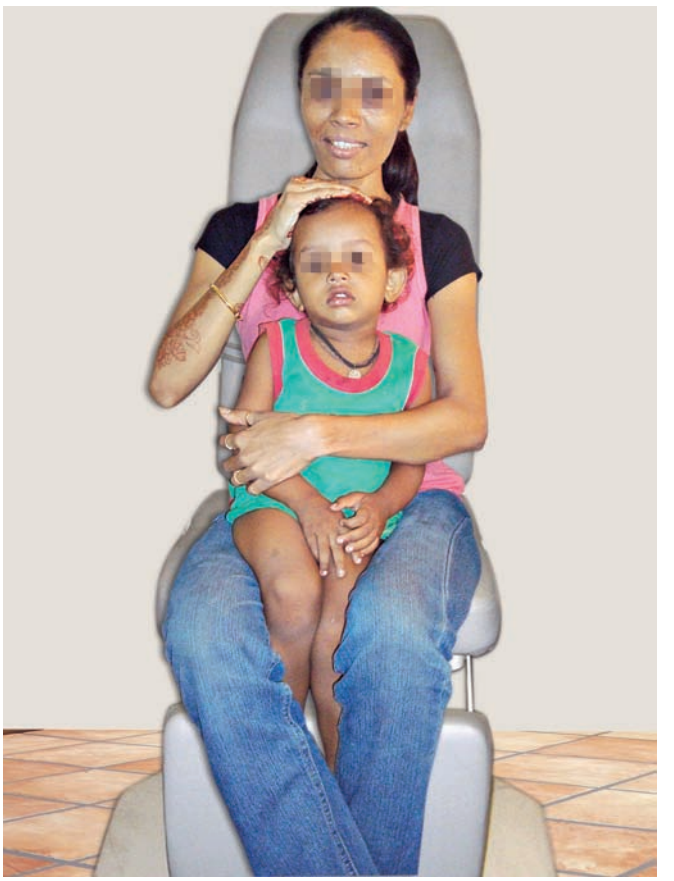
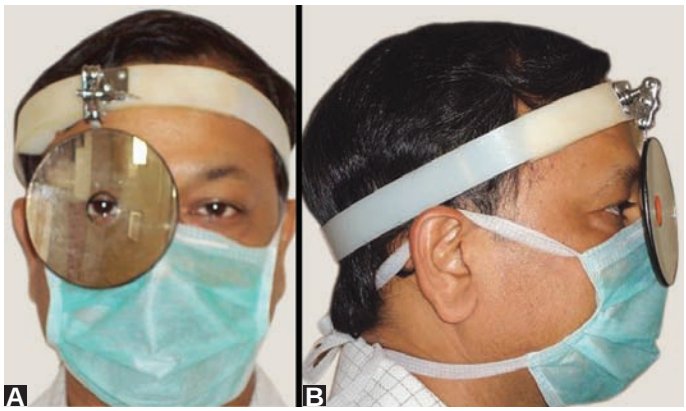


Fig. 2: Child's position. The child is sitting in mother's lap. His feet are held between mother's feet. His hands are hold by mother's left hand. Mother's right hand holds the head of child

Fig. 3: Bull's eye lamp placed on left side of patient at the level of shoulder



Figs 4A and B: An ENT surgeon adjusts a head mirror on right eye. (A) Front view; (B) Side view

Fig. 5: Fungating metastatic neck nodes
Source: Dr Amit Goyal, Shillong

TABLE 3 Characteristic symptoms of some of the swellings and ulcers and their causes

Painful of few days duration	Inflammatory
Swelling without pain for years	Benign neoplastic
Without pain or with mild pain for months	Chronic inflammation or malignancy
Throbbing painful swelling	Acute inflammation or suppuration
Pain before or during the appearance of lesion	Inflammation
Pain months after the appearance of lesion	Malignancy
Sudden increase of preexisting stationary lesion	Benign to malignant change
Fever	Inflammation/suppuration, Hodgkin's disease
Similar lumps at other places	Neurofibromatosis, Hodgkin's disease

2. *Mode of onset*: Sudden or gradual. Ask about the preceding events such as trauma and surgery.
3. *Pain*: Painful or painless. Pain suggests acute inflammation. Malignant lesions are usually painless to start with and pain may appear later on. Syphilitic and tubercular lesions are also painless.
4. *Progress*: Stationary, fluctuating, increasing, or sudden increase of pre-existing stationary swelling. Later changes in the swelling should be inquired such as softening, ulceration, fungation and inflammatory.
5. *Site*: The site from where the swelling started must be inquired in cases of large swellings.
6. *Associated symptoms*: One must inquire about the associated symptoms of anorexia, loss of body weight, dysphagia, dysarthria, hoarseness of voice, respiratory distress, ear pain, and nasal discharge.
7. *Fever*: It suggests inflammatory swelling.
8. *Impairment of function:* Impairment of function after the appearance of swelling should be inquired such as facial palsy after parotid swelling.
9. *Recurrence*: Recurrence of swelling after the initial surgery suggests malignancy. Recurrent ulcers include aphthous and herpes stomatitis.
10. *Past history*: Syphilis and tuberculosis.
11. *Personal history*: Chewing of *Paan* (betel leaf), *Sopari* (betel nut), tobacco, slaked lime, smoking, alcohol, Chutta cancer of hard palate (smoking of reverse cigars with burning end in mouth), Khaini or Masala cancer (mixture of lime and tobacco kept in vestibule of mouth).
12. *Family history*: Tuberculosis, HIV/AIDS, Von Recklinghausen's disease and malignant tumors.

Examination of Swelling

Examination of a swelling mainly consists of inspection and palpation and assess following components:

1. *Site and shape*: The common sites of dermoid cysts are line of fusion of embryonic processes and include outer canthus of eye (between frontonasal process and maxillary process), behind the ear (between mesodermal hillocks) (Fig. 6). Shape may be round, oval, or irregular.

Fig. 6: Cystic swelling in the upper part of retroauricular sulcus

2. *Size:* Size in centimeters is measured in both horizontal and vertical dimensions.
3. *Extent*: The medial-lateral, anterior-posterior and superior-inferior extent must be described using anatomical landmarks.
4. *Color*: Black (nevus and melanoma), red (arterial hemangioma), purple (venous hemangioma), bluish (ranula).
5. *Surface*: Cauliflower (squamous cell carcinoma), irregular multiple branching (papilloma), smooth (cyst) (Fig. 7), lobular with smooth bumps (lipoma), nodular (matted lymph nodes, nodular goiter), irregular and rough (carcinoma).
6. *Skin over swelling*: Red and edematous (inflammation) (Figs 8 and 9), black punctum (sebaceous cyst), pigmentation (moles, nevi, after irradiation), peel of an orange "Peau d' orange" (carcinoma), scar (previous operation, injury, suppuration) and ulceration (fungation of metastatic nodes) (Fig. 5).
    a. Fixity to skin: These swellings include skin swellings (papilloma, epithelioma, sebaceous cyst) and subjacent malignant swelling involving skin such as secondary metastatic neck nodes and carcinomas of oral cavity, larynx, laryngopharynx, thyroid, and salivary glands.
7. *Edge*: Well-defined (neoplasms), irregular (malignant), smooth (benign neoplasms and cysts), diffuse or indistinct (acute inflammation) (Fig. 8).
    a. Slip sign: It helps in differentiating between lipoma and cystic swelling. On palpation of the edge of swelling cyst margin slips away whereas lipoma margin yields.
8. *Nature:* Pedunculated (papilloma) or sessile.
9. *Number*: Lipoma and dermoid cysts are usually solitary. Neurofibromatosis and Hodgkin's disease have multiple swellings. Inflammatory lesions such as herpetic, tubercular and syphilitic can be multiple.
10. *Pulsation*: Pulsatile (aneurysm, carotid body tumor) and nonpulsatile.
11. *Impulse on coughing/crying*: It is present in swellings which are in continuity with cranial cavity or spinal canal such as meningocele.
12. *Movement on deglutition*: It is present in swellings which are fixed to larynx and trachea such as goiter (Fig. 10), thyroglossal cyst, subhyoid bursitis, enlarged pre- and paratracheal lymph nodes.
13. *Movement on protrusion of tongue*: Thyroglossal cyst moves up on protrusion of tongue (Fig. 11).
14. *Pressure effects*: Neck venous engorgement (retrosternal extension of goiter) and facial palsy (parotid malignancy).

Fig. 7: Congenital frontal cyst

Fig. 8: Furuncle nose. Note the diffuse (indistinct edge) red color tender swelling of tip of the nose

Fig. 9: Left side cold abscess in the parotid region and upper part of neck. It is extending from the angle of mandible to the mastoid and elevating the ear lobule

Fig. 10: Thyroid nodule left side. The swelling moved with deglutition

Fig. 12: Eliciting fluctuation and fluid thrill in a right side nasolabial cyst

Fig. 11: Thyroglossal cyst. The swelling moved up on protruding out of tongue

Fig. 13: Eliciting fluctuation and fluid thrill in a left side parotid abscess

15. *Temperature and tenderness:* They indicate inflammation.
16. *Consistency*: Soft (lipoma), cystic (cyst, hematoma, abscess), firm (fibroma), hard (bony, cartilaginous and malignant tumors), variable (malignancy), crepitus (surgical emphysema), molded due to inside putty-like (pultaceous) material (sebaceous and dermoid cysts), pits on pressure (inflammatory).
17. *Fluctuation and fluid thrill*: It is present in cystic swellings such as cyst (Fig. 12), hematoma, and abscess (Fig. 13). False fluctuation may be elicited in lipoma, myxoma and soft fibroma.
18. *Translucent*: Translucency is present in the cysts that contain clear fluid such as cystic hygroma and laryngocele.
19. *Reducible*: Meningocele completely disappears on pressure and reappears after coughing.
20. *Compressible*: Compressible swellings include hemangiomas and lymphangiomas.
21. *Pulsatile*: Pulsatile swellings include aneurysms (expansile pulsation) and carotid body tumor (transmitted pulsation).
22. *Depth of swelling*: Swellings deeper to sternocleidomastoid may become inconspicuous when the patient is asked to rotate his/her head in opposite direction against resistance whereas superficial swellings would become more prominent. If the tumor is fixed to prevertebral fascia of deeper muscles tumor may be moved sideways but cannot be moved in the axis of muscles. Similarly tumors of vessels and nerves may be moved side ways but cannot move in the line of particular vessel or nerve. Bony tumors are fixed.
23. *Bruit and murmurs*: These are the features of vascular swellings.
24. *Lymphatic drainage*: The draining lymph nodes must invariably be palpated. If the swelling is at the levels of neck lymph nodes (Fig. 14) then the drainage area of that particular group of lymph nodes must be examined to know the primary site of infection and/or carcinoma.

Examination of Ulcers

It mainly consists of inspection and palpation and assesses following components:

1. *Site:* Rodent ulcers (Fig. 15) occur in the upper part of face above the level of angle of mouth. Tuberculous ulcers are common in neck. Lupus, a cutaneous tubercular ulcer is common on face. Malignant ulcers are common in oral cavity, pharynx and larynx.

Fig. 14: Secondary metastatic neck nodes on right side involving levels Ib to V. Primary lesion was carcinoma of supraglottic larynx. Note the healed tracheostomy opening and widening of thyroid cartilage

Fig. 15: An early rodent ulcer (basal cell carcinoma) of nasomaxillary skin
Source: Dr Amit Goyal, Shillong

2. *Shape:* Shape may be round (Fig. 16), oval, or irregular. Tubercular ulcers are oval shape to start with but later on coalesce and give irregular crescent edge. Syphilitic ulcers also unite and give Serpiginous shape. Malignant ulcers have irregular shape (Fig. 17).
3. *Number*: Inflammatory lesions such as herpetic, tubercular and syphilitic can be multiple.
4. *Size:* Size in centimeters is measured in both horizontal and vertical dimensions.
5. *Extent*: The medial-lateral, anterior-posterior, and superior-inferior extent must be described using anatomical landmarks.
6. *Edge*: The edge may be of the following types:
    a. Inflamed and edematous: Spreading ulcer
    b. Blue and whitish zones: Healing ulcer

Fig. 16: Aphthous ulcer on uvula of soft palate. Note the round shape and sharp erythematous margin

Fig. 17: Carcinoma hard palate left side crossing midline and encroaching opposite side palate. Note the irregular and cauliflower lesion. Note the separate lesion involving uvula and soft palate with normal looking in between intermittent mucosa

    c. Undermined edge: Tuberculosis
    d. Punched out edge: Gummatous and deep trophic ulcers
    e. Sloping edge: Healing ulcers
    f. Raised and beaded edge: Rodent ulcers
    g. Rolled out or everted edge: Squamous cell carcinoma (Fig. 18) and ulcerated adenocarcinoma
    h. Induration: Hardness is the feature of malignancy
7. *Floor*: The exposed surface of ulcer may be covered with following structures:
    a. Red granulation tissue: Healthy and healing ulcer
    b. Pale and smooth granulation tissue: Slowly healing ulcer
    c. Wash leather slough: Gummatous ulcer
8. *Discharge*: It may be of following types:
    a. Scanty serous: Healing ulcer
    b. Purulent: Spreading and inflamed ulcer
    c. Sero-sanguineous: Tubercular and malignant ulcers
    d. Greenish: *Bacillus pyocyaneus* infection.
9. *Surrounding area*: It may have following features:

Fig. 18: Carcinoma anterior two-third tongue left lateral border

Fig. 20: Healed herpes lesions. Note the involvement of the lower third of facial skin supplied by the mandibular division of trigeminal nerve. The skin of angle of mandible is not involved

   a. Glossy red and edematous: Acute inflammation
   b. Scar or wrinkling skin (Fig. 19): Tuberculosis
   c. Healed herpes lesions (Fig. 20)

10. *Tenderness*: Severe tenderness suggests acute inflammation. Mild tenderness may be seen in chronic infections such as tuberculosis and syphilis. Carcinomatous ulcers are not tender to begin with but can become tender later.
11. *Base:* The marked induration (hardness) of the base on which floor of the ulcer rests is the feature of malignancy.
12. *Depth*: The depth of the indurated base must be recorded in millimeters. Trophic ulcers may reach up to the bone.
13. *Bleeding*: Bleeding on touch is the feature of malignant and infected ulcers.
14. *Fixity with deeper structures*: It is the feature of malignant ulcers and deep lying osteomyelitis.
15. *Surface*: Cauliflower (squamous cell carcinoma), irregular multiple branching (papilloma), irregular and rough (carcinoma) (Fig. 18).
16. *Surrounding skin*: Red edematous tender and warm skin indicates inflammation. Peel of an orange "Peau d'orange" suggest carcinoma, scar (previous operation, injury, suppuration), and ulceration (fungation of metastatic nodes).
17. *Lymphatic drainage*: The draining lymph nodes must invariably be palpated. The enlarged tender regional lymph nodes suggest acutely inflamed ulcer. Tubercular ulcers may cause matted lymphadenopathy of regional lymph nodes. In rodent ulcers lymph nodes are not involved because of the early obliteration of the lymphatics by the malignant cells. In other carcinomas patient may have hard regional lymph nodes. In cases of malignancy regional lymph nodes may be enlarged due to secondary infection. Therefore in cases of suspicion fine needle aspiration cytology (FNAC) or frozen section is required.

Fig. 19: Healed tubercular lesion (ruptured cold abscess)

EXAMINATION OF CRANIAL NERVES

Table 4 shows the components, muscles innervated, nuclei with their site, symptoms and tests of all the twelve cranial nerves.

HEADACHE

Headache is one of the commonest symptoms of medicine. In addition to the long list of medical causes, there is wide range of ear, nose and throat diseases that can cause headache.

Definition

Headache refers to the pain or discomfort between the orbits and occiput and arises from pain-sensitive structures (Table 5).

Etiology

Headache can arise from psychological, otolaryngological, ophthalmological, neurological, dental and systemic diseases (Table 6). Tension type headache is most common.

Some of the common causes are as follows:

1. *Raised intracranial tension*
   a. Space occupying lesions: Cerebellopontine angles (CPA) tumors such as acoustic neuroma, pituitary tumors, otogenic brain abscesses (cerebellar and temporal lobe).

TABLE 4 Nuclei and tests of cranial nerves*

Cranial nerves and muscles' innervated	*Nuclei*	*Site*	*Components and symptoms*	*Tests*
I Olfactory	Olfactory bulb and tract > primary and secondary olfactory cortex**		Special sensation; Anosmia	Smell
II Optic	Optic chiasma and tracts > lateral geniculate body and visual cortex**		Special sensation; Blindness	Visual acuity, field, fundus, light reflex
III Oculomotor All extraocular muscles (except lateral rectus and superior oblique), sphincter pupil and ciliary muscles (Fig. 25)	Oculomotor	Midbrain	Motor; Ptosis (Fig. 21), diplopia, squint	Medial (Fig. 22), upward (Fig. 23) and downward (Fig. 24) rotations of eye ball
	Edinger-Westphal		Parasympathetic	Light and accommodation reflexes; dilated pupil
IV Trochlear Superior oblique	Trochlear	Midbrain	Motor; Diplopia, squint	Looking down and medially
V Trigeminal All the muscles of mastication (temporalis, masseter, medial and lateral pterygoids), tensor palati and tympani; mylohyoid and anterior belly of digastric	Motor	Pons	Motor; Difficulty in chewing	Jaw movements (deviates ipsilaterally) & jerk
	Sensory		Sensory; Facial numbness	Touch and pressure of face; conjunctival and corneal reflexes
	Spinal			Pain and temperature of face
VI Abducent Lateral rectus	Abducent	Pons	Motor; Diplopia, squint	Lateral rotation of eye ball (Figs 25 and 26)
VII Facial All the muscles of facial expression (except levator palpabri superioris), auricular, posterior belly of digastric, stylohyoid and stapedius	Motor	Pons	Motor; Facial palsy	Facial asymmetry, inability to close eye (Fig. 27)
	Tractus solitarius		Special sensation; Loss of taste	Taste of anterior 2/3 tongue
	Superior salivatory		Parasympathetic	Submandibular salivary flow
	Lacrimatory		Parasympathetic; dryness of eye	Lacrimation
VIII Auditory	Two cochlear	Junction of pons and medulla	Special sensation; Hearing loss	Sensorineural hearing loss (Hearing tests)
	Four vestibular		Vertigo	Jerky nystagmus
IX Glossopharyngeal Stylopharyngeus	Nucleus ambiguus	Medulla	Motor; Dysphagia	Stylopharyngeus
	Tractus solitarius		Sensory	Gag reflex (sensory)
	Inferior salivatory		Parasympathetic	Parotid salivary flow
X Vagus	Nucleus ambiguus	Medulla	Motor: Nasal regurgitation, dysphagia, hoarseness of voice; dysarthria	Gag reflex (motor); Uvula deviation to opposite side; vocal cord palsy
XI Accessory–cranial part Muscles of soft palate, pharynx & larynx except stylopharyngeus and tensor palati		Medulla		
XI Accessory–spinal part (Trapezius and sternocleidomastoid)	Spinal	Spinal cord	Motor	Elevation of shoulder and turning of head
XII Hypoglossal (All the muscles of tongue except palatoglossus)	Hypoglossal	Medulla	Motor; Dysarthria, dysphagia	Tongue wasting, fasciculation, protrusion (ipsilateral deviation)

*Proprioceptive mesencephalic nucleus of trigeminal nerve, taste fibers of IX and X are not shown as they are not clinically tested; ** Olfactory and optic nerves do not have nuclei.

TABLE 5 Pain sensitive structures for headache

Intracranial	Venous sinuses, cortical veins, basal arteries, dura of base of skull
Extracranial	Vessels and muscles of scalp, orbital contents, mucosa of nose and paranasal sinuses, external and middle ear, teeth and gums

TABLE 6 International classification of headache type (1988)

Type	*Percentage*
Tension type headache	45
Migraine	30
Eye/sinus diseases	< 8
Systemic infections	7
Head injury	3
Drugs	2
Cerebrovascular diseases	< 1
Other intracranial diseases	< 1
Metabolic diseases	< 1
Cluster headache	< 1
Neuralgias	< 1

Fig. 21: Left oculomotor palsy. Note: Left eye ptosis due to the paralysis of levator palpabri superioris

Fig. 22: Left oculomotor nerve palsy. Left eye cannot be rotated medially (adduction) due to medial rectus palsy

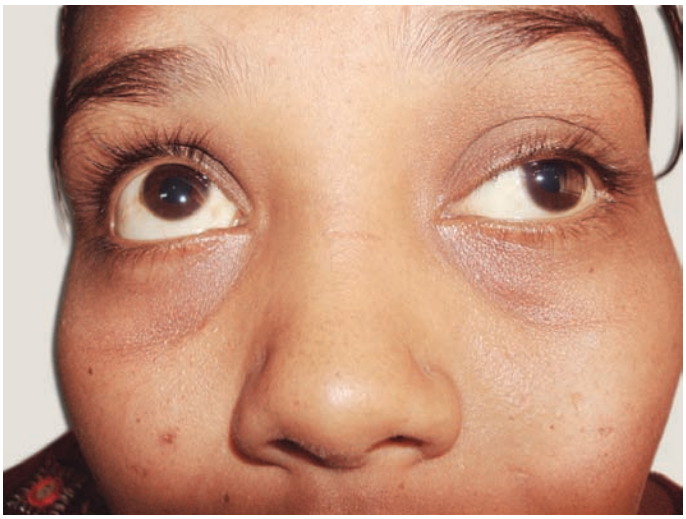


Fig. 23: Left oculomotor nerve palsy. Left eye cannot be rotated upward due to superior rectus palsy

Fig. 24: Left oculomotor nerve palsy. Left eye cannot be rotated downward due to inferior rectus palsy

Fig. 25: Left oculomotor nerve palsy. Note the ptosis and left eye can be rotated laterally (abduction) because the lateral rectus is innervated by the abducent nerve

Fig. 26: Left abducent nerve (CN VI) palsy. Left eye is not rotating laterally (abduction) due to left lateral rectus palsy. Note right eye rotates medially

Fig. 27: Left facial nerve palsy. Note patient cannot close his left eye and move left angle of mouth laterally

   b. Infections: Encephalitis and meningitis caused by suppurative otitis media and sinusitis.
   c. Intracranial venous sinuses: Otogenic sigmoid sinus thrombophlebitis and rhinogenic cavernous sinus thrombophlebitis block the outflow of intracranial venous blood.
   d. Hypertension: Severe arterial hypertension.
2. *Intracranial aneurysms*: They can put pressure and involve cranial nerves. Rupture will result in subarachnoid hemorrhage.
3. *Migraine* (See chapter Central Vestibular Disorders).
4. *Temporalis muscle*:
   a. Grinding teeth at night
   b. Impacted wisdom teeth
   c. High bite
   d. Temporomandibular joint disorders
   e. Clenching of jaws
5. *Cervical spondylosis*: Prolapse of cervical intervertebral disc.
6. *Temporal arteritis*.
7. *Tension headache*: Anxiety or depression symptoms are often present.
8. *ENT causes*:
   a. Nose and paranasal sinuses: Rhinosinusitis, deviated nasal septum (touching middle turbinate), vacuum headache (negative pressure in sinuses) and tumors
   b. Nasopharynx: Adenoiditis, infection, cysts and tumors.
   c. Oral cavity: Infections, ulcers and malignancy.
   d. Oropharynx: Peritonsillar abscess, parapharyngeal abscess, retropharyngeal abscess and tumors.
   e. Larynx and laryngopharynx: Malignant tumors.
   f. Ear: Intracranial complications of suppurative otitis media and tumors.
9. *General*: Headache is common constitutional symptoms in cases of fevers.

Headache in children: The commonest causes are febrile illness and rhinosinusitis. In children with fever and neck stiffness, consider meningitis, encephalitis and cerebral abscess.

History and Examination

1. *Histor***y**: Inquire about the following information
   a. First or the previous attacks.
   b. Onset is acute or gradual (days or weeks).
   c. Duration and frequency: Know whether chronic or recurrent.
   d. Site of headache.
   e. Accompanying features.
   f. Factors: Aggravating, relieving and precipitating.
2. *Examination*: In addition to the general and ENT head and neck examination one should perform ocular (acuity, tenderness, strabismus, tension, fundus), psychological and neurological examination.
3. *Characteristic features*: See Table 7 for the common causes of headache and their characteristic features.
   a. Migraine: See chapter Central Vestibular Disorders.
   b. Raised intracranial pressure: Headache is generalized and aggravated by bending and coughing. Headache is worse in morning on awakening. It may awaken patient from sleep. The severity gradually progresses. In later stages it is associated with vomiting and transient loss of vision with sudden change in posture.

Tension-type Headache

This is the most common type of primary headache. Exploration of the underlying cause of chronic anxiety is important.

- *Clinical features*: Headache is nonpulsatile, diffuse, dull, aching and band-like. Headache is usually more in occipital and cervical region. Patients often complain of poor concentration and other vague nonspecific symptoms.
- *Aggravating factors*: Headache is constant daily and may be exacerbated by emotional stress, fatigue, noise or glare. It gets worse on touching scalp (pericranial tenderness).
- *Treatment:* Antimigrainous agents are prescribed when simple analgesics are not effective.
  - Relaxation techniques: The massage, hot baths and biofeedback are helpful.
  - Botulinum toxin type A: Local injection may be useful.

TABLE 7 Common causes of headache and their features

Cause	*Acute (A), Subacute (S), Chronic (C), Recurrent (R)*	*Accompanying diagnostic features*	*Investigations*
Tension type headache	C and R	Anxiety, depression	
Migraine	A and R	Visual/neurological aura, nausea, vomiting	
Ocular eye strain	C and R	Impaired visual acuity	Refractive error
Glaucoma	A and R	Misting of vision, haloes around objects	Intraocular tension
Retrobulbar neuritis	A	Unilateral blindness	Visual evoke response
Sinusitis	A, R	Cold and purulent nasal discharge	CT scan PNS
Post-traumatic	A	Head injury	CT scan head
Drugs	A and C	Vasodilator drugs	
Benign intracranial hypertension	S and R	Vomiting, papilloedema	CSF pressure monitoring
Intracranial hemorrhage	A (Instant onset)	Vomiting, neck rigidity, impaired consciousness	CT scan brain
Meningitis and encephalitis	A and S (gradual onset)	Vomiting, neck rigidity, impaired consciousness, fever	
Intracranial tumor and chronic subdural hematoma	S and R	Vomiting, papilloedema, impaired consciousness, focal features	MRI/CT Brain
Cluster headache	A and R	Lacrimation, rhinorrhea	
Temporal arteritis	S	Thickened, tender scalp arteries, jaw claudication	ESR, temporal artery biopsy
Cervical spondylosis	C and R	Pain in neck, shoulder and arm	X-ray cervical spine

Depression Headache

- *Clinical features*: Headache is usually worse on arising in the morning. Patients may have other symptoms of depression and somatic delusions.
- *Treatment:* It consists of psychiatric consultation and antidepressant drugs.

Cluster Headache

It is mainly the disorder of middle-aged men.

- *Clinical features:* Episodes of severe unilateral pain around one eye occur daily (more in night) for 4–8 weeks. Episodes awaken the sleeping patient and last for 15 minutes to 3 hours. It is associated with ipsilateral conjunctival congestion (red eye), lacrimation, nasal congestion and rhinorrhea. Occasionally, patient may develop Horner's syndrome, which may be transient or longstanding. Spontaneous remission may last for weeks or months. In cases of chronic cluster headache there is no period of remission.
- *Triggers*: They include alcohol, stress, glare, or specific foods.
- *Treatment of acute episode:* Oral drugs are usually not effective. Following measures are often effective.
  - Sumatriptan: Subcutaneous (6 mg) or intranasal (20 mg/spray).
  - Oxygen: Inhalation of 100% oxygen (12–15 L/min for 15 minutes) via a non-rebreather mask.
  - Zolmitriptan: 5–10 mg nasal spray.
  - Dihydroergotamine: 0.5–1 mg IM or IV.
  - Xylocaine viscous: 1 mg of 4–6% solution intranasally.
- *Prophylactic agents*: They include cyproheptadine, lithium carbonate, verapamil (240–960 mg daily), topiramate (100–400 mg daily) and methysergide (2–12 mg daily).
- *Transitional therapy*: Prophylactic agents are not immediately effective therefore following drugs are used in transitional therapy.
  - Ergotamine tartarate: Rectal suppository (0.5–1 mg) at night or twice daily, 2 mg daily orally, or 0.25 mg subcutaneously 3 times daily for 5 days per week.
  - Prednisone**:** 60 mg daily for 5 days followed by gradual withdrawal.
  - Dihydroergotamine**:** 9.25 mg IV over several days or 0.5 mg IM twice daily.
  - Xylocaine and corticosteroid: Greater occipital nerve block with local injection of xylocaine and corticosteroid.

Post-Traumatic Headache

- *Clinical features*: This constant dull headache develops within a day or so following closed head injury, may worsen over weeks and then gradually subsides. Throbbing sensation may be localized, lateralized, or generalized. Headache developing after 2 weeks of injury is usually not related to head injury.
  - Associated features: Patients may have associated nausea, vomiting, or scintillating scotomas. The headache is associated with lightheadedness, irritability,

difficulty in concentration and in coping with work. Other associated complaints which may be present include impaired memory, emotional instability and increased irritability.
  - Vertigo: The associated disequilibrium may have rotatory component and aggravated by postural change or head movement.
- *Investigations*: They are usually not helpful. CT/MRI scans are usually normal. EEG may show nonspecific changes. ENG may suggest peripheral or central vestibulopathy.
- *Treatment*: Reassurance with graduated rehabilitation is important. If simple analgesics fail then amitriptyline, antiseizure drugs, propranolol, or ergot derivatives are used.

Cough Headache

This is usually a self-limiting disease. In some patients it may persist for several years.

- *Clinical features*: This transient (few minutes or less) but severe headache is produced by coughing, straining, sneezing and laughing.
- *CT/MRI scans*: They should be done in all patients to rule out space occupying intracranial lesions and Arnold-Chiari malformation. CT/MRI may be repeated annually as a small lesion may not show initially.
- *Treatment*: Lumbar puncture may alleviate the symptoms completely. Indomethacin (75–150 mg daily) is effective in some patients.

Giant Cell (Temporal or Cranial) Arteritis

In addition to the superficial temporal, other arteries which are often affected are vertebral, ophthalmic and posterior ciliary arteries. Early diagnosis and treatment can prevent blindness which is usually permanent.

- *Clinical features*: This unilateral severe throbbing headache occurs in elderly people. The headache is often associated or preceded by myalgia, malaise, anorexia and weight loss. The superficial temporal artery becomes thickened, tender and nonpulsatile. Jaw claudication (pain during chewing and talking) is pathognomonic. Many patients present with blindness (transient or permanent) and diplopia. Some patients may develop stroke, hearing loss, myelopathy and neuropathy.
- *Biopsy:* Prompt temporal artery biopsy (at least 2 cm in length) is important. Biopsy shows lymphocytes, histiocytes, plasma cells and giant cells in media and adventitia.
- *Treatment*: On clinical suspicion immediately initiate prednisone (60 mg/day for 1 month before tapering). In cases of blindness intravenous pulse Methylprednisolone (1 gm daily for 3 days) is given. When tapering prednisone, erythrocyte sedimentation rate (ESR) or C-reactive protein is a useful but not absolute guide.
  - Aspirin: Low dose aspirin (up to 81 mg/day orally) may prevent blindness and stroke.
  - Methotrexate: It is modestly effective.
  - Anti-tumor necrosis factor (TNF) therapies: They are not effective.
- *Complications:* Thoracic aortic aneurysm can result in aortic regurgitation, dissection and rupture.

FACIAL PAIN

Facial pain can occur due to many different disorders of nose and paranasal sinuses, teeth and gums, eyes and orbits, temporomandibular joints, oral cavity and salivary glands, jaws and cranial nerves.

Trigeminal Neuralgia (Tic Douloureux)

It is most common in middle and later life. Women are more affected than men.

- *Clinical features*: Momentary episodes of sudden sharp, stabbing and shooting facial pain occur in the region of one or more divisions (ophthalmic, maxillary and mandibular) of the trigeminal nerve. Pain may radiate towards ear, eye, mouth or nostril. Precipitating factors include chewing, speaking, face washing, tooth-brushing, cold winds, or touching of trigger spots such as upper lip or gum. Spontaneous remissions for several months or longer are not uncommon. Neurological examination is normal.
- *Differential diagnosis*: In a young patient multiple sclerosis must be suspected. Evoked potential testing and CSF examination may be corroborative. CT/MRI scan is done to rule out posterior cranial fossa tumor.
- *Treatment*
  - Carbamazepine (600–1200 mg/day in 3–4 divided doses): It is usually very effective. They need monitoring of CBC and liver function tests. Oxcarbazepine also offers good results.
  - Phenytoin (200–400 mg/day): It is used when Carbamazepine or oxcarbazepine are not effective or not tolerated.
  - Baclofen (10–20 mg 3 or 4 times a day): It may be used either alone in combination with above molecules.
  - Gabapentine (900–2400 mg/day in 3 divided doses) or pregabalin (up to 300 mg/day): It is used when above medicines do not work or patient is suffering from multiple sclerosis.
  - Noninvasive procedures:
    - Radiofrequency rhizotomy: This simple procedure is preferred in elderly patients with a limited life expectancy.
    - Trigeminal root gamma radiosurgery: This noninvasive procedure is successful in 80% of patients.
  - Exploration of posterior cranial fossa: Simple decompression and separation of anomalous vessel from the nerve root usually produce lasting relief. The anomalous artery or vein impinging on trigeminal nerve root is usually not seen on CT, MRI or arteriograms. It is not indicted in patients with multiple sclerosis.

Atypical Facial Pain

- *Clinical features:* Diffuse, dull, persisting pain occurs in depressive young or middle aged woman. It may be unilateral or bilateral. Initially the burning pain has restricted distribution but soon spreads to rest of the face. In some patients pain may involve other side, neck or occipital region.

- *Treatment:* Patients may be given analgesics, tricyclic antidepressants, carbamazepine, oxcarbazepine or phenytoin but the response is often not satisfactory.

Glossopharyngeal Neuralgia

This disorder is uncommon.

- *Etiology*: Usually no structural abnormality is found. Multiple sclerosis may be the cause in some patients.
- *Clinical features*: This neuralgic pain (similar in quality to trigeminal neuralgia) occurs usually in tonsillar fossa. It may occur deep in the ear or at base of the tongue. The precipitating factors may be swallowing, chewing, talking, or yawning. It may be accompanied by syncope.
- *Treatment*: Pharmacological treatment is similar to trigeminal neuralgia and should be given a trial. Microvascular decompression is indicated when medical treatment fails.

Post-herpetic Neuralgia

About 15% patients of shingles (Herpes zoster) develop postherpetic neuralgia.

- *Risk factors*: They include old age, severe rash (long duration, painful and scarring), involvement of ophthalmic division of trigeminal nerve and delayed acyclovir therapy.
- *Clinical features*: Dull, burning, persisting (occasionally paroxysm) pain occurs after herpetic lesions. It affects the trigeminal area especially ophthalmic division. Healed lesion with scars is associated with sensory loss.
- *Treatment*: It is essentially medical. In addition to the medical treatment (carbamazepine, phenytoin, gabapentine and pregabalin) mentioned in the trigeminal neuralgia it includes the following:
  - Amitriptyline and perphenazine: If simple analgesics are not effective tricyclic antidepressant (amitriptyline up to 100–150 mg/day) in conjunction with phenothiazine (perphenazine 2–8 mg/day) is usually effective.
  - Gabapentin and morphine: The combination of gabapentin and morphine taken orally is more effective than taken individually.
  - Topical applications: Capsaicin cream (0.025%) and lidocaine (5%) are worthy of trial.
- *Prevention*: Live-attenuated zoster vaccine to elderly patients (> 60 years) prevents occurrence of herpes zoster and markedly reduces morbidity.

TEMPOROMANDIBULAR (CRANIOMANDIBULAR) DISORDERS

TMD consists of not only the internal derangement of temporomandibular joint (TMJ) but also areas extrinsic to TMJ. Temporomandibular (craniomandibular) disorders (TMD) patients present with musculoskeletal symptoms such as diffuse facial pain with jaw movements, limitation of mandibular movement and masticatory muscle and TMJ tenderness. Patients with internal derangement of TMJ (disc displacement, osteoarthrosis, inflammation and congenital, developmental, traumatic and neoplastic disorders) present with well localized pain.

Etiology

The causes of TMD and TMJ disorders are shown in Table 8. They occur due to the misalignment of one TMJ. TMD may be caused by malocclusion, abnormal bite, or faulty dentures.

Clinical Features

Patients usually have dental loss with altered bite. Other features are following:

- Aching pain occurs around the ear that is aggravated by chewing.
- Limitation in jaw movements.
- Deviation in mandibular motion.
- Locking of jaw in open or closed position.
- Clicking sound/crepitus within TMJ.
- Masticatory muscles pain and tenderness.

Myogenic (Masticatory Muscles) Disorders

Temporomandibular disorder patients with myogenic pain usually have normal TMJ. The limitation of jaw movement is due to muscle pain and stiffness. The pain is usually due to hyperfunction. Myogenic disorders include myofascial pain, fibrositis, muscle splinting (trismus), spasm and swelling (myositis), contracture, bruxism, hypertrophy and dyskinesia. There is usually no evidence (clinical and radiographic) of internal derangement of TMJ.

- *Predisposing factors*: Nocturnal or diurnal bruxism (grinding or clenching) is very common. It is more during physical or emotional stress. This hyperactivity results in muscle fatigue and spasm. This strain which is due to overuse or improper use must be treated as such.
- *Clinical features*: The most common complaint of the patients is diffuse facial pain. According to the involvement of muscles, patients with myogenic disorders may have following symptoms.

TABLE 8 Common causes of temporomandibular disorders

TMJ disorders (internal derangement)	Displacement, dysfunction and perforation of disc, adhesions and degenerative joint disease
Myogenic (masticatory muscles) disorders	Muscle trismus or splinting
Chronic mandibular hypomobility	Muscular and capsular fibrosis, bony ankylosis and fusion, adhesions and elongation of coronoid process
Chronic mandibular hypermobility	Subluxation and dislocation
Inflammatory disorders	Synovitis, capsulitis, hyperuricemia and degenerative, infectious, traumatic and rheumatoid arthritis

- Masseter muscle: It is most commonly involved and produces jaw pain.
- Temporalis muscle: This is the second most commonly involved muscle and produces headache.
- Medial pterygoid: It produces odynophagia. Patient develops sensation of painful swollen gland just beneath the angle of mandible.
- Lateral pterygoid: It produces referred ear pain or retrobulbar pain.

- *Differential diagnosis*: In jaw (masticatory) claudication, pain develops progressively with mastication. It is a feature of giant cell arteritis.

Meniscal Displacement

Disk displacement can be a cause or result of osteoarthritis. Both the disorders can be present independently. About 20% of the population may have some form of disk displacement without pain or dysfunction.

Imaging Studies

1. *OPG:* Panoramic view shows only gross disease (degenerative and ankylosis) and general joint configuration.
2. *CT scan*: It is ideal for fine osseous and fibrous evaluation.
3. *MRI:* It is ideal for evaluating disk morphology and position and early degenerative bone changes.

Treatment

It consists of the treatment of the underlying TMJ disorder. Most TMD patients respond to nonsurgical therapy that includes rest, and medical, physical and splint therapies. Surgical treatment is indicated only for internal derangement of the TMJ.

- *Restricted mouth opening and soft diet*: Limitation of function reduces painful loads on the inflammatory tissues.
- *Analgesics*: Non-narcotic analgesics (acetaminophen) and nonsteroidal anti-inflammatory drugs (NSAIDs) constitute the initial therapy. When a particular NSAID is not effective then a different one should be used. Long-term or high doses of NSAIDs may cause gastrointestinal bleeding, ulcers, or nephrotoxicity. The commonly used NSAIDs include ibuprofen (900–2400 mg/day in divided doses), diclofenac (50 mg bd/tds), naproxen (275–550 mg/day), ketoprofen (75 mg/day), piroxicam (20 mg/day), celecoxib (100–200 mg bd) and valdecoxib (10 mg/day).
- *Muscle relaxants*: Centrally acting agents are usually prescribed. They have general sedating effect on the central nervous system (CNS). The commonly used agents include methocarbamol (1500–3000 mg/day in divided doses), tizanidine (2–4 mg tds), carisoprodol (100–1400 mg/day in divided doses), cyclobenzaprine (5–30 mg/day) and metaxalone (800 mg tds).
- *Anxiolytics*: Benzodiazepines are sedatives and reduce anxiety and muscle hyperactivity. The commonly used agents include clonazepam (0.5–1 mg 1–4 times/day), diazepam (2–10 mg 1–4 times/day) and alprazolam (0.5 mg 1–3 times/day).
- *Antidepressants*: Tricyclic and serotonin-reuptake inhibitors are useful in the management of chronic pain. The commonly used agents include amitriptyline (10–25 mg/day), imipramine (10–25 mg/day), fluoxetine (5 mg/day), sertraline (50 mg/day), citalopram (20 mg/day), venlafaxine (75 mg/day), nortriptyline (10–25 mg/day), bupropion (100 mg bd) and paroxetine (20 mg/day).
- *Narcotics*: They are used in chronic pain patients. The caution is required as they are habit forming agents.
- *Bite-appliance (splint) therapy*: The splint is worn on the teeth and stabilizes the occlusion (in bruxism) and reduces the load. It relieves acute myogenic and TMJ pain.
- *Other therapies*: Physical therapy, behavioral training and stress management are effective in many patients.
- *Surgery*: Surgical correction is indicated only when internal derangement symptomatic TMJ patients do not respond to above mentioned modalities of treatment. The details of surgical options, which include arthrocentesis, arthroscopy, arthrotomy (arthroplasty) and meniscal repair (plication), are beyond the scope of this chapter.

Clinical Highlights

1. **ENT examination:** Keep a specific order with clinical examination and within each area such as ear, nose, mouth, pharynx and larynx, face and scalp, neck and neurologic exam. It ensures complete examination and avoids missing anything.
2. **Inadvertent injury**: When examining ear, nose and throat, your hand that is holding the instrument should be stabilized against the patient. This practice prevents inadvertent injury upon sudden patient movement.
3. **Assurance**: Before examination always describe to the patient what you intend to do. Assure the patient that the procedure will not hurt him/her. Again assure the patient that if anything bothers then s/he should tell you.
4. **Height**: Adjust the patient's chair height. Patient's head should be little higher than the examiner.
5. **Oral and pharyngeal examination:** It also includes transoral palpation. It is important because certain lesions such as submucosal mass/nodule of carcinoma tongue can be felt and not seen.
6. **Headache**: The common causes in women are migraine, tension and trigeminal.
7. **Diurnal headache:** It is a feature of acute frontal sinusitis.
8. **Vacuum headache:** It is associated with frontal sinus.

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10 Otologic Symptoms and Examination

Man begins to struggle and fight against nature. He makes many mistakes, he suffers. But eventually, he conquers nature and realizes his freedom. When he is free, nature becomes his slave.

—Swami Vivekananda

Points of Focus

- **EAR SYMPTOMS**
- **EAR EXAMINATION**
  - ✓ Examination Findings and their Causes
  - ✓ Otoscopy: External auditory canal, Otoscope, Tympanic membrane
  - ✓ Siegelization: Siegel's Pneumatic Speculum
  - ✓ Eustachian Tube: eustachian Catheterization, Valsalva Maneuver
  - ✓ Functional Examination
- **OTALGIA:** Etiology, History and Examination, Management
- **OTORRHEA:** Etiology, History, Physical Examination, Investigations, Treatment
- **EAR POLYP:** Etiology, History and Examination, Investigations, Treatment
- **TINNITUS:** Classification, Etiopathology of Nonpulsatile Subjective Tinnitus, History Physical Examination, Pulsatile Tinnitus, Radiological Imaging, Tinnitus Management Program, Acoustic Therapies, Lifestyle Changes, Surgical Management
- **HYPERACUSIS**
- **CLINICAL HIGHLIGHTS**

EAR SYMPTOMS

The common otologic symptoms are otorrhea, otalgia, hearing loss, vertigo and tinnitus (Table 1). This chapter reviews the diagnoses to be considered when evaluating these symptoms. The pain and discharge indicate inflammatory disorders while auditory disorders present with hearing loss and tinnitus. Vertigo occurs in vestibular lesions. The symptoms of hearing loss, vertigo and facial palsy are discussed in other chapters.

TABLE 1 Ear symptoms

Common	Otorrhea, otalgia, hearing loss, vertigo and tinnitus
Other	Itching, deformities, swelling, facial palsy, injury/ Foreign body (FB)
Associated	Fever, headache, vomiting, convulsions

EAR EXAMINATION

Examination consists of both physical (pinna and the surrounding area, external auditory canal, tympanic membrane (TM), middle ear, mastoid, Eustachian tube, facial nerve and other cranial nerves) and functional examination (hearing and vestibular). See Table 2 for the general format of ear examination, which should be followed for the each side of ear. The table also shows the findings of clinical examination as well as their causes.

The pinna needs inspection as well as palpation of its surfaces and the surrounding area including postauricular sulcus (Figs 1 to 3). The irregularities of the mastoid are "ironed out" in periosteal inflammation.

The external auditory canal can be examined by pulling the pinna upward (in children, downward), backward and laterally while the tragus is pulled forward (Fig. 4). This procedure not only spreads open the meatus but also straightens it and shows tympanic membrane.

TABLE 2 Examination of ears: findings and their causes

Physical Examination

- *Pinna and surrounding area* (Chapter Diseases of External Ear)
  - Size, shape and position: Microtia/macrotia; cauliflower ear; bat ear
  - Swellings: Furuncle/hematoma/neoplasm; perichondritis, Abscesses (Mastoid or zygomatic abscess), lymph nodes
  - Ulcer: Malignancy, trauma, herpes zoster
  - Scar: Trauma/operation (endaural/postaural)
  - Sinus and fistula: Preauricular sinus, postauricular (mastoid) fistula
  - Palpation: Raised temperature and tenderness (perichondritis or abscess); thickness of tissues (perichondritis); fluctuation (hematoma, seroma or abscess). Painful movement of pinna (furunculosis of the external canal)
- *Examination of external auditory canal* (See chapter Diseases of External Ear)
  - Size of meatus: Atresia/narrow/wide
  - Contents of lumen: Wax, debris, discharge, granulations, polyp, and foreign body (FB)
  - Swelling: Furuncle, sagging of posterosuperior area (coalescent mastoiditis), exostosis, neoplasm (benign or malignant)
- *Examination of tympanic membrane (TM)* (See chapter of Otitis Media)
  - Color: Red and congested (acute otitis media), bluish (secretory otitis media or hemotympanum) and a chalky plaque (tympanosclerosis)
  - Retraction: General retraction (tubal occlusion), retraction pockets attic or posterosuperior region/collection of epithelial flakes, deeply retracted, fixed to promontory (adhesive otitis media)
  - Bulging: Acute otitis media, hemotympanum or neoplasm
  - Thickness and transparency: Opaque/thick; translucent/semitransparent/ transparent/very thin
  - Vesicles or bullae: Herpes zoster or myringitis bullosa
  - Perforation (acute or chronic suppurative otitis media): Size (small, medium, subtotal or total in pars tensa), shape (oval, round or kidney), single or multiple (tuberculosis), site, central (Safe chronic suppurative otitis media) or marginal (at the periphery involving the annulus) or attic (in pars flaccida) with cholesteatoma
  - Mobility (siegelization): Mobile (normal), restricted (presence of fluid), fixed (adhesions in the middle ear), or hypermobile (an atrophic segment of tympanic membrane)
- *Examination of middle ear* (through the perforation, transparent TM, exploration):
  - Mucosa: Normal/atrophic/hypertrophic/polypoidal/granulations/in-growth/ flaques of squamous epithelium (cholesteatoma)
  - Contents: Ossicles, growth, FB
  - Normal structures: Oval and round windows, Eustachian tube
- *Examination of mastoid (Chapter complicatory of Suppurative Otitis Media)*
  - Swelling: Abscess or enlarged nodes, furuncle (obliteration of retroauricular groove), fistula (burst abscess), scar (previous operation)
  - Palpation: Smooth (periosteal inflammation/subperiosteal abscess)/irregular/ tenderness (mastoiditis) over antrum (just above and behind the meatus), tip, and between mastoid tip and antrum
- *Examination of Eustachian tube (See chapter Disorders of Eustachian Tube)*
  - Eustachian catheterization
  - Posterior rhinoscopy
  - Valsalva maneuver
- *Examination of facial nerve* (See chapter Facial Nerve Disorders)
  - Lower/upper motor neuron
  - Grade: Complete/partial

Functional Examination

- *Auditory function* (See chapter on Hearing Evaluation)
  - Voice test
  - Tuning fork tests
  - Rinne test
  - Weber test
  - Schwabach test
  - Absolute bone conduction test
  - Audiometry**:** Pure tone, speech and impedance
- *Vestibular function* (See chapter on Evaluation of Vertigo)
  - Spontaneous nystagmus
  - Fistula test
  - Positional tests

Otoscopy

For the OPD ear instruments, see chapter on Instruments.

- *Aural speculum*: This is inserted into the cartilaginous portion of the external auditory canal (EAC) after retracting the pinna (Fig. 5). It is used for examination and operations of the EAC, TM, and middle ear. The use of the largest ear speculum that can easily enter the canal is safe and provides a better view.

Ear speculum should not reach into the bony part of the EAC because that is painful and patient will not allow.

- *External auditory canal*: Look for the abnormalities of walls of EAC and its contents such as discharge, polyp or any other mass (Fig. 6).

Fig. 1: Postauricular dermatitis

Fig. 4: Pulling the pinna upwards, backwards and laterally and tragus forwards spread and straighten external auditory canal (EAC)

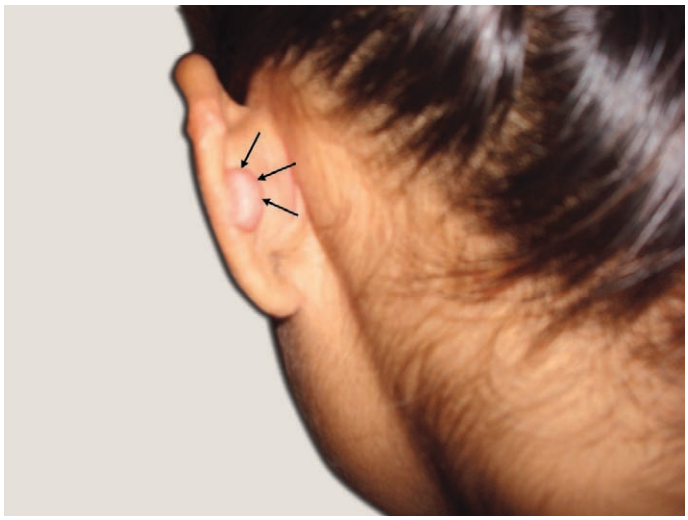


Fig. 2: Keloid. Medial surface of left pinna after ear piercing

Fig. 5: Ear speculum insertion. With left hand external auditory canal (EAC) is straightened and with right hand ear speculum is inserted in the cartilaginous part of EAC

Fig. 3: Cystic swelling in the upper part of retroauricular sulcus

Fig. 6: Nevus of external auditory canal
Source: Dr Amit Goyal, Shillong.

Otoscope

It has its own illumination and magnification and facilitates examination of external auditory canal and tympanic membrane. It is especially useful in examining the ears and nose of infants and bedridden patients (Figs 7A and B).

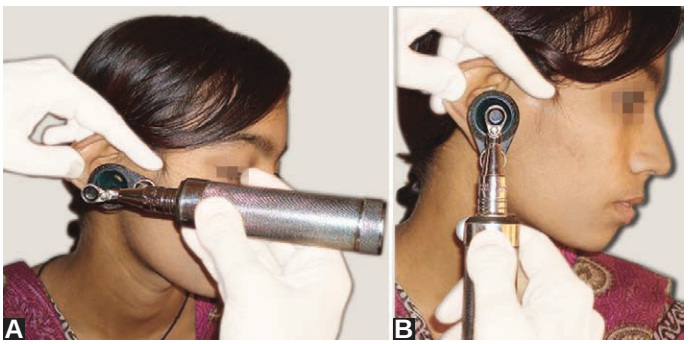
- *Tympanic membrane*: The normal TM (Fig. 8) is obliquely set at the medial end of the meatus. It is semitransparent and pearly white in color. It consists of two parts: pars tensa and pars flaccida. The site and size of perforations, retraction, and granulations must be recorded. When TM is semitransparent or perforated, some structures (ossicles, windows, eustachian tube, tumors) of the middle ear can be seen through it.

Otoscope is especially useful in examining both the ear as well as the nose of infants.

Siegelization

Mobility of TM is tested with Siegel's speculum (Chapter on "Instruments"). The other uses of this pneumatic speculum are following:

- Fistula test for labyrinthine fistula.
- Suction of middle ear secretions in cases of chronic suppurative otitis media (CSOM).
- Pushing of medicines through the central perforation of TM.



Figs 7A and B: Otoscopy examination. Two ways (A and B) of holding and introducing the otoscope. Note the retraction of pinna by other hand

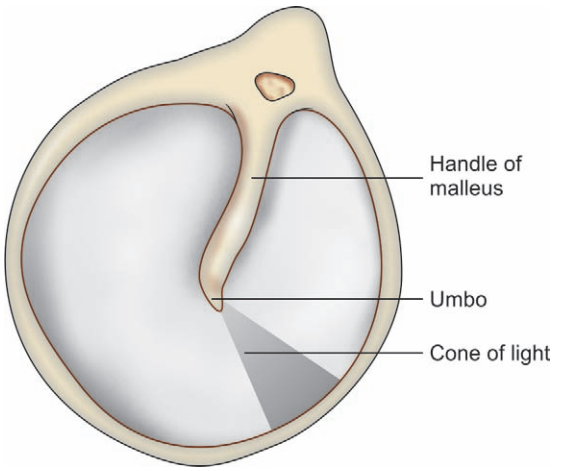


Fig. 8: Tympanic membrane showing attic, malleus handle, umbo and cone of light

Eustachian Tube

- *Inspection*: The pharyngeal opening of Eustachian tube can be seen by posterior rhinoscopy mirror, nasal sinuscope and nasopharyngoscope.
- *Patency*: (Chapter on "Disorder of Eustachian Tube")
  - *Eustachian catheterization*: Eustachian catheter, which is mainly used to test patency of the Eustachian tube, is sometimes also helpful in removing FBs from the nose. After anesthetizing the nasal cavity with 4% lignocaine, Eustachian catheter is passed into the nasopharynx along the floor of nose. Then it is turned medially and slightly withdrawn till the posterior free border of the nasal septum is felt. Now it is rotated 180° laterally to lie against the opening of Eustachian tube. A bulb is attached and then air is insufflated. An auscultation tube, which connects patient's ear to that of the examiner, is employed. Hearing of the sound of air entering the middle ear indicates the patency of Eustachian tube.
  - *Valsalva maneuver*: In the presence of a hole in TM, air can be heard escaping from the ear when patient tries to blow with mouth and nose closed.

Functional Examination

Functional examination consists of auditory function (voice test, tuning fork tests, Rinne test, Weber test, Schwabach test, absolute bone conduction test, and audiometry) and vestibular function (spontaneous nystagmus, fistula test, and positional tests). The most commonly used tuning fork is 512 Hz but forks of other frequencies such as 256, 1,024 and 2,048 Hz are also desirable. All these functional tests are discussed in detail in chapters of Hearing Evaluation and Evaluation of Vertigo. It is important to test facial nerve and know the grade of palsy (chapter of Facial Nerve Disorders).

OTALGIA (EARACHE)

Otalgia refers to pain in and around the ear. Ear pain is a very common otological complaint. For the treatment it is essential to find its cause.

Etiology

- *Primary otalgia*: Pain in and around ear can be caused by inflammatory, traumatic, and neoplastic conditions of the ear (primary otalgia).
- *Secondary otalgia*: The secondary otalgia is referred from the head and neck regions, which are innervated by the nerves that also supply to ear (Box 1). These nerves are as following (Fig. 9) (See also the chapter on Anatomy and Physiology of Ear):
  1. Cranial nerve V (trigeminal nerve): The auriculotemporal nerve, which is a branch of posterior division of mandibular division of trigeminal nerve supply to the external ear (tragus, anterior pinna, anterior lateral surface of tympanic membrane, anterosuperior wall of external auditory canal). Trigeminal nerve also supplies to face, temporomandibular joint (TMJ), oral cavity (lips, cheeks, anterior two third tongue, floor of mouth, palate), salivary glands, nose and paranasal sinuses and nasopharynx.

Box 1: Causes of otalgia

A. Local causes (Primary otalgia)

- Auricle: Skin lesion, perichondritis/chondritis, trauma
- External auditory canal: Furuncle*, impacted wax*, otitis externa*, trauma*, foreign bodies especially live insects*, otomycosis*, myringitis bullosa, herpes zoster oticus (Ramsay Hunt Syndrome), and malignant neoplasms
- Middle ear: Acute otitis media*, Eustachian tube obstruction*, cholesteatoma*, mastoiditis*, Barotrauma, and malignancy middle ear
- Intracranial complications of otitis media: Extradural abscess

B. Referred causes (Secondary otalgia)

1. *Area supplied by CN V (trigeminal nerve)*
   - Dental and periodontal diseases*: Caries tooth, apical abscess, impacted 3rd molar, malocclusion. Erupting dentition in children.
   - Oral cavity: Infection, trauma, aphthous* or malignant ulcers of oral cavity*
   - Salivary glands: Parotid and submandibular inflammatory and malignant diseases
   - Periauricular lymphadenopathy from scalp or neck infections
   - Temporomandibular joint*: Myofascial pain dysfunction. bruxism, osteoarthritis, recurrent dislocation, ill-fitting denture, malocclusion, Costen's syndrome
   - Nose and paranasal sinuses: Trauma, infection, tumors and contact points between turbinates and septal spur
   - Nasopharynx: Infection and tumors and after adenoidectomy*
   - Sphenopalatine (Sluder's) neuralgia
   - Trigeminal neuralgia
   - Headache: Tension type, traction and inflammatory headaches
   - Atypical facial pain
2. *Area supplied by CN IX (glossopharyngeal nerve)*
   - Oropharynx*: Acute tonsillitis, peritonsillar abscess, Post tonsillectomy, Benign and malignant ulcers of soft palate, tonsil and its pillars and base of tongue. Tuberculosis
   - Elongated styloid process (Eagle's syndrome)
   - Glossopharyngeal neuralgia
3. *Area supplied by CN X (vagus nerve)*
   - *Vallecula, larynx, laryngopharynx, esophagus*: Malignancy* or ulcerative lesions
   - *Thyroid*: Thyroiditis
   - *Cardiac/pulmonary*: Coronary Artery Disease (CAD), aneurysmal dilation of great vessels
   - *Esophagus*: Hiatus hernia with gastroesophageal reflux
4. *Area supplied by C2 and C3 spinal nerves*
   - Cervical arthritis/disc disease
   - Cervical spondylosis*, injuries of cervical spine, caries spine
5. *Facial nerve*: Geniculate neuralgia, Bell's palsy*, and herpes zoster oticus
6. *Psychogenic*

*Common causes of otalgia

2. Cranial nerve VII (facial nerve): It innervates the skin of lateral concha and antihelix, lobule, mastoid, posterior EAC, and posterior portion of TM.
3. Cranial nerve IX (glossopharyngeal nerve): Its tympanic branch (Jacobson's nerve) carries preganglionic secretomotor fibers to parotid gland. It goes into middle ear and forms tympanic plexus (supplies to medial surface of TM, mucosa of middle ear, eustachian tube, and mastoid air cells). Glossopharyngeal nerve also supplies to oropharynx, which include tonsil, soft palate, posterior pharyngeal wall and base of tongue.
4. Cranial nerve X (vagus nerve): Its auricular branch (Arnold's nerve) supplies to external ear (concha, infero-posterior EAC and TM) and postauricular skin. The vagus nerve also supplies to vallecula, epiglottis, larynx or laryngopharynx, trachea, esophagus, and thyroid gland and cardiopulmonary system.
5. C_2 and C_3 spinal nerves: The lesser occipital and greater auricular nerves of cervical plexus supply to the postauricular region of the ear and cervical spines.

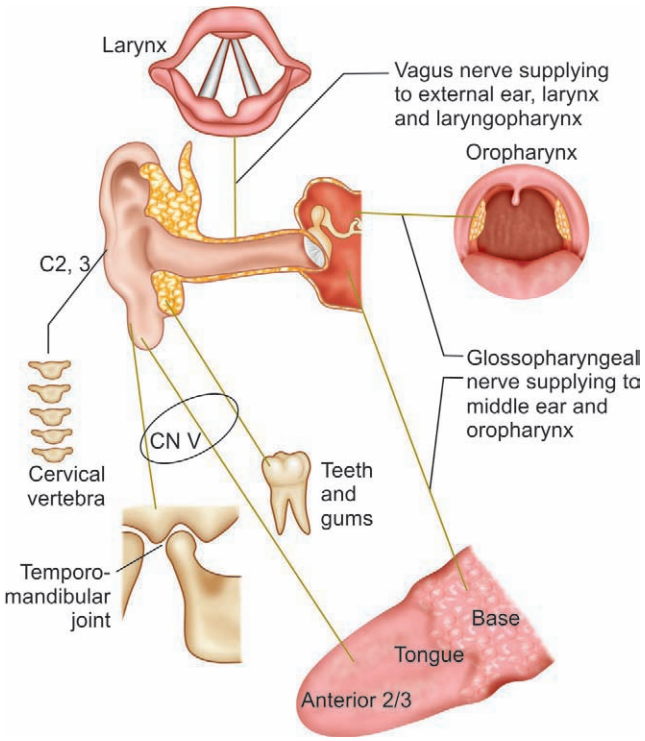


Fig. 9: Referred otalgia. Common causes of referred otalgia and nerve supply of ear

History and Physical Examination (Box 2)

History and physical examination will usually reveal the source of ear pain and indicate whether the otalgia is local or referred in origin (Fig. 10 and Box 1).

- *Dental eruptions*: Ear pulling in preverbal children may indicate erupting dentition.
- *Infection and neoplasms within nose and paranasal sinuses*: They can irritate vidian nerve and result in referred otalgia.
- *Temporomandibular joint dysfunction*: In cases of TMJ dysfunction or Costen's syndrome (resulting from dental malocclusion), there occurs masticator muscle spasm that causes referred ear pain. Such patients usually give history of excessive gum chewing, malocclusion and bruxism.
- *Sluder's (sphenopalatine) neuralgia*: Lancinating pain in lower face that radiates to orbit, temple, forehead, and upper neck suggest Sluder's neuralgia.
- *Herpes zoster oticus*: The earache can occur even in the absence of vesicular eruption.
- *Bell's palsy:* These patients can have ear pain before the onset of facial paralysis.

- *Eagle's syndrome:* The ear pain is secondary to stretching and irritation of glossopharyngeal nerve from elongated styloid process.
- *Glossopharyngeal neuralgia*: The patient presents with sharp and lancinating pain that radiates to ear and originates in the tongue base, soft palate, or tonsillar fossa. The trigger zone may be the Rosenmuller fossa.
- *Exertional left earache*: In cases of exertional earache on left side, patient must have cardiac consultation to rule out coronary artery disease.

Box 2: Otalgia: History, physical examination and investigations

A. *History*
- Onset and duration. Sudden, gradual, preceding events
- Quality and localization: Vague and nonlocalizing
- Associated ear symptoms: otorrhea, hearing loss and ear fullness, swelling, trauma and foreign body
- Whether ear pain is associated with: mastication, swallowing, voice change, purulent rhinorrhea or physical exertion
- Past history: Ear surgery

B. *Physical examination*
- Ear, nose, throat, head and neck examination
- Dental examination
- Craniomandibular: Palpation of temporomandibular joint (TMJ) and surrounding musculature
- Cervical spines
- Neurologic examination: Cranial nerves

C. *Investigations*
- Flexible fiberoptic examination of nose, pharynx and larynx (Nasopharyngolaryngoscopy)
- Esophagogram
- X ray cervical Spines
- Orthopantomogram (OPG)
- X-rays for TMJ and styloid process

- *Functional*: When no cause is found, ear pain may be functional in origin. In such cases periodic re-evaluation is important.

If the ear pain appears to be disproportional to otological findings especially in elderly patients, preclude the possibility of referred otalgia especially upper aerodigestive tract malignancy.

Management

- Further investigations are advised to confirm the clinical diagnosis.
- In case of suspected malignancy of oral cavity, pharynx, larynx, nose and paranasal sinuses, biopsy will confirm the diagnosis.
- Once the cause of ear pain is known, treatment is instituted accordingly.

OTORRHEA

Otorrhea[1], which refers to discharge from the ear, is a very common otologic complaint.

Etiology

Infective diseases are the most common causes of otorrhea (Box 3).

- *Otitis externa:* It may not allow middle ear microscopic examination and may be secondary to otitis media.
  - *Hearing aids*: Scanty serous otorrhea in normal ears may represent moisture trapping after washing or may be due to the use of hearing aid or ear plug.
- *Parotitis*: Though rare, infection of parotid gland can extend into EAC through Santorini fissures, which are fibrous channels within the cartilaginous EAC.

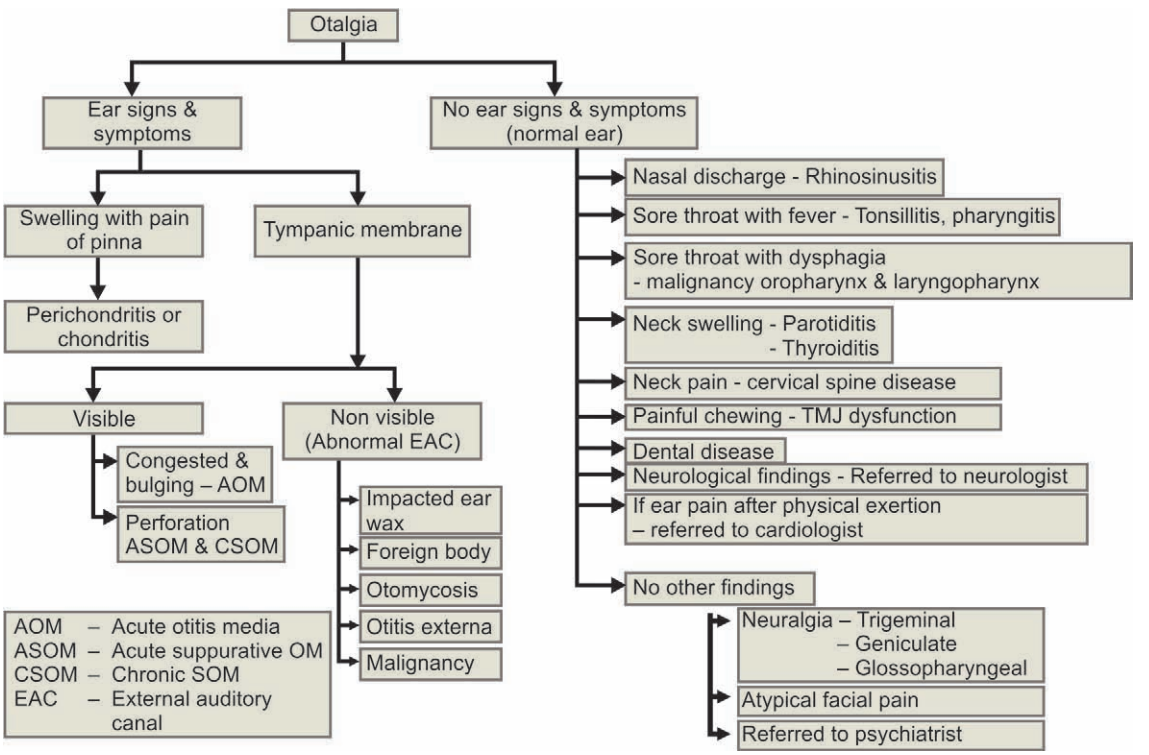


Fig. 10: Flow chart for clinical diagnoses of causes of otalgia

[1] Oto, ear + G. rhoia, flow

Box 3: Causes of otorrhea

- *External auditory canal:* Otitis externa (OE)*, otomycosis*, furunculosis*, keratosis obturans, malignant or necrotizing otitis externa, acute dermatitis (such as seborrheic, eczematous), parotid abscess, canal cholesteatoma, neoplasms (ceruminoma and malignancy)
- *Conditions predisposing to secondary otitis externa:* Osteoma, exostoses, canal dehiscence into the mastoid or glenoid fossa, otitis media with perforation, allergy to ear drops
- *Tympanic membrane*: Granulation tissue*, granular myringitis*, bullous myringitis, retraction pocket, cholesteatoma
- *Middle ear and mastoid*: Acute and chronic suppurative otitis media (ASOM and CSOM)* and mastoiditis, cholesteatoma*, granulomatous disease, neoplasms
- *CSF otorrhea*: Fracture temporal bone, tegmen defect, cochlea deformity, Hyrtl's disease

* Common causes of otorrhea

- *Less common causes:* They are secondary infection of glomus tumors and carcinomas in adults and histiocytosis and rhabdomyosarcoma in children.

In children, the most common causes of ear discharge are ASOM and CSOM while in adults they are CSOM and OE.

ASSESSMENT (BOX 4)

History

Otorrhea may be profuse or scanty and continuous or intermittent. Otorrhea from acute suppurative otitis media may be bloody, mixed with mucus, or mucopurulent and typically short-lived.

Box 4: Otorrhea: History, physical examination and investigations

- *History*
  - Onset and duration: Acute, chronic
  - Character of discharge: Clear, mucoid, purulent, bloody and pulsatile
  - Preceding events: History of trauma, URI
  - Associated ear symptoms: Otalgia, pruritus, hearing loss and vertigo
  - Past history: Dermatologic disease, otitis media or externa, neurotology/ear surgery
- *Physical Examination (Microscopic Ear Examination and Cleaning)*
  - External auditory canal: Edema/inflammation, granulations polyp and growth
  - Tympanic membrane: Perforation, granulations or de-epithelialization, myringotomy tube, retraction and cholesteatoma
  - Tuning fork tests: Conductive or sensorineural hearing loss
  - Systemic: Neurologic deficit or features of associated systemic disease
- *Investigations*
  - Audiometry
  - Tympanometry
  - Culture and sensitivity
  - Chemical analysis: Glucose and chlorides for CSF
  - CT scan
  - Biopsy

Types of Discharge and their Causes

- *Mucopurulent*: Acute suppurative otitis media (ASOM) and chronic suppurative otitis media (CSOM)
  - *Serous*: Eczematous otitis externa (OE)
  - *Sanguineous* (blood tinged): ASOM, granulations, trauma, tumors
  - *Putrid (foul smelling)*: Cholesteatoma
  - *Watery*: Watery otorrhea, which may be copious, subtle or intermittent, suggests a cerebrospinal fluid (CSF) leak
- *Associated Symptoms and their Causes*
  - *Pain*: Acute OE
  - *Pruritus*: Chronic OE and otomycosis
- *Recurrent Otitis Externa*
  - Eczema or psoriasis of ear canal skin.
- Past history
  - *Ear surgery*: Recurrence of middle ear disease or infection in mastoid cavity.
  - *Neurotologic surgery*: Immediate or delayed CSF otorrhea.
  - *Grommets*: Otorrhea is common after myringotomy tube insertion.

ASOM, CSOM and cholesteatoma are the most common sources of purulent or mucoid ear discharge.

Examination

Evaluation requires meticulous suctioning of secretions under microscope to find the source of discharge.

- *Otitis externa*: It tends to delay a thorough ear microscopic examination because of edema, debris and tenderness.
- *Foreign body*: An unsuspected FB may be found.
- *Stenosis*: Refractory OE may result in EAC stenosis.
- *Tympanic membrane*: Perforation, granulation and tympanostomy tube, retraction, tympanic sclerosis and cholesteatoma
- *Postoperative mastoid cavity*: In cases of modified or radical mastoidectomy, infection must be differentiated from recurrent or residual cholesteatoma.

Investigations

- *Audiometry*: If the EAC is obstructed and tuning fork tests reveal conductive hearing loss, audiometry is deferred.
- *Tympanometry*: It is not done in suspected case of CSF otorrhea as it can result in pneumocephalus. It may be painful in cases of otitis externa.

3. *Culture and sensitivity*:
   a. *Cautions:*
      i. Topical antibiotic should be stopped before taking the sample for culture and sensitivity as they will affect the culture growth.
      ii. In refractory cases of infective otorrhea and myringotomy tube, sample should be taken from deep in the EAC or from a perforation.
   b. *Microorganisms*:
      i. *Most common microorganisms causing OE* are *Pseudomonas aeruginosa* (in malignant or necrotizing OE) and *Staphylococcus aureus*.

ii. *Actinomyces israelii*: It is an anaerobic gram-positive bacterium that can cause OE from primary dental or parotid infection. This refractory OE presents with granulation tissue in EAC and thick yellow ear discharge and needs surgical debridement and prolonged antibiotic therapy.

4. *CT imaging*: Preoperative CT imaging is important in cases of EAC stenosis (congenital, traumatic or neoplasms) with cholesteatoma.
5. *Biopsy*: It is required to know the nature of neoplasms.
   a. *Contraindication:* In cases of glomus tumors it should be avoided as it can result in profuse bleeding.
6. *Immunodeficiency and allergy*: Identifying underlying immunodeficiency or allergy should be considered in refractory cases.

Treatment

The treatment begins once the cause of the otorrhea is determined. Antibiotic, topical steroid, and repeated ear toilet may be required. Aminoglycosides ear drops are not used as they are ototoxic. Ear drops are not prescribed in case of CSF otorrhea as they can enter into cranium.

1. *Topical isopropyl alcohol*: In normal ears it can dry the scanty serous otorrhea.
2. *Pre-existing perforation and grommet*: Water should not go into the ear. A short course of topical antibiotic usually controls the otorrhea. Tube removal and repeated debridement is needed for adequate drainage.
3. *Granular myringitis*: It usually responds to topical steroid or 5-Fluorouracil. Some cases may need debridement of the granulations.
4. *De-epithelialization of tympanic membrane:* If does not respond to drying agent (Burrow's solution) it may need spilt-thickness skin graft.
5. *Acute otitis media:* Systemic antibiotic is indicated. Topical antibiotic will prevent secondary OE. In refractory cases, antibiotic is changed as per the report of culture and sensitivity. A pinpoint perforation is enlarged to establish adequate drainage.
6. *Cholesteatoma and neoplasms:* Surgery is indicated in these cases.
7. *Open mastoid cavity otorrhea:* It needs debridement and treatment with topical antifungal or antimicrobial eardrops or acidifying and drying agents such as boric acid and alcohol solution. Granulation tissue and mucosalized epithelium in the mastoid bowl requires chemical cauterization.

EAR POLYP

An ear polyp is a pedunculated mass, which lies in the EAC. It may arise from EAC or middle ear. Granulomatous and neoplastic lesions may present as ear polyp. It is usually associated with otorrhea and hearing loss. Other associated symptoms include bleeding, pain and itching.

Etiology

- *External auditory canal disorders*: Furuncle, trauma, FBs and keratosis obturans may trigger formation of granuloma (Box 5). The tumors of EAC are rare.

Box 5: Causes of polyp

- *External auditory canal*: Unhealed furuncle, trauma, keratosis obturans, foreign body, tumors (ceruminoma, exostosis, malignancy)
- *Middle ear*: CSOM (tubotympanic and cholesteatoma), glomus tumors (tympanicum and jugulare), carcinoma

- *Chronic suppurative otitis media especially cholesteatoma*: Middle ear mucosa gets prolapsed and present as pedunculated granuloma.
- *Glomus tumors*: They are rare and present with red fleshy polyp that bleeds easily.
- *Malignancy of middle ear*: It is rare.

History and Examination

- *Associated symptoms*: Ear polyp is usually associated with otorrhea and hearing loss. Other associated symptoms include bleeding, pain and itching.
- *Consistency*: Inflammatory polyps are soft whereas tumors are firm, friable and bleed readily.
- *Probing*: It is useful in determining the site of origin of the polyp. A probe can be passed all around if it is arising from the middle ear but it cannot if polyp is arising from the EAC. Glomus and malignant tumors bleed readily on probing.
- *Tuning fork tests*: They will reveal the type of hearing loss.

Investigations

- *Culture*: Discharging material is submitted for culture and cytological examination.
- *Audiometry*: It is done if tuning fork shows sensorineural hearing loss.
- *CT scan*: It is advised in cases of glomus and malignant tumors and complications of suppurative OM
- *Biopsy*: It will reveal the nature and cause of polyp.

Treatment

The predisposing and causative disorders should be treated.

- *Antibiotics*: Inflammatory polyps usually respond well to antibiotics.
- *Surgery*: Polyp may need removal and histopathology study.
  - Polypectomy facilitates better visualization and drainage of middle ear.

Caution: Middle ear polyps should not be avulsed as it can damage middle ear structures such as ossicles and facial nerve.

TINNITUS

Tinnitus[2] stands for the perception of sound (ringing or noise), which has no external stimulus. Approximately one-third people experience tinnitus sometime in their lives. The etiology of this common ear symptom is poorly understood. The severity of sound ranges from nearly undetectable to severe and debilitating. Though there is no cure for chronic tinnitus, there are various modalities of management, which significantly improve the quality of patients' life.

[2] Tinnire (Latin word) means "to ring"

Auditory hallucination: It is a psychiatric disorder in which patient hears voices or other organized sounds like music or conversation.

Classification

Tinnitus can be categorized into different groups (Box 6).

1. Subjective and Objective: Subjective tinnitus, which is the most common, is perceived only by the patient. Objective tinnitus can be perceived by the examiner also.

Box 6: Classification and causes of tinnitus

1. Pulsatile (Noncontinuous)–Subjective or objective
   a. Vascular
      i. *Arterial*
         ◊ *Atherosclerosis*: Atherosclerotic Carotid Artery Disease (ACAD) and atherosclerosis of external carotid or subclavian
         ◊ *Glomus Tumors*: Glomus tympanicum, Glomus jugulare
         ◊ *Vascular malformations*: Congenital arteriovenous malformations (AVM), Acquired arteriovenous fistulae (AVF)
         ◊ *Arterial dissection*: Carotid, vertebral.
         ◊ *High cardiac output*: Pregnancy, anemia, exercise, thyrotoxicosis
         ◊ Otosclerosis
         ◊ Hypertension
      ii. *Venous*
         ◊ Benign Intracranial Hypertension (BIH) Syndrome (Pseudotumor Cerebri)
         ◊ Venous hum
         ◊ Jugular bulb anomalies
   b. Nonvascular
      i. *Myoclonus* (multiple sclerosis, brainstem infarct)
         ◊ Palatal
         ◊ Tensor tympani
         ◊ Stapedius
      ii. *Vascular neoplasm*
         ◊ Skull base
         ◊ Temporal bone
      iii. *Temporomandibular joint disease*
   c. Idiopathic
2. Nonpulsatile (Continuous—Subjective–Most common
   a. *With Hearing Loss*
      i. *External ear*: Wax, hair, foreign bodies, live insects
      ii. *Middle ear*: Otitis media, patulous Eustachian tube, hemotympanum (head injury)
      iii. *Labyrinth*: Meniere's disease, noise exposure, ototoxic agents, presbyacusis, and other causes of sensory hearing loss
      iv. *CN VIII*: Acoustic neuroma and other causes of neural hearing loss
   b. *Without Hearing Loss*
      i. Psychogenic
      ii. Hypotension
      iii. Hypoglycemia
      iv. Epilepsy
      v. Migraine
      vi. Idiopathic

2. Pulsatile and Nonpulsatile
3. Continuous and Noncontinuous.
4. Vascular and nonvascular
5. Aureum and Capitum.
6. Unilateral and Bilateral.
7. Mild, Moderate and Severe.
8. Acute and Chronic.
9. With hearing loss and without hearing loss.
10. Vascular, myogenic, external ear, middle ear, cochlear, neural, and central.
11. Central and peripheral

Etiopathology of Nonpulsatile Subjective Tinnitus

The exact etiology and pathophysiology of nonpulsatile subjective tinnitus are poorly understood. Whether the tinnitus is generated by cochlea, cochlear nerve or central nervous system (CNS) is not known. The current evidence suggests that it is generated in the CNS.

Following two factors are suggested:

- Deprivation of input, or
- Abnormal input from the ear

Neuronal plasticity and changes in neuronal input may result to following structural or functional changes:

- Changes in the nuclei of the ascending auditory pathways.
- Redirection of auditory input to extra auditory pathways.

History and Physical Examination

History better defines the patient's tinnitus (Box 7). Patient should describe the nature of the sound. Any change in the tinnitus that has occurred during the course of disease should be noted. Eighty-five percent of tinnitus patients have hearing loss. Most patients have normal neuro-otologic examination except for the hearing loss.

- *Otoscopy*: In cases of pulsatile tinnitus, otoscopy may reveal a middle ear mass.
- *Audiogram*: It may reveal otosclerosis, Meniere's disease, or noise induced hearing loss.
  - *Cerebellopontine angle tumors*: The cerebellopontine angle (CPA) lesions should be suspected in cases of unilateral or significant asymmetrical sensorineural hearing loss (SNHL). Ten percent of acoustic neuroma patients present with tinnitus. This rate increases to 83% with passage of time.
- *Abnormally patent Eustachian tube*: Tinnitus synchronous with respiration may occur in cases of abnormally patent Eustachian tube.

Pulsatile Tinnitus

The pulsatile or throbbing tinnitus is rare. Table 3 shows the key clinical features of some common causes of pulsatile vascular tinnitus.

- *Pulsatile vascular tinnitus*: The sound of pulsatile vascular tinnitus correlates with the patient's arterial pulse.
- *Venous pulsatile tinnitus*: In cases of pseudotumor cerebri and venous pulsatile tinnitus, turning of head and pressure over internal jugular vein will decrease or stop the tinnitus. Light pressure over ipsilateral jugular vein and turning of head on one particular side diminish or obliterate jugular venous tinnitus.

Box 7: History and physical examination

A. *History*
- Site and side: Ear or head, unilateral or bilateral
- Mode of onset, duration, progression, frequency, continuous/intermittent associated events
- Quality (Loudness and pitch): Popping, clicking or cricket, swishing, pulsating or throbbing, ringing, buzzing, hissing, rustling, or roaring
- Severity: How it impacts patient's life?
- Factors aggravating and relieving
- Other otologic symptoms such as hearing loss, vertigo, otalgia, and otorrhea
- H/o noise exposure, ototoxic agents, head injury, CNS infections
- Medical, psychological, and social histories and conditions
- Past history of Otologic surgery
- Family history of Otologic disease

B. *Physical Examination*
- EN.. Head and neck examination
- Neurotologic examination including otoscopy, hearing tests, cranial nerve tests
- In cases of pulsatile tinnitus
  - ◊ Auscultation: Assess for bruits in and around ear, orbits and neck
  - ◊ Compare the rate of objective pulsatile tinnitus with patient's pulse
  - ◊ Effects of light pressure over internal jugular vein and turning the head on either side
  - ◊ Observe the concurrent contractions of soft palate and rhythmic movements of tympanic membrane
  - ◊ Flexible nasopharyngoscopy in cases of palatal myoclonus
- Medical review of systems
- Neurologic consultation in cases of myoclonus and pseudotumor cerebri.

C. *Audiologic Evaluation*
- Tuning fork tests
- Pure tone audiometry
- Speech perception in quiet and noise
- Tympanometry

D. *Tinnitus Evaluation*
- Matching tinnitus to sounds
- Determine minimum masking levels
- Measure residual inhibition

E. *Radiological Imaging*

- *Venous hum*: It is an idiopathic pulsatile tinnitus, which is attributed to turbulent flow within the transverse-sigmoid-jugular venous system. This unilateral pulsatile tinnitus usually occurs on right side in women between the ages of 30 and 50 years. The transverse-sigmoid-jugular venous system are right-sided dominant (superior sagittal sinus continues as right-sided transverse sinus) in 75% of individuals.
- *Myoclonus*: In cases of myoclonus, pulsatile tinnitus sound (clicking or snapping) does not correlate with heartbeat.

Benign intracranial hypertension (BIH) syndrome, atherosclerotic carotid artery disease (ACAD), glomus tumors and dural vascular malformations such as congenital arteriovenous malformations (AVM) and acquired arteriovenous fistulae (AVF) are among the most common causes of pulsatile vascular tinnitus.

TABLE 3 Common causes of pulsatile tinnitus and their key clinical features

Causes	*Clinical features*
Benign Intracranial Hypertension (BIH) Syndrome (pseudotumor cerebri)	Young obese females complaining of headache, visual changes and dizziness, CSF pressure > 200 mm H_2O
Glomus Tumors, Persistent stapedial artery, Dehiscent carotid, Aberrant carotid	Reddish pulsatile mass medial to tympanic membrane
Vascular malformations	Normal otoscopy
Atherosclerotic Carotid Artery Disease	Risk factors of atherosclerosis, Carotid bruit
Dehiscent High-riding Jugular Bulb	Bluish mass medial to tympanic membrane and conductive hearing loss
Arterial dissection (carotid, vertebral)	Cerebral Ischemic Symptoms
Venous hum	Right-sided pulsatile tinnitus in women between the ages of 30 and 50 years

Radiological Imaging

Table 4 shows the various modalities of imaging studies and their indications in cases of tinnitus.

Most patients of bilateral nonpulsatile subjective tinnitus with symmetrical SNHL need no imaging studies.

- Tinnitus: It is a symptom not a disease. It can be so disabling that the patient can have suicidal tendencies.
- Subjective tinnitus is most common and is heard only by the patient.
- Objective tinnitus can be heard by both patient as well as examiner.

Tinnitus Management Program

In spite of treating every reasonable medical cause for tinnitus, majority of patients report little improvement. The patients, who

TABLE 4 Radiological imaging and their indications

Imaging modalities	*Indications*
Carotid Ultrasound	Vascular pulsatile tinnitus in atherosclerotic disease
Contrast-enhanced CT	Vascular pulsatile tinnitus such as paraganglioma (jugular fossa erosion and middle ear mass), aberrant carotid or persistent stapedial artery (absent foramen spinosum)
Magnetic Resonance Imaging (MRI)	Myoclonus to evaluate for multiple sclerosis.
Gadolinium-enhanced (MRI)	Unilateral tinnitus or asymmetrical sensorineural hearing loss (SNHL) to evaluate IAC and CPA
Magnetic Resonance Angiography and Venography (MRA/MRV)	Vascular abnormalities including dural AVFs
Conventional Angiography	Presence of bruit in cases of normal CT/MRI

consider their tinnitus to be a significant debilitating problem, need specialized management program (Box 8). The protocols of Tinnitus Management Program (TMP) can be customized to address the particular circumstances of each patient.

- The tinnitus and its coincident symptoms such as insomnia, anxiety/stress, depression and fatigue form a vicious circle of symptoms. These coincident symptoms need effective treatment (medical and psychotherapy) because their increase would worsen the tinnitus.
- Multimodal strategies are designed as per the need of specific patient.
- Patient education, reassurance, and counseling should be the part of patient's evaluation.

In many cases of chronic tinnitus, there is no "cure" however there is a wide array of strategies, which helps patients in getting control over their tinnitus.

Acoustic Therapies

Acoustic therapies (Box 9) constitute a vital component of an effective tinnitus management program. This noninvasive therapy provides not only immediate relief but there also occurs tinnitus suppression or temporary disappearance of tinnitus (residual inhibition).

Box 8: Elements of comprehensive tinnitus management program

- Team members: Otolaryngologist, audiologist, neurologist, psychiatrist, psychologist, sleep or pain specialist
- History and physical examination along with patient education, reassurance, and counseling
- Evaluation of acoustic therapies: Use of external sounds to make tinnitus less noticeable
- Treatment of medical conditions
- Avoidance of factors aggravating tinnitus and lifestyle changes
- Psychological counseling and psychiatric management
  - Cognitive therapy, relaxation therapy, biofeedback, hypnosis, stress reduction techniques, psychotherapy (to reduce insomnia, anxiety, depression, and obsessive-compulsive tendencies
  - Medications for insomnia, anxiety, depression, obsessive-compulsive tendencies, phobias

Box 9: Examples of acoustic therapies

- *Pleasant environmental sounds*:
  - Music, ocean waves, rain forest, summer night, television, or relaxation CDs
  - Tabletop sound machines
  - Cassette tapes and CDs
  - Tabletop water fountains
  - Fans and air purifiers
- *Sounds close to the ears*
  - Sound pillow and pillow speakers
  - Headphones and earphones
- *Sounds directly into the ear canal*
  - In-the-ear sound generators or maskers
  - Hearing aids
  - Combination instruments (combination of hearing aid and sound generator)

- *In-the-ear devices*: For many patients direct sounds to their ears are more effective than environmental sounds. In-the-ear devices, though comparatively expensive are most portable, inconspicuous and efficient in obtaining relief from the tinnitus.
- *Hearing aids*: They are beneficial for the patients who have significant hearing loss. Better hearing may relieve some patients of the frustration, isolation and depression.
- *Combination instruments*: They are indicated in the patients who are already using the hearing aid and feel that sound generators will provide additional tinnitus relief.

Lifestyle Changes (Box 10)

Many patients observe that their tinnitus worsens when they are tired. They should get enough restful sleep.

Though exercise may increase tinnitus problem in patients, it should be done regularly as it helps in stress reduction and improve cardiovascular health, muscle tone, mood and sleep patterns.

- Earplugs and earmuffs protect against harmful sounds and noises, which increase not only loudness of tinnitus but also produce noise-induced hearing loss.
- Tinnitus is less bothersome when patients are busy in their work. Retired persons should be encouraged to keep themselves busy in enjoyable and rewarding activities and volunteer work.
- Patients are encouraged to participate in social functions and positive interactions with family members, friends and relatives.

Surgical Management

It may be possible, though in minority of the cases, to eliminate the tinnitus by operating on the underlying cause of the tinnitus (Table 5).

HYPERACUSIS

Hyperacusis can be defined as "increased or painful sensitivity to every day sounds that would not trouble a normal person". These patients may develop phonophobia and overprotect their ears by wearing earplugs and earmuffs. After thorough ENT examination these patients are initiated into auditory

Box 10: Factors aggravating and relieving tinnitus (lifestyle changes)

A. *Factors aggravating tinnitus*
- Stress/fatigue
- Excessive noise exposure
- Upper respiratory tract infections and allergy
- Moving jaws/clenching teeth
- Head and neck trauma
- Changes in altitude
- Too much use of aspirin, alcohol, caffeine, acetaminophen, or ibuprofen
- Changes in body position
- Tobacco and marijuana

B. *Factors relieving tinnitus*
- Regular Exercise
- Avoidance of noise exposure (use earplugs and earmuffs)
- Keeping oneself busy
- Personal relationships and socialization

TABLE 5 Surgical management of vascular tinnitus

Disease	*Surgery*
Carotid artery stenosis	Carotid endarterectomy
Glomus tumors	Surgical excision, Stereotactic radiosurgery
Dural AVFs	Arterial compression exercises, surgical obliteration, transarterial or transvenous embolization, stereotactic radiosurgery
High-riding jugular bulb	Surgical correction
Venous hum	Ligation of internal jugular vein
Pseudotumor cerebri	Weight loss (weight-reduction surgery), reduction of ICP (acetazolamide, furosemide, lumbar-peritoneal shunt CSF diversion technique)
Otosclerosis	Stapes surgery
Acoustic neuroma	Surgical removal
Meniere's disease	Variety of transtympanic treatments, endolymphatic shunt placement, CN VIII or cochlear nerve section
Severe to profound sensorineural hearing loss (SNHL)	Cochlear implantation

Contd...

Contd...

desensitization program and tinnitus retraining therapy (TRT), which consists of patient education, directive counseling, reassurance and sound therapy.

Clinical Highlights

1. **Siegel's pneumatic speculum**: It provides magnified view of tympanic membrane (TM). It is used to test the mobility of the tympanic membrane. We can perform fistula test and introduce medicine into middle ear.
2. **Nerve supply of tympanic membrane**: Lateral surface is supplied by auriculotemporal nerve (CN V_3) and auricular branch of vagus. The medial surface is supplied by glossopharyngeal nerve.
3. **Nerve supply of the middle ear**: Glossopharyngeal nerves.
4. **Jacobson's nerve**: This branch of glossopharyngeal nerve supplies middle ear and mastoid air cells, and supplies secretomotor fibers to parotid. The section of this nerve relieves gustatory sweating.
5. **Referred otalgia**: The common causes of ear pain in the absence of ear disease include peritonsillar abscess, cancer of the pyriform fossa, temporomandibular joint (TMJ) dysfunction, and ulcer oral tongue. In cases of carcinoma base of tongue referred ear pain is through the glossopharyngeal nerve and otic ganglion.
6. **Costen's syndrome**: Patient usually presents with referred otalgia, which may be associated with vertigo and tinnitus.

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11 Hearing Evaluation

He who grumbles at the little thing that has fallen to his lot to do will grumble at everything. Always grumbling, he will lead a miserable life and everything will be a failure. But that man, who does his duty as he goes, putting his shoulder to the wheel, will see the light, and higher and higher duties will fall to his share.

—Swami Vivekananda

Points of Focus

- **AUDIOLOGY AND ACOUSTICS**
  - ✓ Sound
  - ✓ Frequency and Pitch
  - ✓ Pure Tone and Complex Sound
  - ✓ Overtones and Timbre
  - ✓ Intensity and Loudness
  - ✓ Decibel (dB): Audiometric Zero, Hearing Level (HL), Sound Level Meter, dB A Scale,
  - ✓ Sensation Level (SL)
  - ✓ Loudness Discomfort Level
  - ✓ Dynamic Range
  - ✓ Noise: White Noise, Narrow Band Noise, Speech Noise
- **TYPES OF HEARING LOSS**
  - ✓ Conductive Hearing Loss
  - ✓ Sensorineural Hearing Loss (SNHL)
  - ✓ Mixed Hearing Loss
- **NEED OF HEARING EVALUATION**
- **METHODS OF HEARING EVALUATION**
  - ✓ Traditional Screening Tests
  - ✓ Voice Tests (Conversation and Whisper)
- **TUNING FORK TESTS (TFT)**
  - ✓ Tuning Forks
  - ✓ Methods
  - ✓ Rinne Test
  - ✓ Masking
  - ✓ Weber Test
  - ✓ Absolute Bone Conduction (ABC) Test
  - ✓ Schwabach's Test
  - ✓ Bing Test
  - ✓ Gelle's Test
- **PURE TONE AUDIOMETRY (PTA)**
  - ✓ Method
  - ✓ Masking
  - ✓ Uses of PTA
  - ✓ Recruitment
  - ✓ Fowler's Alternate Binaural Loudness Balance Test (ABLB)
  - ✓ Short Increment Sensitivity Index (SISI)
  - ✓ Carhart's Tone Decay Test
- **SPEECH AUDIOMETRY**
  - ✓ Speech Reception Threshold (SRT)
  - ✓ Speech Discrimination Score (SDS)
  - ✓ Modifications: Optimum Discrimination Score (ODS), Half Peak Level (HPL), Rollover Curve
- **IMPEDANCE AUDIOMETRY**
  - ✓ Tympanometry
  - ✓ Eustachian Tube and Grommet Testing
  - ✓ Acoustic Reflex
- **ELECTROCOCHLEOGRAPHY**
- **BRAINSTEM EVOKED RESPONSE AUDIOMETRY (BERA)**
- **OTOACOUSTIC EMISSIONS**
- **AUDITORY STEADY STATE RESPONSE (ASSR)**
- **CLINICAL HIGHLIGHTS**

AUDIOLOGY AND ACOUSTICS

Audiology refers to the study of hearing disorders through the hearing evaluation as well as the rehabilitation of the patients with hearing impairments. Acoustic pertains to hearing and perception of sounds. Acoustics is the science that deals with the sounds and their perception.

Sound

Sound is a form of energy, which is produced by any vibrating object. A sound wave is produced by compression and rarefaction of molecules of the medium in which it travels such as air, liquid, solid. Velocity of sound in the air, at 20°C, at sea level, is 344 meters (1,120 feet) per second. Velocity of sound is faster in liquid and fastest in solid medium.

Frequency and Pitch

Frequency refers to the number of cycles per second. Its unit Hertz (Hz) is named after a German scientist Heinrich Rudolf Hertz. A sound of 500 Hz means 500 cycles per second. Pitch is a subjective sensation that is produced by frequency of sound. Higher the frequency, greater is the pitch.

Pure Tone and Complex Sound

A single frequency sound is called a pure tone such as 250, 500 or 1,000 Hz. In pure tone audiometry (PTA), thresholds of hearing in decibels for various pure tones from 250 to 8,000 Hz are measured. The sound, which has more than one frequency, is called a complex sound such as voice and speech.

Overtones and Timbre

Overtones are frequencies which are above the fundamental frequency of the vibrating source. Overtones determine the quality or the timbre of sound.

Intensity and Loudness

The intensity is the strength of sound and determines loudness. The unit of intensity of sound is decibel (dB). Loudness is a subjective sensation produced by intensity. Greater the intensity of sound more is the loudness. At 1 m distance following are the intensities of different types of speech (Table 1).

Decibel

Decibel is 1/10th of a bel[1]. Decibel represents a logarithmic ratio between two sounds (sound being described and the reference sound). Sound can be measured in following two ways:

1. *As power (watts/cm^3)*: It is used in music systems and theaters.
2. *As pressure (dynes/cm^3)*: It used in audiology.

The reference sound, which has sound pressure level (SPL) of 0.0002 dynes/cm^3, roughly corresponds to the threshold of normal human hearing at 1,000 Hz. Decibel notation avoids dealing with large figures of SPL.

[1]Alexander Graham Bell, the inventor of telephone

TABLE 1 Intensities of different types of speech

Whisper	30 dB
Normal conversation	60 dB
Shouting	90 dB
Discomfort in the ear	120 dB
Pain in the ear	130 dB

If a sound has an SPL of 1,000 (10^3) times the reference sound, it is expressed as 20 x 3 = 60 dB. In the same way, a sound of 1,000,000 (10^6) times the reference sound SPL is expressed as 120 dB.

Several dB scales, which have different reference scale, are used to measure sounds and hearing. In audiometry, hearing is measured on a biologic scale in decibels hearing level (dB HL). Environmental sounds are measured on a physical scale in decibels sound pressure level (dB SPL). The decibel A scale (dB A) is noise level scale.

- *Audiometric zero*: According to the International Standards Organization (ISO), audiometric zero is the mean value of minimal audible intensity in a group of normally hearing healthy young adults.
- *Hearing level*: Hearing level (HL) is SPL, which is produced by an audiometer at a specific frequency. It is measured in dB with reference to audiometric zero. The 70 dB sound of audiometer is represented as 70 dB HL.
- *Sound level meter*: This instrument measures level of noise and other sounds. It has different weighting networks (A, B, C) for different sensitivities at different frequencies for describing the sound.
- *dB A scale*: Noise levels are expressed as *dB A*. The dB A scale is weighted toward high frequency noises (1,000–5,000 Hz), which cause more sensorineural hearing loss (SNHL) than equivalent levels of low-frequency noise. Hearing protection for workers are required if noise exposure at the workplace is more than 90 dB A for 8 hours a day.

Sensation Level

The sensation level (SL) refers to the sound level above the threshold of hearing which will produce a sensation. 40 dB SL means patient was tested at 40 dB above this threshold, which could be either 0 dB (normal persons) or more in cases of hearing loss.

In cases of normal hearing it means 0 + 40 = 40 dB SL while in a case of hearing loss of say 50 dB, it would be 50 + 40 = 90 dB HL.

The discrimination scores in speech audiometry are tested at 30–40 dB SL. Stapedial reflex is elicited at 70–100 dB SL.

Loudness Discomfort Level

This level of sound produces discomfort in the ear. It is important while prescribing a hearing aid. Normally it is 90–150 dB SL.

Dynamic Range

Dynamic range is important while prescribing hearing aids. It refers to the difference between the most comfortable level and the loudness discomfort level. The recruitment in cochlear type of hearing loss reduces the dynamic range.

Noise

Noise is an aperiodic complex sound. Three types of noises are described—white, narrow band and speech.

1. *White noise:* It is similar to white light, which contains all the colors of the visible spectrum. White noise contains all frequencies in audible spectrum. This broadband noise is used for masking.
2. *Narrow band noise*: In this narrow band white noise certain frequencies, above and below the given noise, are filtered out. So the frequency range becomes smaller than the broadband white noise. The narrow band white noise is used for masking in PTA.
3. *Speech noise*: The speech noise has only speech range frequencies from 300–3,000 Hz. Other frequencies are filtered out.

Masking of non-test ear is always done during the bone conduction tests. For air conduction tests, masking is required only when difference of hearing between two ears exceeds 40 dB.

TYPES OF HEARING LOSS

Clinically hearing loss is traditionally divided into the three types—conductive, sensorineural and mixed.

Conductive Hearing Loss

The diseases, which interfere with the conduction of sound from the external ear to the stapediovestibular joint, result in conductive hearing loss (CHL). CHL is mostly managed successfully with medical and surgical treatment. Example includes the following (chapter Conductive Hearing Loss).

1. *External auditory canal obstructions*: Wax and inflammatory swelling.
2. *Tympanic membrane perforation*: Traumatic, acute suppurative otitis media (ASOM) and chronic suppurative otitis media (CSOM).
3. Eustachian tube obstruction
4. *Middle ear fluid*: Due to otitis media with effusion.
5. *Ossicles fixation*: Otosclerosis, tympanosclerosis and congenital.
6. *Ossicles disruption*: Due to disease necrosis and trauma.

Sensorineural Hearing Loss

The diseases of cochlea (sensory type) and CN VIII and its central connections (neural type) result in sensorineural hearing loss (SNHL). Retrocochlear HL refers to lesions of 8th nerve and central auditory connections. Some of the common examples include the following:

a. *Sensory (Cochlear)*: Presbycusis, mumps, noise-induced.
b. *Neural:* Neural presbycusis and acoustic neuroma.

Mixed Hearing Loss

It has elements of both conductive and SNHL. The air-bone gap in PTA indicates conductive element while impairment of bone conduction indicates SNHL. The most common causes of mixed hearing loss (MHL) are otosclerosis and CSOM.

NEED OF HEARING EVALUATION

The hearing assessment facilitates in finding out the following features of hearing loss:

- *Type of hearing loss*: Conductive, sensorineural or mixed.
- *Severity of hearing loss*: Mild, moderate, moderately severe, severe, profound or total.
- *Site of lesion*: The diseases of external ear, tympanic membrane and middle ear result in CHL. Clinical examination and different types of audiometry are helpful in finding the site of lesion. Special hearing tests can differentiate between different types of SNHL such as cochlear, retrocochlear and central.
- *Cause of hearing loss:* Detailed history and laboratory investigations will indicate towards the causes of hearing loss which may be congenital, traumatic, infective, neoplastic, degenerative, metabolic, ototoxic, vascular or autoimmune process.

METHODS OF HEARING EVALUATION

- The history taking and complete ear, nose, throat, head and neck examination and general evaluation of other systems are mandatory before performing the different types hearing tests (Box 1).

Box. 1: Different types of hearing tests

1. *Traditional Screening Tests*
   a. Watch test
   b. Finger friction tests
   c. Voice Tests (conversation and whisper)
2. *Tuning Fork Tests (TFT)*
   a. Air conduction
   b. Bone conduction
   c. Rinne test
   d. Weber test
   e. Absolute bone conduction (ABC)
   f. Schwabach's test
   g. Bing test
   h. Gelle's test
3. *Audiometric Tests*
   a. Pure tone audiometry (PTA)
      i. Air and bone conduction thresholds
      ii. Recruitment
         - Alternate binaural loudness balance test for recruitment
         - Short increment sensitivity index (SISI) test
      iii. Tone decay test
   b. Speech audiometry
      i. Speech reception threshold (SRT)
      ii. Discrimination score
   c. Impedance audiometry
      i. Tympanometry
      ii. Acoustic reflex measurement
4. *Evoked Response Audiometry*
   a. Brainstem Evoked Response Audiometry (BERA)
   b. Electrocochleography (EcoG)
5. *Otoacoustic emissions*
6. *Central auditory tests*
7. *Hearing evaluation in infants and children* (chapter Hearing Impairment in Infants and Young Children)

- The most commonly employed methods of hearing evaluation are tuning fork tests (TFT: Rinne, Weber, absolute bone conduction and air conduction) and PTA. Other methods, which are routinely done in otology clinics, include speech audiometry, impedance audiometry, brainstem evoked response audiometry (BERA) and otoacoustic emission.
- Usually methods of hearing evaluation are classified into two groups subjective and objective.
  - Subjective methods which need patient's co-operation and reliability include TFT, PTA and speech audiometry.
  - Objective methods which can be done even in children include impedance, electrocochleography (EcoG), BERA and otoacoustic emission.
- Evoked response audiometry measures the electrical activity in the auditory pathways in response to auditory stimuli. It requires special equipment with an averaging computer. There are several types of evoked response audiometry, but the two, which have gained clinical acceptance, are EcoG and BERA.
- Central auditory tests find the defects in the central auditory pathways and the temporal cortex. The test signals can be delivered to either one ear (monotic) or both the ears (dichotic). "Staggered spondaic words" test is widely employed. These tests are not used routinely and are beyond the scope of this book. The hearing evaluation methods employed only in infants and children are discussed in chapter Hearing Impairment in Infants and Young Children.

Traditional Screening Tests

These tests were popular for screening before the audiometric era but are now practically obsolete.

- Finger friction test is a rough but quick method of screening. Rubbing or snapping the thumb and a finger close to patient's ear give rough idea of patient's hearing.
- In watch test a clicking watch is brought close to the ear and the distance is measured.

Voice Tests

Conversation and Whisper

A normal person hears conversational voice at 12 meters (40 feet) and whisper (with residual air after normal expiration) at 6 meters (20 feet).

- *Method*: The test is done in quiet surroundings. The patient stands at a distance of 6 meters while his test ear faces toward the examiner. Patient's eyes are closed to avoid lip-reading. The non-test ear is occluded with intermittent pressure on the tragus by an assistant. The spondee words (baseball, sunlight, day-dream) or numbers with letters (AB4, X3B) are spoken by the doctor while s/he is gradually walking towards the patient. The distance at which conversational voice and the whispered voice are heard is measured.
- *Disadvantage*: Lack of standardization in intensity and pitch of voice and the level of ambient noise.
- *Interpretations*: 6 meters (20 feet) distance is taken as normal for both conversation and whisper.

TUNING FORK TESTS

Tuning Forks

A tuning fork (TF) consists of a base (footplate), stem and two prongs. The tuning forks of 256, 512, 1,024 and 2,048 Hz[2] (cycles/second) are used. The tuning fork of 512 Hz is most important. The lower frequency TF produce sense of bone vibration while higher frequencies have shorter decay time.

Methods

1. *Charging*: A tuning fork may be charged by striking against the examiner's elbow, rubber heel of the shoe or special rubber pad. TF should be sounded gently. Overtones with excessive vibration lead to inaccurate results.
2. *Air conduction*: For testing air conduction (AC), a charged TF is placed vertically, about 2 cm away from the opening of EAC. The sound waves are conducted via EAC, tympanic membrane and middle ear ossicles to the inner ear. Thus, the AC tests function of both the conducting mechanism and the cochlea.
3. *Bone conduction*: For testing bone conduction (BC), the base of charged TF is placed firmly on the mastoid bone. In this way cochlea is stimulated directly by vibrations, which are conducted via skull bones. Thus, the BC tests only the cochlear function.
4. *Tests*: Normally AC is louder and heard twice longer than the BC. The most important TFT are Rinne, Weber and ABC (Table 2).

Rinne Test

- *Principle*: The Rinne test compares AC with BC.
- *Method*: The base of a charged TF is placed on the mastoid and when patient stops hearing, TF is brought near the EAC. If patient still hears then AC is said to be more than BC. Alternatively, patient is asked to compare the loudness of sound of AC and BC (Figs 1A and B).

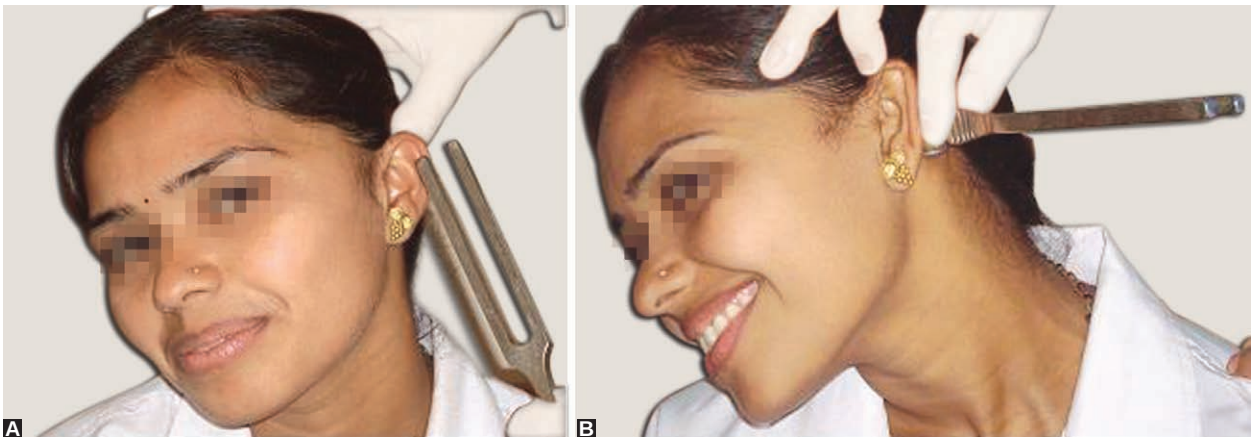
Interpretations

1. *Rinne positive*: Rinne test is said to be positive when AC is longer or louder than BC. The positive Rinne (AC>BC) is seen in cases of normal hearing and SNHL.

TABLE 2 The interpretation of tuning fork tests (TFT)

Tests	*Normal*	*Conductive hearing loss*	*Sensorineural hearing loss*
Rinne	AC > BC (Rinne +)	BC > AC (Rinne -)	AC > BC (Rinne +)
Weber	Center	Lateralized to affected ear	Lateralized to better ear
Absolute bone conduction	Normal	Normal	Reduced
Schwabach	Normal	Lengthened	Shortened
Bing	Positive	Negative	Positive
Gelle	Positive	Negative (ossicular lesions)	Positive

[2] Hertz (a German scientist) is a unit of frequency of sound. 1 Hz = 1 cycle/second



Figs 1A and B: Rinne Test. It compares the loudness of sound of air conduction (A); Bone conduction (B)

A negative Rinne with 256, 512 and 1,024 Hz shows air-bone gap of approximately 15, 30, 45 dB respectively.

2. *Rinne negative*: A negative Rinne (BC > AC) is seen in cases of conductive hearing loss.
3. *Rinne equivocal*: If AC is equal to BC, it indicates mild CHL.
4. *False Rinne negative*: In false negative Rinne (seen in profound SNHL), patient does not hear any sound of AC but responds to only BC. This response to only BC can be either from the opposite normal ear (through transcranial transmission) or patient considers vibration as sound. The masking of the non-test ear with Barany's noise box while testing for bone conduction usually solves the problem. Weber test will help in detecting the false Rinne negative.

If Rinne is negative in one ear and Weber does not lateralize to that ear it means that test ear has severe to profound SNHL.

Masking

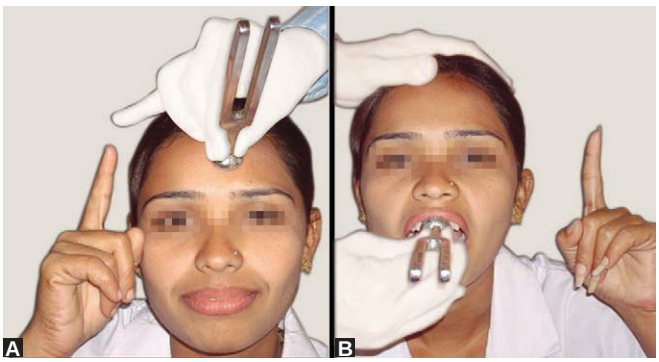
In masking, a continuous noise is presented to the better hearing ear (non-test ear) to prevent crossing over of sound from the worst ear (test ear) to the better ear. The masking of the non-test ear avoids the problem of false Rinne negative. Following types of noise may be used for masking:

1. Barany's noise box generates noise and masks the better ear while testing the hearing loss.
2. Noise with piece of paper.
3. Narrow-band noise during audiometry.

Weber Test

- *Principle*: BC of both ears is compared simultaneously. The sound travels directly to the cochlea via bone.
- *Method*: The base of a charged TF is placed in the center of the forehead or upper central incisors (Figs 2A and B). The patient is asked to tell on which side s/he hears better. The Weber test is said to be lateralized on the side on which patient hears well.

Weber test is quite sensitive as difference of only 3–5 dB hearing level can result in lateralization. Weber test readily detects the false Rinne negative.



Figs 2A and B: Weber Test compares bone conduction of both ears simultaneously. The base of a charged TF is placed in the center of the forehead (A); or upper central incisors (B). Patient raises finger on the side of better hearing (lateralization)

Interpretations

1. *Normal*: Normal person hears equally in both ears or s/he may not hear in either ear.
2. *Conductive hearing loss*: The test is lateralized toward the affected ear. If both the ears have CHL, Weber is lateralized to worst ear.
3. *Sensorineural hearing loss*: The test is lateralized towards the better ear.
4. *Degree of hearing loss*: Lateralization with TF 512 Hz implies a CHL of 15–25 dB in ipsilateral ear or a SNHL in the contralateral ear.

Tuning fork tests: Weber and Rinne tests are important and confirm the diagnosis of CHL.

Weber test: It detects a difference in hearing levels between two ears. It is more sensitive in diagnosing CHL.

Absolute Bone Conduction (ABC) Test

- *Principle*: The ABC test compares the duration of patient's bone conduction with that of the examiner (presuming doctor's hearing to be normal).
- *Method*: EAC of both the patient and examiner are occluded to prevent ambient noise entering into the EAC. The charged TF is put on the mastoid. When patient stops hearing, TF is transferred to the examiner's mastoid.

Interpretations

1. *Normal and conductive hearing loss*: Equal ABC of both patient and doctor.
2. *Sensorineural hearing loss*: Examiner's ABC is longer than the patient's ABC.

Schwabach's Test

In this TFT, BC of patient is compared with that of the normal hearing examiner but EAC is not occluded. In comparison to examiner, patient's duration of hearing is reduced in SNHL and increased in CHL.

Bing Test

Bing test examines the effect of occlusion of ear canal on the BC hearing. The base of a charged TF is kept on the mastoid while the examiner alternately closes and opens the EAC by pressing on the tragus.

In cases of normal hearing and SNHL, patient hears louder when EAC is occluded and softer when the EAC is open (Bing positive). In case of CHL, patient will appreciate no change in hearing (Bing negative).

Gelle's Test

Gelle's test examines the effect of increased air pressure in ear canal on the BC hearing. The TF is placed on the mastoid. The air pressure is increased in the EAC by Siegel's speculum, which pushes the ossicles inward and raises the middle ear pressure that results in decreased hearing (Gelle's positive) due to the immobility of basilar membrane.

There is no change in hearing (Gelle's negative) when ossicular chain is fixed (otosclerosis, tympanosclerosis) or disconnected (traumatic). Positive Gelle's test is seen in cases of normal hearing and SNHL. It is a good test when the facility of impedance audiometry is not available.

PURE TONE AUDIOMETRY

This non-invasive subjective test is a graphic recording of hearing level both quantitatively and qualitatively. An audiometer is on electronic device that generates pure tones. The intensity of these tones are either increased or decreased in 5 dB steps. The audiometer is so calibrated that the hearing of a normal person is at zero dB[3] level.

The AC thresholds are measured usually for tones of 250, 500, 1,000, 2,000, 4,000 and 8,000 Hz. The BC thresholds are measured usually for 250, 500, 1,000, 2,000 and 4,000 Hz. The speech frequencies range from 500–2,000 Hz[4].

- *Pure tone average:* The pure tone average is an average of the AC thresholds at 500, 1,000, and 2,000 Hz speech frequencies.
- *Air-bone gap*: The BC thresholds are the measure of cochlear function. The difference between the thresholds of AC and BC called air-bone gap (A-B gap) is a measure of CHL.
- *Decibel*: A decibel is an arbitrary logarithmic unit of intensity of sound. Hearing is measured on a biologic scale in dB HL. The environmental sounds are measured on a physical scale in dB SPL. In humans, normal hearing at 125 Hz is about 45 dB SPL while at 1,000 Hz it is about 7 dB SPL. That means normal human ear is able to hear high frequencies better than the low frequencies. A "0 dB HL" on the audiogram is simply a reference level, which represents normal hearing across the entire frequency spectrum 250–8,000 Hz.

Method

Audiometry is done in a soundproof room (ideal) or a quite room. First AC and then BC is recorded separately for each ear. The pure tones are presented to the ears by headphone (for AC) and vibrator (for BC). The graph on which these thresholds are charted is called audiogram. For right ear red color and for left ear blue color pencils are used. For AC continuous line and for BC interrupted (broken) line is used for recording (Figs 3 and 4).

Masking

Air conduction sounds "crossover" occurs when a 50 dB difference exists between AC thresholds of two ears. BC sounds "crossover" may occur even at 0 dB difference between the BC thresholds of the two ears. The narrow-band noise in the non-test ear (better hearing) is employed for the masking. When difference between the AC of two ears is 40 dB or more, the better ear is masked. Masking avoids getting a shadow curve from the better ear. Masking of the non-test ear is necessary during the BC studies.

Uses of PTA

Uses of the PTA include the following:

1. Measure of thresholds of AC and BC tells the degree and type of hearing loss (mild-profound, CHL/SNHL).
2. The progress of the disease and response to the treatment can be documented (improving/progressive).
3. The type and necessary setting of hearing aids can be determined.

Fig. 3: Pure tone audiometry (PTA). See the green colored audiometer on the table

[3]The intensity of sound is measured in logarithmic units called decibel (dB), which is the smallest change in the intensity of the sound that can be appreciated by a normal human ear.

[4]The audible range of frequency is 20–20,000 Hz

Fig. 4: Pure tone audiometry (PTA) in a soundproof audiometry room

4. The degree of handicap for medicolegal purposes can be determined.
5. Speech reception threshold (SRT) can be predicted.

Interpretations

1. *Normal*: AC and BC threshold closely follow in the range of 0–20 dB.
2. *Conductive hearing loss*: BC is normal but AC is reduced (especially in lower frequencies). It gives wide gap between AC and BC called air-bone gap (A-B gap) (see Fig. 3 of ch. 12).
3. *Sensorineural hearing loss*: Both AC and BC are reduced specially in higher frequencies (see Figs 2 to 5 of ch. 13).
4. *Mixed hearing loss*: Both AC and BC are reduced but AC is more reduced than BC that results in air-bone gap.
5. *Severity of hearing loss*
   a. Normal: Hearing between 10–20 dB
   b. Mild: Hearing loss between 20–40 dB
   c. Moderate: Hearing loss between 40–60 dB
   d. Severe: Hearing loss between 60–80 dB
   e. Profound: Hearing loss more than 80 dB

Recruitment

- In this phenomenon of abnormal appreciation of loud sounds, a loud sound which is tolerable in normal ear may grow to abnormal levels of loudness in the recruiting ear and thus becomes intolerable. So though the patient has hearing loss he cannot tolerate loud sounds.
- These patients are poor candidates for hearing aids.
- Recruitment is a feature of cochlear hearing loss (Meniere's disease, presbycusis). It is absent in normal, CHL (external and middle ear diseases) and nerve hearing loss (acoustic neuroma).
- The common tests for recruitment are Fowler's alternate binaural loudness balance test (ABLB) and short increment sensitivity index (SISI) test.

Fowler's Alternate Binaural Loudness Balance Test

It is done in a case of unilateral hearing loss. A tone of 1,000 Hz is played alternately to the normal and the affected ear. The intensity in the affected ear is adjusted to match the loudness in normal ear. The test begins at 20 dB above the threshold and is repeated at every 10 dB rise until the loudness is matched or the limits of audiometer reached.

Interpretations

- *Normal hearing, conductive and nerve hearing loss*: The initial difference is maintained throughout
- *Cochlear lesions*: They show partial, complete or over-recruitment

Short Increment Sensitivity Index Test

Due to the recruitment patients of cochlear lesions can distinguish smaller changes in intensity of pure tone better than patients of normal hearing, conductive and nerve hearing loss.

- *Method*: A continuous tone is delivered 20 dB above the threshold and sustained for about 2 minutes. At every 5 seconds the tone is increased by 1dB. Twenty such blips are delivered. Patient is asked to indicate the blips heard. SISI score is presented in percentage.

Interpretations

- *Score less than 20%*: Normal hearing and conductive and nerve hearing loss.
- *Score more than 70%*: Cochlear hearing loss.

Carhart's Tone Decay Test

This simple test is a measure of nerve fatigue, which is a feature retrocochlear hearing loss.

- *Principle*: A normal person can hear a tone continuously for 60 seconds. In nerve fatigue, patient stops hearing earlier.
- *Method*: A tone of 4,000 Hz is delivered at 5 dB above the patient's threshold for 60 seconds. When patient stops hearing, intensity is increased each time by 5 dB. The procedure is continued till patient hears the tone continuously for 60 seconds or tone's upper limit is reached.
- *Interpretation*: A tone decay of more than 25 dB is diagnostic of a retrocochlear hearing loss.

SPEECH AUDIOMETRY

The patient's ability to hear and understand the speech is measured in speech audiometry. The two parameters studied are: SRT and speech discrimination (SD) score (Table 3).

Speech Reception Threshold

Speech reception threshold is the minimum intensity at which 50% of spondee words are repeated correctly.

TABLE 3 Relation between the Speech Discrimination (SD) Score and Ability to Understand Speech.

SD score	*Ability to understand speech*
90–100%	Normal
76–88%	Slight difficulty
60–74%	Moderate difficulty
40–58%	Poor
< 40%	Very poor

- *Spondee words*: These are two syllable words with equal stress on each syllable such as oatmeal, popcorn, shipwreck, blacknight, blackboard, football, eardrum, sunset, and daydream.
- *Method*: A set of spondee words is delivered (in the form of either recorded tapes or monitored voice) to each ear of the patient. The word lists are delivered through the headphone of an audiometer. The intensity of spondee words are changed in 5 dB steps till half of them are correctly heard.
- *Interpretations*: The SRT is normally within the range of 10 dB of the average of pure tone threshold of three speech frequencies (500, 1,000 and 2,000 Hz).
  - In cases of hearing loss, SRT is more than 10 dB better than pure tone average.

Speech Discrimination Score or Speech Recognition Score

Discrimination score is a measure of patient's ability to understand speech.

- *Phonetically balanced words*: Phonetically balanced (PB) words are single syllable words such as fish, dish, pin, sin.
- *Method*: The PB words are delivered through the headphone to each ear at an intensity 30–40 dB above the SRT. A list of 50 PB words is presented and the number correctly heard is multiplied by 2 (Table 3).
  - DS is the percentage of words correctly heard by the patient.

Interpretation:

1. *Normal hearing*: SD score of 95–100%
2. *Conductive hearing loss*: SD score 90–100% but at higher intensities.
3. *Sensorineural hearing loss*: SD score is less. Nerve hearing loss has very poor score in comparison of cochlear hearing loss.

Modifications

There is another better method of speech audiometry, in which percentage of PB words correctly heard by the patient at different intensity levels are charted on a graph. The following three parameters are ascertained:

a. *Optimum discrimination score*: Optimum discrimination score (ODS) is the highest score irrespective of the intensity at which PB words are delivered.
b. *Half peak level*: Half peak level (HPL) represents the intensity at which 50% of the words are expected to be heard (half ODS). This is a derived figure from the above graph.
c. *Roll over curve*: This shape of the speech audiogram is typical of retrocochlear lesion. The ODS is maintained as a horizontal line or show a drop with increase in intensity levels.

Uses

The following are some of the uses of speech audiometry:

- Differentiate between an organic hearing loss and the functional one.
- The intensity level for the best discrimination score is useful for fitting and setting of hearing aid.
- Differentiate between cochlear and retrocochlear lesions.

IMPEDANCE AUDIOMETRY

It measures the impedance (resistance), which is offered by the conducting mechanism of tympanic membrane and middle ear and their compliance (suppleness) to sound pressure transmission. This objective method of audiometry consists of tympanometry and acoustic reflex measurements.

Tympanometry

- *Principle*: When a sound hits tympanic membrane, some of the sound energy is absorbed while the rest is reflected. A stiffer tympanic membrane reflects more of sound energy than a compliant one. The pressure in a sealed EAC is changed. The reflected sound energy is measured to find the compliance or stiffness of the tympano-ossicular system. The compliance of tympano-ossicular system against various pressure changes is charted. In this way the status of healthy or diseased middle ear is found.
- *Equipment*: It mainly consists of a probe which has three channels (Figs 5 and 6) with following functions. The probe snugly fits into the EAC.
  1. Deliver a tone of 220 Hz
  2. A microphone, which picks up the reflected sound

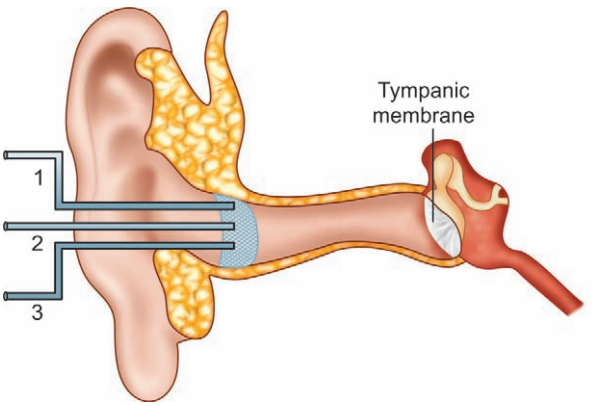


Fig. 5: Probe of impedance audiometer showing three channels. 1, Oscillator producing tone; 2, Air pump for increasing and decreasing of air pressure; and 3, Microphone that picks up and measures reflected sound pressure level

Fig. 6: Impedance audiometry in process. Note impedance audiometer lying on the table and the patient with ear probe with headband

3. Bring changes in air pressure in EAC from positive to normal and then negative

- *Interpretations*: Tympanometry results are represented by air pressure/compliance graphs, which are called tympanograms that are diagnostic of certain middle ear pathologies (Table 4). The compliance of ear drum is maximized when air pressure on both sides is equal. The peak air pressure of tympanogram is equal to middle ear pressure.
- The middle ear pressure with normal Eustachian tube function is between 0–150 mm H_2O.
- The middle ear pressure more negative than 150 mm H_2O suggests poor Eustachian tube function.
- *Eustachian Tube and Grommet Testing (Inflation Deflation test)*: In cases of intact or perforated tympanic membrane, tympanometry can be done for testing function of Eustachian tube and patency of the grommet. Grommet is placed in the tympanic membrane for the treatment of serous otitis media.

A negative and a positive pressure (200 or +200 mm of H_2O) is created while patient is asked to swallow 5 times in 20 seconds. The ability to equilibrate the pressure suggests normal Eustachian tube function and patency of the grommet.

Acoustic Reflex

- *Principle and method*: A loud sound (70–100 dB above the threshold of hearing) causes bilateral (ipsilateral and contralateral) contraction of the stapedial muscles which are detected by tympanometry. Tone is delivered to one ear and the reflex is detected from the same and the contralateral ear.
- *Acoustic reflex neural pathways* (Fig. 7): Though the majority of neurons run through ipsilateral pathway, some crosses the brainstem to continue to the opposite cochlear nucleus. The pathway begins at cochlea and proceeds through CN VIII, cochlear nucleus, trapezoid body, superior olivary complex, and motor facial nucleus to the stapedial muscle.

TABLE 4 Different types of tympanogram

Type	*Characters of graph*	*Diseases of middle ear*
Type A	Normal Curve	Normal middle ear function
Type As	Normal curve with low compliance at or near normal ambient air pressure	Fixation of ossicles such as otosclerosis, malleus fixation
Type Ad	High compliance at or near normal ambient pressure	Ossicular discontinuity, thin and lax tympanic membrane
Type B	A flat or dome-shaped curve. No change in compliance with pressure changes	Middle ear fluid
Type C	Maximum compliance (peak) at negative pressure range < 150 mm of H_2O	Retracted tympanic membrane, negative pressure or some fluid in middle ear

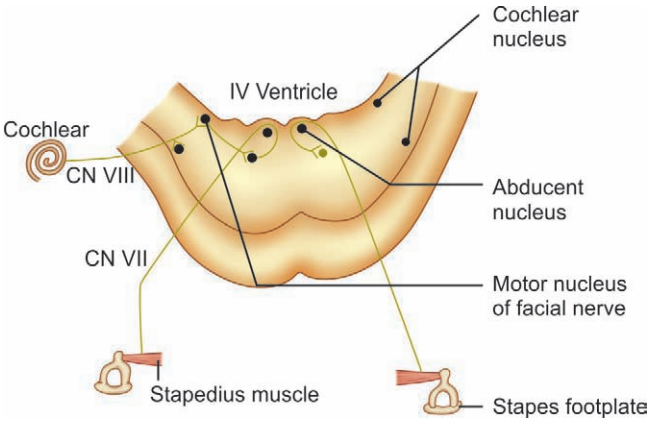


Fig. 7: Cut section of the pons showing acoustic reflex pathways

- *Uses*: The following are some of the uses of this objective method of acoustic reflex measurement:
  a. Evaluation of hearing in infants and young children.
  b. Malingerers can be identified as they will show positive stapedial reflex.
  c. Differentiation between cochlear and nerve hearing losses.
     1. *Cochlear lesions*: Presence of stapedial reflex at lower intensities 40–60 dB (normally 70 dB) above the threshold of hearing due to recruitment.
     2. *8th nerve lesion*: A sustained tone of 500 or 1,000 Hz, 10 dB above acoustic reflex threshold, for 10 seconds, brings the reflex amplitude to 50%. This abnormal adaptation is due to stapedial reflex decay.
  d. Facial nerve palsy: Absence of stapedial reflex in normal hearing ear indicates that the site of lesion of the facial nerve palsy is proximal to the nerve to stapedius. The appearance of reflex in cases of facial nerve palsy indicates return of function and a favorable prognosis.

ELECTROCOCHLEOGRAPHY

Electrocochleography (EcoG) measures electrical potentials, which arise in cochlea and CN VIII in response to auditory stimuli within first 5 milliseconds. It consists of following three types of responses (see chapter Anatomy and Physiology of Ear)

1. Cochlear microphonics
2. Summating potentials
3. Action potential of 8th nerve

Method

The recording electrode (a thin needle) is placed on the promontory through the tympanic membrane. The test can be done under local anesthesia however children and anxious uncooperative adults need sedation or general anesthesia, which has no effect on EcoG responses.

Uses

- Detection of hearing threshold in young infants and children to within 5–10 dB
- Differentiate cochlear lesions from of the 8th nerve lesions.

BRAINSTEM EVOKED RESPONSE AUDIOMETRY

[Auditory Brainstem Response (ABR) or Auditory Nerve and Brainstem Evoked Potentials (ABEP)]

Brainstem evoked response audiometry is a non-invasive technique that finds the integrity of central auditory pathways, which consists of 8th nerve, pons, midbrain, and forebrain. The electrical potentials are generated in response to several click stimuli. They are picked up from the vertex by surface electrodes. BERA measures hearing in the range of 1,000–4,000 Hz. Seven waves are generated in the first 10 milliseconds (Table 5).

The first, third and fifth waves (among 5 waves shown in roman numbers I, II, III, IV, and V) are most stable and studied for absolute latency, inter-wave latency and the amplitude (Figs 8 and 9). The exact anatomic areas of origin of waves though disputed are, shown in Table 5. BERA finds out the threshold of hearing in infants, which are in high risk groups. It aids in the diagnosis of retrocochlear lesions.

For its role in the hearing impairment in infants and children see chapter on Hearing Impairment in Infants and Children.

OTOACOUSTIC EMISSIONS

Otoacoustic emissions (OAE) are low-intensity sounds, which are produced by movements of the outer hair cells of the cochlea. They are produced spontaneously and in response to the acoustic stimuli. OAE are picked up by a miniature microphone, which is placed snugly in the EAC. Absence of OAE indicates disorders of outer hair cells.

This non-invasive objective test can diagnose damage to the outer hair cells due to acoustic trauma and ototoxic drugs. It aids in the assessment of hearing in infants. Sedation does not interfere with OAE.

The OAE travels through basilar membrane, perilymph, oval window, ossicles, tympanic membrane, and ear canal. OAE are present in nerve hearing loss as the outer hair cells are normal.

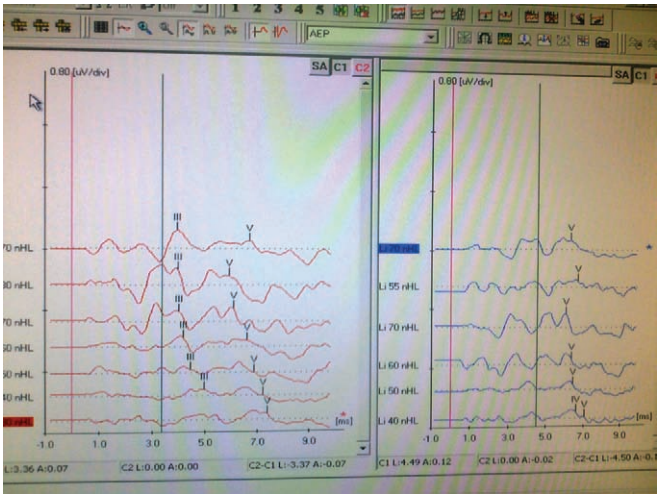


Fig. 8: Normal BERA with normal latency and click stimulus from 80 to 40 dBnHL
Source: Dr Amit Anand and Dr Hemant Shah, Consultant Audiologists, Anand and Ahmedabad, Gujarat

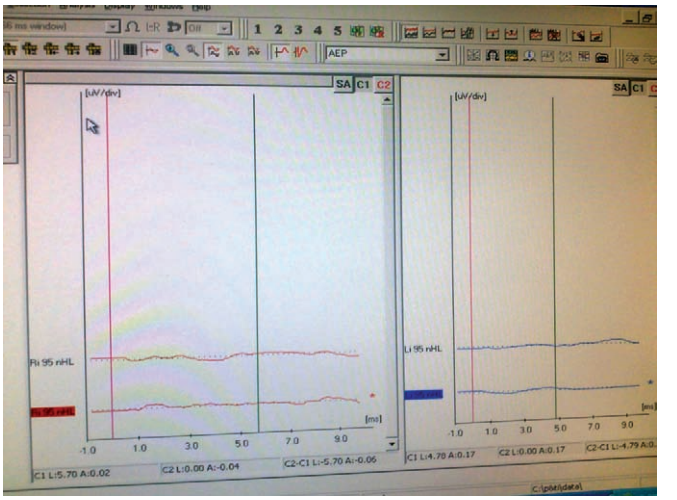


Fig. 9: BERA Severe hearing loss. No peaks seen at 95 dBnHL
Source: Dr Amit Anand and Dr Hemant Shah, Consultant Audiologists, Anand and Ahmedabad, Gujarat

TABLE 5 Brainstem evoked response audiometry (BERA): Waves and their sites of origin

Wave I	VIII CN
Wave II	Cochlear nuclei (pons)
Wave III	Superior olivary complex (pons)
Wave IV	Lateral lemniscus (pons)
Wave V	Inferior colliculus (mid brain)

Types

1. *Spontaneous otoacoustic emissions*: They are present in normal hearing ear or when hearing loss does not exceed 30 dB. It may be absent in about 50% of the normal persons.
2. *Evoked otoacoustic emission*: Evoked otoacoustic emission (EOAE) are acoustic signals generated by cochlear outer hair cells in response to auditory stimulation. EOAE measures only cochlear status and is independent of neural activity and CNS status. EOAE takes lesser time and uses broader frequency range than click evoked ABR.
   - Transient evoked otoacoustic emission: Transient evoked otoacoustic emission (TEOAE) provides information over a broad frequency range (500–6000 Hz) that occurs after a brief stimulus. A series of click stimuli are presented at 80–85 dB SPL. TEOAE are observed in neonatal ears in the absence of external and middle ear disorders.
   - Distortion evoked otoacoustic emission: Distortion evoked otoacoustic emission (DEOAE), which provides frequency specific information, occurs in response to simultaneous presentation of two pure tones. The screening algorithms are robust in neonates and infants and use "DP grams."

Uses

- Screening test of hearing in neonates, uncooperative or mentally challenged patients.
- Distinguish between cochlear (acoustic trauma and ototoxic drugs) and retrocochlear hearing losses (auditory neuropathy).

AUDITORY STEADY STATE RESPONSE (ASSR)

This is multiple auditory steady evoked response (MASTER) acquisition system. ASSR is an objective PTA, which assesses only AC. It generates multiple frequency modulated auditory stimulus and acquires electrophysiological responses to these stimulus.

Method

- The continuous sinusoid stimulus, which is amplitude or frequency modulated at relatively slow rates, is delivered to the ear via insert phones.
- The electrodes on the patient's head record the electroencephalographic activity, which is averaged by the system.
- The ASSR is converted to the frequency domain using fast Fourier transform (FFT) techniques. The physiological audiogram obtained by MASTER shows air-conduction hearing threshold at each frequency (Fig. 10).
- It needs quite patient and good control of noise. In cases of uncooperative patients and children it is done under sedation/general anesthesia because similar to ABR/BERA the patient's movements are not allowed.

Advantages

1. The frequency spectrum of the stimulus is considerably narrower than the tone burst of ABR/BERA.
2. Due to continuous stimulus it is possible to achieve higher stimulation level than clicks or tone bursts (transient stimulus), which are used in ABR/BERA. Therefore it is easier to distinguish between severe and profound hearing loss. So it is possible to record ASSR in ears (especially in cochlear implant candidates) with no measurable ABR.
3. Analysis is easier than BERA and does not need much training.
4. ASSR frequency-specific thresholds correlate with audiometric thresholds in both children and adults.

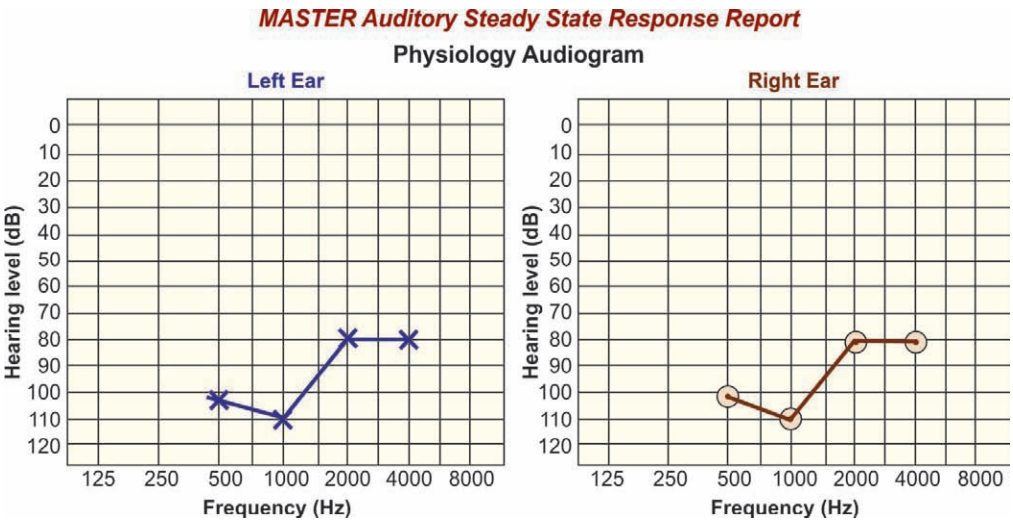


Fig. 10: ASSR Study. Severe to profound sensorineural hearing loss (SNHL)
Source: Dr Amit Anand and Dr Hemant Shah, Consultant Audiologists, Anand and Ahmedabad, Gujarat

Clinical Highlights

1. **Overtones:** Quality of sound depends on overtones.
2. **Speech frequencies:** They include 500, 1,000, and 2,000 Hz. Vowels are low frequency sounds while consonants are high frequency sounds.
3. **dB and sound energy:** dB SL is a unit for threshold of hearing in an audiogram. A sound of 20 dB is 100 fold increase in sound energy.
4. **Recruitment:** Patients with recruitment have intolerance (increased sensitivity) to loud sounds. It is seen in cochlear lesions.
5. **Tone decay (Auditory fatigue):** This is the change in auditory threshold when a continuous tone is presented. It is seen in retrocochlear lesions such as acoustic neuroma.
6. **Retrocochlear hearing loss:** Lesions of cochlear nerve and nucleus will cause ipsilateral profound sensorineural hearing loss. A lesion at other part of central hearing pathway (superior olive, lateral lemniscus, inferior colliculus, medial geniculate body, auditory radiations, and primary auditory cortex in temporal lobe), which represent higher level of auditory processing, would result in mild bilateral hearing loss and decreased ability to localize a source of sound.
7. **Weber test:** External auditory canal obstruction and middle ear diseases cause conductive hearing loss. The tuning fork (usually of 512 Hz) placed on the bridge of the nose or center of forehead will be heard better (lateralization) on the side of conductive hearing loss. In Weber test external and middle ears are bypassed. Air and bone conduction interfere with each other on the normal side, making that ear less sensitive.

8. **Weber's test lateralization:** In external ear occlusion and middle ear diseases Weber's test will be lateralized towards the diseased ear. Right side lateralization denotes right ear conductive hearing loss or left ear sensorineural hearing loss.
9. **Rinne's test:** If Rinne's test is negative with a tuning fork (TF) of 512 Hz but positive for 1024 Hz, minimum predicted AB gap on audiometry would be 30 dB.
10. **Gelle's test:** It examines the effect of increased air pressure in ear canal on the BC hearing. The TF is placed on the mastoid. The air pressure is increased in the EAC by Siegel's speculum. Normally and in SNHL, increased middle ear pressure results in decreased hearing (Gelle's positive). There is no change in hearing (Gelle's negative) when ossicular chain is fixed (otosclerosis, tympanosclerosis) or disconnected (traumatic).
11. **Unilateral sensorineural hearing loss:** With 512 Hz tuning fork, the Rinne's test (without masking) may be false or negative but Weber's test will be lateralized towards normal ear.
12. **Decreased bone conduction:** In an audiogram it indicates the involvement of cochlea.
13. **Maximum conductive hearing loss:** It is approximately 54 dB and is caused by ossicular disruption with intact tympanic membrane. Ossicular disruption with perforated tympanic membrane results in about 38 dB hearing loss.
14. **Phonetically balanced words:** These are used to measure speech discrimination score.
15. **Spondee words:** These are used to measure speech reception threshold.
16. **Otoacoustic emissions:** They arise from outer hair cells.
17. **Brainstem evoked response auditory:** Interpretation is affected by the age of the child. It tracks electrical conductivity of hearing up to the brain-stem.
18. **Tuning fork tests:** They should confirm the results of audiometry. Never trust the audiogram alone when surgical intervention is considered. The inconsistency must be resolved with the audiologist.
19. **Types of hearing losses:** In addition to mixed hearing loss there are two major types of hearing loss: 1) conductive, and 2) sensorineural.
20. **Discrimination test:** It measures patient's ability to understand speech.

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12 Conductive Hearing Loss and Otosclerosis

It is better to do something; never mind even if it proves to be wrong; it is better than doing nothing. Purity, patience, and perseverance are the three essentials to success, and above all–love.

—Swami Vivekananda

Points of Focus

- **CLASSIFICATION OF HEARING LOSS**
  - ✓ Differences Between Conductive and Sensorineural Hearing Losses
- **CONDUCTIVE HEARING LOSS**
  - ✓ Etiology
  - ✓ History and Physical Examination
  - ✓ Treatment
- **OTOSCLEROSIS**
  - ✓ Pertinent Anatomy
  - ✓ Etiology
  - ✓ Types
  - ✓ Pathology
  - ✓ Clinical Features
  - ✓ Audiometry
  - ✓ Differential Diagnoses
  - ✓ Treatment
- **STAPEDECTOMY**
  - ✓ Selection Criteria
  - ✓ Contraindications
  - ✓ Anesthesia
  - ✓ Operative Steps
  - ✓ Postoperative Care and Follow-Up
  - ✓ Complications
- **CLINICAL HIGHLIGHTS**

CLASSIFICATION OF HEARING LOSS

The two broad categories (Box 1) of hearing loss are conductive hearing loss (CHL) and sensorineural hearing loss (SNHL). SNHL is further divided into sensory (cochlear) and neural (CN VIII and central auditory connections).

Sensorineural hearing loss and CHL have their own characteristic features. On the basis of history, examination, tuning fork tests and audiometry, usually they can be easily differentiated from each other (Table 1).

CONDUCTIVE HEARING LOSS

The disorders of external and middle ear up to stapediovestibular joint interfere with the conduction of sound and cause CHL (Table 2).

The ossicular disorders with intact tympanic membrane cause more hearing loss than ossicular diseases with tympanic membrane perforation.

Box 1: Classification of different types of hearing losses

A. Non-organic
   1. Malingering
   2. Psychogenic
B. Organic
   1. Conductive Hearing Loss (CHL)
   2. Sensorineural Hearing Loss (SNHL)
      a. Peripheral
         i. Cochlear (Sensory)
         ii. CN VIII (Neural)
      b. Central
         i. Brainstem (medulla, pons, midbrain)
         ii. Thalamus
         iii. Temporal lobe
   3. Mixed Hearing Loss

Etiology

The causes may be congenital, traumatic, infectious/inflammatory, neoplasms and miscellaneous. They may lie in external ear, tympanic membrane, middle ear space, ossicles or in Eustachian tube (Box 2).

TABLE 1 Differences between conductive hearing loss and sensorineural hearing loss

Features	*Conductive hearing loss*	*Sensorineural hearing loss*
Speech understanding	Good	Poor
Intolerance to loud sounds	Absent	Present in cochlear lesions
Speech of the patient	Low voice	Loud voice
Paracusis willisii	Common	Absent
Common associated symptom	Otorrhea/earache	Tinnitus
Profound hearing loss	Never	Common
Rinne test	Negative (BC > AC)	Positive (AC > BC)
Weber test	Lateralized toward worst ear	Lateralized toward better ear
Absolute bone conduction	Normal	Reduced
Pure Tone Audiometry (PTA) Air-bone gap	Present	Absent
PTA: Recruitment	Absent	Present in cochlear lesions
PTA: Tone decay	Absent	Present in CN VIII lesion
PTA: Frequencies	Usually low tones involved	Usually higher tones involved
PTA: Thresholds	Never > 60–70 dB	Can be > 60–70 dB
Speech discrimination	Not affected	Poor
Site of lesion	External and middle ear	Internal ear, CN VIII and central auditory connections

TABLE 2 Common causes of conductive hearing loss and their approximate hearing loss in dB

Occlusion of external auditory canal (EAC)	30–40 dB
Perforation of tympanic membrane. Hearing loss depends on the size and site of perforation	10–40 dB
Ossicular interruption without eardrum perforation (Fixity and discontinuity)	54 dB
Ossicular interruption with eardrum perforation (Fixation and discontinuation)	38 dB
Malleus fixation	10-25 dB
Closure of oval window	60 dB
Pure and complete conductive hearing loss	60–70 dB

Box 2: Causes of conductive hearing loss

- *External auditory canal*: Wax, foreign bodies, otitis externa, congenital and acquired stenosis, exostoses, osteomas, tumors, cyst
- *Tympanic membrane*: Perforations (traumatic, ASOM, CSOM), tympanosclerosis, retraction
- *Ossicles*: Fixation (otosclerosis, tympanosclerosis, adhesive otitis media, congenital); discontinuity (traumatic, inflammatory, cholesteatoma)
- *Middle ear*: Otitis media with effusion, hemotympanum, cholesteatoma, tumors (benign and malignant); Eustachian tube dysfunction

The most common causes include wax, otitis media [Acute suppurative otitis media (ASOM) and chronic suppurative otitis media (CSOM)], tympanosclerosis, and otosclerosis. In children, the most common cause is otitis media with effusion.

History and Physical Examination

In the evaluation of a CHL, following points should be considered during the history taking and physical examination keeping in mind the various causes of CHL (Box 2):

- *History*: Mode and age of onset; duration and progression of hearing loss; side of ear; associated symptoms of ear fullness, ear pain, otorrhea, tinnitus, vertigo/dizziness; preceding history of upper respiratory tract infection (URI), trauma, medications; visual, speech, or other neurologic deficit; family history.
- *Physical examination* (Fig. 1): Local examination of external ear, tympanic membrane, middle ear; pneumatic otoscopy; tuning fork tests; complete nose, throat, head and neck examination.
- See chapters (1) Otologic Symptoms and Examination; and (2) Hearing Evaluation.

Treatment

Most cases of CHL can be managed by medical and surgical treatment, which depend upon the cause of deafness. The details of the treatment of these conditions are given in their respective chapters but Table 3 briefly provides different modalities and their indications.

Different types of mastoid and tympanoplasty operations are described in chapter Middle Ear and Mastoid Surgeries.

OTOSCLEROSIS

Otosclerosis is a common disorder of bony labyrinth with normal tympanic membrane. It is characterized by gradually progressive CHL as a result of stapes fixation. Though there is no cure, the treatment of choice is stapedectomy.

Pertinent Anatomy

Labyrinth consists of three parts: membranous labyrinth, perilymphatic labyrinth and bony labyrinth. The bony labyrinth has three layers: endosteal, bony (enchondral) and periosteal (See chapter Anatomy and Physiology of Ear).

1. *Membranous labyrinth (Otic labyrinth or endolymphatic labyrinth):* Otic labyrinth consists of utricle, saccule, cochlear duct (scala media), semicircular ducts and endolymphatic duct and sac. It is filled with endolymph.

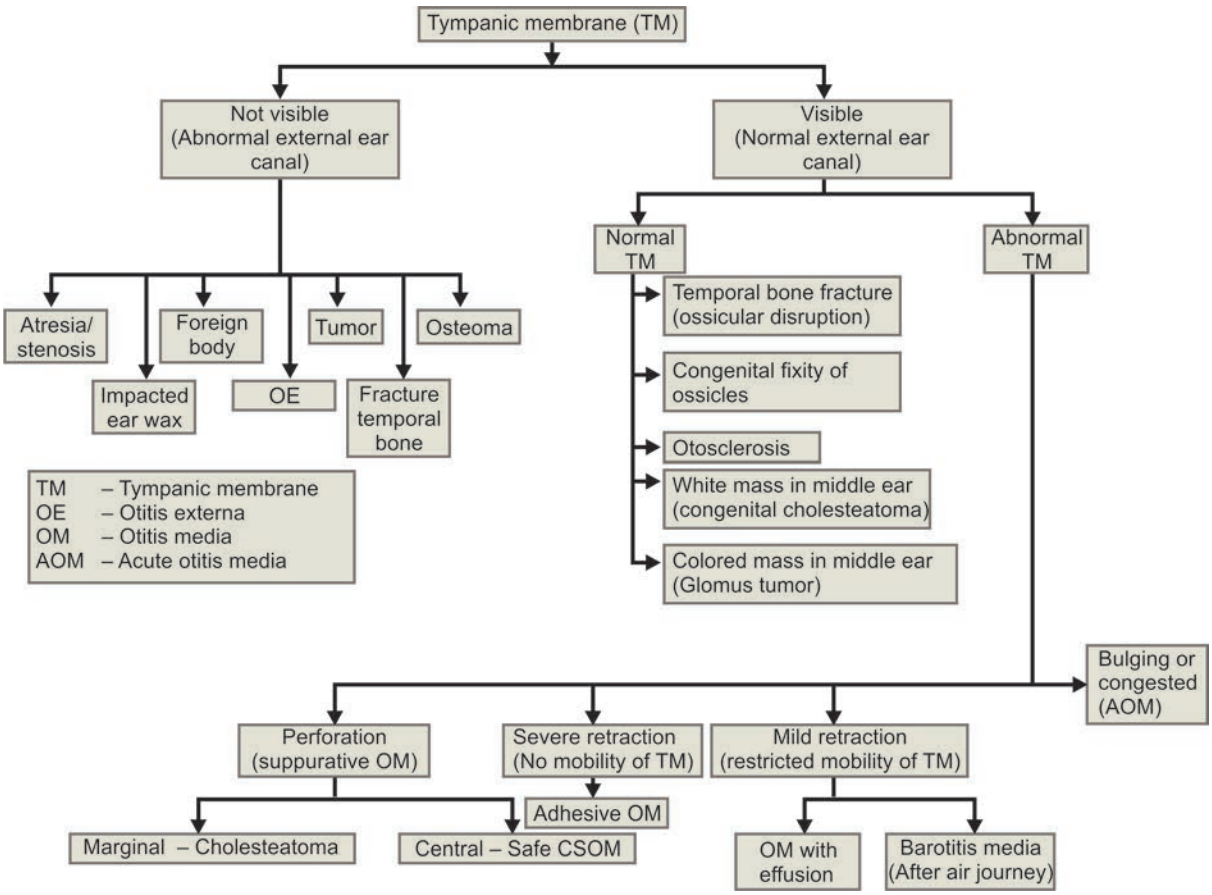


Fig. 1: Flow chart for clinical diagnoses of causes of conductive hearing losses

TABLE 3 Different modalities of treatment of conductive hearing loss and their indications

Removal of external auditory canal *canal occlusions*	Impacted wax, foreign body, osteoma, exostosis, keratosis obturans, tumors, meatal stenosis
Myringotomy	Acute otitis media
Grommet	Otitis media with effusion
Stapedectomy	Otosclerosis (fixation of stapes footplate)
Tympanoplasty	Tympanic membrane perforations and ossicular disruptions
Hearing aids	When surgery is not possible, refused or failed

2. *Perilymphatic labyrinth or space (Periotic labyrinth):* Periotic labyrinth surrounds the otic labyrinth and filled with perilymph. It consists of vestibule, scala tympani, scala vestibuli and perilymphatic spaces of semicircular and endolymphatic ducts.
3. *Bony labyrinth (Otic capsule):* It consists of three layers: endosteal, enchondral and periosteal.

The bony labyrinth ossifies from 14 centers. The first center appears in the cochlea at 16 weeks. The last center appears in the posterolateral part of posterior semicircular canal at 20th week.

I. Endosteal, which is the innermost layer, lines the internal surface of bony labyrinth.
II. Bony (enchondral) layer, which is subject to little change in life, develops from the cartilage.
   - Pathogenesis

- Otosclerosis is a primary disease of this enchondral bony labyrinth. In this hard enchondral bone, some islands of cartilage remain unossified. This cartilage rests due to certain nonspecific factors, are activated to form new spongy bone (otospongiosis).
- These irregular foci of spongy bone replace normal dense enchondral bony labyrinth. Therefore, many call this disease as otospongiosis. The otosclerotic focus usually involves the stapes region and results in stapes fixation and conductive deafness. The fissula ante fenestram, which lies in front of the oval window, is the site of predilection for stapedial type of otospongiosis. The otospongiosis process can involve other areas of bony labyrinth and can cause SNHL or remains asymptomatic.

III. Periosteal, which is the outermost layer, covers the external surface of bony labyrinth that is temporal bone.

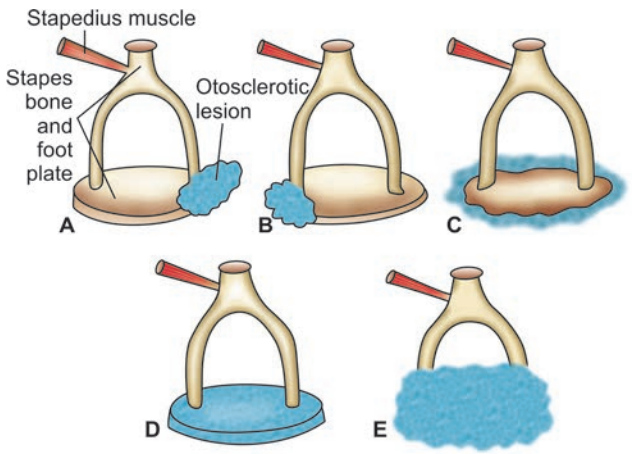
Etiology

The exact cause of the disease is yet not known. The following factors have been documented in the literature:

- *Heredity*: About 50% of the cases give positive family history. Remaining cases are sporadic. An autosomal dominant inheritance with penetrance in range of 20–40 has been reported. Other studies report heterogenetic, with more than one gene defect. Some cases have been suggested to be related with COL1A1 gene, which is one of the two genes that code for type I collagen (predominant collagen of bone).
- *Osteogenesis imperfecta*: About 50% cases of type I osteogenesis imperfecta develop hearing loss, histological changes and COL1A1 expression that are indistinguishable from otosclerosis. Patients of osteogenesis imperfecta have history of multiple fractures. The van der Hoeve syndrome presents with the triad of osteogenesis imperfecta, otosclerosis and blue sclera.
- *Viral*: Many reports suggest that otosclerosis may be related to a persistent measles virus infection of otic capsule. Perhaps it is similar to Paget's disease of bone, which is related with defective paramyxovirus.

Types

1. *Stapedial otosclerosis*: Stapedial otosclerosis is the most common variety. It causes stapes fixation and presents with conductive deafness (Figs 2A to E).
   a. Anterior focus: The fissula ante fenestram, which lies in front of the oval window, is the site of predilection.
   b. Posterior focus: Otosclerosis may start behind the oval window.
   c. Circumferential: Disease process spreads around the margin of the stapes footplate.
   d. Biscuit type: Disease process involves the footplate but annular ligament is free.



Figs 2A to E: Types of otosclerotic lesions causing fixation of stapes footplate. (A) Anterior focus, the site of predilection "fissula ante fenestram"; (B) Posterior focus the lesion begins behind the oval window; (C) Circumferential, lesion begins around the margins of footplate; (D) Biscuit type lesion involves only footplate sparing the annular ligament; (E) Obliterative lesion completely obliterates the oval window niche

   e. Obliterative type: The disease process completely obliterates the oval window niche.
2. *Cochlear otosclerosis*: It involves region of round window and areas in the bony labyrinth and petrous part of temporal bone. It presents with irreversible SNHL, which is probably caused by toxic materials liberated into the inner ear fluid.
3. *Histologic otosclerosis:* Histologic otosclerosis is diagnosed only on histological examination. Patient remains asymptomatic.

Pathology

1. *Gross appearance*
   a. Otosclerotic lesions appear chalky white, grayish or yellow.
   b. The red color lesions indicate increased vascularity, which is the feature of active and rapidly progressive otosclerotic focus.
2. *Histology*: A wave of abnormal bone remodeling occurs with resorption of enchondral bony labyrinth, which is replaced with hypercellular woven spongy bone that further remodels and results in sclerotic mosaic architecture.
   a. Immature active lesions: Numerous marrow and vascular spaces (increased vascularity) with plenty of histiocytes, osteoblasts and osteoblast precursor cells, and mononuclear cells indicate active remodeling phase. A lot of cement substance is present which stains blue with hematoxylin-eosin stain. Acute inflammatory cells are absent.
   b. Mature lesions: Less vascular spaces and laying of more bone and fibrillar substance than cementum and stains red with hematoxylin-eosin stain.

Clinical Features

Otosclerosis is characterized by gradually progressive CHL with normal tympanic membrane. In most cases, the disease is bilateral.

1. *Race*: White races are affected more than blacks. It is common in Indians but rare among Chinese and Japanese.
2. *Age of onset*: Patients are usually between 20–30 years of age. Disease is rare before 10 and after 40 years.
3. *Hormonal effect*: In females, deafness seems to worsen or manifest during pregnancy and menopause.
4. *Trauma*: Some patients try to correlate deafness with an accident or a major operation.
5. *Hearing loss*: The presenting feature is painless bilateral gradually progressive CHL.
   - Paracusis Willisii: In this phenomenon the patient's hearing improves in noisy background. It happens because a normal person raises his voice in noisy surroundings and patient takes advantage of that. The speech discrimination is not affected in pure conductive hearing loss.
   - Tuning fork tests and audiometry They show CHL.
6. *Tinnitus*: It is usually present in cochlear otosclerosis and active lesions.
7. *Vertigo*: It is an uncommon symptom. The cause of it is

not well understood. Hypertension and metabolic disorders are usually present in these cases. Some otologists consider vertigo as a contraindication to stapedectomy surgery because they feel the results are poor because of associated endolymphatic hydrops.

8. *Speech*: Low, monotonous, well modulated soft speech.
9. *Otoscopy*: Tympanic membrane is normal and mobile.
   - Schwartz sign: It is a reddish hue seen through the tympanic membrane on the promontory. It indicates active focus, which is vascular.
10. *Eustachian tube:* Its functions are normal.

Audiometry

- *Conductive hearing loss* more for lower frequencies (Fig. 3).
- *Carhart's notch*: There is a dip (from 500–4,000 Hz) in bone conduction curve, which is maximum (15 dB) at 2,000 Hz (5 dB at 500 Hz, 10 dB at 1,000Hz, 15 dB at 2,000 Hz and 5 dB at 4,000 Hz). The Carhart's notch disappears after successful stapedectomy surgery.
- *Air-bone gap*: The degree of footplate fixation is estimated by the size of air-bone gap. Audiometry does not predict the pattern and extent of oval window involvement. It is determined on exploratory tympanotomy during the stapedectomy surgery.
- *Mixed hearing loss* with SNHL element indicates cochlear otosclerosis.

Differential Diagnoses

They include following causes of CHL. They can be differentiated with the help of ear microexamination, siegalization, impedance audiometry and exploratory tympanotomy:

1. Serous otitis media
2. Adhesive otitis media
3. Tympanosclerosis
4. Attic fixation of head of malleus
5. Ossicular discontinuity
6. Congenital stapes fixation

Treatment

There is no curative treatment. The treatment of choice is stapedectomy. Other modalities of management include sodium fluoride therapy and hearing aids. Stapes mobilization and fenestration operations are performed occasionally.

- *Sodium fluoride:* Sodium fluoride hastens the maturity of active focus and arrests further SNHL.
- *Stapedectomy:* Stapedectomy operation consists of removal of the fixed stapes and insertion of prosthesis between the incus and oval window. Various types of prosthesis include Teflon piston, stainless steel piston, Tefwire or fat and stainless steel wire. In 90% cases, hearing improves.
- *Stapes mobilization:* About 1% of otosclerotic ears have fibrous fixation of stapes. Stapes mobilization provides good permanent hearing in these cases. Simple mobilization of stapes is not indicated in most of the cases as it commonly results in refixation.
- *Fenestration operation:* In fenestration operation, which is almost abandoned, an alternative window is created in the lateral semicircular canal. The main disadvantage is a postoperative mastoid cavity and an inherent hearing loss of 25 dB.
- *Hearing aid:* Hearing aids offer good hearing results and are indicated in patients who refuse surgery or are unfit for surgery.

- Otosclerosis: Fifty percent of patients give positive family history. Patient has CHL with normal tympanic membrane and impaired acoustic reflexes. The patients with negative Rinne (BC > AC) are candidate for stapedectomy, which provides very gratifying results.

STAPEDECTOMY

An ideal case for stapedectomy surgery is also an ideal candidate for hearing aid. So the patient should be fully informed of the results and risks of the stapedectomy.

Selection Criteria

- Firmly fixed stapes is indicated by an air-bone gap of minimum 30 dB for the speech frequencies and a negative Rinne for 256- and 512-Hz magnesium tuning forks and speech discrimination score of 60% or more.

The successful stapedectomy and stapes mobilization correct the CHL, remove Carhart's notch and often lead to over-closure of air-bone gap.

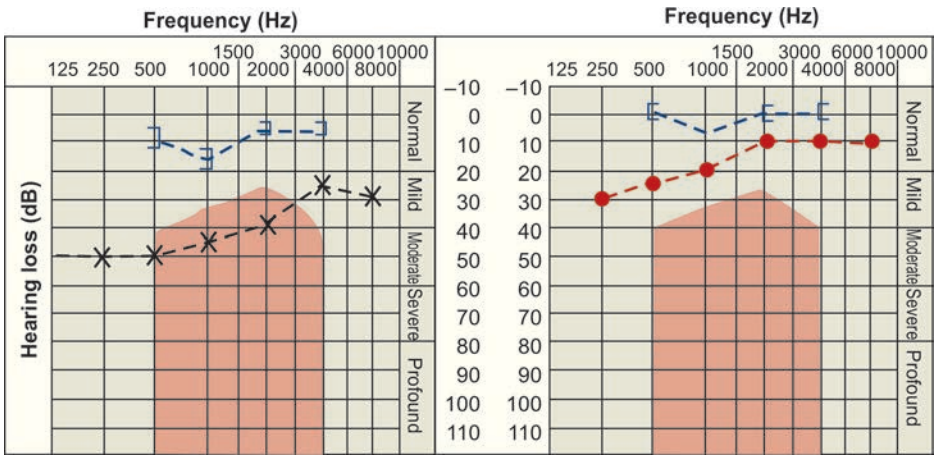


Fig. 3: Audiogram otosclerosis. Bilateral predominantly low-frequency conductive hearing loss. Left ear stapes surgery planned

- Mixed profound hearing loss with sufficient speech discrimination: Stapedectomy improves hearing aid use.

Contraindications

- *The only hearing ear*: There are about 1% chances of developing dead ear.
- *Vertigo*: History of vertigo in recent months is usually associated with Meniere's disease. There is heightened risk of postoperative SNHL.
- *Young children*: Recurrent Eustachian tube dysfunction commonly causes AOM in children and can displace the prosthesis. The otosclerotic focus is usually active and progresses rapidly in children and can close the oval window.
- *Certain occupations*:
  a. Postoperative vertigo can interfere the working in some professions such as athletes and high construction workers.
  b. In divers and frequent fliers, air pressure changes can damage the hearing and induce severe vertigo.
  c. Industrial workers who work in noisy surroundings are more vulnerable to occupational SNHL.
- *Local diseases*: Otitis externa, tympanic membrane perforation and exostosis should be treated before the stapedectomy.
- *Pregnancy*: Stapedectomy is avoided.

Anesthesia

Surgery is preferably done under local anesthesia so that hearing can be tested on the table.

Operative Steps

1. Infiltration of ear canal with lidocaine and epinephrine.
2. Obtaining of the tissue graft to cover oval window: vein, temporalis fascia, perichondrium or fat.
3. Endomeatal curved or triangular skin incision.
4. Elevation of the posterior deep meatal skin and fibrous annulus from sulcus tympanicus.
5. Removal of 2–4 mm posterosuperior bony overhang (Fig. 4) of the canal rim for an adequate exposure of oval window, stapes, facial nerve canal and pyramid.
6. Removal of stapes superstructure.
7. Making a hole in the stapes footplate (stepedotomy) or remove a part of footplate (stapedectomy).
8. Tissue seal of oval window.
9. Placement of prosthesis between the long process of incus and oval window (Figs 5 and 6): Shea platinum Teflon cup piston, Robinson stainless steel prosthesis, Shea Teflon piston, McGee piston, Fisch platinum Teflon piston and House wire prosthesis.
10. Repositioning the tympanomeatal flap.

Postoperative Care and Follow-Up

- Analgesics and antibiotics if needed.
- Avoid straining and blowing of nose.
- Outpatient surgery and discharged after several hours.
- First follow-up the day after surgery.

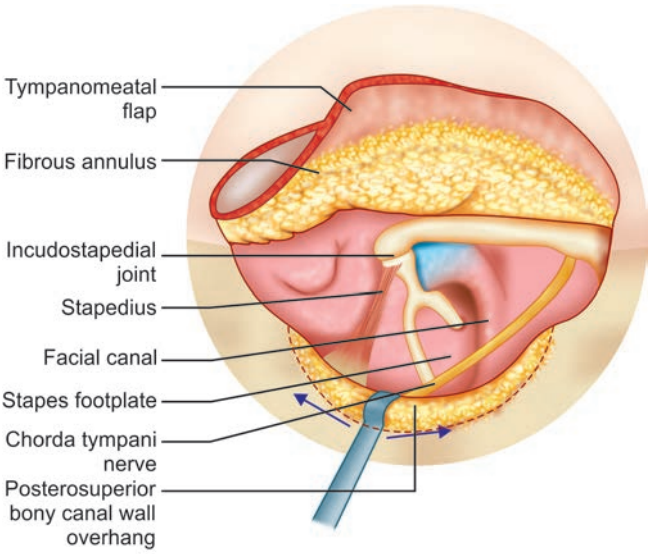


Fig. 4: Stapedectomy. After elevating the tympanomeatal flap, posterosuperior bony canal wall is removed to get better view of stapes

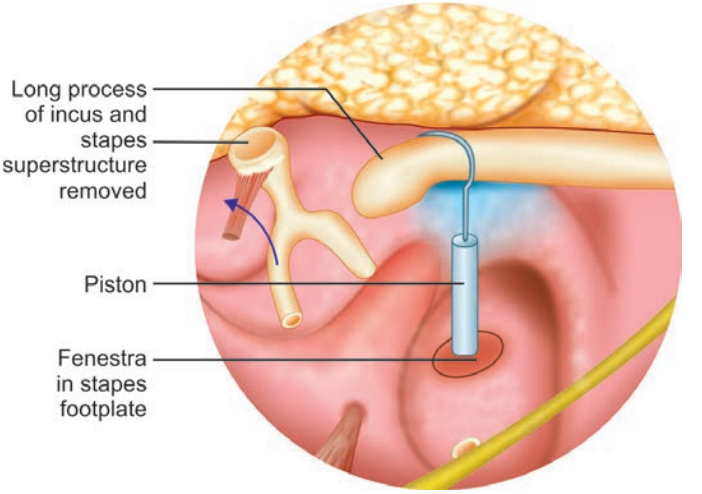


Fig. 5: Stapedectomy piston. After creating a fenestra in the stapes footplate, piston is placed from long process of incus to the footplate fenestra. Note stapes superstructure is removed

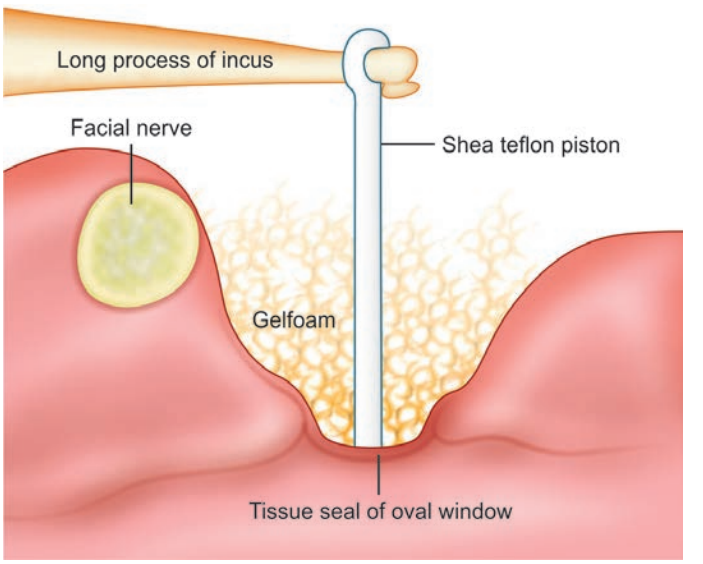


Fig. 6: Shea teflon piston

- The patients can begin their office work 1 week after the surgery.
- Second follow-up 2 weeks to a month.
- Annual follow-up for audiometry. SNHL requires sodium fluoride therapy.
- Imbalance, ear fullness, tinnitus and hearing loss indicate perilymph fistula (requiring tympanotomy) or endolymphatic hydrops (responding to medical treatment).

Complications

- Sensorineural hearing loss: 2% patients develop SNHL. Slowly progressive high frequency loss has been seen in long term follow-up. 0.5% patients get a "dead" ear.
- Vertigo.
- Loss or distortion of taste sensation.
- Unhealed perforation of tympanic membrane.
- Facial paresis/palsy.
- Perilymphatic fistula.

Clinical Highlights

1. **Otosclerosis**: Patients present with gradually progressive bilateral conductive hearing loss (CHL), which increases during pregnancy. Tympanic membrane and Eustachian tube are normal. Rinne's test is negative.
   a. **Prevalence**: The prevalence of clinical otosclerosis is highest in Caucasians.
   b. **Inheritance**: Otosclerosis has autosomal dominant inheritance.
   c. **Pathogenesis**: The disease process starts in bone of otic capsule which develops from cartilage.
   d. **Fissula ante fenestram**: This is the most common site of stapedial otosclerosis and is located at the anterior edge of oval window.
   e. **Schwartz sign**: This sign of active otosclerosis (such as during pregnancy) is seen as a pink reflex (reddish hue seen over the promontory) through intact tympanic membrane in the area of oval window.
   f. **Carhart's notch**: This characteristic feature of otosclerosis is seen in the audiograms. Bone conduction (sensorineural hearing loss) dips maximum at 2000 Hz (2 kHz).
   g. **Impedance audiometry (Tympanogram)**: Otosclerosis patients show 'A' type of curve with absent acoustic reflex.
   h. **Treatment**: Treatment of otosclerosis includes hearing aid, stapedectomy, and sodium fluoride therapy. Treatment of choice for a young stapedial otosclerosis office going patient is stapedectomy. Sodium fluoride (NaF) is being used in the treatment of cochlear otosclerosis.
2. **Habenula perforata***:* The openings, through which branches of cochlear nerves enter the cochlea. If wide, they can lead to a perilymph gusher in stapes surgery. This X-linked disease can be diagnosed on CT. It is associated with congenitally enlarged internal acoustic meatus and stapes fixation.
3. **Blue mantle***:* It is associated with otospongiosis.
4. **Van der Hoeve syndrome**: It is the triad of osteogenesis imperfecta, CHL, and blue sclera.
5. **Conductive hearing loss**: The most common causes are ear wax, otitis media, otomycosis and otosclerosis.

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13 Sensorineural Hearing Loss

Free! We, who cannot, for a moment, govern our minds, nay, cannot hold our minds on a subject, and focus it on a point to the exclusion of everything else, for a moment! Yet we call ourselves free. Think of it! The mind uncontrolled and unguided will drag us down, down, for ever—rend us, kill us; and the mind controlled and guided will save us, free us.

—Swami Vivekananda

Points of Focus

- **SENSORINEURAL HEARING LOSS**
  - ✓ Etiology
  - ✓ History, Examination and Investigations
  - ✓ Treatment
  - ✓ Prophylaxis
- **LABYRINTHITIS**
- **SYPHILIS**
  - ✓ Pathology
  - ✓ Clinical Features
  - ✓ Diagnosis
  - ✓ Treatment
  - ✓ Prognosis
- **OTOTOXICITY**
  - ✓ Pathology
  - ✓ Clinical Features
- **CISPLATIN**
- **AMINOGLYCOSIDE ANTIBIOTICS**
  - ✓ Risk Factors
  - ✓ Prophylaxis
  - ✓ Histopathology
  - ✓ Clinical
- **NOISE TRAUMA**
  - ✓ Acoustic Trauma
  - ✓ Noise-Induced Hearing Loss (NIHL)
  - ✓ Factors Affecting Noise Trauma
  - ✓ Pathology
  - ✓ Clinical Features
  - ✓ Pure Tone Audiogram (PTA)
  - ✓ Prophylaxis
- **SUDDEN SENSORINEURAL HEARING LOSS (SSNHL)**
  - ✓ Etiology
  - ✓ Clinical Features
  - ✓ Investigations
  - ✓ Treatment
  - ✓ Prognosis
- **PRESBYCUSIS**
  - ✓ Predisposing Factors
  - ✓ Pathology and Audiometry
  - ✓ Clinical Features
  - ✓ Treatment
- **GENETIC SNHL**
  - ✓ Familial Progressive SNHL
  - ✓ Immune-Mediated SNHL
- **NON-ORGANIC HEARING LOSS**
  - ✓ Malingering
  - ✓ Stenger Test
  - ✓ Diagnosis
- **DEGREE OF HEARING LOSS**
  - ✓ WHO Classification
  - ✓ Degree of Hearing Handicap
- **THE ONLY HEARING EAR**
- **CLINICAL HIGHLIGHTS**

SENSORINEURAL HEARING LOSS

Sensorineural hearing loss (SNHL) results from lesions of the cochlea, 8th nerve and central auditory pathways. There are characteristic features of SNHL, which differentiate it from conductive hearing loss (CHL) (see chapter Hearing Evaluation). The central auditory transmission has bilateral pathways from each ear. So the central defects, which cause subtle findings (such as impaired sound localization), are difficult to detect.

Etiology

The causes of SNHL lie in cochlea, cranial nerve (CN) VIII, brainstem and temporal lobe. They may be congenital, traumatic, infectious, inflammatory, iatrogenic, neoplastic, senile or miscellaneous (Box 1).

The congenital hearing loss is due to the anomalies of the inner ear or damage to the hearing apparatus by prenatal or perinatal factors. The causes may be divided into genetic and nongenetic. The genetic cause may have delayed onset and affect only the hearing. Other genetic causes are part of a larger syndrome affecting other systems of the body as well.

Box 1: The common causes of sensorineural hearing loss

1. Congenital: Genetic and nongenetic
2. Infections (viral, bacterial or spirochetal): Labyrinthitis and meningitis
3. Trauma to labyrinth and cranial nerve (CN) VIII in fractures of temporal bone and ear surgery.
4. Ototoxic drugs: Streptomycin, gentamicin
5. Endolymphatic hydrops: Primary or idiopathic (Meniere's disease) and secondary
6. Tumors: CN VIII acoustic neuroma
7. Systemic diseases: Diabetes, multiple sclerosis, syphilis, hypothyroidism, kidney disease, autoimmune disorders, blood dyscrasias
8. Miscellaneous: Sudden idiopathic sensorineural hearing loss (SNHL), familial progressive SNHL, noise-induced hearing loss (NIHL), presbycusis

History, Examination and Investigations (Fig. 1)

The essential elements of history and physical examination, which help in determining the cause and site of lesion (cochlear, nerve, central) include following:

- *History*: Congenital or acquired; unilateral or bilateral side involved; age and mode of onset; duration and progression (stationary/progressive/fluctuating); special features (recruitment and discrimination) and severity (mild, moderate, severe, profound) of hearing loss; events preceding hearing loss such as infection, trauma, strain, medication, surgery, noise exposure; associated ear symptoms of tinnitus, otorrhea, pain, vertigo/dizziness; family history.
- *Physical examination*: Complete ear (including otoscopy and tuning fork tests), nose, throat, head and neck examination including CN, neurologic and other systems to see any association with other syndromes.
- *Audiometry*: Pure tone audiometry (PTA) to know the severity (mild, moderate, moderately severe, severe, profound or total) and nature (high-frequency, low-frequency, mid-frequency or flat type); speech; impedance, evoke response, and otoacoustic emission.
- *Laboratory tests*: Depending upon the suspected etiology, following investigations may be ordered: CT/MRI (congenital cholesteatoma, glomus tumor, malignancy, acoustic neuroma); CBC (leukemia); blood sugar (diabetes); serology (syphilis); T3, T4, thyroid stimulating hormone(TSH; hypothyroidism); kidney function tests.

Treatment

Hearing is an essential part of communication. Early detection of SNHL and immediate attention towards its management are of paramount importance. Measures are taken to stop and reverse the progress and for rehabilitation program. Following are some of the treatment options of different causes of SNHL:

1. Syphilis is treatable with high doses of penicillin and steroids.

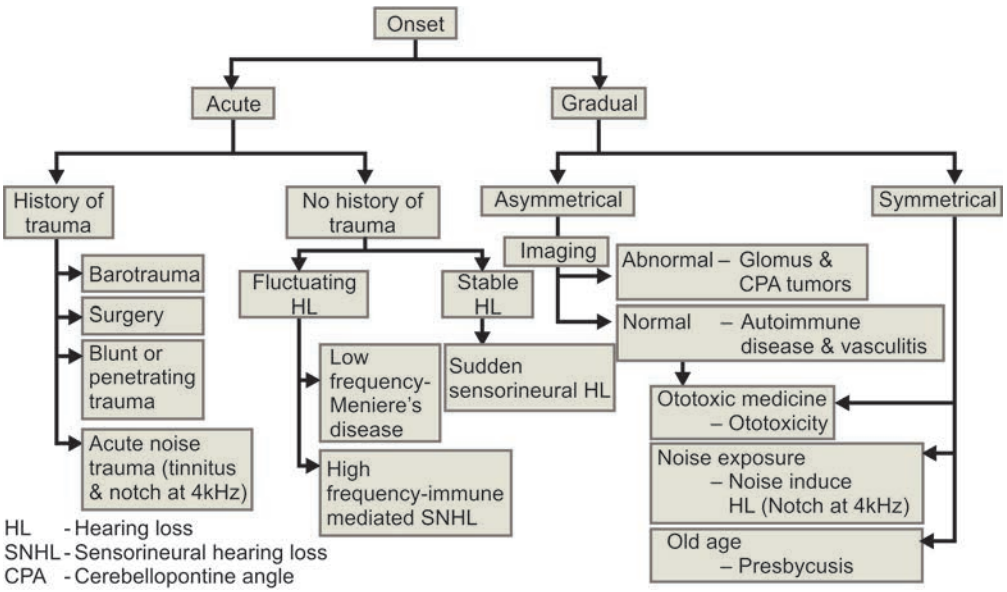

Fig. 1: Flow chart for the differential diagnoses of sensorineural hearing loss (SNHL) in adults

2. Hypothyroidism needs thyroxin replacement therapy.
3. Serous labyrinthitis can be reversed by active treatment of middle ear infection.
4. Perilymph fistula needs surgical correction with sealing the fistula in the oval or round window with fat or other material.
5. Discontinuation of ototoxic drugs.
6. Rehabilitation with hearing aids and other devices are discussed in chapter Hearing Aids and Cochlear Implants.

Prophylaxis

1. Ototoxic drug administration should have regular monitoring. Drug should be discontinued at the right time.
2. Avoidance of noisy surroundings.

LABYRINTHITIS

The infection of labyrinth may be viral or bacterial. (See chapter Peripheral Vestibular Disorders)

- *Viral labyrinthitis*: Viruses can reach the labyrinth via blood stream and affect stria vascularis, endolymph and organ of Corti. The viruses which have been documented to cause labyrinthitis are measles, mumps and cytomegalovirus. Other viruses, which are known to cause hearing loss but lack direct proof, are rubella, herpes zoster, herpes simplex, influenza and Epstein-Barr.
- *Bacterial labyrinthitis*: Bacterial infections can reach inner ear through the middle ear (tympanogenic) and CSF (meningogenic). Suppurative otitis media and meningitis are common causes of labyrinthitis and SNHL.

SYPHILIS

Hearing loss (6.5% of unexplained SNHL) occurs in secondary, tertiary and congenital syphilis. Seven percent patients of Meniere's disease (chapter Vestibular Disorders) have syphilis.

Pathology

Obliterative endarteritis and mononuclear infiltration produce periostitis and gummatous (central necrosis with surrounding lymphocytic infiltration and vascular occlusion) osteitis/periostitis. Narrowing of endolymphatic duct and sac due to atrophy and fibrosis leads to endolymphatic hydrops.

Clinical Features

- *Secondary Syphilis*
  - Hearing loss: Abrupt onset, bilateral, progressive.
  - Vestibular symptoms: They are usually uncommon. But when occur, episodes of acute vertigo are similar to Meniere's disease.
  - Hennebert's sign: The positive fistula test in the absence of fistula may be present.
  - CNS features: Patients may have headaches, stiff necks, cranial nerve palsies and optic neuritis.
  - Secondary syphilitic features: Skin rashes and lymphadenopathy.
  - CSF: Lymphocytic pleocytosis, elevated proteins and normal glucose.
- *Tertiary Syphilis*
  - Otologic features are similar to late congenital syphilis.
  - CSF may have minimal pleocytosis and elevated or normal protein.
  - Electrocochleography: Features similar to Meniere's disease.
- *Congenital Syphilis*
  - *Early (birth to 3 years)*
    - Hearing Loss: Rapid bilateral profound symmetric SNHL.
    - Vestibular features: They are relatively few and may vary from mild imbalance to protracted vertigo with vegetative features lasting days.
    - Meningoneurolabyrinthitis: Systemic features such as meningitis overshadow otologic symptoms.
    - Multi-system: Extensive damage of other organs proves fatal.
  - *Late (8–20 years)*
    - Hutchinson's triad: It consists of SNHL, interstitial keratitis and notched incisors. It is an exclusive feature of congenital syphilis.
    - Hutchinson's teeth are peg-shaped and notched permanent upper central incisors and mulberry molars (first lower molar grinding surface has many tiny cusps).
    - Hearing loss: It is bilateral, asymmetric and progressive but fluctuating flat SNHL. Speech discrimination scores are disproportionately low to pure tone threshold. Recruitment is present.
    - Tinnitus may appear intermittently.
    - Vestibular features: Episodes of acute vertigo are similar to Meniere's disease.
    - Hennebert's sign: It is due to the softened gummatous otic capsule.
    - Tullio's phenomenon: Nystagmus and vertigo are often caused by loud noise.
    - Caloric responses: They are often decreased.
    - Interstitial keratitis: It occurs in 90% cases.

Diagnosis

- Treponemal tests (Free Treponemal Antigen Absorption (FTA-abs) and microhemagglutination *Treponema pallidum* (MHATP)):They detect organism and are more sensitive (95%). Non-treponemal tests (screening VDRL and rapid plasma reagin (RPR)) detect only 70% patients of otosyphilis.
- Other tests: Serum rapid plasma region (RPR) test and fluorescent treponemal antibody-absorption (FTA-ABS) test and CSF-VDRL test.
- *Lumbar puncture*: It is usually not required as otosyphilis is occasionally associated with neurosyphilis.
- Identification of spirochetes by darkfield examination of perilymph obtained by stapes footplate (labyrinthotomy.)

Treatment

Penicillin and prednisone is the mainstay of treatment.

1. *Early syphilis*: Injection benzathine penicillin 2.4 million units intramuscular (IM).
2. *Late syphilis*: Injection procaine penicillin 1.8 million units IM OD with Tab, Probenecid 500 mg QID for 17–21 days, or injections aqueous crystalline penicillin G, 2–4 million

units IV every 4 hours for 10 days, or IV Ceftriaxone in neurosyphilis in HIV-infected patients.

- Tab. Prednisone, 30–60 mg OD or EOD for first week. Recurrence of hearing loss requires long-term maintenance of prednisone (10–20 mg OD).

Prognosis

It is poor in early congenital and better in others. Hearing improvement especially speech discrimination occurs in 35–50% of patients.

OTOTOXICITY

There is a long list of drugs and chemicals, which can cause SNHL, tinnitus and dizziness (Box 2). The clinical features of ototoxicity (Fig. 2), which may manifest during treatment or after completion of (delayed ototoxicity) the treatment, include SNHL, tinnitus and/or dizziness/vertigo.

Pathology

Table 1 shows the target sites of action of common ototoxic agents.

Clinical Features

- *Erythromycin*: Transient blowing tinnitus and loss of hearing (usually flat type of SNHL) and in some cases vertigo.
  - Some patients have confusion, fear, psychiatric disturbances, visual changes, blurred speech, sensation of being drugged, or lack of control.
- *Loop diuretics*: They block transport of sodium and chloride ions in the ascending loop of Henle. Usually the effect is reversible but permanent damage can occur.
- *Salicylates*: Patient of salicylate ototoxicity presents with tinnitus and bilateral SNHL predominantly affecting higher frequencies. Probably they interfere at enzymatic level. Hearing loss due to salicylates is reversible on cessation of the drug.
- *Chloroquin*: Ototoxic symptoms, which are reversible, include tinnitus and SNHL. The symptoms generally appear with smaller doses in the susceptible patients.
- *Quinine toxicity (Cinchonism)*: Deafness (reversible/permanent), vertigo, tinnitus, headache, visual loss and nausea. If mothers receive quinine during the first trimester of pregnancy, congenital SNHL and hypoplasia of cochlea can occur.
  - *Sensorineural hearing loss:* High frequency loss, 4 kHz notch, speech discrimination less than 30%.
- *Deferoxamine (Desferioxamine)*: It is an iron-chelating substance used in the treatment of thalassemic patients who receive repeated blood transfusions and in turn have high iron-load. The onset of SNHL may be sudden or delayed. SNHL is usually permanent. In some, SNHL is reversible on the cessation of drug.
- *Topical ear drops*: The damage to the cochlea is due to the absorption of ototoxic eardrops through oval and round windows. SNHL has been documented with the topical use of chlorhexidine and aminoglycoside antibiotics such as neomycin, framycetin and gentamicin.

TABLE 1 The site of action of ototoxic agents

Ototoxic agent	*Site of action*
Aminoglycoside antibiotics	Destroy type1 hair cells of the crista ampullaris; destruction of outer hair cells
Vancomycin	Damage to cochlear hair cells initially of basal turn (high frequency)
Loop diuretics	Edema and cystic changes in the stria vascularis
Salicylates	Probably interfere at enzymatic level
Chloroquin and quinine	Vasoconstriction in the small vessels of the cochlea and stria vascularis
Cytotoxic drugs	Affect the outer hair cells

Box 2: List of ototoxic agents

- Aminoglycoside antibiotics: Streptomycin, gentamicin, tobramycin, neomycin, kanamycin, amikacin, sisomycin
- Analgesics: Salicylates, indomethacin, phenylbutazone, ibuprofen
- Antimalarials: Quinine, chloroquin
- Loop diuretics: Furosemide, ethacrynic acid
- Cytotoxic drugs: Nitrogen mustard, cisplatin, carboplatin
- Chemicals: Alcohol, tobacco, marijuana, carbon monoxide poisoning
- Topical applications: Chlorhexidine and aminoglycoside ear drops
- Miscellaneous: Erythromycin, ampicillin, vancomycin, propranolol, propylthiouracil, deferoxamine

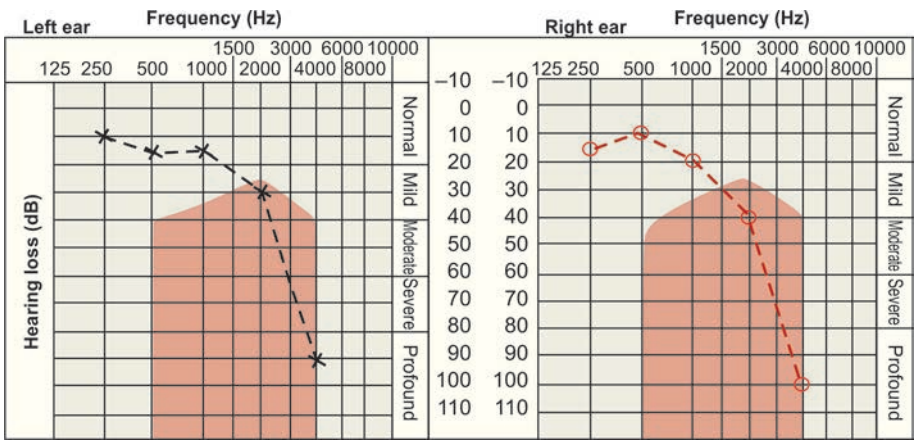

Fig. 2: Audiogram of a patient with ototoxicity. Note bilateral symmetrical profound high-frequency sensorineural hearing loss (SNHL)

- *Cytotoxic drugs*: Nitrogen mustard, cisplatin and carboplatin can cause cochlear damage. They affect the outer hair cells of cochlea.

The patients taking ototoxic medicines must be instructed to report development of tinnitus, hearing loss, imbalance and vertigo.

CISPLATIN

- *Indications*: Carcinomas of ovary, testis, bladder, lung, head and neck.
- *Side effects*: Nausea, vomiting, neurotoxicity, ototoxicity and nephrotoxicity.
- *Hearing loss*: It is permanent and bilaterally symmetrical. High frequencies are affected first and may extend into middle frequencies range. Hearing loss is related to dose, age, noise exposure, low serum albumin, anemia, other ototoxic drugs and cranial irradiation.
- *Ear pain*: In some cases.
- *Tinnitus:* It may be transient or permanent.
- *Vertigo and disequilibrium*: It occurs especially in patients with preexisting vestibular problems.
- *Prophylaxis:* Audiometric monitoring (especially high-frequency audiometry) before beginning of each cycle.

AMINOGLYCOSIDE ANTIBIOTICS

They include streptomycin, kanamycin, tobramycin, amikacin, netilmycin, sisomycin, gentamicin and neomycin.

- They can damage kidneys and inner ear.
- Ototoxicity usually occurs after days or weeks of exposure.
- Though some are more toxic to either cochlea or the vestibule, their toxicity is not selective. Auditory toxicity occurs in 20%, while vestibular may occur in 15%.
  - Streptomycin, gentamicin and tobramycin are primarily vestibulotoxic. So patient presents with dizziness/vertigo. In large doses, they can damage the cochlea also.
  - Neomycin, kanamycin, amikacin, sisomycin and dihydrostreptomycin are cochleotoxic.

Risk Factors

The following patients are particularly at risk:

1. Bacteremia and fever.
2. Hepatic and renal dysfunction.
3. Elderly patients above 65 years of age.
4. Combination of other ototoxic drugs—Cisplatin, ethacrynic acid and furosemide, amphotericin-B and cyclosporine.
5. Past history of receiving ototoxic agents.
6. *Genetic susceptibility*: Mitochondrial RNA mutation may sensitize the auditory system even to single dose of drug. In genetic susceptible cases aminoglycoside antibiotic binds to the ribosome and interferes with protein synthesis, thus causing death of the cochlear cells.

Prophylaxis

- Measurement of peak and trough serum levels of aminoglycoside help in monitoring ototoxicity, which may continue even after cessation of therapy.
- Measurement of peak and trough serum levels of gentamicin help in monitoring nephrotoxicity; however, ototoxicity can develop when all of these are controlled.
- Audiometric monitoring (especially high-frequency audiometry) is done before and after treatment and weekly in cases of long treatment.

Histopathology

- Damage to outer hair cells of basal turn spreads to apical region.
- Progressive destruction of spiral ganglion cells.
- Thinning of stria vascularis.
- Hair cell damage in apex of cristae and in striola regions of maculae may extend to type I hair cells.

Clinical

- High-frequency hearing loss is detectable, which progresses to lower frequencies and speech range.
- Onset of vestibular ototoxicity is unpredictable.
  - Imbalance and ataxia worsened by motion or ambulation.
  - May lead to oscillopsia and complete inability to walk without assistance.
- Clinical improvement, which is rarely complete, may begin after 2 months.

Incidence of ototoxicity is highest with neomycin. Netilmycin is the weakest ototoxic.

NOISE TRAUMA

Sensorineural hearing loss caused by excessive noise is divided into two groups: acoustic trauma and noise-induced hearing loss (NIHL). Noise trauma, an occupational hazard is well-known in boiler makers, iron and coppersmiths and artillery men. The compensations asked for and the responsibilities thrust upon the employer and the employees are well-known.

Acoustic Trauma

A single brief exposure to a very intense sound (such as an explosion, gunfire or a powerful cracker) can result in permanent damage to hearing. Sudden loud sound has the potential to damage outer hair cells, organ of Corti, Reissner's membrane, tympanic membrane and ossicular chain. Noise level in rifle or a gun fire may reach 140–170 dB sound pressure level (SPL). Gunfire is a common nonoccupational cause of noise trauma. The hearing loss is in the range of 4,000 Hz.

Noise-Induced Hearing Loss

Noise-induced hearing loss is a major cause of preventable SNHL. SNHL may follow chronic exposure of noisy occupations, which are less intense sounds than the acoustic trauma.

- *Temporary threshold shift*: The hearing, which is impaired on the exposure to noise, recovers after an interval of time ranging from few minutes to few hours.
- *Permanent threshold shift*: The hearing loss becomes permanent and does not revert back.

Factors Affecting Noise Trauma

The SNHL caused by noise trauma depends on the following factors, which should be kept in mind for the safety of hearing:

- *Frequency*: The noise of 2,000–3,000 Hz frequencies causes more SNHL than lower and higher than these frequencies.
- *Intensity and duration*: As the intensity of noise increases, the permissible time for exposure is reduced. Table 2 shows the permissible limits of time for various intensities of noises. The "*5 dB rule of time-intensity*" maintains that any rise of 5 dB noise level will reduce the permitted noise exposure time to half. A noise level of 90 dB (A) SPL, 8 hours a day for 5 days per week is the highest safe limit in the factories. The exposure of more than 115 dB (A) is not permitted. The impulse noise, which is greater than 140 dB (A) is not permitted.
- *Continuous interrupted*: The continuous noise is more harmful than the interrupted one.
- *Susceptibility*: Some persons are genetically susceptible to noise trauma.
- *Pre-existing ear disease*: They can affect the impact of noise on the inner ear.

Pathology

Noise-induced hearing loss damages hair cells, which begin at the basal turn of cochlea. Outer hair cells are affected earlier than the inner hair cells.

Clinical Features

- Shouting to converse at workplace.
- Aural fullness, tinnitus, or muffled hearing after the work.

Pure Tone Audiogram

The PTA (Figure 3) shows following characteristic findings in NIHL:

TABLE 2 The permissible limits of time for various intensities of noise levels

Noise level A dB	*Permitted daily exposure in hours*
90	8
95	4
100	2
105	1
110	½
115	¼

1. *Early stage*: A typical notch at 4 kHz is seen in both for air and bone conduction. It is usually symmetrical in both the ears. Patient complains of high-pitched tinnitus and difficulty in hearing only in noisy surroundings.
2. *Late stage*: The notch deepens and widens to involve lower and higher frequencies. Involvement of speech frequencies (500, 1,000 and 2,000 Hz) result in hearing difficulty even in calm surroundings also.

Prophylaxis

Persons who work in factories where noise is above 85 dB (A) should have following precautions:

- Pre-employment and then annual audiograms for early detection.
- Use of ear protectors (ear plugs or ear muffs) provides protection up to 35 dB.
- Rehabilitation is required in cases of NIHL.

SUDDEN SENSORINEURAL HEARING LOSS

Sudden sensorineural hearing loss (SSNHL) develops over a period of hours or a few days and hearing loss may be partial or complete. Though mostly unilateral (Fig. 4), it may affect both the ears. Patient may have associated symptoms of tinnitus and temporary spell of vertigo.

Etiology

- In most of the cases, it is idiopathic as the cause remains obscure in 85–90% of the cases.
- Three etiological factors speculated are viral, vascular and the rupture of cochlear membranes.
- Spontaneous formation of perilymph fistulae in the oval or round window can occur.
- The causes of sudden hearing loss, which must be ruled out in idiopathic SSNHL, are listed in Box 3.

Clinical Features

- Sudden sensorineural hearing loss develops over a period of hours or a few days.
- Hearing loss may be partial or complete.
- Though mostly unilateral, it may affect both the ears.
- Patient may have associated symptoms of tinnitus and temporary spell of vertigo.

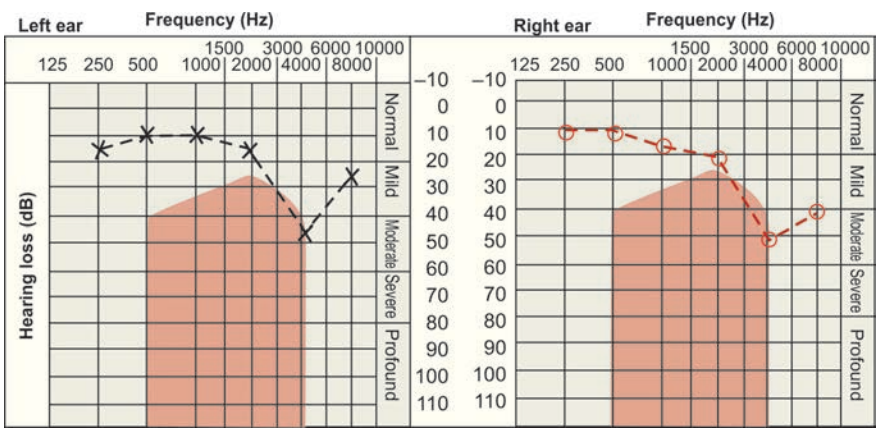

Fig. 3: Audiogram of a patient with noise-induced hearing loss. Note bilateral symmetrical sensorineural hearing loss (SNHL) with notch at 4,000 Hz

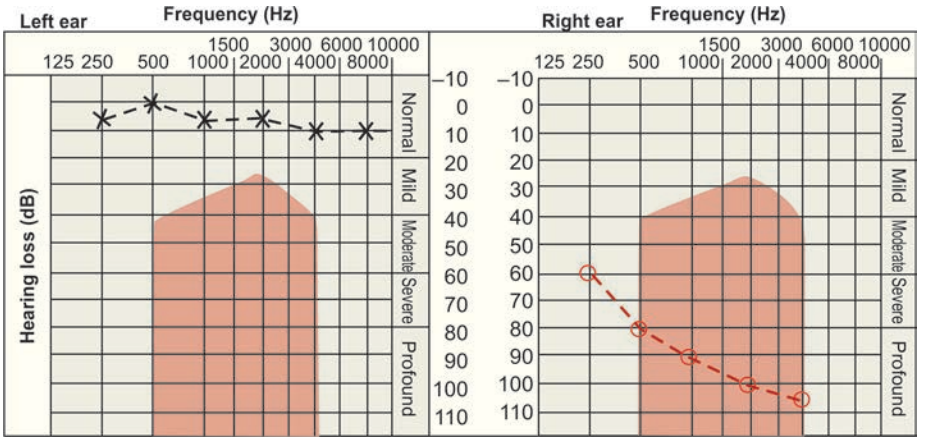

Fig. 4: Audiogram of a patient with SSNHL. Note high-frequency profound sensorineural hearing loss (SNHL) in right ear

Box 3: Causes of sudden sensorineural hearing loss (SSNHL)

1. Infections: Mumps, herpes zoster, meningitis, encephalitis, otosyphilis, Lyme disease, otitis media
2. Trauma: Head injury, ear operations, noise trauma, barotraumas (diving and ascending) and spontaneous rupture of cochlear membranes.
3. Vascular: Hemorrhage (leukemia), embolism, thrombosis, and spasm of labyrinthine or cochlear artery. Risk factors include diabetes, hypertension, polycythemia, macroglobulinemia and sickle cell trait
4. Ototoxic drugs: Section "Ototoxicity"
5. Tumors: Acoustic neuroma, metastases in cerebellopontine angle, carcinomatous neuropathy
6. Miscellaneous: Meniere's disease, Cogan's syndrome, multiple sclerosis, hypothyroidism, sarcoidosis
7. Psychogenic

- Upper respiratory infections (URI) within 1 month of hearing loss.

The cause of sudden hearing loss should be discovered by detailed history and physical examination including audiometry.

Investigations

1. *Laboratory*: The following laboratory investigations may be ordered to confirm the cause, which is suspected on the bases of history and physical examination and audiometry findings: CBC, ESR, tests for syphilis, diabetes, hypothyroidism, blood disorders and lipid profiles, and vestibular tests.
2. *Gadolinium-enhanced MRI*: It is indicted on the suspicion of acoustic neuroma. Though only 1–3% of SSNHL is due to acoustic neuroma, about 10% of acoustic neuroma patients present with SSNHL.
3. *Exploratory tympanotomy*: It may be done where perilymph fistula is strongly suspected.

Treatment

As the cause remains often obscure, treatment also remains empirical. Though the corticosteroid therapy is the mainstay vasodilator, diuretics, anticoagulants and thrombolytic agents have also been tried.

In addition to bed rest, treatment consists of the following elements:

1. *Steroid therapy*: Prednisolone 40–60 mg in a morning dose for 1 week and then tapered off in a period of 3 weeks. Steroids relieve edema as they have anti-inflammatory effect. They are of particular use in hearing loss of moderate degree.
2. *Antivirals* such as acyclovir.
3. *Inhalation of carbogen* (5% CO_2 + 95% O_2): It increases cochlear blood flow and provides better oxygenation.
4. *Low molecular weight dextran*: The infusions decrease blood viscosity. It is contraindicated in cardiac failure and bleeding disorders.
5. *Hyperbaric oxygen therapy*: If given in the first month of onset of hearing loss, some benefits have been seen.

Prognosis

About 50% patients recover spontaneously within 15 days. Chances of recovery are less after one month.

- *Good prognostic factors*: They include young adults, moderate and low-frequency hearing loss, lack of vertigo and early treatment.
- *Poor prognostic factors*: They include old age, profound deafness, presence of vertigo, vascular risk factors and delayed treatment.

PRESBYCUSIS

Sensorineural hearing loss, which occurs due to the aging process, is called presbycusis. More than one-third of elderly persons over the age of 75 have this senile hearing loss.

Predisposing Factors

Though the exact cause is yet not known, following factors have been implicated: hereditary predisposition, chronic noise exposure, vascular risk factors, ototoxicity, metabolism, arteriosclerosis and diet.

Pathology and Audiometry

The following four pathological types have been described:

1. *Sensory*: The degeneration of the organ of Corti begins at the basal coil and progresses gradually toward the apex. So the higher frequencies are affected first. The speech discrimination remains good.

2. *Neural*: The degeneration of spiral ganglion begins at the basal coil and progresses towards the apex. Neurons of higher auditory pathways may also be affected. The PTA shows high tone loss. Speech discrimination is poor, which is out of proportion to the pure tone loss.
3. *Strial or metabolic*: The atrophy of stria vascularis occurs in all turns of cochlea. The physical and chemical processes of energy production are affected. This type of presbycusis pathology runs in families. PTA shows flat graph. Speech discrimination remains good.
4. *Cochlear conductive*: The stiffening of the basilar membrane affects its own movements. Audiogram is sloping type.

Clinical Features

1. Slowly progressive, symmetric SNHL in people over the age of 60.
2. *Background noise*: Great difficulty in hearing in the presence of background noise. Patients may hear well in quiet surroundings in early stage of the disease.
3. *Discrimination*: Poor speech understanding. Typical complaint "I can hear but cannot understand."
4. *Recruitment*: Intolerance to loud sounds due to the recruitment. The patient has hearing loss but if someone speaks loudly, s/he retorts "why are you shouting? I am not deaf."
5. *Tinnitus*: It may be bothersome and the only complaint.
6. *Audiogram* (Fig. 5): SNHL is greatest in frequencies > 2,000 Hz with significant decrease in speech discrimination.

Treatment

Patients are advised for the following:

- Hearing aid.
- Lessons in speech reading through visual cues.
- Curtailment of smoking and stimulants like tea and coffee for decreasing tinnitus.

GENETIC SENSORINEURAL HEARING LOSS

Genetic causes are responsible for major portion of SNHL.

Features

- May be congenital or late onset.
- Hearing loss may be progressive or stable.
- Mid- or high-frequency SNHL.
- Associated abnormalities of external ear, skin, eye, kidneys, CNS, musculoskeletal and other systems may be present.
- The examples of common syndromic SNHL include Waardenburg's syndrome, Alport syndrome and Usher's syndrome (see chapter Hearing Impairment in Infants and Young Children).

Familial Progressive Sensorineural Hearing Loss

This genetic disorder leads to progressive degeneration of the cochlea. The disease begins in late childhood or early adult life. The SNHL (flat or basin-shaped audiogram) is bilateral with excellent speech discrimination.

Immune-Mediated Sensorineural Hearing Loss

Though poorly understood and rare, the condition is treatable medically. Autoimmune disorders, which have been implicated in SNHL, include Cogan's syndrome, Wegener's granulomatosis (WG), Behçet's disease, systemic lupus erythematosus (SLE) and multiple sclerosis (MS).

Clinical Features

- Unexplained, bilateral, rapidly progressive SNHL in adults (20–50 years).
- Normal otoscopy.
- Features of coexistent autoimmune disease uncommon.
- Poor speech discrimination relative to pure tone thresholds.

Laboratory

- Serologic tests for syphilis.
- Cellular and humoral antigen-specific tests such as lymphocyte transformation testing and Western blot.

Treatment

- Steroids.
- Cytotoxic drugs.
- Plasmapheresis.

NON-ORGANIC HEARING LOSS

Non-organic hearing loss (NOHL) can be either due to malingering or psychogenic.

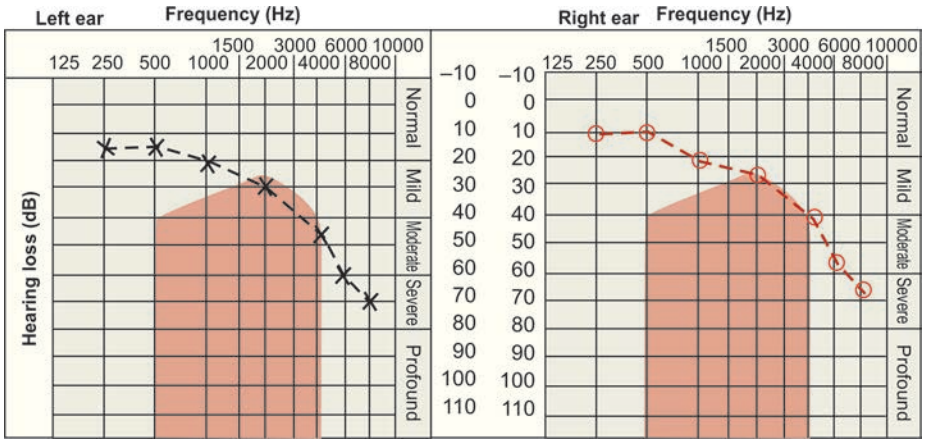

Fig. 5: Audiogram of a patient with presbycusis. Note bilateral symmetrical sensorineural hearing loss (SNHL)

Malingering

- The malingerer usually has motive and claim for some compensation.
- They have history of exposure to industrial noises, head injury or ototoxic medication.
- Patient may present with total hearing loss in both the ears, total loss in only one ear or an exaggerated loss in one or both ears.
- Patient makes exaggerated efforts to hear such as making cupped hand behind the ear.

Stenger Test

- *Principle*: If a tone of two different intensities (one greater than the other) is delivered to two ears simultaneously, only the ear which receives tone of greater intensity will hear it.
- *Method*: Patient is blindfolded during the test. Take two tuning forks of equal frequency. Vibrate and put them simultaneously at an equal distance (say 30 cm) from each ear. Patient will claim to hear it in the said normal ear. Now bring the tuning fork closer (say 10 cm) to the feigned ear (said deaf ear) while keeping the tuning fork on the normal side at the same distance. The patient will deny hearing in both ears though s/he must be hearing in the said normal ear. A patient with true hearing loss will continue to hear on the normal side. This can be done with a double-channel audiometer using pure tone or speech signals.

Diagnosis

In the era of objective hearing tests, it is not difficult to find a malingerer. Following findings will confirm the malingering.

a. *Inconsistent results of repeat PTA and speech audiometry*: The results of repeat tests in malingerer vary greater than 15 dB.
b. *Absence of shadow curve* in a patient of unilateral hearing loss is diagnostic of NOHL. A shadow curve is normally obtained when testing bone conduction without masking the healthy ear. The shadow curve is due to the transcranial transmission of sound to the healthy ear.
c. *Inconsistency in PTA and speech recognition threshold*: The pure tone average of three speech frequencies (500, 1,000 and 2,000 Hz) is normally within 10 dB of speech recognition threshold (SRT); if it is more than 10 dB, it points to NOHL.
d. *Acoustic reflex threshold*: The stapedial reflex is elicited normally at 70–100 dB SL. The presence of reflex in the feigned deaf ear will indicate NOHL.
e. *Brainstem evoke response audiometry*: Brainstem evoke response audiometry (BERA) will establish hearing acuity within 5–10 dB of actual thresholds.

DEGREE OF HEARING LOSS

The term hearing loss is preferred over the deafness. The WHO recommended in 1980 that the term "deaf" should be applied when hearing impairment is so severe that patients are unable to benefit from any type of amplification. So the cases included in the deaf category have hearing loss more than 90 dB in the better ear (profound impairment) or total loss of hearing in both the ears.

WHO Classification

The recommended WHO Classification (1980) of Degree of Hearing Loss (Table 3) is on the basis of average of speech frequencies (500, 1,000, 2,000 Hz) calculated from PTA with reference to ISO: R. 1970 (international calibration of audiometers). Table 3 also shows the disability to understand speech with different degrees of hearing loss.

WHO considers only three speech frequencies (500, 1,000 and 2,000 Hz). Currently, it is felt that frequency of 3,000 Hz is important for hearing in the presence of noise. American Academy of Ophthalmology and Otolaryngology takes into account the average of four speech frequencies 500, 1,000, 2,000 and 3,000 Hz when calculating the handicap.

Department of Personnel, Government of India

The categorization and percentage of hearing impairment recommended by Department of Personnel, Government of India is given in Table 4. The Indian Government has reserved certain percentage of vacancies and extended certain other benefits for group C and D.

Degree of Hearing Handicap

The hearing impairment can affect person's ability to hear and perform certain activities (disability) and is termed as handicap by the society. The degree of hearing impairment and handicap are termed in percentage for the purposes of compensation.

- *Method of calculating in percentage*
  - Calculate the average of thresholds of hearing for speech frequencies of 500, 1,000 and 2,000 Hz for the ear (say = A).
  - Deduct from it 25 dB (as up to 25 dB loss is considered normal) means A-25.
  - Multiply it by 1.5, means (A-25) x 1.5. This is the percentage of hearing impairment for the ear.
  - Percentage handicap of an individual= [(Better ear % x 5) + worse ear%]/6
  - One example is given in Table 5.

TABLE 3 WHO classification of degree of hearing loss and difficulty in hearing speech

Average threshold of speech frequencies in the better ear in dB	*Degree of hearing impairment*	*Difficulty in hearing speech*
0–25 dB	Not significant	Can hear faint speech
26–40 dB	Mild	Difficulty with faint speech
41–55 dB	Moderate	Difficulty with normal speech
56–70 dB	Moderately severe	Difficulty with loud speech
71–91 dB	Severe	Understand only shouted or amplified speech
> 91 dB	Profound	Cannot understand even amplified speech

TABLE 4 Recommended categorization and percentage of hearing impairment as per the Department of Personnel, Government of India

Category	*Hearing impairment*	*PTA speech frequencies threshold in better ear in dB**	*Speech discrimination in better ear***	*Percentage of hearing impairment*
I or A	Mild	26–40 dB	80–100%	Less than 40%
II or B	Moderate	41–55 dB	50–80%	40–50%
III or C	Severe	56–70 dB	40–50%	50–75%
IV or D	a) Total	No hearing	No discrimination	100%
	b) Near total	91 dB and above	No discrimination	100%
	c) Profound	71–90 dB	Less than 40%	75–100%

*Pure tone average of AC thresholds of three speech frequencies 500, 1,000 and 2,000 Hz in better ear; No response to any of the three speech frequencies (500, 1,000, 2,000 Hz.) is equivalent to 130 dB loss; PTA calibrated at ISO R 389-1970 standards; For children, when PTA is not possible, free field testing should be employed.
**Wherever possible, the pure tone audiometric results should be supplemented by the speech discrimination score.

TABLE 5 Example of calculating hearing handicap in percentage

	500 Hz	1,000 Hz	2,000 Hz	Average
Right ear	*50*	65	80	65 dB
Left ear	*20*	35	50	35 dB
Right ear impairment %		(65–25) x 1.5 = 60%		
Left ear impairment %		(35–25) x 1.5 = 15%		
Total handicap (as per formula)		[(15x 5) + 60]/6 = 22.5 = 23% (rounded off)		

THE ONLY HEARING EAR

Even profound unilateral loss of hearing does not produce a serious handicap. Patient may have following complaints:

- Surprisingly many grown-up children are not aware of their congenital unilateral profound hearing loss. This accidental finding during examination comes as a shock to the parents.
- Impaired localization of the sound source.
- Difficulty in speech discrimination in noisy background.
- Difficulty in hearing when the speaker is on the side of affected ear.

Precautions

1. Patient is given instructions for the safety of the only hearing ear regarding noise trauma, ototoxic drugs, vascular risk factors and healthy food and lifestyle changes.
2. Usually surgeons do not operate on the only hearing ear, until unless it has life-threatening disorder such as cholesteatoma or acoustic neuroma.

Clinical Highlights

1. **Hutchinson's triad:** It is the feature of congenital syphilis.
2. **Fluctuating hearing loss:** The common causes include ear wax, otitis media with effusion (serous otitis media), Meniere's disease, perilymph fistula and malingering. Uncommon causes are autoimmune disorder of inner ear and syphilitic labyrinthitis.
3. **Ototoxic drugs:** The common ototoxic drugs are chloroquin, cisplatin and furosemide.
   a. Vestibulotoxic drugs: The common vestibulotoxic drugs are streptomycin, gentamicin and minocycline.
   b. Reversible ototoxicity: The ototoxic effects of quinine, salicylates and furosemide, are reversible if their administration is stopped.
   c. Aminoglycoside ototoxicity: The principal site of damage is outer hair cells of basal turn of cochlea.
   d. Streptomycin ototoxicity: Semicircular ducts and saccule are very sensitive to streptomycin sulfate therapy.
   e. Furosemide ototoxicity affects stria vascularis of scala media.
4. **Noise-induced hearing loss:** Audiogram shows a dip at 4,000 Hz in air conduction curve.
5. **Sudden sensorineural hearing loss:** The causes of sudden sensorineural hearing loss due to hemorrhage into cochlea include leukemia, sickle cell disease and thalassemia. Hypertension does not cause hemorrhage in cochlea.
6. **Presbycusis:** Decreased ability to perceive high-frequency sounds in elderly people due to the loss of hair cells at the base of cochlea.

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14 Hearing Impairment in Infants and Young Children

Men in general lay all the blame of life on their fellowmen, God or fate. We reap what we sow; we are the makers of our own fate. The wind is blowing; those vessels whose sails are unfurled catch it, and go forward on their way, but those which have their sails furled do not catch the wind. Is that the fault of wind?

—Swami Vivekananda

Points of Focus

- **INTRODUCTION**
- **ETIOLOGY**
  - ✓ Prenatal Causes
  - ✓ Perinatal Causes
  - ✓ Postnatal Causes
- **CLINICAL FEATURES**
  - ✓ Meningitis
  - ✓ Syndromes with Genetic Hearing Loss
- **HIGH-RISK REGISTRY**
- **UNIVERSAL NEWBORN HEARING SCREENING (UNHS)**
  - ✓ Interpretations
  - ✓ Limitations
  - ✓ Brainstem Evoked Response Audiometry
  - ✓ Evoked Otoacoustic Emission: Transient and Distortion Evoked Otoacoustic Emission
- **EVALUATION OF UNHS REFER INFANTS**
  - ✓ History
  - ✓ Physical Examination
  - ✓ Hearing Tests
- **OTHER HEARING TESTS**
  - ✓ Screening Procedures: Arousal Test, Auditory Response Cradle
  - ✓ Behavior Observation Audiometry: Moro's Reflex, Cochleopalpebral Reflex, Cessation Reflex
  - ✓ Distraction Techniques
  - ✓ Conditioning Techniques: Play Audiometry
  - ✓ Speech Audiometry
  - ✓ Ancillary Laboratory Testing
  - ✓ Screening for Maternally Transmitted Infection (TORCH and Syphilis)
  - ✓ Clinical Testing
  - ✓ Genetic Testing: Screening for Connexin 26 Mutation
  - ✓ Temporal Bone Imaging
- **TREATMENT**
  - ✓ Hearing Aids
  - ✓ Cochlear Implants
- **REHABILITATIVE MEASURES**
  - ✓ Parental Guidance
  - ✓ Development of Speech and Language: Auditory-Oral Communication, Manual Communication
  - ✓ Total Communication: Vibrotactile Aids
  - ✓ Education and Vocation of Deaf
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

Bilateral severe to profound (70 dB HL or more) hearing loss occurs in 1 per 1,000 live births. These children fail to develop speech. These children though have no defect in their speech producing apparatus become mute (dumb). Because they never hear any speech they do not develop it. The first 5 years of life is critical for the development of speech and language. Therefore, early identification and assessment of hearing loss in infants and children are vital.

Failure to manage hearing loss in infants and early childhood can affect acquisition of speech, language, cognitive and socioemotional development. Progressive, late-onset and acquired hearing loss in childhood is a diagnostic challenge. In children of less severe hearing loss, speech though developed

may be defective. Hearing loss happens not just to a child but the whole family.

Mild hearing loss in children: Address it early to prevent speech delay in children.

ETIOLOGY

Etiology of hearing loss remains obscure in about 50% children. The causes of hearing loss in children are traditionally divided into three groups: before birth (prenatal), during birth (perinatal) and after birth (postnatal).

Prenatal Causes

- *Genetic defects*: They may affect inner ear alone or several organs (Table 1: syndromic deafness). The inner ear pathologies include:
  - Michel aplasia: The inner ear does not develop at all.
  - Mondini aplasia: Cochlea has only 1.5 turns.
  - Scheibe aplasia: In this commonest type of inner ear pathology, the membranous cochlea and saccule are abnormal, but rest of the inner ear is normal.
  - Alexander aplasia: This pathology, which affects only basal turn of cochlea, causes high-frequency sensorineural hearing loss (SNHL).
- *Maternal infections*: They include toxoplasmosis, rubella, cytomegalovirus and herpes, type 1 and 2 (TORCH) and syphilis.
- *Drugs*: They include streptomycin, gentamicin, tobramycin, amikacin, quinine and chloroquin. They cross the placental barrier and damage the cochlea. Thalidomide causes abnormalities of ears, limbs, heart, face, lip and palate.
- *Radiation*: Especially in first trimester.
- *Other factors*: They are nutritional deficiency, diabetes, toxemia and hypothyroidism.

Perinatal Causes

- *Anoxia*: The neonatal anoxia is caused due to placenta previa, prolonged labor, cord round the neck and prolapsed cord. It damages the cochlear nuclei and causes hemorrhage into the ear.
- *Prematurity*: Birth before term and birth weight less than 1,500 gm are risk factors.
- *Birth injuries*: Forceps delivery can result in intracranial hemorrhage with extravasation of blood in the inner ear.
- *Neonatal jaundice*: Bilirubin level more than 20 mg % damages the cochlear nuclei.
- *Ototoxic drugs*: They are used for neonatal meningitis or septicemia.
- *Neonatal meningitis.*

Postnatal Causes

- *Genetic*: Though genetic, it can manifest later in childhood or adult life (Table 1). The child may have hearing loss alone (as in familial progressive sensorineural deafness) or with anomalies of other systems (such as Alport, Klippel-Feil and Hurler syndromes).
- *Nongenetic*: They are similar to adults and include following:
  - Infections: Measles, mumps, varicella, influenza, meningitis, encephalitis and otitis media.
  - Ototoxic drugs: Chapter "Sensorineural Hearing Loss."
  - Trauma: Fractures of temporal bone, middle ear surgery or perilymph leak.
  - Noise trauma: Chapter "Sensorineural Hearing Loss."

TABLE 1 Syndromes with genetic hearing loss

Syndrome	*System*	*Features*
Congenital autosomal recessive		
Onchodystrophy	Nail	Dystrophy of finger and toe nails, defects of teeth, hair, sebaceous glands
Jervell and Lange-Nielson	Heart	Syncopal attacks, prolonged QT interval in ECG
Pendred's	Thyroid	Goiter, hypothyroidism
Klippel-Feil		Short neck (fusion of one or more cervical vertebrae) spina bifida, canal atresia, mixed loss
Congenital autosomal dominant		
Waardenburg's	Skin	White forelock, depigmentation of skin, heterochromia iridis, increased intercanthal distance and antimongoloid slant of eyes
Teacher-Collins (mandibulofacial dysostosis)	Craniofacial and cervical	Hypoplasia of malar bones and mandible, coloboma of lower lids, malformed external and middle ear and conductive loss
Crouzon's		Craniofacial dysostosis (frog eyes), parrot nose, ear malformation, mixed loss
Delayed onset autosomal recessive		
Usher's	Eyes	Progressive retinitis pigmentosa
Delayed onset autosomal dominant		
Alport's	Kidney	Glomerulonephritis and corneal dystrophy
Van der Hoeve's	Skeletal	Fragile bones, blue sclera and deafness (conductive, sensorineural or mixed)
Congenital chromosomal abnormalities		
Trisomy 21(Down's syndrome)		Stenosis of ear canal, high incidence of serous otitis media
Trisomy 13–15		Low-set ears with malformation, cleft lip and palate, congenital heart disease
Trisomy 16–18		Cardiac abnormalities, low-set pinna, atresia of canal

CLINICAL FEATURES

Hearing loss should be suspected in the following events:

- Sudden loud sounds/noises fail to disturb the sleep and startle the child.
- Failure to develop speech in 1–2 years.
- Defective speech and poor school performance can be due to hearing loss. These children may be wrongly labeled as mentally retarded.

Meningitis

- About 33% of all acquired SNHL are caused by meningitis. Hearing loss varies by causative organisms: *Streptococcus pneumoniae*, 20%; *Haemophilus influenzae*, 12%; and *Neisseria meningitidis*, 5%.
- Bacteria, fungi or inflammatory cells reach the basal turn of scala tympani from the subarachnoid space by way of a patent cochlear aqueduct (often patent in small children and closes as the skull grows) or up the internal auditory canal reaching the cochlear modiolus passing through perineural and perivascular spaces.
- *Otological findings*: Deafness (usually profound and permanent involving all frequencies) may be either unilateral or bilateral. It usually occurs early.
  - Vertigo, nausea, vomiting and ataxia often develop regardless of hearing loss.
- *Treatment*: Immediate broad-spectrum antibiotic and then modified as per the report of culture and sensitivity of CSF.
  - Corticosteroids at the earliest for 2 days reduces incidence of hearing losses, which once occur is permanent.

Syndromes with Genetic Hearing Loss (Table 1)

Fifty percent cases of infant hearing loss are genetic in origin. Of them, 75–80% is autosomal recessive, 15–20% is autosomal dominant, and 1–2% is X-linked. Few children have mitochondrial inheritance (Table 1).

HIGH-RISK REGISTRY

Failure to manage hearing loss in infants and early childhood can affect acquisition of speech, language, cognitive and socioemotional development. Two percent to 2–5% neonates with following risk factors have moderate to profound hearing loss (Fig. 1).

- Family history of hereditary childhood SNHL
- Prenatal infections TORCH
- Craniofacial abnormalities
- Ototoxic drugs
- Birth weight less than 1,500 gm
- Stigmata or other findings associated with hearing loss (conductive or sensorineural) syndrome, such as deformed pinna, cleft palate and craniofacial deformities
- Neonatal jaundice: Bilirubin level more than 20 mg % requiring an exchange transfusion
- Bacterial meningitis: Especially *Haemophilus influenzae*
- Neonatal asphyxia with seizures or coma: No spontaneous respiration in first 10 minutes; hypotonia persisting for 2 hours. Mechanical ventilation lasting for 5 days or more
- Apgar score[1]: 0–4 at 1 minute or 0–6 at 5 minutes

About 50% of children with moderate to profound congenital hearing loss have no risk factors. Therefore, the universal newborn hearing screening (UNHS) has evolved.

UNIVERSAL NEWBORN HEARING SCREENING (UNHS)

The UNHS is done within first 3 months of life. It detects the permanent hearing loss at an average age of 3 months. But in the absence of UNHS, average age of diagnosis was found to be 31.25 months. Reliance on physician and parent's observations were not found successful. Behavioral methods of testing

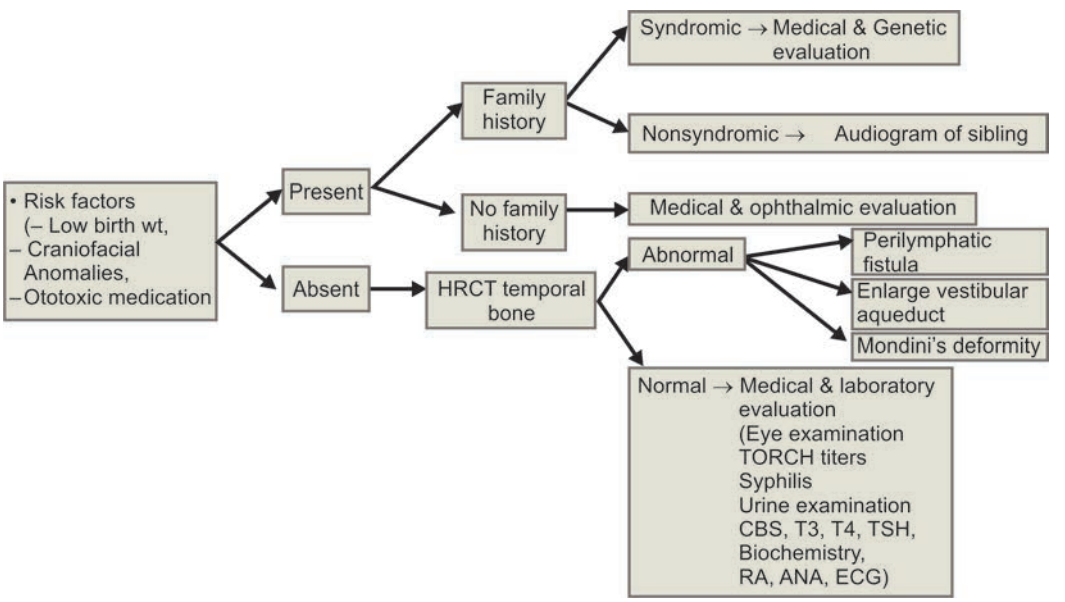

Fig. 1: Flow chart for differential diagnoses of sensorineural hearing loss (SNHL) in children

[1] Evaluation of a newborn's physical status by assigning numerical values (0–2) to each of five criterias: 1) heart rate, 2) respiratory effort, 3) muscle tone, 4) response stimulation, and 5) skin color; a score of 8–10 indicates best possible condition.

do not meet the requirement of UNHS. Currently, brainstem evoked response audiometry (BERA) and evoked otoacoustic emissions (EOAE) are employed either singly or in combination.

Interpretations

The screening outcomes are either pass or refer:

- Pass infants need no further testing at this time but progressive, late-onset and acquired hearing loss in childhood cannot be ruled out at this time.
- Refer infants require a diagnostic hearing evaluation. Electrocochleography measures auditory sensitivity to within 20 dB. It is an invasive procedure and avoided.

The future development of frequency-specific auditory brainstem responses [(ABR, BERA)] or auditory steady state responses (ASSR) will probably replace current two-tiered screening protocol (OAE, ABR/ASSR).

Limitations

No test is perfect. Regular monitoring of hearing, speech and language milestones is important even in infants who pass TEAOE, DPOAE or ABR.

- *Brainstem evoked response audiometry*: Click-evoked auditory nerve and brainstem evoked potentials (ABEP) often misses hearing losses when hearing is normal at some frequencies.
- *Otoacoustic emissions* screening is difficult in noisy environment, presence of vernix or debris in external auditory canal (EAC) and collapsing EAC in first 24 hours after birth.
  - Otoacoustic emissions may miss inner hair cell and auditory nerve hearing losses.

Brainstem Evoked Response Audiometry

Synonyms: Auditory Brainstem Responses

For hearing screening, responses are elicited by air-conducted clicks (having broadband frequency spectrum) at a level that produces response in normal hearing ears. They produce no response in ears with hearing loss of 30–35 dB HL. This evoked potential, which is present in neonates as early as 25 weeks gestational age, is unaffected by sleep, attention and sedation (Chapter Hearing Evaluation).

Auditory brainstem responses is affected in disorders of external, middle and internal ears and auditory nerve and brainstem. The current automated ABR detection algorithms replace the subjective impression of the examiner as to the presence or absence of ABR.

Evoked Otoacoustic Emission

Evoked otoacoustic emissions are acoustic signals generated by cochlear outer hair cells in response to auditory stimulation. Evoked otoacoustic emissions, which measures only cochlear status, is independent of neural activity and CNS status. Evoked otoacoustic emissions takes lesser time and uses broader frequency range than click-evoked ABR (Chapter Hearing Evaluation).

- *Transient evoked otoacoustic emission* (TEOAE) provides information over a broad frequency range (500–6,000 Hz) that occurs after a brief stimulus. Transient evoked otoacoustic emission are observed in neonatal ears in the absence of external and middle ear disorders.
- *Distortion evoked otoacoustic emission* (DEOAE) provides frequency-specific information that occurs in response to simultaneous presentation of two pure tones. The screening algorithms, which are robust in neonates and infants, use "DP grams."

EVALUATION OF UNIVERSAL NEWBORN HEARING SCREENING REFER INFANTS

The infants who fail UNHS should have detailed history, physical examination including ear microscopy and vestibular tests, threshold frequency-specific ABR and OAE and visual reinforcement audiometry (VRA). Newborn hearing screening does not identify progressive hearing loss, which does occur in preschool children and constitute 15–20% of SNHL young children.

The ancillary testing includes laboratory and genetic testing, screening for maternally transmitted infection and temporal bone imaging.

History

- Family history for congenital hearing loss, eye and cardiac abnormalities
- Prenatal maternal history of infection (TORCH and syphilis), diabetes, hypothyroidism, alcohol, smoking and drugs
- Perinatal risk factors

Physical Examination

A complete physical examination is mandatory though majority of infants do no have any positive findings. See for the features mentioned in Table 1 showing hearing loss syndromes.

External and middle ear disorders are the most common source of failed initial hearing screening. A thorough evaluation should include following elements:

- Neurologic and developmental milestones
- Nystagmus: Spontaneous and head shaking
- EAC: Clean it and see for stenosis and atresia
- Tympanic membrane for thickness, appearance and motion and middle ear effusion (MEF)
- Pneumatic otoscopy in older children: In 0–6 months of age, both tympanometry and pneumatic otoscopy can miss MEF The incidence of MEF in the Ist year of life is quite high (61%).

Tympanostomy tube insertion is done for persistent MEF. In infants of acute otitis media, antibiotic therapy is given with close follow-up for the effusion resolution.

Hearing Tests

- *Frequency-specific ABR thresholds*: Both air and bone conductions are evaluated using tone-pip ABR. Significant air-bone gap indicates external and middle ear disorders.
- *Evoked otoacoustic emission*: See other sections and Chapter Hearing Evaluation.
- *Tympanometry using a probe-tone frequency of 1,000 Hz or more*: Provide status of middle ear. The traditional tympanometry (using 226 Hz probe tone) is invalid in 0–6 months of age.
- *Visual reinforcement audiometry*: Visual reinforcement audiometry is a form of conditioning technique. Infants turn their heads in the direction of sound source. If this response is rewarded by activation of a lighted, animated toy, the infant usually continue to respond and an audiogram is obtained.

Visual reinforcement audiometry testing can be done either under earphones or sound field testing. Older infants, who yield fewer false positive responses than 6 months infant, tend to reject earphones.

- Visual reinforcement audiometry thresholds for 6 months infant are within 10 dB of adult thresholds.
- Visual reinforcement audiometry detects even slight threshold elevations seen in MEF.

OTHER HEARING TESTS

The auditory function tests in neonates, infants and young children can be classified into various groups (Box 1). Though currently the gold standards are ABR and OAE, the following tests were done traditionally.

- *Screening procedures*: These are based on infant's behavioral response to the sound signal.
  - Arousal test: A high-frequency narrow band noise presented for 2 seconds to the light sleeping infant normally awakes the infant twice when three such stimuli are presented.
  - Auditory response cradle: The baby is placed in a cradle. The transducers monitor the infant's behavior (trunk and limb movement, head jerk and respiration) in response to auditory stimulation.
- *Behavior observation audiometry*: Auditory signal produces a change in infant's behavior, such as alerting, cessation of an activity, widening of eyes and facial grimacing.
  - Moro's reflex: It consists of sudden movement of limbs and extension of head in response to sound of 80–90 dB.
  - Cochleopalpebral reflex: A child blinks in response to a loud sound.
  - Cessation reflex: In response to a sound of 90 dB, an infant stops activity or starts crying.
- *Distraction techniques*: The children of 6–7 months age turn their heads to locate the source of sound.
  - Method: The child is seated in mother's lap. One person distracts the child's attention while the other produces a sound from behind or from one side. Note whether the child tries to locate the sound by turning head.
  - Sounds: They are high-frequency rattle (8 kHz), low-frequency hum, whispered sound as "S, S, S", xylophone, warbled tones or narrow band noise (500–4,000 Hz).
- *Conditioning techniques (play Aaudiometry)*: The child is conditioned to perform any act on hearing a sound. The acts may be placing a marble in a box, a wooden or plastic block in the bucket, or a ring on a post. The test can be done either in the free-field or by using headphones. A frequency-specific audiogram is possible in a child of 2–4 years of age.

Box 1: Methods of hearing assessment in infants and young children

- Neonatal screening procedures: Arousal test, auditory response cradle, electric response audiometry
- Behavior observation audiometry: Moro's reflex, cochleopalpebral reflex, cessation reflex
- Distraction techniques
- Conditioning techniques: Visual reorientation audiometry, play audiometry
- Objective tests: Evoked response audiometry, impedance audiometry, otoacoustic emissions, heart rate audiometry

- *Speech audiometry*: The spondee words are presented to the child along with the pictures. The child is asked to point the appropriate picture or repeat the word. The intensity level of the spondee words is gradually lowered. The speech reception threshold is determined. To examine the expressive ability, the child is asked to name the toys and objects.

Ancillary Laboratory Testing

They are ordered either to rule out or confirm specific etiologies suspected by hearing tests result, child's age, history, physical examination, routine blood chemistry and complete blood count. For example, in cases of family history of episodes of syncope, ECG is taken to look for the evidence of Jervell and Lange-Nielsen syndrome.

Screening for Maternally Transmitted Infection (TORCH and Syphilis)

- Infant' blood for specific IgM for TORCH and syphilis.
- Cultures for CMV in blood, urine or saliva: Detection in first 3 months of life and early treatment may lessen severity of neurologic sequelae.
- FTA-ABS test for congenital syphilis.

Clinical Features

- *Congenital toxoplasmosis*: Usually associated with chorioretinitis, intracerebral calcification and microcephaly. Early treatment may reduce progress of hearing loss.
- *Herpes simplex type 1* (transplacental transmission) and *type 2* (vaginal transmission).
- *Congenital rubella*: Fifty percent of the patients present with hearing loss alone.

Genetic Testing

- *Chromosomal karyotyping* is indicated when SNHL infant has:
  - Unrecognizable constellation of anomalies
  - Neurologic or cardiac defects
  - Mother gives history of multiple miscarriages
  - Suspected Pendred's syndrome
- *Screening for connexin 26 mutation*: GJB2 gene encodes for connexin 26 and is said to be responsible for about 50% of all autosomal recessive congenital hearing loss.
  - 35delG deletion: It involves a single base pair and is responsible for 70% of all connexin 26 mutations. The screening tests for 35delG deletion are available.
  - DFNB1 (prefix DFNB, signifies autosomal recessive hearing loss): It stands for autosomal recessive nonsyndromic hearing loss mutations in connexin 26 and is associated with prelingual nonprogressive bilateral SNHL (mild to profound) without temporal bone CT abnormality.

Temporal Bone Imaging

- *Noncontrast Temporal Bone CT*:
  - In all children with SNHL especially in progressive SNHL and craniofacial anomalies.
  - The dilated vestibular aqueduct is associated with Pendred's syndrome.

- Mondini dysplasia and semicircular canals abnormalities are also seen.
- Internal auditory canal stenosis and possible auditory nerve aplasia is associated with X-linked form of deafness (DFN3).

- *Magnetic Resonance Imaging*: It is done prior to cochlear implantation to assess patency of the cochlea, position of facial nerve and presence of auditory nerve.

Deaf and dumb: Deaf persons are not intellectually dumb. Psychological tests usually overlook the intellectual capacity of deaf people as they consist of culturally biased material.

TREATMENT

- *Otitis media*: In infants of acute otitis media, antibiotic therapy is given with close follow-up for the effusion resolution.
  - The incidence of MEF in the Ist year of life is quite high (61%). Tympanostomy tube insertion is done for persistent MEF.
- *Meningitis*: Immediate broad spectrum antibiotic and then modified as per the report of culture and sensitivity of CSF.
  - Corticosteroids at the earliest for 2 days reduces incidence of hearing losses, which once occur is permanent.
- *Auditory rehabilitation*: After identifying the type and extent of hearing loss, appropriate rehabilitative measures should be initiated at the earliest, which include amplification and cochlear implantation. Hearing aids help in developing lip reading.

Hearing Aids

- Hearing aid should be fit at the earliest. Regular follow-up appointments with audiologists is important (Chapter "Hearing Aid and Cochlear Implant").
- *Indication*: Permanent bilateral hearing loss (conductive or sensorineural) more than 20 dB HL between 1,000 and 4,000 Hz.
- *Bone conduction hearing aids*: They are indicated in cases of atresia, stenotic ear canals and recurrent otorrhea.
- *Children less than 7 years*: They need behind-the-ear (BTE) or bone conduction hearing aids.
- *Infants with 55 dB or higher hearing losses*: Personal wireless FM systems improve the speech-to-noise ratio.

Cochlear Implants

Each child should be evaluated carefully for the cochlear implant candidacy (chapter "Hearing Aid and Cochlear Implant").

- Younger the child better are the results of cochlear implantation. Infants younger than 1 year of age are also receiving cochlear implants.
- Post meningitis deafness needs early implantation to avoid later difficulties of implantation due to cochlear ossification.
- Cochlear ossification (usually post meningitis) and auditory nerve aplasia preclude this intervention.

REHABILITATIVE MEASURES

Playgroups and parent groups facilitate parents to learn how to accommodate their baby's communication needs. Services of expert professionals who deal with childhood hearing loss and early childhood education are important. It is essential to know not only the degree and type of hearing loss but also other associated handicaps, such as blindness or mental retardation. The prelingual and postlingual hearing loss bears prognostic importance.

The broad groups of communication methodologies include American Sign Language (ASL), spoken and signed language and auditory-oral methods. Aims of rehabilitation include development of speech and language, adjustment in society and useful vocational employment.

Parental Guidance

Hearing loss happens not just to a child but the whole family. The parents should be sympathetically told of child's disability and how to care for it. Parents should know regarding the following requirements:

- Care and periodic replacement of hearing aid
- Change of ear moulds as the child grows
- Follow up visits for reevaluation
- Education at home
- Selection of vocation

Development of Speech and Language

Reception of information occurs through visual, auditory or tactile faculties while expression is through the oral/written speech or manual sign language. The proper communication needs either the improvement in hearing through amplification or the development of visual or tactile means of communication.

- *Auditory-oral communication*: This method of normal people communication can be used for the children of postlingual deafness. Hearing aids augment auditory reception. Training is imparted in speech reading, which encourages attention towards the movements of lips, face and natural gestures of hand and body. Expressive skill should be encouraged through verbal speech.
- *Manual communication*: It uses the sign language and finger-spelling method. The abstract ideas are difficult to express because general public does not understand it.

Total Communication

This form of communication, which is good for children with prelingual severe to profound deafness, employs all the sensory inputs such as auditory, visual, tactile and kinesthetic. These children are taught to develop oral speech, lip-reading and sign language.

- *Vibrotactile aids*: These are useful in children who are both deaf, as well as blind. The aids are fitted to the child's hand or sternum. The vibrations of speech are perceived by the child through tactile sensation (Have you seen film "Black" of Amitabh and Rani Mukherji?).

Education and Vocation of Deaf

- The residential and day schools for the deaf children are available.
- The children with moderate hearing loss can be integrated into normal schools with preferential seating arrangement in the class.

- In radio hearing aids, the microphone and transmitter are with the teacher and the receiver and amplifier are worn by the child. The child can hear the teacher's voice better avoiding the environmental noises.
- Given the opportunity, commensurate with their ability, the deaf people, who are sincere and good workers, can be usefully employed in several vocations.

Clinical Highlights

1. **Antenatal causes**: Inner ear malformation in fetus can occur when pregnant mother is exposed to radiation, German measles, cytomegalovirus and thalidomide.
2. **Michel aplasia**: It is the total lack of inner ear development. MRI, which shows membranous labyrinth, differentiates Michel aplasia from the labyrinthine ossification.
3. **Mondini's malformation**: Cochlea is normal in size but lacks bony partitions between the coils. Cochlea looks empty on CT scan. Vestibule and vestibular aqueduct are dilated. External and middle ears are normal.
4. **Waardenburg's syndrome (WS)**: The features include heterochromia iridis, dystopia canthorum, Hirschsprung's disease, hypopigmented areas of skin and sensorineural hearing loss. (SNHL)
5. **Pendred's syndrome**: This congenital profound sensorineural hearing loss is associated with abnormality of thyroxin synthesis. Goiter, which later becomes nodular, appears in adolescence.
6. **Alport's syndrome**: It is associated with glomerulonephritis.
7. **Klippel-Feil syndrome**: It is associated with congenital spinal deformities.
8. **Usher's syndrome**: It is associated with retinitis pigmentosa.
9. **Brain-stem evoked response audiometry (BERA)**: It is indicated in assessing hearing loss in neonates.

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15 Hearing Aids and Cochlear Implants

It is the coward and fool who say, "This is fate". But it is the strong man who stands up and says, 'I will make my fate'. It is people who are getting old who talk of fate. Young men generally do not come to astrology.

—Swami Vivekananda

Points of Focus

- **INTRODUCTION**
- **TRAINING**
  - ✓ Speech Reading (Lip Reading)
  - ✓ Auditory Training
  - ✓ Speech Conservation
- **HEARING AIDS**
  - ✓ Parts
  - ✓ Types
  - ✓ Electroacoustic Properties
  - ✓ Shapes and Sizes
  - ✓ Contralateral Routing of Signals (CROS)
  - ✓ Monaural/Binaural Amplification
  - ✓ Hearing Aids in Children
  - ✓ Analog, Hybrid and Digital Hearing Aids
  - ✓ Evaluation of Hearing Aid Candidates
- **ASSISTIVE DEVICES**
  - ✓ Assistive Listening Devices and Systems
  - ✓ Alerting Devices
  - ✓ Telecommunication Devices
- **IMPLANTABLE HEARING AIDS**
  - ✓ Advantages
  - ✓ Indications
  - ✓ Classification
  - ✓ Classification
  - ✓ Bone-Anchored Hearing Aid (BAHA)
  - ✓ Vibrant MED-EL Soundbridge
- **COCHLEAR IMPLANTS**
  - ✓ Available Devices Components and their Functioning
    - – External Parts: Microphone, Speech Processor, Transmitter
    - – Internal Parts: Receiver/Stimulator, Magnet, and Antenna, Electrode
  - ✓ Outcomes
  - ✓ Indications
  - ✓ Evaluation of Cochlear Implant Candidates
  - ✓ Selection Criteria: General, Adults, and Children
  - ✓ Surgical Procedure
  - ✓ Postoperative Complications
  - ✓ Postoperative Mapping
- **AUDITORY BRAINSTEM IMPLANT**
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

Hearing impaired patients need auditory rehabilitation for better communication. The auditory rehabilitation includes not only instrumental devices but also training.

The various types of instruments are hearing aids, assistive devices and cochlear implants (Box 1). The heightened volume increases audibility and reduces the strain of understanding sound in daily listening situations.

TRAINING

- *Speech reading (Lip reading)*: This integrated process for understanding the speech studies movements of lips, facial expression, gestures and the probable context of conversation. The skill is useful for hearing impaired individuals who have high frequency loss and difficulty in hearing in noisy surroundings.
- *Auditory training*: Auditory training is useful for those using hearing aids and cochlear implants. It is used along with the

Box 1: Instrumental devices for hearing impaired patients

- *Hearing Aids*
  - Body-worn
  - Behind-the-ear (BTE)
  - Spectacle
  - In-the-ear (ITE)
  - Canal types
    - In the canal (ITC)
    - Completely in canal(CIC)
- *Assistive Devices*
  - Assistive listening devices and systems
  - Alerting devices
  - Telecommunication devices
- *Implantable Hearing Aids*
  - Bone-anchored hearing aid (BAHA): Entific BAHA, RetroX
  - Implantable middle ear hearing aids
    - Piezoelectric actuator: Rion-E, Otologics MET, Implex TICA
    - Piezoelecric sensor and actuator: St. Croix Envoy
    - Magnetic actuator: Vibrant MED-EL Soundbridge, Soundtec DDHS
- *Cochlear Implants*
  - Advanced Bionics Corporation: HiRes 90K
  - Cochlear Corporation: Nucleus 5 System
  - MED-EL Corporation: C40 + device

skill of speech reading. The patient is exposed to various listening situations with different degrees of difficulty. They are taught selectively to concentrate on speech sounds.

- *Speech conservation*: Patients with profound hearing loss loses the ability to monitor their own speech production. The defects appear in articulation, response, pitch and the volume of voice. Speech conservation educates such persons to use their tactile and proprioceptive feedback systems so that they can monitor their speech production.

HEARING AIDS

Any patient of hearing loss, conductive or sensorineural who cannot be helped by medical and surgical means, is a candidate for hearing aid (Figs 1 to 3).

Parts

A hearing aid machine presents amplified sounds to the ear. It has the following parts (Fig.1):

- *Microphone*: It picks up sounds and converts them into electrical impulses.
- *Amplifier*: It amplifies electrical impulses.
- *Receiver*: It converts electrical impulses back to sound.
- *Volume control*
- *Battery power source*
- *Earmold*: The amplified sound is carried to the ear through the earmold. Poor earmold fitting results in annoying acoustic feedback, amplification of background noise and distortion of sound.

Types

There are mainly two types of hearing aids: (1) air conduction and; (2) bone conduction. Most of the hearing aids are of air conduction type.

1. *Air conduction hearing aid*: The amplified sound is transmitted to the external auditory canal (EAC), in which the receiver is situated.
2. *Bone conduction hearing aid*: The amplified sound is transmitted to the mastoid bone through a bone vibrator which snugly fits on the mastoid. So the vibrator acts as a receiver and directly stimulates the cochlea. This type of aid is indicated in patients who suffer from active ear discharge, otitis externa and atresia of the ear canal.

Electroacoustic Properties

The following terms are used to describe the properties of hearing aid:

- *Saturation sound pressure level*: Saturation sound pressure level (SSPL) is the maximum amount of sound pressure output (power), which can be produced by an aid.
- *Acoustic gain*: It is the difference between the output and input of a hearing aid. Example: If a tone is presented at 60 dB and

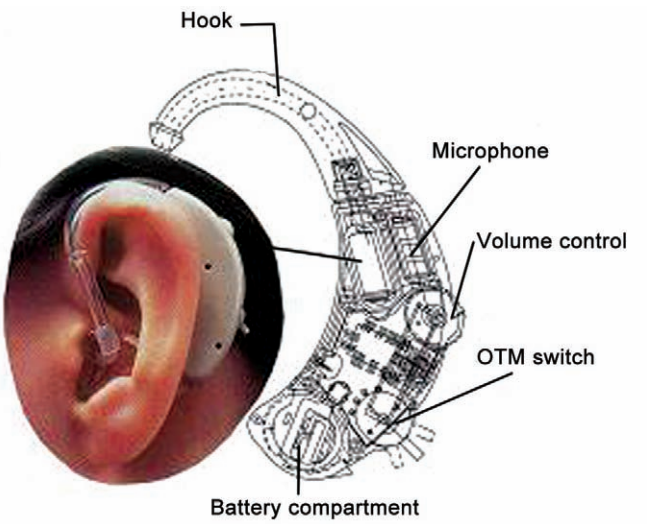
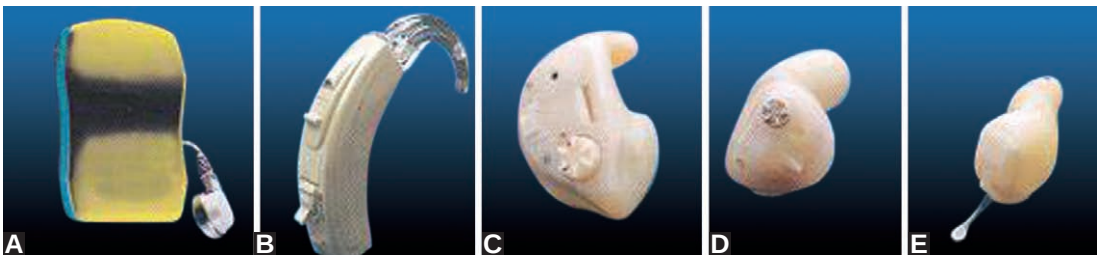


Fig. 1: Behind-the-ear hearing aid in position and its parts
Source: Siemens



Figs 2A to E: Hearing aids. (A) Body-worn; (B) Behind-the-ear; (C) In the ear; (D) In the canal; (E) Completely in canal
Source: Siemens

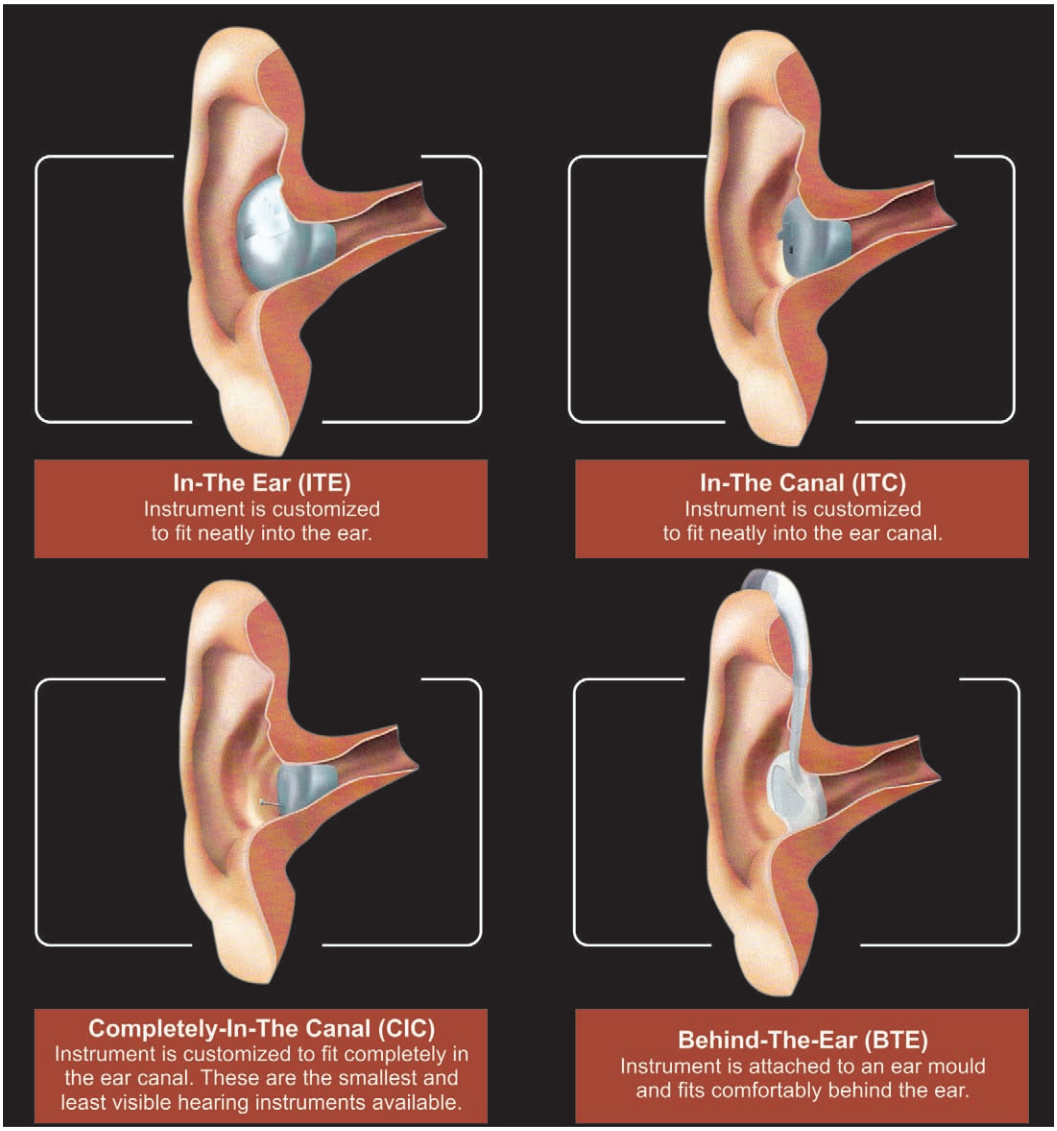


Fig. 3: Hearing aids in position
Source: A&M Hearing Instruments

hearing aid produces output of 100 dB, then the acoustic gain of that hearing aid at that frequency is 40 dB.

- *Frequency response*: It expresses the gain of a hearing aid across a range of frequencies.
- *Distortion*: It is the clarity of sound produced by a hearing aid.
- *Loudness compression*: The amplification of sound is not tolerated beyond a level that is comfortable to the patient. To avoid uncomfortable output level, loudness compression automatically reduces the gain when hearing input increases.
  - Single-band compression applies across the entire frequency range.
  - Multiple-channel compression applies to specific frequencies at which the patient has recruitment or loudness discomfort.
- *Occlusion effect*: Complete occlusion of EAC shifts in the peak of the natural resonance of the EAC. An increase in low frequency amplification occurs, which is not desired in normal low frequency hearing. The occlusion effect can be lessened either with a vent or electronic filtering of low frequency.
- *Acoustic feedback*: It occurs when amplified sound leaks from the receiver back into the microphone. It leads to an unpleasant high-pitched squeal. Acoustic feedback, which is more common in ITE and ITC, results from short microphone-to-receiver distance, wax, vents and poor fitting.

  Poor earmold fitting results in annoying acoustic feedback, amplification of background noise and distortion of sound.

Shapes and Sizes

On the bases of shapes and sizes (Figs 2 and 3), hearing aids can be classified into following groups:

- *Body-worn*: Microphone, amplifier and the battery are in one small box, which is worn at the chest level. The receiver is situated in the EAC. It allows high degree of amplification and minimal feedback. These aids are rarely used now because of their size. Clothes rub against the microphone and cause excessive unusual sounds.
- *Behind-the-ear*: Microphone, amplifier, receiver and battery lie in one small unit which is worn behind the ear (BTE) by a hook-shaped rigid tube that fits over the auricle. Through a

soft polyethylene tube and an earmold, it is coupled to the ear canal. It has been used for mild-to-moderate hearing loss particularly high frequency ones.

- *Eye-glass-aid or spectacle types*: In this modification of BTE, the unit is housed in the auricular part of the spectacle frame. Though not very popular now, it can be used by hearing loss persons who wear eye glasses.
- *In-the-ear*: All the parts of hearing aid are housed in a small box, which resembles earmold in size and shape and is worn in the ear (ITE). It has been used in cases of mild-to-moderate hearing losses with flat configuration.
- *Canal types*: The hearing aid is small and worn in the ear canal. They are liked because of their small size. The ear canal should be large and wide. The patient should have the dexterity to manipulate the minute controls in the aid. Small size limits number of controls for adjustment. Two types of hearing aids are available in this category:
  - ITC: They have limited venting options.
  - CIC: In CIC, which is the smallest and invisible type, only a clear filament protrudes from the ear canal and helps in removing the hearing aid. They provide full or partial resolution of the occlusion effect. Telephones can be used as they do not need a vent. They resolve wind-noise problem. They can develop feedback with jaw movement.
- *Features of BTE and ITE/ITC/CIC*: They are the most commonly used hearing aids.
  - ITE/ITC/CIC are one shell hearing aids and improve localization of sounds.
  - Smaller the size higher is the cost and cosmetic appearance.
  - Increased amplification provided by the pinna boosts gain in higher frequencies.
  - Amount of gain is limited due to problems with acoustic feedback.
  - Patients need to have good vision and manual dexterity to manipulate smaller size hearing aid. This is a problem for older patients.
  - The canal length and bore size and vent size of BTE earmold or ITE shell may be used to alter the acoustic output.
  - A relatively large vent is required if less output at low frequency is desirable as in cases of high frequency hearing loss.

Contralateral Routing of Signals

In this special hearing aid, microphone is fitted in the profoundly deaf ear and the receiver is placed in the better ear. The microphone in the deaf ear picks up sounds, which are passed to the receiver placed in the better ear. This type of arrangement, which is useful in patients with unilateral profound hearing loss, helps in localizing the sound coming from the side of the deaf ear.

The contralateral routing of signals (CROS) can be accomplished through wiring or through radiofrequency transmission. A nonoccluding or open earmold is used in better ear to allow detection of sounds without amplification. If the better ear also has hearing loss, a microphone can be made available on that side.

Monaural/Binaural Amplification

In cases of bilateral hearing loss, binaural hearing aid amplification eliminates head shadow effect and is advantageous. In cases of monaural hearing aid, another ear suffers a reduction in word recognition score. The other benefits of binaural amplification include better speech discrimination and localization. The improved ease of listening avoids sensory deprivation.

- *Head shadow effect*: When sound has to cross the head to reach other side of the ear, 6 dB loss in sound intensity occurs.

Hearing Aids in Children

An infant should be fitted with a hearing aid based on available audiometric information. More reliable hearing tests can be done later on. The amplification is adjusted as the child grows. Overamplification is avoided as it can damage the inner ear. In infants with congenital atresia or microtia, hearing aid can be fitted at the age of 2 months.

- Prelingual deafness is the hearing loss that occurs prior to the development of basic spoken language skills, which usually occurs at 2–3 years of age.
- Postlingual deafness is the hearing loss that occurs after the development of basic spoken language skills. It has prognostic importance in candidates of cochlear implants.

Analog, Hybrid and Digital Hearing Aids

- *Analog hearing aids*: Traditional hearing aids are analog machines. They convert the acoustic energy to electric signals (similar in shape or analogous to actual sound waves), which are amplified and delivered to transducer that converts electric signal back to acoustic energy.
- *Hybrid hearing aids*: They use digital components that control and modify the operation of the analog components in the signal-processing stage.
- *Digital hearing aids*: In these, electric signal is modified to a digital signal, which is composed of discrete signals coded by binary numbers. These have not only the higher fidelity signal but can also modify output in many ways. They can be thus programmed for an individual's specific needs. Though they are expensive, they can contain several frequency responses, which can be selected by the patients depending on their auditory needs.

Evaluation of Hearing Aid Candidates

- *History and physical examination* should note the following elements:
  - Age and dexterity of patient
  - Condition of the outer and middle ear
  - Cosmetic concerns of the patient
  - Type of earmold
- *Audiogram*: It determines following parameters and helps in selecting the type of hearing aids:
  - Degree of hearing loss
  - Frequencies affected
  - Type of hearing loss: Conductive or sensorineural
  - Presence of recruitment
  - Uncomfortable loudness level
  - The type of fitting: monaural, binaural (separate aid for each ear), binaural Y-connection (one aid with two receivers) or the CROS type

ASSISTIVE DEVICES

Assistive devices help the hearing impaired patients in special difficult situations, warn them of danger signals and help in telecommunication. These devices are divided into three groups:

- *Assistive listening devices and systems*: These devices help the hearing impaired to listen efficiently in the presence of background noise, over the telephone, in auditoriums or theaters. They can be used by either individual person or by a group. The technology employed includes hard-wired system, induction loops, amplitude modulation (AM), frequency modulation (FM) or infrared signals.
- *Alerting devices*: They produce an extra loud sound signal so that patients can hear a telephone or a doorbell, a baby crying in another room, an alarm clock or the noise of a smoke detector.
  - Hearing dog: It is trained to bark loudly at the sound of a doorbell and cry of a baby.
  - Light signal or vibrations: Patients with profound or total hearing loss need devices where the sound is changed into a light signal or vibrations. Alarm clock with flashing lights and devices producing strong vibrations can awake the individual and even shake the bed.
- *Telecommunication devices*: An amplifier added to the hand set of a telephone can amplify the sound. A telephone coupler can be connected to the telephone and the signal produced is picked up by the hearing aid.
  - Telecommunication devices for the deaf (TDD): They are meant for the profoundly or totally deaf patients. They convert typed message into sounds, which is transmitted over the standard telephone lines. The other end TDD converts these sound signals back into the written massages.
  - Closed caption television decoder provides cues to the hearing impaired patients so that they can enjoy news, movies and other TV programs.

IMPLANTABLE HEARING AIDS

Implantable hearing aids can be totally or partially concealed. The transducer of the aid is coupled directly to the bone or ossicular chain (malleus or stapes).

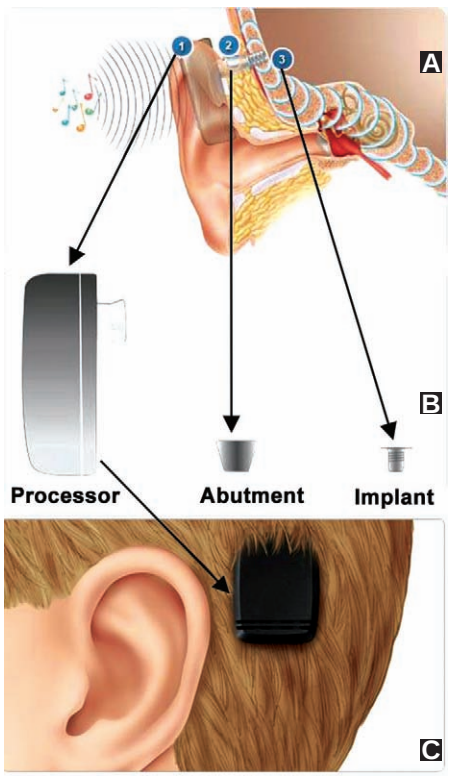
- *Advantages*: These devices offer:
  - Better acoustic gain
  - No feedback
  - Low battery consumption
- *Indications*: In addition to sensorineural hearing losses, they are indicated in conductive hearing losses, (congenital or acquired), which are not amenable to surgical treatment.
- *Classification*: Implantable hearing aids can be classified into following categories:
  - BAHA: Entific BAHA and RetroX
  - Implantable middle ear hearing aids
    - *Piezoelectric*: An electric current is passed into a piezoceramic crystal, which changes its volume and produces vibratory signals. The piezoelectric transducer is coupled to ossicles and drives the ossicular chain by vibration. There are two types of piezoelectric devices:
      - Piezoelectric actuator: Rion-E, Otologics middle ear transducer (MET), Implex totally integrated cochlear amplifier (TICA).
      - Piezoelectric sensor and actuator: St. Croix Envoy.
    - *Magnetic actuator*: An electric current is passed into a coil, which creates a magnetic flux and drives an adjacent magnet. The magnet is attached to the ossicle, which conveys vibrations to the cochlea. The examples of this aid include Vibrant MED-EL Soundbridge and Soundtec DDHS.

Salient features of some of the above will be described in this section.

Bone-Anchored Hearing Aid (BAHA)

Bone-anchored hearing aid is based on the principle of bone conduction. It uses surgically implanted abutment to transmit sound by bone conduction directly to cochlea (Figs 4A to C). Thus, BAHA bypasses EAC and middle ear.

- *Components*: BAHA has following three components:
  - *Titanium fixture*: It is surgically embedded in the skull bone. The process of osseointegration, whose completion usually takes 2–6 months, binds the titanium fixture with the surrounding tissue.
  - *Titanium abutment*: The abutment is attached to the titanium fixture. It remains exposed outside the skin.
  - *Sound processor*: After the completion of osseointegration, it is attached to the abutment.
- *Indications*: BAHA is suited to following category of patients, who cannot use or be benefited by BTE and ITE/ITC/CIC hearing aids (Box 2):



Figs 4A to C: Bone-anchored hearing aid (BAHA). (A) BAHA system; (B) BAHA parts; (C) BAHA processor in position
Source: Cochlear Limited

Box 2: Disadvantages of BTE and ITE/ITC/CIC hearing aids

- Acoustic feedback
- Spectral distortion
- Occlusion of EAC
- Blockage of EAC and insert by ear wax
- Earmold skin reaction
- Not suitable in discharging ears

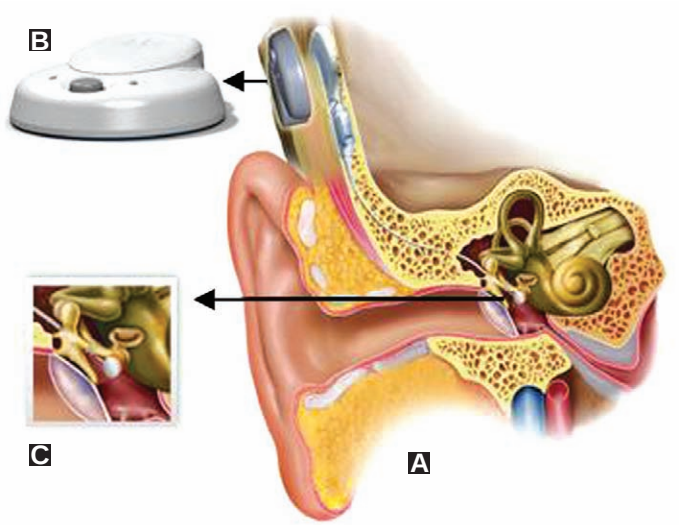
Key: BTE, Behind-the-ear; ITE, In-the-ear; ITC, In the canal; CIC, Completely in canal, EAC, External auditory canal.

- Chronic otitis media and externa: Chronic otorrhea not amenable to treatment.
- Congenital anomalies of ear such as microtia.
- Canal atresia (congenital and acquired) not amenable to treatment.
- Otosclerosis and tympanosclerosis: When surgery is contraindicated or patient is not willing for surgery.
- Unilateral profound hearing loss: BAHA offers better result than CROS and substantially improves speech recognition even in noisy background. BAHA is implanted on the side of hearing loss. It transmits the sound to normal side cochlea through the bone conduction. This process eliminates the head shadow effect.

- *Surgical Technique*: Surgery is performed in single stage in adults but children need two stages. In first stage, the titanium fixture is placed into the skull bone. In second stage, which is performed after the 6 months (period of osseointegration) titanium abutment is connected through the skin to the fixture.
- *Complications*: They are very few and include:
  - Failure of osseointegration
  - Local infection and inflammation

Vibrant MED-EL Soundbridge

- *Components* (Figs 5A to C): This semi-implantable hearing aid has two components—internal and external
  - *Internal component*: The vibrating ossicular prosthesis (VORP) is made up of three parts: the receiver, floating mass transducer (FMT) and a conductor link between receiver and FMT (connected to the incus).
  - *External component*: This is worn behind the ear and called audio processor. The audio processor contains a microphone and transmits sound across the skin by radiofrequency to the internal receiver VORP.
- *Indications/Candidacy*: Adult and elderly patients with moderate-to-severe hearing loss who are not satisfied with BTE and ITE/ITC/CIC hearing aids (Box 2).
- *Surgical procedure*: The receiver of the internal device is positioned under the skin over the mastoid bone. After performing mastoidectomy, the ossicular chain is visualized through a posterior tympanotomy approach. FMT is attached to the long process of incus. The residual hearing is not affected as the middle ear ossicular chain is not disturbed. External audio processor is fitted and programmed 6–8 weeks after the surgery. It attaches magnetically behind the ear.
- *Advantages*: The mechanical energy is directly passed to ossicles. Bypassing of EAC and tympanic membrane eliminates problems of occlusion, feedback, discomfort and wax which are faced in external hearing aids. It provides improved sound quality even in noisy surroundings.



Figs 5A to C: Vibrant MED-EL sound bridge. (A) Implant in position; (B) External audio processor worn on head; (C) Internal implantable part floating mass transducer
Source: MED-EL Corporation

COCHLEAR IMPLANTS

Cochlear implants, which are becoming popular, are relatively recent development. These electronic devices convert sound signals into electrical impulses which directly stimulate the cochlear nerve.

They are indicated for patients of profound binaural sensorineural hearing loss (with nonfunctional cochlear hair cells) who have intact auditory nerve functions and show little or no benefit from hearing aids. Other prognostic findings include general health, level of motivation, expectations and patient's support group.

Depending on whether the child was deafened before or after the acquisition of speech and language, hearing loss is called prelingual or postlingual. The children with congenital and early childhood hearing loss need early intervention with hearing aids or cochlear implants. Auditory deprivation during the early developmental period results in degeneration in central auditory pathways, which limits the speech and language acquisition following cochlear implantation.

Available Devices

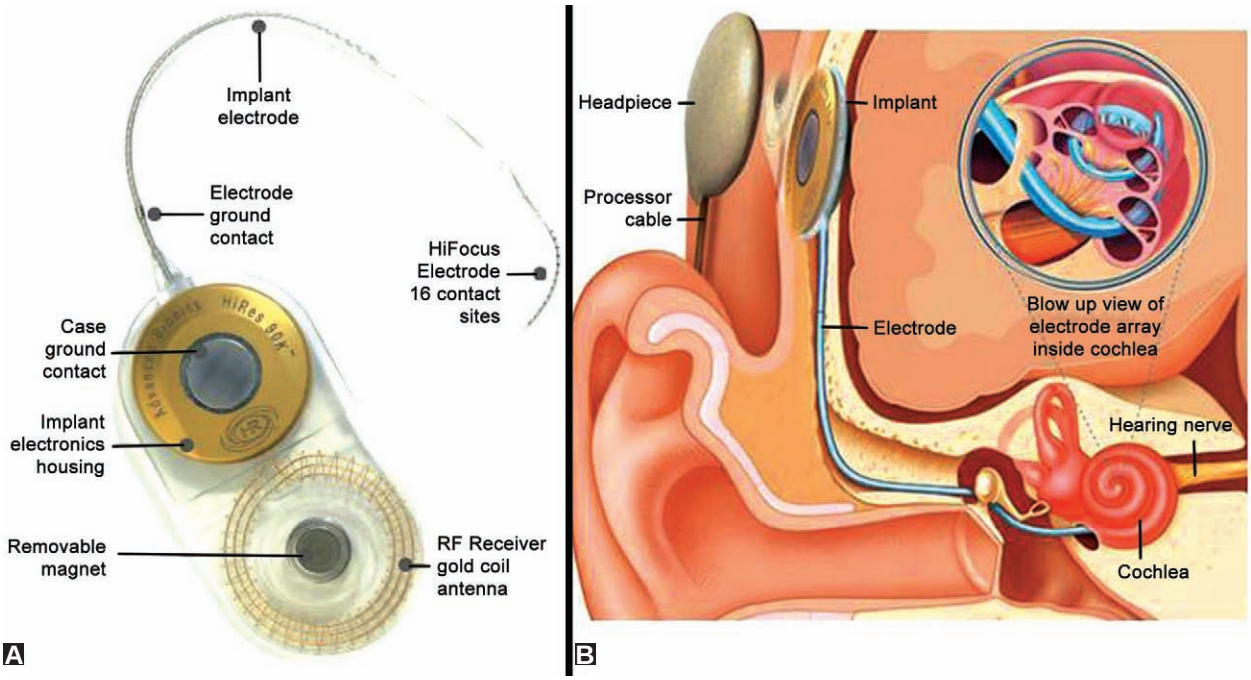
The three devices, which are approved by the US Food and Drug Administration, are manufactured by following companies:

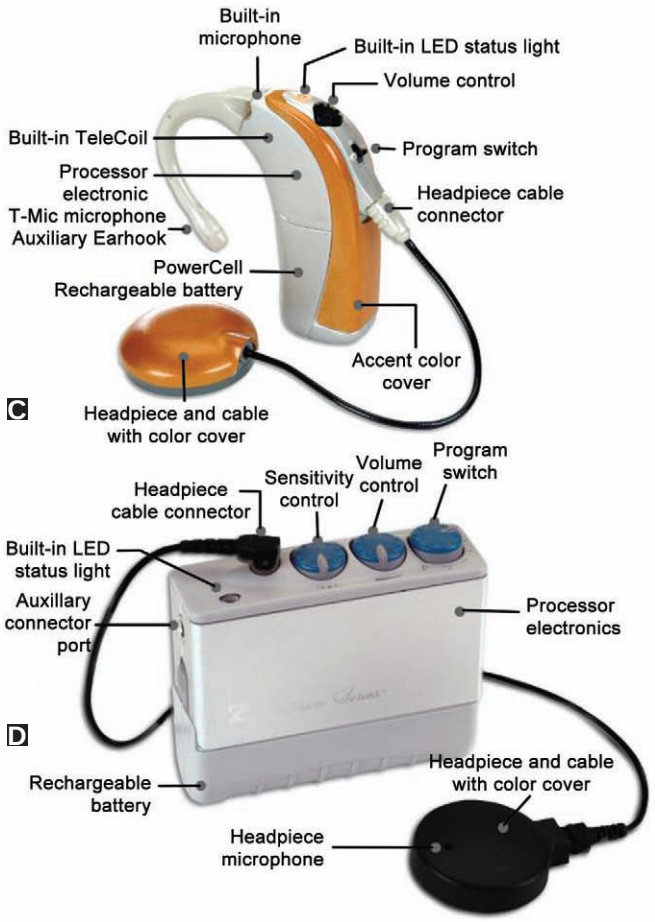
- *Advanced Bionics Corporation*: Clarion C-II/HiRes 90K (Figs 6A to D)
- *Cochlear Corporation*: Nucleus 5 System (Figs 7A to D)
- *MED-EL Corporation*: Pulsar and Sonata (Figs 8A and B)

Each device has its own advantages and disadvantages but no differences in performance between devices have been reported. So selection among them is difficult and final decision usually depends on patient's preference.

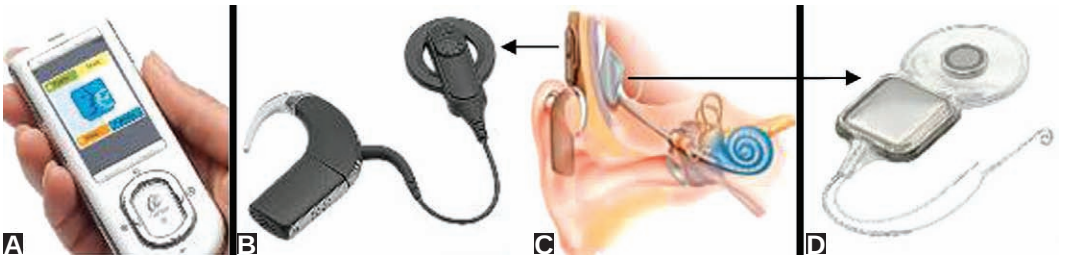
Components and their Functioning

Cochlear implant consists of following internal and external components:

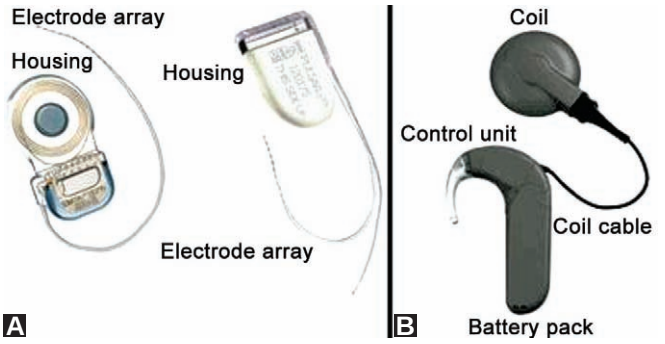




Figs 6A to D: HiRes 90K cochlear implant. A. Internal parts; and B. Implant in position; (C) External Parts (Harmony Behind-the-ear); (D) External parts (Platinum Body-worn)
Source: Advanced Bionics Corporation



Figs 7A to D: Nucleus 5 cochlear implant. (A) Bidirectional remote assistant, which monitor; control and manage the functions of implant; (B) External parts (behind-the-ear); (C) Implant in position; (D) Internal parts
Source: Cochlear Corporation



Figs 8A and B: MED-EL Cochlear Implant. (A) Internal part (Sonata and Pulsar); (B) External parts (Opus 2)
Source: MED-EL Corporation

- *External component*: Microphone, speech processor and transmitter are connected by wires. Current models have one integrated piece.
  - Microphone: Microphone, placed behind the ear (BTE) picks up the acoustic signals and converts it into electrical signals, which are delivered to the speech processor.
  - Speech processor: Speech processor can be body-worn or BTE. It modifies the signal and delivers it to the transmitter. It uses specific speech coding strategy to translate acoustic information into electric stimulation.
    Speech coding strategies: They include:
    - Simultaneous analogue strategy (SAS)
    - Continuous interleaved sampling (CIS)
    - Spectral peak (SPEAK)
    - Advanced combination encoder (ACE)
  - Transmitter: Transmitter is secured over the mastoid to the magnet in the implanted portion of machine. It delivers the signal to the implanted receiver/stimulator via FM radio frequency or magnetic coil.
- *Internal component*: It is surgically implanted and comprises following:
  - Receiver/Stimulator, magnet and antenna: Receiver/stimulator has a magnet and antenna. It is placed under the skin. It decodes and further modifies the signal and delivers it to electrode.
  - Electrode: Electrode is placed into the scala tympani through round window cochleostomy. The electrode has multiple channels. The electrodes stimulate the remaining cochlear nerve tissue, usually spiral ganglion cells in cochlea.

Outcomes

The sound produced by cochlear implants are perceived as auditory sensation (not like normal hearing), which vary in pitch and loudness. The speech processor selects specific characteristics of sound, which are important for understanding the speech. Some patients even become able to understand speech without visual cue. Some patients even enjoy music and can talk on phone. Children can attend normal schools.

Children and adults with postlingual short duration deafness achieve very good results. If implanted early (12 months of age) even prelingual deafened children develop good understanding of speech and acquisition of language over couple of years. They need constant auditory-verbal training. Prelingually deafened adults obtain very limited benefit from cochlear implants. They just develop sound awareness.

- *Factors improving outcomes of cochlear implants in children*: They include:
  - Postlingual hearing loss in children
  - Early implantation (1 year of age) in prelingual children
  - Shorter duration of postlingual hearing loss
  - Neural plasticity within central auditory pathway and area

Indications for Cochlear Implant Evaluations

The following referral criteria are for evaluating the patient to know whether he/she will be benefited by the cochlear implant:
Adults

- Thresholds of 70 dB or more at 1,000 Hz and above in the better ear
- Word discrimination less than 70%
- Communication difficulties even with appropriate hearing aid use

Children: There is no lower age limit for evaluation.

- Thresholds of 90 dB HL or more at 2,000 Hz and above in better ear
- Aided levels poorer than 35 dB HL (especially at 4,000 Hz) in better ear
- BERA:
  - No response in both ears
  - No response in one ear and elevated responses in other ear
- Poor development of auditory or communication skills
- Severely impairing auditory neuropathy/dyssynchrony

Evaluation of Cochlear Implant Candidates

The cochlear implant candidates are assessed carefully for the following elements:

- *History and examination* in detail to determine the medical fitness and predict expected outcomes.
- *Audiologic evaluation* to determine suitability for the implant and depending on the age include–
  - Pure tone audiogram
  - Speech discrimination tests
  - Tympanometry
  - Otoacoustic emissions
  - Auditory brainstem responses
  - Auditory steady state responses
- *Imaging* of temporal bone and brain especially for cochlea and auditory nerve: CT and MRI.
- *Hearing aid fitting and performance*: A hearing aid trial and evaluation is mandatory. To determine the performance, it may include:
  - Aided free-field sound detection thresholds
  - Aided speech perception and discrimination scores
- *Communicative status*: Speech and language evaluation to determine any developmental language or articulation disorders. During the postoperative mapping of the device, it helps in identifying areas of deficit in speech perception and necessary programming of the implant.
- *Psychological evaluation*: To determine the cognitive status or mental functioning disabilities other than hearing loss.
- Comparison between the performance of the patient and implant recipients.
- Decision for or against the implant.
- Selection of the ear.
- Expectations of prospective patients/parents.
- All candidates are vaccinated against meningitis especially *Haemophilus influenzae* type B, *Pneumococcus* and in some areas, *Meningococcus*.

Selection Criteria

The guidelines vary and change over time. The current generally accepted selection criteria include:

General

- No evidence of central auditory lesions or absence of auditory nerve
- No contraindications for surgery
- Realistic expectations by the patient and family members
- Willingness to comply with follow-up procedures

Adults

- Hearing loss with pure tone average of 70 dB HL
- Hearing aid fitting and performance: 1–3 months of hearing aid use
- Aided scores on open-set sentence tests less than 50%

Children

- Age: 1–17 years
- Pure tone average thresholds of 90 dB HL or more
- Minimal benefit from hearing aids: 3–6 months of hearing aid use

Surgical Procedure

The surgery is done under general anesthesia. The position of the device is marked and the incision is planned. With the postaural/endaural approach, the site is prepared for the location of receiver/stimulator. Flaps are elevated using a two-layered approach. First layer flap is of skin and subcutaneous tissue and second layer is of musculoperiosteal flap. After creating a pocket under the second flap, a "well" is drilled in the skull bone that would house the receiver/stimulator.

Depending upon the type of electrode, 1.0–1.6 mm cochleostomy opening is made anteroinferior to the round window in the basal turn of cochlea. The electrode array is gently and gradually entered into the scala tympani through the cochleostomy till its complete insertion. The receiver/stimulator is fixed in the created "well" with ties.

Electrophysiological tests are performed to determine the electrode impedance and telemetry responses.

- *Introduction of Electrode*: The two popular methods of introducing the electrode array are posterior tympanotomy facial recess approach and Varia technique.
  - *Posterior tympanotomy facial recess approach*: A simple mastoidectomy is performed. Round window is exposed by opening the facial recess, through which the electrode array is passed to cochleostomy.
  - *Varia technique*: The posterior tympanomeatal flap is elevated to expose the round window. A tunnel is drilled in the posterior bony meatal wall directed from 11 o'clock position for right ear and 1 o'clock position for left ear. The electrode array is passed to the cochleostomy through the tunnel created in posterior meatal wall.
    Advantages: The advantages of this technique include:
    - No need of mastoidectomy
    - Takes less time
    - Minimal chances of injury to facial nerve

Postoperative Complications

Though occurs in less than 1–2% of cases, facial nerve injury during the surgery is feared most. The alteration of taste due to the irritation or injury to chorda tympani is quite common after surgery. Postoperative bleeding or hematoma formation occurs occasionally. The extrusion or exposure of the device though uncommon is one of the most feared late complications. Other early and late complications are listed in Table 1.

Postoperative Mapping

Activation of the device is done 3–4 weeks after the surgery and the implant is programmed or "mapped". Mapping continues

TABLE 1 Early and late complication of cochlear implant surgery

Early complications	*Late complications*
• Facial palsy	• Extrusion or exposure of device
• Taste disturbances	• Pain
• Bleeding or hematoma	• Device migration or displacement
• Infection	• Late device failure
• Wound dehiscence/flap necrosis	• Otitis media
• Early device failure	
• CSF leak	
• Vertigo and dizziness	
• Meningitis	

during the period of rehabilitation and finely tunes the processor to get best hearing with the implant. All patients need auditory-verbal therapy, in which emphasis is laid on making the patient listen and speak without using lip reading and visual cues. The rehabilitation period is a task that needs concerted efforts of patient, family and the professionals concerned.

AUDITORY BRAINSTEM IMPLANT

Auditory brainstem implant (ABI) directly stimulates the cochlear nuclei in the brainstem and bypasses the auditory nerve.

- *Indications*: It is indicated in cases of bilateral auditory nerve lesions. This implant is ideal for the cases of bilateral acoustic neuroma (Neurofibromatosis type 2) when auditory nerve is severed during the surgery.
- *Device*: It is placed in the lateral recess of the fourth ventricle. The multielectrode array of "Nucleus" ABI, which is similar to cochlear implant, is attached to a decron mesh that is placed on the brainstem. The facility of removable magnet of receiver/stimulator makes the device MRI compatible.
- *Outcomes*: ABI helps in communication, awareness and recognition of environmental sounds. Its efficiency is not similar to cochlear implant.

Clinical Highlights

1. **Hearing aids**: A hearing aid consists of microphone, amplifier, and receiver. It does not have speech processor and an electrode array.
2. **Cochlear implant**: A cochlear implant consists of microphone, speech processor and electrode array.
   a. **Indications***: Cochlear implant is indicated in children with bilateral severe sensory hearing loss who are not benefited by hearing aids. So it can be used in cases of bilateral severe to profound hearing loss caused by meningitis, otosclerosis, ototoxic drugs and mumps.*
   b. **Contraindications***: Cochlear implant is contraindicated in cases of retrocochlear hearing loss.*
3. **Brain stem implant**: Auditory brainstem implant is indicated in cases of bilateral profound neural hearing loss following surgery for bilateral acoustic schwannoma.

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16 Diseases of External Ear and Tympanic Membrane

It is our own mental attitude which makes the world what it is for us. Our thoughts make things beautiful, our thoughts make things ugly. The whole world is in our own minds. Learn to see things in the proper light.

—Swami Vivekananda

Points of Focus

- **INTRODUCTION**

DISORDERS OF AURICLE

- **CONGENITAL:** Bat Ear, Preauricular Appendages, Macrotia, Anotia, Microtia
  - ✓ Preauricular Sinus and Cyst
- **TRAUMATIC: LACERATIONS, AVULSION, HEMATOMA**
  - ✓ Frostbite
- **ERYSIPELAS**
- **PERICHONDRITIS AND CHONDRITIS**
- **CHONDRODERMATITIS NODULARIS CHRONICA HELICIS**
- **RELAPSING POLYCHONDRITIS**

DISORDERS OF EXTERNAL AUDITORY CANAL

- **CONGENITAL:** Atresia, Collaural Fistula
- **TRAUMA**
- **FOREIGN BODIES**
- **EAR MAGGOTS**
- **OTITIS EXTERNA (OE)**
  - ✓ Chronic OE, Externa Allergic OE, Contact Dermatitis, Seborrhea Granular OE
- **OTOMYCOSIS:** Aspergillus Niger, Candida Albicans, Fungal Infection of Temporal Bone
- **FURUNCULOSIS**
- **KERATOSIS OBTURANS**
- **EAR WAX**
- **EAR SYRINGING**
- **BULLOUS OTITIS EXTERNA AND MYRINGITIS**

DISORDERS OF TYMPANIC MEMBRANE (TM)

- **GRANULAR MYRINGITIS**
- **MALIGNANT OE**
- **RETRACTED TM**
  - ✓ Atrophic TM
  - ✓ Retraction Pockets and Atelectasis TM
- **TYMPANOSCLEROSIS**
- **PERFORATION OF TM**
- **TRAUMATIC RUPTURE OF TM**
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

The diseases of external ear may be categorized into congenital, traumatic, inflammatory, neoplastic and miscellaneous groups (Table 1).

DISORDERS OF AURICLE

CONGENITAL DISORDERS

The developmental abnormalities of the external ear may be minor or major.

- Bat ear (lop ear) is an abnormally protruding ear, which has large concha and poorly developed antihelix and scapha (Fig. 1). If the child or parents are concerned, the deformity can be corrected surgically after the age of 6 years.
- Preauricular appendages are skin-covered tags that appear between the tragus and the angle of mouth (Fig. 2). They usually contain small piece of cartilage.
- Macrotia is an abnormally large pinna.
- Anotia refers to complete absence of auricle. The anomaly is usually a part of the first arch syndrome.
- Microtia is a major developmental anomaly, which varies

TABLE 1 Diseases of external ear and tympanic membrane

Congenital	Bat or lop ear*, preauricular appendages, preauricular pit or sinus*, anotia, microtia, macrotia, atresia of external auditory canal (EAC), collaural fistula, Treacher-Collins syndrome
Trauma	Hematoma*, lacerations*, avulsion of pinna, frostbite*, foreign bodies*, perforation of tympanic membrane (TM)*
Inflammatory	Perichondritis*, chondritis, chondrodermatitis nodularis chronica helicis, acute otitis externa*, furuncle*, cellulitis, erysipelas, chronic otitis externa* (allergic, contact dermatitis, seborrheic or psoriasis, neurodermatitis, granulations), otomycosis*, otitis externa hemorrhagica or bullous myringitis, herpes zoster oticus (Ramsay Hunt syndrome), malignant or necrotizing otitis externa, perforation of TM*
Tumors	– Benign: Preauricular cyst, sebaceous cyst, dermoid cyst, keloid, hemangiomas, papilloma, cutaneous horn, keratoacanthoma, neurofibroma, osteoma, exostoses, ceruminoma, sebaceous adenoma – Malignant: Squamous cell carcinoma, basal cell carcinoma, melanoma, adenocarcinoma, malignant ceruminoma, malignant melanoma (See chapter Tumors of Ear and Cerebellopontine Angle)
Miscellaneous	Ear wax*, keratosis obturans, stenosis of EAC, retracted TM* (general, retraction pockets, atelectasis), tympanosclerosis*, atrophic TM, healed otitis media*, cholesteatoma

* Common Disorders

Fig. 1: Protruding pinna with large concha and poorly developed antihelix

Fig. 2: Preauricular appendages. Skin-covered tags between tragus and angle of mouth

in severity (Fig. 3). It may be unilateral or bilateral. It is frequently associated with deafness and anomalies of external, middle and internal ears.

Preauricular Sinus and Cyst

Preauricular sinus is a commonly seen congenital anomaly (Fig. 4). The faulty union of hillocks of the first and second branchial arches during the development of pinna results in a sinus at the root of helix. The sinus has a branching tract, which is lined by squamous epithelium. When sinus gets blocked, it results in a retention preauricular cyst.

- *Clinical feature:* Preauricular cyst usually is infected and presents as a painful swelling.
  - The sinus presents as a small opening in front of the crus of helix.
  - The sinus may get repeatedly infected (abscess formation) and present with painful swelling and purulent discharge.

Fig. 3: Microtia

Fig. 4: Preauricular sinus/pit

- *Treatment:* Surgery is indicated if there is unsightly swelling or infection.
  - The cyst or sinus track needs complete surgical excision to avoid recurrence.
  - If abscess does not respond to antibiotics, incision and drainage is required.

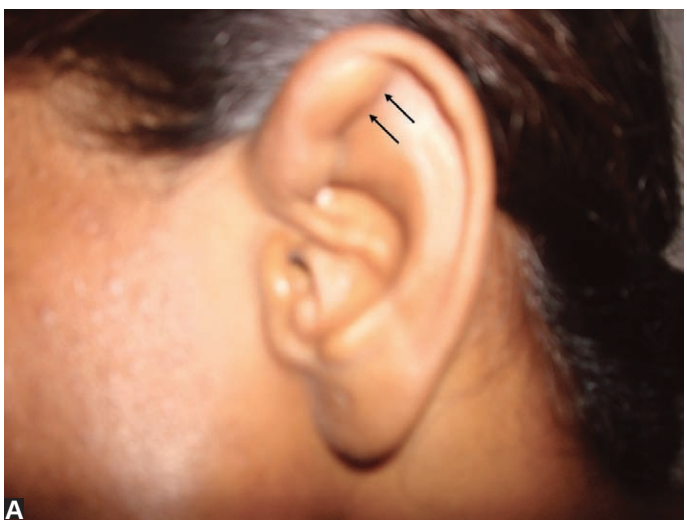
TRAUMATIC DISORDERS

- *Lacerations*: They are repaired at the earliest and the perichondrium is stitched with absorbable sutures under the cover of broad-spectrum antibiotics.
  - The stripping of perichondrium from cartilage must be avoided as it results in avascular necrosis. The fine nonabsorbable sutures are employed for the skin closure.
- *Avulsion of pinna* (Fig. 5): When the pinna remains attached to the head by a small skin pedicle, primary reattachment is usually successful. Completely avulsed pinna may be either reimplanted by the microvascular surgery or the skin of the avulsed segment of pinna is removed and the cartilage implanted under the postauricular skin for later reconstruction.
- *Hematoma of auricle* (Figs 6A and B): It refers to collection of blood between the auricular cartilage and perichondrium due to blunt trauma.
  - Occupations: Common in boxers, wrestlers and rugby players.
  - Cauliflower ear: This auricular deformity occurs when extravasated blood clots and gets organized.
  - Complication: Infection of hematoma leads to severe perichondritis. Prophylactic antibiotic course is given.
  - Treatment: Aspiration of hematoma under strict aseptic conditions followed by a pressure dressing packing all concavities of the auricle to prevent reaccumulation of blood. Aspiration may need to be repeated.
    - Incision and drainage is done when aspiration fails. Pressure is applied by dental rolls, which are tied with through-and-through sutures.

Fig. 5: Avulsion of upper part of pinna

Frostbite

Depending upon the severity, patient may present with erythema, edema, bullae, necrosis of skin and subcutaneous tissue or complete loss of the affected part.



Figs 6A and B: Auricular hematoma

Treatment: It consists of the following:

- Rewarming of pinna at a temperature of 38°–42°C with moist cotton pledgets.
- Application of 0.5% silver nitrate soaks: Superficial infection is controlled.
- Analgesics: Considerable pain occurs during the rapid rewarming of the ear.
- Bullae: They are protected from rupturing.
- Systemic antibiotics take care of deep infection.
- Surgical debridement: It is considered after several months when true demarcation between the dead and living tissues appears.

ERYSIPELAS

This dermal infection is caused by beta-hemolytic streptococcus.

- *Clinical features*: The affected area becomes erythematous, indurated and tender (Fig. 7).
  - Hallmark: Infection spreads along a well-demarcated border.
- *Treatment:* Oral systemic antibiotics, such as penicillin.

PERICHONDRITIS AND CHONDRITIS

The condition is the result of secondary infection. The formation of abscess between the perichondrium and cartilage cuts of the blood supply of the cartilage and result in its necrosis.

Etiology

- *Infected hematoma*: Blunt trauma in boxers, wrestlers, and fist-fighting
- *Penetrating trauma*: Ear piercing, assaults, bites and after ear surgery
- *Otitis externa*: Direct extension from an otitis externa (OE)
- *Causative organism*: Pseudomonas

Fig. 7: Erysipelas

Figs 8A and B: Perichondritis. Tender and erythematous swelling. (A) Lateral surface showing pus coming out; (B) Medial surface showing healed suture site of injury

Clinical Features

- Tender, erythematous, indurated pinna (Figs 8A and B).
- Fluctuance occurs in presence of abscess and chondritis.

Treatment

- *Antibiotics*: Oral quinolone antibiotic (such as ciprofloxacin) and local application of 4% aluminium acetate compresses.
- *Surgical*: Elimination of any piercing foreign body (FB).
  - Prompt incision and drainage of abscess: Pus is submitted for culture and sensitivity. Wound is packed with antibiotic-impregnated ribbon gauze.
  - Resection of necrotic cartilage and placement of through-and-through Penrose or catheter drain.
  - Postoperative course of antibiotic for 7–10 days.

Complications

- Cauliflower ear deformity of the cartilage

CHONDRODERMATITIS NODULARIS CHRONICA HELICIS

Elderly men about the age of 50 years are the common victims of this auricular disease.

- *Clinical features*: The condition presents with small painful nodules, which appear near the free border of helix.
  - Nodules are so tender that the patient is not able to sleep on the affected side.
- *Treatment*: Excision of the nodule with its skin and cartilage.

RELAPSING POLYCHONDRITIS

It is a rare autoimmune disease, which involves cartilages of ear, nose, larynx, trachea and chest. In cases of recurrent bilateral auricular swelling, rheumatologic workup for relapsing polychondritis must be done at the earliest.

- *Clinical features*: The auricle becomes inflamed and tender. External Auditory Canal (EAC) becomes stenotic.
  - Surprisingly, lobule of the ear is spared as it does not have any cartilage.
- *Treatment*: High doses of systemic steroids.

DISORDERS OF EXTERNAL AUDITORY CANAL

CONGENITAL DISORDERS OF EXTERNAL AUDITORY CANAL

- *Atresia of external auditory canal*: It may occur alone or in association with microtia.
  - Atresia with microtia: It is more common and may be associated with anomalies of the middle and internal ears and other structures.
  - Isolated atresia: It is due to failure of canalization of the first branchial cleft. The outer meatus is obliterated with fibrous tissue or bone. The deep meatus and the tympanic membrane (TM) are normal in these isolated cases of atresia.
- *Collaural fistula*: It is an abnormality of the first branchial cleft. The one opening of fistula is situated in the neck just below and behind the angle of mandible while second opening is in the external canal or the middle ear. The track of the fistula passes through the parotid gland in close relation to the facial nerve.

TRAUMA OF EXTERNAL AUDITORY CANAL

- Minor lacerations of EAC occurs while (Q-tip injury) scratching the ear with hair pins, needles or match stick or unskilled instrumentation. They generally heal without any complications.
- Major lacerations of EAC usually result from gun shot wounds, automobile accidents or fights. In some cases, the condyle of mandible injures the anterior wall of EAC.
  - Treatment: These cases of major lacerations are managed surgically to have skin-lined EAC of adequate diameter as the stenosis of the canal is the most common complication.

FOREIGN BODIES OF EAR

Children may insert a variety of nonliving items in their ears (Box. 1).

Types and Features of Foreign Bodies

- Adults, who are fond of scratching their ears, can have a broken end of matchstick or an overlooked cotton swab.
- Vegetable FBs swell up with time and get impacted in the ear canal. They may even result in suppuration.
- Living FBs cause intense irritation and pain. No attempt should be made to catch them alive.
- Injury to TM, middle ear structures, inner ear (subluxation of stapes into vestibule) can occur due to the introduction of hairpin or a bullet.

Box 1: Types of ear foreign bodies

- *Nonliving*:
  - Hygroscopic: Grain seeds such as rice, peas, wheat, maize
  - Non-hygroscopic: Slate pencil, chalk, metallic ball bearings, matchstick, cotton swab, paper, sponge, plastic
- *Living*: Flying or crawling insects such as mosquitoes, beetles, cockroaches and ants

Treatment

- *Antibiotics*: Antibiotics facilitate in controlling infection and edema.
- *Ear drops*: Hygroscopic FB can be shrunk with glycerin and absolute alcohol drops.
- *Removal of foreign body*: The best way is to remove them under ear microscope. Unskilled attempts may lacerate not only the meatal lining but can also damage the TM and the ear ossicles. Methods of removing a foreign body include:
  - Forceps removal: Soft and irregular FBs, such as piece of paper, swab or a piece of sponge, can be removed with forceps. Smooth and hard objects, such as steel ball bearing, tend to slip from the forceps and move inwards and may injure the TM while grasping with forceps.
  - Syringing: Seed grains and smooth objects can be removed with syringing.
  - Hooking out: Wax hook or vectis is passed beyond the FB and pressed against either floor or posterior wall and FB is hooked out.
  - Insects: First, they are killed by instilling oil, spirit, chloroform or water and then removed.
  - Postaural approach: FBs impacted in deep meatus, medial to the isthmus and pushed into the middle ear may need postaural incision for their removal.
  - Removal under general anesthesia: In cases of impacted FB, uncooperative children, and FB that failed to come out with earlier attempts are removed under general anesthesia.

Complications

They include the following:

- Injury to TM and middle ear structures
- Secondary infection: Otitis externa and otitis media

Live insect in external ear: Never remove live insect from external ear. First either drown or anesthetize the insect. Live angry insect can further damage the canal.

EAR MAGGOTS

Ear maggots are mostly seen in the months of August, September and October. Flies are attracted to the foul-smelling ear discharge and lay their eggs, which hatch out into larvae (maggots).

- *Clinical features*: Severe pain with swelling round the ear and blood-stained discharge.
  - Maggots in the EAC.
  - Perforation of the TM.
- *Treatment*: Chloroform water eardrops kill the maggots, which are then removed by forceps.

OTITIS EXTERNA

Otitis externa is an inflammation of the EAC.

Microorganisms

- Most common microorganism: *Pseudomonas aeruginosa*.
- Other common organisms: *Staphylococcus aureus* and *Staphylococcus epidermidis*.

- Uncommon microorganisms: Coryneform (diphtheroids), Gram-negative rods (*Enterobacter Klebsiella, Proteus, Escherichia coli*), *Streptococcus* and *Enterococcus*.
- *Actinomyces israelii* is an anaerobic Gram-positive bacterium that can cause OE from primary dental or parotid infection.
  - This refractory OE presents with granulation tissue in EAC and thick yellow ear discharge and needs surgical debridement and prolonged antibiotic therapy.
- Cultures for bacteria and fungus are indicated in persistent or refractory infection.

Treatment

- Debridement of EAC clears the infected material and allows topical antibiotics to reach the infected site.
- Acidification of EAC (acetic acid solution and antibiotics in acidic suspensions and solutions) kills many bacteria and fungus.
- Topical antibiotics provide a concentration of many times greater than systemic antibiotics. Thus usually it is not necessary to start systemic antibiotics even in resistant infection.
  - Antibiotic impregnated ear wicks (cotton, merocel, ribbon gauze) offer better absorption through an edematous canal skin.
  - Quinolone antibiotic drops (ofloxacin, ciprofloxacin) with dexamethasone (hydrocortisone avoided as it leaves a precipitate) in an acidic vehicle are the best as they have no ototoxicity and no potential for allergic reactions.
- Oral antibiotics are reserved for complications of OE such as cellulitis, perichondritis, chondritis and malignant OE.

Acute Otitis Externa

An abrogation of hydrophobic ceruminous coating of EAC exposes the underlying epithelium to water, infection and other contaminants and lead to progressive erythema and edema.

Predisposing Factors

- *Warm humid environment*: Excessive sweating changes the canal pH to alkaline, which favors bacterial growth.
- *Obstruction of EAC*: Stenosis, exostoses, impacted wax.
- *Ear trauma*: Use of cotton-tip swabs, irrigators for wax removal, in-the-canal hearing aids, digital ear thermometers, unskilled instrumentation.
- *Water contamination of EAC*: Usually swimming related.

Clinical Features

- *Presenting complaints*: Otalgia and otorrhea
  - Ear pain may be aggravated with jaw movements
  - Itching is present in some cases
- *Early stage*: An erythematous canal with scant discharge
- *Later on*: Edematous and exquisitely tender canal occluded with purulent squamous debris
- *In severe cases*: Enlarged and tender regional lymph nodes and periauricular cellulitis
- No additional systemic sign and symptoms

Chronic Otitis Externa

- No prior history of trauma or water contamination.
- Usually not painful.
- Examination may reveal fungal hyphae in the canal or keratin debris from chronic dermatitis.
- Acute exacerbation is usually preceded by itching.

Types

The four different disease states with distinct etiologies are allergic, contact dermatitis, seborrhea and granular OE.

1. *Allergic OE*
   - Clinical features: Edema and purulent discharge secondary to topical antibiotic (most commonly neomycin) ointment or drops.
     - Maculopapular eruption in concha and EAC.
     - A thickened and erythematous canal.
   - Treatment: Elimination of offending agent, debridement and topical steroid drops.
2. *Contact Dermatitis*
   - Clinical features: It can result from a variety of agents, such as hairsprays, shampoos and hearing aid molds.
     - A thickened and erythematous canal.
   - Treatment: Elimination of offending agent.
     - Debridement and topical steroid drops.
3. *Psoriasis or Seborrhea*
   - Clinical features: Hyperkeratosis and lichenification of EAC skin.
     - Seborrheic OE is associated with seborrheic dermatitis of the scalp. Itching is the presenting complaint. Greasy yellow scales may be seen in the canal, over the lobule and in postauricular sulcus.
     - Secondary bacterial or fungal infections may occur.
   - Treatment: Treatment of scalp seborrhea.
     - Ear toilet, salicylic acid and sulphur cream
4. *Granular OE*
   - Clinical features: It results from chronic bacterial or fungal infection of EAC.
     - Granulation and excoriation seen on TM and EAC skin.
   - Culture for bacteria and fungus may show causative organisms.
   - Treatment
     - Repeated debridement.
     - Cauterization of granulations.
     - Topical antibiotic or antifungal creams.
     - Topical gentian violet facilitates at drying the EAC.
     - Topical steroid or 5-fluorouracil.
     - De-epithelialization of TM: If does not respond to drying agent (Burow's solution) it may need spilt-thickness skin graft.

Complications of Otitis Externa

- *Cellulitis*: It present as an erythematous ear. There is no induration like perichondritis.
  - Treatment is with systemic antistaphylococcal agent.
- *Perichondritis or Chondritis*
- *Medial canal fibrosis*: In chronic otitis externa (COE), thick fibrous scar may obstruct deep aspect of EAC. TM appears lateralized with absence of typical landmarks.
  - Surgical treatment consists of excision of the fibrous scar followed by canaloplasty with split thickness skin graft and if needed tympanoplasty.
- *Malignant OE*: See next section.

Otitis externa treatment: First line treatment for otitis externa is topical antibiotics. Systemic antibiotics are given only for severe infections.

OTOMYCOSIS

Fungal infection of ear is frequently seen in hot and humid climate of tropical and subtropical countries.

- Superficial fungal infection is the most common and limited to EAC. Invasive type is rare and involves temporal bone.
- In cases of associated bacterial infection, the fungal infection may not be immediately evident.
- Common fungal species are *Aspergillus niger* and *Candida albicans*.

Clinical Features

- Intense itching, discomfort, or ear pain
- Discharge with musty odor and ear blockage
- Sodden, red and edematous meatal skin

Treatment

- *Antifungal agents*: Povidine iodine, 2% salicylic acid in alcohol.
  - Antifungal treatment should be continued for a week after the apparent cure.
- *Dry ear*: Ear should be kept dry.
- *Antibiotic/steroid ear drops* in cases of associated bacterial infection helps in reducing edema and inflammation and better penetration of antifungal agents.

Aspergillus Niger

- *Clinical features*: Pigmented fungal tufts atop a tangle of hyphal threads resembling a cotton ball or moist white plug dotted with black debris ("wet newspaper").
  - Aggressive infection involves epithelial and subcutaneous tissues and may result in TM perforation, which heal spontaneously following medical treatment.
- *Treatment*: Frequent debridement and acidic drops (acetic acid) and gentian violet painting of ear canal and TM.
  - Persistent infection needs oral itraconazole.

Candida Albicans

- *Clinical features*: This opportunistic fungal infection is common in patients getting prolong course of antibiotic ear drops.
  - The EAC appears wet and macerated and is filled with soft, curd-like debris sprouting hyphae.
- *Treatment*: Antifungal solutions or creams, such as clotrimazole, nystatin (100,000 units/ml of propylene glycol).

Fungal Infection of Temporal Bone

- Coccidiomycosis of EAC and mastoid resemble eczema or an allergic dermatitis.
- Secondary mycoses of temporal bone may be caused by *Cryptococcus, Candida, Blastomycoses* and mucor infection involving meninges or paranasal sinuses.
- Primary invasive mycoses of mastoid and temporal bone can occur in HIV infection, diabetic and elderly patients and result in rapidly progressive hearing loss and facial paralysis. *Aspergillus fumigatus* and *Aspergillus flavus* result in serious morbidity and mortality.
- Systemic invasive mycoses (*Aspergillosis, Mucormycosis, Candidiasis, Cryptococcosis, Coccidioidomycosis and Histoplasmosis*) involve temporal bone in cases of defective host defenses (diabetic ketoacidosis, chemotherapy, corticosteroids, AIDS).
- Chronic fungal or bacterial middle ear infections spread to inner ear through oval or round window or osteomyelitis of petrous bone.

FURUNCULOSIS

- This localized form of acute otitis externa (AOE) is a staphylococcal infection of the hair follicles, which are present only in the outer cartilaginous part of EAC.
- Nasal vestibule may harbor staphylococci and fingering in the nose transfer the infection to other sites of body.

Clinical Features

- Though usually single, the furuncles may be multiple.
- Severe pain and tenderness are out of proportion to the size of the furuncle.
- A small nodular swelling proceeds to fluctuance.
- Movements of the pinna and lower jaw (chewing) become painful.
- Posterior meatal wall furuncle causes edema over the postauricular region with obliteration of the retroauricular groove.
- Periauricular lymph nodes (anterior, posterior and inferior) may get enlarged and tender.
- In cases of recurrent infection, blood should be tested for diabetes.

Treatment

- Medical: Oral antistaphylococcal antibiotics, analgesics, local heat and antibiotic ointment.
- Ten percent ichthamol glycerin ear wick provides splintage and reduces pain. Hygroscopic action of glycerin reduces edema, while ichthamol is mildly antiseptic.
- Incision and drainage is done if the abscess has formed.

KERATOSIS OBTURANS

In this EAC disorder, a dense plug of keratin, which gets accumulated in the deep bony ear canal, completely occludes the EAC. A faulty epithelial migration pattern of EAC has been documented.

Clinical Features

- Both ears are frequently involved.
- Adults between 30 years and 60 years are affected.
- Severe otalgia from an aggressive secondary OE.
- Deafness, tinnitus and in some cases, ear discharge.
- Pearly white mass of keratin material filling the ear canal.
- Thickening and mucosalization of tympanic membrane.
- Blunting of periannular canal skin from scarring.
- The pressure absorption of canal bone results in widening of the canal and facial nerve palsy.

Treatment

- Treatment of secondary OE.
- Removal of the epithelial debris under ear microscopy.

Prophylaxis: Cleaning of the canal before the occurrence of inflammation and infection.

- Use of keratolytic agent, such as 2% salicylic acid in alcohol.

EAR WAX

Anatomy and Physiology

Sebaceous and ceruminous (modified sweat glands) glands open into the space of the hair follicle. Secretion of both these glands mixes with the desquamated epithelial cells and keratin and form wax. This lubricates the ear canal and entraps the foreign material that enters into the canal.

Usually a small amount of wax is formed which dries up and is expelled from the meatus by movements of the jaw. Excessive wax secreted by the glands is deposited as a plug in the outer EAC.

Components of Ear Wax

The ear wax is made up of the following components:

- Sebaceous gland's secretion, which is rich in fatty acids
- Ceruminous gland's secretion, which is rich in lipids and pigment granules
- Hairs
- Desquamated epithelial debris and keratin, which are shed from the TM and bony meatus
- Dirt

Factors Facilitating Ear Wax Problem

Following are the factors, which facilitate wax problem:

- Narrow and tortuous ear canal, stiff hair and exostosis facilitate retention of wax, which may dry up and form a hard impacted mass.
- Excessive secretion of wax and dusty occupations result in increased amount of wax.
- Self-cleaning of ear wax may push wax into the deeper bony EAC.

Clinical Features

- *Hearing loss or sense of blocked ear*: Sudden hearing loss may occur when water enters into the EAC (wax swells up) while bathing or swimming.
- *Tinnitus and giddiness* due to impaction of wax against the TM.
- *Reflex cough* can result from the stimulation of auricular branch of vagus nerve.
- *Wax granuloma*: The impacted wax ulcerates the meatal skin and results in granuloma formation.

Treatment

It consists of removal of wax by either syringing or other ear instruments.

- *Wax softening agents*: Hard impacted mass usually needs prior softening with any of the following wax softening agents.
  - Five percent sodabicarb in equal parts of glycerin and water
  - Hydrogen peroxide
  - Liquid paraffin
  - Olive oil
  - Paradichlorobenzene 2%
- *Removal*: Instrumental manipulation should always be done by skilled hands preferably under ear microscopy. Cerumen hook, scoop or Jobson-Horne probe are usually employed. Wax hook and vectis are used for removing wax, foreign bodies and debris from the external ear canal. It may be used for removing foreign body nose. In cases of impacted wax, a space is created between the wax and canal wall. The instrument is then passed beyond the wax and the lump of wax is then dragged out. If the lump breaks, syringing or suctioning may be used to remove the fragment.

EAR SYRINGING

- *Aural Syringe*: This metal syringe consists of a cylinder with a well fitting piston and a nozzle (Fig. 9).

Method

Patient is seated comfortably and the diseased ear faces towards the doctor. A towel is placed on the shoulder. Patient's head is slightly tilted toward the shoulder. A kidney tray is held snugly well below the ear to collect the return fluid. Boiled tap water cooled to body temperature (or normal saline) is used.

The auricle is pulled upwards and backwards while the direction of the stream of ear syringe is towards the postero-superior wall of the meatus. Pressure of water that builds up deeper to the wax expels the wax out (Fig. 10). In cases of impacted wax, some space is created between the wax and the meatal wall so that stream of water passes through that. Otherwise wax would be pushed deeper. After the procedure, ear canal and tympanic membrane are dried with cotton.

Cautions

- Too much force of the syringing can rupture the TM and lead to intense pain and dizziness and fainting.
- Past history of ear discharge or an existing perforation must be asked before venturing for the syringing. A quiescent otitis media can get reactivated after syringing.

Fig. 9: Ear syringing. The auricle is pulled upwards and backwards and the direction of the ear syringe is posterosuperior

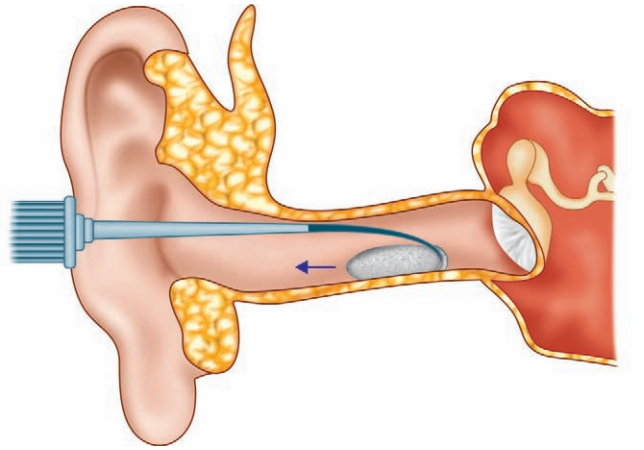


Fig. 10: Ear syringing. Water jet of ear syringe taking out lump of wax

- Too cold or too hot water would stimulate the labyrinth and result in vertigo.
- Ulceration in meatal wall needs application of broad-spectrum antibiotic ointment.

HERPES ZOSTER OTICUS-RAMSAY HUNT SYNDROME (VARICELLAZOSTER VIRUS)

See chapter Facial Nerve Disorder.

BULLOUS OTITIS EXTERNA AND MYRINGITIS

This inflammatory/infectious disease is believed to be primarily viral.

- *Cause*: Viral infection, *Mycoplasma pneumoniae* and *Haemophilus influenzae*.
- *Predisposing Factors*: Upper respiratory tract infection and winter season.

Clinical Features

- Exquisite (severe) ear pain with bloody or serous ear discharge.
- Hemorrhagic vesicle(s) or bulla(e) on the deep bony ear canal and TM. They rupture spontaneously and result in serosanguineous discharge.
- Associated middle ear effusion is common.
- *Hearing loss*: Conductive or mixed with significant sensorineural element.

Treatment

- Analgesics
- Topical antibiotic/steroid ear drops to prevent bacterial superinfection
- Macrolide or quinolone antibiotics for associated mycoplasma
- Rupturing of the bullae may relieve pain

DISORDERS OF TYMPANIC MEMBRANE

GRANULAR MYRINGITIS

- An idiopathic inflammatory process of TM.
- Patches of granulation tissue and mucosalized epithelium on the TM.
- In advanced case, the complete TM gets thickened. Granulation tissue exudes thin transudate and may become secondarily infected.
- Secondary granulations may be associated with impacted wax, long-standing foreign body (such as myringotomy tube) or external ear infection.

MALIGNANT OR NECROTIZING OTITIS EXTERNA

This rare OE of immunocompromised patients is an aggressive and potentially fatal infection, which progressively spreads to skull base and intracranial structures.

- *Causative microorganisms: Pseudomonas aeruginosa* is most common.
  - *Staphylococcus aureus* and *Staphylococcus epidermidis* are rare.
- *Immunocompromised patients*: Elderly diabetics, HIV/AIDS, myeloid malignancies, anticancerous drugs and organ transplant recipients.

Clinical Features

- *Severe otalgia and otorrhea.*
- *Hallmark finding*: Granulations in the floor of EAC at the bony-cartilaginous junction.
- *Facial nerve* is the most commonly involved cranial nerve.
- *Advanced stage*: IX, X and XI CN palsy.
- *Intracranial spread*: Headache, fever, neck stiffness and altered levels of consciousness.
- *Spread of infection to neighboring structures*: Such as temporomandibular joint, mastoid, middle ear and petrous bone.
- *Children*: Acute onset of painful otorrhea in children with immunosuppression, diabetes or Stevens-Johnson syndrome and poor general health.
  - Prognosis is better than adult form of disease.

Investigations

- Bacterial and fungal culture
- High-resolution computed tomography (HRCT) scan
- Single-photon emission tomography (SPECT) with radionuclide tracers

Treatment

The success of aggressive medical treatment will predict the prognosis.

- *Antipseudomonal antibiotics*: Oral or parenteral (depending upon the severity) ciprofloxacin for 6–8 weeks or longer. In ciprofloxacin resistance cases other antibiotics such as tobramycin, ticarcillin and third generation cephalosporins may be used.
- *Hyperbaric oxygen therapy.*
- *The treatment of the cause of immunosuppression,* such as diabetes and HIV/AIDS.
- *Surgical debridement of nonviable sequestra of bone* is required when bone involvement is resistant to antibiotic therapy.

RETRACTED TYMPANIC MEMBRANE

The retraction of TM is the result of negative middle ear pressure, which occurs due to the blockage of Eustachian tube (See

chapters Disorders of Eustachian Tube; and Chronic Suppurative Otitis Media and Cholesteatoma).

- *Otoscopy*: Following otoscopic findings indicate that TM is retracted towards the middle ear.
  - Dull and lusterless TM
  - Absent or interrupted cone of light
  - Apparent foreshortening of the malleus handle
  - Extra prominent lateral process of malleus
  - Sickle shaped anterior and posterior malleal folds

Atrophic Tympanic Membrane

- This thin membrane easily gets collapsed with Eustachian tube insufficiency (See chapter Disorders of Eustachian Tube). It can be present in following conditions:
  - In cases of otitis media with effusion, the middle fibrous layer of TM gets absorbed.
  - When a central perforation of TM heals, only epithelial and mucosal layers grow and the intervening fibrous layer remains absent in healed drum.

Retraction Pockets and Atelectasis Tympanic Membrane

- A segment of thin and atrophic TM or the entire membrane may collapse inwards due to Eustachian tube insufficiency (See chapter Disorders of Eustachian Tube).
- The membrane may get plastered onto promontory and also wrap round the ossicles.
- A deep retraction pocket (usually in posterosuperior and attic region) may accumulate keratin debris and form a cholesteatoma.
- *Sade classification*: See chapter Chronic Suppurative Otitis Media and Cholesteatoma.

TYMPANOSCLEROSIS

This hyalinization and later calcification in the fibrous layer of TM appears as chalky white plaque.

- *Causes*: Generally, it is result of safe chronic suppurative otitis media (CSOM), serous otitis media and ventilation tube.
- *Clinical features*: If the ossicles and middle ear are not involved, the condition may remain asymptomatic.
  - It generally affects TM. It can involve ligaments, joints of ossicles, muscle tendons and submucosal layer of middle ear cleft and results in conductive hearing loss.

PERFORATION OF TYMPANIC MEMBRANE

They may be central, attic or marginal and are usually seen in cases of CSOM. (see chapter Chronic Suppurative Otitis Media and Cholesteatoma).

TRAUMATIC RUPTURE OF TYMPANIC MEMBRANE

- *Etiology*: The injuries are following:
  - Q-tip injury: Scratching the ear with hair pins, needles or match stick or unskilled instrumentation.
  - Air pressure: An open handed slap, a kiss on the ear, blast and forceful valsalva. Explosive blasts can produce more than 200 dB sound pressure level (SPL).
  - Fluid pressure: Diving, water sports and forceful syringing.
  - Fracture of temporal bone.
- *Clinical features*: Sudden deafness, tinnitus or dizziness after the injury.
  - Ear microscopic examination: Perforation with the edges turned either outward or inward.
- *Treatment*: The edges of perforation are repositioned and splinted under ear microscopic examination.
- *Complications*: Rupture of the TM may be associated with following complications:
  - Facial paralysis
  - Subluxation of stapes
  - Vertigo and nystagmus
  - Sensorineural hearing loss

Clinical Highlights

1. **BAT ear:** This is the most common congenital deformity of the ear.
2. **Polyotia**: This is the rarest congenital anomaly of the ear.
3. **Preauricular sinus**: Asymptomatic (no discharge or inflammation) patients need only education and observation.
4. **Swimmer's otitis externa**: The treatment of this recurrent busts of otitis externa after swimming includes 2% acetic acid ear drops after swimming.
5. **Otomycosis**: Commonest causative fungus is *Aspergillus niger*. The treatment includes 2% salicylic acid, 1% gentian violet, and clotrimazole.
6. **Ear wax**: It is a mixture of secretions of sebaceous and apocrine glands and contains a bactericidal enzyme.
7. **Reflex cough** response while cleaning the ear canal is mediated by stimulation of the mastoid branch (Arnold nerve) of CN X (vagus).
8. **Necrotizing otitis externa (Malignant otitis externa)**: This *Pseudomonas aeruginosa* infection of external auditory canal occurs in elderly patients who are diabetic or immunocompromised. The lesion can spread to skull base and involves facial nerve at stylomastoid foramen.
9. **Blue drum**: Tympanic membrane appears blue in cases of hemotympanum (temporal bone fracture), glue ear, glomus tumor and hemangioma of middle ear.
10. **Dry traumatic rupture of tympanic membrane**: The treatment is protection of ear against water and ear drops.

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17 Disorders of Eustachian Tube

Even the least work done for others awakens the power within; even thinking the least good of others gradually instills into the heart the strength of a lion. Great men are those who build highways for others with their heart's blood.

—Swami Vivekananda

Points of Focus

- **ANATOMY**
  - ✓ Parts and Openings
  - ✓ Muscles of Eustachian Tube (ET)
  - ✓ Mucosa of ET
  - ✓ Nerve Supply
  - ✓ Differences between Infant and Adult ET
- **PHYSIOLOGY**
  - ✓ Ventilation and Regulation of Middle Ear Pressure
  - ✓ Protective Functions
  - ✓ Clearance of Middle Ear Secretions
  - ✓ Ventilation of Middle Ear Cleft
- **EXAMINATION OF ET**
- **TESTS FOR ET FUNCTION**
  - ✓ Maneuver Building Positive Pressure: Valsalva and Politzer Tests, Catheterization
  - ✓ Maneuver Building Negative Pressure: Toynbee's Test
  - ✓ Test for Mucociliary Drainage/Clearance
  - ✓ Sonotubometry
- **OBSTRUCTION OF ET**
  - ✓ Etiology
  - ✓ Adenoids
  - ✓ Cleft Palate
  - ✓ Down's Syndrome
  - ✓ After-Effects
  - ✓ Clinical Features
  - ✓ Retraction Pockets and Atelectasis TM
- **PATULOUS EUSTACHIAN TUBE**
  - ✓ Etiology
  - ✓ Clinical Features
  - ✓ Treatment
- **CLINICAL HIGHLIGHTS**

ANATOMY

Eustachian[1] tube (auditory or pharyngotympanic tube) connects nasopharynx with the tympanic cavity. It is about 36 mm long and runs downwards, forwards and medially from its tympanic end, forming an angle of 45° with the horizontal.

Parts

Eustachian tube (ET) consists of two parts—bony and fibrocartilaginous (Fig. 1). The area where two parts meet is the narrowest part of the tube and called isthmus.

1. *Posterolateral bony part*: It forms one third of the (12 mm) of the total length.
2. *Anteromedial fibrocartilaginous part*: It forms two third of the (24 mm) of the tube. This is made of a single piece of cartilage, which is folded upon itself and forms the medial lamina, roof and a part of the lateral lamina (Figs 2A and B). The rest of the lateral lamina is made of fibrous membrane.
   - Elastin hinge: It is rich in elastin fibers and is situated in the roof at the junction of medial and lateral lamina. By its recoil it keeps the tube closed when dilator of ET (tensor veli palatini) is not in action.

[1] Bartolomeo Eustachio (1524-1574)

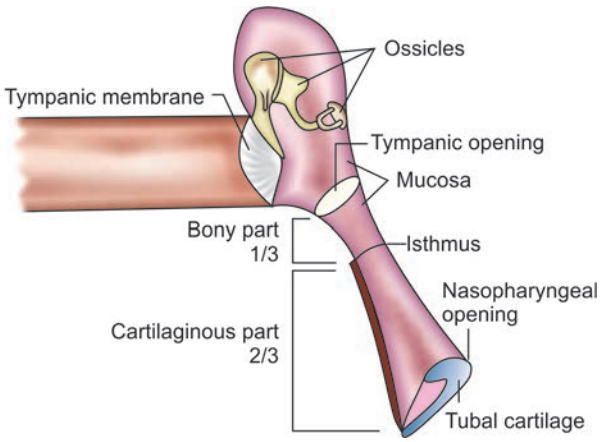
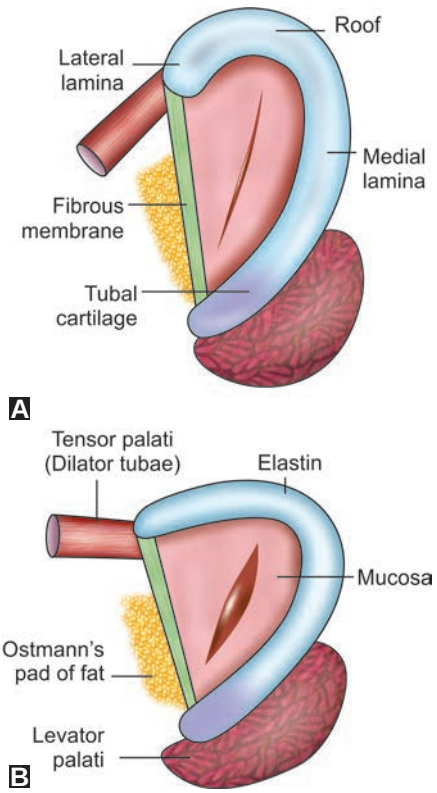


Fig. 1: Eustachian tube anatomy. Horizontal section



Figs 2A and B: Eustachian tube physiology. Vertical section of Eustachian tube. (A) Closed in resting position; (B) Contraction of tensor veli palatini (dilator tubae) opens the tube. Elastin roof helps in regaining resting closure position

- Ostmann's pad of fat: It keeps the tube closed and is related laterally to the membranous part of the cartilaginous tube. The closed tube protects itself and middle ear from the reflux of nasopharyngeal secretions.

Openings

Eustachian tube has two openings—tympanic and pharyngeal.

1. *Tympanic opening*: This is bony and measures 5 x 2 mm, is situated in the anterior wall of middle ear, a little above the middle ear floor.
2. *The pharyngeal end*: This is slit-like, is situated in the lateral wall of the nasopharynx, about 1.25 cm behind the posterior end of inferior turbinate.
   - Torus tubarius: Posteriorly, the cartilage produces an elevation, which is called torus tubarius.
   - Fossa of Rosenmuller: Posterior to torus tubarius is situated the fossa of Rosenmuller, which is a common site for malignancy of nasopharynx.

Muscles in Relation to Eustachian Tube

The three muscles, which are related to the ET, are tensor veli palatini (tensor palati), levator veli palatini (levator palati) and salpingopharyngeus (Fig. 3).

1. *Tensor veli palatini muscle*: The medial fibers of the tensor veli palatini muscle, which are attached to the lateral lamina of the tube, when contract open the tube.
2. *Levator veli palatini muscle*: It runs inferior and parallel to the cartilaginous part of the tube and forms a bulk under the medial lamina. This muscle during contraction pushes the tube upward and medially thus assisting in opening the tube.

Mucosa of Eustachian Tube

- The pseudostratified ciliated columnar epithelium is interspersed with mucous secreting goblet cells.
- The submucosa, especially in the cartilaginous part, is rich in seromucinous glands.
- The cilia beat in the direction of nasopharynx.
- The ciliary movements drain middle ear secretions and fluid into the nasopharynx.

Nerve Supply

- Tympanic branch of CN IX carries sensory and parasympathetic secretomotor fibers to the tubal mucosa.
- Mandibular branch of trigeminal (CN V_3) supplies tensor veli palatini and tensor tympani.
- Cranial part of CN XI through vagus supplies the levator veli palatini and salpingopharyngeus.

Differences between the Infant and Adult Eustachian Tube

In infants, infections from the nasopharynx can easily reach the middle ear because the ET in them is wider, shorter and more horizontal (Table 1). The milk can also regurgitate into the middle ear, when infants are not fed in head-up position. The tube attains adult morphology and functions by the age of 7–10 years.

- In infant ET is 13–18 mm long at birth that is about half the size of adult tube.
- The tube is more horizontal in infants. At birth it forms an

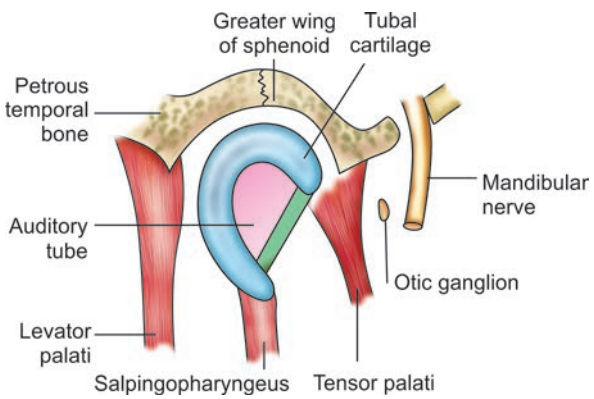


Fig. 3: Relations of the cartilaginous part of eustachian tube

TABLE 1 Differences between infant and adult Eustachian tube

	Infant	*Adult*
Length	13–18 mm at birth	31–38 mm
Direction	More horizontal. Forms an angle of 10^0	Forms an angle of 45^0 with the horizontal
Angulation at isthmus	Absent	Present
Bony part	Relatively longer and wider	Relatively short and narrow
Tubal cartilage	Flaccid	Comparatively rigid and remains closed
Density of elastin at the hinge	Less dense	More dense
Ostmann's pad of fat	Less in volume	Large in volume

angle of 10° with the horizontal while at age 7 and later it is 45°.

- In comparison to adults, bony part is relatively longer and wider in infants.
- Retrograde reflux of nasopharyngeal secretion occurs in infants because tubal cartilage is flaccid, whereas it is comparatively rigid in adults and remains closed and protects the middle ear from the reflux.
- In infants elastin hinge is less dense and does not efficiently keep the tube closed by recoil.
- Ostmann's pad of fat is large in adults and helps in keeping the tube closed.

PHYSIOLOGY

- Eustachian tube usually remains closed. It opens during swallowing, yawning and sneezing.
- The poor tubal function in infants and young children are responsible for more middle ear disorders in them.
- Posture can affect the tubal function. The tubal mucosa gets congested in recumbent position and during sleep due to venous engorgement.
- The three main functions of the ET are: (1) ventilation and regulation of middle ear pressure; (2) protection of middle ear against nasopharyngeal sound pressure and reflux of nasopharyngeal secretions and; (3) middle ear clearance of secretions.

Ventilation and Regulation of Middle Ear Pressure

During normal hearing, the pressure on two sides of the tympanic membrane remains equal. The increase and decrease in middle ear pressure affects hearing. Eustachian tube opens periodically and equilibrates the air pressure in the middle ear with that of ambient pressure.

Protective Functions

- The ET remains closed and protects the middle ear against abnormally high sound pressures of the nasopharynx, which if transmitted to the middle ear can interfere with normal hearing.
- A functioning ET protects middle ear from reflux of nasopharyngeal secretions. The reflux occurs if the tube is wide (patulous tube), short (as in infants), or the tympanic membrane is perforated.
- High pressures in the nasopharynx during forceful nose blowing and closed-nose swallowing (big adenoids or bilateral nasal obstruction) force nasopharyngeal secretions into the middle ear.

Clearance of Middle Ear Secretions

- Mucosa of the ET and anterior part of the middle ear are ciliated columnar.
- The cilia beat towards the nasopharynx and thus clear the middle ear secretions and debris into the nasopharynx.
- The clearance function is further enhanced by active opening and closing of the tube.

Ventilation of Middle Ear Cleft

- Ventilation of the middle ear cleft occurs through the ET.
- From the mesotympanum, air passes to attic, aditus, antrum and mastoid air cells.
- Mesotympanum communicates with the attic through the anterior and posterior isthmi, which are situated in membranous diaphragm that lies between the mesotympanum and the attic.
  - Anterior isthmus lies between the tendon of tensor tympani and the stapes.
  - Posterior isthmus lies between the tendon of stapedius, pyramid, and the short process of incus.
  - Retrofacial air cells: The middle ear can communicate directly with the mastoid air cells through the retrofacial air cells.

EXAMINATION OF EUSTACHIAN TUBE

Eustachian tube can be examined with the following means:

- Posterior rhinoscopy.
- Rigid nasal endoscope (Sinuscope).
- Flexible nasopharyngolaryngoscope.
- Eustachian tube endoscopy or middle ear endoscopy: It is performed with a very fine flexible endoscope that is passed into ET.
- Microscope/endoscope: Retraction pockets, fluid in the middle ear, movements of tympanic membrane with respiration may be seen.
  - In cases of preexisting perforation of tympanic membrane, the tympanic end of ET tube can be examined by a microscope, as well as an endoscope.
- Allergy testing may reveal the allergic cause.
- Imaging: CT scan of temporal bones and of paranasal sinuses may reveal the cause of ET obstruction.
  - MRI excludes the multiple sclerosis in cases of patulous Eustachian tube.

- Tests for the assessment of function of the tube are described in the next section.

TESTS FOR EUSTACHIAN TUBE FUNCTION

Tests for the function of Eustachian are listed in Box 1.

Maneuver Building Positive Pressure in Nasopharynx

- *Principle*: The principle of this method is to build positive pressure in the nasopharynx so that air enters the ET. These methods can be employed therapeutically to ventilate the middle ear.
- *Method*: Air is pushed into the middle ear and tympanic membrane moves outwards which can be seen by an otoscope/microscope.
  - In cases of tympanic membrane perforation, a hissing sound and in cases of middle ear discharge, a cracking sound can be heard with the help of stethoscope.
    - An auscultation tube can connect the patient's ear under test to that of the examiner.
- *Complications/Contraindications*: The following are the relative contraindications for this method as they can result in complications:
  - Atrophic scar: Healed or thin tympanic membrane can rupture.
  - Infection of nose and nasopharynx: Infected secretions are pushed into the middle ear and can cause otitis media.
- *Valsalva test*: About 65% of the patients can successfully perform this test.
  - Method: Patient pinches his nose.
    - Patient takes a deep breath through mouth and then closes his mouth.
    - Patient tries to blow his cheeks and pushes air into the ears.
- *Politzer test*: This test is good for children who cannot perform Valsalva test (Fig. 4).
  - Method: An olive-shaped tip of the Politzer's bag is introduced into the child's ipsilateral nose (side of ET to be tested).
    - Other nose is also closed/pinched.
    - Politzer bag is compressed while the child swallows sips of water or says "ik, ik".
- *Eustachian tube catheterization*:
  - Method: Nose is anesthetized by topical spray of lignocaine.

Box 1: Tests for Eustachian tube function

- Maneuver building positive pressure in nasopharynx
  a. Valsalva test
  b. Politzer test
  c. Catheterization
- Maneuver building negative pressure in nasopharynx
  a. Toynbee's test
- Tympanometry (Chapter Hearing Evaluation)
- Mucociliary drainage/clearance
  a. Saccharine
  b. Methylene blue
  c. Antibiotic/steroid ear drops
- Sonotubometry

Fig. 4: Politzer test. An olive-shaped tip of the Politzer's bag is introduced into the ipsilateral nose. Other nose is also pinched. Politzer bag is compressed while patient swallows sips of water or says "ik, ik"

    - Eustachian catheter (see chapter Instruments) is passed along the floor of nose till it reaches the nasopharynx. The catheter is than rotated 90° medially and then gradually pulled outward so that it touches the posterior border of nasal septum. The catheter is then again rotated 180° laterally so that tip lies in the pharyngeal opening of the ET (Fig. 5).
    - A Politzer bag is connected to the catheter and air is insufflated.
  - Complications: The procedure should be done gently. Complications include:
    - Injury to Eustachian tube opening which can result in scarring.
    - Bleeding from the nose.
    - Perforation of the ear drum is possible if air is squeezed with a great force.
    - Vasovagal syncope may occur.
  - Difficulties: Adenoids and nasal obstructions (deviated septum and nasal mass) prevent the tip of the catheter from reaching the ET.
  - Therapeutic indications of eustachian catheter:
    - Relieving the blockage of the ET.

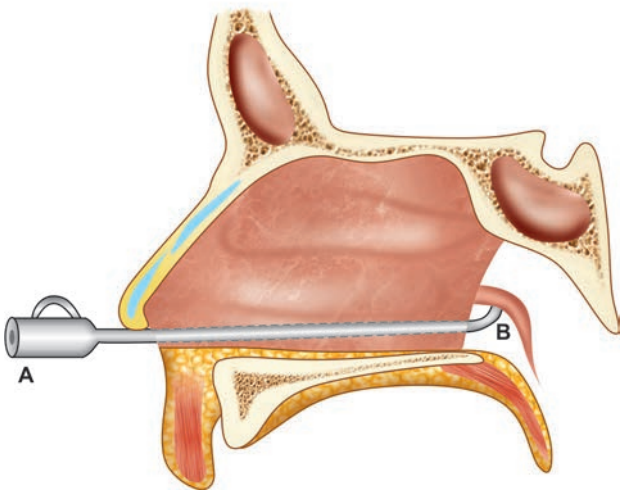


Fig. 5: Eustachian catheter in Eustachian tube opening

- Insufflations of medicine into the middle ear.
- Removal of nasal foreign bodies.

Maneuver Building Negative Pressure in Nasopharynx

- *Toynbee's test*: It is a physiological test.
  - *Method*: The patient is asked to swallow while he/she keeps the nose pinched.
  - The maneuver draws air from the middle ear into the nasopharynx and causes inward movement of tympanic membrane, which can be seen through an otoscope/microscope.

Tests for Mucociliary Drainage/Clearance

These tests are conducted in cases of tympanic membrane perforation. The time taken by materials to reach the pharynx and impart its taste provides an idea about the functioning of the Eustachian tube and a measure of clearance function. The following materials can be used:

- Saccharine solution drops are instilled in the ear. The time taken by saccharine to reach the pharynx and impart a sweet taste is noted.
- Methylene blue dye is instilled into the ear. The time taken by the dye to stain the pharyngeal secretions is noted.
- Antibiotic/steroid ear drops are instilled into the ear. The time taken by drops to impart its bad taste is noted.

Sonotubometry

This noninvasive technique provides the information on active tubal opening. A tone, which is delivered to the nose, is recorded from the external auditory canal (EAC). The tone is heard louder if the ET is patent. The duration for which the tube remains open can also be noted. The accessory sounds produced during swallowing interfere with the test results.

OBSTRUCTION OF EUSTACHIAN TUBE

The hallmark of ET obstruction is middle ear effusion. The child may remain asymptomatic.

Etiology

The common causes of tubal obstruction can be divided into two groups mechanical and functional.

1. *Mechanical obstruction:* It can due to intrinsic and extrinsic causes.
   - Intrinsic causes: Upper respiratory infection (viral or bacterial), allergy, sinusitis and barotrauma.
   - Extrinsic causes: Hypertrophic adenoids, nasopharyngeal tumors/mass, deviated nasal septum (DNS), nasal polyp.
2. *Functional obstruction*: It can be due to increased cartilage compliance and poor function of tensor veli palatini. The common causes in this category include cleft palate, submucous cleft palate and Down syndrome.
   - Increased cartilage compliance resists opening of the tube.
   - Poor function of tensor veli palatini results in the failure of active tubal-opening.

Adenoids

- Big adenoids can result in otitis media with effusion or recurrent acute otitis media because of the following elements:
  - Mechanical obstruction of ET.
  - Reservoir for pathogenic organism.
  - Mast cells of the adenoid tissue release inflammatory mediators that lead to tubal obstruction.
- Treatment consists of adenoidectomy.

Cleft Palate

- Otitis media with effusion is common in cases of cleft palate. The poor tubal functions are due to the following reasons:
  - High elastin density in the abnormalities of torus tubarius makes the tube opening difficult.
  - In 40% cases of cleft palate, tensor veli palatini muscle has poor function and does not insert into the torus tubarius.
- Treatment: Even after the repair of cleft palate, many children need grommets to ventilate their middle ear.

Down's Syndrome

The poor tone of tensor veli palatini muscle and abnormal shape of nasopharynx result in defective function of ET in children suffering from Down's syndrome.

After-effects of Eustachian Tube Obstruction

The ET opens intermittently during swallowing, yawning and sneezing through the active contraction of tensor veli palatini muscle. Air (mixture of oxygen, carbon dioxide, nitrogen and water vapor) fills the spaces in middle ear and mastoid. If the ET remains blocked for long time, first oxygen and later other gases diffuse out into the blood and creates negative middle ear pressure, which results in retraction of tympanic membrane. It leads to "locking" of the ET.

The ET can be blocked suddenly and may remain blocked for a long time.

- Acute tubal blockage can result into collection of transudate and later exudate (acute otitis media) and even hemorrhage (barotitis media) in the middle ear.
- Prolonged tubal blockage/dysfunction can lead to a chain of events beginning from otitis media with effusion (OME), atelectatic ear, retraction pocket, which may lead to erosion of incudostapedial joint and cholesteatoma formation.

Clinical Features

Symptoms and signs of tubal occlusion vary depending upon the acuteness of the condition and severity.

- Otalgia, which may be mild to severe.
- Hearing loss.
- Popping sensation or tinnitus.
- Disturbances of equilibrium or vertigo.
- Retracted tympanic membrane.
- Congestion along the handle of malleus and pars tensa.
- Transudate behind the tympanic membrane imparts an amber color and may show fluid level.
- Conductive hearing loss.
- In severe barotraumas: Markedly retracted tympanic membrane with hemorrhages in subepithelial layer, hemotympanum. Perforation is uncommon.

Retraction Pockets and Atelectasis

The obstruction in different pathways of middle ear cleft ventilation can result in retraction pockets or atelectasis of the

tympanic membrane (see chapters Diseases of External Ear and Tympanic Membrane). A retraction pocket is an invagination of the tympanic membrane. The negative middle ear pressure, which is the cause of retraction pocket, is often secondary to otitis media.

- *Sade classification*: See chapter Chronic Suppurative Otitis Media and Cholesteatoma.
- *Sites*: The common sites are pars flaccida and posterosuperior quadrant of pars tensa. Retraction pockets or atelectasis of the following parts of tympanic membrane can occur:
  - Total atelectasis of tympanic membrane occurs due to the obstruction of ET.
  - Retraction pocket in posterior part of middle ear occurs due to obstruction in middle ear while the anterior part is ventilated.
  - Attic retraction pocket occurs due to the obstruction of isthmi.
- *Sequelae*: As a retraction pocket deepens, desquamated keratin cannot be cleared and a cholesteatoma is formed. Other changes, which depend on the location of pathologic process, include thin atrophic tympanic membrane (due to the absorption of middle fibrous layer), ossicular necrosis and tympanosclerosis.
  - Cholesterol granuloma and collection of mucoid discharge in mastoid air cells occur due to the obstruction at aditus. Middle ear and attic appear normal.
- *Treatment*: It consists of correction/repair of the cause and pathologic process and establishment of ventilation.

PATULOUS EUSTACHIAN TUBE

Eustachian tube becomes undue patent.

Etiology

Mostly it is idiopathic but following predisposing factors may be present:

- Rapid weight loss
- Pregnancy especially third trimester
- Multiple sclerosis

Clinical Features

- *Presenting complaints*: Hearing of own voice (autophony) and breath sounds.
- *Otoscopy/Microscopy*: The movements of tympanic can be seen with inspiration and expiration especially when patient breathes after closing the opposite nostril. Due to undue patency of ET, pressure changes in the nasopharynx are easily transmitted to the middle ear.

Treatment

Acute condition is generally self-limiting. In chronic cases treatment consists of the following:

- Weight gain
- Oral administration of potassium iodide
- Cauterization of the tubes
- Insertion of a grommet

Clinical Highlights

1. **Eustachian tube**: It provides communication between middle ear and nasopharynx and is 36 mm in length. Its lateral one-third is bony and medial two-third is cartilaginous. It remains closed at rest. The opening of the tube, which is an active process, occurs due to contraction of tensor veli palatini muscle while closure occurs due to recoil of the cartilaginous part.
   a. **Gerlach tonsil**: This is lymphoid tissue of eustachian tube.
   b. **Development**: Mucosa of auditory tube and middle ear is derived from an outgrowth of the endoderm of the 1st pharyngeal pouch.
2. **Frenzel maneuver**: It opens the eustachian tube and ventilates the middle ear. The muscles of the floor of mouth and pharynx are contracted while nose, mouth and glottis are closed. In comparison to Valsalva maneuver it is difficult to learn.
3. **Patulous eustachian tube**: It is common in third trimester of pregnancy. The woman becomes aware about their own sounds. On otoscopy one can see movements of tympanic membrane that are synchronous with respiration. The movements are exaggerated when patient breaths only through the ipsilateral side of nose.

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18 Acute Otitis Media and Otitis Media with Effusion

Come, do something heroic. The remedy for weakness is not brooding over weakness, but thinking of strength that is already within you. Whatever you do, devote your whole mind, heart, and soul to it.

—Swami Vivekananda

Points of Focus

- **INTRODUCTION**
  - ✓ Risk Factors

ACUTE OTITIS MEDIA (AOM)

- **ETIOPATHOLOGY**
  - ✓ Routes of Infection
  - ✓ Predisposing Factors
  - ✓ Causative Microorganisms
- **CLINICAL FEATURES**
  - ✓ Pathology and Clinical Features (5 Stages)
- **DIAGNOSIS**
  - ✓ Differential Diagnoses
- **TREATMENT**
  - ✓ Medical Treatment
  - ✓ Surgical Treatment
- **RECURRENT AOM**
  - ✓ Treatment
- **ACUTE NECROTISING OTITIS MEDIA**

OTITIS MEDIA WITH EFFUSION (OME)

- **ETIOLOGY**
- **CLINICAL FEATURES**
- **DIAGNOSIS**
- **TREATMENT**
  - ✓ Medical
  - ✓ Surgical
- **SEQUELAE AND COMPLICATIONS**
  - ✓ Adverse Effects of Chronic Hearing Loss in Children
- **UNILATERAL OME IN ADULTS**
- **OTITIC BAROTRAUMA / AERO OM**
  - ✓ Etiopathology
  - ✓ Clinical Features
  - ✓ Treatment
  - ✓ Prophylaxis
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

Otitis media (OM) refers to inflammation of middle ear cleft. OM is not only the most common bacterial infection in children but also a leading cause of hearing loss in children. OM is the most common disease of childhood, with the exception of viral upper respiratory infection (URI). Most of the children have at least one incidence of middle ear infection. Fifty percent of the children get three or more episodes of OM and 25% have six or more episodes. The highest prevalence occurs in the first 2 years and decreases thereafter. OM is more frequent in winter months.

The two major classes of OM are acute otitis media (AOM) and chronic otitis media with effusion (OME). AOM consists of middle ear effusion (MEE) and features of acute infection (fever, ear pain and red bulging eardrum). OME is MEE without features of inflammation. AOM and OME represent the two stages of the same disorder.

Risk Factors

Most of the causes of OM are related to Eustachian tube (ET), which are described in detail in Chapter Disorders of Eustachian Tube. The risk factors, which are same for AOM and OME, include following:

- Male children.
- Bottle feeding (breast feeding is protective).
- Allergy.
- Crowded living conditions and poor socioeconomic status.
- Smoking by family members in the home.

- Siblings having OM.
- Viral infections in the home and daycare centers.
- Heredity and genetic factors.
- Associated conditions:
  - Cleft palate
  - Immunodeficiency
  - Ciliary dyskinesia
  - Cystic fibrosis
  - Down's syndrome.

Prophylactic Measures

Factors, which can reduce the morbidity of OM, include:

- Proper vaccination.
- Breast feeding.
- Better general health and nutrition.
- Public awareness.

Eustachian tube dysfunction: It is the key to the pathogenesis of otitis media.

ACUTE OTITIS MEDIA

Acute otitis media (AOM) implies an acute pyogenic inflammation of the middle ear cleft, which includes ET, middle ear, attic, aditus, antrum and mastoid air cells.

Though AOM can occur in all ages, it is mainly the disease of children as the ET is shorter, wider and more horizontal and opens at the lower level in children. Both ears may be involved.

ETIOPATHOLOGY

Routes of Infection

- *Eustachian tube*: The infection in the middle ear usually reaches through the ET. Reflux from the nasopharynx into the middle ear occurs during the swallowing; nose blowing and closed-nose swallowing (Toynbee's maneuver). It is the result of negative middle ear pressure (sniffing). Following conditions may be present in cases of AOM:
  - Anatomical obstruction: Big adenoids and nasopharyngeal tumors.
    Currently it is believed that it is not the obstruction but the bacterial entry into the middle ear due to the failure of protection (abnormally patent tube), which is more important. Big adenoids, which are elevated during swallowing, may obstruct the posterior choanae and increase nasopharyngeal pressure that results into reflux.
  - Infections: Adenoiditis, tonsillitis, rhinitis, sinusitis, pharyngitis.
    Adenoids serve as a bacterial reservoir in nasopharynx in children with AOM.
  - Forceful blowing of nose: The forcible blowing of the nose can push the infection into middle ear through the ET.
  - Swimming: Especially during diving, water enters in the nose under pressure. If water is infected it can spread infection to nose, sinuses and the middle ear.
  - Iatrogenic: Postnasal packing and after adenoidectomy.
  - Feeding bottle: In the supine position, bottle feeding may lead milk to enter middle ear via ET.
- *Pre-existing tympanic membrane perforation*: Trauma while cleaning the external auditory canal (EAC) an open hand slap on the ear and past chronic suppurative otitis media (CSOM).
- *Fracture of temporal bone*: In cases of head injury, middle ear may be involved with the fracture of the temporal bone.
- *Blood borne infection*: Rare.

Predisposing Factors

- *Reduced immunity*: Malnourishment, poor dietary habits, too much physical and mental exertion, exposure to extremes of climate and temperatures, can affect the overall resistance of the persons and infections can occur easily.
- *Barotrauma*: Atmospheric pressure changes especially during flying and deep water diving can affect ET.
- *Exanthematous fevers*: Measles, diphtheria, whooping cough.
- *Palatal disorders*: Cleft palate and palatal palsy.
- *Nasal allergy*: Inhalants and foods.

Causative Microorganisms

Generally viral nasal infection precedes the ear bacterial infection. The adenoids of children with recurrent AOM contain pathogenic bacteria in clinically significant amounts.

- Most common: *Streptococcus pneumoniae*, *Haemophilus influenzae, Branhamella catarrhalis (Morexella catarrhalis)*.
- Other common: *Streptococcus pyogenes*, *Staphylococcus aureus*.
- Uncommon: Gram-negative bacilli from skin (after trauma) such as *Bacillus proteus*, *Bacillus pyocyneous*, *Bacillus coli*.
- In some cases no organisms are found.

CLINICAL FEATURES

Pathology and Clinical Features (5 Stages)

The course of disease is usually divided into the five stages, which begin from tubal occlusion, presuppuration and suppuration and ends with either resolution or complications. If the proper antibiotic therapy is started early during the course of AOM, disease process may revert back from any stage. The resolution may start even without the rupture of TM.

Most of the children will have a preceding history (cold and cough) of upper respiratory tract infection. Infants become fussy, sleep poorly and often pull or tug at the affected ear and fever heralds the onset of AOM.

I. *Stage of tubal occlusion*: The edema and hyperemia of nasopharynx and Eustachian tube occludes ET, which leads to absorption of air and creation of negative middle ear pressure. Some middle ear effusion may occur but is not clinically appreciable.

- Symptoms
  - Mild deafness.
  - Ear fullness and ear pain.
  - No fever.
- Signs
  - Retracted tympanic membrane: Findings include relative shortening and more horizontal position of malleus handle, prominent lateral process of malleus and loss of light reflex
  - Conductive hearing loss.

II. *Stage of presuppuration*: Prolonged tubal occlusion facilitates invasion of pyogenic organism into middle ear and results in mucosal hyperemia. Inflammatory exudates appear in the middle ear.

- Symptoms
  - Marked throbbing ear pain, which can awake the child from sleep in night.
  - High degree of fever and restlessness.
  - Bubbling sound in the ear.
  - Deafness though present does not get child's attention due to the severe ear pain.
- Signs
  - Pars tensa congested and bulging out with cart-wheel appearance (leash of blood vessels along the handle of malleus and at the periphery of TM) and loss of light reflex.
  - Later on complete TM including pars flaccida get uniformly congested and red.
  - Tuning fork tests show conductive hearing loss.

III. *Stage of suppuration:* There occurs formation of pus in the middle ear and somewhat in mastoid air cells. Tympanic membrane starts bulging.

- Symptoms
  - Excruciating ear pain.
  - Increasing deafness.
  - Constitutional symptoms due to absorption of toxins include rising fever, which may be accompanied with vomiting, diarrhea and even convulsions.
- Signs
  - Tympanic membrane appears red and bulging to the point of rupture with loss of landmarks.
  - Handle of malleus engulfed by the swollen and protruding TM.
  - A yellow spot on the TM where rupture is imminent (a nipple like protrusion of TM with a yellow spot on its summit).
  - Tenderness over the mastoid antrum in the region of suprameatal triangle due to mastoidism.
  - Clouding of air cells in X-ray of mastoid because of exudates.

IV. *Stage of resolution*: The TM ruptures (due to pressure necrosis) and results in otorrhea and subsidence of other symptoms. Inflammatory process begins resolving.

- Symptoms
  - Otorrhea: Blood tinged ear discharge (serosanguinous) later becomes mucopurulent.
  - Ear pain and fever subside.
- Signs
  - External auditory canal filled with blood tinged or mucopurulent discharge, which may be pulsatile (lighthouse sign: pus coming out under pressure and synchronizing with each arterial dilatation of heartbeat).
  - Perforation of pars tensa usually in anteroinferior quadrant.

V. *Stage of complication:* In majority of the children AOM is self limiting and responds well to medical treatment. If the virulence of organism is high and resistance of the child is poor, infection may spread beyond the middle ear space. Complications occur in the second week and constitutional and infectious sign and symptoms reappear.

The complications (Chapter Complications of Suppurative Otitis Media) include acute mastoiditis, subperiosteal abscess, facial paralysis (Fig. 1), labyrinthitis, meningitis, extradural abscess, brain abscess and lateral sinus thrombophlebitis. Infants are at greater risk because AOM and meningitis usually coexist.

DIAGNOSIS

Acute otitis media is a clinical diagnosis.

- *Tests for hearing*: They show conductive hearing loss.
- *CT temporal bone*: It is indicated only in cases of refractory mastoiditis. The clouding of air cells (because of exudates) and their pressure necrosis (coalescent mastoiditis) may be seen. Demineralization of the air cell septa is the key radiographic sign of mastoid osteitis.
- *Bacteriological examination*: The ear discharge is submitted for the culture and sensitivity to know the type of causative microorganism and the antibiotic to which they are sensitive.

Differential Diagnoses

- *Causes of otalgia*: See Chapter Otologic Symptoms and Examination.
- *Causes of otorrhea*: See Chapter Otologic Symptoms and Examination.
- *Otitis externa, myringitis and bullous myringitis*: See Chapter Diseases of External Ear and Tympanic Membrane.
- *Crying child*: The TM may look red and congested but is never edematous. The light reflex is always present.

TREATMENT (FIG. 2)

Medical Treatment

There has been considerable debate on the usefulness of antibiotics for the treatment of AOM. There are no evidences which show that antihistamine, decongestants, or any other form of adjunct medical therapy result in shortening the course of AOM.

- *Antibiotics*: Traditionally antibiotic therapy is continued till TM and hearing become normal. Different antibiotic regimes have been suggested such as:

Fig. 1: Left side facial palsy in a case of acute suppurative otitis media. Note the blood tinged discharge from left ear. Otoscopy showed subtotal central perforation

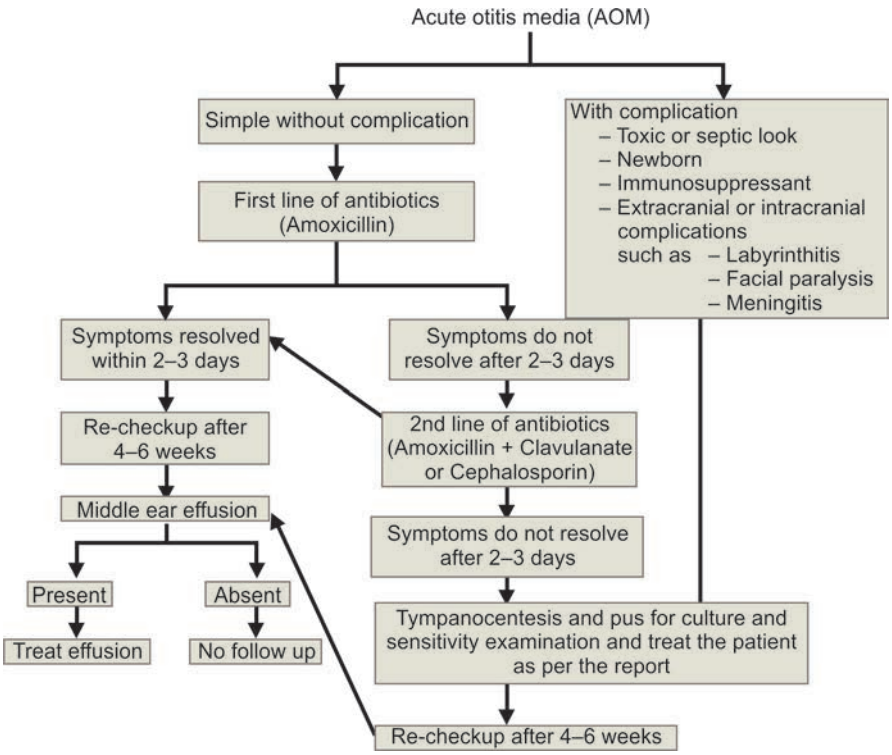

Fig. 2: Flow chart showing management of acute otitis media

- A single intramuscular dose of ceftriaxone.
- A 5-day course or 10–14 days course of oral antibiotics.
- The first line of antibiotic is amoxicillin (40 mg/kg/day in three divided doses).
- The children who are allergic to penicillin can be given cefaclor, cotrimoxazole and erythromycin.
- Beta-lactamase-producing *H. influenzae* or *Morexella catarrhalis* need amoxicillin-clavulanate, cefuroxime axetil or cefixime.

- *Decongestants*: Topical ephedrine (1% in adults and 0.5% in children), oxymetazoline (nasivion) and xylometazoline (otrivin) nasal drops and oral pseudoephedrine 30 mg bid and phenylephrine hydrochloride with or without antihistaminic are said to relieve ET edema and promote ventilation of middle ear.
- *Analgesics and antipyretics*: Paracetamol take care of pain and fever.
- *Ear drop and aural toilet*: Ear discharge must be cleaned. Water is prevented from entering the ear.
  - Quinolone/steroids ear wick/drops take care of local infection and inflammation.
- *Dry local heat*: It relieves pain.

Surgical Treatment

- *Tympanocentesis*: It is needle aspiration of fluid from middle ear. The culture and sensitivity of ear fluid for knowing the organism and selecting the antibiotics is indicated in following conditions:
  - Premature newborns.
  - Immunocompromised patients.
  - Failure of previous antibiotic therapy.
  - Intracranial complications.
- *Myringotomy*: An incision is put in the TM to evacuate middle ear fluid. It is usually preceded by tympanocentesis. The indications are following:
  - Bulging eardrum.
  - Acute excruciating pain.
  - Unresponsive to antibiotics.
  - Facial palsy.
  - Intracranial complications.
- *Mastoidectomy*: Diagnosis of osteitis on CT warrants mastoidectomy to remove the necrotic and infected bone.
- *Incision and drainage* of Subperiosteal postauricular abscess.
- *Tympanoplasty*: In cases of permanent tympanic perforation and ossicular necrosis.

RECURRENT ACUTE OTITIS MEDIA

Infant and children may get recurrent episodes of AOM, which may be 4–5 or more in a year.

- Usually, episodes of AOM follow acute upper respiratory tract infections.
- The recurrent AOM may be superimposed upon pre-existing maximal expiratory flow (MEF).
- Feeding the babies in supine position without propping up the head can cause the milk to enter into the middle ear.

Treatment

Try to find the predisposing factor and cause of the recurrent infection and treat them. Generally treatment includes following:

- *Antimicrobial prophylaxis*: Many use single daily dose of amoxicillin (20 mg/kg) for 3–6 months to prevent recurrent attacks of AOM. Prophylaxis with long-term use of an antibiotic is currently discouraged.
  Surgical therapy better reduces the number of new episodes of AOM and removes persistent effusion.

- *Surgical prophylaxis with tympanostomy tubes (Grommet):* In recurrent and chronic cases, ventilation of middle ear is provided by inserting a ventilatory tube (grommet) through the eardrum.
  Indication: Four bouts of AOM in 6 months or 6 bouts in 1 year.
- *Adenoidectomy*: With or without tonsillectomy in cases of adenoid hypertrophy and infection.
- *Management of allergy*: In cases of inhalant or food allergy.

ACUTE NECROTISING OTITIS MEDIA

This type of AOM is seen in children, who are suffering from measles, scarlet fever and influenza.

- Causative organism is *beta-haemolyticus streptococcus*.
- Rapid destruction of complete TM along with its annulus, mucosa of promontory, ossicular chain and mastoid air cells occur.
- Profuse otorrhea.
- Sequel: Secondary acquired cholesteatoma. Ingrowth of squamous epithelium from the EAC.
- Treatment
  - Antibacterial therapy is started at the earliest and continued for minimum 7–10 days.
  - Cortical mastoidectomy is needed in refractory cases and acute mastoiditis.

OTITIS MEDIA WITH EFFUSION

(Synonyms: Chronic Secretory OM; Chronic Serous OM; Mucoid OM; Glue Ear)

In OME, there occurs collection of non-purulent nearly sterile effusion in the middle ear cleft. The effusion is usually thick and viscid but may be thin and serous.

In younger children, OME is most often the unresolved stage of AOM. In older children, OME has a silent onset without a clinically evident antecedent AOM.

ETIOLOGY

The ET dysfunction is nearly always present in children with OME. The obstruction, which is usually functional and due to edema and viscous secretions, is believed to be secondary to disease process rather than the cause of OME.

Current evidence suggests that effusion is not sterile. Polymerase chain reaction (PCR) studies have found metabolically active bacteria in culture-negative middle ear effusions.

- *Malfunctioning of Eustachian tube*: It fails to ventilate and drain the middle ear. Factors affecting the middle ear clearance mechanism include following:
  - Ciliary dysfunction.
  - Mucosal edema and hyperplasia.
  - Viscous secretions.
  - Middle ear/nasopharyngeal pressure gradient.
  - Adenoid hyperplasia and infection.
  - Chronic rhinitis, sinusitis, tonsillitis.
  - Benign and malignant tumors of nose, paranasal sinuses and nasopharynx.
  - Cleft palate and palatal paralysis.

  Currently it is felt that tubal obstruction along with failure of clearance, which are common findings in children with OME, may be secondary rather than the primary process.
- *Allergy*: Allergy to inhalants and food stuffs.
- *Unresolved* acute otitis media: Inadequate antibiotic course just inactivates infection but does not resolve it completely. Low grade infection acts as stimulus for mucosa to secret more fluid.
- *Viral infections*: Adeno and rhinoviruses of upper respiratory tract.
- *Increased secretory activity of middle ear mucosa*: Increased number of mucous and serous secreting cells results in more secretions.

Persistent unilateral otitis media in adults: Nasopharyngoscopy must be done to rule out neoplastic lesion of the nasopharynx which can result in blockage of Eustachian tube opening.

CLINICAL FEATURES

- May be completely asymptomatic.
- The child turns up the volume of television and is not attentive during normal conversation.
- Insidious conductive hearing loss (rarely exceeds 40 dB), which may be unnoticed by the parents and is accidentally discovered during audiometry.
- Delayed and defective speech in children due to the hearing loss.
- Mild earache.

Otoscopy/Microscopy (Fig. 3)

- Retracted, less mobile or immobile TM; outward brisk movement of TM on reducing pressure in EAC with pneumatic otoscope.
- Dark, fluid-filled middle ear obscures the vision of long process of incus.

Pneumatic otoscopy: It is the gold standard for the diagnosis of otitis media with effusion.

DIAGNOSIS

In infants and young children otoscopy and tympanometry findings are sufficient for the diagnosis.

- *Tympanometry*: When the middle ear is filled or impacted with effusion, compliance is low and the tympanogram is

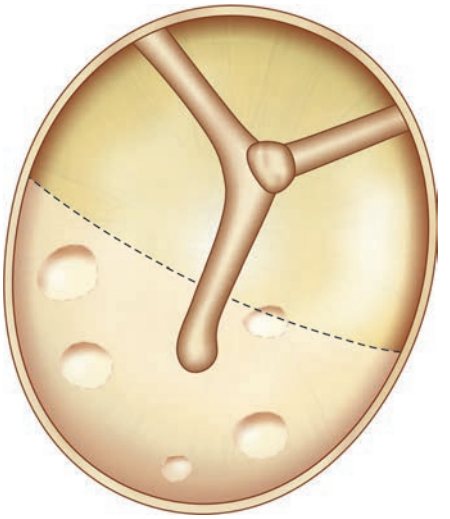

Fig. 3: Otitis media with effusion. Otoscopy shows fluid level and air bubbles

flat because reflected energy does not vary with the pressure change.
Caution: The ear canals of infants younger than 7 months are hypercompliant and normal tympanograms are possible in cases of OME.

- *Audiometry*: Mild to moderate conductive hearing loss is the most common finding. Because of excellent cochlear sensitivity in children, bone conduction thresholds are usually less than audiometric zero.

TREATMENT

If the predisposing and causative factors are known they should be addressed. Considering the long-term impact, surgical therapy is more cost effective than the medical for the severe cases.

Medical

- *Antibiotic therapy*: The efficacy of antibiotic therapy has been shown by many reports. Like AOM, OME is a bacterial disease. Following antibiotics have been suggested:
  - Combination of erythromycin ethylsuccinate and sulfisoxazole.
  - Trimethoprim-sulfisoxazole.
  - Amoxicillin.
  - Amoxicillin-clavulanate.
- *Antihistamines and decongestant*: Currently, routine use of decongestants in children with OME has been abandoned.
- *Inflation of middle ear* (Valsalva's maneuver, Politzerization or Eustachian catheterization): Though popular these maneuvers are not generally used in USA.
- *Corticosteroids*: The effects of steroids are equivocal and not clear.

Surgical

Surgical treatment is considered when effusion persists and is associated with hearing loss. The insertion of tympanostomy tube (grommet) with or without adenoidectomy is preferred over myringotomy alone.

- *Myringotomy*: Aspiration of "glue" or middle ear effusion (Fig. 4).
- *Tympanostomy tube (Grommet):* Most widely used treatment option for OME when present for more than 3 months with associated hearing loss of more than 30 dB in better ear.

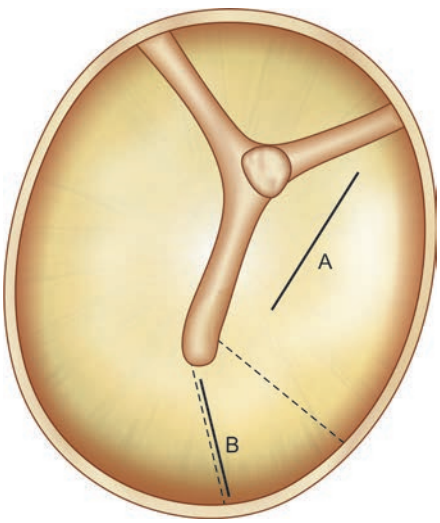

Fig. 4: Incisions for middle ear effusion. Two incisions: one in anterosuperior (A), and other in anteroinferior (B) quadrant

- *Adenoidectomy*: This surgery is being increasingly used for the management of OME in children 4 years of age or older.

SEQUELAE AND COMPLICATIONS

A chronically and frequently ill, irritable, painful, or inattentive child can lead to disruption of family life because of extra cost and the time lost from work. The sequelae include:

- Atrophic TM.
- Atelectasis of the middle ear.
- Tympanic membrane perforation.
- Ossicular necrosis.
- Tympanosclerosis.
- Adhesive OM.
- Retraction pockets.
- Chronic suppurative OM and cholesteatoma.
- Cholesterol granuloma.
- Sensorineural hearing loss.

When middle ear pressure rapidly becomes lower than atmospheric pressure, a clear, watery transudate may form in the middle ear. This serous OM is sporadically seen in children and adults and may be caused by viral infection, URI and acute tubal dysfunction.

Adverse Effects of Chronic Hearing Loss in Children

The following adverse effects on cognitive, linguistic, auditive and communication skills may occur:

- Articulation development.
- Receptive vocabulary development.
- Phonologic awareness.
- Academic development.
- Language delays and lower IQ scores.

UNILATERAL OME IN ADULTS

In cases of unilateral OME in adults and elderly patients, nasopharynx must be examined with the help of endoscope for the presence of nasopharyngeal mass, which may be obstructing the ET.

AERO OTITIS MEDIA (OTITIC BAROTRAUMA)

In this type of OME, ET fails to maintain middle ear pressure at ambient atmospheric level.

Etiopathology

- *Pressure changes*: When the pressure is relatively high in the middle ear (during ascent), air escapes via the ET passively. But when the pressure is low the equalization of pressure may not occur due to the locking of the tube. ET is actively opened by swallowing and yawing. A descent (during a flight and deep water diving) produces a relative negative middle ear pressure.
- *Eustachian tube dysfunction*: Edema or obstruction of the ET due to adenoids, rhinitis or deviated nasal septum aggravates the problem of locking of the tube.

Clinical Features

- Deafness may get relieved by swallowing and yawning.
- Earache may be severe.

- Tinnitus may be present.
- Vertigo is not common.
- Middle ear may have air bubbles or effusion.

Treatment

- Repeated swallowing, yawning and Valsalva maneuver.
- Antibiotics, analgesics, decongestants (topical nasal drops and oral tablets).
- Myringo puncture with injection of air into the middle ear.
- Myringotomy with grommet insertion in refractory cases.

Prophylaxis

- Avoid flying and diving during rhinitis.
- Decongestion of the nose before the flight especially before the descent. Take decongestant nasal drops/spray and tablets.
- Repeated swallowing during descent, e.g. sipping of water/drinks; sucking of sweets/chocolates/chewing gum.
- Never sleep during the descent.
- Perform intermittently Valsalva maneuvers.
- Treatment of the cause of ET dysfunction such as nasal polyps, septal deviation, adenoids, allergy, chronic rhinosinusitis.

Clinical Highlights

1. **Acute otitis media**: The common causative organisms of this disease of children (in descending order) are *Streptococcus pneumoniae*, *Haemophilus influenzae* and *Moraxella catarrhalis*. Antibiotics for 10 days control pain and fever. The conductive hearing loss may persist for long time and needs just observation for 3 months for the fluid to drain spontaneously.
2. **Otitis media with effusion**: Otitis media with effusion (OME) is the most common cause for bilateral conductive hearing loss in children. The predisposing conditions include cleft palate, Down's syndrome, carcinoma nasopharynx and adenoidal hypertrophy.
   a. Tympanometry shows evidence of negative pressure and fluid in the middle ear.
   b. Treatment: The treatment of choice for glue ear is myringotomy with grommet insertion.
   c. Complication**s**: They include retraction pockets, middle ear atelectasis, cholesterol granuloma and cholesteatoma. The persistent negative pressure can lead to deposition of cholesterol crystals and granuloma formation in the middle ear and mastoid.
   d. Unilateral OME in adults: Rule out nasopharyngeal pathology especially the carcinoma in elderly patients.

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19 Chronic Suppurative Otitis Media and Cholesteatoma

Be strong! You will understand the Gita better with your biceps, your muscles, a little stronger. It is truth alone that gives internal strength. Strength is the medicine for the world's disease.

—Swami Vivekananda

Points of Focus

- **INTRODUCTION**
- **MASTOID PNEUMATIZATION**
- **ATELECTASIS AND ADHESIVE OM**
  - ✓ Sade Classification
  - ✓ Sequelae
  - ✓ Treatment
- **CHRONIC SUPPURATIVE OTITIS MEDIA (CSOM)**
  - ✓ Epidemiology
  - ✓ Types of CSOM
  - ✓ Differences between Safe and Unsafe CSOM
  - ✓ Microbiology
- **ATTICOANTRAL CSOM/CHOLESTEATOMA**
  - ✓ Structure of Cholesteatoma
  - ✓ Types of Cholesteatoma
  - ✓ Pathogenesis of Acquired Cholesteatoma: Invagination Theory, Migration Theory, Basal Cell Hyperpalsia Theory and Squamous Metaplasia Theory
  - ✓ Spread of Cholesteatoma
  - ✓ Destruction of Bone
  - ✓ Clinical Features
  - ✓ Complications
  - ✓ Investigations
  - ✓ Surgical Treatment: Types of Surgery and Factors Determining Extent and Type of Surgery
  - ✓ Conservative Treatment
- **TUBOTYMPANIC CSOM**
  - ✓ Etiology
  - ✓ Clinical Features
  - ✓ Investigations
  - ✓ Treatment
    - – Medical therapy
    - – Surgical treatment
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

Chronic suppurative otitis media (CSOM) is the most common cause of otorrhea. In most children, acute otitis media (AOM) and otitis media with effusion (OME) subside either spontaneously or after treatment. Sequelae and complications of AOM and OME, some of which will be discussed in this chapter, are following:

- Middle ear atelectasis
- Adhesive otitis media
- Chronic perforation of tympanic membrane
- Chronic otomastoiditis
- Cholesteatoma
- Ossicular erosion or fixation
- Sensorineural hearing loss
- Tympanosclerosis
- Petrositis
- Labyrinthitis
- Facial palsy
- Intracranial complications

Some of the above topics are discussed in other chapters.

MASTOID PNEUMATIZATION

In comparison to healthy subjects, chronic OME patients have more sclerotic mastoids and decreased pneumatization. The two explanations have been forwarded:

1. *Hereditary theory*: The children with decreased pneumatization of the mastoid are prone to OME.
2. *Environmental theory*: Chronic OME leads to decreased pneumatization of the mastoid.

A cause and effect relationship between chronic OME and hypopneumatization of the mastoid is not clear. A recent study suggests that progressiveness of cholesteatoma is related to ventilatory conditions (Figs 1 and 2) of mastoid rather than Eustachian tube (ET) function.

ATELECTASIS AND ADHESIVE OTITIS MEDIA

See chapters Disorders of Eustachian Tube; and Diseases of External Ear.

- Atelectasis refers to absence of gas from a part or whole of the middle ear, due to the failure of ventilation and resorption of gas.
- Adhesive otitis media is an inflammation of the middle ear caused by prolonged ET dysfunction resulting in permanent retraction of tympanic membrane and obliteration of middle ear space.

The important factors in middle ear atelectasis are following:

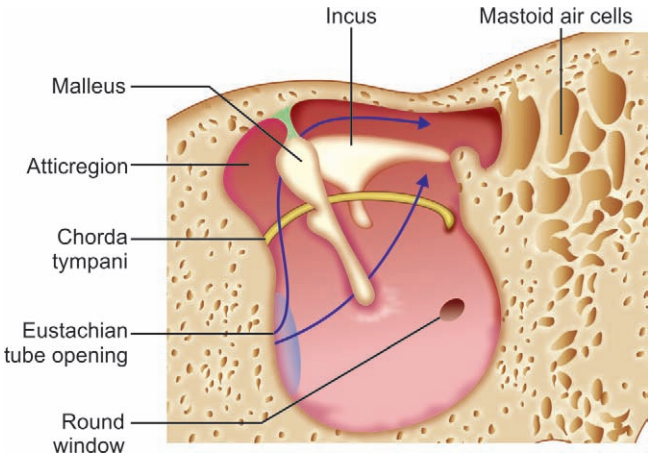


Fig. 1: Ventilation of mastoid antrum and air cells. Narrow openings for airway around the ossicles within the attic region

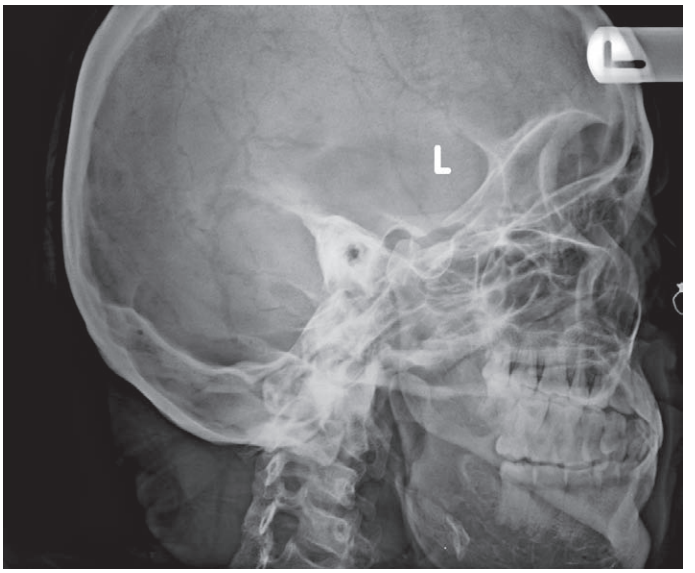


Fig. 2: X-ray mastoid left showing loss of pneumatization with sclerosis
Source: Dr Jayesh Patel, Consultant Radiologist, Anand, Gujarat

- Long standing ET dysfunction.
- Bilateral diffusion (gas exchange) between middle ear and blood: The middle ear gas composition resembles venous blood and differs from air.
- Nonpneumatized mastoid.

Sade Classification

Sade described four stages of tympanic membrane retraction (Figs 3A to E):

1. *Stage I retraction of tympanic membrane:* See chapter Diseases of External Ear.
2. *Stage II retraction of tympanic membrane with contact onto incus*.
3. *Stage III middle ear atelectasis*: Tympanic membrane is retracted onto the promontory and the ossicles but not adherent to these structures. The mucosal lining of the middle ear remains intact. Middle ear is obliterated partially or completely. It may be reversible with ventilating tubes. The administration of nitrous oxide during anesthesia can restore the tympanic membrane to its normal position.
4. *Stage IV adhesive otitis media:* Middle ear space is totally obliterated. Tympanic membrane is adherent to ossicles and promontory. The mucosal surface is absent.

Sequelae

- Destruction of the collagen-containing fibrous layer of tympanic membrane. It occurs due to repeated episodes of AOM and results in weakening, thinning and tympanosclerosis of tympanic membrane.
- Retraction of tympanic membrane may cause erosion of long process of incus and stapes superstructure.
- Deep retraction pockets in attic and posterosuperior region are considered precursors of cholesteatoma.
- Acquired primary cholesteatoma.

Treatment

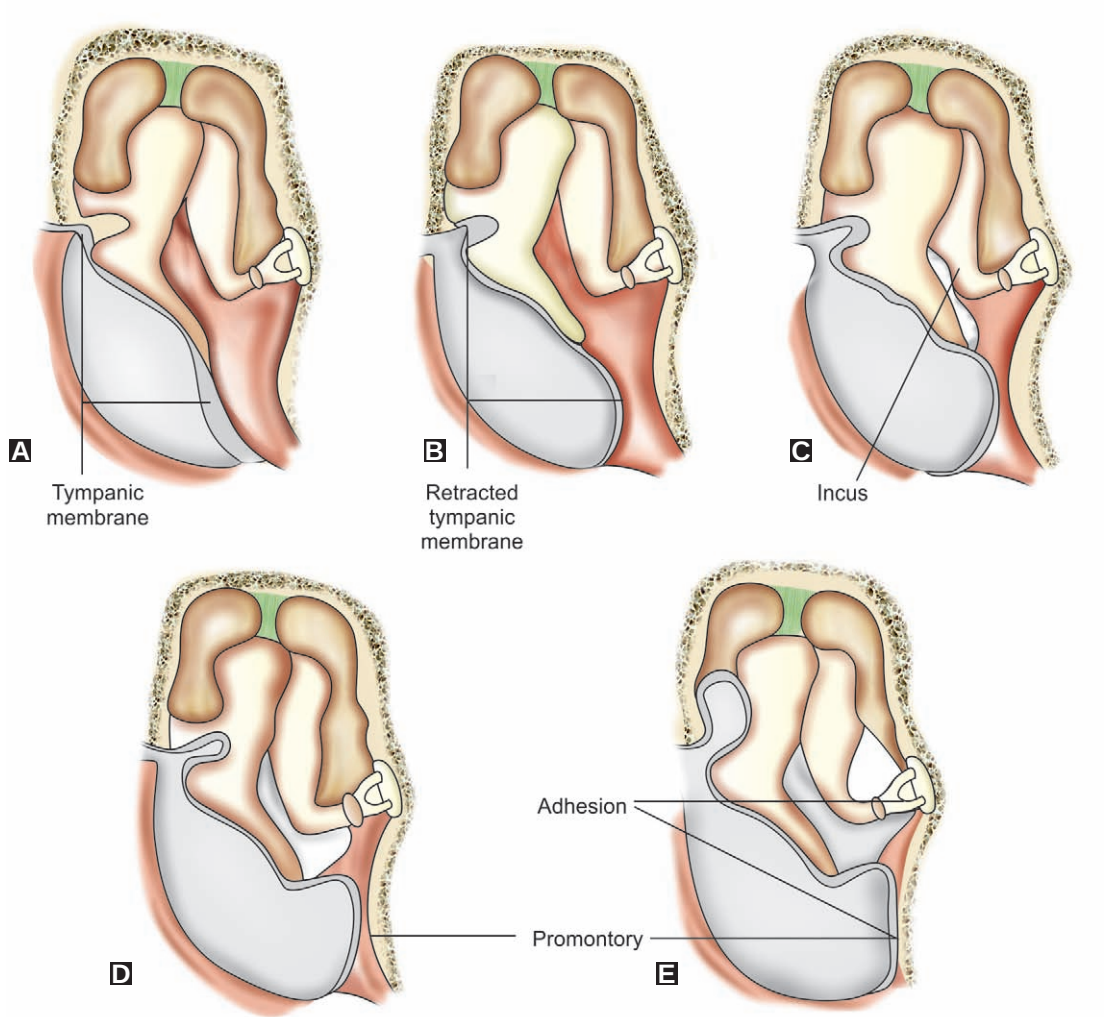
- *Tympanoplasty*: Hearing loss and ossicles erosion may need tympanoplasty for the reinforcement of atelectatic tympanic membrane and the improvement of hearing.
- *Hearing aids.*

CHRONIC SUPPURATIVE OTITIS MEDIA

Chronic suppurative otitis media is a long-standing infection of the middle ear cleft. It is characterized by ear discharge and a permanent perforation of tympanic membrane. The perforation's edges are covered by squamous epithelium. It does not heal spontaneously and become, a sort of an epithelium-lined fistulous track.

Epidemiology

- Incidence is higher in poor socioeconomic standards, poor nutrition and lack of health education.
- It affects both sexes and all age groups.
- In India prevalence rate is higher in rural area (46/1,000 persons) and lesser in urban area (16/1,000 persons).
- CSOM is the leading cause of hearing impairment in rural population.



Figs 3A to E: Sade Classification. Four stages of tympanic membrane retraction. (A) Normal; (B) Stage I retracted tympanic membrane; (C) Stage II retracted tympanic membrane touching incus; (D) Stage III middle ear atelectasis; (E) Adhesive otitis media

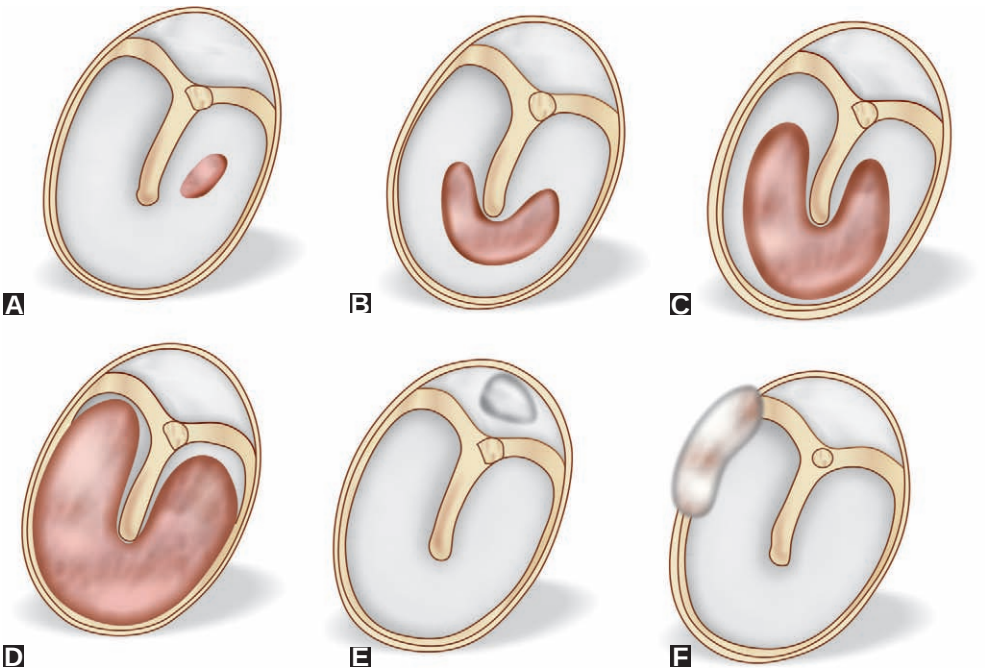
Types of CSOM

Traditionally CSOM is divided into two types: tubotympanic and atticoantral. Table 1 shows the differences between atticoantral and tubotympanic types of CSOM.

1. *Tubotympanic type (safe or benign)*: This type of CSOM involves anteroinferior part of middle ear cleft and is associated with a permanent central perforation. As there is no risk of serious complications it is called safe or benign CSOM.
   - Active (wet perforation): In the presence of inflammation of mucosa and mucopurulent discharge, the disease is called active.
   - Inactive (dry perforation): In the absence of inflammation of mucosa and mucopurulent discharge, the disease is called in active.
   - Permanent perforation: Long duration nonhealing dry perforation indicates that external squamous epithelium is fused with the internal mucosa at the margins of the perforation.
   - Healed chronic otitis media: Healing of perforation leads to its closure with thin membrane (fibrous layer absent). It may be associated with tympanosclerosis or some conductive hearing loss.
2. *Atticoantral type (unsafe, dangerous, posterosuperior lesion or cholesteatoma)*: This type of CSOM involves attic and

TABLE 1 Differences between atticoantral and tubotympanic types of CSOM

	Tubotympanic CSOM	*Atticoantral CSOM*
Nature	Safe and benign	Unsafe and dangerous
Otorrhea (ear discharge)		
• Odor	Odorless	Malodorous (putrid)
• Amount	Usually profuse	Usually scanty
• Type	Usually mucoid	Usually purulent
• Periodicity	Usually intermittent	Usually continuous
Perforation (Figs 4A to F)	Central	Attic or marginal
Granulations	Uncommon	Common
Polyp (Fig. 5)	Pale	Red and fleshy
Cholesteatoma	Absent	Present
Intracranial complications	Never	Not uncommon



Figs 4A to F: Types of perforations of tympanic membrane in cases of chronic suppurative otitis media (CSOM). (A) Small central perforation in anterosuperior quadrant; (B) Medium size kidney shaped central perforation; (C) Subtotal central perforation; (D) Total perforation with destruction of fibrous annulus; (E) Attic perforation of pars flaccida; (F) Posterosuperior marginal perforation. Note: A, B and C occur in tubotympanic CSOM. D, E and F seen in CSOM with cholesteatoma

Fig. 5: A ear polyp

posterosuperior regions of the middle ear cleft. It is associated with an attic or marginal perforation in posterosuperior quadrant of the pars tensa. As it is often associated with risk of serious complications due to the bone erosion nature of cholesteatoma, it is called unsafe or dangerous CSOM. Granulations and osteitis are present in many cases.

- Inactive: Self-cleaning retraction pocket in posterosuperior pars tensa or attic region with potential chances of cholesteatoma is called inactive disease.
- Active: Active cholesteatoma erodes bone, forms granulations and presents with putrid continuous ear discharge.
- Examination: Suction clearance and examination under operating microscope forms an important part of the clinical examination and assessment of any type of CSOM.

Microbiology

The incriminating microorganisms are identical in both the types of CSOM. The culture (aerobic and anaerobic) may show multiple organisms.

- Aerobic organisms: *Pseudomonas aeruginosa* (most common), *Proteus*, *Escherichia coli* and *Staphylococcus aureus*.
- Anaerobes: *Bacteroides fragilis* (most common) and anaerobic streptococci.

ATTICOANTRAL CSOM OR CHRONIC OM WITH CHOLESTEATOMA

The term cholesteatoma is a misnomer as it is neither a tumor nor contains cholesterol. It has destructive nature and erodes bone. Infected cholesteatoma causes rapid bone destruction.

The ciliated columnar epithelium in the anterior and inferior part, cuboidal epithelium in the middle part and pavement-like epithelium in the attic region line the middle ear cleft. In normal persons there is no keratinizing squamous epithelium in the middle ear cleft and its presence (skin in the wrong place) is called cholesteatoma (epidermosis or keratoma).

The cholesteatoma is seen in sclerotic mastoid but whether the latter is the cause or the effect of the former is not yet clear.

Structure of Cholesteatoma (Fig. 6)

Cholesteatoma is an epidermal inclusion cyst, which opens into the external auditory canal (EAC). It contains desquamated debris (mainly white-yellow keratin flakes resembling

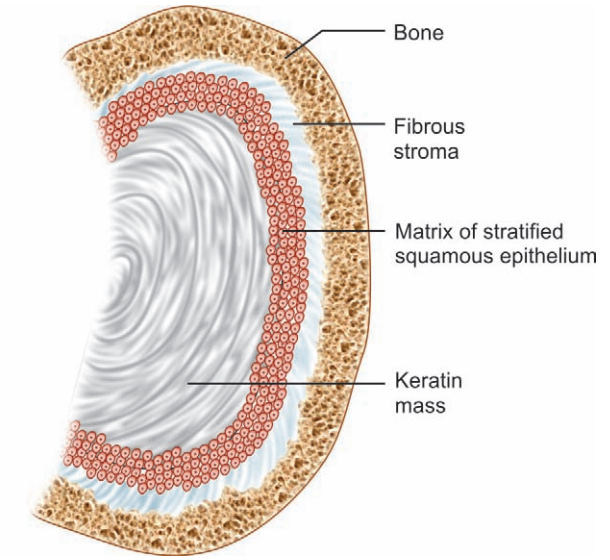


Fig. 6: Cholesteatoma structure. Stroma, matrix and keratin mass

cholesterol crystals) from its keratinizing squamous epithelial lining.

Cholesteatoma has two parts—matrix and central white mass.

1. *Matrix*: It is made up of keratinizing squamous epithelium, which rests on a thin stroma of fibrous tissues.
2. *Central white mass*: It consists of keratin debris, which is produced by the matrix.

Types of Cholesteatoma

Cholesteatoma of temporal bone is classified into two categories: congenital and acquired (Box 1).

1. *Congenital cholesteatoma*: It arises from the embryonic epidermal cell rests (keratinizing epithelium) entrapped in the middle ear cleft or temporal bone. The three important sites include middle ear, petrous apex and the cerebellopontine angle.

   Clinical features: A middle ear congenital cholesteatoma presents with conductive hearing loss and a white mass that can be seen behind an intact tympanic membrane. It may rupture through the tympanic membrane and present with a discharging ear. Then it becomes indistinguishable from CSOM.
2. *Acquired cholesteatoma*: They are the most common varieties of cholesteatomas and result from AOM and OME (Fig. 7). Acquired cholesteatoma is also called unsafe CSOM. Acquired cholesteatomas are further divided into two types: primary and secondary (Box 1).
   a. Primary acquired cholesteatoma: In primary acquired cholesteatoma, there is neither history of previous otitis media, a pre-existing perforation, nor otorrhea.

Box 1: Types of cholesteatoma

- Congenital
  - Middle ear
  - Petrous apex
  - Cerebellopontine angle
- Acquired
  - Primary
    - Attic region
    - Posterosuperior region
  - Secondary

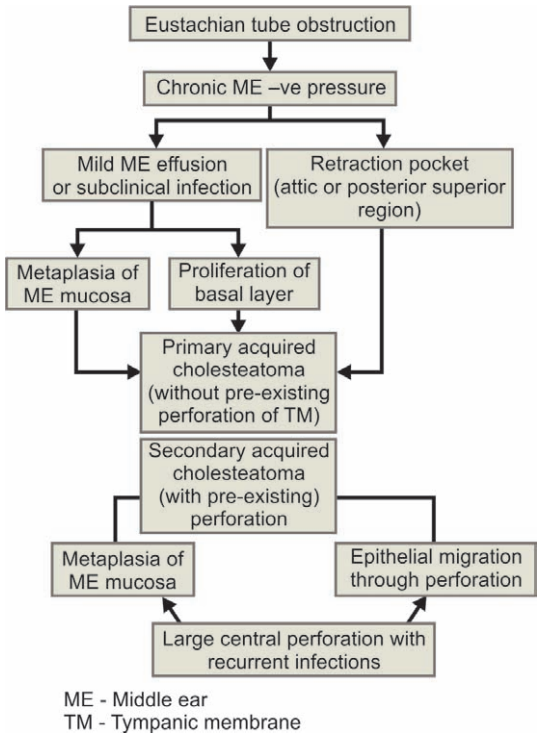


Fig. 7: Etiopathogenesis of primary and secondary acquired cholesteatoma

   b. Secondary acquired cholesteatoma: This cholesteatoma occurs in pre-existing perforation of pars tensa, which is usually posterosuperior marginal perforation or sometimes large central perforation.

Pathogenesis of Acquired Cholesteatoma

The pathogenesis of acquired cholesteatoma is yet a matter of debate. The four basic theories are invagination, hyperplasia, migration and metaplasia (Fig. 8). Attempts have been made to explain the pathogenesis on the basis of combination of these theories.

1. *Invagination theory (Wittmaack)*: This theory explains primary acquired cholesteatoma. Invagination of tympanic membrane from the attic or posterosuperior part of pars tensa occurs in the form of retraction pockets. The outer surface of tympanic membrane is lined with stratified squamous epithelium, which after invagination forms the matrix of cholesteatoma and lays down keratin in the pocket.

   As the retraction pocket deepens because of negative middle ear pressure and repeated inflammation, desquamated keratin cannot be cleared from the recess and results in cholesteatoma. Bacteria can infect the keratin matrix, forming biofilms resulting into chronic infection and epithelial proliferation.

   The most common sites for this primary acquired cholesteatoma are pars flaccida or attic (being less fibrous and less resistance to displacement) and posterosuperior quadrant of pars tensa. The attic perforation is simply the proximal end of an expanding invaginated sac.
2. *Epithelial invasion or migration theory (Habermann)*: This theory explains secondary acquired cholesteatoma. The keratinizing squamous epithelium of tympanic membrane or deep canal wall migrates into the middle ear through a tympanic membrane perforation. The pre-existing

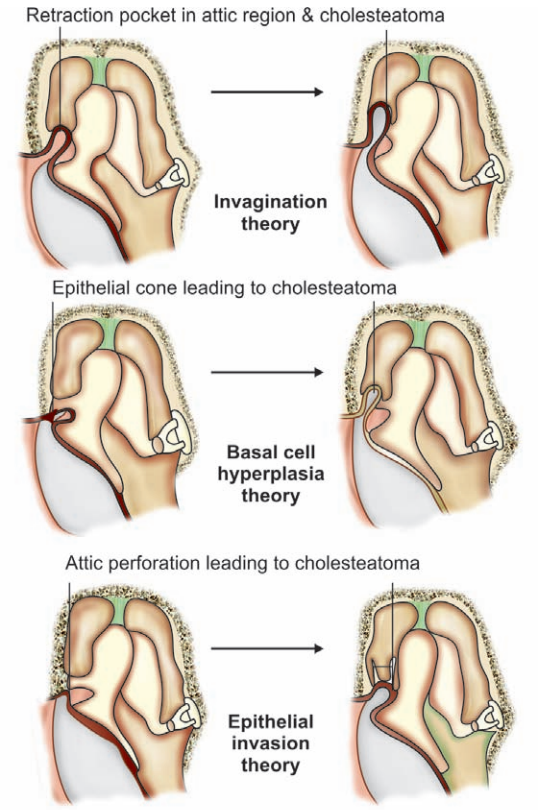


Fig. 8: Pathogenesis of cholesteatoma: Invagination, basal cell hyperplasia and epithelial ingrowth theories

perforation is especially of the marginal type where part of annulus tympanicus has been destroyed.

The damaged (due to inflammation) inner mucosal lining of tympanic membrane, allows the outer keratinizing squamous epithelium to migrate inward and produce this secondary acquired cholesteatoma. Cholesteatomas, which arise after temporal bone fractures, may result from this migration.

3. *Basal cell hyperpalsia theory (Lange and Ruedi)*: Under the influence of infection, basal cells of germinal layer of skin can proliferate and lay down keratinizing squamous epithelium. Prickle epithelial cells of pars flaccida can invade the subepithelial tissue by means of proliferating columns of epithelial cells. Basal lamina disruptions have been documented. These basal lamina breaks allow invasion of epithelial cones into the subepithelial connective tissue and the formation of microcholesteatomas, which may enlarge and perforate an intact tympanic membrane and present as primary acquired cholesteatoma.
4. *Squamous metaplasia theory (Wendt and Sade)*: Middle ear mucosa, like respiratory mucosa elsewhere, can undergo metaplasia due to repeated infections and transform into squamous epithelium. Such a change has also been reported in OME. Middle ear mucosa can undergo metaplasia due to repeated infection through a pre-existing perforation and result in secondary acquired cholesteatoma.

The simple squamous or cuboidal epithelium of middle ear cleft can undergo a metaplastic transformation into keratinizing epithelium. The pluripotent epithelial cells, stimulated by inflammation can become keratinizing, which would enlarge because of accumulated debris and contact with tympanic membrane. With infection and inflammation, cholesteatoma results in perforation of the tympanic membrane and present as primary acquired cholesteatoma.

Spread of Cholesteatoma

In the middle ear cleft, cholesteatoma follows the path of least resistance and causes enzymatic bone destruction. The growth of attic cholesteatoma is limited by the mucosal folds and suspensory ligaments of the ossicles.

Attic cholesteatoma first invades Prussak's space (lateral most portion of epitympanum) and then into the recesses of epitympanum posteriorly, lateral to the body of incus. Inferiorly it goes into the middle ear via pouch of von Troltsch. Anteriorly cholesteatoma enters into the protympanum. An attic cholesteatoma thus extends posteriorly into the aditus, antrum and mastoid, inferiorly into the mesotympanum and medially surrounds the incus and head of the malleus.

Destruction of Bone

Cholesteatoma destroys the bones, which come in its way such as ear ossicles, bony labyrinth, canal of facial nerve, sinus plate and tegmen tympani. This bone destruction results in several complications.

Formerly bone destruction was believed to be due to pressure necrosis. Currently, bone destruction has been attributed to enzymes. They are liberated by osteoclasts and mononuclear inflammatory cells (associated with cholesteatoma) and include as collagenase, acid phosphatase and proteolytic enzymes.

Clinical Features

- *Symptoms*
  - No symptoms: Patients may remain asymptomatic in initial stages of disease.
  - Ear discharge: The persistent malodorous (putrid due to anaerobic bacteria) ear discharge is usually purulent and scanty in amount. It can be so scanty that the patient may not be aware of it.

    The cessation of discharge in a continuously discharging ear is an ominous sign. The perforation might be sealed by crusted discharge, mucosa or polyp. The obstruction of the free flow of purulent discharge has the potential to result in complications.
  - Slowly progressive deafness: The severity of hearing loss, which is conductive, varies. Hearing is normal when ossicular chain is intact. The cholesteatoma, which have destroyed the ossicles, may bridge the gap caused by destroyed ossicles. The sensori- neural element may be added to hearing loss.
  - Bleeding: It can occur from granulations and red fleshy polyp while cleaning the ear.
  - Symptoms of sequelae: Pain, vertigo, facial palsy, headache, vomiting, ataxia and fever.
- *Otoscopy/Microscopy/Endoscopy*
  - Perforation: The most common sites of the perforation, which is marginal, are attic and posterosuperior region (Fig. 4). An attic perforation may be hidden behind a small amount of crusted discharge.
  - Retraction pocket: The degree of an invagination and retraction in the attic and posterosuperior pars tensa varies from shallow and self-cleansing pocket to deep pocket with accumulation of keratin and infected debris.

- Cholesteatoma: Pearly white flakes of cholesteatoma can be seen in the retraction pockets. The most common sites are attic and posterosuperior region but cholesteatoma may extend and present into other parts of middle ear cleft.
  - Primary acquired cholesteatoma: Defect of variable size in attic or posterosuperior region containing keratin debris.
  - Secondary acquired cholesteatoma: Keratinizing epithelium migrating through a pre-existing perforation into middle ear.
  - Congenital cholesteatoma: Pearl-like mass appears behind an intact tympanic membrane.
- Granulation tissue: Granulation tissues surround the area of osteitis especially attic and posterosuperior region and may be present in the attic, antrum, posterior tympanum and mastoid.
- Polyp: A fleshy red polyp may be seen filling the meatus. An ear polyp in a chronic case of otorrhea should be considered due to cholesteatoma until proven otherwise.
- Ossicular necrosis: Bony destruction may involve the long process of incus, stapes and handle of malleus or the entire ossicular chain.
- Cholesterol granuloma: This mass of granulation tissue, which appears blue in color, may be present in association with cholesteatoma or in the mesotympanum behind an intact drum. Cholesterol granuloma is a mass of granulation tissue with foreign body giant cells surrounding the cholesterol crystals. It is a reaction to the retained secretions and hemorrhages.

- *Fistula test:* The positive fistula test indicates erosion of lateral semicircular canal.

Congenital cholesteatoma: A whitish mass behind a child's intact tympanic membrane with hearing loss indicates congenital cholesteatoma.

Complications

The sign and symptoms of complications, which need immediate referral to specialist for urgent management, include (Table 2), pain, vertigo, headache, facial palsy, listless, drowsiness, loss of appetite, fever, nausea and vomiting, irritability, neck rigidity, diplopia, ataxia and painful swelling around the ear.

Pain in cases of CSOM is an ominous sign and needs special attention to rule out intracranial complications.

Investigations

- *Tuning fork test and audiogram*: They are essential for preoperative assessment and to confirm the degree and type of hearing loss.
- *Imaging*: The extent of bone destruction, degree of mastoid pneumatization or sclerosis, a low lying dura and an anteriorly placed sigmoid sinus can be seen. These findings help in the planning of surgery. High-resolution computed tomography (HRCT) and magnetic resonance imaging (MRI) are recommended for revision mastoid operations. They help in:
  - Establishing presence of cholesteatoma.
  - Operative planning.
- *Culture and sensitivity* for microorganisms

TABLE 2 Sign and symptoms of complications of CSOM and their causes

Signs	*Symptoms*
Pain	Otitis externa, extradural abscess, perisinus abscess
Vertigo	Labyrinthitis, cerebellar abscess
Facial weakness	Facial palsy due to erosion of facial canal
A listless child with insomnia and anorexia	Extradural abscess
Headache, nausea and vomiting	Raised intracranial pressure
Fever, irritability and neck rigidity	Meningitis
Diplopia	Gradenigo's syndrome
Drowsiness and bradycardia	Brain abscess
Ataxia	Cerebellar abscess
Nominal aphasia	Temporal lobe abscess
Painful swelling around the ear	Mastoid abscess

Surgical Treatment

The mainstay of treatment is surgical removal of cholesteatoma and rendering the ear safe. The secondary part of the surgery includes preservation and reconstruction of hearing system. In presence of complications surgery should be performed at the earliest possible.

A. *Types of surgery*: The two types of surgical procedures are done to remove the cholesteatoma. They are canal wall down and canal wall up procedures (Table 3). See chapter Middle Ear and Mastoid Surgeries.

- Atticotomy: Transcanal.
- Canal wall-up procedure (intact posterior meatal wall or closed procedure): This procedure employs combined approach. Cholesteatoma is approached through the EAC and mastoid cavity. The posterior canal wall remains intact and keeps the EAC and mastoid cavity separate (closed). Th ough it gives dry ear, better hearing and less postoperative care, patients need long postoperative follow-up. The chances of residual and recurrent cholesteatoma are high. The approaches for this procedure include:
  - With facial recess approach
  - Without facial recess approach
- Canal wall-down procedure (open procedure): The cholesteatoma is fully exteriorized and mastoid cavity and EAC become one big cavity. The commonly performed procedures include:
  - Radical mastoidectomy
  - Modified radical mastoidectomy
  - Bondy procedure
- Tympanoplasty: Reconstructive surgery
  - During the primary surgery
  - Second stage

B. *Factors determining extent and type of surgery*: These are:

- Hearing status of both the ears.

TABLE 3 Differences between canal wall up and canal wall down procedures

	Canal wall-up	*Canal wall down*
Examples	Combined approach tympanoplasty, posterior tympanotomy	Mastoidectomy: Simple, Radical, Modified radical, Bondy
Posterior canal wall	Not removed	Removed
Mastoid and ear canal	Remains separate	Merge with each other
Meatoplasty	Not required	Required
Postoperative regular cleaning under microscope	Usually not needed	Usually required
Rate of recurrence/ residual cholesteatoma	High	Low
Second look surgery	Required after 6 months	Not required
Swimming	Allowed	Usually discouraged
Hearing aid fitting	Easy	Problematic

- Extent of cholesteatoma.
- Mastoid pneumatization.
- Function of ET.
- Presence of complications.
- Patient factors: Age, occupation and general medical status.

Conservative Treatment

- Though has a limited role it should be considered in the critically ill patient and only-hearing ear, where risk of surgery may not outweigh benefits.
- In small and limited cholesteatoma or having fairly large opening in the EAC, entrapped keratin may be removed directly or by irrigation with saline or 1:1 distilled white vinegar and 70% isopropyl alcohol for stabilization.

TUBOTYMPANIC CSOM OR CHRONIC OM WITHOUT CHOLESTEATOMA

The tubotympanic disease remains localized to the mucosa of anteroinferior part of the middle ear cleft. The processes of healing and destruction go together and depend upon the virulence of organism and resistance of the patient.

Acute exacerbations occur frequently. The cochlea may be damaged due to absorption of toxins from the oval and round windows and hearing loss becomes mixed type.

Etiology/Predisposing Factors

- *Permanent perforation*: This benign type of CSOM is mostly the sequela of a large central perforation of childhood AOM, which usually follows after exanthematous fever. The permanent perforation allows repeated infection through the external ear canal causing otorrhea. The middle ear mucosa, which is exposed to the environment, gets sensitized to dust, pollen and other aeroallergens from the environment.
- *Eustachian tube*: Ascending infection from the tonsils, adenoids and sinuses may result in persistent or recurring otorrhea.
- *Allergy*: Allergy may be from foods (such as milk, eggs, fish) and inhalants (pollen, fungi, dusts).

Clinical Features

- *Symptoms*
  - Ear discharge: The odorless ear discharge may be mucoid or mucopurulent, profuse or scanty and constant or intermittent. The ear discharge is common at the time of upper respiratory tract infection and when water enters into the ear.
  - Hearing loss: The severity of hearing loss varies but is rarely profound. The hearing may improve in the presence of discharge due to "round window shielding effect", because the discharge helps to maintain the phase differential. In the dry ear sound waves reach both the oval and round windows simultaneously and cancel each other's effect.
- *Otoscopy/Microscopy/Endoscopy*
  - Central perforation of pars tensa (Fig. 4): Note following points:
    - Size: small, medium, large, or subtotal (extending up to annulus).
    - Shape: round, oval or kidney shaped.
    - Position: anterior, posterior, or inferior to the handle of malleus.
  - Middle ear mucosa: It is normal (pale pink and little moist) when the disease is inactive but looks inflamed red edematous and velvety when disease is active.
  - Polyp (Fig. 9): An edematous and inflamed mucosa protrudes through the perforation and presents in the external canal as polyp, which is usually pale (pink and fleshy polyp in cases of atticoantral cholesteatoma).
  - Tympanosclerosis: The hyalinization and subsequent calcification occurs in the subepithelial connective tissue of tympanic membrane and middle ear mucosa. It appears as white chalky deposits on the promontory, ossicles, joints, tendons and oval and round windows and may interfere with the mobility of these structures.
  - Ossicular chain: Ossicular chain usually remains intact and mobile but may show some degree of necrosis, particularly of the long process of incus.
  - Fibrosis and adhesions: They result from the healing process and may impair mobility of ossicular chain or block the ET.
  - Granulations: They may be seen over the remnant of tympanic membrane.
  - Rule out in-growth of squamous epithelium from the edges of perforation. Though rare, cholesteatoma can coexist with a central perforation.

Investigations

- *Hearing tests (tuning fork and audiogram)*: They give an idea regarding the degree and type of hearing loss. The severity of hearing loss, which is usually conductive, varies but rarely exceeds 50 dB. The cochlea may get damaged due to absorption of toxins from the oval and round windows and results in sensorineural element in conductive hearing loss (mixed type).

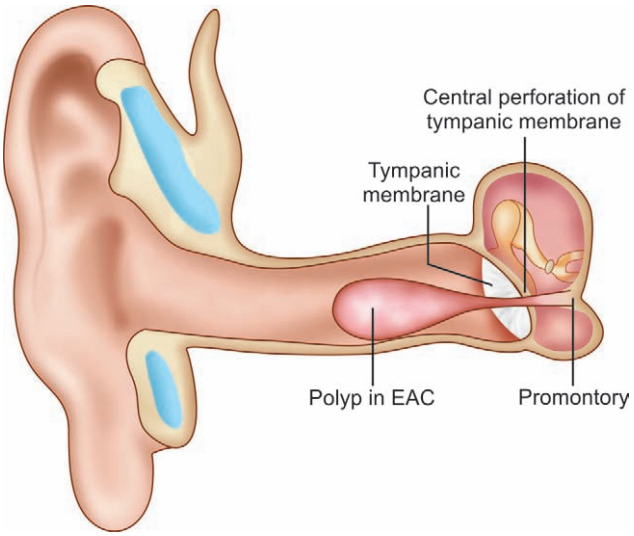


Fig. 9: Ear polyp in tubotympanic CSOM. Polyp arising from the promontory and coming to EAC through the central perforation of tympanic membrane

- *Culture and sensitivity*: Culture and sensitivity of ear discharge helps in identifying the microorganisms and selecting the proper antibiotics.
- *Imaging*: Mastoid is usually found sclerotic. The pneumatized mastoid may show clouding of air cells. There is no evidence of bone destruction, which is a feature of attico-antral cholesteatoma.

Treatment

In the presence of otorrhea, the aim is to clean the ear discharge and control infection. In cases of dry ear, perforation may be repaired by tympanoplasty, which not only improve hearing but also prevents recurrent infection and acute exacerbations.

- *Medical therapy*
  - Aural toilet: Removal of discharge and debris from the EAC can be done by dry mopping with absorbent cotton buds and suction clearance under microscope.
  - Antibiotic/steroid ear drops: Antibiotic/steroid ear drops 3–4 times a day in wet and running ears have local antimicrobial and anti-inflammatory effects.
    - An acid pH of ear drops helps in eliminating pseudomonas infection.
    - Ear drops of 1.5% acetic acid are useful for acidifying ear canal.
    - Ototoxic antibiotics ear drops are not used.
    - The most preferred antibiotics and steroids are quinolone group (ciprofloxacin and ofloxacin) and dexamethasone respectively.
    - After instilling drops, the ear is kept upward when the patient is in supine position. Intermittent pressure on the tragus facilitates the drops to reach in the middle ear.
    - The ear drops are stopped once the ear becomes dry. They should never be used in dry ears as it can make the ear wet and discharging.
    - The ear should be examined regularly as ear drops can cause maceration of canal skin, local allergy, growth of fungus and resistance of organisms.
  - Systemic antibiotics: They are prescribed only in cases of acute exacerbation. There is no role of systemic antibiotics in the treatment of uncomplicated CSOM.
  - Patient's instructions:
    - Water should not enter in the ear while bathing, swimming and hair-wash. The ear plugs and rubber inserts may be employed.
    - Avoid hard nose-blowing as it can push the infection from nasopharynx to middle ear.
    - Avoid self-cleaning of the ear.
    - Stop the ear drops once the ear becomes dry.
    - Take treatment of upper respiratory infections at the earliest.
  - Treatment of source of infection: Treatment of contributory diseases (infections of tonsils, adenoids, nose and paranasal sinuses and allergy) is important.
- *Surgical treatment*
  - Removal of ear polyp or granulations: They facilitate ear toilet and treatment with local antibiotics. An ear polyp is never avulsed because it may be attached to the stapes, facial nerve and horizontal semicircular canal and can lead to complications (such as facial paralysis and labyrinthitis).

2. Tympanoplasty: In a dry ear, myringoplasty/tympanoplasty restore hearing and check repeated infection from the external ear canal. (See chapter Middle Ear and Mastoid Surgeries).

Clinical Highlights

1. **Cholesteatoma**: It erodes bone. Attic perforation is associated with cholesteatoma. Primary treatment of middle ear cholesteatoma is canal wall-down mastoidectomy, which may be combined with tympanoplasty.
2. **Primary cholesteatoma:** The posterosuperior retraction pocket can progress and lead to primary cholesteatoma.

FURTHER READING

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20 Complications of Suppurative Otitis Media

Strength is life; weakness is death. Strength is felicity, life eternal, immortal; weakness is constant strain and misery, weakness is death. I faced the brutes and they fell back and finally fled. That is the lesson for all life--face the terrible, face it boldly.

—Swami Vivekananda

Points of Focus

- **INTRODUCTION**
- **FACTORS INFLUENCING COMPLICATIONS**
- **PATHWAYS OF SPREAD**
- **CLINICAL FEATURES**
- **ACUTE MASTOIDITIS**
- **MASKED (LATENT) MASTOIDITIS**
- **EXTRATEMPORAL COMPLICATIONS**
  - ✓ Postauricular Abscess
  - ✓ Zygomatic Abscess
  - ✓ Luc's Abscess
  - ✓ Citelli's Abscess
  - ✓ Parapharyngeal and Retropharyngeal Abscesses
  - ✓ Bezold's Abscess
- **PETROSITIS**
- **FACIAL PARALYSIS**
- **LABYRINTHITIS**
- **EXTRADURAL ABSCESS**
- **SUBDURAL ABSCESS**
- **MENINGITIS**
- **OTOGENIC BRAIN ABSCESS**
- **LATERAL SINUS THROMBOPHLEBITIS**
- **OTITIC HYDROCEPHALUS**
- **CLINICAL HIGHLIGHTS**

Acute ear pain with continued high grade fever is to be dreaded for the patient may become delirious and die.

—Hippocrates in 460 BC

INTRODUCTION

The otitis media is one of the most commonly treated infections. In acute otitis media (AOM) and cholesteatoma, disease process is usually limited only to the mucoperiosteal lining of the middle ear cleft. When it breaks into the bony walls of the cleft, various complications can arise (Fig. 1). Complications can occur as the result of bony destruction from either hyperemic decalcification in AOM or chronic bioenzymatic activity in cholesteatoma.

In pre-antibiotic era, AOM frequently led to intratemporal and intracranial complications. In the pre-antibiotic era, complications of suppurative otitis media (SOM) were very common and led to high morbidity. The three major intracranial complications were brain abscess, meningitis and infective thrombosis of lateral sinus. Purulent meningitis was the most frequent cause of death, followed by sinus thrombosis second and brain abscess last. The two intratemporal complications of SOM purulent labyrinthitis and petrositis often led to fatal intracranial extension.

After the introduction of sulfonamide for AOM in 1935 and penicillin in 1942, the incidence of complications decreased abruptly.

Now purulent meningitis and brain abscesses (temporal lobe more common than cerebellar) are the most frequent intracranial complications but sinus thrombosis has nearly disappeared as a cause of death. In majority of the patients the cause is cholesteatoma.

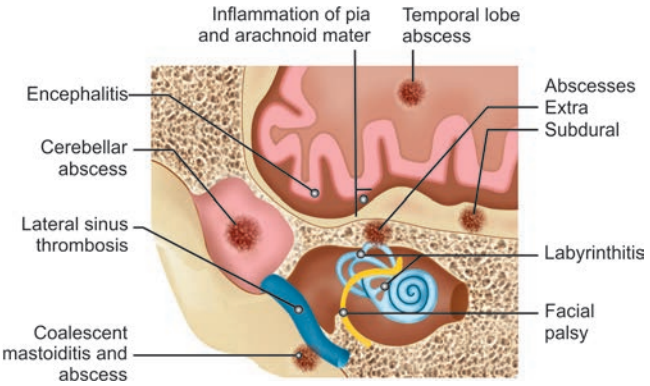

Fig. 1: Complications of suppurative otitis media involving structures in relation with middle ear

Appearance of ear pain in cases of chronic otorrhea indicates an impending intracranial complications until unless prove otherwise. Foul-smelling, creamy ear discharge indicates a fulminant, destructive process. Headache and drowsiness are danger signs. One of the earliest signs of brain abscess is a visual field defect. Fever is mostly due to either meningitis or sinus thrombosis.

In the current era of new range of antibiotic, though the incidence of complications has significantly declined they do occur and must be kept in mind as the early diagnosis and prompt treatment reduce mortality. Complications of SOM can be categorized into three groups: intratemporal, extratemporal and intracranial (Box 1).

Box 1: Complications of suppurative otitis media

- Intratemporal
  - Persistent tympanic membrane perforation
  - Mastoiditis
    - ◊ Acute coalescent mastoiditis
    - ◊ Masked (latent) mastoiditis
  - Acute petrositis
  - Facial paralysis
  - Labyrinthine fistula
  - Labyrinthitis
    - ◊ Labyrinthine fistula
    - ◊ Serous labyrinthitis
    - ◊ Suppurative labyrinthitis
- Extratemporal
  - Postauricular abscess
  - Zygomatic abscess
  - Bezold's abscess
  - Meatal abscess (Luc's abscess)
  - Behind the mastoid (Citelli's Abscess)
  - Parapharyngeal and retropharyngeal abscesses
- Intracranial
  - Extradural (epidural) abscess
    - ◊ Middle cranial fossa
    - ◊ Posterior cranial fossa
  - Subdural empyema
  - Meningitis
  - Brain abscess
    - ◊ Temporal lobe of cerebrum
    - ◊ Cerebellum
  - Lateral sinus thrombophlebitis
  - Cerebrospinal fluid otorrhea
  - Otitic hydrocephalus

FACTORS INFLUENCING DEVELOPMENT OF COMPLICATIONS

Complications usually occur when ear infections are either uncontrolled or inadequately controlled. The factors which influence complications of SOM, include:

- *Microorganisms*: High virulence of organisms.
- *Host*: Poor nutrition and resistance and immunosuppression.
- *Antibiotics*: Inadequate course and dose of antibiotic in treating acute middle ear and mastoid infections.
  - Resistance of organisms to antibiotics.
- *Chronic systemic diseases*: Diabetes mellitus, tuberculosis, nephritis, leukemia, etc.
- *Pneumatization of temporal bone*: Intracranial complications of AOM are common in poorly pneumatized temporal bone.

PATHWAYS OF SPREAD

The uncontrolled or poorly controlled middle ear cleft infections can cause complications through any of the following three routes.

1. *Bone erosion*: This is the most common route of spread. The process of hyperemic decalcification occurs in acute infections. Chronic infection results in osteitis and granulation tissue and in some cases osteomyelitis. Bony erosion is caused by chronic bioenzymatic activity in cases of cholesteatoma.
2. *Progressive retrograde thrombophlebitis of small venules*: The rich network of veins within the temporal bone is in direct communication with extracranial, intracranial and cranial diploeic veins. Haversian canal veins communicate with intracranial dural venous sinuses and superficial veins of brain. The mastoid bone infection can result in thrombophlebitis of venous sinuses (usually sigmoid sinus) and even cortical vein thrombosis. The bony walls of middle ear and mastoid air cells are found intact during surgery. This type of spread is common in acute infections.
3. *Preformed pathways*:
   - Congenital dehiscence: Dehiscence in Fallopian facial canal and floor of hypotympanum over the jugular bulb
   - Patent sutures: Petrosquamous suture
   - Temporal bone fractures: The fibrous scar permits infection
   - Surgical defects: Stapedectomy, fenestration and exposure of dura
   - Perilymphatic fistula: Congenital or acquired
   - Normal anatomy openings: Infection of labyrinth and from labyrinth to the meninges
     - Oval and round windows
     - Internal acoustic meatus
     - Cochlear aqueduct
     - Endolymphatic duct and sac

CLINICAL FEATURES

The symptoms of impending intracranial complications include:

- *Ear pain*: Appearance of ear pain (deep boring nature) in cases of chronic otorrhea indicates impending intracranial complications until unless prove otherwise.
- Severe headache that patient never had is again an ominous sign.

- *Persistence of otorrhea*: Foul-smelling, creamy (viscous and thick) ear discharge, which may be blood stained, indicates a fulminant, destructive process. Scanty discharge is indicative of poor drainage.
- Other clinical features of impending intracranial complications are as following:
  - High-grade fever, toxic look, photophobia, irritability and altered consciousness.
  - One of the earliest signs of brain abscess is a visual field defect.
  - Stiff neck and generalized malaise are mostly due to meningitis.

ACUTE MASTOIDITIS

This is inflammation of mucosal lining of antrum and mastoid air cells invariably occurs in AOM. However, in "mastoiditis," which is now relatively rare, infection involves bony walls of the mastoid air cells.

Ear pain and tenderness extending into the postauricular region and fever are hallmark symptoms. Protrusion of pinna and postauricular erythema and swelling are classic findings.

Etiology and Predisposing Factors

- Acute otitis media especially in cases of measles, exanthematous fevers, poor nutrition and diabetes.
- Microorganisms: Beta-hemolytic streptococci (most common) and other organisms causing AOM. Anaerobic organisms are also common.
- Mastoids with well-developed mastoid air cells.
- Children are affected more.

Pathology

Acute mastoiditis can manifest in two ways:

1. *Acute periostitis* (inflammation of periosteum): Spread of infection occurs via venous channels.
2. *Acute osteitis* (coalescent mastoiditis): Destruction of mastoid air cells trabeculae.

The pathological processes include production of pus under tension and hyperemic decalcification and osteoclastic resorption of bony walls.

- *Production of pus under tension*: The inflammatory process to mucoperiosteal lining increases the amount of pus. The drainage of pus through a small perforation of tympanic membrane (TM) and Eustachian tube (ET) cannot keep pace with the amount of pus production. Swollen mucosa of the antrum and attic impede the drainage and result in further accumulation of pus.
- *Hyperemic decalcification and osteoclastic resorption*: Hyperemia causes dissolution of calcium from the mastoid air cells (hyperemic decalcification).

The destruction and coalescence of mastoid air cells convert mastoid into a single large cavity, which is filled with pus and also called "empyema mastoid." The mastoid cortex may be broken leading to subperiosteal abscess which can burst the overlying skin and result in discharging fistula.

Clinical Features

The change in the character and reappearance of clinical features during the resolving AOM points to the development of acute mastoiditis.

- *Pain*: Behind the ear in mastoid region.
- *Fever*: Usually low grade. In children, fever is high with a rise in pulse rate.
- *Ear discharge*: Profuse and increases in purulence.

The persistence of otorrhea beyond 3 weeks in a case of AOM indicates mastoiditis.

The discharge may cease (due to obstruction in its drainage) with progressive worsening of clinical features. Mucopurulent or purulent discharge is often pulsatile (light-house effect). It comes through the perforation of pars tensa.

- *Mastoid tenderness*: Tenderness may be present over the middle of mastoid process, mastoid tip, posterior border or the root of zygoma. Tenderness over the suprameatal triangle may be seen in AOM due to inflammation of mastoid antrum. Tenderness may be compared with that of the healthy side.
- *Sagging of posterosuperior meatal wall*: It indicates periostitis of bony wall that lies between the antrum and deep bony canal.
- *Perforation of tympanic membrane*: A small central perforation or nipple like protrusion can be seen in pars tensa with congestion of remaining TM. The TM may remain intact (inadequate antibiotics treatment) but looks dull and opaque.
- *Swelling over the mastoid region*: Edema of periosteum imparts a smooth "ironed out" feeling over the mastoid. If pus bursts bony cortex, a subperiosteal fluctuant abscess is seen. The pinna is pushed forward and downwards.
- *Conductive hearing loss.*
- *General appearance*: Ill and toxic look.

Investigations

- *CBC*: Polymorphonuclear leukocytosis and raised ESR.
- *CT temporal bone*: Loss of bony trabeculae indicates coalescent mastoiditis. CT can also identify other intracranial or neck abscesses.
- *Culture and sensitivity of ear discharge*: To know the organism and the antibiotics to which they are sensitive.

Differential Diagnoses

- *Suppuration of mastoid lymph nodes*: Scalp infection can lead to postauricular lymphadenopathy, which though rarely suppurates and leads to abscess formation. There is no history of preceding otitis media and TM is normal. The abscess is quite superficial.
- *Furunculosis of external auditory canal (EAC)*: It should be differentiated from acute mastoiditis (Table 1).

Treatment

Majority of the cases of acute mastoiditis usually respond to culture-directed intravenous antibiotics and myringotomy with or without tympanostomy tube.

- *Hospitalization and antibiotics*: These patients need intravenous antibiotics in high doses. Till the report of culture and sensitivity, amoxicillin or ampicillin is given. As the anerobic organisms are often present, metronidazole is added.
- *Myringotomy*: Small TM perforation is insufficient and wide myringotomy facilitates pus drainage.

TABLE 1 Differentiating features between acute mastoiditis and furuncle of external auditory canal

	Acute mastoiditis	*Furuncle of external auditory canal*
Preceding history	Ear discharge and acute otitis media	Self cleaning of ear
Ear discharge	Mucoid or mucopurulent	Pain and swelling subside after purulent (mixed with blood) discharge
Conductive hearing loss	Always present	Only on canal occlusion
Tenderness	On mastoid region	Anterior or inferior to pinna
Auricle movement	Usually painless	Always painful
Displacement of auricle	Inferior, anterior and lateral	Only anterior
Retroauricular sulcus	Not obliterated	May be obliterated
Tympanic membrane	Perforation	Normal
External auditory canal swelling	Sagging of posterosuperior deep canal wall	In outer cartilaginous part
Pre-or postauricular lymph nodes	May be present and tender	Absent
Imaging temporal bone	Destruction of mastoid air cells	Normal

- *Cortical mastoidectomy*: The cortical mastoidectomy exenterates all mastoid air cells along with the removal of pockets of pus. Adequate antibiotic treatment is continued for 5–10 days postoperatively. The indications include the following:
  - Subperiosteal abscess.
  - Sagging of posterosuperior meatal wall.
  - Positive reservoir sign: external auditory canal fills with pus immediately after it has been cleaned.
  - No improvement or worsening of condition in spite of adequate medical treatment for 48 hours.
  - Complications: Facial paralysis, labyrinthitis and intracranial complications.

Complications

They include subperiosteal abscess, labyrinthitis, facial paralysis, petrositis, extradural abscess, subdural abscess, meningitis, brain abscess, lateral sinus thrombophlebitis, and otitic hydrocephalus (Table 1).

MASKED (LATENT) MASTOIDITIS

It consists of slow destruction of mastoid air cells without the acute sign and symptoms of acute mastoiditis, such as pain, ear discharge, fever and mastoid swelling.

- *Etiology*: Inadequate dose, frequency and duration of antibiotic therapy are the most incriminating factors. Oral penicillin in AOM controls acute symptoms but smoldering infection continues in the mastoid.
- *Pathology*: Mastoidectomy in latent mastoiditis shows extensive destruction of the air cells with granulation tissue and dark gelatinous material filling the mastoid. Erosion of the tegmen tympani and sinus plate can result in an extradural and perisinus abscess respectively.
- *Clinical features*: The child does not do entirely well.
  - Mild mastoid pain and tenderness.
  - Persistent conductive hearing loss.
  - Tympanic membrane looks thick and loses its translucency.
- *Imaging*: Mastoid will reveal clouding of air cells with loss of cell outline.
- *Treatment*: Treatment consists of cortical mastoidectomy with full doses of antibiotics.

EXTRATEMPORAL COMPLICATIONS (ABSCESSES)

About 50% patients of mastoiditis develop subperiosteal abscess (Fig. 2).

- *Postauricular abscess*: This common abscess forms over the Macewen's triangle of the mastoid. The pus travels along the vascular channels of lamina cribrosa. Auricle is displaced forward, outward and downward. It is common in children.
- *Zygomatic abscess* (Figs 3A and B): Infection of zygomatic air cells, which are situated at the posterior root of zygomatic arch, results in zygomatic abscess. The pus can present either superficial or deep to the temporalis muscle. Swelling occurs in front of and above the auricle. Associated edema of the upper eyelid is common.
- *Meatal abscess (Luc's abscess)*: Mastoid abscess breaks the bony wall that lies between the antrum and external bony canal and may burst into the meatus. Swelling becomes visible in the deep part of bony meatus.
- *Behind the mastoid (Citelli's abscess)*: Pus can travel either along the mastoid emissary vein or occipitotemporal (also called occipitomastoid) suture. Citelli's abscess is situated

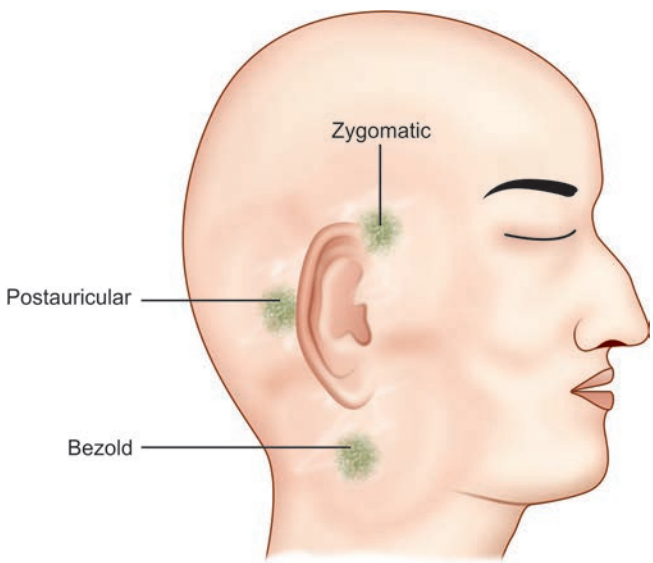

Fig. 2: Type of mastoid abscesses

Figs 3A and B: Zygomatic abscess. (A) Front view; (B) Lateral view

posterior to the mastoid. The post-auricular mastoid abscess forms over the mastoid.

- *Parapharyngeal and retropharyngeal abscesses*: Infection of the peritubal cells in cases of acute coalescent mastoiditis can result into these abscesses. (See chapter Deep Neck Abscesses).

Bezold's Abscess (Fig. 4)

- In acute coalescent mastoiditis, pus can break the thin medial side of the tip of the mastoid and present as upper neck swelling. The abscess may follow any of the following courses:
  - Lie deep to sternocleidomastoid muscle and pushing it laterally.
  - Lie in the digastric triangle: Follow the posterior belly of digastric and present between the tip of mastoid and angle of jaw.
  - Lie in the upper part of posterior triangle.
  - Lie in the parapharyngeal space.
  - Lie along the carotid vessels.
- *Clinical features*: Sudden onset of pain, fever, a tender swelling in the neck and torticollis in cases of purulent otorrhea.
- *Differential diagnoses*:
  - Acute upper deep cervical lymphadenitis
  - Abscess of the lower part of the parotid gland
  - Infected branchial cyst
  - Parapharyngeal abscess
  - Jugular vein thrombosis
- *CT scan*: It will establish the diagnosis.
- *Treatment*: It is surgical. Intravenous antibiotics are given as per the report of culture and sensitivity of the pus taken at the time of surgery.
  - Cortical mastoidectomy with exploration of mastoid tip for a fistulous opening into the soft tissues of the neck.
  - Incision and drainage of the neck abscess is done through a separate incision and putting a drain.

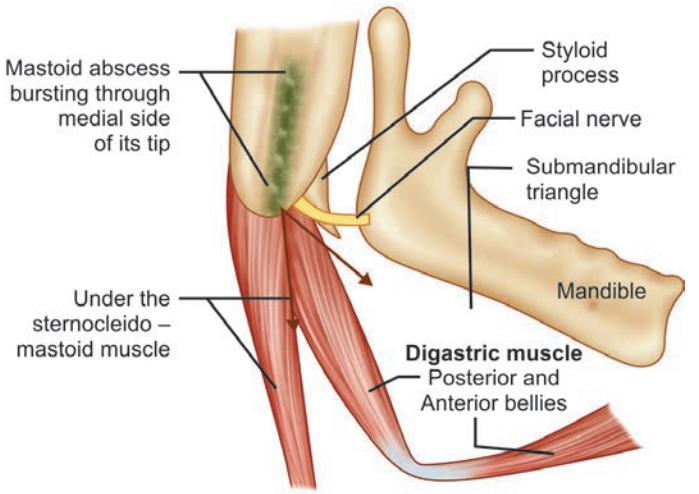

Fig. 4: Bezold's abscess. Through the mastoid tip abscess extends either deeper to sternocleidomastoid muscle or into the submandibular triangle

PETROSITIS OR PETROUS APICITIS

Infection of the petrous part of temporal bone is called petrositis, which is usually associated with acute coalescent mastoiditis, latent mastoiditis and cholesteatoma.

Pertinent Anatomy

The petrous bone is pneumatized in about 30% individuals. The cranial nerve CN VI (abducens) and CN V trigeminal ganglion are closely related to petrous apex. There are two groups of air cells' tracts that communicate mastoid and middle ear to the petrous apex. Infection may pass through these cell tracts and reach petrous apex.

- *Posterosuperior tract*: From the attic and antrum the tract passes around semicircular canals to petrous apex. This tract begins in the Trautmann's triangle or the attic.
- *Anteroinferior tract*: From the hypotympanum the tract passes around the ET and cochlea to the petrous apex. Tract is situated near the tympanic opening of ET anterior to the cochlea and passes above the carotid artery.

Clinical Features

- Gradenigo's syndrome or triad (1904)
  - Cranial nerve VI palsy
  - Deep seated ear or retro-orbital pain (due to the involvement of CN V ganglion)
  - Persistent ear discharge
- Fever, headache, vomiting and neck rigidity may be present.

Persistent ear discharge in cases of cortical or modified radical mastoidectomy may be due to petrositis.

- CT scan of temporal bone: Confirm the diagnosis.

Gradenigo's triad: It consists of otorrhea, abducens palsy and retrobulbar pain, and is the classic presentation of petrous apicitis.

Treatment

- *Antibiotic therapy*: It should precede and follow the surgery. Most patients respond well to antibacterial therapy alone, which is given in high doses and continued for 4–5 days after the disappearance of symptoms.
- *Radical or modified radical mastoidectomy*: The fistulous tract is curetted and enlarged to provide free drainage. The anteroinferior tract cells need radical mastoidectomy.

FACIAL NERVE PARALYSIS

It can occur in cases of both AOM as well as cholesteatoma.

Acute Otitis Media

- *Clinical features*: In cases of dehiscent fallopian bony canal, facial nerve lies just under the middle ear mucosa. Inflammation of middle ear easily spreads to epineurium and perineurium and results in facial paralysis, which usually manifests within 10 days.
  - Facial palsy beyond 2 weeks of AOM indicates erosion of bony facial canal.
- *Treatment*: Facial nerve function usually recovers with systemic antibiotics and wide-field myringotomy or tympanostomy tube. Some prefer concomitant corticosteroids.
  - Cortical mastoidectomy is required occasionally.

Chronic Suppurative Otitis Media

Facial paralysis can result from either cholesteatoma or penetrating granulation tissue. Cholesteatoma destroys bony facial canal and involves facial nerve.

- *Clinical features*: Facial nerve paralysis is insidious and progressive.
- *High-resolution computed tomography of temporal bone*: It is essential in the evaluation of facial nerve palsy. It delineates extent of disease and identifies cholesteatoma and neoplasms.
- *Treatment*: It is urgent modified or radical mastoidectomy. Facial canal should be examined from the geniculate ganglion (processus cochleariformis) to the stylomastoid foramen.
  - In cases of granulation tissue and cholesteatoma, bony facial canal is uncapped in the area of involvement. Granulation tissue surrounding the nerve is removed without damaging the nerve sheath. Granulation tissue invading the nerve sheath is left in place.
  - The nerve segment destroyed by the granulation tissue needs resection. The nerve grafting is usually preferred in the second stage when infection is controlled and fibrosis has matured.

LABYRINTHITIS

It is described in detail in the chapter of peripheral Vestibular Disorders.

EXTRADURAL (EPIDURAL) ABSCESS

This collection of pus between the bone and dura can occur in both AOM, as well as cholesteatoma. An extradural abscess frequently coexists with sinus thrombophlebitis and precedes brain abscess.

Pathology

- The overlying dural bone can be destroyed by hyperemic decalcification in AOM and bioenzymatic activity cholesteatoma. Pus collects in epidural space.
- In venous thrombophlebitis mode of spread overlying dural bone remains intact.
- The affected dura gets covered with granulations and appears unhealthy and discolored.
- An extradural abscess may lie in
  - Middle cranial fossa
  - Posterior cranial fossa
  - Perisinus abscess: Outside the dura of lateral venous sinus in posterior cranial fossa

Clinical Features

- Asymptomatic cases are discovered incidentally during cortical or modified radical mastoidectomy.
- Persistent headache, which disappears with free flow of pus from the ear (spontaneous abscess drainage) occurs on the side of otitis media.
- Severe ear pain.
- General malaise with low-grade fever.
- Pulsatile purulent otorrhea.

Diagnosis

- *Contrast enhanced CT or MRI*: Dural elevation is indicative of extradural abscess.

Treatment

- *Antibiotic therapy*: An antibiotic cover before and after the surgery is mandatory.
- *Mastoidectomy (cortical, radical or modified radical):* The overlying bone is removed until the healthy dura appears. In cases of strong suspicion overlying intact tegmen tympani or sinus plate is deliberately removed to evacuate any collection of pus.

Follow-Up

Patient should be closely observed for further intracranial complications, such as sinus thrombosis, meningitis or brain abscess.

SUBDURAL ABSCESS OR EMPYEMA

It refers to collection of pus in subdural space that lies between dura and arachnoid mater. The subdural space is divided into several large compartments, which are anatomically confined, by foramen magnum, tentorium cerebelli, base of the brain and falx cerebri.

The subdural empyema is the most rare complication now. It used to be fatal in pre-antibiotic era.

Pathology

Infection can spread either by erosion of bone and dura or by thrombophlebitic process (intervening bone remains intact).

Pus in the subdural space lies against the convex surface of cerebral hemisphere and results in pressure symptoms. Later on the pus gets loculated at various places in subdural space. The developing subdural empyema quickly evolves into a fatal mass.

Clinical Features

Sudden unusually severe headache, which is usually associated with fever and vomiting and rapid deterioration of patient, points towards subdural abscess.

The clinical features are due to meningeal irritation, thrombophlebitis of cortical veins of cerebrum and raised cerebrospinal fluid (CSF) pressure.

- *Meningeal irritation*: Headache, fever (102° F or higher), malaise, progressive drowsiness, neck rigidity and Kernig's sign.
- *Thrombophlebitis of cortical veins of cerebrum*: Aphasia, contralateral hemiplegia and hemianopia. The Jacksonian type of epileptic fits may progress to status epilepticus.
- *Raised CSF pressure*: Papilledema, ptosis and dilated pupil due to CN III involvement. Other CNs may also be involved.

Diagnosis

- *MRI*: MRI is superior to CT. It can distinguish between epidural and subdural abscesses. Multiple, discrete, lobulated subdural collections may be seen. MRI allows differentiation of sterile, bloody and infected collections. The advantages of MRI include absence of bone artifact, increased contrast between bone, CSF brain parenchyma and multiplanar imaging capabilities.
- *CT*: Loculated subdural abscess can be seen.
- Lumbar puncture is contraindicated as it can result in herniation of cerebellar tonsil.

Treatment

It is a neurosurgical emergency.

- Series of burr holes to drain subdural space abscess.
- High dose intravenous antibiotics.
- Radical mastoidectomy: It is done once the subdural abscess is treated.

MENINGITIS

It refers to the inflammation of leptomeninges (pia-arachnoid) and CSF of subarachnoid space, which surrounds brain, spinal cord and optic nerves. Mortality rate is very high.

> Meningitis is the most common intracranial complication of SOM.
>
> In infants and children, meningitis is often the complication of AOM, while in adults it occurs due to cholesteatoma. The one-third cases of meningitis are otogenic in origin.

Causes

- Suppurative otitis media: AOM and cholesteatoma
- Temporal bone fracture
- CSF leak
- Middle ear and mastoid surgery

Pathways of Infection

The middle ear and mastoid infection can reach meninges via following pathways:

- Preformed pathways
  - Patent petrosquamosal suture
  - Perineural spaces to the internal auditory canal (uncommon via endolymphatic ducts) via labyrinth through round and oval windows
- Retrograde venous thrombophlebitis
- Direct erosion of bone (mastoiditis and petrositis) and dura

Microbiology

- Major pathogens are *Hemophilus influenzae* and *Streptococcus pneumoniae*.
- Anaerobic organisms indicate interventricular rupture of brain abscess.
- Polymicrobial infection of CSF occurs in less than 1% of cases.

Clinical Features

The severity of symptoms and signs depends on severity of infection. The raised intracranial tension and meningeal and cerebral irritations vary with the extent of disease.

- Earliest symptoms include–
  - Fever (102–104° F) often with chills and rigors
  - Headache
  - Vomiting (sometimes projectile)
  - Photophobia
  - Irritability and restlessness
- Seizures: Infants may have seizures.
- Neck rigidity
- Kernig's sign: In this sign extension of leg with thigh flexion on abdomen is painful.
- Brudzinski's sign: In this sign flexion of neck results in flexion of hip and knee.
- Motor deficit: Cranial nerve palsies and hemiplegia.
- Deep tendon reflexes: Exaggerated initially but later become sluggish or absent.
- Papilledema: Late feature.
- Drowsiness: It may progress to delirium or coma.

Diagnosis

- *High-resolution computed tomography temporal bone*: It is the imaging modality of choice. It provides not only the bony details but also rules out congenital ear malformations, which permit CSF leak through an inner ear fistula.
- *MRI*: It shows middle ear fluid and inflammatory changes in brain and meninges.
- *Fundoscopy*: Indistinct disk margin and choking of vessels.
- *Lumbar puncture*: Imaging usually precedes lumbar puncture (LP) to identify mass effect that can lead to herniation on LP. CSF shows following features:
  - Cloudy (turbid) or yellow (xanthochromic)
  - Raised cell count with predominance of polymorphs
  - Raised protein level
  - Low level of sugar and chlorides
  - Gram-staining to identify organisms
  - Culture and sensitivity: To find the causative organisms and their antibiotic sensitivity
- *Ear pus for culture and sensitivity*: To find the causative organisms and their antibiotic sensitivity.

Treatment

Medical treatment always takes precedence over surgery. Surgery is done when general condition of patient permits. If there is no satisfactory response to medical treatment, early surgery is required.

- *Antibiotics*: Crystalline penicillin, ampicillin, chloromycetin or third generation cephalosporin (drugs of choice) intravenously for 7–10 days. Antibiotics are changed according to the culture and sensitivity report.
- *Surgical*:
  - Acute otitis media: Myringotomy and/or cortical mastoidectomy.
  - Cholesteatoma: Radical or modified radical mastoidectomy.

OTOGENIC BRAIN ABSCESS

Brain abscess is a focal suppurative process of brain parenchyma. It is surrounded by an area of inflammation (encephalitis)

- Fifty percent cases of brain abscesses in adults and 25% in children are otogenic.
  - In adults, cholesteatoma and in children AOM are the most common causes.
- Cerebral abscess occurs more frequently than cerebellar abscess.
  - Most cerebellar abscesses are due to SOM.

- Bimodal Age Distribution: Peak incidences are in pediatric age and 4th decade of life.
- Sex: Male-to-female ratio is 3:1.
- Mortality is greater in cerebellar abscesses (about 25%) in this antibiotic era.

Route of Infection

Otogenic brain abscesses are usually the result of venous thrombophlebitis rather than direct dural extension. Retrograde thrombophlebitis of dural vessels, which terminate in white matter, is usually caused by the osteitis and granulation tissue.

- *Cerebral abscess*: Cerebral abscess is often associated with extradural abscess that occurs due to erosion of the tegmen. In cases of retrograde thrombophlebitis the tegmen remains intact.
- *Cerebellar abscess*: Cerebellar abscess usually develops through the Trautmann's triangle. It is often associated with extradural abscess, perisinus abscess, sigmoid sinus thrombophlebitis or labyrinthitis. The retrograde thrombophlebitis is uncommon.

Bacteriology

Multiple organisms with high incidence of anaerobes are usually seen. Polymicrobial cultures are influenced by host immune status. *H. influenzae* is very rare.

- *Aerobic*:
  - Gram-positive: *Pyogenic staphylococci, Streptococcus pneumoniae* and *Streptococcus haemolyticus*
  - Gram-negative: *Proteus mirabilis, Escherichia coli, Klebsiella and Pseudomonas aeruginosa*
- *Anaerobic*: *Peptostreptococcus* and *Bacteroides fragilis*.

Pathology and Four Stages

There are four stages of brain abscess development:

1. *Stage of invasion (Initial encephalitis) of 1–3 days*: The mild symptoms include headache, low grade fever, malaise and drowsiness. They often pass unnoticed.
2. *Stage of localization (Latent abscess) of 4–10 days*: In this asymptomatic stage, which may last for several weeks, pus is getting localized by the formation of a capsule.
3. *Stage of enlargement or early capsule formation (manifest abscess) of 10–13 days*: The enlarging abscess surrounded by a zone of edema aggravates the severity of clinical manifestations, which are due to raised intracranial tension and the focal involvement of brain.
   a. Raised CSF pressure.
   b. Cerebrum and cerebellum: Focal symptoms and signs.
4. *Stage of termination or late capsule formation (rupture of abscess) of 14 days*: The enlarging abscess ruptures into either the ventricle or subarachnoid space (Fig. 5) and results in fatal meningitis.

Clinical Features

The patient often appears toxic and drowsy and complains of deep cranium pain. Dizziness, ataxia, vomiting and nystagmus indicate cerebellar abscess. Temporal lobe abscess results in seizures.

As the brain abscess is often associated with extradural abscess, perisinus abscess, meningitis, sinus thrombosis and labyrinthitis, the clinical picture may be overlapping.

Clinical features are mainly due to raised intracranial tension and the area of brain affected.

A. *Raised intracranial tension*
   - Headache is severe and generalized, worse in the morning.
   - Nausea and vomiting (usually projectile): It is more common in cerebellar abscess.

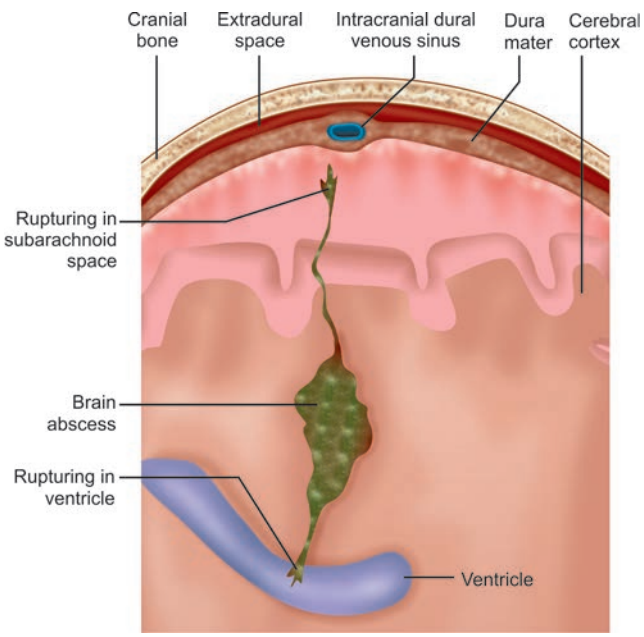

Fig. 5: Brain abscess rupture in ventricle and subarachnoid space. Rupture usually occurs in the ventricle because encapsulation of abscess is more on outer side

- Level of consciousness: Lethargy progresses to drowsiness, confusion, stupor and finally coma.
- Papilledema: Usually appears after 2–3 weeks. Appears early in cerebellar abscess.
- Slow pulse.
- Subnormal temperature.

B. *Localizing features of temporal lobe abscess*
- Nominal aphasia: It occurs if the lesion is of dominant cerebral hemisphere, which is left side in right handed persons. Patient cannot name the common objects, such as key, pen and phone. However patient explains the utility of that object.
- Contralateral homonymous hemianopia: It indicates pressure on the optic radiations. The visual field defect is usually in the upper quadrants. It can be recorded by perimetry.
  - Confrontation test: The examiner stands in front of the patient and compares his visual field with that of the patient.
- Contralateral motor paralysis: They are:
  - Upward spread: Facial palsy is followed by palsy of arm and leg.
  - Inward spread towards internal capsule: Paralysis of leg is followed by the paralysis of arm and face.
- Epileptic fits: Small and involuntary smacking movements of lips and tongue. Generalized fits can also occur.
- *Pupillary changes and oculomotor palsy*: It suggests transtentorial herniation.

C. *Localizing features of cerebellar abscess*
- Suboccipital headache associated with neck rigidity.
- Spontaneous nystagmus
- Ipsilateral ataxia: Patient staggers to the side of lesion.
- Finger nose test: Past pointing and intention tremor.
- Dysdiadokokinesia: Rapid pronation and supination movements become slow and irregular on the affected side.

Investigations

- *CT scan*: It reveals not only the site and size of an abscess but also other associated complications, such as extradural abscess, sigmoid sinus thrombosis.
  "Ring" sign: Brain abscess appears hypodense area surrounded by an area of edema. Temporal bone is better evaluated by CT than MRI.
- *MRI*: MRI is superior to CT. It detects not only the subtle changes in brain parenchyma but also spread of abscess into the subarachnoid space or into the ventricle.
- *Lumbar puncture*: There is risk of coning and usually not done. CSF shows following features:
  - Rise in pressure
  - Increase in protein with normal glucose
  - White cell count is raised (polymorphs or lymphocytes depending on the acuteness of infection) but is much less than meningitis.

Treatment

It is controversial.

A. *Medical*: The different medical treatments are:
- Antibiotics: Parenteral chloromycetin, penicillin or its derivatives. Anaerobes respond to metronidazole. Aminoglycoside, such as gentamicin, covers pseudomonas and proteus. As per the report of culture and sensitivity, antibiotics are changed. Newer and more effective antibiotics can obviate the neurosurgical intervention.
- Raised intracranial tension: Dexamethasone, 4 mg intravenous 6 hourly or mannitol 20% in doses of 0.5 g/kg body weight.
- Suction clearance of ear and use of topical ear drops.

B. *Neurosurgical*: Life saving neurosurgical intervention takes precedence over the otologic management. Aspiration (pus sent for culture and sensitivity) is followed by repeat CT or MRI scans to see whether it is diminishing in size. Excision is required if abscess is expanding or not decreasing in size. Penicillin is instilled into the abscess after its aspiration. Neurosurgeon may consider any of the following procedures:
- Burr hole: Repeated aspiration of pus
- Excision of abscess
- Open incision of the abscess and evacuation of pus

C. *Otologic*: Only neurologically stabilized patients are taken for tympanomastoid surgery. Ear surgery is planned only after the abscess has been managed by antibiotics and neurosurgery. Cholesteatoma needs radical mastoidectomy, which removes the irreversible disease and exteriorizes the infected area.

LATERAL SINUS THROMBOPHLEBITIS

(Synonyms: Sigmoid sinus thrombosis or otogenic suppurative thrombophlebitis)

It refers to inflammation of inner wall of lateral venous sinus with formation of a thrombus, which gets infected. With the advent of new range of antibiotics, the incidence of this complication has declined yet mortality remains high.

Etiology

It is a complication of acute coalescent mastoiditis, masked mastoiditis and cholesteatoma.

Pathology

The pathological process is divided into the following four stages:

1. *Perisinus abscess*: Extradural abscess forms outer to the dural wall of the sinus.
2. *Endophlebitis and mural thrombus*: Inflammation of the inner wall of the venous sinus results in thrombus formation (deposition of fibrin, platelets and blood cells) within the lumen of sinus.
3. *Obliteration of sinus and intrasinus abscess*: Expanding mural thrombus occludes the sinus lumen. Organisms invade the thrombus and form intrasinus abscess that releases infected emboli into the blood stream (septicemia).
4. *Extension of thrombus*: Thrombotic process continues both proximally and distally. It may spread to confluence of sinuses, superior sagittal sinus, cavernous sinus, mastoid emissary vein, jugular bulb or internal jugular vein.

Bacteriology

- *Acute otitis media*: *Hemolytic streptococcus*, *Pneumococci.*
- *Cholesteatoma*: *Bacillus proteus*, *Pseudomonas pyocynea*, *Escherichia coli* and *Staphylococci*.

Clinical Features

High-grade fever, toxic look, deep ear pain, restlessness, neck stiffness and papilledema are typical features of suppurative thrombophlebitis.

- *Fever*: Hectic type of fever with chills and rigors coincides with the release of septic emboli into blood stream. Fever has one or more peaks a day. Profuse sweating follows during the fall of temperature and patient becomes alert with a sense of well-being.
  - This fever pattern resembles malaria but lacks regularity. Antibiotics may mask this classic pattern.
- *Headache*: During perisinus abscess it is mild. Headache becomes severe when venous obstruction increases the intracranial pressure.

Griesinger's sign: Tenderness and edema over the posterior part of mastoid, which is due to thrombosis of mastoid emissary vein, are pathognomonic for thrombophlebitis of sigmoid sinus.

- *Tenderness along jugular vein*: It indicates thrombophlebitis along the jugular vein.
- *Enlarged and tender jugular lymph nodes*: It may lead to torticollis.
- *Anemia and pallor*: Due to beta-hemolytic streptococci. They are progressive.
- *Papilledema*: It is seen when clot extends to superior sagittal sinus, which is in the continuation of right sigmoid and transverse sinus. Fundus shows blurring of disc margins, retinal hemorrhages or dilated veins.
  - *Crowe-Beck test*: Pressure on opposite side internal jugular vein produces engorgement of retinal veins and supraorbital veins and subside on release of pressure.
- *Cavernous sinus thrombosis*: Proptosis, ptosis, chemosis and ophthalmoplegia.

Investigations

- *Blood culture:* It helps in finding the causative organisms.
  - Blood sample is taken when patients have chills and rigor because at that time organisms enter the blood stream.
  - Repeated cultures are required to identify the organisms.
- *Peripheral blood smear*: Absence of malarial parasites rules out malaria.
- *Culture and sensitivity*: Of ear discharge.
- *Lumbar puncture*: CSF examination normal except for rise in pressure. It helps to exclude meningitis.
  - Queckenstedt's test or Tobey-Ayer test: In this test CSF pressure is recorded by manometer when one or both jugular veins are compressed manually. Compression of affected (thrombosis) side of vein produces no effect. Compression of opposite side internal jugular vein produces rapid rise in CSF pressure.
- *Contrast enhanced CT scan:* Though not seen always "delta sign" in axial cuts is typical of sinus thrombosis. It is an empty triangular area having rim enhancement and central low density area that is seen at the level of sigmoid sinus.
- *MRI*: MRI is more sensitive than CT. MRI shows blood flow, sinus obstruction, subsequent reversal of flow and higher resolution in detailing nerve tissue. On gadolinium-enhanced MRI, thrombus appears as a soft tissue signal associated with a vascular and bright appearance of dural walls (delta sign).

Complications

- Septicemia.
- Pyemic abscesses in lung, bone, joints or subcutaneous tissue.
- Meningitis and subdural abscess.
- Cerebellar abscess.
- Thrombosis of jugular bulb and jugular vein with involvement of IX, X and XI CNs.
- Cavernous sinus thrombosis: Chemosis, proptosis, fixation of eyeball and papilledema.
- Otitic hydrocephalus: Thrombus extends to superior sagittal sinus through the transverse and confluence of sinuses.

Treatment

- *Antibiotic therapy*: Injection crystalline penicillin 1 million units IM 6 hourly cover pyogenic cocci. Antibiotic can be changed as per the report of culture and sensitivity. Antibiotics are continued postoperatively for a week.
- *Mastoidectomy*: A complete cortical mastoidectomy in cases of AOM and modified radical mastoidectomy in cases of cholesteatoma is performed. Sinus bony plate must be removed to expose the dura and drain the perisinus abscess.

  Destruction of sinus dura, unhealthy and discolored dura with granulations on its surface indicate infected clot or intrasinus abscess, which must be drained. After packing the sinus above and below with a pack that lies between the bone and dura of sinus to control bleeding, dura of the sinus is incised and the infected clot and abscess is drained.

  Healthy red clot beyond the abscess at either end of sinus is not disturbed. Pack is removed after 5–6 days. The wound is closed secondarily.
- *Ligation of internal jugular vein*: Though rarely required, it is needed when antibiotic and surgical treatment fails to control embolic phenomenon and rigors and tenderness and swelling of internal jugular vein is spreading into the heart.
- *Supportive treatment*: Repeated blood transfusions combat anemia and improve patient's general condition.

OTITIC HYDROCEPHALUS

Otitic hydrocephalus consists of raised intracranial pressure and normal CSF. It is seen in both children as well as adolescents. Some prefer the term "benign raised intracranial tension" because there is no associated ventricular dilation.

In this rare complication spontaneous recovery occurs in some cases.

Mechanism

Following two factors result in raised intracranial tension:

1. The thrombosis of the sigmoid sinus, which continues as internal jugular vein, causes obstruction to venous return.
2. When thrombosis extends to superior sagittal sinus, which generally continues as right sigmoid sinus, it will impede the functioning (absorption of CSF) of arachnoid villi.

Clinical Features

The following clinical sign and symptoms are due to raised intracranial tension.

- *Severe headache*: It is the presenting complaint and may be associated with nausea and vomiting.
- *Diplopia*: It is due to paralysis of lateral rectus, which is supplied by CN VI.
- *Blurring of vision*: It is because of papilledema and optic atrophy.
  - Papilledema: The 5–6 diopters papilledema may be accompanied with patches of exudates and hemorrhages.
- Nystagmus.

Diagnosis

- *MRI*: MRI allows better evaluation of venous sinuses.
- *Lumbar puncture*: It should be done with caution lest herniation of cerebellar tonsil occur. The pressure of CSF, which is sterile and normal in cell, protein and sugar content, exceeds 300 mm of water (normal 70–120 mm H_2O).

Treatment

A. *Medical*
   - Medical treatment includes corticosteroids, mannitol, diuretics and acetazolamide.
   - Middle ear infection requires antibiotic therapy.

B. *Surgical*
   - The raised CSF pressure must be reduced to prevent optic atrophy and blindness. This can be done with the following procedures:
     - Repeated lumbar puncture or placement of a lumbar drain.
     - Lumboperitoneal shunt: It drains CSF into the peritoneal cavity.
   - Mastoidectomy is done for managing the sinus thrombosis. Decompression of sigmoid sinus is recommended.
   - Optic sheath decompression: It prevents optic atrophy.

Clinical Highlights

1. **Hyrtl's fissure (Tympanomeningeal hiatus)**: This embryonic remnant runs parallel to cochlear aqueduct. It connects subarachnoid space to middle ear just anterior and inferior to the round window. It can be a source of congenital CSF otorrhea or meningitis in cases of middle ear infections. Normally it is obliterated.
2. **Imaging**: MRI is the investigation of choice in extradural, Bezold's and cerebral abscesses. CT is the investigation of choice in cases of coalescent mastoiditis.
3. **Extracranial complications**: They include facial nerve palsy, hearing loss and labyrinthitis.
4. **Coalescent mastoiditis**: Two weeks later the patient of an acute SOM develops a swelling over the mastoid, pain in the ear, fever and pulsatile ear discharge. The treatment includes intravenous antibiotics and cortical mastoidectomy.
5. **Gradenigo's syndrome (triad)**: Mastoid air cells responsible for Gradenigo's syndrome are petrous apex cells. Patient presents with the triad of ear discharge, diplopia (CN VI paralysis) and retro-orbital pain (involvement of trigeminal ganglion). It is seen in cases of petrositis (an abscess in the petrous apex), which is a complication of CSOM (coalescent mastoiditis and cholesteatoma). Egleton, Almoor, Ramadier and Frenker operations are done for drainage of petrositis abscess.
6. **Postauricular mastoid abscess**: The treatment includes incision and drainage, antibiotics and mastoidectomy.
7. **Bezold's abscess**: Features include torticollis, a tender swelling behind the angle of mandible and fever, chronic ear discharge with granulations in the ear canal.
8. **Sigmoid sinus thrombosis**: It is confirmed by mastoid exploration.
   a. **Griesinger's sign**: This is the edema over the mastoid seen in patients of lateral sinus thrombosis. The thrombosis of mastoid emissary vein impedes venous drainage and causes edema over the mastoid.
   b. **Picket-fence graph of temperature**: It is the diurnal spikes of fever of 104^0 or 105^0 F and is the feature of sigmoid sinus thrombosis. The antibiotics use will alter the pattern.

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21 Evaluation of Dizzy Patient

Stand up and fight! Not one step back that is the idea. Fight it out, whatever comes. Let the stars move from the spheres! Let the whole world stand against us! Death means only a change in garment. What of it? Thus fight! You gain nothing by becoming cowards. Taking a step backward, you do not avoid any misfortune.

—Swami Vivekananda

Points of Focus

- **INTRODUCTION**
  - ✓ Indications for Immediate Referral to Otoneurologist
  - ✓ Diseases Needing Urgent Treatment
- **EVALUATION–GENERAL OUTLINE**
- **DESCRIPTION OF DIZZINESS**
- **ONSET, DURATION AND PROGRESSION**
- **PROVOKING FACTORS**
- **ASSOCIATED SYMPTOMS**
- **PERSONAL, FAMILY AND PAST HISTORY**
- **SPONTANEOUS NYSTAGMUS**
  - ✓ Method of Eliciting Nystagmus
  - ✓ Interpretations
  - ✓ Degree of Nystagmus (Alexander Law for Peripheral Vestibular Nystagmus)
  - ✓ Ewald's Law
- **DYNAMIC OCULAR EXAMINATION**
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  - ✓ Saccades
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  - ✓ Vestibulo-ocular Reflex
  - ✓ Vestibulo-ocular Reflex Cancellation
- **FISTULA TEST**
  - ✓ Principle
  - ✓ Method
  - ✓ Interpretation
- **VALSALVA MANEUVER**
- **DIX-HALLPIKE MANEUVER**
  - ✓ Method for Posterior Canal BPPV
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  - ✓ Modified Dix-Hallpike Maneuver for Lateral Canal BPPV
- **OPTOKINETIC TEST**
- **ROTATION TEST**
- **COLORIC TEST**
  - ✓ Gitzgerald-Hallpike Bithermal Caloric Test
  - ✓ Interpretations
- **TANDEM WALKING**
- **ROMBERG'S TEST**
- **CEREBELLAR TESTS**
- **HYPERVENTILATION**
- **ORTHOSTATIC HYPOTENSION**
- **SPECIAL VESTIBULAR TESTS**
- **DIFFERENCES BETWEEN CENTRAL AND PERIPHERAL VERTIGO**
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

One of the most common and frustrating complaints patients bring to their doctors is dizziness. Dizzy patients often have difficulty describing their symptoms, so determining the cause can be challenging. Often the patients use the term dizziness to describe different sensations, which are categorized in four groups:

1. Vertigo: False sense of motion, such as rotation (spinning or whirling)
2. Disequilibrium: Unsteadiness, or loss of balance
3. Lightheadedness: Floating feelings, head fullness, or out of body sensations
4. Presyncope: Near faint

Vertigo accounts for 54% of cases of dizziness in primary care. The causes of vertigo are categorized into two groups: peripheral (labyrinth and vestibular nerve) and central (vestibular nuclei and their central nervous system connections brainstem and cerebellum).Clinical, otological and vestibular examinations are important to differentiate between central and peripheral vestibular disorders. Identifying the central vestibular diseases is critical because some of them are life-threatening or produce severe neurologic sequelae. The most important step in the diagnosis is an unhurried and detailed history.

Dizziness can occur in normal persons (psychophysiologic dizziness) who have extreme sensitivity to unnatural motions (motion sickness such as during sea and car travel), anxiety and panic, depression and phobic behavior (psychiatric disorders).

Indications for Immediate Referral to Otoneurologist

The "red flag" indicators, suggested in UK's National Health Service for immediate referral to balance specialist are the following:
(*Source*: www.wise.nhs.uk/sites/clinicalimprovcollab/ENT/pages)

- Sudden, new, unilateral development or progression of hearing loss
- Incapacitating dizziness for more than 6 weeks
- Severe tinnitus
- Any neurological symptoms or signs
- Discharging ears
- Ear pain
- Progressive unsteadiness or falls
- Extreme cases of social, occupational or emotional stress

Diseases Needing Urgent Treatment

Cerebrovascular strokes (hemorrhages and infarcts) need urgent treatment from the neurologists, which can save certain lives. The distinguishing features of these conditions are sudden onset with rapid progression and deterioration in patient condition. The patient cannot walk and stand even with support. Other conditions, which are not urgent but need immediate treatment include, trauma to the labyrinth or temporal bone, bacterial labyrinthitis, otosyphilis and perilymphatic fistulas.

EVALUATION–GENERAL OUTLINE

The history (Box 1) and examination (Box 2) provide important clues to the diagnosis (Table 1). A patient's history, which is of paramount importance in cases of balance disorders, often provides enough information to diagnose a cause for various types of dizziness (Table 1). Vertigo in the absence of otological, neurological or medical disorders is a diagnostic dilemma for the clinicians.

In many cases, investigations are not required at all except audiogram. Audiogram provides a baseline for the hearing, which may later become relevant in dizzy patients. Occasionally, it exposes an unexpected hearing loss the patient had not noticed, but which may be significant. Audiometry is usually performed in all the cases of vertigo and helps in establishing the diagnoses of peripheral vestibular disorders, which are accompanied with hearing loss, such as Ménière's disease.

Box 1: Elements in the history of dizzy patients

- Description of the dizziness
  - Lightheadedness
  - Disequilibrium and unsteadiness
  - Vertigo and oscillopsia
  - Presyncope
- Episodic or continuous
- Duration of the individual attack: Seconds, minutes, hours, days or months
- Effect of head movements: Worse, better or no effect
- Specific positions that induce vertigo
- Preceding history of
  - Trauma: Physical, barotraumas, surgical
  - Medicines: For infections, hypertension, hyperglycemia and cardiac arrhythmias and CNS disorders
- Medical conditions: Hypothyroidism, diabetes mellitus, anemia, autoimmune diseases and hypoperfusion of brain from postural hypotension or cardiac arrhythmia
- Psychogenic disorders: Anxiety and panic disorders and agoraphobia
- Triggering events: Stimuli (sound, pressure, or movements), rolling onto the side in bed
- Associated/concomitant symptoms
  - Ear: Discharge, pain, hearing loss, tinnitus, ear fullness
  - Eye: Diplopia, vision loss
  - Headache
  - CNS: Paralysis and paresthesias, dysarthria, dysphagia
  - Sweating, dyspnea and palpitation

There is no indication for routine caloric testing, imaging or blood analysis. The positive findings of history and examination are the main indications of any necessary investigations. Blood tests for vascular risk factors, and inflammatory, infectious and hypercoagulation disorders should be done in presence of signs or symptoms, which suggest these conditions. In cases of Meniere's disease, one should exclude syphilis, which can present like Meniere's disease.

Neuroimaging studies are considered in patients who have neurologic manifestations or cerebrovascular risk factors. Magnetic resonance imaging (MRI) is superior to computed tomography for the diagnosis of vertigo because of its superior ability to visualize the posterior fossa. A unilateral or asymmetrical hearing loss, in absence of known cause (such as trauma, exposure to gunfire) is an absolute indication for MRI to exclude acoustic neuroma. Magnetic resonance angiography of the posterior fossa vasculature may assist in diagnosing vertebrobasilar insufficiency, thrombosis of the labyrinthine artery, anterior or posterior inferior cerebellar artery insufficiency and subclavian steal syndrome.

DESCRIPTION OF DIZZINESS

One must determine whether the patient has true whirling sensation or another type of dizziness, such as lightheadedness, imbalance or presyncope.

Box 2: Elements of examination in dizzy patient

- Ear, nose, throat, head and neck examination
- Assessment of hearing
- CNS examination especially cranial nerves, gait, strength, reflexes and sensation in the legs and cerebellar functions for postural instability and disequilibrium
- Static ocular examination-Spontaneous nystagmus, effect of vergence, gaze-holding abilities
- Dynamic ocular examination: Saccades, smooth pursuit, optokinetic, cancellation of vestibulo-ocular reflex (VOR, Fig. 1).
- Low frequency VOR, head thrust test and dynamic visual acuity
- Induced nystagmus: Fistula test, head-shaking nystagmus, positional and positioning tests, valsalva-induced nystagmus, Tullio's phenomenon, optokinetic, rotatory chair and caloric test
- Hyperventilation and orthostatic hypotension
- Vestibulospinal tests: Romberg testing, tandem walking, past pointing, Unterberger stepping test and eyes-closed turning test
- Quantitative tests: Eye movement recordings (electronystagmography, Infrared video image analysis; static and dynamic tests) and posturography, motor control and sensory organization tests

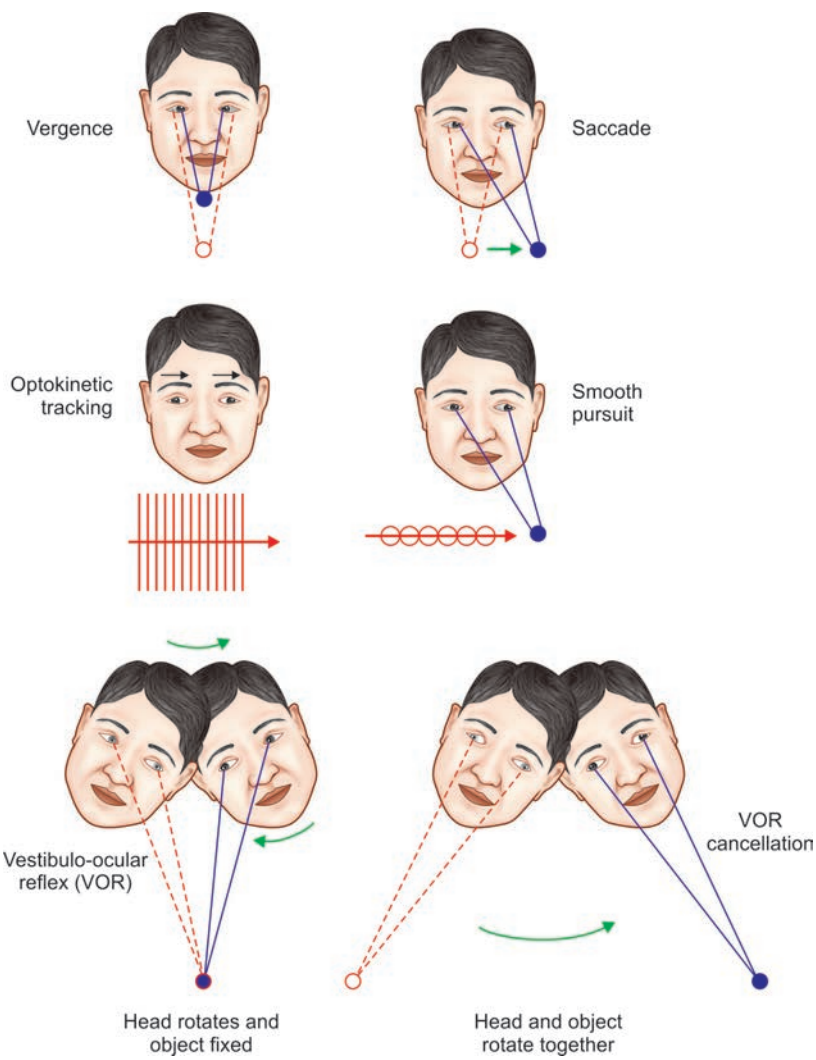


Fig. 1: Classes of eye movements. Saccades rotate the eye to bring an image onto fovea. Pursuit maintains the moving image on the fovea. Nonconjugate vergence movements make eyes to move in opposite direction to place the moving object on both fovea.

Certain patients with vestibular dysfunction, such as ototoxicity and acoustic neuroma, feel only a sensation of imbalance or disequilibrium in the head. Imbalance that is felt in the legs usually results from a cerebellar or proprioceptive deficit or from weakness of the legs.

Reduction in blood flow to entire brain leads to faintness or presyncope (before fainting or losing consciousness), which clinically presents as lightheaded sensation that may be accompanied with giddiness, generalized weakness and pallor. The causes of presyncope include postural hypotension,

TABLE 1 Common differential diagnoses of vertigo and their key features

Group/cause	*Key features*
Peripheral causes	
Suppurative labyrinthitis	Findings of middle ear infections (pain, discharge, deafness) or meningitis (headache, fever, neck rigidity)
Vestibular neuritis	Acute prolonged isolated vertigo without any otological and neurological findings. Nystagmus is of peripheral type
Benign positional paroxysmal vertigo	Transient (for seconds) episodes of vertigo triggered by certain head positions, such as rolling in bed
Cholesteatoma	Chronic foul smelling ear discharge. Ear microexamination will reveal the cholesteatoma
Herpes zoster oticus (Ramsay Hunt syndrome)	Vesicular eruption in the posterior part of external ear and facial palsy
Ménière's disease	Recurrent episodes of vertigo, hearing loss, tinnitus or aural fullness
Perilymphatic fistula	Dizziness of variable duration after trauma, surgery, excessive straining. Spontaneous in profound congenital deafness. Episodes of dizziness triggered by pressure and noise
Central causes	
Cerebellopontine angle tumor	Unilateral gradually progressive sensorineural hearing loss with imbalance (usually not spinning vertigo)
Cerebrovascular disease such as transient ischemic attack or stroke	Sudden spontaneous attack of vestibular symptoms associated with motor or sensory deficits and central type of vertigo. Vascular risk factors present
Migraine	Episodic throbbing headaches, usually unilateral and accompanied by nausea, vomiting, photophobia or phonophobia
Multiple sclerosis	Sensory or motor deficits. Sudden hearing loss, facial palsy, blindness. CNS dysfunctions are disseminated in time and space
Other causes	
Cervical vertigo	Vertigo triggered by somatosensory input from head and neck movements
Drug-induced vertigo	H/o adverse reaction to medications, alcohol abuse
Psychological	Associated mood, anxiety, somatization, personality disorders

cardiac dysarrhythmia, such as in Adams-Stokes disease, low cardiac output resulting from valvular disease or ventricular dysfunction.

The term lightheadedness includes vague terms, such as a fear of falling, a swimming sensation. The common etiology of lightheadedness is usually emotional disturbance. Psychiatric and emotional problems are a common cause of dizziness. Chronic continuous dizziness may be a sign of depressive illness. Emotional factors may influence symptoms of organic origin, for example stress may provoke bouts of Meniere's disease as it does asthma and migraine.

ONSET, DURATION AND PROGRESSION

Preceding history of recent upper respiratory viral illness is present in many cases of acute vestibular neuritis. Herpes zoster oticus is more common in cases of immunosuppression (immunosuppressive medications, advanced age, stress).

In acute vestibular neuritis, suppurative labyrinthitis and Meniere's disease, initial symptoms typically are severe but lessen over the next few days. In strokes symptoms are severe and do not lessen over the next days, even additional symptoms appear and patient may deteriorates with the passage of time. Patients presenting with constant vertigo lasting for weeks may have a psychological cause.

The duration of episodes of vertigo helps in distinguishing certain causes of dizziness (Table 2). Longer the vertigo lasts, greater are the chances that there is a central cause. The presence of vertigo upon awakening in the morning was found suggestive of peripheral vestibular disorders. Peripheral vertigo usually has a sudden onset while vertigo of central nervous system origin is not sudden with the exception of acute cerebrovascular events.

PROVOKING FACTORS

Provoking factors and the events preceding the onset of vertigo usually prove useful in narrowing the differential diagnoses (Table 3). Vertigo, which occurs only with positional changes,

TABLE 2 Typical duration of symptoms for different causes of vertigo

Duration	*Suggested diagnoses*
Seconds	Benign paroxysmal positional vertigo
Minutes to hours	Transient ischemic attack, Ménière's disease and migraine
Days	Vestibular neuritis, suppurative labyrinthitis, strokes, multiple sclerosis
Weeks	Psychogenic

TABLE 3 Provoking or aggravating factors for different causes of vertigo

Provoking factors	*Suggested diagnoses*
Changes in head position	Benign positional paroxysmal vertigo (BPPV)
Neck movements	BPPV, cervical vertigo, Arnold-Chiari syndrome
Stress	Psychiatric or psychological causes; migraine
Changes in ear pressure and loud noises	Perilymphatic and labyrinthine fistula, superior semicircular canal dehiscence, syphilis
Excessive straining	Superior semicircular canal dehiscence, Chiari malformations
Spontaneous episodes (no triggers)	Acute vestibular neuritis, Ménière's disease, cerebrovascular disease (stroke or transient ischemic attack); multiple sclerosis

such as turning over in bed, bending over at the waist and then straightening up, or hyperextending the neck, indicates benign positional paroxysmal vertigo (BPPV).

Factors, which provoke migraine headaches (such as food triggers, stress and improper sleep) can precipitate episodes of vertigo.

Perilymphatic fistula is a breach between the inner ear and middle ear. It may be caused by a direct blow, scuba diving (barotrauma), heavy weight bearing, or excessive straining with bowel movements. Sneezing or movements that place the affected ear downward also can provoke vertigo in perilymphatic fistula.

The Tullio's phenomenon (nystagmus and vertigo caused by loud noises or sounds) suggests a peripheral cause for vertigo, which includes perilymphatic fistula, labyrinthine fistula, dehiscence of superior semicircular canal and syphilis.

One should ask about psychological and social stressors or psychiatric history. A history of anxiety or panic attacks associated with vertigo in the absence of physical causes may point to hyperventilation.

Tullio's phenomenon or Hennebert's sign: They are present in patients with labyrinthine fistula and perilymphatic fistula.

ASSOCIATED SYMPTOMS

Hearing loss, pain, nausea, vomiting, headache or neurologic symptoms help in differentiating various causes of vertigo. Most causes of vertigo with hearing loss are peripheral. Pain usually occurs in acute middle ear and mastoid infections and intracranial complications of cholesteatoma. Nausea and vomiting are often present in acute vestibular neuritis and episodes of Ménière's disease and sometimes in BPPV. In cases of central vertigo, nausea and vomiting tend to be less severe. Weakness, dysarthria, vision disorders, paresthesia, altered level of consciousness, ataxia or other sensory and motor deficits are present in central causes of vertigo, such as cerebrovascular strokes, brain tumors and multiple sclerosis.

Migraine patients experience typical headache (often throbbing, unilateral, sometimes proceeded by an aura), nausea, vomiting, photophobia and phonophobia. Twenty-one to thirty-five percent of patients with migraine have vertigo.

In most cases of visual or joint impairment, such as refractive errors, cataract or arthritis, the patients are usually aware of the cause of dizziness and do not consult ENT surgeons. One should keep in mind certain cardiovascular conditions, which can present with dizziness especially syncope. Few early cases of Parkinsonism may present with isolated dizziness, otherwise very rarely patient of general illnesses come to vertigo clinic.

PERSONAL, FAMILY AND PAST HISTORY

Medical history should include medications, trauma, or exposure to toxins. Older patients, especially those with vascular risk factors (smoking, diabetes, hypertension and hyperlipidemia) are at higher risk of developing cerebrovascular strokes. Family history should include hereditary conditions, such as migraine and risk factors, for cerebrovascular disorders. Treatment of hypertension causes more imbalance than the hypertension, which per se, is an extremely rare cause of balance problem.

SPONTANEOUS NYSTAGMUS

Nystagmus consists of involuntary, rhythmical, oscillatory movement of eyes. It may be horizontal, vertical or rotatory. Vestibular jerky nystagmus has a slow and a fast component. The direction of nystagmus is indicated by the direction of the fast component.

The past-pointing and falling occur toward the slow component of nystagmus, which indicates the side of vestibular weakness. In case of right side acute vestibular failure, fast component of the nystagmus is to the left but the past-pointing and falling will be towards the right (side of the slow component).

Method of Eliciting Nystagmus

The patient is either seated or lies supine on the bed. The examiner's finger is about 30 cm from the patient's eye in the central position. The finger is moved to the right or left, up or down 30° from the central position, while the patient's eyes follow the finger. If the eye is moved more than 30° from central position it results in gaze nystagmus.

Interpretations

- Presence of spontaneous nystagmus usually indicates an organic lesion.
- Peripheral vestibular nystagmus is due to lesions of labyrinth and eight nerve while central vestibular nystagmus is due to lesions in the central neural pathways (vestibular nuclei, brainstem, and cerebellum).
- In irritative lesions or stimulation of the labyrinth (serous labyrinthitis and BPPV) the direction of the nystagmus is towards the side of lesion.
- In paretic lesions (purulent labyrinthitis, trauma to labyrinth, and damage of eight nerve) nystagmus is in the opposite direction.
- Nystagmus of peripheral origin is suppressed by optic fixation (by looking at a fixed point) and enhanced in darkness and on closing the eyes. The use of Frenzel glasses (20 diopter glasses) abolishes optic fixation.

- The vertical, rotatory, dysconjugate and monocular nystagmus indicates central vestibular lesions. A down beating vertical nystagmus is a characteristic sign of Arnold Chiari malformation.

Degree of Nystagmus (Alexander Law for Peripheral Vestibular Nystagmus)

The intensity of nystagmus is indicated by its degree (Figs 2A to C.

- *1st degree*: This weak nystagmus is present only when patient looks in the direction of fast component.
- *2nd degree*: This moderate nystagmus is present when patient looks straight ahead.
- *3rd degree*: This strong nystagmus is present even when patient looks in the direction of slow component.

Ewald's Law

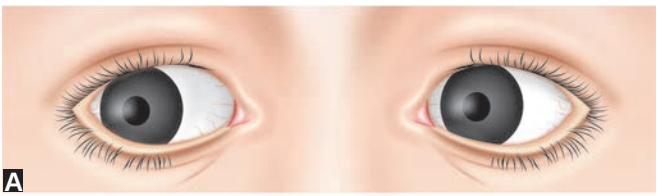
It has following three major components:

1. Head and eye movements in the plane of canal being stimulated decide direction of endolymph flow.
2. Ampullopetal flow of endolymph is excitatory in horizontal (lateral) semicircular canal.
3. Ampullofugal flow of endolymph is excitatory in vertical (anterior and posterior) semicircular canals.

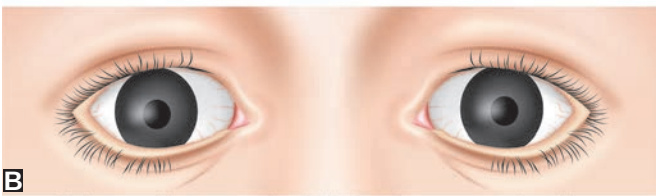
DYNAMIC OCULAR EXAMINATION (FIG. 1)

Vergence

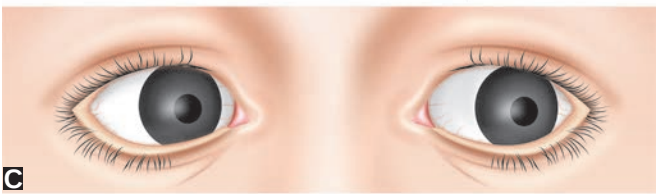
The nonconjugate eye movements of vergence make the eyes move in opposite directions to place the image of an object simultaneously on both fovea.



I degree: Nystagmus appears only when the patient looks in the direction of fast component (opposite the side hypolabyrinth)



II degree: Nystagmus is visible in primary neutral position but not in the direction of slow component



III degree: Nystagmus can be seen even in the direction of its slow component

Figs 2A to C: Alexander's law. When the eyes look toward the direction of fast phage of nystagmus; (A) intensity of slow phase of nystagmus increases, while it decreases when patient looks toward the direction of slow phase of nystagmus (C)

Saccades

Keeping the head still, patient alternately looks towards examiner's nose and then finger, which are held at different locations at about 15° away from primary positions. Saccades rotate the eye to bring an image onto the fovea. Saccade velocity, accuracy and initiation time are observed.

Smooth Pursuit

Patient's eyes follow a slowly moving object, no faster than 20°/sec. Pursuit maintains the image on the fovea as it moves across the visual field. The presence of more corrective saccades in one direction is noted.

Optokinetic Tracking

Optokinetic tape is used to elicit a number of alternating pursuit movements and saccades in succession. It detects asymmetries or dissociation between two eyes.

Vestibulo-ocular Reflex

From the primary neutral position, brief, high-acceleration horizontal head thrusts (head thrust test) are applied while patient is instructed to look at the examiner's nose. In normal persons, compensatory eye movements keep the patient's eyes remain stable on the examiner's nose. When the head is rotated on the side of hypoactive labyrinth it results in a delayed catch-up saccade to maintain gaze.

Vestibulo-ocular Reflex Cancellation

Patient's eyes fixate on examiner's finger that moves with patient's head. It tests the patient's ability to maintain steady gaze on an object moving with the head.

FISTULA TEST

Principle

The pressure changes in the external canal are transmitted to the labyrinth. It stimulates labyrinth and results in nystagmus and vertigo. When pressure is increased in external canal, ampullopetal flow of endolymph or displacement of cupula results in nystagmus of the same side.

If pressure is decreased in external ear canal, ampullofugal displacement of cupula or endolymph occurs. The quick component of induced nystagmus would be directed to the opposite side.

Method

It can be performed in two ways:

1. *Pressure on tragus*: Sudden inwards pressure on the tragus increases air pressure in the ear canal and stimulates the labyrinth.
2. *Siegel's speculum*: The increased pressure in the ear canal produces vertigo and nystagmus.

Interpretation

- *Negative fistula test*: Normally, the pressure changes in the external auditory canal do not stimulate the labyrinth. It is absent in following conditions:
  - Normal persons
  - Dead labyrinth

- *Positive fistula test*: A positive fistula test also implies that the labyrinth is still functioning. It produces vertigo and nystagmus and is present in the following conditions:
  - Labyrinthine fistula: Erosion of horizontal semicircular canal as in cholesteatoma.
  - Perilymphatic fistula: Abnormal opening in the oval window (post-stapedectomy fistula) or the round window (rupture of round window membrane).
  - Fenestration operation: Surgically created window in the horizontal canal.
- *False negative fistula test*: It occurs when cholesteatoma covers the site of fistula and does not allow pressure changes to be transmitted to the labyrinth.
- *A false positive fistula test (Hennebert's sign)*: It is the positive fistula test in the absence of a fistula. It is seen in following conditions:
  - Congenital syphilis: Stapes footplate becomes hypermobile.
  - Meniere's disease: It occurs in about 25% cases due to fibrous bands connecting utricular macula to the stapes footplate. Movements of stapes result in stimulation of the utricular macula.

VALSALVA MANEUVER

- Forced exhalation with closed nose and mouth increases pressure in the middle ear through Eustachian tube. It causes vertigo in some cases of perilymphatic fistulae.
- Forced exhalation with closed glottis raises intracranial pressure, which can cause dizziness in cases of anterior semicircular canal dehiscence and Chiari malformations.

DIX-HALLPIKE MANEUVER

Dix-Hallpike maneuver has a positive predictive value of 83% and negative predictive value of 52% for the diagnosis of BPPV.

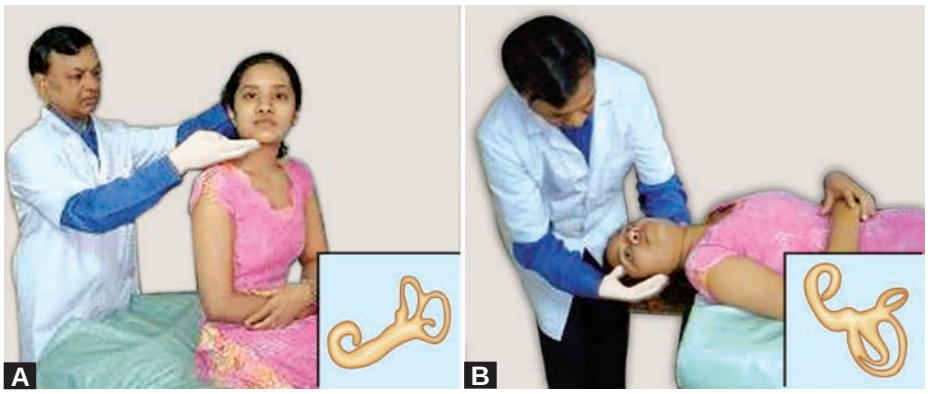
Method for Posterior Canal BPPV

From the sitting position with head turned 45° to one side, patient is rapidly lowered and placed in supine position and head extends over the edge of the examination couch (Figs 3A and B). Eye movements are observed with Frenzel glasses for 45 seconds and patient is brought back to upright position. The process is repeated with the head turned opposite side and then with the head extended supine. The classic eye movements are enhanced by the use of Frenzel Lenses.

Interpretations

The intensity of induced vertigo and nystagmus wanes with repeated maneuvers in peripheral vertigo but is not affected in central vertigo. The patients with positive Dix-Hallpike maneuver and vertigo or vomiting suggest a peripheral vestibular disorder. Purely vertical (usually downbeat) or torsional nystagmus without a latent period, unfatigable (does not wane with repeated maneuvers) suggests central causes for vertigo, such as a posterior fossa tumor or hemorrhage.

- *Features of peripheral vestibular lesions*: Presence of latency of onset of nystagmus, duration less than 1 minute, associated with vertigo, nystagmus disappears with repeated testing (fatigable), and vertigo may recur with nystagmus in the opposite direction on return of head to upright position.
  - Combined vertical upbeating and rotary (torsional) component beating toward downward eye (superior poles of eyes beat toward the downward ear): It is the most common finding and indicates:
    - BPPV–Canalithiasis of posterior semicircular canal.
  - Combined vertical downbeating and rotary (torsional) component beating toward upward eye (superior poles of eyes beat toward the upward ear)
    - BPPV–Canalithiasis of superior semicircular canal. It is very rare.
  - Nystagmus present with patients head turned to both right and left sides
    - Bilateral BPPV.
    - Head is not positioned correctly in the plane of posterior canal during testing of the normal side.
    - Head injury or brainstem ischemia.
- *Features of central vestibular lesions*: In head hanging position nystagmus begins without a latent period or persists with a constant slow-phase velocity for as long as the head position is maintained. Subjective vertigo is much less than expected from the intensity of nystagmus.
  - Pure vertical nystagmus usually downbeating with respect to the head. It is the most common type in central vestibular lesions. Other ocular motor abnormalities, which are present in this type of central nystagmus, include gaze-evoked nystagmus, impaired smooth pursuit, impaired vestibulo-ocular reflex (VOR) suppression and rebound nystagmus. The causes are following:
    - Multiple sclerosis (MS).



Figs 3A and B: Dix Hallpike test for right side posterior canal BPPV. (A) First sitting position (inset shows debris near ampulla of posterior canal); (B) Second supine head hanging position (Inset shows debris moving away from the ampulla in the posterior canal)

- Cerebellar lesion–Chiari malformations (cerebellar tonsils displaced caudally through foramen magnum).
- Lesion at craniocervical junction.

Modified Dix-Hallpike Maneuver for Lateral Canal BPPV

(Horizontal Positional Nystagmus)

- *Method:* Patient lies in supine position with head resting on the couch (not extended on the edge of the couch). First head is turned rapidly on one side so that ear rests on the couch. Eye movements are seen for 30 seconds with Frenzel Glasses. Then head is brought back to central supine position. Lastly the process is repeated and head is turned to opposite side. The classic eye movements are enhanced by the use of Frenzel lenses.
- *Interpretations:* They can be in the form of peripheral and central
  - Peripheral lesions: Horizontal nystagmus may beat toward (geotropic) or away (ageotropic) from the downward ear. Shorter latency and increase in magnitude while maintaining the test position are characteristic features. Less susceptible to fatigue with repetitive position.
    - Lateral canal BPPV (17% of cases). The pathogenesis is usually cupulolithiasis with or without canalithiasis. Particles are usually in the long arm of the lateral canal (canalithiasis) far from ampulla (geotropic nystagmus) or near the ampulla/on the opposite side of cupula (ageotropic nystagmus), floating within the endolymph or embedded in the cupula (cupulolithiasis).
  - Central lesions: Sustained, large amplitude nystagmus, which is present during visual fixation. Nystagmus occurs in more than one head position. Nystagmus has an associated vertical (especially downbeating) component.
    - Central vestibular disorder
    - Alcohol intoxication

OPTOKINETIC TEST

- *Method*: Patient's eyes follow a series of vertical stripes on drum, which moves first from right to left and then left to right.
- *Interpretations*: They can be–
  - Normal: Nystagmus with slow phase in the direction of moving stripes.
  - Abnormalities: Brainstem and cerebral hemisphere lesions.

ROTATION TESTS

- *Method*: Patient sits with head flexed 30° in Barany's revolving chair, which is rotated 10 times in 20 seconds first in clockwise direction and then in anticlockwise direction. The rotating Barany's chair is stopped suddenly and nystagmus is observed (Fig. 4).
- *Advantage*: It can be done in cases of atresia of external auditory canal.
- *Disadvantge*: Both the labyrinths are stimulated simultaneously.
- *Interpretations*: Twenty-five to forty seconds nystagmus is normal.

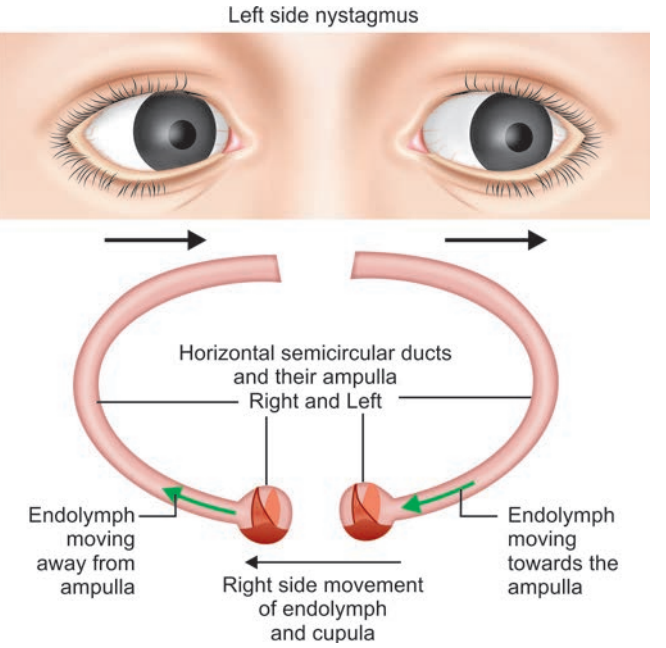


Fig. 4: Nystagmus in rotation test. At the end of rotation to right, endolymph continues to move to the right in both lateral semicircular canals and produce left direction nystagmus because of the left-side ampullopetal and right-side ampullofugal movements of cupula

CALORIC TEST

It compares the labyrinthine (lateral SCC) functioning of two ears, however there are no absolute figures for normality. In rare cases only when genuinely needed, a strong stimulus (2 ml ice water) though unpleasant to patient can be used to know the presence or absence of vestibular function.

Gitzgerald-Hallpike Bithermal Caloric Test

The patient lies in supine position with head flexed 30° (to bring horizontal SCC in vertical position) (Fig. 5). Ears are alternately irrigated for 40 seconds with water that is 7°C above and below the normal body temperature, i.e. 30° C and 44° C. The eyes are observed for nystagmus. A gap of 5 minutes is kept between each ear test. The time is recorded noted from the start of

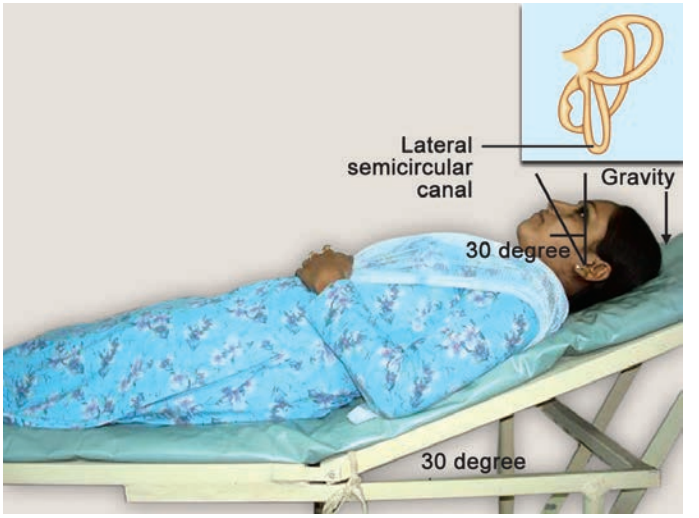


Fig. 5: Caloric testing. In reclined position, 30° elevation of head end of table brings the lateral semicircular canal in vertical plane (inset)

irrigation to the end of nystagmus on a calorigram. It is easy to remember the mnemonic COWS, which says that **c**old water will produce nystagmus on the **o**pposite side while **w**arm water on the **s**ame side.

Interpretations

- *Canal paresis*: Lesser response than normal indicates depressed function of labyrinth as occurs in Meniere's disease.
- *Dead labyrinth*: No response indicates absence of labyrinthine functions as occurs in purulent labyrinthitis, ototoxicity and acoustic neuroma.
- *Directional preponderance*: The total duration of right side of nystagmus (not the side of ear), which is produced by cold water in left ear and warm water in right ear, is compared with the left side nystagmus that is generated by cold water in right ear and warm water in left ear. Twenty five to thirty percent more nystagmus on any one side is called directional preponderance to that particular side, which indicates side of the central lesion that is away from the side in a peripheral lesion. It does not localize the site of lesion in central vestibular pathways.
  - Ipsilateral canal paresis and contralateral directional preponderance: It occurs in Meniere's disease.
  - Ipsilateral canal paresis and directional preponderance: It occurs in acoustic neuroma.

TANDEM WALKING

The patient walks (first with eye open and then closed) along a straight line up to a fixed point. In uncompensated peripheral vestibular lesions, with eyes closed, the patient deviates to the affected side. Patients with peripheral vertigo may have some impaired balance but are still able to walk. But patients with central vertigo often cannot walk or even stand without falling.

ROMBERG'S TEST

Although Romberg's sign is usually done in cases of vestibular and proprioceptive disorders, it was found only 19% sensitive for peripheral vestibular disorders.

- *Method*: The patient stands with feet together and arms by the side. First the eyes are open and then close. In "Sharpened Romberg test," the patient stands with one heel in front of toes and arms folded across the chest.
- *Interpretations*: With the eyes open, compensated cases can compensate the imbalance but with eyes closed patient sways to the side of lesion. In central vestibular disorder, patient shows instability even with eyes open.

CEREBELLAR TESTS

Gaze evoked, rebound and abnormal optokinetic type nystagmus are present in cerebellar lesions. Depending on the site of lesion, cerebellar disorders can have following features:

- *Cerebellar hemisphere*
  - *Asynergia*: Finger nose test becomes abnormal.
  - *Dysmetria:* Patients is not able to control range of motion.
  - *Adiadochokinesia*: Inability to perform rapid pronation and supination.
  - *Rebound phenomenon*: Patient is not able to control movement when opposing force is suddenly withdrawn.
- *Vermis*
  - Gait: Wide base and inability to make sudden turns on walking.
  - Falls
  - Truncal ataxia

HYPERVENTILATION

Hyperventilation for 30 seconds helps in ruling out hyperventilation syndrome in psychogenic causes of vertigo. One should be aware that though rarely it can cause true vertigo in patients with perilymphatic fistula and acoustic neuroma.

ORTHOSTATIC HYPOTENSION

Orthostatic hypotension upon standing (systolic blood pressure drops of 20 mm Hg or more and pulse increase of 10 beats per minute) is seen in cases of dehydration or autonomic dysfunction.

SPECIAL VESTIBULAR INVESTIGATIONS

A wide range of vestibular investigations (such as electronystagmography, caloric and rotating chair tests) is available but they have few clinical uses. The following tests are rarely of value clinically and are usually not done.

- *Rotating chair test*: The rotating chair stimulates both sides of labyrinth concurrently and can test each pair of SCCs separately in each of the three different planes.
- *Electronystagmography*: It records eye movements and detects nystagmus. The torsional component of nystagmus cannot be recorded with electronystagmography.
- *Computerized tests*: The computerized system such as force plate, records all changes in the position of the center of gravity of the patient, who stands on a small platform. The amount of sway in various situations such as with eyes closed and standing on one leg is recorded. It is difficult to justify the use of these tests for routine clinical work.

DIFFERENCES BETWEEN CENTRAL AND PERIPHERAL VERTIGO

Key information from the history can be used to make the distinction between the peripheral and central causes of vertigo. They include the timing and duration of vertigo, provoking or aggravating factors, and associated features, particularly neurologic symptoms and hearing loss. Characteristics distinguishing key features of peripheral and central causes of vertigo are shown in Table 4. The associated symptoms help in differentiating central (brainstem, cerebellum) from peripheral (inner ear, cranial nerve VIII) vestibular disorders.

Whirling, spinning or rotatory sensations are usually associated with peripheral vestibular disorders and are usually accompanied with nausea or vomiting. Nystagmus in peripheral vestibular disorders is usually horizontal and rotational, lessens when the patient focuses the gaze and is usually triggered by some provoking factor. In cases of central vertigo, nystagmus is purely horizontal, vertical or rotational; does not lessen when patient focuses the gaze, and persists for a longer period. The duration of vertiginous episode has important diagnostic value.

TABLE 4 Distinguishing characteristics of peripheral and central vertigo

Feature	*Peripheral vertigo*	*Central vertigo*
Nystagmus		
• Type	Combined horizontal and torsional	Purely vertical (most common), horizontal, or torsional
• Direction	One direction	May change direction
• Visual fixation	Inhibits	No change
• Fatigable	Yes	No
• Latency	Present	Absent
Imbalance	Mild to moderate but able to walk	Severe and unable to stand or walk
Nausea and vomiting	Usually present and severe	Varies
Hearing loss, tinnitus	Common	Rare
Neurologic symptoms (Motor and sensory deficiencies, ataxia, Horner's syndrome)	Absent	Common
Recovery	Begins within days	Slow
Head thrust sign	Present	Absent
Common causes	BPPV, vestibular neuritis, Meniere's disease, trauma to labyrinth, infection and drugs	Migraine, vertebrobasilar insufficiency, cerebrovascular accidents, multiple sclerosis, brain tumors and psychogenic

Clinical Highlights

1. **Utricle**: Utricle is the 1st part of membranous labyrinth to appear during the intrauterine life.
2. **Semicircular canals**: They connect with the utricle via five openings. Semicircular canals of two sides are paired synergistically (horizontal canals of both sides and one side posterior with opposite side superior).
3. **Vestibular function tests**: Some of the common tests include fistula test and cold caloric test.
4. **Tullio's phenomenon**: In this vertigo is induced by loud sounds. The causes include perilymphatic and labyrinthine fistulae congenital syphilis and additional third labyrinthine window in cases of fistula of semicircular canal and fenestration operation in the presence of mobile footplate of stapes.
5. **Fistula test**: It can be performed by applying intermittent pressure on the tragus or by using Siegle's speculum. Fistula test is positive in cases of labyrinthine fistula, hypermobile stapes footplate, following fenestration surgery and erosion of the horizontal semicircular canal in cholesteatoma. Positive fistula indicates that labyrinth is still active and alive. The test is negative when labyrinth is dead.
6. **Hennebert's sign**: This is positive fistula test in the absence of fistula. The causes include congenital syphilis (utricular adhesions to stapes) and some cases of Meniere's disease.
7. **Romberg's sign**: It is indicative of not the cerebellum lesions but the dorsal column (somatosensory) lesions.
8. **Frenzel glass**: Nystagmus is best observed in the darkened room by illuminated Frenzel glass, which is nothing but a 20 diopters lens.
9. **Causes of ipsilateral (same direction) nystagmus**: Irrigation of ear with warm water and serous labyrinthitis.
10. **Causes of contralateral (opposite direction) nystagmus**: Purulent labyrinthitis, labyrinthectomy and irrigation of ear with cold water.
11. **Dix-Hallpike maneuver**: This test is used in patients with episodic positional vertigo. On Dix-Hallpike testing, central nystagmus appears immediately without a latent period as soon as head is in critical position.
12. **Fitzgerald Hallpike Bithermal caloric test**: The lateral (horizontal) semicircular canal (SCC) is stimulated (tested) by irrigating cold (30°C) and warm (44°C) water in the external auditory canal. Cold water induces opposite side nystagmus while warm water results into the same side nystagmus (COWS (Cold, opposite; Warm, same)). In a sitting position with head tilted 60° backward, lateral semicircular canal is stimulated during caloric testing. To bring the lateral SCC in vertical position, patient's head is raised 30° forward if s/he is in supine position but in a sitting position the head is tilted 60° backward.

FURTHER READING

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2. Biswas A. Clinical Audiovestibulometry for Otologists and Neurologists, 3rd Edition. Mumbai: Bhalani Medical Book House; 2001.
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22 Peripheral Vestibular Disorders

Take up one idea. Make that one idea your life—think of it, dream of it, and live on that idea. Let the brain, muscles, nerves, every part of your body, be full of that idea, and just leave every other idea alone. This is the way to success.

—Swami Vivekananda

Points of Focus

- **INTRODUCTION**
- **BENIGN PAROXYSMAL POSITIONAL VERTIGO**
  - ✓ Incidence
  - ✓ Etiological Factors
  - ✓ Pathogenesis
- **ACUTE VESTIBULAR NEURITIS**
- **MENIERE'S DISEASE**
  - ✓ Pathogenesis
  - ✓ Nystagmus
  - ✓ Audiogram
- **DELAYED ENDOLYMPHATIC HYDROPS**
- **RECURRENT VESTIBULOPATHY**
- **MIDDLE EAR EFFUSION**
- **LABYRINTHINE FISTULA**
- **SEROUS LABYRINTHITIS**
- **SUPPURATIVE LABYRINTHITIS**
  - ✓ Routes of Infection
  - ✓ Predisposing Factors
- **PERILYMPHATIC FISTULA**
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

Benign paroxysmal positional vertigo (BPPV), acute vestibular neuritis and Ménière's disease are the most common causes of vertigo. It is estimated that more than 25% of patients who present to their general practitioner with vertigo suffer from BPPV. Otitis media is the most common cause of vertigo in children. However, some of the other common causes are enumerated in Box 1.

BENIGN PAROXYSMAL POSITIONAL VERTIGO

This is the most common cause of vertigo.

Incidence

- Benign paroxysmal positional vertigo is the most common cause (20–40%) of peripheral vertigo.
- Age: 11–84 years; mean age of onset fourth to fifth decades. Incidence increases with age.
- Pediatric BPPV patients have association with migraine.
- Slightly increased incidence in females.

Etiological Factors

In about 52% of cases, one or more of the following etiological factors are present.

- Most common are closed head injury and vestibular neuronitis (vertigo lasting days).
- Infections
- Old age
- Surgery (stapedectomy or nonotologic)
- Prolonged bed rest and inactivity
- Benign paroxysmal positional vertigo can develop in cases of Ménière's disease, viral labyrinthitis and recurrent vestibulopathy.

Pathogenesis

Otoconia gets displaced from utricle to semicircular canal (SCC) usually posterior.

- *Cupulolithiasis*: Deposition of otoconia on the cupula of posterior SCC.

Box 1: Peripheral vestibular disorders

- Infectious
  - Viral labyrinthitis–Measles, mumps, infectious mononucleosis, varicella-zoster virus oticus (Ramsay Hunt syndrome)
  - Toxic labyrinthitis
  - Serous labyrinthitis
  - Suppurative labyrinthitis
  - Labyrinthine fistula
  - Syphilis
- Traumatic
  - Labyrinthine concussion
  - Temporal bone fracture
  - Ear surgery–Stapedectomy
  - Collagen vascular disorders
  - Autoimmune inner-ear disease
  - Nonsyphilitic interstitial keratitis (Cogan's syndrome)
- Ischemic
  - Occlusion of labyrinthine artery–Sudden hearing loss and vertigo
- Drug-induced
  - Aminoglycoside antibiotics
  - Antineoplastic agents: Cisplatin
- Endolymphatic hydrops
  - Idiopathic: Ménière's disease
  - Secondary: Delayed endolymphatic hydrops, recurrent vestibulopathy, syphilis, Cogan's disease
- *Neoplastic*
  - Acoustic neuroma, meningioma, epidermoid
- Miscellaneous
  - Ear wax
  - Familial vestibulopathy
  - Recurrent vestibulopathy
  - Vestibular neuritis
  - Sudden sensorineural hearing loss
  - Perilymphatic fistula
  - Benign paroxysmal positional vertigo
  - Superior semicircular canal dehiscence syndrome
  - Congenital cholesteatoma

- *Canalithiasis*: Free floating material (debris) within the lumen of posterior SCC.
- *Lateral canal BPPV* (17% of cases): The pathogenesis is usually cupulolithiasis with or without canalithiasis. Particles are usually in the long arm of the lateral canal (canalithiasis) far from ampulla (geotropic nystagmus) or near the ampulla/on the opposite side of cupula (ageotropic nystagmus), floating within the endolymph or embedded in the cupula (cupulolithiasis).

Clinical Features

- Sudden brief (seconds) spells of severe vertigo associated with change in head position, such as
  - Rolling over in the bed
  - Getting into bed and assuming a supine position
  - Arising from a bending position
  - Extending the neck
  - Turning rapidly
- Vertigo spell lasts for seconds and never more than a minute. |However patients usually complain of longer subjective feeling of dizziness.
- Bouts of vertigo are clustered in time. Remissions may last for months or more.
- The active spells may be associated with the feeling of light headedness or mild imbalance, which is worsened by head movement.
- Some patients have chronic balance problem, which may be worse at the time of awakening from the sleep.

Types

Though following three types of BPPV are described but the most common is posterior canal BPPV.

1. Posterior semicircular canal BPPV
2. Lateral semicircular canal BPPV
3. Superior semicircular canal BPPV

Diagnosis

Dix-Hallpike test: The BPPV is classified as active if rotatory nystagmus is elicited by Dix-Hallpike test. Patients with a typical history of BPPV, but a negative Dix-Hallpike test and whose symptoms had settled are diagnosed as resolved BPPV. For detail of the test method and interpretations, see chapter Evaluation of Vertigo.

Differential Diagnoses

- Vascular compression of cranial nerve VIII complex
- Multiple sclerosis
- Acoustic neuroma (see table 1 of chapter 25)

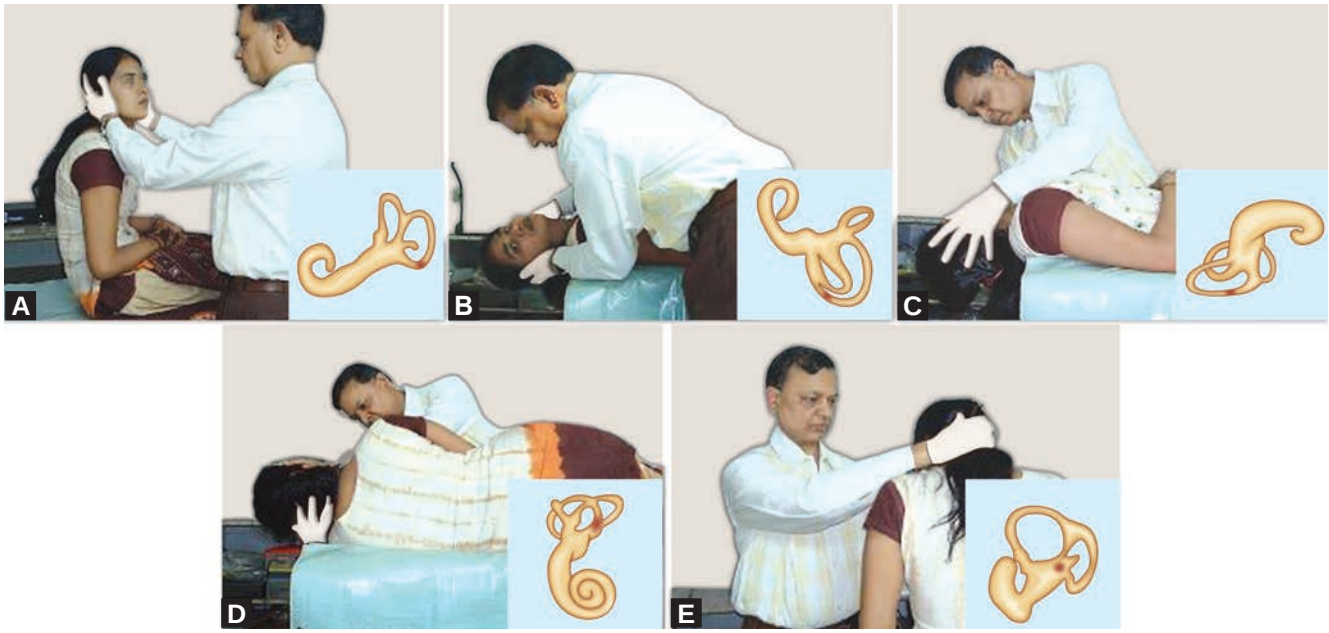
Treatment

Treatment consists of repositioning maneuvers. There is no role of any medicine. The following maneuvers are effective in majority of the BPPV patients.

Repositioning Maneuvers

After the maneuver, patients are told to sleep with 45°elevated head and avoid bending. En bloc movement of the head and body is preferred during these maneuvers. Mastoid vibrator gives slightly better results. During these maneuvers, posterior canal BPPV can be converted to lateral canal BPPV (resolves in several days), superior canal BPPV (vertigo worsens when the affected ear is up) or canalith obstructs common crus and simultaneously excite both superior and posterior canals. Thus, the opposing vertical components are canceled and the torsional components are added.

- For Posterior SCC
  - *Epley maneuver (Figs 1A to E)*: It is effective in 90% cases of posterior canal BPPV. Meclizine or benzodiazepine 1 hour before may be given in anxious patients. First two positions (upright and supine position with extended neck turned 45°) are similar to Dix-Hallpike maneuver (see chapter Evaluation of Vertigo). Then head is rolled



Figs 1A to E: Canalith repositioning for right posterior canal benign paroxysmal positional vertigo. (A) Patient sitting with head turned right and debris near ampulla (inset); (B) Patient in supine position with extended neck and debris move away from the ampulla in the posterior canal (inset); (C) Head moved 90° to left and debris comes near the common crus (inset); (D) Patient rolled onto left and head faces down and debris starts entering into the utricle from the common crus (inset); (E) Patient in upright position and debris collects in utricle (inset)

180° in two 90° increments so that the offending ear is up. From this position, patient is brought to the sitting position. The repositioning maneuver is repeated again and again until no nystagmus is produced. Head remains in that particular position until any nystagmus resolves.

- *Semont maneuver (Figs 2A to D)*: The patient is turned rapidly into the position that provokes vertigo and kept in that position for 4 minutes. Then patient is moved quickly to opposite position with normal ear down. After some time, patient is asked to sit slowly in upright position.

- For Lateral SCC
  - *Geotropic nystagmus*: Lying on the normal side with affected ear up for 12 hours—effective in 92% cases.
  - *Modified Epley (Figs 3A to E)*: Patient lies on the affected side with normal ear up. Then, patient is moved in 90° increments to supine, affected ear up, prone and at last affected ear down.
- For Superior SCC
  - The head is positioned into a dependent position so that ampulla of superior canal is superior, then turned by 180°, and finally brought back to the original position.
  - The patient initially lies on the normal side, then head is turned downwards 45°, then horizontally, upwards 45° and at last patient sits upright. The patient remains in each position for 30 seconds.

Surgical Treatment

The following surgical procedures are considered in cases that are refractory to the above mentioned maneuvers and the symptoms are disabling to the patient:

- Singular neurectomy
- Posterior SCC occlusion

ACUTE VESTIBULAR NEURITIS

(Synonyms: Vestibular neuronitis, Acute viral labyrinthitis)

This is the second most common cause of peripheral vestibular vertigo after the BPPV.

> Acute vestibular neuronitis is the second most common cause of peripheral vestibular vertigo.

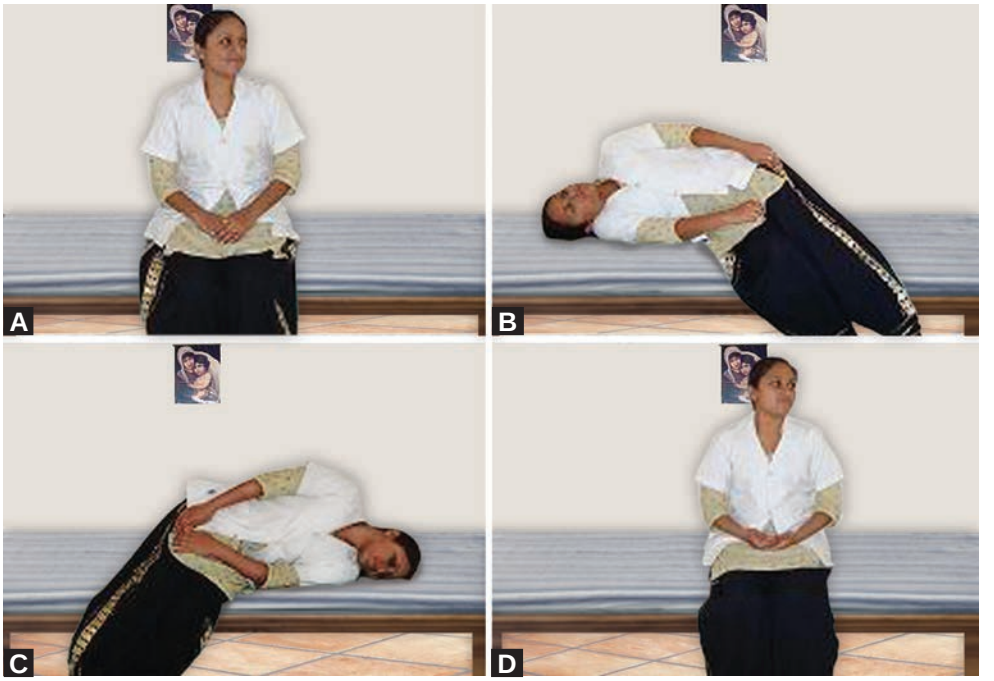
Etiology

The exact cause of the disease is yet not known.

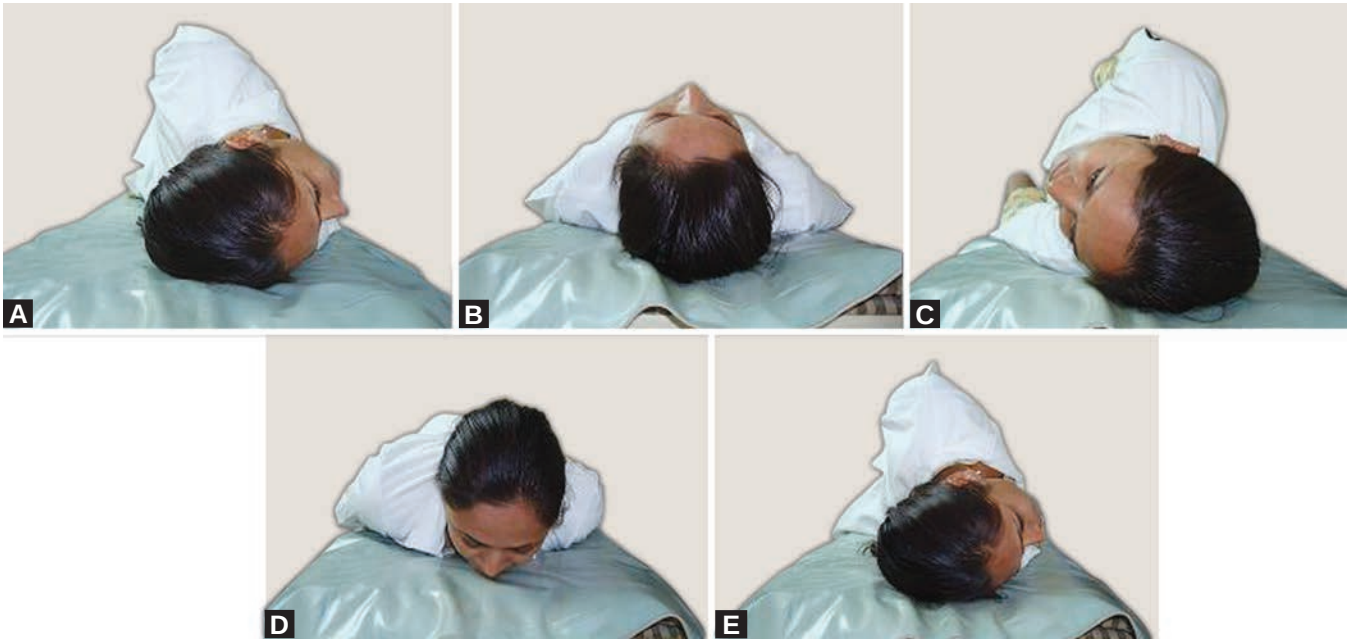
- Fifty percent patients have preceding history of upper respiratory infections and sinusitis.
- Following microorganisms, which cause vertigo, have been suggested: mumps virus, varicella-zoster virus (VZV), influenza, Epstein-Barr virus (infectious mononucleosis), herpes simplex virus (HSV), *Borrelia burgdorferi* (Lyme disease) and *Treponema pallidum*.
- Many patients have preceding infection of herpes virus (increased serum titer).

Clinical Features

- Often patients have dramatic sudden onset of vertigo and attendant vegetative symptoms (such as nausea, vomiting and sweating) with gradual, definite improvement throughout the course.
- Occasionally, patients have bouts of attacks over several weeks. Few patients complain of stuffiness in their ear.
- The direction of horizontal or horizontal-rotary nystagmus is towards uninvolved side.
- Caloric responses are decreased or absent.
- Cerebrospinal fluid (CSF) and electroencephalogram are normal.



Figs 2A to D: Semont maneuver for right posterior canal benign paroxysmal positional vertigo. (A) Upright sitting position; (B) Patient is turned rapidly into the position that provokes vertigo and kept in that position for 4 minutes; (C) Then patient is moved quickly to opposite position with normal ear down; (D) After some time, patient is asked to sit slowly in upright position



Figs 3A to E: Modified Epley maneuver for right lateral canal benign paroxysmal positional vertigo. (A) Patient lies on the affected side with normal ear up. Then patient is moved in 90° increments to (B) supine; (C) affected ear up; (D) prone; and at last (E) affected ear down

- Bilateral disease though rare must be considered in differential diagnoses of bilateral vestibular loss.
- Reduction or absence in caloric response identifies the side of involvement.
- Though there is no subjective hearing loss (HL) audiograms usually show HL in high frequencies (10–15 kHz).

Prognosis

- Vertigo usually lasts many hours to 1–2 weeks and then spontaneously disappears over weeks to months.
- Imbalance caused by rapid head movement may last for months after the resolution of acute disease.
- Paroxysmal positional vertigo occurs subsequently in some patients. Some patients have recurrent attacks (transient positional vertigo triggered by sudden head movements) for years, which are usually less intense.

Differential Diagnoses

- First episode of recurrent vestibulopathy
- Hemorrhage parietoinsular cortex or superior or medial temporal gyri (vestibular cortex)

- Multiple sclerosis.
- Acoustic neuroma (Table 1 of chapter 25)

Treatment

The treatment is nonspecific and includes–

- Hospitalization and intravenous (IV) fluids
- Diazepam, 5–10 mg IV
- Promethazine IV, per rectum or oral 25–50 mg 4–6 hourly

Prolong antivertiginous therapy delays the recovery time.

MÉNIÈRE'S DISEASE (IDIOPATHIC ENDOLYMPHATIC HYDROPS)

Prosper Ménière first described this disorder in 1861. Symptom complex consists of spontaneous episodic vertigo, fluctuating sensorineural hearing loss (SNHL), tinnitus and often a sensation of fluctuating ear fullness.

Incidence

- Varies from 7.5–157 per 100,000 persons.
- Affects primarily Caucasians with a slight female preponderance.
- About 10% of dizziness patients have Ménière's.
- Age of onset is 4–90 years but peak incidence is in 40–60 year age group.
- Bilateral disease develops in 47% of cases followed up for 20 years.

Etiology

Exact cause of the disease is not yet known.

- Viral: The causative roles of viruses (HSV, cytomegalovirus or VZV) remain uncertain.
- *Hereditary*: Familial occurrence in 10–20% cases. There is an autosomal dominant mode of inheritance.
- *Autoimmune*: Certain genetically acquired major histocompatibility complexes specifically human leukocyte antigens (HLA) B8/DR3 and Cw7 have been associated with Meniere's disease.

Pathogenesis

- Perisaccular ischemia and fibrosis lead to inadequate absorption of endolymph by the endolymphatic sac.
- Overaccumulation of endolymph at the expense of perilymphatic space results in the distortion of membranous labyrinth (Fig. 4). Alterations in the size of endolymphatic duct and sac along with reductions in tubular specializations of the lining of these structures have been observed.
- Endolymphatic hydrops mainly occurs in pars inferior (cochlea and saccule) and changes in pars superior (utricle and SCC) are usually less obvious.
- The basilar membrane gets distended into scala tympani.
- Saccular distension can distort not only utricle and SCCs but can also come in contact with stapes footplate. That can cause Hennebert's sign.
- Hair cells and their neurons are usually spared.
- Membranous rupture, which can occur in any part of the inner ear, allows leakage of potassium-rich endolymph into perilymph. The high concentration of K^+ depolarizes the neurons and inactivates both vestibular and auditory neurons that result in vertigo (paralytic nystagmus) and deafness.
- Healing of membranes allows restitution of normal chemical and clinical status.
- Repeated membranous rupture and potassium exposure lead to chronic deterioration in the functions of inner ear.

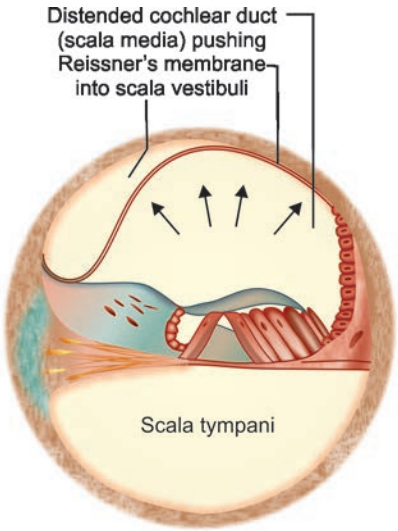


Fig. 4: Ménière's disease. Distended cochlear duct pushing the Reissner's membrane into scala vestibuli

Clinical Features

- The typical presentation consists of recurring attacks (3–11 per year) of spinning vertigo (96.2%) in horizontal axis (usually for minutes to 2–3 hours) with tinnitus (91.1%) and ipsilateral HL (87.7%). Attack lasting more than a day is inconsistent with the diagnosis.
- Vertigo attacks are usually preceded by an aura consisting of fullness in the ear, increasing tinnitus and decrease in hearing. Attacks may be sudden without any warning or may awaken patient from sleep.
  - Attacks may cease spontaneously after 2 years (57%), 8.3 years (71%) or may occur for 20–40 years.
  - In the early phases of disease, patient may have either vestibular (recurrent vestibulopathy) or auditory symptoms. The terms vestibular Ménière or cochlear Ménière are considered inappropriate by American Academy of Otolaryngology-Head and Neck Surgery Committee on Hearing and Equilibrium.
  - Cluster of attacks are separated by long remissions. Patients can have single, sporadic attacks or periods of unrelenting recurring attacks. Patient may be minimally inconvenienced or completely incapacitated. Emotional impact may be equivalent to major medical problem.
  - Attacks are often accompanied by nausea, vomiting, diarrhea or sweating. Vertigo is exacerbated with any head movement.
  - Between the attacks patient may be totally symptom-free or feel disequilibrium, light headedness or tilt.
- Hearing loss is fluctuating and progressive. Diplacusis (a difference in the perception of pitch between the ears) and recruitment (intolerance to loud sounds) may be present.
- Tinnitus is nonpulsatile and may be whistling or roaring,

continuous or intermittent. It becomes louder or changes pitch as an attack approaches and improves after the attack.

- In Lermoyez attack, increased tinnitus and HL precede the vertiginous episode and dramatically resolve with onset of vertigo.
- Drop attacks (otolithic crises of Tumarkin)
  - Occasionally sudden unexplained falls without loss of consciousness or associated vertigo occur. Patient feels pushing or moving during this short-lived spells.
  - Acute utriculosaccular dysfunction leads to inappropriate postural adjustment via vestibulospinal pathway.
  - Other causes of drop spells include cardiogenic, vertebral basilar insufficiency and migraine.

Nystagmus

The diseased ear cannot be determined just on the basis of direction of nystagmus. The direction of horizontal nystagmus, which is the cardinal finding, varies over the course of the attack.

- Early irritative nystagmus: Ipsilateral
- Later paralytic nystagmus: Contralateral
- Late recovery nystagmus: Ipsilateral

Audiogram (Fig. 5)

- Low frequency fluctuating and progressive SNHL and a coincident nonchanging high frequency loss. Average pure tone loss is of 50 dB. Profound deafness is rare.
- Peaked at 2 kHz (tent-like audiogram).
- Overtime HL flattens.
- Recruitment present in 56% cases.
- A mean speech discrimination score of 53%.
- *Staging*: On the basis of pure tone average (500, 1,000, 2,000 and 3,000 Hz) in dB of the worst HL (audiogram) in the interval of 6 months before treatment, Ménière's disease has been graded into following four stages:
  1. *Stage I*: = Less than 25 dB
  2. *Stage II*: 26–40 dB
  3. *Stage III*: 41–70 dB
  4. *Stage IV*: More than 70 dB

Differential Diagnoses

Ménière's disease, which is an idiopathic lesion, is a clinical diagnosis (Box 2). The following conditions, which are included in Ménière's syndrome or secondary Ménière's disease, can mimic the clinical features of Ménière's disease and should be kept in mind.

- Migraine and basilar migraine
- Autoimmune disease of inner ear and otosclerosis
- Syphilis and Cogan's syndrome
- Cardiogenic
- Vertebral basilar insufficiency
- Trauma: Head injury or ear surgery
- Acoustic neuroma (Table 1 of chapter 25)

Investigations

Findings observed in different investigations include:

- *Glycerol test*: Ingestion of glycerin 1.5 g/kg mixed with equal volume of juice is followed by serial audiograms over 3 hours.
  - Positive test: 25 dB shift at three consecutive frequencies, or 16% improvement in speech discrimination.

Box 2: Classification and diagnosis of Ménière's disease. Committee of hearing and equilibrium. American Academy of Otolaryngology-Head and Neck Surgery

- Certain: Confirmed by histopathology
- Definite: Two or more episodes of spontaneous whirling vertigo lasting for 20 minutes or more
  - Documented sensory hearing loss (HL) on audiometry on at least one occasion
  - Tinnitus or aural fullness
  - Other causes excluded
- Probable:
  - One episode of spontaneous whirling vertigo
  - Documented sensory HL
  - Tinnitus or aural fullness
  - Other causes excluded
- Possible:
  - Vestibular variant: Episodes of spontaneous whirling vertigo without documented HL.
  - Cochlear variant: Fluctuating or fixed with disequilibrium but without episodes of spontaneous whirling vertigo
  - Other causes excluded.

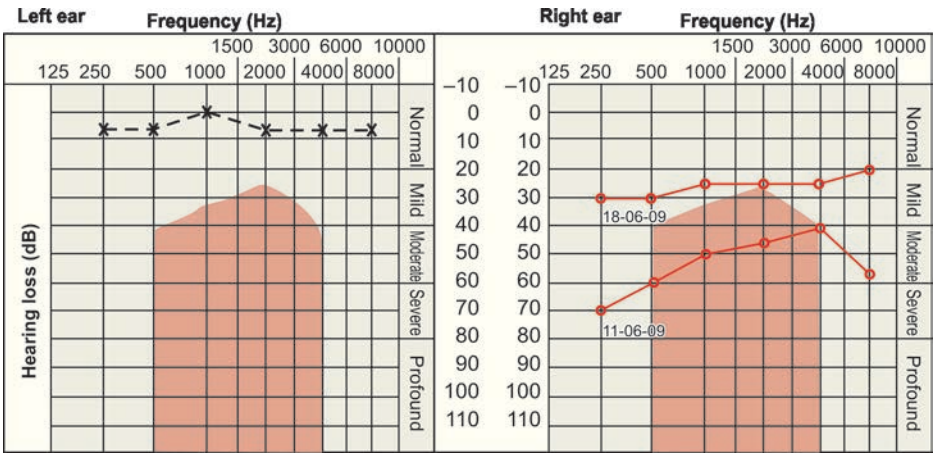


Fig. 5: Audiogram Ménière's disease right ear. Low-frequency fluctuating (improvement after 1 week) sensorineural hearing loss in right ear

- *Dehydrating agents*: Urea, glycerol and furosemide produce measurable improvement in audiometric score, reduction in summating potential (SP) negativity (electrocochleography) and change in gain of vestibulo-ocular response to rotational stimulation. Sensitivity and specificity vary widely.
- *Electronystagmography*: Caloric response is significantly reduced in 48–73.5% cases. Complete loss occurs in 6–11%.
- Electrocochleography
  - Infrequently used.
  - Summating potential: It is larger and more negative due to the distension of basilar membrane into scala tympani.
  - Action potential (AP) of CN VIII: Reduction in amplitude.
  - SP/AP ratio: Increases (most commonly used value).
- *Imaging*: It is usually not required for the diagnosis. Following findings may be seen.
  - CT scan: Hypoplasia of the endolymphatic sac and duct results in decreased visualization of vestibular aqueduct and reduction in periaqueductal pneumatization. Reduced retrolabyrinthine bone might be predisposing factor to disease.
  - MRI: Significantly smaller and shorter endolymph drainage system.
  - Gadolinium-enhanced MRI: Enhancement of endolymphatic sac reflects inflammation of sac.

Medical Treatment

There is no curative therapy for Ménière's disease. Following conservative measures are usually taken:

A. *General Measures*
   - Reassurance: Explaining the nature of disease relieves the patient's anxiety.
   - Lifestyle changes
     - Cessation of smoking.
     - Avoid excessive intake of water and salt.
     - Avoid too much tea, coffee and alcohol.
     - Avoid stress: Practice mental relaxation exercises, meditation and yoga.
     - Avoid flying, diving and working at great heights to prevent accident in cases of sudden episode.

B. *Acute Attack*
   - Reassurance and bed rest
   - Vestibular sedatives: They are given parenteral if vomiting precludes oral use. They take care of vertigo and anxiety and include prochlorperazine (Stemetil), promethazine theoclate (Avomine), dimenhydrinate (Dramamine), and diazepam. Atropine can also be very effective.
   - Vasodilators:
     - Inhalation of carbogen (5% CO_2 with 95% O_2) causes cerebral vasodilation and improves labyrinthine circulation.
     - Histamine diphosphate 2.75 mg in 500 ml glucose slow IV drip helps in managing acute episodes but is contraindicated in asthmatics and should not be given empty stomach. The adverse side effects include tachycardia, disturbed cardiac rhythm, hypotension, hyperthermia and bronchospasm.

C. *Chronic Phase*
   - Dietary modifications
     - Salt restriction
     - Intermittent dehydration
     - Hyperosmolar dehydration
   - Diuretics
     - Hyperosmotic diuretic isosorbide.
     - Acetazolamide: Carbonic anhydrase inhibitor.
     - Furosemide 40 mg on alternate day with potassium supplement may help in some cases.
   - *Vasodilators*: Quantitative improvement was not seen consistently with these medications.
     - Histamine, betahistine, papaverine analog, eupavarine, nicotinic acid, adenosine triphosphate and dipyridamole.
   - *Symptomatic medications*: They are beneficial in reducing the symptoms and in improving tolerance.
     - Antivertiginous, antiemetics, sedatives, antidepressants and psychiatric treatment.
   - *Propantheline bromide*: Pro-Banthine 15 mg three times a day is found effective in some cases.
   - *Elimination of allergen*: In cases of allergy, the causative allergen should be eliminated or desensitized.
   - *Hormones*: Endocrinal hypofunctioning, such as hypothyroidism, need appropriate replacement therapy.
   - Other therapies
     - Acupuncture
     - Herbal remedies
   - Pulsed positive pressure in external auditory canal and pressure equalization tube.
   - Bilateral Ménière's disease.
     - Parenteral streptomycin: Streptomycin is primarily vestibulotoxic and usually leaves hearing unchanged. Indicated only for incapacitating bilateral Ménière's disease.
     - 1.5–3.5 gm/day until ice water response is absent (for 17.5 days average; 39 gm mean total dose).
     - 20 gm total dose preserves some vestibular function, which decreases the disabling oscillopsia and vestibular ataxia.
   - Immunosuppressant therapy
     - Chronic steroids
     - Nonsteroidal immunosuppressant: Methotrexate

Surgical Treatment

The patients who do not respond to medical treatment, following surgical procedures should be considered.

- Hearing-conservative nonvestibular ablative surgery
  - *Endolymphatic sac decompression*: The procedure may be combined with insertion of Silastic sheeting, tubing, one-way valve, tubed sponge, gelatin or tissue stent. The drainage pathway is routed to the epidural space by incision of the back wall of the sac. Fifty to seventy-five precent patients have complete resolution of the vertigo.
  - *Cochleosacculotomy*: When mechanical blockage of utricular and saccular ducts does not allow endolymph to reach endolymphatic sac. Perforation of the saccule is done behind the oval window with a pick through the round window. Seventy percent of patients have complete control of vertigo. Major drawbacks include recurrence and high incidence of HL.

- *Fick and Cody Tack procedures*: They create fistulas in the saccule via the oval window.
- *Transtympanic corticosteroid infusion of middle ear*.

- Hearing-conservative vestibular ablative surgery
  - *Vestibular neurectomy*: About 95% patients have complete resolution of vertigo. The following approaches may be employed:
    - Middle fossa approach
    - Suboccipital approach
    - Retrosigmoid approach
    - Retrolabyrinthine approach
  - Intratympanic injections of aminoglycoside
    - Streptomycin: not used now due to high risk of HL
    - Gentamicin: Angular vestibulo-ocular reflex (rapid head thrusts) by excitation of SCC on treated side is reduced. Sevent to ninety percent patients have complete control of vertigo. Profound HL occurs in 0–4%. It is considered first line of treatment among the invasive vestibular procedures when medical measures fail.
    - 17 mg/dose three times daily with a 12 dose total.
    - Weekly
    - Single dose
    - Injections into perilymph/endolymph via horizontal SCC (Shea): Generally not used.
  - Cryosurgery
  - Ultrasound
  - Cochlear dialysis
  - Cochlear implantation: Bilateral severe-to-profound HL
- Non-hearing conservative vestibular ablative surgery
  - *Labyrinthectomy*: When serviceable hearing is absent: Pure tone average more than 60 dB, speech discrimination less than 50%.
    - Ablation with hypertonic saline
    - Transcanal
    - Transmastoid
  - Translabyrinthine cochleovestibular neurectomy
  - Destruction of Scarpa's ganglion.

DELAYED ENDOLYMPHATIC HYDROPS

- *Clinical features*: Ménière's like attack of vertigo in cases of preexisting profound HL in one or both ears (1–74 years) due to trauma (physical and acoustic), idiopathic sudden SNHL, labyrinthitis (mumps, influenza, mastoiditis, meningitis, diphtheria, measles) and idiopathic childhood deafness.
- *Lasix test*: Dehydrating agents, such as furosemide (Lasix) improve the vestibular response of the affected ear.
- *Treatment*: Similar to Ménière's disease.

RECURRENT VESTIBULOPATHY

The disorder, which is relatively common, was formerly called as vestibular Ménière's. Cause is not yet known but some postulate viral due to similarities with vestibular neuritis.

Clinical Features

- Mean age of onset 37 years with equal sex distribution.
- Attacks of vertigo (sudden, lasting for minutes to hours and recurring at variable interval) similar to Ménière's disease without audiologic symptoms.
- Nine percent patients have additional features of BPPV.
- Diagnosis may change to Ménière's disease because about 14% patients develop audiologic symptoms.
- No neurological features.
- Caloric response is reduced only in 22% (50% in Ménière's).
- Seventy percent patients recover completely over 9.5 years.

Treatment

- It is mainly symptomatic and supportive.
- There is no role of diuretics.
- Vestibular neurectomy may be considered in certain patients.

MIDDLE EAR EFFUSION

- Otitis media with effusion is the most common cause of vestibular disturbance in children.
- The dizziness may develop from transient negative middle ear pressure changes associated with displacement of round window membrane, leading to secondary movement of perilymph.
- Glue ear children may walk clumsily or trip easily.
- Dizziness clears with myringotomy and grommets.
- Serous labyrinthitis can occur secondary to the superadded infection of middle ear effusion.

LABYRINTHINE FISTULA

(*Synonyms: Circumscribed labyrinthitis, paralabyrinthitis and perilabyrinthitis*)

There occurs loss of endochondral bone without loss of perilymph usually due to cholesteatoma (mainly horizontal SCC). Inflammation of a discrete portion of bony labyrinth and endosteum occurs without the involvement of membranous labyrinth.

Causes

- Chronic otitis media: Cholesteatoma—most common
- Congenital syphilis
- Carcinoma
- Glomus jugulare tumors
- Fenestration operation for otosclerosis: No inflammation

Clinical Features

- Patient may remain asymptomatic.
- Brief periods of imbalance, disequilibrium, or vertigo. Patient has normal equilibrium most of the time.
- *Fistula sign*: Momentary imbalance on pushing external ear canal with washing cloth. Transient vertigo induced by washing the ear or by pushing on the tragus.
- *Tullio's phenomenon*: Loud sounds provoke brief imbalance.

Investigations

- *Fistula test positive*: See chapter Evaluation of Vertigo.
- *CT scanning in bone algorithm*: Bone erosion of lateral SCC.

Treatment

Surgery: Complete removal of cholesteatoma matrix is done from the fistula site. Labyrinthine fistula (LF) is usually accompanied by erosion of bony fallopian canal and tegmen. So

surgeon should be careful. The blue line of actual fistula is identified along with the adjacent thinned layer of bone. A small piece of tissue is placed over the site. Large fistula may be left covered by cholesteatoma matrix, which would form the lining of mastoid cavity. Fistula sign persists until the regrowth of bone.

SEROUS LABYRINTHITIS

This refers to a transient nonpurulent inflammation or chemical irritation of the inner ear, which usually does not result in any permanent damage.

- If diagnosed and treated successfully before the development of suppurative labyrinthitis, the prognosis is excellent.
- Hearing and vestibular dysfunction are entirely reversible.
- The perilymph is involved but not the endolymph.

Causes

- Causes of LF
- Bacterial or viral toxin invasion through round or oval window: All types of otitis media and mastoiditis
- Blood-borne infection
- Meningeal inflammation
- Ear surgery: Fenestration and stapedectomy

Clinical Features

- Spontaneous vertigo in cases of middle ear infection usually indicates serous labyrinthitis. Vertigo is more severe than LF but less severe than suppurative labyrinthitis.
- Irritative type of nystagmus (hyperactive labyrinthitis): Quick component is directed toward the affected ear.
- If the condition is secondary to LF, fistula sign will be positive. Though there occur worsening of LF symptoms. Vertigo is less severe because some compensation develops during the inflammation of LF.
- Hearing may be impaired but not markedly. If useful hearing is present, then suppurative labyrinthitis has not developed. Insidious high-tone SNHL is a frequent accompaniment of chronic suppurative otitis media (CSOM) even without vertigo.
- Caloric tests usually reveal diminished vestibular response.

Treatment

- Active and prompt treatment prevents development of suppurative labyrinthitis (permanent and complete damage of the labyrinth) and spread to meninges.
- If vertigo develops in cases of acute otitis media (AOM), a wide myringotomy and culture and sensitivity should be done quickly.
- In cases of CSOM, modified radical mastoidectomy (without entering into the labyrinth) under the full cover of antibiotics should be done at the earliest.

SUPPURATIVE (PURULENT) LABYRINTHITIS

It is characterized by complete hearing loss (HL) (permanent) and acute vertigo. Vertigo slowly resolves over weeks to months. Patients usually give history of ear discharge, ear pain, preceding cold and cough or meningitis.

Routes of Infection

- *Tympanogenic*: Otitis media, mastoiditis, petrositis, temporal bone fracture, penetrating injury, middle ear surgery and tumors.
- *Meningogenic*: Meningitis.
- *Hematogenic*: From distant or systemic infection or from adjacent areas, such as meningitis, encephalitis or brain abscess.

Etiology

All the causes of serous labyrinthitis can result in suppurative labyrinthitis. The most common causes are AOM, cholesteatoma and meningitis.

Predisposing Factors

- Mondini deformity
- Enlarged vestibular aqueduct
- Stapes surgery
- Weak or dehiscent internal auditory canal opening or foramina into the medial aspect of labyrinth
- Patent cochlear aqueduct
- Labyrinthine fistula.

Clinical Features (Three Stages)

I. *Acute (Manifest) Stage (1–2 Weeks)*:
Rapid onset (0.5–1 hour) of tinnitus, whirling vertigo, pallor, diaphoresis (sweating), nausea and vomiting occur. Symptoms remain unrelenting and may not respond to any treatment for 8–12 hours. Dizziness improves during next few days but head motion evokes severe vertigo and nausea. CNS compensation occurs over next 2-3 weeks.
   - Brisk jerky nystagmus: Horizontal rotary nystagmus has the quick component on affected side for the first day (irritative) but than toward the opposite side (paralytic).
   - Vestibular symptoms peaking during the first few days gradually subside after the first week.
   - Patient lies quietly on the affected ear and cannot stand or sit.
   - Even slight head movements produce vomiting.
   - Complete HL, past pointing and fall occur toward the diseased side.
   - Unilateral labyrinthitis produces more severe vestibular upset than bilateral meningogenic labyrinthitis.

II. *Chronic (Latent or Fibrous) Stage (1–6 Weeks)*:
   - Milder vestibular upset and positional vertigo.
   - Absence of both cochlear and vestibular function.
   - Bilateral lesions lead to difficulty in walking especially in the dark or on a soft surface.

III. *Healed (Compensated or Ossification) Stage (Labyrinthitis Ossificans)*:
   - New bone forms over several months.
   - Many years are required for the development of complete labyrinthitis ossificans.

PERILYMPHATIC FISTULA

(*Synonyms*: *Inner ear fistula, round or oval window fistula*)
These inner ear fistulae provide communication between the perilymphatic space and the middle ear or intramembranous

communication between endolymphatic and perilymphatic spaces. The common sites are oval and round windows.

Etiology

- Barotrauma
- Penetrating trauma
- Surgery: Stapedectomy
- Head trauma
- Explosive blast
- Physical exertion (heavy lifting or straining)
- Spontaneous in cases of congenital HL (congenital malformations such as Mondini's deformity).

Fistula should be ruled out in vertigo cases of congenital SNHL.

Clinical Features

Auditory and vestibular symptoms are variable.

- *Dizziness* variable and include episodic incapacitating vertigo like Ménière's, positional vertigo, motion intolerance, to occasional disequilibrium. Spontaneous inner ear fistula in cases of congenital HL.
- *Tullio's phenomenon*: Vertigo occurs after exposure to loud noises.
- *Hearing loss* (fluctuating/progressive) is usually present but not must.

Investigations

- *Audiogram*: Variable SNHL, high frequency loss to a low frequency loss or flat one. Fluctuating pure tone thresholds and speech discrimination scores. Isolated mild conductive HL.
- *Electrocochleography*: Larger summating potential (nonspecific).
- *Caloric response*: Reduced (nonspecific).

Diagnosis

Diagnosis is confirmed by the following tests:

- *Fraser test*: Improvement in pure-tone threshold or speech discrimination after the patient has been in Trendelenburg position for 30 minutes.
- *Fistula test*: Positive (chapter Evaluation of Dizzy Patients.
- *Surgical exploration*: It is done in cases of positive fistula test.
- *Fiberoptic exploration*: It may be done with rigid scope through myringotomy or flexible endoscope through Eustachian tube.
- *Free amino acid content* of fluid and CSF/perilymph specific protein tau Transferrin differentiate between perilymph and mucosal secretions.

Differential Diagnoses

The clinical picture of inner ear fistulae mimic the following conditions:

- Ménière's disease
- Infection labyrinthitis
- Acoustic neuroma
- Superior SCC dehiscence syndrome
- CNS lesions: Chiari malformations (cerebellar tonsils displaced caudally through foramen magnum): Confirmed by positional downbeat nystagmus and ocular motor findings localizing to vestibulocerebellum.

Treatment

- *Conservative*: Monitoring of vestibular and hearing functions is done along with following measures:
  - Bed rest
  - Head elevation
  - Laxatives to prevent straining and rise in CSF pressure.
- *Surgical*: It is indicated when HL worsens and dizziness persists. Accumulation of fluid in either window requires patching with blood clot, fat or temporal fascia.
  - Outcomes: Dizziness improves much better (49–100%) than hearing (24–49%).

Clinical Highlights

1. **Benign paroxysmal positional vertigo:** BPPV is the most common cause of peripheral vertigo. Predisposing factors for BPPV include age more than 40 years, vestibular neuronitis, head trauma and ear surgery.
2. **Ménière's disease**: It occurs due to raised endolymphatic pressure. The typical features include fluctuating sensorineural hearing loss (SNHL), episodic vertigo, roaring tinnitus and ear fullness. Pulsatile tinnitus and unconsciousness are not the features of Ménière's disease. Because of recruitment, patient develops intolerance to loud sounds.
   a. *Audiogram*: In early Ménière's disease, it shows a rising curve.
   b. *Treatment*: The treatment of the patients, who have failed all medical treatment but retain a serviceable hearing, includes endolymphatic sac decompression, intratympanic gentamicin and vestibular nerve section.
   c. *Cody Tack operation*: It is one of the surgical options in the treatment of Ménière disease.
   d. *Lermoyez syndrome*: In this variant of Ménière's disease, patient first gets hearing loss and tinnitus, which are relieved following the episode of vertigo.
3. **Cogan's syndrome**: The features include episodic vertigo, interstitial keratitis, SNHL and negative serology for syphilis.
4. **Hennebert's phenomenon**: In cases of perilymph fistula, patient develops disequilibrium after nose blowing or heavy weight lifting.
5. **Leukemia**: Hemorrhage into the inner ear can occur in cases of leukemia.

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23 Central Vestibular Disorders

The more this power of concentration, the more knowledge is acquired, because this is the one and only method of acquiring knowledge. In doing anything, the stronger the powers of concentration, the better will that thing be done. This is the one call, the one knock, which opens the gates of nature and lets out floods of light.

—Swami Vivekananda

Points of Focus

- **INTRODUCTION**
- **MIGRAINE**
  - ✓ Basilar Migraine
  - ✓ Treatment of Acute Attacks
  - ✓ Treatment of Vestibular Symptoms
  - ✓ Prophylactic Medications
  - ✓ Lifestyle Changes
- **VERTEBROBASILAR INSUFFICIENCY**
  - ✓ Vascular Risk Factors
  - ✓ Vertebrobasilar Angiography
- **SUBCLAVIAN STEAL SYNDROME**
- **WALLENBERG'S SYNDROME**
  - ✓ Characteristic Ocular Motor Abnormalities
- **CEREBELLAR INFARCTION**
  - ✓ Isolated Cerebellar Infarction
- **CEREBELLAR HEMORRHAGE**
- **MULTIPLE SCLEROSIS**
- **MOTION SICKNESS**
- **PHOBIC POSTURAL VERTIGO**
- **HYPERVENTILATION**
- **AGORAPHOBIA**
- **CERVICAL VERTIGO**
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

Labyrinthine disorders coexisting with central neurological or cardiovascular cause are classified as multifactorial. An idiopathic diagnosis is only made after other causes have been excluded and the patient's dizziness has resolved. Patients with neck pain or occipital headache but no features of otological or neurological disease in whom vertigo is provoked by changes in neck position are diagnosed musculoskeletal (cervical) vertigo.

Migraine is the most common cause of central vestibular disorders. However, some of the other common causes are enumerated in Box 1.

MIGRAINE

Migraine has been reported the most common cause of dizziness in many neuro-otologic clinics.

- Usually begins in first three decades and prevalence peaks in fifth decades.
- Family history is often present.
- Migraine without aura is the most common variety.
- The usual migraine headache is unilateral, throbbing, moderate to markedly severe and aggravated by physical activity. Sensitivity to light and noise are common.
- Migraine headache may be associated with episodic vertigo, motion sensitivity and nonspecific dizziness, such as swimming or rocking sensation inside the head.
- Episodic vertigo (rotational or to-and-fro) usually lasts for few minutes to several hours.
- Vertigo is associated with postural imbalance and unsteadiness with motion sensitivity or visual hallucinations.
- Vertigo after taking certain food.
- Multiple attacks may be several per month.
- Vertigo usually occurs during the headache but may occur during the headache free interval or preceding the headache.

Box 1: Central vestibular disorders

- *Vascular*
  - Ischemic stroke/transient ischemic attack
  - Subclavian steal syndrome
  - Vertebrobasilar transient ischemic attacks
  - Wallenberg's syndrome
  - Lateral pontomedullary infarction
  - Cerebellar infarction
  - Hemorrhage: Brainstem and cerebellar
  - Migraine: Basilar migraine and vestibular migraine
- *Inflammatory*
  - Multiple sclerosis
  - Cerebellitis
  - Susac syndrome
- *Infectious*
  - Intracranial complications of otitis media
  - Meningitis
  - Cerebellar abscess (fungal or bacterial)
- *Craniocervical Junction Disorders*
  - Chiari malformation
- *Metabolic*
  - Wernicke's encephalopathy
  - Diabetes
  - Vitamin B_{12} deficiency
  - Hypothyroidism
  - Hypoparathyroidism
  - Hypoglycemia
  - Hyperventilation
- *Toxic*
  - Medications (methotrexate) and alcohol
- *Degenerative*
  - Parkinson's disease
- *Trauma*
  - Post-concussion syndrome
- *Physiologic*
  - Motion sickness
- *Psychophysiologic*
  - Chronic anxiety
  - Panic disorder
  - Phobic postural vertigo
- *Multisensory Disturbances*
  - Peripheral neuropathy and visual loss
  - Aging: Cerebellar atrophy and diffuse small vessel ischemia

Basilar Migraine

It is a subtype of migraine with aura.

- Recurrent headache usually occipital.
- This is associated with minimum two of the multiple neurologic symptoms of brain stem, cerebellum and occipital lobe (supplied by basilar artery).
- Consciousness is impaired quite often.
- Symptoms of vertigo, fluctuating hearing loss and tinnitus may make basilar migraine difficult to distinguish from Ménière's disease and vertebrobasilar transient ischemic attacks (TIAs).
- Tinnitus is common.
- Low frequency fluctuating hearing loss is common. Sudden profound unilateral sensorineural hearing loss occur occasionally. Migraine related vasospasm can cause cochlear infarction.

Treatment of Acute Attacks

- *Nonspecific*:
  - Analgesics: Aspirin, acetaminophen, nonsteroidal anti-inflammatory drugs and combination of analgesics.
  - Side effects: Medication overuse headache and analgesic rebound headache.
- *Migraine specific:*
  - Triptans: Sumatriptan, naratriptan, rizatriptan, zolmitriptan, almotriptan, frovatriptan and eletriptan.
  - Disadvantages: High cost and contraindications in cardiovascular disorders. Because of the theoretical risk of vasospasm and stroke, triptans are contraindicated in basilar migraine.

Treatment of Vestibular Symptoms

The following antiemetic and antivertiginous agents are used:

- Promethazine 25 mg or 50 mg daily or via suppositories (sedative) for more severe cases.
- Meclizine and dimenhydrinate are less sedative and useful for milder episodes.
- Metoclopramide improves gastric motility, aid absorption of oral drugs and reduces nausea and vomiting.

Prophylactic Medications

Start with a low dose and increase gradually to prevent side effects.

- *Beta-blockers*: Propranolol and metoprolol are useful for preventing migraine vertigo.
- *Tricyclic amines*: Amitriptyline and nortriptyline, which are effective and well tolerated, are advocated as first line medical treatment.
- *Calcium channel blockers*: Verapamil and nimodipine prevent vasospasm of inner ear arteries. They may also stabilize metabolic defect in cases of abnormal calcium channel, such as familial hemiplegic migraine and episodic ataxia type 2 (EA-2).
- *Antiepileptic drugs*: Valproic acid, gabapentine (effective and well-tolerated) and topiramate.
- *Carbonic anhydrase inhibitor*: Acetazolamide presumably changes cerebellar pH and stabilizes defective calcium channels and dramatically effective in EA-2.
- *Benzodiazepines with clonazepam*: Modulate central and peripheral vestibular gamma-aminobutyric acid receptors.
- *Selective serotonin-reuptake inhibitors*: Fluoxetine and venlafexine treat psychiatric symptoms (depression, anxiety, panic attacks and psychophysiologic vertigo), which may develop in vestibular migraine.

Lifestyle Changes

- Regular sleep
- Regular meals
- Exercise:
  - Regular aerobic exercise
  - Tai Chi exercises in cases of mild imbalance
- Increasing the level of physical activity
- Avoiding peaks of stress and troughs of relaxation
- Avoid dietary triggers
- Relaxation training and biofeedback
- Physical therapy and vestibular rehabilitation when daily activities are limited due to dizziness.

Vertigo spells in children: Migraine is not uncommon in children. It must be considered when vertigo spells are not associated with ear malformations or middle ear infections.

VERTEBROBASILAR INSUFFICIENCY

This is a common cause of central vertigo. Transient ischemic attacks (TIA) are common in elderly and usually last for minutes and completely resolve within 24 hours. Presentation can mimic acute peripheral vestibular disorder.

Etiology

- *Common*: Atherosclerosis, embolism and dissection of the vertebral artery (in young persons).
- *Uncommon*: Arteritis and other inflammatory conditions, hypercoagulation disorders, hyperviscosity syndromes.

Atherosclerosis of subclavian, vertebral and basilar arteries can cause thrombosis and infarction in the brainstem. Cerebellum is usually spared due to anastomoses. An embolus from heart or aortic arch can get lodged in distal branch and cause isolated cerebellar infarction.

Vascular Risk Factors

In the presence of following vascular risk factors, vascular causes of vertigo must be ruled out.

- Diabetes
- Hypertension
- Smoking
- Hyperlipidemia

Clinical Features

Vertigo lasts more than a minute but less than several hours.

- Abrupt isolated vertigo is a common presentation of TIAs of labyrinth and/or brain.

Isolated spells of vertigo continuing for more than 3 months usually rule out TIA.

- Accompanied features
  - Visual symptoms (diplopia, visual field defects, blindness and visual illusions and hallucinations) may be followed by drop-attacks, unsteadiness-incoordination and weakness in limbs.
  - Confusion, headache, hearing loss, loss of consciousness, numbness in limbs, dysarthria, tinnitus, perioral numbness are rarely accompanied with episodes of vertigo.

Vertebrobasilar Angiography

The most common site of abnormality is near the origin of anteroinferior cerebellar artery (AICA), which affects the blood supply to labyrinth and flocculus.

Caution

- Isolated episodes of vertigo in elderly patients who have vascular risk factors should be evaluated without delay as an OPD patient.
- The patients of vertigo episodes accompanied with weakness of limbs, blindness and altered consciousness should be urgently evaluated for impending basilar artery thrombosis.
- Vasculitis may simulate vertebrobasilar arteriosclerosis. They include cranial arteritis, polyarteritis nodosum, systemic lupus erythematosus and syphilis. Though rare, they should be kept in mind.

Treatment

- Aspirin alone or in combination with dipyridamole
- Clopidogrel: Newer platelets inhibitor
- Percutaneous stenting in cases who are refractory to medical treatment.

SUBCLAVIAN STEAL SYNDROME

In this syndrome, stenosis of proximal part of subclavian artery occurs prior to the origin of vertebral artery. Siphoning of the blood into the arm occurs due to retrograde blood flow down the vertebral artery.

Clinical Features

- Vertigo exacerbated by exercising the arm
- Signs and symptoms of brain stem ischemia
- Bruit in axilla or supraclavicular region
- Difference in pulse and blood pressure between two arms (20 mm Hg).

WALLENBERG'S SYNDROME

(Lateral Medullary Infarction)

Causes

- Occlusion of vertebral artery is more frequently seen than the posteroinferior cerebellar artery (PICA).
  - Dissection of the vertebral artery in young persons
  - Hypertension and atherosclerosis in elderly
- Demyelinating disease–rare

Though the cerebellum is usually spared, the damage to spinocerebellar tract in inferior cerebellar peduncle causes ataxia.

Clinical Features

- *Partial syndrome*: Only disequilibrium and a tendency to fall to one side.
- *Complete syndrome*: It presents with vertigo, nausea, vomiting, diplopia, severe gait and ipsilateral limb ataxia; ipsilateral Horner's syndrome and facial anesthesia and contralateral hemianesthesia.
  - Bizarre sensations of body and environment
  - Localizing features: Dysphagia, hoarseness and dysphonia; decreased gag and ipsilateral vocal cord weakness

Characteristic Ocular Motor Abnormalities

- *Spontaneous nystagmus*: Mixed horizontal and torsional, usually directed toward normal side.
- *Lateropulsion*: Overshooting (deviation) of eyes to the side of lesion with a blink or closing lids.
- *Smooth pursuit*: Patient follows slowly moving target, no faster than 20°/sec. It is impaired when target moves away from the side of lesion.
- *Saccades*: Keeping the head still, patient alternately fixates physician's nose and then finger, which are held at different positions approximately 15° away from primary position. Vertical saccades have inappropriate ipsilateral horizontal component and appear oblique.

- Torsi pulsion: Inappropriate torsion of eye with horizontal saccades.

- *Skew deviation and ocular tilt reaction*: Skew deviation with ipsilateral hypotropia, ipsilateral head tilt, cyclodeviation (rolling of superior pole of cornea) and subjective visual vertical deviation.
  - Vertical diplopia: One image above the other when one eye is covered with red glass (Maddox rod).
  - Torsional diplopia: One image is tilted with respect to other.
  - Alternate cover test: Look for vertical corrective movement.

CEREBELLAR INFARCTION

Twenty-five percent elderly people suffering from acute isolated vertigo patients have a cerebellar infarction.

Clinical Features

- Vertigo and vomiting, ipsilateral limb ataxia, Horner's syndrome and facial hemianesthesia and contralateral body anesthesia.
- Brainstem features help in localizing feeding vessels PICA, AICA or superior cerebellar artery.
  - *Superior cerebellar artery*: Due to the involvement of medial lemniscus, there is impaired vibration and position sense of contralateral side.
  - *Anteroinferior cerebellar artery*: Unilateral paresis of muscles of mastication.
  - *Posteroinferior cerebellar artery*: Dysphagia, dysphonia.
- *Ocular motor abnormalities*: Saccadic dysmetria is opposite of Wallenberg's syndrome
  - *Contrapulsion*: Contralateral overshooting of saccade and ipsilateral undershooting of saccade.
  - *Vertical saccade*: Inappropriate horizontal component directed toward healthy side.

Isolated Cerebellar Infarction

- Vertigo, vomiting and severe gait ataxia (profound gait imbalance) with relatively little limb ataxia.
- Direction changing or spontaneous downbeat nystagmus, which is not suppressed by visual fixation.

Differential Diagnosis

- *Acute peripheral vestibulopathy*: Unilateral head thrust sign.

Management

- *Monitoring*: First few days monitoring for swelling, brainstem compression and hydrocephalus are vital as they may require neurosurgical intervention.
- *Thrombolysis*: Emergency evaluation for thrombolysis therapy is lifesaving.
- *Neurosurgical*: Management of cerebellar infarction needs prompt neurosurgical intervention. Large infarct produces progressive swelling, which compresses brainstem. Surgical decompression and ventriculostomy relieve hydrocephalus and is lifesaving. Swelling and herniation of cerebellar tonsils can lead to quadriplegia, coma and death.

CEREBELLAR HEMORRHAGE

Headache or stiffness of the neck in cases of brain stroke suggests hemorrhage. Sudden vertigo with multiple neurological findings (localizing signs and symptoms) often rapidly progress to coma and death. Pontine (5%) and cerebellar (10%) hemorrhage constitute 15% of all intracerebral hemorrhages.

Causes

They include hypertension (most common), vascular malformations, intracranial tumors, anticoagulation, bleeding diathesis, amphetamine or cocaine use, trauma and hemorrhagic infarction.

Clinical Features

- Often begins with vertigo, headache, vomiting and an inability to stand and walk.
- Severe ataxia, dysmetria, gaze-evoked nystagmus, stiff neck. Absence of head thrust sign will differentiate it from acute peripheral vestibular disease.
- Abducens and facial palsies can occur from hematoma compression.
- Compression of brainstem due to expanding hematoma can result in rapid coma and death either immediately or after some period of stability.

Management

- *Noncontrast CT or MRI*: Shows acute intraparenchymal hemorrhage. MRI reveals acute infarction not shown on CT.
- Stabilize vital signs and protect airway.
- *Coagulation studies*: Immediate reversal of anticoagulation to prevent expansion of hematoma if patient is taking any anticoagulant.
- Control raised intracranial pressure.
- *Neurosurgical intervention*: Evacuation of hematoma.

MULTIPLE SCLEROSIS

Multiple sclerosis (MS) is a demyelinating disorder of CNS, where myelin is formed from oligodendroglia. Demyelinating plaque affects vestibular nuclei, cerebellum and its peduncles and cranial nerves.

Etiology

- *Cause*: It is not yet known but autoimmunity, infection and heredity may play a role.
- *Age*: 15–50 years. Peak incidence is at age 24 years.
- *Sex*: Females are more affected than males (2:1).

Clinical Features

- Usually manifests as relapsing and remitting or progressive disease. CNS dysfunctions are disseminated in time and space.
- The most common symptoms are vision loss (optic neuritis) and diplopia (bilateral internuclear ophthalmoplegia). Other common findings include weakness, ataxia or sudden hearing loss.
- Myriad neurological features may include

- *Pyramidal signs:* Weakness, hyper–reflexia and Babinski sign.
- *Sensory tracts*: Loss of vibration or joint position sense.
- *Cerebellum*: Ataxia and intention tremor.
- *Brainstem*: Cranial nerves involvement.
- *Partial transverse myelitis*: Paraparesis.

- Vertigo occurs in 50% cases but is initial symptom in only 5% cases.
  - Sustained over days to weeks.
  - *Paroxysmal or positional*: Positional nystagmus with vertigo may be the first manifestation of MS. Confirm with Dix-Hallpike maneuver (chapter Vertigo Evaluation).
  - *Involvement of intrapontine portion of vestibular nerve or nucleus may resemble vestibular neuritis*: Vertigo (hours to days), vomiting, imbalance, direction-fixed (toward normal side) horizontal-torsional nystagmus and canal paresis (caloric test). Though mimicking peripheral lesion, this central nystagmus is not suppressed by visual fixation.
- Sudden hearing loss may accompany vertigo.
- Demyelinating plaque involving vestibular nuclei, cerebellum and its peduncles: Severe ataxia, direction-changing nystagmus, intention tremor or pyramidal signs.
- Eyes
  - Pendular nystagmus often causes oscillopsia and poor vision. It is a common feature.
  - Depending on the horizontal and vertical components, eye movements may be oblique, elliptical, or circular.

Differential Diagnoses

- The inflammatory disorders of labyrinth, which can present with recurrent vertigo, include syphilis, Cogan's syndrome and sarcoidosis and Susac's syndrome (retinocochleocerebral vasculopathy).
- The connective tissue lesions, which can have recurrent vertigo, include systemic lupus erythematosus, rheumatoid arthritis, Behcet's disease and antiphospholipid syndrome.

Supportive Investigations

There is no diagnostic test.

- *MRI*: It shows demyelinating plaques.
- *CSF*: Oligoclonal bands or a raised IgG index are seen.
- *Evoked potential*: Slowing along visual, auditory or somatosensory pathways due to demyelination.
- *Brainstem evoked response audiometry*: Slowing between cochlea and inferior colliculus.

Treatment

- Injection methylprednisolone 500–1,000 mg/day for 3–5 days hastens recovery from acute exacerbations.
- Immunomodulating drugs reduce the relapse rate and slow down the progression of MS.
  - Interferon-beta 1a and 1b
  - Copolymer 1
- Paroxysms of vertigo are sometimes precipitated by hyperventilation and are managed with membrane stabilizing drugs, such as carbamazepine or phenytoin, or with acetazolamide.

MOTION SICKNESS

This common form of physiological dizziness occurs in susceptible individuals usually with prolonged vestibular stimulation. It may also be caused by visual stimulation. This may be induced by real or apparent motion. The condition is the result of the mismatch of information that is reaching the vestibular nuclei and cerebellum from the visual, labyrinthine and somatosensory systems. It is usually managed effectively by labyrinthine sedatives.

Complete bilateral vestibular loss makes the patient resistant to motion sickness.

Migraine and Motion Sickness

- Migraine patients are more prone to motion sickness
- Motion sickness in children may be the starting feature of migraine

Clinical Features

Dizziness, fatigue, pallor, cold sweats, salivation, nausea and vomiting develop when person is aboard a ship, in a car, on an airplane, or in space.

Aggravating Factors (Visual Vestibular Conflict)

- Visual vestibular conflict occurs when person is not able to visualize movement, viz. sitting in an enclosed cabin of a ship.
- Reading in a car and riding in the back seat of a car.
- Migraine patients are more prone to motion sickness.

Relieving Factors (Minimizing Visual Vestibular Mismatch)

- Standing on deck of the ship and focusing on the horizon or land.
- Sitting in the front seat of a car and looking off in the distance.
- Minimizing head movements by resting the head against the headrest of a vehicle.

Treatment

- *Physical Measures*
  - Repeated gradual exposure to motion produces protective habituation.
- *Antihistamines (Block H_1 Receptor in Emetic Center)*
  - Dimenhydrinate (Dramamine), diphenhydramine (Benadryl) and promethazine (Phenergan) though sedative are most effective.
  - Meclizine and cyclizine are less sedative and less effective.
  - Oral administration is advisable an hour before motion exposure.
  - Injection or rectal suppository of promethazine is very effective once vomiting begins.
- *Antiemetics*: Following agents have little or no effect:
  - D_2 receptor antagonists: Metoclopramide and domperidone
  - $5HT_3$ receptor antagonists: Ondansetron.
- *Scopolamine (Acetylcholine Muscarinic Receptor Antagonist)*
  - It is a good prophylactic agent particularly for prolonged exposure at sea.
  - Transdermal preparation is applied to mastoid region four hours before motion exposure. It releases drug gradually for 72 hours.
- *Amphetamine and Ephedrine*
  - Though effective their use is restricted because of addictive potential.

PHOBIC POSTURAL VERTIGO

Phobic postural vertigo patients are obsessively preoccupied with their psychophysiologic dizziness, which is predominantly subjective postural imbalance without falls.

Clinical Features

The following are the characteristic features:

- Subjective imbalance while standing and walking.
- Episodes of fluctuating unsteadiness lasting seconds to minutes.
- Momentary perceptions of illusory body perturbations.
- Though can be spontaneous, following perceptual provoking factors (develop rapid conditioning, generalization and avoidance behavior) are often present:
  - *Places*: Bridge, staircase, empty room or street.
  - *Social situations*: Crowd, shopping malls, concert or restaurant.
- Distressing vegetative symptoms develop with or without anxiety.
- Personality: Obsessive-compulsive type. Labile affect and mild depression are common.
- Begins after particular emotional stress, serious illness and organic vestibular lesion.

Differential Diagnosis

Panic disorder or agoraphobia.

Treatment

- Reassurance
- Encourage regular physical activity
- Avoid too much worrying about symptoms
- *Suggestive and behavioral therapy*: Exposure to feared situations, cognitive restructuring and relaxation therapy.
- Self-controlled desensitization by repeated exposure to situation that triggers vertigo.

HYPERVENTILATION

Though hyperventilation usually induces dizziness in anxious or phobic individuals, it can also cause dizziness in peripheral or central vestibular disorders.

Rapid drop in Pco_2 is presumed to result in cerebral vasoconstriction.

- Symptoms include giddiness, lightheadedness, feelings of suffocation, perioral and acral (extremities such as limbs, fingers, or ears) paresthesias (sensation of burning, pricking, tickling, numbness or tingling).
- Voluntary hyperventilation can reproduce symptoms but may also provoke symptoms from peripheral or central vestibular disorders.

AGORAPHOBIA

(*agora*, marketplace; *phobos*, fear)

- Irrational fear of leaving the familiar setting of home, or venturing into the open, often associated with panic attacks.
- Patients of this type of panic disorder avoid or endure with great distress the feared situation.

CERVICAL VERTIGO OR WHIPLASH VERTIGO

(Musculoskeletal Vertigo)

Vertigo provoked by changes in neck position in patients with neck pain or occipital headache without any otological or neurological disease is diagnosed musculoskeletal vertigo.

Vertigo is usually provoked with movements of neck to the side of injury. It may develop 7–10 days after the accident. Examination reveals tenderness of neck, spasms of cervical muscles and limitation of neck movements.

Exact mechanism of cervical vertigo is not known. It is said to be due to disturbed vertebrobasilar circulation, involvement of sympathetic vertebral plexus or alteration of tonic neck reflexes.

Currently it is believed that the vertigo in these patients is due to following conditions:

- Post-traumatic benign paroxysmal positional vertigo
- Concussion
- Deceleration injury
- Direct trauma to labyrinth

Cervical spondylosis is common in old people and too often blamed for symptoms. It should be considered only when vertigo is clearly associated with movements of the neck (not the head) and appreciable radiological changes.

After ruling out the cervical fracture, Dix-Hallpike examination and audiogram usually clinch the diagnosis. Some patients may need ENG and MRI studies.

Clinical Highlights

1. **Migraine**: Headache may last for hours or days. Females are affected more than males. It can cause neuralgic symptoms.
2. **Lateral medullary syndrome (Posterior inferior cerebellar artery occlusion) or Wallenberg syndrome**: The patients with thrombosis of posterior inferior cerebellar artery (ischemia of lateral part of medulla) presents with vertigo, dysphagia, dysphonia, ataxia, ipsilateral Horner's syndrome and loss of pain and temperature sensation on ipsilateral face and contralateral limbs. The syndrome consists of following features:
   a. Ipsilateral sensorineural hearing loss and vertigo (vestibular and cochlear nuclei lesions).
   b. Analgesia and thermal anesthesia of opposite side of body (spinothalamic tract lesion) and ipsilateral side of face (spinal nucleus of trigeminal nerve).
   c. Dysphagia and drooping of palate (nucleus ambiguus lesion).
   d. Ipsilateral Horner's syndrome (lesion of descending hypothalamic tract): Dryness of face and constriction of pupil.

e. Tendency to fall on same side.
f. Horizontal nystagmus with fast component on opposite side (vestibular nuclei lesion).

3. **Multiple sclerosis**: Optic nerves are most commonly affected because they are direct outgrowths of the CNS and their axons have myelin sheaths formed by oligodendrocytes. Patients may present with other symptoms such as vertigo, sudden hearing loss or facial palsy.

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24 Facial Nerve Disorders

The world is ready to give up its secrets, if we only know how to knock, how to give it the necessary blow. The strength and force of the blow come through concentration. There is no limit to the power of the human mind.

—Swami Vivekananda

Points of Focus

- **PERTINENT ANATOMY**
- **SURGICAL LANDMARKS**
- **CLINICAL EVALUATION**
  - ✓ House-Brackmann System of Grading
- **PATHOPHYSIOLOGY OF NERVE INJURY**
- **SUNDERLAND CLASSIFICATION**
- **DIFFERENCES BETWEEN UPPER AND LOWER MOTOR NEURON PALSY**
  - ✓ Upper Motor Neuron Facial Palsy
  - ✓ Lower Motor Neuron Facial Palsy
- **INVESTIGATIONS**
  - ✓ Electrical Tests: Nerve Excitability Test, Maximal Stimulation Test, Electroneurography, Electromyography, Nerve Conduction Velocity
  - ✓ Topodiagnostic Tests: Schirmer's Test, Stapedial Reflex, Electrogustometry, Submandibular Salivary Flow
  - ✓ Imaging: HRCT and MRI
- **CAUSES OF FACIAL NERVE PALSY**
- **SEQUELAE/COMPLICATION OF FACIAL NERVE PALSY**
- **BELL'S PALSY**
- **RECURRENT FACIAL PALSY**
- **MELKERSSON SYNDROME**
- **RAMSAY HUNT SYNDROME**
- **TEMPORAL BONE FRACTURE**
- **LYME DISEASE**
- **SARCOIDOSIS**
- **MOBIUS SYNDROME**
- **IATROGENIC OR SURGICAL TRAUMA**
- **HYPERKINETIC DISORDERS OF FACIAL NERVE**
- **SURGICAL TREATMENT OF FACIAL PALSY**
- **CLINICAL HIGHLIGHTS**

PERTINENT ANATOMY

The facial nerve (CN VII) is the nerve of second branchial arch. Its sensory root is also called nerve of Wrisberg and carries both secretomotor as well as gustatory fibers.

Functional Divisions

It is a mixed nerve and consists of following types of fibers:

1. *Branchial motor fibers*: They supply muscles of facial expression (develop from the 2nd branchial arch) and arise from the motor facial nucleus that is situated in the medial side of floor of the IV ventricle in the pons.
2. *Parasympathetic preganglionic secretomotor fibers*: They supply to the lacrimal gland, submandibular and sublingual salivary glands and the glands in the nose and palate. They arise from the superior salivary and lacrimatory nuclei that are situated near the motor facial nucleus in pons.
3. *Gustatory fibers*: They carry taste sensation from the anterior two-third of tongue and go to the nucleus of solitary tract.
4. *General sensations*: They are carried from the concha and retroauricular skin.

Motor Nucleus of Facial Nerve (Fig. 1)

Motor nucleus of facial nerve is situated in the pons near the nucleus of abducens. Upper part of the nucleus which innervates

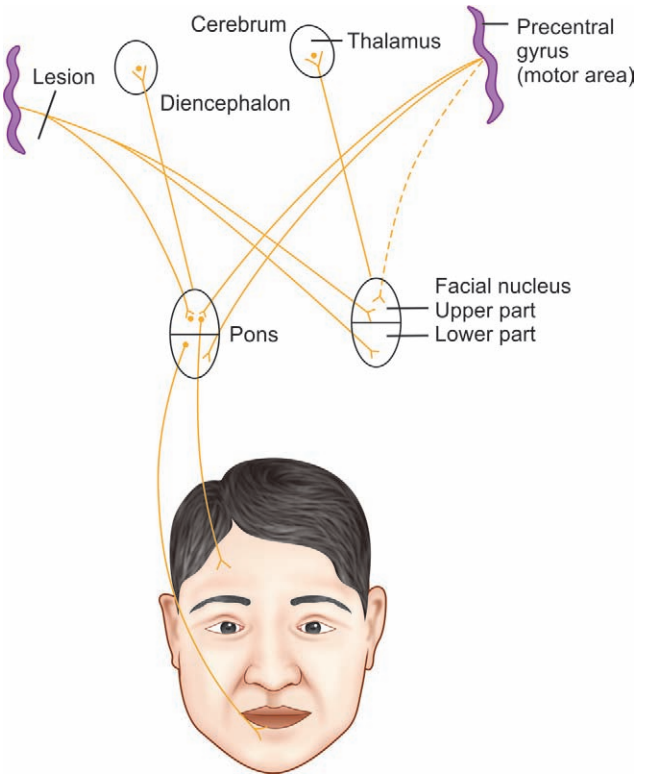


Fig. 1: Facial nerve nucleus. Upper part of nucleus, which supplies frontalis muscle of forehead, receives supranuclear corticonuclear fibers from both the sides. Therefore in unilateral supranuclear facial palsy frontalis muscle is spared

forehead muscles receives corticonuclear fibers from both the cerebral hemisphere, while the lower part of nucleus that supplies lower face gets only crossed fibers from opposite side of cerebral hemisphere. Facial motor nucleus also receives fibers from the thalamus by alternate routes and provides involuntary emotional expressions such as happy, sad, depressed and angry faces. This explains why frontalis muscle and emotional facial expressions remain intact in supranuclear lesions.

Course of Facial Nerve

The course of facial nerve is divided into three parts: intracranial, intratemporal and extracranial.

1. *Intracranial course*: Motor fibers first hook round the nucleus of 6th nerve and then are joined by the sensory root (nerve of Wrisberg). Facial nerve along with the vestibulocochlear and abducens nerves leaves the brainstem at pontomedullary junction. It travels through the cerebellopontine angle (CPA) and along with vestibulocochlear nerve enters the internal auditory canal (IAC). At the fundus of IAC facial nerve enters the bony fallopian canal.
2. *Intratemporal course* (Fig. 2): The part of the facial nerve, from internal acoustic meatus to stylomastoid foramen, is further divided into four segments: meatal, labyrinthine, tympanic and mastoid.
   a. Meatal segment: It lies within IAC. The meatal foramen is the narrowest aperture of facial canal. The length of facial nerve from brainstem to IAC is 23–24 mm.
   b. Labyrinthine segment (3–5 mm): It extends from IAC fundus (meatal foramen) to the geniculate ganglion (may be dehiscent) where facial nerve takes a posterior turn forming the first "genu". The bony fallopian canal in the labyrinthine segment is narrowest and more prone to compression in Bell's palsy.
   c. Tympanic or horizontal segment (8–11 mm): It extends from geniculate ganglion to just above the pyramidal eminence, where it turns inferiorly and makes the second genu. The tympanic segment lies above the oval window (dehiscence in 15–30%) and below the lateral semicircular canal.
   d. Mastoid or vertical segment (10–14 mm): It extends from the pyramid to the stylomastoid foramen.
3. *Extracranial course*: The facial nerve comes out of the temporal bone through the stylomastoid foramen. Here it crosses the styloid process and enters into the parotid gland. This extracranial part from stylomastoid foramen to the termination of its peripheral branches (Fig. 3) is situated in the substance of parotid gland.

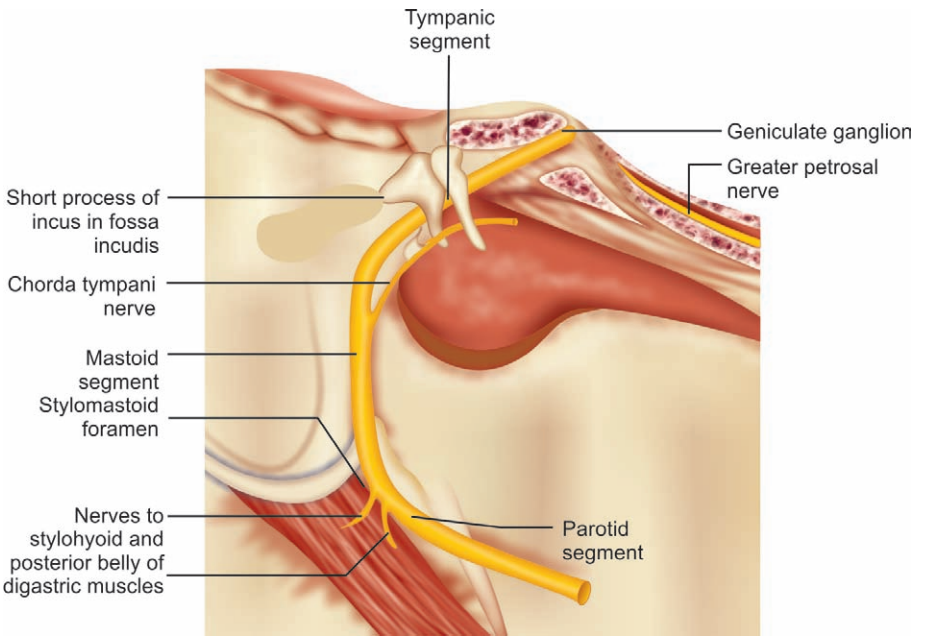


Fig. 2: Intratemporal course of facial nerve

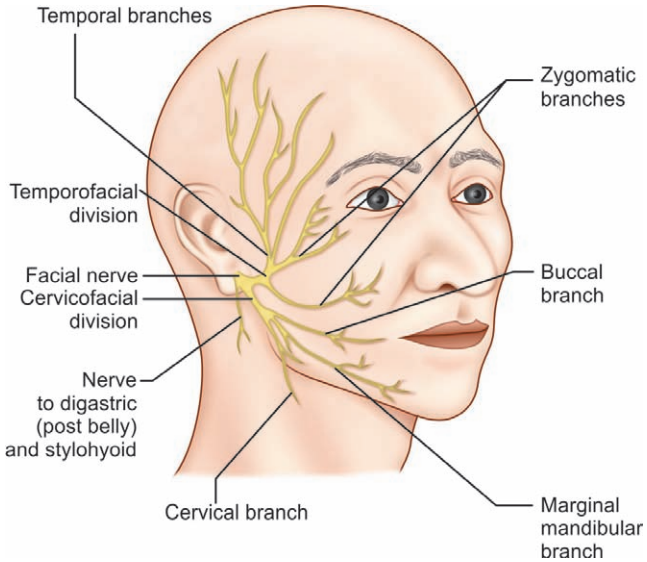


Fig. 3: Branches of facial nerve after emerging from stylomastoid foramen

Branches of Facial Nerve

1. *Greater superficial petrosal nerve*: This first branch of the facial nerve arises from geniculate ganglion and carries preganglionic secretomotor fibers to lacrimal gland and the glands of nasal mucosa.
2. *Nerve to stapedius*: This branch arises at the level of second genu and carries motor fibers to the stapedius muscle.
3. *Chorda tympani*: It arises from the middle of mastoid vertical segment and passes between the incus and neck of malleus. It leaves the middle ear cavity through petrotympanic fissure. It carries preganglionic parasympathetic secretomotor fibers to submandibular and sublingual salivary glands and gustatory fibers (taste sensation from anterior two-third of tongue).
4. *Communicating branch*: It joins auricular branch (Arnold's nerve) of vagus and supplies the concha, retroauricular groove, posterior meatus and the outer surface of tympanic membrane.
5. *Posterior auricular nerve*: It supplies muscles of pinna and occipital belly of occipitofrontalis muscle.
6. *Muscular branches*: They supply to stylohyoid and posterior belly of digastric, which are developed from the 2nd branchial arch.
7. *Terminal branches*: The facial nerve, after crossing the styloid process, divides into two terminal divisions upper temporofacial and a lower cervicofacial. The smaller terminal branches from these two divisions include temporal, zygomatic, buccal, mandibular and cervical. This network of terminal divisions and branches of facial nerve supply all the muscles of facial expression (except levator palpabri superioris) and form pes anserinus (goose-foot). Table 1 shows the terminal branches of facial nerve in parotid gland, the muscles they supply and the methods of their testing.

Blood Supply

Following are the arteries and areas of facial nerve supplied by them:

1. *Labyrinthine artery, branch of anterior inferior cerebellar artery*: Meatal segment within the IAC.
2. *Petrosal artery, branch of middle meningeal artery*: Perigeniculate area.
3. *Stylomastoid artery, branch of posterior auricular artery*: Mastoid and tympanic segments.

TABLE 1 The terminal branches of facial nerve in parotid gland, the muscles they supply and methods of their testing

Branch	*Muscles supplied*	*Facial movement*
Temporal	Frontalis, corrugator supercilii, procerus and upper orbicularis oculi	Raising of eyebrows
Zygomatic	Lower orbicularis oculi	Tight shutting of eye
Buccal	Zygomaticus major and minor, levator anguli oris, buccinator and upper orbicularis oris	Showing of teeth
Ramus mandibularis	Lower orbicularis oris, depressor anguli oris, depressor labii inferioris and mentalis	Whistle or puckering lips
Cervical	Platysma	Contraction of platysma

SURGICAL LANDMARKS

Middle Ear Cleft

- *Processus cochleariformis*: It presents the site of the geniculate ganglion which lies just anterior to it. It shows the beginning of horizontal tympanic segment.
- *Oval window and horizontal semicircular canal*: The tympanic segment is situated above the oval window and below the horizontal semicircular canal.
- *Short process of incus*: Facial nerve lies medial and inferior to the short process of incus at the level of aditus ad antrum.
- *Pyramid*: Facial nerve is situated posterior to the pyramid and the tympanic sulcus.
- *Tympanomastoid suture*: The vertical mastoid segment of facial nerve lies about 6–8 mm deep to this suture and always runs behind the level of this suture.
- *Digastric ridge*: Facial nerve leaves the mastoid through the stylomastoid foramen, which is situated at the anterior end of digastric ridge.

Parotid Gland

- *Tragal cartilage pointer*: The tragal cartilage ends on a point. It is a sharp triangular extension of tragal cartilage, which seems to point towards facial nerve. Facial nerve lies 1–1.5 cm medial and inferior to tragal point surrounded by a small aggregation of fat and overlain by a small vessel.
- *Tympanomastoid suture*: Facial nerve lies 6–8 mm deep to the suture.
- *Styloid process*: Facial nerve lies on the posterolateral aspect of the styloid process near its base.
- *Posterior belly of digastric*: Follow the posterior belly of digastric up to 5 mm below the bony meatal edge. The facial nerve lies between the mastoid and the posterosuperior part of the posterior belly of digastric muscle. The facial

nerve passes downwards, forwards and laterally immediately above the upper border of digastric posterior belly.

- *Mastoid process*: Follow the anterior border of mastoid process up to the vaginal process of tympani bone. Facial nerve bisects the tympanomastoid angle at the tympanomastoid suture.
- *Peripheral branches*: The following branches may be followed proximally:
  - *Temporal*: It bisects a line drawn from tragus to lateral canthus of eye.
  - *Buccal:* It runs 1 cm above and parallel to Stensen's duct over the masseter.
  - *Ramus mandibularis*: It travels superficial to the facial vessels 2 cm below inferior border of mandible and 1 cm anterior to angle of mandible.

CLINICAL EVALUATION OF FACIAL PALSY

The following elements of history and physical examination are important in evaluating a patient of facial nerve palsy:

- Onset: Sudden or gradual; events preceding the onset of facial nerve palsy such as trauma, surgery, ear pain, exposure to extreme of temperature.
- Unilateral/bilateral.
- Duration and progression.
- Associated symptoms: Ear discharge, vertigo, dysphagia, dysphasia.
- Past history of facial palsy.
- Major medical illness.
- Complete ear, nose, throat, head and neck examination including ear microexamination, hearing tests and audiometry.
- Central nervous system examination: Especially cranial nerves, cerebellum and motor system.
- Determine whether the palsy is complete or incomplete; segmental or uniform involvement.
- Topodiagnostic tests such as Schirmer.

House-Brackmann System of Grading Facial Nerve Palsy

Facial weakness can be subtle, moderate, near total or total. House-Brackmann system of grading facial nerve palsy (Table 2) has been widely used (endorsed by the American Academy of Otolaryngology-Head and Neck Surgery).

PATHOPHYSIOLOGY OF NERVE INJURY

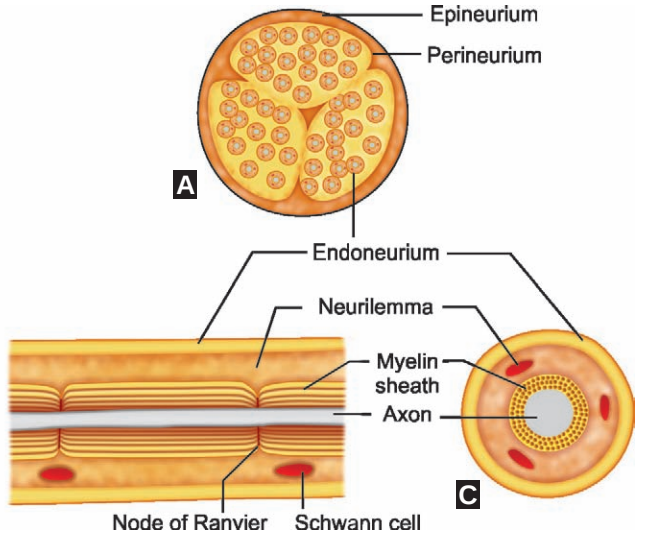
A nerve fiber consists of axon, myelin sheath and neurilemma and is covered by endoneurium (Figs 4A to C). A bundle of nerve fibers forms a fascicle, which is enclosed in a sheath called perineurium. The fascicles are bound together by epineurium.

The degree of nerve injury determines the degeneration and regeneration of nerve and its function. Traditionally nerve injuries are divided into three types: neuropraxia, axonotmesis and neurotmesis.

SUNDERLAND CLASSIFICATION (FIG. 5)

Based on anatomical structure of the nerve, Sunderland classified the severity of nerve injuries into five degrees. This classification is now widely accepted. The third, fourth and fifth degrees are types of neurotmesis.

1. *First degree (obstruction to axoplasm) neuropraxia*: In this conduction block, flow of axoplasm through the axons is obstructed. No morphological changes are seen. In this type of injury, recovery of function is complete. Nerve excitability test (NET) and maximum stimulation test (MST) are normal. Electromyography (EMG) fails to show voluntary motor action potential as they are not conducted across the blockade.
2. *Second degree (injury to axon) axonotmesis*: There is loss of axons, but endoneurial tubes remain intact. Wallerian



Figs 4 A to C: Nerve structure. (A) Cross-section of nerve; (B) Longitudinal section of nerve fiber; (C) Cross-section of nerve fiber

TABLE 2 House-Brackmann system of grading facial nerve palsy

Grade	*Dysfunction*	*Symmetry at rest*	*Frontalis motion*	*Effort in eye closure*	*Motion of mouth angle*
I	Absent	Normal facial functions in all areas			
II	Mild	Normal	Moderate to good	Minimum	Slight asymmetry
III	Moderate	Normal	Slight to moderate	Moderate	Moderate asymmetry
IV	Moderately severe	Normal	None	Incomplete closure	Moderate with great effort
V	Severe	Absent	None	Incomplete closure	Slight
VI	Total	No movement in any area			

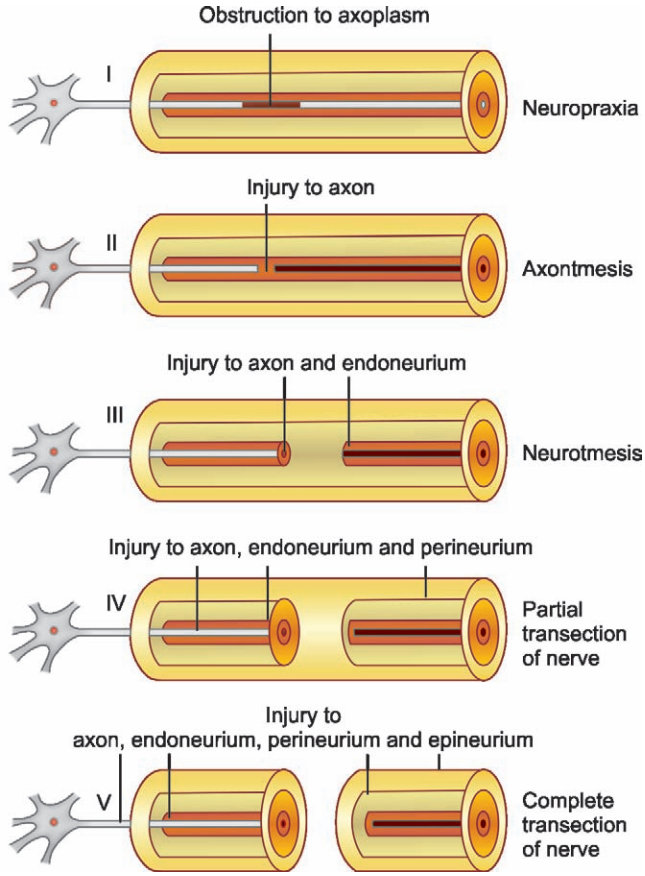


Fig. 5: Sunderland histologic classification of peripheral nerve injury is helpful in understanding the results of electrical tests

degeneration occurs distal to the lesion. Electrical tests show rapid and complete degeneration, with loss of voluntary motor units. During regeneration, axons will grow into their respective tubes. Recovery is good.

3. *Third degree (injury to endoneurium) neurotmesis*: It is an injury to nerve fiber along with both Wallerian degeneration and loss of endoneurium. Electrical tests show complete nerve degeneration. During regeneration, axon of one tube can grow into another. There are chances of synkinesis.
4. *Fourth degree (injury to perineurium)*: Partial transaction of nerve occurs. Scarring occurs and impairs regeneration of axons.
5. *Fifth degree (injury to epineurium)*: There occurs complete nerve transaction.

The first three degrees are seen in viral and inflammatory lesions of the nerves. The fourth and fifth degrees happen in surgical and accidental traumas and in neoplasms.

Facial nerve regeneration: The regeneration and degree of return to normal is dependent on the degree of initial injury (neuropraxia vs. neurotmesis). The most important factor in history is whether the palsy developed slowly over days or immediately at the time of the injury.

DIFFERENCES BETWEEN UPPER AND LOWER MOTOR NEURON PALSY

Upper Motor Neuron Facial Paralysis

The paralysis is of only the lower half of face (Fig. 6) on the contralateral side. Frontalis muscle movements are retained due to bilateral innervation of upper part of motor facial nucleus.

Fig. 6: Lower half left facial palsy. Note patient can close her both eyes. This patient also had associated left side abducent nerve lateral rectus palsy

- Involuntary emotional expressions and the tone of facial muscles remain intact.
- Central facial paralysis is caused by cerebrovascular accidents (hemorrhage, thrombosis or embolism), tumor or an abscess.

Lower Motor Neuron Facial Paralysis

In the peripheral facial paralysis, all the ipsilateral muscles of the face become paralyzed. Patient is unable to frown, close the eye, purse the lips and whistle.

- *Nuclear palsy*: It is identified by associated paralysis of 6th nerve, the nucleus of which is situated near to the motor nucleus of facial nerve.
- *Cerebellopontine angle lesions*: They are usually associated with vestibular and auditory defects such as vertigo and sensorineural hearing loss (SNHL).
- *Lesion in the bony fallopian canal*: From internal acoustic meatus to stylomastoid foramen the lesions can be localized with the help of topodiagnostic tests.
- *A lesion in the parotid area*: It affects only the terminal branches of the nerve, which may be involved by the tumor or injury.

INVESTIGATIONS

Electrical Tests

These tests are useful not only in differentiating between the neuropraxia and degeneration of the nerve but also help in predicting prognosis and indicating the time for surgical decompression.

Nerve Excitability Test (NET)

This test measures the threshold of stimulation of both sides of facial nerve and compares them. A 1/sec^2 wave pulse which is 1 m/sec in duration is applied. The steadily increasing intensity till facial twitch appears tells the threshold.

- *Interpretations*: They are:
  - Conduction block: There is no difference between the normal and paralyzed side.

- Degeneration: The difference between normal and paralyzed sides exceeds 3.5 mA.

- *Limitations*: Degeneration of nerve fibers cannot be detected earlier than 3 days. When degeneration sets in, nerve excitability is gradually lost. So the patient should come after 3 days and before 3 months.

Maximal Stimulation Test (MST)

Instead of measuring the threshold of stimulation, it measures the current level which gives maximum facial movement and compares it with the normal side. Maximal stimulation indicates degeneration and incomplete recovery.

- *Advantage*: MST becomes abnormal earlier than NET. So it is a better prognostic indicator.
- *Disadvantage*: Response is subjectively graded as equal, decreased or absent.

Electroneurography

In this evoked electroneurography (ENoG), the facial nerve is stimulated and the compound action potentials (amplitudes of the summation potentials) from the facial muscles are recorded and measured objectively. Supra-maximal level of current is applied over the main trunk of facial nerve. The readings are compared with the normal side.

The peak-to-peak amplitude is directly proportional to the number of intact motor axons. So the test assesses extent of neuronal degeneration. The response of paralyzed side is reported as a percentage of response on normal side, thus telling the proportion of fibers that have degenerated.

- *Interpretation*: Fall of summating potential to 10% of the normal value is an indication (90% degeneration) for the surgical decompression.
- *Limitation*: It must be done within the two weeks of the onset of palsy.

Electromyography

It records spontaneous activity of facial muscles at rest and on voluntary contraction. Electrode is directly inserted into the muscle. The test provides information regarding intact motor units in acute phase and detects reinnervation potentials.

- *Interpretations* : They are–
  - Normal: At rest, normal muscle does not show any electrical activity. On voluntary contraction, normal volitional motor unit potentials are observed.
  - Denervated muscle: Fibrillation potentials appear within 14–21 days after denervation.
  - Earliest signs of recovery: Reinnervation potentials can be seen much before (up to 12 weeks) any visible facial movement.
- *Limitation*: It cannot assess the degree of degeneration or prognosis for recovery.

Nerve Conduction Velocity (NCV)

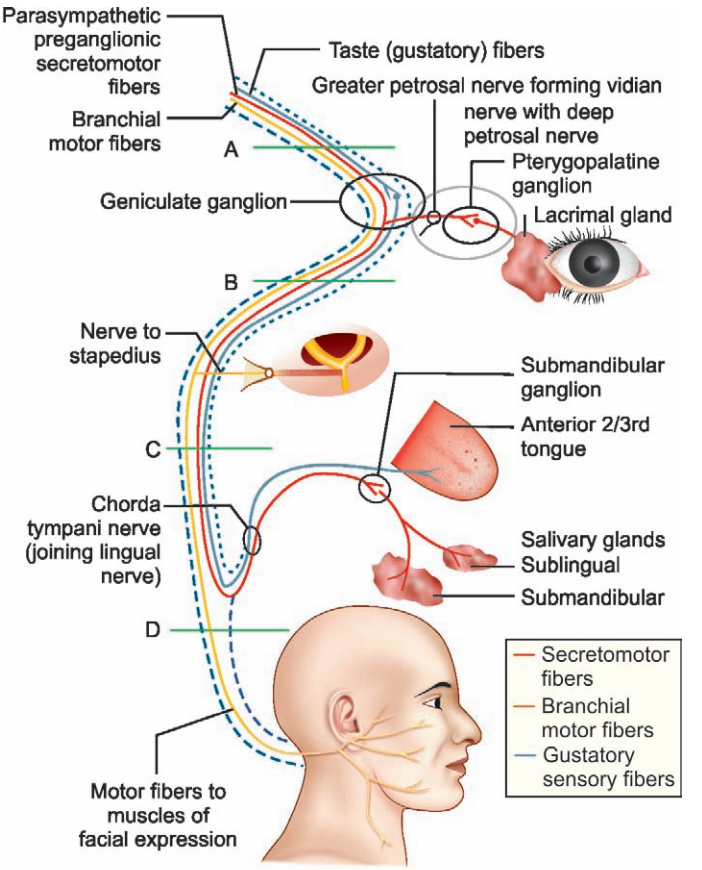
NCV can be measured between the stylomastoid foramen and the ramus mandibularis. A strong correlation exists between decrease in NCV and decrease in compound nerve action potential EnoG during the first 2 weeks of facial palsy.

- *Interpretations:* They are as follows:
  - 37–58 m/sec: Normal
  - 20–30 m/sec: 50% chance of residual paresis
  - < 10 m/sec: Poor prognosis

Topodiagnostic Tests

Topodiagnostic tests are done to find the site of lesion in the intratemporal segment of facial nerve (Figs 7A to D).

1. *Schirmer's test* (Fig. 8): A strip of filter paper (5 mm x 35 mm) is hooked in the lower fornix of each eye and kept for 5 minutes. The length of wetting of strip is measured. The lacrimation of the two sides is compared.
   Interpretations: 25–30% decrease in lacrimation indicates that lesion is proximal to the geniculate ganglion. The greater superficial petrosal nerve carrying secretomotor fibers to lacrimal gland arises from the geniculate ganglion.
2. *Stapedial reflex*: Stapedial reflex is lost in the lesions that lie above the nerve to stapedius. It is tested by tympanometry (see chapter on Hearing Evaluation).
3. *Taste test or Electrogustometry*: It measures function of chorda tympani. A drop of salt or sugar solution is placed on one side of the protruded tongue. The electrogustometry is another method of testing taste sensations. Impairment of taste sensation indicates that lesion is above the origin of chorda tympani.



Figs 7A to D: Topographical lesions of facial nerve. (A) Lesion above the geniculate ganglion damages the motor fibers to facial muscles and stapedius, secretomotor fibers to lacrimal and submandibular salivary glands, and taste fibers; (B) Lesion between the geniculate ganglion and nerve to stapedius spare secretomotor fibers to lacrimal gland; (C) Lesion between the nerve to stapedius and chorda tympani nerve spares stapedial reflex and lacrimation; (D) Lesion below chorda tympani nerve affects only muscles of facial expression

Fig. 8: Schirmer's Test. A strip of filter paper hooked in the lower fornix of each eye. The length of wetting of strip measures lacrimation of the two sides

4. *Submandibular salivary flow*: Polythene tubes are passed into both sides of Wharton's ducts. The drops of saliva are counted during 1 minute period. It also measures function of chorda tympani. Decreased salivation indicates that lesion is above the origin of chorda tympani.

Imaging

It is not indicated in each and every case of facial nerve palsy. If required high resolution computerized tomography (HRCT) or MRI, is the study of choice.

1. *High resolution computerized tomography*: It is the test of choice in cases of skull base trauma/temporal bone fracture. The common sites of injuries are following:
   a. Perigeniculate ganglion area: In longitudinal fracture of temporal bone
   b. Tympanic segment: In middle ear surgeries
   c. Mastoid segment: In mastoid surgeries
2. *Gadolinium-enhanced MRI*: It is the imaging of choice in cases of inflammatory and neoplastic lesions. Labyrinthine segment and meatal foramen is the site of lesion in cases of Bell's palsy and Ramsay Hunt syndrome.

CAUSES OF FACIAL NERVE PARALYSIS

The cause may be central or peripheral. The peripheral lesion may involve the nerve in its intracranial, intratemporal or extratemporal parts. Peripheral lesions are more common and about two-third of them are of the idiopathic variety. The various causes of facial nerve palsy are listed in Box 1.

Box 1: Causes of facial palsy

- *Central*: Brain abscess, pontine gliomas, poliomyelitis, multiple sclerosis and cerebrovascular strokes
- *Cerebellopontine Angle Tumors*: See chapter on Tumors of Ear and CPA
- *Intratemporal Part*
  - *Idiopathic*: Bell's palsy, recurrent facial palsy and Melkersson's syndrome

Contd...

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  - *Infection*: Acute and chronic suppurative otitis media, Herpes zoster oticus, tuberculosis and malignant otitis externa
  - *Surgical trauma*: Mastoidectomy and stapedectomy
  - *Accidental trauma*: Fractures of temporal bone
  - *Neoplasms*: Malignancy of external and middle ear, rhabdomyosarcoma, histiocytosis, leukemia, glomus tumors, facial nerve neuroma, metastasis to temporal bone (from cancer of breast, bronchus, prostate)
- *Parotid*: Malignancy, surgery, accidental injury and birth trauma
- *Congenital*: Compression injury, Mobius syndrome, lower lip paralysis
- *Systemic Diseases*: Diabetes mellitus, hypothyroidism, uremia, polyarteritis nodosa, Wegener's granulomatosis, sarcoidosis (Heerfordt's syndrome), leprosy, Guillain-Barre syndrome, Lyme's disease, AIDS, infectious mononucleosis and autoimmune diseases

Bilateral Facial Nerve Paralysis

The causes of bilateral facial nerve paralysis include:

- Guillain-Barre syndrome
- Lyme disease
- Mobius syndrome
- Sarcoidosis
- Temporal bone fractures

SEQUELAE/COMPLICATION OF FACIAL NERVE PALSY

Peripheral facial paralysis may result in following complications:

- *Incomplete recovery*: It can result in facial asymmetry, epiphora (due to inability to close eye), drooling of saliva and difficulty in chewing food (due to weak oral sphincter). Drooling during chewing and drinking and impairment of speech may result in social problems.
- *Exposure keratitis*: Evaporation of tears from the opened eye results in dryness that leads to keratitis and corneal ulcer. The condition becomes worst when lesion is above the geniculate ganglion. The findings of corneal irritation are redness, itching, foreign body sensation and visual blurring.
- *Synkinesis (mass movement)*: While closing the eye corner of mouth twitches. It may happen vice versa. It occurs due to cross innervation of fibers.
- *Tics and spasms*: Involuntary movements of the facial muscles occur, which are due to the faulty regeneration of fibers.
- *Contractures*: The fixed contraction or fibrosis of atrophied muscles affects movements of face. The facial symmetry is maintained at rest.
- *Crocodile tears (gustatory lacrimation)*: The condition is characterized by the ipsilateral lacrimation during the mastication. The faulty regeneration of parasympathetic fibers occurs and they supply lacrimal gland instead of the salivary glands. Treatment consists of the section of greater superficial petrosal nerve or tympanic neurectomy.
- *Frey's syndrome (gustatory sweating)*: This condition is characterized by the ipsilateral sweating and flushing of skin over the parotid during mastication. It happens in some cases of parotid surgery.
- *Psychological and social problems.*

BELL'S PALSY

Bell's palsy is an idiopathic demyelinating disease. It is characterized by an acute isolated unilateral lower motor neuron facial paralysis.

- It accounts for over 50% of acute facial palsies.
- Both sexes are equally affected.
- There is no age bar but incidence rises with increasing age.
- Risk factors include diabetes (angiopathy) and pregnancy (retention of fluid).

Etiology

- *Viral infection*: Many reports suggest viral infections due to *Herpes simplex*, *Herpes zoster* and *Epstein-Barr virus*. Some consider Bell's palsy a part of the polyneuropathy. Other cranial nerves may also be involved.
- *Vascular ischemia*: Primary ischemia may be induced by cold or emotional stress. It causes increased capillary permeability that leads to exudation of fluid, edema and compression of microcirculation of the nerve (secondary ischemia).
- *Hereditary*: About 10% of patients give positive family history. The narrow fallopian canal (may be due to hereditary predisposition) can make the nerve susceptible to early compression at the slightest edema.
- *Autoimmunity*: T-lymphocyte changes have been seen.

Clinical Features

This sudden onset of complete or incomplete isolated unilateral lower motor neuron facial palsy may present (Figs 9A and B) with following features:

- Inability to close eye.
- Bell's phenomenon: When patient tries to close the eye, eyeball turns up.
- Dribbling of saliva from the angle of mouth.
- Asymmetrical face.
- Epiphora: Tears flowing down from the eye.
- Earache: Ear pain may precede or accompany the facial palsy.
- Hyperacusis: Sensitivity to loud sounds due to stapedial palsy.
- Diminished taste sensation: It may occur due to the involvement of chorda tympani.
- Recurrence: Bell's palsy is recurrent either ipsilateral or contralateral in 12% of patient.

Diagnosis

Diagnosis of Bell's palsy is usually clinical and by exclusion. Patient requires careful history and examination to exclude other known causes of facial paralysis.

- *Laboratory tests*: Some patients need radiological studies, complete blood count (CBC), peripheral blood smear, sedimentation rate, blood sugar and serology.
- *Nerve excitability tests*: They are done daily or on alternate days to monitor nerve degeneration.
- *Topodiagnostic tests*: They help in establishing the cause and site of lesion.

Differential Diagnoses

The presence of any of the following features rule out the diagnosis of Bell's palsy:

- Preceding history of temporal bone trauma or surgery.
- Presence of acute suppurative otitis media (ASOM), cholesteatoma, vesicles in and around ear, glomus tumors, malignant tumors of ear and CPA, CNS diseases (such as strokes).
- Multiple cranial nerve palsies.
- Bilateral facial nerve paralysis.
- Facial palsy at birth.
- Gradual onset of facial paralysis.
- Failure to recover within 6 months.

Treatment

Regular electrophysiological assessment is important to know the extent of nerve damage and determine the need of surgical decompression.

A. General Measures.

1. Reassurance.
2. *Analgesics*: For the relief of ear pain.
3. *Eye Care*: Eye must be protected against exposure keratitis. The preventive measures include–
   a. Artificial tears (methylcellulose drops) every 1–2 hours and 4–5 times per day.
   b. Eye ointment followed by patching or taping the eye.
   c. Cover for the eye in night.
   d. Protect the eye from wind, foreign bodies and drying with glasses and moisture chambers.
   e. Temporary tarsorrhaphy may be needed in some cases.
4. *Physiotherapy*: The facial muscles massage though does not influence recovery, gives psychological support. Active facial movements should be encouraged.

B. Medical Treatment

1. *Steroids*: Though their utility has been doubtful they are used by many, if patient reports within 1 week. Steroids have been reported to prevent incidence of synkinesis and crocodile tears and shorten the recovery time. Prednisolone, 1 mg/kg/day divided into morning and evening doses for 5–10 days depending upon whether the paralysis

Figs 9A and B: Right Bell's facial palsy. (A) Note the inability to close the eye and move the angle of mouth and upward rolling of eye ball (Bell's phenomenon); (B) Recovery of facial palsy

is incomplete or is recovering. Thereafter the doses are tapered in next 5–10 days.

Contraindications: Pregnancy, diabetes, hypertension, peptic ulcer, pulmonary tuberculosis and glaucoma.

2. *Acyclovir*: Steroids are generally combined with acyclovir.
3. *Other drugs*: Vasodilators, vitamins, mast cell inhibitors and antihistaminics have not been found useful.

C. Surgical Facial Nerve Decompression

It relieves pressure on the nerve fibers and improves their microcirculation. Both the vertical and tympanic segments of nerve are decompressed. Some favor decompression of the whole fallopian canal including labyrinthine segment. The approaches include postaural and middle fossa.

Prognosis

- Majority of the patients (85–90%) recover fully.
- Ninety five percent patients of incomplete Bell's palsy recover completely.
- The chances of complete recovery are better when clinical recovery begins within three weeks of onset.
- Some of the patients (10–15%) do not recover completely and some stigmata of regeneration remain.
- Recurrent facial palsy may not recover fully.

RECURRENT FACIAL PALSY

- In 3–10% cases of Bell's palsy, they get recurrent episodes of facial palsy.
- In cases of unilateral recurrent facial palsy, facial nerve neuroma should be ruled out.
- Other causes include Melkersson's syndrome, diabetes, sarcoidosis and tumors.

MELKERSSON'S SYNDROME

- This idiopathic disorder has a triad of facial paralysis, swelling of lips and fissured tongue.
- Patients get recurrent attacks of facial palsy.
- Treatment is similar to Bell's palsy.

RAMSAY HUNT SYNDROME OR HERPES ZOSTER OTICUS (VARICELLA-ZOSTER VIRUS)

The disease is caused by Varicella-Zoster Virus (VZV) that remains latent in geniculate ganglia for decades after an attack of acute chickenpox. VZV starts replicating first in geniculate ganglia and then travels down and affect facial nerve, inner ear, spiral and vestibular ganglia. Ganglia of trigeminal and VIII CN have also been implicated (Figs 10A and B).

Clinical Features

- Prodrome of severe otalgia.
- Stimuli for viral reactivation: Immunosuppression, physical and psychological stressors.
- Painful vesicles with erythematous base appear in the canal and concha, behind pinna and or soft palate. Later vesicles rupture and form crusts.
- Unilateral facial palsy and deep ear pain manifest 1–2 days later. Facial paresis usually recovers over weeks.
- About 25% patients have vertigo, nystagmus, tinnitus and hearing loss.

Figs 10A and B: Recovering facial palsy and herpes lesions. (A) Front view is showing left facial paresis and healing herpes lesions; (B) Lateral view showing herpes lesions of left external ear

- Sensorineural hearing loss occurs only in 6% of cases.
- Caloric responses are either decreased or absent.

Diagnosis

- Gadolinium-enhanced MRI of temporal bone: Enhancement of geniculate ganglion of facial nerve.
- HPE: Multinucleated giant cells in vesicle scrapings.
- VZV from vesicle fluid.
- VZV DNA by polymerase chain reaction (PCR) assay in vesicle fluid or cerebrospinal fluid (CSF).

Treatment

- Tab. Acyclovir 800 mg 5 times a day or Famcyclovir 500 mg tds or Valacyclovir 1 gm tds for 7 days.
- Tab. Prednisone tapering dose 6 days course beginning usually with 1 mg/kg.
- Topical antibiotic/steroid ear drops.
- Eye care.

TEMPORAL BONE FRACTURE

Facial nerve palsy is due to intraneural hematoma, compression by a bony spicule or transection of nerve. Penetrating wounds of cheek, face and parotid gland can lacerate facial nerve trunk and its branches.

Vertigo with vegetative symptoms and SNHL herald labyrinthine damage. Hearing loss varies from mild and transient to profound and permanent. Vertigo usually subsides gradually over days to weeks. Nystagmus beats toward normal side. Fall and past pointing are in the direction of the affected ear (toward the slow phase of nystagmus).

Types

1. Traditionally temporal bone fractures are divided into three categories:
   a. Transverse
   b. Longitudinal
   c. Oblique/mixed
2. They can be categorized in the following groups:
   a. Otic capsule sparing
   b. Otic capsule disrupting

The differences are enumerated in Table 3.

TABLE 3 Differences between longitudinal and transverse temporal bone fractures

	Longitudinal	*Transverse*
Frequency	More common (80%)	Less common (20%)
Site of blow	Parietal	Occipital
Otic capsule	Usually otic capsule sparing	Usually disrupting
Fracture line	From squamous part of temporal bone to foramen lacerum, parallel to long axis of petrous pyramid	From foramen magnum to foramen spinosum through jugular foramen, across the petrous pyramid
Injury to external and middle ear	Common	Absent
Ear bleeding	Common	Absent because tympanic membrane is intact
Hemotympanum	Absent	Common
CSF otorrhea	Present	Absent (CSF may pass through Eustachian tube)
Labyrinth or CN VIII injury	Uncommon	Common
Hearing loss	Usually conductive	Usually sensorineural
Vertigo	Uncommon (may be present due to concussion)	Generally severe due to labyrinth or CN VIII injuries
Facial paralysis	Uncommon (20%)	Common (50%)
Onset of facial palsy	Usually delayed	Usually immediate
Site of facial nerve injury	Horizontal tympanic segment	Meatal or labyrinthine segment geniculate ganglion

Transverse Temporal Bone Fracture (Usually Otic Capsule Disrupting)

- Blow is to the occipital region.
- Fracture line involves foramen magnum, jugular foramen, IAC, petrous pyramid and otic capsule and foramen lacerum.

Clinical Features

- Rupture of membranous labyrinth: Acute vertigo, SNHL, nystagmus beats towards healthy ear. Fall and past pointing towards affected ear. Vertigo gradually subsides over days to weeks.
- Laceration of VIII CN.
- Facial nerve paralysis.
- CSF fistula.
- Greater risk of intracranial injuries.
- Higher risk of delayed meningitis because otic capsule enchondral bone does not remodel and heal.

Longitudinal Temporal Bone Fracture (Usually Otic Capsule Sparing)

- Blow is to the temporoparietal region
- Fracture line involves squamosal portion, posterosuperior wall of external auditory canal (EAC), mastoid air cells, middle ear and tegmen tympani and mastoideum and tegmen of facial hiatus

Clinical Features

- Conductive or mixed hearing loss.
- External auditory canal: Fracture along the roof, CSF otorrhea, hemorrhage, presence of brain herniation.
- Labyrinthine concussion.
- Benign paroxysmal positional vertigo (BPPV)

Management of Temporal Bone Fracture

- Secure airway, control bleeding, examine neurological status, stabilize and evaluate cervical spine.
- Assess facial nerve function at the earliest.
- Auricle: Lacerations and hematoma if not treated can result in chondropathy or cauliflower ear.
- Examine EAC for fracture along the scutum and roof, CSF otorrhea, degree of hemorrhage, presence of brain herniation. Blood and cerumen may be aspirated but never irrigated.
- Look for any tympanic membrane perforations and hemotympanum.
- Ear packing is discouraged and done only in cases of significant bleeding. Profuse bleeding not controlled by packing is managed by carotid ligation or angiography for balloon occlusion.
- Early ear canal stenting is required in cases of severely traumatized ear canals. It is not done acutely.
- Vertigo and nystagmus (peripheral or central) usually resolves spontaneously.
- BPPV needs repositioning maneuver.
- Positive fistula test, which is not done in acute setting (for the fear of introduction of air and infection in the ear) and in cases of CSF fistula, indicates perilymph fistula that present with vertigo and nystagmus for more than a week and fluctuating or progressive SNHL.
- Take audiogram in cases of CSF fistula and facial palsy.
- CT head assesses intracranial hemorrhage. HRCT of temporal bone (Fig. 11) is indicated in cases of facial palsy, CSF leak, fracture roof of EAC and suspected vascular injury.
- The timings of facial palsy are important. Delayed paralysis is treated conservatively.
- Immediate facial paralysis usually requires surgery (decompression, reanastomosis of cut ends or cable nerve graft), which is performed 3 weeks after injury.

Temporal bone fracture: Clinical features include hearing loss, dizziness, facial weakness, ear bleeding, hemotympanum, raccoon eyes, and/or bruising over the mastoid cortex (Battle's sign).

1. Radiological investigation: The best radiologic examination is a fine cut, axial and coronal, temporal bone CT scan.

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2. Indications for surgical exploration of the facial nerve: They include–
   a. Immediate onset of complete facial paralysis
   b. Delayed onset of complete facial paralysis associated with
      i. Radiologic evidence of a fracture through the fallopian canal of facial nerve.
      ii. Poor prognostic testing with electroneuronography (ENoG) or electromyography (EMG).

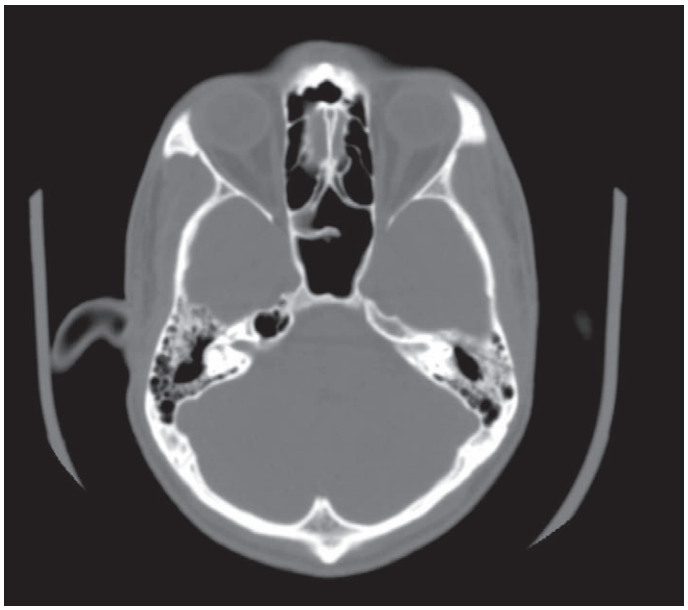

Fig. 11: HRCT temporal bone axial section showing longitudinal fracture of temporal bone
Source: Dr Amit Goyal, Shillong

LYME DISEASE (BANNWARTH'S SYNDROME)

- This multisystem spirochete (*Borrelia burgdorferi*) disease involves skin, nervous system, heart and joints.
- It is transmitted by Ixodes ticks from the primary reservoirs (white-footed mice and white-tailed deer).
- In USA, the disease was initially recognized in 1975 in Lyme, Connecticut. In Europe it is called Bannwarth's Syndrome.
- It affects all ages and both sexes.

Clinical Features

- Acute facial nerve palsy, which is usually unilateral, recovers completely and spontaneously after weeks to few months.
- There may be preceding history of ear pain, facial pain or paresthesias.
- Intensely red and violet nodules (lymphocytoma) on the ear lobe.
- Sudden SNHL, positional vertigo and Meniere's like symptoms have been reported.

Diagnosis

- Clinical in endemic regions.
- Enzyme linked immunosorbent assay and Western blotting: Detection of specific antibody to B. burgdorferi.

Treatment

- Highly sensitive to doxycycline.
- Other antibiotics are amoxicillin, erythromycin, cefuroxime, ceftriaxone and imipenem.

SARCOIDOSIS

- It is a chronic multisystem disease of unknown etiology.
- It most frequently affects lungs and occurs in 3rd–4th decade of life.
- Blacks are 10 times more affected than whites.

Clinical Features

- Most common: Cough and granulomatous skin rashes and bilateral hilar adenopathy.
- Triad of uveoparotid fever (Heerfordt's syndrome): Parotitis, uveitis, facial nerve palsy and mild fever.
- Otologic: SNHL, vestibular dysfunction and occasionally granulomatous lesion in EAC, middle ear and mastoid.
- CN VII palsy**:** Often bilateral, sudden and resolve spontaneously.
- Eye: Iridocyclitis and keratoconjunctivitis.
- Nervous system: Peripheral mononeuritis or polyneuritis.
- Lymphadenopathy and hepatosplenomegaly.
- Myalgia and arthralgia.
- Cardiac failure.

Investigations

1. *X-ray chest*: Hilar adenopathy
2. *Serum*: Hypercalcemia, elevated serum angiotensin converting enzyme
3. *Biopsy*: Noncaseating epithelioid granulomas

Treatment

- Spontaneous resolution occurs
- *First line agent*: Corticosteroids in progressive disease with involvement of eye, heart and CNS
- *Second line agents*: Cytotoxic drugs such as methotrexate, azathioprine, and cyclophosphamide
- *Infliximab*: It inhibits the release or blocks the effect of a key cytokine, which appears to be tumor necrosis factor

MOBIUS SYNDROME

1. This congenital facial palsy, which may be unilateral or bilateral, complete or incomplete, is associated with bilateral abducens palsy.
2. Concomitant tongue weakness and clubfoot are common.

IATROGENIC OR SURGICAL TRAUMA

Facial nerve may be damaged accidentally during stapedectomy, tympanoplasty or mastoid surgery. The paralysis may be immediate (needs earliest surgical decompression and repair) or delayed (treated conservatively). The exposed nerve may be pressed by the pressure of ear packing. This just needs removal of ear pack.

Surprisingly even if the facial nerve is surgically removed secondary to a parotid tumor, facial nerve functions can

spontaneously return. The precise mechanism of this rare occurrence is not known.

Prophylaxis

Thorough anatomical knowledge and temporal bones dissection under good quality operative microscope trains the surgeon and avoids facial nerve injuries during the ear surgeries. Other tips, which help in avoiding damage to the nerve, are following:

- Always work along the course of facial nerve and never across the nerve.
- Constant irrigation during the drilling avoids thermal injury to the facial nerve.
- Diamond burr must be used when working near the nerve.
- If nerve is exposed, do not get scared. Just handle the nerve gently. Avoid unnecessary handling and instrumentations.
- Do not remove those granulations which penetrate facial nerve.

HYPERKINETIC DISORDERS OF FACIAL NERVE

The involuntary twitching of facial muscles on one or both sides occurs in cases of hemifacial spasm and blepharospasm.

Hemifacial Spasm

There occurs involuntary unilateral repeated twitching of facial muscles. There are two types of hemifacial spasms idiopathic and secondary:

1. *Essential or idiopathic*: Cause is not known.
   Treatment consists of the following:
   a. Selective section of the branches of facial nerve in the parotid
   b. Puncturing the facial nerve with a needle in its tympanic segment
   c. Injection of botulinum blocks the neuromuscular junction by preventing release of acetylcholine in the affected muscle
2. *Secondary*: The irritation of facial nerve can be caused by following CPA disorders
   a. Acoustic neuroma
   b. Congenital cholesteatoma
   c. Glomus tumor
   d. Vascular loop at the cerebellopontine angle: Treated by microvascular decompression through posterior fossa craniotomy

Blepharospasm

Twitching and spasms of the orbicularis oculi muscles on both sides results in closure of both the eyes causing functional blindness.

- *Etiology*: The cause, which is yet not certain, perhaps lies in the basal ganglia.
- Treatment includes:
  - Selective section of nerves supplying muscles around the eye on both sides.
  - Injection botulinum A into the peripheral muscles gives relief for 3–6 months. It can be repeated if required.

SURGICAL TREATMENT OF FACIAL NERVE PALSY

1. *Suppurative otitis media*: In addition to the aggressive antibiotic therapy following surgical measures are taken in cases of otitis media:
   - Myringotomy: It is indicated in cases of acute otitis media, otitis media with effusion and coalescent mastoiditis.
   - Tympanomastoid surgery: In cases of coalescent mastoiditis and cholesteatoma.
2. *Decompression*: The nerve may be compressed by intraneural edema and hematoma and a fractured bone in the fallopian canal. The compressed nerve is exposed and uncapped in surgical decompression. The facial nerve sheath is slit to relieve pressure.
   When electrical tests indicate progressive nerve weakness (> 90%) facial nerve decompression should be done at the earliest in cases of Bell's palsy, Ramsay Hunt syndrome and longitudinal temporal bone fracture.
   a. The labyrinthine and meatal segments are best approached via middle fossa technique.
   b. The tympanic and mastoid segments are best approached via transmastoid technique (Fig. 12).
   c. A retrolabyrinthine approach is best for the CPA. (Chapter on Tumors of Ear and CPA)
   d. For traumatic facial nerve palsy with total deafness, a translabyrinthine approach is considered.
3. *End to end anastomosis*: It is a suitable procedure for extratemporal part of facial nerve, when the gap between the severed ends of nerves is few millimeters, so that ends could be approximated without any tension. A 9.0 or 10.0 monofilament suture is used to tie the nerve ends.
4. *Nerve graft (cable graft)*: It is indicated when the gap between severed ends is more and cannot be closed without tension by end to end anastomosis (physically damaged or missing portion of nerve). Nerve graft is usually taken from greater auricular nerve, lateral cutaneous nerve of thigh or the sural nerve. In the fallopian canal, graft may not need any suturing.
5. *Hypoglossal-facial anastomosis*: It is indicated when proximal facial nerve stump cannot be identified. Anastomosis of hypoglossal nerve to the severed peripheral end of the facial nerve improves the muscle tone and permits some movements of facial muscles. It leads to unilateral atrophy of tongue muscles. However, patient

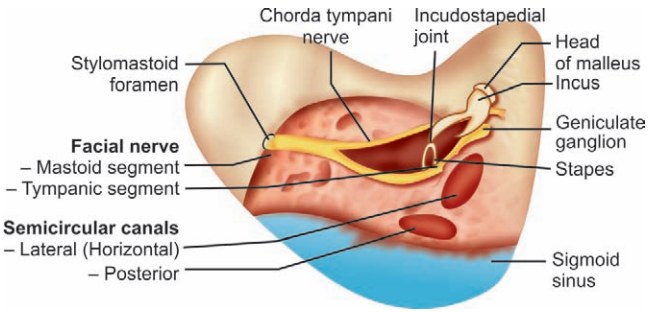

Fig. 12: Transmastoid exposure of left facial nerve. Posterior tympanotomy through facial recess shows incus and stapes

adjusts to the difficulty in chewing and articulation within few weeks.

6. *Plastic procedures*: The procedures which include facial slings, face lift operation or slings of masseter and temporalis muscle, improve cosmetic appearance in cases where nerve grafting is not possible or has failed. The slings of masseter and temporalis muscle provide not only the facial symmetry but also some movement to face.

Clinical Highlights

1. **Endoneurium**: In the nerves, endoneurium helps promote regeneration of severed axons. It is produced mainly by Schwann cells and forms a sleeve for regenerating axons.
2. **Bill's bar**: In the upper part at the lateral end (fundus) of internal auditory canal, this vertical crest separates facial nerve canal from the superior vestibular nerve canal.
3. **Facial nerve segments**: Labyrinthine segment is the shortest segment (4 mm) of facial nerve. The longest segments (approximately 16 mm) are intracranial and parotid. Intracanalicular and tympanic are approximately 8 mm, while mastoid is 12 mm.
4. **Facial nerve supplies**: All the muscles derived from second branchial arch such as muscles of facial expression and stylohyoid, posterior belly of digastric and stapedius muscles.
5. **Chorda tympanic nerve**: This branch of facial nerve carries efferent preganglionic parasympathetic secretomotor fibers to submandibular and sublingual salivary glands. It joins lingual nerve in infratemporal fossa.
6. **Greater petrosal nerve**: Preganglionic secretomotor fibers for lacrimation leave the geniculate ganglion of facial nerve in this branch. So the tearing is lost in supragenicular or transgeniculate lesions of facial nerve.
7. **Parotid relations**: During superficial parotidectomy, the most reliable landmark to identify main trunk of facial nerve is tympanomastoid suture. Other landmarks include mastoid tip, styloid process and auricular cartilage. Facial nerve exits anterior to diagnostic ridge and lies above the posterior belly of digastric.
8. **Middle ear relations**: Landmark used for identification of geniculate ganglion of facial nerve is processus cochleariformis. Tympanic segment of facial nerve lies above the oval window and below the horizontal semicircular canal. Mastoid segment of facial nerve always lies behind the tympanomastoid suture and can be located 6–8 mm deeper to tympanomastoid fissure.
9. **Upper motor neuron facial palsy**: The corticobulbar fibers unilaterally innervate the facial motor neurons to the lower face. Upper face has a bilateral corticobulbar innervation. Lesions of corticobulbar fibers (in brain stem and genu of internal capsule) result in upper motor neuron facial palsy such as in pseudobulbar palsy. Blink reflex remains intact bilaterally. Patient can shut both eyes with equal power and can wrinkle his forehead scalp bilaterally. Liquid drips out from the corner of mouth.
10. **Lesion distal to geniculate ganglion**: This lower motor neuron facial palsy patient will have phonophobia and loss of stapedial reflex and taste sensation.
11. **Bell's palsy**: It is an idiopathic isolated lower motor neuron facial palsy. In Bell's palsy hyperacusis is due to the paralysis of stapedius muscle.
12. **Bell's phenomenon**: In cases of lower motor neuron facial nerve paralysis, eyeball turns up and out while closing the eyes.
13. **Recurrent facial paralysis**: The common causes include acoustic neuroma, diabetes, Melkersson-Rosenthal syndrome and sarcoidosis.
14. **Melkersson-Rosenthal syndrome**: The features include recurrent facial paralysis, fissured tongue and circumoral edema.
15. **Ramsay Hunt syndrome**: This *Herpes zoster* virus infection involves geniculate ganglion of facial nerve. Vesicular eruptions are seen in concha, posteromedial surface of pinna and soft palate.
16. **Pre-eclampsia**: Facial paralysis during pregnancy can occur in cases of preeclampsia.
17. **Temporal bone fracture**: Facial nerve palsy is common in cases of transverse temporal bone fracture.
    a. **Battle's sign**: Ecchymosis seen over the mastoid in cases of temporal bone fracture.
    b. **Transverse fractures of temporal bone**: They result from frontal or occipital blow to head and are more likely to cause injury to labyrinth and facial nerve. They are less common than longitudinal fractures.
    c. **Longitudinal fracture of temporal bone**: They occur due to a blow from the side of head and have less chances of facial palsy. They cause conductive hearing loss.
18. **Concussion of labyrinth**: The most common cause of high frequency sensorineural hearing loss (SNHL) in cases of head injury is labyrinthine concussion.
19. **Iatrogenic facial nerve palsy**: Mastoidectomy is the most common cause of iatrogenic facial nerve palsy.

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25 Tumors of the Ear and Cerebellopontine Angle

Truth, purity, and unselfishness–wherever these are present, there is no power below or above the sun to crush the possessor thereof. Equipped with these, one individual is able to face the whole universe in opposition.

—Swami Vivekananda

Points of Focus

- **BENIGN TUMORS OF EXTERNAL EAR**
  - ✓ Cysts
  - ✓ Hemangiomas
  - ✓ Papilloma
  - ✓ Keratoacanthoma
  - ✓ Neurofibroma
  - ✓ Osteoma
  - ✓ Exostoses
  - ✓ Ceruminoma
  - ✓ Keloid of Auricle
- **MALIGNANT TUMORS OF EXTERNAL EAR**
  - ✓ Squamous Cell Carcinoma of Auricle
  - ✓ Basal Cell Carcinoma of Auricle
  - ✓ Melanoma of Auricle
- **TUMORS OF MIDDLE EAR AND MASTOID**
  - ✓ Glomus tumors (Paraganglioma)
    - – Jugular Foramen Syndrome
  - ✓ Carcinoma of Middle Ear/External Auditory Canal (EAC)
  - ✓ Rhabdomyosarcoma
  - ✓ Leukemia
- **INTERNAL AUDITORY CANAL (IAC) AND CEREBELLOPONTINE ANGLE (CPA)**
  - ✓ Acoustic Neuroma
  - ✓ Surgery for CPA Tumors
- **CLINICAL HIGHLIGHTS**

BENIGN TUMORS OF EXTERNAL EAR

Osteomas/exostoses of external auditory canal (EAC) are very common benign tumors, which usually need no treatment. Ear polyps, which are usually not true neoplasms, are (Chapter Otologic Symptoms) commonly associated with chronic suppurative otitis media.

Cysts

- *Sebaceous cyst*: The common site for sebaceous cyst is post-auricular sulcus below and behind the ear lobule.
- *Sebaceous adenoma*: It arises from sebaceous glands of the meatus and presents as a smooth, skin-covered swelling in the outer meatus.
- *Dermoid cyst*: It is usually present as a rounded mass over the upper part of mastoid behind the pinna.
- *Treatment*: These swellings require total surgical excision.

Hemangiomas

Hemangiomas are congenital tumors seen in children and may involve other parts of face and neck.

- Capillary hemangioma is a mass of capillary-sized blood vessels and present as a: port-wine stain, which does not regress spontaneously.
- Cavernous hemangioma appears as strawberry tumor. It consists of endothelial lined spaces filled with blood. It increases rapidly during the first year of childhood but regresses thereafter. It disappears by the fifth year.

Papilloma

- Papilloma *(wart)* is a viral disease. It presents as a tufted growth or flat grey plaque and has rough surface. It needs surgical excision or curettage with cauterization of its base.
- Cutaneous horn is a form of papilloma with heaping up of

keratin. It presents as horn-shaped tumor at the rim of helix in elderly people.
- Treatment consists of surgical excision.

Keratoacanthoma

- Keratoacanthoma, though a benign tumor, clinically resembles malignant one and presents as a raised nodule with a central crater.
- To begin with it grows fast but then slowly regresses and leaves a scar.
- Treatment is excision biopsy.

Neurofibroma

- Neurofibroma, which may be associated with von Recklinghausen's disease presents as a non-tender, firm swelling.
- If the tumor occludes EAC or presents a cosmetic problem it needs surgical excision.

Osteoma

- Osteoma arises from cancellous bone of EAC and presents as a single, smooth, bony, hard, pedunculated tumor.
- It usually arises from the posterior meatal wall near the outer end of bony EAC.
- Surgical removal is done by fracturing through its pedicle or removal with a drill.

Exostoses

- *Types*: Exostoses arise from compact bone. They are multiple and bilateral.
- *Occupations*: It is common in divers and swimmers, as their ear canals are frequently exposed to cold water.
- *Sex*: Males are affected three times more often than females.
- *Clinical features*: They present as smooth, sessile, bony swellings in the deeper part of the bony meatus near the tympanic membrane.
- *Treatment*: They need surgical removal only if they are large enough to impair hearing or cause retention of wax and debris. They are removed with high speed drill to restore normal sized meatus. The deep exostoses lie in close relation to the facial nerve. Therefore, the gouge and hammer are not used.

Ceruminoma

Ceruminoma is a tumor of modified sweat glands which secrete cerumen.

- *Clinical features*: This benign tumor presents as a smooth, firm, skin-covered polypoid swelling. It is usually attached to the posterior or inferior wall and seen in the outer part of EAC. The tumor can obstruct the EAC and results in retention of wax and debris.
- *Treatment*: Wide surgical excision is required to prevent recurrence. If malignancy is suspected on histology, postoperative radiotherapy should be given. Malignant ceruminoma is more common than benign (2:1 ratio).

Keloid of Auricle

- The common sites for keloids, which occur after trauma or piercing of the ear for ornaments, are the lobule or helix. Black races are more often affected. Postauricular keloids (Fig. 1) are occasionally seen after mastoid surgery.

Treatment: Surgical excision usually results in recurrence (Fig. 2). Recurrence can be avoided by local injection of triamcinolone into the surgical site and pre and postoperative radiation with a total dose of 600–800 rads, which are delivered in four divided doses.

MALIGNANT TUMORS OF EXTERNAL EAR

These neoplasms are uncommon. The three most common neoplasms of the auricle are as follows:

1. Basal cell carcinoma.
2. Squamous cell carcinoma.
3. Melanomas.
   - *Etiology*: The radiation and chronic ear inflammation are the two important etiologic factors.
   - *Most common site of ear carcinoma*: It is auricle (85%). EAC (10%) and middle ear (5%) are rare sites of ear carcinoma.
   - *Most common true neoplasms of the temporal bone and EAC*: They are fibrous dysplasia, Langerhans cell histiocytosis (LCH), leukemia and sarcomas.

Fig. 1: Keloid after postauricular mastoid surgery

Fig. 2: Recurrent keloid. Recurrence after surgery of keloid that occurred after ear piercing

Squamous Cell Carcinoma of Auricle

Clinical Features

- *Predisposing factor*: Prolonged exposure to direct sunlight.
- *Age and sex*: Common in fair-complexioned people especially in males who are in their fifties.
- *Most common site*: Helix.
- *Lesion*: A painless nodule or an ulcer with raised everted edges and indurated base.
- *Metastases:* Its spread to regional lymph nodes occurs very late.

Treatment

- In small lesion with no nodal metastasis, local excision with 1 cm of safety margin is sufficient.
- In large tumor coming within 1 cm of EAC with nodal metastases, total amputation of the pinna with en bloc removal of parotid gland and cervical lymph nodes is done.

Basal Cell Carcinoma of Auricle

- *Common sites*: They are the helix and the tragus.
- *Age*: It is more common in elderly men (> 50 years).
- *Lymphatic metastasis*: It is usually absent.
- *Clinical features:* Nodular ulcer with raised or beaded edge and central crust, which on removal result in bleeding.
  - Lesion usually extends circumferentially into the skin but may penetrate deeper cartilage or bone.

Treatment

- Superficial skin lesions: Radiotherapy.
- Lesions involving cartilage: Surgical excision similar to squamous cell carcinoma.

Melanoma of Auricle

- Predisposing factor: Direct exposure to sun.
  - Common in men of light complexion.
- Metastases: It occurs in 16–50% of the cases.

Treatment

- *Early lesion*: Wedge resection and primary closure in cases of superficial melanoma which is less than 1 cm in diameter and situated over the helix.
- *Advanced lesion*: Superficial melanoma (larger than 1 cm), infiltrative melanomas, melanoma of posterior auricular surface or concha and recurrent melanomas need resection of pinna, parotidectomy and radical neck dissection.

TUMORS OF MIDDLE EAR AND MASTOID

Tumors of middle ear and mastoid can be divided into benign and malignant. Malignant tumors are further divided into primary and secondary (Box 1).

- Tumors of the ear may masquerade as chronic ear infection. While treating these chronic EAC and middle ear infections, the suspicion of neoplasm should be kept in mind.
- Glomus tumors are the most common true neoplasms of the middle ear.
- Hemangiomas, squamous cell carcinoma and rhabdomyosarcoma can also occur.

Box 1: Tumors of the middle ear and mastoid

- Benign
  - *Glomus tumor*: Glomus jugulare and glomus tympanicum
- Malignant
  - Primary
    - ◊ *Carcinoma*: Squamous cell carcinoma, adenocarcinoma
    - ◊ *Sarcoma*: Rhabdomyosarcoma, osteosarcoma, lymphoma, fibrosarcoma and chondrosarcoma
  - Secondary
    - ◊ *Metastatic*: Carcinoma of bronchus, breast, kidney, thyroid, prostate and gastrointestinal tract
  - From adjacent areas:
  - Nasopharynx, external auditory canal and parotid

Glomus Tumors (Paragangliomas)

- This most common benign neoplasm of middle ear arises from the glomus bodies, which resemble carotid body in structure. The paraganglionic cells of the tumor are derived from the neural crest.
- Females are affected five times more than the males. It is often seen in the middle age (40–50 years). In such cases search should be made for other glomus tumors.

Pathology

1. This benign non-encapsulated tumor is extremely vascular neoplasm.
2. Though the tumor is locally invasive, its growth is very slow, extending over several years.
3. Histology shows of masses or sheets of epithelial cells which have large nuclei and a granular cytoplasm. The thin walled blood sinusoids without any contractile muscle coat are in abundance and account for profuse bleeding.
4. They carry < 3% malignancy conversion rate.
5. Less than 1% tumors are associated with catecholamine secretions.

Sites

The glomus bodies are present in the dome of jugular bulb and along the course of tympanic branch of 9th cranial nerve (Jacobson's nerve) on the promontory. So there are mainly two types of glomus tumors:

1. *Glomus Jugulare*: This tumor arises from the dome of jugular bulb and invades the hypotympanum and jugular foramen. It involves cranial nerve, CN IX, X, XI and XII and may compress jugular vein and invade its lumen.
2. *Glomus Tympanicum*: This tumor arises from the promontory and may cause facial paralysis.

Spread

- *Local*: The tumor first fills the middle ear and then invades the tympanic membrane and present as an ear polyp, which bleeds readily. It may later on invade following structures:
  - Labyrinth, petrous pyramid and the mastoid.
  - Jugular foramen and the 9th to 12th cranial nerves.
  - Through Eustachian tube, tumor may enter into the nasopharynx.
  - Posterior and middle cranial fossa.

- *Metastatic*: Lungs and bones metastases are rare. Metastatic lymph node involvement can occur.

Clinical Features

a. *Intratympanic Tumor*
   - Earliest features are deafness and tinnitus.
     - Hearing loss: Conductive and slowly progressive.
     - Tinnitus: Pulsatile, swishing character and synchronous with pulse. It can be stopped by carotid pressure.
     - Audible systolic bruit over mastoid.
   - *Otoscopy*
     - Red reflex through intact tympanic membrane.
     - Rising sun appearance: When tumor arises from the floor of middle ear.
     - Tympanic membrane may appear bluish and bulging.
     - Pulsation sign (Brown's sign): On increasing the ear canal pressure with Siegle's speculum, tumor pulsates vigorously and then blanches. Reverse happens with release of pressure.

b. *Ear Polyp*
   - Profuse bleeding: Either spontaneously or on attempts to clean EAC.
   - Dizziness or vertigo.
   - Facial paralysis.
   - Earache is less common than in carcinoma of the external and middle ear.
   - Otorrhea: It is due to secondary infection and simulates chronic suppurative otitis media polyp.
   - Red, vascular polyp filling the meatus, which bleeds readily and profusely on manipulation. Biopsy is contraindicated.

c. *9th to 12th Cranial Nerve Palsies*
   This is a late feature appearing several years after aural symptoms.
   - Dysphagia.
   - Dysarthria and hoarseness of voice.
   - Unilateral paralysis of soft palate, pharynx and vocal cord.
   - Weakness of trapezius and sternocleidomastoid muscles.

d. *Catecholamine features*: Headache, sweating, palpitation, hypertension and anxiety.

d. Mass over the mastoid or in the nasopharynx.

Pulsatile tinnitus: In cases of pulsatile tinnitus, always first rule out the paraganglioma (glomus tympanicum or jugulare).

Differential Diagnoses

Because of their appearance, glomus tumors may be mistaken for:
- High-riding jugular bulb or dehiscent jugular bulb.
- Aberrant carotid artery.

Jugular Foramen Syndrome

This syndrome consists of the palsies of the CN IX, X and XI, which pass through the jugular foramen. The causes include:
- Tumors
  - Glomus tumors (Paraganglioma)
  - Nerve sheath tumors
  - Sarcomas
- Lymphadenopathy
- Skull fracture

Investigations

- *Catecholamines*:
  - Serum levels of catecholamines.
  - Break-down products of catecholamines in urine: Vanillylmandelic acid, metanephrine, etc.
- *CT scan-head*: Bone window, 1 mm thin sections help to distinguish glomus tympanicum from the glomus jugulare. Caroticojugular spine is eroded in the glomus jugulare. The aberrant carotid artery, high or dehiscent jugular bulb can also be diagnosed.
- *Magnetic resonance imaging*: Magnetic resonance imaging (MRI) provides soft tissue extent of the tumor.
- *Magnetic resonance angiography and venography*: Magnetic resonance angiography (MRA) and venography delineate invasion of jugular bulb and vein or compression of the carotid artery.
- *Four vessel angiography*: It provides following information:
  - Extent of tumor
  - Compression of internal carotid artery
  - Finding other carotid body tumors
  - Decision for embolization of tumor
  - Brain perfusion studies and adequacy of contralateral internal carotid artery and circle of Willis.
- *Biopsy*: It is contraindicated because the tumor is highly vascular and bleeds profusely.

Treatment

Treatment modalities include surgical removal, radiation, embolization and combination of these techniques.
- *Surgical removal*: Depending upon the extent of tumor it can be removed through transmeatal, transmastoid or skull base approach.
- *Radiation*: It does not cure but reduces the vascularity of the tumor and arrests its growth. Its indications are as follows:
  - Inoperable tumors
  - Residual tumors
  - Recurrences after surgery
  - Older patients who cannot withstand extensive skull base surgery.
- *Embolization*: Embolization reduces the vascularity of tumor before surgery. It may become the sole treatment in inoperable and irradiated patients.

Carcinoma of Middle Ear/EAC

Carcinoma of middle ear and mastoid, though the commonest primary middle ear malignancy is rare tumor. It arises either primarily from middle ear or is an extension of carcinoma of deep bony meatus.

Etiology

1. *Age*: Patients are in the age group of 40–60.
2. *Sex*: It is slightly more common in females.
3. *Chronic suppurative otitis media*: Most patients (75%) have associated chronic ear discharge that mimics chronic suppurative otitis media (CSOM). Chronic irritation is perhaps the causative factor. It can develop in radical mastoid cavities.

4. *Radium dial painters*: Primary carcinoma of mastoid air cells is seen.

Pathology

- Types of carcinoma of EAC in order of decreasing frequency include squamous cell carcinoma, basal cell carcinoma and adenoid cystic carcinoma.
- Adenocarcinoma, which is less common, arises from the glandular elements of middle ear.

Spread

- The tumor destroys ossicles, facial canal, internal ear, jugular bulb, carotid canal, deep bony meatus and mastoid.
- Medially it spreads toward the petrous apex. Dura is usually resistant to the spread.
- Anteriorly it spreads and involves parotid gland, temporomandibular joint and infratemporal fossa.
- Through the Eustachian tube it can enter into the nasopharynx.
- Lymph nodes are involved at an advanced stage.

Clinical Features

Appearance of persistent and inordinate pain or blood stained discharge in chronic cases of painless mucopurulent or purulent otorrhea (CSOM or chronic otits externa) should raise the suspicion.

- Foul-smelling blood-stained discharge.
- Severe ear pain especially in night.
- An ulcerated area in the meatus or a bleeding friable polypoid mass or granulations.
- Meaty or polypoid mass in EAC (squamous cell carcinoma) (Fig. 3).
- Small pimple with significant pain (adenoid cystic carcinoma).
- Serpiginous ulceration (basal cell carcinoma).
- Facial nerve palsy can occur because of local extension (Fig. 3).
- Labyrinthine involvement causes SNHL and vertigo.
- Late features: Parotid mass, CN IX X XI and XII palsy, cervical lymphadenopathy.

Fig. 3: Carcinoma left middle ear and external auditory canal (EAC). Mass in left EAC. Inset shows left facial palsy of the same patient

- Tumor may originate from nasopharynx.
- Enlargement of regional lymph nodes: Preauricular, postauricular, infra-auricular and upper deep cervical.

Investigations

- *CT scan and angiography*: They show the extent of tumor and bone invasion (Figs 4A and B).
- *Biopsy*: Biopsy should be taken in cases of intractable case of chronic OE. Histopathology will confirm the nature of lesion.

Treatment

It consists of en bloc wide surgical excision with postoperative radiation. The combination of surgery and radiotherapy offers better prognosis.

- *Surgery*: Depending on the extent of tumor, it may consist of radical mastoidectomy, subtotal/total petrosectomy.
- *Radiotherapy*: Radiotherapy alone is a palliative measure and is considered when the growth involves CN IX, X, XI and XII or spreads to the cranial cavity or nasopharynx.

Rhabdomyosarcoma

This rare tumor arises from the embryonic muscles, tissue or the pluripotential mesenchyme. It mostly affects children.

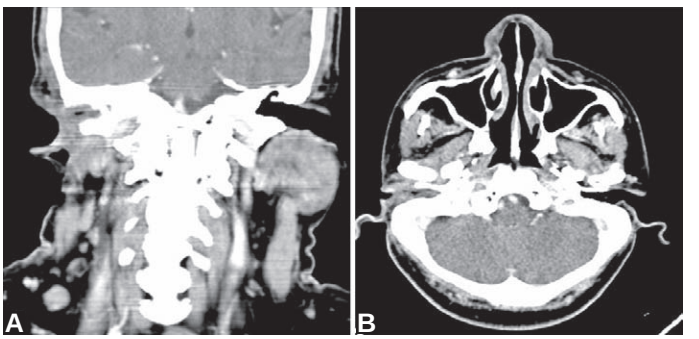
Clinical Features

a. Early cases mimic CSOM and have ear discharge, polyp or granulations.
b. Facial palsy occurs early.
c. Swelling in the surrounding region of the ear.

- *Biopsy*: It establishes the diagnosis.
- *Treatment*: A combination of radiation and chemotherapy is the best modality of treatment. Surgery is considered in selected localized tumors.
- *Prognosis*: It is poor.

Leukemia

- Twenty percent of leukemia patients eventually involve ear/temporal bone.
- They present as mucosal ulceration or bleeding from EAC.
- The tympanic membrane and mucosa of middle ear become irregular and thick.
- The disease may eventually involve facial and auditory nerves.



Figs 4A and B: HRCT Temporal bone of carcinoma right external ear. (A) Coronal section; (B) Axial section
Source: Dr Swati Shah, Associate Professor, Radio-diagnosis, GCRI, Ahmedabad

INTERNAL AUDITORY CANAL AND CEREBELLOPONTINE ANGLE (FIG. 5)

Contents of Internal Auditory Canal

In addition to internal auditory vessels, the internal auditory canal (IAC) contains following nerves:

1. Facial nerve in anterosuperior quadrant.
2. Superior vestibular nerve in posterosuperior quadrant.
3. Cochlear nerve in anteroinferior quadrant.
4. Inferior vestibular nerve in posteroinferior quadrant.

Boundaries of Cerebellopontine Angle

The cerebellopontine angle (CPA). Figure 6 is a potential space in posterior cranial fossa. It has following boundaries:

1. Anterior: Petrous part of temporal bone.
2. Posterior: Cerebellum.
3. Inferior: Cerebellar tonsil.
4. Superior: Pons and cerebellar peduncles.

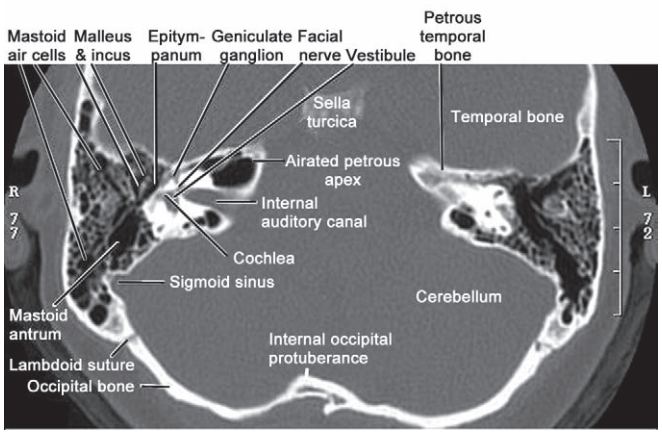


Fig. 5: HRCT temporal bone axial section showing internal auditory canal

Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

Relations of CPA

Facial and vestibulocochlear nerves pass superiorly and laterally through the CPA and enter into the IAC. Other cranial nerves have following relations with CPA:

1. *Superior*: Trigeminal nerve.
2. *Inferior*: Glossopharyngeal, vagus and accessory cranial nerves.

Acoustic Neuroma

(Synonyms: Vestibular Schwannoma or Neurilemmoma)

Acoustic neuroma (AN) is the most common (80%) CPA tumor and constitutes 10% of all the brain tumors.

This benign, encapsulated eighth nerve tumor is extremely slow growing and consists of elongated spindle cells with rod-shaped nuclei lying in rows or palisades. The unilateral tumors are more common. In patients of neurofibromatosis bilateral tumors are seen.

Growth

Acoustic neuroma arises from the Schwann cells of the vestibular nerve twice as often as from the cochlear nerve. It reaches the CPA (Figs 7A to D) after eroding and widening IAC. Anterosuperiorly it involves the CN V and inferiorly involves the IX, X and XI CN, which lie in jugular foramen. The big AN can displace of brainstem and put pressure on cerebellum and raise intracranial tension. Seventy percent of ANs grow slowly over years but 30% remain stable.

Genetic Association and von Recklinghausen Disease

Five percent of AN patients have neurofibromatosis (NF), which is also called as von Recklinghausen's disease (Fig. 8). The NF associated tumors are not capsulated and may be multiple or bilateral. These AN tumors are more aggressive. Though rarely, they can undergo malignant change.

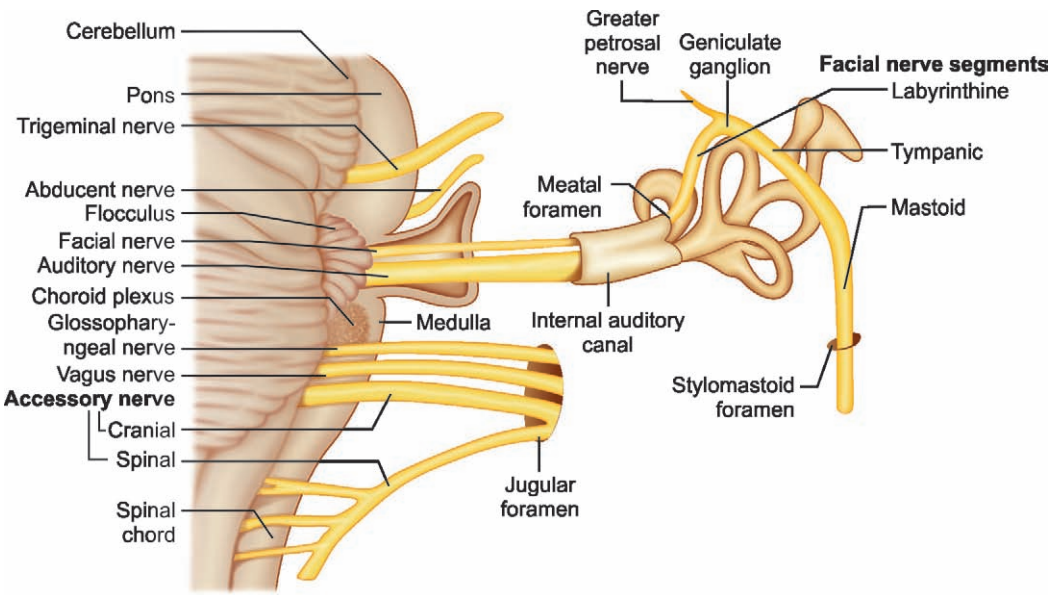
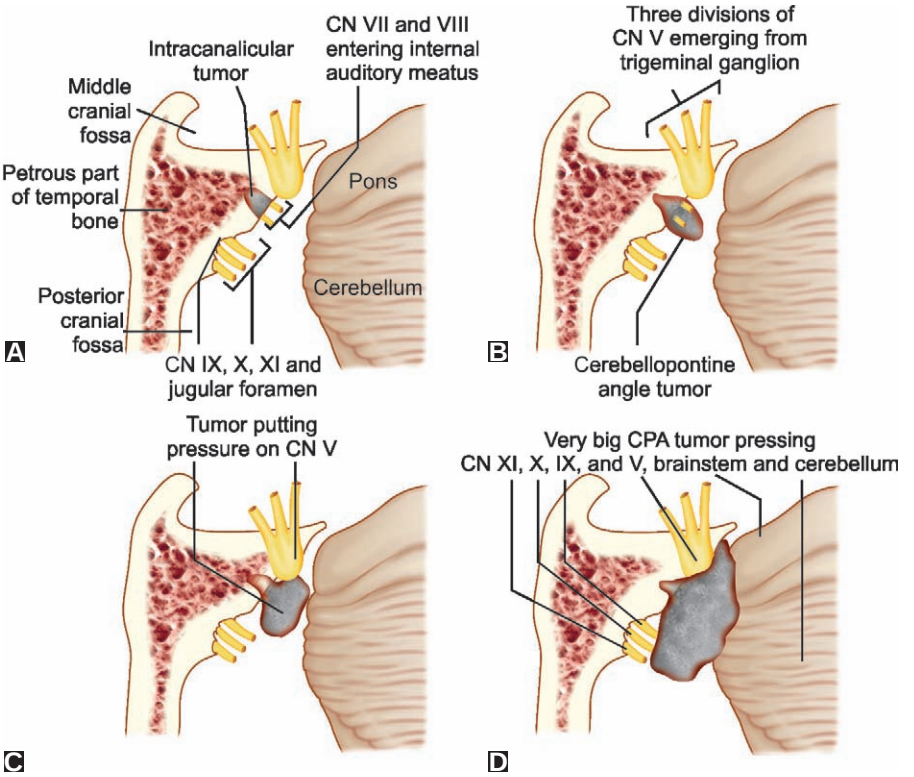


Fig. 6: Cerebellopontine angle (CPA) angle and cranial nerves VII, VIII, IX, X and XI



Figs 7A to D: Acoustic neuroma. (A) Intracanalicular tumor; (B) Cerebellopontine angle tumor; (C) Tumor pressing trigeminal nerve; (D) Tumor pressing glossopharyngeal, vagus, and accessory nerves, brainstem and cerebellum

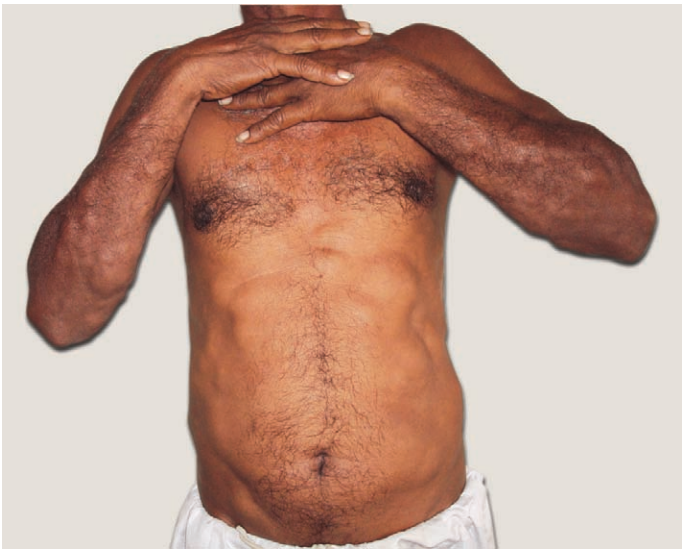


Fig. 8: Multiple neurofibromatosis (von Recklinghausen's disease)

Clinical Features

Asymptomatic AN occurs in about 2% of the population.

The cases of progressive unilateral sensorineural hearing loss (SNHL) and unilateral tinnitus must be investigated to rule out AN.

- *Onset*: The growth of AN is extremely slow and patient's history may extend over several years.
- *Age and sex*: Patients are usually in age group of 40–60 years. Female to male ratio is 3:2.
- *Intracanalicular tumor*: It is limited to IAC and can put pressure on cochlear and vestibular nerves and internal auditory artery. Progressive unilateral SNHL, which is often accompanied by tinnitus, is the most common presenting symptom.
  - Hearing loss: Progressive unilateral SNHL. Difficulty in understanding speech is out of proportion to the pure tone hearing loss. Some patients present with sudden SNHL.
  - Tinnitus
  - Vestibular symptoms: Imbalance or unsteadiness occurs in 50% of patients. Sudden-onset rotatory vertigo is rare because slow growth of tumor results in vestibular compensation.
- *Facial nerve*: Sensory fibers are more sensitive and affected early. Motor fibers are more resistant so are affected late.
  - Histeliberger's sign: Numbness of posterior aspect of concha, which is supplied by the sensory fibers of the facial nerve.
  - Electrogustometry: Loss of taste anterior 2/3 tongue.
  - Schirmer's test: Reduced lacrimation.
  - Blink reflex: It is delayed may be an early manifestation.
- *CN V*: The first extracanalicular nerve to be involved is trigeminal. The tumor is in CPA and of about 2.5 cm size.
  - Reduced corneal sensitivity.
  - Numbness or paresthesia of face.
- *CN IX and X*
  - Dysphagia, hoarseness of voice and nasal regurgitation of fluid.
  - Palatal, pharyngeal and laryngeal paralysis.
- *CN XI, XII, III, IV and VI:* They are involved when tumor is very large.
- *Brainstem*: Long motor and sensory tracts are involved.
  - Weakness and numbness of the arms and legs with exaggerated tendon reflexes.

- *Cerebellum*
  - Finger-nose test, knee-heel test, dysdiadochokinesia, ataxic gait, inability to walk along a straight line with tendency to fall to the affected side.
- *Raised intracranial tension*
  - Blurring of vision, headache, nausea, vomiting, and diplopia (CN VI involvement).
  - Fundus examination: Papilloedema (blurring of disc margins).

Audiometry

Retrocochlear hearing loss occurs. The differences between the cochlear and retrocochlear hearing losses are described in chapter Hearing Evaluation.

1. Hearing loss is more marked in high frequencies.
2. Poor speech discrimination disproportionate to pure tone hearing loss. Roll over phenomenon (reduction of discrimination score when loudness is increased beyond a particular limit) is commonly observed.
3. Recruitment absent.
4. Threshold tone decay test present.
5. Impedance: Stapedial reflex absent.
6. Brainstem evoked response audiometry: A delay of >0.2 m/sec in wave V between two ears is significant. Wave V may be absent or prolonged.

Caloric test

Diminished or absent response in 96% of patients.

Gadolinium-enhanced MRI

Thin section gadolinium-enhanced MRI (Figs 9A and B) is the gold standard for diagnosis of acoustic neuroma and is superior to CT scan. Intracanalicular tumor of a few millimeters size can be diagnosed. On the basis of size AN is classified into four groups (Fig. 6):

1. Intracanalicular (limited to IAC).
2. Small size (up to 1.5 cm).
3. Medium size (1.5–4 cm).
4. Large size (over 4 cm).

Vertebral Angiography

It is helpful in differentiating acoustic neuroma from other CPA tumors.

Differential Diagnoses

Acoustic neuroma is the most common CPA tumor. It should be differentiated from the following CPA tumors that are listed in order of descending frequency:

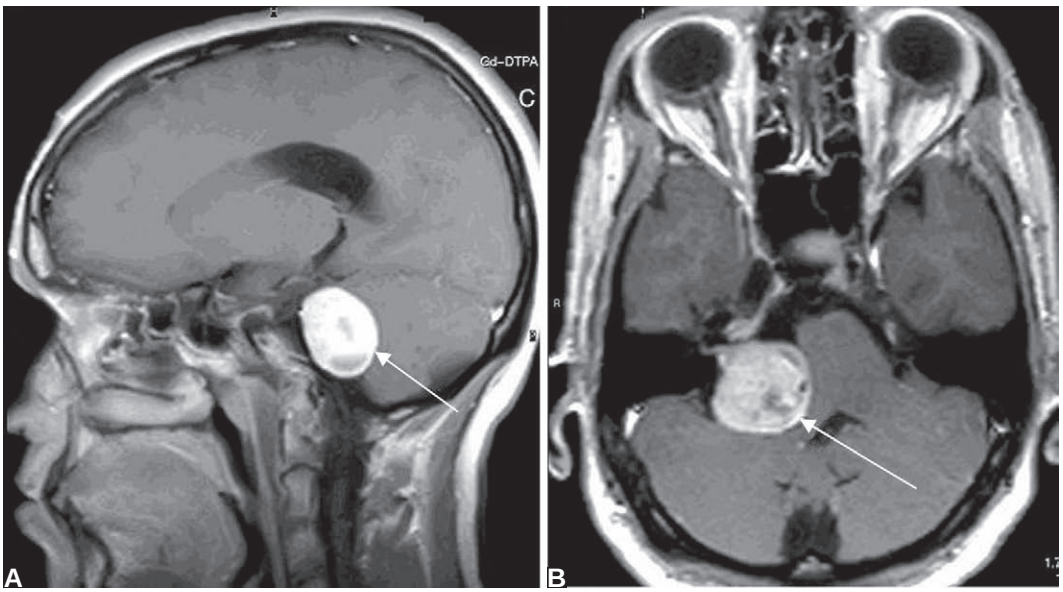
- Meningioma
- Lipomas
- Epidermoid
- Cholesterol granulomas
- Cholesteatoma
- Arachnoidal cyst
- Aneurysm
- Metastasis
- Schwannoma of other cranial nerves such as CN V, VII, IX, X, and XI

Acoustic neuroma should be differentiated from other common causes of dizziness (Table 1), which are described in detail in chapters Central and Peripheral Vestibular Disorders.

Treatment

The treatment of first choice is surgical removal. The second choice is gamma knife surgery. Conventional external radiotherapy (low tolerance of the central nervous system to radiation) and chemotherapy do not have any role in the treatment.

- *Surgery*: The surgical approach depends upon the size and extent of the tumor. The following approaches are employed:
  - Middle cranial fossa or transtemporal supralabyrinthine.
  - Translabyrinthine.
  - Retrolabyrinthine.
  - Suboccipital/retrosigmoid.



Figs 9A and B: MRI of brain with gadolinium contrast in a case of cerebellopontine angle (CPA) tumor. (A) Sagittal section; (B) Axial section
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

TABLE 1 Differential diagnoses of common causes of dizziness

Clinical features	*Meniere's disease*	*Vestibular neuritis*	*Benign paroxysmal positional vertigo*	*Acoustic neuroma*
Type of dizziness	Whirling vertigo	Whirling vertigo	Whirling vertigo	Unsteadiness
Duration of dizziness	Recurrent episodes (lasting for minutes to hours)	Acute prolonged attack (lasting for days)	Short spells (lasting for seconds) in certain positions	Recurrent Momentary or chronic
Tinnitus	Present	Absent	Absent	Present
Vomiting	Very common	Always	Uncommon	Rare
Sensorineural hearing loss	Moderate and fluctuating sensory type with recruitment	Absent	Absent	Progressive neural type with tone decay
Cranial nerve involvement (V, VII, VIII, IX, X and XI)	Absent	Absent	Absent	Present in late stages

- *Gamma knife surgery*: This stereotactic gamma-irradiation therapy or linear accelerator (LINAC) delivers high dose of ionizing radiation to the tumor and minimizes its effect on the surrounding neural tissue. It facilitates arrest of the tumor growth and reduction in its size. It can be used in following patients:
  - Patient is not willing for surgery.
  - Tumor < 3 cm.
  - Contraindications to surgery.
  - Residual tumor after surgery.
- *Conservative approach*: Patients who are elderly or have serious medical problems may be kept under wait and watch policy because ANs are slow growing.

Surgery for CPA Tumors

Translabyrinthine Approach

- *Indication*: CPA tumor < 3 cm size.
- *Operation*: Postauricular mastoidectomy and labyrinthectomy are performed. Dura is exposed by removing bone overlying posterior and middle cranial fossa and sigmoid venous sinus. The IAC is entered and posterior fossa dura is incised. The tumor is removed and dura is closed. Fat graft is used to obliterate the space.
- *Advantages*: Direct approach to entire IAC and its contents; preserve facial nerve; and minimal cerebellar retraction; neurotologists well familiar with surgical anatomy of the area.
- *Disadvantages*: Complete hearing loss, sigmoid sinus, facial nerve and middle fossa dura limit the exposure.

Retrolabyrinthine Approach

- *Indication*: Small CPA tumor not extending into lateral part of IAC.
- *Operation*: It is an extension of a mastoidectomy. Labyrinth is not damaged. CPA medial to the porus of IAC is visualized. Vestibular nerve may be cut. The dura between sigmoid sinus and labyrinth is opened and tumor is removed.
- *Advantages*: Neurotologists well familiar with surgical anatomy of the area.
- *Disadvantages*: No access to lateral part of IAC; otic capsule, sigmoid sinus, middle fossa dura and cerebrum limit the exposure.

Middle Cranial Fossa or Transtemporal Supralabyrinthine Approach

- *Indications*: Small intracanalicular tumor < 1 cm size, facial nerve decompression and repair and vestibular neurectomy.
- *Operation*: A temporal craniotomy is performed through a scalp incision, which is made above the auricle. Temporal lobe is minimally retracted extradurally. The superior surface of petrous bone is drilled. Labyrinth remains intact. The meatal and labyrinthine parts of facial nerve are identified. IAC is entered and tumor is removed.
- *Advantages*: It may preserve hearing. Limited dural incision.
- *Disadvantages*: Neurotologists less familiar with surgical anatomy of the area; Cochlea, semicircular canals and temporal lobe limit the exposure.

Suboccipital/Retrosigmoid Approach

- *Indications*: Large CPA tumor > 3 cm size.
- *Operation*: A craniotomy is made below the sigmoid sinus. Posterior fossa dura is opened and cerebellum is retracted. The CPA tumor becomes evident and majority of which is debulked. The posterior surface of petrous bone is drilled preserving the contents of IAC and posterior semicircular canal. Now IAC tumor is removed preserving the facial nerve. The part of the tumor, which lies anterior and medial to the IAC is also removed.
- *Advantages*: It preserves hearing; large tumor can be removed.
- *Disadvantage*: Large exposure of brain; neurotologists less familiar with surgical anatomy of the area; cerebellum, brainstem and posterior semicircular canal limit the exposure.

Perioperative Care

Cranial nerve and somatosensory monitoring; EMG technique for facial nerve; and BERA or electrocochleography for cochlear nerve during the skull-base surgery are essential.

Postoperative Sequelae/Complications

- *Common*: SNHL, temporary facial nerve palsy and balance problem, CSF leak, meningitis.
- *Rare*: Stroke, intracranial bleeding, pneumocephalus, cerebellar ataxia, and even death.

Clinical Highlights

1. **Exostosis of external auditory canal**: These multiple bony swellings are common in swimmers. Osteoma of external auditory canal is usually single and occurs at suture lines such as tympanomastoid.
2. **Carcinoma of middle ear**: Features include elderly persons, chronic ear discharge, facial palsy, pain in the ear and friable ear polyp with tendency to bleed.
3. **Glomus tumors (paraganglioma)**: The tumors grow very slowly and are multicentric in origin. Common sites are carotid bifurcation, jugular foramen and promontory of middle ear. Diagnostic biopsy is contraindicated.
4. **Glomus jugulare**: The features include ear bleeding, pain, tinnitus and progressive deafness and a red swelling behind the intact tympanic membrane which blanches on pressure with pneumatic speculum. Other characteristic features are rising sun appearance, Aquino's sign, Brown's sign and Phelps sign. Management includes preoperative embolization, surgery and radiotherapy.
5. **Aquino's sign**: Otoscopy shows blenching of glomus jugulare tumor on compression of carotid artery.
6. **Phelps sign**: It is seen in cases of glomus jugulare tumor and consists of destruction of bone between the carotid canal and jugular foramen.
7. **Brown's sign (Pulsation sign)**: It is seen in glomus jugulare tumor. During otoscopy with pneumatic speculum, on raising the pressure red mass behind the drum pulsates vigorously while reverse occurs on releasing the pressure.
8. **Glomus tympanicum**: Treatment of choice if the tumor is restricted to promontory is surgical removal.
9. **Acoustic neuroma**: The most common site of origin is superior vestibular nerve.
   a. **Clinical features**: Unilateral tinnitus and sensorineural hearing loss are the earliest features. Facial palsy is a late feature. The earliest extracanalicular cranial nerve to be involved is trigeminal (especially sensory fibers affecting the corneal reflex).
   b. **Hitzelberger sign**: Numbness in the posterosuperior wall of external auditory canal (supplied by CN VII) in cases of acoustic neuroma.
   c. **Differential diagnoses**: They include endolymphatic hydrops and cerebellopontine angle meningioma.
   d. **Audiogram**: The patient can have any type of asymmetrical sensorineural hearing loss that may be high frequency, low frequency, flat, and sudden hearing loss. Difficulty in understanding speech is out of proportion to the hearing loss.
   e. **MRI scan with Gadolinium**: This is the most sensitive diagnostic modality for acoustic neuroma.

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26 Nasal Symptoms and Examination

Many are known to do great works under the stress of some strong emotion. But a man's true nature is known from the manner in which he does his insignificant daily task.

—**Holy Mother Sarada Devi**

Points of Focus

- **INTRODUCTION**
- **HISTORY TAKING:** Nasal Obstruction and Discharge, Epistaxis, Headache and Disturb Smell
- **EXAMINATION**
- **EXTERNAL NOSE**
- **VESTIBULE**
- **ANTERIOR RHINOSCOPY:** Thudicum Nasal Speculum, Septum, Inferior and Middle Turbinates and Meatuses, and Floor of Nose
  - ✓ Topical Nasal Decongestant
  - ✓ Probe Test
  - ✓ Posture Test
  - ✓ Infant's Examination
- **POSTERIOR RHINOSCOPY**
- **PATENCY OF NASAL CAVITIES:** Spatula Test, Cotton-Wool Test and Alae Nasi Movements
- **PARANASAL SINUSES**
  - ✓ Tenderness
  - ✓ Transillumination
  - ✓ Endoscopic Examination
- **SPECIAL INVESTIGATIONS OF NASAL COMPLAINTS**
- **SMELL**
  - ✓ Factors Affecting Olfactory Testing
  - ✓ Causes of Olfactory Problems
  - ✓ Tests for Smell: Electro-Olfactogram (Eog)
- **MEASUREMENT OF MUCOCILIARY FLOW**
  - ✓ Indigo-Carmine Test/Saccharin Sodium Test
- **NASAL OBSTRUCTION**
  - ✓ Unilateral Nasal Obstruction
  - ✓ Bilateral Nasal Obstruction
  - ✓ Differential Diagnosis
  - ✓ Measurement of Nasal Obstruction: Acoustic Rhinometry and Rhinomanometry (Rhinometry)
- **NASAL VALVES DISORDERS**
  - ✓ Internal and External Nasal Valves
  - ✓ Cottle Test
  - ✓ Treatment: Spreader Grafts, Nasal Flaring
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

External nose does not need any special instrument for examination. Anterior rhinoscopy allows assessment of nasal cavity, septum and inferior turbinate. A child's nose can be examined with a wide speculum otoscope. Nasal endoscopy allows a thorough evaluation of intranasal anatomy and pathology that is not possible with standard anterior rhinoscopy and head mirror/light examination. For general scheme of case taking and general set up of Bull's eye lamp light source and head mirror see chapter History and Examination in section Basic Sciences.

HISTORY TAKING

History must include the details of all the complaints mentioned in the chief complaints (Table 1) that begin with the appearance of first symptom and extend up to the time of consultation. It usually consists of the mode of onset

TABLE 1 Common complaints of nose and paranasal sinuses

Main complaints	*Associated complaints*
• Nasal stuffiness/obstruction • Nasal discharge: anterior/postnasal drip • Sneezing • Itching • Nose bleed (epistaxis) • Nasal crusting • Disturb smell • Emitting foul smell to others • Swelling nose and paranasal sinuses • Nose deformities: congenital or acquired • Injury/foreign body (FB)	• Headache • Vomiting • Fever • Facial fullness/pain • Exophthalmos • Change in voice (hypernasal/hyponasal) • Snoring/obstructive sleep apnea (OSA) • Cough • Epiphora • Deafness(conductive)

(sudden/gradual), preceding events causing onset, course of symptoms (progressive/constant/fluctuant and continuous/intermittent), factors aggravating/relieving, other accompanying complaints and the treatment taken. In cases of unilateral disorder, note the side and site of affliction. If both the sides of nose are affected, then it should be mentioned which is worst affected. Many a time, negative answers are equally important in arriving at a diagnosis. Inquiries are made of any systemic diseases, which patient might be suffering from such as diabetes, hypertension, coronary artery disease, liver or kidney disease, tuberculosis, HIV/AIDS and a bleeding disorder.

- *Nasal obstruction*: It may be unilateral or bilateral. In patients with bilateral nasal stuffiness, it is important to know whether obstruction occurs simultaneously or alternately. The common causes of unilateral nasal obstruction include foreign body (especially in children), deviated nasal septum, hypertrophied inferior turbinate, polyp, and growth, whereas the causes of bilateral nasal obstruction include adenoids (in children), allergic rhinitis, large antrochoanal polyp and nasopharyngeal angiofibroma (exclusively in young adult males).
- *Nasal discharge*: Watery discharge associated with paroxysmal sneezing/nasal itching is typical of allergic rhinitis. Purulent nasal discharge indicates sinusitis (Fig. 1).

In children, a unilateral purulent blood stained nasal discharge is usually due to foreign body unnoticed by parents.

- *Epistaxis*: In children, the most common site of nose bleed is Little' area. It is due to the frequent picking of nose. Bouts of profuse bleeding in cases of anemic young adolescent males indicate nasopharyngeal angiofibroma. In cases of elderly people, the most common cause is hypertension though malignant lesions should also be kept in mind.
- *Headache*: In cases of sinusitis, headache is worst in the morning and then subsides gradually towards the evening. Frontal headache is common in frontal sinusitis. Occipital headache is associated with sphenoidal sinus infections. Patients with maxillary sinusitis usually complain pain/fullness sensation in maxillary region and upper dental arch. Pain in cases of ethmoidal sinusitis is near the medial canthus and side of the nose.
- *Disturb smell*: In cases of disturb smell, note whether it is anosmia, hyposmia, parosmia, hyperosmia, olfactory hallucinations, or cacosmia. Cacosmia (perception of putrid odor) is common in cases of empyema of maxillary sinus. Hyposmia (partial loss of smell sensation) is common in cases of nasal obstruction. Anosmia (total loss of smell sensation) indicates neural pathology. Parosmia (perversion of sense of smell) and hyperosmia is common in hysterical ladies.

EXAMINATION

Nasal examination usually includes physical examination of external nose, vestibule, anterior rhinoscopy, posterior rhinoscopy and functional examination of nose. Box 1 shows the general format of examination of nose and paranasal sinuses and causes of common findings.

EXTERNAL NOSE

The skin and osteocartilaginous framework of nose needs proper inspection as well as palpation for the skin lesions and the osteocartilaginous framework deformities. Examine for the following components (Box 1):

1. *Injuries* with or without nasal/skull fracture.
2. *Acne rosacea*: It has typical butterfly rash over nose and cheeks.
3. *Healed lupus*: Patient has loss of tissue around the edges of nares. It can be due to trauma.
4. *Swelling and cysts*:
   a. Dermoids: They occur over nasal bones and columella.
   b. Dermoid cyst: It can present as a discharging sinus over the osteocartilaginous junction of nasal bridge.
   c. Furuncle (Figs 2 and 3): It presents as a tender red swelling near the tip of nose.
   d. Dental cysts/abscess: They present as swelling near the nasal alae.
   e. Rhinophyma: It presents as enlargement of the lower part of nose.
5. *Superficial ulcers and inflammation*: It can be secondary to nasal discharge or due to herpes simplex.
   a. Herpes zoster: Skin lesions are restricted to the area of maxillary nerve (CN V_2) distribution.
   b. Rodent ulcer: Basal cell carcinoma (Fig. 4)

Palpation of external nose differentiates between bony, cartilaginous and soft tissue swelling and diagnose deformity, fracture (crepitus) and other lesions.

VESTIBULE

Vestibule is an anterior skin lined part of nasal cavity having vibrissae (hairs in nasal vestibule). It can be easily evaluated

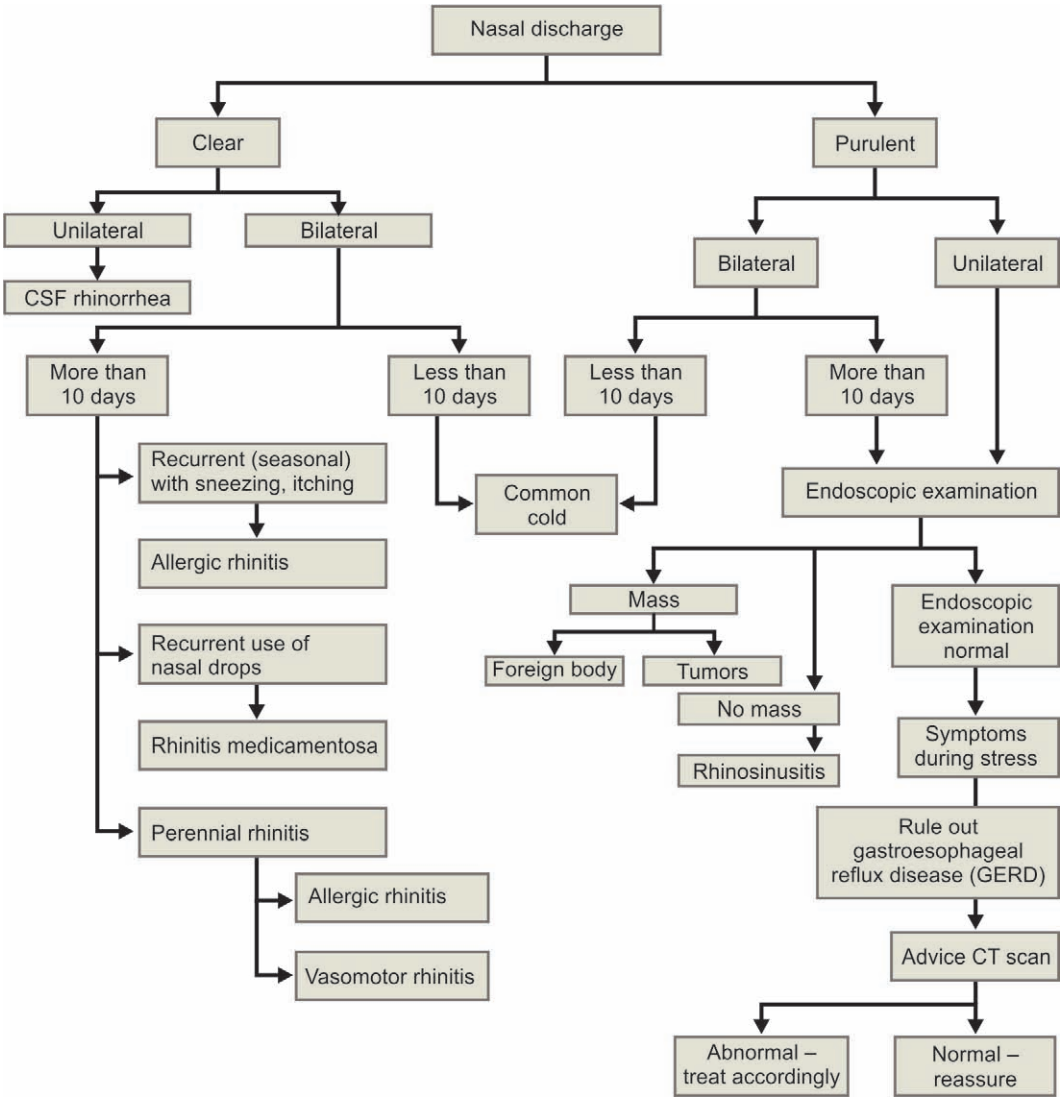


Fig. 1: Flow chart for clinical diagnoses of rhinorrhea

by lifting the tip of nose. For further details, refer to chapter Diseases of External Nose. Examine for the following components (Box 1):

1. *Metaplasia*: Frequent picking of nose produces metaplasia that stands out as whitish dry areas.
2. *Ulcerations and perforation*: Septal perforations can be due to frequent picking and industrial dusts and fumes especially nickel and chrome.
3. *Caudal deflection of septum*: Dislocation of septal cartilage from its groove in the maxillary crest.
4. *Furuncle or fissures (Fig. 3)*: Vestibular examination becomes painful.
5. *Papilloma*: It is usually pedunculated.
6. *Cyst*: Nasolabial cyst presents as a smooth bulge in the floor and lateral wall of left nasal vestibule and upper part of upper lip. Cyst obliterates the alar facial fold (Fig. 5).

ANTERIOR RHINOSCOPY (EXAMINATION OF NASAL CAVITY)

Thudicum Nasal Speculum

A Thudicum or Vienna type of nasal speculum is held in the left hand and assists in widening the vestibule (Figs 6A and B). The blades of speculum are inserted into the less sensitive skin line vestibule and should not touch the septal mucosa which is very sensitive and vascular. The nasal speculum is closed while introducing and opened during examination and remains partially open when removing from the nose (avoid picking vibrissae). The size of the nasal speculum should be chosen according to the age of patient and size of the nose.

Examination

Patient's head needs to be tilted in different directions to examine different sites in the nose: septum, inferior turbinate and meatus, middle turbinate and meatus and floor of the nose (Box 1).

1. Septum: It is rarely straight and mild deviations are not significant. In some cases, even gross looking deviations do not cause any functional problem. Note for any perforations, granulations, deviations.
2. Inferior and middle turbinates: Compensatory hypertrophy of middle and inferior turbinates is common on the concave side of deviated nasal septum. In cases of chronic and allergic rhinitis, there can occur hypertrophy of inferior margins and posterior ends of middle and inferior

Box 1: Examination of nose and paranasal sinuses: findings and their causes

Physical examination of nose

- *External nose*:
  - Swelling: Furuncle, septal abscess, dermoid, glioma, sebaceous cyst
  - Scars: Operation or trauma
  - Sinus: Congenital dermoid
  - Ulcer/neoplasm: Rhinophyma, basal cell/squamous cell carcinoma or melanoma, Herpes simplex/zoster
  - Deformity: Deviated or twisted nose, hump, depressed bridge, bifid or pointed tip, destruction of nose (trauma, syphilis, and cancer), and longer nose appearance in aged persons due to drooping of the nasal tip
  - Palpation: Raised temperature, tenderness, fluctuation, fixity of skin, thickening of soft tissues, crepitation (fracture)
  - Enlargement of bony skeleton: Paget's disease or fibro-osseous dysplasia
  - Cartilaginous enlargement: Chondroma or chondrosarcoma
- *Vestibule*: Furuncle, fissure (chronic rhinitis), crusting, dislocated caudal end of the septum, and tumors (cyst, papilloma, carcinoma or melanoma).
- *Anterior rhinoscopy*
  - Nasal cavity: Narrow (septal deviation or hypertrophy turbinates, polyp, growth), wide (atrophic rhinitis); discharge (mucoid, mucopurulent, purulent, blood); crusting; foreign body.
  - Septum: Deviation or spur, ulcer, perforation, swelling (hematoma or abscess), growth (rhinosporidiosis, hemangioma, fibroangioma), bony destruction (syphilis), or cartilaginous destruction (lupus vulgaris).
  - Floor of nose: Defect (cleft palate or fistula), swelling (dental cyst), neoplasm (hemangioma), or granulations (foreign body or osteitis).
  - Inferior and middle turbinates: Enlarged and swollen (hypertrophic rhinitis), small and rudimentary (atrophic rhinitis). Congested in inflammation and pale in allergy.
  - Inferior and middle meatuses: Discharge/polyps in the middle meatus (infection of maxillary, frontal or anterior ethmoidal sinuses) and between the middle turbinate and the septum (posterior group of ethmoid sinuses).
  - Mass: Polyp, rhinosporidiosis, or carcinoma.
  - Probing: Site of attachment, consistency, mobility and sensitiveness of the mass, bleeding on touch.
- *Posterior rhinoscopy*
  - Posterior choanae: Atresia, polyp
  - Posterior ends of inferior turbinates: Hypertrophy
  - Discharge: In middle meatus (infection of maxillary, frontal or anterior ethmoidal sinuses); above middle turbinate (infection of the posterior ethmoid or the sphenoid sinuses).

Functional examination of nose

- *Patency of nose*: Spatula test/Cotton-wool test
- *Sense of smell*: Clove oil, peppermint, coffee, and essence of rose.

Examination of paranasal sinuses

- *Inspection and palpation*
  - Swelling or fullness: Cheek, lip, lower and upper eyelids and the malar region, forehead and root of nose.
  - Orbit and eyeball: Margins, swelling, proptosis, eye movements, vision (acuity and field), convergence, pupil reactions, fundus, redness, and displacement of eye ball.
  - Vestibule of mouth: Obliteration/fullness/swelling in gingivolabial/gingivobuccal sulcus
  - Upper alveolus and palate: Swelling, ulcer, loose teeth, or carries teeth
  - Swelling and tenderness: Maxillary/canine fossa/frontal/between nose and eye (anterior ethmoid)
- *Transillumination*: For maxillary, frontal and ethmoidal sinuses

turbinates. Middle turbinate concha bullosa (contains air cells), paradoxical shape (convex lateral surface) and polypoidal changes are common findings.

3. Inferior and middle meatuses: Inferior meatus is rarely visible. Most nasal polyps first appear in middle meatus. Purulent secretions in middle meatus can come from infections of anterior paranasal sinuses (maxillary, anterior ethmoid and frontal) as they all drain in it.
4. Floor of nose: A swelling in the floor of nose may extend from hard palate and alveolar process of maxilla. A floor ulcer may communicate with oral cavity. Foreign bodies are usually seen between septum and inferior turbinate.
   - *Topical nasal decongestant*: The solution of cocaine or xylocaine with adrenaline brings about vasoconstriction (shrinkage of mucous membrane) and local anesthesia and facilitates the proper nasal examination.

Fig. 2: External nose swelling due to nasal furuncle

Fig. 3: Nasal furuncle in the vestibule of right nose

Fig. 4: An early rodent ulcer (basal cell carcinoma) of nasomaxillary skin

Fig. 5: Right side nasolabial cyst. Note the smooth bulge in the floor of right nasal vestibule and upper part of upper lip. Cyst is obliterating the alar facial fold

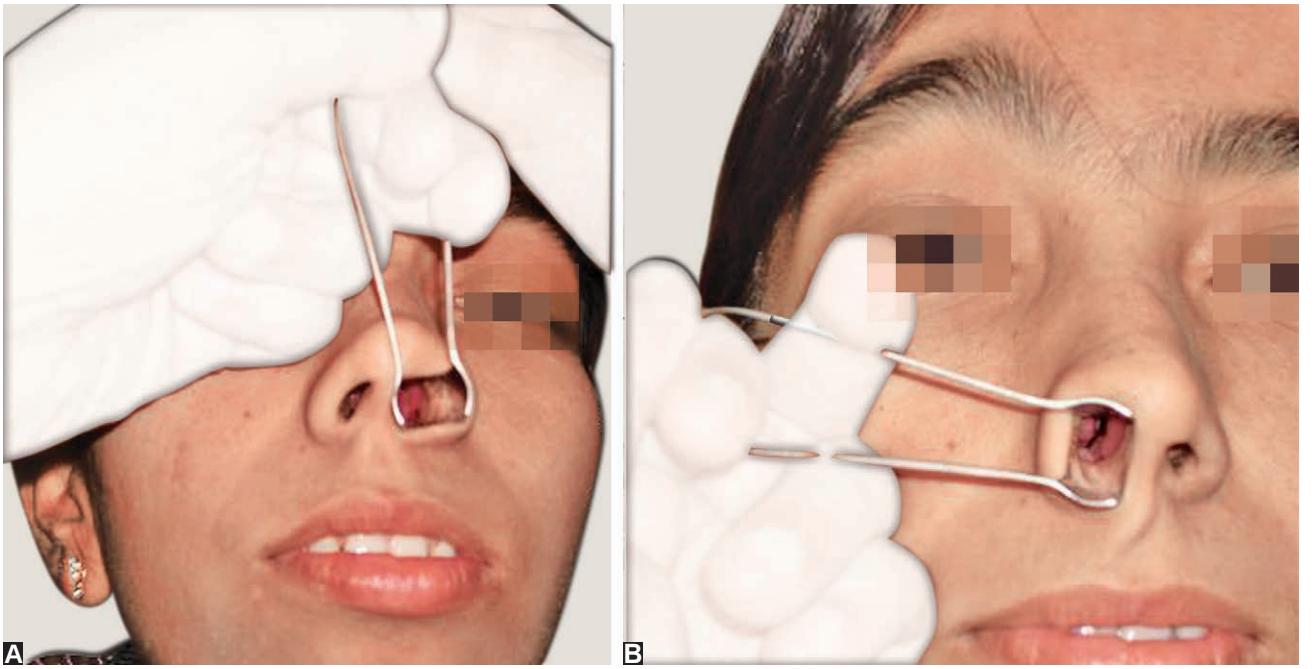
The hypertrophied inferior turbinate does not reduce in size much.

- *Probe test*: It is done under topical anesthesia. A probe is passed on all surfaces of mass and helps in ascertaining the site of attachment, consistency, mobility, vascular nature and sensitivity of the growth. Ulcers should be probed to know the exposed underlying bone. Probing of an ulcer in the floor of nose can exclude or confirm its oral cavity communication. Rhinolith gives a grating sensation on probing.
- *Posture test*: Drainage of purulent discharge from various sinuses depends upon the posture of patient. After wiping out, the purulent discharge from the middle meatus note the timings of discharge and the posture of patient.
  - Frontal sinus: Pus reappears immediately if the patient is sitting in upright position (Head forward chin down position).
  - Ethmoidal sinus: Pus reappears after some time (10–15 minutes) if the patient is sitting in upright position.
  - Maxillary sinus: Pus reappears if the head is so bent that the affected maxillary sinus is in upward position.

Infant's Examination

The infant (including its arms) should be wrapped in a cloth and a nurse must control its head movements.

- Nasal discharge: Slight discharge is considered normal during the first few days of life. Persistent snuffles indicate trauma and infection from difficult delivery and need swab for culture.
- Nasal obstruction: Infants do not know mouth breathing and severe or complete nasal obstruction (bilateral choanal atresia) can be life-threatening. Nasal airway is adequate if baby breaths quietly with mouth shut. Moderate nasal obstruction may interfere with feeding and baby becomes restless after a few sucks. Bubbling nasal mucus and clouding of bright surface indicate partial patency of nasal airway.



Figs 6A and B: Anterior rhinoscopy. (A and B) Thudicum nasal speculum held in left hand widening the vestibule. Two ways of holding and introducing of the nasal speculum

- Testing: Nasal airway patency can be tested with a blunt-ended soft rubber or plastic catheter. In cases of choanal atresia catheter will not appear in the oropharynx. Diagnosis is confirmed with contrast (radiopaque solution into each nostril) X-ray under general anesthesia.

POSTERIOR RHINOSCOPY

It consists of examining the nasopharynx and posterior part of nasal cavity by the postnasal mirror (Fig. 7). The patient opens his mouth and breathes quietly. Postnasal mirror is warmed but should not be hot. It is always better to test on the back of hand before introducing. The examiner depresses the patient's tongue with a tongue depressor that is held in left hand and introduces posterior rhinoscopic mirror (postnasal mirror). The mirror should be held in right hand like a pen and carried behind the soft palate, along the tongue but without touching the posterior third of tongue (to avoid gag reflex). The reflected light from the head mirror illuminates the area of nasopharynx and the examiner sees the reflected image of the postnasal space in the postnasal mirror. If the patient is quiet and relaxed, then usually soft palate does not contract and hide the view. This procedure especially needs concentration, patience and practice.

PATENCY OF NASAL CAVITIES

- *Spatula test (Fig. 8)*: A clean cold tongue depressor is held below the nose while patient exhales. Each area of mist formation on either side is compared.
- *Cotton-wool test*: A fluff of cotton is held against each nostril and its movements indicate the nasal blow of air while the patient inhales or exhales.
- *Alae nasi movements*: In cases of inspiratory obstruction, alae nasi collapse onto the septum.
- *Cottle test*: See section of nasal valves disorders of this chapter.

SENSE OF SMELL

See section smell of this chapter.

Ammonia, which stimulates the fibers of trigeminal nerve, is not used for testing smell.

PARANASAL SINUSES

They are examined by inspection, palpation and transillumination. The anterior group of sinuses (maxillary, frontal and anterior ethmoid) drains in middle meatus. The posterior ethmoid drains into superior meatus. The sphenoid sinus opens into sphenoethmoidal recess.

All the structures, which are adjacent to the different walls of these sinuses, need attention of the examiner. Sphenoid sinus, which opens in the sphenoethmoidal recess, lies deep and is not easy to examine directly. Frontal sinus has three walls: anterior, posterior and floor but only the anterior wall can be examined externally.

Tenderness

Tenderness of the sinuses can be elicited by pressure or percussion with a finger on their walls.

- *Frontal sinus*: Anterior and inferior walls above the medial part of eyebrow and above the medial canthus (Fig. 9).
- *Maxillary sinus*: Anterior wall over the cheek lateral to nose (Fig. 10).
- *Anterior ethmoids*: Medial wall of orbit just behind the root of nose.

Transillumination

- *Maxillary sinus*: A specially made light source is placed in the mouth and the mouth is closed. Normally, a crescent of light in the inferior fornix and glow in the pupil, which are equally bright on either side can be seen. The affected

Fig. 7: Posterior rhinoscopy. The examiner depresses the tongue and introduces posterior rhinoscopic mirror held in right hand and carried behind the soft palate.

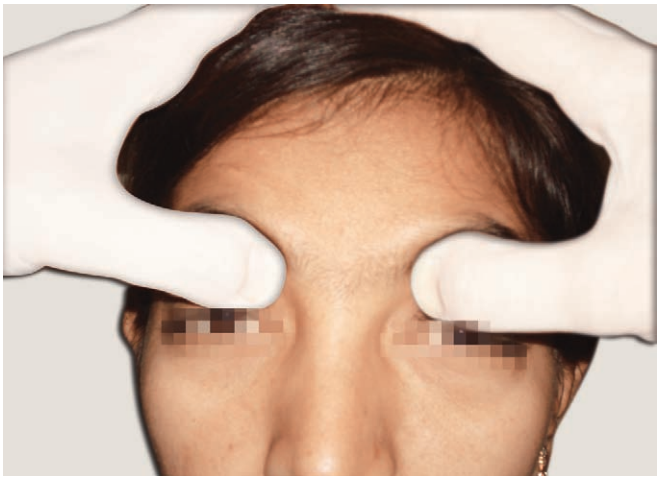


Fig. 9: Tenderness of the frontal sinuses elicited by pressure with finger on their walls

Fig. 8: Spatula test for patency of nose. A clean cold tongue depressor held below the nose while patient exhales. Mist formation on either side is compared

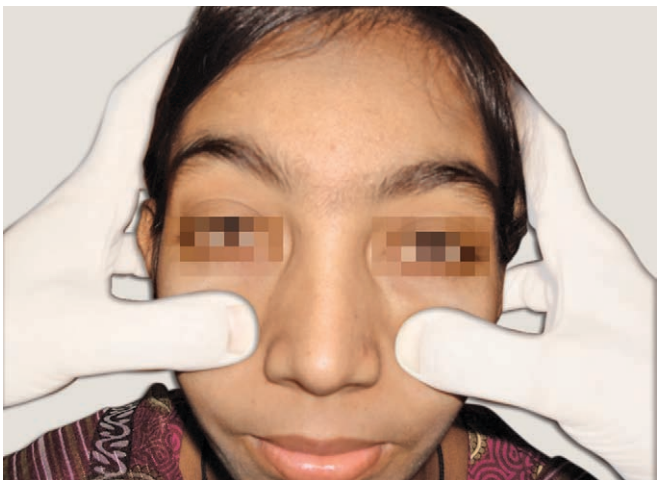


Fig. 10: Tenderness of the maxillary sinuses elicited by pressure with finger on their walls

side maxillary sinus will not transmit light if there is pus, thickened mucosa or a neoplasm.

- *Frontal sinus*: A small light source is placed in the superomedial angle of the orbit. The transmission of light from the anterior wall of the both side frontal sinuses is compared.

Endoscopic Examination

See chapter Operations of Nose and Paranasal Sinuses.

SPECIAL INVESTIGATIONS OF NASAL COMPLAINTS

SMELL

The odorous substance should be volatile and reach the olfactory area. Any lesion anywhere in the olfactory pathway (olfactory mucosa, olfactory nerves, olfactory bulbs and tract and the cortical center of olfaction) will affect smelling power of the person.

Factors Affecting Olfactory Testing

They include age, satiety, gender, adaptation and habituation and odor mixtures.

Causes of Olfactory Disorders

They include idiopathic, olfactory dysfunction, parosmia and phantosmia, decreased and distorted olfactory ability, congenital anomalies, obstructive sinonasal diseases, upper respiratory tract infections, HIV-AIDS, head injuries, iatrogenic (medications and surgeries), neoplasms, senile changes, toxic exposure, metabolic and endocrinal disorders, epilepsy and psychiatric disorders.

- *Loss of smell*: It can result from–
  - Nasal obstruction due to nasal polyps, enlarged turbinates, edema of mucous membrane as in common cold, allergic and vasomotor rhinitis.
  - Atrophic rhinitis.
  - *Peripheral neuritis*: Toxic or influenzal.
  - *Injury to olfactory nerves and olfactory bulb*: Fractures of anterior cranial fossa.

- *Intracranial lesions pressing olfactory tracts*: Abscess, tumor and meningitis

- *Parosmia*: It refers to perversion of smell. Patient interprets the odors incorrectly, which is usually disgusting odors. The causes of parosmia include:
  - *Recovery phase of postinfluenzal anosmia*: Misdirected regeneration of nerve fibers.
  - Intracranial tumor.

Anosmia: The three most common causes are sinonasal disease, post-upper respiratory tract infection and trauma (injury to olfactory nerves at cribriform plate or brain injury).

Tests for Smell

In routine testing, patient is asked to close eyes and smell common odors such as lemon, peppermint, rose, garlic, coffee, and cloves. Each side of the nose is tested separately. Quantitative estimation (quantitative olfactometry) needs special equipment.

- *Electro-olfactogram*: Electrode, which is placed on the olfactory epithelium, records a slow, negative and monophasic potential in response to odorants. *Electro-olfactogram* (EOG) represents a "generator potential".

MEASUREMENT OF MUCOCILIARY FLOW

- *Indigo carmine test/saccharin sodium test*: A drop of indigo carmine (8 mg/ml) and a drop of saccharin sodium are put at the anterior part of the floor of inferior meatus. After 3 minutes, patient is asked to swallow for 30 minutes and tell any taste of sweet (saccharin sodium). The examiner simultaneously inspects the posterior pharyngeal wall at these intervals for the blue dye of indigo carmine. The time lag between the perception of sweet taste and appearance of blue dye in the pharynx is noted. This time lag is called mucous transport time (MTT).
  - The normal value of MTT is 12–15 minutes (for ciliary beat frequency of 10/sec and transit time of 6 mm/minute).
  - More than 30 minutes MTT is considered grossly abnormal.
  - Other methods: They include photoelectric registration device and radioactive particle transit observed by gamma camera and multicollimator detectors.

NASAL OBSTRUCTION

Causes of Nasal Obstruction

1. *Unilateral Nasal Obstruction*
   a. *Infectious*: Furuncle, hypertrophic turbinate, concha bullosa, antrochoanal polyp and unilateral sinusitis.
   b. *Congenital*: Atresia and stenosis of nares, unilateral choanal atresia, nasoalveolar cyst.
   c. *Traumatic*: Foreign body, rhinolith, deviated nasal septum (DNS) and synechia
   d. *Neoplasms*: Papilloma, bleeding polyp of septum, benign and malignant tumors of nose and paranasal sinuses, and nasopharynx.
2. *Bilateral Nasal Obstruction*
   a. *Infectious*: Bilateral vestibulitis, rhinosinusitis (infectious, allergic and others), nasal polyps, atrophic rhinitis, septal abscess and large choanal polyp.
   b. *Congenital*: Congenital atresia of nares, bilateral choanal atresia and Thornwald's cyst.
   c. *Structural*: Collapsing nasal alae, stenosis of nares, DNS, adhesions between soft palate and posterior pharyngeal wall.
   d. *Traumatic*: Septal hematoma.
   e. *Neoplasms*: Large benign and malignant tumors.
   f. *Miscellaneous*: Hypertrophic turbinates and adenoid hyperplasia.

Differential Diagnosis (Fig. 11)

Observe the nose during normal and exaggerated nasal breathing and watch for the collapse of internal and external nasal valves. The site and causes of nasal obstruction include:

1. *Vestibule*: Caudal septal dislocation, synechia or stenosis
2. *Nasal valve*: Post-rhinoplasty synechia
3. *High septal deviation*: Examine the upper part of nasal septum.
4. *Turbinates*: Hypertrophic turbinates or concha bullosa
5. *Choanal*: Choanal atresia, choanal polyp.
   a. Unilateral choanal atresia, which is usually asymptomatic, may be missed in children.
   b. Choanal polyp, which may not be visible on the anterior rhinoscopy, can be seen with posterior rhinoscopy and nasal endoscopy examination.
6. *Polyps and septal hematoma*
7. *Tip ptosis*: It contributes to decreased nasal airflow.

Measurement of Nasal Obstruction

Resistance to airflow is the force that impedes flow of air. The diameter of nasal cavity is the most important variable in nasal airflow. In normal cases, airflow resistance is reduced one third after topical decongestion and further two third after wide alar retraction.

The measurement of obstruction is helpful in treatment planning, outcomes analysis and documentation for insurance and medicolegal reasons.

1. *Acoustic rhinometry*: It is an easy, minimally invasive, accurate and expeditious method of measuring dimensions of nasal airway. The sound waves are delivered to nasal cavity and the reflected sound waves are measured (rhinogram), which include calculation of minimal cross-sectional area of nose and nasal airway resistance. The first and second dips in rhinogram are caused due to nasal valve and anterior tips of turbinates (inferior and middle) respectively. The effect of topical decongestion helps in differentiating anatomic from physiologic causes of obstruction.
2. *Rhinomanometry or rhinometry*: This computerized electronic technique measures transnasal pressure and resultant nasal airflow. Initially, water columns and mechanical devices were used in rhinomanometry. The measurement of nasal airflow is a sort of an inverse measurement of nasal obstruction.

   This sensitive airflow rhinometry is a dynamic test of resistance to nasal airflow both before and after decongestion. It provides a numeric value of resistance. More than 3 cm H_2O/L/s combined bilateral resistance of adult nose at rest indicates nasal obstruction.

   The rhinometry measurement of airflow poorly correlates with subjective perception of nasal stuffiness and obstruction.

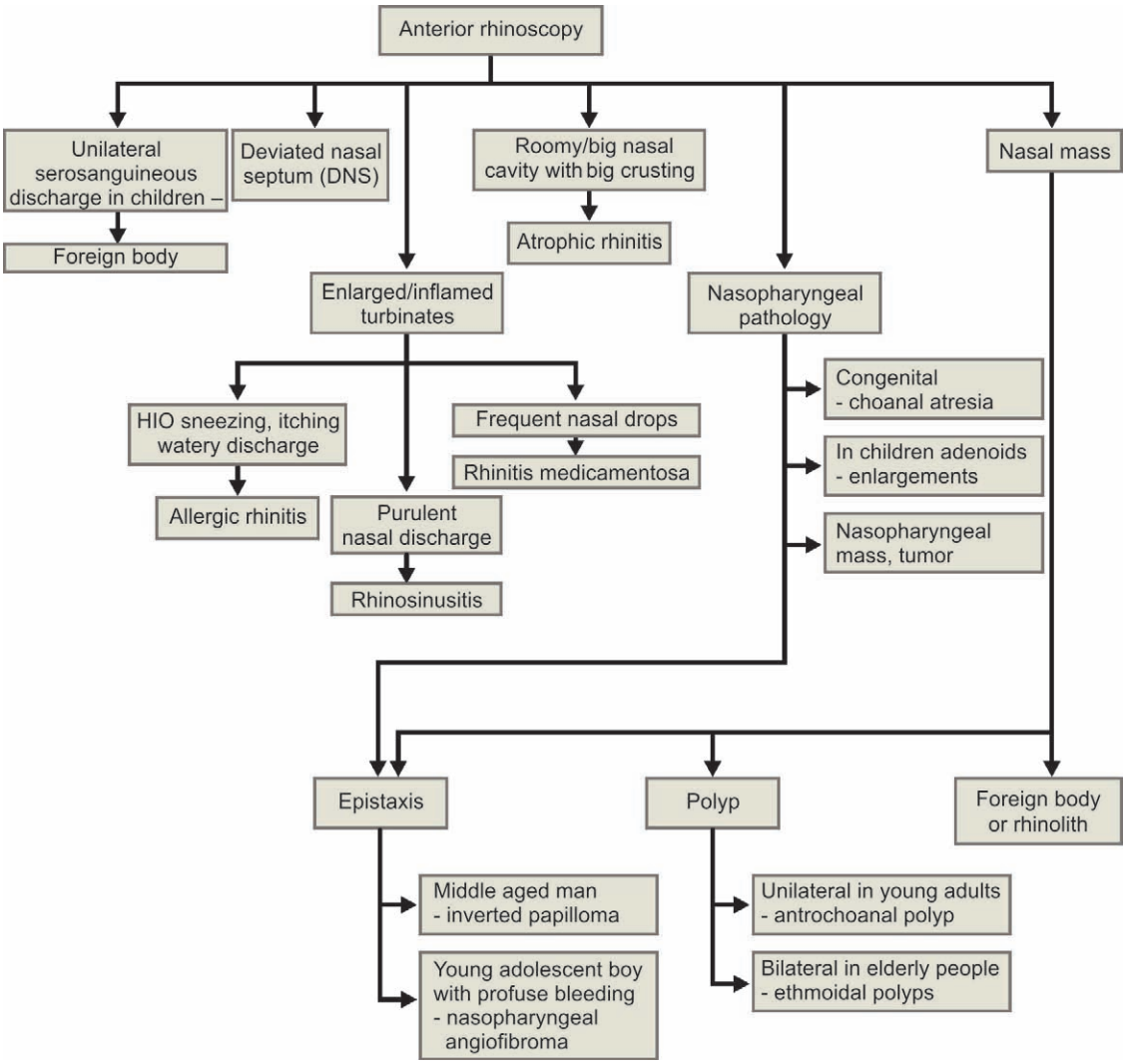
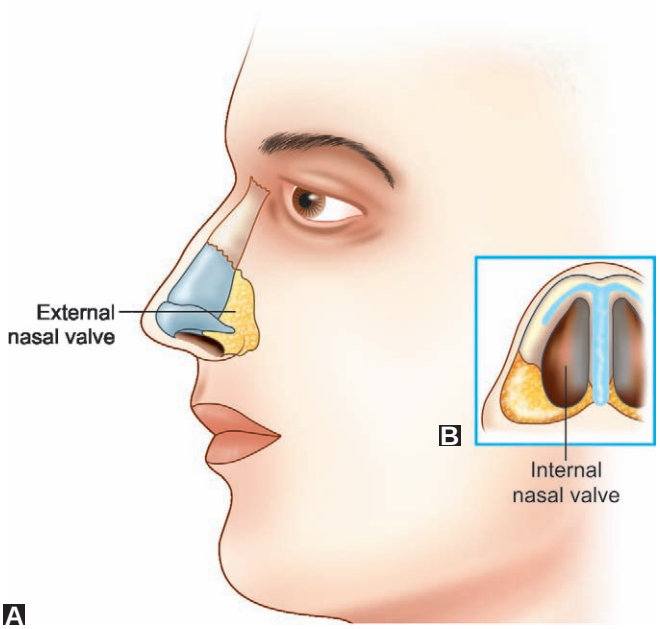


Fig. 11: A flow chart showing anterior rhinoscopy findings of various causes of nasal obstruction

NASAL VALVES DISORDERS (FIGS 12A AND B)

- *Internal nasal valve*: It is the narrowest area of airway. It is bounded by septum, lower margin of ULC and anterior aspect of inferior turbinate. The septal deformities and loss of ULC support can lead to nasal obstruction. The ULC may be thickened, twisted and concave or absent because of prior surgery.
- *External nasal valve*: It is at the level of caudal septum. This laterally based space is bounded by bony pyriform nasal aperture, lateral margin of ULC and lateral crura of lower lateral cartilage (LLC). The reasons of external valve compromise includes rhinoplasty, aging and caudal septal dislocation or trauma.
- *Cottle test (Fig. 13)*: In this test, which is done for the abnormality of the nasal valve, cheek is drawn laterally while patient breathes quietly. If there is subjective improvement in nasal airway, the test is positive, which indicates nasal valve compromise. The test also can be performed by lateralizing the ULC with a cotton-tipped applicator or cerumen curette.
- *Treatment*:
  - Spreader grafts: The angle between the ULCs and septum



Figs 12A and B: Nasal valves. (A) External nasal valve area (bounded by pyriform notch of maxilla and upper and lower lateral cartilages); (B) Internal nasal valve (inset) is bounded by septum, inferior edge of lower lateral cartilage, and anterior aspect of inferior turbinate

Fig. 13: Cottle test

can be widened with spreader grafts. The lateral wall of the nose can be stiffened with batten or umbrella graft.

- Nasal flaring: Nasal flaring 4-0 sutures are placed through the inferior lateral margin of the ULCs bilaterally and tied across the nasal dorsum. It acts as a fulcrum and opens internal valve angle.

RADIOLOGICAL IMAGING

See chapter Diagnostic Imaging.

DIAGNOSTIC ANTRUM PUNCTURE

See chapter Operations of Nose and Paranasal Sinuses.

ALLERGIC TESTS

See chapter Allergic and Nonallergic Rhinitis.

Clinical Highlights

1. **Transillumination test:** It was commonly performed to diagnose maxillary and frontal sinusitis.
2. **Ammonia:** It is used not for testing sense of smell but is used for testing lacrimation (greater petrosal nerve of facial nerve). In nose, ammonia stimulates fibers of trigeminal nerve supplying the nasal mucosa. A patient with complete anosmia will respond to inhalation of ammonia.
3. **Kallmann syndrome:** It consists of anosmia and congenital hypogonadism (amenorrhea in females).

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27 Diseases of External Nose and Epistaxis

Three things are necessary to make every man great, every nation great: (1) Conviction of the powers of goodness; (2) Absence of jealousy and suspicion; and (3) Helping all who are trying to be and do good.

—**Swami Vivekananda**

Points of Focus

DISEASES OF EXTERNAL NOSE

- **INFECTIONS**
  - ✓ Cellulitis
  - ✓ Furuncle
  - ✓ Vestibulitis
- **DEFORMITIES**
  - ✓ Saddle Nose
  - ✓ Hump Nose
  - ✓ Crooked and Deviated Nose
  - ✓ Stenosis and Atresia of Nares
- **TUMORS**
  - ✓ Dermoid Cyst
  - ✓ Encephalocele or Meningoencephalocele
  - ✓ Glioma
  - ✓ Nasoalveolar Cyst
  - ✓ Rhinophyma or Potato Tumor
  - ✓ Papilloma
  - ✓ Basal Cell Carcinoma (Rodent Ulcer)
  - ✓ Squamous Cell Carcinoma (Epithelioma)
  - ✓ Melanoma

EPISTAXIS

- **PERTINENT ANATOMY**
  - ✓ Little's Area/Kiesselbach's Plexus
- **CAUSES**
- **EVALUATION:** History and Examination
- **SITES OF EPISTAXIS:** ANTERIOR AND POSTERIOR
- **INVESTIGATIONS**
- **TREATMENT**
  - ✓ General measures
  - ✓ Nasal cautery
  - ✓ Anterior nasal packing
  - ✓ Posterior Nasal Packing
  - ✓ Arterial embolization
  - ✓ Arterial ligation
  - ✓ Surgical treatment
- **CLINICAL HIGHLIGHTS**

DISEASES OF EXTERNAL NOSE

INFECTIONS

People usually rub their nasal vestibule and surrounding skin. Minor abrasions and hair follicles are common sites of both acute and chronic infections. Coagulase-positive staphylococci are the most common pathogens. Nose is free of pathogens in majority of normal people.

Cellulitis of Nose

- *Etiology*: The causative organisms are streptococci or staphylococci. The infection may be an extension from the nasal vestibule.
- *Clinical features*: A red, swollen and tender nose.
- *Treatment*: Systemic antibacterial, hot fomentation and analgesics.

Furuncle or Boil of Nose

- *Etiology*: An acute infection of the hair follicle by *Staphylococcus aureus*.
- *Predisposing factors*: These are:
  - Picking of the nose
  - Plucking the nasal vibrissae
- *Clinical features*: A small, red, exquisitely painful and tender swelling, which may spread to the tip and dorsum of nose (Fig. 1) and rupture spontaneously in the nasal vestibule (Fig. 2).
- *Treatment*: Medical treatment includes warm compresses, analgesic and topical and systemic antibiotics.
  - The incision and drainage is done if fluctuation appears.
  - Furuncle should not be squeezed or prematurely incised because infection can spread to cavernous sinus through venous thrombophlebitis.
- *Complications*:
  - Cellulitis of the upper lip
  - Septal abscess
  - Cavernous sinus thrombosis

Fig. 1: Furuncle nasal vestibule. An acute red, exquisitely painful swelling spreading to the tip and dorsum of nose

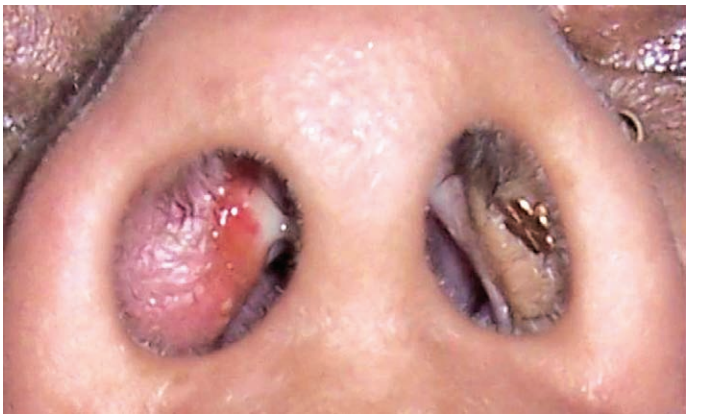


Fig. 2: Furuncle right nasal vestibule. Burst open and discharging pus

Nasal Vestibulitis

- *Etiology*: This diffuse dermatitis is caused by *S. aureus*.
- *Predisposing factors*: They are:
  - Nasal discharge due to rhinitis or sinusitis or nasal allergy.
  - Frequent picking or wiping of nose with handkerchief.
- *Clinical features*: Nasal vestibulitis can be of two types—acute and chronic.
  - Patients present with red, painful and tender swelling.
  - Crusts and scales and painful fissure, erosion or excoriation (Fig. 3).
  - The infection may involve upper lip.
- *Treatment*: They include:
  - Treat the cause of nasal discharge.
  - Clean all the crusts and scales with cotton soaked in hydrogen peroxide and apply antibiotic-steroid ointment, which is continued for few days after the apparent cure to prevent likely relapse.
  - A chronic fissure is cauterized with silver nitrate.

Fig. 3: Nasal vestibulitis

DEFORMITIES OF EXTERNAL NOSE

The appearance of the external nose is frequently the subject of concern. These nasal deformities can be either congenital or acquired. The acquired defects are usually due to injuries (Fig. 4). Plastic surgery is required for the restoration of appearance. The commonest deformity is the familial hump nose.

Saddle Nose (Depressed Nasal Dorsum)

It may be bony, cartilaginous or both.

- *Etiology*:
  1. Depressed nasal fracture is the most common cause.
  2. Excessive removal of septum in submucous resection.
  3. Septal hematoma.
  4. Septal abscess.
  5. Granulomatous lesions of nose: Leprosy, tuberculosis and syphilis.

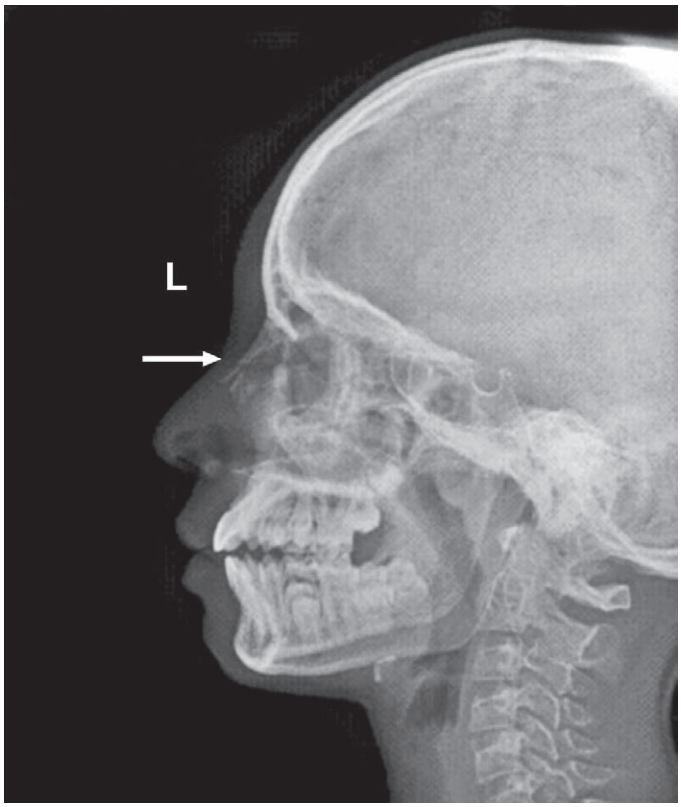


Fig. 4: X-ray nasal bone lateral view showing fracture of nasal bone
Source: Dr Jayesh Patel, Consultant Radiologist, Anand, Gujarat

- *Treatment*: Augmentation rhinoplasty, which fills the dorsum with cartilage, bone or a synthetic implant. Autografts are usually preferred over allografts. The chances of extrusion are more with synthetic implants, (silicone and teflon).
  - In cases of cartilaginous depression, septal or auricle cartilage is laid in a single or multiple layers.
  - In case of both cartilaginous and bony deformity, cancellous bone from the iliac crest is used as graft.

Hump Nose

- Hump may involve bone, cartilage or both.
- *Treatment*: Reduction rhinoplasty.

Crooked and Deviated Nose

In crooked nose, the dorsum of the nose (from frontonasal angle to the tip) is curved in either C or S shaped (Fig. 5). In deviated nose, dorsum of the nose (from frontonasal angle to the tip) is deviated to one side.

- *Etiology*: Trauma is the most common cause. As the nose grows, injuries, which may be sustained during birth, neonatal period or childhood, can manifest into these deformities.
- *Treatment*: Rhinoplasty or septorhinoplasty corrects deformity and nasal obstruction.

Stenosis and Atresia of Nares

- *Etiology*
  1. Web formation and stenosis may occur after trauma or surgery of nasal tip or vestibule. In Young's operation, which is done in atrophic rhinitis, nares are deliberately closed with vestibular skin flaps.
  2. Destructive inflammatory lesions of nose.
  3. Congenital atresia due to noncanalization of epithelial plug. It is rare.
- *Treatment*: Reconstructive plastic surgery.

TUMORS OF EXTERNAL NOSE

They can be divided into three categories—congenital, benign or malignant (Table 1).

Dermoid Cyst of Nose

- *Types*: It is of two types—simple and with sinus (Figs 6A to C).
  - *Simple dermoid*: It presents as a midline swelling over the nasal bones and does not have any external opening.
  - *Dermoid cyst with sinus*:
    - External pit: Infants and children present with a pit or a sinus over the dorsum of nose. Hair may be protruding out from the sinus.
    - Intracranial connection: The sinus track communicate intracranially. Dermoid cyst lies between nasal bones and upper part of septum.
- *Treatment*: Surgery consists of splitting of the nasal bones and removal of cyst along with its extension in the upper part of the nasal septum. Intracranial extension needs associated neurosurgical approach.

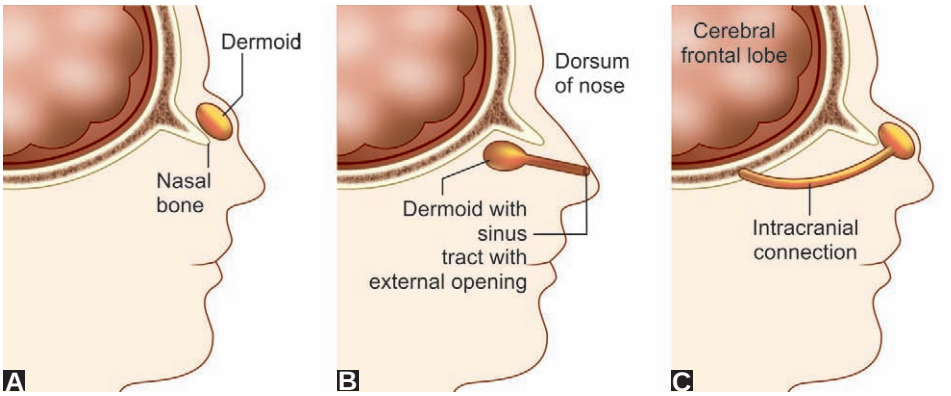
Extranasal Encephalocele or Meningoencephalocele

Herniation of brain tissue with meninges occurs through a congenital bony defect, which may be intranasal or extranasal. For intranasal meningoencephalocele**,** see chapter Tumors of Nose, Paranasal Sinuses and Jaws.

Fig. 5: Crooked nose "S" shaped

TABLE 1 Classification of swellings of external nose and vestibule

Congenital	*Benign*	*Malignant*
Dermoid	Rhinophyma or potato tumor	Basal cell carcinoma (rodent ulcer) Squamous cell carcinoma (epithelioma).
Encephalocele or meningoencephalocele	Papilloma Hemangioma	
Glioma	Pigmented nevus	
Nasoalveolar cyst	Seborrheic keratosis	Melanoma
	Neurofibroma	
	Tumors of sweat glands	



Figs 6A to C: Nasal dermoid. (A) Simple dermoid lies superficial to nasal bones; (B) Dermoid with sinus. Dermoid situated deep to nasal bones and sinus tract has an external pit in the midline of dorsum of nose; (C) Dermoid with intracranial connection

- *Clinical features*: An extranasal meningoencephalocele presents with pulsatile swelling. This cystic swelling is reducible and shows cough impulse. Swelling may be present in the following sites:
  1. Root of nose (nasofrontal variety).
  2. Side of nose (nasoethmoidal variety).
  3. Anteromedial aspect of the orbit (naso-orbital variety).
- *Treatment*: It needs neurosurgery, which includes severing the stalk from the brain and repairing the bony defect.

Extranasal Glioma

Most of the gliomas, which are nipped off portions of encephalocele (during embryonic development), are extranasal (60%). Other types include intranasal (30%) and both intra and extranasal (10%). For intranasal glioma, see chapter Tumors of Nose, Paranasal Sinuses and Jaws. Extranasal gliomas are encapsulated.

- *Clinical feature*: Firm subcutaneous swellings may be seen on nasal bridge, side of nose or near the inner canthus.
- *Treatment*: Glioma is removed by external nasal approach.

Nasolabial (Nasoalveolar, Klestadt's) Cyst

This squamous epithelium-lined cyst arises from the epithelial rests situated at the junction of globular, lateral nasal and maxillary processes. It lies on the bone and causes an excavation. It is closely attached to the floor of the nose. Globulomaxillary cyst is intrabony.

- *Clinical features*: It presents as a smooth and soft bulge in the lateral wall and floor of nasal vestibule anterior to inferior turbinate (Fig. 7). Large cyst obliterates the alar facial fold (nasolabial sulcus). The fluctuation and fluid thrill is best felt with one finger in vestibule and second under the lip.
- *Treatment*: The cyst is excised through sublabial approach from gingivobuccal sulcus near the midline. A portion of nasal mucosa may be removed because cyst is adherent to it.

Rhinophyma or Potato Tumor

This slow growing benign tumor occurs due to the hypertrophy of sebaceous glands in the region of nasal tip. Most of the patients are men past middle age.

- *Clinical features*: It is usually seen in long-standing cases of acne rosacea.
  - Patient presents with pink and lobulated mass over the nose with superficial vascular dilation, which gives unsightly appearance to the face.

Fig. 7: Left side nasolabial cyst. Note the smooth bulge in the floor and lateral wall of left nasal vestibule and upper part of upper lip. Cyst is obliterating the alar facial fold

  - The big tumors can cause nasal obstruction and obstructed vision.
- *Treatment*: Paring down of the bulk of tumor may be done with sharp knife or carbon dioxide laser. The raw area may be allowed to re-epithelialize. The complete excision of tumor is usually followed by skin grafting.

Papilloma or Wart of Nose

- *Clinical features*: This may be single or multiple and pedunculated or sessile.
- *Treatment*: It consists of surgical excision, which is usually done under local anesthesia.

Basal Cell Carcinoma (Rodent Ulcer) of External Nose

This most common malignant tumor of nose skin (87%) equally affects either sex and occurs in the age group of 40–60 years.

- *Clinical features*:
  1. Common sites are tip and ala of nose.
  2. This slow growing lesion may present as a cyst, pearly papule, nodule or an ulcer with rolled edges.
  3. The lesion, which remains confined to the skin for a long time, may invade underlying cartilage or bone.
  4. Lymph node metastasis is extremely rare.
- *Treatment*: The extent of surgery depends on the size, location and depth of the tumor.

- Early lesion: It may be treated with cryosurgery, irradiation or surgical excision. The latter includes 3–5 mm of normal skin around the tumor mass.
- Recurrent and extensive lesions: The wide resection of recurrent and extensive lesions involve cartilage and bone. The large surgical defect is closed by local or distant flaps or a prosthesis.

Squamous Cell Carcinoma (Epithelioma) of External Nose

This second most common malignant tumor of nose skin (11%) equally affects either sexes and occurs in the age group of 40–60 years.

- *Clinical features*:
  1. It presents as an infiltrating nodule or an ulcer with rolled out edges.
  2. The common sites are lateral wall of the vestibule and columella. It which may extend into nasal floor and upper lip.
  3 Nodal metastases to the parotid and submandibular nodes are seen in 20% cases.
- *Treatment*:
  1. Early lesion: It responds well to radiotherapy.
  2. Advanced lesions: Advanced lesions, which involve bone or cartilage, need wide surgical excision and plastic repair of the defect.
  3. Metastatic cervical lymph nodes: They require block dissection.

For squamous cell carcinoma of nose and paranasal sinuses, see chapter Tumors of Nose, Paranasal Sinuses and Jaws.

Melanoma Nose

- *Clinical features*: This slow growing, rare lesion may present as superficially spreading type or nodular invasive type.
- *Treatment*: It is treated with surgical excision.

For intranasal malignant melanoma, see chapter Tumors of Nose, Paranasal Sinuses and Jaws.

EPISTAXIS

Bleeding from the nose is called epistaxis (synonym: nosebleed). Epistaxis is a symptom and one should try to find its local and systemic causes.

- Approximately 50% population experience nosebleed in their life but it is serious enough to seek medical consultation in less than 10%. Some patients present as an emergency. There is no age bar.
- Children usually have mild anterior nasal bleeding while elderly have profuse posterior nose bleeding.
- Males are affected more than females but after the age of 50 years both the sexes are affected equally.

PERTINENT ANATOMY

Little's Area/Kiesselbach's Plexus

Little's area is situated in the anterior inferior part of nasal septum and is supplied by branches of both external and internal carotid arteries.

The four arteries, the branches of which anastomose richly and form a vascular plexus (Kiesselbach's plexus) in this region are:

1. Anterior ethmoidal.
2. Septal branch of superior labial.
3. Septal branch of sphenopalatine.
4. Greater palatine.

This vascular area is the most common site of nosebleed in children and young adults. It gets dried due to the effect of inspiratory current and easily traumatized due to frequent picking (fingering) of nose.

Retrocolumellar Vein

It runs vertically downward just behind the columella and crosses the floor of nose. It is a common source of venous bleeding in young people.

CAUSES

Causes of epistaxis can be divided into local and general (Box 1). The most common cause of nosebleed in children is repeated fingernail trauma to Little's area. In elderly people, most common cause is atherosclerotic changes and hypertension. In many cases, no obvious cause is ascertained (idiopathic). In vicarious menstruation, nosebleed occurs at the time of menstruation.

EVALUATION

The initial evaluation of nosebleed patients includes assessment of hemodynamic stability, airway compromise and vital parameters.

History

Complete history includes following questions:

1. Side of bleeding; worst side if bleeding is bilateral.
2. Symptoms of posterior nasal bleeding such as expectoration of blood and hematemesis.
3. Precipitating events such as trauma, acute infection, nasal drops and sprays and surgery.
4. Duration and amount of bleeding.
5. Syncopal or near syncopal attacks.
6. Risk factors such as hypertension, leukemia, hemophilia, purpura, congestive heart failure, renal failure and liver dysfunction and their medications.
7. Drugs such as aspirin, NSAIDs (ibuprofen), heparin, antiplatelet drugs, high doses of vitamin E.
8. Past history of bleeding and its treatment.
9. Family history of bleeding tendencies such as hemophilia.

Examination

1. Examination should include vital parameters, complete ear, nose and throat examination, general features and systemic examination.
2. The examination is often treatment oriented and should try to locate the cause and site of bleeding. Patient needs reassurance.
3. The local anesthesia (4% xylocaine) and decongestants (phenylephrine) and mild sedation make the nasal examination as well as patient comfortable.

Box 1: Causes of epistaxis

1. Local causes (Nose, Paranasal Sinuses and Nasopharynx)
   a. *Trauma*: Finger nail trauma (obsessive compulsive disorder), injuries of nose (accidental, homicidal, surgery), maxillofacial trauma, head injuries, nasal intubation, foreign bodies, rhinolith, blowing of nose too hard and violent sneezing.
   b. *Infections*: Rhinitis, nasal vestibulitis, sinusitis, adenoiditis, diphtheria, pyogenic granuloma, rhinosporidiosis, granulomatous lesions (tuberculosis, syphilis, sarcoidosis, Wegener's granuloma), atrophic rhinitis and rhinitis sicca, maggots and leeches, neglected foreign body.
   c. *Neoplasms*:
      i. Benign: Hemangioma, inverted papilloma, juvenile angiofibroma, aneurysms, encephalocele.
      ii. Malignant: Epidermoid carcinoma, adenocarcinoma, sarcoma, esthesioneuroblastoma.
   d. *Environmental*: High altitudes, sudden decompression (Caisson's disease), chemicals, pollution.
   e. *Drugs*: Nasal sprays and drops of antihistaminics and steroids, sniffing of cocaine, snuff.
   f. Miscellaneous: Septal deformities and perforation.
2. General causes
   a. *Cardiovascular*: Hypertension, mitral stenosis, congestive heart failure, eclampsia of pregnancy, tumors of mediastinum (raised venous pressure).
   b. *Hemopoietic*: Aplastic anemia, leukemia, thrombocytopenic and vascular purpura, coagulopathies (congenital and acquired), hemophilia, Christmas disease, polycythemia vera, multiple myeloma.
   c. *Nutritional*: Malnutrition, scurvy, alcohol abuse and vitamin A, D, C, E and K deficiencies and high doses of vitamin E.
   d. *Blood vessels*: Arteriosclerosis, collagen diseases and hereditary hemorrhagic telangiectasia (HHT).
   e. *Liver disease*: Hepatic cirrhosis (deficiency of factor 2, 7, 9 and 10).
   f. *Kidney disease*: Chronic nephritis and renal failure.
   g. *Drugs*: Aspirin, nonsteroidal anti-inflammatory drugs (NSAIDs) such as ibuprofen and diclofenac and anticoagulant therapy such as heparin and coumadin.
   h. *Acute infections*: Influenza, measles, chickenpox, whooping cough, rheumatic fever, infectious mononucleosis, typhoid, pneumonia, malaria, dengue fever.

4. Clots need evacuation by forceps and suction. Patient can be asked to blow the nose, which helps in cleaning the nose and bringing out large clots. Patient should be encouraged to cough out clots.
5. Vital signs and patient's mental status need constant attention during the physical examination.
6. Some patients may need examination under general anesthesia.
7. Posterior nasal examination in postnasal bleeding needs nasal endoscope.
8. Loupe magnification helps in localizing the site of anterior nasal bleeding.

SITES OF EPISTAXIS

1. The most common site of bleeding in children and young people is Little's area.
2. Other sites include–
   - Above the level of middle turbinate from anterior and posterior ethmoidal branches of ophthalmic artery (branch of internal carotid artery).
   - Below the level of middle turbinate from sphenopalatine branch of internal maxillary artery (branch of external carotid artery).
3. Sometimes, the bleeding site may be hidden by middle and inferior turbinates.
4. Local decongestants help in localization of the bleeding site and packing of the nose.
5. In posterior nasal and nasopharynx, blood flows directly into the pharynx.
6. In general systemic disorders, there occurs diffuse bleeding from septum and lateral nasal wall.

Anterior Epistaxis

- In anterior epistaxis, blood flows from anterior nasal opening (Table 2).
- It is more common than posterior nasal bleeding. The common sites are Little's area and anterior part of lateral nasal wall.
- It is usually mild and may be controlled by local pressure (pinching of nose) or anterior packing.
- It mostly affects children and young adults and the most common cause is trauma.

Posterior Epistaxis

- Posterior nasal bleeding, which is less common but more severe, occurs spontaneously. Most of the patients are more than 40 years of age. Bleeding is so severe that most patients require hospitalization and postnasal packing (Table 2).
- The blood, which can be seen in the pharynx, is swallowed by the patient and then later on vomited out as a coffee colored vomitus that may erroneously be considered as hematemesis.
- The bleeding site, which is difficult to localize, is mostly in posterosuperior part of nasal cavity.
- The most common cause is hypertension and arteriosclerosis.

INVESTIGATIONS

Recurrent and profuse nosebleed cases need following investigations. These must be ordered as per the patient's requirement:

1. *Complete blood count*: Anemia, leukemia and thrombocytopenia.
2. Bleeding time, clotting time, prothrombin time, partial thromboplastin time.
3. *Radiological*: X-ray chest, computed tomography (CT), magnetic resonance imaging (MRI) and angiography.

TREATMENT

The treatment includes general measures, nasal cautery, anterior and posterior nasal packing, arterial embolization and ligation and surgery.

A. General Measures

The following general measures are taken in cases of epistaxis:

1. Home care (Box 2).
2. Reassurance and mild sedation.
3. Position of the patient: Sitting with a backrest.
4. Recording of blood loss: Direct and through spitting and vomiting.

TABLE 2 Differences between anterior and posterior epistaxis

	Anterior nasal bleeding	*Posterior nasal bleeding*
Incidence	More common	Less common
Common sites	Little's area	Posterosuperior part of nasal cavity
Localization	Easy	Difficult
Common age	Children and young people	Elderly people, more than 40 years
Common cause	Trauma (fingernail)	Arteriosclerosis and hypertension
Nasal endoscope	Not required	Required
Severity	Mild bleeding	Profuse bleeding
Management	Comparatively easy and in OPD	Difficult and in operation theater
Anesthesia	Usually local	Usually general

Box 2: Home care of epistaxis patients

Patients and their relatives are given following advises if bleeding occurs at home:

- Prevention
  1. Avoid frequent cleaning (obsessive-compulsive tendencies) of nose with tissue paper or finger.
  2. Maintain proper nasal hydration with saline, gels and ointment.
  3. Increase ambient humidity with a bedroom humidifier.
- Treatment
  1. Pinch the nose with thumb and index finger for about 5 minutes. It usually stops the bleeding from the Little's area, which is the most common site of bleeding.
  2. Place a small piece of cotton soaked in decongestant nasal drops.
  3. Lean back no further than 45°.
  4. In Trotter's method, patient sits and leans little forward over a basin to spit any blood. Patient breathes quietly from the mouth.
  5. Cold compresses over the nose results in reflex vasoconstriction. Drops of ice-cold water directly in the nose.
  6. Report to doctor if bleeding does not stop.

5. Monitoring of vital parameters: Pulse, temperature, blood pressure and respiration, hemodynamic stability (blood transfusion if needed) and airway compromise.
6. Intermittent oxygen: Bilateral nasal packing increases pulmonary resistance due to nasopulmonary reflex.
7. Antibiotics in cases of infection and nasal packing.
8. Investigation and treatment of the cause.

B. Nasal Cautery

Light anterior nasal packing should be done after cauterization.

1. *Chemical cautery* with a bead of silver nitrate is helpful in cases of mild bleeding. The tip of silver nitrate stick should be held in contact of bleeding site for several seconds. The cauterized area becomes grayish-white in color.
2. *Electrocautery (monopolar, bipolar or suction cautery)*: It is employed in cases of failure and severe bleeding.
3. *Topical anesthesia* with 4% xylocaine and then local infiltration with xylocaine adrenaline are used before cauterization.
4. *Endoscopic nasal cautery*: Posterior bleeding points are seen and cauterized better under endoscopic vision. With the help of suction cautery, the procedure may be done successfully under local anesthesia and sedation.
5. *Complication*: Avoid deep and bilateral cautery as they can cause septal perforation.

C. Anterior Nasal Packing (Fig. 8)

1. *Indications*: Anterior nasal packing is done in cases of active anterior epistaxis. Cauterization of the bleeding area is tried first. But if bleeding is profuse and the site of bleeding cannot be localized, anterior nasal packing is done.
2. *Method*: Nose must be cleared of blood clots by suction and forceps. One meter long ribbon gauze (width 2.5 cm in adults and 12 mm in children), which is soaked in liquid paraffin, is packed tightly in each nasal cavity by layering the gauze from floor to the roof and from before backwards. The initial few centimeters of ribbon gauze are folded upon it and introduced along the floor. Some consultants prefer vertical layers of ribbon gauze from back to the front. If bleeding starts from another nose then posterior nasal bleeding must be suspected. Either one or both nasal cavities may be packed.
3. *Removal*: Pack can be removed after 24 hours or after 2–3 days.
4. *Packing materials*: In addition to ribbon gauze, other packing materials include Merocel Pope or Kennedy nasal sponges, prefashioned anterior nasal balloons, Gelfoam, and oxidized cellulose (Surgicel).
5. *Systemic antibiotics* are started to prevent infection and toxic shock syndrome.

D. Posterior Nasal Packing

Postnasal packing usually requires general anesthesia and patient needs hospitalization.

1. *Indications*: Posterior nasal packing is done when cauterization fails and bleeding site cannot be determined.

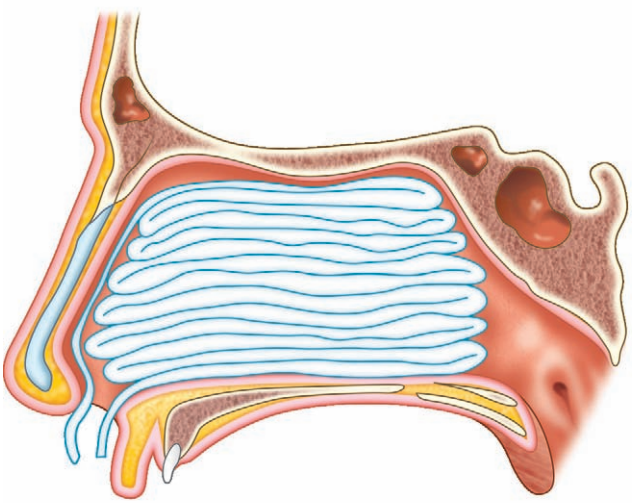


Fig. 8: Anterior nasal packing in horizontal layers with nonresorbable vaseline gauze strip

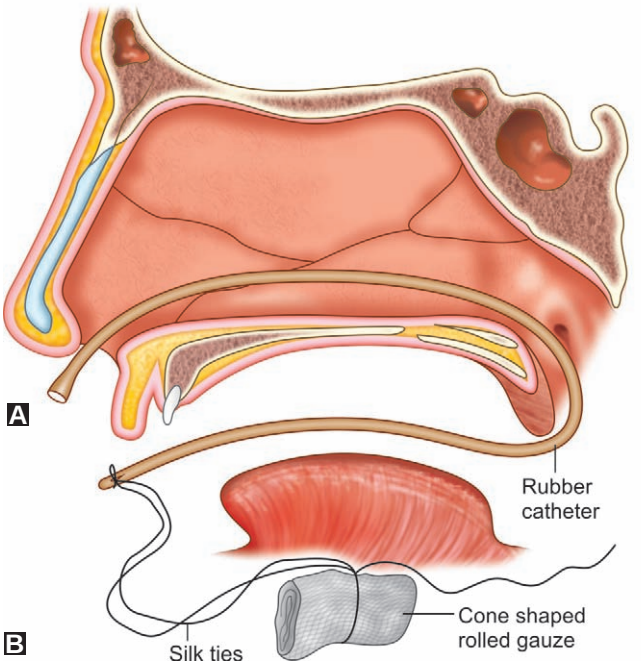
2. *Methods*:
   a. *Gauze*: A piece of gauze is rolled into the shape of a cone. Then three silk ties are tied to this cone-shaped gauze (Figs 9A and B). A rubber catheter is passed through the nose. Its pharyngeal end is brought out from the mouth and the silk threads of postnasal pack are tied to it. The catheter along with the silk threads is gradually withdrawn from nose and postnasal pack tied with silk threads is guided into the nasopharynx with the index finger. Anterior nasal cavity is now also packed. The silk threads are tied over a dental roll. The third silk thread, which is cut short, hangs in the oropharynx and helps in easy removal of the postnasal packing.
   b. *Foley's catheter* (Fig. 10): Some ENT surgeons use Foley's catheter, the bulb of which is inflated with saline. The catheter is pulled out and choana is blocked by the inflated bulb. Then anterior nasal packing is done.
   c. *Nasal balloon* (Fig. 11): The current variety of nasal balloon has two bulbs, one lies in postnasal space while another remains in nasal cavity.

Posterior epistaxis: One to two percent of patients die within 1 year.

E. Arterial Embolization

Arterial embolization is done in refractory cases of epistaxis. It is an invasive process and performed in angiography suite by an experienced neuroradiologist.

1. *Method*: First diagnostic angiography of bilateral carotid system is done. Then in ideal cases, catheter is guided into internal maxillary artery. Gelfoam and/or polyvinyl alcohol particles are used for embolization.
2. *Contraindications*: It is contraindicated in cases of severe atherosclerotic disease, anomalous anastomosis and allergy to contrast.



Figs 9A and B: Posterior nasal packing. (A) Procedure; (B) Post-nasal pack: gauze piece rolled into cone with three silk ties

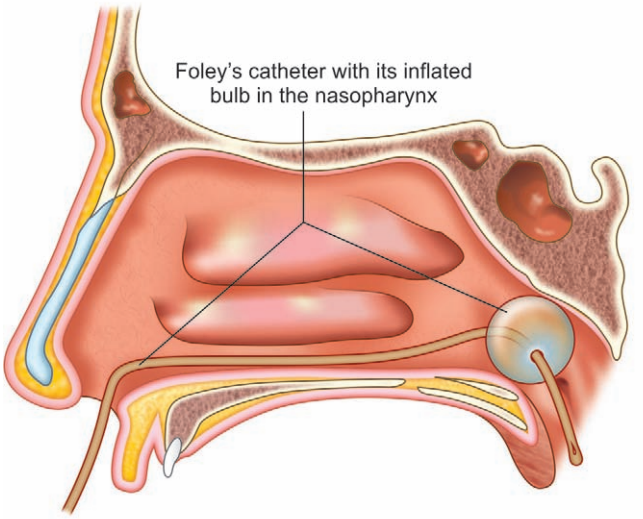


Fig. 10: Posterior nasal packing with Foley's catheter

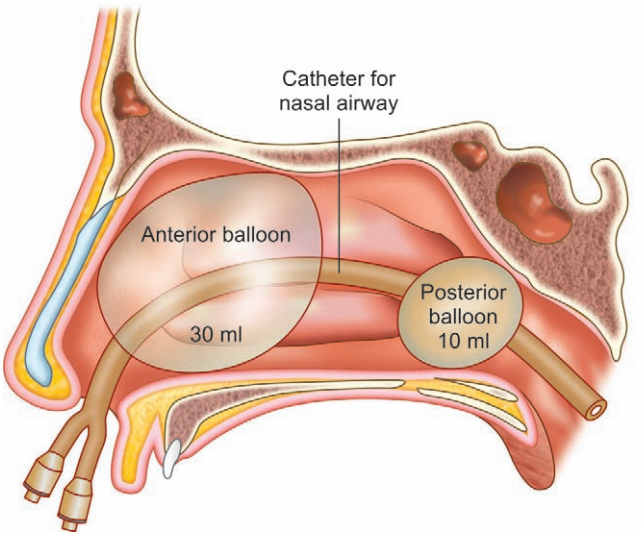


Fig. 11: Epistaxis balloon. Smaller (10 ml) posterior balloon and bigger (30 ml) anterior balloon are inflated. Channel of catheter provides airway for nasal breathing

F. Arterial Ligation

1. *External carotid artery*: Ligation of external carotid artery above the origin of its first branch (superior thyroid artery) is done, when the area of profuse bleeding (supplied by external carotid system) is not controlled by packing and cauterization. The facilities of embolization and ligation of its more peripheral branches have obviated its need.
2. *Maxillary artery*: The internal maxillary artery can be approached in various ways. It is ligated in cases of uncontrollable posterior epistaxis.
   a. Transantral: In Caldwell-Luc approach, posterior wall of maxillary sinus is removed and clips are applied to the maxillary artery or its branches. In pterygopalatine fossa, the maxillary artery is usually anteroinferior to the maxillary and vidian nerves.
   b. Endoscopic ligation of the maxillary artery can also be done through nose.
3. *Ethmoidal arteries*: The bleeding area above the level of middle turbinate if not controlled by cautery and packing

is managed by ligation of anterior and posterior ethmoidal arteries, which are exposed in the medial wall of the orbit. The foramina of anterior and posterior ethmoidal arteries, which are about 18 mm and 28 mm respectively away from anterior lacrimal fossa crest, are situated in frontoethmoid suture. The external ethmoid incision is employed.

G. Surgical Treatment

1. *Persistent nasal septal bleeding*: The elevation of mucoperichondrial flap from the nasal septum results in fibrosis and constriction of blood vessels. Submucous resection operation with removal of septal spur also achieves the same end result.
2. *Hereditary hemorrhagic telangiectasia or Osler-Weber-Rendu Disease*: Hereditary hemorrhagic telangiectasia involves the anterior part of nasal septum and causes recurrent episodes of profuse bleeding.
   a. *Laser*: It can be managed by Argon, potassium titanyl phosphate (KTP) or neodymium-doped yttrium aluminium garnet (Nd: YAG) lasers. The recurrence, which is well known in the surrounding mucosa, needs repeated application of lasers.
   b. *Septodermoplasty*: The anterior part of septal mucosa is excised and replaced by a split skin graft.

Clinical Highlights

1. **Augmentation rhinoplasty**: Nasal tip deformity is corrected by augmentation rhinoplasty.
2. **Rhinophyma**: This hypertrophy of sebaceous glands of nasal tip is associated with acne rosacea.
3. **Little's area**: It is the anteroinferior part of nasal septum and is the most common site of epistaxis. It constitutes about 90% cases of nosebleed.
4. **Kiesselbach's plexus**: Arteries which participate in Kiesselbach's plexus include anterior ethmoidal, greater palatine and superior labial.
5. **Woodruff's plexus**: This plexus of veins, which is situated inferior to the posterior end of inferior turbinate, is a site of posterior epistaxis in adults.
6. **Ethmoidal arteries**: Anterior and posterior ethmoidal arteries are the branches of ophthalmic artery, which is the branch of internal carotid artery.

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28 Infectious Rhinosinusitis

In doing evil we injure ourselves and others also. In doing good we do good to ourselves and to others as well. If I do an evil action, I must suffer for it; there is no power in this universe to stop or stay it. Similarly, if I do a good action, there is no power in the universe which can stop its bearing good results.

—**Swami Vivekananda**

Points of Focus

- **INTRODUCTION**
- **CLASSIFICATION**
  - ✓ Viral Rhinosinusitis
  - ✓ Acute Bacterial Rhinosinusitis
  - ✓ Chronic Rhinosinusitis
  - ✓ Pediatric Rhinosinusitis
- **COMPLICATIONS OF RHINOSINUSITIS**
  - ✓ Mucocele/Pyocele
  - ✓ Orbital Complications
  - ✓ Cavernous Sinus Thrombosis
  - ✓ Osteomyelitis
  - ✓ Intracranial Complications
- **NASAL POLYPS**
  - ✓ Antrochoanal Polyp and Ethmoidal Polyps
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

Rhinosinusitis (RS) is an inflammation of the nose and paranasal air sinuses. It accounts for about 5% of visits to primary care physicians. Its common causes include viral, bacterial and allergic. RS in children is a multifactorial disease and the importance of predisposing factors changes with increasing age.

There is almost always involvement of nose in all the inflammatory sinus condition. Therefore Task Force of the Rhinology and Paranasal Sinus Committee in 1997 suggested the term rhinosinusitis (RS), instead of sinusitis. Formerly, sinusitis and rhinitis were described separately.

CLASSIFICATION

The RS can be divided into different types on the basis of duration. Patients whose symptoms never disappear completely for more than 12 weeks are termed chronic rhinosinusitis (CRS), while patients who have complete recovery between the episodes of RS, which lasts for more than 7 days, are considered to have recurrent rhinosinusitis.

1. Acute Rhinosinusitis (< 4 weeks duration)
   1. Viral Rhinosinusitis (VRS)
   2. Acute Bacterial Rhinosinusitis (ABRS)
2. Subacute Rhinosinusitis (4–12 weeks duration)
3. Chronic Rhinosinusitis (> 12 weeks duration)
   a. With Polyps
   b. Without Polyps
4. Recurrent Acute Rhinosinusitis (3 episodes in 6 months or 4 or > 4 episodes of acute RS in 1 year)

For the sake of clinical management and description we can classify them as following:

1. Acute and subacute bacterial rhinosinusitis
2. Chronic rhinosinusitis
3. Fungal sinusitis
4. Rhinosinusitis in children
5. Complications of bacterial sinusitis

Sinusitis

In the anatomical classification, the sinusitis can be maxillary, ethmoidal, frontal, sphenoidal, multisinusitis (involvement of >1 sinus), unilateral, bilateral and pansinusitis (involvement of all the sinuses). The commonly involved sinuses are maxillary and ethmoid. A sinusitis is termed open if pus can drain through its natural ostium while it is not possible in close type. The latter causes more severe symptoms and can cause complications.

Acute RS, which is primarily an infectious disease, can range from acute viral rhinitis (the common cold) to ABRS.

VIRAL RHINOSINUSITIS (COMMON COLD)

The average attacks of VRS in a year in an adult can be two to four. Approximately 20–30% cases of acute rhinosinusitis are viral.

Etiology

1. The common viruses are: rhinovirus (most common in adults) and parainfluenza viruses.
2. Other viruses are adenovirus, picorna virus and its subgroups such as rhinovirus, coxsackie and ECHO viruses.
3. Respiratory syncytial virus (RSV) and influenza virus are more destructive to respiratory cilia. These have many serotypes, which vary in potency and severity of infection.
4. The infection is usually contracted through airborne droplets.
5. Incubation period is 1–4 days and illness lasts for 2–3 weeks.

Clinical Features

Viral rhinosinusitis and ABRS have many common clinical presentations.

1. Usually VRS begins with sore throat, which lasts for 1 or 2 days and then followed by cough and nasal discharge.
2. In beginning there may be burning sensation at the back of nose, nasal stuffiness, rhinorrhea and sneezing.
3. Patient has chill and low-grade fever.
4. Nasal watery discharge is profuse and may become mucopurulent due to secondary bacterial invasion (*Streptococcus haemolyticus*, *Pneumococci*, *Staphylococcus*, *Haemophilus influenzae*, *Klebsiella pneumoniae* and *Morexella catarrhalis*).

Treatment

There is no approved treatment. The following ancillary therapies have little supporting data.

1. *Efficient in symptom control*: Decongestants, anticholinergics, and first generation antihistamines (diphenhydramine, chlorpheniramine, clemastine).
   a. Antihistamines: Symptoms can be easily controlled with antihistaminics and nasal decongestants. Antihistamines may impair psychomotor performance often without sedation or other noticeable symptoms. Patients should not drive or operate heavy machinery. In elderly patients (> 65–70 years of age) they are avoided as the risks of delirium and cognitive decline are high. As the rhinorrhea is diminished due to anticholinergic property newer less sedating antihistamine are less effective.
   b. Oral decongestants: They may cause insomnia and agitation. Use with caution in cases of hypertension, ischemic heart disease, glaucoma, prostatic hypertrophy, diabetes mellitus and geriatric patients. They are contraindicated in patients using monoamine oxidase (MAO) inhibitors or having uncontrolled hypertension or severe coronary artery disease.
   c. Ipratropium bromide nasal spray: It reduces rhinorrhea.
   d. Topical decongestants: For nasal congestion, use should be limited to 3 days.
2. *Possible efficacy*: Zinc gluconate lozenges, vitamin C (2–3 g/day in divided doses), Echinacea extract and saline irrigation are helpful in some cases.
3. *No significant benefit*: Guaifenesin, saline spray, steam, "nonsedating" antihistamines (loratidine, fexofenadine, cetrizine).
4. *Analgesics*: Non-aspirin containing analgesics should be preferred because aspirin causes increased shedding of viruses.
5. *Others*: Chicken soup.
6. Antibiotics have no role in VRS.
7. *General:* Bed rest and plenty of fluids are encouraged.

Course

1. The VRS resolves within 3 weeks in most of the adults regardless of bacterial colonization (*Streptococcus pneumoniae*, *M. catarrhalis* and *H. influenzae*), which are significantly more symptomatic.
2. Less than 2% in adults and 30% in children, VRS progress to ABRS.

Complications

1. Complications are occasional and include sinusitis, pharyngitis, tonsillitis, bronchitis, pneumonia and otitis media.
2. In influenzal rhinitis (influenza viruses A, B or C) complications due to bacterial invasion are common.
3. In rhinitis associated with exanthemas (measles, rubella, chicken pox) secondary infection and complications are more frequent and severe.

PANDEMIC INFLUENZA A H1N1 (SWINE FLU)

See chapter Fungi and Viruses

ACUTE BACTERIAL RHINOSINUSITIS

Distinguishing a common cold (VRS) from ABRS though not easy is important. In ABRS, symptoms of VRS after several days suddenly start worsening instead of slowly improving.

Microbiology

1. *Most common bacteria*: The three bacteria, which account for the vast majority of cases of ABRS in adults and children, are *S. pneumoniae*, *H. influenzae* and *M. catarrhalis*.
2. *Other*: They include anaerobes, *Streptococcal* species and *Staphylococcus aureus*.
3. *Anaerobic organisms and mixed infections*: They are seen in sinusitis of dental origin.

Predisposing Factors

They are following:

1. Viral rhinosinusitis
2. Allergic rhinosinusitis
3. Trauma
4. Physical stigmata such as DNS
5. Swimming and diving: Infected water enters the sinuses through the ostia. High content of chlorine gas in swimming pools causes chemical inflammation.
6. Barotraumas.
7. Dental infections and extraction of upper molars and premolars: They involve maxillary sinus.
   a. Periapical dental abscess may burst into the sinus.
   b. The root of a tooth, during extraction, may be pushed into maxillary sinus.

All the etiologic factors mentioned in CRS can also contribute in inciting ABRS.

Pathology

1. Acute inflammation of mucosa causes hyperemia, exudation of fluid, outpouring of polymorphonuclear cells and increased activity of serous and mucous glands.
2. The exudate, which is serous in beginning, becomes mucopurulent and causes destruction of mucosal lining. Failure of ostium to drain pus results in empyema of the sinus with destruction of its bony walls. Dental infections are very fulminating.
3. The progress of disease depends on the–
   a. Virulence of organisms
   b. Defense system of the host
   c. Structural features of osteomeatal complex

Clinical Features

Fewer than 5 in 1,000 colds are followed by ABRS. Symptoms of upper respiratory tract infection which last for more than 10 days or worsen after 5 to 7 days indicate ABRS.

1. Predictors of ABRS
   a. Best predictors of ABRS, which differentiate it from VRS, include–
      i. Maxillary toothache
      ii. Poor response to decongestants
      iii. Colored/purulent nasal discharge
      vi. Abnormal transillumination
   b. Other predictors include unilateral facial pain, pain with bending and mildly elevated ESR.
   c. Findings having little predictive value include headache, difficulty in sleeping, sore throat, sneezing, malaise, itchy eyes, fever or sweats and painful chewing.
   d. Severe symptoms include both high-grade fever and purulent nasal discharge for 3–4 consecutive days.
2. Rhinorrhea (nasal discharge) may be thick, thin, clear or purulent.
3. Cough may remain throughout the day and may be more at night.
4. Headache usually comes up on waking, gradually increases and reaches its peak at about mid-day and then starts subsiding (office headache).
5. Ethmoidal pain, which may be aggravated by movements of the eyeball, is localized over the bridge of the nose, medial and deep to the eye.
6. Halitosis, periorbital edema and nasal congestion may be present.
7. Additional adult features may be facial pain, headache and dental and gum pain (especially in maxillary sinus involvement). Pain may be aggravated by stooping, coughing or chewing.
8. Nasal examination shows edema, hyperemia and purulent debris. The most specific feature is purulent exudates in middle meatus. Nonspecific features are facial tenderness on palpation over sinuses or the maxillary teeth.
9. Postural test is done if no frank pus is seen in the middle meatus.
   Method: Middle meatus is decongested with a pledget of cotton soaked in topical decongestant. The patient is instructed to sit with the affected sinus turned up. Purulence in the middle meatus may be seen after 10–15 minutes.

Investigations

1. *Transillumination test*: The affected sinus is found opaque. It is performed in a darkroom using a bright light. For maxillary sinuses light is placed in oral cavity and light transmission is compared over maxillary sinuses. For frontal sinuses light is placed in the superior medial corner of the anterior orbit. Frontal sinuses often develop asymmetrically. Interpretations:
   a. Normal: Typical light transmission
   b. Dull: Reduced light transmission
   c. Opaque: No light transmission
2. *Nasal endoscopy examination*: See chapter Operations of Nose and Paranasal Sinuses.
3. *CT imaging:* Indications include refractory and recurrent cases of ABRS and in suspected cases of complications to determine extent of orbital, bony and CNS involvement. CT is done after the completion of antibiotic course. The findings may include obstruction of osteomeatal complex, polypoid changes and bony abnormalities.
   a. Plain sinus X-rays are now not recommended.
   b. A limited sinus CT scan (coronal sections only) provides excellent imaging detail. It cuts 50% of radiation exposure.
   c. New low dose CT scanners have only 10–15% of radiation exposure.
   d. Red flag findings are unilateral disease, sinus expansion and bony erosion.
   e. Abnormal findings are sinus opacification, air-fluid level, marked mucosal thickening, polyps.
   f. Findings not generally concerning are small retention cysts, concha bullosa, minimal mucosal thickening.
4. *Antral puncture*: Aspiration of fluid from sinus is done for the identification of infecting organism. Gram's staining, culture and sensitivity are done especially in refractory cases and complications. The recovery of bacteria in a density of at least 104 colony-forming units/ml is considered significant.
5. *Plain radiographs*: In the present era of CT the role of plain radiographs are debatable. Radiographs show presence of fluid or clouding of the sinuses (Fig. 1). Completely opaque sinuses usually (about 85%) have pus in them, while 50% sinuses that show thickened mucous membrane with central aeration have pus in them.

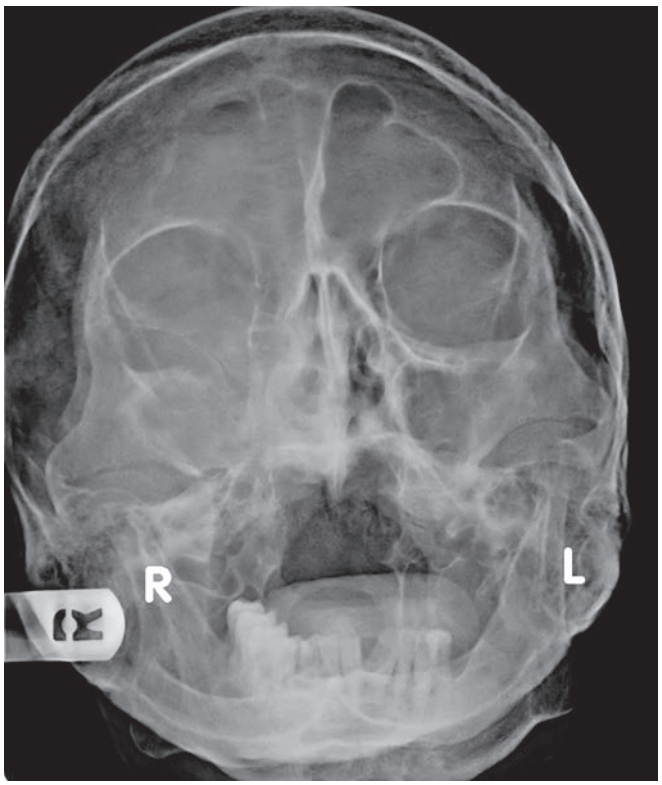

Fig. 1: X-ray paranasal sinus Water's view showing right side haziness in frontal, ethmoidal and maxillary sinuses (pansinusitis)
Source: Dr Jayesh Patel, Consultant Radiologist, Anand, Gujarat

Course

1. The following causes are considered in patients who fail to respond to appropriate antibiotics: Tumors, granulomatous diseases, allergic rhinitis and invasive (immunocompromised patients) and allergic fungal sinusitis. CT imaging and biopsy will confirm the diagnosis.
2. Patients with recurrent ABRS are evaluated for predisposing factors, which include anatomical abnormalities especially of osteomeatal complex, immunodeficiency, allergies, irritants and cystic fibrosis. Recurrent ABRS cases need endoscopy and culture and sensitivity of the sinus/nasal fluid.

Differential Diagnosis

1. Mucocele and neoplasms of the sphenoid sinus. They clinically mimic features of acute sphenoiditis. They should always be excluded in any case of isolated sphenoid sinus involvement that is rare.
2. Diphtheritic rhinitis: Diphtheria, which is rare these days, may be primary or secondary to faucial diphtheria. A grayish tenacious membrane, removal of which causes bleeding, covers the inferior turbinate and the floor of nose. Excoriation of anterior nares and upper lip are present in some cases. Treatment consists of isolation of the patient, systemic penicillin and diphtheria antitoxin.
3. Irritative rhinitis is caused by exposure to dust, smoke or irritating gases (ammonia, formalin, acid fumes), foreign body and intranasal manipulation. Patient develops an immediate catarrhal reaction with sneezing, rhinorrhea and nasal congestion, which usually pass off rapidly with removal of the offending agent. The symptoms may persist if nasal epithelium has been damaged and the infection supervenes.

Treatment

The reasonable strategy for many patients is to treat symptomatically and recommend antibiotics only if symptoms do not begin to improve. The incidence of severe complications and progression from acute to chronic rhinosinusitis is extremely low. If the symptoms are severe or worsening and clinical suspicion of ABRS is high include antibiotics in the treatment regime.

1. *Antibiotics:* They have not been shown to decrease the risk of complication or progression to CRS. Expensive antibiotics are often prescribed when equally effective and less expensive alternatives are available.
   - First line of antibiotics in uncomplicated ABRS, are amoxicillin and trimethoprim-sulfamethoxazole for 10–14 days. The course may be extended to 3 weeks in cases of partial improvement.
     - Amoxicillin (40 mg/kg divided twice daily) 500 mg 8 hourly or 875 mg 12 hourly.
     - Trimethoprim-sulfamethoxazole 160 mg/800 mg 12 hourly.
   - First line alternatives (only for patients allergic to both first line antibiotics) are doxycycline and azithromycin.
     - Doxycycline hyclate 100 mg 12 hourly.
     - Azithromycin 500 mg daily for 3 days.
     - Cefuroxime axetil 250–500 mg 12 hourly.
     - Cefdinir 300 mg 12 hourly or 600 mg daily.
     - Others: Loracarbef, clarithromycin and cefprozil.
   - Second line of antibiotics in cases of first line failure (or severe symptoms, suspected complications and resistance). Combination of amoxicillin and clavulanate, cefpodoxime and cefuroxime (have activity against beta-lactamase-producing bacteria and resistant *S. pneumoniae*). In adults fluoroquinolones are also recommended.
     - Amoxicillin high dose 875–1,000 mg 8 hourly.
     - Amoxicillin/clavulanate potassium 875/125 to 2,000/125 12 hourly.
     - Levofloxacin 500 mg daily (increased risks of tendon rupture in those over age 60, in kidney, heart and lung transplant patients and with use of simultaneous corticosteroid).
   - Intravenous antibiotics in cases of complications: For suspected penicillin-resistant pneumococci: Cefotaxime (50–75 mg/kg per day in 4 divided doses), with or without vancomycin (40 mg/kg/day in 4 divided doses). Then antibiotic based on the report of culture and sensitivity of aspirated fluid from sinus must be started. If needed, subspecialty consultation and surgical drainage must be considered.
   - Metronidazole: In cases of anaerobic organisms seen in sinusitis of dental origin.
2. *Adjunctive therapy*: They are not well proved but do give symptomatic relief (see treatment of VRS) and include–
   - Nasal steroid spray: For recurrent acute RS or acute RS superimposed on CRS addition of following high dose nasal corticosteroids for 3 weeks may decrease duration of symptoms and improve rate of clinical success.
     - Flunisolide 25 mcg/spray—8 sprays (200 mcg) each nostril 12 hourly.
     - Mometasone furoate or Fluticasone 50 mcg/spray—4 sprays (200 mcg) each nostril 12 hourly.

- Oral and topical decongestant
  - Nasal decongestant drops: 1% ephedrine or 0.1% xylo-or oxymetazoline (not for > 5 days because of the potential for rebound congestion and development of rhinitis medicamentosa)
- Steam inhalation alone or medicated with menthol or Tr. Benzoin Co. is done 15–20 minutes after nasal decongestion.
- Hot fomentation and analgesics.

3. *Surgery:* The surgery is reserved for patients with threatened intraorbital or intracranial complications. The reported success of endoscopic sinus surgery is about 80–90%. It is indicated in following conditions:
   - Persistent disease despite medical therapy.
   - Recurrent RS with identifiable and related anatomical or acute pathological abnormalities in the osteomeatal complex.
   - Empyemas, which do not respond to antibiotics, need surgical drainage. Endoscopic sinus surgeries are successfully employed in managing empyema of paranasal sinuses.
     - Antrum puncture in maxillary sinus empyema: Decongestants (combined with xylocaine 4%) nasal packing cover mucosa of middle and inferior turbinates and middle and inferior meatuses. Inferior meatus antrum puncture with trocar and cannula is done. The purulent secretions of maxillary sinus are displaced with normal saline irrigation. The mucopurulent secretions flow out through normal maxillary ostium. Children will need general anesthesia.
     - Trephining for empyema of frontal sinus: It is done through an incision on the medial part of upper eyelid. It exposes the floor of frontal sinus, the bone of which is removed with a cutting drill. The pus is collected for culture and sensitivity and sinus is irrigated. A plastic catheter is inserted and fixed and sinus is irrigated with normal saline two or three times daily. The patency of frontonasal duct can be determined by adding a few drops of methylene blue to the irrigating fluid and its exit can be confirmed in the nose. Drainage tube is removed when frontonasal duct becomes patent.
     - Empyema of ethmoid or sphenoid sinuses also needs surgical drainage, which can be performed either intranasal or external approach (sphenoethmoidectomy).

Antibiotic resistance patterns in ABRS: Penicillin resistant *S. pneumoniae* (25–40%), beta lactamase producing *H. influenzae* (30–40%) and beta lactamase producing *M. catarrhalis* (92%).

CHRONIC RHINOSINUSITIS

Several noninfectious factors important in the pathogenesis of RS are patency of sinus ostia, nasal airflow, mucociliary activity, immunocompetence and the nature and quantity of secretions.

Bacterial Causes

The microbial pathogens present in CRS include *S. aureus*, coagulase-negative *Staphylococcus*, anaerobic and Gram-negative bacteria.

Etiologic and Predisposing Factors

Many CRS patients do not have bacterial cause but have hyperplastic mucosal disease such as nasal polyposis. There occurs hyperemia and edema of mucous membrane with hypertrophy of seromucinous glands and increase in goblet cells. Blood sinusoids especially over the turbinates are distended.

The underlying etiological and predisposing factors and associated conditions of CRS are following:

1. *Infections*: Viral, bacterial, fungal and parasitic infections of neighboring organs such as pharynx (tonsils and adenoids) and upper teeth.
2. *Traumatic*: Maxillofacial injuries, foreign bodies, surgery and nasal packing.
3. *Hypersensitivity*: Allergy IgE mediated and non IgE mediated hypersensitivities, aspirin hypersensitivity with asthma and polyps and vasomotor factors.
4. *Environmental*: Cold and wet climate and overcrowding.
5. *Immunodeficiency*: Congenital and acquired (AIDS).
6. *Endocrinal*: Diabetes, rhinitis of pregnancy and hypothyroidism.
7. *Mucociliary abnormalities*: Cystic fibrosis and primary ciliary dysmotility.
8. *Mass*: Neoplasms of nose, sinuses and nasopharynx (benign and malignant), retention cysts, antrochoanal and ethmoidal polyps.
9. *Autoimmune or idiopathic*: Granulomatous (sarcoid and Wegner's), vasculitis (systemic lupus erythematosus) (SLE) and Churg Straus Syndrome) and pemphigoid.
10. *Pollution*: Chronic irritation from dust, smoke, cigarette smoking and snuff.
11. *Structural defects*: Deviated nasal septum (DNS), synechia, choanal atresia and osteomeatal complex abnormalities.
12. *Lifestyle*: Excessive intake of carbohydrates and lack of exercise.

Clinical Features

1. *Duration*: Chronic rhinosinusitis is diagnosed if patients have symptoms for more than 12 weeks.
2. *Predisposing factors*: The features of predisposing factors must be sought for.
3. *Nasal obstruction*: It may worsen on lying and affects the dependent side of nose.
4. *Nasal discharge and post-nasal drip*: It may be mucoid or mucopurulent, thick and viscid. Patient can have a constant desire to blow the nose or clear the throat.
5. *Headache*: The swollen turbinates especially middle impinging on the nasal septum results in headache.
6. *Anterior rhinoscopy*: It reveals dull red nasal mucosa and swollen turbinates, which pit on pressure and shrink with topical decongestants (compared from hypertrophied inferior turbinate).
7. *Sinuscopy*: Nasal endoscopy may show allergic or cobble stoned mucosa (sarcoid), presence and location of purulent discharge and polyps.
8. *Samter's triad*: Polyps in adults are frequently associated with asthma especially aspirin-sensitive (aspirin triad syndrome).

The sign and symptoms may be divided into two types: major and minor. The presence of two or more major factors or one major factor and two minor factors (Table 1) or purulent nasal discharge make the diagnosis of CRS.

Investigations

- *Nasal endoscopy*
- *Culture and sensitivity*: Aspiration of purulent discharge near the ostium (not from nasal floor) under endoscopic guidance with a sinus secretion aspirator is done. For culture calginite swabs are good but discharge may not easily adhere to the swab.
- *CT scan*: CT scan shows the extent of disease, complicating anatomy, asymmetry of skull base and orbital blowout fracture. CT scan is done before surgery and in cases of antibiotic failure. The presence of just only sinus mucosal abnormalities does not mean much and is not an indication for surgery as these findings may be incidental in asymptomatic patients also (Fig. 2).

TABLE 1 Major and minor factors in chronic rhinosinusitis

Major factors	*Minor factors*
• Facial pain/pressure	• Headache
• Facial congestion/fullness	• Fever
• Nasal obstruction/blockage	• Halitosis
• Nasal discharge/purulence	• Fatigue
• Discolored postnasal discharge	• Maxillary dental pain
• Hyposmia	• Cough
• O/E: Purulent nasal discharge	• Ear pain/pressure/fullness

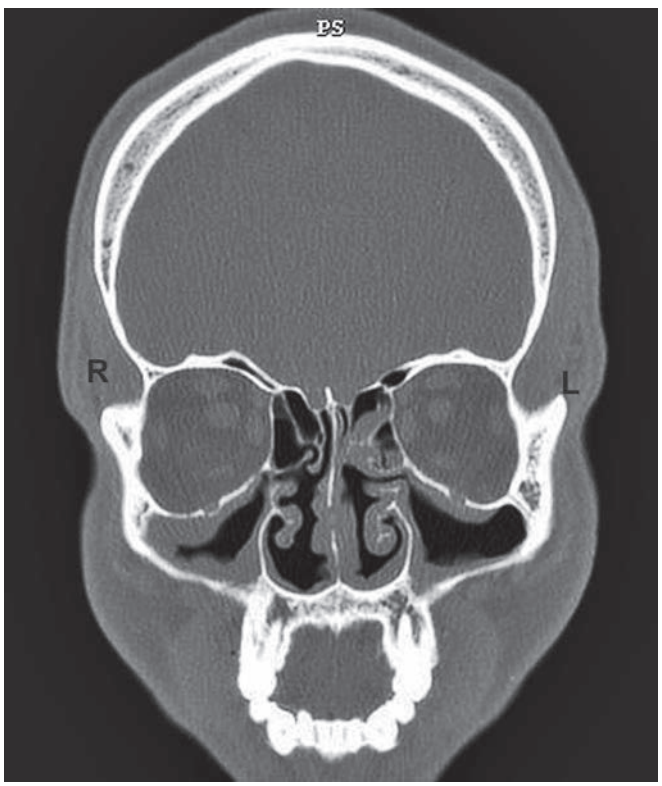

Fig. 2: Plain CT scan paranasal sinuses coronal section. Mucosal thickening in left ethmoidal sinus. Circumferential mucosal thickening and polyp formation in bilateral maxillary sinuses
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

  - If CT scan suggests no inflammatory disease that means patient is not having RS. The alternative diagnoses include allergic rhinitis, atypical facial pain, migraine or tension headache, nasal drying, gastroesophageal reflux disease (GERD), atrophic rhinitis, temporomandibular joint (TMJ) or dental pain.
- *Allergy evaluation*: It is almost mandatory before the consideration of surgery as approximately 60% patients have evidence of allergy.

Differential Diagnoses

1. *Cystic fibrosis or primary ciliary dyskinesia*: If the polyps are present in children then cystic fibrosis or primary ciliary dyskinesia syndrome must be ruled out.
2. *Fungal infection*: In cases of very sticky, rubbery, yellow, tan or green mucus, special stains for fungus should be performed.
3. *Systemic diseases*: In cases of antibiotic and surgical failure cases, systemic diseases (sarcoid, Wegener's granulomatosis) and mucosal and ciliary abnormalities should be considered.
4. *Gastroesophageal reflux disease*: If the presenting complaint is postnasal discharge then patient must be evaluated for GERD.

Chronic hyperplastic rhinosinusitis: Eosinophilic infiltration is the hallmark in most of the patients and about 50% of patients have asthma.

Medical Treatment

Treat the cause with particular attention to tonsils, adenoids, allergy, personal habits (smoking or alcohol indulgence), environment or work situation (smoky or dusty surroundings).

1. Antibiotics and steroids usually broad-spectrum antibiotic is started immediately and then changed as per the report of culture and sensitivity. A 28-day antibiotic course is used along with a nasal steroid spray (NSS), which may be combined with short course of tapering oral steroids, before the case is considered for surgery. The NSS is tapered and started again if symptoms recur.
   - Culture directed topical antibiotic:
     a. Mupirocin, 5 g in 45 ml saline, as nasal irrigation in cases of *S. aureus*.
     b. Gentamicin, 80 mg in 500 ml saline, as nasal irrigation, in cases of *Pseudomonas*.
2. Saline nasal douching daily before NSS helps in some refractory cases. One to three tablespoons of saline (1–2 teaspoons of salt and 1 teaspoon of baking soda per 1 quart of water) should be used per sitting. They help to keep the nose free from viscid secretions and also remove superficial infection.
3. Allergy, if present must be managed.
   a. Topical antihistamines
   b. Anticholinergics (ipratropium)
   c. Leukotriene modifier (montelukast, zafirlukast and zileuton) may be helpful in some cases.
   d. In cases of associated allergic fungal sinusitis, desensitization and antifungal agents may be helpful.
4. Nasal decongestants relieve nasal obstruction and improve sinus ventilation. Excessive and longtime use is avoided

because it may lead to rhinitis medicamentosa and affect mucociliary clearance.

Surgical Treatment

Endoscopic Sinus Surgery

- In antibiotic failures when CT scan is positive, endoscopic sinus surgery (ESS) is considered. Massive polyps are usually not cured with antibiotics and recurrence usually occurs after surgery. Long term NSS and oral steroids (especially perioperative) do prevent or delay recurrence after the removal. Noninclusion of sinus ostium, postoperative scarring and systemic diseases (sarcoid, Wegener's granulomatosis), mucosal and ciliary abnormalities can result in failure.
- With endoscopic sinus surgery, almost all the paranasal sinus lesions can be managed successfully. See chapter Operations of Nose and Paranasal Sinuses.

Maxillary Sinus (See chapter Operations of Nose and Paranasal Sinuses)

- Antral puncture and irrigation.
- Intranasal antrostomy is done if antral puncture and sinus irrigations fail. A window, which provides aeration and free drainage, is made in the inferior meatus.
- Caldwell-Luc operation: The antrum is approached through the anterior wall with sublabial incision. After removing the irreversible diseases, another window is created between the antrum and inferior meatus.

Frontal Sinus

- Intranasal drainage procedures facilitate free drainage through the frontonasal duct and include correction of deviated septum, removal of polyps or anterior part of middle turbinate, intranasal ethmoidectomy and treatment of maxillary sinusitis.
- Trephination of frontal sinus.
- External frontoethmoidectomy (Howarth's or Lynch operation): The frontal sinus is approached through the floor. A curvilinear incision is made around the inner margin of the orbit. Diseased mucosa and ethmoid cells are removed and a new frontonasal duct is created.
- Osteoplastic flap operation (unilateral or bilateral): Through a coronal or a brow incision, the anterior wall of frontal sinus is elevated as an osteoplastic flap, which is based inferiorly. The diseased mucosa and purulent material are removed and the sinus drained through a new frontonasal duct. In some cases it is desirable to obliterate the sinus after removing all diseased and normal mucosa of the sinus.

Ethmoid Sinuses

- Intranasal ethmoidectomy: It is done for ethmoid infection and polyps, which are removed between the middle turbinate and the medial wall of orbit (lamina papyracea). The frontal and sphenoid sinuses are also drained by this approach. ESS is better technique. (See chapter Operations of Nose and Paranasal Sinuses).
- External ethmoidectomy: Ethmoid sinuses can be approached through medial orbital incision.
- Frontosphenoethmoidectomy: This surgery also provides access to sphenoid and frontal sinuses.

Sphenoid Sinus

- Access to the sphenoid sinus is obtained by removing its anterior wall. It can be accomplished by external ethmoidectomy, trans-septal approach, and ESS.

PEDIATRIC RHINOSINUSITIS

Pediatric RS is a multifactorial disease and the importance of the predisposing factors changes with increasing age. Depending on the course of disease it is divided into acute (10 days to < 3 weeks), subacute (3 weeks to < 12 weeks) and chronic (> 12 weeks). In recurrent disease complete resolution occurs between the episodes, which are 3 or more in 6 months or more than 4 in 1 year.

Pathophysiology

1. Obstruction in the drainage pathways of the sinuses results in stasis of secretions that lead to sinus disease. The obstruction may be anatomic, physiologic or both.
2. The anatomic obstruction can be due to turbinate (hypertrophic or enlarged, concha bullosa or pneumatized and paradoxical bend), septum (deviation and spur), polyps, enlarged adenoids and mucosal inflammation from allergy, infection and irritation.
3. Other conditions, which may increase the incidence of sinus disease includes allergy, GERD, air pollution, first- or second-hand smoke and day care environment.

Clinical Features

- *Acute rhinosinusitis*: It is often preceded by viral upper respiratory infection (URI) with clear nasal discharge, which often recedes within 5 days. If the symptoms of purulent discharge, facial pain, nasal obstruction and daytime cough persist longer than 10 days, a diagnosis of acute sinusitis should be considered. The features of severe infection include high fever (> 40°C) and periorbital edema.
- *Chronic sinusitis*: Clinical features include night time cough, nasal discharge and obstruction and postnasal drip. Headache wax and wane symptoms persist for more than 12 weeks. Other features include facial pain, ocular or dental pain, sore throat, low grade fever and asthma.

Differential Diagnosis

- *Allergic rhinitis*: Table 2 provides the differentiating features of allergic rhinitis and bacterial rhinosinusitis. Detailed description of allergic rhinitis can be found in chapter Allergic and Nonallergic Rhinitis.
- *Recalcitrant rhinosinusitis*: Consider and evaluate for immunodeficiency, cystic fibrosis, ciliary immotility disorders and GERD.

Diagnosis

1. Clinical.
2. Nasal endoscopic examination.
3. Culture of sinus secretions.
4. Allergic assessment: Skin prick test, nasal smear, RAST and trial of treatment.
5. CT scan (Limited coronal cuts) PNS: Though gold standard for

TABLE 2 Differences between allergic and bacterial rhinosinusitis

Clinical features	*Allergic rhinitis*	*Bacterial sinusitis*
Nasal discharge	Clear thin rhinorrhea	Purulent drip
Nasal obstruction	Bilateral	Unilateral/bilateral
Headache/facial pain	Bilateral	Localized/generalized
Paroxysms of sneezing	Significant	Not significant
Itching of nose and eyes	Present	Absent
Specific findings	Pale, swollen, purple inferior turbinate, allergic nasal crease/allergic salute	Mucopurulence from middle meatus, periorbital swelling, facial tenderness
Personal history of asthma, food intolerance, hay fever, infantile eczema, family history of atopy in a first degree relative	Present	Absent
Response to antihistamines	Present	Absent
Nasal smear—eosinophilia	Present	Absent.
Diagnosis	Skin prick test and radioallergo-sorbent testing (RAST)	Culture of sinus secretions

evaluating sinonasal disease CT is not necessary to confirm the diagnosis of rhinosinusitis in children. It is indicated only when the sinus surgery is considered.

Treatment

1. *Medical therapy*: It is the initial treatment modality. Antibiotics, which are indicated in following children, allow for earlier resolution and may prevent complications.
   - Severe acute RS
   - Acute RS with protracted course.
   - Suspected or proven suppurative complications

   a. Acute rhinosinusitis: Antibiotics for 10–14 days; prolonged to 1 month when symptoms do not resolve completely.
      - First line antibiotic: Amoxicillin at the normal or high dose.
      - First line alternatives (Children allergic to penicillin): Cefdinir, cefuroxime, cefpodoxime, azithromycin (for 3 days) or clarithromycin.
      - Second line antibiotics (unresponsive, more severe disease or resistant S. *pneumoniae*): High-dose amoxicillin/clavulanate (90 mg/kg amoxicillin).
      - Children with vomiting: Parenteral ceftriaxone.

   b. Chronic rhinosinusitis:
      - Beta lactam stable antibiotic for 3 weeks
      - Topical nasal sprays:
        - Mometasone furoate for children: 2 years
        - Fluticasone propionate for children: 4 years
        - Other topical nasal steroids for children: 6 years
      - Antihistamines: Newer nonsedating
      - Mucolytics: Guaifenesin
2. *Surgical treatment:* It is rarely needed except in cases of orbital or intracranial complications.
   a. Adenoidectomy: In presence of adenoid hypertrophy in cases of CRS.
   b. Functional endoscopic sinus surgery: Controversial in pediatric rhinosinusitis.

COMPLICATIONS OF RHINOSINUSITIS

The complications of sinusitis can broadly be divided into acute and chronic and local and distant (Box 1). The acute local complications such as orbital, intracranial and bony are more common.

The alarming signs and symptoms for intracranial or intraorbital extension of rhinosinusitis include:

- High fever
- Severe pain
- Worsening headache
- Meningeal signs
- Infraorbital hypesthesia
- Altered mental status
- Significant facial swelling
- Diplopia
- Ptosis
- Chemosis
- Proptosis
- Abnormal pupillary or extraocular movements

MUCOCELE/PYOCELE

Mucocele, the chronic cysts of sinuses are lined with pseudostratified or low-columnar epithelium containing few goblet cells. These round or oval cysts grow concentrically and expand very slowly over 10 years or more.

Box 1: Complications of rhinosinusitis

1. *Acute*
   a. Orbital
      i. Inflammatory edema
      ii. Orbital cellulitis
      iii. Subperiosteal abscess
      iv. Orbital abscess
      v. Cavernous sinus thrombosis
   b. Intracranial
      i. Abscess: Extradural, subdural, cerebral
      ii. Meningitis
      iii. Encephalitis
      iv Cavernous and superior sagittal sinus thrombosis
   c. Bony
      i. Osteitis/Osteomyelitis (Pott's puffy tumor)
   d. Dental
   e. Toxic shock syndrome
2. *Chronic*
   a. Mucocele/Pyocele
3. *Associated diseases*
   a. Otitis media
   b. Adenotonsillitis
   c. Bronchiectasis

Pyocele is similar to mucocele but its contents are purulent. It can result from infection of a mucocele.

Etiology

The suggested etiologies are:

1. Obstruction of sinus ostium
2. Obstruction of minor salivary gland duct present within the mucosal lining of sinuses.

Maxillary Mucocele

- It is an incidental finding on radiographs and rarely requires specific treatment.
- If needed, it can be aspirated through puncture of either inferior meatus or canine fossa.

Frontoethmoidal Mucocele

This is the most common type of mucocele.

- *Clinical features*: It presents with frontal headache, proptosis, deep nasal or periorbital pain and diplopia. The latter is caused due to inferior and lateral displacement of eyeball.
  - The swelling is cystic and non-tender; eggshell crackling may be elicited.
- *Imaging*: Radiograph shows clouding of sinus with sclerosis of surrounding skull and loss of scalloped outline of frontal sinus.
- *Treatment*: It requires surgical removal (frontoethmoidectomy) or endoscopic marsupialization into the nasal cavity.
  - Ethmoidal mucocele causes a bulge in the middle meatus of nose and can be drained by uncapping the ethmoidal bulge (or with external ethmoid operation) and establishing free drainage.

Sphenoethmoidal Mucocele

- *Clinical features*: They present with headache (occipital and vertex) or deep nasal pain, diplopia, visual field disturbance and eyeball displacement.
  - Exophthalmos is always present and the pain is localized to the orbit or forehead.
  - In superior orbital fissure syndrome there is involvement of CN III, IV, VI and ophthalmic division of CN V.
  - Orbital apex syndrome: Involvement of CN II, III, IV, V_1, V_2, VI.
- *Imaging:* Radiographic findings confirm the diagnosis. The slow expansion leads to destruction of sphenoid and posterior ethmoid sinuses.
- *Treatment*: It includes opening it widely into the nasal cavity.
  - Endoscopic sinus surgery: Anterior wall of the sphenoid sinus is removed, cyst wall uncapped and its fluid contents evacuated.
  - External: Ethmoidectomy with sphenoidotomy is performed.

ORBITAL COMPLICATIONS

Purulent frontal and ethmoidal sinusitis may result in orbital complications.

The infection can travel into the orbit through thin lamina papyracea and thrombophlebitis.

Clinical Features

1. An inflammatory edema of eyelids, which is the first indication of orbital involvement, progresses to cellulitis, erythema, proptosis and temperature (101°F).
2. Later on chemosis increases, ophthalmoplegia occurs and fundus shows mild vascular congestion (Table 3).
3. Fever may increase to 102°F–104°F and an abscess may form along medial wall of orbit or within periorbita.
4. Superior orbital fissure syndrome presents with deep orbital pain, frontal headache and progressive paralysis of CN VI, III and IV.
5. Orbital apex syndrome consists of features of superior orbital fissure syndrome and involvement of the optic nerve and maxillary division of the trigeminal.

Diagnosis

CT scan and ultrasound will confirm the abscess formation.

Treatment

1. Orbital inflammation and cellulitis need sinus drainage and intravenous antibiotics.
2. An abscess needs surgical drainage. The failure to drain can lead to permanent orbital sequelae and intracranial complications.

OSTEOMYELITIS/OSTEITIS

Osteomyelitis is an infection of bone marrow while the osteitis means infection of the compact bone. Osteomyelitis can be of either the maxilla or the frontal bone.

Osteomyelitis of Maxilla

It is more common in infants and children than adults because of the presence of spongy bone in the anterior wall of the maxilla. Infection usually starts in the dental sac and spreads to the maxilla. The primary infection of the maxillary sinus rarely causes osteomyelitis.

1. *Clinical features*: They include erythema, swelling of cheek, edema of lower lid, purulent nasal discharge and fever. Subperiosteal abscess and fistulae form in infraorbital region, alveolus or palate, or in zygoma. Sequestration of bone can occur.
2. *Treatment*: It includes large doses of antibiotics, drainage of abscess and removal of the sequestra.

TABLE 3 Chandler classification of orbital involvement

Inflammatory edema	Lid edema present, normal visual acuity and extraocular movements
Orbital cellulitis	Diffuse edema of orbital contents but no discrete abscess formation
Subperiosteal abscess	Pus collection along lamina papyracea; inferior and lateral eyeball shift
Orbital abscess	Pus within orbit, proptosis, chemosis, ophthalmoplegia, dim vision
Cavernous sinus thrombosis	Bilateral involvement of eyes, toxic look and findings of meningismus

3. *Complications*: Damage to temporary or permanent tooth-buds, maldevelopment of maxilla, oroantral fistula, persistently draining sinus and epiphora.

Osteomyelitis Frontal Bone

It is more common in adults because frontal sinus is not developed in infants and children.

- *Etiology*: It usually results from acute infection of frontal sinus, which may be direct or through thrombophlebitis. Other causes are trauma and surgery.
- *Pott's puffy tumor*: A subperiosteal abscess develops over anterior surface of frontal sinus and results in the swelling of overlying soft tissue.
- *Extradural abscess*: Pus may form internally as an extradural abscess.
- *Treatment*: It consists of large doses of antibiotics, drainage of abscess and trephining of frontal sinus through its floor.
  - It may require removal of sequestra and necrotic bone with osteoplastic flap.

CAVERNOUS SINUS THROMBOSIS

The infection of paranasal sinuses especially ethmoid and sphenoid (less commonly the frontal) and orbit can cause thrombophlebitis of the cavernous sinuses (Fig. 3).

- *Clinical features*: The clinical features include bilateral orbital involvement, rapidly progressive chemosis and ophthalmoplegia, retinal congestion, high fever (105°F) and prostration (Table 4).
  - The condition progresses so rapidly that even immediate aggressive treatment usually cannot save vision and patient's life.
- *Diagnosis*: CT scan confirms the diagnosis.
- *Treatment*: It includes intravenous antibiotics, drainage of abscess and orbital decompression. Heparinization slows progression of thrombosis.

INTRACRANIAL COMPLICATIONS

Frontal, ethmoid and sphenoid sinuses form floor of the anterior cranial fossa and infection from these can cause following intracranial complications:

1. Meningitis
2. Encephalitis
3. Extradural abscess
4. Subdural abscess
5. Brain abscess

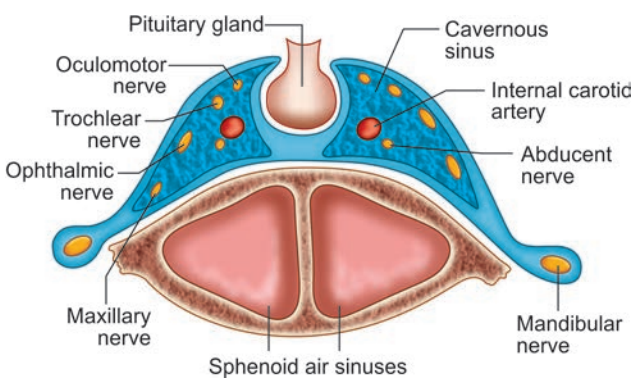

Fig. 3: Cavernous sinus relations and contents

TABLE 4 Differences between orbital cellulitis and cavernous sinus thrombosis

	Orbital cellulitis	*Cavernous sinus thrombosis*
Source	Commonly ethmoid sinuses	Nose, sinuses, orbit, ear and pharynx
Onset and progress	Slow	Abrupt
Cranial nerve involvement	Involved concurrently with complete ophthalmoplegia	Involved individually and progressively
Side	Usually involve affected side eye	Involves both eyes
Toxemia	Absent	Present
Fever	Present	High temperature with chills
Mortality	Less	Very high

6. Cavernous sinus thrombosis.

The infection spreads directly through the defect in posterior wall of frontal sinus or through diploic frontal vessels and causes subdural abscess.

- *Clinical features*: Thrombosis of dural vessels results in focal cerebral abscess, seizures and neurologic deficits.
  - Neck rigidity is an alarming sign in cases of sinusitis.
  - Other features of raised CSF pressure include headache, intractable vomiting and deterioration in consciousness.
- *Management*: It needs close collaboration with neurosurgeon. Treatment usually consists of intravenous antibiotics and proper drainage of involved sinus.

HYPERTROPHIED TURBINATES

See chapter Nasal Septum

NASAL POLYPS

Nasal polyps are traditionally divided into antrochoanal polyp and bilateral ethmoidal polyps. They are non-neoplastic masses of edematous sinonasal mucosa.

Etiology

The exact etiology of nasal polyps is not well understood. They are usually manifestations of:

1. *Rhinosinusitis*: Allergic and non-allergic origin and non-allergic rhinitis with eosinophilia syndrome.
2. *Cystic fibrosis*: Disorders of ciliary motility and abnormal composition of nasal mucus.
3. *Allergic fungal sinusitis*.
4. *Samter's triad*: It is the triad of nasal polyps, asthma and aspirin intolerance.
5. *Kartagener's syndrome*: Bronchiectasis, sinusitis, situs inversus and ciliary dyskinesis.
6. *Young's syndrome*: Sinopulmonary disease and azoospermia.
7. *Churg-Strauss syndrome*: Asthma, fever, eosinophilia, vasculitis and granuloma.
8. *Nasal mastocytosis*: Nasal mucosa is infiltrated with mast cells with few eosinophils. Skin tests for allergy and IgE levels are normal.

9. *Neoplasms*: Simple nasal polyp may be associated with malignancy, which is common in people above 40 years and should be excluded by histological examination.

Pathology

1. There occurs collection of extracellular fluid and edema of nasal mucosa (polypoidal change), especially of middle meatus and middle turbinate. The polyps are sessile in the beginning and later become pedunculated due to the effect of gravity and the excessive sneezing.
2. The polyps are usually lined with ciliated columnar epithelium, which on exposure to atmospheric irritation may undergo metaplastic change to transitional and squamous type. Submucosa contains large intercellular spaces filled with eosinophils and round cells.

Common Sites of Origin

They are uncinate process, bulla ethmoidalis, ostia of sinuses and medial surface and edge of middle turbinate. Nasal polyps usually never arise from the septum or the floor of nose.

Clinical Features

1. Nasal polyps can occur at any age.
2. They usually present with nasal stuffiness/obstruction.
3. Patients may have partial or total anosmia (loss of sense of smell) and headache.
4. Paroxysmal sneezing, watery nasal discharge and itching indicate allergy.
5. Nasal purulence indicates associated infective cause.
6. Nasal polyps can be sessile or pedunculated, solitary or multiple, unilateral or bilateral.
   a. Polyps usually appear as pale, smooth, glistening, grape-like masses.
   b. They do not bleed on touch while infective or neoplastic lesions may bleed.
   c. They are insensitive to probing. This feature differentiates solitary polyp from hypertrophy of the turbinate or cystic middle turbinate.
   d. Big polyps, which protrude out from the nostril (Fig. 4), appear pink and vascular (simulating neoplasm).
   e. Their gradual progression may result in broadening of nose and increased intercanthal distance.

Staging

Polyps can be staged as following according to their size.

- Stage I: Limited to the extent of middle turbinate.
- Stage II: Extending beyond the limit of middle turbinate.
- Stage III: Approaching to inferior turbinate.
- Stage IV: Going up to the floor of nose.

Antrochoanal Polyp

1. It arises from the maxillary antrum near its accessory ostium and comes out and grows towards posterior choana and anterior nasal cavity (Table 5).

Fig. 4: Big nasal polyp protruding out from right nostril appears pink and vascular and simulating neoplasm

TABLE 5 Differences between antrochoanal and ethmoidal polyp

	Antrochoanal polyp	*Ethmoidal polyps*
Age	Common in children	Common in adults
Etiology	Usually infection	Allergy or multifactorial
Number	Solitary	Multiple
Side	Unilateral	Bilateral
Origin	Maxillary sinus	Ethmoidal sinuses
Growth	Grows towards posterior choana	Grow towards anterior nares
Lobes	Two (Dumbbell) or three (trilobed)	Multiple small and grape-like masses
Obstruction	Usually unilateral	Usually bilateral
History	Relatively short	Relatively long
Recurrence	Uncommon	Common
Treatment	Usually surgical	Usually medical

2. It consists of three parts:
   - Antral: a thin stalk.
   - Choanal: round and globular.
   - Nasal: flat from side to side.
3. It is usually single and unilateral and is seen in children and young adults.
4. Unilateral nasal obstruction may become bilateral, when polyp grows into the nasopharynx and obstructs both sides choanae.
5. Voice becomes thick and dull (hyponasality).
6. Mucoid nasal discharge unilateral or bilateral is usually seen.
7. A soft, smooth, grayish and mobile mass covered with nasal discharge may be seen through both anterior and posterior rhinoscopy.

8. A large antrochoanal polyp may be seen hanging down in the oropharynx and/or protruding out from the nostril, which look pink and congested.
9. If an antrochoanal polyp grows only posterior, it may be missed on anterior rhinoscopy.
10. Table 5 shows the differences between antrochoanal and ethmoidal polyps.

Differential Diagnoses

a. *Blob of mucus*: It may look like a polyp but disappears on blowing the nose.
b. *Hypertrophied middle turbinate*: It is pink in color and hard like bone and can be easily differentiated from polyp with probe testing.
c. *Angiofibroma*: It usually occurs in adolescent males and present with profuse recurrent episodes of epistaxis. If suspected, it should not be probed because that can cause brisk bleeding.
d. *Malignancy:* They are fleshy pink in appearance, friable in nature and have tendency to bleed on touch. A red and fleshy, friable and granular mass presenting with epistaxis and orbital complications should arouse the suspicion of malignancy. Some time a polyp may masquerade a malignancy, so all polyps are subjected to histology.
e. *Pediatric masses*: Mucoviscidosis, gliomas and encephalocele. Encephalocele, which should be aspirated (CSF), presents like a polyp.

Investigations

a. CT scan of paranasal sinuses (Fig. 5) shows not only the extent and nature of lesion but also helps to plan surgery. They have replaced plain X-rays.
b. X-ray of paranasal sinuses show opacity of the involved antrum.
c. X-ray lateral view soft tissue nasopharynx shows globular swelling of antrochoanal polyp, which is differentiated from angiofibroma by the presence of a column of air behind the polyp.

Treatment

a. *Medical*: Early polypoidal changes usually revert to normal with antihistaminics, control of allergy and infection and steroids (local and systemic).
   - Steroids also prevent recurrence after surgery. Contraindications for steroids include hypertension, peptic ulcer, diabetes, pregnancy and tuberculosis.
b. *Surgical*: Currently almost all the polyps, which do not respond to medical treatment, are managed by ESS.
   - Snare: Solitary pedunculated polyp can be removed with snare.
     - Krause's nasal snare is used for removal of nasal polyps and partial turbinectomy. The polyp mass is engaged in the wire loop and avulsed.
   - Intranasal ethmoidectomy: Multiple ethmoidal polyps need uncapping of the ethmoidal air cells through intranasal route. It is best performed with functional endoscopic sinus surgery (FESS), which facilitates drainage and ventilation to the other involved sinuses such as maxillary, sphenoidal or frontal.
   - External ethmoidectomy: It is done through the medial wall of the orbit by an external incision, which is medial to medial canthus. This external approach is considered in cases of recurrence, where surgical landmarks are distorted and ill-defined.
   - Transantral ethmoidectomy (Caldwell-Luc approach): It is indicated when maxillary antrum is also involved. The maxillary antrum is opened and the ethmoid air cells are exenterated through the medial wall of the antrum. This procedure can be avoided with the use of angled sinuscopes.
   - An antrochoanal polyp can be avulsed either through the nasal or oral route. Recurrence, which is uncommon after complete removal, may need Caldwell-Luc operation.

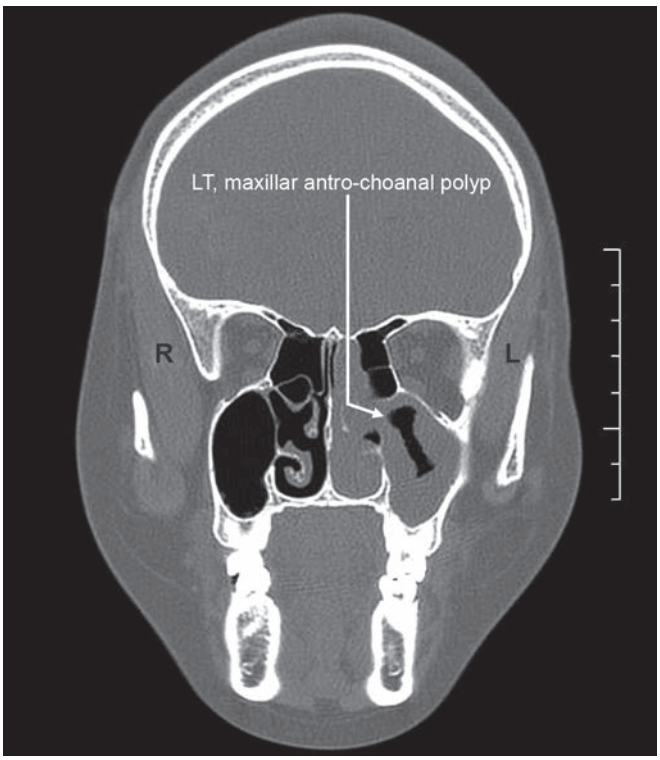

Fig. 5: Plain CT scan paranasal sinuses coronal section. A case of left antrochoanal polyp obliterating left nasal cavity and showing mucosal thickening in left maxillary sinus
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

FUNGAL SINUSITIS

See chapter Nasal Manifestation of Systemic Diseases

ATROPHIC RHINITIS (OZENA)

See chapter Nasal Manifestation of Systemic Diseases

Clinical Highlights

1. **Common cold** is most often caused by Rhinovirus.
2. **Acute bacterial rhinosinusitis**: Most common causative organisms include *Streptococcus pneumoniae* followed by *Haemophilus influenzae*. Anaerobic and mixed infections occur in sinusitis of dental origin. The first line of antibiotic treatment is amoxicillin for 10 days. The second line of antibiotic in refractory cases is amoxicillin with clavulanic acid.
3. **Cystic fibrosis**: The clinical features include chronic sinusitis and multiple nasal polyps, recurrent chest infections and malabsorption syndrome. Sweat chloride test confirms the diagnosis.
4. **Kartagener's syndrome (immotile cilia syndrome)**: It is caused by deficiency of ATPase motor protein dynein and is characterized by respiratory infections and sterility. Clinical features include recurrent/chronic sinusitis, bronchiectasis, sterility and situs inversus. Ciliary motility is disturbed. Electron microscope shows absence of dye inside the arms in A-tubules.
5. **Cavernous sinus**: The cavernous sinus is situated lateral to the body of sphenoid bone in the floor of middle cranial fossa. The cranial nerves pass through it include CNS III, IV, VI and V (ophthalmic and maxillary divisions).
6. **Cavernous sinus thrombophlebitis/infection**: Patient is unable to move eye in any direction (ophthalmoplegia) and has ptosis, dilated pupil (loss of pupillary light reflex) and altered skin sensation over the maxilla and frontal bones and loss of corneal blink reflex. The abducent is the first cranial nerve to be affected resulting in medially deviated eyeball.
7. **Pott's puffy tumor**: It develops in the pyogenic infection of frontal sinus (Osteomyelitis).
8. **Sluder's neuralgia**: The clinical features include rhinorrhea, increased lacrimation and nasal stuffiness.
9. **Antrochoanal polyp**: It arises from maxillary sinus and it grows posteriorly. It is often unilateral. The treatment of choice in children as well as in adults is functional endoscopic sinus surgery (FESS). Caldwell-Luc operation is considered in cases of recurrence.
10. **Ethmoidal polyp**: These commonly seen polyps are usually bilateral and multiple in number. Patients are usually middle aged. They arise from uncinate process and middle turbinate. Allergy is an etiological factor. They are often associated with bronchial asthma.
11. **Samter's triad**: It consists of nasal polyps, bronchial asthma and aspirin sensitivity.
12. **Common causes and treatment of polyps**: The common causes of nasal polyps are aspirin intolerance, fungal sinusitis and chronic rhinosinusitis of allergic and nonallergic origin. Topical steroid is the most effective medical treatment of nasal polyps. The next line of treatment is FESS.
13. **Rhinolalia clausa**: The common causes include allergic rhinitis, adenoids and nasal polyps.
14. **Syndromes with infertility and sinopulmonary disease:** They include cystic fibrosis, Kartagener's syndrome and Young's syndrome.

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29 Nasal Manifestation of Systemic Diseases

None will be able to resist truth and love and sincerity. Are you sincere? Unselfish even unto death? And loving? Then fear not, not even death. Have faith that you are brave, born to do great things! Let not the barks of puppies frighten you—no, not even the thunderbolts of heaven—but stand up and work!

—Swami Vivekananda

Points of Focus

- **INTRODUCTION**
- **WEGENER'S GRANULOMATOSIS**
- **PERIPHERAL T-CELL NEOPLASM (NONHEALING MIDLINE GRANULOMA)**
- **ATROPHIC RHINITIS (OZENA)**
  - ✓ Secondary Atrophic Rhinitis
- **RHINITIS SICCA**
- **RHINITIS CASEOSA**
- **SARCOIDOSIS**
- **CHURG-STRAUSS SYNDROME**
- **RHINOSCLEROMA**
- **TUBERCULOSIS**
- **LUPUS VULGARIS**
- **NONTUBERCULOUS MYCOBACTERIA**
- **LEPROSY**
- **SYPHILIS**
- **HISTOPLASMOSIS**
- **RHINOSPORIDIOSIS**
- **FUNGAL SINUSITIS**
  - ✓ Acute Invasive Fungal Sinusitis
  - ✓ Chronic Invasive Fungal Sinusitis
  - ✓ Fungus Balls
  - ✓ Saprophytic Fungal Infection
  - ✓ Allergic Fungal Sinusitis
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

Formerly several granulomatous lesions, which are described in this chapter, were called lethal midline granuloma. Various systemic diseases involving the nose, which are discussed in this chapter, can be classified into four major groups: vasculitides, bacterial infection, fungal infections and lymphomas (Table 1).

These lesions are usually the manifestations of systemic diseases, which always have other systemic features that help in making the diagnosis. Biopsy from nasal lesions not only establishes the correct diagnosis but also excludes a neoplasm. Many of these diseases simulate neoplasms.

TABLE 1 Systemic diseases involving nose

Group	*Diseases*
Vasculitides	Wegener's granulomatosis, sarcoidosis and Churg-Strauss syndrome
Bacterial	Rhinoscleroma, syphilis, tuberculosis, lupus and leprosy
Fungal	Rhinosporidiosis, aspergillosis, mucormycosis, histoplasmosis and blastomycosis
Lymphoma	Peripheral T-cell neoplasm (Non-healing midline granuloma)

WEGENER'S GRANULOMATOSIS

Wegener's granulomatosis (WG), which is now viewed as vasculitis and an autoimmune disease, was first described by Friedrich Wegener in 1939 as necrotizing granulomas and vasculitis of upper and lower respiratory tract. In addition to respiratory tract, WG also involves kidneys and skin. It should be differentiated from peripheral T-Cell neoplasm and large septal perforation in cases of drug abuse (cocaine).

Clinical Features

There are three main types of WG: Types 1, 2 and 3.

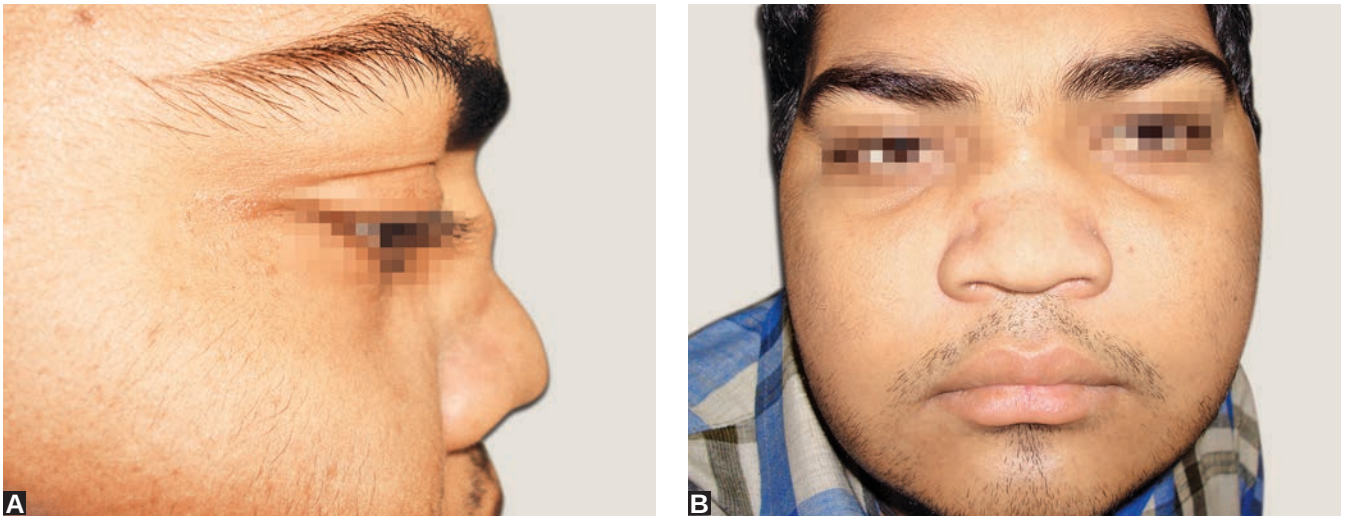
1. *Type 1 limited form*: Patient presents with clear, purulent or blood stained nasal discharge (chronic rhinosinusitis), which does not respond to medical treatment. Other nasal findings include severe pain over dorsum of nose, very large nasal crusts and granulations and later on septal perforation and saddle nose deformity (Figs 1A and B). The general symptoms of systemic vasculitis are weakness, malaise, fatigue, night sweats and migratory arthralgias.
2. *Type 2 with pulmonary involvement*: Cough, hemoptysis and single or multiple cavity lesions in X-ray chest.
3. *Type 3 with wide dissemination*: Involvement of multiple organs such as airway, lungs, renal and skin. Destruction can involve eyes, ear, orbit, palate, oral cavity or oropharynx.
   a. Eye and Orbital findings (Either alone or with nose). Nasolacrimal duct obstruction, bilateral nonspecific episcleritis, bilateral or unilateral proptosis, pseudotumor of orbit.
   b. Otological findings: Unilateral or bilateral serous otitis media and profound sensorineural hearing loss.
   c. Oropharyngeal: Diffuse gingival lesions, minor salivary gland involvement and oral ulcers.
   d. Subglottic involvement: It is common and may be either alone or with nose. Symptoms are of asthma but flow volume studies are normal. In a case of very short segment of upper tracheal stenosis, WG should be ruled out.
   e. Kidneys: Renal failure is usually the cause of death.

Diagnosis

1. *CBC*: Anemia and raised erythrocyte sedimentation rate (ESR).
2. *Urine*: Red cells, casts and albumin in urine and raised serum creatinine levels.
3. *X-ray chest*: Single or multiple cavity lesions in X-ray chest.
4. *Biopsy*: Seven or eight pieces of tissue from all turbinates should be taken after removing the crusts. They are sent for stains and culture including acid-fast bacillus (AFB) and fungi. WG shows necrosis and ulceration of mucosa, epithelioid granuloma and necrotizing vasculitis, which may involve small arteries or veins such as arterioles, capillaries and venules.
5. *Antineutrophilic cytoplasmic antibody*: Antineutrophilic cytoplasmic antibody (ANCA) is of two types:
   a. Cytoplasmic antineutrophilic cytoplasmic antibody (c-ANCA): It is so specific for WG that it may preclude the need for biopsy such as in subglottic stenosis. But a negative c-ANCA does not rule out WG. The raised titer of c-ANCA is an ominous sign.
   b. Perinuclear antineutrophilic cytoplasmic antibody (p-ANCA): It is not as specific as c-ANCA but is useful in inflammatory bowel disease, autoimmune liver disease and rheumatoid arthritis.

Treatment

1. *Immunosuppressive therapy*: Oral cyclophosphamide (2 mg/kg/day for 6 months to 1 year and then after the disappearance of symptoms tapered gradually) and prednisone (1 mg/kg/day for 1 month and then gradually tapered following 2 months). In cases of relapse this standard protocol is started again.
2. *Trimethoprim-sulfamethoxazole* (Bactrim/Septran): It cures the early limited type 1 WG. It may be continued indefinitely after the discontinuation of standard protocol because it may prevent relapse.
3. *Plasma exchange and intravenous immunoglobulin*: Intravenous immunoglobulin (IVIG) is tried in cases, which do not respond to standard protocol. Plasma exchange is useful in dialysis-dependent patients.



Figs 1A and B: Saddle nose deformity due to Wegener's granulomatosis. Anterior rhinoscopy revealed large nasal crusts, granulations and big septal perforation. The puffiness of face is due to steroid therapy. (A) Lateral view; (B) Front view

4. *Surgical*: Reconstructive surgeries such as tympanoplasty, rhinoplasty and tracheoplasty are considered when WG is in remission.

PERIPHERAL T-CELL NEOPLASM (NONHEALING MIDLINE GRANULOMA, POLYMORPHIC RETICULOSIS)

This slowly destructive disease of the nose and midfacial region is differentiated from WG by the absence of pulmonary and renal involvement (Table 2). The disease is a malignant lymphoma or its variant. Rate of destructive process depends on the patient's immunological response.

Clinical Features

1. Unilateral lesions in nose extending to soft tissue of nose, upper lip, oral cavity, maxillary sinus and orbit.
2. Lesions are explosive and rapidly progressive.
3. Secondary infection of lesions by gram-negative and anaerobic organisms.

Diagnosis

1. *Biopsy*: It will show mixed population of cells (mature lymphocytes, plasma cells and large lymphoreticular cells), which resembles picture of lymphoma.
2. *Immunohistochemical studies*: By using antibodies to leukocytes-common antigen CD45RB, B-cell lineage markers CD20, T-cell lineage markers CD3, CD43, CD45RO and natural killer marker CD57.
3. *EBV-RNA*: Detected by in situ hybridization.

Treatment

1. *Localized lesion*: Curative radiotherapy followed by surgical debridement and a nasal prosthesis.
2. *Multiorgan disease*: Standard leukemia protocol.

TABLE 2 Comparison of Wegener's granulomatosis and peripheral T-cell neoplasms

	Wegener's granulomatosis	*Peripheral T-cell neoplasms*
Distribution of lesion	Focal and localized	Diffuse
Onset	Gradual	Explosive
Progress	Gradual	Rapid
Ear, tracheal, & renal	Involvement common	Involvement very uncommon
Histology	Vasculitis	Polymorphic lymphoid infiltrate
c-ANCA	Diagnostic	Negative
EBV-RNA	Absent	Detected
Immuno-histochemical study	No role	Diagnostic
Treatment	Immuno-suppression	Radiotherapy

Abbreviations: c-ANCA, cytoplasmic antineutrophilic cytoplasmic antibody; EBV-RNA, Epstein-Barr virus-ribonucleic acid

ATROPHIC RHINITIS (OZENA)

This chronic inflammation of nose is characterized by the roomy nasal cavities and foul-smelling large crusts in posterior nasal cavity. There occurs atrophy of nasal mucosa and turbinate bones.

Atrophic rhinitis can be of two types: primary and secondary. The primary atrophic rhinitis is more common and will be described in more detail.

Etiology

The exact cause of primary atrophic rhinitis is not known but various theories and predisposing factors, which have been proposed, are following:

a. *Hereditary:* More than one family member can be affected.
b. *Hormonal*: It usually starts at puberty. Females are affected more than males. Spontaneous regression of symptoms occurs after menopause.
c. *Racial*: White and yellow races are more affected than natives of equatorial Africa.
d. *Dietary*: Deficiency of vitamin A, D or iron or other dietary factors are seen. Patients are usually from low socioeconomic level.
e. *Infective*: The various organisms cultured from cases of atrophic rhinitis are *Klebsiella ozaenae (Perez bacillus), diphtheroids, Proteus vulgaris, Escherichia coli, Staphylococci* and *Streptococci*. They are not considered primary causative organisms but are said to be secondary invaders, which are responsible for foul smell.
f. *Autoimmune*: Some unspecified agents are said to trigger antigenicity of nasal mucosa that leads to production of antibodies which destruct nasal mucosa.

Pathology

The respiratory ciliated columnar epithelium of the nasal cavity is replaced by stratified squamous type. There occur atrophy of seromucinous glands, venous blood sinusoids and nerve elements. Obliterative endarteritis, which causes resorption of turbinates and widening of nasal chambers, can be seen in the mucosa, periosteum and bone. The disease persists for years and usually recovers spontaneously in middle age.

Clinical Features

1. Patients are usually females around puberty. They emit foul smell from the nose, which make them social outcast. Patients themselves are not aware of this foul smell because of marked anosmia (merciful anosmia).
2. Nasal obstruction due to large crusts filling the nose is present in spite of unduly wide nasal chambers.
3. Epistaxis usually occurs when the crusts are removed.
4. Greenish or grayish black dry crusts.
5. Roomy nasal cavities and atrophy of turbinates allow easy visibility of posterior wall of nasopharynx and ostium of sinuses.
6. Nasal mucosa looks pale.
7. Septal perforation and nasal saddle deformity are not uncommon.
8. In atrophic pharyngitis, atrophic changes occur in the pharyngeal mucosa, which look dry and glazed with crusts.
9. Obstruction to Eustachian tube may result in middle ear effusion, which present with deafness.

10. In atrophic laryngitis, patient has cough and hoarseness of voice.
11. X-ray of paranasal sinuses show small, underdeveloped and thick walled sinuses, which appear opaque. Due to arrested development, paranasal sinuses remain small. Antral puncture becomes difficult due to thick sinus walls.

Treatment

Complete cure is not yet possible. Treatment consists of both medical and surgical management.

A. *Medical*

It aims at maintaining nasal hygiene and removal of crusts, which take care of putrefying smell and further crust formation.

a. *Warm normal saline or alkaline nasal irrigation*: It facilitates removal of crusts. One teaspoonful of alkaline powder (sodium bicarbonate 1 part, sodium biborate 1 part, and sodium chloride 2 parts) is dissolved in 280 ml of water. It is used for irrigating the nasal cavities 2 or 3 times a day. Later on just once every 2 or 3 days is sufficient. The alkaline solution loosens the crusts and removes thick tenacious discharge. Hard crusts need to be removed with forceps or suction.

b. *25% glucose in glycerin*: Application of this paint, after the removal of crusts, inhibits the growth of proteolytic organisms responsible for foul smelling.

c. *Antibiotics*: Spraying or painting of antibiotics may eliminate secondary infection.

i. *Kemicetine antiozaena solution, which contains Chloromycetin, estradiol and vitamin D_2, has been found useful.*

ii. *Systemic use of streptomycin (effective against Klebsiella), 1 g/day for 10 days, has shown to reduce crusting and bad odor.*

d. *Estradiol nasal spray*: It is said to increase vascularity of nasal mucosa and regenerate seromucinous glands.

e. *Placental extract*: This submucosal intranasal injection provides symptomatic relief.

f. *Potassium iodide*: This oral preparation has been shown to promote and liquefy nasal secretions.

B. *Surgical*

1. *Young's operation*: In this plastic surgery both the nostrils are completely closed within the nasal vestibule with flaps. Nasal mucosa may revert to normal and crusting is reduced. The nostrils are opened again after about 6 months or later.
2. *Modified Young's operation*: The nostrils are partially closed that avoids the discomfort of nasal obstruction.
3. *Narrowing the nasal cavities*: Narrowing of the nasal airway helps in decreasing the crusting. The techniques include:
   a. Submucosal intranasal injection of teflon paste.
   b. Fat, cartilage, bone or teflon grafts under the mucoperiostium of the floor and lateral wall of nose and the mucoperichondrium of the septum.
   c. Medial displacement of lateral nasal wall.

Secondary Atrophic Rhinitis

1. Syphilis, lupus, leprosy and rhinoscleroma cause atrophic changes and destruction of the nasal structures.
2. The long-standing purulent rhinosinusitis, radiotherapy and excessive surgical removal of turbinates can result in atrophic rhinitis.
3. Unilateral atrophic changes on the wider side nasal cavity are seen sometimes in cases of marked deviation of nasal septum.

RHINITIS SICCA

The respiratory ciliated columnar epithelium of anterior part of nose undergoes squamous metaplasia with atrophy of seromucinous glands.

1. *Clinical features*: This crust-forming lesion of anterior third of nose (especially nasal septum) is seen in workers (bakers, iron and goldsmiths), who work in hot, dry and dusty surroundings.
   a. The removal of crusts from the anterior part of septum causes ulceration, nosebleed and later on even septal perforation.
2. *Treatment*: Patient is instructed to correct the occupational surroundings and use masks and filters. Following measures help immensely:
   a. Application of bland or antibiotic-steroid ointments.
   b. Avoidance of frequent nose pricking and removal of crusts.
   c. Nasal saline douche.

RHINITIS CASEOSA

In this uncommon condition, granulomatous sinus mucosa destroys bony walls of sinus.

- *Clinical features*: It usually involves one side of nose and mostly affects males.
  - Examination reveals offensive purulent discharge and inspissated cheesy material, which is possibly the result of chronic rhinosinusitis.
  - This condition needs differentiation from malignancy.
- *Treatment*: It consists of removal of debris and granulation tissue, which restore free drainage of the affected sinus.
- *Prognosis*: It is usually good.

SARCOIDOSIS

This granulomatous disease resembles tuberculosis. There is absence of caseation. Although the cause is not known, sarcoidosis is associated with abnormalities of cell-mediated and humoral immunity.

Clinical Features

The symptoms may indicate involvement of lungs, lymph nodes, skin, eyes (episcleritis), salivary glands (parotid swelling), oropharynx (tonsillar hypertrophy), larynx (epiglottic swelling and subglottic swelling) or neuropathy (sudden deafness and unilateral or bilateral facial nerve palsy).

1. *External nose*: Raised papular lesions may coalesce and form bluish-red swellings. These firm and elastic lesions extend deeply and involve entire dermis.
2. *Lupus pernio*: This nasal sarcoidosis presents with chronic violaceous cutaneous lesions over nose, cheeks, ears and fingers.
3. *Nasal symptoms*: Nasal obstruction, pain and sometimes epistaxis.
4. *Nasal findings*: Crusting and diffuse mucosal swelling involving septum, inferior turbinate, nasal vestibule or skin of face.

Diagnosis

In addition to the positive findings of X-ray chest and biopsy, it is necessary (before labeling the case as sarcoidosis) to exclude other causes of granulomatous changes.

1. *X-ray chest*: Diffuse pulmonary infiltrate with hilar adenopathy.
2. *Biopsy*: Noncaseating granuloma establishes the diagnosis.
3. *Bronchoalveolar lavage*: Greater than 28% T lymphocytes indicate high intensity alveolitis.
4. *Gallium-67 scanning*: The inflammatory tissue of involved organs takes up and accumulates the isotope.
5. *Angiotensin-converting enzyme*: Elevated.
6. *Serum and urinary calcium levels*: Elevated.
7. *Interleukin (IL)-2 receptor (sIL-2R)*: Increased levels.
8. *Erythrocyte Sedimentation Rate:* Raised
9. *Hypergammaglobulinemia*: In some patient.
10. *Alkaline phosphatase*: Increased in some cases.
11. *ECG*: Prolonged PR interval, bundle-branch block, arrhythmias, nonspecific sinus tachycardia and (ST)-segment changes.
12. *CSF*: Increased lymphocytes and protein if leptomeninges are involved.

Treatment

1. Systemic and topical steroids:
   a. Cortisone containing eye solutions for ocular involvement.
   b. Aerosol containing cortisone for the nasal, epiglottis and subglottis involvement
   c. Neuropathies respond well to oral steroids.
2. Methotrexate: Low doses of methotrexate in progressive and cases unresponsive to steroids have shown improvement.

CHURG-STRAUSS SYNDROME

The characteristic feature of Churg-Strauss syndrome (CSS) is eosinophil-rich granulomatous inflammation and necrotizing vasculitis. It presents with asthma and eosinophilia.

- Three phases of CSS–
  - *Prodromal phase*: It may persist for years and it consists of allergic rhinitis, polyposis and asthma.
  - *Second phase*: Blood and tissue eosinophilia and eosinophilic pneumonia or gastroenteritis is present.
  - *Final phase*: It has features of life-threatening systemic vasculitis.
- Treatment: It responds well to corticosteroids.

RHINOSCLEROMA

Von Hebra coined the term rhinoscleroma in 1870. Rhinoscleroma is endemic in several parts of the world. In India, northern parts are affected more than the southern states. There is no age and sex bar.

Etiopathology

The causative microorganism, *Klebsiella rhinoscleromatis* (Frisch bacillus), is a Gram-negative bacillus. Mode of infection is not clear. The disease begins in the nose but extends to nasopharynx, oropharynx, larynx (mostly subglottic region), trachea and bronchi.

Clinical Features

There are four stages of this disease: catarrhal, atrophic, granulomatous and cicatricial.

1. *Catarrhal:* Foul smelling purulent nasal discharge for weeks to months.
2. *Atrophic stage*: This stage presents with crusting, which resembles atrophic rhinitis.
3. *Granulomatous stage*: Multiple granulomatous nodules, which enlarge and coalesce, are seen in nasal mucosa. Subdermal infiltration of lower part of external nose and upper lip gives "woody" feel. These painless nodules are non-ulcerative and can be found in pharynx, larynx, trachea and bronchi.
4. *Cicatricial stage*: Fibrosis leads to stenosis of nares, distortion of upper lip and adhesions in the nose, nasopharynx, oropharynx and larynx. The subglottic stenosis manifests as respiratory distress.

Diagnosis

1. *Biopsy*: Submucosa is infiltration with plasma cells, lymphocytes, eosinophils, Mikulicz cells and Russell bodies. The vacuolated Mikulicz cells (almost diagnostic), large foamy histiocytes with central nucleus stains well with hematoxylin and eosin. Cytoplasm contains causative bacilli. Russell bodies, homogeneous eosinophilic inclusion bodies (accumulation of immunoglobulins secreted by the plasma cells), are found in the plasma cells.
2. *Cultures of infected tissue*: The causative organisms are cultured and are diagnostic.

Treatment

1. *Antibiotics*: Both streptomycin (1 g/day) and tetracycline (2 g/day) for 4–6 weeks. Repeat if necessary after 1 month. Treatment is stopped only when two consecutive cultures from the biopsy material are negative.
2. *Steroids*: They can be combined to reduce fibrosis.
3. *Surgery*: In fourth stage of fibrosis and stenosis, surgery is required to establish the airway and correct nasal deformity. A silastic stent facilitates re-epithelialization.
4. *Radiotherapy*: It is not effective.

TUBERCULOSIS

1. Tuberculosis of nose secondary to lung tuberculosis is more common than primary tuberculosis of nose.
2. Anterior part of nasal septum and anterior end of inferior turbinate are the common sites of involvement.
3. Nodular infiltration is usually followed by ulceration and perforation of cartilaginous part of nasal septum.
4. Diagnosis: Biopsy and special staining of sections for acid-fast bacilli, culture of organisms and animal inoculation confirm the diagnosis.
5. Treatment is antitubercular therapy.

LUPUS VULGARIS

This low-grade tuberculous infection commonly affects nasal vestibule or the skin of nose and face.

1. *Skin lesions*: Brown and gelatinous (apple-jelly) nodules.
2. *Nasal vestibule*: Chronic vestibulitis and perforation in the cartilaginous part of nasal septum.
3. *Diagnosis*: Biopsy confirms the diagnosis. It is difficult to isolate tubercle bacilli by culture or animal inoculation in lupus vulgaris.
4. *Treatment*: Antitubercular therapy.

NONTUBERCULOUS MYCOBACTERIA

1. This usually represents primary infection and is less susceptible to standard AKT.
2. The organisms most commonly responsible for cervical lymph node infection are *Mycobacterium scrofulaceum, Mycobacterium szulgai* and *Mycobacterium xenopi*. Mycobacteria are acid-fast, nonmotile, weakly Gram-positive rods of Actinomycetales class.

LEPROSY

Leprosy is common in the tropics and caused by *Mycobacterium leprae*. It is widely prevalent in India. The nose is involved more commonly in lepromatous type in comparison to tuberculoid or dimorphous forms of leprosy. Common sites of involvement are anterior part of nasal septum and anterior end of inferior turbinate.

Clinical Features

1. It presents initially with excessive nasal discharge with red and swollen mucosa.
2. Crusting and bleeding occurs later.
3. Nodular lesions on the septum ulcerate and cause perforation of cartilaginous part of septum.
4. Late sequelae: Atrophic rhinitis, depression of nasal bridge, destruction of anterior nasal spine with retrusion of the columella.

Diagnosis

- *Scrapings of nasal mucosa and biopsy*: Acid fast lepra bacilli are present in the foamy appearing histiocytes called lepra cells.

Treatment

1. *Antibiotics*: Dapsone, rifampicin and isoniazid.
2. *Reconstruction procedures*: They are performed when disease is inactive.

SYPHILIS

Syphilis is usually classified into two types: acquired and congenital.

Acquired Syphilis

It is further divided into three types—primary, secondary and tertiary.

1. *Primary*: Primary chancre of the vestibule is rare.
2. *Secondary*: Present in nose with simple rhinitis with crusting and fissuring in the nasal vestibule. The presence of mucous patches in the pharynx, skin rash, fever and generalized lymphadenitis suggest the diagnosis.
3. *Tertiary*: Nose is commonly involved in this tertiary stage. Typical manifestation is nasal septum gumma, which destroys both bony and cartilaginous parts of nasal septum. Other findings include offensive nasal discharge with crusts, bony or cartilaginous sequestra, saddle nose deformity and perforation of the hard palate.

Congenital Syphilis

It has two forms: early and late.

1. *Early form*: In the first 3 months of life, it manifests as "snuffles" and subsequently other findings appear such as purulent nasal discharge, fissuring and excoriations of nasal vestibule and skin of upper lip.
2. *Late form*: In puberty, clinical features of tertiary syphilis manifest such as gumma and perforation of nasal septum. Other stigmata of syphilis (corneal opacities, deafness and Hutchinson's teeth) are also present.

Complications

They include vestibular stenosis, perforations of nasal septum and hard palate, secondary atrophic rhinitis and saddle nose deformity.

Diagnosis

1. *Serological tests*: VDRL
2. *Biopsy of the tissue*: Special stains demonstrate *Trepenoma pallidum*.

Treatment

1. *Penicillin*: Benzathine penicillin 2.4 million units IM every week for 3 weeks with a total dose of 7.2 million units.
   a. Nasal alkaline wash
   b. Removal of nasal crusts
2. *Surgery*: Removal of bony and cartilaginous sequestra and correction of deformities are done when disease becomes inactive.

HISTOPLASMOSIS

1. This granulomatous fungus, which affects infants and elderly, primarily involves larynx and tongue.
2. Humans and animals are infected by inhalation of infected dust such as soil from chicken houses. Feces of bat or birds are rich in *Histoplasma capsulatum*.
3. Nasal findings are accompanied by pulmonary symptoms. They include nodules or ulcers, cough, chest pain and hoarseness of voice.
4. Diagnosis
   a. Positive skin tests and microscopic identification of fungus confirm the diagnosis.
   b. Complement fixation test: 1:32 titer is diagnostic.
5. Treatment: Amphotericin B 1–10 mg/day progressing to 1 mg/kg for a total dose of 2 g for 2–3 months period.

RHINOSPORIDIOSIS

This fungal granuloma, which is caused by *Rhinosporidium seeberi*, is seen in southern states of India, Pakistan and Sri Lanka. The fungus, which is acquired through contaminated water of ponds frequented by animals, usually involves nose, nasopharynx, lip, palate, conjunctiva, epiglottis, larynx, trachea, bronchi, skin, vulva and vagina. Yet it is not possible to culture the fungus or transfer the disease to experimental animals.

Clinical Features

1. *Symptoms*: Blood-tinged nasal discharge, nasal stuffiness and frank epistaxis.
2. *Nose findings*: A leafy, pink to purple color polypoidal mass attached to nasal septum or lateral wall may extend into the nasopharynx and hang behind the soft palate. The surface of the mass is studded with white dots (sporangia of fungus). The mass is very vascular and bleeds easily on touch.

Diagnosis

- *Biopsy*: Sporangia, oval or round in shape are filled with spores which are bursting through its chitinous wall.

Treatment

1. *Surgery*: Complete excision of the mass with diathermy knife and cauterization of its base. Recurrence is not uncommon.
2. *Medical*: Drugs are not effective but dapsone has been tried with some success.

FUNGAL SINUSITIS

a. Types: *Fungal rhinosinusitis has five distinct forms:*
   - Invasive:
     - Acute Invasive Fungal Sinusitis
     - Chronic Invasive Fungal Sinusitis
   - Noninvasive:
     - Fungus Balls
     - Saprophytic Fungal Infection
     - Allergic Fungal Rhinosinusitis

b. Immunology*:* The evaluation of the immunologic status of the patient is very important because fungal sinusitis is related to host immunocompetency.

c. Clinical features*:* The chronic symptoms in noninvasive fungus include facial pain, nasal obstruction, cacosmia, nasal polyps, proptosis and allergic mucin casts.

Acute Invasive Fungal Sinusitis

- *Etiology*: The most common fungi are aspergillus species and mucorales **(**mucormycosis).
  a. Mucormycosis: It is more common in diabetics. Fungal elements are broad, ribbon-like (10–15 mm), irregular and rarely septated.
  b. Aspergillus: This species have narrow hyphae with regular septations and 45° branching.
- *Clinical features*: The immunocompromised patients (Diabetic, Transplant, Leukemia, and AIDS) are most at risk for invasive fungus. These cases need quick evaluation and aggressive management. Steroids administration may be one of the precipitating factors.
  a. Acute clinical picture: It includes palatal erosion, impairment of vision, limitations in extraocular movements, fever, nasal and facial anesthesia and nasal necrosis.
  b. Mucopurulence: It is variable depending on the neutropenia of the host.
- *Investigations*: The investigations are as follows:
  a. CBC in immunosuppressed patients may show neutropenia or evidence of left shift.
  b. Coagulation profile includes platelets, bleeding time, prothrombin time and partial thromboplastin time.
  c. Blood sugar is found raised in invasive fungus especially mucormycosis.
  d. In culture an endoscopic aspirate culture is done with immediate fungal stains.
  e. In biopsy if fungal stains are negative and suspicion is strong then biopsy with immediate frozen section and fungal stains are done. Coagulation abnormalities if any are corrected before biopsy.
  f. CT scan: A coronal CT scan is the minimum requirement of any fungal sinus surgery.
     - Axial scans provide additional information for the funguses of frontal and sphenoidal sinuses. Contrast studies are ordered if invasive fungus shows bony erosion, intracranial or orbital extension.
- *Treatment*: Attempts must be made to reverse the immunocompromised status by controlling its cause.
  a. Conservative debridement: Repeated conservative debridement (removal of necrotic tissue) may be needed in operative candidates, whose immunocompromised state can be reversed. Orbital exenteration should be avoided, when patient is not blind, even if there is involvement of the orbit. In cases of bone marrow transplant, when ingraft fails, immunocompromised state cannot be reversed and heroic surgeries are futile and must not be done.
  b. Antifungal agents: The systemic and topical antifungal agents appropriate for the cultured fungus must be started immediately.
     - Intravenous amphotericin B is usually started while awaiting fungal report.
     - Ketoconazole and itraconazole: If *Pseudallescheria boydii* grows on culture.
- *Prevention*: The use of high-efficiency particulate air filter systems and elimination of potted plants can reduce fungal infection in high-risk neutropenic patients.

Chronic Invasive Fungal Sinusitis

- *Causative fungi*: *A. flavus* (most common), *A. fumigatus, Alternaria, P. boydii, Sporothrix schenckii.*
- *Types*: There are two types of chronic indolent invasive fungal sinusitis and both progress over weeks to months to years.
  a. Granulomatous: It is common in Sudan and infects immunocompetent patients.
  b. Nongranulomatous: It occurs in immunocompromised patients.
- *Diagnosis*: Fungal culture and biopsy with special fungus stains confirm the diagnosis.
- *Treatment*: It consists of–
  - Repeated courses of antifungal therapy.
  - Surgical debridement.
  - Attempts to control the cause of immunocompromise.

Fungus Balls

a. *Causative fungal species*: The fungi presenting with balls are *A. flavus, A. fumigatus, Alternaria* and *Mucor*.

b. *Sites*: Maxillary sinus is most commonly involved followed in descending frequency by sphenoid, ethmoid and the frontal.

c. *Clinical features*: This noninvasive fungus may remain asymptomatic for months to years but may become invasive if the patient becomes immunocompromised.
   - It may present with facial pain and cacosmia.

d. *CT scan:* Fungus balls show total or partial sinus opacification (hyperdense area) and rarely bony erosion.

e. *Diagnosis*: It is usually diagnosed at surgery and suspected on CT scans.
   - Culture may be negative but fungus can be seen with special stains.

f. *Treatment*: Usually no antifungal therapy is required. The recurrence after surgical removal is uncommon.

Saprophytic Fungal Infection

- *Clinical*: The fungus grows on mucocrusts, seen after endoscopic sinus surgery (ESS).
- *Treatment*: It includes removal of crusts.
- *Prevention*: The patient is instructed to irrigate the nose weekly with sterile saline and wear a mask in moldy environments.

Allergic Fungal Rhinosinusitis

The manifestations of this noninvasive fungus occur because of a hypersensitivity response by the patient to the fungus. It responds to systemic steroids.

- *Etiology:* Dematiaceous (darkly pigmented) fungal species include *Alternaria, Bipolaris*, and *Curvularia*.
- *Clinical features*: Allergic fungal rhinosinusitis (AFRS) is found in cases of allergy, polyps and allergic mucin nasal casts.
- *Laboratory*:
  - Eosinophilia.
  - Total immunoglobulin IgE is raised along with fungus-specific IgE and IgG, which are difficult to evaluate as many fungi cause AFRS.
  - In quiescent AFRS, total IgE may be normal.
- *CT scan*: AFRS shows heterogeneity of tissue densities within the sinuses. Post contrast MRI shows fungal growth with changes of sinusitis (Fig. 2).
- *Histopathology*: Hyphae are seen in eosinophil-rich mucin without any evidence of tissue invasion. So the pathologist should assess the mucin not the polyp.

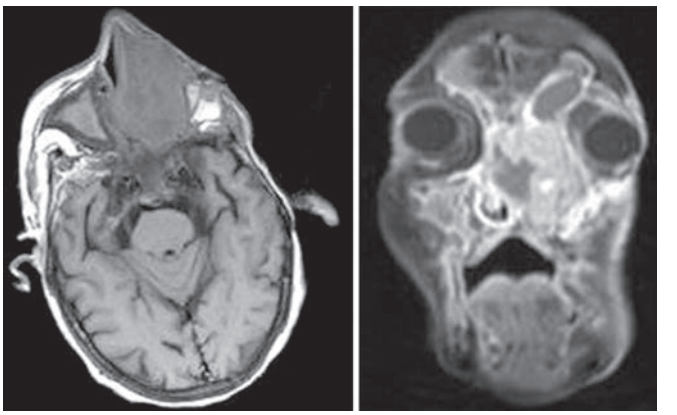


Fig. 2: Post-contrast coronal MRI showing fungal growth with changes of sinusitis. Moderate contrast enhancement in left nasal cavity mass
Source: Dr Swati Shah, Professor, Radiodiagnosis, GCS Medical College.

  - An allergic mucin contains necrotic inflammatory cells, eosinophils and Charcot-Leyden crystals (byproduct of eosinophil degranulation).
- *Treatment*: It includes surgery, steroids, immunotherapy and antifungal agents.
  - Surgery: Surgical removal of all allergic mucin.
  - Steroids: If it is difficult to remove allergic mucin completely, then remission is achieved with systemic steroids. In cases of recurrence, repeated surgical removal of mucin combined with perioperative steroids (prednisone 60 mg for several days and tapered off over 2 to 4 weeks) is beneficial.
  - Immunotherapy: It contains fungal agents, started within 4–8 weeks of surgery, and prevents recurrence.
  - Oral antifungal agents: Their role is controversial. A long course of oral itraconazole (400 mg daily for 1 month and tapered over 3 months) needs regular monitoring of liver functions. It has shown good results.

Clinical Highlights

1. **Atrophic rhinitis**: Patients cannot perceive offensive smell emanating from their own noses. Degenerative changes in olfactory mucosa results in anosmia.
2. **Alkaline nasal douche**: This mixture contains sodium chloride (2 parts), sodium bicarbonate (1 part) and sodium borate (1 part). It does not contain glucose.
3. **Rhinoscleroma**: This Gram-negative coccobacillus (*Klebsiella rhinoscleromatis*) infection has three stages—catarrhal, granulomatous and cicatricial. Woody infiltration of the upper lip is common clinical feature. Usually the patients are adults and from UP. They present with nasal obstruction and crusting of nose. An infiltrating lesion that involves nasal vestibule and upper lip, results in broadening of nasal dorsum. Involvement of larynx (subglottic region) and trachea leads to airway obstruction. **Mikulicz' cells** and **Russell bodies** are characteristic features seen on histology. Drugs used in the treatment of rhinoscleroma include streptomycin, ciprofloxacin and tetracycline.
4. **Russell bodies**: These are plasma cells, which contain rounded eosinophilic structures and are seen in rhinoscleroma.
5. **Syphilis and tuberculosis**: Syphilis involves nasal bones while tuberculosis affects nasal cartilage.
6. **Congenital syphilis**: The features of congenital syphilis include perforation in the nasal septum, saddle nose deformity, snuffles in the newborn and atrophic rhinitis.
7. **Rhinosporidiosis**: This fungal infection of nose presents as nasal polyp.
8. **Acute fulminant fungal sinusitis**: This invasive fungal infection is commonly seen in immunosuppressed patients of diabetes, HIV, transplant and chemotherapy. Management includes earliest treatment of the incriminating factors, aggressive surgical debridement and amphotericin-B.
9. **Mucormycosis**: This acute invasive fungal infection of nose and paranasal sinuses involves lateral nasal walls and turbinates and quickly spreads to orbit, palate, face and cranium. Fungal hyphae invade blood vessels and cause ischemic necrosis. Treatment includes surgical debridement and amphotericin-B.
10. **Noninvasive fungal sinusitis**: They include fungal ball and fungal allergic sinusitis (presenting with polyps) and do not require antifungal treatment.
11. **Aspergillus species**: This is the common etiological agent in paranasal sinus mycoses.

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30 Allergic and Nonallergic Rhinitis

What we want is this Shraddha (faith). What makes the difference between man and man is the difference in the Shraddha (faith) and nothing else. What makes one man great and another weak and low is this Shraddha (faith). Give up the awful disease, that idea of ridiculing everything. Be strong and have this Shraddha (faith), and everything else is bound to follow.

—Swami Vivekananda

Points of Focus

ALLERGY AND IMMUNOLOGY

- ✓ Humoral Immunity: IgE, IgG, IgA, IgM and IgD
- ✓ Mediators Released by Sensitized Mast Cells and their Effects: Histamine, Prostaglandins, Leukotrienes
- ✓ Cell-Mediated Immunity
- ✓ Natural Killer Cells
- ✓ Complement

TYPES OF IMMUNOLOGIC (HYPERSENSITIVITY) MECHANISM

- ✓ Type I—Immediate (IgE-Mediated Hypersensitivity): Atopy and Anaphylaxis
- ✓ Type II—Cytotoxic (Antibody-Mediated Hypersensitivity)
- ✓ Type III—Immune Complex-Mediated Hypersensitivity
- ✓ Type IV—Delayed (T Cell-Mediated Hypersensitivity)

ALLERGIC RHINITIS

ETIOLOGY: Allergens (Pollens, Molds, Insects, Animals, House Dust, Dust Mite, Ingestants, Drugs

- ✓ Predisposing Factors
- ✓ Pathogenesis
- ✓ Mediators Released by Sensitized Mast Cells and their Effects

CLASSIFICATION: Seasonal (Hay Fever, Summer Colds and Rose Fever) and Perennial Allergic Rhinitis

CLINICAL FEATURES

- ✓ Symptoms: Pruritus, Sneezing, Rhinorrhea and Stuffiness
- ✓ Examination: Allergy Salute, Edematous Inferior Turbinate, Thin Watery Discharge, Polyps and Superadded Infection
- ✓ Complications/Associated Conditions

INVESTIGATIONS

- ✓ Specific-IgE Antibody Tests
- ✓ Skin Tests
- ✓ In Vitro Tests of IgE Antibody

TREATMENT

- ✓ Avoidance Therapy
- ✓ Drug Therapy: Antihistamines, Sympathomimetic Drugs (Nasal Decongestants), Corticosteroids, Mast Cell Stabilizers (Cromolyn and Nedocromil), Anticholinergic Agents (Nasal Topical Ipratropium Bromide), Leukotriene Modifiers, Anti-IgE Antibody Therapy, Macrolides, and New Therapies
- ✓ Immunotherapy
- ✓ Surgery

NONALLERGIC RHINITIS (VASOMOTOR RHINITIS)

PATHOPHYSIOLOGY

CLASSIFICATION

- ✓ Nonallergic Rhinitis with Eosinophilia (NARE)
- ✓ Drug-Induced Rhinitis
- ✓ Rhinitis Medicamentosa
- ✓ Honeymoon Rhinitis
- ✓ Emotional Rhinitis
- ✓ Hormone-Related Rhinitis
- ✓ Gustatory Rhinitis
- ✓ Non Airflow Rhinitis
- ✓ Idiopathic or Vasomotor Rhinitis

CLINICAL FEATURES

INVESTIGATIONS

TREATMENT

- ✓ Medical: Avoidance of Inciting Factors, Antihistaminics and Oral Decongestants, Topical Steroids (Beclomethasone Dipropionate, Budesonide or Fluticasone), Systemic Steroids, Psychological Counseling for Emotional Adjustment, Exercise, and Tranquilizers
- ✓ Surgical

CLINICAL HIGHLIGHTS

ALLERGY AND IMMUNOLOGY

Immunology

Immune system differentiates self from nonself antigens and eliminates nonself (foreign elements), such as pathological microorganisms. Immune cells (lymphocytes) are derived from stem cells in the bone marrow. B cells (humoral immunity) mature in bone marrow itself. T cells (cell-mediated immunity) mature in thymus. Genetic control plays an important role in most immune functions.

Allergy

Allergy is an altered immune reaction to an allergen (foreign protein). These genetically predisposed allergy patients have an overzealous immunologic response on re-exposure to that allergen to which they are allergic. This immunologically mediated disease may be local or systemic. It is manifested with tissue inflammation and organ dysfunction. The organs involved most commonly are skin and respiratory system. Anaphylaxis is the dreaded form of systemic allergy. The localized form of allergy may involve vasculature, gastrointestinal tract or other visceral organs.

Excluding trauma and tumors, allergy factor plays a role in about 50% of the otolaryngology practice.

Humoral Immunity

This is antibody-mediated immunity. B-lymphocytes make five types of immunoglobulins (antibodies): IgA, IgM, IgG, IgD and IgE. B-lymphocytes in association with T-helper lymphocytes produce specific antibodies to any particular antigen. Antibodies have their own specific functions. IgE attaches to surface antigen of mast cell, and release histamine. Disorders of B-lymphocytes and immunoglobulins result in preponderance of bacterial infections.

IgE (< 1% of Serum Immunoglobulins)

It is involved in immediate (type I) hypersensitivity, and can be detected in cord blood. It may be elevated in parasitic diseases. IgE attaches to mast cell, and is bridged by an antigen for allergic reaction.

- Allergic rhinitis: In the allergic rhinitis (AR), IgE is produced by plasma cells which are regulated by T suppressor lymphocytes and T helper cells. In genetically predisposed persons, allergens produce specific IgE antibodies which are "Y" shaped, and have Fc and Fab portions. Fc end of IgE becomes fixed to tissue mast cells and blood basophils. On subsequent exposure, allergen combines with Fab end of IgE antibodies which are already fixed to mast cells. Two such IgE antibodies, which bridge the allergen and mast cell, activate the mast cell. The disruption of mast cells leads to the release of chemical mediators which are of two types—preformed and newly synthesized (Table 1). Histamine, leukotrienes, prostaglandins and others stimulate H_1 receptors in the nasal mucosa and blood vessels, and produce rhinorrhea and mucosal edema. Itching and sneezing are produced by the stimulation of nerve endings.

IgG (75% of Serum Immunoglobulins)

There are four subclasses. Three can bind to complement. IgG4 blocks antibodies stimulated by immunotherapy. Neutrophils and killer cells have IgG receptors. IgG deficiency can cause recurrent sinus and pulmonary infections.

IgA (15% of serum immunoglobulins)

It is found in secretory and mucosal tissue (respiratory, GIT, genitourinary) and secretions (milk and saliva). Its level rises slowly in the Ist year of life. Two IgA molecules (dimer) join with epithelial secretions, and form immunocompetent secretory IgA molecule which provides surface protection against antigens to mucous membranes.

IgM (8% of Serum Immunoglobulins)

This large immunoglobulin consists of five monomer units, and rises rapidly in the Ist year of life. This is body's earliest defense against bacteria, and first to rise. It is later supplanted by IgG in subsequent exposures.

IgD (< 1% of Serum Immunoglobulins)

This is similar to IgG. Its functions are yet not known.

Chemical Mediators

Histamine

It is present in mast cells and basophils. It causes increased vascular permeability, edema and vasodilation. Histamine is chemical mediator of skin test whealing and type-I (IgE)

TABLE 1 Mediators released by sensitized mast cells and their effects

Name of mediator	*Effects*
Preformed	
Histamine	Vasodilatation and bronchospasm
Eosinophilic chemotactic factor for anaphylaxis	Attracts eosinophils
Neutrophil chemotactic factor	Attracts neutrophils
Heparin	Enhances phagocytosis
Newly formed	
Prostaglandins	Vasoactive and bronchospastic
Leukotriene	Vasoactive and bronchospastic
Platelets aggregating factor	Platelets release histamine & serotonin
Thromboxane A	Spasmogenic
Tumor necrosis factor	Attracts neutrophils and eosinophils

hypersensitivity. It may be a factor in late-phase allergic response. Chocolate and red wine stimulate histamine production even in normal person. There are two types of histamine receptors on cell surfaces:

a. H_1 receptors: They are present on smooth muscles of vessels, bronchi and mucosa of gastrointestinal tract (GIT).
b. H_2 receptors: They are present on suppressor T cells, basophils, mast cells, neutrophils and gastric cells.

Prostaglandins

They are formed from arachidonic acid. They are of different types. Prostaglandins D2 (PGD2) is active in immediate type hypersensitivity. It is spasmogenic and vasoactive, and inhibits platelet aggregation. Dietary change (more of olive oil, avocados, nuts, and flax seeds; and less of saturated animal fats) can moderate overproduction of PGD2. These dietary changes are recommended in allergy patients.

Leukotrienes

These mediators of immediate hypersensitivity response are derived from the action of 5-lipoxygenase on arachidonic acid. These vasopermeable factors are responsible for delayed anaphylactic reactions, especially to aspirin. Leukotriene inhibitors are now frequently prescribed for the treatment of allergy patients.

Cell-Mediated Immunity

It has an integral role in delayed hypersensitivity response. This involves T lymphocytes (T-helper and T-suppressor cells), which produce lymphokines (soluble factors) to enhance or suppress immune response. T-helper cells increase production of B lymphocytes and antibodies to a particular antigen. T-suppressor cells suppress antibody production to antigen. Immunotherapy activates the T-suppressor cells. Disorder of T cells causes increase in fungal, viral and protozoal infections.

Natural Killer Cells

These front-line soldiers are derived from lymphocytes. Natural killer cells constitute about 10% of lymphocytes population. They have no immunologic memory. They find and destroy mutant cancer and virus-infected cells. They attach to these abnormal cells, and release cytokines to kill them.

Complement

This group of plasma proteins acts through enzymatic activity to kill the cell. It is important in type II and III hypersensitivity. Antigen-antibody complex activates the complement cascade resulting in cellular swelling and lysis. There occurs intracellular water retention and electrolytes depletion.

TYPES OF IMMUNOLOGIC (HYPERSENSITIVITY) MECHANISM

Gell and Coombs described four types of immunologic mechanism involved in hypersensitivity reactions: immediate, cytotoxic, immune complex, and delayed. Others have suggested two more types: stimulating antibody and antibody-dependent, cell-mediated cytotoxicity. Classification serves not only as rational basis for diagnosis and treatment but also elucidates the underlying pathogenic mechanism of clinical symptoms.

Type I–Immediate (IgE-Mediated Hypersensitivity)

Allergen links adjacent IgE antibodies which occupy receptors sites on mast cells. The activating and degranulating mast cells generate mediators which cause vasodilation, visceral smooth muscle contraction, mucous secretory gland stimulation, vascular permeability, and tissue inflammation. Arachidonic acid metabolites, cytokines and other mediators lead to a late-phase inflammatory response which manifests several hours later. The two clinical subgroups of type I allergies are: atopy and anaphylaxis.

Atopy

The allergic reaction is localized to one or more susceptible target organs. The examples are allergic rhinitis, allergic asthma, atopic dermatitis and allergic gastroenteropathy. These patients have an inherited tendency to develop IgE antibodies to multiple allergens, and have strong familial tendency.

There are certain clinical conditions that resemble allergic hypersensitivity in the absence of an immunologic mechanism. Nonallergic (intrinsic) asthma is triggered by inhaled dusts and fumes, weather changes, viral respiratory infections and stress rather than by aeroallergen-induced IgE-mediated mast cell degranulation. Irritant dermatitis results from physical or chemical damage to skin rather than by sensitized lymphocytes. "Red man syndrome", an anaphylactoid reaction, secondary to vancomycin, occurs from nonimmunologic release of mast cell mediators. Therefore, before the diagnosis of allergy, we must know the type of allergens and the nature of disease caused by an IgE-mediated mechanism.

Anaphylaxis

Certain allergens can cause a generalized release of mediators from mast cells, resulting in systemic anaphylaxis. They are drugs, insect venoms, latex and foods. The clinical features of this potentially fatal condition are:

- Hypotension or shock due to vasodilatation
- Bronchospasm
- Muscle contractions: Gastrointestinal and uterine
- Urticaria or angioedema: Isolated urticaria and angioedema are common presentations of anaphylaxis, and have better prognosis.

Management: The proper treatment of anaphylactic shock includes:

- Intubation and ventilation with 100% oxygen
- Epinephrine: For laryngeal edema and hypotension
- Tracheostomy is required if laryngeal edema does not respond to epinephrine
- Glucagons: For hypotension
- IV fluids: For volume expansion
- Beta-agonists: Albuterol or metaproterenol inhalation for resistant bronchospasm

Type II–Cytotoxic (Antibody-Mediated Hypersensitivity)

The cytotoxic reaction of either IgE or IgM antibody to cell bound antigens results in activation of the complement cascade and the destruction of the cell. Examples of this allergy include immune hemolytic anemia and Rh hemolytic disease.

Type III–Immune Complex-Mediated Hypersensitivity

Complexes of allergen and IgG or IgM antibodies get deposited in tissues, and trigger complement cascade. The best examples of this type of hypersensitivity are Arthus reaction and serum sickness (characterized by fever, arthralgia and dermatitis).

Type IV–Delayed (T Cell-Mediated Hypersensitivity)

The reaction manifests after a latent period of 1–2 days. Allergic contact dermatitis and hypersensitivity pneumonitis (extrinsic allergic alveolitis) are some of the examples of cell-mediated hypersensitivity.

ALLERGIC RHINITIS

Definition: *A symptomatic disorder of nose induced by an IgE mediated inflammation after allergen exposure (ARIA).*

Rhinitis is the commonest chronic disease of human beings. One in six people suffer from rhinitis. Allergic rhinitis (AR) constitutes more than 50% of all allergies in India, and its incidence is steadily increasing worldwide. Surprisingly, it is more common in developed countries where the pollution is lesser in comparison to developing countries. Allergic rhinitis and asthma frequently coexist; in fact AR appears first in about 45% of patients.

The essential symptoms of this IgE-mediated atopic allergic disease are: nasal pruritus, congestion, rhinorrhea or paroxysms of sneezing. Demonstration of allergic hypersensitivity by in vivo or in vitro testing of specific-IgE antibody to aeroallergens is mandatory for confirming the diagnosis of AR.

ETIOLOGY

Allergens

Allergens are the causal substances of AR. They are capable of making the body produce IgE antibodies. The following groups of allergens can cause AR:

Pollens

An important aspect in the context of pollen allergy is the seasonal variation of the prevalence of various pollens. This is important, as correlation of symptoms and prevalent pollen has to be done; the clinical relevance of the pollen can then be more optimally assessed. The pollen prevalence varies depending on the climatic condition, increasing encroachment on agrarian land, and other factors. Palynological surveys have been in progress in India, and the following common pollens have been found in different cities:

- *Delhi:* Helianthus, Amaranthus, Cassia, Cenchrus, Morus and Imperata.
- *Jaipur:* Cynodon, Holoptelea and Prosopis.
- *Kolkata:* Argemope, Cannabis and Brassica.
- *Kolkata:* Lantana, Cucurbita, Cassia and Cocos nucifera.
- *Andhra Pradesh:* Cassia, Ageratum, Salvadora and Ricinus.
- *Bengaluru:* Albizia, Parthenium and Artemisia.

Molds

The common fungal spores in India are Cladosporium, Fusarium, Alternaria, Curvularia, Aspergillus, Mucor, Phoma, Penicillium and Trichoderma. Fungi migrate indoors, and can cause AR. The common sites where fungal spores proliferate are food storage areas, indoor houseplants, leaves, soiled upholstery, garbage collection bins, damp walls, poorly maintained air-conditioners, refrigerator drip pans and grain storage. In "Sick building syndrome," fungal allergens form a component of indoor pollution. Outdoor mold levels drop in freezing weather but indoor mold levels remain fairly constant and rise in warmth and moisture.

Insects

Indoor allergens from cockroach antigens play an important role in aggravating asthma in sensitive subjects. Indoor allergies to bedbugs, house flies, mosquitoes and fleas have also been reported. Other occupations, which can be exposed to different types of insect allergens, are silk workers (moth larvae and silkworms), laboratory workers (beetles, locusts, flies and butterflies) and agricultural workers (grain mites, moths and bees).

Animals

Among the animal allergens (dander, saliva, hair and feathers), most common are cats and dogs. Other mammals causing sensitization among humans are horses, rabbits, guinea pigs, monkeys and mice. Cat dander is very light and clings stubbornly to clothing, bedding and upholstered furniture.

House Dust

Dust of household origin is a complex mixture of animals, fungi, algae, insect debris, human epithelial scales, plants and food remnants.

Dust Mite

It has been proved that mites (*Dermatophagoides*) in mattresses can sensitize people. In India, *D. farinae* is the more dominant species and the other is *D. pteronyssinus*. Thick carpets, humid (50–70% relative humidity) and warm (65–80°F) climate favor mite growth. The human epithelium is the substrate for mites. Dusting of rooms increases mite dissemination and causes acute exacerbations in sensitive cases. The common reservoirs include bedding, mattresses, pillows, carpets, upholstered furniture and stuffed toys.

Ingestants

Foods are especially important in children. There may be one obvious substance, such as eggs, strawberries, nuts or fish; but in nasal allergy, it is more common to have a less obvious cause, such as milk or wheat. Citrus fruits and pulses are also common allergens.

Drugs

Aspirin and iodides, hypotensive and other cholinergic drugs, and insulin injections are known to cause allergy reactions.

Predisposing Factors

Though exactly they do not cause AR, they are important and are frequent coexisting factors.

Hereditary

Multiple gene interactions are responsible for allergic phenotype. Genes on chromosomes 5, 6, 11, 12 and 14 seem to control inflammatory process in atopy. A family history of similar or allied complaints is common. Perhaps, atopy is inherited in a dominant fashion; though may not be constantly manifested clinically.

Endocrine

Pubertal, marital, natal and menopausal conditions have the potential to influence the nose significantly.

Psychological

Many psychological factors can cause functional disorders of the nose. It is one of the main etiological factors in cases of vasomotor rhinitis.

Physical

Changes in the humidity, temperature and pollution of air can contribute to the development of allergic rhinitis.

Infection

Viral and bacterial infection may increase the permeability of the tissue to allergens.

Contacts

Contacts to skin or nasal mucosa are not common causes of nasal allergy, but coexistent dermatitis should be excluded. Long time use of nasal drops or sprays for the relief of nasal stuffiness can produce adverse effects (rhinitis medicamentosa). Penicillin and sulpha drugs when used locally can also cause reaction.

Irritants

Fumes (e.g. diesel oil), pepper, tobacco smoke and pollution are also predisposing factors.

Pathogenesis

It is not only IgE mediated specific allergen but also nonspecific stimuli (such as, weather changes, emotional stimuli, medicines, pollution and infections), which act on mast cells and drop cAMP/cGMP ratio. The released mediators from mast cells (Table 1) result in increased vascular permeability and vasodilatation (tissue edema and nose block), smooth muscle contraction (bronchospasm) and stimulation of gland secretions (rhinorrhea).

Due to the "priming effect," the nasal mucosa, which is already sensitized to a specific allergen, reacts to smaller doses of that particular allergen and even to other nonspecific allergens.

Types of Allergic Response

There are two types of clinical allergic response—immediate and delayed. The clinical manifestations of early and late phases are overlapping in cases of recurrent and continuous exposure of allergens.

1. *Immediate (early phase)*: It occurs within 5–30 minutes of exposure and leads to release of vasoactive amines, such as histamine, which clinically manifest as sneezing, discharge, blockage and bronchospasm. The stimulated mast cells secrete chemical mediators (histamine, prostaglandins and leukotrienes).
2. *Late (delayed phase)*: The late phase, which occurs 2–8 hours after exposure, is due to infiltration of eosinophils, neutrophils, basophils, monocytes and CD4+ T cells at the site of allergen deposition. This delayed phase clinically manifests as swelling, congestion and thick secretion, which subside slowly.

CLASSIFICATION

There are two clinical types of AR—seasonal (hay-fever) and perennial. Patients of AR who have a decreased resistance to acute rhinitis, upper respiratory tract infection (URI), sinusitis and otitis media, may develop asthma (Fig. 1).

1. **Seasonal Allergic Rhinitis**: Hay fever and summer colds are common terms for seasonal AR which produce stuffy/runny nose, paroxysm of sneezing and itchy nose/eyes/throat and excess mucus in nose/throat. The condition may be a mere nuisance, or interferes with work and recreation. Pollens of common trees often cause early springtime hay fever while late springtime pollens come from the grasses.
   - Hay fever: It is a misnomer because neither it is caused by hay nor it produces fever.
   - Summer cold: It should not be confused with acute rhinitis (coryza); that is caused by virus and not by allergens.
   - Rose fever: It is also a misnomer because colorful or fragrant flowering plants rarely cause allergy, as their pollens are too heavy to be airborne.
2. **Perennial Allergic Rhinitis**: Perennial rhinitis is caused by allergens that are present through all seasons, and they

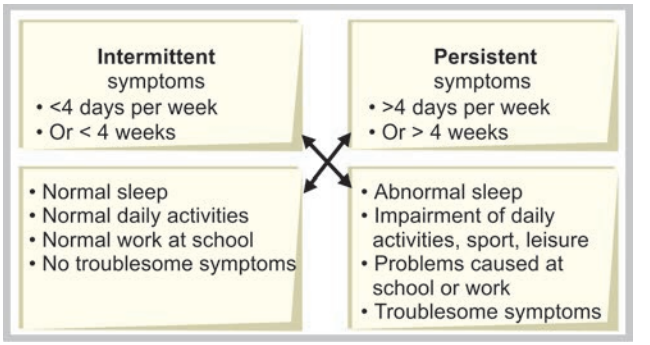

Fig. 1: Classification of Allergic Rhinitis ARIA (Allergic rhinitis and its impact on asthma)

include animal dander (cats, dogs, horses and other pets, wool and feathers), cosmetics, molds, foods and house dust. Allergies that become worse in wintertime, when the hot air furnaces are turned on, are due to house dust. Molds spoil bread, rot fruit and mildew clothing. These fungi also grow on dead leaves, grass, hay, straw, grains and houseplants and in the soil. Mold spores may be in the outside air all year except in mountain area when snow covers the ground. Damp places, such as basements and laundry rooms, are ideal places for the fungal growths. Molds can also be found in cheeses and fermented alcoholic beverages.

CLINICAL FEATURES

Symptoms

Diagnostic

The diagnostic symptoms include the following:

- Nasal pruritus: Itching may also involve eyes, palate, or pharynx.
- Paroxysms of sneezing: Some patients have a "tickling" sensation without sneezing while others are exhausted with sneezing.
- Rhinorrhea: This is a clear watery discharge, which may be extraordinarily profuse. A postnasal "drip" may occur, though less often than in infective rhinitis.

Bilateral Nasal Stuffiness

It is due to venous stasis of the inferior turbinates and mucosal edema. Obstruction from polyps tends to be constant. It occurs more commonly in vasomotor rhinitis than AR. Some patients complain of anosmia intermittently or continuously, even in the absence of obstruction.

Severity

Symptoms vary in severity from day to day, or even from hour to hour. The severity of symptoms is more in seasonal AR in comparison to perennial AR.

Age

Though there is no age and sex bar, AR usually affects school going children. A common sequence is eczema in infancy, then rhinitis followed by asthma. Nasal allergy is less common after 50 years of age.

Associated Symptoms

Allergic rhinitis may be associated with–

- Lower respiratory symptoms: Cough, wheezing, chest tightness, or dyspnea.
- Eye: Eye irritation.
- Skin: Pruritus; or eczematous dermatitis.

Allergens

The history must include a survey of allergen exposure associated with home, work, hobbies, and habits as well as medications (Box 1).

Examination

Allergy Salute

The external nose examination may show a transverse nasal crease across the middle of nasal dorsum (Dennie-Morgan line), which occurs due to repeated upward rubbing of nose simulating a salute.

Inferior Turbinate

Examination reveals edematous or inflamed submucosa but in severe conditions, mucosa may look pale, boggy or blue-tinged due to vascular engorgement and venous congestion. Vascular dilatation and stasis lead to a purplish discoloration of inferior turbinates. Anterior and posterior ends of inferior turbinates may become much enlarged. There occurs intercellular transudation of tissue fluid (edema) due to damage to capillary endothelium and loosening of cellular cement.

Thin Watery Discharge

It occurs from increased activity of the seromucinous glands.

Polyps

They are pedunculated portions of edematous mucosa and may be single o multiple. They usually develop in the ethmoidal sinuses and from the middle turbinate or antral lining.

Superadded Infection

It is not uncommon. The mucosa becomes reddish in color and the secretions become more viscid (jelly-like) and purulent.

Complications/Associated Conditions

Sinuses

Involvement of the sinuses occurs as AR progresses. The common manifestations are generalized thickening of the lining mucosa, development of polyps (single or multiple) in the sinuses and fluid effusion into the sinuses. The fluid is sterile and clear, but may become thick and gum-like in some chronic cases.

Eyes

Ocular features include edema of lids, congestion, cobblestone conjunctiva, and allergic shiners (dark circles under the eyes).

Ears

Serous otitis media due to Eustachian tube block manifests with retracted tympanic membrane and fluctuating and conductive deafness.

Pharynx

Hyperplasia of submucosal lymphoid tissue manifests as granular pharyngitis. Persistent AR in children can result in "adenoid faces" and orthodontic problems.

Larynx

Edema of vocal cords present with hoarseness of voice.

Box 1: Case history of allergic rhinitis patient

Patient's Name: Age: Sex:

Occupation: Date:

Address:

Nose complaints and duration: Sneezing/itching/stuffiness/watery discharge/other (specify)

Are the symptoms: Seasonal/perennial

Severity of symptoms: Mild/moderate/severe

Other associated complaints: Asthma/eczema/conjunctivitis/GIT/others

Events preceding complaints

Time and mode of onset

Event preceding the first attack: Change of occupation/infective illness/change of home/others (specify)

Likely causative allergens

Do the symptoms most often/more severe

1. All the year round: Yes/No Worst months:
2. In the open air/indoors/both
3. At home/at work/both; Is the home: In country/in town; old/new/damp
4. Weather: Dry/humid
5. During day/at night: In bed on waking/in bedroom/when dust is disturbed/on bed making
6. Is the bedroom heavily carpeted/curtained
7. Does the bed have: An old mattress or pillow/a feather pillow or eiderdown/a flock mattress, pillow or eiderdown/a kapok mattress pillow or eiderdown
8. Any contact with: Birds/horses/cats/dogs/rabbits/sheep/other pets or animals (specify)
9. Work or hobbies involving: Wood/wool/hay/straw/grain/dust (other than house dust) or other substances (specify)
10. Contact with particular: Plants/detergents/other substances (specify)
11. Certain foods: Milk/wheat/egg/other (specify)
12. Do the symptoms occur immediately or delayed
13. Any known drug allergy
14. Any evidence of fungus infections
15. Current medications
16. Any other relevant information

Past allergic history: Infantile eczema/rhinitis/bronchitis/asthma/eczema/urticaria/edema/migraine/food allergy/anaphylaxis

Family history of allergy: Mother/father/sibling/grand parents/others (specify)

Allergens for which testing is required (Suspected allergens from history)

Bronchial Tree

AR patients have fourfold risk of developing bronchial asthma.

INVESTIGATIONS

Complete Blood Count

The eosinophil count of the blood is raised, especially in the morning, and always in the presence of an extrinsic allergen.

Nasal Smear

Eosinophils may be found in great numbers in the nasal secretions, or on microscopic examination of the nasal mucosa or polyps.

Intranasal Provocation Test

It is a crude method, which is occasionally used these days. A drop of test solution can induce rhinorrhea, and sometimes lacrimation.

Elimination Tests

It can occasionally be helpful, especially in suspected food allergies.

Specific-IgE Antibody Tests

Allergy tests reveal an immune response to either one or more allergens. These positive test results must be correlated with the history before concluding the incriminating allergen(s). IgE antibodies are detected by in vivo (skin tests) or in vitro methods. The type of immune response should be in harmony with the nature of the disease. IgE antibodies cause allergic rhinitis and not delayed-type allergic contact dermatitis.

Skin Tests

Prick or intradermal skin tests generate a localized pruritic wheal (induration) and flare (erythema) which is maximal at 15–20 minutes. It is a very useful tool for the diagnosis of allergic

respiratory disease (rhinitis and asthma). While selecting the allergen extracts, which are available commercially for the common allergens (pollens, fungi, animal dander, and dust mites), patient's geographic area should be kept in mind. The negative and positive (histamine) controls are important for valid results and accurate interpretation.

- Skin testing detects the presence of IgE antibody in tissue, and shows biologic activity. It is more sensitive, specific, and rapid than in vitro radioallergosorbent Testing (RAST).
- Precautions
  - Antihistamines (H_1 antagonists, tricyclic antidepressants, phenothiazines) must be withdrawn prior to skin testing.
- Disadvantages
  - It is difficult to perform skin testing in the cases of extensive dermatitis.
  - There is very small risk of inducing a systemic reaction in prick type of testing.
- The selected intradermal tests to allergens negative by prick testing can be performed.
- Intradermal skin testing is more sensitive for detecting IgE-mediated anaphylaxis, and is often employed for testing low-molecular-weight drug penicillin and high-molecular-weight protein drugs (e.g., heterologous serum and insulin).

In Vitro Tests of IgE Antibody

Allergen-specific IgE antibodies are detected in serum by *radioallergosorbent test (RAST)* or *enzyme-linked immunosorbent assay (ELISA)*. Protein allergens are covalently linked to the immunosorbent (solid) phase. Patient's serum is allowed to incubate with the allergen-coated solid phase. The system is washed, and a second incubation is done employing labeled anti-IgE antibody (rabbit anti-human IgE). The allergen-specific IgE is quantified by comparing colorimetric changes or radioactive counts against a standard curve. Common allergens are available commercially for RAST or ELISA testing. A negative response of allergen-specific IgE is sufficient to rule out significant inhalant allergy.

- Advantages
  - The main advantages of in vitro testing are: (1) no risk of a systemic reaction, and (2) test results are not affected by antihistaminics.
  - The in vitro technique is useful for detecting IgE antibodies to certain occupational chemicals or potentially toxic allergens.
  - Serial measurement of total IgE is useful in allergic fungal sinusitis.
- Disadvantages
  - In vitro tests are less sensitive than skin tests because they estimate circulating IgE. Atopic allergy is caused by IgE antibodies bound to mast cells. It can give false-positive results in patients with high total serum IgE levels.
  - False-negative results are observed in cases treated with immunotherapy.
  - The test results are not immediately available.
  - The total serum IgE level is higher in atopic patients than the nonatopic individuals. The measurement of total IgE alone has little value. In some cases, high total IgE (> 100 IU/mL) is not associated with true allergy while some significant allergy cases may show low total IgE level.

TREATMENT

The main treatment modalities of allergy rhinitis are avoidance of the allergen, pharmaco therapy, immunotherapy and local surgical interventions.

Avoidance Therapy

Though it may not reduce the underlying immunologic sensitivity, avoidance of allergen exposure cures the clinical manifestations. So, if it is possible, then it should always be considered in addition to other modalities of treatment. Accurate diagnosis of the causative allergens is mandatory for the success of avoidance therapy (Box 2).

Pollens

Pollens release occurs in the early morning and they can travel significant distances. Closing windows and remaining in air-conditioned environments can significantly decrease pollen exposure.

Animal Dander

The patient may benefit from keeping away from their pet animals. Sometimes, however, it is essential to remove the animal from the home altogether. Vacuum cleaning with high-efficiency particulate arresting (HEPA) filter vacuum helps in removing dander.

House Dust Mites

The mattress and pillows should be encased in dust-proof material. The bedroom should not have carpets, upholstered furniture and stuffed animals. Control the relative humidity (< 50%). The room should be cleaned frequently. Electronic air purifiers have yet not proved their effectiveness. Acaricides (benzyl benzoate) to eliminate dust mites are still under investigation.

Mold Spores

Though outdoor mold spores are unavoidable during certain seasonal activities (gardening and fanning) they should be avoided. Indoor mold spores can be avoided by repairing leaks and cleaning mold buildup in sinks, shower curtains and pipes.

Drug Therapy

Pharmacotherapy can help not only in the reversal of the vascular and inflammatory responses in the target tissues but can also inhibit release of mediators from mast cells and the action of mediators on their target cells.

Antihistamines

Though not in all the cases, antihistamine therapy is helpful in allergic rhinitis and in urticaria. Antihistamines rarely alleviate symptoms of asthma but they are not contraindicated in treating associated rhinitis or pruritus. Histamine released from mast cells by reaction of allergen with IgE antibody can be effectively blocked by antihistamines, which are competitive inhibitors of the histamine receptors.

- First-generation: The use of first-generation antihistamines (chlorpheniramine, brompheniramine, diphenhydramine, clemastine, hydroxyzine) is limited due to their side effects,

Box 2: Instructions for the patients of allergic rhinitis

1. Patients, who live in hill areas, feel better when they come to plains and vice versa.
2. Patients allergic to pollens feel better when they live on some beach where they breathe only sea breezes or in an extremely dry climate where almost nothing can grow.
3. Use a pollen mask while mowing grass or cleaning house.
4. Keep windows and doors closed during the pollination seasons.
5. Rooms should be dusted with a moist duster. Vacuum cleaner is better than broom.
6. Change air filters monthly in the heating and air conditioning system. An air purifier has the potential to help patients. Install a good humidifier if you enjoy indoor heating in the winter. Be careful, mold can grow even in the humidifier.
7. Patients living in damp, dark and humid climatic region feel improvement when they shift to dry climatic region.
8. Get rid of home-plants and pet animals (such as cats, dogs, horses), which produce dander, wool and feathers.
9. Replace feather pillows, woolen blankets, and woolen clothing to cotton or synthetic materials.
10. Avoid the use of stuffed beddings, carpets, and sofas. Put the beddings in the sun, and get them cleaned regularly. Cover mattress and box springs with plastic sheets.
11. If necessary, use antihistamines and decongestants that suit you most.
12. Raise the head-end of bed by placing a brick or two under bedposts at head-side of bed.
13. Daily exercise and avoidance of smoking and pollution make significant difference.
14. Take a balanced diet supplemented with vitamins, especially C, and avoid too much carbohydrates.
15. False food allergy in children includes chocolate, pork products, canned foods, fish, fermented cheese, egg white, shell fish, strawberries, tomatoes, pineapple, papaya, alcohol, pulses, cereals, peanuts (lectins), cured and smoked products.
16. Other food items, which can cause problems, are fruits (lemons, orange and banana), dry fruits (ground nuts, cashew nuts and almonds), and spices (coriander, clove and black pepper).
17 Remain neither empty stomach for too long nor overload yourself with food. Maintain regularity about the meals and always take fresh things
18. Take light dinner and that too about 3 hours before going to bed. Excessive hot, cold, and spicy foods should be avoided if they cause problems.

such as sedation and dry mucous membranes. Occasionally, they can cause seizures and tachyarrhythmias.

- Second-generation: Non-sedating H_1 receptor-blocking second-generation antihistamines include loratidine, cetrizine (derived from hydroxyzine) and acrivastine. They do not readily cross blood-brain barrier, and are now the drugs of choice. Terfenadine and astemizole have been removed from the market in India and USA.
- Third-generation: Third-generation antihistamines, which are metabolites and congeners of existing drugs, include fexofenadine (active metabolite of terfenadine), desloratadine (derived from loratadine), levocetrizine (from cetrizine), and tecastemizole (from astemizole). Rupatadine is a new selective histamine H_1 receptor and platelet activating factor antagonist.
- Topical: To avoid the systemic side effects, azelastine and levocabastine, topical antihistamines are used intranasally.

Sympathomimetic Drugs (Nasal Decongestants)

Alpha-adrenergic agonists (vasoconstricting) are used orally and topically as nasal decongestants.

- Oral (pseudoephedrine and phenylephrine): The common side effects of oral decongestants are insomnia, tremor and tachycardia. Though freely available and widely used in India, phenylpropanolamine (PPA) has been removed from USA market due to the reports of hemorrhagic strokes.
- Topical (phenylephrine, naphazoline, oxymetazoline and xylometazoline): Daily use of topical nasal decongestants can lead to rhinitis medicamentosa due to the rapid development of rebound vasodilation.

Corticosteroids

These drugs play an extremely effective role in the treatment of AR because of their anti-inflammatory action and not immunosuppressive effects. Corticosteroid drugs are available in oral, intramuscular, intravenous and intranasal forms.

- Systemic: Short-term systemic burst therapy is indicated for treatment of severe AR, severe asthma, allergic contact dermatitis, acute exacerbations of hypersensitivity pneumonitis, and allergic bronchopulmonary aspergillosis. While using steroids, due attention must be paid to their side effects and toxicity, such as hypothalamic-pituitary-adrenal suppression.
- Topical (steroid nasal spray): Long-term topical corticosteroid (flunisolide, beclomethasone, mometasone, budesonide, fluticasone and triamcinolone) nasal therapy is an effective, comparatively safe and an essential aspect of management of the inflammatory phase of the AR.
  - Their potential side effects are epistaxis, nasal irritation, crusting and nasal septal perforation and potential risk of growth inhibition in children. It is desirable to break their use for 1–2 weeks every 2–3 months.
  - All commercially available steroids' nasal sprays are more or less similar in their efficacy, often requiring only a single daily dose after an initial period of therapy with four doses daily for 1 week.

Mast Cell Stabilizers (Cromolyn and Nedocromil)

These drugs help in prophylaxis and prevent the response to allergen by stabilizing the mast cell. They are more effective

when applied directly to the involved organ. Because of their short duration of action, they must be given three or four times a day.

- Sodium chromoglycate, 2% solution, is available as nasal drops or spray, or as an aerosol powder. Though cromolyn is not effective in all the patients, but due to its very few side effects and wide margins of safety, it can be safely used. A high-dose oral form of cromolyn sodium is also available, but poor oral absorption limits its effectiveness.
- Sodium nedocromil is available as bronchial inhaler for patients of allergic asthma.

Anticholinergic Agents (Nasal Topical Ipratropium Bromide)

It is effective in cases of rhinitis. The anticholinergic effect of ipratropium inhibits mucous membrane glandular secretion of nose. First-generation antihistamines also have systemic anticholinergic activity. Ipratropium is preferred as adjunctive treatment of allergic as well as nonallergic rhinitis because it does not cause rhinitis medicamentosa. Though ipratropium does not alleviate sneezing, pruritus, or nasal congestion, it is useful for treatment of postnasal drip and rhinorrhea.

Leukotriene Modifiers

They act by inhibiting formation of leukotrienes or blocking their effect. Zileuton is 5-lipoxygenase inhibitor. Cysteinyl-leukotriene inhibitors include zafirlukast and montelukast. The montelukast got FDA approval in 2003 for the treatment of seasonal AR. Montelukast specifically targets the cysteinyl leukotriene receptor, and is a viable alternative for the treatment of seasonal rhinitis. Combination therapy (montelukast 10 mg and levocetrizine 5 mg) is reported to be a more effective strategy than monotherapy, in cases of moderate-to-severe symptoms of persistent AR.

Anti-IgE Antibody Therapy

The omalizumab, a recombinant monoclonal antibody to IgE, got FDA approval in 2003 for treatment of moderate to severe asthma. Studies have shown its efficacy in allergic rhinitis but the use is limited due to its high cost and parenteral administration.

Macrolides

Macrolides have shown direct anti-Th-2 inflammatory effect that is independent of their antimicrobial effect. The clarithromycin was found equivalent to prednisolone in reducing the concentration of proinflammatory cytokines interleukin 5 and interleukin 8.

New Therapies

The new medicines, which are under development, are tryptase inhibitors, cytokine modulators, chemokine inhibitors, cell-adhesion blockers and costimulation inhibitors.

Immunotherapy

Treatment of AR by the repeated long-term injection of allergen has been shown to be an effective method for reducing or eliminating clinical manifestations.

Indications

The best candidates for immunotherapy are the cases of severe allergic rhinitis who respond poorly to drug and avoidance therapy, or when allergens are unavoidable. Immunotherapy has shown good results in AR and allergic conjunctivitis, while the poor results in asthma have been attributed to the multifactorial nature of the disease. Food or drug hypersensitivity is best treated by avoidance.

Immunologic Effects

Circulating levels of IgE antibodies increase slightly during the first few months but later on, decrease to substantially lower levels than prior to injections. Seasonal rises in the level of IgE antibodies to specific pollens are blunted or eliminated. Immunotherapy produces IgG blocking antibody. Studies have reflected changes in regulatory T cells favoring suppression of IgE antibody production. Higher thresholds for release of inflammatory mediators and decreases in late phase allergic reactions have been postulated to the reduction in biologic sensitivity of nose.

Clinical Effects

Majority of the immunotherapy patients better tolerate natural pollen exposure during the specific pollen seasons. While on immunotherapy, there is significant decrease in symptoms and medication usage. Some patients become completely asymptomatic. The beneficial effect may persist even after the cessation of treatment.

Procedure

A sterile aqueous solution of incriminating allergen(s) is administered by subcutaneous injection in gradually increasing doses once or twice a week until a maintenance dose is reached. Then the interval is increased to every 4 weeks. The maintenance dose is usually 1–10,000 times the starting dose. Gradually increasing doses minimize the risk of systemic allergic reactions during immunotherapy. Three to five years is the usual course of therapy.

Adverse Effects

Reactions to treatment may be local or systemic. Localized skin reactions occur at injection sites that can be immediate and late phase. Though these local reactions are not harmful, the dose must be adjusted to avoid them. Anaphylaxis is a potential problem and must be prevented by careful monitoring of dosage. The patient must remain at the hospital for 30 minutes after each injection so that if needed, systemic reaction can be managed promptly and efficiently.

Surgery

Surgery can be an adjuvant management in some refractory cases of AR when conservative treatment fails. It is not indicated in children, though a simple antral wash can be performed if there is associated sinusitis. The adjuvant surgical procedures employed for the management of AR include the following:

- *Antral wash out*: It is done when AR is complicated by chronic maxillary sinusitis. Sinus openings are obstructed due to edema around the osteomeatal complex (OMC).

Edema may be pronounced in presence of deviated nasal septum (DNS), hypertrophied turbinate, and adenoids in children. Sinusitis in AR cases usually presents with exaggeration of nasal obstruction, change in the character of rhinorrhea (watery to mucoid or mucopurulent) and additional symptoms, such as headache, hyposmia and epistaxis. X-ray shows ground glass opacity of affected sinuses. The antral wash may have to be combined with procedures, such as septoplasty, turbinectomy and adenoidectomy. Currently, functional endoscopic sinus surgery (FESS) has been replacing antral wash.

- *Endoscopic sinus urgery*: If polyps are obstructive, they are removed. The sinus operations are performed when gross infection is present.
- *Cauterization*: The cauterization produces fibrosis by coagulation (chemical/thermal), which subsequently destroys the nerve endings, blood vessels and mucous glands. Symptoms of sneezing, nasal obstruction and rhinorrhea are minimized.
  - Chemocautery by silver nitrate.
  - Electrocautery by red hot galvanocautery probes.
  - Submucous diathermy of inferior turbinate: It destroys the erectile tissue and mucous glands. After introducing a long lumber needle (LP) within the submucous space of inferior turbinate, the diathermy is done intermittently for 2–3 times that results in the shrinkage of inferior turbinates, and relieves nasal obstruction. Submucous diathermy, unlike chemocautery or electrocautery, does not destroy mucosal surface epithelium which performs mucociliary clearance of the mucus.
- *Septoplasty:* The correction of DNS not only relieves the nasal obstruction but also facilitates drainage and intranasal medications. The associated hypertrophied turbinates need turbinectomy or submucous diathermy along with septoplasty.
- *Inferior turbinectomy*: Total inferior turbinectomy, an effective operation for establishing the nasal airway, has shown some adverse effects in long-term follow up, such as oozing, hyposmia and atrophic rhinitis. Submucosal removal preserves overlying mucosa and prevents adverse effects of total inferior turbinectomy.
- *Laser and cryosurgery*: Laser and cryosurgery are found to be useful surgical treatments for AR and vasomotor rhinitis (VMR).
- *Vidian neurectomy*: Vidian neurectomy, which is now no more popular, destroys the preganglionic parasympathetic secretomotor nerve fibers that supply to the nasal mucosa. It used to be done for intractable VMR, AR and recurrent polyps.
- *Tonsillectomy and adenoidectomy*: These operations are done in children only if there are obvious evidences of recurrent or chronic infection or significant hypertrophy.

NONALLERGIC RHINITIS (VASOMOTOR RHINITIS)

Approximately, 10% of the population suffers from chronic or recurrent nasal symptoms. The prevalence of nonallergic rhinitis (NAR) in otolaryngology practice is on, and average about 40%. In many cases, it is difficult to differentiate NAR from AR. AR is more common than NAR. Quite commonly, patients have mixed features of both NAR and AR. Both conditions have similar presentations, manifestations, and treatment. Nonallergic rhinitis presents with all the symptoms of AR (such as rhinorrhea, congestion and sneezing) but these patients have negative allergic history, skin testing and nasal cytology. Nasal itching and paroxysmal sneezing are usually lesser with NAR in comparison to AR. Other common terms used for NAR are vasomotor rhinitis and perennial rhinitis.

PATHOPHYSIOLOGY

Nasal mucosa has rich blood supply, which is similar to the erectile tissues that have venous sinusoids or "lakes" surrounded by smooth muscle fibers. These smooth muscle fibers act as sphincters and control the filling or emptying of sinusoids. Sympathetic stimulation causes vasoconstriction and shrinkage of mucosa (decongestion of nose), while parasympathetic stimulation causes not only excessive secretion from the nasal glands (rhinorrhea) but also vasodilation and engorgement (congestion of nose).

Autonomic nervous system, which supplies the nasal mucosa, is under the control of hypothalamus. Therefore, emotions play a significant role in NAR. Autonomic nervous system is not stable in cases of NAR. Nasal mucosa becomes hyperreactive and responds unduly to several non-specific stimuli, such as change in temperature, humidity, blasts of air, small amounts of dust or smoke.

CLASSIFICATION

In cases of NAR, nasal mucosa responds unduly to various stimuli and produces symptoms of NAR, such as rhinorrhea, congestion and sneezing. Some of these conditions have been given specific terms, which are categorized under the NAR (vasomotor rhinitis). The etiological classification of NAR is given in Table 2.

Nonallergic Rhinitis with Eosinophilia

Patients of nonallergic rhinitis with eosinophilia (NARE) have perennial symptoms with episodes of watery discharge, itching, sneezing and epiphora. They have negative or irrelevant reports by skin or in vitro allergy testing. The specific allergy triggers are absent, but aggravating factors usually include weather changes and exposure to chemical irritants. Nasal smear shows marked eosinophilia.

Drug-Induced Rhinitis

- Several anti-hypertensive drugs, such as beta-blockers, alpha-blockers, angiotensin-converting enzyme (ACE) inhibitors and vasodilators can result in nasal congestion and stuffiness.
- The anticholinesterase drug (neostigmine) used in myasthenia gravis have acetylcholine like action, which may affect the nose mucosa.
- Aspirin and NSAID are well known to cause sinusitis and asthma.
- Contraceptive pills (estrogen) can cause nasal obstruction.

TABLE 2 Etiological classification of nonallergic rhinitis

Medicines: Local and systemic	Topical decongestants (rhinitis medicamentosa), antihypertensive, beta-blockers, aspirin, nonsteroidal anti-inflammatory drug (NSAID), oral contraceptive, psychotropic
Hormonal	Puberty, pregnancy, menstruation, postmenopausal, hypothyroidism, acromegaly
Irritant and corrosives	Irritant rhinitis due to dusts, fumes, smoke, gases, chemicals, pollution, acids, organophosphates
Smell	Perfumes, cosmetics, fragrances, deodorizers, cleaning agents
Taste	Gustatory rhinitis due to hot and spicy foods
Occupational	Occupational NAR due to pyrethrum (insecticide), acid anhydrides (adhesive industry), toluene (body spray), latex, grains, paint industries
Environmental	Temperature, humidity, barometric changes, weather, altitude, aviation
Local: Structural, infections, tumors and trauma	DNS, septal perforation, nasal valve collapse, choanal atresia, turbinate and adenoid hypertrophy, cleft palate, rhinosinusitis, rhinoscleroma, polyps, inverting papilloma, malignancies.
Substance abuse	Cocaine, alcohol and nicotine
Emotional	Anxiety, tension, hostility, humiliation, resentment and grief
Exercise	Sedentary lifestyle and recumbency
Decreased nasal airflow	Laryngectomy, tracheostomy
Atrophic changes	Surgery, aging, infections, atrophic rhinitis
Systemic diseases	Autoimmune systemic lupus erythematosus, Sjogren's, vasculitides (Wegener's granulomatosis, sarcoid), Ciliary dyskinesia (Young's and Kartagener's syndromes), cystic fibrosis, superior vena cava and Horner's syndrome
Idiopathic	Vasomotor rhinitis and nonallergic rhinitis with eosinophilia

Rhinitis Medicamentosa

The long-term use of cocaine and topical nasal decongestants (cause rebound congestion) leads to rhinitis medicamentosa.

- *Treatment:* It consists of withdrawal of topical decongestant, oral and/or topical steroid therapy, and surgical reduction of hypertrophied turbinates are usually effective.

Honeymoon Rhinitis

Too much sexual activity can lead to nasal congestion and rhinorrhea.

Emotional Rhinitis

Nose mucosa may react to several emotional conditions, such as anxiety, tension, hostility, humiliation, resentment and grief.

- *Treatment* includes counseling for emotional adjustment. Imipramine, an antidepressant which has anticholinergic effect, has been found useful.

Hormone-Related Rhinitis

- Hypothyroidism*:* In hypothyroidism, there occurs hypoactivity of the sympathetic system with predominance of parasympathetic activity, which causes nasal congestion and discharge. These patients need supplement thyroid hormone.
- Menstruation*:* Fluctuating levels of hormones during menstruation can cause nasal symptoms.
- Puberty*:* The changing blood hormone concentrations during puberty can result in rhinitis.
- Pregnancy*:* Pregnant women may develop edema of the nasal mucosa due to hormonal changes. The common nasal symptoms are watery discharge and stuffiness. The severity of these nasal symptoms parallels blood estrogen levels. Nasal symptoms decrease towards term because the blood is shunted away from nose towards the growing uterus. The persistent congestion may lead to secondary infection and rhinosinusitis.
  - Treatment: Usually, these women respond to limited local measures, such as saline drops and topical steroids. Limited surgery (cryosurgery) to hypertrophied turbinate is done in refractory cases.

Gustatory Rhinitis

The ingestion of hot and spicy food leads to mucoid or watery nasal discharge. The onset is immediate and lasts for as long as the food is ingested. Stimulation of afferent sensory nerves activates the parasympathetic nerves, which lead to nasal gland secretion, sweating and epiphora.

Non Airflow Rhinitis

In laryngectomy and tracheostomy cases, there is no nasal airflow. The turbinates are swollen due to loss of vasomotor control. Similar changes also occur in cases of choanal atresia or adenoidal hyperplasia. Stagnation of discharge in the nasal cavity can lead to infection.

Idiopathic or Vasomotor Rhinitis (VMR)

When no cause of NAR is found, the condition is termed VMR which is usually said to be due to imbalance of autonomic nerve fibers that supply to nasal mucosa. It can be either increased parasympathetic, or decreased sympathetic activity.

CLINICAL FEATURES

It is important to note the pattern and timing of symptoms, exacerbating and relieving factors, and environmental triggers. Exclusion of systemic diseases and hormonal imbalances must be done.

Symptoms

Paroxysmal Sneezing

Especially in morning while getting out of the bed.

Excessive Rhinorrhea

In some cases, profuse and watery discharge is the presenting and only symptom. The nose may start dripping when the patient bends forward. This needs to be differentiated from CSF rhinorrhea.

Nasal Obstruction

The bilateral nasal stuffiness usually alternates from one side to other. This is generally more marked at night, when the dependent side of nose is often blocked.

Postnasal Drip

Though uncommons it can be the only symptom.

Examination

Nose

The mucosa is usually boggy and edematous with clear mucoid secretions. The turbinates are usually congested and hypertrophic.

Pharynx

Mucosal injection and lymphoid hyperplasia involving tonsils, adenoids, and base of tongue may be seen.

Complications

Nonallergic rhinitis cases may subsequently develop polyps, turbinate hypertrophy and sinusitis.

INVESTIGATIONS

- Absolute eosinophil count
- Nasal smear
- Skin and in vitro allergy tests to rule out allergic rhinitis
- Acoustic rhinometry for measuring nasal patency
- Smell testing
- CT scan in cases of sinus disease
- MRI in cases of mass lesions

TREATMENT

Medical

- *Avoidance of inciting factors* such as sudden change in temperature, humidity, blasts of air or dust.
- *Antihistaminics and oral decongestants:* They help in relieving nasal obstruction, sneezing and rhinorrhea.
- *Topical steroids* (beclomethasone dipropionate, budesonide or fluticasone) are useful in controlling symptoms.
- *Systemic steroids* are reserved for complicated and refractory cases.
- *Psychological counseling* for emotional adjustment.
- *Exercise* is an important adjunct to treatment.
- *Tranquilizers* help in some patients.

Surgical

It is more or less similar to allergic rhinitis.

- Reduction of hypertrophied turbinates.
- Correction of deviated nasal septum.
- Removal of polyps.
- Sectioning of the parasympathetic secretomotor fibers to nose (vidian neurectomy) for controlling refractory excessive rhinorrhea.

Clinical Highlights

1. **Angioedema:** Deficiency of C1 esterase inhibitor, an inherited condition causes angioedema. The increased production of C1 esterase leads to anaphylatoxins that causes capillary permeability and edema.
2. **Cromolyn sodium:** It provides protection against nasal allergy when used just before exposure to allergen.
3. **Surgical treatment of allergic rhinitis:** In some of the refractory and selective cases the surgical procedures include radiofrequency ablation, laser ablation or surgical removal of the inferior turbinate.
4. **Vidian neurectomy:** Vidian nerve section has been used in cases of vasomotor rhinitis with excessive watery rhinorrhea.
5. **Nonallergic rhinitis with eosinophilia (NARE):** Nasal smear shows increased number of eosinophils in nonallergic eosinophilic rhinitis.
6. Rhinitis medicamentosa: The topical nasal decongestant that can cause rhinitis medicamentosa is xylometazoline.

FURTHER READING

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31 Nasal Septum

Arise, awake, sleep no more; within each of you, there is the power to remove all wants and all miseries. Think that infinite power, infinite knowledge, and indomitable energy lie within you. Believe this, and that power will be manifested.

—Swami Vivekananda

Points of Focus

- **FRACTURE OF NASAL SEPTUM**
  - ✓ Features
  - ✓ Types: Jarjavay and Chevallet Fractures
  - ✓ Treatment
  - ✓ Complications
- **DEVIATED NASAL SEPTUM (DNS)**
  - ✓ Types: Caudal, C-shaped, S-shaped, Spurs, Thickening
  - ✓ Clinical Features: Nasal Obstruction, Headache, External Deformity, Epistaxis, Hyposmia/Anosmia
  - ✓ Complications: Mouth Breathing, Sleep Apnea, Recurrent or Chronic Rhinosinusitis, Middle Ear Infection, Atrophic Rhinitis, Asthma
  - ✓ Treatment: Submucous Resection, Septoplasty
- **SEPTAL HEMATOMA**
  - ✓ Clinical Features: Bilateral Nasal Obstruction and Septal Swelling, Frontal Headache
  - ✓ Treatment: Aspiration, Incision and Drainage
  - ✓ Complications: Thickened Septum, Septal Abscess
- **SEPTAL ABSCESS**
- **PERFORATION OF NASAL SEPTUM**
- **HYPERTROPHIED TURBINATES**
  - ✓ Compensatory Hypertrophic Rhinitis
- **NASAL SYNECHIA**
- **CHOANAL ATRESIA**
- **CLINICAL HIGHLIGHTS**

FRACTURE OF NASAL SEPTUM

Facial trauma can result in fracture of nasal septum. It may be inflicted on the nose from the front, side or below. The nasal septum can get buckled or fractured vertically or horizontally (Figs 1A and B). For nasal bone fracture, see chapter Maxillofacial Trauma.

Features

- In a smashed nose, septum is crushed to pieces, which may overlap each other or project into the nasal cavity.
- A crushing blow from the front may cause buckling, twisting, fractures and duplication of nasal septum with telescoping of its fragments.
- Trauma to the lower nose can cause fracture of the septal cartilage or its dislocation from the vomerine groove without associated fractures of nasal bones.
- Septal injuries usually lead to profuse epistaxis.
- If the septal mucosa is intact, septal trauma can cause septal hematoma, which should be drained early to prevent absorption of the septal cartilage and saddle nose deformity.

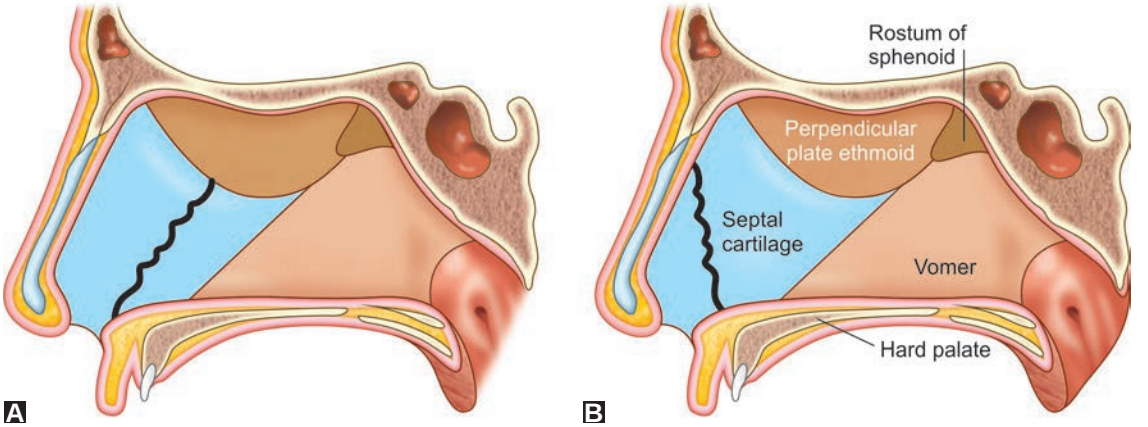
Types

"Jarjavay" Fracture (Fig.1A)

This horizontal fracture starts just above the anterior nasal spine and runs backwards parallel and above the junction of septal cartilage with the vomer. It results from blows from the front.

"Chevallet" Fracture (Fig. 1B)

This vertical fracture starts from anterior nasal spine and runs upwards towards the junction of bony and cartilaginous dorsum of nose. It results from blows from below.



Figs 1A and B: Types of nasal septum fractures. (A) Jarjavay horizontal fracture just above the junction of septal cartilage with vomer; (B) Chevallet vertical fracture of septal cartilage extending from anterior nasal spine to the junction of bony and cartilaginous dorsum of nose

Treatment

They need early treatment.

- Hematoma is drained at the earliest.
- Dislocated or fractured septum is repositioned and supported between mucoperichondrial flaps. The mattress sutures are taken and nasal packing is done.
- The associated fractures of nasal pyramid are treated concomitantly.

Complications

- Deviation of the cartilaginous nose.
- Asymmetry of nasal tip, columella and the nostril.
- Septal hematoma

DEVIATED NASAL SEPTUM

Deviated nasal septum (DNS) is quite a common condition, which usually presents with nasal obstruction. There is no age and sex bar but usually males are affected more than females. Deviated nasal septum is more common in Caucasians in comparison to the Negroes. In some hereditary cases, it runs in families.

Etiological Factors

Accidental Trauma

A lateral blow displaces the septal cartilage from the vomerine groove and maxillary crest. A crushing frontal blow results in buckling, fractures and telescoping of its fragments. Childhood injuries are often forgotten.

Natal Trauma

Trauma inflicted to the fetus at difficult delivery (forceps) during its passage through the narrow birth canal should be immediately attended. They can lead to septal deviation in later life.

Antenatal

Abnormal intrauterine postures can compress nose and upper jaw.

Developmental

Nasal septum develops from the tectoseptal process that descends and meets the two halves of the developing palate, which usually descends and widens further to accommodate the teeth. Buckling of the nasal septum can result from unequal growth between the palate and the skull base. If septum starts growing at a more rapid rate than face, it becomes buckled.

Mouth Breathers

Buckling of nasal septum in children can result from highly arched palate, which occurs in mouth breathers such as in adenoid hypertrophy.

Cleft Lip and Palate

These conditions may be associated with dental abnormalities.

Mass in Nose

Tumors and polyps of nose can result in deviation of the nasal septum to opposite side.

Types

Deviation may be bony/cartilaginous, anterior/posterior, horizontal/vertical and superior/inferior (Fig. 2).

Anterior Dislocation (Caudal Septal Deviation)

Septal cartilage dislocates from anterior nasal spine into one of the nasal cavity (Figs 3A and B). It is better seen at the base of nose (Fig. 4).

C-Shaped Deformity

The nasal septum becomes like an arc. On the concave side, nasal cavity is wider and can lead to compensatory hypertrophy of turbinates or atrophic changes. The septal blood vessels get stretched on the convex surface of DNS.

S-Shaped Deformity

It can be either in vertical or anteroposterior plane and may cause bilateral nasal obstruction.

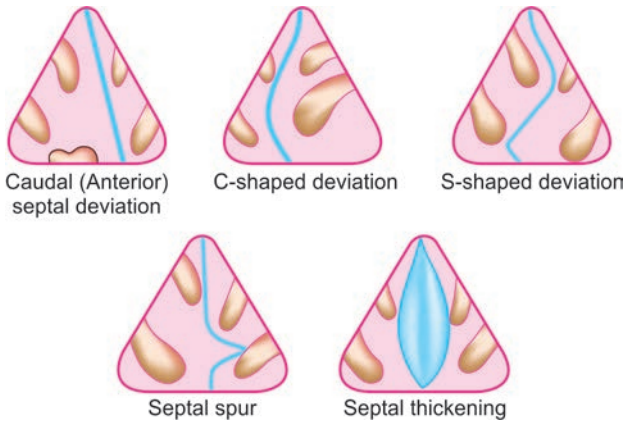
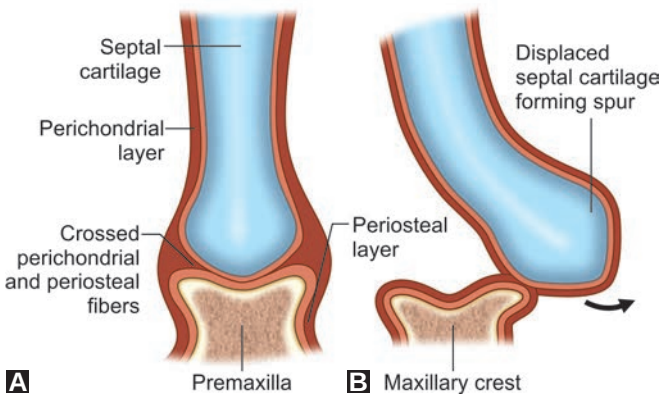


Fig. 2: Types of deviated nasal septum



Figs 3A and B: Inferior aspect of septum and maxillary crest. (A) Normal; (B) Displaced septal cartilage shifted from the crest and forming spur

Fig. 4: Caudal septal deviation better seen at base of nose. Septal cartilage dislocates from anterior nasal spine into left nasal cavity

Spurs

Spur is a shelf-like projection of cartilage, bone or both, which may press on the lateral wall of nose. It is usually found at the junction of vomer bone and cartilage. A spur can give rise to headache and predispose the patient to repeated epistaxis.

Thickening

It occurs due to organization of hematoma, overriding of fractured or dislocated septal fragments.

Clinical Features

Surprisingly, many patients who have gross DNS do not have any complaint whereas some patients with minor DNS complain the most.

Nasal Obstruction

Nasal obstruction, which may be unilateral or bilateral, is the most common symptom of DNS. High septal deviation causes more nasal obstruction than lower ones because respiratory currents pass through upper part of nasal chamber.

Headache

A spur, pressing on the lateral wall of nose, can present with neuralgic headache. Sinusitis and negative pressure in sinuses can also present with headache. It can result from improper working of sinus ostium due to DNS.

External Deformity

Septal deflections may be associated with external nasal deformities (cartilaginous, bony or both) or deformities of the nasal tip or columella (See chapter Diseases of External Nose and Epistaxis).

Epistaxis

Mucosa over the convex part of deviated septum gets exposed to the dry air currents, which can lead to the formation of crusts. The removal of these crusts can cause bleeding.

Hyposmia/Anosmia

Severe DNS may not allow the inspired air to reach the olfactory region.

Complications

- *Mouth breathing*: It can lead to recurrent infection of pharynx, larynx and lungs.
- *Sleep apnea*
- *Recurrent or chronic rhinosinusitis*: Obstruction of sinus ostia due to marked DNS, can result into poor ventilation and drainage of the sinuses.
- *Middle ear infection*: Deviated nasal septum can affect the Eustachian tube and predisposes patients to middle ear infection.
- *Atrophic rhinitis*: Atrophic changes may occur on the concave side.
- *Asthma*: Deviated nasal septum may act as a trigger for bronchospasm.

Treatment

The treatment is surgical. It is required, only when deviated septum produces nasal obstruction or other clinical features. For details of the septal surgery, see chapter Operations of Nose and Paranasal Sinuses.

Submucous Resection

The mucoperichondrial and mucoperiosteal flaps on either side of the septum are elevated. The deflected parts of the bony and cartilaginous septum are removed.

Septoplasty

It is a conservative septal surgery. Only the most deviated part of nasal septum is removed. Rest of the septal framework is corrected and repositioned. Mucoperichondrial and mucoperiosteal flap is generally raised only on one side of the septum. Septoplasty is replacing submucous resection (SMR) operation. These operations are dealt in chapter Operations of Nose and Paranasal Sinuses.

- Septal surgery in children: It can interfere with the growth of nasal skeleton and is usually avoided in children. However, if a child has severe septal deviation and causing marked nasal obstruction a very conservative surgical correction (septoplasty) can be considered.

SEPTAL HEMATOMA

It refers to collection of blood under the mucoperichondrium or mucoperiosteum of the nasal septum.

Etiology

- Nasal trauma
- Septal surgery
- Spontaneous in bleeding disorders

Clinical Features

- Bilateral nasal obstruction is usually presenting symptom.
- Frontal headache and a sense of pressure over the nasal bridge.
- Bilateral septal swelling, which is soft, fluctuant, smooth and round.

Treatment

- *Aspiration*: Small hematoma can be evacuated with a wide bore needle.
- *Incision and drainage*: Larger hematoma needs incision and drainage. It is done through a small anteroposterior incision that is parallel to the nasal floor. A small piece of mucosa is excised, which facilitates drainage.
  - Nasal cavities are packed to prevent reaccumulation of blood.
- *Systemic antibiotics* prevent septal abscess.

Complications

- *Thickened septum*: Organization of hematoma into fibrous tissue.
- *Septal abscess*: It leads to necrosis of cartilage and depression of nasal dorsum.

SEPTAL ABSCESS

Etiology

- Secondary infection of septal hematoma.
- Furuncle: Vestibule of the nose or upper lip.
- Acute infection such as typhoid or measles.
- Diabetic patients are more prone.

Clinical Features

- Bilateral nasal obstruction.
- Pain and tenderness over the bridge of nose.
- Fever with chills.
- Frontal headache.
- Red and swollen skin over the nose.
- Smooth bilateral swelling of the nasal septum with fluctuation and congestion of septal mucosa.
- Enlarged and tender submandibular lymph nodes.

Treatment

Incision and Drainage

It is performed at the earliest. Pus should be sent for culture and sensitivity examination. The preferred site of incision is dependent part of the abscess. Small piece of septal mucosa is excised, which facilitates drainage. Necrotic pieces of septal cartilage must be removed. In some cases, incision needs to be reopened daily for 2–3 days for draining pus and removal of necrotic pieces of septal cartilage.

Systemic Antibiotics

They may need to be changed as per the report of culture and sensitivity and continued for a period of 10 days.

Complications

- Saddle nose deformity: That necrosis of septal cartilage can result in depression of the nasal dorsum in the supratip area. It needs augmentation rhinoplasty, which is performed after 2–3 months.
- Septal perforation due to necrosis of septal flaps.
- Meningitis.
- Cavernous sinus thrombosis.

PERFORATION OF NASAL SEPTUM

Etiology

- *Trauma*: Habitual nose picking and perforation of septum for putting ornaments
- *Septal surgery*: SMR and cauterization of septum (chemicals or galvanocautery)
- *Septal abscess*: Infected hematoma
- *Nasal myiasis*: Maggots nose
- *Foreign body*: Rhinolith or neglected foreign body

Chronic granulomatous conditions: Lupus, tuberculosis and leprosy cause perforation in the cartilaginous part of septum. Syphilis leads to perforation in the bony part

- *Wegener's granuloma*: This midline destructive lesion may cause total septal destruction
- *Occupational*: Chrome platters and painters
- *Drug induced*: Snuff and cocaine addicts and long-term use of topical corticosteroids
- *Idiopathic*

Clinical Features

- Whistling sound during inspiration or expiration in small anterior perforations.

- Nasal obstruction due to crust formation.
- Epistaxis occurs when crusts are removed.

Diagnosis

- Biopsy in cases of granulomatous lesions will reveal the exact nature of disease.
- Hemogram and serological tests help in diagnosing systemic diseases.

Treatment

Medical

Treatment of the cause of perforation.

Small Perforations

Surgically closed with plastic flaps.

Larger perforations

Difficult to close surgically. Alkaline nasal douches and application of a bland ointment keep the nose crust-free. A thin silastic button, which can be worn by the patients, relieves the symptoms.

Polychondritis: Seventy percent to eighty percent patients have involvement of the nasal septum.

HYPERTROPHIED TURBINATES

In hypertrophic rhinosinusitis, there occurs thickening of mucosa, submucosa, seromucinous glands, periosteum and bone.

Clinical Features

- The nasal obstruction is the predominant symptom.
- Turbinal mucosa is thick and does not pit on pressure and no shrinkage with topical decongestant due to presence of underlying fibrosis.
- The inferior turbinate is hypertrophied (mulberry appearance) in its entirety including anterior end, posterior end, and inferior border.

Treatment

It includes surgical reduction in size of turbinates, which can be achieved with various methods such as:

- Linear cauterization
- Submucosal diathermy
- Cryosurgery of turbinates
- Submucous resection of turbinate bone
- Lasers have also been employed
- Total or partial turbinectomy (partially removed at its anterior end, inferior border or posterior end)
  - Turbinectomy scissors: These long and blunt tipped scissors are bent at an obtuse angle. They are used for turbinectomy and cutting the nasal septal cartilage
- Hypertrophied middle turbinate can be removed partially or totally.
- Caution: Excessive removal of turbinates can result in persistent crusting and atrophic changes.

Compensatory Hypertrophic Rhinitis

In marked deviation of septum, the roomier side of the nose shows hypertrophy of inferior and middle turbinates to reduce the wide space (to overcome the ill effects of drying and crusting). Hypertrophic changes may or may not reverse with the correction of nasal septum and may require reduction of turbinates.

NASAL SYNECHIA

Etiology

Injury to opposing surfaces of nasal mucosa between the nasal septum and turbinates can result in adhesion formation by scar tissue. The common causes include intranasal surgical procedures such as:

- Septal surgery.
- Polypectomy.
- Removal of foreign bodies.
- Reduction of nasal fractures.
- Endoscopic sinus surgery.
- Anterior nasal packing.
- Infective ulcerative lesions.

Clinical Features

- Nasal synechia can impede drainage from the sinuses and cause recurrent sinusitis.
- Patient presents with nasal obstruction, headache and nasal discharge.
- Minor synechia may remain asymptomatic.

Treatment

It consists of cutting of synechia and prevention of their recurrence by placing a thin silastic, plastic, and cellophane sheets or gelatin sponge.

CHOANAL ATRESIA

It refers to closure of posterior nasal choana. This congenital anomaly is caused by persistence of primitive bucconasal membrane.

Clinical Features

Choanal atresia may be classified as unilateral and bilateral; complete and incomplete; bony (90%) and membranous (10%).

Unilateral

It is more common and may remain undiagnosed until adult life.

- Presence of unilateral mucoid discharge, which is devoid of air bubbles.

Bilateral

Newborns are natural nose breather and bilateral choanal atresia causes respiratory obstruction, which if not managed immediately, can prove fatal.

- Asphyxia endangers newborn's life.
- Cyclic asphyxia: Newborn breathes intermittently through mouth.
- Difficulty in suckling: Bilateral nose block makes suckling difficult.

Diagnosis

- Catheter cannot be passed from nose to pharynx.
- Drops of methylene blue into the nose do not come into the pharynx.
- X-ray lateral view after putting radio-opaque dye into the nose will show the atresia.
- CT scan shows extent and nature of atresia.

Treatment

Tracheostomy or Endotracheal Intubation

Emergency management for airway is required in bilateral choanal atresia.

McGovern's Technique

A feeding nipple with a large hole can obviate the need for tracheostomy. It provides a good oral airway.

Correction of Atresia (Recanalization)

It is done at the age of one and a half years. The different approaches include transnasal, nasal endoscopic and transpalatal.

Clinical Highlights

1. **Deviated nasal septum (DNS)**: Patients can present with recurrent sinusitis compensatory hypertrophy of turbinate, nasal stuffiness and epistaxis.
2. **Complications of septal abscess**: They include depression of nasal bridge, meningitis and cavernous sinus thrombophlebitis.
3. **Anterior ethmoidal neuralgia**: Concha bullosa of middle turbinate pressing on the nasal septum can result into anterior ethmoidal neuralgia.
4. **Nasal synechia**: To prevent synechia formation after nasal surgery, packing of ribbon gauze with liquid paraffin is helpful.
5. **Choanal atresia**: It is usually unilateral and more common in females (2:1). It is usually seen on the right side. The bony atresia is more common than membranous (9:1).

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8. Give the anatomical description of inner valve of the nasal vestibule. How do you treat its collapse? Do you advise any precautions during septorhinoplasty to avoid its occurrence? Reader's Forum-45. Indian J Otolaryngol Head Neck Surg. 2007;59:401-2.
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10. In a child who has marked anterior dislocation with DNS, how early will you do septoplasty and if so, any special technique or precautions? What nasal packing material will you use? If a child has huge adenoids, would you stage the surgery or do it simultaneously. Reader's Forum-39. Indian J Otolaryngol Head Neck Surg. 2006;58:213-4.

32 Maxillofacial Trauma

Develop the power of concentration and detachment, and then with a perfect instrument you can collect facts at will. Pay as much attention to the means of work as to its end.

—**Swami Vivekananda**

Points of Focus

- **INTRODUCTION**
- **ETIOLOGY**
- **CLASSIFICATION**
- **GENERAL PRINCIPLES**
  - ✓ Airway
  - ✓ Breathing
  - ✓ Circulation
- **EVALUATION**
  - ✓ History and Examination
  - ✓ Radiology
  - ✓ Laboratory
- **SOFT TISSUE INJURIES**
  - ✓ Facial Lacerations
  - ✓ Parotid Gland
  - ✓ Facial Nerve
- **FRONTAL SINUS**
  - ✓ Anterior Wall Fractures
  - ✓ Posterior Wall Fractures
  - ✓ Injury to Frontonasal Duct
- **SUPRAORBITAL RIDGE**
- **FRONTAL BONE**
- **NASAL BONES AND SEPTUM**
- **NASO-ORBITAL ETHMOID**
- **ZYGOMA (TRIPOD FRACTURE)**
- **ZYGOMATIC ARCH**
- **ORBIT (BLOW-OUT FRACTURE)**
  - ✓ Pure Blowout Fracture
  - ✓ Impure Blowout Fracture (Rim Fracture)
- **NASO-MAXILLARY COMPLEX (LEFORT FRACTURES)**
- **MANDIBLE**
  - ✓ Factors Affecting Displacement
  - ✓ Mode of Injury
- **OROANTRAL FISTULA**
  - ✓ Sites and Pathways
  - ✓ Localization of CSF Leak (CSF Tracers)
- **CSF RHINORRHEA**
- **FOREIGN BODY NOSE**
- **RHINOLITH**
- **NASAL MYIASIS**
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

Injuries of the maxillofacial region involve both soft tissues and bones. In this era of fast life and expressways the incidence of automobile accidents, which results in multiple injuries including maxillofacial (Fig. 1), are significantly increasing. The management of facial trauma is usually preceded by general management, which includes maintenance of airway and management of hemorrhage and associated injuries.

ETIOLOGY

The maxillofacial trauma can be the result of following incidences:

1. Motor vehicle accidents
2. Industrial injuries
3. Sports injuries
4. Altercations-fistfights
5. Gunshot and other penetrating injuries
6. Animal bites
7. Burns

CLASSIFICATION

For the sake of description and management, the injuries of maxillofacial region can be divided into following:

1. Soft tissue injuries of face
2. Maxillofacial fractures
3. Orbital fractures
4. Fractures of the mandible
5. Maxillofacial trauma in children

The maxillofacial region (Fig. 2) can be divided into three regions: upper, middle and lower one-third (Box 1). The upper third lies above the supraorbital ridge while lower third consists of mandible and lower teeth. The middle third lies between the supraorbital ridge and angle of mouth.

GENERAL PRINCIPLES

The trimodal distribution of death after trauma consists of deaths within seconds to minutes (overwhelming head and brain injury), from minutes to hours (golden hour for trauma care) and days to weeks (progressive organ failure and infection).

Airway, breathing and circulation (ABC) need immediate consideration and action. Resuscitation measures start simultaneously along with the initial survey (such as pupil size and reactivity and level of consciousness) and consists of intravenous fluids, nasogastric tube (not in fractures of cribriform plate and midface), cardiac monitors, urinary catheter (not in urethral injury), radiological and laboratory investigations. Injection tetanus toxoid 0.5 ml is given to patients, who are not recently immunized.

In blunt trauma cases X-ray chest posteroanterior (PA) view, lateral cervical spines and anteroposterior (AP) pelvis are usually ordered. The secondary survey includes head-to-toe examination of completely undressed patient. The respective specialists should attend the associated injuries of head, cervical spines, neck, larynx, chest, abdomen and limbs. The eyeball injuries are managed by ophthalmologists.

Life-threatening injuries: Most of them can be identified during the primary survey.

A. Airway

Maintenance of airway is given the highest priority. While assessing and stabilizing airway, maintain control of cervical spines. The reasons of airway obstruction can be many such as loss of skeletal support, aspiration of foreign bodies, blood and gastric contents and swelling of tissues.

1. In addition to displacing the mandible and chin forward, other nonsurgical ways to manage airway include oropharyngeal and nasopharyngeal airways and orotracheal and nasotracheal intubations (avoid in extensive midface fractures). While intubating the patient spinal immobilization must be maintained.
2. Needle (high pressure oxygen through a large bore IV catheter 12–16 gauge) and surgical cricothyrotomy (avoid in children under 12) should be considered.
3. The tracheostomy may be needed to secure the airway.

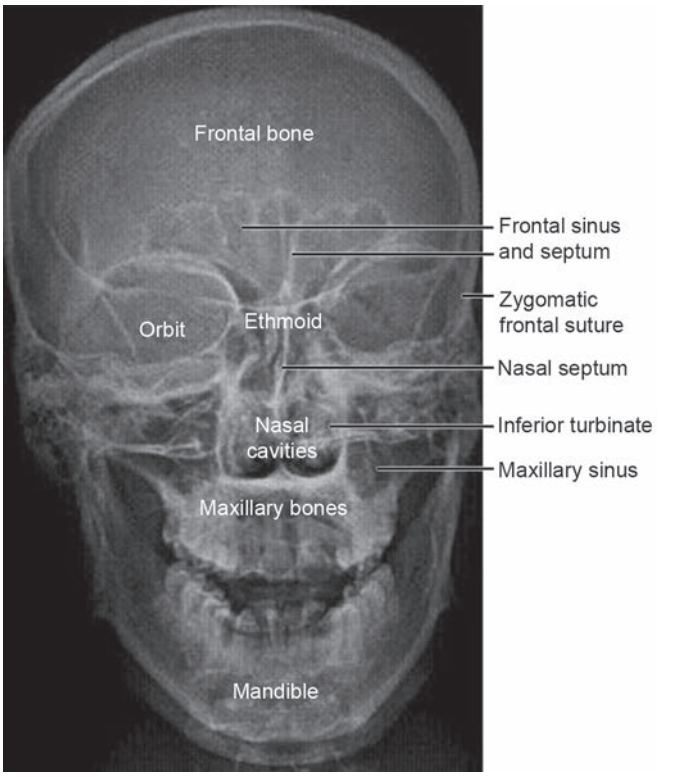


Fig. 1: X-ray skull AP view showing different maxillofacial bones
Source: Dr Jayesh Patel, Consultant Radiologist, Anand, Gujarat

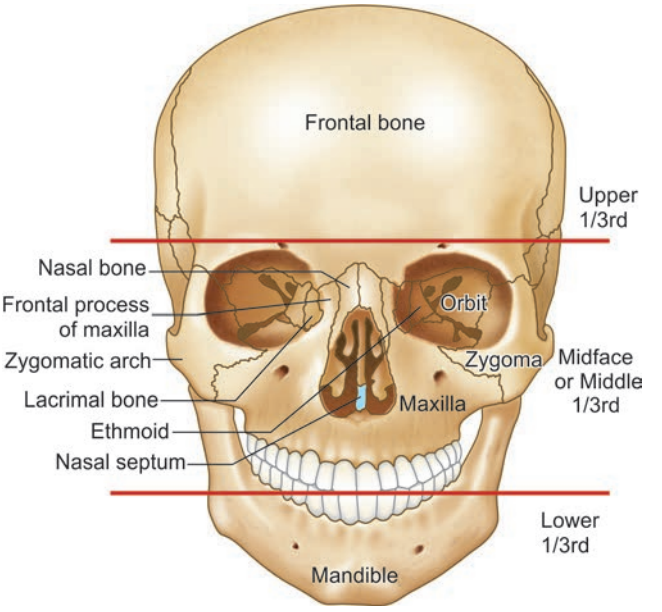


Fig. 2: Norma frontalis. Maxillofacial region showing three regions and common fracture sites. Upper third lies above the supraorbital ridge and lower third consists of mandible and lower teeth. Middle third lies between the supraorbital ridge and angle of mouth

Box 1: Classification of fractures of maxillofacial region

A. Upper third of the face
  a. Frontal sinuses
    i. Anterior wall
    ii. Posterior wall
    iii. Floor
    iv. Frontonasal duct
  b. Supra-orbital ridge
  c. Frontal bone
B. Middle third of the face
  a. Nasal bones and septum
    i. Depressed
    ii. Angulated
  b. Naso-Orbital Ethmoid (NOE)
  c. Zygomatic arch
  d. Orbit (Blow-out fracture)
    i. Floor
    ii. Medial wall
  e. Maxillary and malar region
    i. Central (naso-maxillary complex)
      – Le Fort I
      – Le Fort II
      – Le Fort III
    ii. Lateral (malar-maxillary complex)
C. Lower third of the face
  a. Temporomandibular joint
  b. Mandible
    i. Symphyseal or parasymphyseal
    ii. Body
    iii. Angle
    iv. Ramus
    v. Coronoid
    vi Condylar
    vii Alveolar

- Intubation: It is required if the patient has Glasgow Coma Scale score of 8 or less.
- Cervical spine immobilization in trauma patients: It is vital to maintain cervical spine immobilization while managing the airway.
- Airway: Before aggressive ventilation, always confirm the placement of an endotracheal or tracheotomy tube by observing:
  – Humidification in the tube upon expiration,
  – Affirm CO_2 return,
  – Auscultation for bilateral breath sounds to rule out mainstem bronchi placement).

B. Breathing

Maintaining and aiding ventilation and oxygenation (100% oxygen with a tight-fitting reservoir mask) are of paramount importance. The factors, which can prevent proper ventilation, include chest injuries such as pneumothorax, flail chest segments, hemothorax, pulmonary contusion and rupture of diaphragm.

C. Circulation

The recognition of the cause and treatment of shock needs priority. The profuse hemorrhage (keep in mind the undiagnosed abdominal source) can result in inadequate perfusion of end organs. Injuries of face, which has rich blood supply, can lead to profuse bleeding that can result in shock.

- *Classification of blood loss*: Depending on the percentage of the total blood volume, the blood loss can be classified into four classes: I (15%), II (15–30%), III (30–40%) and IV (> 40%).
- *Monitoring*: The condition of organ perfusion can be judged from pulse, blood pressure, temperature, urine output, skin color and mental status.
  - Electrocardiogram (ECG) may show evidence of cardiac contusions such as dysrhythmia, fibrillations, and ST segment changes. Tension pneumothorax, cardiac tamponade and severe hypovolemia can result in electromechanical dissociation.
- *Treatment*: The treatment of shock includes:
  - Rapid infusion of warm, isotonic, crystalloid fluids through two large-bore (18 gauge or larger) intravenous lines.
  - The class III and IV blood loss needs blood transfusion.
  - Bleeding of vessels is stopped by pressure or ligation.

Blood loss: Young healthy patients can lose up to 30% of blood volume with minimal symptoms.

EVALUATION

Evaluation of maxillofacial trauma includes history, physical examination, imaging and laboratory investigations.

History and Examination

The history and physical examination should include following elements:

1. Airway assessment, hemorrhage and neurovascular status (loss of consciousness).
2. Associated injuries: Cervical spines, eyes and head, major systems and life-threatening injuries.
3. Time and mechanism of injury, wound mechanism; blunt trauma (high speed/low speed), penetrating trauma and spontaneous fracture.
4. Associated medical history, tetanus vaccine status and malignancy.
5. Focused head and neck examination including orbit and globe (enophthalmos and diplopia, hyphema, corneal and scleral lacerations, movements, vision, visual field, convergence, fundus), facial skeleton and nose (fractures nondisplaced/displaced), point of tenderness and crepitus, pain on function, cranial nerves, hypoesthesia/paresthesia, malocclusion, intraoral structures and soft tissues, facial burns (deep/superficial).

Radiology

The radiological investigations include:

1. Plain radiographs: Open-mouthed Towne's and lateral oblique (for mandible).
2. CT scan axial and coronal 3 mm cuts.
3. Panorex.

The wide availability of spiral high-resolution CT scanners has replaced X-ray imaging for the assessment of craniomaxillofacial injuries. The exception is an isolated nasal bone fracture. The axial cuts are best for frontal, naso-orbital-ethmoid (NOE), zygomatic arch and vertical orbital wall fractures, while coronal sections are good for orbital roof and floor and pterygoid plates. Three-dimensional reconstruction (of < 1.5 mm slices) displays

overall facial architecture. For mandible most surgeons prefer plain X-rays and panoramic tomography.

Facial trauma: Thin cut (1–3 mm) facial CT with coronal reformatting is the ideal imaging study.

Laboratory

The laboratory investigations include:

1. Complete blood count (CBC)
2. Coagulation studies
3. Liver function tests
4. Alcohol level
5. Toxicology screen

SOFT TISSUE INJURIES

Facial Lacerations

Facial lacerations are thoroughly cleaned of any dirt and foreign materials and are closed by accurate approximation layer by layer.

Parotid Gland

The exposed parotid tissue needs suturing. Injuries of parotid duct are more serious. Both ends of the injured parotid duct (Stensen's) should be identified and sutured with fine suture over a polyethylene tube, which is left for 3 days to 2 weeks.

Facial Nerve

If the facial nerve is severed, the cut ends of the nerve are identified (needs superficial parotidectomy) and approximated with 8–0 or 10–0 silk under operative microscope.

FRONTAL SINUS

The fractures of frontal sinus can be linear horizontal, linear vertical and comminuted of anterior and posterior walls.

Anterior Wall Fractures

They may be depressed (cosmetic defect) and comminuted. In these fractures, frontal sinus is approached either through an existing external skin wound or a brow incision. While elevating the bone fragments, care must be taken not to strip them from the periosteum. The frontal sinus is inspected to rule out posterior wall fracture.

Posterior Wall Fractures

These fractures are commonly accompanied with dural tears, brain injury and cerebrospinal fluid (CSF) rhinorrhea and require neurosurgical management. Dural tears are repaired with temporalis fascia and small frontal sinuses are obliterated with fat.

Injury to Frontonasal Duct

The chances of obstruction to sinus drainage are high and may result in a mucocele. To avoid this complication, a large communication between the sinus and the nose is created. Small frontal sinuses need complete removal of the sinus mucosa and obliteration with fat.

SUPRAORBITAL RIDGE

Clinical Features

Supraorbital ridge fractures present with following sign and symptoms:

1. Periorbital ecchymosis (purplish patch caused by extravasation of blood into the skin).
2. Flattening of the eyebrow.
3 Proptosis or downward displacement of eye.
4. Impacted bone fragment in the orbit.

Treatment

These fractures require open reduction through an incision either in the brow or in transverse skin line of the forehead.

FRONTAL BONE

Frontal bone fractures, which often extend into the orbit, can be depressed and linear and with or without separation. They are commonly associated with brain injury and cerebral edema and require neurosurgical management.

NASAL BONES AND SEPTUM

Because of the projection of nose on the face, nasal bone fracture is the most common fracture of maxillofacial region. Nasal bone fractures are the third most common fractures of the body (first clavicle and second wrist).

Nasal fractures often involve nasal septum which may be buckled, dislocated or fractured into several pieces (Figs 3A to H). Septal hematoma can occur. For the details of nasal septal fracture see chapter on Nasal Septum.

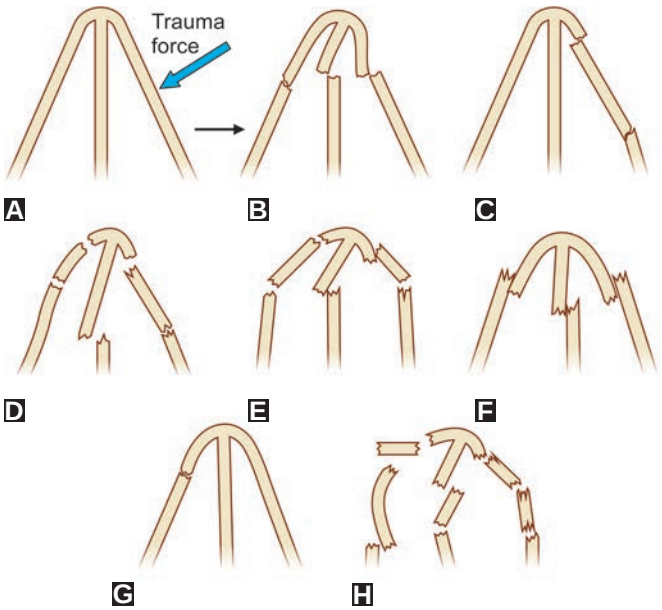
Types

Magnitude and direction (front or side) of traumatic force determine the depth and type of injury. There are two types of nasal fractures: depressed (frontal blow) and angulated (lateral blow).

1. *Depressed*: The lower parts of nasal bones are thinner and easily give way. A frontal blow can cause an open book fracture, where nasal septum collapses and nasal bones splay out. Greater force can result in comminuted fractures of not only nasal bones but also frontal processes of maxillae, which flatten and widen the nasal dorsum.
2. *Angulated*: A lateral blow can cause either unilateral depression of nasal bone on the side of injury or fracture at both the nasal bones and the septum. It results in deviation of nasal bridge.(Fig. 4)

Clinical Features

1. Pain.
2. Swelling appears within few hours and obscures defect.
3 Nasal obstruction.
4. Bleeding from nose and external wound.
5. Nasal deformity (depressed from the front or side) and nasal pyramid deviation.



Figs 3A to H: Nasal fracture patterns. (A) Traumatic force from lateral side of nasal bone; (B) Results in fracture of perpendicular plate of ethmoid; (C) Unilateral fracture of nasal bone; (D) Bilateral fracture of nasal bones; (E) Open book-splayed; (F) Impacted fracture; (G) Greenstick; and (H),Comminuted fracture

6. Periorbital ecchymosis (Purplish patch caused by extravasation of blood into the skin).
7. External lacerations, exposure of nasal bones and cartilage in compound fractures.
8. Tenderness, crepitus and mobility of fractured fragments.
9. Septal deviation/hematoma.

Diagnosis

1. *Clinical*: Diagnosis is usually clinical.
2. *X-rays*: Waters', right and left lateral and occlusal views which though usually do not change the line of management, are important for documentation and medicolegal matter.

Treatment

The active treatment is needed only in cases of deformity and septal deviation. It consists mainly of closed and open reduction. The presence of swelling not only hides deformity but also interferes with accurate reduction. The best time for reduction is either immediate (before the appearance of edema) or when swelling has subsided (after 5–7 days). The reduction should be done within 2 weeks otherwise fracture heals, which is faster in children.

- *Closed reduction* (Figs 5 and 6): Depressed fractures can be reduced by a straight blunt elevator along with outside digital manipulation. Laterally displaced nasal bridge can be reduced only with external firm finger pressure. Impacted fracture fragments require disimpaction, which is performed with the help of Walsham and Asche's forceps (see chapter on Instruments). Septal hematoma is drained. Though simple fractures do not need intranasal packing, unstable fractures require both intranasal packing as well as external splint.
  - Walsham's forceps: They are used for the disimpaction and reduction of the fractures of nasal bones.
  - Asch's septum forceps: They are used for reducing fractures of nasal septum by lifting the nasal septum forwards (Fig. 6).

Fig. 4: Nasal bone fracture. Note right side displacement of bridge of nose

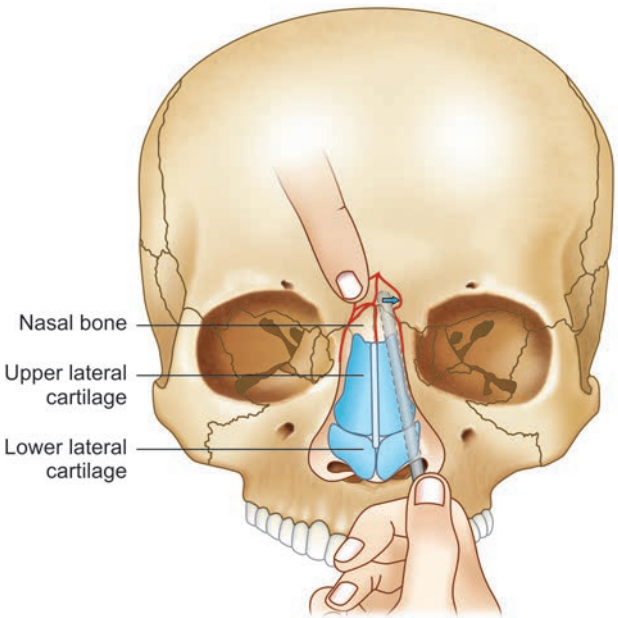


Fig. 5: Closed reduction of nasal bone fracture

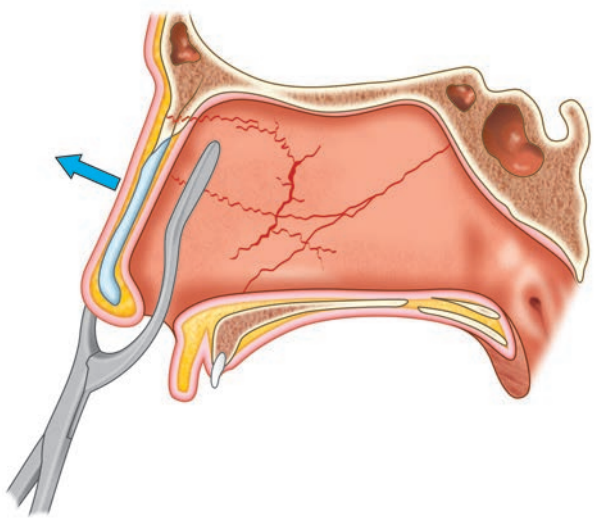


Fig. 6: Closed reduction of septal fracture with Asch forceps

- *Open reduction*: It is required occasionally when closed methods fail and in some cases of septal injuries deformities.
- *Rhinoplasty and septorhinoplasty*: Healed nasal deformities are corrected by rhinoplasty and septorhinoplasty.

Septal hematoma: Before reducing the nasal bone fractures, a septal hematoma must be ruled out because failure to drain it may result in a septal abscess, septal perforation and/or saddle nose deformity.

NASO-ORBITAL ETHMOID (NOE)

The fractures of NOE region occur when the force directly hits over the nasion. The fractures can involve nasal bones, perpendicular plate of ethmoid, ethmoidal air cells and medial orbital wall. They get displaced posteriorly. Medial canthal ligament can be avulsed. Injury can extend to cribriform plate, frontal sinus, frontonasal duct, extraocular muscles, eyeball and the lacrimal apparatus.

Clinical Features

1. *Signs of medial canthal tendon disruption:* They are:
   a. Telecanthus: The widening (> 35 mm) of the intercanthal distance (normal is half of the interpupillary distance) occurs due to lateral displacement of medial orbital wall.
   b. Narrowing of palpebral fissure: The distance between the medial and lateral canthus decreases.
   c. Epiphora: There occurs overflow of tears upon cheek.
2. *Pug nose*: There occur depression of nasal bridge and elevation of nose tip.
3. *Periorbital ecchymosis*: It is the purplish patch caused by extravasation of blood into the skin of periorbital region.
4. *Orbital hematoma*: It occurs due to bleeding from anterior and posterior ethmoidal arteries.
5. *CSF leakage*: It can occur due to fracture of cribriform plate and dura.
6. *Displacement of eyeball*.

Diagnosis

CT scan evaluates NOE region.

Treatment

- *Closed reduction*: An uncomplicated NOE fracture is reduced with Asche's forceps and is stabilized with a wire that is passed through fractured fragments and septum. The wire is then tied over the lead plates. Intranasal packing is done and splinting is kept for about 10 days.
- *Open reduction*: The indications of open reduction include extensive comminuted nasal and orbital bones fracture and injuries to lacrimal apparatus, medial canthal ligaments and frontal sinus.
  - Method: An H-type incision, which can be extended to the eyebrows for accessing frontal sinuses, provides good exposure of the fractured area. The repair of medial canthal ligaments and lacrimal apparatus is done first. Medial canthal ligaments are repaired with a wire. The fractures of nasal bones and medial orbital walls are reduced and nasal bridge height is achieved. Intranasal packing helps in restoring the contour.

ZYGOMA (TRIPOD FRACTURE)

- Zygoma fracture (caused by direct trauma) is the second most common fracture (after nasal bones) of maxillofacial region.
- Fracture line involves zygomaticofrontal suture, orbital floor, infraorbital margin and foramen, anterior wall of maxillary sinus and the zygomaticotemporal suture (Fig. 7).
- The flattening of the malar prominence and a step-deformity of infraorbital margin results from the inferior and posterior displacement of the lower segment of zygoma, which is separated from its three processes (tripod fracture).
- Orbital contents may get herniated into the maxillary sinus.

Clinical Features

1. Ecchymosis of periorbital region (including conjunctiva and maxillary buccal sulcus) within 2 hours of injury is pathognomonic.
2. Flattening of malar eminence with step-deformity of inferior and lateral margins of orbit.
3. Hypesthesia or anesthesia over anterior portion of face occurs due to injury of infraorbital nerve.
4. Trismus (difficulty in opening mouth) occurs due to impaction of fractured zygoma fragment and coronoid process of mandible.
5. Oblique slant of palpebral fissure occurs due to inferior displacement of lateral palpebral ligament (lateral canthal tendon).
6. Lateral canthus and pupil of eye are at lower level in comparison to the normal eye.
7. Diplopia (double vision) and restricted upward ocular movements (entrapment of inferior rectus muscle).
8. Periorbital emphysema may occur due to escape of air from the maxillary sinus on nose blowing.

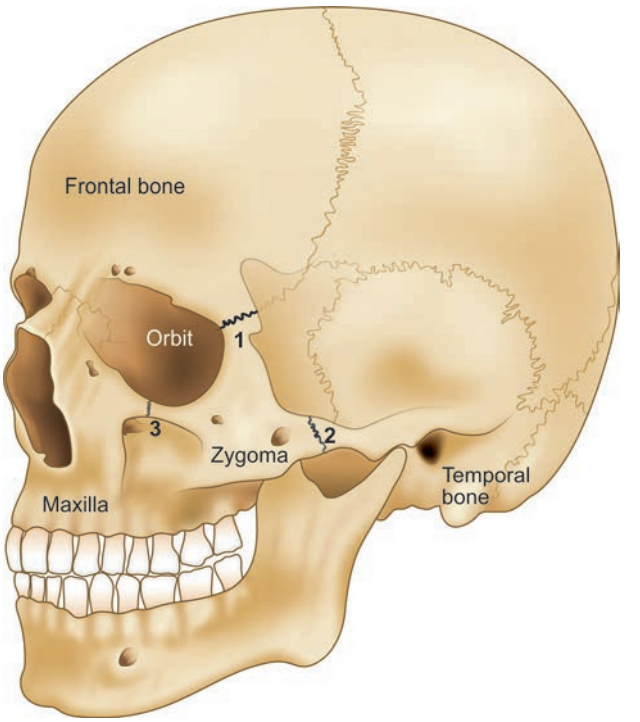


Fig. 7: Left zygoma (tripod) fracture showing three sites of fracture. (1) Zygomaticofrontal; (2) Zygomaticotemporal; (3) Infraorbital

Diagnosis

1. Waters' and exaggerated Waters' views show the fracture and displacement and clouding of maxillary sinus (presence of blood).
2. CT scan shows comminuted fractures and depression of orbital floor and herniation of orbital contents.

Treatment

Open reduction and internal wire fixation offer good results in displaced fractures.

1. The frontozygomatic suture is approached through lateral brow incision.
2. Displaced fracture is reduced by passing an elevator behind the zygoma.
3. A separate incision in the lower lid exposes infraorbital margin. The fracture of orbital floor can also be repaired through this incision.
4. Transantral approach: The antrum is exposed as in Caldwell-Luc operation. Maxillary sinus blood is aspirated and then fracture is reduced. Antral packing helps in stabilizing the reduced fracture. Antral pack is removed in about 10 days either through the buccal incision or through the intranasal antrostomy.

ZYGOMATIC ARCH

- The two fractured fragments of zygomatic arch fracture have three fracture lines. There occurs impingement of fragments on the condyle or coronoid process.
- *Clinical features*: Characteristic depression is seen in the area of zygomatic arch.
  - Local pain and tenderness: They are aggravated during talking and chewing and limit the movements of mandible.
- *Diagnosis*: Submentovertical view of the skull shows zygomatic arch fracture. Waters' view is also helpful.
- *Treatment*: Open reduction.
  Method: A vertical incision in the hair-bearing area above and in front of the ear is used. Temporal fascia is incised and an elevator is passed deep to it. The elevator is passed under the depressed arch fragments and elevated. Fixation is usually not needed because the fragments remain stable.

ORBIT (BLOWOUT FRACTURE)

There are two types of blowout fractures: pure and impure.

Pure Blowout Fracture

It is an isolated fracture of orbital floor and occurs when a nonpenetrating blunt object (ball, fist, hockey puck and cork) strikes the globe. The increased intraorbital pressure results in blowing out of thin walls (especially floor) of orbit. The orbital contents get herniated into maxillary antrum. There is no damage to orbital rim.

- *Clinical features*: The patient may present with:

1. Pain confined to orbit.
2. Ecchymosis of lid, conjunctiva and sclera.
3. Enophthalmos: It is a good indicator of blowout fracture.
4. Inferior displacement of the eyeball.
5. Diplopia: The entrapment of inferior rectus and inferior oblique muscles result in restricted up and down movement of the eyeball.
6. Hypoesthesia or anesthesia over anterior portion of face (infraorbital nerve injury).
7. Forced duction test is done after topical conjunctival anesthesia. The episcleral tissue in the region of inferior oblique insertion is grasped with fine-toothed forceps. By passively rotating, restrictions of the eyeball movements are checked. The complete restriction of passive movements indicates entrapment of muscle. This test is done in operation room after the reduction of blow out fracture. The free passive movements of eyeball indicate success of surgery.

Orbital blowout fractures: The simple forced duction testing detects extraocular muscle entrapment.

Impure Blowout Fracture (Rim Fracture)

In contrast to pure blowout fracture orbital rim gets involved along with the floor of the orbit. The zygomatic and Le Fort II maxillary fractures are usually accompanied by fractures of orbital floor. The repair of zygomatic component of injury may unmask undiagnosed impure blowout fracture by revealing enophthalmos.

Diagnosis

- Waters' view shows a teardrop shaped opacity (herniating orbital fat) hanging from the roof of maxillary antrum (teardrop sign) and loss of double cortical lines (orbital rim and floor).
- CT scan is replacing plain Waters' view.

Treatment

- Indications for surgery are enophthalmos and persistent diplopia.
- *Transantral approach*: Through this approach, orbital floor fracture is easily reduced with a finger after opening the maxillary antrum. The packing kept in the antrum supports the reduced fragments.
- *Infraorbital approach*: It can be used either alone or in combination with transantral approach. An incision in a skin crease of the lower eyelid is made. In badly comminuted fractures of orbital floor, an autogenous bone graft (iliac crest, nasal septum, outer table of calvarium and anterior wall of antrum) or cartilage (septal and conchal) is used for reconstruction of the floor of the orbit. Inorganic implants such as gelfilm, silicon and teflon sheets, Marlex mesh and titanium mesh, have also been used.

NASO-MAXILLARY COMPLEX

Types

In 1901, the French surgeon Rene Le Fort described three typical patterns of the fractures of naso-maxillary complex (in fresh cadavers after using low-velocity impact forces), which are now called Le Fort I, II and III (Fig. 8). These Le Fort fractures, which are usually bilateral, may be mixed.

1. *Le Fort I fracture (Transverse)*: The fracture line runs above and parallel to the palate and crosses lower part of nasal septum, maxillary antra and the pterygoid plates.
2. *Le Fort II fracture (Pyramidal)*: The fracture line runs through the root of nose, lacrimal bone, floor of orbit, upper part of maxillary sinus and pterygoid plates. It has some clinical features of zygomatic fractures.
3. *Le Fort III fracture (Craniofacial disjunction)*: There occurs complete separation of facial bones from the cranial bones. The fracture line runs through the root of nose, ethmo-frontal junction, superior orbital fissure, lateral wall of orbit, frontozygomatic and temporozygomatic sutures and the upper part of pterygoid plates.

Clinical Features

a. Mobile palate: It is a pathognomonic sign of Le Fort fracture. After stabilizing the forehead with one hand, examiner tries to see the mobility of palate and upper teeth with other hand.
b. Facial edema and ecchymosis.
c. Malocclusion of teeth with anterior open bite.
d. Elongation or compression of midface.
e. Epistaxis.
f. Step-offs or movement of facial skeleton at fracture site.
g. CSF rhinorrhea, blindness and airway obstruction in Le Fort II and Le Fort III fractures.

Diagnosis

- X-rays skull: Water's, posteroanterior, and lateral views are helpful.

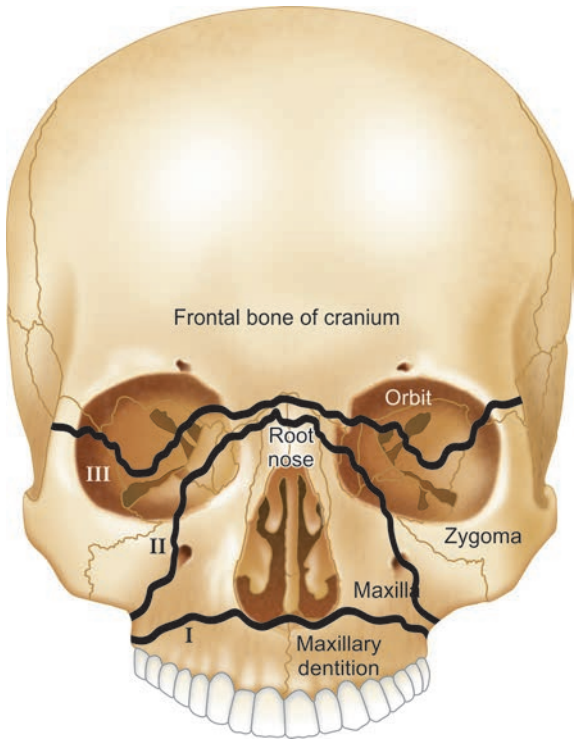


Fig. 8: Le Fort classification of fractures of naso-maxillary complex crossing nasal septum and pterygoid plates. (I) Transverse (separating maxillary dentition); (II) Pyramidal (fracture of root of nose, medial wall and floor of orbit and maxilla), (III) Craniofacial disjunction (separating face from the cranium)

- CT scans better delineate fracture lines and the displacement of fragments and is replacing plain X-rays.

Treatment

Airway restoration and control of severe hemorrhage (maxillary artery and its branches) are the first priorities. Good cosmetic and functional results need early treatment. But associated intracranial and cervical spine injuries may delay reduction and fixation, which include:

a. Open reduction and interosseous wirings.
b. Interdental and intermaxillary wiring using arch bars.
c. Wire slings from frontal bone, zygoma or infraorbital rim to the teeth or arch bars.

MANDIBLE

Classification

Fractures of mandible have been classified according to their location. Figure 9 show Dingman's classification of mandible fractures. The incidences of multiple and single fractures are almost equal.

1. Condylar fractures are the most common and others in descending frequency are fractures of the angle, body and symphysis.
2. Fractures of the ramus, coronoid and alveolar processes are uncommon.

Factors Affecting Displacement

Factors affecting the displacement of mandibular fractures are:

1. Pull of mandible muscles on the fragments.
2. Direction of fracture line.
3. Bevel of the fracture.

Mode of Injury

Though condylar fractures are caused by indirect trauma to the chin or opposite side of the mandible body, the most common cause of other mandibular fractures is direct trauma.

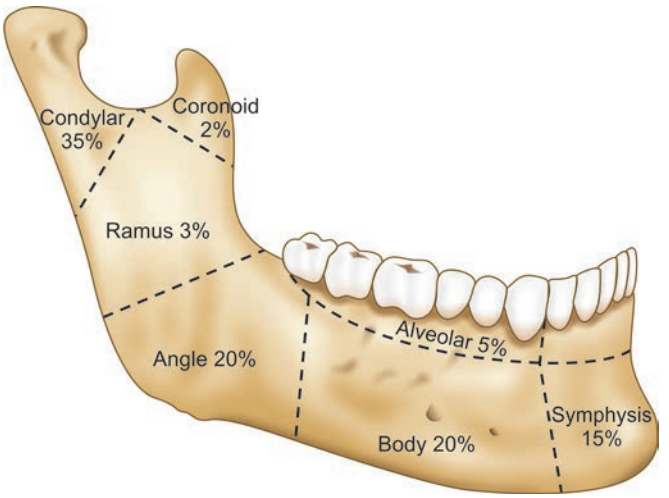


Fig. 9: Dingman's classification of mandible fractures. In descending order common fractures are condylar, angle, body, and symphysis (CABS)

Clinical Features

1. *Fractures of condyle*: Pain, trismus, tenderness at fracture site, malocclusion of teeth and deviation of jaw to the opposite side on opening the mouth (due to displacement of fragments).
2. *Fractures of angle, body and symphysis*: The intraoral and extraoral inspection and palpation reveal malocclusion of teeth, ecchymosis of oral mucosa, and step-deformity, tenderness and crepitus at fracture site.

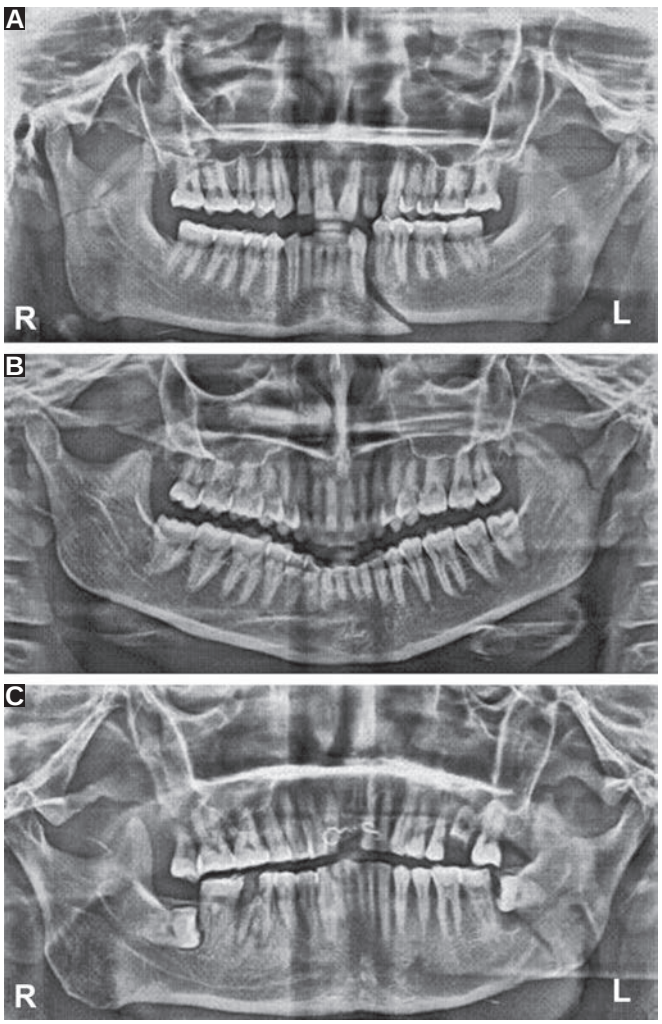
Diagnosis

1. X-rays PA view of the skull for condyle.
2. Right and left oblique views of mandible are helpful.
3. Panorex view is gold standard (Figs 10A to C).

Treatment

Both closed as well as open methods can be employed for reduction and fixation.

1. *Closed methods*: They include interdental wiring and intermaxillary fixation and external pin fixation. In condylar fractures, intermaxillary fixation with arch bars and rubber bands are used. The intermaxillary wires are usually removed in 3 weeks and jaw exercises encouraged. Immobilization beyond 3 weeks in condylar fractures can result in ankylosis of temporomandibular joints. The intermaxillary wires may be reapplied for another week if occlusion is not good. The process may need repetition till the bite and jaw movements are satisfactory.
2. *Open methods*: After exposing the fracture site, fragments are fixed by direct interosseous wiring, which is strengthened by a figure of eight wire tie. Compression plates, which avoids prolong immobilization and intermaxillary fixation, are widely used to fix the fragments. The indications include edentulous patients with bilateral condylar fractures and children.



Figs 10A to C: Orthopantogram (OPG) of mandibular fracture. (A) Parasymphyseal fracture; (B) Subcondylar fracture; (C) Angle fracture

Source: Dr Jaydeep Doshi, Consultant Radiologist, Anand, Gujarat

Mandibular fractures: Open reduction and internal fixation (ORIF) with bone plates and lag screws is the most accepted treatment. It provides stable rigidity and early patient function. Condylar fractures: Severe fractures are problematic even with ORIF.

OROANTRAL FISTULA

Oroantral fistula is situated on the upper alveolus gingivolabial sulcus. It is a communicating track between the maxillary antrum and oral cavity.

Etiology

a. Dental extraction of the upper second premolar and molars is the most common cause because the roots of these teeth are closely related to the cavity of maxillary antrum. Apical tooth abscess is also important predisposing factor.
b. Postoperative complications of Caldwell- Luc operation: Sublabial incision may fail to heal.
c. Carcinoma of maxilla: Erosion of the floor of maxillary antrum.
d. Penetrating injuries and fractures of maxilla.
e. Infection of maxilla: Osteitis and syphilis.

Clinical Features

1. Nasal regurgitation: Semisolid food and liquids pass through fistula and come into antrum and then to nose.
2. Foul smelling discharge from fistula and nose: It occurs due to infection of maxillary antrum.
3. Inability to create positive and negative pressures in the oral cavity: Patient finds it difficult to blow musical instruments and suck through a straw as air gets leaked through the oroantral fistula.
4. A probe can be passed through the fistulous track from oral cavity into the antrum.

Treatment

a. *Recent fistula*: Immediately after tooth extraction, make sure that there is no infection and retained tooth in the antrum. Conservative treatment includes suturing of gum margins and a course of antibiotics.

b. *Chronic or large fistula*: After treating maxillary sinusitis with antibiotics and repeated antral irrigations, the surgical repair is done with a palatal or a buccal flap. The fistulous track lined by squamous epithelium is excised and bony edges of fistula are smoothened and prepared for the flaps.
c. *Caldwell-Luc operation*: It provides a nasoantral window for free drainage. It is needed to remove a retained tooth root, foreign body (FB) and diseased mucosa.
d. *Cancer maxilla*: In these cases fistula is closed by a dental obturator. Fistula permits observation into antral cavity.

CEREBROSPINAL FLUID RHINORRHEA

As the name suggests, CSF rhinorrhea means CSF leaks come out through nasal cavity.

Etiology

a. *Traumatic (immediate and delayed)*:
   - Head injuries, temporal bone fracture and maxillofacial traumas.
   - Surgeries of frontal, ethmoid and sphenoid sinus, hypophysectomy, endoscopic sinus surgery.
b. *Tumors*: Intracranial neoplasms such as tumors of the pituitary and olfactory bulb, large osteomas of frontoethmoid region, skull base neoplasms such as sinonasal and nasopharyngeal cancers.
c. *Congenital defects*: Encephalocele.
d. Hydrocephalus and benign intracranial hypertension.
e. *Infection*: Sinus mucocele and osteomyelitis.
f. Idiopathic.

Sites and Pathways

1. Anterior cranial fossa CSF: It reaches the nose through the cribriform plate, ethmoid air cells or frontal sinus.
2. Middle cranial fossa CSF: It reaches the nose through sphenoid sinus.
3. Otorhinorrhea: In transverse temporal bone fracture CSF enters into the middle ear and then comes to nasopharynx and nose through the Eustachian tube.

Diagnosis

Patient presents with dribbling of clear fluid from the nose on bending or straining. CSF rhinorrhea should be differentiated from the usual nasal discharge of rhinitis and sinusitis (Table 1).

1. *Physical*: CSF remains clear when stand in a test tube, whereas nasal discharge leaves a sediment because of mucus and other proteins.
2. Nasal discharge stiffens the handkerchief while CSF does not.
3. *Glucose content*: Oxidase-peroxidase paper strip or biochemical tests will show the glucose content of CSF. Compare with sugar in CSF obtained from lumbar puncture as sugar is reduced in cases of meningitis.
4. *β 2 transferrin*: It is specific for CSF and confirms the diagnosis of CSF leak.
5. *Electronic nose*: This novel strategy can distinguish between CSF and serum.
6. *Otorhinorrhea*: Middle ear may show fluid and conductive hearing loss.
7. *Double ring sign or target sign*: In traumatic cases, when CSF is mixed with blood, *double ring sign (or target sign)* helps in diagnosis. Discharge is placed on a piece of filter paper, which shows a central spot of blood and peripheral halo of CSF that spreads more and faster than blood.

Complications

1. Meningitis and intracranial infections.
2. Pneumocephalus and secondary brain compression.

Localization of CSF Leak (CSF Tracers)

Site of leak can be seen in high resolution CT, thin coronal cuts with bone window. HRCT shows the area of bony defect. Intrathecal agents, which are used for localizing the site of CSF leaks, include visible dyes, radionuclide markers and radiopaque dyes (Table 2).

a. *Intrathecal fluorescein*: Injection fluorescein 10% 0.1 ml diluted in patient own CSF is infused slowly over 30 minutes. Nasal endoscopy identifies the characteristic green color of the fluorescein in nose and sinuses at the site of CSF leak.
b. *Radionuclide cisternography*: Radionuclide is injected intrathecally and pledgets of cotton are placed close to the suspected sites of CSF leak such as olfactory slit (cribriform plate), middle meatus (frontal and ethmoidal sinuses), sphenoethmoidal recess (sphenoid sinus) and Eustachian tube (temporal bone). Monitoring of the distribution of tracer is done with a scintillation camera. These pledgets are assayed for tracer 12–24 hours later with a gamma counter.
c. *CT cisternography*: CT is done after intrathecal administration of radiopaque contrast (metrizamide), which shows the presence of CSF leak.

TABLE 1 Differences between CSF rhinorrhea and nasal secretions

	CSF rhinorrhea	*Nasal secretions*
History	Surgery, injury and tumor	Sneezing, stuffiness, itching
Discharge flow	A few drops or a stream of fluid	Continuous
Aggravation	Straining and forward bending	Allergy, infection, crying
Sniffing back	Impossible	Possible
Character	Thin, watery and clear	Mucus, mucopurulent, tears
Taste	Sweet	Salty
Sugar content	Normally more than 30 mg/dl	Less than 10 mg/dl
β 2 transferrin	Present	Absent

TABLE 2 CSF tracers

Visible dyes	*Fluorescein*
Radionuclide markers	Radioactive iodine (I^{131}) serum albumin (RISA), technetium (99mTc)-labeled serum albumin and diethylene-triamterene-penta-acetic acid (DTPA) and Indium (In^{111})-labeled DTPA
Radiopaque dyes	Metrizamide

d. *MRI cisternography*: It is a non-invasive and non-ionizing technique. Heavily T2 weighted image with fat suppression and video reversal provides a means to image CSF.

Treatment

The cause of CSF rhinorrhea needs attention and needful treatment.

A. *Early traumatic CSF rhinorrhea*: These are managed conservatively by:
1. Placing the patient in strict bed rest and head elevation (semi-sitting position).
2. Patient is advised to avoid nose blowing, sneezing, coughing and straining.
3. Stool softeners.
4. Prophylactic antibiotics prevent meningitis.
5. If needed subarachnoid drainage through a lumbar catheter.
6. Acetazolamide and mannitol decrease intracranial pressure.
7. Nasal packing and drops are not used.

B. *Persistent CSF rhinorrhea*: These patients need surgery. The repair can be done by–
1 Intracranial approach: Done by neurosurgeons.
2. Nasal endoscopic approach: It is usually preferred in leaks from the frontal sinus, cribriform plate, ethmoid and sphenoid sinuses.

FOREIGN BODY NOSE

Foreign bodies are mostly seen in children. They may be living (maggots) and nonliving. Non living consist of organic (seeds and grams) and inorganic such as piece of paper, chalk, button, pebbles, beads and cell-battery. Iatrogenic FBs include sponges, cotton swabs and gauge packing.

Food and vomiting material can enter the nose through the incompetent nasopharyngeal sphincter. Bullets enter directly through penetrating wound. Sequestration of bone in situ can also occur.

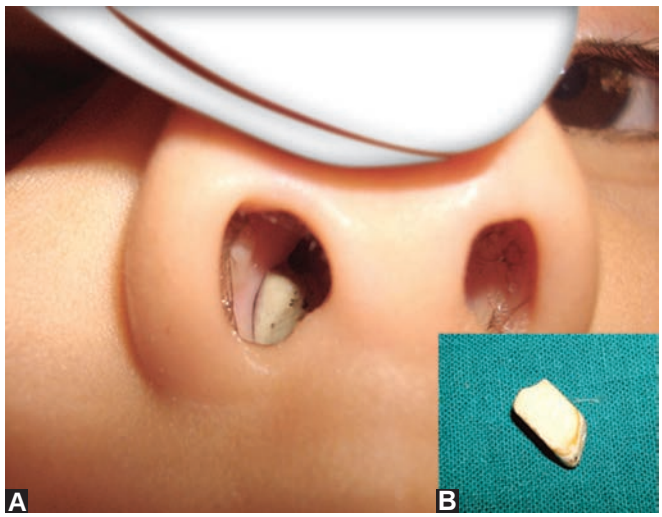
Clinical Features

1. Patient or the parents of children come with clear history of FB in nose.
2. Sneezing, nose block and bleeding are common symptoms.
3. Many children do not tell or simply forget to report to their parents. These children present with unilateral foul smelling nasal discharge with or without blood staining. In all the children who have unilateral foul-smelling nasal discharge, foreign body nose must be ruled out.
4. On examination: FB is visible (Figs 11A and B)

Causes of Unilateral Blood Stained Discharge

1. Forgotten FB nose.
2. Rhinolith and sequestra.
3. Nasal diphtheria.
4. Nasal myiasis (maggots).
5. Acute and chronic unilateral sinusitis.
6. Malignancy of nose and paranasal sinuses.

Foreign body nose in children: A unilateral foul smelling purulent nasal discharge in a child is mostly due to forgotten foreign body nose.



Figs 11A and B: (A) Foreign body (chalk writing stick) in right nose; (B) Removed piece of chalk writing stick.

Complications

A FB nose if not removed can result in–

1. Nasal infection and sinusitis.
2. Rhinolith formation.
3. Swallowing and sticking in esophagus.
4. Inhalation into the tracheobronchial tree.

Treatment

FBs need removal:-

1. *Paper piece and cotton swab*: They are removed with a pair of forceps.
2. *Solid and rounded foreign bodies*: A blunt hook or Eustachian catheter is passed beyond the limit of FB. Then the FB is dragged out along the nasal floor.
3. *Children and apprehensive patients*: General anesthesia with cuffed endotracheal tube should be used. Patient should be kept in Rose's position and nasopharynx is packed.
4. *Nasal endoscopes*: They facilitate the removal of FB lodged far posterior in the nose, which may need to be pushed into nasopharynx.

RHINOLITH

Rhinolith, calculus or concretions (single or multiple) in nose, forms around FB, blood clot or inspissated secretion. Deposition of carbonates and phosphates of calcium and magnesium occurs gradually but progressively. It grows into a large irregular mass (hard or friable), which may result in pressure necrosis of septum and lateral wall of nose.

Clinical Features

Rhinoliths, which are seen commonly in adults, present with:

1. Unilateral nasal obstruction and foul-smelling discharge which may be blood-stained.
2. The mucosal ulceration and granulations can cause epistaxis and neuralgic pain.
3. A grey brown/greenish-black mass, which feels stony hard on probing, is seen in the nose.

Treatment

- Rhinolith should be removed under general anesthesia.
- While manipulating it may break off because it is usually brittle and friable.
- Large rhinolith, which can be broken into pieces, usually needs lateral rhinotomy.

NASAL MYIASIS (MAGGOTS NOSE)

This condition is common in India and mostly seen in months of August, September and October.

Etiopathology

1. Maggots, which are larvae of flies (Genus *Chrysomya*), can infest nose, nasopharynx and paranasal sinuses and cause extensive destruction.
2. Foul smelling nasal discharge attracts flies, which lay their eggs (about 200) that hatch into larvae within 24 hours and secondary infection follows.
3. The common causes of foul smelling nasal discharge include atrophic rhinitis, syphilis and leprosy. These diseases are characterized by nasal crusting and loss of nasal sensation. Poor hygiene is an important contributing factor.

Clinical Features

1. Initially patients may have irritation, sneezing and lacrimation and later on headache.
2. Patient presents with epistaxis (thin blood-stained nasal discharge), puffy eyelids and lips, fever, toxemia and cellulitis of nose and face.
3. Later on the maggots start crawling out of the nose.
4. Patient emits foul smell.
5. Maggots lead to destruction of nose and paranasal sinuses, soft tissue of face, palate (perforation) and even eyeball.
6. Fistulae may form around the nose.
7. Intracranial complications can kill the patient.

Treatment

1. Removal of all the maggots by the forceps: While removing, maggots go away from light into darker cavities.
2. Topical liquid paraffin, diluted chloroform or ether and turpentine oil nasal drops: They are used to irritate and stupefy the maggots so they come out of the nose.
3. Nasal douche with warm saline: It facilitates removal of slough, crusts and dead maggots.
4. Antibiotics: They take care of secondary infection.
5. Personal hygiene: It should be observed to prevent the recurrence.

Clinical Highlights

1. **Le Fort's classification:** It is the classification of fractures of maxilla.
2. **Mandibular fracture:** The posterior part of mandible may be displaced superiorly by the contraction of masseter muscle. Patient will have inability to close the mouth, bloodstained saliva from mouth, intense pain in the ipsilateral jaw, and anesthesia of chin or ipsilateral lower lip (laceration of the inferior alveolar nerve of mandibular division of trigeminal nerve). Angle's classification refers to fracture of the mandible.
3. **Oroantral (Orodental) fistula:** First molar is most commonly incriminated tooth.
4. **CSF rhinorrhea:** The most common site of CSF rhinorrhea is cribriform plate. Next common site is ethmoidal sinuses. Beta-2 transferrin is the diagnostic test for CSF rhinorrhea.
5. **Nasal myiasis:** Maggots nose are best treated by chloroform diluted with water.

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33 Tumors of Nose, Paranasal Sinuses and Jaws

Does man make money, or does money make man? Does man make name and fame, or name and fame make man? Try to understand this. Be a man first, and you will see how all those things and the rest will follow of themselves after you. Give up that hateful malice, that dog-like bickering and barking at one another, and take your stand on good purpose, right means, righteous courage, and be brave. When you are born a man, leave some indelible mark behind you.

—**Swami Vivekananda**

Points of Focus

TUMORS OF NOSE AND PARANASAL SINUSES

- **INTRODUCTION**
  - ✓ Histopathology
- **NEOPLASMS IN CHILDREN**
- **DIAGNOSIS**
- **ANGIOFIBROMA**
- **INTRANASAL MENINGOENCEPHALOCELE**
- **GLIOMAS**
- **NASAL DERMOID**
- **MONOSTOTIC FIBROUS DYSPLASIA**
- **SQUAMOUS PAPILLOMA**
- **OSTEOMAS**
- **PLEOMORPHIC ADENOMA**
- **CHONDROMA**
- **SCHWANNOMA AND NEUROFIBROMA**
- **OSSIFYING FIBROMA AND CEMENTOMA**
- **ODONTOGENIC TUMORS**
- **RINGERTZ INVERTED PAPILLOMA**
- **MENINGIOMAS**
- **HEMANGIOMAS**
- **HEMANGIOPERICYTOMA**
- **PLASMACYTOMA**
- **MALIGNANT NEOPLASMS**
- **MALIGNANCY OF MAXILLARY SINUS**
- **MALIGNANCY OF ETHMOID SINUS**
- **MALIGNANCY OF FRONTAL SINUS**
- **MALIGNANCY OF SPHENOID SINUS**
- **ADENOCARCINOMA**
- **ADENOID CYSTIC CARCINOMA**
- **MALIGNANT MELANOMA**
- **OLFACTORY NEUROBLASTOMA**
- **SARCOMAS**
- **RHABDOMYOSARCOMA**

TUMORS AND RELATED JAW LESIONS

- **INTRODUCTION**
- **MANAGEMENT OF JAW SWELLINGS**
  - ✓ Clinical evaluation
  - ✓ Investigations: Orthopantomography (OPG), CT Scan, Fine-needle Aspiration Cytology (FNAC), Open Biopsy
  - ✓ Treatment: Enucleation and Curettage, Marginal or Segmental Resection, Composite Resection, Mandibular Reconstruction
- **FISSURAL CYSTS**
- **PERIAPICAL CYSTS**
- **FOLLICULAR (DENTIGEROUS) CYSTS**
- **ODONTOGENIC KERATOCYST (OKC)**
- **BASAL CELL NEVUS SYNDROME**
- **RETENTION CYST**
- **AMELOBLASTOMA**
- **OSSIFYING FIBROMA**
- **FIBROUS DYSPLASIA**
- **CHERUBISM**
- **ADENOMATOID ODONTOGENIC TUMOR**
- **CLINICAL HIGHLIGHTS**

TUMORS OF NOSE AND PARANASAL SINUSES

INTRODUCTION

Both benign and malignant tumors of the nose and paranasal sinuses are uncommon. The malignant neoplasms are more common than benign. The separation of nasal tumors from tumors of paranasal sinuses is difficult except in early stages. In addition to primary tumors, these areas can be encroached with growths of nasopharynx, cranial and oral cavity.

Squamous cell carcinoma is the most common (80%) malignant tumor of nose and paranasal sinuses.

Benign tumors are usually smooth, localized and covered with mucous membrane while malignant masses are usually friable, have a granular surface and tend to bleed easily.

Tumors of nose and paranasal sinuses can be divided into three categories—benign, intermediate and malignant (Table 1).

Histopathology

Squamous cell carcinoma is the most common (80%) malignant tumor of nose and paranasal sinuses. Other malignant tumors include adenoid cystic carcinoma and adenocarcinoma. The other non-epithelial tumors include neoplasms of lymphoid tissue, soft tissue, cartilage and bone. Osteogenic sarcoma, chondrosarcoma, rhabdomyosarcoma, angiosarcoma, malignant histiocytoma are rare sarcomas.

NEOPLASMS IN CHILDREN

- The common benign neoplasms are fibro-osseous and odontogenic while malignant includes embryonal rhabdomyosarcoma.
- *Biopsy*: Generous tissue sample is usually taken for biopsy because special stains and electron microscopy may be required.
- *Treatment*: Tumors are usually treated surgically and not with irradiation because—
  - Most tumors are not radiosensitive.
  - Radiation affects growth of facial skeleton.
  - Risk of radiation-induced malignancy.

DIAGNOSIS

1. *Endoscopy*: Endoscopy of the nose provides not only the detailed examination but also facilitates an accurate biopsy.
2. *CT scan (Coronal and Axial)*: It helps in assessing bony involvement and soft tissue extension especially retro-orbital and orbital apex region and infiltration of naso-pharynx.
   - *Limitations*: Poor delineation in areas of dental filling, orbital floor and intracranial extension in isodense avascular lesions.
   - *Contrast study*: For evaluation of intracranial extension.
3. *MRI*: T1 and T2 weighting with gadolinium accurately defines soft tissue extent of disease. MRI is able to accurately differentiate between tumor mass and retained secretions in the sinuses.
4. *Angiography*: The indications of angiography include: Enhancing CT lesions.
   - Tumor near internal carotid artery.
   - Tumor involving sphenoid sinus and skull base.
   - Delineation of feeding vessels for embolization.
5. *Digital-Enhancement Angiography*: Its advantages are following:
   - Rapidly performed.
   - Less need for selective catheterization.
   - Requires less amount of contrast.
6. *Ultrasound*: B-mode scanning can assess orbital masses but is not as precise as CT scanning.
7. *Positron emission tomography*: The indications of positron emission tomography (PET) include—
   - Assessing the presence of distant metastases especially before extensive cancer surgery.
   - Follow-up after concomitant chemoradiation.
8. *Biopsy*: Biopsy from the tumor presenting in nose or extending to oral cavity is usually taken with biting punch forceps under local anesthesia. Biopsy from the intrasinus tumor should preferably be taken transnasal with endoscope because canine fossa or external approaches breach the margins of a later en bloc resection.

TABLE 1 Classification of tumors of nose and paranasal sinuses

Benign	*Intermediate*	*Malignant*
Squamous papilloma	Inverted papilloma	Squamous cell carcinoma
Encephalocele*	Meningioma	Adenocarcinoma
Glioma*	Hemangioma	Adenoid cystic carcinoma
Dermoid*	Hemangiopericytoma	Malignant melanoma
Neurofibroma	Ameloblastoma	Olfactory neuroblastoma
Schwannoma	Plasmacytoma	Lymphoma
Angiofibroma*		Osteogenic sarcoma
Osteoma		Chondrosarcoma
Chondroma		Fibrosarcoma
Ossifying fibroma		Rhabdomyosarcoma*
Cementoma		
Fibrous dysplasia*		
Odontogenic tumor*		

*These tumors are seen in children

Contraindications: Vascular tumors and encephalocele.
- Expansion of soft and cystic mass after coughing or valsalva maneuver indicates intracranial connection or major venous connection.
- Fine needle aspiration returns cerebrospinal fluid (CSF) or active bleeding, which indicates vascular tumor.

9. *Aspiration cytology*: It is helpful in cases of intra-antral tumor and tumors that cause proptosis and present along medial aspect of the orbit.

ANGIOFIBROMA

The primary site of origin of this tumor is posterior part of nasal cavity near the sphenopalatine foramen. This is discussed in detail in the chapter of Tumors of Nasopharynx.

INTRANASAL MENINGOENCEPHALOCELE

Meningoencephalocele is usually seen in infants and young children. It is herniation of brain tissues and meninges through foramen cecum or cribriform plate.

- *Clinical*: A smooth polyp like mass in the upper part of nose between the septum and middle turbinate can be seen, which increases in size on crying or straining.
- *Differential diagnosis*: It can be easily misdiagnosed as a polyp, which if avulsed results in CSF rhinorrhea and meningitis.
- *Biopsy*: It is contraindicated.
- *CT scan*: It demonstrates the defect in the base of skull.
- *Treatment*: Frontal craniotomy with severing the stalk from the brain is done along with the repair of dural and bony defect. After sealing the cranial defect, intranasal mass is removed.
- *Extranasal meningoencephalocele*: See chapter on Diseases of External Nose.

GLIOMAS

- Thirty percent gliomas are intranasal and 10% both intra and extranasal.
- Patients are infants and children.
- It presents as a firm polyp and may be seen protruding at the anterior nares.
- Extranasal Glioma: See chapter Diseases of External Nose.

NASAL DERMOID

- There occurs widening of upper part of nasal septum with splaying of nasal bones and hypertelorism.
- A pit or a sinus in the midline of nasal dorsum with hair protruding can be seen.

For more detail see chapter Diseases of External Nose.

MONOSTOTIC FIBROUS DYSPLASIA

In this disease, bone is replaced by fibrous tissue.

- *Site*: The most common site is maxilla followed by ethmoid and frontal sinuses.
- *Clinical features*: Patient develops disfigurement of the face (Fig. 1), nasal obstruction and displacement of the eye.
- *Treatment*: Wide removal (surgical resculpturing) provides good cosmetic and functional results. See also the last section of this chapter.

Fig. 1: Fibrous dysplasia. Disfigurement of face due to the fullness on the right side of nose

SQUAMOUS PAPILLOMA

- This verrucous growth is similar to skin warts. It can arise from the vestibule and lower part of nasal septum.
- These papillomas may be single/multiple and pedunculated/sessile.
- Treatment: They need local excision. The cauterization of the base prevents recurrence.
  - They can be managed even with cryosurgery or laser.

OSTEOMAS

Osteoma is formed of mature lamellar bone.

- The most common site is frontal sinus followed by ethmoid and maxillary.
- They usually remain asymptomatic and are seen incidentally on X-rays.
- The obstruction to the sinus ostium can lead to mucocele and pressure symptoms in the orbit, nose and cranium.
- Treatment: Symptomatic osteoma needs surgical removal.

PLEOMORPHIC ADENOMA

This rare tumor, which usually arises from the nasal septum, needs wide surgical excision.

CHONDROMA

The various types include mixed, and fibro, osteo and angiochondromas. Histological differentiation between benign and malignant tumors is not completely defined.

- The common sites of origin are ethmoid, nasal cavity and nasal septum.
- They present as smooth, firm and lobulated growth.
- Treatment is wide excision because of their tendency to malignant transformation.

SCHWANNOMA AND NEUROFIBROMA

These indolent tumors have slow progressive growth and become symptomatic by obstructing sinus ostium. They arise from components of peripheral nerve.

- *Schwannoma*: It is an isolated encapsulated tumor.
- *Neurofibroma*: It is woven into the nerve and may be multiple.
- *CT scan*: Neuroma and neurilemmoma show irregular patchy appearance.
- *Treatment*: These tumors need conservative local excision. Malignant change can occur in cases with multiple neurofibromatosis (von Recklinghausen's disease).

OSSIFYING FIBROMA AND CEMENTOMA

- *Ossifying fibroma*: Histologically it looks similar to fibrous dysplasia.
  - Age: It is seen in young adults.
  - Radiology: The sclerotic bony margin can be seen.
  - Treatment: It can be shelled out easily.
- *Cementoma*: It is a variant of ossifying fibroma and needs local excision.

ODONTOGENIC TUMORS

These rare tumors account for only 1% of all jaw tumors.

- *Types*: Ameloblastoma (adamantinoma) and calcifying epithelial tumor of Pindborg.
- *Origin*: These locally aggressive tumors arise from the odontogenic tissue and usually involve maxillary sinus.
- *Treatment*: It is surgical excision.

INVERTED PAPILLOMA

(Transitional Cell Papilloma or Ringertz Tumor)

The neoplastic epithelium of this papilloma grows towards underlying stroma rather than on the surface.

- *Clinical features*: It is mostly seen in males between 40–70 years of age.
  - This is unilateral and arises from the lateral wall of nose. Epistaxis is not uncommon.
  - It presents as red or grey masses, which may be translucent, edematous or simulate nasal polyps.
- *Treatment*: It is wide surgical excision. Inadequate removal results in recurrence.
  - *Lateral rhinotomy*: Medial maxillectomy and en bloc ethmoidectomy.
  - *Endoscopic sinus surgery*: Some surgeons have obtained good results with endoscopic sinus surgery (ESS).
- *Prognosis*: Inverted papillomas are known for their recurrence.
  - In 10–15% cases they convert into squamous cell carcinoma.

Inverted papilloma: The excision should be aggressive because this aggressive nasal benign tumor has frequent coexisting carcinoma.

MENINGIOMAS

1. *Intracranial meningioma*: It can invade sinuses and orbit.
   - Radiology: It usually shows hyperostosis of ethmoidal region.
   - Treatment: It needs surgical excision. Radiotherapy is able to stabilize inoperable tumors.
2. *Extracranial meningioma*: It arises from ectopic arachnoid tissue and needs electron microscopy for identification.

HEMANGIOMAS

1. *Capillary hemangioma*: This bleeding polyp of the anteroinferior part of nasal septum is a soft, dark red and pedunculated/sessile tumor.
   a. It presents with recurrent epistaxis and nasal obstruction.
   b. This smooth growth may become ulcerated.

   Treatment: It needs local excision with a cuff of surrounding mucoperichondrium.
2. *Cavernous hemangioma*: It arises from the turbinates.
   - *Treatment*: It is treated by surgical excision with preliminary cryotherapy. Extensive lesions may need combined radiotherapy and surgical excision.

HEMANGIOPERICYTOMA

This rare vascular tumor arises from the pericytes, which surround the capillaries. It is common in nose but may also involve sinuses.

- *Clinical features*: Most of the patients are in the age group of 60–70 years.
  - Patient presents with epistaxis.
  - Tumor may be slowly enlarging or aggressively infiltrating.
  - Tumors are of varying appearance.
- *Biopsy*: Brisk bleeding occurs on biopsy. The nature of the tumor, whether benign or malignant, cannot be distinguished histologically.
- *Treatment*: It consists of wide surgical excision. Delayed recurrence is likely.
  - Radiotherapy is for inoperable and recurrent lesions.

PLASMACYTOMA

- Solitary plasmacytoma of nose without generalized osseous disease predominantly affects males over 40 years.
- *Treatment*: If total regression does not occur with radiotherapy, surgery is done.
- *Prognosis*: These patients need long term follow-up because there are chances of developing multiple myeloma.

MALIGNANT NEOPLASMS

- *Incidence*: Malignancy of paranasal sinuses accounts for 15% of all upper respiratory tract neoplasms.
- *Sites*: The most common site is the maxillary sinus followed in descending order by ethmoids frontal and sphenoid.
  - In nose, squamous cell carcinoma may arise from the vestibule, anterior part of nasal septum (Nose-picker's cancer) or the lateral wall of nasal cavity.
- *Histology*: More than 80% of the malignant tumors of nose and paranasal sinuses are squamous cell carcinomas. Other in descending order, are adenoid cystic carcinoma and adenocarcinoma. Melanoma and sarcomas are rare.

Etiology

The exact cause of sinus malignancy is not yet clear but following are some predisposing factors:

1. *Industrial workers*: Workers of hardwood furniture industry, nickel refining, leather work and mustard gas manufacturing have higher incidence of sinonasal cancer.

a. Adenocarcinoma of the ethmoids and upper nasal cavity is more common in workers of furniture industry.
b. Workers of nickel refining are more prone to develop squamous cell and anaplastic carcinoma.

2. *Geographical*: Bantus of South Africa who use locally made snuff, which is rich in nickel and chromium, have higher incidence of sinonasal cancer.
3. *Aflatoxin*: It is found in certain foods and dust.
4. *Polycyclic hydrocarbons*
5. *Mesothorium (Thorotrast)*: It is a radiopaque dye used in antrum.

Clinical Features

a. *Early*: Facial pain, nasal obstruction and epistaxis.
b. *Eye and Orbit*:
   - Diplopia (double vision) and squint due to involvement of oculomotor nerves and extraocular muscles.
   - Loss of vision due to the involvement of optic.
   - Proptosis due to tumor compression of periorbita (orbital fascia).
   - Epiphora due to involvement of lacrimal duct.
c. Facial numbness due to involvement of branches of infraorbital nerve (CNV_2).
d. Facial swelling due to involvement of facial soft tissue (Fig. 2).
e. Malocclusion, widening of upper alveolus, loose and nonvital teeth due to involvement of upper alveolus.
f. Trismus due to involvement of pterygoid muscles.(Fig. 3)
g. Neck swelling due to involvement of jugular chain lymph nodes.
h. Deafness. Serous otitis media due to involvement of nasopharynx.
i. Mass: Necrotic intranasal mass and/or alveolar/palatal mass or ulceration.
j. Cranial nerves involvement: CN II, III, IV, V_1 and V_2, VI.

Diagnosis

1. *Radiographs*: CT scan has replaced plain X-rays, which show opacity of sinus with expansion and destruction of the bony walls.
2. *CT scan*: Axial and coronal planes show bony and soft tissue extent of tumor and help in the staging of disease (Figs 4 and 5). Figures 6 to 8 show MRI findings.
3. *Endoscopy*: Endoscopy of the nose and maxillary sinus provides not only detailed examination but also facilitates an accurate biopsy.
4. *Biopsy*: Biopsy is taken from the growth in the nose or mouth. In early suspected cases, sinus may be explored through Caldwell-Luc operation but preferred approach is endoscopic intranasal. Direct visualization of the tumor site helps in staging.

Classification and Staging

The most accepted staging is tumor, nodes and metastases (TNM). Other classifications also provide valuable insight in planning the treatment and predicting the prognosis.

1. *Ohngren's classification*: An imaginary line, which extends between medial canthus and the angle of mandible, divides the maxilla into two regions anteroinferior and posterosuperior (Fig. 9). Anteroinferior growths are easy to manage and have better prognosis than posterosuperior tumors.
2. *Lederman's classification*: Two horizontal lines of Sebileau, one passing through the orbit floors and other through antral floors, divide the area into three regions (Fig. 10):
   a. Suprastructure: Ethmoid, sphenoid and frontal sinuses and the olfactory area of nose.
   b. Mesostructure: Maxillary sinus and the respiratory part of nose.
   c. Infrastructure: Containing alveolar process.
   The vertical line at the plane of medial wall of orbit separates ethmoid sinuses and nasal fossa from the maxillary sinuses. Suprastructure and infrastructure of Lederman's classification is not similar to Ohngren's classification.
3. *TNM classification*: According to American Joint Committee on Cancer (AJCC), TNM classification is for squamous cell carcinoma (Tables 2 and 3). The CT scan and MRI are usually required for staging these tumors.

Fig. 2: Malignancy maxillary sinus. Anterior spread towards face. Note swelling of infratemporal fossa region and mass in right nasal cavity

Fig. 3: Carcinoma maxillary sinus with trismus indicating retromolar extension

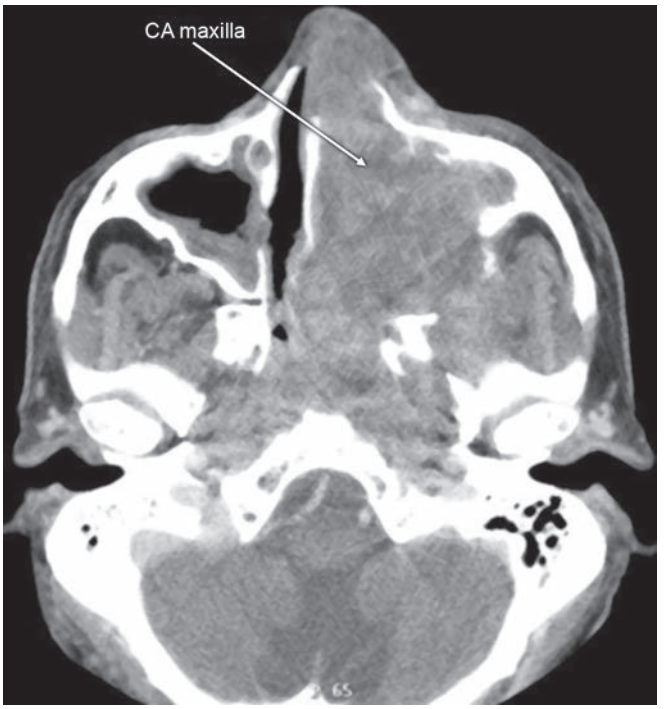


Fig. 4: Contrast CT scan of nose and paranasal sinuses axial section. A large growth of left maxillary sinus. It is destructing the medial wall of maxillary sinus and extending into the left nasal cavity. It is destructing the posterolateral wall of maxillary sinus and pterygoid plates and extending into pterygopalatine fossa and pterygomaxillary fissure and involves lateral pterygoid muscle. Posteriorly the mass is extending into the nasopharynx through the choana with complete obliteration of nasopharyngeal lumen

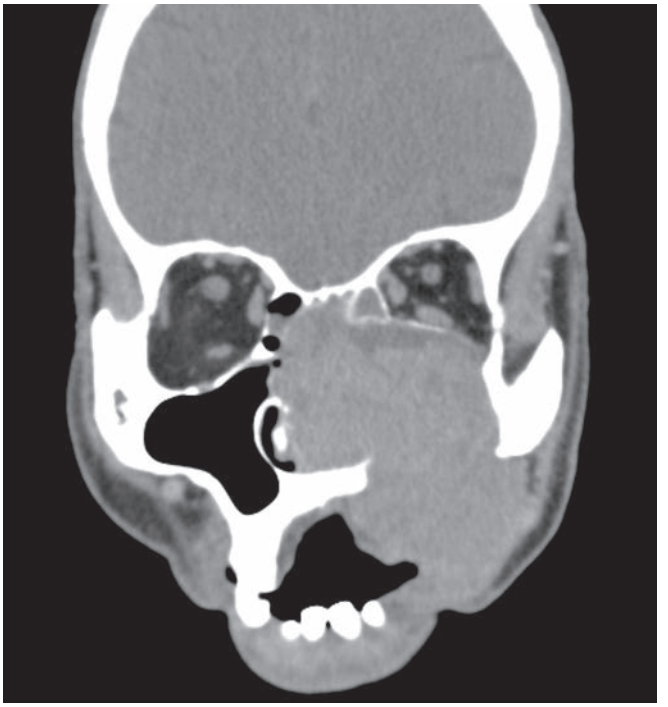
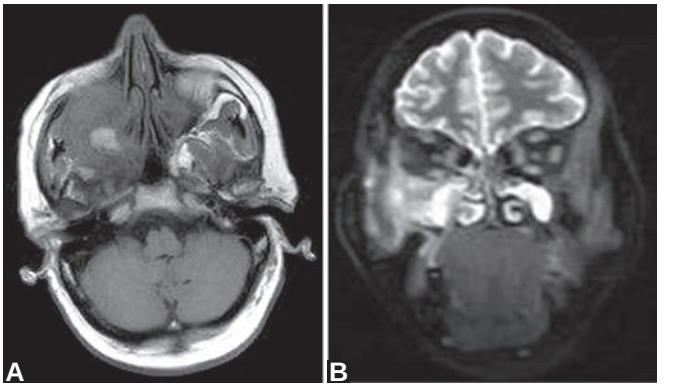
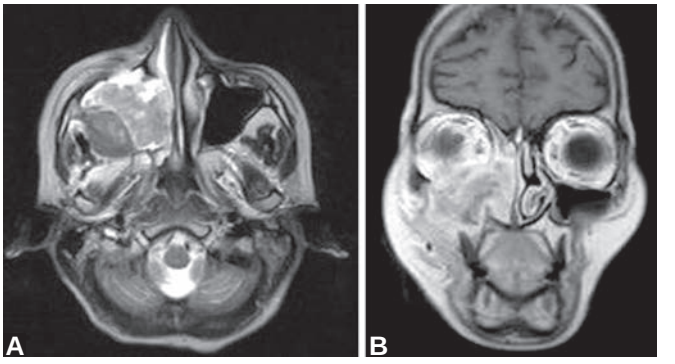


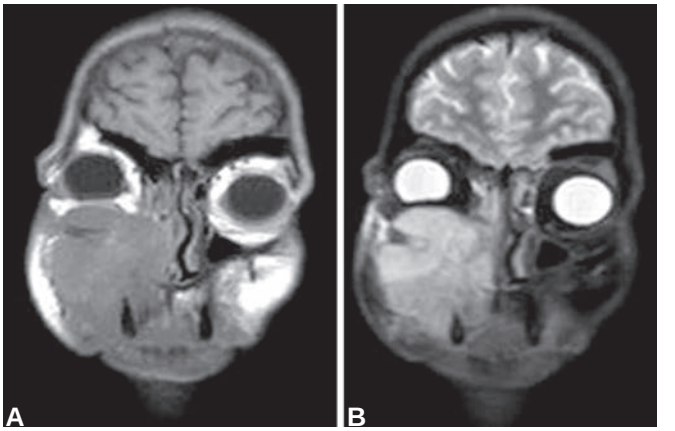
Fig. 5: CT scan coronal section CA Maxilla with infraorbital extension. Note tumor is reaching up to skull base
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat



Figs 6A and B: MRI CA hard palate and maxilla right side. (A) Hypointense mass on T1W axial image; (B) Hyperintense mass on T2W coronal image
Source: Dr Swati Shah, Professor, Radiodiagnosis, GCRI Medical College, Ahmedabad



Figs 7A and B: MRI Contrast. Moderately enhancing malignant mass in right maxillary sinus (A) T2W axial image; (B) TIW coronal image (Post contrast)
Source: Dr Swati Shah, Professor, Radio-diagnosis, GCRI Medical College, Ahmedabad



Figs 8A and B: MRI coronal sections. Carcinoma right maxilla with intraorbital extension. Mass is hypointense on T1W (A)and hyperintense on T2W (B)
Source: Dr Swati Shah, Professor, Radio-diagnosis, GCRI Medical College, Ahmedabad

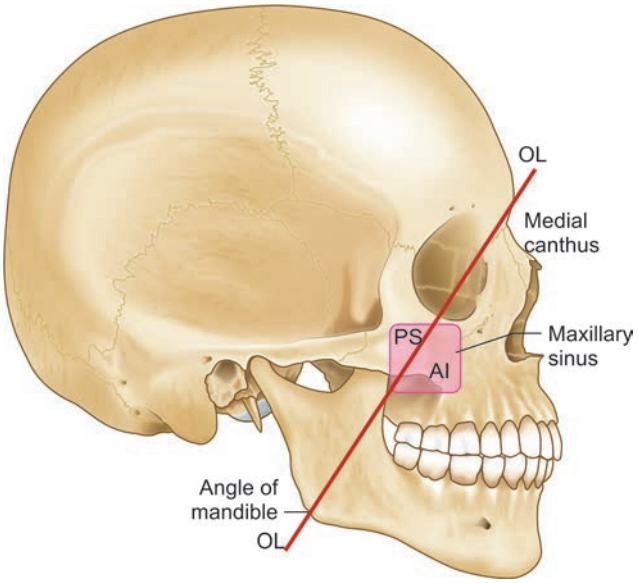


Fig. 9: Ohngren's classification. Ohngren's line is an imaginary line (OL), which extends between medial canthus and the angle of mandible, divides the maxilla into two regions anteroinferior (AI) and posterosuperior (PS). AI growths are easy to manage and have better prognosis than PS tumors

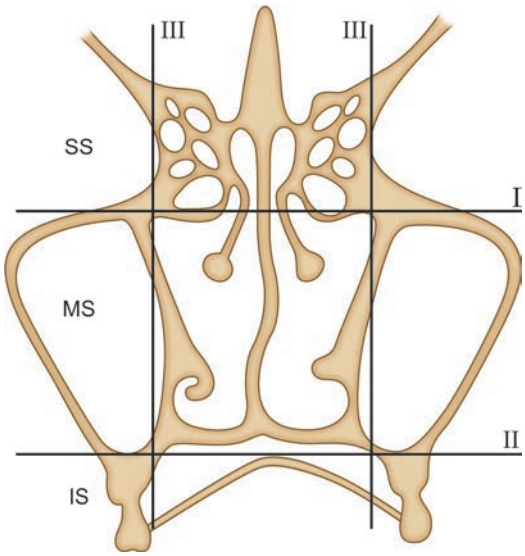


Fig. 10: Lederman's classification. Two horizontal lines of Sebileau, one passing through the orbit floors (I) and other through antral floors (II), divide the area into three regions: Suprastructure (SS), Mesostructure (MS), and Infrastructure (IS). The vertical line (III) at the plane of medial wall of orbit separates ethmoid sinuses and nasal fossa from the maxillary sinuses

TABLE 2 Primary tumor (T) staging (American Joint Committee on Cancer (AJCC)) of carcinoma nasal cavity, maxillary and ethmoid sinuses

Maxillary sinus	
T_1	Tumor limited to maxillary sinus mucosa with no involvement of bone
T_2	Tumor involving bone of hard palate and/or middle nasal meatus. No extension to posterior wall of maxillary sinus and pterygoid plates
T_3	Tumor involving bone of the posterior wall of maxillary sinus, floor or medial wall of orbit; ethmoid sinuses, subcutaneous tissues
T_{4a}	Tumor involving any of the following: anterior orbital contents, skin of cheek, pterygoid plates, infratemporal fossa, cribriform plate, sphenoid or frontal sinuses
T_{4b}	Tumor involving any of the following: orbital apex, dura mater, brain, middle cranial fossa, cranial nerves (other than maxillary division of trigeminal nerve), nasopharynx, or clivus
Nasal cavity and ethmoid sinus	
T_1	Tumor restricted to any one subsite
T_2	Tumor involving two subsites in a single region or extending to involve an adjacent region within the nasoethmoidal complex
T_3	Tumor involving medial wall or floor of the orbit, maxillary sinus, palate, or cribriform plate
T_{4a}	Tumor involving any of the following: anterior orbital contents, skin of nose or cheek, minimal extension to anterior cranial fossa, pterygoid plates, sphenoid or frontal sinuses
T_{4b}	Tumor invades any of the following: orbital apex, dura, brain, middle cranial fossa, cranial nerves (other than maxillary division of trigeminal nerve), nasopharynx, or clivus

TABLE 3 Staging of cancer of nose and paranasal sinuses

Stage 1	$T_1\ N_0\ M_0$
Stage 2	$T_2\ N_0\ M_0$
Stage 3	$T_3\ N_0\ M_0$; $T_{1-3}\ N_1\ M_0$
Stage 4 A	$T_4\ N_{0-1}\ M_0$
Stage 4 B	$T_{1-4}\ N_{2-3}\ M_0$
Stage 4 C	$T_{1-4}\ N_{1-3}\ M_1$

4. *Histopathological classification*: In addition to the location and extent of tumor, histological nature of malignancy is also important in deciding the line of treatment.
   a. Grades of squamous cell carcinoma:
      - Well differentiated.
      - Moderately differentiated.
      - Poorly differentiated.
   b. Cell types: They include:
      - Squamous cell carcinoma.

- Undifferentiated carcinoma.
- Transitional cell carcinoma.
- Carcinoma with inverted papilloma.

c. Vascular or perineural invasion: Their presence or absence should be noted.

MALIGNANCY OF MAXILLARY SINUS

Most antral malignancies are anterior ethmoidal because they usually invade anterior and posterior ethmoid sinuses.

Clinical Features

Most patients are males in the age group of 40–60 years.

a. *Early features*: The lesion arises from the sinus mucosa and may remain silent for a long period, during which patient has vague symptoms of rhinosinusitis such as nasal stuffiness, blood stained nasal discharge, facial paresthesia or pain and epiphora.

b. *Late features*: Later on other features appear when tumor spreads and destroys the bony confines and involves surrounding structures.

1. Medial spread towards nasal cavity: Nasal obstruction, discharge and epistaxis.
2. Anterior spread towards face: Swelling of the cheek (Fig. 2) and later invasion of the facial skin.
3. Inferior spread towards alveolus: Expansion of alveolus, dental pain, loosening of teeth, poor fitting of dentures, ulceration of gingiva and swelling in the hard palate.
4. Superior spread towards orbit: Facial paresthesia/anesthesia, proptosis, diplopia, ocular pain and epiphora.
5. Posterior spread into pterygomaxillary and infratemporal fossa: Trismus due to involvement of pterygoid plates and the muscles (Fig. 3).
   - Spread to nasopharynx, sphenoid sinus and base of skull.
6. Intracranial spread: Through ethmoids, cribriform plate or foramen lacerum.
7. Lymphatic spread: Neck swelling. Cervical node metastases (submandibular and upper jugular nodes) are uncommon and occur in the advanced stages.
   - Maxillary and ethmoid sinuses drain primarily into retropharyngeal nodes, which are inaccessible to palpation.
8. Distant metastases: Though rare they mostly occur in lungs and occasionally in bones.

Treatment

1. *Surgery*: It is the mainstay therapy. Complete radical maxillectomy includes removal of maxilla along with the nasal bone, the ethmoid sinus, and in some cases pterygoid plates. It is adequate when tumor is confined to maxilla, or extends to facial soft tissues, palate, or anterior orbit but without invasion of the ethmoidal roof, posterior orbit, or pterygoid region. Figure 11 shows Weber-Ferguson incision.
2. *Radiotherapy*: It is given either before or after surgery. Curative radiotherapy or chemoradiation may make the inoperable tumors operable. Neutron beam irradiation is most suited to adenoid cystic carcinomas.
   a Preoperative radiotherapy: A full course of preoperative radiotherapy is followed 4–6 weeks later by total or extended maxillectomy.

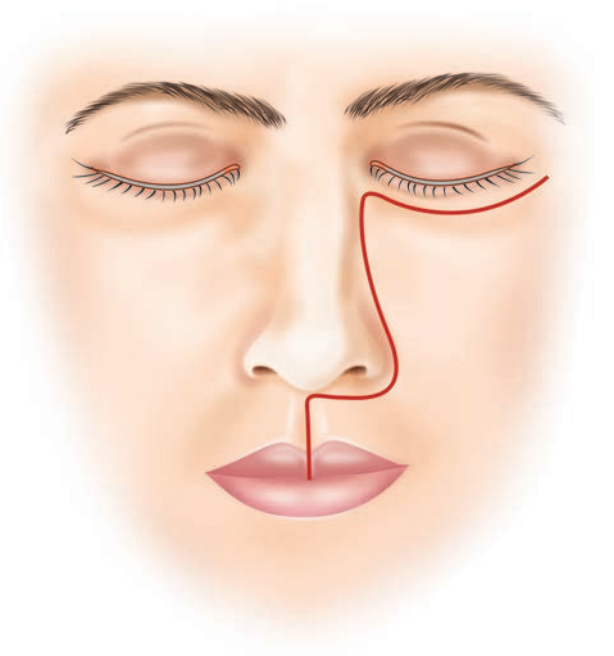


Fig. 11: Weber-Ferguson's incision for maxillectomy starts at the upper lip philtrum on the operative side and goes up to the columella. It continues round the margin of the ala and up the lateral border of the nose. Near the medial canthus of eye it turns laterally in a rounded fashion to go 5 mm below the lower lid margin

   b. Indications of postoperative radiotherapy
      - Large tumors.
      - Positive margins.
      - Perineural or perivascular invasion.
      - Lymph node metastasis.
   c. Adverse affects of radiotherapy
      - 5,800 rad: Severe panophthalmopathy with severe corneal ulceration in 100% cases.
      - 2,800–5,400 rad: Cataracts and visual disturbances in 86% cases.
3. *Eye ball*: Some surgeons prefer to spare the eye whereas others intentionally include the eye during surgery and irradiation. Orbital exenteration in patients with ethmoid tumor has been reported to increase survival.
4. *T_1 and T_2 tumors*: They may be treated by maxillectomy or radiotherapy.
5. *T_3 and T_4 tumors*: Usually a combination of radiotherapy and surgery is employed.

Prognosis

Overall, 5-year survival is about 30%. The multimodal treatment, which is combination of chemotherapy, radiation and surgery, improve the results.

MALIGNANCY OF ETHMOID SINUS

Ethmoid sinuses are usually involved from extension of maxillary sinus growths.

Clinical Features

A. Early features include nasal obstruction, blood stained nasal discharge and retro-orbital pain.

B. Late features: Broadening of the nasal root, lateral displacement of eyeball and diplopia.

1. Intracranial: Meningitis due to invasion of cribriform plate.
2. Cervical lymph node involvement: Rare, upper nodes may be involved.

Imaging

1. *CT scan*: It shows the extent of disease.
2. *MRI*: It reveals intracranial spread.

Treatment

1. *Early cases*: Preoperative radiotherapy, followed by total ethmoidectomy through lateral rhinotomy approach.
2. *Late stages*: Craniofacial resection, when cribriform plate is involved, exposes anterior cranial fossa and facilitates total exenteration of the growth in one piece.

Prognosis

Five years survival is about 30%.

MALIGNANCY OF FRONTAL SINUS

It is uncommon and usually seen in males of 40–50 years age group.

1. *Presenting features*: Pain, swelling of the frontal region and swelling above the medial canthus (erosion of floor of frontal sinus).
   a. Orbital features: When growth extends to ethmoids.
   b. Involvement of dura of anterior cranial fossa: Growth invades the posterior wall of the sinus.
2. *Treatment*: Preoperative radiation followed by surgery, which includes removal of frontal and ethmoid sinuses and orbital exenteration. This neurosurgical approach resects the dura of anterior cranial fossa.

MALIGNANCY OF SPHENOID SINUS

1. Very rare.
2. Clinical features similar to the inflammatory lesions of sphenoid sinus.
3. CT and MRI scan and biopsy through sphenoidectomy confirm the diagnosis and extent of disease.
4. Treatment: Radiotherapy is the mainstay of treatment.

ADENOCARCINOMA

1. *Common sites*: It is usually seen in upper nasal cavity and ethmoid.
2. *Occupations*: It is common in persons who are associated with woodworking, furniture making and leather work.
3. *Treatment*: They have aggressive local progression and need surgically aggressive en bloc resection.

ADENOID CYSTIC CARCINOMA

1. *Spread*: These tumors, which are usually seen in antrum, spread to skull base along the neural sheath.
   - Distant metastases are more common than regional metastasis.
2. *Treatment*: It usually includes surgical resection with irradiation (preferably neutron beam).
   - Three modality therapy: Some recommend combination of chemotherapy (regional infusion with 5-fluorouracil), surgery (maxillary resection) and irradiation.

MALIGNANT MELANOMA

- Most of the patients are of 50 years of age and both the sexes are equally affected.
- Immunological defenses play an important role in the control of this tumor.

Clinical Features

1. *Black colored nasal mass and discharge*: A slaty-grey or bluish-black polypoid mass and black nasal discharge.
   - Amelanotic varieties are non-pigmented.
2. *Sites*: Common site is anterior part of nasal septum followed by middle and inferior turbinate.
3. *Metastases*: Regional (cervical lymph node) metastases and distant (blood stream) metastases are common.

Treatment

1. *Wide surgical excision*: A five-year survival rate after surgical excision is about 30%.
2. *Radiotherapy and chemotherapy*: They suppress the immune processes.

OLFACTORY NEUROBLASTOMA

1. *Origin*: Tumor of olfactory placode.
2. *Sex and age*: Seen in either sex and in any age group.
3. *Clinical*: A cherry red, polypoidal mass in the upper third of the nasal cavity (Fig. 12).
4. *Biopsy*: Very vascular and bleeds profusely on biopsy.
5. *Spread*: Lymph node or systemic metastases can occur.
6. *Treatment*: Though moderately radiosensitive it is treated with radiation.
   a. Some favor surgical excision followed by radiation.
   b. Craniofacial resection is suggested for the tumors of cribriform plate.

SARCOMAS

- Osteogenic sarcomas and chondrosarcoma, which have relentless local progression, are more common in mandible than maxilla.

Fig. 12: Dark brown colored polypoidal mass covered with clotted blood coming out of right side nose. Biopsy: Small cell undifferentiated carcinoma/neuroblastoma. Immunohistochemistry advised to confirm the origin

- Treatment includes en bloc resection and/or irradiation (preferably neutron beam).
  a. Some prefer induction chemotherapy.

RHABDOMYOSARCOMA

Rhabdomyosarcoma is the most common pediatric malignancy of upper respiratory tract.

- This aggressive tumor shows rapid progression and dissemination.
- Histological subtypes: They include alveolar, botryoid and embryonal.
- Treatment: It is usually multimodality and includes clear surgical margins supplemented by irradiation and chemotherapy.

TUMORS AND RELATED JAW LESIONS

INTRODUCTION

The jaws (maxilla and mandible) contain teeth. Masses related to teeth are called odontogenic, while masses not related to the teeth are called nonodontogenic. In addition, the masses of the jaw can be further categorized into neoplastic (benign and malignant), infectious/inflammatory, and congenital groups (Table 4). The two most common cysts of the jaw are periapical and follicular cysts. Salient features of some of the common jaw swellings will be mentioned in this chapter.

MANAGEMENT OF JAW SWELLINGS

Clinical Evaluation

As in all the surgical cases, a complete history and physical examination are of paramount importance.

1. *Benign*: A slow-growing, painless, nonspecific swelling is the usual clinical picture.
2. *Bruit*: A bruit over the mass or in the common carotid is often present in a vascular malformation and tumor.
3. *Malignancy*: Pathologic fractures, malocclusion and trismus are indicators of malignancy.
   a. Pain: Pain or paresthesias usually indicate neural invasion or compression and raise the suspicion of a malignancy.
      – Pain is also common in cases of recent infection of a benign lesion.
   b. Complications: The complications include recurrence, infection and pathologic fracture. The rapid increase in size should raise the suspicion of malignancy.

Investigations

1. *Orthopantomography*: The panoramic (Panorex) radiograph is very important.
   a. It displays not only the location and density of the lesion but also presence or absence of septa or loculations and reaction of surrounding bone and teeth. It helps in knowing the cause of swelling.

TABLE 4 Classification of jaw masses

Odontogenic	*Nonodontogenic*
Benign tumors	
• Ameloblastoma • Calcifying epithelial odontogenic tumor (Pindborg tumor) • Odontogenic adenomatoid tumor • Calcifying odontogenic cyst • Ameloblastic fibroma • Odontoma and fibro-odontoma • Odontogenic fibroma or myxoma • Cementoma • Dentinoma	• Ossifying fibroma • Fibrous dysplasia • Cherubism • Osteoma • Osteoblastoma
Malignant tumors	
• Primary intraosseous carcinoma • Ameloblastic: – Fibrosarcoma – Dentinosarcoma – Odontosarcoma	• Osteosarcoma • Chondrosarcoma • Metastatic malignancies
Inflammatory/Infectious swellings	
• Radicular cyst (periapical or lateral periodontal cysts) • Residual cyst	• Retention cyst • Traumatic bone cyst • Idiopathic bone cavity • Aneurysmal bone cyst and • Stafne's mandibular lingual cortical defect
Developmental swellings	
• Follicular (or dentigerous) cyst • Odontogenic keratocyst • Eruption cyst • Alveolar cyst of infants • Gingival cyst of adults • Developmental lateral periodontal cyst	• Nasopalatine duct cyst • Midpalatal cyst of infants • Nasolabial cyst • Globulomaxillary cyst • Medial mandibular and palatal cysts

   b. Well demarcated lesions surrounded by sclerotic bone are often benign and slow growing masses.
   c. Odontogenic masses are usually near teeth, which may be diseased.
   d. Ill-defined lytic lesions with resorption of bone and neighboring teeth are locally aggressive masses and can be malignant.
2. *CT scan*: It accurately displays cortical thinning and local invasion.
3. *Fine-needle aspiration cytology*: Though fine-needle aspiration cytology (FNAC) may not identify the tumor in some cases it may be helpful prior to open biopsy. FNAC can be very crucial in identifying vascular malformation or tumor, where open biopsy can give rise to disastrous outcome.
4. *Open biopsy*: Final diagnosis and treatment include an open biopsy**,** which can be combined with curettage, enucleation and resection.

Jaw swellings: The panoramic OPG (Panorex) X-ray is indispensable for the diagnosis.

Treatment

The basic treatment modalities for the odontogenic cysts and tumors are curettage, enucleation and resection.

1. *Enucleation and curettage*: Simple enucleation**,** with or without curettage, is often used to treat almost all odontogenic cysts and many odontogenic tumors such as odontoma, ameloblastic fibroma and fibro-odontoma, adenomatoid odontogenic tumor, calcifying odontogenic cyst, cementoblastoma, and central cementifying fibroma. The odontogenic keratocyst cannot be adequately treated with enucleation and curettage.
2. *Marginal or segmental resection*: A marginal or segmental resection of the mandible and surrounding mass is often sufficient for the treatment of benign locally invasive tumors and cysts. This modality is not adequate for persistent and locally invasive lesions such as odontogenic keratocyst (recurrent), ameloblastoma, Pindborg tumor, odontogenic myxoma, and squamous odontogenic tumor.
   a. *Marginal resection*: The resection of only alveolar margin of the mandible is performed.
   b. *Segmental resection*: It involves the resection of a complete segment (full height of the mandible) of certain portion of mandible. This type of resection may require reconstruction and some type of bone graft.
3. *Composite resection*: Composite resection of bone and neighboring soft tissues is used for malignant tumors such as malignant ameloblastoma, ameloblastic carcinoma, ameloblastic fibrosarcoma and odontosarcoma, and primary intraosseous carcinoma. A CT scan helps not only in delineating the extent of tumor but also in preoperative planning.
4. *Mandibular reconstruction*: Generally a patient can wear false teeth after the mandible reconstruction. But whether patient would be able to chew food depends on the size and location of the defect and the type of reconstruction. Controversy is about the time and type of mandibular reconstruction. Some surgeons feel that there is no need of reconstruction if the resected segment of the mandible is short and located at the angle. Others argue that though the patient may wear a denture but s/he will not be able to chew with force. Even the types of bone graft at the site of lesion vary from surgeon to surgeon.

FISSURAL CYSTS

They arise from the epithelium, which is trapped between the fusing embryonic processes.

- *Median palatal cysts*: Formerly number of nonodontogenic developmental cysts (such as nasopalatine duct cysts, globulomaxillary cysts, and nasoalveolar cysts) was thought to be fissural. But now median palatal cyst is said to be the only true fissural cyst of the jaw, which is formed by the growth of epithelium, trapped between embryonic palatal shelves. Median palatal cysts are uncommon.
  - *Clinical feature*: They present as a prominent midline palatal swelling.

PERIAPICAL CYSTS

Periapical cysts are inflammatory and odontogenic in origin. They are the most common and constitute >50% of jaw cysts.

- *Clinical features*: They develop secondary to inflammation at the apex of a non-vital tooth.
  - Patients usually have poor dental hygiene and teeth show other dental diseases too.
  - The condition is often asymptomatic but may present with pain either on biting or percussion.
- *Diagnosis*: Radiographs show area of radiolucency at the root apex.

FOLLICULAR (DENTIGEROUS) CYSTS

Follicular (dentigerous) cysts are developmental and odontogenic in origin. They account for about 10% of jaw cysts.

- *Clinical features*: Follicular cysts may be quite large (5 cm diameter). They are formed around the crown of an unerupted fully formed tooth, which is usually a third molar or canine.
- *Diagnosis*: Radiographs show radiolucency at the crown of an unerupted tooth.

ODONTOGENIC KERATOCYST

Odontogenic keratocyst (OKC) is known to recur. It is often difficult to diagnose as its radiologic and/or histologic features are similar with other cysts.

- *Clinical features*: Like a follicular cyst, it may occur in association with the crown of an unerupted tooth.
  - When OKC occurs in association with a tooth root, it may resemble a periapical cyst.
- *Histopathology*: Only an expert histopathologist can diagnose OKC because it has some specific criteria. Otherwise like a follicular cyst OKC shows a thin connective tissue wall, which is lined by a thin layer of stratified squamous epithelium.
- *Prognosis*: Its differentiation from other types of cysts is important because OKC has a very high recurrence rate (10–60%).
- *Treatment*: It consists of enucleation and curettage. Due to their high recurrence rate they need long postoperative follow-up.

BASAL CELL NEVUS SYNDROME

Basal cell nevus syndrome is a genetic syndrome that presents with multiple odontogenic keratocysts and basal cell carcinomas.

- *Clinical features*: This autosomal dominant syndrome affects young persons. It is associated with bifid ribs, a wide nasal bridge, mandibular prognathism, calcification of the falx cerebri or palmar pitting.

RETENTION CYST

Retention cyst is an inflammatory nonodontogenic cyst and is iatrogenic in origin.

- *Clinical feature*: A past history often of a Caldwell-Luc procedure may be elicited from the patient. The surgery can cause entrapment of sinus epithelium within the incision tract, which may result into formation of a retention cyst.
- *Histopathology*: The respiratory ciliated columnar epithelium lines the cyst.
- *Treatment*: Simple excision is often curative.

AMELOBLASTOMA

Ameloblastomas are benign neoplasms of odontogenic origin. They are rare tumors (about 1% of jaw tumors and cysts).

- *Origin*: These benign epithelial odontogenic tumors arise from odontogenic epithelium of the enamel organ. There are several types of ameloblastomas.
- *Age*: Though there is no age bar those associated with a dentigerous cyst or impacted tooth typically occur before the age of 40.
- *Clinical features*: These tumors are benign but locally invasive. The clinical course of an ameloblastoma is like that of basal cell carcinoma of skin in local growth, invasion and limited metastatic potential.
  - Often patient's presenting symptom is a painless swelling.
  - About 20% of ameloblastomas are associated with impacted teeth or dentigerous cysts.
- *Imaging*: A multiloculated, radiolucent area resembling "soap bubbles" or "honeycomb" is pathognomonic radiological sign.
- *Histopathology*: Follicles lined by tall columnar cells with reversed nuclear polarity are the characteristic patterns. The ameloblastic cells resemble the basal cells of basal cell carcinoma. The epithelium is supported by a mature collagenous stroma.
- *Treatment*: Treatment of choice is wide local excision.

Malignant Ameloblastoma

Malignant ameloblastoma as well as ameloblastic carcinoma have a very poor prognosis.

- *Malignant ameloblastoma*: In a malignant ameloblastoma though the cells retain their benign histologic pattern, they metastasize to lung and lymph nodes.
- *Ameloblastic carcinoma*: It also metastasizes to lung and lymph nodes but the cells appear cytologically malignant.

OSSIFYING FIBROMA

An ossifying fibroma is a slow-growing nonodontogenic benign tumor.

- *Site*: Mandible is involved in majority of the cases and less frequently maxilla.
- *Age and sex*: Patients are often women in their third to fourth decades.
- *Clinical features*: This expansile tumor though grows slowly but destroys the normal bone. The substantial size of it produces facial asymmetry.
  - Examination: Well-circumscribed and marble-like mass can be seen in the bone.
- *Radiology*: It can be radiopaque or radiolucent and the normal radiologic landmarks are distorted. Large masses can show the evidence of both bone destructions as well as bone formation.
- *Histopathology*: It shows collagenous stroma and cementoid deposits.
- *Treatment*: It includes excision and curettage.

FIBROUS DYSPLASIA

This benign nonodontogenic jaw neoplasm is a hamartomatous lesion. Normal bone is replaced by fibrous tissue, which calcifies in an abnormal pattern. Polyostotic fibrous dysplasia can be a feature of Alport's syndrome.

- *Site*: Maxilla is more commonly affected than the mandible.
- *Age*: It often presents in the first or second decade of life.
- *Clinical features*: Unilateral facial deformity due to slow growing diffuse and painless swelling is usually the only clinical finding (Fig. 1).
- *Radiology*: Diffuse margins with "ground-glass" appearance are evident.
- *Prognosis*: The disease usually has a good prognosis.
- *Juvenile aggressive type of fibrous dysplasia*: It is a rapidly destructive lesion and obliterates tooth buds. It is refractory to treatment.
- *Malignant transformation*, although rare, can occur.
- *Treatment*: It is cosmetic conservative surgery and involves shaving and recontouring of the bone. There is no need to be radical in removing the diseased bone because distinct border is not found.
  - There is controversy not only about the need of surgery but also about its timing because studies have shown that fibrous dysplasia "bums itself out" around the age of puberty.

CHERUBISM

- Cherubism, a benign nonodontogenic neoplasm is a self-limited disease of the jaw bones.
- *Age*: Patient usually presents before the age of 5 years.
- *Sex*: This rare congenital disease is more common in males.
- *Etiology*: It displays an autosomal dominant inheritance pattern.

- *Clinical features*: The child usually present with premature tooth displacement and loss.
  - Symmetric mandibular enlargement gives these children a round, cherub-like face.
- *Radiological examination*: It shows bilateral, multiple, multilocular, well-defined radiolucencies with a thin or absent cortex.
- *Prognosis*: It is usually good as the features generally regress spontaneously by puberty.
- *Treatment*: Some children may need facial contouring.

ADENOMATOID ODONTOGENIC TUMOR

- *Age and sex*: The patients of adenomatoid odontogenic benign tumors are young females (< 20 years of age).
- *Site*: Two-third of cases involve mandible, and most are anterior to the permanent molars.
- *Prognosis*: A rapid life cycle of these tumors makes them end in amyloid and calcific material. Even if some portions of the tumor, difficult to access are left in maxilla, there is no need for revision because recurrence is rare.

Clinical Highlights

1. **Fibrous dysplasia:** The most common site of fibrous dysplasia is maxilla.
2. **Osteoma:** The most common site of osteoma is frontal sinus.
3. **Bleeding polyp of the nose:** It is the hemangioma / fibroangioma of nasal septum.
4. **Glioma:** It is the commonest congenital tumor of the nose in children.
5. **Inverted papilloma (Ringertz tumor):** This unilateral nasal papilloma arises from the lateral wall of nose. Polypoidal masses may resemble allergic nasal polyps. Inward growth of squamous or transitional cell epithelium towards fibrovascular stroma lends the name of inverted papilloma to it. It is well-known for its recurrence and malignant change (squamous cell carcinoma in 10–15% of patients).
6. **Malignancy paranasal sinuses:** Nearly 80% are squamous cell carcinoma. Maxillary sinus is the most common site. Other sites in decreasing order are nasal cavity, ethmoid sinuses, frontal and sphenoid sinus.
7. **Risk factor for adenocarcinoma of sinonasal tract:** Wood workers.
8. **Risk factors for squamous cell carcinoma of paranasal sinuses:** Smoking, nickel and chromium plating industry, leather industry, polycyclic volatile hydrocarbons, mustard gas, and isopropyl oil.
9. **Adenoid cystic carcinoma:** Perineural invasion is most commonly seen in adenoid cystic carcinoma.
10. **Ethmoidal cancer:** Most of the ethmoidal cancers are extensions of carcinoma maxilla and patients die of meningitis. Patients with adenocarcinoma of ethmoid usually give history of wood-dust exposure.
11. **Ohngren's line:** It is useful in assessing the prognosis of carcinoma of maxillary sinus.
12. **Superior orbital foramen (fissure):** In cases of tumor compressing the structures traversing the superior orbital foramen, patient experiences pain and altered sensation on the skin of the anterior scalp and dorsum of the nose. Branches of the ophthalmic division of trigeminal nerve (CN V), which carry general sensation from frontal region and dorsum of the nose, traverse through superior orbital foramen (fissure). Tumor of the maxillary sinus, which lies inferior to the orbit, erodes the superior orbital fissure through the floor of the orbit.
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34 Oral Symptoms and Examination

One must not speak unpleasant truths unnecessarily. By indulging in rude words one's nature becomes rude. One loses sensitivity without control over one's words.

—**Holi Mother Sarada Devi**

Points of Focus

- **ORAL CAVITY**
  - ✓ Symptoms: Ulcers, Growth, Pain, Xerostomia, Excessive Salivation, Dysgeusia, Trismus
  - ✓ Examination: Findings on Examination
  - ✓ Tongue Depressor
- **EVALUATION OF CANCER LESIONS**
  - ✓ History
  - ✓ Examination
  - ✓ Investigations
  - ✓ Imaging
- **SALIVARY GLANDS**
  - ✓ Clinical Features
- **DIAGNOSTIC IMAGING**
  - ✓ Sialography
  - ✓ Radiosialography
  - ✓ Ultrasonography
  - ✓ Computed Tomography
  - ✓ CT Sialography
  - ✓ Magnetic Resonance Imaging
  - ✓ MRI Sialography
- **FINE-NEEDLE ASPIRATION CYTOLOGY**
- **CLINICAL HIGHLIGHTS**

For general scheme of case taking and general set up of Bull's eye lamp light source and head-mirror *see* chapter History and Examination in section of Basic Sciences.

ORAL CAVITY

Symptoms

Patients may come to doctor after observing some findings in their mouth, such as an abnormal growth, coating of tongue, cleft lip, cleft palate or oroantral fistula. Patient may see their circumvallate papillae of tongue in the mirror or feel by finger and develop cancer phobia.

- *Pain*: Pain may be referred to the ear. It can occur in any part of the oral cavity.
- *Xerostomia*:[1] Dryness of mouth can result from mouth breathing, radiotherapy, or generalized lesions of the salivary glands.
- *Excessive salivation*: The common causes of excessive salivation are: ulcers of mouth and pharynx, poor orodental hygiene, ill fitting denture and iodide therapy.
- *Dysgeusia*:[2] Taste buds on the anterior two third of tongue appreciate sweet, sour and salt tastes. Patient can have unilateral or bilateral perverted, diminished or loss of taste. It may be associated with lesions, such as heavily coated tongue, or injury to chorda tympani or the facial nerve.

[1] G. xeros, dry + G. stoma, mouth
[2] Dys, bad, difficult + G. geusis, taste

- *Trismus*:[1] The common causes of difficulty in opening the mouth, which are related to the local oral cavity are oral submucous fibrosis, ulcerative lesions, dental abscess (Figs 1 and 2), trauma to mandible or maxilla, and malignant lesions of buccal mucosa and retromolar trigone.
- *Other*: Other oral cavity complaints include ulcers, swelling, ankyloglossia, cleft lip, cleft palate, injury, halitosis, toothache and gums swelling and bleeding.

Sense of taste: It is highly redundant due to its innervation. It is nearly impossible to lose all sense of taste.
Taste and flavor: They are usually confused with each other. The taste includes only the ability to sense sweet, salty, bitter and sour tastes. Flavor includes both taste and smell (80%). Patients with taste problems may be having flavor and a smell disorder.

Fig. 1: Masticator space abscess. Tender red swelling extending over ramus of mandible and obliterating subangular depression

Fig. 2: Trismus in a patient of masticator space abscess

Examination

Examine all the different parts of oral cavity by both inspection, as well as palpation (Box 1). Tongue depressors (Fig. 3) are used in the examination of oral cavity and oropharynx and are available in different sizes for children and adults.

- *Lips*: Lips have an outer (cutaneous), an inner (mucosal) surface and a vermilion border.
- *Buccal mucosa:* Is examined by asking the patient to open the mouth and then retracting the cheek with a tongue depressor.
- *Vestibule of mouth*: Examine the complete vestibule of mouth. Look for not only the change in color but also change in surface appearance. Parotid duct opening may be red, swollen and discharging. It can be seen opposite the crown of upper second molar tooth. Examine the skin of the cheek because carcinoma of buccal mucosa can invade the same (Fig. 4).

Box 1: Findings on examination of oral cavity

- *Lips (upper and lower):* Swellings, growths, vesicles, ulcers, crusts, scars, unilateral or bilateral clefts
- *Buccal mucosa*: Change in color, ulceration, vesicles or bullae (pemphigus), white stria (lichen planus), blanched appearance with submucosal fibrous bands (submucous fibrosis), leukoplakia, erythroplakia, pigmentation, atrophic change in mucosa, swelling or growth
- *Opening of parotid duct*: Red, swollen, secretions (viral or suppurative parotitis)
- *Gums (upper and lower jaws):* Red and swollen gums (gingivitis), ulcerated gums covered with membrane (viral ulcers or Vincent's infection), hyperplasia (pregnancy or dilantin therapy), growths (benign or malignant neoplasms of maxilla or mandible), periodontitis
- *Teeth*: Number, tartar, loose teeth, carious, malocclusion (fractures of mandible or maxilla, abnormalities of temporomandibular joint), impacted last molar
- *Hard palate*: Cleft palate, oronasal fistula (trauma or syphilis), high arched palate (mouth breathing in adenoids), swelling (tumors of palate and nose), bony growth in midline of hard palate (torus palatinus), ulcers/growths (benign or malignant)
- *Tongue*: Macroglossia (hemangioma, lymphangioma, cretinism, edema or abscess), ankyloglossia (congenital tongue tie, cancer tongue or floor of mouth, painful ulcer, abscess), deviation on protrusion (hypoglossal paralysis on the side of deviation), bald or smooth tongue (iron deficiency anemia, median rhomboid glossitis, geographical tongue), fissures (Melkersson's syndrome, syphilis), ulcers (aphthous, traumatic due to jagged tooth or denture, malignant, syphilitic or tubercular), red/white lesions (leukoplakia, erythroplakia), proliferative growth (malignancy)
- *Floor of mouth*: Short frenulum (tongue-tie), scar (trauma or corrosive burn), ulcer (trauma, erosion of submandibular duct stone, aphthous ulcer, malignancy), swelling (ranula, sublingual dermoid, calculus of submandibular duct, benign or malignant tumors, Ludwig's angina)
- *Opening of submandibular duct*: Red, swollen, secretions

[1] L. fr. G. trismus, a creaking, rasping

Fig. 3: Tongue depressor used in the examination of oral cavity and oropharynx. Note submucosal cleft palate with bifid uvula

Fig. 4: Skin involvement in buccal mucosa carcinoma

- *Teeth and gums*: Examine gums and teeth of both upper and lower jaws. Cheeks and lips are retracted with the help of tongue depressor for examining the outer surface of gums while tongue is pushed away for examining the inner surface of gums.
- *Hard palate*: See for any swelling (Fig. 5), ulcer and cleft.
- *Anterior two-third tongue*: Only anterior two-third tongue, which consists of the tip, dorsum, lateral borders and under-surface, is included in the oral cavity. Tongue should be examined in its natural position and then patient is asked to protrude it and move it in different directions (Figs 6 to 11).
- *Floor of mouth*: The floor of mouth consists of the area that lies under the tongue and two lateral gutters (Fig. 12). The latter are examined by two tongue depressors that retract tongue and cheek. The submandibular duct opens on the summit of raised papilla on either side of the tongue frenulum. The swellings in the floor of mouth are examined by bimanual palpation, which help in differentiating between submandibular salivary gland and submandibular lymph nodes.

Fig. 5: Hard palate hemangioma
Source: Dr Amit Goyal, Shillong

Fig. 6: Hairy leukoplakia tongue

Fig. 7: White patches on the dorsum of tongue. Note on the left side one big patch and two small patches on right side

Fig. 8: Black colored tongue due to the ingestion of iron syrup

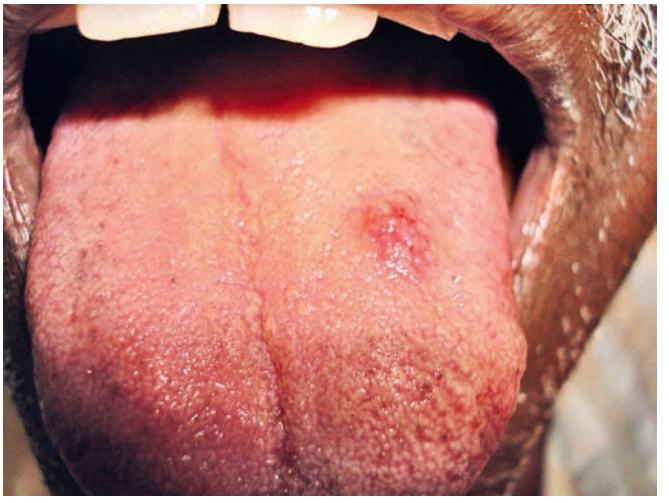


Fig. 9: Tongue bite during chewing food

Fig. 10: Glossitis involving bilateral margins of the tongue. A 38-year-old male patient with complaints of burning sensations on the margins of tongue for 5 days

Fig. 11: Carcinoma anterior two-third tongue left lateral margin. Note two separate infiltrative lesions involving tip and middle one-third tongue

Fig. 12: Ulcerative lesions involving under surface of tongue and floor of mouth

Tongue Depressor

One blade of Lack's tongue depressor is slightly bent at the end. The bent end is used for holding the depressor and supports the little finger of the examiner. The other blade depresses the tongue and is used like a lever to depress anterior two-thirds of the tongue with the fulcrum over the lower teeth.

Caution: Touching of the posterior one third of the tongue usually leads to the gag reflex and not tolerated by the patient.

- *Uses*: It is used for examining the oral cavity and the pharynx. In addition to the depressing of tongue it can also be used for:
  - Squeezing the tonsil
  - Retraction of cheek
  - Test for gag reflex
  - Checking nasal air blast (cold spatula test)
  - Spatula test for suspected case of tetanus

EVALUATION OF CANCER LESIONS

The diagnosis of oral cancer is primarily clinical and confirmed by histopathology. A detailed history, physical examination including palpation (Fig. 13) and investigations should include the following elements:

History

- *Symptoms*: They include changes in the fit of existing denture, otalgia, oral-dental pain, odynophagia, facial numbness, trismus, dysarthria, dysphagia, bleeding, halitosis or weight loss.
- *Personal and past history*: Patient is inquired about medications, allergies, medical illnesses, previous surgeries, tobacco and alcohol use.

Examination

- *Local lesion*: Dimension and extension including crossing midline, adjacent structures involved; fixation to underlying periosteum of mandible (Fig. 14) or maxilla, regional lymphatic spread

Fig. 13: Bimanual palpation of floor of mouth and submandibular region

Fig. 14: Carcinoma left lower alveolus (premolar to retromolar region) extending to buccal mucosa, floor of mouth and cheek skin. Patient is retracting her cheek

- *Need for reconstruction*: Explore need for reconstruction and their options, and available free and pedicled flap.
- *Dental evaluation*: Dental evaluation before radiation treatment, prosthodontic evaluation for surgical obturator in cases of maxillectomy.
- *Speech and swallowing*: Consultation and counseling with speech and swallowing pathologist.

Investigations

- *Routine*: ECG, X-ray chest (preferably both posteroanterior and lateral views) and basic laboratory profile, liver profile in alcoholics
- *Biopsy*: Punch/incisional biopsy of the perimeter of the tumor including some normal adjacent mucosa confirms the tissue diagnosis. Areas of necrosis and infection should be avoided as this tissue may confuse the diagnosis.
- *Fine needle aspiration cytology*: Fine needle aspiration cytology (FNAC) is indicated in cases of suspicious nodes in the presence of known primary carcinoma.
- *Synchronous second primary cancer*: Search for synchronous upper aerodigestive tract cancers. About 15% patients of the oral cavity cancer have multiple primary cancers, which are present in the upper aerodigestive tract. Risk factors (such as smoking and alcohol) are common for all these cancer sites.
- *Panendoscopy*: It includes bronchoscopy, esophagoscopy and direct laryngoscopy, and has been advocated by many in all head and neck cancer patients.

Imaging

They help in knowing the extent of primary tumor and regional lymphadenopathy.

- *Computerized tomography*: CT is best for demonstrating cortical bone erosion and lymph node metastases.
- *Magnetic resonance imaging*: MRI is best for seeing soft tissue invasion by tumor and extension into medullary bone.
- *Orthopantogram*: In patients with suspected mandibular invasion, panorex or orthopantogram facilitates dental evaluation.
- *Ultrasound*: Abdominal ultrasound detects liver metastasis.
- Positron emission tomography *and Single-photon emission computed tomography*: In stage IV patients, PET and single photon emission computed tomography identify occult distant metastasis.

SALIVARY GLANDS

Clinical Features

Parotid[1] swelling fills the retromandibular hollow region and elevates the lobule of ear (Fig. 15). Large swellings involve the preauricular region also.

- *Acute painful swelling*: An acute painful swelling of the parotid gland usually indicates an inflammatory process.

[1] Surrounding the ear

Fig. 15: Left parotid swelling in a 10-year-old girl. Note the infraauricular fullness in the area between the angle of mandible and mastoid

- Acute bacterial parotitis: It usually occurs in association with sialolithiasis of the parotid gland or in patients who are elderly, malnourished, dehydrated, or immunocompromised.
- Mumps: Acute viral parotitis is mostly due to mumps.
- Calculus: An obstruction of the Stensen's duct with a stone may lead to an acute painful swelling of the parotid gland.

- *Bilateral parotid swelling*: Sjogren's syndrome and other forms of autoimmune parotitis usually present with bilateral parotid swelling, which may be asymmetrical.
- *Painless slow growth*: The tumors of the parotid gland usually present as a painless swelling for a long duration and have a slow growth rate. Patients incidentally notice the appearance and consider it a "recent" lump.
- *Site and extent of swelling*: Benign tumors of the parotid are commonly located in its tail. They are usually well defined, nontender and freely mobile. The parotid tumor may be present either in the superficial or in the deep lobe. Tumors can arise from the deep lobe or extend from the superficial to the deep lobe (dumbbell-shaped tumors) in the parapharyngeal space (through the narrow stylomandibular tunnel) and displace oropharyngeal wall medially.
- *Rapid increase in size of a longstanding mass*: It should raise the suspicion of malignant transformation. It may be due to inflammation and cystic degeneration. They are most commonly associated with Warthin's tumor.
- *Metastatic*: In cases of parotid mass, ask about a history of cancer of the scalp or facial skin. Metastasis to the parotid gland can occur from skin cancer including melanoma.
- *Malignant*: Presence of following features indicate malignancy:
  - Facial nerve paresis or paralysis
  - Pain
  - Fixation of mass to the overlying skin or underlying structures
  - Cervical lymphadenopathy.

DIAGNOSTIC IMAGING

CT and MRI are often used in a complementary fashion when evaluating lesions in and around bone. Both CT and MRI are able to detect extrinsic and intrinsic masses of a major salivary gland and determine whether the mass is well circumscribed or infiltrating.

- *CT versus MRI*: For salivary gland tumors, gadolinium-enhanced MRI is equal or superior to contrast enhanced CT. They may even be able to approximate the relative position of the facial nerve in the parotid gland. The approximate course of the facial nerve can be estimated on the axial views because nerve exits the stylomastoid foramen and curves around the ascending ramus at a distance of about 8 mm.
- *Indications*: They are indicated in following conditions:
  - Suspicion of malignancy
  - Tumors of deep lobe of parotid or parapharyngeal space
  - Tumors of submandibular and minor salivary glands

A small well-defined mass of superficial lobe of parotid does not need imaging studies.

Sialography

- *Indications*: It is used in the evaluation of following conditions:
  - Radiolucent calculi
  - Obstructive disease
  - Inflammatory lesions
  - Penetrating injuries
  - Mass lesions
- *Contraindications*: They include the following:
  - Iodine allergy
  - Acute sialadenitis
- *Contrast*: Water-soluble media (such as meglumine diatrizoate) is preferred over oil-soluble media.
- *Technique*: The ostium is gently dilated after topical anesthesia. The duct is cannulated and contrast is injected gently until patient experiences pain. For each gland (parotid or submandibular), anteroposterior, lateral and oblique X-rays are taken to eliminate all bony overlapping. If contrast medium is retained, the abnormalities provide the diagnosis. Several variations include simultaneous xeroradiography, use of pneumography with tomography, secretory sialography and CT sialography.

Radiosialography

This radioactive scanning is most commonly used for parotid gland. It usually uses technetium. It is used to evaluate parenchymal function and to detect mass lesions. Radioisotope scanning is of no use in lesions of the ductal system. The scan should be performed in resting state because uptake in the parotid is greater.

- *Findings*: In a normal study glands are symmetric. Warthin's tumor and rare oncocytoma are the only radiopositive tumors. Postradiation and chronic sialadenitis show decreased and delayed uptake.

CT scanning and MRI are superior in the evaluation of mass lesions.

Ultrasonography

Ultrasonography (US) has been found good at distinguishing glandular from extraglandular masses. Ultrasound is quite good at delineating space occupying lesions of the major salivary gland. It differentiates between cystic and solid lesions.

- *Normal gland*: The normal gland is of homogenous echo texture, appearing more echogenic (reflects the fat interspersed within the glandular tissue) than the adjacent muscle (Figs 16 and 17).
- *Neoplasms*: They are usually hypoechoic to normal glandular tissue.
  - Malignant tumors: They have a low reflectivity with poorly defined borders.
  - Pleomorphic adenomas: They have a variable reflectivity with well-defined borders.
- *Inflammatory lesions*: They have high reflectivity with diffuse borders.
- *US-guided procedure*: Ultrasonic imaging has been used to direct needle aspiration of parotid abscesses and localizing calculi (but less accurate than CT).
- *Color Doppler ultrasound*:
  - Malignant tumors: They show a higher grade of vascularity than benign tumors.
  - Pleomorphic adenoma: Peripheral vascularity with a hypovascular center.
- *Limitations*: US is unable to evaluate
  - Deep lobe parotid masses,
  - Masses obscured by the mandible, and
  - Masses with parapharyngeal or retropharyngeal extension.

Like radioisotope scanning and sialography, US is being supplanted by CT scanning for the evaluation of masses.

Computed Tomography

- *Findings*
  - *Low grade malignant and benign tumors*: They tend to have regular and smooth borders.
  - *High grade neoplasms*: They tend to have irregular infiltration into the parenchyma (irregular outlines and diffuse borders) and nodal metastases.
  - *Pleomorphic adenoma and Warthin's tumor*: They usually have sharp borders.
- *Characteristics*
  - Good for intrinsic and extrinsic parotid masses but of little use in evaluating generalized parenchymal disease or ductal architecture.
  - Far superior in detecting calculi (Figs 18A and B) and in evaluating deep lobe and parapharyngeal space lesions.
  - Excellent at separating cystic lesions from solid masses and lymphoma from other neoplastic masses.

CT Sialography

It is found excellent for:

- Differentiating intrinsic from extrinsic masses

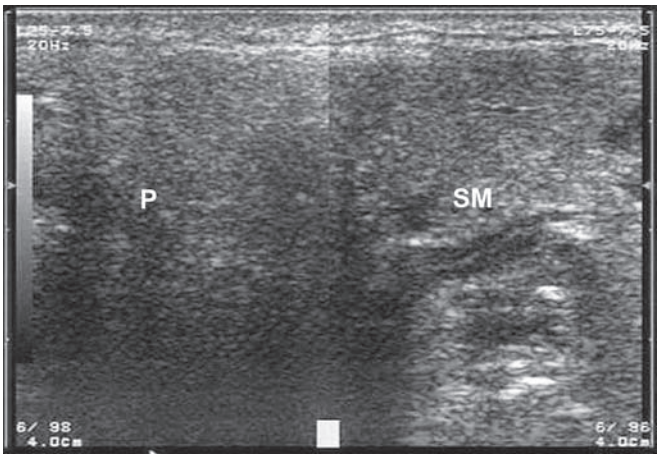


Fig. 16: Ultrasonography neck showing echogenicity of normal parotid (P) and submandibular (SM) salivary glands
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

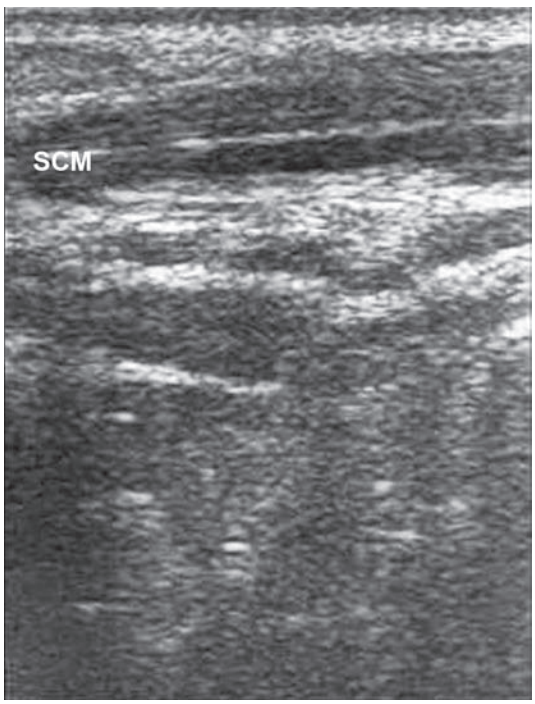
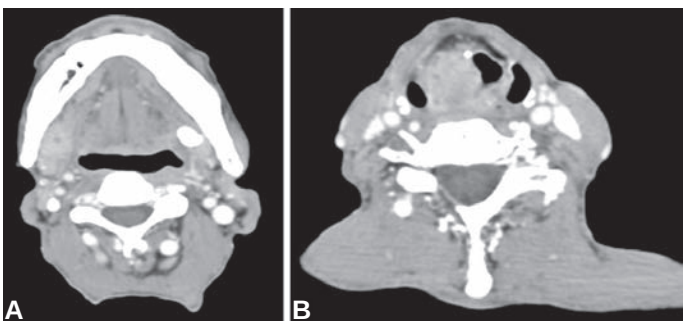


Fig. 17: Ultrasonography neck showing echogenicity of normal sternocleidomastoid (SCM) muscle
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat



Figs 18A and B: CT scan neck axial sections showing incidentally detected left submandibular calculus (A) in a case of right aryepiglottic fold mass (B)
Source: Dr Swati Shah, Professor, Radio-diagnosis, GCRI Medical College, Ahmedabad

- Differentiating benign from malignant parotid neoplasms
- Showing the relationship of the mass to the facial nerve and adjacent bone involvement.
- Differentiating superficial from deep lobe tumors
- Separating parapharyngeal masses from deep lobe parotid tumors

Equally good results are seen using intravenous contrast enhancement.

Malignant and inflammatory salivary gland lesions including abscesses are enhanced by contrast CT. Calculi are best detected without contrast material because small blood vessels may simulate small calculi.

Magnetic Resonance Imaging

MRI is excellent at separating adjacent soft tissues. MRI is quite sensitive to the presence of masses within the gland (outlining the margins of an intraglandular mass).

- *Normal appearance*: The signal is heterogeneous on the T1 weighted signal (the variable fat content of the parotid gland gives it an intermediate to bright intensity). On T2 weighted images also the gland is heterogeneous (because of the serous secretions and water content of the gland). Fast T2 weighted MRI with thin sections can accurately evaluate the ductal architecture.
- *Abnormalities*: They are:
  - Pleomorphic adenomas: They usually have a homogenous or heterogenous appearance, showing intermediate to low signal intensity on T1 weighting and appearing hyperintense on T2 weighting (because of myxoid tissue) and shows homogenous enhancement following gadolinium.
  - Warthin's tumors: They demonstrate the microcysts on T2 weighting.
  - High grade malignant tumors: They are often hypointense on both T1 and T2 weighted images. MRI is again good at evaluating the extension of neoplasms beyond the parotid bed.
  - Spread: Contrast enhanced T1 weighted images are used to see perineural spread, bone invasion and tumor mapping.
    - At the skull base, where there is abundant fat around the bony foramina, the hyperintense enhancing tumors show up quite well.
- *Limitations:*They are:
  - Less sensitive to subclinical inflammation and cystic lesions (such as first arch branchial cleft cysts within the parotid gland) and not sensitive at all to calcification.

MRI is unable to distinguish between a benign and a malignant salivary gland lesion, except in extreme cases (those with infiltrating borders).

MRI Sialography

It assesses the ductal changes in Sjögren's syndrome.

FINE-NEEDLE ASPIRATION CYTOLOGY

Overall accuracy (sensitivity and specificity) of FNAC in salivary gland lesions, which can be compared with frozen section depending on the ability of the pathologists, is around 91.1%. Cellular elements present in cystic lesions, often allow a definitive diagnosis.

Either ultrasound, CT or MRI, which differentiate solid from cystic lesions before the needle biopsy, should be used in conjunction with FNAC.

Clinical Highlights

1. **Saliva:** 1,000–1,500 ml of saliva is secreted in 24 hours. Major amount of saliva, when salivary glands are not stimulated is secreted by submandibular glands.
2. **Excessive salivation:** Some of the causes are oral iodides, poorly fitting denture, ulcers in oral cavity and peritonsillitis.
3. **Sialography:** It is used to diagnose stones, chronic inflammation and tumors in parotid and submandibular glands. It is contraindicated in acute inflammation and acute sialectasis.

FURTHER READING

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35 Oral Mucosal Lesions

We want that education by which character is formed, strength of mind is increased, the intellect is expanded, and by which one can stand on one's own feet.

—Swami Vivekananda

Points of Focus

- **INTRODUCTION**

RED/WHITE LESIONS

- **ORAL SUBMUCOUS FIBROSIS**
  - ✓ Potential for Malignant Change
- **LEUKOEDEMA**
- **ORAL LEUKOPLAKIA**
  - ✓ Histopathology
  - ✓ Potential for Malignant Change
  - ✓ Molecular Biology
  - ✓ Management
- **ORAL HAIRY LEUKOPLAKIA**
- **ORAL LICHEN PLANUS**
- **CHRONIC DISCOID LUPUS ERYTHEMATOSUS**
- **CANDIDIASIS**
  - ✓ Acute Pseudomembranous Candidiasis (Thrush)
  - ✓ Chronic Hypertrophic (Hyperplastic) Candidiasis or Candidal Leukoplakia
  - ✓ Median Rhomboid Glossitis
  - ✓ Other Clinical Forms of Candidiasis 476
- **FORDYCE'S SPOTS**
- **NICOTINE STOMATITIS**

VESICULOBULLOUS/ULCERATIVE LESIONS

- **PEMPHIGUS VULGARIS**
- **MUCOUS MEMBRANE PEMPHIGOID**
- **HERPES SIMPLEX VIRUS**
  - ✓ Primary Herpes Simplex Infection
  - ✓ Laboratory Investigations
  - ✓ Secondary Herpes Simplex Infection or Recurrent Herpes Simplex Infection
- **HAND, FOOT AND MOUTH DISEASE**
- **HERPANGINA**
- **ACUTE NECROTIZING ULCERATIVE GINGIVITIS**
- **RECURRENT APHTHOUS STOMATITIS**
- **BEHÇET'S SYNDROME**
- **ERYTHEMA MULTIFORME**
- **TRAUMATIC (EOSINOPHILIC) GRANULOMA**
- **TRAUMATIC ULCERS**
- **RADIATION MUCOSITIS**
- **BLOOD DISORDERS**
- **DRUG-INDUCED ORAL LESIONS**

PIGMENTED LESIONS

- **MELANOTIC MACULES**
- **MELANOMA**
- **AMALGAM TATTOO**
  - ✓ Lesion and its Site

LESIONS OF TONGUE

- **GEOGRAPHICAL TONGUE**
- **HAIRY TONGUE**
- **FISSURED TONGUE**
- **TONGUE TIE**
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

The oral cavity works to keep a person hydrated, nutritionally healthy and well communicated. It protect upper aerodigestive tract. Impairment to oral health can lead to malnutrition, infection, impaired communication, pain and an impaired quality of life. The oral cavity acts as a window into a person's body as several systemic diseases manifest initially through oral cavity. The various oral mucosal lesions are enumerated in the Box 1. The purpose of this chapter is to provide an overview of the oral cavity disorders.

RED/WHITE LESIONS

ORAL SUBMUCOUS FIBROSIS

It is an insidious painless oral cavity disease, which is characterized by juxta epithelial deposition of fibrous tissue that sometimes even extends to the pharynx. Joshi in 1953 first described this condition in India. The disease is prevalent (2–5 per 1,000) throughout the Indian subcontinent.

Etiology

Several factors operate together and cause this disorder. Exact etiology of this condition is not known but the following factors have been incriminated:

- Prolonged local irritation:
  - Most of these patients have habit of *chewing paan (a specially prepared leaf),* betel nut (*sopari*) and tobacco. The hard and rough surface of betel nut causes mechanical irritation. *Alkaloids* in betel nut (such as *arecoline*) cause chemical irritation and stimulate collagen synthesis and the proliferation of buccal mucosa fibroblasts. *Tannins* in betel nut stabilize the collagen fibrils and render them resistant to degradation by the collagenase.
  - *Smoking* of cigarettes/*Bidies* also leads to local irritation.
  - *Excessive amount of chilies and spices* in the daily food may also be an additional factor.
- *Dietary deficiency*: As there occurs recurrent vesicle formation and ulceration of the oral mucosa a dietary deficiency of iron, vitamins B-complex and A has been proposed.
- *Cell mediated immune process*: Some consider it a cell-mediated immune reaction to arecoline. Arecanut chewing causes collection of activated T-lymphocytes and macrophages in subepithelial layers of oral mucosa, which result in reduced production of antifibrotic cytokines (less collagenase) and increased production of fibrinogenic cytokines (act on mesenchymal cells and proliferate fibroblasts). These lead to increased production of collagen.
- *Localized collagen disease*: As the histopathological changes seen in submucous fibrosis are similar to the collagen diseases such as rheumatoid arthritis and scleroderma, some scientists think it to be a localized collagen disorder.
- *Racial*: Disease usually affects Indians or people of Indian origin living abroad. Sporadic cases are also seen in Nepal, Thailand, South Vietnam and Sri Lanka.
- *Genetic*: As the disease usually affects Indians and not all the people who chew *paan*, *sopari* and tobacco, some authorities strongly feel it to be genetic disorder.

Pathology

- Early cases show polymorphonuclear leukocytes, eosinophils and few lymphocytes while lymphocytes and plasma cells appear in advanced cases. The higher population of activated T-lymphocytes mainly T-helper/inducer lymphocytes (minor population of B-cells), macrophages and high CD4+ and CD8+ lymphocyte ratio in subepithelial tissue suggest main role of cellular immune response and minor role of humoral immunity.
- There occurs a fibroelastotic transformation of connective tissues in lamina propria associated with epithelial atrophy, which is sometime preceded by vesicle formation.
- Juxta-epithelial fibrosis occurs with atrophy or hyperplasia of overlying epithelium, which shows areas of epithelial dysplasia.

Potential for Malignant Change

- Leukoplakia and squamous cell carcinoma are some time associated with this condition as the predisposing factors for all these disorders are common.
- The malignant transformation (Fig. 1) has been observed in 3–7.6% of cases.

Clinical Features

- *Age*: Though there is no age bar, the disease mostly affects 20–40 years of age group.

Box 1: Oral mucosal lesions

- *Red/white lesions*: Oral submucous fibrosis, leukoedema, oral leukoplakia, oral hairy leukoplakia, oral lichen planus, chronic discoid lupus erythematosus, candidiasis, fordyce spots, nicotine stomatitis
- *Vesiculobullous/ulcerative lesions*: Pemphigus vulgaris, mucous membrane (cicatricial) pemphigoid, primary herpes simplex infection, recurrent herpes simplex infection, herpes simplex infection, hand, foot and mouth disease, herpangina, acute necrotizing ulcerative gingivitis, recurrent aphthous stomatitis, Behcet's syndrome, erythema multiforme, traumatic (eosinophilic) granuloma, traumatic ulcers, radiation mucositis, blood disorders, drug-induced oral lesions
- *Pigmented lesions*: Melanotic macules, melanoma, amalgam tattoo
- *Systemic diseases*: Cardiovascular, endocrine, gastroenterology, neurological, renal, hematological (leukemia, agranulocytosis, pancytopenia, cyclic neutropenia, sickle cell anemia)
- *Collagen-vascular and granulomatous disorders*: Sjögren's syndrome, systemic lupus erythematosus, scleroderma, dermatomyositis-polymyositis, sarcoidosis, Wegener's granulomatosis
- *Lesions of tongue*: Geographical tongue, hairy tongue, fissured tongue, tongue tie

- *Habits*: History of chewing of *paan, sopari* and *tobacco* is almost always present.
- *Trismus:* The majority of the patients present with gradually progressive painless difficulty in opening the mouth (Fig. 2).
- *Ankyloglossia*: The disease may advance and cause difficulty in protruding out the tongue.
- *Soreness and burning mouth*: Some patients have soreness of mouth with constant burning sensation, which worsens during meals especially of pungent spicy type. In later stages, patient develops insidious, painless and progressive trismus and ankyloglossia.
- *Vesicles/Ulcers*: Few patients complain of repeated vesicular eruption on the palate and pillars. Initially there occurs patchy redness of mucous membrane with formation of vesicles, which rupture to form superficial ulcers.
- *Fibrous bands*: The most common sites of white fibrotic bands (Fig. 3) are soft palate, faucial pillars, retromolar area and buccal mucosa. In later stages, fibrosis develops in the submucosal layers along with the blanching of mucosa with loss of suppleness. Fibrosis and scarring, which can be seen and felt has also been demonstrated in the underlying muscle that lead to further restrictive mobility of soft palate, tongue and jaw.

Fig. 1: Malignancy of tongue right lateral margin in case of oral submucous fibrosis

Fig. 2: Oral submucous fibrosis. Severe trismus with 0.73 cm mouth opening

Fig. 3: Oral submucous fibrosis. White fibrous bands involving soft palate, faucial pillars and retromolar area

Treatment

- *Medical*
  - *Local steroids/hylase*: Topical injection of steroids, which may be combined with hylase, into the area of fibrous bands (injection dexamethasone 4 mg and hylase 1500 IU in one ml intraoral submucosal biweekly at different sites for 8–10 weeks) is more effective than their systemic use. This brings significant improvement in symptoms and relieves trismus.
  - *Avoidance of irritant factors* (areca nuts, pan, tobacco, pungent foods) is of paramount importance.
  - *Vitamins and minerals:* Treatment of existent anemia or vitamin deficiencies. Vitamin A, zinc and antioxidants therapy has shown some beneficial effect.
  - *Jaw opening exercises*: They are encouraged.
- *Surgical*: Number of surgical procedures have been reported (Box 2) but more common include:
  - *Surgical incision of fibrous bands*: Severe trismus associated with marked fibrous bands can be treated by surgical excision and grafting. It gives immediate dramatic improvement in opening of the mouth but usually results in rebound trismus.
  - *Lasers* have also been used to cut the fibrous bands.
  - *Coronoidectomy and temporal muscle myotomy*.
  - *Reconstruction*: Several types of grafts and flaps have been tried after cutting the fibrous bands (Box 2).

LEUKOEDEMA

- This common asymptomatic incidental finding is characterized by a diffuse and generalized mild surface opacification (milky alteration) of oral mucosa (usually buccal) which is a normal variation of oral mucosa.

Box 2: Grafts and flaps tried for reconstruction in the management of oral submucous fibrosis

- Bilateral tongue flap
- Nasolabial flaps
- Island palatal mucoperiosteal flap
- Bilateral radial forearm free flap
- Buccal pad of fat graft
- Temporalis fascia graft
- Split skin graft

- Stretching of mucosa dissipates milky alteration and differentiates it from other white lesions of oral mucosa such as leukoplakia, hyperkeratotic conditions, white spongy nevus, homogenous type of lichen planus and hereditary benign intraepithelial dyskeratosis.
- Exact cause is not known but the factors that may have a role include smoking, alcohol, bacterial infections and electrochemical interactions.
- It does not require treatment but must be differentiated from leukoplakia.

ORAL LEUKOPLAKIA

Leukoplakia clinically present as a white patch. It should be differentiated from other white lesions of oral mucosa such as leukoedema, lichen planus, discoid lupus erythematosus, white spongy nevus and candidiasis.

Risk Factors

The exact cause is not known but the risk factors include:

- *The incriminating factors*, which are seen along with this lesion, are:
  - Tobacco smoking
  - Smokeless tobacco: Tobacco chewing
  - Alcohol abuse: It is especially harmful if combined with smoking.
  - Areca nut and betel
  - Chronic trauma (frictionally-induced hyperkeratosis): It occurs due to ill-fitting dentures and cheek bites. This is not a true leukoplakia but frictionally-induced hyperkeratosis. Reversal occurs on elimination of traumatic influence. Chronic friction trauma does not cause dysplastic or malignant disease.
    - *Linea alba*: In the buccal mucosa it occurs at the occlusal line.
    - *Morsicatio*: Lip biting habit can cause a slightly granular hyperkeratotic surface over labial mucosa.
  - *Chronic sun exposure (Actinic cheilitis)*: Patches of leukoplakia interspersed with patchy melanotic pigmentation may develop along the lower lip vermilion surface.
  - *Sanguinaria*: It is herbal root extract present in some mouth washes and toothpastes. A translucent to slightly opaque white keratosis patch with well-defined margin and smooth surface develops along the upper labial alveolar mucosa. Reversal occurs on withdrawal of Sanguinaria product.
- *Associated diseases*: The lesion is some time associated with:
  - Submucous fibrosis.
  - Hyperplastic candidiasis.
  - Plummer-Vinson syndrome.
- *Sex*: Males are affected two to three times more often than females.
- *Age*: Mostly it is seen in the fourth decade of life.

Clinical Features

- *Site*: Though the most common sites are buccal mucosa (especially in India) and oral commissures, it may also be seen over floor of mouth, tongue, gingivobuccal sulcus and lip.
- *Lesion*: Widely variable clinical lesions include homogeneous and smooth, focal or diffuse, or heterogeneous and multifocal with variable texture.
  - The white, yellowish or gray surface alteration with ill-defined margins. Plaques may be small circumscribed or extensive and soft or thicker, which feel crusty.
  - Surface texture can be finely granular or slightly papillary, ulcerative, erosive, nodular, or verrucous.
- *Induration:* Induration indicates malignant change and immediate biopsy should be taken.

Clinical Forms

There are different clinical types of leukoplakia. Nodular and erosive types have higher incidence of malignant transformation.

- *Homogenous leukoplakia (Thin leukoplakia)*: There is a smooth or wrinkled white patch, which is less often associated with malignancy. Macular lesion may gradually progress to more opaque elevated thickened and furrowed, leathery or wrinkled appearance (Fig. 4). Some lesions may disappear over time.
- *Nodular (Speckled) leukoplakia*: There occur nodular white patches with erythematous base (Fig. 5).
- *Erosive leukoplakia (Erythroleukoplakia)* presents white patches, which has erosions and fissures and is interspersed with red patches (erythroplakia).
  Erythroplakia:
  - Erythroplakia presents as a bright red velvety red patch (Fig. 6) or plaque usually over lower alveolar mucosa, gingivobuccal sulcus and the floor of the mouth.
  - The lesion is irregular and clearly demarcated from adjacent normal epithelium.
  - Red vascular connective tissue of the submucosa shines through the mucosa due to decreased keratinization of mucosal epithelium.
  - Clinically lesion may look like granular and/or interspersed with areas of leukoplakia, which is usually indistinguishable from erythroleukoplakia type of leukoplakia.
  - Most of the erythroplakia lesions show severe dysplasia, carcinoma in situ or frank invasive carcinoma. The malignant potential is 17 times higher than in leukoplakia.
  - Treatment of this lesion needs excision biopsy either surgically or CO_2 laser and regular follow-up.
- *Proliferative verrucous leukoplakia*: This uncommon variant of leukoplakia is multifocal and persistent and occurs

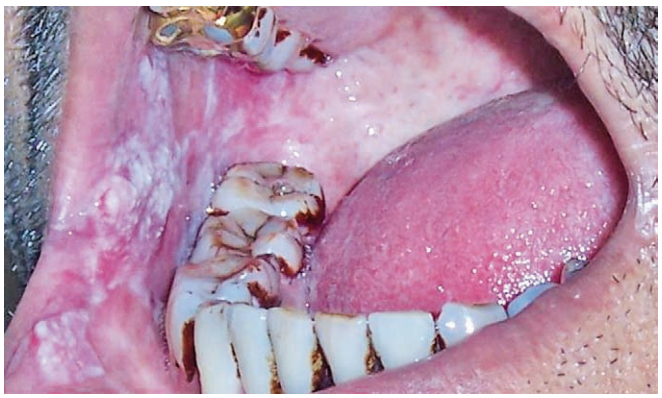

Fig. 4: Leukoplakia of buccal mucosa. Smooth and wrinkled white patches. Macular lesions and more opaque elevated thickened and furrowed with leathery and wrinkled appearance

Fig. 5: Leukoplakia tongue anterior two-third lateral margin without induration. Note nodular white patches with erythematous base

Fig. 6: Erythroplakia tongue. Bright red velvety red patch on the lateral margin of tongue

more often in women. A thin flat white patch progresses to leathery thickened and papillary to verrucous quality. Recurrence rate is high and 70% cases develop squamous cell carcinoma.

Histopathology

- It ranges from hyperkeratosis and acanthosis to dysplasia (disordered cell growth and architectural distortion) or carcinoma in situ to invasive squamous cell carcinoma. The dysplasia is traditionally graded as mild, moderate, or severe.
- Carcinoma in situ shows dysplasia in the entire epithelial compartment (top-to-bottom effect) or severe epithelial dysplasia. Cellular atypia, which is component of dysplasia, refers to abnormal cellular features.
- About 25% of leukoplakias show epithelial dysplasia that may be from mild to severe grades. Higher grade of dysplasia indicates increased chances of malignant change.
- A clinical shift from homogeneous to heterogeneous, speckled, or nodular form is an indication for rebiopsy.

Potential for Malignant Change

- Leukoplakia is the most common premalignant oral mucosal lesion.
- The chances of malignant change are from 1–17.5% (average 5%), which varies according to the site, type and duration of the lesion and age of the patient.
  - *Age and duration*: More the age and duration of the lesion greater are the chances of malignant change.
  - *Site:* Leukoplakia of floor of the mouth and ventral surface of tongue have higher incidence of malignant change.

Molecular Biology

- *Loss of heterozygosity*: Loss of heterozygosity (LOH) at 3p and 9p with additional losses at 4p, 8p, 11q or 17p indicates 33-fold increase in risk of cancer development.
- *Aneuploidy*: Eighty four percent of precancers having aneuploidy develop carcinoma.

Management

- *Biopsy*: An incisional biopsy (multiple biopsies in extensive lesions) must be taken from suspicious areas (such as erythematous, granular, ulcerated and indurated) to know the grades of dysplasia and rule out malignancy.
- *Benign or minimal dysplasia*: Observation or excision. Spontaneous regression is not uncommon in homogenous variety if incriminating factors are removed.
- *Premalignant lesions of moderate to severe dysplasia*: Excision.
  - Methods of complete removal: Scalpel excision, laser ablation, electrocautery or cryoablation.
- *Chemoprevention*: It is indicated in treated cases or mild dysplasia. The agents include retinoids, antioxidants, cyclooxygenase (COX)-2 inhibitors.

ORAL HAIRY LEUKOPLAKIA

- *Etiology*: This asymptomatic white lesion of oral cavity, which is considered to be caused by Epstein-Barr virus, has a relationship with immunosuppression.
- *Most common site*: Bilateral tongue margins. Less common sites include dorsum of tongue, buccal mucosa and floor of mouth.

- *Lesions*: Range from macular smoothly textured lesions, subtle white keratotic vertical streaks to thick corrugated ridges and shaggy surface.
- *Diagnosis*: Histology and demonstration of Epstein-Barr virus.
  - *Histology*: Hyperkeratosis and irregular surface projections and irregularities.
  - *Demonstration of Epstein-Barr virus*:
    - In situ hybridization
    - Southern blot procedure
    - Polymerase chain reaction (PCR)
    - Ultrastructural study
- *Treatment*: It is not required. Lesion disappears with antiretroviral therapy in HIV.

ORAL LICHEN PLANUS

This mucocutaneous immunologic disorder is relatively common (0.2–2% of population). There occurs T-cell lymphocytic reaction to surface epithelial antigen.

Clinical Features

- Oral lesions may be associated with skin lesions, which consist of pruritic, purple, polygonal papules that are seen on the forearms and medial side of thigh.
- The multifocal and bilateral nature of lesion differentiates lichen planus from other oral mucosal disorders.

Clinical Forms

Oral lesions present in various forms such as reticular, plaque, atrophic, erosive and bullous. Concomitant presence of various forms is not uncommon. In cases of erosive lichen planus or atrophic lichen planus, there is risk of malignant change.

- *Reticular lichen planus*: Symmetrical bilateral asymptomatic buccal lesions often in lower mucobuccal folds are seen in middle-aged population. White keratotic striae form lace-like pattern over a normal or erythematous mucosa. Other less common sites include dorsum and lateral portion of tongue, gingiva and vermilion surface of lip. No active treatment except reassurance is required.
- *Erosive lichen planus*: It is characterized by painful ulcer on the buccal mucosa, gingivae or lateral tongue, which is surrounded by a keratotic periphery. Treatment consists of topical steroids.
- *Atrophic or erythematous lichen planus*: Thinned edematous glossy reddened mucosa with loss of surface keratinization dominates faint white striae.
- *Bullous lichen planus*: In this rare variant the bullae: Size range from few millimeters to over 1 cm. They rupture and result in painful ulceration.

Treatment

- *Corticosteroids*: Topical, intralesional injections and systemic.
- *Other alternatives*: Hydroxychloroquine, azathioprine and retinoids.

CHRONIC DISCOID LUPUS ERYTHEMATOSUS

- Oral lesions are similar to those of erosive form of lichen planus. They are always associated with skin lesions.
- Oral lesions consist of circumscribed and little elevated white patches with surrounding telangiectatic halo.
- Malignant change usually occurs in labial lesion near vermilion border in males. These patients should avoid bright sunlight by the application of ultraviolet barrier cream to the lips.

CANDIDIASIS (MONILIASIS)

This infection is caused by *Candida albicans and has* two forms: thrush and chronic hypertrophic candidiasis.

Risk Factors

They include following:

- *Systemic:* Diabetes, antibiotics, age and immunosuppression.
- *Local:* Corticosteroids, xerostomia, heavy smoking and dentures.

Clinical Features

Symptoms range from none to burning, dysgeusia, sensitivity and generalized discomfort. Odynophagia indicates involvement of pharynx.

Acute Pseudomembranous Candidiasis (Thrush)

- *Age*: This condition can be seen in infants, children and adults.
- *Lesion*: Thrush presents as white/gray patches on the oral mucosa and tongue, which when wiped off, leave an erythematous mucosa.
- *Predisposing conditions*: Adults are usually affected when they are either immunocompromised or are dehydrated and suffering from diabetes, AIDS or some systemic malignancy/taking broad spectrum antibiotics, cytotoxic drugs, steroids or radiation.
- *Treatment* includes:
  - Topical application of nystatin or clotrimazole.
  - Systemic antifungal agents are fluconazole, itraconazole and ketoconazole.
  - Management of predisposing condition.

Chronic Hypertrophic (Hyperplastic) Candidiasis or Candidal Leukoplakia

This invasive *C. albicans* infection has high incidence of malignant change.

- *Most common site*: The lesion mostly affects anterior buccal mucosa often placed posterior to labial commissure along the occlusal line. A triangular pattern is seen with its apex directed posteriorly. Other less common sites include palate and lateral tongue surface.
- *Appearance*: The dense chalky plaques of keratin cannot be wiped off. They are thicker and more opaque than noncandidal leukoplakia.

- *Treatment*
  - *Surgery*: This condition usually requires excisional surgery.
  - *Antifungal*: Long-term (many months) antifungal (such as nystatin, amphotericin or miconazole) therapy eliminates candidal infection and reduces the risk of malignant transformation.

Median Rhomboid Glossitis

- The exact cause of this condition is not well understood. Some believe it to be a persistence of tuberculum impar. Recent studies have revealed chronic candida infection.
- As the name suggests a red rhomboid area, devoid of papillae is seen on the dorsum of tongue in front of foramen cecum.
- The condition is asymptomatic and an incidental finding and does not need treatment.

Oral thrush in adults: The common risk factors are corticosteroid and broad spectrum antibiotics, pregnancy, diabetes mellitus, nutritional deficiency and human immunodeficiency virus.

Other Clinical Forms of Candidiasis

- Acute erythematous candidiasis
  - A painful erythematous lesion on the hard palate can develop after a course of broad-spectrum antibiotics.
- Chronic erythematous/atrophic candidiasis
  - This is a type of denture sore mouth which can remain asymptomatic.
- Angular cheilitis
  - Fissured, macerated or erythematous lesion involves angle of mouth (oral commissure) and extends on to the adjacent skin of the face. Associated staphylococcal infection may be present.

FORDYCE'S SPOTS

- The aberrant sebaceous glands may be seen as yellowish or yellow-brown spots, which shine through the buccal or labial mucosa.
- They are seen equally in both the sexes and are considered normal.

NICOTINE STOMATITIS

- The condition is a misnomer as nicotine is not the cause of this disease.
- In smokers (especially reverse smoking) palatal mucosa shows pin point red spots in the center of umbilicated papular lesions, which are due to inflammation of the minor salivary glands. The openings of the ducts of minor salivary glands react to the heat of the smoke.
- *Treatment*: Patients are advised to give up the habit of smoking.

VESICULOBULLOUS/ULCERATIVE LESIONS

PEMPHIGUS VULGARIS

This autoimmune mucocutaneous life threatening disorder is characterized by intraepithelial cleavage and affects older age group of 50–70 years.

Etiopathology

Adhesion molecule desmogleins (transmembrane glycoproteins) DSG1 (skin) and DSG3 (oral and oropharyngeal mucosa) are pemphigus antigens, which are targeted by autoantibodies of IgG class that are deposited within intercellular space and produces direct damage to the desmosomes.

Clinical Features

Upper aerodigestive tract lesions precede skin lesions by months to years in more than 70% of the patients.

- *Lesions*: The initial vesiculobullous lesions produce erosions, blisters, ulcers and pain that tend to run a chronic course. In contrast to pemphigoid, pemphigus ulcers heal faster and without scarring. Healing is followed by formation of new lesions.
- *Sites*: Predominantly oropharynx, soft palate and buccal and labial mucosa. Erythematous and friable gingival marginal lesions bleed easily on slightest provocation. They extend to alveolar mucosa.

Diagnosis

- *Clinical*
- *Histological:* Clinically intact mucosa near the pemphigus ulcer shows separation of suprabasal layer (parabasal and superficial epithelium) from basal layer of the overlying epithelium.

3. *Direct immunofluorescence examination*: Fluorescence of intercellular space regions with anti-IgG antibody is diagnostic for pemphigus vulgaris.

Treatment

Treatment includes systemic steroids and cytotoxic drugs.

1. *Initial treatment*: Prednisone 1 mg/kg supplemented by azathioprine or mycophenolate moftil.
2. *Severe cases*: Cyclophosphamide in conjunction with plasmapheresis.
3. *Recalcitrant cases:* Intravenous IgG .

MUCOUS MEMBRANE PEMPHIGOID OR CICATRICIAL PEMPHIGOID

This is a heterogeneous cluster of autoimmune subepithelial disorder.

Etiopathology

Autoantibodies of IgG and C_3 are directed against the molecular components of basement membrane zone. Linear deposition of IgG and C_3 at the mucosal basement zone is characteristic.

Clinical Features

- *Head and neck sites*: Oral mucosa is most commonly involved followed by ocular (conjunctiva), nasal, nasopharyngeal, laryngeal and esophageal areas. Keratinized tissue of palatal and gingival area is more commonly affected than buccal.
- *Lesions*: Patchy distribution of vesicles and bullae and erythematous features. Bulla filled with clear or hemorrhagic fluid ruptures to form superficial ulceration, which

are covered with shaggy collapsed mucosa. Intraoral scarring is less frequent than ocular scarring that can lead to symblepharon, ankyloblepharon, corneal opacification, entropion and trichiasis.

Diagnosis

- *Clinical*
- *Histopathology*: Separation of mucosal epithelium from the underlying lamina propria and absence of significant inflammation. Biopsy should be taken from an area near the inflamed, erosive, or bullous lesion.
- *Direct immunofluorescence*: Linear IgG fluorescence along the basement membrane is characteristic.

Treatment

- *Mild disease*: This includes:
  - Dapsone for 12 weeks
  - Tetracycline/nicotinamide
- *Oral mucosa involvement*: Topical corticosteroids of moderate to high potency.
- *Gingival involvement*: Similar to pemphigus, skin lesions may be absent and treatment consists of steroids.
- *Severe and rapidly progressive disease*: Systemic prednisone and cyclophosphamide.

HERPES SIMPLEX VIRUS: HERPETIC GINGIVOSTOMATITIS OR OROLABIAL HERPES

The herpes simplex virus infection has two types of clinical presentation: primary and secondary.

Etiopathology

Human herpes simplex virus (HHV-1) binds to keratinocytes and neurons and result in vesicles and migration of viruses to nerve ganglion (trigeminal, vagus, dorsal root and sympathetic), where replication occurs between 2 and 10 days of provocation of recurrent infection.

Primary Herpes Simplex Infection

- *Prevalence*: Affects 60–90% of population. Common in children and less common in adults.
- *Spread*: Direct contact and saliva of recovering or distantly infected persons.
- *Incubation period*: It is 5–7 days.
- *Prodrome* (48 hours): Focal mucosal erythema and tenderness.
- *Lesion*: Group of thin-walled, delicate and short-lived clusters of multiple small vesicles which like herpangina rupture and form ulcers surrounded by inflammation.
- *Site*: Any part of the oral cavity both keratinized and nonkeratinized can be involved.
- *Marginal gingivitis*: Classically gingiva appears erythematous, boggy and tender with wide spread vesicles and ulcers. The disease can involve oropharynx and perioral skin (Fig. 7).
- *Resolution*: Clinical resolution occurs within 7–14 days.

Fig. 7: Herpetic gingivostomatitis involving perioral skin

Laboratory Investigations

Though generally not required as the diagnosis is mainly clinical following are the valid techniques for confirmation of diagnosis:

- Viral isolation and culture.
- Cytological analysis of vesicle content.
- Serum antibody titers.

Treatment

- Symptomatic and supportive.

Secondary Herpes Simplex Infection or Recurrent Herpes Simplex Infection

In recurrent human herpes simplex virus (HHV-1) infection, virus lies dormant in the trigeminal ganglion. Once reactivated, they travel along peripheral sensory nerves and involve oropharyngeal mucosa.

Clinical Features

- *Age*: It usually affects adults and is milder in form as adults develop some immunity to herpes virus.
- *Provocations*: Some of the common precipitating factors are emotional stress, fatigue, fever, pregnancy or immune deficiency states.
- *Prodrome*: Painful, tingling, or burning with subsequent vesicles at the site.
- *Lesions:* Pinhead size clustered vesicles occur over erythematous and edematous background. After 1–2 days vesicles rupture and form tender ulcers and ultimately crusting. Crusting phase is of 5–7 days. Ulcers heal without scarring. In immunosuppressed patients ulcers are big and scarring occurs.

Clinical Forms

- *Herpes labialis*: This is the most common clinical form of recurrent herpes (Fig. 8). The frequency ranges from 5–23%. The site of affection is the vermilion border of the lip, skin vermilion junction and adjacent skin. The site remains same in repetitive episodes.

Fig. 8: Herpes labialis

- *Recurrent intraoral herpes simplex*
  - *Typical*: Multiple, small, closely cropped tender erosions and ulcers occur within the keratinized epithelium of hard palate and attached gingiva. On the hard palate lesions are seen unilaterally along the distribution of greater palatine nerve particularly in the first molar and premolar areas. In mandibular gingiva also the site of predilection is molar and premolar regions.
  - *Unusual presentation*: Widely scattered vesicles and ulcers in association with pain, tenderness and fever in adults.

Diagnosis

- *Biopsy*: Intraepithelial vesicle (filled with serum and free floating, virally infected keratinocytes) in association with mixed inflammatory infiltrate.
- *Smear preparation by unroofing vesicle*: Enlarged infected keratinocytes with multilobulated viral inclusions (Tzanck cells).
- *Viral culture*: Vesicular fluid.

Treatment

- *Topical:* Topical docosanol cream and penciclovir cream for herpes labialis.
- *Systemic*: Immunocompetent adults usually do not require the specific treatment, which includes acyclovir, 200 mg, five times in a day for 5 days that helps in cutting down the course of recurrent herpes labialis.

HAND, FOOT AND MOUTH DISEASE

In this viral infection, which usually affects children, vesicles occur not only in oral cavity (palate, tongue and buccal mucosa) but also on the skin of hands, feet and sometimes even buttocks.

HERPANGINA

- *Causatives organism*: Coxsackie viral infection.
- *Age*: It mostly affects children.
- *Lesion*: There occurs multiple, small vesicles which rupture to form small ulcers. These ulcers are usually 2–4 mm in size and have a yellow base and red areola around them.
- *Most common sites:* Movable mucosa of the faucial pillars, tonsils, soft palate and uvula.
- *Treatment*: No special treatment is needed. Ulcers usually heal by themselves within a week time.

ACUTE NECROTIZING ULCERATIVE GINGIVITIS

- *Causative microorganisms*: Vincent's infection, anaerobic fusiform bacilli and spirochete (*Borrelia vincentii*).
- *Age*: Usually affects young adults and middle-aged persons.
- *Lesions*: Lesion starts at the interdental papillae and then spreads to free margins of the gingivae. Gingivae become red and edematous.
- *Vincent's angina*: Ulcers get covered with necrotic slough. They can be seen not only over the gingivae but also on the tonsil.
- *Diagnosis* can be confirmed by smear from the affected area.
- *Treatment*: It includes
  - Systemic antibiotics which also cover the anaerobes (penicillin or erythromycin and metronidazole).
  - Frequent mouth washes with sodium bicarbonate solution.
  - Attention to dental hygiene.

RECURRENT APHTHOUS STOMATITIS

This most common nontraumatic form of oral ulcerative disease chiefly affects oral and oropharyngeal mucosa.

Etiology

The exact etiology is not well understood. The multifactorial etiology may be at play, include following factors:

- *Autoimmune disease*
  - Both T-cell mediated and antibody-mediated processes
- *Predisposing factors*
  - Local physical trauma
  - Ultraviolet light
  - Psychological stress
  - Hormonal influences
  - Professional groups
  - Higher socioeconomic status
  - Nonsmokers and nonusers of smokeless tobacco
- *Diseases having oral aphthous ulcerations*
  - Crohn's disease/ulcerative colitis.
  - Behçet's syndrome.
  - Celiac disease/nontopical sprue: Malabsorption/Gluten-sensitive enteropathy.
  - Food hypersensitivity: Nuts (walnuts, hazelnuts, Brazil nuts), spices, tomatoes, and chocolate.
  - HIV.
  - Nutritional: Hematinic and other deficiency states such as vitamin B_{12}, folic acid and iron.
  - Sweet's syndrome: Acute febrile neutrophilic dermatosis.
  - Periodic fever, aphthous stomatitis, pharyngitis and adenitis syndrome.
- Drug-induced aphthous-type oral ulcerations
  - Non-steroidal anti-inflammatory drugs
  - Beta-blockers
  - Potassium channel blockers.

Clinical Features

- *Lesions and their sites*: It is characterized by recurrent, painful and superficial ulcers on the movable mucosa of oral cavity (lips, cheeks, tongue and floor of mouth) and oropharynx (soft palate and tonsillar pillars). It spares fixed mucosa of the hard palate and gingivae. Absence of vesicles and blistering and involvement of only nonkeratinized mucosa differentiate it from herpes infection.
- *Clinical forms*: The clinical forms are divided into three classes: minor, major and herpetiform aphthous ulcers (Table 1).
  - *Minor*: Most common form. Small multiple ulcers occur in anterior mouth (Figs 9 and 10).
  - *Major*: Major ulcers are deeply created, very big (2–4 cm) and sharply marginated (Fig. 11). In immunocompromised patients major ulcers are more severe, deeper and painful and last for longer than 6 weeks time and may serve as a marker for HIV progression.
  - *Herpetiform*: The disproportionate pain, adult onset and absence of vesicles differentiate herpetiform ulcers from herpes ulcerations (Fig. 12).

Differential Diagnoses

Aphthous ulcers should be differentiated from recurrent herpes simplex infection (Table 2).

Treatment

- *Mild and infrequent episodes*: Symptomatic treatment.
  - Lignocaine viscous helps in relieving local pain.
  - Topical application of steroids and cauterization with 10% silver nitrate help many patients.
  - Tetracycline (250 mg) dissolved in 50 ml of water four times a day as mouth rinse and then to be swallowed.
- *Severe and continuous episodes*: Short-term systemic steroids.
- *Major ulcers*: Intralesional steroids.

TABLE 1 Clinical forms of recurrent aphthous stomatitis

	Minor aphthous ulcers	*Major aphthous ulcers*	*Herpetiform-type aphthous ulcer*
Frequency	Most common (85%)	Less common (10%)	Least common (5%)
Size of ulcer	Up to 10 mm	More than 10 mm	Pinhead to 2 mm
Site of ulcer	Nonkeratinized mucosa of anterior oral cavity	Posterior oral cavity/oropharynx	Wide spread, rarely keratinized mucosa
Type of ulcer	Central necrotic area surrounded by red halo	Deep with sharp margins	Shallow crater form
Number of ulcers	Multiple	Usually single	Multiple
Duration	7–10 days	6 weeks	7–10 days
Pain	Moderate	Very painful	Moderate
Odynophagia	Usually absent	Present	Usually absent
Healing	Without scarring	With scarring	Without scarring

Fig. 9: Minor aphthous ulcers on nonkeratinized labial mucosa of anterior oral cavity

Fig. 10: Minor aphthous ulcers on nonkeratinized mucosa of anterior oral cavity. Small multiple ulcers on tongue, anterior tonsillar pillar and soft palate

Fig. 11: A major aphthous ulcer. Deeply crated very big ulcer with sharp margins involving right anterior tonsillar pillar and soft palate

Fig. 12: Herpetiform aphthous ulcers (very small and multiple) on nonkeratinized mucosa of soft palate and absence of vesicles

BEHÇET'S SYNDROME

This oculo-oro-genital syndrome is characterized by a triad of:

- Aphthous-like ulcers in the oral cavity. The edge of the ulcer is characteristically punched out.
- Genital ulcerations.
- Uveitis.

The syndrome can also involve other systems of the body such as joints and central nervous system.

ERYTHEMA MULTIFORME

Erythema multiforme (EM) is a self-limiting, mucocutaneous disease of unknown etiology. It is usually associated with either herpes simplex infection or drug ingestion (antiseizures and sulphonamides).

Types

The disease may be acute (more common), self limiting or chronic, mild or severe, mucosal, cutaneous or both. It can have overlapping spectrum of following three degrees:

1. EM minor: Self limited, mild disease of skin with minimal oral involvement.
2. EM major or Stevens-Johnson syndrome (SJS): Fulminant, progressive, epithelial necrosis of skin and mucosa.
3. Toxic epidermal necrolysis (TEN).

Triggers

They include:

- *Viral infections*: Recurrent HHV-1 (strongest and most common trigger), Epstein-Barr virus, chronic hepatitis C and parvovirus B19.
- *Drug-induced EM*: Antiseizure drugs (carbamazepine and phenytoin) and sulfonamides are most common. Others include antibiotics and analgesics.
- *Photosensitivity.*

Clinical Features

It has rapid onset and involves skin and/or mucous membranes. About 25% of patients have only oral lesions. Mucosal and cutaneous bullae or ulceration occur in symmetrical distribution.

- Oral mucosal lesions
  - *Lesions*: Oral mucosal vesicles or bullae soon rupture and form irregular size and shape ulcers, which are covered with pseudomembrane (fibrinous plaque) and bleed easily.

TABLE 2 Differentiation between recurrent intraoral herpes simplex and recurrent aphthous stomatitis

	Recurrent intraoral herpes simplex	*Recurrent herpetiform aphthous stomatitis*
Cause	HHSV I/II	Exactly not known
Ultraviolet light and foods	No relation	Triggers
Mucosa involved	Keratinized	Nonkeratinized
Prodrome	Common	Uncommon
Vesicle phase	Present	Absent
Pain	Relatively less	Disproportionately greater
Biopsy	Viral cytopathic effect	Nonspecific
Treatment	Antiviral (topical and systemic)	Steroids (topical and systemic)

- *Site*: Any part of oral mucosa can be involved but the common sites are lips, buccal mucosa and tongue.

Diagnostic feature: Hemorrhagic crusts on the vermilion portion of lips with edema and severe tenderness are the distinctive feature.

- *Oral and oropharyngeal dysfunctions*: Sialorrhea, pain, odynophagia, dysarthria, inability to chew and swallow.

- Other mucosal sites
  - Eyes, genitalia, esophagus and pulmonary tract.
- Skin lesions
  - Target or iris lesions (concentric erythematous to pigmented patches) on the palms, soles and extensor surfaces of the extremities can be seen if the skin is involved.

Differential Diagnoses

In the absence of iris or target lesions, other mucocutaneous diseases (severe aphthous stomatitis, pemphigus, pemphigoid and erosive lichen planus) should be ruled out. If necessary, biopsy should be taken.

Treatment

- *Specific* treatment is controversial.
- *Symptomatic treatment*: Analgesics, oral hygiene, bland mouth rinses, topical steroids, antifungal, and anesthetics.
- *Short-course of corticosteroids*: In EM minor cases as the disease is self limiting.
- *Antivirals*: In cases of prior HHV-1 infection.

TRAUMATIC (EOSINOPHILIC) GRANULOMA

- *Etiology*: Though it is said to be due to deep mucosal injury, many consider it of unknown etiology as these patients never have an event of trauma.
- *Lesion*: It is characterized by benign, large (1–2 cm), self-limiting and chronic (weeks to months) oral painful ulcer, which occurs in and after fifth decade of life. Ulcer has craterform center, sharp margins and milky white firm periphery.
- *Site*: This rapid onset ulcer usually develops along the lateral and ventral surface of tongue. Occasionally, ulcer may be seen on the dorsum of the tongue.

Treatment

- Observation.
- Topical or intralesional corticosteroids.
- Excisional biopsy.

TRAUMATIC ULCERS

- An ulcer on the lateral border of tongue may be due to jagged tooth or ill-fitting denture.
- Cheek bite causes ulcer on the buccal mucosa.
- Injury with a pencil or tooth brush can lead to ulcer on the palate.
- Accidental ingestion of acids or alkalis or hot fluids presents with acute ulcerative lesions of oral and oropharyngeal mucosa.
- A tablet of aspirin, kept against a painful tooth to get relief from toothache may lead to aspirin burn, which is seen in the gingivobuccal sulcus.

RADIATION MUCOSITIS

- Radiation therapy can affect the oral and pharynx mucosa. The mucosa initially becomes red and later on forms spotty areas of mucositis which coalesce to form large ulcerated areas that are covered by slough.
- Mucositis of cancer chemotherapy (such as methotrexate, 5-fluorouracil and bleomycin) manifests as erythema, edema and ulceration.

BLOOD DISORDERS

- *Acute leukemia*: Acute lymphoblastic leukemia occurs in young children while acute myeloid leukemia affects middle aged or elderly people. It can cause hypertrophy of gums with ulceration and bleeding.
- *Agranulocytosis*: It may present as ulcerations in throat with severe neutropenia.
- *Cyclical neutropenia* (periodic falls in neutrophil count): Patients are prone to infections and oral ulceration.
- *Pancytopenia:* There occurs a drop in RBC count, white cell count and platelets. CBC and peripheral blood films usually indicate the diagnosis, which further needs the study of bone marrow aspiration.

DRUG-INDUCED ORAL LESIONS

- Drugs like penicillin, tetracycline, sulfa drugs, barbiturates and phenytoin may cause erosive, vesicular or bullous lesions in the oral cavity.
- Contact stomatitis can also occur due to local reaction to mouth washes, lozenges, chewing gum, tooth paste or to prosthetic dental materials. Oral lesions vary from erythema to vesicles and bullae formation.

PIGMENTED LESIONS

Benign pigmented lesions (Fig. 13) have the potential of changing into malignant melanomas. About one-fourth of mucosal melanomas resemble benign lesions therefore biopsy becomes mandatory.

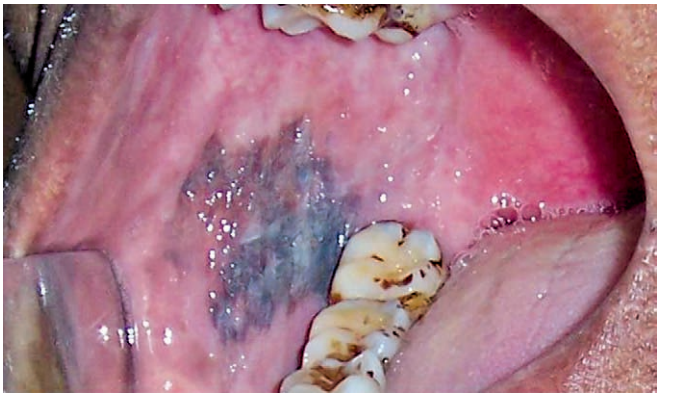

Fig. 13: Benign pigmented lesion buccal mucosa. Macular zone of homogeneous hyperpigmentation with well-defined margins

MELANOTIC MACULES

The most common sites are the vermilion portion of lower lip (30%) and gingiva and alveolar mucosa (23%). Other sites include buccal (16%) and labial mucosa (9%).

Lesion

- *Discrete* uniformly pigmented macules.
- *Mucosal melanotic macule*: About 10 mm macular zone of homogeneous hyperpigmentation with well-defined margins. Focal melanosis (smoker's melanosis) present over buccal mucosa as scattered macular evenly pigmented patches.
- *Mucosal melanotic nevi*: Macular to papular hyperpigmented lesions can appear in young or at birth.

Treatment

- Excisional biopsy.

MELANOMA

Oral mucosal melanoma is a rare neoplasm.

- The most common sites are hard and soft palate (40%) and gingiva (30%).
- Preexisting melanosis presents lateral spread or superficial spreading melanoma in adults. Initially the lesion is heterogeneously pigmented and flat with irregular margin. As it increases in surface area, the degree of pigmentation increases to deeper brown to gray-brown.
- Nodular melanoma arises ab initio as a rapidly growing nodule.

Treatment

- Extent of surgical excision depends upon the depth, size and site of the melanoma.

Prognosis

- The prognosis is poor. Five year survival rate is 15%.

AMALGAM TATTOO

This is an extrinsic pigmentation of oral mucosa which occurs due to traumatic implantation of dental silver amalgam into neighboring oral mucosa.

Lesion and Its Site

- Well defined, macular grayish-black focal usually 4 mm size (1 mm to 1.5 cm) mucosal discoloration in lower jaw gingiva, mucobuccal fold or buccal mucosa.

Treatment

- Simple surgical excision.

LESIONS OF TONGUE

GEOGRAPHICAL TONGUE OR MIGRATORY GLOSSITIS

- This asymptomatic condition is characterized by erythematous area, which is devoid of papillae and surrounded by an irregular keratotic white outline (Fig. 14).
- The shape of lesions keeps on changing.
- It does not require any treatment.

HAIRY TONGUE

- Clinical Features: The excessive formation of keratin causes elongation of the filiform papillae on the dorsum of the tongue. Due to chromogenic bacteria, they look like brown or black color hair.
- *Smoking*: It could be one of the factors.
- *Treatment*
  - Scraping of the lesions with a tongue cleaner.
  - Application of half strength hydrogen peroxide.
  - Improving the general nutritional status by vitamins.
  - Smoking is prohibited.

FISSURED TONGUE (FIG. 15)

It may be congenital or acquired.

- *Acquired*: It may be due to syphilis, deficiency of vitamin B complex or anemia.
- *Congenital:* In Melkersson Rosenthal syndrome, congenital fissuring of tongue (scrotal tongue) is associated with recurrent attacks of facial palsy.

Fig. 14: Geographical tongue. Note the erythematous area that is devoid of papillae and surrounded by an irregular keratotic white outline

Fig. 15: Fissured tongue

TONGUE TIE (ANKYLOGLOSSIA)

A mobile tongue helps not only in speech but also in maintaining orodental hygiene. It cleans the debris and prevents formation of dental plaques.

- True tongue tie, which produces symptoms, is not common. Once the tongue can be protruded beyond the lower incisors it should not cause speech defects.
- Treatment: Thick significant tongue tie needs transverse surgical release with vertical closure. Thin mucosal fold is simply incised.

Clinical Highlights

1. **Fordyce's spots:** They represent normal variants of ectopic sebaceous glands. Fordyce's spot present as granules in oral cavity.
2. **Precancerous lesions:** Oral lesions having malignant potential are erythroplakia (erythroplasia), leukoplakia, lichen planus and submucous fibrosis.
3. **Wickham's striae:** It is a feature of lichen planus.
4. **Leukemia:** Oral findings include pale mucous membrane, gingival hypertrophy and petechial hemorrhages.
5. **Hand, foot and mouth disease:** It is caused by Coxsackie A virus.
6. **HIV:** Oral manifestations include oral candidiasis, hairy leukoplakia and recurrent aphthous ulcers.
7. **Taste buds:** They are in highest number (250) in circumvallate papillae and least (1–18) in fungiform papillae. There are practically no buds in filiform papillae. Foliate papillae have second highest number (100) of taste buds.

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36 Disorders of Salivary Glands

We need to have three things: the heart to feel, the brain to conceive, the hand to work. Make yourself a dynamo. If you are pure, if you are strong, you, one man, are equal to the whole world.

— **Swami Vivekananda**

Points of Focus

- INTRODUCTION

INFLAMMATORY DISORDERS

- MUMPS
- ACUTE SUPPURATIVE SIALADENITIS
- PAROTID ABSCESS
- NEONATAL SUPPURATIVE PAROTITIS
- RECURRENT PAROTITIS OF CHILDHOOD
- CHRONIC SIALADENITIS
  - ✓ Benign Lymphoepithelial Lesion
  - ✓ Kuttner's Tumor
- TUBERCULOUS MYCOBACTERIAL DISEASE
- NONTUBERCULOUS MYCOBACTERIAL DISEASE
- ACTINOMYCOSIS
- CAT SCRATCH DISEASE
- TOXOPLASMOSIS
- HIV-ASSOCIATED SALIVARY GLAND DISEASE

OBSTRUCTIVE DISORDERS

- SIALOLITHIASIS

NEOPLASMS OF SALIVARY GLANDS

- HISTOGENESIS OF NEOPLASMS
- PLEOMORPHIC ADENOMA
- WARTHIN'S TUMOR
- ONCOCYTOMA
- HEMANGIOMA
- LYMPHANGIOMAS
- MUCOEPIDERMOID CARCINOMA
- ADENOID CYSTIC CARCINOMA
- ACINIC CELL CARCINOMA
- SQUAMOUS CELL CARCINOMA
- MALIGNANT MIXED TUMOR
- ADENOCARCINOMA
- LYMPHOEPITHELIAL CARCINOMA

XEROSTOMIA

- SJÖGREN'S SYNDROME
- DIFFUSE INFILTRATIVE LYMPHOCYTOSIS SYNDROME
- FREY'S SYNDROME
- CLINICAL HIGHLIGHTS

INTRODUCTION

Salivary gland disorders can be broadly divided into following categories:

- Congenital
- Inflammatory
- Obstruction and trauma
- Neoplasms
- Degenerative conditions

Congenital disorders, which include aplasia/agenesis, ductal atresia, fistula and ectopic salivary tissue, are rare.

INFLAMMATORY DISORDERS

MUMPS

Mumps is derived from Danish word "mompen" that means mumbling like an old man, which occurs in mumps due to pain and trismus.

Etiology

Mumps is an acute viral parotitis, which is caused by the paramyxovirus (RNA virus). Other viruses, which can cause acute viral parotitis, are: coxsackie viruses A and B, enteric cytopathic human orphan virus, cytomegalovirus and lymphocytic choriomeningitis virus.

Epidemiology

- It is the most common cause of nonsuppurative acute sialadenitis. It mostly affects children.
- Mumps is highly contagious.
- The peak incidence occurs in the spring in temperate climates (little variation in tropics).
- The paramyxovirus is endemic in the community. It is disseminated by means of airborne droplets from salivary, nasal and urinary secretions.
- This paramyxovirus enters through the upper respiratory tract and then localizes in glandular and central nervous system tissue.
- It has an incubation period of 2–3 weeks.
- Viral infection of salivary glands may be locally asymptomatic. The transmission from blood to saliva occurs without localizing signs in many systemic viral infections such as rabies, hepatitis, influenza and poliomyelitis.

Clinical Features

- *Viral prodome*: Low-grade fever, headache, myalgia, anorexia, arthralgia and malaise just before parotid swelling.
- Mumps is characterized by localized pain, which is exacerbated by chewing.
- Parotid gland swelling is tense and firm.
- Painful swelling of the gland causes displacement of the pinna, otalgia, trismus and dysphagia.
- There is bilateral parotid gland swelling in 75% of cases but submandibular gland might be affected in rare cases. Usually one side parotid will swell first followed by enlargement of the other gland in 1–5 days.
- The overlying parotid skin is stretched with a glazed appearance, but there is usually no erythema or warmth.

Investigation

- *Viral serology*: Complement fixing soluble (S) antibodies against the nucleoprotein core of the virus is associated with active infection and their levels peak at 10 days to 2 weeks and disappear within 8–9 months. A fourfold increase in antibody titer is diagnostic for acute infection. Complement fixing viral (V) antibodies against outer surface hemagglutinin appear later than S antibodies and persist at low levels for many years.
- A leukocyte count may show leukopenia.
- There is an elevation in the serum salivary type amylase.

Treatment

Supportive measures include

- Bed rest
- Oral hygiene
- Hydration
- Dietary modifications to minimize glandular secretory activity.

Fever usually subsides before the resolution of glandular edema, which may take several weeks.

Complications

Orchitis, aseptic meningitis, pancreatitis, nephritis and sensorineural hearing loss.

Prevention

Subcutaneous injection of vaccination (live attenuated Jerry Lynn vaccine), usually in combination with measles and rubella vaccines is given after 12 months of age. The antibodies produced by vaccine, persist for at least 5 years. The vaccine is contraindicated in pregnancy, immunocompromised states and allergies to neomycin.

ACUTE SUPPURATIVE SIALADENITIS

The parotid is most commonly involved salivary gland. The parotid gland's serous saliva, unlike mucinous saliva of other salivary glands, is deficient in lysosomes, IgA antibodies and sialic acid, which have antimicrobial properties. The saliva from other glands (submandibular and sublingual glands) contains high molecular weight glycoproteins that competitively inhibit bacterial attachment to the epithelial cells of the salivary ducts.

Predisposing Factors

- *Age*: It usually affects 50 and 60 years old people (equal incidence among men and women).
- *Debilitating conditions*: Malignant lesion and pre-existing infection.
- *Postoperative period*: Major abdominal and hip repair surgery. It occurs within the first two postoperative weeks.
- *Local:* Stenosis and sialolithiasis.
- *Systemic diseases*: Diabetes mellitus, hypothyroidism, renal failure and Sjögren's syndrome.
- *Dehydration or significant hemorrhage*: The retrograde bacterial contamination of the salivary ducts from the oral cavity occurs due to the stasis of salivary flow. It is secondary to dehydration or significant hemorrhage.
- *Medications.*

Causative Microorganisms

- Penicillin resistant *Staphylococcus aureus* in hospitalized patients.
- *Streptococcus pyogenes*, *Streptococcus viridans*, *Streptococcus pneumoniae* and *Haemophilus influenzae* (community-acquired cases).
- Anaerobic bacteria: *Peptostreptococcus*, *bacteroides* species and *fusobacterium*.

Clinical Features

- It usually presents with rapid onset of pain and swelling over the affected salivary gland, fever, chills and malaise.
- Dehydration with dry mucous membranes and local tenderness, warmth and induration.
- Bimanual palpation results in suppurative discharge from the duct orifice.

Investigations

- Leukocytosis with neutrophilia and normal serum amylase.
- Computed tomography (CT) or ultrasound (US) is indicated to look for abscess formation if patient does not respond to medical treatment.
- Cultures of purulent drainage from the duct orifice. Percutaneous needle aspiration limits the amount of contamination.

Differential Diagnoses

Usually the diagnosis is apparent. Differential diagnoses include lymphoma, Bezold's abscess, cervical adenitis, dental abscesses presenting as buccal or masseteric space abscesses, infected branchial cleft or sebaceous cysts.

Treatment

It begins with aggressive medical treatment and includes:

- Prompt fluid and electrolyte replacement, oral hygiene, reversal of salivary stasis and antimicrobial therapy.
- Salivary flow should be stimulated by sialogogues such as lemon drops and orange juice.
- Regular external and bimanual massage, starting from the distal bed of the gland and working in the direction of duct drainage helps greatly in drainage.
- Analgesics and local heat application alleviate discomfort.
- Antimicrobial therapy, which might need change after the culture results, should be continued for 1 week after resolution of symptoms. Antibiotics include augmented penicillin (Beta-lactamase producing bacteria in 75% cases) and antistaphylococcal penicillin or a first-generation cephalosporin, vancomycin or linezolid (for methicillin-resistant *S. aureus* infection) and metronidazole (for anaerobes).
- Surgical drainage of a loculated abscess is done if conservative measures fail.

PAROTID ABSCESS

- Multiple small abscesses may coalesce and form large abscess in an advanced case of suppurative parotitis (Fig. 1).

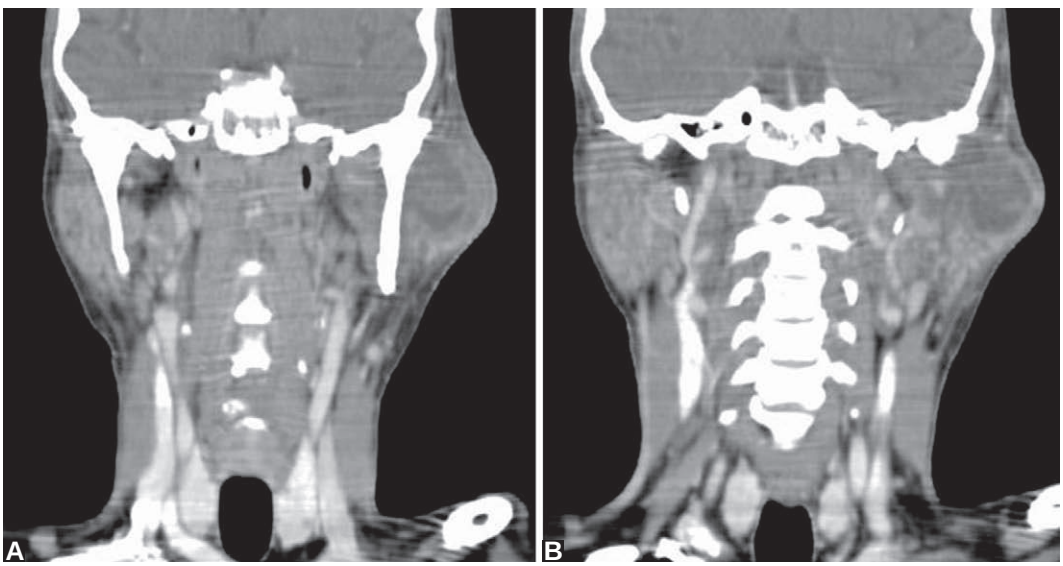
An US examination of the swelling will reveal the abscess. The usual fluctuation may not be elicited due to the dense fibrous capsule (derived from investing layer of deep cervical fascia) of the parotid gland.

- Figures 2A and B show contrast CT scan left parotid abscess.

Treatment

- *Incision and drainage*: In addition to the treatment of acute parotitis abscess needs incision and drainage. It is done under the cover of antibiotics. An anterior based facial flap

Fig. 1: Left side parotid abscess. 57 year-old-male diabetic irradiated patient of right side carcinoma tonsil. Patient had painful swelling that reduced in size with antibiotics and abscess got localized



Figs 2A and B: Contrast CT scan left parotid abscess. Coronal images show peripherally enhancing hypodense lesion in left parotid gland
Source: Dr Swati Shah, Professor, Radiodiagnosis, GCR Medical College, Ahmedabad

is elevated. With blunt dissection and radial incisions in the parotid fascia abscess is drained. Radial horizontal incisions prevent injury to the facial nerve branches which run in same direction. Drain should be placed and the central aspect is left to heal by secondary intention.

Complications

- Suppuration of potential spaces of the face, neck and mediastinum.
- Rupture through the floor of the external auditory canal or spontaneous drainage through the cheek.
- Rare complications are osteomyelitis, thrombophlebitis of the jugular vein, septicemia, respiratory obstruction and even death.
- In cases of facial nerve paralysis, underlying malignancy must be ruled out.

NEONATAL SUPPURATIVE PAROTITIS

This uncommon condition mostly affects parotid gland. Neonatal suppurative parotitis, common in preterm and male neonates, is usually caused by *S. aureus* (most common), *Escherischia coli*, *Pseudomonas aeruginosa* and group B streptococci. Infections are either through oral cavity (most common) or blood borne (usually gram-negative bacteria). Fever, anorexia, irritability and failure to gain weight are present in affected child. Erythema of the skin overlying the involved gland might be observed. The swelling, which is tender, firm or fluctuant often becomes bilateral. Gram's stain and culture of pus from the duct or from fine needle aspiration (FNA) reveals the causative organisms. Drainage is done when prompt clinical improvement does not occur with parentral antibiotics or fluctuance of the gland increases.

RECURRENT PAROTITIS OF CHILDHOOD

It is the second most common inflammatory salivary gland disease of childhood (8 months to 16 years) after mumps. Boys are affected more than girls. This disease of unknown etiology is characterized by periodic episodes of swelling and pain.

Recurrent parotitis of childhood is the second most common inflammatory salivary gland disease of childhood (8 months to 16 years) after mumps.

Proposed Etiologies

- Congenital ectasia of portions of the secondary ductal system predisposes children to *S. aureus* and *S. viridans* colonization
- Autosomal inheritance
- IgG3 and IgA deficiencies
- Juvenile onset primary Sjögren's syndrome
- Viruses are Mumps parotitis, Epstein-Barr virus (EBV) and HIV.

Clinical Features

- Recurrent episodes of acute or subacute unilateral parotid gland swelling (Figs 3 and 4) along with fever, malaise and pain after a meal.
- Exacerbations occur every 3–4 months and last for days to weeks.
- *Sialography*: Sialectasis appears as numerous scattered punctate pools of contrast.
- *Ultrasound*: An enlarged gland with multiple small hypoechoic areas.
- *MR Sialography*: This noninvasive study may be used during acute episodes.
- Pus must be sent for culture.

Treatment

Treatment includes

- Adequate hydration
- Gland massage
- Local heat
- Sialagogues
- Appropriate intravenous penicillinase-resistant antistaphylococcal antibiotics.

Most cases resolve spontaneously in late adolescence.

Fig. 3: Left side acute parotid gland swelling in a 10-year-old girl

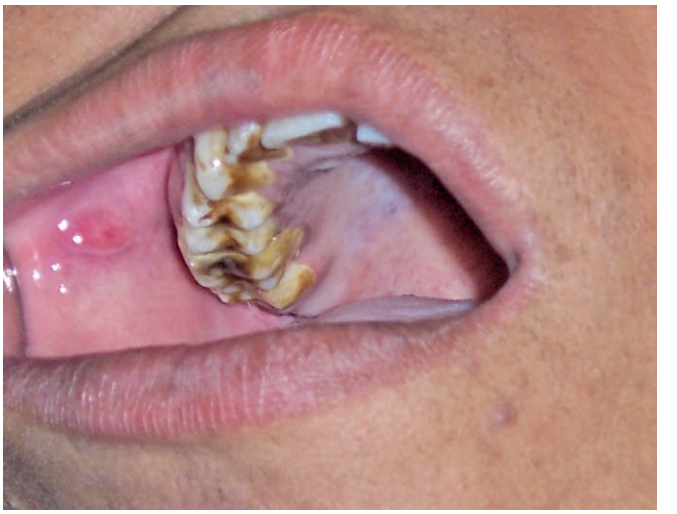


Fig. 4: Stensen's duct opening of right side parotid gland is congested

CHRONIC SIALADENITIS

Salivary stasis, caused by salivary duct obstruction, predisposes to episodes of infection and inflammation. Parotid is the most frequently affected salivary gland.

Inciting Factors

They include sialolithiasis, stricture duct, extrinsic compression by tumor, stenosis secondary to scar, congenital dilation and foreign bodies.

Clinical Features

- Recurrent swelling and tenderness of the affected gland associated with eating.
- Minimal saliva can be milked from the duct orifice.
- Usually the condition is preceded by an attack of acute suppurative sialadenitis.

Treatment

Surgical removal of the gland when conservative management fails to control symptoms.

Complications

They include a benign lymphoepithelial lesion, Kuttner's tumor and ductal carcinoma.

Benign Lymphoepithelial Lesion

- Women in the fifth to sixth decade of life are more commonly affected.
- An asymptomatic mass associated with Sjögren's syndrome is also termed as *Mikulicz's disease*.
- FNA cytology and follow-up are important because there are chances of development of malignancy.

Kuttner's Tumor

- It occurs exclusively in the submandibular gland.
- Middle-aged adults present with painless mass.
- There are chances of developing malignancies, so close follow-up is needed.

TUBERCULOUS MYCOBACTERIAL DISEASE

The most common manifestation of *Mycobacterium tuberculosis* infection in the head and neck is cervical lymphadenopathy. Older children and adults are affected more.

Routes of Infections

- Primary salivary tuberculosis is an uncommon entity.
- Primary infection evolves from a focus in the tonsils or gingival sulcus ascending to the glands by way of their ducts. This may then spread to the cervical nodes through the lymphatic drainage. This most frequently affects the parotid gland.
- Secondary infection of the salivary glands occurs by way of hematogenous or lymphatic spread from the lungs.
- The submandibular gland is the more commonly involved gland after systemic tuberculous infection.

Clinical Features

- *Constitutional signs:* They include fever and night sweats. Weight loss might be absent.

  Clinically, there are two different forms.
- *An acute inflammatory lesion with diffuse glandular edema*: It may be confused with an acute sialadenitis or an abscess.
- *A chronic tumorous lesion*: It is seen as a discrete slow growing mass that mimics a neoplasm.

Differential Diagnoses

Differential diagnoses must include the inflammatory and neoplastic diseases.

Investigations

- Chest X-ray is usually negative.
- CT images show three patterns:
  - Involved lymph nodes are seen as nonspecific homogenous enhancement.
  - A nodal mass is seen with central lucency and thick rims of enhancement and minimally effaced fascial planes.
  - Fibrocalcified nodes are seen in patients previously treated for tuberculosis.
- *Mantoux test*: Purified protein derivative skin test should be positive.
- *Fine needle aspiration cytology*: Characteristic cytologic features include granulomatous inflammation and epithelioid histiocytes.
- *Culture and acid fast smears* of aspiration material.

Treatment

- *Anti-Koch therapy*: Multiple drug therapy.
- *Complete surgical excision*: In cases in which the diagnosis is uncertain or the lesion is resistant to medical therapy. Complete surgical excision is both diagnostic and curative.

NONTUBERCULOUS MYCOBACTERIAL DISEASE

More than 92% of mycobacterial cervicofacial infections in children younger than 5 years of age are a result of nontuberculous mycobacterial (NTM) disease.

Causative Organisms

Mycobacterium kansasii, *Mycobacterium scrofulaceum* and *Mycobacterium avium intracellulare.*

Portal of Entry

It is believed to be through the mouth. Tonsils have been implicated in particular.

Clinical Features

- Rapidly enlarging and persistent neck mass that fails to respond to antibiotic therapy in a pediatric patient.

- The skin becomes adherent to the surrounding tissues and develops a characteristic violaceous discoloration.
- The infection might progress to fluctuation and the development of a draining sinus.
- Associated cervical lymphadenopathy is more commonly unilateral and located in the high jugular nodes or preauricular areas.

Investigations

- CT images shows asymmetric cervical lymphadenopathy with contiguous low density, necrotic, ring enhancing masses involving the subcutaneous fat and skin. Inflammatory stranding of the subcutaneous fat characteristic of bacterial inflammation is minimal or absent.
- NTM-specific antigens have been developed and are reported to be extremely successful for diagnosis.
- Polymerase chain reaction techniques to detect mycobacterial RNA in tissue and gastric aspirates.
- Traditional culture of NTM might take up to 6 weeks and is frequently negative.

Treatment

- Medical treatment consists of prolonged courses of clarithromycin.
- Complete surgical excision of the involved salivary gland and nodes is the treatment of choice.

ACTINOMYCOSIS

The most common clinical infection is cervicofacial (55%). Isolated parotid involvement can occur by means of either retrograde ductal migration or of direct spread of an invasive cervicofacial infection.

Causative Organism

Actinomycosis species is a Gram-positive, anaerobic, non- acid-fast bacillus. They are normal commensal in the tonsils and carious teeth. *Actinomyces israelii* is the most common. Others include *Actinomyces bovis* and *Actinomyces naeslundii*.

Predisposing Factors

Poor oral hygiene, trauma to the mucosa, diabetes, immune suppression, long-term steroid use and malnutrition.

Clinical Features

- Painless, indurated enlargement of the involved gland: It might mimic a neoplasm.
- Multiple draining cutaneous fistulas: It is quite common. A chronic purulent drainage might occur with granulomatous involvement and spread to adjacent tissue. The periphery of the lesion is densely fibrotic and avascular.
- A history of recent dental disease and manipulation is common.

Diagnosis

- CT scans typically demonstrate obliteration of the normal tissue planes and extensive soft tissue destruction.
- Anaerobic cultures are obtained for species identification and to confirm the diagnosis. The recovery rate in culture is < 50%.
- Smears and stains for sulfur granules and the organisms: Needle aspiration of the mass or a fistula swab. Sulfur granules have also been described for nocardiosis.
- Biopsy specimens show firm fibrous encasement of multiloculated abscesses containing whitish yellow purulent discharge.

Treatment

- Penicillin 6 weeks parenteral course followed by an additional 6 months of oral course completely eradicates the organism.
- Other acceptable alternatives include clindamycin, doxycycline, or erythromycin.
- Surgical excision is necessary to remove extensive fibrosis and sinus tracts, when antibiotics fail. It also helps in diagnosis.

Prognosis

Prognosis is generally favorable. Cure rates approach 90% despite a delayed diagnosis in most instances.

CAT SCRATCH DISEASE

Cat scratch disease is a granulomatous lymphadenitis. It results from cutaneous inoculation caused by scratch trauma from a domestic cat.

Causative Organism

Bartonella henselae, Gram-negative bacilli.

The reservoir for *B. henselae* is kittens. The major vector for cats is the cat flea.

Clinical Features

- A papule or pustule at a scratch or bite site followed in 1–2 weeks by the development of lymphadenopathy in the region of inoculation.
- Erythema and fluctuance of the involved nodes with spontaneous suppuration occur in 10–30% of patients.
- Fever and mild systemic symptoms occur in about one-third of patients.
- Preauricular lymphadenopathy might be confused with a parotid neoplasm.

Diagnosis

- *Antibodies to B. henselae*
  - Indirect fluorescent antibody
  - Enzyme immunoassay
- *Polymerase chain reaction*: Bartonella polymerase chain reaction hybridization assay with an aspirate or biopsy specimen.
- *Histological examination with Warthin Starry silver staining*: Lymph node shows reticular cell hyperplasia, granuloma formation and widening of arteriolar walls. In more advanced stages, stellate areas of necrosis coalesce to form multiple microabscesses.

- *Culture*: Bartonella is a slow growing organism and culture requires a 6 weeks incubation period.

TOXOPLASMOSIS

Toxoplasmosis is an uncommon disease.

Causative Organism

Toxoplasma gondii: The usual host for this organism is the domestic cat.

Clinical Features

Both disseminated and lymphadenopathic forms of the diseases have been described.

- Parotid gland disease might involve singular or multiple intraparotid or periparotid lymph nodes.
- *Disseminated form*: Immunocompromised individuals are most at risk for the disseminated form. The features include myalgia, lethargy and anorexia combined with hepatosplenomegaly, pericarditis and myocarditis.
- *Lymphadenopathic form*: It occurs commonly. Most patients present with isolated cervical lymphadenopathy.

Biopsy

Definitive diagnosis can only rarely be provided by isolation of the organism. The characteristic histopathologic findings in affected lymph nodes include following:

- The lymph node architecture is preserved.
- The hyperplastic follicles and germinal centers show abundant mitoses and necrotic nuclear debris.
- Epithelioid cells, with abundant, pale eosinophilic cytoplasm. They occur singly or in groups and are found in cortical and paracortical zones and sinuses.

Confirmation of a presumptive histological diagnosis is made by acute and convalescent serologic testing.

Treatment

Chemotherapy (combined administration of pyrimethamine and trisulfapyrimidines) is generally reserved for obviously progressive infections or those involving pregnant or immunocompromised individuals.

HIV

HIV lesions of salivary glands include Kaposi's sarcoma, lymphoma and lymphoproliferative and cystic enlargement of the major salivary glands with accompanying salivary dysfunction.

HIV-Associated Salivary Gland Disease

It refers to diffuse enlargement of the salivary glands due to HIV. Parotid gland is the most commonly affected. Salivary secretions contain low concentrations of HIV.

Clinical features: Patients usually presents with gradual, nontender enlargement of one or more of the salivary glands. Decreased salivary gland function results in xerostomia and sicca symptoms.

CT demonstrates multiple cysts that appear as low attenuation, thin walled masses and diffuse lymphadenopathy.

MRI reveals homogenous masses of intermediate signal intensity on proton density and T2-weighted images.

Medical treatment consists of zidovudine, maintenance of good oral hygiene and the use of sialogogues.

Excision biopsy: Controversial.

> Sialadenitis: Mumps is the most common parotid viral infection. Less common viral infections are cytomegalovirus, coxsackie and Epstein-Barr viruses. Bacterial sialadenitis is usually caused by coagulase positive *S. aureus. S. pneumoniae*, *E. coli, H. influenzae* and oral anaerobe infections may also occur.

OBSTRUCTIVE DISORDERS

SIALOLITHIASIS

The formation of stones in the salivary ductal system is called sialolithiasis.

Etiology

Exact etiology is uncertain. Salivary stasis and ductal inflammation and injury are important contributing factors.

Eighty to ninety percent of calculi develop in Wharton's duct of submandibular gland. Stensen's duct of parotid constitutes 10–20% and sublingual duct only 1%. The reasons are the following:

- Wharton's duct is longer and has a larger caliber. It is angulated against gravity as it courses around the mylohyoid muscle.
- Submandibular secretions are more viscous and have a higher calcium and phosphorus concentration.

Parotid stones are mostly located at the hilum or parenchyma, while in the submandibular gland, they tend to develop in the duct. Elderly people are more affected than children. Most of the patients are male.

Composition

They are composed mainly of calcium phosphate and carbonate in combination with an organic matrix of glycolproteins and mucopolysaccharides and small amounts of other salts such as magnesium, potassium and ammonium.

Clinical Features

- Recurrent episodes of postprandial salivary colic pain and swelling.
- Past history of recurrent attacks of acute suppurative sialadenitis might be present.
- Bimanual palpation reveals the presence of a stone in most cases.
- Parotid stones might be seen just at the orifice of Stensen's duct or along the course of the duct.

Investigations

- *Plain radiographs* (intraoral or occlusal views) identify radiopaque stones but in the submandibular gland 80% of stones are radiolucent.
- *Ultrasound*: It detects 90% of stones if they are > 2 mm.
- *CT scanning* with fine cuts is very accurate at detecting salivary stones.

- *Digital subtraction sialography*, which lessens the interference of surrounding bony structures, can detect 95–100% of radiolucent stones.
- *MR sialography* of the submandibular duct with evoked salivation is superior to US and has accuracy similar to digital subtraction sialography.

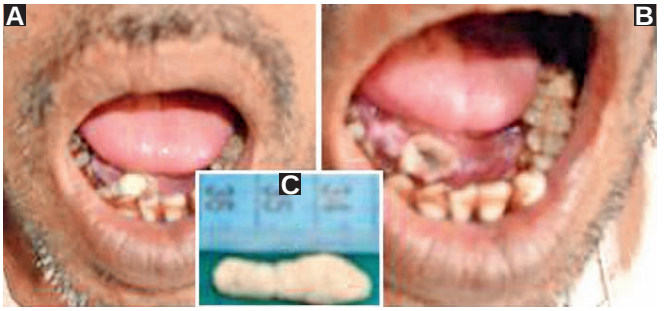
Treatment

- *Nonsurgical management*: It consists of the use of:
  - Sialogogues
  - Local heat
  - Hydration
  - Massaging of the involved gland
  - Antibiotic coverage is started in cases of infection.
  - Manually milking out: Submandibular stones nearer the duct orifice may be manually milked out through the duct opening (Figs 5A to C).
- *Surgical management*: It consists of:
  - *Incision of duct*: Submandibular stones, which are no more than 2 cm from the duct orifice, may be either manually milked out through the duct opening or the duct is incised directly over the stone. There is no need for closure of Wharton's duct after the procedure.
  - *Sialadenectomy*: Submandibular stones located more proximal and near gland will require sialadenectomy, which may be performed either through transcervical or transoral approach. Parotid stones are more difficult to manage because of the anatomy of Stensen's duct.
- *Recent advances*: Use of various combination of baskets, graspers and intracorporeal lithotripsy have been employed to treat sialolithiasis in both the parotid and submandibular glands.
  - Extracorporeal shock wave lithotripsy reduces stones to small fragments, which are then flushed out of the duct with spontaneous salivation or the use of a secretagogue.
  - Sialoendoscopy: Rigid endoscopes are used to visualize and remove salivary duct stones.

NEOPLASMS OF SALIVARY GLANDS

Salivary gland tumors, majority of which are benign, constitute just 3–4% of all head and neck neoplasms. Seventy percent of the salivary gland tumors arise in the parotid gland (Fig. 6). The chances of a tumor being benign are more in major salivary glands (80% of parotid and 50–60% of submandibular) while less in minor salivary glands (25%). Therefore, majority of the minor salivary glands tumors are malignant. The sign and symptoms of malignancy are: rapid growth, restricted mobility, fixity of overlying skin, pain and facial nerve involvement.

The tumors of salivary glands are either of epithelial or mesenchymal origin (Table 1). Pleomorphic adenoma is the most common salivary gland tumor and the number two is mucoepidermoid carcinoma. Other tumors in series of common frequency are: adenoid cystic carcinoma, adenocarcinoma, malignant mixed tumor and Warthin's tumor (second most common benign tumor). The pleomorphic adenoma of the parotid gland needs surgical excision that provides both definitive diagnosis and adequate treatment. Management of other types of salivary neoplasms is challenging because of their relative infrequency and variable biologic behavior.



Figs 5A to C: Wharton's duct calculus. This patient had right submandibular abscess. (A) Right side Wharton's duct calculus coming out from its opening; (B) Appearance of Wharton's duct opening (too much dilated) after the milking out of calculus. The submandibular abscess drained through this opening; (C) Gross appearance and measurement of the removed Wharton's duct calculus

HISTOGENESIS OF NEOPLASMS

The two main theories of tumorigenesis, proposed for salivary gland neoplasms, are multicellular and bicellular reserve cell theories.

Fig. 6: Swelling of the parotid gland

TABLE 1 Tumors of salivary glands

Benign	*Malignant*
Epithelial (Adenomas)	Epithelial
• Pleomorphic adenoma • Adenolymphoma (Warthin's tumor) • Oncocytoma • Monomorphic adenoma	• Mucoepidermoid carcinoma • Adenoid cystic carcinoma (cylindroma) • Acinic cell carcinoma • Adenocarcinoma • Malignant mixed tumor • Squamous cell carcinoma • Undifferentiated carcinoma
Mesenchymal	Mesenchymal
• Vascular: Hemangioma • Lymphatic: Lymphangioma • Lipoma • Neurofibroma • Benign cyst	• Lymphoma • Sarcoma

1. *Multicellular cell theory*: Each neoplasm is thought to originate from a distinctive cell type. All differentiated salivary cell types retain the ability to undergo mitosis and regeneration.
   a. Warthin's and oncocytic tumors arise from striated ductal cells,
   b. Acinic cell tumors arise from acinar cells, and
   c. Mixed tumors arise from intercalated duct and myoepithelial cells.
2. *Bicellular reserve cell theory*: According to this theory, various types of salivary neoplasms originate from the basal cells (pluripotential cell populations) of either the excretory or the intercalated duct, which act as a reserve cell with the potential for differentiation into a variety of epithelial cells. Hence, all the heterogeneity salivary tumors are thought to arise from one of these two cells. Some molecular evidence supports this reserve cell theory.
   a. Adenomatoid tumors, including pleomorphic adenoma and oncocytic tumors arise from reserve cell of intercalated duct.
   b. Epidermoid tumors, such as squamous cell carcinoma and mucoepidermoid carcinomas arise from the reserve cell of the excretory duct.

Etiology

The exact etiology remains unknown but certain environmental factors (such as radiation, viruses, diet and certain occupational exposures) and specific genetic abnormalities may increase the risk of developing tumors (Box 1). Some patients of salivary gland cancer were found to have past history of skin cancer.

Box 1: Factors associated with high risk of primary salivary malignancy

- Radiation exposure
- Full mouth dental X-ray
- Skin cancer
- Rubber industry
- Nickel compound/alloy
- Hair dye
- Silica dust
- Kerosene cooking fuels
- Vegetables preserved in salt.

1. *Radiation*: Exposure to ionizing radiation (diagnostic, therapeutic, accidental and atomic explosions) may increase the risk of developing both benign and malignant salivary gland tumors. Risk was higher for malignant tumors especially mucoepidermoid carcinoma. Warthin's tumor showed the highest dose-response related risk. The risk of salivary gland neoplasia was not found influenced by duration of cellular telephone use.
2. *Viral*: Epstein-Barr virus has been found associated with lymphoepithelial carcinoma in the Asian population but there is no evidence of its causal role in other primary benign and malignant neoplasms of salivary glands. Other viruses including human papillomavirus, human herpesvirus 8 and cytomegalovirus do not have any etiologic role.
3. *Smoking*: Warthin's tumor is found associated with cigarette smoking.
4. *Occupational factors*: Exposure to silica dust, nickel alloys and nitrosamines (rubber workers) and use of kerosene as cooking fuel have been reported associated with increase the risk of malignancy in salivary glands.
5. *Hormonal factors*: Women with a history of early menarche and nulliparity were found to have increased risk of developing cancer of salivary glands.
6. Hair dye and alcohol intake in women have been reported to increase the risk of developing cancer of salivary glands.
7. *Dietary factors*: Vegetables preserved in salt were found associated with twofold risk of salivary malignancy.
8. *Genetic factors*: Genetic aberrations, which are found associated with the salivary gland neoplasia, include allelic loss and point mutation, structural rearrangement of chromosomal units (most commonly translocations), the monosomy and the presence of polysomy.
   - Allelic loss: Loss at 12q in pleomorphic adenomas and multiple losses at 9p, 3p and 17p in carcinoma ex-pleomorphic adenoma have been reported. Loss of heterozygosity occurs at 8q, 12q and 17p in carcinoma expleomorphic adenoma (17p in high disease stage and increased proliferative rate).
   - Monosomy and polysomy: Monosomy (absence of one chromosome) of chromosome 17 and polysomy (an extra chromosome) of chromosomes 3 and 17 in salivary gland tumors are higher in adenoid cystic carcinoma compared with pleomorphic adenoma.
   - Structural rearrangement: Cytogenetically, monoclonal and high frequency of tumor specific chromosome bands abnormalities 3p21, 8q12 and 12q14-15 are seen in pleomorphic adenomas of the salivary glands. Ectopic expression of the PLAG1 gene occurs in ple morphic adenomas with 8q12 aberrations.

Prophylaxis

The polyunsaturated fatty acids, dark yellow vegetables (carrots and sweet potatoes), live stock liver and vitamins A and C have been found to offer some protection.

PLEOMORPHIC ADENOMA

- This most common benign slow growing tumor of salivary glands, usually arise from the tail of parotid (Fig. 7) and submandibular glands. It can also arise from minor salivary glands and deep lobe of the parotid, which presents as a parapharyngeal tumor in the oropharynx (Figs 8 and 9).
- The tumor may be quite large at first presentation. It is usually seen in the third or fourth decade and has propensity for females.
- Figures 10A and B show CT scan of pleomorphic adenoma of superficial lobe of right parotid gland with peripheral enhancement and well-defined margins.
- These "mixed tumors" have both epithelial and mesenchymal elements in variable amount. The stroma may be mucoid, fibroid, vascular, myxochondroid or chondroid.
- This encapsulated tumor sends pseudopods into the surrounding glands, therefore it is essential that surgical excision of the tumor should include surrounding normal gland tissue. These pseudopods may be left behind if the tumor is simply "shelled out". Superficial parotidectomy is done for superficial parotid tumor.

Fig. 7: Swelling of the tail of parotid gland

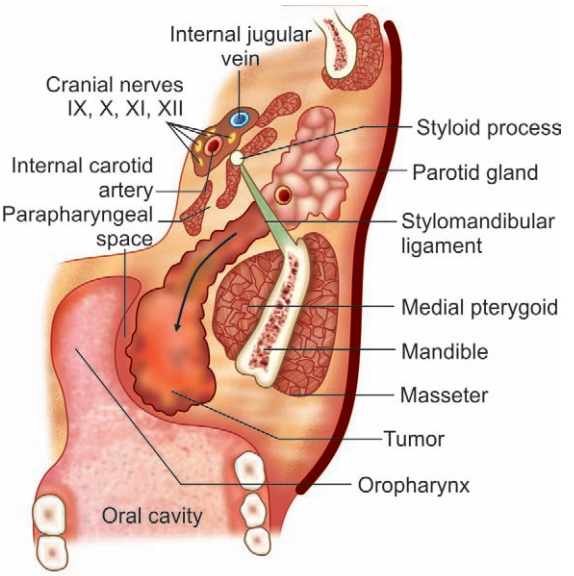


Fig. 8: Parotid tumor axial view. Tumor extending into parapharyngeal space posterior to stylomandibular ligament

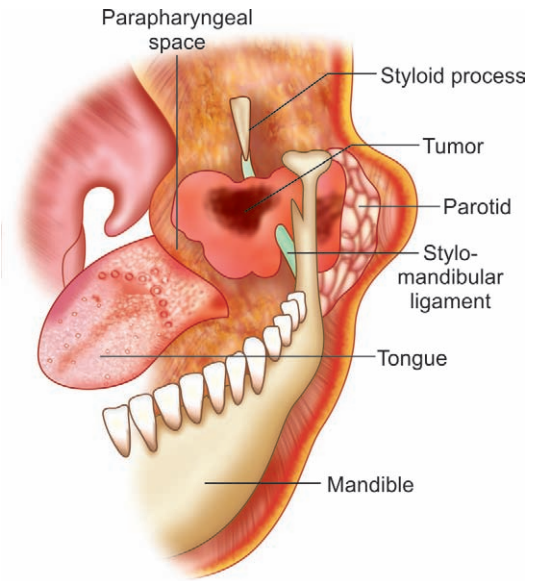


Fig. 9: Parotid tumor coronal view. Dumb-bell parotid tumor extending into parapharyngeal space anterior to stylomandibular ligament through the stylomandibular membrane between the stylomandibular ligament and mandible

WARTHIN'S TUMOR OR ADENOLYMPHOMA (PAPILLARY CYSTADENOMA LYMPHOMATOSUM)

- Warthin's tumor is commonly seen between fifth and seventh decade with male preponderance (5:1).
- The most common site is the tail of the parotid and is bilateral in 10% cases. They may be multiple.
- It is a rounded encapsulated tumor, which may be at times cystic with mucoid or brownish fluid.
- Histopathology shows its epithelial and lymphoid elements.
- Usually superficial parotidectomy is performed. However, they can be enucleated without danger of recurrence.

ONCOCYTOMA

- It comprises less than 1% of all salivary gland tumor and usually does not grow more than 5 cm in size.
- This oxyphil adenoma arises from acidophilic cells (oncocytes). It is mostly seen in the superficial parotid lobe of the elderly people.
- Oncocytoma shows increased uptake of technetium-99. It may be malignant, benign or cystic in nature.
- Treatment is superficial parotidectomy.

HEMANGIOMAS

- Hemangiomas are the most common benign parotid tumors in children and predominantly affect females.
- Congenital hemangioma grows rapidly in the neonatal period and then involutes spontaneously. Cutaneous hemangioma coexists in 50% of the cases.
- Characteristically, they are soft and painless and increase in size with crying or straining. The overlying skin shows bluish discoloration.
- Treatment is surgical excision if they do not regress.

LYMPHANGIOMAS

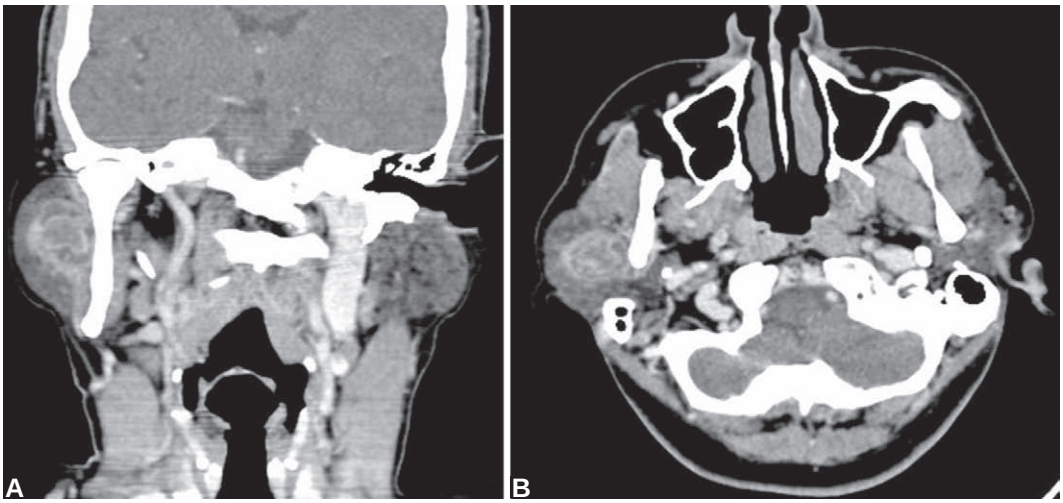
- These less common tumors feel soft and cystic and involve parotid and submandibular glands.
- They do not regress in size spontaneously and need surgical excision.

MUCOEPIDERMOID CARCINOMA

- This slow growing malignant tumor of parotid can metastasize and involve facial nerve. Mucoepidermoid tumors of minor salivary glands are more aggressive while in major salivary glands they behave like pleomorphic adenoma.
- The mucoepidermoid tumor has both the areas of mucin producing cells as well as squamous cells.
- The tumors have been classified as low grade and high grade. The tumors, which have greater epidermoid element, are more malignant.
  - Low grade tumors are more common in children. They have good prognosis and 90%, 5 years survival rates.
  - High grade tumors are more aggressive. They have poor prognosis and 30%, 5 years survival rate.

Staging

Tumor, nodes and metastases (TNM) classification of major salivary gland malignancy is shown in Table 2.



Figs 10A and B: CT scan coronal (A) and axial (B) sections. Pleomorphic adenoma right parotid lesion in superficial lobe with peripheral enhancement and well-defined margins
Source: Dr Swati Shah, Professor, Radiodiagnosis, GCRI Medical College, Ahmedabad

Treatment

- Low grade parotid tumors are managed by superficial or total parotidectomy. Surgery depends upon the location and extent of the tumor (Table 2). Facial nerve is preserved.
- The aggressive high grade tumors need total parotidectomy and facial nerve is sacrificed if invaded by tumor. Facial nerve is grafted in same sitting.
- If needed radical neck dissection is also combined.

ADENOID CYSTIC CARCINOMA (CYLINDROMA)

- This is a slow growing tumor, which infiltrates widely into the tissue planes and muscles.
- It spreads through perineural spaces and lymphatics and causes pain and facial nerve palsy.
- Distant metastases can occur in lungs, brain and bones.

TABLE 2 American Joint Committee on Cancer (AJCC) Cancer Staging (2002) and UICC TNM classification of malignant tumors of major salivary glands

T_x	Primary tumor cannot be assessed
T_0	No evidence of primary tumor
T_1	Tumor <= 2 cm size
T_2	Tumor 2–4 cm
T_3	Tumor more than 4 cm or extraparenchymal extension
T_{4a}	Tumor invades skin, mandible, ear canal, or facial nerve
T_{4b}	Tumor invades skull base, pterygoid plates, or encases carotid artery
N_x	Regional lymph nodes cannot be assessed
N_0	No regional lymph node metastasis
N_1	Metastasis in a single ipsilateral lymph node 3 cm or less
N_{2a}	Metastasis in single ipsilateral lymph node 3–6 cm
N_{2b}	Metastasis in multiple ipsilateral lymph nodes < 6 cm
N_{2c}	Metastasis in bilateral or contralateral lymph nodes < 6 cm
N_3	Metastasis in lymph node > 6 cm
M_x	Distant metastasis cannot be assessed
M_0	No distant metastasis
M_1	Distant metastasis
Stage grouping	
Stage I	$T_1\ N_0\ M_0$
Stage II	$T_2\ N_0\ M_0$
Stage III	$T_3\ N_0\ M_0$; $T_{1\text{-}3}\ N_1\ M_0$
Stage IVA	$T_{4a}\ N_{0\text{-}1}\ M_0$; $T_{1\text{-}4a}\ N_2\ M_0$
Stage IVB	$T_{4b}\ N_{0\text{-}3}\ M_0$; $T_{1\text{-}4b}\ N_3\ M_0$
Stage IVC	$T_{1\text{-}4b}\ N_{0\text{-}3}\ M_1$

Treatment

- Treatment is by radical parotidectomy, which includes large surrounding normal tissue.
- Radical neck dissection is done if nodal metastases are present.
- Postoperative radiation is given if the margins of the tumor are not clear.
- Local recurrences after surgical excision, which may be as late as 10–20 years, are common.

ACINIC CELL CARCINOMA

This low grade tumor appears similar to a benign mixed tumor. It presents as a small, firm, movable and encapsulated tumor. Bilateral tumors are also seen. Metastases are rare.

Treatment is superficial or total parotidectomy.

SQUAMOUS CELL CARCINOMA

This rapidly growing painful tumor infiltrates and ulcerates through the skin, and metastasizes to neck nodes.

Treatment is by radical parotidectomy, which includes surrounding part of muscle, mandible, temporal bone and the involved skin. Radical neck is combined if nodal metastases are present. Postoperative radiotherapy is given.

MALIGNANT MIXED TUMOR

This tumor can develop in old benign mixed tumor. Rapid growth and appearance of pain in a slow growing benign tumor indicates malignant change. A "de novo" tumor has much shorter history.

Treatment is radical parotidectomy.

ADENOCARCINOMA

This highly aggressive tumor mostly arises in minor salivary glands and sends distant metastases.

LYMPHOEPITHELIAL CARCINOMA OR UNDIFFERENTIATED CARCINOMA

This rare aggressive painful tumor has a tendency to spread rapidly. It becomes fixed to skin and ulcerates. It causes facial paralysis and cervical nodal metastasis. Treatment is wide excision combined with radical neck dissection and postoperative radiotherapy.

Malignancy of salivary glands: Their presentation may be similar to benign tumors and can lead to delay in diagnosis.

XEROSTOMIA

Xerostomia refers to dryness of mouth resulting from diminished or arrested salivary secretion. Xerostomia causes difficulty in chewing, swallowing and phonation, adherence of food to the buccal mucosa and multiple dental caries.

- Diabetes and cystic fibrosis should be assessed.
- Sedatives, antipsychotics, antidepressants, antihistamines and diuretics are most often associated with oral dryness.
- Salivary gland exposure to therapeutic irradiation > 4,000 cGy will result in severe and permanent secretory hypofunction.
- Xerostomia and xero-ophthalmia, the most common presentation of Sjogren's s syndrome, is also called Mikulicz disease.

SJÖGREN'S SYNDROME

Sjögren's syndrome is a chronic autoimmune disorder of the exocrine glands. The salivary and lacrimal glands are primarily affected. The lymphocytic infiltration results in glandular hypofunction leading to dryness of the mouth and eyes. The disease might even evolve into a malignant lymphoid process.

Types

1. *Primary Sjögren's syndrome*: This type of Sjögren's syndrome is confined to the exocrine glands.
2. *Secondary Sjögren's syndrome*: Patients have the characteristic signs and symptoms of primary Sjögren's syndrome associated with features of other autoimmune disease.

Secondary Sjögren's syndrome is the triad of: keratoconjunctivitis sicca (involvement of lacrimal gland), xerostomia (involvement of salivary glands and mucous glands of the oral cavity) and autoimmune connective tissue disorders such as rheumatoid arthritis.

Epidemiology

- The estimated prevalence of Sjögren's syndrome is believed to be 1–3%.
- The disease is most commonly seen in patients during their fourth to fifth decade of life.
- More than 90% of patients are women.

Etiology

Etiology is uncertain and is believed to involve genetics, immune system and environmental exposures.

Clinical Features

- Predominant clinical presentation: Dryness of the mouth and eyes.
- Salivary gland enlargement is most common in the parotid glands and occurs in 25–66% of patients. The bilateral enlargement may be recurrent and episodic or chronic and fixed.
- Patients with persistent unilateral or bilateral parotid gland enlargement are at higher risk for the development of lymphoma.

Oral Findings

- Xerostomia causes difficulty in chewing, swallowing and phonation, adherence of food to the buccal mucosa and multiple dental caries.
- Intolerance to acidic and spicy foods is a common complaint.

- Dry and sticky oral mucosal surfaces.
- Absence of pooled saliva in the floor of the mouth.
- The tongue is typically smooth with fissures and atrophy of the filiform papillae.
- Intraoral fungal overgrowth with *Candida albicans*.
- Expression of scant or cloudy saliva from the salivary ducts. Objective evaluation of salivary flow rate can be performed with Lashley cups that fit over the opening of Stensen's duct and collect saliva.

Eye Findings

- Most common ocular complaint: Foreign body sensation in the eye ("gritty" or "sandy" feeling).
- Chronic irritation and destruction of the corneal and conjunctival epithelium causes keratoconjunctivitis sicca.
- Dilation of the bulbar conjunctival vessels, pericorneal injection, irregularity of the corneal image and occasionally enlargement of the lacrimal gland.
- Schirmer's test: For tear secretion rate assessment.
- Staining of damaged corneal and conjunctival epithelia by rose Bengal dye is specific for keratoconjunctivitis sicca.

Systemic Manifestations

Generalized malaise, low grade fever, myalgia and arthralgia, bronchitis or pneumonia, renal tubular acidosis, vasculitis (Raynaud's phenomenon and recurrent urticaria-like lesions), peripheral sensory and motor polyneuropathies.

Laboratory Investigations

Laboratory investigations show raised erythrocyte sedimentation rate, positive rheumatoid factor and positive antinuclear antibodies. Biopsy from the lower lip shows evidence of involvement of minor salivary glands.

- *Sialography*: Sialectasis is seen in 85–97% of patients.
- *Testing for autoantibodies* to the ribonuclear proteins Ro (SS-A) and La (SS-B) is done by use of enzyme-linked immunosorbent assay.
- *Biopsy*: The biopsy of labial accessory salivary glands is taken from areas with normal overlying mucosa. The histopathology shows a lymphocytic infiltrate producing a chronic focal sialadenitis and multiple focal mononuclear aggregates that are adjacent to and replace the normal acini.

Treatment

The treatment consists of symptomatic therapy and prevention of irreversible damage to the teeth and eyes.

- Saliva substitutes and chewing of sugarless gum or candies.
- *Fluoride*: for treating and preventing dental caries.
- Eradicating fungal overgrowth.
- *Systemic sialogogues*: Pilocarpine (muscarinic cholinergic agonist) 5 mg three to four times daily, side effects include sweating, flushing and increased urination.
- *Treatment for keratoconjunctivitis*: Eye lubricants and eye patching if corneal ulceration develops.
- *Systemic corticosteroids or cytotoxic drugs*: They are reserved for the severe extraglandular complications such as glomerulonephritis or necrotizing vasculitis.

Sjögren's syndrome: A positive ANA, RF, SS-a, SS-b and an elevated ESR are indicative of Sjögren's syndrome. Biopsy from lip confirms the diagnosis and shows atrophy of minor salivary glands with an abundance of lymphocytes and histiocytes.

DIFFUSE INFILTRATIVE LYMPHOCYTOSIS SYNDROME

- The sicca symptom complex clinically mimics Sjögren's syndrome. It is characterized by the presence of persistent circulating CD8 lymphocytosis and visceral CD8 lymphocytic infiltration (predominantly in the salivary glands and lungs).
- Autoantibodies present in Sjögren's syndrome are absent in diffuse infiltrative lymphocytosis syndrome (DILS).
- Findings of needle aspiration and CT or MRI are sufficiently typical to provide a presumptive diagnosis.
- This condition is treated with corticosteroids and immunosuppressive therapy.

FREY'S SYNDROME (GUSTATORY SWEATING)

- *Gustatory sweating* manifests several months after the parotid operation. It is characterized by sweating and flushing of the preauricular skin during mastication.
- This condition is the result of aberrant innervation of sweat glands by parasympathetic secretomotor fibers which were destined for the parotid. Thus, these postganglionic fibers from the otic ganglion carried by auriculotemporal nerve, instead of causing salivary secretion cause secretion from the sweat glands.
- The placement of a sheet of fascia lata between the skin and the underlying fat may prevent secretomotor fibers reaching the sweat glands.

Treatment

- Usually no treatment other than reassurance is required.
- In cases of significant nuisance and social embarrassment, the condition is treated by tympanic neurectomy of Jacobson's nerve, which carries preganglionic parasympathetic secretomotor fibers from the inferior salivary nucleus through the glossopharyngeal nerve.

Clinical Highlights

1. **Complications of mumps:** They include unilateral sensorineural hearing loss, thyroiditis, pancreatitis, and orchitis.
2. **Sarcoidosis:** The clinical features include parotid swelling, facial paralysis, cervical lymphadenopathy, and diabetes insipidus.
3. **Salivary calculi:** Eighty percent of salivary calculi are seen in submandibular gland. Twenty percent of submandibular gland calculi are radiolucent.
4. **Nonneoplastic and noninflammatory parotid swellings:** The common causes are obesity, hypothyroidism, diabetes mellitus, and malnutrition.
5. **Parotid tumor:** Most common parotid tumor in children is lymphoma. Parotid gland is the most common site of pleomorphic adenoma.
6. **Hemangioma:** It is present in neonates with an isolated unilateral parotid swelling with bluish overlying skin. Swelling increases when child cries.
7. **Adenoid cystic carcinoma:** It is the most common malignant tumor of the submandibular salivary gland. Adenoid cystic carcinoma has a tendency for perineural invasion.
8. **Acinous cell carcinoma:** Among the malignancies of parotid gland, this has the best prognosis.
9. **Superficial parotidectomy:** This surgical treatment is adequate in cases of oncocytoma, pleomorphic adenoma, basal cell adenoma, and acinic cell carcinoma of parotid gland. Because of the section of greater auricular nerve, superficial parotidectomy is followed by anesthesia of the lower part of pinna.
10. **Xerostomia:** Some of the causes are antihistamines, uremia, Sjögren's syndrome, and mouth breathing.
11. **Sjogren's syndrome:** This autoimmune disease associated with collagen disorder is predominantly seen in women (9:1) between the ages of 40 and 60 years. Dryness of the eyes and dry mouth are the most common features. Parotid enlargement, which may be chronic or relapsing, develops in one-third of patients. Three to ten percent patients develop lymphoma.
12. **Mikulicz disease:** Treatment of choice is steroid therapy.
13. **Frey's syndrome (Gustatory sweating):** This is the flushing and sweating of skin of parotid region during eating in parotidectomy patients. Parasympathetic postganglionic secretomotor fibers supplying the parotid gland are misdirected and innervate (to postganglionic sympathetic fibers supplying the sweat glands) skin of the parotid area. Sectioning of the Jacobson's nerve (tympanic branch of glossopharyngeal nerve carrying preganglionic secretomotor fibers for parotid gland) on the promontory of middle ear (tympanic neurectomy) alleviate the symptoms.

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37 Neoplasms of the Oral Cavity

Go, all of you, wherever the people are in distress, and mitigate their sufferings. At the most, you may die in the attempt–what of that? How many like you are being born and dying like worms every day? What difference does that make to the world at large? Die you must, but have a great ideal to die for and it is better to die with a great ideal in life.

— **Swami Vivekananda**

Points of Focus

- **INTRODUCTION**

BENIGN TUMORS OF ORAL CAVITY

- **PAPILLOMA**
- **PLEOMORPHIC ADENOMA**
- **HEMANGIOMA**
- **LYMPHANGIOMA**
- **GRANULAR CELL TUMOR**
- **AMELOBLASTOMA**
- **TORUS**
- **PYOGENIC GRANULOMA**
- **IRRITATION FIBROMA**
- **MUCOCELE**
- **RANULA**
- **DERMOID CYSTS**

CARCINOMA OF ORAL CAVITY

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- ✓ Risk Factors
- ✓ Molecular Biology
- ✓ Premalignant Lesions
- ✓ Subsites of Oral Cavity
- ✓ Clinical Evaluation and Investigations
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- ✓ Histopathology
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- **CARCINOMA LIPS**
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- **CARCINOMA ALVEOLAR RIDGE**
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- **CARCINOMA FLOOR OF MOUTH**
- **CARCINOMA BUCCAL MUCOSA**
  - ✓ Verrucous Carcinoma
- **CARCINOMA HARD PALATE**
- **CARCINOMA RETROMOLAR TRIGONE**
- **MINOR SALIVARY GLAND TUMORS**
- **MELANOMA**
- **LYMPHOMA**
- **KAPOSI'S SARCOMA**
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

The tumors of oral cavity can be classified into two major categories: benign and malignant. Benign tumors and tumor-like lesions can be further divided into two groups: solid and cystic. Tori and dermoid cysts are congenital lesions. Malignant tumors have two histopathological types: carcinoma and sarcoma (Box 1).

BENIGN TUMORS OF ORAL CAVITY

PAPILLOMA

- Squamous papilloma is usually associated with human papillomavirus (HPV)-6 and HPV-11 virus subtypes. It is one of the most frequently occurring conditions.

Box 1: Tumors of the oral cavity

- Benign
  - Solid
    - ◊ Papilloma
    - ◊ Pleomorphic adenoma
    - ◊ Hemangioma
    - ◊ Lymphangioma
    - ◊ Granular cell tumor
    - ◊ Ameloblastoma
    - ◊ Torus (congenital)
      - Torus palatinus
      - Torus mandibularis
    - ◊ Inflammatory
      - Pyogenic granuloma
      - Irritation fibroma
  - Cystic
    - ◊ Mucocele
    - ◊ Ranula
      - Simple
      - Plunging
    - ◊ Dermoid cysts (congenital)
      - Sublingual
      - Submental
- Malignant
  - Carcinoma
    - ◊ Squamous cell carcinoma
    - ◊ Nonsquamous cell carcinoma
      - Minor salivary gland tumors
      - Melanoma
      - Lymphoma
  - Sarcoma
    - ◊ Kaposi's sarcoma

- The peak incidence is in the third to fifth decades.
- Most common sites are soft and hard palate, uvula, tongue, lips and buccal mucosa.
- Mostly, they are pedunculated and white in color and less than 1 cm in size. Their surface is usually wart like but in some cases it is smooth.
- *Treatment*: Surgical excision or ablation with CO_2 laser.

PLEOMORPHIC ADENOMA

- It is the most common variety of minor salivary gland benign neoplasms.
- It usually involves soft and hard palate but can involve any part of oral cavity.
- It presents as a painless progressive submucosal tumor.
- *Treatment*: It needs wide excision because the recurrence rate is high.

HEMANGIOMA

- Oral cavity mucosal hemangiomas (Fig. 1) represent 14% of all hemangiomas. They are mostly seen in children.
- Most common site is lip.
- Present as a soft, painless, red or blue mass of usually < 2 cm size. Extensive lesions involve significant portions of oral cavity and oropharynx.
- An infected hemangioma may look similar to a pyogenic granuloma.
- *Treatment*: Congenital hemangioma usually does not need any treatment as the spontaneous regression is wellknown.

Fig. 1: Hemangioma tongue

Hemangiomas that are large and persistent and continue to grow need treatment. The sclerotherapy, cryosurgery and laser have not been found useful. Microembolization alone or as a preoperative adjunct to surgery has been found useful.

- Phlebectasias are dilated veins, which occur on the oral or lingual mucosa in 40–50 years patients.

LYMPHANGIOMA

- The most common site of lymphangioma is anterior two-thirds of tongue.
- The lesion may be either diffuse (macroglossia) or localized compressible soft swelling.
- *Treatment*: Small lesions are totally excised. In large diffuse lesions total excision is not advised. Partial excision reduces the bulk.

GRANULAR CELL TUMOR

- Earlier it was thought to arise from the muscle and was called myoblastoma. Now it has been reported to be derived from Schwann cells.
- Most common site is tongue. Other sites include soft palate, uvula and labial mucosa.
- It presents as a firm, painless, relatively immobile sessile less than 1.5 cm submucosal nodule.
- Congenital epulis, which is a granular cell tumor, involves the gums of future incisors in female infants.
- *Treatment*: Excision biopsy. Recurrence is less than 10%.

AMELOBLASTOMA

- It is the most common neoplasm of odontogenic origin.
- It is believed to arise from rests of primitive dental lamina, which is related to the enamel organ in alveolar bone.
- Patients are usually in third decade of life.
- Most common site is molar/ramus area of mandible.
- CT scan shows unilocular or multilocular radiolucency with cortical bone expansion.
- *Treatment*: En bloc resection with at least 1 cm margins of normal appearing tissue.
- Recurrence rate is 22%.
- Malignant transformation is rare.

TORUS

Torus is a frequently observed developmental anomaly. It presents as a bony outgrowths in the second decade of life. It continues to grow slowly throughout the life.

Clinical Features

- Tori are more common in females.
- These pedunculated or multilobulated broadly based smooth bony masses are usually asymptomatic.
- In later life, they may interfere with denture placement and get repeatedly injured while eating.
- Torus palatinus is found in the midline of hard palate.
- Torus mandibularis is found on the lingual surface of mandible in the premolar region.

Treatment

- Removal from the underlying cortex with osteotomes or cutting burrs.

PYOGENIC GRANULOMA

- This reactive granuloma usually occurs in response to trauma or chronic irritation.
- It mostly involves anterior gingivae but can be seen over tongue, buccal mucosa or lips.
- It presents as soft smooth reddish to purple raised or pedunculated mass, which bleeds on touch.
- Pregnancy granuloma or Epulis gravidarum is similar to pyogenic granuloma. It starts in the first trimester of pregnancy and regresses after the pregnancy.
- *Epulis granulomatosa* occurs after tooth extraction.
- *Treatment*: Excisional biopsy and removal of potential traumatic or infective factor. Recurrence is uncommon. Pregnancy granuloma needs excision if persists after pregnancy. During pregnancy, it is not removed as the recurrence rate is very high.

IRRITATION FIBROMA

- This common tumor-like condition of oral cavity is found in 1.2% of adults.
- It usually becomes apparent during or after fourth decade.
- Asymptomatic solitary sessile or pedunculated firm mass which is seldom larger than 1.5 cm.
- Sites are buccal, labial or tongue mucosa.
- History of chronic irritation is present.
- *Treatment*: Conservative excisional biopsy.

MUCOCELE

- This is a soft cystic bluish color retention cyst of minor salivary gland.
- Though it can occur anywhere in oral cavity, its most common site is the lower lip (Fig. 2).
- *Treatment*: Surgical excision.

RANULA

- Ranula, a cystic grayish translucent swelling occurs in the lateral part of the floor of mouth and pushes the tongue up.
- Ranula is the result of obstruction of the ducts of sublingual salivary gland.

Fig. 2: Mucous retention cyst of lower lip

- Plunging ranula is quite big and extends into the neck.

Treatment

- *Excision*: Small ranula may be completely excised.
- *Marsupialization*: Large ranula needs marsupialization. It is difficult to excise the ranula completely. The thin wall and ramifications of ranula go in various tissue planes.

DERMOID CYSTS

- Dermoid cysts are lined by keratinized squamous epithelium. They are formed from epithelial rests that are found along embryonic fusion lines. They contain elements of epidermal appendages such as hair follicles, sweat glands and connective tissue.
- Head and neck accounts for about 7% of total dermoid cysts; of this, 6.5–23% are found in floor of the mouth.
- As they enlarge, difficulties in deglutition, speech and respiration occur.
- There are two types of dermoids in this region: sublingual and submental.
  - *Sublingual dermoid* is situated above the mylohyoid.
    - It can be either median or lateral.
    - It shines as a white mass through the mucosa.
  - *Submental dermoid* develops below the mylohyoid muscle.
    - It presents as a submental swelling.
- *Treatment*: Complete excision of the cyst.

CARCINOMA OF ORAL CAVITY

Epidemiology

The five most frequent cancers in Indian males (in descending order) are mouth/oropharynx (Fig. 3), trachea/bronchus/lungs, lymphomas/multiple myeloma, esophagus, and leukemia. In women, they are (in descending order) breast, cervix, ovary, mouth/oropharynx, esophagus, and lymphomas/multiple myeloma (*Source*: ICMR, 2004). The incidence rate of oral cancer in India is very high (44.8 males and 23.7 females in 100,000 population) in comparison to 11.2 of USA. This preventable disease is caused by tobacco, alcohol, *paan*, reverse smoking, areca nut and betel quid.

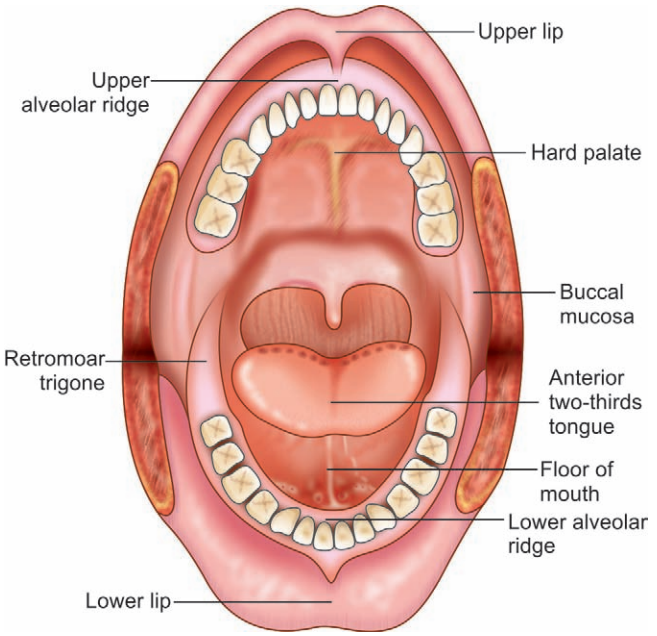


Fig. 3: Oral cavity overview

Risk Factors

The risk factors, which are associated with the development of oral cavity cancers, include several Ss such as smoking, spirit, *sopari* (areca nut), sharp and septic tooth, syphilitic glossitis and syndrome Plummer-Vinson. Tobacco and alcohol are the most common preventable factors.

- *Smoking*: Incidence of oral cancer is six times more in smokers. Reverse smoking, where burning end of the "churat" (rolled tobacco leaf) is put in the mouth, gives still higher incidence of cancer of the hard palate. Pipe smoking has been associated with lip cancer. Forty percent of patients who continue smoking after definitive treatment develop recurrence or second head and neck malignancy.
- *Alcohol*: Cancer of upper aerodigestive tract occurs six times more in heavy drinkers. Individuals who both smoke and drink have 35 times more risk. Alcoholic mouthwashes have also been implicated.
- *Chewing of paan, sopari and tobacco*: *Paan* (specially prepared leaf), *sopari* (betel nut, product of Areca catechu tree), quid (powdered tobacco mixed with lime) are placed in the mouth and carcinoma develops at the site of their lodgment. This bad habit is largely responsible for oral cancer in Indians. Betel nut is a mild stimulant similar to that of coffee.
- *Avitaminosis and malnutrition*: Riboflavin deficiency is proposed to be responsible for cancer in alcoholics.
- *Dental caries, sharp jagged teeth and ill fitting dentures*: They cause chronic irritation, which may result in malignant change.
- *Human papillomavirus*: The role of HPV has been in a subset of head and neck squamous cell carcinoma.
- *Environmental ultraviolet light exposure*: It has been associated with lip cancer.
- *Long-term immunosuppression*: There is 30 fold increased risk with renal transplant.
- *HIV infection*: Kaposi's sarcoma may occur in oral cavity.
- *Other carcinogenic factors proposed in the etiology are following*:
  - Hot spicy food
  - Xeroderma pigmentosa
  - Chronic glossitis
  - Cirrhosis.

Molecular Biology

- The cases of oral verrucous carcinoma that expressed H-ras gene also demonstrated evidence of HPV DNA (deoxyribonucleic acid).
- Human papillomavirus is a mucosotropic virus. HPV oncoproteins E6 and E7 can degrade tumor suppressor gene products of p53 and pRB, respectively. Significant increase in HPV DNA was found within oral cavity carcinoma cases. HPV-6 and HPV-16 were found risk factors for oral cavity carcinoma independent of age and tobacco and alcohol use. HPV (OR-3.7) was found a risk factor for carcinoma, independent of tobacco (OR-2.63) and alcohol (OR-2.57) use.
- Tumors negative for glutathione S-transferase (GST-π) biomarker were found more responsive to cisplatin and 5-fluorouracil.
- Lack of bel-2 expression found consistent with an improved 3-year disease-free survival.
- Studies suggest role of p53, EGFR, transforming growth factor (TGF)-α, and cyclin D1 in predicting prognosis of head and neck carcinoma.

Premalignant Lesions

- *Lichen planus*: It has been associated with lip cancer. In cases of erosive lichen planus or atrophic lichen planus, there is risk of malignant change (*see* Chapter 35: Oral Mucosal Lesions).
- *Syphilis*: The syphilitic interstitial glossitis with an endarteritis causes atrophy of the overlying epithelium that is more vulnerable to carcinogenic irritants. Patients need treatment of syphilis and regular follow-up.
- *Plummer-Vinson syndrome or Paterson-Kelly syndrome or Sideropenic dysphagia*: It consists of achlorhydria, iron deficiency anemia, and mucosal atrophy of mouth, pharynx and esophagus. The oral mucosa becomes thin, pale and atrophic. The disease is particularly common in women. It is said to be a premalignant condition and may be responsible for cancer of the oral cavity and post cricoid region. The epithelial atrophy is extremely vulnerable to carcinogenic irritants. Anemia responds to iron supplements.
- *Dyskeratosis congenita*: This syndrome consists of
  - Reticular atrophy of skin with pigmentation
  - Nail dystrophy
  - Oral leukoplakia with thickened, fissured and white mucosa.
- *Leukoplakia* (*see* Chapter 35: Oral Mucosal Lesions).
- *Erythroplakia* (*see* Chapter 35: Oral Mucosal Lesions).
- *Chronic hyperplastic candidiasis* (*see* Chapter 35: Oral Mucosal Lesions).
- *Discoid lupus erythematosus* (*see* Chapter 35: Oral Mucosal Lesions).
- *Submucous fibrosis* (*see* Chapter 35: Oral Mucosal Lesions).

Subsites of Oral Cavity

The oral cavity extends from the lips to the level of anterior tonsillar pillar. Subsites of cancer in oral cavity are (*Source*: AJCC, 1997) (Fig. 3):

- Lip (Vermilion surface)
  - Upper lip
  - Lower lip
  - Commissures
- Buccal mucosa (includes mucosa of cheek and inner surface of lips up to line of contact of opposing lip): Mucosal surfaces of
  - Upper and lower lips
  - Cheeks
  - Retromolar area
  - Upper and lower buccoalveolar sulci
- Anterior two-third of tongue (anterior to vallate papillae)
  - Dorsal and ventral surfaces
  - Lateral borders
  - Tip
- Hard palate
- Lower alveolar ridge (alveolus and gingiva)
- Upper alveolar ridge (alveolus and gingiva)
- Floor of mouth
- Retromolar trigone.

Clinical Evaluation and Investigations

For history taking, clinical evaluation, investigation and imaging studies, see chapter Oral Symptoms and Examination in the section of Oral Cavity and Salivary Glands.

Staging

Table 1 shows TNM classification.

Histopathology

Most of the oral cavity malignant growths (85%) are squamous cell carcinoma and are mainly managed surgically. Other malignancies in decreasing frequency are minor salivary gland tumors, lymphomas, malignant melanomas, sarcoma and metastatic tumors.

Treatment

Multidisciplinary tumor conference, which consists of head and neck and plastic surgeons and radiation and medical oncologists, decides the best line of management. The treatment options include following:

- *Primary tumor including mandible/maxilla*
  - Early lesions T_1/T_2: Transoral complete removal of tumor with adequate margin followed by primary closure, healing by secondary intention, or skin-dermal graft.
  - Larger or more posterior lesions: They need pull through technique for exposure and reconstruction. Lip splitting and mandibulotomy facilitates safe resections of tumors in tongue and floor of mouth in continuity with neck lymphatics.
  - Mandible involvement: Marginal, segmental or hemimandibulectomy. Rim resection avoids mandibular discontinuity without risking local recurrence.
  - Reconstruction: Primary reconstruction is now the rule. The current techniques of reconstruction include
    - Muscle flaps: Pectoralis major, trapezius, and latissimus dorsi
    - Microvascular techniques: Free tissue transfer.
- *Management of neck*
  - N_0: A function sparing elective neck dissection, in which accessory nerve, internal jugular vein (IJV) and sternocleidomastoid muscle are preserved, is advocated in tumors of floor of mouth, lower alveolus and tongue. In presence of 2 or more positive nodes or extracapsular spread postoperative radiotherapy is given. Over 96% of histologically positive nodes can be removed by supra-omohyoid dissection. Alternatively, elective irradiation of clinically negative neck prevents subsequent nodal disease.
  - $N_1/N_{2a}/N_{2b}$: Radical neck dissection followed by postoperative radiotherapy. In inoperable cases, radical external beam irradiation.
  - N_{2c}: In young operable cases, full radical neck dissection on the ipsilateral side and IJV sparing neck dissection on contralateral side followed by postoperative radiotherapy. Radical radiotherapy is advocated in inoperable cases.
  - N_3: External radiotherapy.
  - Lymph node metastasis after primary treatment: After confirming by fine-needle aspiration cytology, radical neck dissection is done followed by postoperative radiotherapy.
- External beam radiation and brachytherapy
- Adjunctive chemotherapy
- Photodynamic therapy
- Organ preservation protocols
- Chemoprevention
- Speech and swallowing rehabilitation
- Pain management
- Palliative care.

- Skin cancers: Surgical excision is the treatment of choice.
- Healing: Primary closure of a facial defect is best if possible. Other options include healing by secondary intention, local flaps, skin grafts and regional or free flaps.

The factors, which help in deciding the optimum management for a patient, includes following:

- *Site and TNM stage*: An alveolar lesion and other sites small lesions (T_1) and neck node metastasis (N_1 and N_2) are best managed by surgery. The management of most T_2 and early T_3 tumors is controversial.
- *Histology*: Adenocarcinoma and melanoma are relatively radioresistant and are preferably treated surgically. Early verrucous carcinoma is suitable for local excision. Advanced verrucous carcinoma is treated as squamous carcinoma.
- *General*: Age, medical condition and life style of the patient.

Prognosis

In spite of advancing technology and surgical procedures during the last 40 years the 5-year cure and survival rate has remained about 55%.

- *Second primary*: The longer a patient survives greater is the risk of developing second primary tumor in the oral cavity, larynx, bronchus or esophagus.

TABLE 1 AJCC Cancer Staging and UICC TNM classification of oral cancer

Primary tumor (T)	
T_x	Unable to assess primary tumor
T_0	No evidence of primary tumor
T_{is}	Carcinoma in situ
T_1	Tumor is < 2 cm in greatest dimension
T_2	Tumor > 2 cm and < 4 cm in greatest dimension
T_3	Tumor > 4 cm in greatest dimension
T_4 (lip)	Primary tumor invading cortical bone, inferior alveolar nerve, floor of mouth, or skin of face (e.g., nose or chin)
T_{4a} (oral)	Tumor invades adjacent structures (e.g. cortical bone, into deep tongue musculature, maxillary sinus) or skin of face
T_{4b} (oral)	Tumor invades masticator space, pterygoid plates, or skull base or encases the internal carotid artery
Regional lymphadenopathy (N)	
N_x	Unable to assess regional lymph nodes
N_0	No evidence of regional metastasis
N_1	Metastasis in a single ipsilateral lymph node, <= 3 cm in greatest dimension
N_{2a}	Metastasis in single ipsilateral lymph node, > 3 cm and < 6 cm
N_{2b}	Metastasis in multiple ipsilateral lymph nodes, all nodes < 6 cm
N_{2c}	Metastasis in bilateral or contralateral lymph nodes, all nodes < 6 cm
N_3	Metastasis in a lymph node > 6 cm in greatest dimension
Distant metastases (M)	
M_x	Unable to assess for distant metastases
M_0	No distant metastases
M_1	Distant metastases
TNM staging	
Stage 0	$T_{is}\ N_0\ M_0$
Stage I	$T_1\ N_0\ M_0$
Stage II	$T_2\ N_0\ M_0$
Stage III	$T_3\ N_0\ M_0$; $T_{1-3}\ N_1\ M_0$
Stage IVA	$T_{4a}\ N_{0-1}\ M_0$; $T_{1-4a}\ N_2\ M_0$
Stage IVB	$T_{4b}\ N_{0-2}\ M_0$; $T_{1-4b}\ N_3\ M_0$
Stage IVC	$T_{1-4b}\ N_{1-3}\ M_1$

Source: American Joint Committee on Cancer (AJCC) Staging Manual, 6th edition. Chicago; 2002

- *Metastasis via blood stream*: It is relatively an early event in oral cancer.
- *Factors affecting prognosis*: They include:
  - Infiltrative growths have comparatively poorer prognosis.
  - Depth of invasion: More the depth of invasion higher the risk of regional metastasis and poorer the 5-year survival.
  - Stage: Higher the stage, poorer is the 5-year survival.
  - Histopathology: The following histopathologic findings have prognostic significance:
    - Keratin production
    - Level of differentiation
    - Nuclear appearance
    - Mitoses
    - Inflammation
    - Desmoplastic reaction
    - Patterns of invasion
    - Vascular invasion

CARCINOMA LIPS

Lips form the anterior boundary of the vestibule. Medial portion of lower lip drains into submental and lateral portion to submandibular nodes. Upper lip drains into preauricular, infraparotid and submandibular nodes.

Clinical Features

- *Sex*: Males in the age group of 40–70 years are more affected.
- *Lesion*: Exophytic or ulcerative type.
- *Most common site*: Between the midline and commissure of the lower lip.
- *Local spread*: Initially tumor spreads laterally and later infiltrates deeply and spread into anterior triangle of neck and invade mandible.
- *Regional lymphatic metastasis*: It is late and develops into submental and submandibular nodes. At later stage, deep cervical group of lymph nodes may get involved.
- *Histopathology*: Mostly, squamous cell carcinoma.

Treatment

Either radiation therapy or surgery for stage I tumors gives cure rates greater than 85%. Surgery for large tumors consists of excision with adequate safety margin and repair. Block dissection is done if lymph nodes are involved. Postoperative radiotherapy to chin and neck region is planned in high-risk advanced stage tumors.

- *Leukoplakia lip or actinic cheilitis*: Leukoplakia or actinic changes on the vermilion border of lip due to ultraviolet radiations can be managed with either lip shave or laser. The resection is reconstructed either by advancing labial or buccal mucosal flaps or use of pedicled anteriorly based tongue flap.
- *2 cm lesion of lower lip*: Up to one-third of lower lip can be removed with a V or W shaped excision with primary closure in three layers (mucosa, muscle and skin) under local anesthesia as an OPD procedure. Initially lip looks tight but improves after 3 months.
- *Reconstruction*: More extensive lip resection needs local flaps for reconstruction.
  - For large central defects of lower lip: Step ladder approach of Johnson advances symmetrical bilateral flaps from lower third of face.
  - For large lateral defects of lower lip, upper lip involving commissure: Fries's "universal procedure", in which lateral flaps are developed following full-thickness incisions in the cheeks parallel to the branches of facial nerve. The flaps are then advanced into the lip defect. The Fries modifications of the Bernard flap can be used for total reconstruction of both lower and upper lips.

CARCINOMA GINGIVA/ALVEOLAR RIDGE

Gums (gingivae) surround the teeth and cover the upper and lower alveolar ridges.

Buccal aspect of mucosa drains into submental and submandibular nodes. Lingual aspect of upper alveolus drains into upper deep cervical and lateral retropharyngeal nodes. Lingual aspect of lower alveolus drains into submandibular nodes.

Clinical Features

- The history of failure of socket healing after tooth extraction is common.
- The history of sudden difficulty in wearing denture is common.
- Superficial ulceration or proliferative tissue at the gingival margin.
- Most common site is the lower jaw in the premolar region and just behind the first molar (Figs 4A and B).
- Local spread: To cheek, floor of mouth, retromolar trigone and hard palate. After invading underlying bone, it spreads rapidly along the neurovascular bundle.
- Nodal metastases: The enlargement of submandibular and upper jugular nodes is common at the time of presentation.

Treatment

Surgery is preferred over radiotherapy. Radiations lead to radio-osteonecrosis.

- Early lesion on the lower alveolus is excised (marginal resection of the mandible).
- Extensive tumor of lower alveolus requires segmental or hemimandibulectomy. Access is achieved via lip split approach. Block dissection is needed when lymph nodes are involved.
- Tumors confined to hard palate, upper alveolus and floor of maxillary antrum needs partial maxillectomy with block dissection if nodes are enlarged.

Figs 4A and B: (A) Carcinoma left lower alveolus (premolar to retromolar region) extending to buccal mucosa, floor of mouth and cheek skin; (B) Note the lower jaw swelling and pus from skin lesion

- Upper alveolar tumor involving inferior structure of maxilla requires total maxillectomy followed by postoperative radiotherapy.
- Primary reconstruction should always be undertaken.

CARCINOMA ORAL TONGUE

Anterior two-third of tongue is the content of the oral cavity. Posterior one-third (base of tongue) is situated behind the circumvallate papillae and forms anterior wall oropharynx. Oral tongue can be divided into different regions: tip, lateral borders, dorsum and the undersurface.

Tip of tongue drains into submental and jugulo-omohyoid nodes. Lateral portion drains into ipsilateral, submandibular and deep cervical nodes. Central portion and base drain into deep cervical nodes of both sides.

Gross Pathology

- *Pre-existing lesions*: Some patients have pre-existing leukoplakia, dental ulcer or syphilitic glossitis.
- *Common site*: Middle third of the lateral border. Figures 5 to 8 show three different types of lesions. The lesion extends early on to the ventral surface of the tongue and floor of mouth. The tip and dorsum are uncommon sites (Fig. 9).

Fig. 5: Carcinoma tongue right lateral margin. Nodular lesion with induration of the surrounding tissue

- *Local spread*: Deeply into the musculature (causes ankyloglossia), inferiorly into the floor of mouth and laterally into alveolus and mandible.
- *Lymph node metastases*: Lateral border of tongue drains into submandibular and upper jugular nodes (Fig. 8). Tip of tongue drains to the submental and jugulo-omohyoid group. Bilateral and contralateral lymph node involvements can occur.
- *Histopathology*: Squamous cell carcinoma is most common.

Clinical Features

- This is commonly seen in 50–70 years of men.
- Early lesions are painless and may remain asymptomatic. Lesion arises in an atrophic depapillated area with an erythroplakia patch with peripheral streaks or areas of leukoplakia.
- Painful ulcer/growth/lump is a late feature.
- Referred earache occurs due to common nerve supply (branches of mandibular division of trigeminal) of the tongue (lingual nerve) and ear (auriculotemporal).
- Enlarged lymph node presents as mass in the neck (Fig. 8). About 50% patients have palpable nodes at the time of presentation. About 12% of the patients present with only lump in neck.
- Late features: Dysphagia, odynophagia (leads to drooling of saliva), ankyloglossia, slurred speech, bleeding, cachexia.
- Lesion: Often, the lesion is exophytic with areas of ulceration.
  - Exophytic: An exophytic lesion like a papilloma.
  - Ulcerative: A nonhealing ulcer with rolled edges (Fig. 6) and grayish white shaggy base with induration. The ulcer may be present superficially or in depths of a fissure infiltrating underlying muscle (Fig. 7).
  - A submucous nodule with induration of the surrounding tissue.
  - The lesion may be associated with leukoplakia.

Treatment

The main modalities of treatment, which depends on the staging of the cancer (Table 1), are surgery and radiotherapy. One should try to preserve function of the tongue along with the total eradication of cancer as far as possible.

Fig. 6: Carcinoma tongue right lateral margin. Ulcerative lesion with rolled edges and grayish white shaggy base with induration

Fig. 7: Malignancy tongue. Nonhealing ulcer with infiltration. The ulcer may be present superficially or in depths of a fissure infiltrating underlying muscle

- *Stage I (T_1 N_0)*: Equal results with radiotherapy or surgery. Simple intraoral excision is preferred. Formal reconstruction is not required if less than one-third of the tongue is excised.
- *Stage II (T_2 N_0)*: Treated by radiotherapy (including tumors and regional lymph nodes) or by surgical excision (Hemiglossectomy) with prophylactic neck dissection.
- *Stage III or IV*: Hemiglossectomy (depending on the extent of the tongue lesion) combined with marginal, segmental or hemimandibulectomy (depending upon the extent of involvement of mandible) and radical neck dissection of lymph nodes (commando operation) followed by postoperative radiotherapy. This strategy gives better results than either

Fig. 8: An advanced case (Stage IV: T_4 N_3) of right side carcinoma tongue (tumor invades into deep tongue musculature) with secondary neck nodes (lymph node > 6 cm in greatest dimension)

Fig. 9: Carcinoma tip of tongue with trismus due to oral submucous fibrosis

surgery or radiotherapy alone. A rim mandibular resection is advised when lesion reaches but does not invade alveolus (Figs 10 and 11).

- *Approach*: In cases of extensive tongue lesions involving floor of mouth and alveolus, access is best via a lip split and mandibulotomy. The traditional pull-through procedure does not allow adequate excision.
- *Reconstruction*:
  - Radial forearm free flap with microvascular anastomosis: When tongue defect does not exceed two-third of the original tongue.
  - Pectoralis major muscle flap: For very large defects, total glossectomy or deeply infiltrating lesions when resection extends to hyoid bone.

Fig. 10: Intraoral view—Operated case of carcinoma tongue right lateral margin involving middle and posterior one-thirds with secondary neck node at level II. The local wide excision included right tongue margin, floor of mouth, tonsillar pillar, and alveolar marginal mandibular resection with modified neck dissection

Fig. 11: External neck incision of the patient shown in Figure 11

CARCINOMA FLOOR OF MOUTH

Floor of mouth is a crescent-shaped area that lies between the gingivae and undersurface of tongue. Anterior portion of the floor (frenulum, sublingual papillae with openings of submandibular ducts) is examined when patient raises the tip of tongue towards the hard palate. Lateral portion of floor of mouth is examined by displacing the tongue in medial direction with the help of a tongue depressor.

Anterior portion of floor of mouth drains into submental and submandibular nodes. Lymphatics from this area also cross the midline. Posterior portion drains into upper deep cervical nodes.

Clinical Features

- *Male/Female ratio*: Males are affected more than females (ratio of 4:1).
- *Early features*: Lesion may remain asymptomatic or cause soreness or irregularity in the floor of the mouth.
- *Submandibular swelling*: It may be due to either enlarged submandibular salivary gland or lymph node metastases.
- *Lesion*: Ulcerative or infiltrative lesion.
- *Leukoplakia*: It is more common in the floor of mouth than at other sites.
- *Common site*: Begins anteriorly near the opening of submandibular duct that get obstructed and lead to enlargement of submandibular gland.
- *Local spreads*: Ventral aspect of the tongue, lingual gingiva, mandibular periosteum or deeply into the floor of mouth and submental space.
- *Lymphatic metastases*: Submandibular and jugulodigastric nodes and may be bilateral
- Histopathology: Usually squamous cell carcinoma.

Treatment

- *Stage I*: If there is no involvement of tongue, lingual gingiva or nodes than small lesions are treated with wide excision or radiotherapy, which offer equal results.
- *Stage II*: Prophylactic neck dissection or irradiation is done for stage II cancer because there is high incidence of micrometastases (40%).
- *Stage III and IV*: These patients need surgery and radiotherapy. Large tumor with involvement of tongue, gingiva, mandible and nodes require hemiglossectomy, marginal or segmental mandibular resection, and radical neck dissection. Resultant defect is reconstructed with either a local or distant flap.

CARCINOMA BUCCAL MUCOSA

Buccal mucosa lines the inner surface of cheeks and lips and extends between the superior and inferior gingivobuccal sulcus. It extends posteriorly up to pterygomandibular raphe and anteriorly to the meeting line of lips. Buccal mucosa drains into submental and submandibular nodes.

Carcinoma of buccal mucosa is the second most common cancer of oral cavity after the tongue.

Pathology

- *Most common sites*: Angle of mouth and the plane of occlusion (Fig. 12) to retromolar region. Gingivobuccal sulcus where *paan* or tobacco quid is kept is not uncommon.
- *Multicentric*: Buccal mucosa cancer may be multicentric as the entire buccal mucosa is "condemned".
- *Lesion*:
  - Exophytic lesions are associated with erythroleukoplakia (Fig. 12).
  - Ulceroinfiltrative lesions may infiltrate deeply.
  - Verrucous carcinoma.
- *Local spread*:
  - The ulcer may spread deeply and involves different layers of cheek such as submucosa, muscle, subcutaneous fat and skin.

Fig. 12: Malignancy buccal mucosa in pre-existing erythroleukoplakia lesion at angle of mouth and in the plane of occlusion

  - Patient develops trismus if buccinator, masseter, or medial pterygoid muscle is involved.
  - If the lesion spreads radially, it involves angle of the mouth and lip anteriorly; retromolar trigone and medial pterygoid muscle posteriorly; upper gingivobuccal sulcus and maxilla superiorly; and lower gingivobuccal sulcus and alveolar ridge and gums inferiorly.
- *Lymphatic spread*: Submandibular, submental, and parotid nodes. Later on upper jugular nodes are affected. In some cases, there occurs direct involvement of upper jugular nodes skipping the submandibular group. Nodal involvement develops in about 50% cases.
- *Histopathology*: Squamous cell carcinoma is most common but salivary gland tumors are also seen.

Clinical Features

- Early lesions are asymptomatic.
- Pain and bleeding.
- Trismus: Involvement of buccinator, masseter or pterygoid muscles leads to trismus (Fig. 13).
- Late feature: Fungating foul smelling bleeding mass over the cheek or in the oral cavity (Figs 14 and 15).

Verrucous Carcinoma

Verrucous carcinoma appears as soft white velvety area of superficial proliferative exophytic lesion (like a white papillary growth), which has considerable keratinization.The deep invasion and induration are minimal. The tumor behaves like a low-grade squamous cell carcinoma. The lymph node metastasis is late.

Treatment

- *Stage I (T_1 N_0)*: Wide excision with safety margin including underlying buccinator muscle followed by quilted split-skin graft.
- *Stage II (T_2 N_0)*: Radiotherapy to the primary lesion and regional lymph nodes if bone is spared. If bone (maxilla and

Fig. 13: Malignancy buccal mucosa right side with trismus

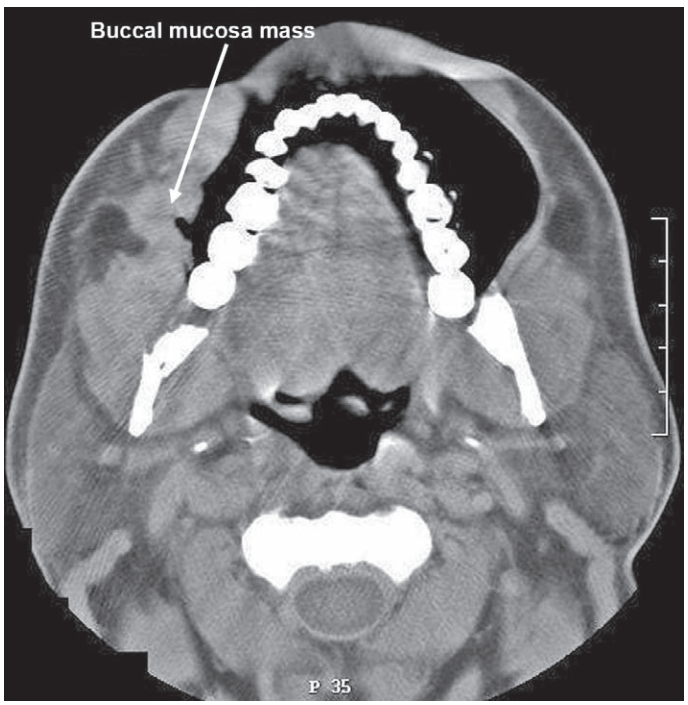
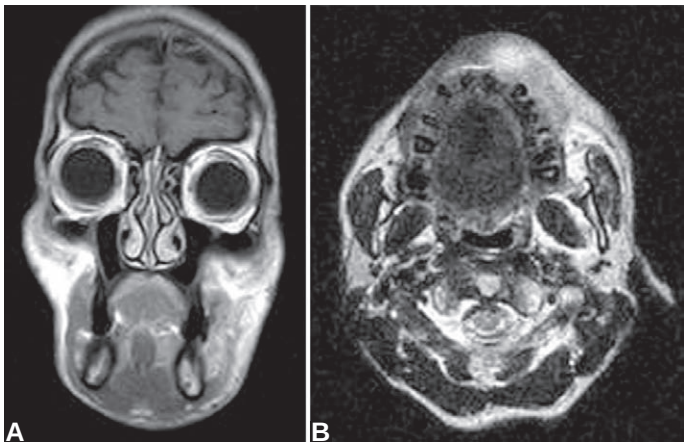


Fig. 14: Carcinoma buccal mucosa. Contrast CT scan neck axial section. Enhancing thickening of right buccal mucosa (12–14 mm thickness) with mild stranding of adjacent buccal planes
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

mandible) is involved or growth infiltrates the muscle, excision of the growth combined with marginal or segmental mandibulectomy or partial maxillectomy along with the reconstruction.

- *Stage III and IV*: Surgical resection combined with neck dissection and reconstruction and postoperative radiotherapy.
- *Reconstruction*:
  - Free radial forearm flap: Lesion extending to retromolar area, maxillary tuberosity or tonsillar fossa.



Figs 15A and B: Coronal (A) and Axial (B) T2W MR images. Hyperintense mass in left buccal mucosa along upper and lower alveolus. Underlying bone shows normal marrow signal intensity
Source: Dr Swati Shah, Professor, Radiodiagnosis, GCRI Medical College, Ahmedabad

  - Buccal fat pad with or without temporalis muscle flap: For reconstruction of maxillary defects, hard and soft palate defects, and cheek and retromolar defects.

CARCINOMA HARD PALATE

It is common in India as some people have the bad habit of reverse smoking (keeping the burning end of *bidi* or cigar in the mouth). Hard palate forms the roof of the oral cavity. Hard palate drains into the upper deep cervical and lateral retropharyngeal nodes. Anterior part of palate drains into submandibular nodes.

Hard palate contains high number of minor salivary glands. Minor salivary glands tumors (adenoid cystic, mucoepidermoid and adenocarcinoma) are more common than squamous cell carcinoma. Most of the squamous cell carcinoma of upper alveolus and hard palate arises from maxillary antrum.

Clinical Features

- *Lesion*: Superficial ulcer with rolled out edges, which may be felt by tongue as painless irregularity on the palate.
- *Local spread*: To gingiva, lip, soft palate. The invasion of hard palate can involve floor of the nasal cavity and the maxillary antrum (Fig. 16). This condition should be differentiated from cancer of maxillary antrum and nose, which could spread to and involve hard palate.
- *Lymphatic metastasis*: It is late and involves submandibular and upper jugular nodes and indicates poor prognosis.

Treatment

- Small tumors are excised along with the hard palate.
- Tumors confined to hard palate, upper alveolus and floor of maxillary antrum needs partial maxillectomy with block dissection if nodes are enlarged.
- Surgical defect in the palate needs closure or a suitable prosthesis.

CARCINOMA RETROMOLAR TRIGONE

Retromolar trigone is a triangular area of mucosa that covers anterior surface of the ascending ramus of mandible. Its base

Fig. 16: Carcinoma left side hard palate crossing midline and extending to alveolus and floor of nasal cavity

lies posterior to the last molar while its apex is adjacent to the maxillary tuberosity.

- It may be either primary or secondary (extension of growths from the gingiva, floor of mouth, buccal mucosa or the palate).
- Depending on the extent of lesion, wide surgical excision with block dissection is done.

MINOR SALIVARY GLAND TUMORS

Minor salivary gland tumors are mostly malignant (adenoid cystic carcinoma 40%, adenocarcinoma 30% and mucoepidermoid carcinoma 20%).

- Most common site is palate. Other sites are tongue, cheek, lip, gums and floor of mouth.
- *Treatment*: Wide surgical excision along with block dissection, if nodes are involved.

MELANOMA

- They are rare in oral cavity and oropharynx.
- Most common sites are palate (50%) and upper gingiva (25%).
- Males are affected double than females.
- The most common age of presentation is sixth decade.
- Initial asymptomatic melanomas are usually raised or nodular lesions. They may be flat.
- Lesion: The area of higher pigmentation varies from black to brown and present for many years. It ulcerates, bleeds and becomes painful. Amelanotic (nonpigmented) variety is red in color and rare (15%).
- The rapid growth is associated with destruction of underlying bone.
- Both the regional lymph node and distant metastases are seen. About 50% patients have metastases at the time of presentation.

Treatment

Wide surgical excision followed by radiotherapy.

Prognosis

- Recurrence is quite common.
- Prognosis is very poor (5-year survival only 5%). Indicators of poor outcome are following:
  - Main guide to prognosis: Histological tumor thickness (Breslow) in mm from the granular cell layer to the deepest identifiable melanocyte.
  - Malignant melanocytes in blood vessels.
  - Multiple or atypical mitoses.
  - Destruction of underlying bone.
  - Presence of metastases.

KAPOSI'S SARCOMA

- This vascular tumor is seen in acquired immune deficiency syndrome (AIDS) patients.
- It is multifocal in origin and primarily affects skin but may also be seen in the oral cavity.
- It presents as a reddish purple nodule or a plaque mostly on the palate.
- The tumor consists of spindle cells with hemorrhagic cleft-like spaces.
- Chemotherapy may offer good result in non-AIDS patients but its response in AIDS patients is poor.

Clinical Highlights

1. **Minor salivary gland tumor:** The most common site of minor salivary gland tumor is hard palate.
2. **Cancer tongue:** Lateral border is the most common site for carcinoma of oral tongue. Patient develops referred earache because the anterior two-third tongue receive its somatosensory innervation from the lingual nerve, which is branch of mandibular division of the CN V. The best treatment for stage III carcinoma of oral tongue is wide excision with supraomohyoid neck dissection and postoperative radiotherapy.
3. **Stage III T2N1M0 carcinoma of buccal mucosa:** The best treatment will be surgical excision of growth with supraomohyoid neck dissection and postoperative radiotherapy.
4. **Abbey-Estlander flap:** It is used to reconstruct defects in cases of carcinoma of lip angle (commissure). Such defects require partial commissure reconstruction.
5. **Melanoma:** Hard palate and maxillary gingiva are the most common sites of melanoma in the oral cavity.

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38 Pharyngeal Symptoms and Examination

The injunctions of Destiny are canceled if one takes refuge in God. Destiny has to strike off with her own hand what she has decreed about such a person.

— **Holy Mother Sarada Devi**

Points of Focus

EVALUATION OF PHARYNX
- **NASOPHARYNX**
  - ✓ Symptoms
  - ✓ Examination
- **OROPHARYNX**
  - ✓ Symptoms
  - ✓ Examination
- **LARYNGOPHARYNX**

EVALUATION OF ESOPHAGUS
- **BARIUM ESOPHAGOGRAPHY**
  - ✓ Fluoroscopy
  - ✓ Cineradiography
  - ✓ Video Esophagography
  - ✓ Modified Barium Swallow
  - ✓ Esophagogram
  - ✓ Oral Contrast Agents
- **ESOPHAGEAL MANOMETRY**
  - ✓ **Indication**
- **AMBULATORY 24-HOURS ESOPHAGEAL PH RECORDING**
  - ✓ **Indication**

DYSPHAGIA
  - ✓ Odynophagia
  - ✓ Causes
- **EVALUATION**
  - ✓ History
  - ✓ Physical Examination
  - ✓ Investigations
- **CLINICAL HIGHLIGHTS**

EVALUATION OF PHARYNX

For general scheme of case taking and general set-up of Bull's eye lamp light source and head mirror, see history and examination chapter in section of Basic Sciences. Pharyngeal symptoms include sore throat, ulcers, growth, pain, odynophagia (painful swallowing), stridor, voice change, nasal voice, dysphagia (difficulty in swallowing), snoring, cough, sputum, injury and foreign body. The associated complaints include fever, headache, earache, conductive hearing loss, vomiting, nasal regurgitation, loss of weight and anorexia. Patient must be inquired about the habits of chewing paan, sopari or tobacco, smoking and alcohol.

NASOPHARYNX

This upper part of pharynx lies behind the nasal cavity, and above the soft palate. Laterally, it communicates with the middle ear through the Eustachian tube opening.

Symptoms

Symptoms of nasopharyngeal disorders include the following:

- Nasal obstruction
- Nasal discharge anterior or posterior
- Epistaxis
- Conductive hearing loss
- Symptoms of cranial nerve palsies
- Metastatic neck nodes.

Examination

Box 1 shows the methods of examination and nasopharyngeal structures, and pathological findings seen during examination and their causes. Examination of cranial nerves and regional lymph nodes is important. The tumors of nasopharynx can involve any of the cranial nerve from second through twelfth (often CN IX, X and XI), and cervical lymph nodes (upper internal jugular, and along the accessory nerve).

- *Anterior rhinoscopy*: Some part of the nasopharynx can be seen in decongested nose (with vasoconstrictors); even on anterior rhinoscopy.
- *Posterior rhinoscopy:* Posterior rhinoscopy provides fragmented view of nasopharynx, which is mentally reconstituted by the examiner. The examiner has to tilt the mirror in different directions to visualize the structures present on different walls of the nasopharynx.
  - Retraction of soft palate with catheters*:* It facilitates postnasal mirror examination in some difficult cases and requires good local or general anesthesia. In this method, a soft rubber catheter is passed through each nostril and then taken out from the mouth through the oropharynx. Both ends of catheter are held together, and pulled forward. Retraction of soft palate makes the mirror examination easy. This method has been becoming obsolete with the advent of sinuscope and flexible nasopharyngolaryngoscopy.
- *Digital examination*: Digital examination though uncomfortable for the patient, is a simple method to palpate the nasopharynx. Examiner, standing behind and right to the patient invaginates patient's cheek with his left finger and introduces right index finger behind and above the soft palate into the nasopharynx. This method is usually avoided in cases of angiofibroma.
- *Endoscopy:* Endoscopy gives a bright and magnified view of the nose and nasopharyngeal structures. It can be performed by either rigid or flexible fiber-optic scope.
  - Rigid nasal endoscope (sinuscope or rhinoscope*):* They are available in different sizes and angles and introduced through the nose after instilling or spraying local anesthetic and decongestant. See details of sinuscopy examination in chapter Operations of Nose and Paranasal Sinuses.
  - Flexible nasopharyngolaryngoscope*:* It offers views of nose, pharynx and larynx. See chapter Laryngeal Symptoms and Examination in section of larynx, trachea and bronchus.

OROPHARYNX

Oropharynx lies behind the oral cavity, and includes tonsils and their pillars, soft palate, base of tongue, and posterior pharyngeal wall (Fig. 1). It lies behind the anterior tonsillar pillars, hard palate and V-shaped row of lingual circumvallate papillae.

Box 1: Examination of nasopharynx: Methods of examination, nasopharyngeal structures, and pathological findings and their causes

- *Methods:*
  - Anterior rhinoscopy
  - Posterior rhinoscopy
    - ◊ Retraction of soft palate with catheters
  - Digital examination
  - Endoscopy: Sinuscope, flexible nasopharyngolaryngoscope
  - Examination of cranial nerves especially CN IX, X and XI
  - Examination of neck nodes especially upper internal jugular, and along the accessory nerve
- *Nasopharyngeal structures:*
  - Anterior: Posterior border of nasal septum, choanae, posterior ends of turbinates and their meatuses
  - Lateral: Opening of Eustachian tube, torus tubarius, and pharyngeal recess (Fossa of Rosenmuller)
  - Floor: Upper surface of soft palate
  - Roof and posterior wall: Adenoids, median recess of nasopharyngeal bursa within the adenoid mass, Rathke's pouch dimple above the adenoids (junction of nasal septum and roof of nasopharynx)
- *Pathological findings and their causes:*
  - Discharge: Middle meatus (infections of anterior group of sinuses), superior meatus (infections of posterior group of sinuses)
  - Crusting. Atrophic rhinitis or nasopharyngitis
  - Mass:
    - ◊ Smooth pale mass (antrochoanal polyp)
    - ◊ Pink lobulated mass (angiofibroma)
    - ◊ Irregular bleeding mass (carcinoma)
    - ◊ Smooth swelling in the roof (Thornwald's cyst or abscess)
    - ◊ Irregular mass with radiating folds (adenoids)
    - ◊ Mulberry irregular mass filling the lower part of choana (hypertrophy of inferior turbinate)
  - Bleeding

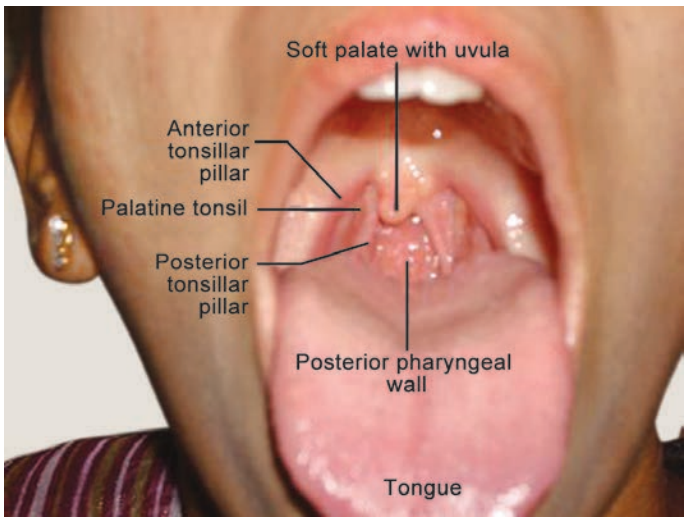

Fig. 1: Oropharyngeal structures seen through opened mouth

Symptoms

The lesions of oropharynx can disturb swallowing, phonation, respiration and hearing. The common symptoms and their causes are mentioned in Box 2.

- *Middle ear symptoms*: A conductive hearing loss can occur due to Eustachian tube malfunction, which may result from enlarged tonsils (interfere movements of soft palate), cleft palate and palatal paralysis. The infections of recurrent pharyngitis and tonsillitis can travel to the middle ear through Eustachian tube.
- *Cancer phobia*: These patients may be concerned with some structure in the oropharynx, which is noticed while looking in the mirror. Hypertrophic circumvallate or foliate papillae can generate cancer fear in these patients who need reassurance.

Examination

The examination begins by asking the patient to open the mouth widely. Tongue depressor is used to examine tonsillolingual sulcus, and to express contents of tonsillar crypts. The base of tongue is examined by laryngeal mirror. The structures of oropharynx and their common lesions are mentioned in Box 3.

- *Tonsils and pillars*: For expressing the material from tonsil crypts, pressure on the anterior pillar is applied with the edge of tongue depressor. Palpation should always be performed with a gloved finger to know the consistency of the mass. There is uniform congestion of the pillars, tonsils and pharyngeal mucosa in acute tonsillitis; however, congestion of only anterior pillars indicates chronic tonsillitis. Ulcer and proliferative growth may extend to or from the tonsil, base of tongue, and the retromolar trigone.
- *Soft palate*: In cases of peritonsillar abscess, uvula becomes edematous, and displaced to the opposite side. To note the movement of soft palate, patient is asked to say "AA". Deviation of the uvula and soft palate occurs to the healthy side in cases of vagus palsy, which may be associated with paralysis of posterior pharyngeal wall that manifests as a "curtain effect" (the paralyzed side moves like a sliding curtain to the healthy side). In cases of submucous cleft palate, in addition to bifid uvula, a notch can be palpated in the midline of the posterior part of hard palate.
- *Base of tongue*: Posterior one-third of tongue is best examined by indirect laryngoscopy and finger palpation. It lies between the V-shaped row of circumvallate papillae and the valleculae. Valleculae are two shallow depressions that lie between the base of tongue and the epiglottis.
- *Palpation*: Palpation of oropharynx including base of tongue is very important, as it helps in locating the infiltrative growth and its extension which is usually missed during inspection. If the patient fails to relax, and does not cooperate even after 4% xylocaine spray, palpation must be conducted under general anesthesia. The examiner must insert his/her finger in patient's cheek (especially in children) between the upper and lower teeth to prevent biting on the examiner's finger.

Box 2: Oropharyngeal symptoms and their causes

- *Sore throat and/or odynophagia (painful swallowing):* Tonsillitis, pharyngitis, aphthous ulcers, abscesses (peritonsillar, parapharyngeal, or retropharyngeal) and lingual tonsillitis
- *Dysphagia (difficulty in swallowing):* Tonsillar enlargements, benign or malignant tumors of tonsils, base of tongue, posterior pharyngeal wall and parapharyngeal region
- *Nasal regurgitation of fluid*: Paralysis of soft palate, cleft palate, palatal perforation
- *Change in voice:* Hypernasality (paralysis of palate), muffled or hot potato voice (space occupying lesions of the oropharynx)
- *Referred earache:* Ulcers of the base of tongue, tonsil, pillars and palate
- *Snoring or sleep apnea syndrome:* Large tonsils and tongue
- *Halitosis (bad smell from the mouth):* Infected tonsils, malignant tumors.

LARYNGOPHARYNX

For the evaluation (Symptoms and examination) of laryngopharynx, kindly see the chapter Laryngeal Symptoms and Examination in the section of Larynx, Trachea and Bronchus.

EVALUATION OF ESOPHAGUS

The symptoms of heartburn, dysphagia and odynophagia almost always suggest a primary esophageal disease.

- *Heartburn (Pyrosis)*: This is sensation of substernal burning that often radiates to the neck. It is highly indicative of gastroesophageal reflux disease (GERD). It is the reflux of acidic (rarely alkali) material into the esophagus, and discussed in detail in chapter Disorders of Esophagus.
- *Odynophagia*: It is sharp retrosternal pain on swallowing indicative of esophagitis due to candida, herpes viruses, Cytomegalovirus (CMV), especially in immunocompromised patients. The traumatic causes include caustic ingestions and pill-induced ulcers (see chapter Disorders of Esophagus).
- *Dysphagia*: The difficulty in swallowing is discussed in detail in other section of this chapter.

Box 3: Findings of oropharyngeal examination and their causes

- *Tonsils*: Present/absent
  - Size: Large and obstructive; small or embedded
  - Symmetry: Unilateral or bilateral enlargement
  - Crypts: White or yellow spots at the openings (follicular tonsillitis); white excrescences which are not easily wiped off (keratosis); expression of cheesy material (normal) or frank pus (septic tonsil)
  - Membrane: Membranous tonsillitis (diphtheria and Vincent's angina)
  - Ulcer: Malignant, aphthous, Vincent's angina, tuberculosis, ulcerating tonsillolith
  - Mass: Cystic (retention cyst), pedunculated or sessile solid mass (papilloma, fibroma), proliferative growth (cancer)
  - Bulge: Peritonsillitis and abscess, parapharyngeal abscess and tumor
  - Palpation: Hard (malignancy or tonsillolith), pulsation (internal carotid artery aneurysm), and bony (an elongated styloid process)
  - Tonsillar pillars: Congestion, ulcer, proliferative growth
- *Soft palate:* Redness, bulge or swelling (peritonsillitis), fibrous band (submucous fibrosis), vesicles, notch in the midline on posterior part of hard palate (submucous cleft palate), paralysis, ulcers/growths (benign or malignant)
  - Uvula: Edematous, displaced to the opposite side (peritonsillar abscess, palatal palsy), bifid uvula, "curtain effect" (paralysis of posterior pharyngeal wall)
- *Posterior pharyngeal wall:* Lymphoid nodules (granular pharyngitis), post nasal purulent drip (chronic sinusitis), thin glazed mucosa and crusting (atrophic pharyngitis), ulcers/growths (benign or malignant)
- *Base of tongue and valleculae:* (Indirect laryngoscopy and palpation): Color of mucosa (normal or congested); prominent veins, varicosities at the base of tongue or lingual thyroid, ulceration (malignancy, tuberculosis or syphilis), solid swelling (lingual thyroid, lymphoma, carcinoma base of tongue), cystic swelling (vallecular cyst, dermoid or thyroglossal cyst)

BARIUM ESOPHAGOGRAPHY (FIGS 2 AND 3)

Barium evaluation is gold standard for swallowing disorders and esophageal function. It allows evaluation of tongue movement, soft palate elevation, epiglottic tilt, laryngeal closure and peristalsis of pharyngoesophageal segment. It differentiates between mechanical lesions and motility disorders.

- *Contraindications*: Barium swallow is not used when aspiration is present or suspected. Other contraindications include perforation of pharynx and esophagus.
- *Indications and Findings* (Figs 4 to 6):
  1. Malignant lesions: They show irregular narrowing of lumen along with mucosal destruction, ulceration and shouldering. The length of tumor is important for its staging.
  2. Benign strictures: Smooth narrowing (may be at multiple sites) of a short or long part of esophagus.
  3. Cardiac achalasia (cardiospasm): Markedly dilated, elongated, and tortuous esophagus (mega esophagus) having fluid level in lumen.

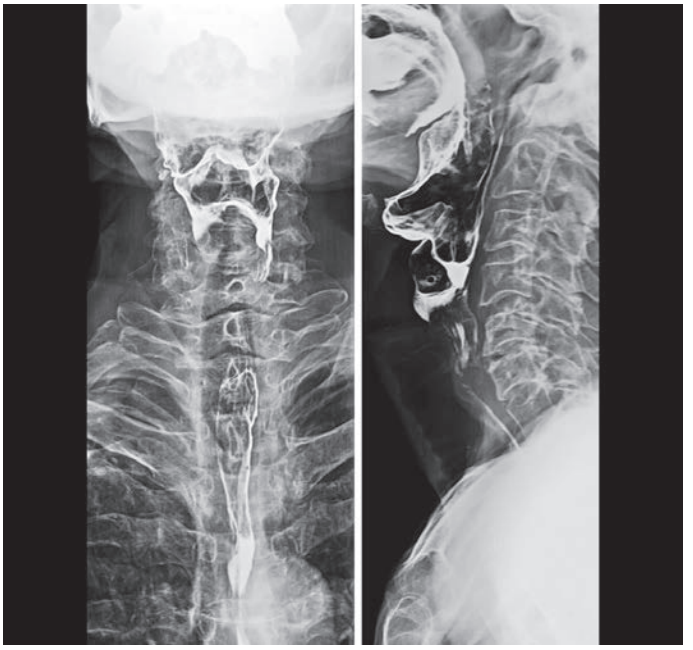

Fig. 2: Normal barium swallow of upper esophagus
Source: Dr Swati Shah, Professor, Radiodiagnosis, GCRI Medical College, Ahmedabad

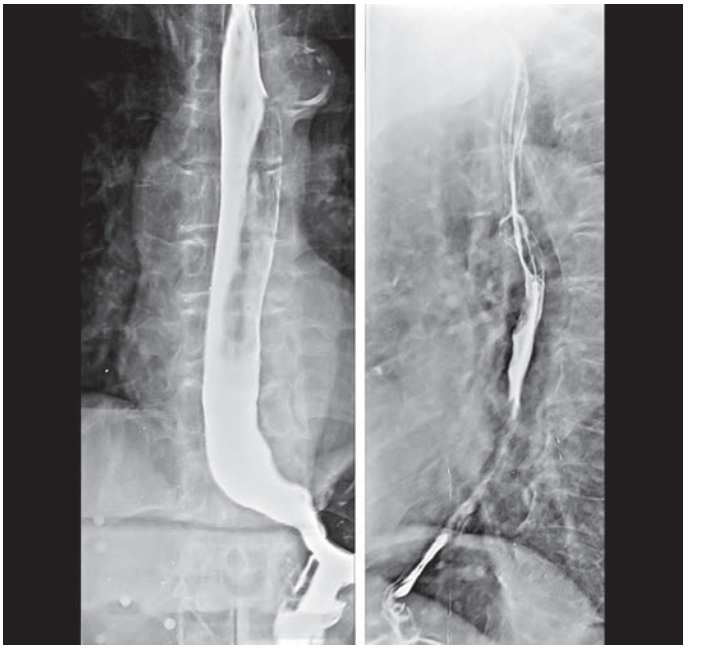

Fig. 3: Normal barium swallow of lower esophagus
Source: Dr Swati Shah, Professor Radiodiagnosis, GCRI Medical College, Ahmedabad

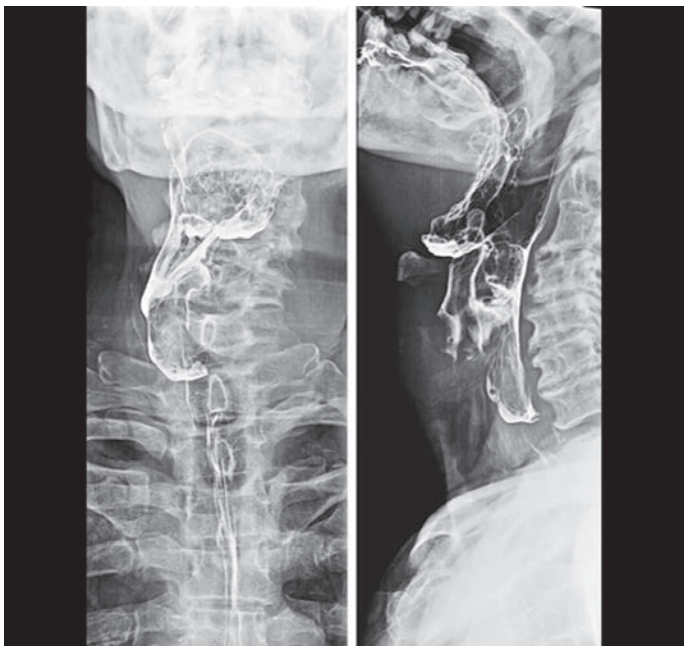

Fig. 4: Barium swallow. Filling defect in left pyriform fossa
Source: Dr Swati Shah, Professor, Radiodiagnosis, GCRI Medical College, Ahmedabad

Fig. 5: Barium swallow. Mucosal irregularity in lower third of esophagus and esophageal-gastric junction
Source: Dr Swati Shah, Professor, Radiodiagnosis, GCRI Medical College, Ahmedabad

4. Esophageal varices: Irregular filling defects (like a string of pearls) in lower portion of esophagus.

Barium swallow vs Esophagoscopy: Many clinicians prefer flexible endoscopy if they strongly suspect mechanical lesions. In motility disorders, barium swallow is done first as it provides more physiologic examination.

Fluoroscopy

In fluoroscopy, imaging intensifier projects emitted light on television screen which allows continuous monitoring of moving process such as barium swallow study.

Fig. 6: Barium swallow. Abrupt obstruction at mid-esophageal level
Source: Dr Swati Shah, Professor, Radiodiagnosis, GCRI Medical College, Ahmedabad

Cineradiography

High resolution images are obtained at a low frame rate. It provides better spatial resolution for mucosal detail, and reveals much of the swallowing dynamics.

Video Esophagography

Oropharyngeal dysphagia is best evaluated with this technique. Low-resolution images are obtained at a high frame rate. It allows longer exposure with less radiation. It evaluates equivocal filling defects. Peristalsis is evaluated with single swallow.

Modified Barium Swallow

This procedure, which is usually conducted by radiologist in conjunction with the speech pathologist, evaluates coordination of swallow reflex and cause, and severity of laryngotracheal aspirations. The patient is given varying thickness of barium suspension, which mimics different food consistencies.

Esophagogram

An esophagogram consists of three phases: Full column, air-contrast and mucosal relief.

- *Full column images*: *Single contrast study*
  - Barium: Thin suspension of barium.
  - Pharynx: Best examined in standing position, and extended neck prevents jaw from obscuring pharynx.
  - Esophagus: Patient is in prone oblique position, and drinks barium from a straw. It evaluates esophageal peristalsis and contour abnormalities. Rapid swallows followed by Valsalva's maneuver dilate the esophagus, and helps in identifying contour defects.
- *Air contrast images: Double contrast study*
  - Patient is upright, and slightly left anterior oblique.

- Swallowing of an effervescent agent (for distending esophageal lumen) precedes swallowing of thick barium suspension.
- Barium coats mucosal surface, and provides its exquisite detail.
- It can evaluate small mucosal tumors, and irregularities.
- Pharynx can be distended with phonation and modified Valsalva's maneuver.

- *Mucosal relief images:*
  - Thick barium suspension without effervescent agent.
  - Good for distal esophagus, gastroesophageal junction, and esophageal varices.
- *Concluding images:*
  - Valsalva and modified Valsalva's maneuvers after esophageal clearance should be done to see the gastroesophageal reflux.

Oral Contrast Agents

- *Barium suspension*: Contraindicated in perforation of pharynx and esophagus.
- *Water-soluble contrast agents used in contrast CT:* They are
  - Ionic contrast agents: Aspiration can lead to pneumonitis and pulmonary edema. Chances of allergic reactions and airway compromise are more.
  - Nonionic contrast agents: Safer but expensive.
- *Oil-based contrast agents*: No more in use.

ESOPHAGEAL MANOMETRY

A small pressure-sensing catheter assembly is passed into esophagus through nose. It performs manometric study of upper esophageal sphincter (UES), lower esophageal sphincter (LES) and esophageal body.

Indications

- To know the exact location of LES for placing a pH probe.
- To study peristaltic function of esophageal body before antireflux surgery.
- To confirm the diagnosis of esophageal motility disorders (achalasia or diffuse esophageal spasm) suggested by endoscopy or barium swallow.

AMBULATORY 24-HOURS ESOPHAGEAL pH RECORDING

A small pH probe is passed nasally and placed 5 cm above LES. It is attached to a portable pH device that continuously record pH for 24 hours. The pH recording tells about the amount of acid esophageal reflux. The temporal correlations between symptoms and reflux can be established.

Indications

It is the best way for documenting acid reflux. It is indicated in GERD only in the following situations:

- *Preoperative*: Before antireflux surgery when endoscopy is normal.
- *Proton pump inhibitor failures*: Symptoms are not responding to proton pump inhibitor (PPI) therapy, and endoscopy is normal.
  - Extraesophageal symptoms: When atypical (extraesophageal) symptoms (chest pain, asthma, cough, hoarseness and sore throat) do not respond to PPI therapy.

ESOPHAGOSCOPY

See chapter Endoscopies.

DYSPHAGIA

Dysphagia refers to difficulty in swallowing. It can be divided into two types: oropharyngeal (difficulty in transferring food bolus from oropharynx to upper esophagus) and esophageal (difficulty in transporting bolus through the body of esophagus).

- *Oropharyngeal dysphagia*: It is characterized by coughing, choking and regurgitation immediately after initiation of swallowing. Dysphagia is more with liquid than soft foods. It may be associated with dysphonia, dysarthria, or other neurological features. The cause may be neurological, muscular, motility or structural disorders.
  - Neurological: Brainstem lesions (pseudobulbar palsy), multiple sclerosis, or myasthenia gravis.
  - Muscular: Myopathies and hypothyroidism.
  - Motility: UES dysfunction.
  - Structural: Malignancy, surgery and Zenker's diverticulum.
- *Esophageal dysphagia*: It may occur due to mechanical obstruction or motility disorders.
  - Mechanical obstruction: The dysphagia is primarily for solids, and is usually continuous and predictable. In progressive lesions, dysphagia worsens with the progress of the lesion. Some of the common causes include:
    - Malignancy: Progressive dysphagia in elderly people.
    - Peptic stricture: Chronic heartburn and progressive dysphagia.
    - Schatzki's ring: Intermittent dysphagia which is not progressive.
  - Motility disorders: The dysphagia is for both solids and liquids. It is episodic, unpredictable and non-progressive. Some of the common causes include:
    - Achalasia: Progressive dysphagia.
    - Diffuse esophageal spasm (DES): Chest pain and intermittent non-progressive dysphagia.
    - Scleroderma: Dysphagia associated with chronic heartburn and Raynaud's phenomenon.

Odynophagia

It is sharp substernal pain on swallowing, and reflects severe erosive disease. The common causes include:

- Infectious esophagitis due to candida, herpes viruses and cytomegalovirus.
- Corrosive injury due to caustic ingestion.
- Pill-induced esophagitis.

Causes

The cause of dysphagia may lie in oral cavity, pharynx or esophagus (Box 4).

Box 4: Causes of dysphagia

- *Oral Causes:* They include disorders of following functions and structures:
  - Chewing: Trismus, fractures of mandible, neoplasms of the upper or lower jaw, and temporomandibular joints disorders
  - Lubrication: Causes of xerostomia such as radiotherapy and Mikulicz disease
  - Tongue: Paralysis, painful ulcers, tumors, abscess and total glossectomy
  - Palate: Cleft palate and oronasal fistula
  - Buccal cavity and floor of mouth: Ulcers and Ludwig's angina
- Pharyngeal Causes
  - Tumors: Tonsil, soft palate, pharynx, base of tongue, supraglottic larynx, or even obstructive hypertrophic tonsils
  - Inflammatory: Acute tonsillitis, peritonsillar abscess, retropharyngeal abscess, parapharyngeal abscess, acute epiglottitis, edema of larynx
  - Spasmodic: Tetanus and rabies
  - Paralytic: Palatal palsy, which present with nasal regurgitation, may be caused by diphtheria, bulbar palsy, or cerebrovascular accidents. Lesions of vagus and bilateral superior laryngeal nerves can lead to aspiration
- *Esophageal Causes:* The lesions may lie in the lumen, on the wall, or outside the wall of esophagus
  - Lumen obstruction: Atresia, foreign body, strictures, and benign or malignant tumors
  - Mucosal inflammation: Acute or chronic esophagitis
  - Motility disorders:
    - ◊ Hypomotility: Achalasia, scleroderma, and amyotrophic lateral sclerosis
    - ◊ Hypermotility: Cricopharyngeal spasm and diffuse esophageal spasm
  - External pressure:
    - ◊ Hypopharyngeal diverticulum
    - ◊ Hiatus hernia
    - ◊ Cervical spines: Osteophytes
    - ◊ Thyroid: Enlargement, tumors, and thyroiditis
    - ◊ Mediastinal: tumor, lymph node enlargement, aortic aneurysm, cardiac enlargement
    - ◊ Vascular rings: Dysphagia lusoria (Fig. 7)

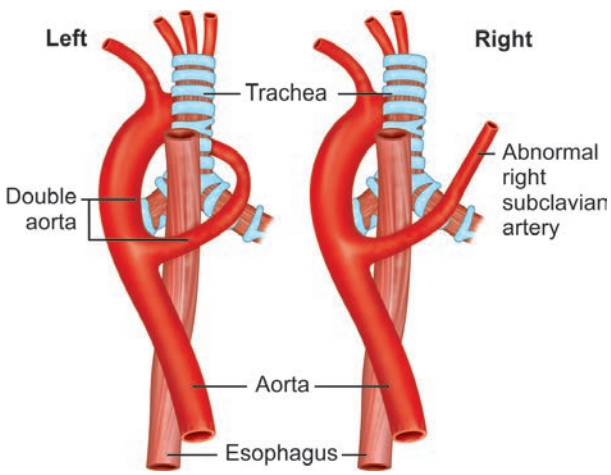

Fig. 7: Pressure on thoracic esophagus from abnormal great vessels as seen from behind. Dysphagia lusoria: (A) Double aorta; (B) Abnormal right subclavian artery

EVALUATION

History

- *Sudden onset:* Foreign body or impaction of food on a stricture or malignancy, and neurological lesions.
- *Gradually progressive:* Malignancy (short history with weight loss) and peptic strictures (long history with heartburn).
- *Chronic, continuous and not progressive:* Stricture.
- *Intermittent:* Spasms or spasmodic episodes over an organic lesion.
- *Liquids/solids:* More with liquids (paralytic lesions), more with solids, and progressing to liquids (malignancy).
- *Intolerance to acid food and fruit juices:* Ulcerative lesions.
- *Associated complaints:*
  - Regurgitation and heartburn: Hiatus hernia.
  - Regurgitation of undigested food in lying down and coughing in night: Hypopharyngeal diverticulum.
  - Nasal regurgitation: Palatal paralysis.
  - Aspiration into lungs: Laryngeal paralysis.

Physical Examination

The physical examination should include oral cavity, oropharynx, hypopharynx, larynx, neck, chest and nervous system including cranial nerves. The lesions of oral cavity and pharynx can be seen during physical examination whereas esophageal causes require investigations such as barium swallow and esophagoscopy.

Investigations

Upper endoscopy, barium swallow, manometry and pH recording are the mainstay of dysphagia evaluation.

- *Hemogram:* Iron deficiency anemia in Plummer-Vinson syndrome.

- *X-ray chest:* For advanced esophageal lesions, cardiovascular, pulmonary and mediastinal diseases or pneumomediastinum.
- *X-ray neck lateral and frontal projections:* For cervical osteophytes, lesions of post-cricoid or retropharyngeal region.
  a. Useful in children.
  b. Shows radiopaque foreign bodies.
  c. Saying "e" during exposure brings the tongue anterior, and shows oropharynx better.
  d. Blowing through closed mouth distends hypopharynx.
- *Computed radiography and digital radiography:* In this filmless technique, image is captured on an array of digital elements, and is read directly into a computer.
  Advantages:
  - Soft tissue differences: It can emphasize subtle soft tissue differences even if the image is suboptimally exposed.
  - Invert image: Radiodense elements appear whiter than radiolucent elements on traditional films but computer technology can invert the image and some aspects of anatomy, and pathology can be visualized better.
- *Barium swallow:* For malignancy, cardiac achalasia, strictures, diverticulum, hiatus hernia or esophageal spasms.
- *Cineradiography:* For motility disorders of esophageal wall or sphincters.
- *Ultrasound:* Transesophageal echosonography can evaluate the depth of malignant ulcer, which helps in staging the disease.
- *Manometry and pH studies:* For motility disorders, gastroesophageal reflux and esophageal spasm (spontaneous or acid induced). A pressure transducer along with a pH electrode and an open-tipped catheter are introduced into the esophagus. They measure the pressures of esophageal wall and sphincters. Gastroesophageal reflux disease is measured by pH electrode, which also measures the effectiveness of esophagus to clear the acid load.
- *Esophagoscopy (flexible or rigid):* For direct examination of esophageal mucosa and biopsy.
- *Associated investigations:* They vary from case to case such as bronchoscopy for bronchial carcinoma, cardiac catheterization for vascular anomalies, and thyroid scan for malignant thyroid.

Clinical Highlight

1. **Dysphagia lusoria**: This dysphagia is caused by the compression of esophagus by an abnormal right subclavian artery, which abnormally arise directly from the aorta and passes anterior or posterior to esophagus.

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39 Pharyngitis and Adenotonsillar Disease

Doing good to others is virtue; injuring others is sin. Strength and manliness are virtue; weakness and cowardice are sins. Independence is virtue; dependence is sin. Loving others is virtue; hating others is sin. Faith in God and in one's own self is virtue; doubt is sin. Knowledge of oneness is virtue; seeing diversity is sin.

— **Swami Vivekananda**

Points of Focus

- **INTRODUCTION**
- **PHARYNGITIS**
  - ✓ Irritative
  - ✓ Bacterial and Viral
- **INFECTIOUS MONONUCLEOSIS**
- **STREPTOCOCCAL TONSILLITIS-PHARYNGITIS**
  - ✓ Membranous Pharyngitis-Tonsillits
  - ✓ White Patches on Tonsils
- **FAUCIAL DIPHTHERIA**
- **TONSILLAR CONCRETIONS/TONSILLOLITHS**
- **INTRATONSILLAR ABSCESS**
- **TONSILLAR CYST**
- **KERATOSIS PHARYNGITIS**
- **DISEASES OF LINGUAL TONSILS**
- **CHRONIC ADENOTONSILLAR HYPERTROPHY**
- **ADENOID FACIES AND CRANIOFACIAL GROWTH ABNORMALITIES**
- **OBSTRUCTIVE SLEEP APNEA**
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

Infections of pharynx, tonsil and adenoids account for a significant proportion of childhood illnesses. They often lead to two of the most common pediatric surgeries—tonsillectomy and adenoidectomy. The predominant symptom of oropharyngeal infection is sore throat, which is one of the most common chief complaints to physicians.

PHARYNGITIS

Pharyngitis refers to inflammation of the pharynx mainly oropharynx. Acute catarrhal or superficial tonsillitis is a part of generalized pharyngitis, and is mostly seen in viral infections. In parenchymatous tonsillitis, which affects tonsil substance, tonsil is uniformly enlarged and red. Pharyngitis is common in adults while tonsillitis is mainly a disease of children. Majority of pharyngeal infections in adults are viral. The *Streptococcus pyogenes* only accounts for 5–10% in adults.

Irritative Pharyngitis

The causes of this common condition include:

- Postnasal drip
- Laryngopharyngeal reflux
- Occupational and environmental exposures.

Bacterial Pharyngitis

- In children, bacterial pharyngitis accounts for 30–40% of cases while in adults, it accounts for only 5–10%.
- Most cases of bacterial pharyngitis are caused by group A β-hemolytic *Streptococcus* (GABHS) *pyogenes*.
- Hoarseness and cough are not suggestive of pharyngitis.
- It is important to confirm the diagnosis of GABHS because majority of acute pharyngitis cases do not have streptococcal infection.
- See the section streptococcal tonsillitis-pharyngitis.

Viral Pharyngitis

- *Coxsackie virus infections*
  - Herpangina: It mostly affects children. Child presents with fever, sore throat, and vesicular eruption surrounded by a zone of erythema on the soft palate and pillars (See Chapter: Oral Mucosal Lesions).
  - Acute lymphonodular pharyngitis: It presents with fever, malaise and sore throat. White yellow, solid nodules on the posterior pharyngeal wall may be seen on examination.
- *Cytomegalovirus:* It affects immunosuppressed transplant patients. The infection mimics infectious mononucleosis but heterophil antibody test is negative.
- *Pharyngoconjunctival fever:* It is caused by *adenovirus,* and presents with sore throat, fever and conjunctivitis. Pain in abdomen mimics appendicitis.
- *Measles:* It is characterized by Koplik's spots (white spots surrounded by red areola) on the buccal mucosa opposite of the molar teeth 3–4 days before the rash.
- *Common cold* (*Rhinovirus*, *Corona virus*, *Parainfluenza virus*): It may affect tonsils and pharynx, and present with sore throat, dysphagia and fever. Tonsils may be enlarged but there is no exudate.
- *Herpes simplex virus*: It may cause exudative or nonexudative pharyngitis, which may be associated with gingivostomatitis (See Chapter: Oral Mucosal Lesions).

Treatment

- Treatment of viral infection is nonspecific and symptomatic.
- Antibiotics are used for secondary bacterial colonization (Gram-negative and *S. aureus*), which increases by about 50% in these viral diseases.

INFECTIOUS MONONUCLEOSIS

Infectious mononucleosis is caused by *Epstein-Barr virus*.

Clinical Features

- Often affects older children and young adults
- Fever, sore throat, exudative pharyngitis
- Both tonsils are enlarged, congested and covered with membrane
- Marked local discomfort.
- Lymphadenopathy: Lymph nodes enlarged in the posterior triangle of neck.
- Hepatosplenomegaly.

Diagnosis

- *Characteristic feature:* Petechiae at the junction of hard and soft palate.
- *CBC:* Fifty percent lymphocytes, of which 10% are atypical. White cell count normal in first week and rises in second week.
- *Serological tests:* Monospot and Paul Bunnell or Ox-cell hemolysis test shows high titers of heterophil antibody.

Treatment

- *Symptomatic* and recovery may take weeks.
- *Antibiotics:* No role except in secondary bacterial infection. Ampicillin causes skin rash in this condition and should be avoided.
- *Management of upper airway obstruction:* Airway obstruction can occur due to significantly enlarged tonsils. It is managed with:
  - Nasopharyngeal airway
  - High dose steroids
  - Tonsillectomy or tracheostomy.

Infectious mononucleosis: Amoxicillin or ampicillin can cause a salmon-colored rash.

STREPTOCOCCAL TONSILLITIS-PHARYNGITI

In acute follicular tonsillitis, infection spreads into the crypts which become filled with purulent material. The openings of crypts present yellowish spots of pus which may coalesce, and form a membrane on the surface of tonsil (acute membranous tonsillitis).

Etiology

- *GABHS* is a precursor of two serious conditions acute rheumatic fever and poststreptococcal glomerulonephritis. It is the most common cause of acute bacterial tonsillitis-pharyngitis. Epidemic forms are seen in recruit camps and daycare facilities. GABHS are Gram-positive cocci that grow in chain.
  - Natural reservoir: Skin, nasopharynx and oropharynx.
  - Spread: Mostly through aerosolized microdroplets; less commonly by direct contact, and rarely through ingestion of contaminated non-pasteurized milk or food.
  - Seasons: Autumn and winter.
- *Non-group A β-hemolytic streptococcal infection* is clinically similar to GABHS. It is comparatively less common.
- *The other bacteria* may primarily infect the tonsil or may be secondary to a viral infection and mimic GABHS infection. They include *staphylococci*, *pneumococci* and *H. influenzae*.

Clinical Features

The disease often affects school going children (peak 5–6 years), but may affect infants and individuals above 50 years of age.

Symptoms

- Throat pain: Dry throat, fullness in throat or sore throat.
- Dysphagia: Difficulty in swallowing or odynophagia.
- Fever: Temperatures 38–40°C may be associated with chills and rigors. The child may present as a case of pyrexia of unknown origin.
- Earache: It may be either referred, or due to acute otitis media.
- Constitutional symptoms: Headache, limb and back pain, malaise and constipation.
- Abdominal pain: It is due to mesenteric lymphadenitis, and simulates acute appendicitis.

Physical Findings

- Tongue: Dry and coated tongue.
- Breath: Fetid breath (halitosis).
- Oropharynx:
  - Hyperemia of pillars, soft palate and uvula.
  - Tonsils red and swollen with yellowish spots of pus at the opening of crypts (acute follicular tonsillitis) which

may coalesce, and form a membrane on the surface of tonsil (acute membranous tonsillitis). This membrane can be easily wiped away with a swab. The enlarged tonsils may meet in midline.
  - Edema of the uvula and soft palate may be present.
- Lymph nodes: Enlarged and tender jugulodigastric lymph nodes.

Diagnosis

Sore throat and fever with cervical adenopathy and pharynx with exudative covering are highly suggestive of *Streptococcus pyogenes*.

- *Rapid strep tests:* Latex agglutination or enzyme-linked immunosorbent assay (ELISA) methods extract antigen (group-A streptococcal) from a swab. It is highly specific (95%), but less sensitive (60–100%) than culture.
- *Throat culture*: Swab the posterior pharynx and tonsillar area when body temperature is greater than 38.3°C or when patient presents only with sore throat, or when rapid strep test is negative in strongly suspected cases.

Treatment

- *General:* Bed rest and plenty of fluids.
- *Symptomatic:* Analgesics and antipyretics such as aspirin or paracetamol.
- *Specific:* Antibiotics for 7–10 days reduces the chances of suppurative complications and acute rheumatic fever, but not poststreptococcal glomerulonephritis.
  - Penicillin or amoxicillin is the drug of choice. If there is no response then suspect beta-lactamase producing organisms and anaerobes, and start
    - Amoxicillin + clavulanic acid or
    - Clindamycin or
    - Erythromycin + metronidazole
- *Asymptomatic carriers:* They usually do not need any treatment except when:
  - Family member is having rheumatic fever.
  - Family members are getting recurrent streptococcal infection.

Group A beta-hemolytic streptococci: It is the most common cause of acute tonsillitis and can also result in rheumatic fever and poststreptococcal glomerulonephritis.

Complications

- Scarlet fever
- Rheumatic fever
- Acute glomerulonephritis
- Acute otitis media: It may coincide with recurrent tonsillitis.
- Subacute bacterial endocarditis (SABE): Acute tonsillitis in cases of valvular heart disease can cause SABE, which is usually due to *streptococcus* viridans.
- Chronic/Recurrent tonsillitis due to incomplete resolution of tonsil infection, which may persist in lymphoid follicles of the tonsil like microabscesses.
- Peritonsillar abscess
- Parapharyngeal abscess
- Retropharyngeal space infection
- Cervical abscess: Suppuration of jugulodigastric lymph nodes.

Differential Diagnosis of Membranous Pharyngitis-Tonsillitis

In membranous tonsillitis an exudative membrane forms over the medial surface of the tonsils. It occurs due to pyogenic organisms.

- *Agranulocytosis*:
  - Ulcerative necrotic lesions in the oropharynx.
  - Patient looks very ill.
  - Diagnosis:
    - Total leukocyte count: 50–2000/cumm.
    - Polymorph neutrophil 5% or less.
- *Leukemia:* In children, acute lymphoblastic leukemia is more common (75%) than acute or chronic myelogenous (25%) leukemia. In adults, 80% are nonlymphocytic, and 20% are acute lymphocytic leukemia.
  - TLC: Greater than 100,000/cumm.
  - Progressive anemia.
  - Bone marrow examination: Blasts cells are seen.
- *Aphthous ulcers:* Small/large, single/multiple painful ulcers on any part of oral cavity or oropharynx. Solitary big ulcer may involve the tonsil and pillars. (See Chapter: Oral Mucosal Lesions).
- *Malignancy tonsil:* (See Chapter: Tumors of Oropharynx).
- *Traumatic ulcer:* Any injury to oropharynx. It may be accidental due to a tooth brush, a pencil or finger. It heals by formation of a membrane which appears within 24 hours.
- *Diphtheria:* See following section and Table 1.
- *Infectious mononucleosis:* See other section.
- *Vincent's angina:* (See Chapter: Oral Mucosal Lesions).
  - *Causative organism*: *Fusiform bacilli* and spirochetes.
  - *Clinical Features*: It has insidious onset with mild fever and discomfort in throat. It presents with membrane, which usually involves one tonsil. This pseudomembrane can be easily removed, and reveal an irregular ulcer.
  - *Diagnosis*: Throat swab shows the causative organisms.

Differential Diagnoses of White Patches on Tonsils

In addition to the above-mentioned causes of membranous pharyngitis-tonsillitis, certain conditions (Box 1), most of which are described in the sections of this chapter, can present with white lesions on the tonsils.

FAUCIAL DIPHTHERIA

The increasing coverage of child population by diphtheria immunization [combined (DPT, DT, dT) and single vaccines] has significantly reduced the incidence of diphtheria. The disease was quite common and feared of. It is now fortunately seen uncommonly. However, it should be kept in mind whenever any membrane is seen on the tonsils of a child (Table 1).

- *Causative organisms:* Gram-positive bacilli, *Corynebacterium diphtheriae.*
- *Spread*: It spreads by droplet infection. Diphtheria carriers harbor organisms in their throat, and have no symptoms.
- *Incubation period*: 2–6 days.

Box 1: Differential diagnoses of white patches on tonsils

- *Trauma:* Foreign body induced ulcer and infection, thermal injuries (hot meal and chemicals) and radiotherapy
- *Bacterial infections*: Acute tonsillitis, diphtheria, Vincent's angina
- *Fungal infections*: Candidiasis
- *Viral infections*: Herpes, infectious mononucleosis
- *Blood dyscrasias*
- *Chronic infections*: Tuberculosis (painful multiple white patches) and syphilis
- *Benign tumors*: Papilloma
- *Malignant:* Irregular white patch with everted margin and induration (carcinoma), smooth nonulcerated swelling with induration (lymphoepithelioma)
- *Tonsillar cyst (Congenital or acquired):* Yellowish white
- *Tonsillar keratosis*: Whitish horny outgrowths
- *Tonsillolith:* Hard whitish lesion
- *Aphthous ulcer*: Intensely painful

Clinical Features

- Children are affected more.
- Diphtheria has slow onset of local discomfort.
- Fever seldom rises above 38°C.
- The dirty gray tenacious membrane extends beyond the tonsils on to the soft palate and posterior pharyngeal wall and after removal, leaves a raw bleeding surface.
- Larynx and nasal cavity can also be affected.
- Cervical lymph nodes (jugulodigastric) get enlarged and tender, and may present as "bull-neck" appearance.
- The child looks ill and toxemic.

Diagnosis

Diphtheria is usually a clinical diagnosis and in cases of doubt, the child should be treated on the line of diphtheria without wasting much time.

- Urine may show albumin.
- Smear and culture of throat swab will reveal *Corynebacterium diphtheriae*.

Complications

Diphtheria exotoxin is toxic to heart and nerves.

- *Heart:* Myocarditis, cardiac arrhythmia and acute circulatory failure.
- *Neurological*: Paralysis of palate, diaphragm and ocular muscles.
- *Larynx*: Airway obstruction.

Treatment

- *Antidiphtheric serum:* Antidiphtheric serum (ADS) is started immediately on clinical suspicion to neutralize the free diphtheria exotoxin. Dose depends on the site, duration and severity of disease.
  - 20,000 to 40,000 units: History of less than 48 hours, or membrane is limited to tonsils only.
  - 80,000 to 120,000 units: History of more than 48 hours, or membrane extends beyond tonsils.
  - Mode of administration: Intravenous infusion in saline in about 60 minutes.
  - Sensitivity test: Horse serum is tested by conjunctival or intracutaneous test with diluted antitoxin. Adrenaline must be handy for any hypersensitivity reaction.
  - Desensitization: Desensitization is required if patient is hypersensitive to ADS.
- *Antibiotics:*
  - Benzyl penicillin 600 mg 6 hourly for 7 days.
  - Erythromycin 500 mg 6 hourly orally in penicillin sensitive individuals.

TONSILLAR CONCRETIONS/TONSILLOLITHS

Tonsillolith (calculus of the tonsil) may be seen in chronic tonsillitis. The blocked tonsillar crypt causes retention of debris, which consists of inorganic salts of calcium and magnesium (formation of stone).

TABLE 1 Distinguishing features of acute follicular tonsillitis and diphtheria

Features	*Acute follicular tonsillitis*	*Diphtheria*
Past history	Recurrent sore throat with fever	Contact with diphtheria patient
Diphtheria vaccination	Taken	Not taken
Age	No age bar	Children
Onset	Acute	Insidious
Throat pain	Severe	Mild
Fever	High grade	Low grade
Hoarseness of voice and respiratory distress	Never	Present in advanced disease
Neck bull due to cervical lymphadenopathy	Absent	Not uncommon
Pulse rate	In proportion to fever	Out of proportion to fever and weak
Toxemia	Absent	Present in advanced disease
Tonsillar membrane	Limited to tonsil and easily removed	May extend adjacent structures, and difficult to remove, and leaves raw bleeding area
Throat swab	Streptococci	*Corynebacterium diphtheriae*
Urine	No albumin	Albumin often present
First line of treatment	Antibiotics	Antidiphtheric serum
Mortality	Nil	High

Clinical Features

- The affected crypt gradually enlarges, and may ulcerate on medial surface of tonsil.
- Retained material may have bacterial growth and then present with halitosis and sore throat.
- Whitish foul-tasting and foul-smelling cheesy material can be expressed from tonsils.
- Tonsilloliths are usually seen in adults. Patient presents with local discomfort or foreign body sensation. They are diagnosed by palpation or probing.

Treatment

- *Conservative:* Expression of concretions/cheesy material and chemical cauterization of crypts with topical silver nitrate application.
- *Tonsillectomy:* In cases of persistent pain, halitosis, or foreign body sensation.

INTRATONSILLAR ABSCESS

Accumulation of pus within the blocked tonsillar crypt can occur in cases of acute follicular tonsillitis.

- *Clinical Features*
  - Marked local pain and dysphagia.
  - Tonsil swollen and red.
- *Treatment*
  - Antibiotics
  - Drainage of the abscess
  - Tonsillectomy

TONSILLAR CYST

Blocked tonsillar crypt may present as a yellowish swelling over the tonsil.

- Usually they are asymptomatic.
- If symptomatic, it can be drained.

KERATOSIS PHARYNGITIS

- White or yellowish dots or horny excrescences on the surface of tonsils, pharyngeal wall or lingual tonsils characterize this benign condition. These excrescences are firmly adherent and cannot be wiped off. They are the result of hypertrophy and keratinization of epithelium.
- Patient does not have features of acute follicular tonsillitis such as sore throat, fever, cervical nodes and exudates.
- Treatment
  - The spontaneous regression does occur so, no specific treatment is required.
  - The concerned patients need reassurance.

DISEASES OF LINGUAL TONSILS

Hypertrophy, infection or abscess can occur in lingual tonsil. Compensatory hypertrophy of lymphoid tissue may occur in response to repeated infections especially in tonsillectomy patients.

Clinical Features

- Discomfort on swallowing
- Feeling of lump in the throat
- Dry cough
- Thick voice
- Lingual tonsil are enlarged, and congested or studded with follicles
- Cervical lymph nodes are enlarged

Treatment

- Antibiotics
- Diathermy coagulation or excision of lingual tonsils (by conventional or laser surgery).

Lingual Tonsillar Abscess

- *Clinical Features:*
  - Severe unilateral dysphagia and excessive salivation.
  - Pain in the tongue
  - Enlarged and tender jugulodigastric nodes.
- *Complications*: Laryngeal edema.
- *Treatment*: Antibiotics, analgesics, proper hydration and incision and drainage of the abscess.

CHRONIC ADENOTONSILLAR HYPERTROPHY

Adenotonsillar hypertrophy (Fig. 1) can cause obstructive sleep apnea (OSA) and affect craniofacial growth and microbiologic flora of tonsils and adenoids (Chronic adenotonsillar hypertrophy).

Etiology

- *Physiological:* In children, there occur physiological enlargement of adenoids and tonsils (associated with generalized lymphoid hyperplasia) in the first to the fourth year of life due to increased immunologic activity.
- *Infection:* β-lactamase producer pathogenic bacteria.
- Recurrent rhinitis and sinusitis.

Fig. 1: Kissing tonsils

- *Allergy* of upper respiratory tract.
- *Irritation:* Second hand smoke exposure.

Clinical Features

The size of the adenoid mass (relative to the available space in the nasopharynx) and infection are important in causing the nasal, aural or general symptoms.

- *Nasal:*
  - Nasal obstruction: Mouth breathing is one of the commonest symptoms. It interferes with feeding or suckling (respiration and feeding cannot take place simultaneously). The child may fail to thrive.
  - Nasal discharge: The choanal obstruction and associated chronic rhinosinusitis may present with wet bubbly nose.
  - Epistaxis: Not common. Acute adenoiditis may cause epistaxis.
  - Rhinolalia clausa: Voice becomes toneless and loses nasal quality.
- *Ear:* Eustachian tube obstruction and infection may result in following features:
  - Conductive hearing loss and retracted tympanic membrane.
  - Recurrent acute otitis media.
  - Chronic suppurative otitis media.
  - Serous otitis media.

ADENOID FACIES AND CRANIOFACIAL GROWTH ABNORMALITIES

The characteristic facial appearance called "adenoid facies" and craniofacial growth abnormalities are caused due to chronic nasal obstruction and mouth breathing. The features, some of which are reversed after adenotonsillectomy, include:

- An elongated face
- Retrognathic mandible
- Dull expression
- Dark circles under the eyes
- Open mouth
- Pinched nose due to disuse atrophy of alae nasi
- Hitched up upper lip
- Open bite, protrusive maxilla and buccal posterior crossbite
- Prominent and crowded upper teeth
- High arched hard palate because of the absence of moulding action of the tongue.

The children with adenotonsillar hypertrophy, which are considered for orthodontic procedures for malocclusion, should have an ENT evaluation.

Airway Obstruction

- *Excessive loud snoring at night*
- *Cor pulmonale*: Chronic upper airway obstruction due to chronic adenotonsillar hypertrophy leads to pulmonary ventilation-perfusion abnormality and chronic alveolar hypoventilation. The chronic hypercapnia and hypoxia result in respiratory acidemia, pulmonary artery vasoconstriction, and right ventricular dilation and then eventually cardiac failure.
- *Obstructive sleep apnea*

Diagnostic Assessment of Tonsils and Adenoids

- *Clinical:* Confirm the features of adenotonsillar hypertrophy mentioned above.
- *Postnasal mirror examination:* An adenoid mass can be seen.
- *X-ray soft tissue nasopharynx lateral view:* Reveal size of adenoids, and extent of nasopharyngeal air space compromise.
- *Rigid or flexible nasopharyngoscope under topical anesthesia:* Shows details of the nasopharynx.
- *Nasal obstruction:* Evaluate for other causes of nasal obstruction such as turbinate hypertrophy.

Assess for overt or submucous cleft of palate: See for bifid uvula and translucent line through the mid soft palate, and palpate for notching of posterior part of hard palate. It is important before adenotonsillectomy.

OBSTRUCTIVE SLEEP APNEA

In children, adenotonsillar hypertrophy is the most common cause of sleep apnea.

Clinical Features

The sleep disturbances may have the following consequences:

- Episodes of apnea
- Too much loud snoring
- Chronic mouth breathing
- Interrupted sleep with frequent awakening and nightmares
- Hypersomnolence
- Poor school/work performance
- Dysphagia
- Hyponasal speech (rhinolalia clausa)
- Failure to thrive: OSA in children may disrupt growth hormone during REM sleep.
- Enuresis: Children with secondary enuresis (develop later in childhood) associated with significant airway obstruction due to adenotonsillar hypertrophy usually respond to adenotonsillectomy. Primary enuresis, which is congenitally present, does not respond to this surgery.
- Obesity: Like adults (classic obesity-related pickwickian syndrome), obesity is not a factor in pediatric OSA. It is due to adenotonsillar hypertrophy (see chapter Obstructive Sleep Apnea).

Diagnosis

- *Polysomnography:* This expensive testing is not must in pediatric OSA, which is associated with confirmed adenotonsillar hypertrophy. But it is required in children where OSA is not associated with adenotonsillar hypertrophy.
- *Sleep sonography:* Tape recording of night sleep patterns at patient's home reliably detects evidence of OSA. It is an easy and most economical method of diagnosing OSA.
- X-ray chest and ECG should be taken preoperatively.

Treatment

- *Medical:* Following measures may cure early disease without resort to surgery:
  - Breathing exercises

- Decongested, saline or steroid nasal drops
- Antihistaminics
- Antibiotics
- Management of any associated nasal allergy.

- *Surgical*: Adenotonsillectomy
  - For indications, contraindications, preoperative assessment, surgical techniques, postoperative care and complications *see* Chapter: Adenotonsillectomy.

Clinical Highlights

1. **Acute tonsillitis:** Most common causative microorganism of acute tonsillitis is *Streptococcus haemolyticus* (Group A beta-hemolytic *Streptococcus pyogenes*).
2. **Pharyngoconjunctival fever:** It occurs in epidemics and causes follicular conjunctivitis and acute pharyngitis with fever.
3. **Herpangina:** This self limiting infection of children causes sore throat and fever.
4. **Infectious mononucleosis:** The sore throat and fever is associated with generalized lymphadenopathy and splenomegaly. Atypical lymphocytes are seen in peripheral smear.
5. **Faucial diphtheria:** It can cause palatal palsy.
6. **Vincent's angina:** The clinical features include gingivitis, stomatitis and ulceration of the tonsils.
7. **Gray white membrane on the tonsils**: Some of the causes are infectious mononucleosis, streptococcal tonsillitis and diphtheria.
8. **Keratosis pharyngitis:** Yellow spots over the tonsil are not easy to wipe off.
9. **Adenoid face**: In children with adenoid hyperplasia, facies show crowded teeth, high-arched palate, open mouth with under slung lower jaw and underdeveloped/pinched nostrils.
10. **Adenoid hypertrophy:** Nasopharyngeal obstruction can cause sinusitis, serous otitis media, and cor pulmonale. Management of adenoid hypertrophy with conductive hearing loss due to otitis media with effusion (OME) is adenoidectomy with grommet insertion.
11. **Rhinolalia clausa:** The common causes include allergic rhinitis, adenoids and nasal polyps.
12. **Cor pulmonale:** It can occur in children with chronic obstruction of upper respiratory tract due to enlarged tonsils and adenoids.

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40 Sleep Apnea and Sleep-disordered Breathing

Isn't it man that makes money? Where did you ever hear of money making man? If you can make your thoughts and words perfectly at one, if you can, I say, make yourself one in speech and action, money will pour in at your feet of itself, like water.

— **Swami Vivekananda**

Points of Focus

- **INTRODUCTION**
  - ✓ Classifications
- **PATHOPHYSIOLOGY OF OBSTRUCTIVE SLEEP APNEA**
  - ✓ Sleep Patterns in OSA
  - ✓ Factors Aggravating OSA
- **DIAGNOSIS AND EVALUATION OF OSA**
  - ✓ History
  - ✓ Physical Examination
  - ✓ Flexible Nasopharyngolaryngoscopy
  - ✓ Radiography
  - ✓ Polysomnography
  - ✓ Home Sleep Studies
  - ✓ Multiple Sleep Lateney Test
- **SEVERITY OF OSA**
- **COMPLICATIONS OF OSA**
- **NONSURGICAL TREATMENT**
  - ✓ Lifestyle Modifications
  - ✓ Nasal Continuous Positive Airway Pressure
  - ✓ Intraoral Devices
- **SURGICAL TREATMENT OF OSA**
  - ✓ Uvulopalatopharyngoplasty
  - ✓ Other Surgical Procedures
- **SURGICAL TREATMENT OF SNORING WITHOUT OSA**
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

Sleep-disordered breathing (SDB) is a type of intrinsic sleep disorder (dyssomnia). It encompasses primary snoring, upper airway resistance syndrome (UARS), obstructive sleep apnea syndrome and obesity-hypoventilation syndrome (Pickwickian syndrome). SDB (apnea-hypopnea index of 5 or more per hour of sleep) occurs in about 9% women and 24% men between the ages of 30 and 60 years. Among the women SDB is more common in postmenopausal age. About 70% SDB patients are obese.

About 25% of adults snore and the prevalence rises with increasing age. About 60% of males over 60 years of age have snoring. Though snoring does indicate some obstruction in upper airway most snorers have no obstructive sleep apnea (OSA). Although OSA patients are typically loud snorers, not all people who snore have OSA. Snoring and OSA probably represent a continuum of a similar pathology. There is difference between the noise of snoring and stridor.

Classifications

- *Snoring*: In this noisy breathing, a rough, rattling inspiratory noise is produced by vibration of pendulous soft palate, or occasionally of vocal cords, during sleep.
  - *Primary snoring*: In these snorers, an apnea-hypopnea index is less than 5 events per hour of sleep.
  - *Upper airway resistance syndrome*: These patients have an apnea-hypopnea index of less than 5 events per hour of sleep but have an elevated arousal index (more than 5 times per hour).
- *Sleep apnea*: It refers to intermittent cessation of breathing for 10 seconds or longer during sleep. The airflow is usually measured at the nose and lips. There are three classes of sleep apnea: central, obstructive and mixed.
  - Obstructive sleep apnea syndrome: In obstructive sleep apnea syndrome (OSAS) transient upper airway obstruction results in intermittent cessation of air flow or

breathing though there is normal respiratory effort. This is the most common type of sleep apnea and is managed by ENT surgeons. An apnea-hypopnea index is more than 5 events per hour of sleep. During the episodes, oxyhemoglobin desaturation becomes less than 90%.
  - Central sleep apnea: Intermittent failure in the respiratory drive centers in central nervous system (CNS) results in cessation in air flow or breathing. The phrenic nerve and diaphragm become temporarily inactive. There is no obstruction in upper respiratory tract. This class of sleep apnea is managed by neurologists and sleep specialists.
  - Mixed sleep apnea: This variant of OSA has components of both central and obstructive sleep apnea.
- *Pickwickian syndrome*: This syndrome is characterized by obesity and hypersomnolence. It was described by Charles Dickens in The Pickwick Papers. In obese persons, weight of neck, redundant soft palate and thick base of tongue contribute to OSA.
- *Stridor*: A high-pitched, noisy respiration, like the blowing of the wind is a sign of respiratory obstruction, especially in the larynx or trachea. See chapter Laryngeal Symptoms and Examination.
  - Congenital stridor: Crowing inspiration occurs at birth or within the first few months of life. It may be without any apparent cause or due to abnormally flaccid epiglottis or arytenoids.
  - Expiratory stridor: A singing sound comes from semi-approximated vocal folds offering resistance to the escape of air.
  - Inspiratory stridor: An inspiratory crowing sound occurs due the pathology of epiglottis or larynx.

PATHOPHYSIOLOGY OF OBSTRUCTIVE SLEEP APNEA

An obstruction at any level of upper respiratory tract (i.e. from nose to the true vocal cords) can result in OSA. During inspiration there occurs a negative pressure within upper respiratory tract. Muscle relaxation of whole body (including muscles of upper respiratory airway) occurs during the deeper stages of sleep (Stages III, IV and REM). These two factors (muscle relaxation and negative pressure) results in collapse and obstruction of the airway in the patients, who have redundant tissue or narrow airway. Obstruction of airway leads to oxyhemoglobin desaturation that results in arousal and brings the patient to a lighter level of sleep. The upper respiratory airway is established again with the loud snorting breathing. The factors mentioned in Box 1 can cause or exacerbate OSA.

Sleep Patterns in OSA

During the deep stages of sleep (Stage III, IV and REM) most obstructive events occur as the muscles are most relaxed. So the OSA patients mainly have less deep sleep of Stage I and II. Due to these restless sleep patterns, these patients are deprived of deep sleep and have daytime somnolence. So the OSA patients have quick onset of sleep and multiple arousals. The successful treatment of OSA results in increase in REM sleep.

Box 1: Factors aggravating OSA

- Alcohol
- Sedatives including antihistamines and cough suppressants
- Allergic rhinitis
- Upper respiratory infections (URI)
- Weigh gain
- Supine position.

Factors Aggravating OSA

Certain factors are known to exacerbate OSA episodes. Most snoring is not pathologic and may be prevented by life style changes by managing factors mentioned in Box 1.

DIAGNOSIS AND EVALUATION OF OSA

History

Though not found totally reliable, history of sleep apnea is elicited from the spouse. The symptoms are listed in Box 2. Though not pathognomonic the typical symptoms are snoring, day time somnolence, morning headache and restless sleep. Periods of apnea result in frequent arousals at night. The neurobehavioral symptoms affect school and work performance.

Physical Examination

- *ENT head and neck examination*: Patients with OSA need complete ENT, head and neck examination and measurement of blood pressure, neck circumference and body mass index. The common findings, which constitute even the etiological factors, are mentioned in Box 3. In children OSA is generally due to tonsil (Fig. 1) and adenoid hypertrophy. For pediatric OSA see chapter Pharyngitis and Adenotonsillar Diseases.
- *Body mass index*: The body weight in kilogram is divided by the height in square meters.
  - Normal: 18.5 to 24.9
  - Over weight: 25 to 29
  - Obesity: 30 to 34.9
- *Collar size*: Neck circumference at the level of cricothyroid membrane should not exceed 42 cm in male and 37.5 cm in females.

Box 2: Symptoms of obstructive sleep apnea

- Snoring
  - Family members complain about patient's loud snoring
- Daytime somnolence
  - Drowsiness at work
  - Fall asleep at work, while driving, or on telephone
- Morning headache and fatigue
- Restless sleep
  - Patient does not feel rested after night sleep
- Periods of apnea (cessation of breathing) resulting in frequent arousals at night
  - Sudden awakening and gasping for air
- Decreased libido and impotence
- Indigestion and gastroesophageal reflux disease
- Hypertension
- Decreased cognitive function including memory loss
- Personality changes, depression and psychosis
- Nocturnal headache, sweating, enuresis.

Box 3: Etiological findings on physical examination in patients with OSA

- *Nose*: Deviated septum, hypertrophic turbinates, allergic rhinitis, nasal valve collapse, polyps/tumors
- *Oral cavity*: Macroglossia, retrognathia, micrognathia, large mandibular tori
- *Oropharynx:* Large tonsils, redundant or large soft palate, large uvula and lateral pharyngeal wall, fullness in base of tongue, lingual tonsillar hypertrophy, banding of posterior pharyngeal wall, narrow skeletal arch
- *Laryngopharynx*: Lateral pharyngeal wall collapse, omega-shaped epiglottis, tumor
- *Larynx*: Vocal cord palsy and tumor
- *Neck:* Full and thick neck (Pickwickian syndrome)
- *Children*: Tonsil and adenoid hypertrophy, nasopharyngeal cyst, encephalocele, choanal atresia, deviated nasal septum and craniofacial or orthodontic malformations
- *General*: Obesity, achondroplasia, chest wall deformity, Marfan's syndrome.

Fig. 1: Kissing tonsils

- *Systemic examination*: Further examination is needed in patients with cor pulmonale or hypertension.

Flexible Nasopharyngolaryngoscopy

This endoscopy examination offers a view of entire nasopharynx, laryngopharynx and larynx.

- *Muller's maneuver*: This test is done before the uvulopalatopharyngoplasty (UPPP) to know whether the patient will benefit from this surgery or not. After passing the scope and during the examination, examiner pinches the patient's nostrils and patient attempts to inhale with closed mouth. If the laryngopharynx and/or larynx collapse, it means test is positive. It indicates that obstruction is below the level of soft palate. These patients will not benefit from UPPP and may need tracheostomy.

Radiography

Imaging studies may involve lateral cephalometric X-rays, fluoroscopy, CT scan, or MRI.

- Standard lateral cephalometric tracings identifies soft-tissue and skeletal obstruction in upper airway. Though limited by two dimensions this simple and cost-effective method measures posterior airway space and evaluates maxillary and mandibular development.

Polysomnography

This is the gold standard test in the evaluation of OSA. It differentiates not only between snoring without OSA, OSA and central sleep apnea but also identifies the severity of the apnea. This expensive test can be done in a sleep lab or in a patient's home. It requires the patient to spend a night.

- *Parameters*: This highly sensitive and specific test measures following elements:
  - Electroencephalogram (EEG) shows brain activity
  - Electromyogram (EMG) shows chin and leg muscle (anterior tibialis) movements
  - Electrocardiogram (ECG) shows cardiac rhythm
  - Electrooculogram (EOG) shows eye movements
  - Pulse oximetry shows blood oxygen saturation
  - Respiratory effort monitoring of chest and abdomen
  - Air movement at nose and mouth
  - Monitoring of body positions
  - Optional: End-tidal CO_2 monitor, esophageal manometer, nasal CPAP/bilevel positive airway pressure.
- *Indications*: Not everyone who snores need sleep study. The snorers, who have following elements, need sleep study.
  - Hypersomnolence, morning headache and restless sleep
  - Socially disruptive loud snoring
  - Prior to any surgery for sleep apnea or snoring.

Home Sleep Studies

These measures are economical. They are not as sensitive or specific as a formal sleep lab polysomnography. Some studies measure only pulse oximetry while others have multichannel recording devices.

Multiple Sleep Latency Test

This test is done during the day in a sleep laboratory. The patient is allowed to take several naps. This test measures the time that patient takes to fall asleep. An average sleep onset of less than 5 minutes indicates excessive daytime sleepiness, which is considered pathologic.

SEVERITY OF OSA

- *Respiratory disturbance index*: Respiratory disturbance index (RDI) is the sum of apneas (cessation of breathing/air flow for > 10 sec) and hypopneas (decrease in air flow by 50%). On the basis of RDI, OSA can be classified into three groups:
  - Mild (10–30)
  - Moderate (30–50)
  - Severe (>50)
- *Degree of oxyhemoglobin desaturation (SaO_2):* It is less than 85% in moderate OSA and less than 60% in severe OSA.
- *Epworth Sleepiness Scale*: It assesses the daytime sleepiness (Table 1). Patients answer the questions on the scale from 0 to 3.

TABLE 1 Epworth Sleepiness Scale

*Answer the chances of dozing/sleeping in the following situations on the scale 0 to 3**

Situation	*Score**
Reading	
Watching TV	
Theater/meeting place	
At traffic light while driving vehicle	
Sitting in passenger seat of a car for 1 hour	
Rest after lunch without alcohol	
On lying down to rest	
**Total Score	

***Score/scale: 0, never doze; 1, slight chance of dozing; 2, Moderate chance of dozing; 3, High chance of dozing.
****Total score less than 8 is normal.**

COMPLICATIONS OF OSA

The chronic OSA patients if not treated can develop significant morbidity and mortality due to following conditions, which can be reversed to normal with the successful treatment of OSA. The decrease psychomotor vigilance results in seven fold increase in the risk of motor vehicle accidents.

- Cardiac arrhythmias
- Cerebrovascular accidents
- Angina and myocardial infarction
- Congestive heart failure
- Cor pulmonale and chronic heart failure
- Hypertension: Systemic and pulmonary
- Peripheral edema
- Polycythemia
- Excessive day time somnolence can result in
  - Accidents during driving vehicle or operating dangerous machines.
  - Less exercise and more weight gain and more severe sleep apnea.

NONSURGICAL TREATMENT

Figure 2 shows the Flow Chart of management of snoring both with and without OSA. Discomfort and poor compliance are major stumbling blocks in the nonsurgical measures in managing snoring and OSA.

Lifestyle Modifications

Behavioral modifications (Box 4) though may reduce snoring and OSA especially in mild cases, patients' compliance is usually a problem.

Positional therapy: The patient is advised to sleep sideway rather than on the back. A rubber ball tied to the back of pajama prevents the patient to adopt supine position, which aggravates the symptoms.

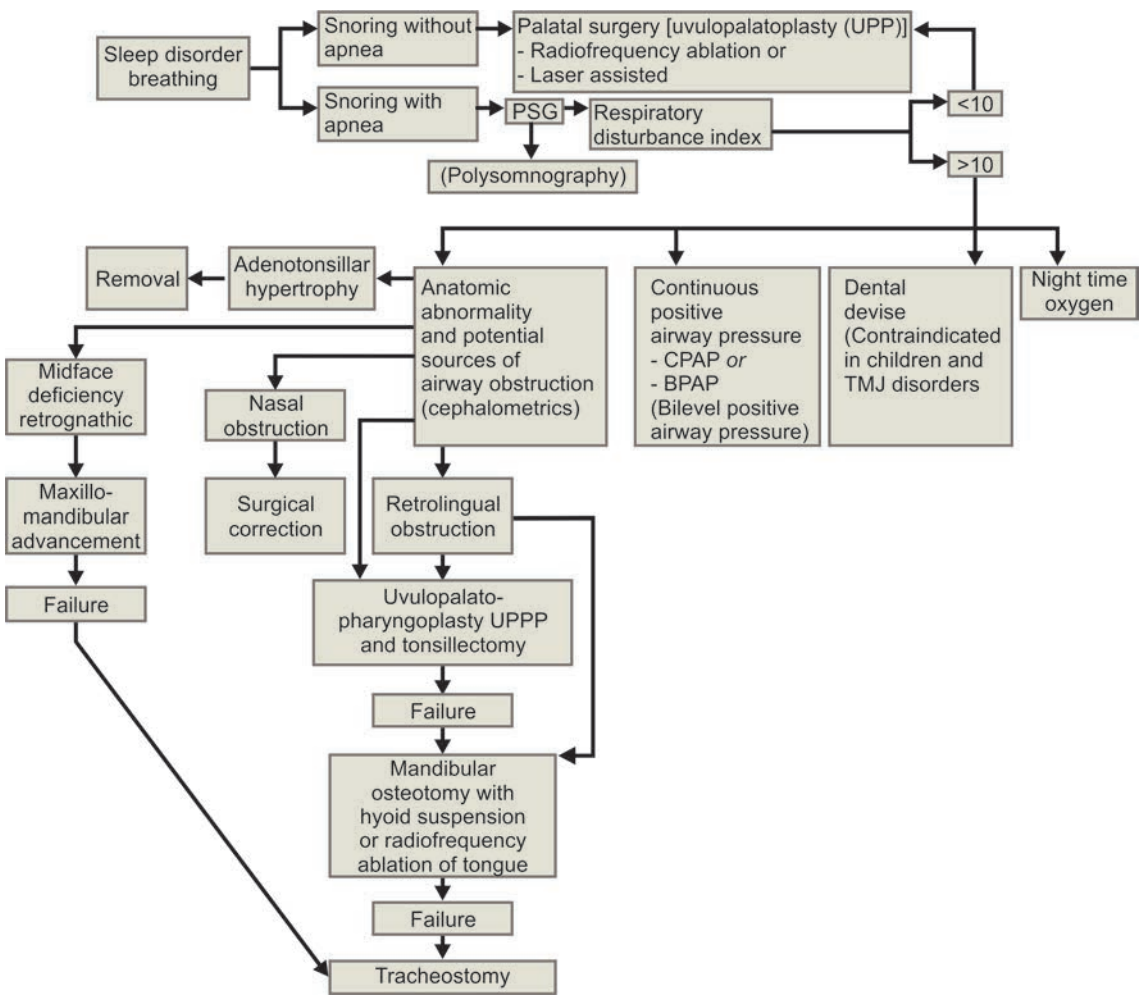

Fig. 2: Flow chart of management of snoring both with and without OSA

Box 4: Lifestyle modifications for the management of snoring and OSA

- Adopt an athletic lifestyle and exercise daily
- Lose weight
- Avoid sedatives (tranquilizers, sleeping pills, antihistaminics and cough suppressants) at bed time
- Avoid alcoholic drinks within 4 hours of retiring to bed
- Avoid heavy dinner within 3 hours of going to bed
- Avoid getting overtired
- Establish regular sleeping patterns
- Sleep sideway rather than on the back
- Raise the head end of the bed about 4" by placing bricks under the bedposts
- Nonsnorer partner should go to sleep first.

Nasal Continuous Positive Airway Pressure

This ventilator type machine is the most effective mean in the nonsurgical management of OSA. Continuous positive airway pressure (CPAP) provides a sort of pneumatic splint to airway and increases its caliber. Though it is 100% effective in treating OSA, patient's compliance is not very good and about 30% patients eventually stop using this. In this nasal CPAP, an airtight mask is kept over the patient's nose with the help of a strap wrapped around the head. The optimum pressure for opening the airway, which is determined during sleep study, is usually kept at 5–20 cm of H_2O.

- *Bilevel positive airway pressure and Auto-titrating PAP*: They have little better patient compliance. Bilevel positive airway pressure (BiPAP) delivers positive pressure at two fixed levels—a higher inspiratory and a lower expiratory pressure. Auto-titrating PAP (APAP) continuously adjusts the pressure.

Intraoral Devices

- Tongue-retaining device (TRD) and mandible advancement device (MAD) hold the tongue and/or mandible forward during the sleep and keep the airway open. Similar to CPAP discomfort and poor compliance are major concerns.

SURGICAL TREATMENT OF OSA

The surgeries are planned according to the location of the obstruction (Table 2).

Uvulopalatopharyngoplasty

This technically easy operation is the most common surgery performed for OSA. The oropharyngeal airway is enlarged in an anterior-superior and lateral dimensions. It is a type of radical tonsillectomy. The tonsils and uvula are removed along with the posterior edge of the soft palate. The anterior and posterior faucial pillars are sutured together. Similarly, the nasal and oral side mucosas of the cut edge of soft palate are sutured together. Patients usually stay for 1 or 2 days in the hospital.

- *Results*: UPPP is very effective in treating snoring but significant improvement (50% or more reduction in RDI) in OSA occurs in about 50% of patients. UPPP alone does not reduce the mortality of OSA. The tracheostomy and CPAP have shown dramatic improvement in OSA mortality. Therefore patient should be told that additional procedures may be required if UPPP fails.
- *Complications*: They are similar to tonsillectomy and includes bleeding (most common), temporary velopharyngeal incompetence (in 5–10% of patients) and nasopharyngeal stenosis (rare). Other minor postoperative complaints, which are temporary, include dry mouth, throat tightness, increased gag reflex and alteration in taste.

TABLE 2 Surgical procedures for the treatment of OSA

Site of obstruction	*Surgical procedures*
Nose and nasopharynx	Septoplasty, turbinectomy, adenoidectomy, nasal valve reconstruction, removal of polyps and tumors
Palate and tonsil	Uvulopalatopharyngoplasty (UPPP), uvulopalatal flap, transpalatal advancement pharyngoplasty, uvulopalatoplasty [Laser-assisted uvulopalatoplasty (LAUP) or Bovie-assisted uvulopalatoplasty (BAUP)], palatal radiofrequency and tonsillectomy
Tongue	Partial glossectomy, lingual tonsillectomy, tongue base radiofrequency and repose tongue suspension, hyoid myotomy and suspension
Jaws	Maxillomandibular osteotomy and advancement, mandibular osteotomy with genioglossus advancement

Other Surgical Procedures

- *Hypopharyngeal and base of tongue procedures*: The findings, which suggest hypopharyngeal obstruction, include
  - Morbid obesity: Body mass index greater than 31 kg/m^2
  - Mandibular skeletal deficiency
  - Lateral cephalometric radiogram: PAS less than 11 mm
  - Flexible laryngoscopy: Narrow retrolingual space
    - Muller's maneuver: Lateral pharyngeal wall collapse
  - *Apnea-hypopnea index*: greater than 30

  The following procedures for advancing base of tongue can be performed in presence of fullness in the base of the tongue.
  - Partial midline glossectomy: This can be performed with the help of either a laser or Bovie cautery.
  - Tongue base radiofrequency: Radiofrequency (RF) needle is inserted submucosally which coagulates tissue and causes scarring. RF can be used in five to six sittings to reduce the size of tongue.
  - Advancement genioplasty with hyoid suspension: In cases of obstruction at the level of base of tongue, retrognathia and micrognathia, the airway is enlarged by advancing genial tubercle of mandible (origin of genioglossus muscle) anteriorly and suspending hyoid bone from mandible by permanent sutures or wire. A rectangular chin portion of mandible including genial tubercles is resected and then rotated 90° and fixed by plates.
  - Hyoid myotomy and suspension: The midline hyoid bone is isolated and then advanced over thyroid cartilage.

- *Mandibular and maxillary advancement*: This procedures is more major than UPPP. It employs sagittal split and LeFort I osteotomies. These osteotomies are then fixed in anterior position with plates and screws. They effectively treat the anatomic anomalies causing OSA and offer long-lasting results.
- *Nasal surgery*: Though rarely a sole cause, nasal obstruction should be managed according to its cause and procedures include septoplasty, septal surgery and removal of polyps and tumors.
- *Tracheostomy*: Though neither preferred by surgeon nor liked by the patient this gold standard treatment of OSA is almost 100% effective. Tracheostomy bypasses the upper airway entirely but needs daily care throughout the life, which is a major stumbling block. It is probably the first line of surgery in very severe OSA or in those who are significantly obese or debilitated.

SURGICAL TREATMENT OF SNORING WITHOUT OSA

- *Uvulopalatoplasty*: Laser-assisted uvulopalatoplasty (LAUP) or Bovie-assisted uvulopalatoplasty (BAUP) can be performed under local anesthesia as an OPD procedure. In this procedure uvula is amputated and 1 cm trenches are created in the soft palate on either side of the uvula. The soft palate elevates and stiffens after healing. It may be performed in two to four stages, each separated by about 1 month. In this way the soft palate resection is titrated to treat snoring without velopharyngeal incompetence.
  - It is highly effective and resolution of snoring occurs in 85–90% of patients. Though it may help in mild or moderate OSA, generally it is contraindicated in cases of OSA.
- UPPP is also very effective in treating snoring.

Clinical Highlights

1. **Causes of snoring:** Some of the causes of snoring are nasopharyngeal angiofibroma, tonsillar enlargement, antrochoanal polyp and adenoid hypertrophy.
2. **Muller's maneuver:** It is performed with flexible nasopharyngolaryngoscopy to find the level and degree of obstruction in cases of obstructive sleep apnea. The patient is requested to make maximal inspiratory effort with nose and mouth closed. The base of tongue, lateral pharyngeal wall and palate are examined for collapsibility. The severity is rated from 0 (minimal collapse) to 4+ (complete collapse).

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41 Tumors of the Nasopharynx

The wise man should sacrifice him for others. I can secure my own good only by doing your good. There is no other way, none so whatsoever. When death is so certain, it is better to die for a good cause.

— **Swami Vivekananda**

Points of Focus

- **INTRODUCTION**
  - ✓ Benign Tumors of Nasopharynx
  - ✓ Malignant Tumors of Nasopharynx
- **JUVENILE NASOPHARYNGEAL ANGIOFIBROMA**
  - ✓ Etiology
  - ✓ Pathology
  - ✓ Site of Origin
  - ✓ Growth and Extensions
  - ✓ Clinical Features
  - ✓ Diagnostic Radiology
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  - ✓ Recurrence
- **NASOPHARYNGEAL CARCINOMA**
  - ✓ Etiology: Genetic, Viral, Environmental, Smoking and Dietary
  - ✓ Pathology
  - ✓ Spread
  - ✓ Clinical Features: Jugular Foramen Syndrome, Collet-Sicard Syndrome, Horner's Syndrome, Trotter's Triad
  - ✓ Endoscopy and Biopsy
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- **THORNWALDT'S DISEASE (PHARYNGEAL BURSITIS)**
- **PROPTOSIS (EXOPHTHALMOS)**
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

Primary tumors of nasopharynx are relatively rare. The most common benign tumor juvenile nasopharyngeal angiofibroma (JNA) and malignant tumor nasopharyngeal carcinoma (NPC) will be discussed in this chapter.

Benign Tumors of Nasopharynx

Majority of the nonepithelial benign tumors in nasopharynx are vascular. Juvenile nasopharyngeal angiofibroma, though rare is the most common benign tumor of nasopharynx. Other benign tumors, which are very rare and arise from the roof or lateral wall of nasopharynx, include teratomas, pleomorphic adenoma, chordoma (derived from the notochord), hamartoma (a focal malformed normal tissue that resembles neoplasm such as hemangioma), choristoma (tissues at an abnormal site) and paraganglioma.

Malignant Tumors of Nasopharynx

Carcinoma is the most common variety of nasopharyngeal malignancy and other rare types include lymphomas, rhabdomyosarcoma in children, plasmacytoma (solitary or part of generalized multiple myelomatosis), chordoma (from remnant of notochord), adenoid cystic carcinoma (from minor salivary glands) and melanoma. Non-Hodgkin's type of lymphoma is more common than Hodgkin's. Almost all are B-cell type. Embryonal rhabdomyosarcoma presents as a polyp in the nasopharynx.

JUVENILE NASOPHARYNGEAL ANGIOFIBROMA

Juvenile nasopharyngeal angiofibroma, though rare is the most common benign tumor of nasopharynx.

Etiology

The exact cause is yet not known.

- A primary aberration of pituitary-gonadal axis is suggested but not proved.
- The tumor is almost exclusively seen in adolescent males in the second decade of life. So it is considered to be testosterone dependent.
- A hamartomatous nidus of vascular tissue in the nasopharynx is said to be activated when male sex hormone appears.

Pathology

Angiofibroma consists of varying amount of vascular and fibrous tissues. Vessels do not have muscle coat and are just endothelium lined. It results into severe bleeding, which cannot be controlled by application of adrenaline. The tumor is covered with nasopharyngeal mucosa and clinically may appear deceptively avascular.

Site of Origin

The site of origin is a matter of dispute. Currently it is believed to arise close to the superior margin of sphenopalatine foramen (bounded by palatine bone, horizontal ala of vomer and root of pterygoid process). In past it was said to arise from the roof of nasopharynx or the anterior wall of sphenoid bone.

Growth and Extensions

Nasopharyngeal fibroma is a locally invasive benign tumor. It destroys the adjoining structures and grows in the following directions and regions:

- Anteriorly into the nasal cavity and produce nasal obstruction, epistaxis and nasal discharge.
- Posteriorly into nasopharynx.
- Laterally into the pterygopalatine fossa (pushing posterior wall of maxillary sinus), pterygomaxillary fissure and thence to infratemporal fossa and cheek (Fig. 1).
- Superiorly the tumor invades floor of middle cranial fossa anterior to foramen lacerum and lies lateral to carotid artery and cavernous sinus. Tumor may go into sella turcica through sphenoid sinus and lies medial to carotid artery. Anterior cranial fossa can be invaded through ethmoid roof cribriform plate.
- Orbit: It enters into the orbit through the inferior orbital fissure. It invades apex of the orbit. It may enter the orbit through superior orbital fissure. It produces proptosis and "frog-face deformity".
- Paranasal sinuses: Maxillary, sphenoid and ethmoid sinuses.

Clinical Features

- *Most common presentation*:
  - Profuse and recurrent episodes of epistaxis and unilateral nasal obstruction occur in more than 80% of cases.
  - It occurs exclusively in males of 10–25 years of age.

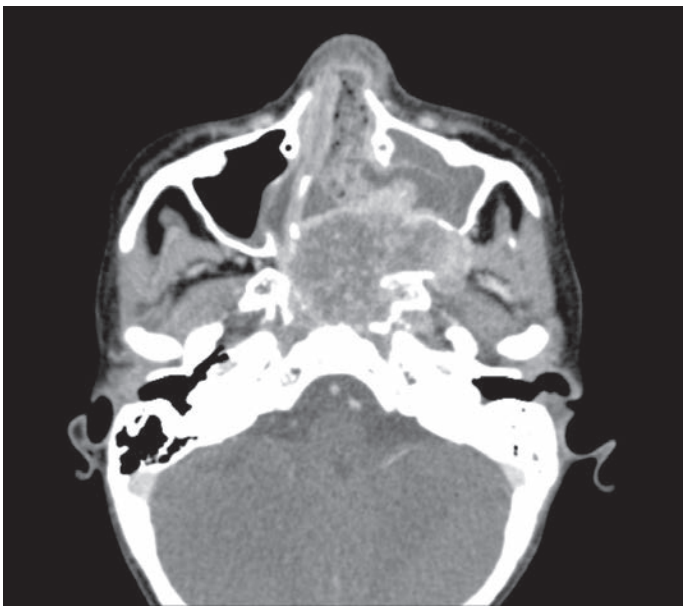

Fig. 1: Contrast CT scan juvenile nasopharyngeal angiofibroma. Note the pterygopalatine fossa and infratemporal fossa extension *Source*: Dr Amit Goyal, Shillong

  - Patient is usually markedly anemic.
  - Denasal speech.
- *Ear*: Conductive hearing loss and serous otitis media occur due to obstruction of Eustachian tube.
- *Local findings*: Pink or purplish nasopharyngeal mass, which is sessile, lobulated or smooth and obstructs one or both choanae. Consistency is firm but digital palpation is never done because it can result in profuse bleeding.
- *Clinical features of advanced disease*
  - Broadening of nasal bridge.
  - Proptosis and superior orbital fissure syndrome.
  - Swelling of cheek or infratemporal fossa.
  - Involvement of 2nd, 3rd, 4th, 5th cranial nerves: Diplopia and visual field defects.

Juvenile nasopharyngeal angiofibroma: In an adolescent male, profuse recurrent episodes nosebleed suggest juvenile nasopharyngeal angiofibroma until proven otherwise.

Diagnostic Radiology

- *Computed tomography (CT) scan with contrast enhancement axial and coronal sections* (Fig. 1): CT is gold standard for JNA and has replaced conventional radiographs. The extent of tumor, bony destruction or displacements can be seen.
- *Magnetic resonance imaging (MRI):* It is indicated when soft tissue extensions are present in following regions:
  - Intracranial
  - Infratemporal fossa
  - Orbit
  - Post-treatment surveillance of residual or recurrent disease.
- *Carotid angiography*: Extension and vascularity of the tumor and its feeding vessels can be seen. It is done for embolization therapy.
- *X-ray soft tissue nasopharynx lateral view*: Soft tissue mass

in the nasopharynx and anterior bowing of posterior wall of maxillary sinus (antral sign) is pathognomic.

- *X-rays of paranasal sinuses and base of skull*: Displacement of nasal septum, opacity of sinuses, destruction of medial antral wall, erosion of greater wing or pterygoid plates of sphenoid or widening of lower lateral margin of superior orbital fissure.

Staging

Sessions's classification, modified by Radkowski reflects the clinical behavior of JNA (Table 1).

Diagnosis

In most cases, clinical features and CT scan will confirm the diagnosis. Biopsy is contraindicated because it results in the profuse bleeding.

Treatment

Open surgical excision is the treatment of choice.

- *Surgical approaches*: Depending upon the origin and extensions of angiofibroma, following approaches may be employed. Occasionally combined approaches are employed especially in significant extension into infratemporal fossa, middle cranial fossa or cavernous sinus. Autologous transfusion preoperatively if the patient status permits should be considered.
  - Transpalatine: Tumors confined to nasopharynx, nasal cavity and sphenoid sinus.
  - Le Fort 1 osteotomy approach: For extension to paranasal sinuses, pterygopalatine fossa and infratemporal fossa.
  - Medial maxillectomy: It provides access to orbit, ethmoid and sphenoid sinuses and anterior skull base.
  - Sardana's approach: Transpalatine + Sublabial.
  - Extended lateral rhinotomy: It gives wide exposure.
    - Via facial incision
    - Via degloving approach
  - Extended Denker's approach
  - Intracranial-extracranial
  - Infratemporal fossa
- *Measures to reduce the vascularity of tumor*: The intraoperative blood loss may be significant (about 2 liters). Following preoperative measures can reduce the vascularity of tumor.
  - Embolization of the feeding vessels.
  - Estrogen therapy: Stilboestrol 2.5 mg three times a day for 3 weeks. Not preferred currently.
  - Preoperative radiation: Generally not favored.
  - Cryotherapy.
- *Endoscopic resection*: For early tumors limited to nose, nasopharynx, ethmoid and sphenoid. Preoperative embolization of feeding vessels and early control of internal maxillary artery facilitate tumor mobilization.
- *Radiation therapy*: For more advanced tumor with significant intracranial extension.

  Complications: Though less with current intensity modulated radiation therapy (IMRT), the feared complications in these young patients are following:
  - Secondary malignancy
  - Abnormal craniofacial development
  - Cataracts
  - Optic atrophy
  - Osteoradionecrosis.

TABLE 1 Radkowski classification of juvenile nasopharyngeal angiofibroma (JNA)

IA	Tumor limited to nose and nasopharyngeal vault
IB	Extension to paranasal sinuses
IIA	Minimal extension to pterygomaxillary fissure (PMF)
IIB	Full extension to PMF and/or erosion of orbital bones
IIC	Extension to infratemporal fossa and/or cheek or posterior to pterygoid plates
IIIA	Erosion of skull base: minimal intracranial
IIIB	Extensive intracranial and/or cavernous sinus extension

Source: Radkowski D, Mcgill T, Healy GB, et al. Angiofibroma. Changes in staging and treatment. Arch Otoalryngol Head and Neck. 1996;122:122-9.

Recurrence

Recurrence of tumor after surgical removal is not uncommon and ranges from 30–50%.

Though well-documented, spontaneous regression with advancement of age is uncommon.

NASOPHARYNGEAL CARCINOMA

Nasopharyngeal cancer (NPC) is most common in southern states of China, Taiwan and Indonesia. In Guangdong Province of Southern China, NPC is the third most common malignancy among men. In India, nasopharyngeal cancer constitutes only 0.5% of all body cancers. It is relatively common in North East states of India where people are predominantly of Mongoloid origin.

Etiology

The exact etiology is yet unknown. NPC is a multifactorial disease. The various factors include genetic susceptibility, environmental factors, dietary and personal habits.

- *Genetic*: Chinese, whether they live in China or elsewhere, have a higher genetic susceptibility to nasopharyngeal cancer.
  - Abnormality in chromosome 1-6, 9, 11, 13, 14, 16, 17, 22 and x.
  - Allelic losses on short arms of chromosomes 3 and 9 lead to inactivation of tumor suppressor genes p14, p15, and p16.
  - EBV: Gains on chromosome 12 and allelic losses on 11q, 13q, and 16q causes invasive carcinoma.
  - Mutations in p53 and aberrant expressions of cadherins cause metastases.
  - Specific haplotypes in human lymphocyte antigen (HLA) on short arm of chromosome 6 increases the risk of NPC.
- *Viral*: Epstein Barr virus (EBV) is usually associated with nasopharyngeal cancer. Specific viral markers are used to screen people in high-risk areas.
- *Environmental*: Air pollution, smoke from burning of incense and wood (polycyclic hydrocarbon).
- *Smoking*: Tobacco and opium.

- *Dietary*:
  - Nitrosamines from dry salted fish and preserved foods.
  - Diet deficient in vitamin C, fresh fruits, carotene, or fiber intake. Vitamin C blocks nitrosification of amines.

Pathology

- Squamous cell carcinoma has various grades of its differentiation or variants such as transitional cell carcinoma and lymphoepithelioma. It is most common (85%).
- Lymphomas constitute 10% and the rest 5% are rhabdomyosarcoma, malignant mixed salivary tumor or malignant chordoma.
- On the basis of histology, WHO has reclassified epithelial growths into three types (Table 2). Type I Keratinizing squamous cell carcinoma can be graded into well, moderately or poorly differentiated.

Spread

- The most common site of origin is fossa of Rosenmuller.
- It may spread into the cranium through foramen lacerum and involve cranial nerves.
- Lymph node involvement is common because nasopharynx has rich lymphatic network.

Clinical Features

- *Age and sex*: It is mostly seen in 4th and 5th decades of life. Incidence starts rising from 2nd decade of life. Males are three times more prone than females.
- *The most common presenting feature* of nasopharyngeal cancer is cervical lymphadenopathy (60–90%) followed in descending order by
  - Hearing loss
  - Nasal obstruction
  - Epistaxis
  - Cranial nerve palsies (CN VI most common)
  - Headache
  - Earache
  - Neck pain
  - Weight loss
- *Local features*: These tumors may present in three forms:
  - Proliferative: Polypoid tumor fills the nasopharynx and causes nasal obstruction.
  - Ulcerative: Epistaxis is common.
  - Infiltrative: Growth infiltrates submucosally.
- *Regional features*: They include following:
  - Nose: Unilateral nasal obstruction and blood stained discharge, denasal speech (rhinolalia clausa).
  - Ear: Unilateral conductive hearing loss, otalgia, serous or suppurative otitis media (due to Eustachian tube obstruction), tinnitus and dizziness. Tumor may enter into the middle ear through the Eustachian tube.

TABLE 2 WHO classification of epithelial carcinoma based on histopathology

Type 1	Keratinizing squamous cell carcinoma
Type 2	Nonkeratinizing carcinoma
	- Without lymphoid stroma
	- With lymphoid stroma
Type 3	Undifferentiated carcinoma
	-Without lymphoid stroma
	-With lymphoid strom

In cases of unilateral serous otitis media in an elderly person, nasopharyngeal cancer must be ruled out.

  - Ophthalmoneurologic: Squint and diplopia due to involvement of CN VI (most commonly involved cranial nerve), ophthalmoplegia (CN III, IV and VI), facial pain and reduced corneal reflex (invasion of CN V through foramen lacerum), exophthalmos and blindness (invasion of CN II at orbital apex)
  - Jugular foramen syndrome: Involvement of IX, X and XI cranial nerves (Fig. 2) due to enlargement of lateral retropharyngeal lymph nodes.
  - Collet-Sicard syndrome: Involvement of IX, X, XI and XII cranial nerves. CN XII may be involved due to extension of growth to hypoglossal canal.
  - Horner's syndrome: Due to involvement of cervical sympathetic chain.
  - Trotter's triad: Conductive deafness, ipsilateral temporoparietal neuralgia (CN V) and palatal paralysis (CN X).
  - Cervical nodal metastases. Patient may present with only cervical lump of nodes. It is usually seen between the angle of jaw and the mastoid. Other nodes include along the spinal accessory in the posterior triangle of neck and supraclavicular nodes. Nodal metastases during first consultation are seen in 75% and about half of them have bilateral nodes. First echelon nodes, the retropharyngeal nodes of Rouviere may be detected on CT.
  - Distant metastases: Bone (thoracolumbar vertebrae, pelvis and femoral heads), lung and liver.

Endoscopy and Biopsy

- *Endoscopy*: Flexible or rigid endoscopes may be employed.
- *Biopsy*: It is essential to show the exact histology of the malignancy. Submucosal disease may need deep biopsies.

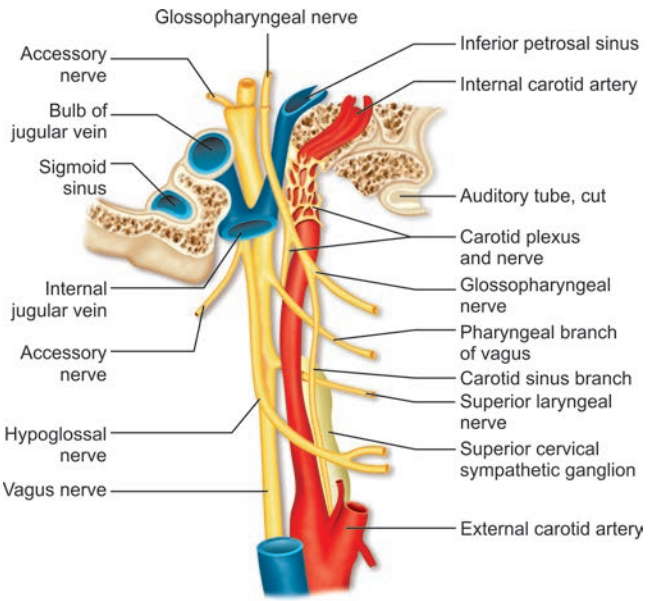

Fig. 2: Right jugular foramen and structures passing through it

Submucosal spread, beneath a normal mucosa, is quite common.

- *Occult primary*: Nasopharyngeal curettage may be performed. In cases of strong suspicion, nasopharynx may be approached transpalatal and a strip of mucosa and submucosa from the region of fossa of Rosenmuller is taken for histology. Nasopharynx is a common site for occult primaries.
- *Fine-needle aspiration cytology:* Fine-needle aspiration cytology of cervical node may be required in some cases.

Serology

- Anti-EBV IgA (Anti-VCA and Anti-EA) anti viral capsid antigen and anti early antigen titers: High IgA anti-VCA titer is very sensitive and high IgA anti-EA titer is very specific for detecting early occult disease. EBV titers may be used for surveillance after treatment.
- EBV DNA titers (anti-EBV-specific DNase) correlate with stage, treatment response, relapse and survival. They may be used for predicting prognosis.

Radiology

- *MRI with gadolinium and fat suppression*: It is the diagnostic imaging modality of choice. Coronal views assess petroclinoid fissure, foramina (lacerum, ovale, rotundum or spinosum) or cavernous sinus. Axial views evaluate retropharyngeal, paranasopharyngeal, pterygomaxillary space and infratemporal fossa.
- *CT* (Fig. 3): Evaluate bone involvement, skull base erosion, and cervical lymphadenopathy particularly in presence of extracapsular spread.
- *Positron emission tomography*: For residual or recurrent disease after treatment.

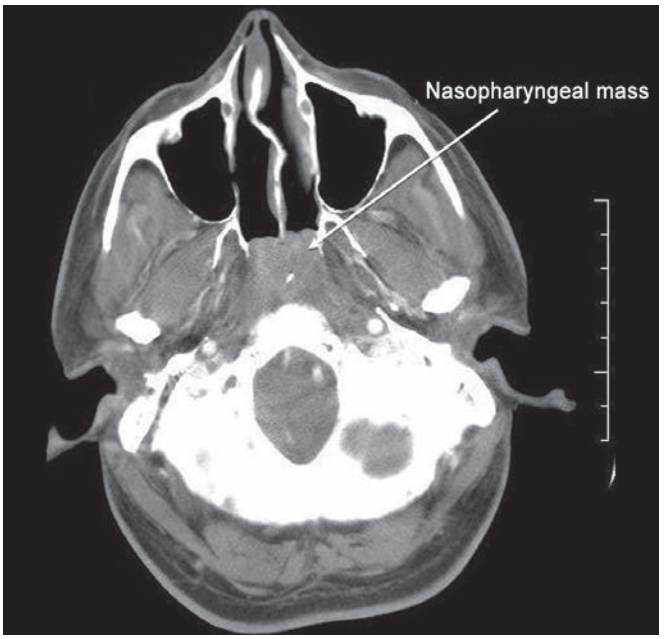

Fig. 3: Contrast CT scan Nasopharyngeal mass axial section. Large soft tissue density mass causing complete obliteration of nasopharyngeal lumen. Laterally it is obliterating lateral recesses and openings of Eustachian tubes. It is also extending into both posterior choanae. Mass shows foci of calcification
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

TNM Classification

In nasopharynx carcinoma, node (N) classification is different from other mucosal carcinoma of head and neck region (Table 3). Less weight is given to Level 1 and 2 nodes (even 6 cm nodes are considered N_1 category). Node enlargement (irrespective of the size) of supraclavicular fossa or Ho's triangle (triangle between medial and lateral ends of clavicle and the point where neck meets the shoulder) is considered N_3 category.

Treatment

- *Irradiation:* It is the primary treatment of choice. External radiation or intracavitary implants (brachytherapy) or both are used.

TABLE 3 AJCC Cancer Staging (2002) and UICC TNM classification of malignant tumors of nasopharynx

Primary tumor (T)	
T_X	Cannot be assessed
T_0	No tumor
T_{is}	Carcinoma in situ
T_1	Tumor is confined to nasopharynx
T_2	Tumor extends to soft tissues of oropharynx or nasal fossa
T_{2a}	No parapharyngeal extension
T_{2b}	With parapharyngeal extension
T_3	Tumor invades bones and/or paranasal sinuses
T_4	Intracranial extension, involvement of cranial nerves, infratemporal fossa, hypopharynx, orbit, or masticator space
Regional lymph nodes (N); Size in greatest dimension	
N_x	Cannot be assessed
N_0	No nodal metastases
N_1	Single ipsilateral node except supraclavicular fossa <= 6 cm
N_2	Bilateral or contralateral nodes except supraclavicular fossa <= 6 cm
N_{3a}	Any node > 6 cm
N_{3b}	Supraclavicular node of any size
Distant metastasis (M)	
M_x	Cannot be assessed
M_0	No distant metastasis
M_1	Distant metastasis
Stage grouping	
Stage 0	$T_{is}\ N_0\ M_0$
Satge I	$T_1\ N_0\ M_0$
Stage IIA	$T_{2a}\ N_0\ M_0$
Stage IIB	$T_{2b}\ N_0\ M_0$ or $T_{1\text{-}2b}\ N_1\ M_0$
Stage III	$T_3\ N_0\ M_0$ or $T_{1\text{-}3}\ N_{1\text{-}2}\ M_0$
Stage IVA	$T_4\ N_{0\text{-}2}\ M_0$
Stage IVB	$T_{1\text{-}4}\ N_3\ M_0$
Stage IVC	$T_{1\text{-}4}\ N_{0\text{-}3}\ M_1$

- External beam radiotherapy is most commonly used. Opposing lateral fields include primary tumor, upper neck and third anterior matched field irradiate lower cervical and supraclavicular nodes. A tumor dose of 6000–7000 radiations is given. The current state-of-art radiation technologies, which are safer and more precise, include stereotactic radiosurgery, three-dimensional conformal (3DCRT) and IMRT.
- Overall survival is in the range of 50–80%.

- *Systemic chemotherapy*: It may be used as palliation for distant metastases or in radiation failures. Single and combination chemotherapy (cicplatinum/5-flurouracil) have been used as an adjunct to radiation. Induction, adjuvant, concomitant, or their combinations have shown improvement in survival.
- *Radical neck dissection* may be done for persistent nodes when primary has been controlled.

Recurrent Disease

Recurrent or residual tumor may be managed with second course of radiation, cryosurgery through a palatal fenestration or skull base surgery in selected cases.

TERATOMAS

- These congenital tumors, seen at birth are six times more common in females.
- *Types:* Dermoid with skin appendages (hairy polyp); True teratomas (have elements of all the three germ layers); and the epignathi (have well developed fetal parts).

THORNWALDT'S DISEASE (PHARYNGEAL BURSITIS)

The pharyngeal bursa represents attachment of notochord to endoderm. It is a median recess that is located in the posterior wall of the nasopharynx in the adenoid mass.

Clinical Features

The infection of pharyngeal bursa (Thornwaldt's Disease) may present with:

- Persistent postnasal discharge and crusting in the nasopharynx.
- Nasal obstruction due nasopharyngeal swelling.
- Serous otitis media due to Eustachian tube obstruction.
- Dull occipital headache.
- Recurrent sore throat.
- Low-grade fever.
- A cystic and fluctuant swelling in the posterior wall of nasopharynx.

Treatment

- Antibiotics for infection.
- Marsupialization of the cystic swelling with removal of the lining membrane.

PROPTOSIS (EXOPHTHALMOS)

Proptosis (*G. proptosis*, a falling forward) refers to protrusion of the eyeball. It can be congenital and familial or acquired and unilateral or bilateral. Unilateral tumors of nasopharynx and paranasal sinuses and retro-orbital region and bilateral is caused by hyperthyroidism exophthalmos.

- *Pertinent anatomy*: The orbit is related to frontal sinus (superiorly), maxillary sinus (inferiorly), ethmoid sinus (medially) and sphenoid sinus (posteromedially).
- *Causes*: They include
  - Congenital and familial
  - Trauma: Maxillofacial injuries, complication of endoscopic sinus surgery.
  - Infection: Complications of rhinosinusitis such as subperiosteal abscess, orbital cellulitis and cavernous sinus thrombosis.
  - Cysts: Mucocele of frontoethmoid sinus.
  - Benign tumors: Inverted papilloma, ethmoidal polyps and fungus, juvenile nasopharyngeal angiofibroma and fibrous dysplasia.
  - Malignant tumors: Carcinoma and sarcomas of nose, paranasal sinuses and nasopharynx.
  - Tumors of retro-orbital region
- *Management*: Clinical features, imaging studies and biopsy will confirm the diagnosis. The cause of proptosis should be treated.

Clinical Highlights

1. **Nasopharyngeal angiofibroma:** Most common site of origin is sphenopalatine foramen. This benign tumor erodes bone and is known for its recurrence on incomplete removal. It can spread intracranially.
   a. **Biopsy**: It is contraindicated. Histological diagnosis of angiofibroma of nasopharynx is made by excision biopsy.
   b. **Antral (Holman-Miller) sign:** It is a feature of nasopharyngeal fibroma.
   c. **Clinical features:** It occurs exclusively in young adolescent boys. Episodes of profuse nosebleed and nasal obstruction are the cardinal symptoms.
   d. **Voice:** Type of voice in nasopharyngeal fibroma is rhinolalia clausa.
   e. **Treatment:** Surgical excision is the treatment of choice in angiofibroma of nasopharynx. If the mass is arising from the lateral wall of nose extending only into the nasopharynx, transpalatal approach is used. Other modalities of treatment include hormones and intensity modulated radiotherapy.
2. **Carcinoma of nasopharynx:** It is related to Epstein-Barr virus infection. The most common site is fossa of Rosenmuller (pharyngeal recess). The squamous cell carcinoma is the most common histological variety. Radiotherapy is usually the first line of treatment.

a. **Clinical features:** The most common presentation is metastatic cervical lymphadenopathy. Chances of metastases are highest in cancer of nasopharynx. It often causes unilateral serous otitis media. Patient can present with squint, facial pain, blindness and mass in the neck.
b. **Extension:** Nasopharyngeal cancer can involve nasal cavity, oropharynx, tympanic cavity and orbit.
c. **Krause's nodes:** These lymph nodes are situated in the jugular foramen. Their enlargement compresses CN IX, X and XI, and produce jugular foramen syndrome.
d. **Trotter's (Sinus of Morgagni) syndrome or triad:** Trotter's triad, which is seen in nasopharyngeal carcinoma, consists of conductive hearing loss (due to Eustachian tube obstruction), immobility of soft palate and neuralgic pain in the distribution of mandibular division of trigeminal nerve. The associated trismus and preauricular fullness may be seen in some cases.
e. **Prognosis:** It is relatively better in keratinizing squamous cell carcinoma of nasopharynx.
f. **Rhabdomyosarcoma:** It is the most common malignant tumor of nasopharynx in children. Orbit is the most common site of rhabdomyosarcoma in the head and neck region.

3. **Nasopharyngeal chordoma:** It originates from the notochord
4. **Luschka pouch:** Persistence of this invagination pouch causes Thornwaldt's cyst which may get infected and form an abscess in the nasopharynx. Notochord, which is attached to the endoderm in the area of nasopharynx, produces this invagination pouch. Pharyngeal bursa is a site for Thornwaldt's cyst.
5. **Rathke's pouch:** It may persist as craniopharyngeal canal in the nasopharynx. It forms anterior pituitary (adenohypophysis).

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42 Tumors of Oropharynx

Character is repeated habit, and repeated habits alone can reform character. The chaste brain has tremendous energy and gigantic will power. We can overcome the difficulty by constant practice. We must learn that nothing can happen to us, unless make ourselves susceptible to it.

—Swami Vivekananda

Points of Focus

MALIGNANT TUMORS

- INTRODUCTION
- HISTOPATHOLOGY
- RISK FACTORS
- EVALUATION
- STAGING
- TREATMENT
- CARCINOMA BASE OF TONGUE
- CARCINOMA TONSIL
- LYMPHOMA
- CARCINOMA OF SOFT PALATE
- CARCINOMA OF POSTERIOR PHARYNGEAL WALL

BENIGN SWELLINGS

- ✓ Papilloma
- ✓ Hemangioma
- ✓ Pleomorphic Adenoma
- ✓ Mucous Cyst

- PARAPHARYNGEAL TUMORS
- STYLALGIA (EAGLE'S SYNDROME)
- CLINICAL HIGHLIGHTS

MALIGNANT TUMORS

INTRODUCTION

The five most frequent cancers in Indian males (in descending order) are mouth/oropharynx, trachea/bronchus/lungs, lymphomas/multiple myeloma, esophagus and leukemia. In women they are (in descending order) breast, cervix, ovary, mouth/oropharynx, esophagus, and lymphomas/multiple myeloma (ICMR, 2004). The incidence rate of mouth and oropharynx cancer in India is very high (12.48 males and 5.52 females in 100,000 population).

The high and increasing incidence of oropharyngeal cancer is becoming a serious health problem. The management of these tumors needs a multidisciplinary team, which consists of head and neck cancer surgeon, reconstructive surgeon, speech and swallowing therapist, prosthodontist, dentist and radiation and medical oncologist. Benign tumors are far less common than the malignant ones.

For the purpose of description of malignant tumors, oropharynx has been divided into four different subsites:

1. Posterior one-third (or base) of tongue
2. Palatine tonsillar area including fossa and pillars
3. Soft palate
4. Posterior pharyngeal wall.

HISTOPATHOLOGY

The tissue of origin for most of the oropharyngeal carcinomas are stratified squamous epithelium, minor salivary glands and lymphoepithelium of Waldeyer's ring. Squamous cell carcinoma (SCC) accounts for 90% of all oropharyngeal malignancies. Some of the varieties are listed below:

- *Squamous cell carcinoma*: Squamous cell carcinoma is the most common variety of carcinoma. It shows various grades of differentiation: well, moderately and poorly differentiated.

- Keratinizing SCC
- Nonkeratinizing SCC
- Basaloid SCC
- Verrucous carcinoma
- Spindle cell carcinoma
- Lymphoepithelial carcinoma: This is a poorly differentiated variant of SCC and shows admixture of lymphocytes. The common sites of this include tonsil, base of tongue and vallecula.

- *Minor salivary gland tumors*: Minor salivary gland malignancy is mostly seen on the palate and fauces.
  - Adenoid cystic carcinoma
  - Mucoepidermoid carcinoma.
- *Lymphoma*: Hodgkin and non-Hodgkin arise from the tonsil and base of tongue in young adults and children. Cervical nodes may be enlarged.
- *Sarcoma*.

Grading

The current WHO grading system is largely based on classic Broders classification. It describes following three grades:

1. Grade 1: Well differentiated.
2. Grade 2: Moderately differentiated.
3. Grade 3: Poorly differentiated.

RISK FACTORS

- Tobacco-smoking
- Alcohol.
- Chewing of betel nut and tobacco
- Human papilloma virus (HPV) type 16 for SCC of tonsil: Tumor specific vaccines for HPV-positive disease are under investigation.

Incidence

- *Geography*: The incidence is high in South Central Asia, South Africa and Europe.
- *Age*: The incidence and mortality increases with age.
- *Gender*: These tumors are three times more common in men than women.
- *Racial-ethnic*: Among men, highest rate is in blacks followed in descending order by whites (non-Hispanic), Vietnamese and native Hawaiians.

Prevention

- *Bad habits*: Stop smoking and chewing of betel nut and tobacco.
- *Diet*: Fresh fruits and vegetables may reduce the relative risk of cancer of upper aerodigestive tract.

EVALUATION

Clinical Features

- *Early*: Throat irritation, burning sensation with acidic food, neck swelling and odynophagia.
- Referred unilateral earache is common.
- Hemoptysis or oral bleeding in tonsil cancer.
- *Late*: Dysphagia, dysarthria or hot potato voice, trismus, obstructive airway symptoms, deafness (serous otitis media due to Eustachian tube obstruction), and weight loss.

Growth Patterns

Three types of growth patterns are described for oral cavity and oropharyngeal tumors and they are:

1. *Exophytic*: Superficially spreading and exophytic types are usually seen in the palatine arch. Metastases are rare.
2. *Ulcerative*: Ulcerative and infiltrative types deeply invade the adjoining structures. They are usually seen in base of tongue and tonsil.
3. *Submucosal.*

Prognosis: Because of marked tendency for regional metastasis, prognosis is poor in ulcerative lesions than exophytic type of growth.

Imaging Studies

- *Computed tomography with contrast*: Computed tomography (CT) shows bony involvement (such as cortical mandibular invasion, cervical spines and skull base) and secondary neck nodes. For the CT anatomy of oral cavity and oropharynx *see* Chapter Diagnostic Imaging.
- *Magnetic resonance imaging (MRI)*: Coronal, transverse and sagittal planes T1W and T1W with gadolinium-diethylene pentaacetate (Gd-DTPA) contrast and fat suppression. It shows soft tissue infiltration such as tongue base, parapharyngeal and pre-epiglottic and bone marrow involvement.
- *X-ray chest posteroanterior (PA) and lateral views*: Assess.
  - Acute or chronic pulmonary diseases (eg. Tuberculosis).
  - Pulmonary metastasis.
  - Synchronous pulmonary primary carcinoma.
- *Positron emission tomography scan (PET)*: Scan indications.
  - High tumor grade and large tumor volume.
  - Recurrent tumor.
  - Neck metastasis of unknown origin.
  - Post-treatment surveillance.
- *Fusion PET-CT*: Provide improved anatomic localization of tumors and metastasis.

Biopsy

- Usually under local anesthesia with a cup forceps but general anesthesia is preferred for base of tongue tumor. Panendoscopy can be performed.
- Fine needle aspiration cytology (FNAC) from neck mass may be done if primary and secondary are clinically evident.

STAGING

The Union for International Cancer Control (UICC) tumor nodes and metastasis (TNM) classification and American Joint Committee on Cancer (AJCC) cancer staging are shown in Table 1.

TREATMENT

The treatment modalities and protocols vary from center to center. They depend upon the site and extent of disease (Table 1), patient's general condition, surgeon's experience and philosophy, and facilities available. The treatment modalities include:

1. Surgery
2. Radiotherapy
3. Chemotherapy

TABLE 1 AJCC cancer staging (2002) and UICC TNM classification of malignant tumors of oropharynx

Primary tumor (T) tumor size in greatest dimension	
T_X	Cannot be assessed
T_0	No tumor
T_{is}	Carcinoma in situ
T_1	Tumor < = 2 cm
T_2	Tumor > 2 cm but < = 4 cm
T_3	Tumor > 4 cm
T_{4a}	Tumor invades any of the following: larynx, deep or extrinsic muscles of tongue, medial pterygoid muscle, hard palate or mandible
T_{4b}	Tumor invades any of the following: lateral pterygoid muscle, pterygoid plates, lateral nasopharynx, skull base or encases carotid artery
Regional lymph nodes (N) For Oropharynx and hypopharynx; Size in greatest dimension	
N_x	Cannot be assessed
N_0	No nodal metastases
N_1	Single ipsilateral node < =3 cm
N_{2a}	Single ipsilateral node >3 cm but < = 6 cm
N_{2b}	Multiple ipsilateral nodes but none > 6 cm
N_{2c}	Bilateral or contralateral nodes but none > 6 cm
N_3	Any node > 6 cm
Distant metastasis (M)	
M_x	Cannot be assessed
M_0	No distant metastasis
M_1	Distant metastasis
Stage grouping for oropharynx and hypopharynx	
Stage 0	T_{is} N_0 M_0
Stage I	T_1 N_0 M_0
Stage II	T_2 N_0 M_0
Stage III	T_3 N_0 M_0 or T_{1-3} N_1 M_0
Stage IVA	T_{4a} N_{0-1} M_0 or T_{1-4a} N_2 M_0
Stage IVB	T_{4b} N_{0-3} M_0 or T_{1-4} N_3 M_0
Stage IVC	T_{1-4} N_{0-3} M_1

4. Combination of surgery and radiotherapy
5. Chemotherapy as an adjunct to surgery or radiotherapy
6. Palliative therapy.

CARCINOMA BASE OF TONGUE

This is very common in India.

Clinical Features

- *Neck swelling*: Mostly patient presents with metastases in cervical nodes.
- *Early symptoms*: They are mostly ignored by the patient for a long time and include sore throat, feeling of lump in the throat and slight discomfort on swallowing.
- *Late features*: They indicate advanced diseases and include referred earache, dysphagia, bleeding from the mouth, and hot potato voice.
- Deeply infiltrative ulcer with induration can be seen and palpated on base of tongue (Fig. 1). The lesion may involve tongue musculature, epiglottis and pre-epiglottic space, tonsil and its pillars, and hypopharynx.

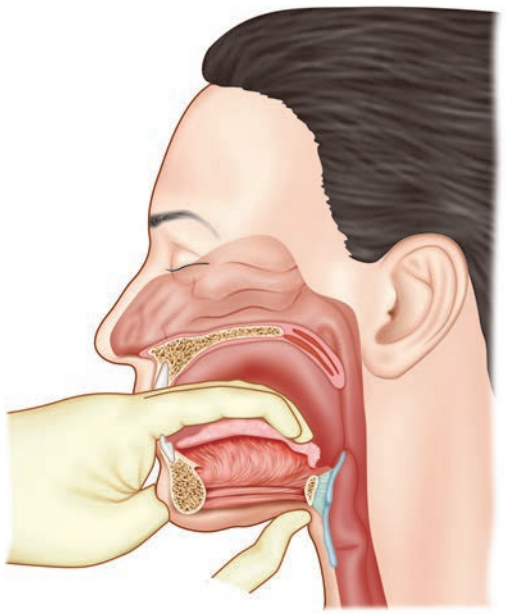

Fig. 1: Bimanual palpation of oropharynx to assess tumor fixation to mandible and submucosal extent in the tongue base. Subtle mucosal abnormalities bleed after palpation

Spread

- *Local*: The deeply infiltrative ulcer involves tongue musculature, epiglottis and pre-epiglottic space, tonsil and its pillars and hypopharynx.
- *Lymphatic*: Most patients (70%) have unilateral or bilateral cervical metastases (usually jugulodigastric nodes) at the time of first consultation.
- *Distant metastases*: Bones, liver or lung.

Diagnosis

- *CT scan*: It shows tumor extension and nodal involvement.
- *Biopsy and examination under general anesthesia*: Examination and palpation under anesthesia provide better idea of the extent of lesion, which is usually far more extensive than it appears. Biopsy will confirm the diagnosis and tell the nature of malignancy.

Treatment

The mode of treatment varies from center to center and doctor to doctor. Tumors, which are radiosensitive (such as anaplastic carcinoma, lymphoepithelioma or lymphoma) are usually treated with radiotherapy.

- *T_1 and T_2 squamous cell carcinoma with N_0 or N_1 neck:* Surgical excision with block dissection followed by radiotherapy if needed.
- *T_3 and T_4 lesions*: Surgical excision with mandibular resection, neck dissection and postoperative radiation.
- *T_4 lesion extending into anterior two-thirds-of tongue or vallecula*: Extensive surgery consists of total glossectomy and laryngectomy in addition to the block dissection. Chemotherapy may be combined with radiotherapy and surgery in such cases.
- *Advanced cases with poor health*:
  - Palliation with radio or chemotherapy.
  - Tracheostomy and gastrostomy in the terminal phases.
  - Strong analgesics for relieving pain.

CARCINOMA TONSIL (FIGS 2 AND 3)

Spread

- *Local*: Tumor may involve soft palate and pillars, base of tongue, pharyngeal wall, parapharyngeal space, hypopharynx, pterygoid muscles or mandible.
- *Lymphatic*: Fifty percent cases show cervical node (usually jugulodigastric) involvement at the time of initial presentation.
- *Distant metastases.*

Clinical Features

- Most common are the persistent sore throat, difficulty in swallowing, pain in the ear or lump in the neck.
- Late features are bleeding from the mouth, halitosis (fetor oris), pain and trismus (due to invasion of pterygoid muscles and mandible).
- Ulcerated lesion with necrotic base on tonsil.
- Lymphomas may simulate indolent peritonsillar abscess and present as unilateral tonsillar enlargement with or without ulceration.
- Palpation of tonsillar area finds the extent of tumor induration.

Fig. 2: Carcinoma of right tonsil involving anterior tonsillar pillar

Fig. 3: Malignancy of tonsil with neck node metastasis on right side

Diagnosis

- *Biopsy*: It will reveal the histological typing.

Squamous cell carcinoma is the most common variety of tonsillar malignancy and second is lymphoma.

Treatment

- *Radiotherapy*: For early and radiosensitive tumors irradiation includes cervical nodes.
- *Surgery*:
  - Excision of the tonsil for early superficial lesions.
  - Wide surgical excision with hemimandibulectomy and neck dissection (commando operation) for larger lesions and bone invasion.
- *Combination therapy*: Surgery and pre- or postoperative radiation.
- *Chemotherapy*: As an adjunct to surgery or radiation.

LYMPHOMA

- Lymphomas are mostly of non-Hodgkin variety.
- Most common site is tonsils.
- Lesion: A smooth submucosal bulky mass of tonsil that may ulcerate.
- Males are affected more.
- Cervical nodes are enlarged in 40–70% of the patients.
- Treatment is radiation and/or chemotherapy.

CARCINOMA SOFT PALATE

Carcinoma in faucial arch, which comprises soft palate, uvula and anterior tonsillar pillar, is often squamous cell variety (usually well-differentiated). It has late tendency for nodal metastases that is similar to oral cavity carcinomas.

The superficially spreading lesions (Fig. 4) may spread locally to the contiguous structures and metastasize to lymph nodes, which are usually upper deep cervical and submandibular.

Clinical Features

- *Most common*: Persistent sore throat, local pain or earache.
- Ulcer or growth with induration.

Fig. 4: Carcinoma of soft palate left side. Superficially spreading infiltrative palatal lesion involving anterior tonsillar pillar

Treatment

- Small tumors can be easily excised.
- Large tumors involving major part of soft palate are treated with radiotherapy. Surgery is reserved for salvage as the morbidity of surgery is significant.
- Radical removal of palate and superior part of lateral pharyngeal wall through mandibulotomy (median, paramedian or lateral angular) need postoperative reconstruction.

CARCINOMA POSTERIOR PHARYNGEAL WALL

Spread

- *Local*: The submucosal spread may involve tonsil, soft palate, tongue, nasopharynx, hypopharynx, parapharyngeal space or anterior spinal ligaments.
- *Lymphatic*: Sixty percent of cases have lymph node metastases, which may be bilateral.

Clinical Features

Local lesion usually remains asymptomatic for a long time and patients present with neck swelling. X-ray soft tissue neck lateral view shows soft tissue fullness in prevertebral space (Fig. 5).

Treatment

- Irradiation.
- Surgical excision combined with block dissection and skin grafting: Approach may be lateral pharyngotomy with or without mandibular osteotomy.

BENIGN SWELLINGS

Some common benign lesions (papilloma, hemangioma and pleomorphic adenoma) are described here. Rare benign tumors include lipoma, fibroma and neuroma.

Fig. 5: X-ray of soft tissue neck lateral view. Posterior pharyngeal mass. Soft tissue fullness in prevertebral space
Source: Dr Jayesh Patel, Consultant Radiologist, Anand, Gujarat

Papilloma

- Usually pedunculated and arises from the tonsil, soft palate or faucial pillars.
- Usually asymptomatic and is noticed accidentally either by patient or doctor.
- Large papilloma may cause local irritation.
- Treatment is surgical excision.

Hemangioma

- *Common sites*: Palate, tonsil, posterior and lateral pharyngeal wall.
- *Types*: Capillary or cavernous type.
- *Treatment*: They are treated only if giving symptoms of bleeding and dysphagia. The different modalities of treatment include:
  - Diathermy coagulation
  - Injection of sclerosing agents
  - Cryotherapy
  - Laser coagulation.

Pleomorphic Adenoma

- Usually seen submucosally on hard or soft palate.
- Potentially malignant.
- Treatment is total excision.

Mucous Cyst

Mucous retention cysts are not uncommon.

- *Most common site*: Vallecula (Fig. 6).
- Yellowish pedunculated or sessile cystic swelling.
- Usually asymptomatic and incidentally seen during examination.
- Large cysts can cause foreign body sensation in the throat.
- *Treatment*: Surgical excision of the pedunculated cyst or incision, drainage and removal of cyst wall (marsupialization).

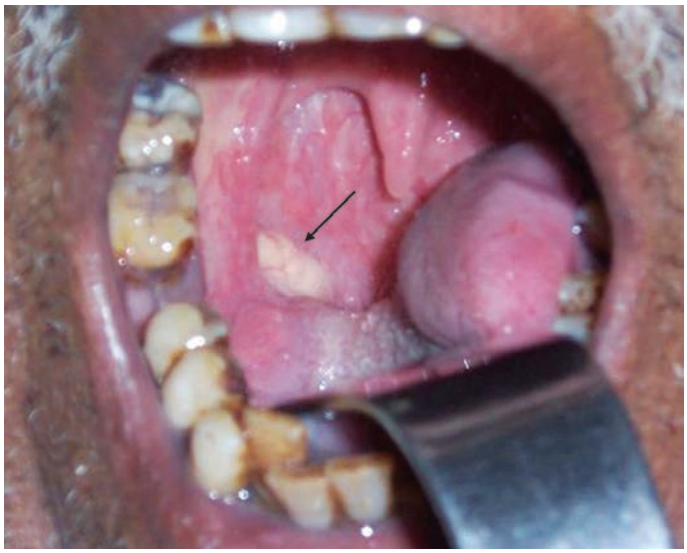

Fig. 6: Mucous retention cyst of vallecula right side. Yellowish cystic swelling visible through the oral cavity on depressing the tongue

PARAPHARYNGEAL TUMORS (FIG. 7)

In parapharyngeal space, which is situated lateral to the pharynx, both benign and malignant tumors are seen. Commonly seen tumors include deep lobe of parotid, neurilemmoma, carotid body chemodectoma, lipoma or aneurysm of internal carotid artery.

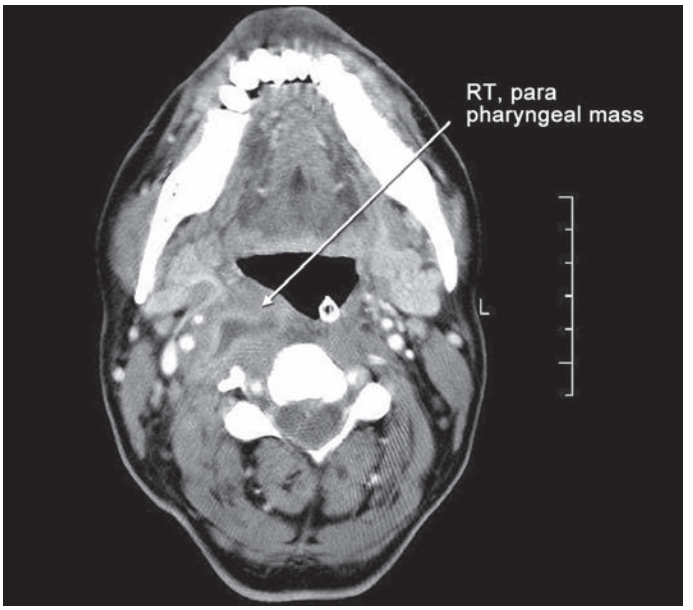

Fig. 7: CT scan of parapharyngeal mass. Contrast CT scan of neck axial section. Soft tissue density mass in right parapharyngeal space, post-styloid compartment. Note the inhomogeneous enhancement with peripheral enhancing component and central nonenhancing necrotic component. Medially it involves pharyngeal mucosal space and bulges into the pharynx. Posteriorly it extends into retropharyngeal wall but not crossing midline. Laterally, loss of fat planes indicates encasing of internal and external carotid arteries and approaching internal jugular vein
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

They usually present with a bulge in lateral pharyngeal wall, which may distort the pillars and soft palate and mimic oropharyngeal neoplasm.

STYLALGIA (EAGLE'S SYNDROME)

- An elongated styloid process or calcification of stylohyoid ligament can cause pain in tonsillar fossa and upper neck. Pain radiates to the ear and gets aggravated on swallowing.
- The elongated styloid process can be palpated transorally in the tonsillar fossa.
- Many persons with elongated styloid process (an incidental and common finding on radiographs skull) do not have any symptoms (Fig. 8).

Diagnosis

- X-ray of skull anteroposterior or lateral view with open mouth show the elongated styloid process.

Treatment

- Symptomatic styloid process may be excised by transoral or cervical approach. Results are equivocal.

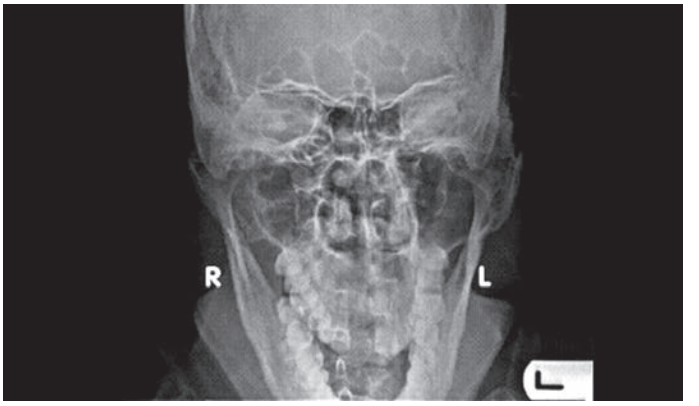

Fig. 8: X-ray of skull for styloid process. Left side styloid process is thick and right side styloid process is very long
Source: Dr Jayesh Patel, Consultant Radiologist, Anand, Gujarat

Clinical Highlights

1. **Parapharyngeal tumor:** Neurogenic tumor is the most common tumor of parapharyngeal space.
2. **Internal carotid artery aneurysm:** This parapharyngeal mass displaces the tonsil and tonsillar fossa medially. Pulsations can be felt on intraoral palpation.

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43 Malignant Tumors of Hypopharynx

Forget not that thy life and wealth are not for sense-pleasure and personal happiness. Forget not that the lower classes, the ignorant, the poor, the illiterate, the cobbler, the sweeper, are thy flesh and blood, thy brothers.

— **Swami Vivekananda**

Points of Focus

- **INTRODUCTION**
  - ✓ Hypopharynx Subsites
- **RISK FACTORS**
- **PATHOLOGY**
- **CLINICAL FEATURES**
- **DIAGNOSIS**
- **STAGING**
- **MANAGEMENT**
  - ✓ Surgery
  - ✓ Radiotherapy
  - ✓ Chemotherapy
  - ✓ Combined Chemotherapy and Irradiation for Larynx Preservation
- **CARCINOMA PYRIFORM SINUS**
  - ✓ Spread
  - ✓ Clinical Features
  - ✓ Diagnosis
  - ✓ Treatment
- **CARCINOMA POSTCRICOID**
  - ✓ Spread
  - ✓ Clinical Features
  - ✓ Diagnosis
  - ✓ Treatment
- **CARCINOMA POSTERIOR PHARYNGEAL WALL**
  - ✓ Spread
  - ✓ Clinical Features
  - ✓ Diagnosis
  - ✓ Treatment
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

Carcinoma of the hypopharynx is very common in India. Most of the tumors are squamous cell carcinoma (SCC), which have various grades of differentiation. The various subsites involved in descending order of frequency are pyriform sinus, postcricoid region and posterior pharyngeal wall.

Benign tumors of laryngopharynx are very rare and include lipoma (relatively more common), papilloma, adenoma, fibroma and leiomyoma. They usually present as smooth well-defined mass. They may be pedunculated and mobile. Treatment consists of endoscopic or open surgical excision.

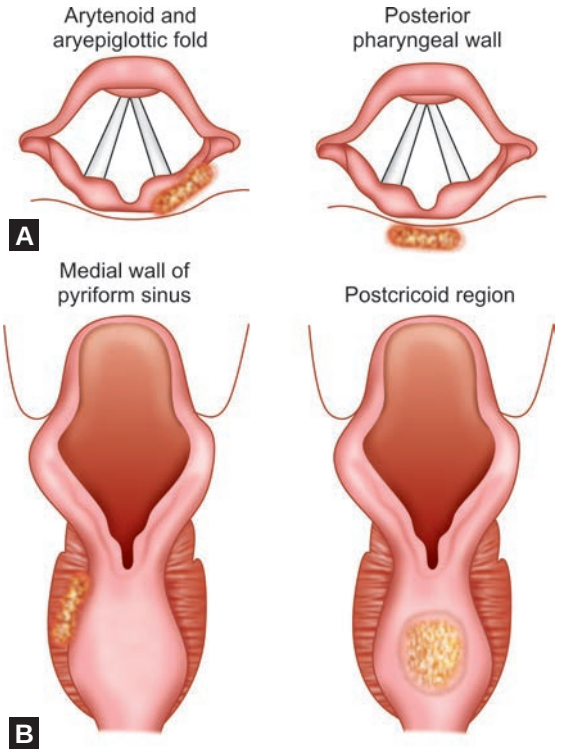
Hypopharynx Subsites

It is subdivided into three regions: pyriform sinus, postcricoid region and posterior pharyngeal wall (Figs 1A and B). For the detail anatomy *see* Chapter Anatomy and Physiology of Oral Cavity, Pharynx and Esophagus.

RISK FACTORS

They include following:

- Alcohol.
- Smoking.
- Chewing of betel nut and tobacco.



Figs 1A and B: Sites of hypopharyngeal cancer. (A) Indirect laryngoscopic; (B) Posterior views of laryngopharynx

- Genetic predisposition.
- Plummer-Vinson or Paterson-Brown-Kelly syndrome or sideropenic anemia, which primarily affects women (85%).
- Nutritional deficiency.
- Low socioeconomic conditions.

PATHOLOGY

Squamous cell carcinoma accounts for 95% of all hypopharyngeal tumors. Less common tumors are adenocarcinoma and lymphoma (especially with AIDS). At the time of presentation more than 75% are in advanced stages III and IV. The subtypes of lymphoma include angiocentric T-cell lymphoma, non-Hodgkin extranodal lymphoma and mucosa associated lymphoid tissue (MALT) lymphomas.

CLINICAL FEATURES

- Most common: Neck mass as most patients come in stage III and IV.
- Common: Shortness of breath and dysphagia are features of advanced disease.
- Early symptoms: Gastroesophageal reflux and sore throat are nonspecific and mostly ignored by the patients.
- Referred otalgia.
- Malnourishment.
- Larynx: Hoarseness of voice due to laryngeal edema and immobility of vocal cord.

DIAGNOSIS

- *Endoscopic examination*: For the evaluation of tumor extension and second primary and ordering imaging studies.
- *Barium swallow*: Though used routinely its role is limited. May show second primary in esophagus. Helps in examining deglutition process, stricture and fistula after surgery.
- *Computed tomography* or *Magnetic resonance imaging*: Computed tomography (CT) and magnetic resonance imaging (MRI) are the primary modality of evaluation before treatment. These imaging techniques upstage clinical tumor stage in up to 90% of the cases. Cartilage invasion is usually better appreciated in CT (Fig. 2). Refer to Chapter Diagnostic Imaging for the CT Anatomy of Laryngopharynx.
- *Biopsy*: For histopathological diagnosis.
- *Positron emission tomography:* For locoregional recurrence, residual or persistent disease and distant metastases.

STAGING

The TNM system of classification (Table 1) allows comparing of end results, determining prognosis and selecting treatment and communicating with other professionals about patients. The main deficiencies of TNM classification are exclusion of the following elements:

- Host factors
- Comorbidities
- Histological grading
- Levels of lymph nodes and their depth and involvement of adjacent structures.

MANAGEMENT

Most patients come in advanced stage and surgery followed by radiotherapy is the standard treatment. The current trend is emerging towards the organ sparing techniques with combined chemotherapy and irradiation. The different modalities of treatment are listed below.

A. Surgery

1. T_{1-2}:
   a. Partial pharyngectomy with primary closure.
   b. Endoscopic CO_2 laser resection with secondary intention healing.

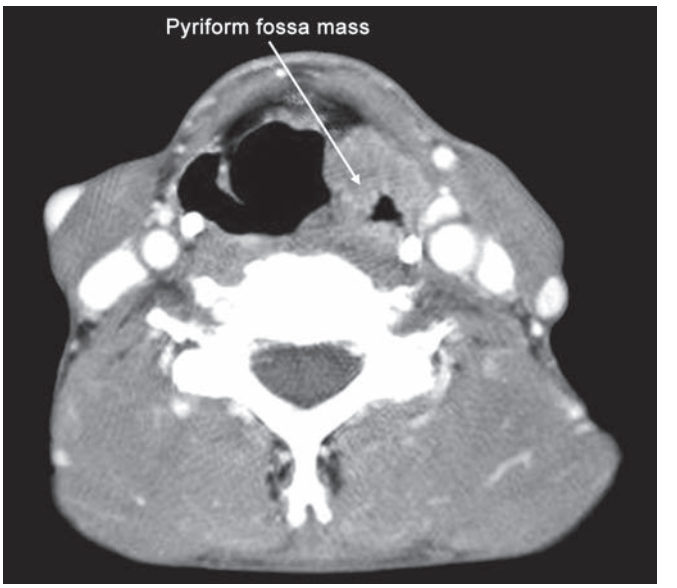


Fig. 2: CT scan of pyriform fossa mass. Contrast CT scan of neck axial section. In homogeneously enhancing mass, which is completely obliterating left pyriform fossa and involving both medial (aryepiglottic fold) and lateral wall and left part of epiglottis
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

TABLE 1 AJCC cancer staging (2010) and UICC TNM classification of carcinoma of laryngopharynx and cervical esophagus

Primary Tumor (T) tumor size in greatest dimension	
T_X	Cannot be assessed
T_0	No tumor
T_{is}	Carcinoma in situ
Hypopharynx	
T_1	Tumor <= 2 cm and limited to one subsite
T_2	Tumor > 2 cm but <= 4 cm or invading >1 subsite without hemilarynx fixation
T_3	Tumor > 4 cm or with hemilarynx fixation cervical esophagus.
T_{4a}	Tumor invades any of the following: thyroid-cricoid cartilage, hyoid bone, thyroid gland, central compartment of soft tissue such as infrahyoid strap muscles and subcutaneous tissue.
T_{4b}	Tumor invades any of the following: prevertebral fascia, mediastinal structure or encasing of carotid artery
Cervical esophagus	
T_1	Tumor limited to lamina propria or submucosa
T_2	Tumor invades muscularis propria
T_3	Tumor invades adventitia
T_4	Tumor invades adjacent structures
Regional lymph nodes (N) size in greatest dimension	
Hypopharynx	
N_x	Cannot be assessed
N_0	No nodal metastases
N_1	Single ipsilateral node <=3 cm
N_{2a}	Single ipsilateral node > 3 cm but <= 6 cm
N_{2b}	Multiple ipsilateral nodes but none > 6 cm
N_{2c}	Bilateral or contralateral nodes but none > 6 cm
N_3	Any node > 6 cm
Cervical esophagus	
N_0	No lymph node metastases
N_1	Lymph node metastasis present
Distant metastasis (M)	
M_x	Cannot be assessed
M_0	No distant metastasis
M_1	Distant metastasis
Stage Grouping	
Hypopharynx	
Stage 0	T_{is} N_0 M_0
Satge I	T_1 N_0 M_0
Stage II	T_2 N_0 M_0
Stage III	T_3 N_0 M_0 or T_{1-3} N_1 M_0
Stage IVA	T_{4a} N_{0-1} M_0 or T_{1-4a} N_2 M_0
Stage IVB	T_{4b} N_{0-3} M_0 or T_{1-4} N_3 M_0
Stage IVC	T_{1-4} N_{0-3} M_1
Cervical esophagus	
Stage 0	T_{is} N_0 M_0
Stage I	T_1 N_0 M_0
Stage IIA	T_{2-3} N_0 M_0
Stage IIB	T_{1-2} N_1 M_0
Stage III	T_3 N_1 M_0 or T_4 N_{0-1} M_0
Stage IV	T_{1-4} N_{0-1} M_1

2 T_{2-3}:
   a. Partial laryngopharyngectomy with regional or free flap.
   b. Supracricoid hemilaryngopharyngectomy with primary closure.
3. T_{3-4}:Total laryngectomy with partial-total pharyngectomy with primary closure or regional or free flap.
4. T_4:Total pharyngo-laryngo-esophagectomy with gastric pull-up.
5. Options for reconstruction
   a. Pectoralis myocutaneous flap.
   b. Gastric pull-up.
   c. Colonic transposition.
   d. Enteric flaps.
      - Jejunal free flap.
      - Gastro-omental.
   e. Fasciocutaneous free flaps.
      - Radial forearm.
      - Lateral thigh.

B. Radiotherapy

1. Primary.
2. Postoperative.
3. Chemoradiation.

C. Chemotherapy.

D. Combined Chemotherapy and Irradiation for Larynx Preservation

- Induction chemotherapy.
- Concomitant chemoradiotherapy.

Prognosis: Hypopharyngeal cancers have poorest 5-year survival (30%) among all the head and neck cancers.

CARCINOMA PYRIFORM SINUS

- It accounts for 60% of all hypopharyngeal cancers.
- Mostly affects males above 40 years of age.
- Usually patients present with metastatic neck nodes, which are very common in pyriform fossa cancer.

Spread

- *Locally*:
  - Superiorly to vallecula and base of tongue
  - Inferiorly to postcricoid region
  - Medially to aryepiglottic folds and ventricles
  - Laterally to thyroid cartilage, thyroid gland or soft tissue mass in the neck.

- *Lymphatic*: Seventy-five percent of cases have cervical nodal metastases (upper and middle group of jugular cervical nodes bilateral in 50% cases) at the time of first consultation. The rich lymphatic network gives rise to early lymphatic metastasis in pyriform fossa SCC. In some cases nodes appear late even after eradication of the primary.
- *Distant metastases*: It is often late and seen in lung, liver and bones.

Clinical Features

The growth usually remains asymptomatic for a long time because of the large size of the pyriform sinus.

- Most common: A mass of lymph nodes in the neck.
- Early symptoms: Sticking in the throat and "pricking sensation" on swallowing.
- Referred otalgia or pain on swallowing.
- Increasing dysphagia.
- Hoarseness and laryngeal obstruction due to laryngeal edema or spread to larynx.
- Growth either exophytic or ulcerative and deeply infiltrative can often be seen on mirror examination. Pooling of secretions may obstruct the view.

Diagnosis

- *Barium swallow*: Filling defect can be seen.
- *CT scan* (Fig. 2): Evaluate primary growth and lymph node metastasis.
- *Endoscopic examination*: For accurate assessment of the extent of primary and panendoscopy to find any other synchronous primary.
- *Biopsy*: For histopathological diagnosis.

Treatment

- *Radiotherapy*
  - Primary treatment for early growth without nodes
  - Planned postoperative radiotherapy usually in all cases of surgery.
- *Surgery*
  - Growth limited to pyriform fossa and not extending to postcricoid region: Total laryngectomy and partial pharyngectomy often combined with elective or prophylactic block dissection of lymph nodes. Pharynx is primarily closed.
  - Growth extending to postcricoid region: Total laryngectomy and pharyngectomy with block dissection and myocutaneous flaps or stomach pull-up for pharyngoesophageal reconstruction.
- *Curative treatment as per the staging:*
  - Stage I: Primary radiotherapy or partial pharyngectomy (PP) or partial pharyngectomy and partial laryngectomy (PPPL).
  - Stage II: Primary radiotherapy or chema radiotherapy or PPPL or total laryngopharyngectomy (TLP).
  - Stage III and IVa: Upfront chemoradiotherapy Total laryngectomy and partial pharyngectomy (TLPP) or total laryngopharyngectomy (TLP) and postoperative radiotherapy/ chemoradiotherapy.

CARCINOMA POSTCRICOID

- It accounts for 30% of all laryngopharyngeal malignancies.
- One-third patients of postcricoid carcinoma suffer from an etiological factor the sideropenic anemia (Paterson-Brown-Kelly or Plummer-Vinson syndrome), which occurs in females and is characterized by hypochromic microcytic iron deficiency anemia.
- An ulcerative type lesion spreads in an annular causes dysphagia.
- Patients are usually females and in their twenties and thirties.

Spread

- *Local*: Cervical esophagus, arytenoids, or recurrent laryngeal nerve at cricoarytenoid joint.
- *Lymphatic*: Paratracheal lymph nodes, which are bilateral and not clinically palpable.

Clinical Features

- *Most common*: Progressive dysphagia in anemic females.
- *Vocal symptoms*: Voice change and aphonia.
- Inspiratory stridor in advanced tumor occurs due to pressure on larynx and bilateral recurrent laryngeal nerves involvement.
- *General symptoms*: Malnutrition and weight loss.
- *Indirect laryngoscopy*: Edema and erythema of the postcricoid region and pooling of secretions in the hypopharynx indicates postcricoid growths, which may not be visible.
  - *Vocal cord mobility*: Restricted vocal cord mobility due to infiltration of recurrent laryngeal nerve or posterior cricoarytenoid muscles.
- *Palpation*: Absence of laryngeal crepitus, which is felt normally while moving larynx over the cervical spine (Fig. 3).

Diagnosis

- *X-ray soft tissue neck lateral view*: Shows widening of prevertebral soft tissue space.

Fig. 3: Post-laryngeal crepitus. On moving the larynx from side to side, normally a post-laryngeal crepitus is felt. It is absent in cases of postcricoid malignancy

- *Barium swallow*: Shows the lower extent of the disease (Fig. 4).
- *Endoscopy*: For assessing the extent of lesion and panendoscopy find other synchronous primary.
- *Biopsy*: For histopathological diagnosis.
- CT scan (neck and thorax): Extent of disease, lymph node and lung metastasis.

Treatment

Some prefer primary surgery while others prefer initial radiotherapy.

- *Radiotherapy*: Primary radiotherapy preserves laryngeal function. Failed cases may be subjected to surgery. Some prefer planned postoperative radiotherapy.
- *Surgery*: Laryngo-pharyngo-esophagectomy with stomach pull-up or colon transposition for reconstructing pharyngo-esophageal segment.
- *Curative treatment as per the staging:*
  - Stage I: Primary radiotherapy or Total laryngophary ngectomy (TLP).
  - Stage II: Primary radiotherapy or chemoradiotherapy or TLP and postoperative radiotherapy.
  - Stage III: TLP or total laryngo-pharyngo-esophagectomy (TLPO) and postoperative radiotherapy/ chemoradiotherapy.
  - Stage IVa: TLPO and gastric pull-up and postoperative radiotherapy/primary chemoradiotherapy.

Prognosis

It is poor with both irradiation and surgical treatment.

Fig. 4: Barium swallow in a case of malignancy postcricoid region. Note the prevertebral soft tissue widening and irregular filling defect

Source: Dr Jayesh Patel, Consultant Radiologist, Anand, Gujarat

CARCINOMA POSTERIOR PHARYNGEAL WALL

This least common lesion accounts for only 10% of all laryngopharyngeal malignancy. It is mostly seen in males above 50 years of age. Growth is either exophytic (more common) or ulcerative and remains localized for long time.

Spread

- *Local*: Prevertebral fascia, muscles and vertebrae.
- *Lymphatic*: Fifty percent patients have nodal metastases (usually bilateral due to midline lesion) on their initial consultation. Retropharyngeal nodes, which are not clinically palpable, are also involved.

Clinical Features

They are following:

- Dysphagia.
- A palpable mass of nodes in the neck.
- Spitting of blood.
- Indirect mirror examination: Growth is either exophytic (more common) or ulcerative.

Diagnosis

- *X-ray soft tissue neck lateral view*: Shows vertical extent and thickness of the tumor and involvement of cervical vertebrae.
- CT scan for tumor and metastasis.
- *Endoscopy*: For accurate assessment of tumor and panendoscopy to find other synchronous primary.
- *Biopsy*: For histopathological diagnosis.

Treatment

- *Radiotherapy*: For early small lesions, particularly exophytic. It preserves function.
- *Surgery*:
  - Early small lesions may be excised via lateral pharyngotomy and primary repair.
  - Advanced lesions may need laryngopharyngectomy and block dissection with repair of the food channel.
- *Curative treatment as per the staging:*
  - Stage I: Primary radiotherapy or Partial Pharyngectomy (PP)
  - Stage II: Primary radiotherapy or chemoradiotherapy PP or total laryngopharyngectomy (TLP).
  - Stage III: Partial pharyngectomy or TLP and postoperative radiotherapy/ chemoradiotherapy.
  - Stage IVa: Total laryngopharyngectomy and postoperative radiotherapy/ chemoradiotherapy.

Prognosis

Gross 5-year cure rate is only 19%.

Clinical Highlights

1. **Plummer-Vinson syndrome: T**his premalignant condition is the disease of females. It is characterized by iron deficiency anemia and dysphagia. Patients usually have koilonychia, atrophic gastritis, and glossitis.
2. **Cancer of pyriform fossa:** The referred ear pain is through CN X (superior laryngeal nerve, branch of vagus nerve).
3. **Laryngeal crepitus:** Post-laryngeal crepitus is present in normal persons and absent in patients with postcricoid malignancy.

FURTHER READING

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44 Disorders of Esophagus

You are not born to live and die one of those caste-ridden, superstitious, merciless, hypocritical, and atheistic cowards. I hate cowardice and political nonsense. God and truth are the only politics in the world, everything else is trash.

—Swami Vivekananda

Points of Focus

- INTRODUCTION
- PERFORATION OF ESOPHAGUS
- CORROSIVE BURNS
- MALLORY-WEISS SYNDROME
- FOREIGN BODIES
- PILL-INDUCED ESOPHAGITIS
- GASTROESOPHAGEAL REFLUX DISEASE (GERD)
- BARRETT'S ESOPHAGUS
- BENIGN STRICTURES
- HIATUS HERNIA
- SCHATZKI'S RING
- PLUMMER-VINSON (PV) SYNDROME
- INFECTIOUS ESOPHAGITIS
- CRICOPHARYNGEAL SPASM
- DIFFUSE ESOPHAGEAL SPASM
- NUTCRACKER ESOPHAGUS
- CARDIAC ACHALASIA
- SCLERODERMA
- ZENKER'S DIVERTICULUM
- GLOBUS HYSTERICUS PHARYNGEUS
- BENIGN NEOPLASMS
- CARCINOMA ESOPHAGUS
- CLINICAL HIGHLIGHTS

INTRODUCTION

The esophagus is subject to a variety of diseases (Box 1). The primary infections of the esophagus are rare except fungal esophagitis, which occurs in patients who are immunocompromised. Esophageal perforation is a surgical emergency. The scar formation and stricture development occur from ingestion of caustic agents. Foreign bodies (FBs) in esophagus are frequently encountered in children. The cricopharyngeal area is a frequent site. Benign tumors of the esophagus are rare. However, malignancies of the esophagus are relatively common.

Esophageal inflammatory disease like gastroesophageal reflux disease (GERD) is quite common. Peristaltic contractions of esophagus pass the food down into the stomach. The upper esophageal sphincter (UES) and lower esophageal sphincter (LES) prevent reflux of food while swallowing. Dysphagia is usually present in esophageal disorders. Odynophagia, painful swallowing indicates esophagitis.

PERFORATION OF ESOPHAGUS

Early diagnosis is essential, as mediastinitis can rapidly prove fatal. Perforation of thoracic esophagus is more serious than cervical esophagus.

Etiology

- *Instrumental trauma*: Esophagoscopy or dilatation of strictures with bougies. The most common site is just above the upper sphincter. The lower esophagus perforation near the hiatus is uncommon.
- *Spontaneous rupture*: This occurs due to vomiting and

Box 1: Esophageal diseases

1. *Congenital*: Atresia, tracheoesophageal fistula, stenosis and dysphagia lusoria
2. *Hiatus hernia*: Sliding and paraesophageal (rolling)
3. *Infectious*: Monilial esophagitis, viral and granulomatous lesions
4. *Inflammatory*: Gastroesophageal reflux disease (GERD), pill-induced esophagitis
5. *Injury*: Foreign bodies, drug-induced, corrosive, penetrating, and Mallory-Weiss syndrome
6. *Perforation*: Spontaneous-Boerhaave's syndrome (Barotrauma), pathological and instrumental
7. *Diverticula*: Zenker's diverticulum (Pharyngeal pouch), mid-esophageal diverticula, epinephric diverticula, and diffuse intramural pseudodiverticulosis
8. *Narrowing*: Webs, strictures and extrinsic compression
9. *Motility Disorders* (upper sphincter, lower sphincter or the body of esophagus)
   a Hypermotility disorder: Cricopharyngeal spasm, diffuse esophageal spasm, nutcracker esophagus, and hypertensive lower esophageal sphincter
   b. Hypomotility disorders: Cardiac achalasia, connective tissue disease (scleroderma and amyotrophic lateral sclerosis) and hypoperistalsis—CREST syndrome
10. *Neoplasms*:
    a. Benign: Leiomyoma and others; Mucosal polyps, lipomas, fibromas and hemangiomas
    b. Malignant: Carcinoma (Squamous cell carcinoma and Adenocarcinoma), malignant melanoma and sarcoma
11. *Miscellaneous*: Schatzki's ring, Crohn's disease, Plummer-Vinson (PV) syndrome, eosinophilic esophagitis and varices

usually involves lower third esophagus. In Boerhaave's syndrome, there occurs postemetic rupture of all the layers of esophagus.
- *Pathological*: Malignancy.
- *Penetrating injuries.*

Clinical Features

- Preceding history of esophagoscopy, vomiting or other injury.
- Cervical esophageal rupture
  - Neck: Pain, local tenderness and surgical emphysema
  - Fever
  - Difficulty in swallowing.
- Thoracic esophageal rupture
  - Chest pain referred to the interscapular region
  - Fever 102–104°F (39–40°C)
  - Signs of shock
  - Surgical emphysema and pneumothorax
  - Hamman's sign: Crunching sound over the heart because of air in the mediastinum.

Diagnosis

- *X-rays of the chest and neck findings*
  - Widening of the mediastinum and retrovisceral space
  - Surgical emphysema, pneumothorax, or gas under the diaphragm
  - Pleural effusion.

Treatment

- *General*
  - Nil by mouth
  - Intravenous fluids for nutrition
  - Antibiotics: To combat infection
  - Management of shock.
- *Perforations of cervical esophagus*: Early cases may be managed conservatively. Drainage is required in cases of suppuration. Retrovisceral space and upper mediastinum are drained.
- *Rupture of thoracic esophagus*: Perforation is surgically repaired and pleural cavity drained within 6 hours. Repair is not possible after 6 hours but infected area needs drainage.

CORROSIVE BURNS

Etiology

Following items may be swallowed accidentally by children or by suicidal, psychotic and alcoholic adults:
- *Acids*: Various cleaners used for toilet, drain, swimming pool and metal.
- *Alkalis*: Drain and oven cleaners, dishwasher detergents and hair relaxers.

Pathology

- *Severity*: Severity of burns depends on the nature, amount and concentration of corrosives and the duration of their contact with the tissue.
- *Extent of lesion*: Alkalis are more destructive and penetrate deep into the esophagus. Lye, a strong alkali (sodium or potassium hydroxide) burns entire esophagus and stomach, which may slough and result in fatal mediastinitis and peritonitis.
- *Stages*: There are three stages of esophageal burns: (1) necrosis, (2) granulations (separation of slough) and (3) stricture. Stricture begins at 2 weeks and continues for 2 months or longer.

Clinical Features

- History of corrosive ingestion and associated burns of face, lips and oral cavity
- Oropharyngeal, retrosternal or epigastric pain
- Dysphagia or odynophagia

- Hypersalivation: drooling
- Laryngeal: Hoarseness and stridor (upper airway obstruction).
- Shock
- Mediastinitis
- Peritonitis
- Acid-base imbalance.

Diagnosis

- History and examination will confirm the diagnosis.
- X-rays of chest and soft tissue neck help in estimating the extent of damage.
- Upper endoscopy will determine the extent of injury and should be done early. Esophagogastroduodenoscopy (EGD) is done within 2 days to know the extent and degree of esophageal and gastric damage for planning further treatment. Endoscope is not passed further than the first severe circumferential burn.

Grading

The grading system (Table 1) helps in predicting prognosis.

- *Grade 1 and 2A*: No stricture formation. Oral liquid diet can be started after 1–2 days.
- *Grades 2B/3A*: Most of the patients develop stricture.
- *Grade 3B*: 65% early mortality rate.

Caustic strictures are longer and tighter and usually refractory and have higher rate of complications during dilation.

Management

- Immediate
  - *ICU*: Patient is kept in ICU.
  - *Wash and irrigation*
    - Mouth: Wash out the mouth with large volume of cold water.
    - Eye: Irrigate eye exposure with large amount of saline or water. Fluorescein staining will determine corneal ulcerations.
    - Never induce emesis because it can worsen the complications.
    - Never neutralize with acid or base because the generated heat would destroy more tissue. Charcoal also is not useful. Use simple water.
  - *Examination*: Examine oropharyngeal injury and airway compromise and watch for perforation.
  - *Tracheostomy or intubation*: For relieving airway obstruction.
  - *Intravenous fluids and electrolytes*: For treating shock and acid-base imbalance.
  - *Maintenance of intake/output chart*: Monitoring of urine output for renal failure.
  - *Analgesic*: For relieving pain.
  - *Antibiotics and steroids:* Parenteral antibiotics for 3–6 weeks and steroids for 4–6 weeks (to prevent stricture) though given often, are not found useful.
  - *Feeding*: Oral liquids can be started after 48 hours if patient can swallow without pain or vomiting. Some patients need nasogastric tube for feeding and maintenance of esophageal lumen.
- Delayed
  - *Esophagogram and EGD* are done every 2 weeks to monitor healing and development of stricture. Patients of corrosive injuries need lifelong follow-up.
  - *Management of stricture* may include:
    - Esophagoscopy and prograde dilatation
    - Gastrostomy and retrograde dilatation or
    - Esophageal reconstruction or bypass.

TABLE 1 Corrosive esophageal injury

Grage 0	Normal
Grage 1	Mucosal edema and erythema
Grage 2A	Superficial ulcers, bleeding and exudates
Grage 2B	Deep focal or circumferential ulcers
Grage 3A	Focal necrosis (brown, black or gray discoloration)
Grage 3B	Extensive necrosis
Grage 4	Perforation

Source: Zargar SA, Kochhar R, Mehta S, et al. The role of fiber-optic endoscopy in the management of corrosive ingestion and modified endoscopic classification of burns. Gastrointest Endosc. 1991;37:165-9.

Caustic strictures are longer and tighter and usually refractory and have higher rate of complications during dilation.

MALLORY WEISS SYNDROME

In this disorder, mucosal tear is caused at squamocolumnar junction (SCJ) by repeated vomiting or retching.

- *Clinical features:* Hematemesis after an episode of emesis or retching.
- *Diagnosis:* EGD will show the site and extent of tear. Barium swallow is not indicated.
- *Treatment:* This self-limiting condition usually responds to conservative measures. Angiographic embolization or vasopressin may be used in refractory cases.

FOREIGN BODIES

An ingested FB may be lodged in tonsil, base of tongue, vallecula, pyriform fossa or esophagus. A sharp fish bone or a needle may go into the tonsillar crypts, base of tongue or vallecula, from where they can be easily removed as an office procedure.

Fish bone, chicken bone, needle or a denture may get lodged in the pyriform fossa, from where they may be removed under local anesthesia or under general anesthesia.

Common Foreign Bodies in Esophagus

Coin, piece of meat, chicken bone, denture, safety pin, or marble piece.

Risk Factors

- *Children* have a tendency to put the things in mouth.
- *Upper denture*: Lack of tactile sensation while chewing.
- *Narrowed esophageal lumen*: Stricture or carcinoma, which may present with sudden obstruction from a piece of meat, fruit or vegetable.
- *Psychotics.*
- *Other factors*: Loss of consciousness, epileptic seizures, deep sleep or alcoholic intoxication.

Common Sites

- Cricopharyngeal sphincter: Commonest site
- Bronchoaortic constriction
- Cardiac end

Sharp or pointed objects can lodge anywhere.

Esophageal foreign body in children: The most common site is just distal to the upper esophageal sphincter.

Clinical Features

- *Choking or gagging* at the time of ingestion.
- *Discomfort or pain* usually increases on swallowing. Foreign body cervical esophagus usually presents with pain and tenderness in the lower part of neck. Constant substernal or epigastric pain indicates esophageal spasm or incipient perforation.
- *Dysphagia*: Obstruction may be total. Partial obstruction can become total due to edema.
- *Drooling of saliva*: It indicates total obstruction. Saliva may be aspirated and cause aspiration pneumonitis.
- *Respiratory distress*: Upper esophagus FB may compress posterior wall of trachea in children. Laryngeal edema can also occur.
- *Indirect laryngoscopy*: Pooling of secretions in the pyriform fossa. Foreign body may be seen protruding from postcricoid region.

Investigations

- *X-rays soft tissue neck lateral view; posteroanterior and lateral view of chest:* X-rays show location of a radiopaque FB (Fig. 1). In children, multiple FBs from nasopharynx to the rectum are not uncommon and should be ruled out.
- *Barium swallow for radiolucent foreign bodies*: Patient is instructed to swallow a piece of cotton soaked in barium or barium filled capsule. The esophagus shows filling defect.

Management

- *Esophagoscopy* and removal of FB under general anesthesia.
- *Cervical esophagotomy*: For impacted FBs or sharp hooks (partial dentures) above thoracic inlet.
- *Transthoracic esophagotomy*: For impacted FBs in thoracic esophagus.
- *Foreign body passed into stomach*: They usually pass with the stools, which should be examined every day. Purgatives should not be used. Indications for operative interference include:
  - Pain and tenderness in abdomen.
  - No progress of FB on serial X-rays taken at a few days interval.
  - Children (< 2 years): FB > 5 cm long usually does not pass through the turns of the duodenum.
  - Pyloric stenosis.

Complications

- Respiratory obstruction in infants and children: Due to tracheal compression by the FB in upper esophagus and laryngeal edema.
- Periesophageal cellulitis and abscess.
- Perforation of esophagus, which may result in
  - Mediastinitis
  - Pericarditis
  - Empyema or
  - Fatal bleeding from aorta.
- Tracheoesophageal fistula.
- Ulceration and stricture.

PILL-INDUCED ESOPHAGITIS

Etiology

- *Factor*: Oral medicines not taken in upright position and without plenty of water or fluid.
- *Common medicines*: Doxycycline, acetylsalicylic acid, nonsteroidal anti-inflammatory drugs (NSAIDs), iron sulfate, quinidine and alendronate.
- *Common sites*: Level of aortic arch and distal esophagus.

Clinical Features

- Chest pain and odynophagia occur due to acute esophagitis.
- Dysphagia indicates development of stricture.

Diagnosis

- *Esophagoscopy*: Mucosal changes include ulcer, plaques resembling candida or stricture.

Treatment

- Repeated dilation for stricture and avoiding of offending drug.
- Prevention is the best treatment.

GASTROESOPHAGEAL REFLUX DISEASE

Gastroesophageal reflux disease refers to damage of esophageal mucosa due to abnormal reflux of gastric content.

Etiology

Inappropriate function of LES permits reflux of gastric contents into esophagus. The conditions, which decrease LES tone, are following:

- High levels of progesterone in pregnancy
- Sliding hiatus hernia
- Fatty foods, chocolate and peppermints
- Tobacco smoking and alcohol
- Drugs, which relax the smooth muscle: Anticholinergic, beta-adrenergic drugs and calcium channel blockers.

Clinical Features

- *Typical*: Heartburn (burning sensation in the epigastric or substernal region) and acid regurgitation usually occur after meal when patient is recumbent.
  - Dysphagia (peptic stricture in esophagus), odynophagia (esophagitis) and belching.
  - Angina-like chest pain, which worsens after sublingual nitroglycerin, while lying in bed at night.

GERD accounts for 70% of noncardiac chest pain.

- *Atypical GERD symptoms or extraesophageal GERD*: Atypical GERD symptoms include nonproductive cough, hoarseness of voice and asthma-like symptoms and dental erosions. They indicate extraesophageal GERD. The diagnosis is

Fig. 1: X-ray soft tissue neck lateral view showing radiopaque foreign body (fishbone) in the laryngopharynx at the level of C_4
Source: Dr Jayesh Patel, Consultant Radiologist, Anand, Gujarat

confirmed when symptoms respond to GERD therapy. The current GERD therapy consists of initial aggressive bid proton pump inhibitors (PPIs) followed by tapering to once daily in responders.

Nonproductive cough: GERD is among the most common causes of chronic cough. Other causes include postnasal drip, asthma and tuberculosis.

- Dental erosions: Repeated long time exposure to acid can lead to loss of enamel and tooth structure.
- Reflux laryngitis or laryngoesophageal reflux (Hoarseness of voice):
  - Symptoms include hoarseness of voice, throat clearing, dysphagia, increased phlegm and globus sensation.
  - Physical findings, which may be secondary to associated smoking, alcohol, allergies, asthma, viral illness and voice abuse, include laryngeal erythema, edema, vocal cord nodules and polyps, granulations, leukoplakia, cancer and pharyngeal ulcerations.
  - Other signs, which again may be secondary to associated nasal congestion and postnasal drip, include interarytenoid bar, arytenoids medial wall erythema and posterior pharyngeal wall cobblestoning.
- Asthma-like symptoms: About 70–80% asthma patients have GERD and treated with PPI. GERD-induced asthma is suspected in following cases
  - Adult onset of asthma
  - No family history of asthma or atopy
  - Heartburn precedes onset of asthma
  - Wheezing exacerbated with meals, exercise or supine position
  - Nocturnal wheezing.

Reflux laryngitis: In cases of chronic hoarseness and cough or globus, first rule out neoplasm and then consider reflux laryngitis.

Diagnosis

- *Clinical:* History and response to therapy confirm the diagnosis.
- *Esophagogastroduodenoscopy*: Erosions and ulcerations at SCJ in reflux esophagitis and findings of Barrett's esophagus.
- *Ambulatory 24-hour pH monitoring*: This gold standard test is done in refractory cases.
- *Bernstein test*: Infusion of 0.1 M HCl in esophagus reproduces chest pain.

Subgroups

The subgroups of GERD include

- *Nonerosive reflux disease*: Nonerosive reflux disease patients have typical GERD symptoms but do not show endoscopic findings of GERD.
- *Reflux esophagitis*: In reflux esophagitis, mucosal changes are confirmed histopathologically.
- *Barrett's esophagus*: This potentially serious GERD complication predisposes to development of adenocarcinoma of esophagus. Normal stratified squamous epithelium of distal esophagus is replaced by intestinal columnar metaplasia (see section of Barrett esophagus).

Treatment

The aims of the treatment include decreasing reflux, improving esophageal clearance and protecting esophageal mucosa.

- *Lifestyle modifications*: Patient's instructions include
  - Weight loss
  - Small and frequent meals
  - Avoid fats, sweets, chocolate, tomatoes, onions, alcohol and caffeine at bedtime
  - Finish dinner 3 hours before going to bed
  - Elevation of head end of bed during sleep.
- *Medical treatment*: It includes
  - Antacids.
  - H_2 receptor antagonists: H_2 receptor antagonist (ranitidine) heals 50% cases of reflux esophagitis.
  - Proton pump inhibitors: PPI (such as omeprazole) heals 80% cases.
  - Prokinetic drugs: Prokinetic drugs (such as metoclopramide), which though do not heal esophagitis, improve esophageal clearance and gastric emptying and raise LES pressure. They should be taken 30 minutes before meal.
- *Antireflux surgery*: In Nissen's fundoplication, fundus of stomach is wrapped around LES.
  - Indication: Patients who do not respond to medical treatment and have normal esophageal peristalsis (confirmed by motility study) otherwise postoperative achalasia can occur.
- Recent developments
  - Endoscopic suturing
  - Injection of biopolymers in LES
  - Radiofrequency delivery to gastroesophageal junction (GEJ)

Complications

- Esophagitis
- Ulcers in esophagus
- Hemorrhage
- Peptic stricture in esophagus
- Aspiration
- Barrett's esophagus.

BARRETT'S ESOPHAGUS

Pathogenesis

In Barrett's esophagus, the normal stratified squamous epithelium of distal esophagus is replaced by intestinal columnar metaplasia. This potentially serious complication occurs in 10–20% of GERD patients. It predisposes to the development of esophageal adenocarcinoma (30% greater incidence), which is more likely in following conditions:

- Barrett is more than 8 cm long
- Presence of dysplasia
- Smoking.

Diagnosis

- *EGD endoscopy:* Pale pink mucosa of distal esophagus is replaced to salmon pink and SCJ is displaced proximal to GEJ. It can be divided into two types, short segment (< 3 cm) and long segment (> 3 cm).
- *Biopsy:* Intestinal columnar metaplasia shows mucin containing goblet cells.

Treatment

- *Antireflux therapy:* As mentioned in GERD.
- *EGD surveillance:* These patients need regular endoscopic surveillance to detect dysplasia and adenocarcinoma at an earlier and curable stage. Current guidelines suggest four-quadrant biopsies at 2 cm intervals along the entire length of Barrett's esophagus.
  - No dysplasia: Every 3 years.
  - Low-grade dysplasia: Every 1 year until no dysplasia.
  - High-grade dysplasia:
    - Focal: Every 3 months.
    - Multifocal: Esophagectomy, ablation therapy, or esophageal mucosal resection.

BENIGN STRICTURES

The esophageal strictures occur when its muscular layer is damaged.

Causes

- Intrinsic
  - *Congenital:* Common in the lower third.
  - *Burns:* Corrosives or hot fluids.
  - *Trauma:* Impacted FBs or external injuries.
  - *Iatrogenic:* Pill-induced, postnasogastric, sclerotherapy, radiation-induced, and sites of surgical anastomosis.
  - *Ulcers:* Reflux esophagitis, diphtheria and typhoid.
- Extrinsic
  - Anomalous vessels and aneurysms.

Clinical Features

- Dysphagia first with solid and then with liquids
- Regurgitation and coughing
- Malnourishment.

Diagnosis

- *Barium swallow*: Shows number, extent and severity of strictures.
- *Esophagoscopy*: Diagnostic and therapeutic.

Treatment

- *Esophagoscopy and prograde dilatation with bougies under direct vision:* Patients usually require repeated dilatation.

  Other types of dilators:
  - *For uncomplicated short and straight strictures*:
    - Mercury-filled rubber Maloney dilators
  - *For long, tight and tortuous strictures*:
    - Wire-guided rigid Savary-Gilliard dilators
    - Balloon dilators: Through-the-scope or wire-guided.
- *Gastrostomy*: It provides feeding to the patient and offers rest to the inflamed mucosa above the strictures. Prograde dilatation may be restored once inflammation subsides and lumen becomes visible. Patient is instructed to swallow a thread, which comes into the stomach, and then prograde or retrograde bouginage are done.
- *Excision and reconstruction:* Strictured segment is excised and plastic reconstruction is done with stomach, colon or jejunum.

HIATUS HERNIA

It refers to displacement of stomach into the chest through the diaphragm. It is common in elderly patients above 40 years of age.

Types

There are two types of hiatus hernia: sliding and paraesophageal.

1. *Sliding hiatus hernia*: Raised intra-abdominal pressure can push the stomach into the thorax in the line of esophagus.
   - Clinical features: Features of reflux esophagitis can cause ulceration, stenosis and hematemesis.
2. *Paraesophageal hiatus hernia or rolling hiatus hernia*: A part of the stomach and its peritoneal covering enters into the thorax by the side of esophagus. There is no reflux esophagitis because GEJ remains below diaphragm and angle between the esophagus and stomach is maintained.
   - Clinical features: The most common symptom is exertional dyspnea, which is caused due to the position of stomach in the thorax. Bleeding occurs in some patients.

Diagnosis

- *Barium swallow*: Shows the extent and type of hiatus hernia.

Treatment

- Conservative measures to reduce reflux esophagitis:
  - Head and chest raised during sleeping.
  - Avoid smoking.
  - Drugs for acidity and reflux esophagitis.
  - Control of obesity: Diet control and exercise.
  - Treatment of the causes of raised intra-abdominal pressure.
- Surgical method is reduction of hernia and repair of diaphragmatic opening.

SCHATZKI'S RING

It is a web-like mucosal ring, which is seen at the SCJ or proximal to LES.

- *Clinical features*: It remains asymptomatic in about 10% people.
  - Young patient presents with episodic dysphagia to solids and sometimes, liquids.
- *Diagnosis*: Barium swallow will confirm the diagnosis.
- *Treatment*: Pneumatic dilation is done in symptomatic patients.

PLUMMER-VINSON (PATTERSON BROWN-KELLY) SYNDROME

Plummer-Vinson (PV) syndrome predominantly affects females past 40 years. It consists of atrophy of the mucous membrane of the alimentary tract, subepithelial fibrosis in lower part of laryngopharynx and iron deficiency anemia.

Clinical Features

- Dysphagia immediately after trying to swallow food
- Iron-deficiency anemia
- Glossitis
- Angular stomatitis
- Koilonychias (spooning of nails)
- Achlorhydria.

Potential of Malignant Conversion

The PV syndrome may be associated with carcinoma of the tongue, buccal mucosa, pharynx, esophagus or stomach. About 10% of the patients develop postcricoid carcinoma.

Diagnosis

- *Barium swallow*: It shows a web (subepithelial fibrosis) in the postcricoid region.
- *Esophagoscopy*: It will show the hypopharyngeal web.

Treatment

- *Oral/parenteral iron:* For correcting anemia. Serum level of iron is more important than hemoglobin.
- *Vitamins B_{12} and B6.*
- *Esophagoscopy* and *dilatation* of webbed area with bougies.

INFECTIOUS ESOPHAGITIS

The three most common causes of infectious esophagitis are candida, cytomegalovirus (CMV) and herpes simplex virus (HSV).

Etiology

- *Candida albicans*: It is the most common form and usually associated with HIV.
- Herpes simplex virus.
- Cytomegalovirus.
- Varicella-zoster virus.
- Human immunodeficiency virus.

Usually, the patients are immunocompromised due to HIV-AIDS, post-transplant treatment, or chemotherapy.

Clinical Features

- *Most common symptom*: Odynophagia
- *Other*: Heartburn, nausea, fever or bleeding.

Candida albicans

- *Predisposing factors*: Advanced age, hypochlorhydria, motility disorders, diabetes mellitus, alcoholism, steroids, oral thrush and HIV.
- *Endoscopy with biopsy and brushing*: Adherent white-pale yellow plaques are seen on endoscopy. Brushing of plaques will show hyphae and budding yeast.

Cytomegalovirus

Cytomegalovirus infects submucosal fibroblasts and endothelial cells and not squamous epithelium.

- *GI symptoms*: Abdominal pain, nausea, and vomiting.
- *Endoscopy findings*: Serpiginous erosions and ulcers that may coalesce and form deep and larger ulcers.
- *Biopsy*: It should be taken from the base of ulcer. Biopsy shows intranuclear and cytoplasmic inclusions and a halo, which surrounds nucleus.
- *Viral culture*: More sensitive than histology alone.

Herpes simplex

Herpes simplex virus infection can be either primary or secondary (reactivation of latent viral infection).

- *Esophagoscopy*: Vesicles, which rupture and form ulcers with raised edges.
- *Biopsy*: It is taken from ulcer margin and shows multinucleated giant cells and ground glass intranuclear inclusion bodies.
- *Viral culture*: More sensitive.

Treatment

- Management of immunocompromised condition.
- Antifungal
  - *Fluconazole*: 100–200 mg/day for 10–14 days.
  - *Clotrimazole* and *nystatin:* Topical 4–5 times a day.
  - *Amphotericin B*: In cases of granulocytopenia to prevent disseminated disease.
- Antiviral
  - *Acyclovir for HSV*: Intravenous 5–10 mg/kg every 8 hours till patient tolerate oral therapy.
  - *Gancyclovir and foscarnet for CMV*: Two-week full dose regimen followed by maintenance therapy for several weeks.

CRICOPHARYNGEAL SPASM

There occurs incoordination between relaxation of the UES and simultaneous contraction of the pharynx. UES fails to relax properly.

Causes

- Cerebrovascular strokes
- Parkinson's disease
- Bulbar polio
- Multiple sclerosis
- Muscular dystrophies.

DIFFUSE ESOPHAGEAL SPASM

In this motility disorder of smooth muscle, there occur spontaneous strong nonperistaltic contractions of the body of esophagus. There occurs degeneration of nerve processes but sphincter relaxation is normal.

Clinical Features

- Sudden dysphagia and substernal chest pain simulating angina pectoris in adults.
- Symptom improves with sublingual nitroglycerin.

Diagnosis

- *Barium swallow*: Shows segmented esophageal spasms, which appear like a rosary bead or a corkscrew.
- *Manometry*: Normal relaxation of the sphincter on swallowing and strong nonperistaltic uncoordinated esophageal contractions.

Treatment

- *Medical:* Calcium channel blockers and nitrates usually give good response.
- *Surgical:* Dilatation of lower esophagus and myotomy of esophagus extending from the arch of aorta to lower sphincter is done in severe refractory cases.

NUTCRACKER ESOPHAGUS

- There occur strong and high amplitude peristaltic esophageal contractions, which can be seen in manometric studies.
- Patients present with dysphagia and substernal pain like diffuse esophageal spasm (DES).
- Treatment is also similar to DES.

CARDIAC ACHALASIA

The characteristic features of this smooth muscle motility disorder are absence of peristalsis in the body of esophagus and high resting pressure in LES, which does not relax during swallowing.

Etiology

Exact primary cause is not known but data suggest following factors:

- Hereditary
- Degenerative
- Autoimmune
- Infectious.

The secondary causes may be following:

- Chagas disease: It consists of cardiomyopathy, megacolon and achalasia.
- Gastric carcinoma
- Lymphoma infiltrating lower esophagus
- Scleroderma.

Pathology

- Patchy infiltration of T lymphocytes, eosinophils and mast cells.
- Loss of ganglion cells.
- Myenteric neural fibrosis.
- Selective loss of postganglionic inhibitory neurons (nitric oxide and vasoactive intestinal polypeptide) in LES results in insufficient LES relaxation. Postganglionic cholinergic neurons, which produces high basal LES pressure, are spared.

Clinical Features

- *Most common symptoms*, which persist and do not progress for years, include
  - Dysphagia for liquids and solids.
  - Regurgitation of swallowed food especially in night. Choking and coughing awake the patient.
  - Chest pain.
- *History of compensatory measures*: Lifting of neck and carbonated drinks.
- *Weight loss*: It is minimal.
- *Physical examination:* It is unremarkable.

Diagnosis

- *Barium swallow with fluoroscopy*: In addition to the characteristic smooth tapering of lower esophagus leading to closed LES resembling rattail or bird's beak, other findings include
  - Loss of primary peristalsis in distal two-third esophagus
  - Poor emptying of esophagus
  - Retained food and saliva produce heterogeneous air-fluid level at the top of barium column
  - Esophagus may be dilated or tortuous like sigmoid
  - Presence of epiphrenic diverticulum.
- *Manometry*: Shows low pressure in the body of esophagus (absent or abnormal peristalsis) and high pressure at LES (increased LES tone), which fails to relax.
- *Flexible endoscopy*: Rule out pseudoachalasia due to GEJ tumor especially in elderly patient with short history and weight loss.

Treatment

- *Endoscopic pneumatic dilatation*: It tears LES muscle fibers and thus reduces LES pressure. Perforation occurs in 5% of patients.
- *Modified Heller's operation*: Myotomy (incision of circular muscle fibers) of narrowed lower portion of esophagus is the definitive treatment.
- *Botulinum toxin injection in LES*: It block's cholinergic nerves and needs to be repeated every 2 years.
- *Medical*: Calcium channel blockers and nitrates.

SCLERODERMA OR PROGRESSIVE SYSTEMIC SCLEROSIS

Scleroderma affects mid and distal esophagus. The atrophy and fibrosis of esophageal smooth muscle result in decrease or absence of peristalsis (similar to achalasia) and incompetent LES (opposite to achalasia).

Clinical Features

The patients usually develop reflux esophagitis and stricture in distal part of the esophagus and may have hiatus hernia. So the clinical features include:

- Progressive dysphagia to solids and liquids like malignancy.
- Features of GERD due to decreased LES tone.
- Cutaneous lesions.

Diagnosis

- *Barium swallow*: Dilation and absence of peristalsis in distal two-third of the esophagus.
- *Motility studies*: Decreased smooth muscle contraction.

Treatment

No effective treatment except antireflux therapy.

ZENKER DIVERTICULUM

This pharyngeal pouch is a pulsion hypopharyngeal diverticulum where hypopharyngeal mucosa herniates through the Killian's dehiscence, which is a weak area between the thyropharyngeal and cricopharyngeal parts of the inferior constrictor muscle (Fig. 2).

Etiology

Exact cause is not clear but spasm of cricopharyngeal sphincter or its incoordinated contractions during the act of deglutition is considered to be an important predisposing factor. Patients are usually old adults.

Pathology

Herniation of pharyngeal mucosa, which extends behind the esophagus, begins in the midline and later on pouch extends to lie on the left. Mouth of the sac becomes wider than the opening of esophagus. Food usually enters into the sac.

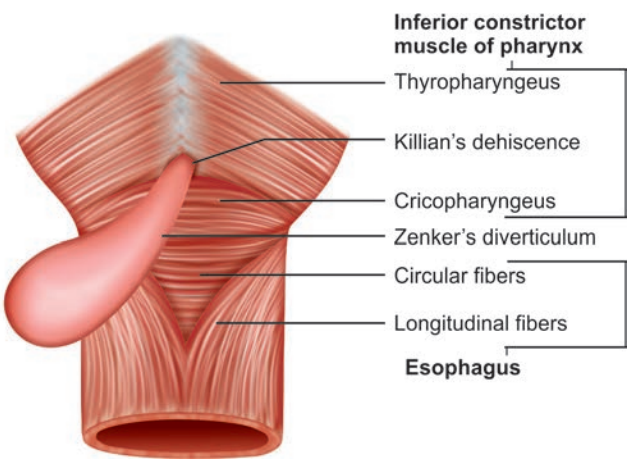

Fig. 2: Zenker's diverticulum of hypopharynx herniating through the Killian's dehiscence between the thyropharyngeal and cricopharyngeal parts of the inferior constrictor muscle

Clinical Features

- Most common symptoms are halitosis, transfer dysphagia (difficulty initiating swallowing) and regurgitation of food days after ingestion.
  - *Dysphagia* may increase after a few swallows, which fill the pouch with the food and then presses on the esophagus.
  - Gurgling sound during swallowing.
  - *Regurgitation of undigested food* at night (due to recumbent position) results in coughing and choking.
- Loss of weight and malnourishment.
- Aspiration pneumonia.

Diagnosis

- Barium swallow will show the site and size of diverticulum.
- EGD and nasogastric intubation are contraindicated because of the risk of perforation of the pouch.

Treatment

- *Diverticulectomy* (Excision of pouch) or *cricopharyngeal myotomy* or both.
- *Dohlman's procedure*: Endoscopic diathermy of the partition wall between esophagus and pouch is preferred in poor risk debilitated patients.

GLOBUS HYSTERICUS PHARYNGEUS

- Functional disorder with no true dysphagia.
- Patient complains of "lump" in the throat. The feeling of lump is more marked between the meals when patient voluntarily and consciously swallows the saliva.
- There is no difficulty in swallowing food.
- Patient usually has fear of throat cancer.
- Clinical examination of the pharynx and larynx is normal.
- *Management:* Rule out any organic cause and reassure the patient.

BENIGN NEOPLASMS

- Neoplasms of esophagus are rare and those found are usually malignant.
- Leiomyoma arises from the smooth muscle of esophagus wall. It accounts for two-third of all benign neoplasms.
- Other benign tumors include mucosal polyps, lipomas, fibromas and hemangiomas.
- They are usually pedunculated and seen in the esophageal lumen.
- *Treatment:* Endoscopic removal is not done because of the fear of perforation. They need surgical excision with external approach.

Leiomyoma

- *Clinical feature*: Tumor causes dysphagia when it exceeds the diameter of 5 cm.
- *Barium swallow*: Ovoid filling defect.
- *Endoscopy*: Submucosal swelling can be seen but biopsy is not taken.
- *Treatment*: Enucleation of the tumor with thoracotomy.

CARCINOMA ESOPHAGUS

The five most frequent cancers in Indian males (in descending order) are mouth/oropharynx, trachea/bronchus/lungs, lymphomas/multiple myeloma, esophagus, and leukemia. In women, they are (in descending order) breast, cervix, ovary, mouth/oropharynx, esophagus, and lymphomas/multiple myeloma (*Source*: ICMR, 2004). Incidence of esophageal malignancy is high in China, Japan, USSR and South Africa. It constitutes 3.6% (in affluent class) and 9.13% (in poor class) of all body cancers in India.

Risk Factors

- *Squamous cell carcinoma*: Usually occurs in proximal two-third of esophagus.
  - Black males
  - Smoking and alcohol consumption
  - Chewing of *Paan, Sopari* and tobacco
  - Diet rich in nitrates and pickled vegetables
  - Existing esophageal lesions:
    - Benign strictures
    - Cardiac achalasia
    - Diverticula
    - Corrosive injury
  - Premalignant conditions
    - Plummer-Vinson syndrome
    - Tylosis
    - Head and neck malignancy
    - Human papillomavirus.
- *Adenocarcinoma*: Usually occurs in distal one-third esophagus and GEJ.
  - White males
  - GERD and Barrett's esophagus
  - Hiatus hernia.

Pathology

Squamous cell carcinoma (SCC) is the most common type (93%) followed by adenocarcinoma (3%). Adenocarcinoma is usually seen in lower esophagus and may be an upward extension of the gastric carcinoma. The incidence of adenocarcinoma is on increase. Other types are very rare.

Spread

- *Direct*: Infiltrate esophageal wall and spread to trachea, left bronchus, aorta or pericardium. Recurrent laryngeal nerves involvement causes aspiration problems.
- *Lymphatic*: Depending on the part of esophagus cervical, mediastinal or celiac nodes may be enlarged. Cervical and thoracic lesions may involve supraclavicular nodes. Spread through submucosal lymphatics may lead to "skip lesions".
- *Blood borne*: Systemic distant metastases may occur in liver, lungs, bone and brain.

Clinical Features

- Early symptoms are substernal discomfort and preference for soft or liquid food.
- Most common symptom is gradually progressive dysphagia first to solids and then to liquids.
- Some patients have odynophagia and iron-deficiency anemia.
- Loss of weight and emaciation.
- Pain referred to the back indicates extension of tumor beyond esophageal walls.
- Coughing, hoarseness of voice, pneumonia and mediastinitis: They occur due to laryngeal paralysis (due to the involvement of recurrent laryngeal nerve) and tracheoesophageal fistula formation.
- Hypercalcemia in some cases of SCC.

- *Squamous cell carcinoma of esophagus*: Prior head and neck cancer increases the risk of this cancer eightfold.
- *Dissemination*: There occurs rapid dissemination of mucosal cancer because esophagus does not have serosal layer.

Diagnosis

- *Barium swallow*: Narrowing and irregular esophageal lumen (uneven ulcerated edges-apple core appearance) without proximal dilatation (Fig. 3).
- *Esophagoscopy with biopsy*: Confirm the diagnosis.
- *Bronchoscopy*: For checking extension of tumor into the trachea and bronchi.
- *CT scan*: Accurately identify metastatic disease.
- *Endoscopic ultrasound*: For assessing depth of invasion.

TNM Staging

Tumor nodes and metastases staging is accomplished after CT scan and endoscopic ultrasound (Box 2).

Treatment

- *T_1 or T_2 with N_0*: Surgery alone.
  - Lower one-third esophagus: The affected segment with proximal margin and fundus of stomach is excised with

Fig. 3: Barium swallow. Carcinoma postcricoid and cervical esophagus. Note the prevertebral soft tissue widening
Source: Dr Jayesh Patel, Consultant Radiologist, Anand, Gujarat

Box 2: TNM classification of carcinoma esophagus

Primary tumor (T)

T_X	Cannot be assessed
T_0	No tumor
T_{is}	Carcinoma in situ
T_{1a}	Tumor involve lamina propria
T_{1b}	Tumor involve submucosa
T_2	Tumor involve muscularis propria
T_3	Tumor involves adventitia
T_4	Tumor involves adjacent structures

Regional lymph nodes (N)

N_x	Cannot be assessed
N_0	Lymph node metastasis absent
N_1	Lymph node metastasis present

Distant metastasis (M)

M_x	Cannot be assessed
M_0	No distant metastasis
M_1	Distant metastasis present

primary reconstruction of the food channel. Surgical mortality is 20%.

- *T_3 or N_1*: Neoadjuvant chemotherapy/radiation before surgical resection.
  - Upper two-third esophagus: Radiotherapy is preferred as the great vessels and involvement of mediastinal nodes, make the surgery difficult.
- *Late stages*: Palliative endoscopic measures include following:
  - Repeated dilation.
  - Esophageal intubation with Celestin or Mousseau-Barbin or a similar tube.
  - Gastrostomy or jejunostomy.
  - Laser [neodymium-doped yttrium aluminium garnet (Nd: YAG)]/photodynamic therapy ablation.
  - Chemotherapy.

Prognosis

- Usually esophageal carcinoma is diagnosed at a late stage and 5-year survival rate is poor (about 5–10%).

Clinical Highlights

1. **Foreign body throat:** Commonest site of impaction of "fishbone" is palatine tonsil.
2. **Caustic ingestion:** Oral burns may not correlate with severity of esophageal lesions. Alkaline agents penetrate deeper tissue layers. Watch for sign of airway obstruction because airway control must be the prime concern. Watch for features of mediastinitis (such as tachycardia, chest pain, fever, sepsis) and peritonitis.
   - **Esophagoscopy:** It is controversial. It may lead to further injury. Early esophagoscopy may help in diagnosis and placements of feeding tube. It is terminated if a significant burn is seen. It is contraindicated after 12 hours.
3. **Boerhaave's syndrome:** Patient develops severe vomiting and chest pain after drinks and heavy dinner. X-ray of chest shows hydropneumothroax.
4. **Cardiac achalasia:** The radiological findings include esophageal dilatation, rat-tail appearance, and failure of lower esophageal sphincter to relax.
5. **Esophageal (Zenker's) diverticulum** occurs through Killian's dehiscence. It arises from posterior part of hypopharynx and causes regurgitation of undigested food. It is treated by diverticulectomy and cricopharyngeal myotomy.
6. **Leiomyoma:** This is the most common benign tumor of esophagus.
7. **Squamous cell carcinoma:** This is the most common type of esophageal cancer.

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45 Laryngeal Symptoms and Examination

Faith, faith, faith in ourselves, faith in God, this is the secret of greatness. If you have faith in all the Gods, and still have no faith in yourselves, there is no salvation for you. Have faith in yourself—all power is in you—be conscious and bring it out. Say, 'I can do everything.' 'Even the poison of a snake is powerless, if you can firmly deny it.'

—Swami Vivekananda

Points of Focus

- **SYMPTOMS**
- **CLINICAL EXAMINATION**
  - ✓ External Examination
  - ✓ Indirect Laryngoscopy with Laryngeal Mirror
- **ENDOSCOPY**
  - ✓ Rigid 90° Fiberoptic Laryngoscope
  - ✓ Flexible Rhinolaryngoscope (Nasopharyngolaryngoscope)
  - ✓ Laryngeal Videoendoscopy
  - ✓ Advantages
  - ✓ Recent Developments in Video Cameras and Recorders
- **LARYNGOSCOPIC PARAMETERS AND PATIENT'S TASK**
  - ✓ Patient's Tasks
  - ✓ Laryngeal Components Examined
  - ✓ Arytenoid and Vocal Fold Motions
  - ✓ Mucous
  - ✓ Vocal Fold Vascularity
  - ✓ Position or Height of Larynx
  - ✓ Supraglottic Activity
  - ✓ Vocal Fold Edges
- **STROBOSCOPY**
  - ✓ Components Examined
  - ✓ Indications

HOARSENESS OF VOICE

- ✓ Etiology
- ✓ History, Examination and Investigations
- ✓ Voice and Speech Disorders

STRIDOR

- ✓ Types
- ✓ Causes
- ✓ Severity of Airway Obstruction
- **ASSESSMENT OF PATIENT WITH STRIDOR**
  - ✓ History and Physical Examination of Upper Airway
  - ✓ Characteristic Features
  - ✓ Investigations
  - ✓ Endoscopy
- **TREATMENT OF STRIDOR**
  - ✓ Acute Airway Obstruction
  - ✓ Chronic Airway Obstruction
- **CLINICAL HIGHLIGHTS**

SYMPTOMS

The common laryngeal and laryngopharyngeal symptoms and their causes are mentioned in Box 1.

CLINICAL EXAMINATION

The general format of clinical examination of larynx, which includes external examination of larynx, indirect laryngoscopy, fiber optic endoscopy (flexible or rigid), assessment of voice and cervical lymph nodes, is given in Box 2. To elicit the post laryngeal crepitus (characteristic grating sound), cricoid is moved from side to side.

External Examination

It includes inspection (Figs 1 and 2) and palpation (Box 2) of area of hyoid bone, thyroid cartilage, thyroid notch, cricoid cartilage

and the tracheal rings for redness of skin, bulging or swelling, widening of larynx, surgical emphysema, change in contour or displacement of larynx, movements of larynx with deglutition and breathing and post laryngeal crepitus.

Indirect Laryngoscopy with Laryngeal Mirror

- *Laryngeal mirror*: It is used for the indirect examination of oropharynx, laryngopharynx and larynx. It is available in various sizes from 6 mm to 30 mm diameter.

The posterior rhinoscopy mirror is smaller and its shaft is bayonet shaped, while the shaft of the laryngeal mirror is straight.

- *Method*: For indirect laryngoscopy (Figs 3 to 5), patient sits erect with the head and chest leaning slightly towards the examiner. Patient protrudes out the tongue, which is wrapped in a piece of gauze cloth and then held by the examiner between the thumb and middle finger. Index finger of the examiner retracts out the patient's upper lip and moustache. To prevent fogging, a laryngeal mirror is always warmed over a spirit lamp or in hot water. It is advisable to test mirror's warmness on the back of hand before inserting into the mouth, because hot mirror can damage the mucosa. The warmed laryngeal mirror is introduced into the mouth and held firmly against the uvula and soft palate while the light is focused on the laryngeal mirror. Patient is asked to breathe quietly. Then the systematic examination begins from the oropharynx, laryngopharynx and larynx. Movements of both the cords are observed when patient takes deep inspiration (abduction of cords) and say "Aa" (adduction of cords) and "Eee" (for adduction and tension).
- Merits
  - Three-dimensional view of larynx with good color resolution.
  - Good visualization during phonation.
- Demerits
  - Epiglottis may obstruct the view during respiration.
  - Examination is not satisfactory in cases of difficult anatomy and strong gag reflex.

Box 1: Larynx and laryngopharynx symptoms and their causes

- *Disorders of voice*: Hoarseness, aphonia, puberphonia, easy fatigability of voice, rough, breathy, bitonal, dysphonic, whispered or feeble.
- *Difficulty in respiration*: Tumors, infection, foreign bodies.
- *Repeated clearing of throat*: Chronic laryngitis, benign or malignant tumors, laryngopharyngeal reflux/ gastroesophageal reflux disease.
- *Pain in throat*: Ulcers, perichondritis, arthritis.
- *Difficulty in swallowing (Dysphagia)*: Epiglottitis and tumors of laryngopharynx.
- *Coughing*: Aspiration of secretions due to laryngeal paralysis.
- *Mass in the neck*: Cervical nodes, direct extension of growth, laryngocele.

- *Failed mirror examination*: Topical anesthesia with 4% lignocaine facilitates laryngeal mirror examination. In some of the patients, laryngeal mirror examination cannot be performed due to anatomical abnormalities such as overhanging of epiglottis. It is also difficult in uncooperative and anxious patient and too much sensitive pharynx. In such cases, endoscopic examination is advised.

Box 2: Examination of larynx and laryngopharynx: Findings and their causes

- *External examination*
  - Inspection and palpation (Area of hyoid bone, thyroid cartilage, thyroid notch, cricoid cartilage, and the tracheal rings). Note the following:
    - ◊ Redness of skin: Abscess, perichondritis
    - ◊ Bulging or swelling: Extension of growth or enlarged lymph nodes
    - ◊ Widening of larynx: Growth of pyriform fossa
    - ◊ Surgical emphysema: Accidental or surgical trauma
    - ◊ Change in contour or displacement of larynx: Trauma or neoplasm
  - Movements of larynx with deglutition and breathing:
    - ◊ Present: Normal
    - ◊ Fixity: Inflammation and malignant infiltration
  - Post laryngeal crepitus:
    - ◊ Present: Normal
    - ◊ Absent: Postcricoid carcinoma, retropharyngeal abscess
- *Techniques of laryngoscopy*
  - Indirect laryngoscopy
  - Rigid endoscopy
  - Flexible endoscopy
- *Structures seen during laryngoscopy*
  - Oropharynx: Base of tongue, lingual tonsils, valleculae, glossoepiglottic folds and pharyngoepiglottic folds.
  - Larynx: Epiglottis, aryepiglottic folds, arytenoids, ventricular bands, ventricles, true cords, anterior commissure, posterior commissure, subglottis and rings of trachea.
  - Laryngopharynx: Pyriform fossa, postcricoid region, posterior wall of laryngopharynx.
- *Cervical lymph nodes*

Fig. 1: Irradiated case of carcinoma vocal cord. Note skin changes due to irradiation, healing tracheostomy opening and swelling in the areas of thyroid and cricoid cartilages

Fig. 2: Secondary metastatic neck nodes on right side involving levels Ib to V. Primary lesion was carcinoma of supraglottic larynx. Note the healed tracheostomy opening and widening of thyroid cartilage

Fig. 3: Laryngeal mirror warming

Fig. 4: Indirect laryngoscopy. Tongue wrapped in a piece of gauze cloth and held by the examiner between the left thumb and middle finger; Left index finger retracts out the upper lip. Laryngeal mirror is firmly against the uvula and soft palate

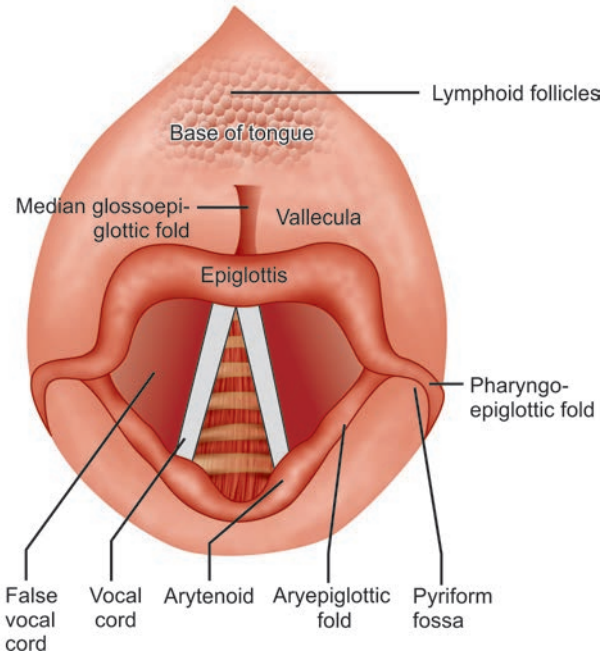

Fig. 5: Structures seen in indirect laryngoscopy

ENDOSCOPY

The continuous light of endoscope helps in studying gross structure and function of larynx, while strobe light assesses mucosal health and vibration pattern. The comparison and an overview of different techniques of indirect laryngoscopic examination are given in Table 1.

Examiner should keep in mind that reactions to topical xylocaine or vasovagal attack can occur. The entire patient's task mentioned in following section should be conducted to examine complete laryngoscopic parameters.

Rigid 90° Fiberoptic Laryngoscope (Telescope)

It gives a clear, wide-angle view of the larynx and laryngopharynx and is used as an outdoor procedure. It is passed through the mouth.

- Merits
  - Higher resolution and brighter picture
  - Better color contrast differentiates subtle lesions and vascular changes
  - More accurate magnification shows small lesions
  - Topical anesthesia usually not required.
- Demerits
  - Size of glottic gap may appear exaggerated.
- Method
  - *Topical anesthesia* **(**topical xylocaine spray**):** It is usually not required, but may be used if needed.
  - *Defogging agents*: Liquid defogger, soap film, surgical wax, or patient's own saliva.
  - *Seventy-degree Endoscope*:
    - Patient's position: Patient bends slightly forward from waist. Back is straight. Neck is extended. Tongue is protruded and held gently by the examiner.
    - Endoscope's position: Endoscope lies under the uvula or between uvula and anterior tonsillar pillar. Endoscope tip is tilted inferiorly. Endoscope is held against cheek or side of tongue.
    - Patient's task: Sustained "ei" lowers the tongue base.

TABLE 1 Comparison of different techniques of indirect laryngoscopic examinations

	Laryngeal mirror	*Rigid telescope*	*Flexible endoscope*
Availability	Free and widely	Uncommon	Uncommon
Cost	Most economical	Expensive	Most expensive
Color of image	Best and natural	Good	Moderate
Patient tolerance	Good	Moderate	Good
Photography quality	Not possible	Excellent	Good
Magnification	Absent	Maximum	Moderate
Stroboscopy quality	Not possible	Excellent	Good
Speech and singing observation	Difficult	Very difficult	Excellent
Vocal mechanics evaluation	Difficult	Not possible	Excellent
Biofeedback voice breathing therapy	Not possible	Possible	Excellent

- *Ninety-degree Endoscope*: There is no need to bend the patient forward and extend the neck and tilt the tip of endoscope.

Flexible Rhinolaryngoscope (Nasopharyngolaryngoscope)

It is passed through the nose under topical anesthesia. This outdoor procedure provides a good view of the larynx, laryngopharynx, subglottis and even upper trachea.

- Merits
  - Larynx can be examined during speech (especially spasmodic dysphonia) and singing
  - Glottic gap size appears more accurate
  - Simultaneous examination of nose and nasopharynx
  - Well-tolerated by patients with strong gag reflex or a young child
  - Sniffing through nose allows assessment of subtle motion changes
  - Topical anesthesia facilitates closure examination of vocal folds, epiglottis and trachea.
- Demerits
  - Light transport and magnification inferior to rigid endoscope
  - Loss of resolution due to fiber bundles (subtle-to-small mucosal changes may be missed)
  - Illumination decreases as the scope is moved away from the tissue
  - Distortion of the periphery of image
  - Seems more invasive than rigid scope
  - Risks of nose bleed.
- Method (Fig. 6)
  - Inform the patient about the procedure
  - Application of topical anesthetic and vasoconstrictor
  - The patient sits straight up and breaths through the nose. The scope is passed along the floor of nose in the inferior meatus. For better examination of the velopharyngeal port, scope should be passed through middle meatus. In the nasopharynx, the tip of the scope is directed downward with the help of positioning lever of the scope. Subsequently scope is advanced downward into the oropharynx and then laryngopharynx and larynx. Once in the oropharynx, the positioning lever of the scope is released and the scope tip straightens out. Take the help of positioning lever of the scope to direct the tip of the scope to view any specific area. Examine the nose, nasopharynx, oropharynx, laryngopharynx and larynx. If secretions obstruct the lens either brush it against the mucosa or ask the patient to swallow when the scope is in oropharynx. Valsalva maneuver facilitates examination of pyriform sinuses.

Laryngeal Videoendoscopy

Laryngeal videoendoscopy and stroboscopy help in studying following:

- Mucosal status and laryngeal pathology
- Vocal fold vibratory patterns during various voicing conditions
- Behavior of glottic and supraglottic structures during phonation
- Alteration of phonatory physiology with treatment probes.

Charged-coupled device (CCD) chips and digital storage facilities have largely replaced the film and videotape (Fig. 7).

Advantages

- Allows analysis in real time and in slow motion
- Determine subtle abnormalities in vocal fold vibrations
- Review by speech language pathologists and voice teachers
- Patients can watch and understand their condition
- Recordings can be stored and edited, enhanced and manipulated for educational or scientific presentations.

Recent Developments in Video Cameras and Recorders

- CCD cameras and chips: Image quality is much better with three-chip cameras than single chip
- Modes: Analog or digital; NTSC, PAL or SECAM
- Integrated scope and camera (eyepiece is replaced with a direct connection to camera)
- CCD chip at distal end of flexible endoscope
- CCD cameras with strobe device that rapidly turn image acquisition on and off
- Videokymography uses rapid scanning properties of CCD
- DVD recorder is fast replacing analog video recorder
- In standard VHS (composite signal), luminance or brightness signal (Y) is combined with color chrominance signal (C) before transmission

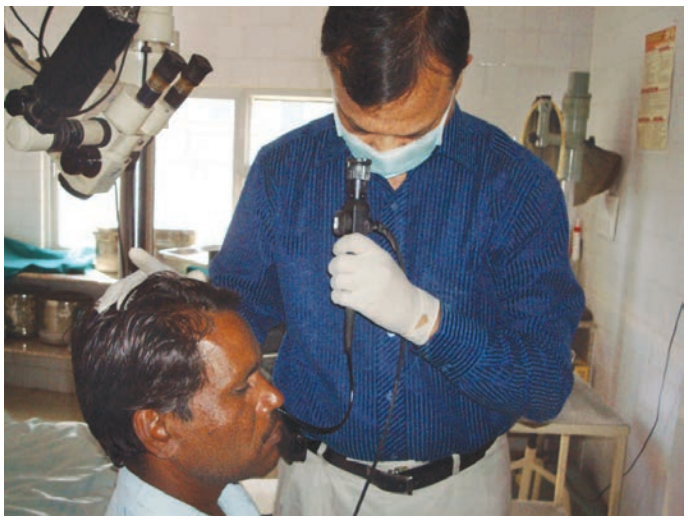

Fig. 6: Flexible nasopharyngolaryngoscopy procedure

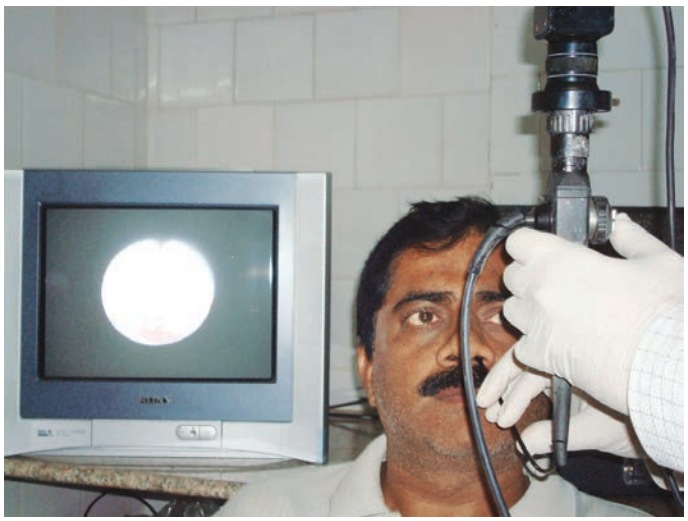

Fig. 7: Flexible video laryngoscopy

- In S-video (S-VHS or Y/C), luminance or brightness signal (Y) and color chrominance signal (C) are transmitted separately and quality is greater.

LARYNGOSCOPIC PARAMETERS AND PATIENT'S TASK

Patient's Tasks

The patient is instructed to perform following tasks during the endoscopic examination of larynx:

- Breathing at rest
- Deep breathing
- Easy coughing or throat clearing
- Laryngeal diadochokinesis: Rapid repetitions of "ee" with glottal stops between productions.
  - Laryngeal diadochokinesis "hee"
- Sustained "ee"
- Quick sniffing through nose
- Speaking
- Singing
- Swallowing
- Valsalva maneuver.

Laryngeal Components of Examination (Fig. 8)

- Abnormalities and asymmetries of laryngeal structures (valleculae, pyriform sinuses, epiglottis, aryepiglottic folds, ventricular folds and posterior glottic rim) are noted.
- Common variants include omega-shaped epiglottis in males, interarytenoid bar, arytenoids medial wall erythema, and posterior pharyngeal wall cobble stoning.
- Features of laryngeal irritation or extraesophageal reflux are edema, erythema and surface irregularities of posterior larynx.

Arytenoid and Vocal Fold Motions

- *Arytenoids*:
  - Upright or rotated
  - Mobile or immobile (Position: median, paramedian, intermediate, or lateral)
  - Symmetric or asymmetric
- *Vocal folds*: Mobile or immobile (paralysis, paresis, arytenoid dislocation, fibrosis, tumor invasion of cricoarytenoid joint)
- *Glottic gap*: Absent or present (size and shape).

Mucous

- *Consistency*: Thick mucous on vocal fold edges or superior surface indicates lack of hydration or chronic irritation.
- *Effect of coughing or swallowing*: They clear the mucous and differentiate from underlying lesion.
- *Pooling*: Mucous pooling in pyriform sinuses can occur in cases of
  - Poor laryngeal sensation
  - Weak lateral pharyngeal wall
  - Inefficient swallow
  - Obstruction in cricopharyngeal region or cervical esophagus.

Vocal Fold Vascularity

- *Normal*: Vocal folds are pearly white

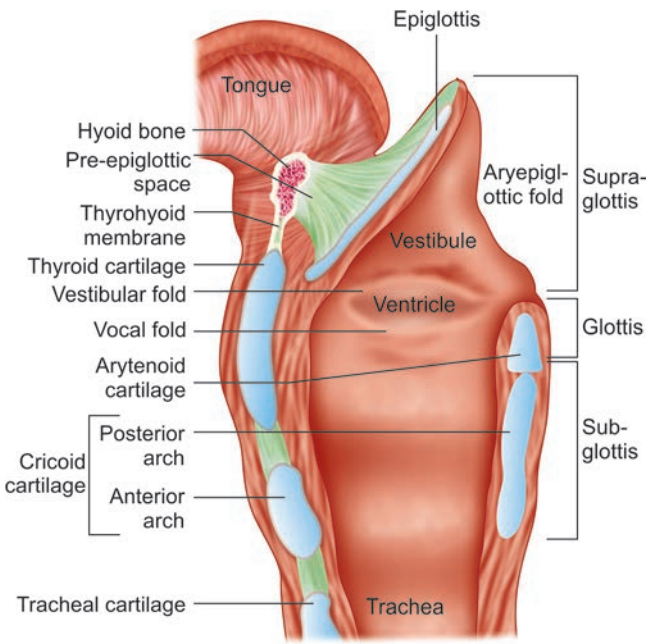

Fig. 8: Structures and parts of larynx

- *Blushing*: It indicates erythema or hyperemia
- *Capillaries*: They may be seen parallel to free edge.
- *Capillary ectasias or microvarices*: They are dilated tortuous vessels. Areas of stiffness or risk of hemorrhage gives diffuse coloring to vocal fold.

Position or Height of Larynx

- *Lateral position or tilting*: They can occur due to mass, trauma, muscle imbalance or injury of superior laryngeal nerve.
- *Vertical movements*: Raising (during ascending pitch glides) and lowering (during descending pitch glides) of larynx occur during talking and singing and in cases of abnormal tension of laryngeal muscle.

Supraglottic Activity

- *Involuntary movements*: Tremor and involuntary contractions (myoclonic movements) occur in neurologic disorders.
- *Compression (constriction)*: May be seen in normal persons.
  - Horizontal: Good singers have anteroposterior constriction without lateromedial compression. Voice therapy maneuvers help in reducing abnormal compressions.
    - *Anteroposterior*: Greater severity in dysphonic subjects
    - *Lateromedial*: Observed with increasing pitch. In extreme cases, ventricular folds touch and may even vibrate and this condition is called *dysphonia plica ventricularis*, which may be a primary disorder or secondary (compensatory) to other lesions such as incomplete glottic closure.
  - Constant or intermittent: Sustained compression during phonation occurs in functional voice disorders.

Vocal Fold Edges

- Straight/smooth
- Convex or concave (bowing)
- Rough/irregular

STROBOSCOPY

Stroboscope determines speed of cyclic motion, which appears slowed (strobe or running phase) or stopped (locked phase). Strobe light illuminates vocal folds at different points of different vibration cycles and creates illusion of slow motion (Table 2). Strobe illumination helps in understanding mucosal scarring and distinguishing cysts from nodules.

Components Examined

This lighting technique helps in examining the following structures and functions:

- Body-cover (thyroarytenoid muscle epithelium and superficial layer of lamina propria) relationship of vocal folds and vibration patterns
- Under surface of vocal fold edges
- Effect of mucosal disease on vocal fold vibrations
- Underlying layered structure of vocal folds
- Glottic closure and mucosal pliability
- Determine tension of vocal folds.

Indications

Stroboscopy is valuable in following vocal fold conditions and may obviate the need for microlaryngoscopy:

- Stiffness
- Scar
- Submucosal injury
- Small vocal fold lesions
- Invasion depth of early cancer lesion
- Asymmetric mass or tension
- Follow-up after phonosurgery.

DIRECT LARYNGOSCOPY (MICROLARYNGOSCOPY) AND BRONCHOSCOPY

See Chapter Endoscopies in section of Operative Procedures and Instruments.

HOARSENESS OF VOICE

Any change in voice quality from harsh, rough or raspy voice to weak voice is usually referred as hoarseness. It is caused by laryngeal dysfunction (variations of periodicity and intensity of consecutive sound waves). In cases of limited lungs or tracheobronchial tree diseases, voice becomes weak and damp.

Etiology

Hoarseness is a symptom and not a disease. The various causes of hoarseness are shown in Table 3. The conditions that interfere with the functions and structure of vocal cords include following:

- *Inappropriate approximation*: Vocal cord paralysis or fixation, tumors of vocal cords.

TABLE 2 Stroboscopic parameters and patient's tasks

Parameters	*Tasks*
1. Closure pattern (complete, posterior gap, incomplete anterior gap, spindle gap, hourglass, irregular or variable)	1. Sustained "ee" a. Normal Pitch Normal Loudness (NPNL) b. Quiet c. Loud d. On inhalation e. NPNL using locked mode
2. Amplitude of vibrations (normal or decreased)	2. Glide a. Mid-range to high and sustaining high note b. Mid-range to low and sustaining low note
3. Mucosal wave (increased or decreased)	3. Voice sentences
4. Adynaminc segment (nonvibrating areas)	4. Singing
5. Vertical closure level (on-plane or off-plane)	5. Humming
6. Phase closure (decrease or increase)	6. Trial therapy or laryngeal manipulation
7. Symmetry (cord lags behind)	
8. Regularity (regular or irregular)	

TABLE 3 Causes of hoarseness of voice (laryngeal disorders)

Infections	Acute and chronic laryngitis: Influenza, exanthematous fever, laryngotracheobronchitis, diphtheria, tuberculosis, syphilis, scleroma, atrophic laryngitis
Neoplasms	Papilloma (solitary and multiple), hemangioma, chondroma, angiofibroma, fibroma, leukoplakia, vocal nodule, vocal polyp, amyloid tumor, contact ulcers, or cancer
Trauma	Submucosal hemorrhage, laryngeal trauma (blunt and sharp), foreign bodies, intubation
Paralysis	Paralysis of recurrent or superior laryngeal or both nerves
Fixation of cords	Arthritis or fixation of cricoarytenoid joints
Congenital	Cysts and laryngocele
Miscellaneous	Dysphonia plica ventricularis, myxedema, gout, hysterical aphonia

- *Abnormal size*: Edema or tumor of vocal cord, partial surgical excision or fibrosis.
- *Abnormal stiffness*: Decrease in paralysis; increase in spastic dysphonia or fibrosis.
- *Improper vibrations*: Congestion, submucosal hemorrhages, nodule or a polyp.

History, Examination and Investigations

- *History*: Patient's occupation and habits, mode of onset and duration, and associated complaints should be noted.

In cases of smokers and elderly people, hoarseness that persists for more than 3 weeks, malignancy vocal cord should be ruled out.

- *Examination*: Indirect laryngoscopy with mirror and endoscopy (rigid or flexible) for structural and functional assessment of the cords and cricoarytenoid joints.
  - Examination of neck, cardiovascular system and cranial nerves.
- *Investigations*:
  - Laboratory investigations
  - Radiological examination
  - Microlaryngoscopy and biopsy of the lesions
  - Bronchoscopy
  - Esophagoscopy.

Voice and Speech Disorders

See chapter Speech and Voice Disorders in section Larynx, Tracheal and Bronchus.

STRIDOR

Stridor is a hallmark of laryngeal obstruction. It is an abnormal (stridulent or harsh) noise that is caused by a turbulent airflow in the impaired airway.

Types

Stridor may be heard during inspiration, expiration or both.

- *Inspiratory stridor*: It is because of obstruction from larynx and pharynx.
- *Expiratory stridor*: Expiratory stridor and prolonged expiratory phase are because of bronchial and low tracheal obstruction.
- *Biphasic stridor*: It is because of obstruction at the level of cervical trachea.

Stertor is a snoring type of noise, which is made by nasopharyngeal and oropharyngeal obstruction.

Severity of Airway Obstruction

The severity of airway obstruction has been categorized in three grades:

I. *Complete obstruction*: No airflow is detectable in or out of lungs.
II. *Partial obstruction*: Dyspnea/stridor is present because of major airway narrowing.
III. *Potential or impending obstruction*: It can be the result of a known anatomical or physical disorder in which respiratory physiology or level of consciousness changes. In these cases, preventing measures must be instituted to prevent further deterioration.

Causes

Stridor may arise from lesions of nose, tongue, mandible, pharynx, larynx, trachea and bronchi (Box 3).

Acute stridor in children is mostly caused by laryngotracheobronchitis (LTB or croup), bacterial tracheitis, acute epiglottitis and laryngeal foreign bodies, while in adult the most common cause is malignancy of the larynx.

- In cases of micrognathia and Pierre-Robin syndrome, stridor is due to falling back of tongue.
- Congenital vascular rings cause both stridor and dysphagia.

ASSESSMENT OF PATIENT WITH STRIDOR

If the situation is not acute and does not require immediate intervention, diagnostic assessment must proceed further. Assessment of stridulous child includes focused history and examination and selected investigations involving endoscopy and imaging. Depending upon the general status of the patient, the assessment can include following elements.

History and Physical Examination of Upper Airway

The clinical manifestations of impaired airway may include dyspnea and stridor, voice change (hoarseness), cough, local pain, restlessness, indrawing of intercostals, suprasternal, and supraclavicular spaces, and drooling. Bleeding and subcutaneous emphysema occur in cases of trauma.

Box 3: Causes of stridor

Infants and Children

1. *Nose*: Congenital choanal atresia, pyriform aperture stenosis, and mid nasal stenosis.
2. *Tongue*: Cretinism, hemangioma, lymphangioma, dermoid and lingual thyroid.
3. *Mandible*: Micrognathia, Pierre-Robin syndrome.
4. *Pharynx*: Congenital dermoid, adenotonsillar hypertrophy, retropharyngeal abscess and tumors.
5. *Larynx*:
   a. Congenital: Laryngeal web, laryngomalacia, cysts, vocal cord paralysis, subglottic stenosis.
   b. Inflammatory: Epiglottitis, laryngotracheobronchitis (LTB), diphtheria, tuberculosis.
   c. Neoplastic: Hemangioma and juvenile multiple papillomatosis.
   d. Traumatic: Physical/chemical/thermal injury, external laryngeal trauma, foreign bodies, iatrogenic (bronchoscopy, or prolonged intubation).
   e. Neurogenic: Laryngeal paralysis.
   f. Miscellaneous: Tetanus and tetany.
6. Trachea and bronchi
   a. Congenital: Atresia, stenosis, tracheomalacia and tracheobronchial malacia.
   b. Inflammatory: LTB and bacterial tracheitis.
   c. Neoplastic: Tumors.
   d. Traumatic: Foreign body, iatrogenic tracheal stenosis (prolonged intubation or tracheostomy).
7. Secondary external compression
   a. Congenital: Vascular rings, esophageal atresia, tracheoesophageal fistula, congenital goiter, and cystic hygroma.
   b. Inflammatory: Retropharyngeal, parapharyngeal, and retroesophageal abscess.
   c. Foreign body: Esophagus.
   d. Tumors: Thyroid.

Adults (In addition to the above causes)

1. *Infections*: Ludwig's angina, peritonsillar abscess, tongue swelling.
2. *Trauma*: Injury of larynx and trachea.
   a. Fractures of mandible or maxillofacial injuries.
   b. Caustic agent ingestion.
   c. Radiation.
3. *Neoplasms*: Malignancy of larynx, pharynx, upper trachea, tongue and thyroid.
4. *Allergy*: Angioneurotic and drug sensitivity.
5. *Obstructive sleep apnea*

- *Severity*: Severity of subcostal, intercostals and suprasternal recession is an indicator of the severity of airway impairment.
- *Cyanosis*: It is a late feature.
- *Nasal patency*: It can be assessed with a mirror, cotton wisp or bell of stethoscope.
- *Jaw and tongue*: Assess the jaw and tongue size.

Avoid examining a child's throat when epiglottitis is suspected.

Characteristic Features

Impairment in airway affects the feeding particularly in infants. Airway impaired babies typically "come up for air" during the breastfeeding. Bottle-fed babies need small hole bottle and thickened feeds. Poor feeding may result in failure to thrive and poor weight gain. The following characteristic features indicate the cause of airway obstruction.

- *Stridor at birth*: It denotes congenital laryngeal web, subglottic stenosis, tracheal narrowing and vocal cord palsy.
- *Effect of position*:
  - Prone position: Stridor of laryngomalacia, micrognathia, macroglossia and innominate artery compression disappears when baby lies in prone position.
  - Supine position: Stridor in supine position occurs with a pedunculated laryngeal mass and micrognathia (results in tongue base occlusion).
- *Effect of crying*: Dynamic stridor evident in first few weeks of life indicates laryngomalacia.
  - Improvement: Airway improvement during crying occurs in gross nasal obstruction, such as bilateral choanal atresia.
  - Worsening: In laryngomalacia, stridor is less at rest and during sleep and becomes worse by crying and feeding.
- *Progress*:
  - Gradual: A gradual increase in severity of stridor implies subglottic hemangioma, mediastinal mass and cancer of upper airway.
  - Rapid: Rapid progression of airway impairment with drooling is hallmark of acute epiglottitis, whereas bacterial tracheitis and laryngotracheobronchitis have relatively prolonged course.
- *Fever*: Associated fever indicates infective condition such as laryngitis, epiglottitis, laryngotracheobronchitis or diphtheria.
- *Cough*: It is present in cases of tracheoesophageal fistula and tracheomalacia.
- *Hoarseness*: It suggests laryngeal papillomatosis and vocal cord palsy.
- *Dying spells*: Spells of apneas with cyanosis are common in severe tracheobronchomalacia.
- *Sequential auscultation*: Sequential auscultation with stethoscope over the nose, open mouth, neck and the chest helps in localizing the site of obstruction.

- *Birthmarks*: They may be associated with subglottic hemangioma.
- *Sound of stridor*:
  - Musical quality: Laryngomalacia
  - Breathy quality: Vocal cord palsy
  - Barking cough: Tracheomalacia.

Aspirations occur in cases of vocal cord palsy, tracheoesophageal fistula and cleft larynx.

Investigations

- Oxygen saturation monitoring in cases of acute airway obstruction.
  - Arterial blood gases estimation.
- *Endoscopy* and *intubation* in operation theater in cases of suspected epiglottitis.
- *X-ray chest PA view*: Ground glass appearance in bronchopulmonary dysplasia and mediastinal shift in cases of obstructive emphysema due to foreign body.
- *X-ray neck lateral view*: Demonstrate subglottis, oropharynx and nasopharynx.
  - Expiratory and inspiratory films (in older children): Diaphragmatic immobility is seen on the side of foreign body obstruction.
- *Videofluoroscopy* (in young children) for diaphragmatic screening and tracheomalacia.
- *Bronchography* with safer nonionic contrast media: Demonstrates tracheobronchial stenosis and malacia.
- *MRI and helical CT*: For tracheal lesions, extrinsic compression and abnormal vasculature.
- *pH probe study*: For pH in upper esophagus and pharynx in cases of gastroesophageal reflux disease.
- *Ultrasound* of vocal cords for vocal cord palsy.

Endoscopy

A careful history, examination and selected needful investigations usually suggest a diagnosis, which needs to be confirmed by endoscopy.

- *Flexible nasopharyngolaryngoscopy*: Ultrathin < 2 mm diameter endoscopes allow examination of even neonates without anesthesia. Vocal cord palsy and laryngomalacia can be seen.
- *Laryngotracheobronchoscopy*: This is gold standard in stridulous child assessment.
- *Microlaryngoscopy*: *See* Chapter Endoscopies in section of Operative Procedures and Instruments.
- *Bronchoscopy*: After bronchoscopy, child is intubated and detailed examination of the larynx and esophagus can be done. Larynx is examined again for active movements of vocal cords when the child is coming out of anesthesia and the tube has been removed (*See* Chapter Endoscopies).

TREATMENT

In acute cases of airway impairment, following things proceed simultaneously:

- *History taking*: Such as duration and foreign body.
- *Examination*: Such as severity of airway obstruction and oxygen saturation.
- *Active resuscitation*: Such as setting up humidified oxygen and preparation for intubation/tracheostomy. In cases of inadequate ventilation, airway must be secured through either medical or surgical means (Box 4). Oral airway helps in managing cases of nasal obstruction due to choanal atresia. Mini tracheostomy handles bronchopulmonary secretions.

Acute Airway Obstruction

- Steroids
  - Dexamethasone 0.15–0.6 mg/kg IM (may be given intravenous and oral) with inhaled steroids in cases of LTB.
  - Nebulized budesonide (2 mg) for home use in cases of recurrent LTB.
- *Nebulized L-Epinephrine* (5 ml of 1:1,000) or racemic epinephrine (0.5 ml) in croup children with moderate-to-severe distress.
- *Continuous positive airway pressure* via nasal cannula: Continuous positive airway pressure (CPAP) takes care of tracheobronchial malacia.
- *Intubation*: Endotracheal intubation is a quick method of establishing airway. A very gentle intubation with a small soft tube just large enough for adequate ventilation and suction of secretions is preferred over pediatric tracheostomy, which brings with it numerous problems. Though oral intubation tends to be easy, nasal intubation is more secured. Severe subglottic stenosis, impacted foreign body, advanced epiglottitis and laryngeal aplasia make intubation impossible.
- *Ventilating bronchoscope*: If the secretions are very thick, viscid and tenacious and airway impairment persists after intubation, a ventilating bronchoscope is passed to examine and see for foreign body.
- *Endoscopy for foreign bodies*: Topical adrenaline use before foreign body removal, decongest mucosa and reduces bleeding.
- *Puncturing trachea or cricothyroid membrane*: The airway can be achieved by puncturing trachea or cricothyroid

Box 4: Technique and devices to secure an airway

- Oral airway
- Orotracheal intubation
- Nasotracheal intubation
- Fiberoptic intubation
- Laryngeal mask airway
- Continuous positive airway pressure (CPAP)
- Ventilating bronchoscope
- Tracheotomy and tracheostomy
- Mini tracheostomy
- Cricothyrotomy (laryngotomy)
- Needle cricothyrotomy
- Transtracheal needle ventilation
- Percutaneous dilational tracheostomy
- Surgical procedures in children to avoid tracheostomy
  - Cricoid split
  - Single stage laryngeal reconstruction

membrane with a 16-gauge plastic-sheathed needle. The needle is withdrawn and cannula is connected to an oxygen supply.

- *Cricoid split*: Cricoid split, which decompress the cricoid ring in cases of subglottic edema or soft immature stenosis, is indicated in neonates who fail extubation and weighs more than 1.5 kg.
- *Single-stage laryngeal reconstruction*: It is indicated in premature neonates who fail extubation and have mature subglottic stenosis.
- *Emergency cricothyrotomy/tracheostomy*: Emergency tracheostomy set with correct size tube should be kept ready. If medical techniques fail, an urgent tracheotomy or cricothyrotomy must be performed, though it is rarely required.

Chronic Airway Obstruction

- Antireflux treatment
- Systemic steroids to treat subglottic hemangioma
- Prophylactic antibiotics in cases of recurrent LTB
- Specific treatment: Interferon, cidofovir and mitomycin C.

Clinical Highlights

1. **Tracheobronchial tree and larynx**: Rate of topical absorption is highest.
2. **Aspiration**: It occurs in cases of vocal cord palsy, tracheoesophageal fistula and cleft larynx.
3. **Pediatric epiglottitis**: Avoid examining a child's throat when epiglottitis is suspected.
4. **Micrognathia and Pierre-Robin syndrome**: Stridor is due to falling back of tongue.
5. **Congenital vascular rings**: They cause both stridor and dysphagia.
6. **Acute stridor in children**: It is mostly caused by laryngotracheobronchitis, bacterial tracheitis, acute epiglottitis and laryngeal foreign bodies.
7. **Stridor in adults**: The most common cause is malignancy of larynx.
8. **Hoarseness in elderly smokers:** If hoarseness that persists for more than 3 weeks, malignancy of vocal cord should be ruled out.

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46 Infections of Larynx

Never say, 'No'; never say, 'I cannot', for you are infinite. Never think there is anything impossible. It is the greatest heresy to think so. You can do anything and everything. You are infinite, deathless, and birthless. Because you are infinite spirit, it does not befit you to be a slave. Arise! Awake! Stand up and fight.

—Swami Vivekananda

Points of Focus

- **INTRODUCTION**
  - ✓ Etiology/Risk Factors
- **ACUTE LARYNGOTRACHEOBRONCHITIS (CROUP OR LARYNGOTRACHEITIS)**
  - ✓ Differences between Croup and Acute Epiglottitis
- **BACTERIAL TRACHEITIS**
- **PEDIATRIC EPIGLOTTITIS**
- **ADULT SUPRAGLOTTITIS**
- **WHOOPING COUGH**
- **DIPHTHERIA**
- **CHRONIC NONSPECIFIC LARYNGITIS**
- **ATROPHIC LARYNGITIS (LARYNGITIS SICCA)**
- **TUBERCULOSIS**
  - ✓ Distinguishing Features of Tuberculosis, Syphilis, Malignancy and Nonspecific Laryngitis
- **LUPUS**
- **SYPHILIS**
- **LEPROSY (HANSEN'S DISEASE)**
- **SCLEROMA**
- **EDEMA OF LARYNX**
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

Acute infections (such as croup and epiglottitis) of larynx are mainly seen in children and develop in hours to days. They present with airway distress and fever. The chronic infections (such as tuberculosis, leprosy and syphilis) of larynx are mainly seen in adults and exist for weeks to months. They usually present with hoarseness and pain and must be distinguished from malignancy. The list of infections of the larynx is given in Table 1.

Etiology/Risk Factors

Following are some etiologic and risk factors of larynx infections:

- Viral infections.
- Bacterial invasion takes place with *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Hemolytic streptococci* and *Staphylococcus aureus*.
- Exanthemata fevers, such as measles, mumps, and chickenpox.
- Vocal abuse
- Allergy
- Thermal or chemical burns of larynx due to inhalation or ingestion of certain substances.
- Laryngeal trauma due to endotracheal intubation, endoscopy or laryngeal surgery.
- Foreign body
- Cricoarytenoid arthritis due to rheumatoid arthritis, systemic lupus erythematosus and Reiter's syndrome.
- Angioneurotic edema.
- Radiation-induced supraglottitis.

TABLE 1 Infections and manifestations of systemic disease of larynx

Acute infections	*Chronic infections*	*Systemic diseases*
Laryngotracheobronchitis (Croup)	Tuberculosis	Rheumatoid arthritis
Bacterial tracheitis	Syphilis	Systemic lupus erythematosus
Pediatric epiglottitis	Leprosy (Hansen's)	Relapsing polychondritis
Diphtheria	Candidiasis	Sarcoidosis
Adult supraglottitis	Histoplasmosis	Wegener's granulomatosis
Whooping cough	Blastomycosis	Amyloidosis
Other: Mumps, measles and chickenpox	Cryptococcosis	
	Actinomycosis	

ACUTE LARYNGOTRACHEOBRONCHITIS CROUP OR LARYNGOTRACHEITIS

Acute laryngotracheobronchitis (LTB) is the most common cause of infectious respiratory obstruction in children.

Etiology/Risk Factors

- Croup is a viral infection (Parainfluenza type I and II and influenza A) and affects usually boys between 3 months and 5 years of age.
- Other viruses, which have been implicated, are adenovirus, respiratory syncytial, influenza, and measles viruses.
- The secondary bacterial infection supervenes soon (by Gram-positive cocci).
- One of the family members usually has history of cold symptoms.

Pathology

The swelling of loose areolar tissue in the subglottic region causes airway obstruction and stridor. Thick tenacious secretions and crusts may completely obstruct the airway.

Clinical Features

- The disease begins with symptoms of upper respiratory infection such as low-grade fever, cold and cough.
- After several days of cold symptoms child develops hoarseness and brassy and barking croupy cough.
- Intermittent inspiratory stridor becomes continuous.
- Signs of upper airway obstruction, such as nasal flaring, and suprasternal, infrasternal and intercostals recession.
- Stridor and dyspnea may lead to hypoxia, hypercapnia, tachycardia, hypoventilation and eventually death.

Diagnosis

X-ray of nasopharynx and neck may show tapered narrowing of the subglottis (steeple sign).

Differential Diagnosis

- *Spasmodic or recurrent croup*: They are usually associated with triggers, such as acid reflux, allergy, or psychological factors and there is usually no infection. They occur repeatedly and usually resolve rapidly. Laryngotracheal stenosis and congenital abnormalities should be ruled out.
- *Bacterial tracheitis*: This condition is the bacterial complication of acute LTB and present with thick purulent respiratory secretions (*see* section Bacterial Tracheitis).
- *Diphtheritic croup*: The child usually has serous nasal discharge and gray-white pharyngeal membrane (*see* section Diphtheria).
- *Epiglottitis*: The characteristic features include abrupt onset of high-grade fever, dysphagia, dyspnea, and hoarseness and toxic appearance without preceding and family history of cold-like symptoms (Table 2).

Treatment

- Following conservative measures prevent laryngeal spasms:
  - Humidification (either hot or cold) softens crusts and tenacious secretions
  - Steam either from a vaporizer or shower
  - Cold steam from a nebulizer.
- *Hospitalization* is needed in cases of airway obstruction, cyanosis, restlessness, depressed sensorium, or toxic appearance.
- *Antibiotics*: Ampicillin (50 mg/kg/day in divided doses) for Gram-positive cocci and *H. influenzae*.
- *Intravenous fluids* for managing dehydration.
- *Steroids*: They reduce edema due to its anti-inflammatory effect, vasoconstriction and reduced vascular permeability.
  - Hydrocortisone 100 mg intravenous
  - Dexamethasone (0.6 mg/kg) intramuscular or oral
  - Nebulized budesonide.
- *Racemic adrenaline* (a mixture of D and L isomer) via nebulizer or respirator helps in reducing edema due to its vasoconstriction action. The child should be monitored for 3 hours (rebound effect) and then allowed to go home. It relieves dyspnea and may avert tracheostomy. Racemic epinephrine has fewer cardiovascular side effects.
- *Intubation/tracheostomy*: In cases of airway obstruction. Tracheostomy is preferred when intubation is expected longer than 72 hours. Assisted ventilation may be needed. The endotracheal tube should be one size smaller.
- Indications for intubation:
  - Rising CO_2 level
  - Worsening neurological status
  - Decreasing respiratory rate.
- Indications for extubation:
  - No fever
  - Diminished secretions
  - Air leak around tube.

TABLE 2 Differences between acute epiglottitis and laryngotracheobronchitis (LTB)

	Acute epiglottitis	*Acute laryngotracheobronchitis*
Pathogen	*H. influenzae* type B	Parainfluenza virus type I and II
Common age group	2–7 years	3 months to 5 years
Site of obstruction	Supraglottic (epiglottis) larynx	Subglottic larynx
Prodromal cold symptoms	Absent	Present
Onset	Abrupt within hours	Gradual within days
Temperature	High-grade	Low grade
Child's appearance	Toxic	Non-toxic
Cough	Usually absent	Barking seal-like
Dysphagia	Severe	Usually absent
Drooling of saliva	Present	Absent
Progression	Rapid	Slow
Family history of cold	Absent	Usually present
X-ray soft tissue neck	Thumb print sign on lateral view	Steeple sign on anteroposterior view

Complications

- Middle ear infections
- Lung infections
- Bacterial tracheitis.

BACTERIAL TRACHEITIS

Etiology/Risk Factors

- This is a bacterial complication of acute LTB.
- The most common pathogen is *Staphylococcus aureus,* but others include *Moraxella (Branhamella) catarrhalis*, nontypable *Haemophilus influenzae* and anaerobes.

Clinical Features

High fever, cough and stridor are most common symptoms.

Diagnosis

- Leukocytosis
- X-ray soft tissue neck lateral view: Shows thick tracheal membrane in 80% cases.
- Bronchoscopy: Aspiration of thick purulent respiratory secretions.
- Culture and sensitivity of purulent secretions.

Treatment

The cornerstones of management are antibiotics and intubation.

Complications

- Toxic shock syndrome
- Secondary acquired pneumonia
- Acute respiratory distress syndrome.

PEDIATRIC EPIGLOTTITIS

In this serious acute inflammatory condition of children, there is marked edema of the epiglottis which obstructs the airway.

Etiology/Risk Factors

- *H. influenzae* type B is the most common causative organism.
- This serious condition usually affects children of 2–7 years of age.
- Immunization against *H. influenzae* type B decreases the prevalence of this disease.

Clinical Features

- Abrupt onset of high-grade fever, dysphagia and odynophagia, drooling of saliva, hoarseness of voice and respiratory distress.
- Child may sit upright (tripod position) with hyperextended neck (Fig. 1).
- Inspiratory stridor and nasal flaring and retractions of suprasternal notch and supraclavicular and intercostals' spaces are seen on examination.
- Air hunger may rapidly progress to cyanosis, coma and death.
- Septicemia.

Fig. 1: Pediatric epiglottitis. Typical positioning (neck is flexed and head extended) of a child with partial upper airway obstruction due to epiglottitis

Diagnosis

- Red and swollen (fiery cherry-red) epiglottis and edema and congestion of other supraglottic structures.
- Tongue depressor and indirect laryngoscopy examination can cause reflex laryngospasm and cardiorespiratory arrest and are not done in these cases. Laryngoscopic examination is done in OT where facilities for intubation/tracheostomy are ready.
- X-ray soft tissue neck lateral view: Shows swollen epiglottis (thumb print sign).
- Throat swab (from epiglottis) & blood culture: *H. influenzae* type B can be seen in blood and epiglottis swab.

Differential Diagnosis

- *Acute laryngotracheobronchitis*: Family members and child usually have preceding history of upper respiratory infections (Table 2).
- *Foreign body respiratory passage*: The child will come with choking or coughing, but no evidence of infection like fever.
- Peritonsillar and retropharyngeal abscesses.

Complications

- Reflex laryngospasm and cardiorespiratory arrest.
- Complete obstruction of airway and death.
- Rare: Meningitis, pneumonia, otitis media.

Treatment

- *Intubation or tracheostomy* under general anesthesia regardless of severity of respiratory distress is the first and topmost priority. Nasotracheal intubation is usually required for 1–3 days.
- *Blood culture*
- *Antibiotics*: Ampicillin + sulbactam or third-generation cephalosporin (cefataxime or ceftriaxone) intravenously is started without waiting for the blood culture report. They are effective against beta-lactamase positive *H. influenzae*.
- *Intravenous fluids*: For adequate hydration.
- *Steroids*: Hydrocortisone or dexamethasone IM or IV relieve edema.
- *Humidification* and *oxygen*: Mist tent or croupette may be needed.

ADULT SUPRAGLOTTITIS

In this acute inflammatory condition, there is marked edema of supraglottic structures: epiglottis, aryepiglottic folds and arytenoids.

Etiology

Most common pathogens are *H. influenzae* and *beta-hemolytic streptococcus*.

Clinical Features

- Less acute and less toxic than pediatric epiglottitis.
- Sore throat and dysphagia are most common symptoms.
- Pale, boggy and edematous epiglottis and other supraglottic structures.
- Some patients may present with stridor, tachycardia or rapid progression.
- Some patients develop epiglottic abscess on its lingual surface.

Diagnosis

- Leukocytosis
- Blood and pharynx culture: Usually negative (positive in cases of *H. influenzae).*

WHOOPING COUGH

Etiology

It is caused by *Bordetella pertussis* and seen in infants, adolescents and adults.

Clinical Features

- Adults have severe and protracted cough.
- Newborns present with catarrhal stage, fever and leukocytosis.

Treatment

Erythromycin estolate in 4 divided doses for 14 days or azithromycin 10 mg/kg on day 1 followed by 5 mg/kg/day on days 2–5.

DIPHTHERIA

- Mostly it affects children below 10 years of age.
- Usually larynx is involved secondary to faucial diphtheria (*see* chapter Pharyngitis and Adenotonsillar Diseases).
- Incidence of diphtheria has significantly declined due to DPT immunization.

Clinical Features

- *General*: Gradual onset, low-grade fever, sore throat, malaise, toxic look, tachycardia and thready pulse.
- *Laryngeal*: Hoarseness, croupy cough, inspiratory stridor, and increasing dyspnea.
- *Diphtheritic membrane*: Gray-white membrane on tonsil, pharynx, soft palate, larynx and trachea. On removal, it leaves a raw bleeding surface. The gray-white pseudomembrane in larynx and trachea obstructs airway.
- *Cervical lymphadenopathy*: "Bull-neck" appearance.

Diagnosis

It is usually a clinical diagnosis.

Throat swab smear and culture of *Corynebacteriun diphtheriae* will confirm the diagnosis.

Treatment

- *Diphtheria antitoxin*: 100,000 units IV as saline infusion neutralize free toxin circulating in the blood.
- *Antibiotics*: Benzyl penicillin, 500,000 units IM every 6 hours for 6 days. Erythromycin to those who are allergic to penicillin.
- *Maintenance of airway*:
  - Direct laryngoscopy removal of diphtheritic membrane and intubation to relieve respiratory obstruction.
  - Tracheostomy may be required.
- *Complete bed rest* for 2–4 weeks to guard against effect of myocarditis.

CHRONIC NONSPECIFIC LARYNGITIS

There are two types of chronic nonspecific laryngitis hyperemic and hypertrophic.

- *Chronic hyperemic laryngitis*: It is a diffuse inflammatory condition, which symmetrically involves the true cords, ventricular bands, interarytenoid region and root of the epiglottis.
- *Chronic hypertrophic (hyperplastic) laryngitis*: It may be either diffuse or localized. Localized varieties include dysphonia plica ventricularis; vocal nodules, vocal polyp, Reinke's edema, and contact ulcer (*see* chapter Benign Tumors of Larynx).

The pseudostratified ciliated epithelium changes to squamous type. There may be hyperplasia and keratinization (leukoplakia) of squamous epithelium of the vocal cords (*see* chapter Benign Tumors of Larynx).

Etiology/Risk Factors

- Infection in paranasal sinuses, teeth, tonsils and lungs.
- Occupational factors: Exposure to dust and fumes, such as in miners, gold or ironsmiths and chemical industries workers.
- Smoking and alcohol
- Chronic coughing or clearing of throat
- Vocal abuse
- Gastroesophageal reflux disease (GERD)
- Inadequate hydration.

Clinical Features

- Hoarseness and easily tired voice.
- Constant hawking, dryness and intermittent tickling, and clearing the throat repeatedly.
- Discomfort in the throat
- Dry and irritating cough
- Hyperemia of larynx; Vocal cords dull red; Flecks of viscid mucus on vocal cords and interarytenoid region.

Differential Diagnosis

- Chronic infections such as tuberculosis and leprosy
- Sarcoidosis and relapsing polychondritis
- Funguses
- Autoimmune disorders
- Acid reflux
- Cancer.

Biopsy

Tissue should be sent for

- Usual tissue stains
- Special fungal stains
- Acid-fast bacilli smears
- Fungal and acid-fast bacilli cultures.

Treatment

- Management of the infections of sinuses, tonsils, teeth, and respiratory system (such as bronchitis, bronchiectasis, tuberculosis).
- Avoid smoking, alcohol or pollution, dust and fumes.
- Voice (speech) therapy.
- Steam inhalations.
- Expectorants
- Management of GERD (*see* chapter Disorders of Esophagus).

ATROPHIC LARYNGITIS (LARYNGITIS SICCA)

Atrophic laryngitis, which is associated with atrophic rhinitis and pharyngitis, refers to atrophy of laryngeal mucosa with crust formation. It is a very rare disease.

Clinical Features

- Hoarseness of voice improves on coughing and removal of crust.
- Dry and irritating cough.
- Dyspnea occasionally due to laryngeal crusts.
- Atrophic mucosa covered with foul smelling crusts. Excoriating and bleeding mucosa on removal of crusts.

Treatment

- Humidification.
- Laryngeal sprays with glucose in glycerin or oil of pine.
- Expectorants containing ammonium chloride or iodides loosen the crusts.

(Atrophic rhinitis is described in chapter Nasal Manifestation of Systemic Diseases).

TUBERCULOSIS

Tuberculosis of larynx is usually secondary to pulmonary tuberculosis and affects middle-aged males. The cases of primary tuberculosis of larynx are also seen. Tubercle bacilli may reach the larynx by bronchogenic, lymphatic or hematogenous routes.

Pathology

- Tuberculosis involves posterior part of larynx and common sites in order of decreasing frequency include interarytenoid region, ventricular bands, vocal cords and epiglottis.
- In bronchogenic spread, the sputum carries the tubercle bacilli, which settle and penetrate the interarytenoid region, from the bronchi.
- Laryngeal mucosa may become red and swollen due to cellular infiltration (pseudoedema). The submucosal tubercles caseate and ulcerate. Stages of perichondritis and cartilage necrosis are uncommon.

Clinical Features

- Weakness of voice followed by hoarseness is a common presentation.
- Severe pain radiating to ears, odynophagia and dysphagia, cough and weight loss.
- Lesions: Nonspecific inflammation to nodular, exophytic lesion or mucosal ulceration.
- Vocal cords: Hyperemia with impaired adduction (early sign), mouse nibbled type ulcers.
- Arytenoid and interarytenoid region: Swelling (mamillated), superficial ragged ulcers, and granulations overlying vocal process of arytenoids.
- Epiglottis: Pseudoedema of the epiglottis (turban epiglottis).
- Ventricular bands and aryepiglottic folds: Swollen.
- Mucosa: Pale.

Differential Diagnoses

This chronic disorder should be differentiated from the syphilis, malignancy, and nonspecific chronic laryngitis (Table 3). Biopsy will confirm the diagnosis.

Diagnosis

- *Mantoux test*: Purified protein derivative is usually positive even if there is no pulmonary involvement.
- *X-ray chest PA view*: Features of pulmonary tuberculosis.
- *Sputum examination*: For acid-fast bacilli and culture and sensitivity.
- *Biopsy*: To exclude carcinoma and distinguish from other condition.

Treatment

It consists of multidrug therapy as per the culture and sensitivity report.

LUPUS

This indolent tubercle infection is usually associated with lupus of nose and pharynx and involves the anterior part of larynx.

Clinical Features

- Painless and usually asymptomatic condition.
- Findings seen incidentally in cases of lupus nose include involvement of epiglottis (may be completely destroyed), aryepiglottic folds and sometimes ventricular bands.
- Pulmonary tuberculosis: Absent.

Treatment

Antitubercular drugs offer good prognosis.

SYPHILIS

Laryngeal syphilis is rare.

- Secondary syphilis may present with diffuse laryngeal hyperemia or coalescing maculopapular rash of supraglottic region.
- Tertiary stage usually presents with diffuse, nodular, gummatous infiltrate. The nodules may ulcerate or coalesce and form larger nodules. The lesions may progress to chondritis, fibrosis and scarring. Laryngeal stenosis may occur.
- Biopsy and serological tests will confirm the diagnosis.

LEPROSY (HANSEN'S DISEASE)

Laryngeal leprosy is a rare condition. The larynx is the second most common site of leprosy involvement in head and neck after the nose (ulceration and perforation).

Clinical Features

- Pain free condition.
- Erythematous or nodular edema of supraglottis may extend to glottis.
- Nodule may enlarge, ulcerate or heal by scar formation, which usually cause laryngeal stenosis.

Diagnosis

Laryngeal biopsy and nasal smear: Acid-fast staining shows *Mycobacterium leprae* (Hansen's bacilli) in the foam cells.

Treatment

Oral dapsone or dapsone and rifampicin are given for 5–10 years (For 1–2 years after the negative biopsy samples).

SCLEROMA

Etiology

This chronic inflammatory disease is caused by *Klebsiella rhinoscleromatis (Frisch bacillus)*. Nose is the most common site of involvement (see chapter Nasal Manifestation of Systemic Diseases).

Clinical Features

- Hoarseness, wheezing and dyspnea are the most common symptoms of larynx scleroma.
- Smooth red swelling in subglottic region.

TABLE 3 Distinguishing features of tuberculosis, syphilis, malignancy and chronic nonspecific laryngitis

Features	*Tuberculosis larynx*	*Syphilis larynx*	*Malignancy larynx*	*Chronic nonspecific laryngitis*
Odynophagia	Severe	Absent	In advanced lesions	Absent
Weight loss	Present	Absent	In advanced lesions	Absent
Voice	Weak aphonic	Rough	Hoarse	Hoarse
Part of larynx affected	Usually arytenoid and interarytenoid region	Anterior one third	Any part	Vocal cords
Lesion findings	Swelling (mamillated), superficial ragged ulcers, and granulations	Tertiary: Large nodules or deep ulcers; Secondary: diffuse laryngeal hyperemia	Ulcer with everted and irregular margins and induration	Congestion and edema
Sputum for AFB	Positive	Negative	Negative	Negative
VDRL	Non-reactive	Reactive	Non-reactive	Non-reactive
X-ray chest	Koch's chest	Normal	Cannon balls in advanced lesions	Normal

Diagnosis

Biopsy

Treatment

- *Antibiotics*: Streptomycin or tetracycline
- *Steroids*: To prevent fibrosis
- *Reconstructive surgery*: For subglottic stenosis.

EDEMA OF LARYNX

Edema glottidis refers to edema of the loose mucosa of supraglottic and subglottic regions. Vocal cords have sparse subepithelial connective tissue (*See* laryngeal causes of stridor and its treatment in chapter Laryngeal Symptoms and Examination).

Etiology

- *Infections*: Acute epiglottitis, LTB, laryngeal tuberculosis and syphilis of larynx, complications of parapharyngeal, peritonsillar and retropharyngeal abscesses and Ludwig's angina.
- *Trauma*: Laryngeal trauma and foreign body, thermal or caustic burns, inhalation of irritant gases or fumes.
- *Iatrogenic trauma*: Surgery of tongue, floor of mouth, endoscopies, intubation.
- *Neoplasms*: Larynx or laryngopharynx malignancy.
- *Allergy*: Angioneurotic edema and anaphylaxis.
- *Radiation*: For larynx or pharynx malignancy.
- *Systemic diseases*: Nephritis, heart failure, or myxedema.

Diagnosis

- Degree of airway obstruction and respiratory distress depend upon the cause.
- Inspiratory stridor.
- Indrawing of suprasternal, supraclavicular and intercostals spaces.
- Complete ear, nose and throat examination usually reveal the cause.
- Indirect laryngoscopy: Edema of supraglottic or subglottic region.
- Children may need direct laryngoscopy.
- X-ray soft tissue neck.

Treatment

- Management of the cause.
- Steroids: In cases of epiglottitis, LTB or edema due to trauma, allergy or radiation.
- Adrenaline (1:1000) 0.3–0.5 ml IM repeated every 15 minutes if necessary for allergic or angioneurotic edema.
- Intubation of larynx or tracheostomy.

Clinical Highlights

1. **Acute epiglottitis:** It is common in children and is caused by *Haemophilus influenzae* type B. It produces a typical "Thumb sign" on lateral X-ray film, which though is usually not ordered. Ceftriaxone (Cephalosporin) is the first line of antibiotic in the management. In a child with suspected diagnosis of epiglottitis, securing an airway by intubation is the first line of treatment.
2. **Acute laryngotracheobronchitis (Croup):** This disease of children is caused by parainfluenza virus type 1, 2, and sometimes 3 and produces subglottic edema of larynx. The child presents with stridor and respiratory distress. Though avoided as any manipulation may precipitate acute obstruction, X-ray soft tissue neck PA. view shows typical "Steeple sign".
3. **Laryngeal tuberculosis:** Mouse nibbled appearance of vocal cords is the characteristic feature of laryngeal tuberculosis.
4. **Turban epiglottis:** It is due to the edema and tuberculous infiltration of the epiglottis. **Lupus**, which is a form of tuberculosis, eats away and destroys the epiglottis.
5. **Pachydermia laryngis:** The only symptom is hoarseness of voice. Diagnosis is made by biopsy, which shows acanthosis and hyperkeratosis. It is not a premalignant condition.

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47 Benign Tumors of Larynx

If there is sin, this is the only sin–to say that you are weak, or others are weak. Whatever you think, that you will be. If you think yourselves weak, weak you will be; if you think yourselves strong, strong you will be.

—Swami Vivekananda

Points of Focus

- **INTRODUCTION**
  - ✓ Classification
  - ✓ Risk Factors of Non-neoplastic Vocal Fold Mucosal Disorders
- **VOCAL NODULES (SINGER'S OR SCREAMER'S NODULES)**
- **VOCAL POLYP**
- **REINKE'S EDEMA (BILATERAL DIFFUSE POLYPOSIS)**
- **CONTACT ULCER OR GRANULOMA**
- **INTUBATION GRANULOMA**
- **LEUKOPLAKIA OR KERATOSIS**
- **AMYLOID TUMORS**
- **DUCTAL CYSTS**
- **SACCULAR CYSTS**
  - ✓ Anterior
  - ✓ Lateral
  - ✓ Laryngopyocele
- **LARYNGOCELE**
  - ✓ Internal
  - ✓ External
  - ✓ Mixed
- **RECURRENT RESPIRATORY PAPILLOMATOSIS**
  - ✓ Juvenile
  - ✓ Adult Onset
  - ✓ Treatment
- **CHONDROMA**
- **HEMANGIOMA**
  - ✓ Infantile Hemangioma (Capillary)
  - ✓ Adult Hemangioma (Cavernous)
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

Classification

Benign tumors of the larynx can be divided into three groups: non-neoplastic and neoplastic tumors, and saccular swellings (Table 1). The benign neoplastic tumors of the larynx are less common than malignant tumors. Laryngeal papillomas account for approximately 80% of all benign neoplasms of the larynx. Others are rare and include pleomorphic adenoma or oncocytoma, rhabdomyoma, neurofibroma, neurilemmoma, lipoma and fibroma.

Risk Factors of Non-neoplastic Vocal Fold Mucosal Disorders

Non-neoplastic vocal fold mucosal tumor-like masses occur as a result of infection, trauma, allergy, reflux, smoking or degeneration. They are seen more frequently than true benign neoplasms. The risk factors of these vocal fold mucosal disorders are following:

- An expressive and talkative persons: Most common
- Occupational: Extreme vocal demands, which may be related to family life, childcare, politics, religion, athletics, musical rehearsal and performance
- Tobacco smoking
- Alcohol
- Insufficient fluid intake
- Infection
- Allergy
- Gastroesophageal reflux disease
- Iatrogenic factors: Medicines (dryness of secretions), endotracheal intubation and laryngeal instrumentations.

TABLE 1 Benign tumors of larynx

Non-neoplastic vocal fold mucosal disorders	*Neoplastic*	*Saccular disorders*
Vocal nodules	Recurrent Respiratory Papillomatosis	Ductal cysts
Vocal polyp	Juvenile type	Saccular cysts
Reinke's edema	Adult-onset type	Anterior
Contact ulcer	Chondroma	Lateral
Intubation granuloma	Hemangioma	Laryngopyocele
Leukoplakia	Granular cell tumor	Laryngocele
Amyloid tumors	Glandular tumors	Internal
	Rhabdomyoma	External
	Lipoma	Combined
	Fibroma	

VOCAL NODULES (SINGER'S OR SCREAMER'S NODULES)

Risk Factors

- Expressive and talkative persons and teachers, actors, vendors or pop or rock singers, stock traders and school going children.
- Vocal trauma occurs when persons speak in unnatural low tones or high intensities.
- They are more common in boys and women.

Pathology

Vocal abuse or misuse leads to submucosal edema and hemorrhage, which undergo hyalinization and fibrosis. Hyperplasia of overlying epithelium appears as a nodule.

- Site: They are bilateral and occur at the area of maximum vibration of the cord. It is at the junction of anterior one-third and posterior two-thirds of the free edge of vocal cord.

Clinical Features

- Chronic hoarseness or repeated episodes of acute hoarseness in women.
- Singers' complaints: Vocal fatigue (reduced vocal endurance), difficulty in singing high notes softly, increased breathiness (air escape), roughness, harshness and increased effort for singing.
- Pain in the neck on prolonged phonation.
- Initially nodules appear soft, reddish and edematous but later on they look grayish or white in color.
- Always bilateral at the junction of anterior one-third and posterior two-thirds of the free edges of vocal cords (Fig. 1).
- Size: Varies from pinhead to half a pea. One side may be bigger than other.

Diagnosis

The subtle nodules affect professional voice users and missed in laryngoscopy. They can be diagnosed with videostroboscopy.

Treatment

- Medical: Proper hydration and management of allergy and acid reflux.

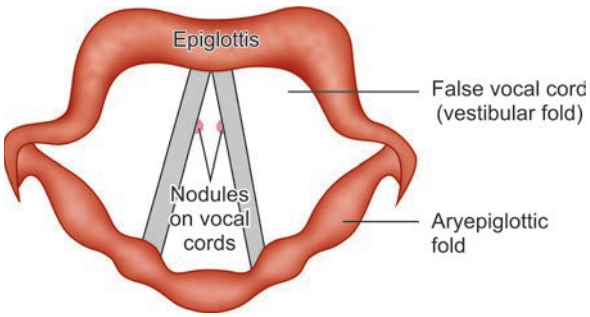

Fig. 1: Bilateral vocal nodules at the junction of anterior one-third and posterior two-thirds of vocal cords

- Voice (speech) therapy: Educating the patient in proper use of voice. Many early nodules disappear. Postoperative speech therapy prevents recurrence.
- Surgery: For refractory or large nodules of long-standing duration. Precise removal with microdissection techniques under operating microscope (microlaryngoscopy). Avoiding trauma to vocal ligament and mucosa. Patient is instructed not to speak for 4 days and gradually progression over 6 weeks to full voice.

VOCAL POLYP

Risk Factors

- Intermittent severe voice abuse, work in noisy environment, and aspirin.
- Sudden shouting results in hemorrhage and submucosal edema.
- Unilateral hemorrhagic vocal cord polyp is more common in men.

Clinical Features

- Sudden onset of hoarseness during extreme vocal abuse is the most common presentation.
- Chronic vocal huskiness or intermittent subtle aberrant sounds.
- Diplophonia (double voice) in some patients due to different vibratory frequencies of the two vocal cords.

- Always unilateral at the junction of anterior one-third and posterior two-thirds of the free edge of vocal cord.
- Soft and smooth (dark and hemorrhagic in early stages) and may become pedunculated, which then flop up and down the glottis during respiration or phonation (Fig. 2).

Treatment

- Medical: Stop anticoagulants and aspirin and manage acid reflux.
- Voice therapy: Early polyps may regress totally.
- Surgery: Microlaryngoscopy superficial surgical excision followed by speech therapy.

REINKE'S EDEMA (BILATERAL DIFFUSE POLYPOSIS)

Risk Factors

They include vocal abuse and long-term smoking. It is common in middle-aged women.

Pathology

Collection of fluid (edema) occurs in the subepithelial space of Reinke.

Clinical Features

- Female patients may complain of male voice
- Hypermasculine voice with large polyps
- Diffuse symmetrical swelling (pale watery bags of fluid) of superior surface and margins of both vocal cords
- Rule out hypothyroidism.

Treatment

- Medical: Cessation of smoking.
- Voice therapy
- Microlaryngoscopy: Polyp reduction with mucosal sparing for epithelialization (vocal cord stripping may lead to aphonia, high and husky voice). Some surgeons prefer to operate one side cord at a time.

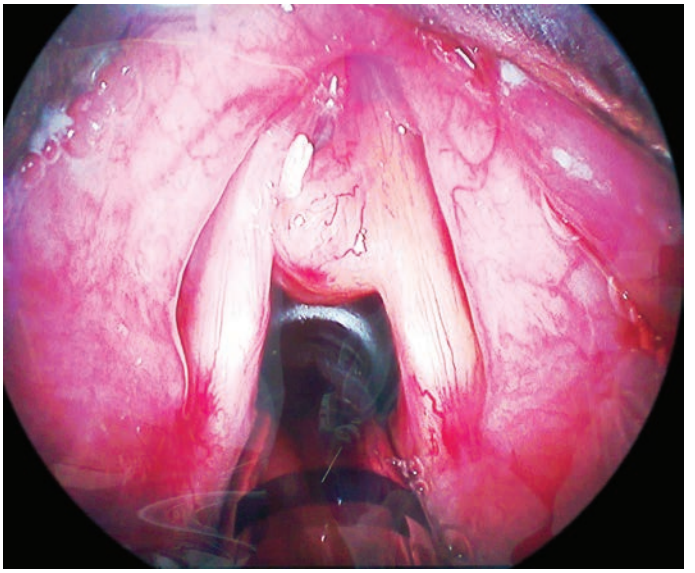

Fig. 2: Microlaryngoscopy right vocal chord polyp (*Courtesy*: Prof. Nitin M Nagarkar, Chandigarh)

CONTACT ULCER OR GRANULOMA

Risk Factors

- Faulty voice production, male lawyers, ministers, teachers and executives.
- Chronic coughing or throat clearing, and psychological stress or conflict.
- Caffeine and alcohol, late-night eating habit and gastric reflux.

Pathology

Vocal processes of arytenoids hammer against each other and result in ulceration and granuloma formation.

Clinical Features

- Most common symptom: Unilateral mild pain over midthyroid cartilage, which may refer to ear.
- Low pitched gravelly morning voice and heartburn.
- Hoarseness occurs when granuloma is large.
- Constant desire to clear the throat or coughing.
- Low frequency range monotone voice which may be husky.
- Usually unilateral ulcer with whitish exudates or bi-lobed granuloma (may become pedunculated) at the vocal process of arytenoids with congestion of arytenoids mucosa.

Treatment

- Antireflux regimen
- Voice therapy
- Injection of depot corticosteroids directly into the lesion
- Surgery should be the last resort because recurrence is predictable. Surgery consists of limited removal and leaving the base or pedicle undisturbed.

INTUBATION GRANULOMA

Risk Factors

They include rough endotracheal intubation, large tube or prolonged presence of tube between the arytenoids, rigid bronchoscopy or other direct laryngeal manipulation.

Pathology

Injury to vocal processes of arytenoids causes mucosal ulceration (perichondrial abrasions) and granuloma formation over the exposed cartilage. Granulomas usually mature and fall off.

Clinical Features

- Hoarseness is uncommon.
- Dyspnea occurs with large granulomas.
- Granuloma may be sessile to large and pedunculated attached to vocal process of arytenoid. Usually bilateral involving posterior thirds of true cords.
- Posterior glottic incompetence in cases of severe tissue loss.
- Arytenoid fixation or interarytenoid synechium may be seen.

Treatment

- Wait and watch with antibiotics for several weeks.
- Speech therapy.

- Surgery: Only if conservative treatment fails, and granuloma (mature) persists. Endoscopic corticosteroid injection into the base of granuloma before removal is suggested. Stalk should be left behind. Topical application of mitomycin C may prevent recurrence.

LEUKOPLAKIA OR KERATOSIS

Risk Factors

Chronic laryngeal irritants

Pathology

The epithelial hyperplasia appears as a white plaque or warty growth and involves upper surface of either one or both vocal cords. It does not affect mobility of cord. In this precancerous condition, "carcinoma in situ" frequently supervenes.

Clinical Features

- Hoarseness
- White plaque or warty growth on the upper surface of either one or both vocal cords.

Treatment

Microlaryngoscopy stripping of vocal cords is needed. Histopathological examination of the tissues rules out any malignant change.

AMYLOID TUMORS

- Most patients are men in the age group of 50–70.
- Presents as a smooth plaque or a pedunculated mass.
- Diagnosis: Histopathology.
- Treatment: Microlaryngoscopic surgical excision.

DUCTAL CYSTS

Blockage of ducts of seromucinous glands of laryngeal mucosa results in these retention cysts.

Common Sites

They include vallecula, aryepiglottic fold, false cords, ventricles and pyriform fossa.

Clinical Features

- Usually remain asymptomatic for long time and seen incidentally during examination.
- Hoarseness, cough, throat pain and dyspnea may occur in cases of very large cysts.
- An intracordal cyst presents with hoarseness similar to an epidermoid inclusion cyst.

Treatment

Marsupialization/removal in symptomatic patients.

SACCULAR CYSTS

Retention of secretion and distension of saccule occurs due to obstruction to the orifice of saccule and presents as a cyst in laryngeal ventricle.

Types

There are two types of saccular cysts—anterior and lateral.

- Anterior saccular cyst is small and present in the anterior part of ventricle and obscure anterior part of vocal cord.
- Lateral saccular cyst is large and may extend into the false cord, aryepiglottic fold and pyriform fossa and may appear in the neck through thyrohyoid membrane.
- Laryngopyocele: It is an infected lateral saccular cyst.

Clinical Features

- Infants with congenital lateral saccular cysts present with weak cry and stridor and cyanosis. Dysphagia may occur.
- In adults: Hoarseness is most common.

Diagnosis

X-ray soft tissue neck lateral view and CT will confirm the diagnosis.

Treatment

- Endoscopic aspiration, marsupialization or CO_2 laser vaporization.
- External approach for large lateral cysts: Midline or lateral thyrotomy approach through thyrohyoid membrane.

LARYNGOCELE

Risk Factors

Trumpet players, glass-blowers or weight lifters.

Pathology

This air-filled cystic swelling is dilatation of the saccule. It is said to arise from raised transglottic air pressure. There are three types of laryngocele; internal, external and combined or mixed (Fig. 3).

- Internal: It remains confined within the larynx and presents as distension of false cord and aryepiglottic fold.
- External: Here distended saccule herniates through the thyrohyoid membrane and presents in neck.
- Combined or mixed: Swellings can be seen both in the larynx, as well as outside in neck.

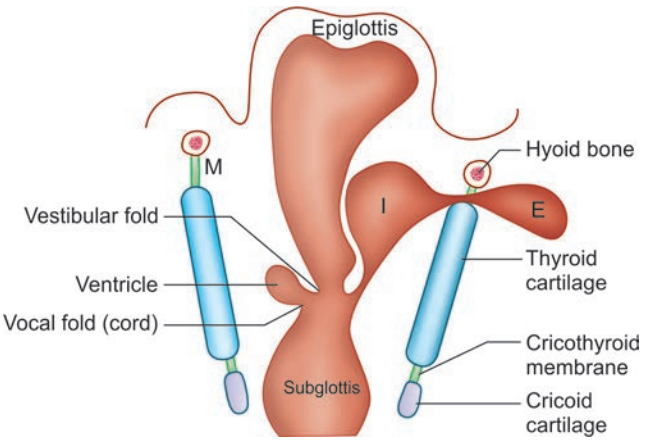

Fig. 3: Mixed type laryngocele showing internal (I) and external (E) components and their relations with thyrohyoid membrane (M)

Clinical Features

- Hoarseness and swelling in neck are common.
- Other symptoms in decreasing order of frequency include airway obstruction, dysphagia, sore throat, snoring and coughing.
- External laryngocele: Reducible swelling in the neck that increases in size on coughing and valsalva.

Diagnosis

- A laryngocele in an elderly person may be caused by saccule carcinoma, which must be ruled out.
- X-ray soft tissue neck PA and lateral views with valsalva and CT scan show the presence and the extent of laryngocele.

Treatment

- Surgical excision through an external approach neck incision.
- Endoscopic marsupialization of an internal laryngocele.

RECURRENT RESPIRATORY PAPILLOMATOSIS

Etiology

The disease is the response to mucosal infection by human papillomavirus of the Papova class. They are of two types—juvenile and adult-onset. Juvenile variety is more common than adult-onset.

Juvenile Papillomas

- Patients are infants and young children.
- Papillomas are multiple and aggressive and involve wide area and rapidly recurrent.
- They are mostly seen on the true and false cords and the epiglottis and may involve trachea and bronchi.
- They may spontaneously regress after puberty. They often recur and need multiple laryngoscopic removals.

Adult-Onset Papilloma

- Papilloma is usually smaller and limited to one site.
- They are usually less aggressive and do not recur.
- Most patients are males in the age group of 30–50 years.
- The most common sites are anterior half of vocal cord and anterior commissure.

Clinical Features

- Hoarseness and stridor (in juvenile type) are the most common presentations.
- Appear as glistening white irregular growths, pedunculated or sessile.

Treatment

- Endoscopic removal under operative microscope can be done with
  - Cup forceps
  - Cryotherapy
  - Microelectrocautery
  - CO_2 laser offers precision and less bleeding
  - Microdebrider: It is helpful as the papillomas are friable and bleed while removing.

Other techniques, which are being tried and giving encouraging results include:

- Interferon therapy
- Indole-3-carbinol, derived from cabbage and broccoli
- Methotrexate
- Intralesional cidofovir
- 585 nm pulsed dye laser
- Photodynamic therapy
- Radiation
- Vaccines.

CHONDROMA

The clinical behaviors of chondroma and low-grade chondrosarcoma are similar.

- Mostly arise from cricoid cartilage and seen in subglottic region.
- Dyspnea common
- Dysphagia and sense of lump in throat when large tumor grows outward from the lamina of cricoid cartilage.
- Most patients are men in the age group of 40–60 years.

HEMANGIOMA

There are two types of hemangiomas: infantile (capillary) and adult (cavernous).

Infantile Hemangioma (Capillary)

- Site: Subglottic area.
- Presentation: Stridor in the first 6 months of life.
- Associated findings: Half of the children have hemangiomas on other parts of body particularly in head and neck region.
- Prognosis: Usually involutes spontaneously.

Treatment

- Steroids
- Tracheostomy in cases of respiratory obstruction.
- This capillary type hemangioma can be vaporized with CO_2 laser.

Adult Hemangiomas (Cavernous)

- Site: Supraglottic larynx.

Treatment

- Asymptomatic: No treatment.
- Symptomatic: Steroid or radiation therapy may be employed. Cavernous hemangioma cannot be treated with laser.

Clinical Highlights

1. **Vocal cord nodules:** The most common site of these bilateral vocal cord nodules is the junction of anterior and middle third, which is the maximum vibratory area during speech.
2. **Reinke's edema:** This disorder is usually associated with smoking and causes diffuse polypoid degeneration of vocal cords.
3. **Contact ulcer:** The causes of this vocal process ulcer include intubation injury, laryngopharyngeal acid reflux, and adductor dysphonia.
4. **Multiple juvenile laryngeal papillomatosis:** This pediatric benign tumor of larynx is caused by human papillomavirus (Papova virus) subtype 6 and 11. Treatment includes microlaryngoscopy and surgical excision preferably with CO_2 laser. CO_2 laser is the best laser for multiple laryngeal papillomas. Recurrence is common and repeated excisions may be required. No modality of treatment can prevent recurrence. Malignant change is uncommon unless radiation has been used as a mode of treatment.
5. **Benign lesions of posterior larynx:** The conditions affecting posterior part of larynx include contact ulcer, pachydermia of larynx, and intubation granuloma.
6. **Laryngeal cartilaginous tumor:** Cricoid cartilage is the most common site of laryngeal cartilaginous tumor.
7. **Laryngocele:** Laryngocele arises as a herniation of laryngeal mucosa from the saccule of the laryngeal ventricle through the thyrohyoid membrane.
8. **External laryngocele:** It produces a swelling in the neck on Valsalva and communicates with laryngeal ventricle. It can be seen on CT. Bryce's sign is the gurgling sound produced on pressing the swelling of laryngocele.

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48 Neurologic Disorders of Larynx

Never mind the struggles, the mistakes. Never mind, these failures, these little backslidings; hold the ideal a thousand times; and if you fail a thousand times make the attempt once more.

—Swami Vivekananda

Points of Focus

- **NEUROLOGICAL DISORDERS OF LARYNX**
  - ✓ Hypofunctional Disorders
  - ✓ Hyperfunctional Disorders
  - ✓ Mixed Disorders
- **CLASSIFICATION OF LARYNGEAL PARALYSIS**
- **POSITIONS OF VOCAL CORDS**
- **CAUSES OF LARYNGEAL PARALYSIS**
- **UNILATERAL RECURRENT LARYNGEAL NERVE PARALYSIS**
- **BILATERAL RECURRENT LARYNGEAL NERVE (ABDUCTOR) PARALYSIS**
- **UNILATERAL SUPERIOR LARYNGEAL NERVE PARALYSIS**
- **BILATERAL SUPERIOR LARYNGEAL NERVE PARALYSIS**
- **UNILATERAL COMBINED (COMPLETE) PARALYSIS OF RECURRENT AND SUPERIOR LARYNGEAL NERVE**
- **BILATERAL COMBINED (COMPLETE) PARALYSIS OF RECURRENT AND SUPERIOR LARYNGEAL NERVE**
- **CONGENITAL VOCAL CORD PARALYSIS**
- **PHONOSURGERY**
  - ✓ Microlaryngeal Surgery
  - ✓ Local Injections
  - ✓ Laryngeal Framework Surgery (Thyroplasty)
  - ✓ Laryngeal Reinnervation Procedures
  - ✓ Reconstructive and Rehabilitative Procedure
- **CLINICAL HIGHLIGHTS**

NEUROLOGICAL DISORDERS OF LARYNX

Hypofunctional Disorders

- *Vocal fold paralysis/Paresis*: The details may be found in other sections of this chapter.
- *Neuromuscular junction disorders*: Myasthenia gravis.
  - Clinical feature: Ptosis or double vision or general fatigue, difficulty in speaking, breathing or swallowing and laryngeal fatigue (repeat ee-ee-ee).
  - Diagnosis: An edrophonium test, electromyography and blood antibodies against acetylcholine.
- *Poliomyelitis*: Acute bulbar polio.
  - Clinical features: Paresis of muscles of pharynx and larynx (except cricopharyngeus), hoarseness, dysphagia, and aspiration.
  - Treatment for aspiration: Cricopharyngeal myotomy.
- *Myopathies*: Dermatomyositis, muscular dystrophy and metabolic myopathies.
- *Medullary disorders*: Amyotrophic lateral sclerosis (ALS), primary lateral sclerosis, postpolio syndrome, Arnold-Chiari malformations, and medullary stroke.
- *Parkinson's disease*.
- *Nonorganic*: Psychogenic, malingering, conversion disorder and hysterical aphonia.

Hyperfunctional Disorders

(*See* Chapter on Voice and Speech Disorders)

- Dystonia
- Pseudobulbar palsy
- Myoclonus
- Essential tremors
- Muscle tension dysphonia

Mixed Disorders

- Multiple sclerosis
- Parkinson's disease

CLASSIFICATION OF LARYNGEAL PARALYSIS

Laryngeal paralysis may be divided into following categories:

- Congenital or acquired
- Unilateral or bilateral
- Complete or incomplete
- Acute or chronic
- Abductor or adductor or both
- Sensory or motor or both
- Organic or functional
- Supranuclear, nuclear or infranuclear (recurrent laryngeal nerve or superior laryngeal nerve or both).
- Idiopathic or secondary.

POSITIONS OF VOCAL CORDS

Though not considered relevant and important now, 5 positions of the vocal cords are traditionally described (Fig. 1). The vocal cord position is not a reliable indicator of lesion site.

- *Median position*: Vocal cord is in midline position such as in phonation. It may occur in recurrent laryngeal paralysis (RLN) paralysis.
- *Paramedian position*: Vocal cord is 1.5 mm away from midline such as in strong whisper. It may occur in RLN.
- *Intermediate (cadaveric) position*: This is the neutral position of cricoarytenoid joint. Vocal cord is 3.5 mm away from the midline. Abduction and adduction take place from this position. This occurs when both recurrent and superior laryngeal nerves are paralyzed.
- *Gentle abduction*: Vocal cord is 7 mm away from midline such as during quiet respiration and paralysis of adductors.
- *Full abduction*: Vocal cord is 9.5 mm away from midline such as in deep inspiration.

CAUSES OF LARYNGEAL PARALYSIS

The causes of laryngeal paralysis may be divided into supranuclear, nuclear, vagal, superior laryngeal, recurrent laryngeal, neuropathies, neuromuscular and muscular. In about 30% of cases no cause (idiopathic) can be ascertained even after thorough investigations.

- Supranuclear (rare): Strokes, tumor or trauma. Diffuse emboli in cerebral cortex may cause sustained abduction (aphonia) or inappropriate adduction (inspiratory stridor).
- Nuclear (nucleus ambiguus in medulla): Usually associated with paralysis of other cranial nerves and neural pathways.
  - Strokes
  - Tumors
  - Motor neuron disease
  - Polio
  - Syringobulbia
- Vagus nerve
  - Intracranial
    - Tumors of posterior fossa
    - Basal meningitis (tubercular)
  - Jugular foramen (skull base)
    - Fractures
    - Nasopharyngeal cancer
    - Glomus tumor
  - Parapharyngeal space (neck)
    - Penetrating injury
    - Parapharyngeal tumor
    - Metastatic nodes
    - Lymphoma
- Recurrent laryngeal nerve (RLN) or low vagal trunk
  - Neck trauma
  - Thyroid diseases and surgery
  - Cervical esophagus cancer
  - Cervical lymphadenopathy
  - Lung apex cancer
  - Cervical pleura tuberculosis
  - Mediastinal lesions (left side only)
    - Bronchogenic cancer
    - Thoracic esophagus cancer
    - Aortic aneurysm
    - Lymphadenopathy
    - Left atrial hypertrophy or dilatation
    - Intrathoracic surgery
- Systemic causes
  - Diabetes
  - Syphilis
  - Diphtheria
  - Typhoid
  - Streptococcal
  - Viral infections
  - Lead poisoning.
- Idiopathic: In about 30% of cases, cause remains obscure.

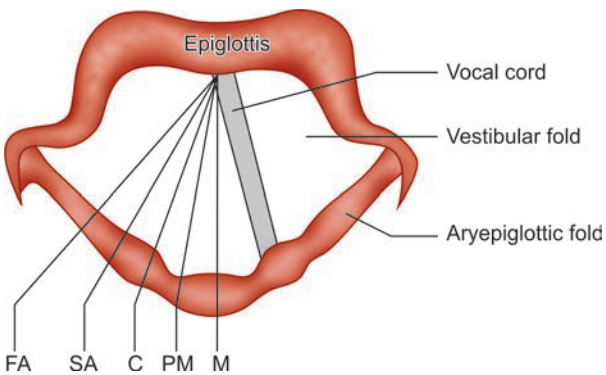

Fig. 1: Vocal cord positions
Abbreviations: M, Median; PM, Paramedian; C, Cadaveric (Intermediate); SA, Slight abduction; FA, Full abduction

UNILATERAL RECURRENT LARYNGEAL NERVE (RLN) PARALYSIS

- It leads to ipsilateral paralysis of all the intrinsic laryngeal muscles except the cricothyroid, which is supplied by the external branch of superior laryngeal nerve.
- The vocal cord may assume a median or paramedian position and does not move laterally (abduction) on deep inspiration.
- Semon's law states that in all progressive organic lesions, abductor fibers of the nerve, which are phylogenetically newer, are more susceptible than adductors that are phylogenetically older. So abductors are first to be paralyzed.
- Wagner and Grossman hypothesis states that cricothyroid muscle (supplied by external branch of superior laryngeal nerve), which has adduction function, keeps the cord in paramedian position.

Etiology

Bronchogenic carcinoma is a common cause. It is excluded by X-ray chest, bronchoscopy and biopsy. It leads to left RLN paralysis. For other causes see etiology section.

Clinical Features

1. About one-third of the patients remain asymptomatic.
2. Changes in voice gradually improve due to compensation. Healthy cord crosses the midline to meet the paralyzed cord.
3. No problems of aspiration or airways obstruction occur.

Treatment

Cases are usually asymptomatic and no treatment is required.

BILATERAL RECURRENT LARYNGEAL NERVE (ABDUCTOR) PARALYSIS

Etiology

The most common causes are neuritis and thyroidectomy operation.

Clinical Features

- The condition is often acute.
- Most common symptoms: Dyspnea and stridor become worse on exertion or infection. They occur due to median or paramedian vocal cords position.
- Voice is good.
- Vocal cords lie in median or paramedian position (due to unopposed action of cricothyroid muscles).

Treatment

For the treatment of dyspnea and stridor, these patients need surgery. Patient is treated with either permanent tracheostomy with a speaking valve or lateralization of the cord. Many patients need emergency tracheostomy when they suffer from upper respiratory tract infection.

- *Tracheostomy*: It relieves stridor and preserves voice but tracheostomy needs regular care.
- *Lateralization of the cord*: In this surgery, vocal cord is moved and fixed in a lateral position. It improves the airway. Though patient does not have tracheostomy but it is at the expense of a good voice. The lateralization of vocal cord may be done with any of the following procedure:
  - Arytenoidectomy: Arytenoid cartilage is excised and the cord is fixed in a lateral position. It may be done with either an external approach or through an endoscope.
  - Thyroplasty Type 2 (see Phonosurgery section).
  - Endoscopic laser cordectomy: This is usually accomplished with CO_2 laser.
  - Nerve muscle implant: Sternohyoid muscle with its nerve supply has been transplanted into the paralyzed posterior cricoarytenoid to bring some movement to the cord. The results are not encouraging.

UNILATERAL SUPERIOR LARYNGEAL NERVE PARALYSIS

Isolated lesions of superior laryngeal nerve are rare. They cause paralysis of cricothyroid muscle and ipsilateral anesthesia of the supraglottic larynx.

Etiology

The involvement of external laryngeal nerve (branch of superior laryngeal nerve) in cases of thyroid surgery, tumors, neuritis or diphtheria causes cricothyroid paralysis.

Clinical Features

- Weak and low pitch voice.
- Occasional aspiration: Unilateral anesthesia of the larynx may pass unnoticed.
- Askew position of glottis (anterior commissure rotated to healthy side).
- Shortening of vocal cord with loss of tension (wavy appearance of paralyzed vocal cord).
- Flapping of paralyzed vocal cord (sags down during inspiration and bulges up during expiration).

BILATERAL SUPERIOR LARYNGEAL NERVE PARALYSIS

In this uncommon condition, both side cricothyroid muscles get paralyzed along with anesthesia of epilarynx (supraglottic).

Etiology

- Surgical or accidental trauma
- Neuritis (mostly diphtheria)
- Pressure by enlarged cervical nodes or neoplasms.

Clinical Features

- Coughing and choking fits during swallowing
- Inhalation of food and pharyngeal secretions
- Weak and husky voice.

Treatment

Chronic and repeated aspiration needs management

- Neuritis cases may recover spontaneously
- Tracheostomy with a cuffed tube
- Esophageal feeding tube
- Epiglottopexy (a reversible procedure) closes laryngeal inlet and protects lungs from repeated aspiration.

UNILATERAL COMBINED (COMPLETE) PARALYSIS OF RECURRENT AND SUPERIOR LARYNGEAL NERVE

There occurs unilateral paralysis of all laryngeal muscles except the interarytenoid, which receives innervation from both the sides.

Etiology

- In thyroid surgery, both recurrent and external laryngeal nerves of one side may be damaged.
- Lesions of nucleus ambiguous may lie in the medulla or posterior cranial fossa.
- Lesions of the vagus nerve proximal to the origin of superior laryngeal nerve in jugular foramen or parapharyngeal space

The healthy cord is unable to approximate the paralyzed cord, which lies in cadaveric position, and results in hoarseness of voice and aspiration.

Clinical Features

- Hoarseness of voice.
- Aspiration of liquids through the glottis.
- Paralyzed vocal cord in cadaveric position (3.5 mm from the midline).

- Glottic incompetence.
- Ineffective cough due to air waste.

Treatment

- *Speech therapy*: The healthy cord may compensate by moving across the midline and approximate paralyzed vocal cord.
- *Medialization of paralyzed vocal cord*: It brings the paralyzed cord towards the midline and than healthy cord can meet it.
  - Injection of teflon paste, collagen, autologous fat, or collagen derivative is injected lateral to the paralyzed cord endoscopically or transcervically.
  - Thyroplasty type 1 (See Phonosurgery section).
  - Muscle or cartilage implant is done through the laryngofissure approach a bipedicled muscle graft or piece of cartilage is inserted between thyroid cartilage and vocal cord. The graft pushes the cord medially.
  - Arthrodesis of cricoarytenoid joint is done by opting for laryngofissure approach where arytenoid cartilage is rotated medially and fixed with a screw.

BILATERAL COMBINED (COMPLETE) PARALYSIS OF RECURRENT AND SUPERIOR LARYNGEAL NERVE

In this rare condition, there occurs total anesthesia of the larynx and all laryngeal muscles are paralyzed and both vocal cords lie in cadaveric position.

Clinical Features

- *Aphonia*: Vocal cords do not approximate
- *Aspiration*: Because off incompetent glottis and laryngeal anesthesia
- *Inability to cough*: It results in retention of secretions in the lungs
- *Bronchopneumonia*: Because of repeated aspirations and retention of secretions.

Treatment

Surgical treatment is preferred to manage the problem of aspiration. The surgical options include following:

- Reversible surgeries
  - Tracheostomy: Facilitates removal of pulmonary secretions and inhaled material.
  - Epiglottopexy: It prevents aspiration into the lungs. Epiglottis is folded backwards and fixed to the arytenoids.
  - Vocal cord plication: Through the laryngofissure approach, true and false cords are approximated with sutures after removing their mucosa. This surgery prevents aspiration and can be reversed.
  - Laryngotracheal separation
  - Tracheoesophageal diversion
  - Endolaryngeal stent
  - Partial cricoidectomy
  - Vertical laryngoplasty
- Irreversible surgeries.
  - Total laryngectomy: It is suggested in cases of progressive and irreversible lesions and when speech is unserviceable
  - Subperichondrial cricoidectomy
  - Glottic closure.

CONGENITAL VOCAL CORD PARALYSIS

Laryngomalacia is the most common cause of neonatal stridor. Vocal fold palsy is the second common cause of neonatal stridor. Congenital unilateral paralysis is more common than bilateral. Bilateral palsy children have features of bilateral abductor paralysis and respiratory obstruction and need tracheostomy.

Etiology

- Idiopathic
- Unilateral
  - Birth trauma: Intracerebral hemorrhage
  - Congenital anomaly of a great vessel or heart
- Bilateral paralysis
  - Hydrocephalus
  - Arnold-Chiari malformation
  - Meningitis
  - Cerebral or nucleus ambiguus agenesis.

Clinical Features

- The high-pitched musical stridor is either inspiratory or biphasic.
- The cry is week.

Diagnosis

1. Awake flexible laryngoscopy will confirm the diagnosis.
2. MRI brain scan in idiopathic cases to rule out Arnold-Chiari malformations.

Treatment

About 70% noniatrogenic unilateral vocal fold palsies and 50% children of bilateral vocal-cord palsies improves spontaneously.

- Tracheostomy is needed in 73% of the patients.
- Feeding can be done through slow-flow nipple or nasogastric feeds.
- For severe aspiration and dysphonia in unilateral palsy.
  - Injection of vocal fold with absorbable gelatin sponge (gelfoam) or polytetrafluoroethylene (teflon)
  - Medialization laryngoplasty (thyroplasty)
- For bilateral palsy
  - Endoscopic lateral cordotomy
  - Arytenoidectomy: Endoscopic or external cervical approach
  - Arytenoidopexy: External cervical approach
  - Expand cricoid cartilage posteriorly with costal cartilage graft.

PHONOSURGERY

These surgical procedures improve the quality of voice and include microlaryngeal surgery, local injections, thyroplasty, reinnervation and reconstructive procedures.

Microlaryngeal Surgery

Excision of vocal cord lesions is done with microlaryngeal surgery or laser. (See Chapter on Endoscopies).

Local Injections

Injection of teflon paste, collagen, autologous fat, collagen derivative or gelfoam can be used to medialize and augment the vocal cord. Injections may be given percutaneous or transorally under the guidance of flexible laryngoscope or through microlaryngoscopy.

Laryngeal Framework Surgery (Thyroplasty)

Isshiki described four types of thyroplasty procedures, which produce functional alteration of vocal cords.

1. *Type I thyroplasty*: Medial displacement of vocal cord to improve voice quality.
2. *Type II thyroplasty*: Lateral displacement of vocal cord to improve the airway.
3. *Type III thyroplasty*: Shortening or relaxation of the vocal cord. Relaxed vocal cord lowers the pitch. The surgery is usually done in cases of mutational falsetto or gender transformation.
4. *Type IV thyroplasty*: Lengthening or tensioning of the vocal cord to elevate the pitch. Male character voice becomes female character. This procedure is done in cases of gender transformation, lax vocal cord, and vocal cord bowing due to aging or trauma.

Laryngeal Reinnervation Procedures

Through a window in thyroid cartilage, a segment of superior belly of omohyoid muscle along with its nerve (ansa hypoglossi) and vessels is implanted into the thyroarytenoid muscle. This procedure is supposed to innervate the paralyzed thyroarytenoid muscle.

Reconstructive and Rehabilitative Procedure

They are needed after tumor resection.

Clinical Highlights

1. **Glottic chink:** In cadaveric, position of the vocal cords vocal chink is 14 mm. In full abduction it is about 19 mm. During the whisper, the position of vocal cord is paramedian.
2. **Vocal cord palsy:** It is commonly caused by lesions of recurrent laryngeal nerve. Because of the intrathoracic course of left recurrent laryngeal nerve left vocal cord palsy is twice more common than right vocal cord palsy.
3. **Combined paralysis of bilateral superior and recurrent laryngeal nerves:** Patient will present with complete loss of speech but there will be no difficulty in breathing because the vocal cords lie in cadaveric position.
4. **Ortner's syndrome:** Paralysis of left recurrent laryngeal nerve in cases of cardiomegaly.
5. **Superior mediastinum tumor:** It impinges upon the arch of aorta and compresses left vagus nerve just before the origin of left recurrent laryngeal nerve. Patient will develop hoarseness of voice due to the left vocal cord palsy.
6. **Aspiration:** Tracheal division and permanent tracheostome is the "Gold standard" surgical treatment for prevention aspiration.
7. **Thyroplasty Type I:** Medialization of vocal cord.
8. **Thyroplasty Type II:** Lateralization of vocal cord.
9. **Thyroplasty Type III:** Shortening (relaxing) in the length of vocal cord by decreasing the distance between the anterior and posterior attachments of vocal cords. It is indicated for gender transformation, mutational falsetto and spasmodic dysphonia. Relaxed vocal cords lower the pitch of voice.
10. **Thyroplasty Type IV:** Lengthening (tightening) of vocal cord. It corrects the androphonia. The male character low pitch voice is converted to female high pitch voice.

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49 Voice and Speech Disorders

Can anything be done unless everybody exerts himself to his utmost? 'It is the man of action, the lion-heart that the Goddess of Wealth resorts to.' No need for looking behind. FORWARD! We want infinite energy, infinite zeal, infinite courage, and infinite patience, and then only will great things be achieved.

—Swami Vivekananda

Points of Focus

- **VOICE AND SPEECH**
  - ✓ Phonation
  - ✓ Resonance
  - ✓ Articulation
- **CLASSIFICATION OF VOICE AND SPEECH DISORDERS**
  - ✓ Disorders of Speech
  - ✓ Disorders of Voice
  - ✓ Hyperfunctional Disorders
  - ✓ Hoarsness
- **DYSPHONIA PLICA VENTRICULARIS**
- **FUNCTIONAL APHONIA**
- **PUBERPHONIA**
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- **HYPONASALITY**
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- **SPASMODIC DYSPHONIA**
  - ✓ Adductor Spasmodic Dysphonia
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  - ✓ Factors Relieving Stuttering
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- **MYOCLONUS**
  - ✓ Branchial or Oculopalatal Myoclonus
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  - ✓ Types
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  - ✓ Abductor Laryngeal Injection
- **CLINICAL HIGHLIGHTS**

VOICE AND SPEECH

Speech is an audible communication and consists of phonation, resonance and articulation.

Expired air and pressure and induced vocal fold vibrations produce phonation, which is modified into voice by the resonance of head, neck, and chest, and motor activity of pharynx and oral cavity. Articulation, which is performed by lips, tongue, palate, pharynx and larynx, shapes the voice into words.

Phonation

Expired air and pressure induce vocal folds vibrations. Factors affecting phonation include:

- *Appropriate Approximation of Vocal Folds*
  - Too tight closure of vocal folds needs excessive expiratory force and makes the voice:
    - Strained
    - Harsh
    - Complete aphonia.

- Too far apart vocal folds need increased expiratory airflow volume and voice becomes
  - Weaker and Whisper (lack of volume)
  - Breathier.
- *Three-Dimensional Shape of Vocal Folds*: Atrophy of vocal folds may lead to
  - *Concavity in axial planes*: Results in incomplete glottal closure.
  - *Concavity in coronal planes*: Results in convergent airflow tract.
- *Length and Tension of Vocal Folds*: The control of length and tension facilitates normal inflections in pitch and tone and lack of control may make the voice:
  - Flat and expression less: Lack of voice inflection.
  - Distorted by uncontrolled pitch breaks.
- *Suppleness of Vocal Fold Mucosa*: It provides free vibrations.

Resonance

The resonance, which is modification of phonation, produces voice. Unmodified phonated sound of vocal folds is strident and unpleasant noise. The principles of modification, which consists of selective amplification and dampening of component frequencies, are used in speech therapy.

- Factors affecting vocal resonance include:
  - Resonance of head, neck and chest
  - Motor activity of pharynx and oral cavity.
- Conditions altering vocal resonance include:
  - Amyotrophic lateral sclerosis
  - Stroke
  - Guillain-Barre syndrome (acute idiopathic polyneuropathy).

Articulation

Articulation, which shapes the voice into words, is done by lips, tongue, palate, pharynx and larynx. The impaired motor output (weakness, palsy, or incoordination), cognitive or language defects may cause dysarthria.

CLASSIFICATION OF VOICE AND SPEECH DISORDERS

Disorders of Speech

- *Dysphasia or Aphasia*: Impaired comprehension or production of speech due to a lesion of dominant cerebral hemisphere.
  - *Receptive (sensory) dysphasia*: Impairment in comprehension of words. Patient appears unaware of deficit. Lesion lies in auditory or visual association areas of cerebral cortex.
    - Auditory aphasia (word deafness): Though hearing is normal, patient cannot comprehend spoken word. Spontaneous speech, reading and writing are normal.
    - Visual aphasia or word (text) blindness or alexia: Though vision is normal patient is unable to comprehend meaning of written word.
    - Jargon aphasia (agrammatism): Severe sensory aphasia and speech is incomprehensible.
  - *Expressive (motor) dysphasia or Broca's aphasia*: Difficulty in speech production or language output (writing or sign). Patient is aware of deficit and can read and hear normally. Lesion lies in cortical motor speech Broca's area. Patient knows what to say but cannot speak. Patient can understand printed words but cannot read aloud. Some patients may write fluently.
  - *Conduction aphasia*: Skipping or repeating words or substituting one word for another (paraphasia). Patient is aware of deficit.
  - *Nominal or amnesic aphasia (anomia)*: Difficulty in naming objects and persons seen, heard, or felt.
  - *Mixed or global or total aphasia*: All aspects of speech and communications are impaired.
- *Dysarthria*: Disturbance of articulation due to faulty working of lips, tongue, palate, pharynx, and larynx.
  - *Supranuclear lesions (corticobulbar tract)*: Pseudobulbar palsy, cerebral diplegia, degenerative or ischemic changes.
  - *Extrapyramidal:* (parkinsonism) Vocal strain, voice arrests, pitch breaks, and pitch instability.
  - *Cerebellar*: Slurring and scanning speech.
  - Nuclear lesions
  - Infranuclear lesions
  - Muscular lesions.

Disorders of Voice

- *Dysphonia*: General change in voice quality.
  - Dysphonia plica ventricularis.
- *Aphonia*: No sound is emanated from vocal folds when there is lack of vocal cord approximation or lack of air passing through the glottis.
  - Functional (Hysterical)
  - Organic.
- *Diplophonia*: When vocal cords are affected differently they vibrate at two different frequencies.
- *Puberphonia* (mutational falsetto voice).
- *Phonasthenia*
- *Disorders of Resonance*
- *Rhinolalia*
  - Rhinolalia aperta (Hypernasality)
  - Rhinolalia clausa (Hyponasality).

Hyperfunctional Disorders

- Stammering (stuttering)
- Spasmodic dysphonia
- Tremors
- Tic disorders
- Myoclonus.

Stridor

It is the noisy inspiration that occurs due to an upper airway obstruction. (*see* chapter Laryngeal Symptoms and Examination).

Hoarseness

Any change in voice quality from harsh, rough or raspy voice to weak voice is usually referred as hoarseness. It is caused by

laryngeal dysfunction (variations of periodicity and intensity of consecutive sound waves). In cases of limited lungs or tracheobronchial tree diseases voice becomes weak and damp. For further details kindly see chapter Laryngeal Symptoms and Examination.

DYSPHONIA PLICA VENTRICULARIS (VENTRICULAR DYSPHONIA)

In this condition ventricular folds (false cords) take over the function of true cords.

Etiology

- *Secondary*: In cases of impaired function of the true cord (paralysis, fixation, surgical excision, or tumors), ventricular bands try to compensate phonatory function of true cords.
- *Functional*: Psychogenic.

Clinical Features

- *Voice*: Rough, low-pitched and unpleasant voice.
- *In functional cases*: In beginning voice is normal but becomes rough later when false cords usurp the function of true cords.

Laryngoscopy

False cords approximate partially or completely on phonation and obscure the view of true cords.

Treatment

- *Secondary*: Ventricular dysphonia is difficult to treat.
- *Functional Type*: Voice therapy and psychological counseling.

FUNCTIONAL APHONIA (HYSTERICAL APHONIA)

- This functional disorder is usually seen in emotionally labile young females.
- The sudden onset aphonia is not associated with other laryngeal symptoms.
- Patient is usually able to whisper.

Laryngoscopy

Vocal cords remain in abducted position and fail to adduct on phonation. The adduction of vocal cords occurs on coughing and sound of cough is good.

Treatment

Reassurance and psychotherapy.

PUBERPHONIA (MUTATION FALSETTO VOICE)

Failure in the change of childhood high-pitched voice to low-pitched male voice after puberty in boys is called puberphonia. However the boy's physical and sexual development is normal.

Pertinent Anatomy

Until puberty, the larynx of male and female have identical dimensions. Childhood voice has higher pitch. After puberty, male larynx grows rapidly. Increase in length of rima glottides (vocal cords lengthen) brings change in character of male voice (voice becomes lower pitch). The female larynx changes a little.

Risk Factors

These boys are emotionally immature, feel insecure and show excessive fixation to their mother or sister. Psychologically, these boys try to avoid male responsibilities.

Treatment

- Speech therapy and psychotherapy.
- Through Gutzmann's pressure test the thyroid prominence is pressed in a backward and downward direction. It relaxes the overstretched cords and low tone voice is produced. In this way boy can learn to produce low tone voice and trains himself to produce syllables, words and sentences.
- Prognosis is usually good.

PHONASTHENIA

This is fatigue of phonatory muscles (especially thyroarytenoid and interarytenoid) and results in weakness of voice.

Risk Factors

Abuse or misuse of voice or following laryngitis.

Clinical Features

- Easy fatigability of voice.
- Indirect laryngoscopy:
  - Elliptical space between the cords (thyroarytenoid weakness).
  - Triangular gap near the posterior commissure (interarytenoid weakness).
  - Keyhole appearance of glottis due to combination of both elliptical space between the cords and triangular gap near the posterior commissure (weakness of both thyroarytenoid and interarytenoid).

Treatment

- Voice rest and vocal hygiene.
- Periods of voice rest after excessive use of voice.

HYPONASALITY (RHINOLALIA CLAUSA)

Blockage of the nose or nasopharynx results in lack of nasal resonance for words and is called hyponasality.

Etiology

- Rhinosinusitis (*see* Chapter on Rhinosinusitis)
- Allergic and non-allergic rhinitis
- Nasal masses such as polyps and tumors
- Nasopharyngeal mass and adenoids
- Familial or habitual speech pattern.

HYPERNASALITY (RHINOLALIA APERTA)

The failure of the nasopharynx to cut off from oropharynx or undue passage between the oral and nasal cavities results in nasal resonance of all the words (even those words, which have little nasal resonance).

Etiology

- Velopharyngeal insufficiency:
  - Short soft palate: Congenital or acquired (submucous fibrosis)
  - Cleft palate or submucous cleft palate
  - Paralysis of soft palate
  - Palatal perforation
  - Large nasopharynx
  - After adenoidectomy
- Oronasal fistula
- Familial or habitual speech pattern.

SPASMODIC DYSPHONIA

Dystonia is characterized by abnormal involuntary movements, which are typically action induced. Spasmodic dysphonia, focal dystonia of larynx is an action induced laryngeal motion disorder.

Synonyms

Spastic dysphonia, spastic aphonia, phonic laryngeal spasm and coordinated laryngeal spasm.

Types

There are three types of spasmodic dysphonia: adductor, abductor and mixed.

1. Adductor is caused by irregular hyperadduction of the vocal folds.
2. Abductor is caused by intermittent abduction of the vocal folds.
3. Mixed type displays a combination of adductor and abductor signs.

Clinical Features

- The vocal cords are usually normal at rest but function abnormally with speaking.
- Reduced loudness of voice.
- Vocal tremor similar to essential tremor.
- Decreased speech intelligibility.
- Symptoms worsen when patient is under stress or on the telephone.
- Associated other cranial forms of dystonia may include spasms of the eyelids and contractures of the pharynx, jaw muscles and muscles of the floor of mouth and tongue.
  - *Factors improving symptom*: Factors, which may improve the symptom temporarily, include following:
  - Pinching of nose
  - Pressing the hand against the back of the head or into abdomen
  - Pulling on ear
  - Yawning, sneezing, singing, yelling, or humming.

Adductor Spasmodic Dysphonia

Abnormal involuntary co-contraction of the vocalis muscle complex results in inappropriate adduction of the vocal folds.

- Choked, strained strangled and monotonal voice.
- Abrupt initiation and termination result in short breaks in phonation.
- Slow and harsh speech.
- Occasionally, compensatory pseudoabductor spasmodic dysphonia compensates for severe adductor laryngeal spasms by whispering.

Abductor Spasmodic Dysphonia

Abnormal co-contraction of the posterior cricoarytenoid muscles results in inappropriate abduction of the vocal cords.

- Breathy and effortful voice
- Abrupt termination results in aphonic whispered segments of speech.

Treatment

- Voice therapy
- Alcohol, sedatives and tranquilizers provide transient improvement. Generally medical treatment is of little avail.
- Sectioning the recurrent laryngeal nerve
- Selective denervation and reinnervation using the ansa cervicalis nerve
- Thyroarytenoid myotomy
- Injections of Botox or collagen.

VOCAL TREMOR

The involuntary, rhythmic and oscillatory movements can affect the muscles of speech production and generate rhythmic alterations in pitch and loudness. It has been described as tremulous or wavy voice or tremulous quavering speech.

Clinical Features

- Vocal tremor can result in rapid decreases and increases in loudness and pitch.
- Tremulous or wavy voice.
- Tremulous quavering or slow speech.
- Pitch breaks (octave breaks to a lower frequency) affects intelligibility of the speech and may lead to aphonia.
- Symptoms increases with emotional stress and fatigue.
- Vocal tremor may be associated with essential tremor, Parkinson's disease, cerebellar ataxia, or flaccid dysarthria.

Treatment

- *Medications*: Propranolol, primidone, clonazepam and diazepam may reduce vocal tremor but are not well proved.
- Thalamotomies
- Chronic electric stimulation of the thalamic nucleus through surgically implanted neuropacers has been reported.
- Local injection of botulinum toxin in the sternohyoid and sternothyroid muscles (5 units in 2.5 unit aliquots are injected on each side). Vocal folds if tremulous are injected at a second session.

STUTTERING

Stuttering is a neurologic, movement disorder in which abnormal, involuntary, and inappropriate use of the speech muscles results in dysfluency. Stuttering is a result of increased muscle tension in the three subsystems of speech (respiratory, phonatory and articulatory). Muscles move too quickly and too far. Other cranial

musculature (such as eyelids and muscles of facial expression) may also inappropriately contract.

Risk Factors

Too much attention or reprimands to childhood dysfluency between 2–4 years.

Prevention
Educate parents not to overreact to child's dysfluency in early stages of speech development.

Clinical Features

- This disorder of speech fluency is characterized by hesitation to initiate, repetitions, prolongations or blocks in speech flow.
- The patient later on may develop secondary mannerisms, which include facial grimacing, eye blink or abnormal head movements.

Factors Relieving Stuttering

Factors, which may increase fluency for a period of time, are:

- Emotional arousal or sensory stimuli
- Motor actions such as walking
- Use of rhythmic patterns such as a metronome or monotone.
- Singing or speaking in a sing-song voice
- Shouting
- Foreign accent or slurred articulation.

Factors Aggravating Stuttering

The factors, which may increase stuttering, include communicative pressures such as public speeches, personal interviews, counseling and meaningful negotiations.

Treatment

- Speech therapy and training
- Antidepressants though given are of no value.
- Small doses of injection botulinum toxin (1 unit or less, bilaterally) produce improvement in 50% cases.
- "SpeechEasy": Using both delayed and frequency altered auditory feedback in the ear devices (a frequency shift of +500 Hz with delayed auditory feedback of 60 m/sec) has shown significant improvement in fluency and normalcy of speech in both youth and adult subjects. It is marked under the trademark "SpeechEasy".

MYOCLONUS

Myoclonus is a disease of central nervous system. It consists of sudden, brief, shock like involuntary movements, which are caused by either muscular contractions (positive myoclonus) or inhibitions (negative myoclonus, asterixis).

Laryngeal Features

- Broken speech pattern and respiratory dysrhythmia (ventilatory dysfunction).
- Vocal cords often show slow rhythmic adduction and abduction at the same timing and frequency as the palatal, pharyngeal and occasionally diaphragmatic contractions.

Treatment

- *Pharmacotherapy*: Though given serotonin, carbamazepine, clonazepam, tetrabenazine, and trihexyphenidyl are generally unresponsive.
- Local injection of botulinum toxin into the thyroarytenoid muscles has been tried successfully.

Branchial or Oculopalatal Myoclonus

It presents with involuntary movement of the soft palate and pharynx, jerks affecting the eyes, face, palate, larynx, diaphragm, neck, shoulder and arm.

Tinnitus

Clicking in the ears is caused by contractions of tensor palati muscle.

TOURETTE'S SYNDROME

This tic disorder is characterized by involuntary vocalizations of articulate words or inarticulate sounds. It may be associated with multiple tics of several body parts. Lingual tics present as hisses and nasal tics as sniffs and snorts.

Clinical Features

- Onset may be in childhood or adolescence.
- Obsessive-compulsive behavior.
- Laryngeal tics: Inappropriate coughing, barking, throat clearing, hooting and grunting.
- The chronic voice abuse may make the voice harsh and results in polypoid changes of the vocal mucosa.

Treatment

- Phenothiazines such as haloperidol.
- Alpha-2-adrenergic agonist such as clonidine.
- Benzodiazepine such as clonazepam.
- Local injections of botulinum toxin to manage rapid facial tics and dystonic tics such as refractory loud barking sounds.

BOTULINUM TOXIN THERAPY

Injection botulinum toxin is said to be a safe and effective therapy for laryngeal dystonia. The graded weakening can be achieved using low doses. Repeating injections achieve optimal weakness.

Botulinum toxin acts at the neuromuscular junction by inhibiting the release of acetylcholine and causes a flaccid paralysis. The steps involved in toxin-mediated paralysis are binding, internalization and inhibition of neurotransmitter release.

The clinical effect of toxin is caused mainly by the peripheral effect. The degree of improvement correlates with weakness due to blockade of neuromuscular transmission. There is 24–72

hour delay between administration of toxin and onset of clinical effect. This time is necessary for the adequate enzymatic disruption of the synaptosomic release process of botulinum toxin.

Types

The currently available, Botulinum toxins are of two types:

1. Type A (Botox), and
2. Type B (Mycobloc).

Frozen lyophilized toxin is reconstituted with sterile saline to various concentrations, as per the need and the indication.

Needle

Injections are given using a tuberculin syringe with a 27-gauge monopolar polytef coated hollow electromyography (EMG) recording needle.

Adductor Laryngeal Injections

Encouraging results (90%) are reported in adductor spasmodic dysphonia but duration of effect lasts between 3–4 months.

- *Percutaneous*: They are performed percutaneously through the cricothyroid membrane and into the thyroarytenoid-vocalis muscle complex using EMG guidance for optimum placement.
- *Indirect Laryngoscopy*: Through an indirect laryngoscopic approach, toxin may be injected into the vocal fold. It does not require special EMG equipment or training.
- *Side Effects*: Patients report improvement in voice within 24 hours followed by–
  - A breathy, hypophonic period lasting 1–2 weeks (45%).
  - Occasionally causing hyperventilation and dizziness when trying to speak.
  - Mild choking on fluids for the first several days (22%).
  - Sore throat or coughing up blood-tinged sputum may occur in the first 1–2 days after injection.

Abductor Laryngeal Injections

- *Percutaneous*: They are managed with an EMG-guided percutaneous injection. The larynx is rotated away from the side of intended injection. A hollow EMG needle with syringe is placed posterior to the posterior edge of the thyroid lamina. The needle is advanced to the cricoid cartilage and moved out under EMG guidance to the optimum position in the posterior cricoarytenoid muscle. Sniffing maximally uses the posterior cricoarytenoid muscle and shows a burst of activity on the EMG. Then toxin is injected.
- *Transcricoid*: This technique is used in the patients who cannot tolerate manual rotation of the larynx. A small amount of anesthetic agent is injected in the subglottic airway via the cricothyroid membrane. The needle is placed through the cricothyroid membrane, traversing the airway until it engages the posterior lamina of the cricoid cartilage. The needle is slowly advanced slightly laterally through the cartilage under EMG control. The posterior cricoarytenoid muscle is encountered and sniffing shows a burst of electric potentials. Then toxin is injected.
- *Dosages*: 2.5–10 units of Botox in 0.1 mL of saline per posterior cricoarytenoid muscle.
- *Unilateral/Bilateral*: Unilateral injections minimize the risk of airway obstruction. Flexible fiberoptic examination shows the degree of abduction on the treated side. If there is no impairment in motion, repeat the injection. If impairment in motion is observed without clinical response, bilateral injections may be needed. The contralateral injection is given only after the return of function on the injected side (usually after 3–4 months) and lower dosages (1.0–2.0 units/0.1mL) administered.

Clinical Highlights

1. **Broca's aphasia (motor or nonfluent aphasia):** This apparent language problem is localized adjacent to the primary motor cortex in the frontal lobe of left (usually dominant) cerebral hemisphere. Patient's speech lacks rhythm and is reduced to the use of nouns and verbs in the wrong tense. Patient can hear and understand the speech. Patient can repeat single words but not a full sentence. Patient may have associated weaknesses in right lower face and right upper limb.
2. **Wernicke aphasia (sensory or fluent aphasia):** Patient cannot comprehend the spoken word. The patient can hear but his complaint is hard of hearing. Patient looks at the doctor as if s/he does not understand what the doctor is saying. Though patient's speech is intact but usually it does not make any sense because of the misuse of words. Patient is not aware of his speech or comprehension problems.
3. **Conductive aphasia:** This language problem involves an impaired ability to repeat an examiner's commands. The speech is fluent and comprehension remains intact. The lesion of this limited language deficit is in the arcuate fasciculus, which connects the Wernicke and Broca areas.
4. **Stuttering:** It is the most common type of speech disorder.
5. **Dysphonia plica ventricularis:** Phonation is produced by the false cords.
6. **Functional aphonia:** It always affects both cords, which fail to adduct on phonation. But vocal cords adduct on coughing.
7. **Phonasthenia:** Clinical features include throat pain, easy fatigability of voice, and a triangular gap on phonation in the interarytenoid.
8. **Gutmann's pressure test:** Pressure on the thyroid prominence in a backward and downward direction improves the voice in cases of puberphonia. The pressure relaxes the over-stretched vocal cords and low-pitched voice can be produced.

FURTHER READING

1. Management of spastic dysphonia. Readers Forum 56. Indian J Otolaryngol Head Neck Surg. 2010;62:335-7.
2. Vaidya S, Vyas G. Puberphonia: A novel approach to treatment. Indian J Otolaryngol Head Neck Surg. 2006;58:20-1.
3. How do you investigate a case of aphonia and how would you treat the same? Reader's Forum Indian J Otolaryngol Head Neck Surg. 2005;57:371-2.

50 Malignant Tumors of Larynx

Say, 'This misery that I am suffering is of my own doing, and it will have to be undone by me alone.' Stand up, be bold, and be strong. Take the whole responsibility on your own shoulders, and know that you are the creator of your own destiny. All the strength and succor you want is within you.

—Swami Vivekananda

Points of Focus

- **INTRODUCTION**
- **RISK FACTORS**
  - ✓ Premalignant Lesions
  - ✓ Preventive Measures
- **EVALUATION**
  - ✓ Clinical
  - ✓ Imaging Study
  - ✓ Detection of Recurrent/Residual Disease
  - ✓ Differential Diagnoses
  - ✓ Indirect Laryngoscopy
  - ✓ Direct Laryngoscopy Biopsy Under General Anesthesia
  - ✓ Histopathology
- **STAGING**
- **MANAGEMENT**
  - ✓ Radiotherapy
  - ✓ Surgery
  - ✓ Combined Therapy
  - ✓ Palliation Therapy
- **GLOTTIC CANCER**
- **SUPRAGLOTTIC CANCER**
- **SUBGLOTTIC CANCER**
- **VERRUCOUS CARCINOMA**
- **ORGAN PRESERVATION THERAPY**
  - ✓ Chemotherapy Protocols
- **PHOTODYNAMIC THERAPY (PDT)**
- **POST-LARYNGECTOMY VOCAL REHABILITATION**
  - ✓ Esophageal Speech
  - ✓ Artificial Larynx
  - ✓ Tracheoesophageal Fistula Speech
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

Cancer of the larynx is quite common in India. Indians are affected more than the people in Western countries. Indian Council of Medical Research (ICMR), National cancer registry reports (April 2005), state that laryngeal cancer accounts for 2.63% of the whole body cancers. It reports incidence of 3.61 new cases per 100,000 of the populations. Male-to-female ratio is 10:1. Incidence in females is increasing as more women are taking to smoking. Patients are mostly in the age group of 40–70 years, but younger people may also be affected. Tumors of larynx can extend to hypopharynx and vice versa. Evaluation of persistent hoarseness is very important, and made easy with in-office endoscopic examination of larynx.

RISK FACTORS

Alcohol and smoking together increase the risk many-fold, which is otherwise just 2–3 times riskier with alcohol or smoking alone.

- *Smoking:* It is a major and the most common risk factor. Cigarette smoke contains benzopyrene and other hydrocarbons, which are carcinogenic. Risk increases with increasing tobacco use. Filtered and light (air-cured) tobacco has 50% lower risk than flue-cured or black tobacco. Rolled cigarettes are more riskier than commercially packaged cigarettes. Cigarettes with greater than 22 mg tar have double the risk. The risk decreases with time, after cessation of smoking. Passive smoking has also been implicated.

- *Alcohol:* Beer and hard liquor are riskier than wine.
- *Radiation:* Previous radiation to neck such as radiation for laryngeal papilloma may induce laryngeal carcinoma.
- *Familial*: Japanese and Russian workers have familial laryngeal malignancy that suggests some incriminating genetic factors.
- *Occupational and Chemical*: Exposure to asbestos, nickel compounds, certain mineral oils, mustard gas, petroleum products, glass-wool, or other chemicals have been incriminated in the development of laryngeal cancer.
- *Racial*: Blacks are twice at risk than white Americans.
- *Genetic*: Either inactivation of tumor suppressive genes or activation of proto-oncogenes occurs. Aneuploidy and genetic alterations of chromosomal region 9p21 (affect p16 gene) and 17p13 (affect p53) have been incriminated. Activation of a proto-oncogene 11q13 is said to amplify oncogene cyclin D1.
- *Human Papillomavirus :* About 40% laryngeal cancer specimens were found Human papillomavirus (HPV)-positive. HPV (oncogenic Type 16) has been associated with head and neck cancer.
- *Head and Neck Cancer*: High incidence of second primary.
- *Dietary*: Phenols in drink and forms of tea such as mate (Latin America) and chimera (Brazil) have been recognized risk factors.
- *Gastroesophageal Reflux:* Majority of cancer patients have associated reflux. Gastric acid reflux leads to chronic laryngeal irritation.
- *Alkaline Reflux:* Achlorhydria and presumed alkaline reflux (identified as laryngeal carcinogen) after gastric resection.
- *Premalignant Conditions:* Leukoplakia, hyperkeratosis with atypia, and carcinoma in situ.

Premalignant Lesions

The five laryngeal squamous abnormalities, which run from benign to malignant, are as follows:

- Hyperkeratosis
- Hyperkeratosis with atypia
- Carcinoma in situ (CIS)
- Superficially invasive carcinoma
- Invasive carcinoma.

Hyperkeratosis with atypia and CIS are managed satisfactorily by removing the strip of cord. Differentiation between severe hyperkeratosis with atypia and CIS is often difficult.

Preventive Measures

They include the following:

- Smoking cessation
- Reduced alcohol drinking
- Toxin free work environment
- Healthy diet.

EVALUATION

Clinical

- Evaluate the hoarseness that persists for longer than 4 weeks. Large supraglottic or subglottic tumors may present with airway obstruction without any history of hoarseness of voice.
- Check for vocal cord mobility. Impairment or fixation occurs due to infiltration of thyroarytenoid muscle, cricoarytenoid joint or recurrent laryngeal nerve. It is an important finding for staging, management and prognosis of laryngeal cancer.
- Evaluate the extent of disease. Vallecula, base of tongue and pyriform fossa should be examined.
- A drawing showing the extent of disease must be made.
- Examine the neck for (Figs 1 and 2)
  - Perichondritis and involvement of thyroid gland and strap muscles.
  - Lymph node metastasis: Note the size and number; whether mobile or fixed and unilateral, bilateral or contralateral.

Imaging Study

- *X-ray Chest PA View:* See for tuberculosis, pulmonary metastasis, mediastinal nodes or bronchogenic carcinoma.

Fig. 1: Carcinoma supraglottic larynx with secondary metastasis in neck involving levels I-B, II, III, IV and V. Note the healed tracheostomy opening

Fig. 2: Carcinoma supraglottic larynx with secondary metastasis in neck nodes. Note the widening of thyroid cartilage and healed tracheostomy opening

- *Computed Tomography or Magnetic Resonance Imaging* (Fig. 3): See for the extent of tumor, invasion of preepiglottic or paraepiglottic space, destruction of cartilage, and lymph node involvement. Nodes greater than 1 cm or nodes with central necrosis are noted. Thyroid or cricoid cartilage destruction is better seen in CT. MRI T2W better highlights submucosal extension into the preepiglottic and paraglottic spaces. For the CT anatomy of larynx, see chapter Diagnostic Imaging.
  - *Whole-lung CT* is needed when X-ray chest shows any abnormality.

Imaging of the neck and airway before biopsy and operative endoscopy recognizes the need for emergent tracheostomy. A debulking biopsy with carbon dioxide laser or forceps may prevent the need for tracheostomy.

Detection of Recurrent/Residual Disease

Early recognition becomes difficult due to mucositis and edema of supraglottic laryngeal structures. Biopsy increases the chances of chondritis. The following imaging methods help:

1. *PET Scan*: It is an accurate and effective means of showing recurrent and residual disease, and distant metastases.
2. *CT Scan*: It is more accurate than clinical examination.

Differential Diagnoses

The gross appearance of following conditions may be mistaken for malignancy. Biopsy will confirm the diagnosis.

- Fungal laryngitis
- Sarcoidosis
- Tuberculosis
- Wegener's granulomatosis

Indirect Laryngoscopy

- *Mirror examination* offers best color and depth perception. Saliva and secretions should be swallowed. An intense gag reflex and difficult patient and anatomy may require supplemental examination.

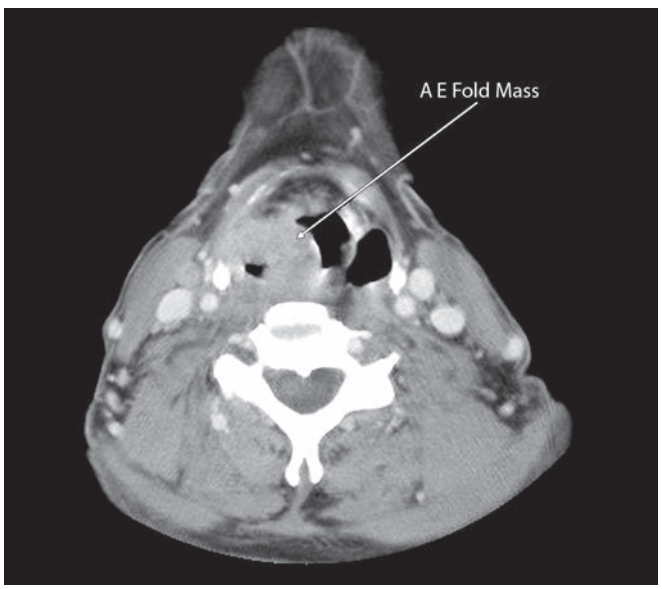


Fig. 3: CT scan neck. AE fold mass.
Source: Courtesy Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

- *Endoscopy*: Mirror examination is usually supplemented by *fiber-optic rigid* (70° or 90° telescope) or *nasal flexible laryngoscopy*; the large channel of which even allows small biopsy without general anesthesia. If needed, the preparations for emergency tracheostomy should be kept ready. *Any cartilage removed at tracheostomy should be submitted for histopathologic examination.*
- *Photo or Video Documentation*: It allows dynamic study of laryngeal function.
- *Stroboscopy*: Normal mucosal wave indicates that lesion is superficial, and does not spread to the vocal ligament.

Direct Laryngoscopy Biopsy under General Anesthesia

It provides better view of the extent of disease and certain hidden areas, which are not seen during mirror examination such as infrahyoid epiglottis, anterior commissure, subglottis and ventricle. Telescopes may be passed through the scope for assessing subglottic and anterior commissure regions.

- *Microlaryngoscopy:* Small lesions of vocal cords are better visualized under operative microscope. It provides more accurate biopsy specimens without damaging the cord. Cord fixation is differentiated from arytenoid fixation by palpating the vocal process. The ventricular extension of growth is assessed with a laryngeal probe.

  The complete excisional biopsy of small suspected lesion with border of healthy tissue helps in determining the depth of invasion, which is difficult with small biopsy specimen of early lesions.
- *Supravital Staining and Biopsy:* Toluidine blue or Lugol's solution applied to the lesion is washed with saline. The dye is taken up in cases of CIS and superficial carcinomas. Leukoplakia does not take up dye. So, it helps in selecting the area for biopsy in cases of leukoplakia.
- *Autofluorescence Laryngoscopy and Lung Imaging Fluorescence Endoscope (LIFE):* For evaluating laryngeal lesions.

Histopathology

Cordal carcinomas are mostly well-differentiated whereas supraglottic lesions are anaplastic. Other variants of squamous cell carcinoma include the following:

- Verrucous carcinoma
- Pseudosarcoma
- Basaloid squamous cell carcinoma.

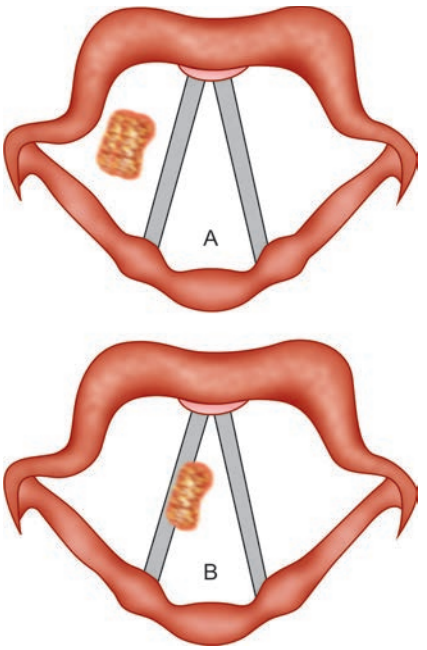
Rare laryngeal malignancies are the following:

- Sarcomas
- Lymphoma
- Adenocarcinoma
- Neuroendocrine carcinoma
- Extramedullary plasmacytoma.

Laryngeal cancer: The most common malignancy of the larynx is squamous cell carcinoma.
Squamous cell carcinoma with various grades of differentiation accounts for 95% of all laryngeal malignancies.

STAGING

For the purpose of staging cancer larynx American Joint Committee on Cancer (AJCC) divides larynx into three regions, which are further subdivided into several subsites (Fig. 4).



Figs 4A and B: Carcinoma larynx. (A) Supraglottic; and (B) Glottic

- *Supraglottis:* Suprahyoid epiglottis (both lingual and laryngeal surfaces), infrahyoid epiglottis, aryepiglottic folds (laryngeal aspect only), arytenoids, ventricular bands (or false cords).
- *Glottis*: True vocal cords including anterior and posterior commissure.
- *Subglottis*: Below the glottis up to lower border of cricoid cartilage.

Transglottic tumors involve supraglottis and glottis across the ventricle, and usually cause fixation of the vocal cord.

The clinical staging is supplemented with diagnostic radiography including CT or MRI (Table 1). The staging and classification help to analyze outcomes of different modalities of treatment, and assists in predicting prognosis of disease.

MANAGEMENT

The treatment depends upon the site and extent of lesion, presence or absence lymph node metastasis and distant metastases. The modalities of treatment include radiotherapy, surgery (conservation and total laryngectomy), chemotherapy and combined therapy.

Radiotherapy

Curative radiotherapy is given for early lesions T1–2. The cords are mobile, and there is no involvement of cartilage and cervical nodes. The main advantage is preservation of voice.

In cases of vocal cord cancer, radiotherapy gives 90% cure rate.

In cases of superficial exophytic lesions of the tip of epiglottis and aryepiglottic folds, it gives 70–90% cure rate.

The results are not good in cases of fixed cords, subglottic extension, cartilage invasion, and nodal metastases. These cases are candidates for surgery.

Surgery

To prove negative surgical margins, the specimen should be submitted for histopathological examination. Pathologist must be oriented by the surgeon.

TABLE 1 AJCC cancer staging (2002) and UICC TNM classification of malignant tumors of larynx and laryngopharynx

Primary tumor (T-tumor)	
T_X	Cannot be assessed
T_0	No tumor
T_{is}	Carcinoma in situ
Supraglottis	
T_1	Tumor limited to only one subsite of supraglottis with normal vocal cord mobility
T_2	Tumor invades mucosa of more than one subsite of supraglottis or mucosa of any of the following structures without vocal cord fixation: glottis, medial wall of pyriform sinus, vallecula or base of tongue
T_3	Tumor limited to larynx with vocal cord fixation and/or involvement of any following structures: postcricoid area, preepiglottic or paraglottic space, or inner cortex of thyroid cartilage.
T_{4a}	Tumor invades thyroid cartilage and any of the following extralaryngeal structures: esophagus, trachea, thyroid, infrahyoid strap muscles, or tongue muscles
T_{4b}	Tumor involves prevertebral space, mediastinum or encasing of carotid artery
Glottis	
T_1	Tumor involves only vocal cords, anterior or posterior commissure with normal vocal cord mobility
T_{1a}	Tumor involves only one vocal cord with normal vocal cord mobility
T_{1b}	Tumor involves both vocal cords with normal vocal cord mobility
T_2	Tumor spreads to supraglottis or subglottis with or without impaired vocal cord mobility
T_3	Tumor limited to larynx with vocal cord fixation and/or involvement of any following structures: paraglottic space, or inner cortex of thyroid cartilage
T_{4a}	Tumor invades thyroid cartilage and any of the following extralaryngeal structures: esophagus, trachea, thyroid, infrahyoid strap muscles, or tongue muscles
T_{4b}	Tumor involves prevertebral space, mediastinum or encasing of carotid artery
Subglottis	
T_1	Tumor limited to subglottis with normal vocal cord mobility
T_2	Tumor spreads to glottis with or without impaired vocal cord mobility
T_3	Tumor limited to larynx with vocal cord fixation
T_{4a}	Tumor invades cricoid or thyroid cartilage and any of the following extralaryngeal structures: esophagus, trachea, thyroid, infrahyoid strap muscles, or tongue muscles
T_{4b}	Tumor involves prevertebral space, mediastinum or encasing of carotid artery

Contd...

Contd...

Hypopharynx tumor size in greatest dimension	
T_1	Tumor ≤ 2 cm and limited to one subsite
T_2	Tumor > 2 cm but ≤ 4 cm or invading > 1 subsite and without hemilarynx fixation
T_3	Tumor > 4 cm or with hemilarynx fixation.
T_{4a}	Tumor invades any of the following: thyroid-cricoid cartilage, hyoid bone, thyroid gland, esophagus, central compartment of soft tissue such as infrahyoid strap muscles and subcutaneous tissue.
T_{4b}	Tumor involves prevertebral fascia, mediastinum or encasing of carotid artery
Regional lymph nodes (N) size in greatest dimension	
N_x	Cannot be assessed
N_0	No nodal metastases
N_1	Single ipsilateral node ≤ 3 cm
N_{2a}	Single ipsilateral node > 3 cm but ≤ 6 cm
N_{2b}	Multiple ipsilateral nodes but none > 6 cm
N_{2c}	Bilateral or contralateral nodes but none > 6 cm
N_3	Any node > 6 cm
Distant metastasis (M)	
M_x	Cannot be assessed
M_0	No distant metastasis
M_1	Distant metastasis
Stage grouping	
Stage 0	$T_{is}\ N_0\ M_0$
Satge I	$T_1\ N_0\ M_0$
Stage II	$T_2\ N_0\ M_0$
Stage III	$T_3\ N_0\ M_0$ or $T_{1-3}\ N_1\ M_0$
Stage IVA	$T_{4a}\ N_{0-1}\ M_0$ or $T_{1-4a}\ N_2\ M_0$
Stage IVB	$T_{4b}\ N_{0-3}\ M_0$ or $T_{1-4}\ N_3\ M_0$
Stage IVC	$T_{1-4}\ N_{0-3}\ M_1$

- *Conservation Surgery:* Conservation laryngeal surgery preserves voice. and avoids a permanent tracheal opening. Cases should be carefully selected for conservation surgery, which includes:
  1. Cordectomy: Excision of vocal cord via laryngofissure or endoscopy.
  2. Partial frontolateral laryngectomy: Excision of vocal cord and anterior commissure.
  3. Partial horizontal laryngectomy: Excision of supraglottis, which include epiglottis, aryepiglottic folds, false cords and ventricle.
- *Total Laryngectomy*: The entire larynx is removed along with hyoid bone, preepiglottic space, strap muscles, and one or more rings of trachea. Pharyngeal wall is closed primarily, and lower tracheal stump is sutured to the skin (Fig. 5). The indications include T3–4 lesions and failure after radiotherapy or conservation surgery. It is combined with block dissection when nodal metastasis is present. It is not done in patients with distant metastasis.
- *Hemithyroidectomy or Subtotal Thyroidectomy*: The

Fig. 5: Narrowing of the tracheostome in an irradiated case of total laryngectomy with partial pharyngectomy and radical neck dissection of right side. Note the incision and skin changes after postoperative radiotherapy

associated hemithyroidectomy or subtotal thyroidectomy is indicated in following conditions:

- Palpable thyroid abnormality
- Subglottic extension or tumors
- T_4 glottic tumors
- T_4 pyriform sinus tumors
- Positive Delphian nodes
- Cartilage destruction

Combined Therapy

Surgery may be combined with pre or postoperative radiation in a planned way to decrease the incidence of recurrence. Preoperative radiation may render fixed nodes resectable. Overall survival is better with postoperative radiotherapy, which is delivered within 7 weeks of surgery. Complications rate is higher with preoperative radiotherapy.

Palliation Therapy

- Endoscopic significant debulking with laser or surgical forceps temporarily relieves the airway.
- Tracheostomy.
- Gastrostomy.

GLOTTIC CANCER

This is the most common laryngeal cancer with good prognosis because patients report early, and lymphatic spread is late. The most common sites are free edge and upper surface of vocal cord in its anterior and middle third.

Spread

- *Local:* Fixation of vocal cord is a bad prognostic sign and indicates involvement of thyroarytenoid muscle.
  - *Anterior*: Anterior commissure and then to the opposite cord
  - *Posterior*: Vocal process and arytenoid region
  - *Upward*: Ventricle and false cord
  - *Downward*: Subglottic region
- *Lymphatic:* There are not much lymphatic in vocal cords.

Lymph node metastasis occurs when cordal lesions spread beyond the region of membranous cord.

Clinical Features

- *Hoarseness of Voice:* An early and most common presentation, because of which glottic cancer is detected early. Cord cancer affects vibratory capacity.
- *Airway Obstruction:* Large growth, accompanying edema or cord fixation may result in stridor.
- *Lesion:*
  a. Vocal cord thickening, raised nodule or an ulcer.
  b. Anterior commissure granulation tissue.

Treatment

- *Carcinoma in situ:* Endoscopic CO_2 laser application or stripping of vocal cord and tissue is sent for biopsy. If biopsy reveals invasive carcinoma, radiotherapy is given. CIS needs regular follow-up.
- *Stage I:* $T_1\ N_0\ M_0$
  - *Middle third of vocal fold lesion*: Radiotherapy (95% cure) or surgery (100% cure): endoscopic, CO_2 laser resection (cordotomy or cordectomy) or open cordectomy (laryngofissure). Surgery failures are given radiotherapy. But, radiotherapy failures may not be amenable to conservation surgery.
  - *Extension to anterior commissure*: Radiotherapy or frontolateral partial laryngectomy. In case of failure, total laryngectomy is indicated.
  - *Extension to arytenoids*: Surgery is preferred.
- *Stage II*: $T_2\ N_0\ M_0$ (glottic tumor with extension to supraglottic or subglottic region with normal or impaired mobility of vocal cord). Normal cord mobility suggests growth limited to the surface whereas impaired mobility indicates invasion into intrinsic laryngeal muscles or paraglottic space. Invasion of paraglottic or subglottic space may be associated with undetected invasion of laryngeal cartilages. With normal cord mobility, radiation gives 86% cure rate while it drops to 63% if cord mobility is impaired. In a case of postradiation edema, which persists for longer than 6 months, deep invasive recurrence must be ruled out.
  - Cord mobile and anterior commissure or arytenoids not involved: Radiotherapy gives good results (upper neck nodes are included in the radiation field). In cases of radiation failure, laryngectomy or partial vertical laryngectomy is indicated.
  - Anterior commissure and/or arytenoids involved or cord mobility impaired: Vertical hemi-laryngectomy or frontolateral laryngectomy. In cases of failure or recurrence, total laryngectomy is done.
- *T_2 and early T_3*: Supracricoid laryngectomy with cricohyoidepiglottopexy.
- *T_3 and T_4*: Total laryngectomy combined with neck dissection (for palpable nodes).
- *More advanced T_4*: These lesions may be treated by combined therapy or palliative treatment.

SUPRAGLOTTIC CANCER

It is less common than glottic cancer. Prognosis is poor because patients report late, and lymphatic spread occurs early. Most common sites are epiglottis, false cords followed by aryepiglottic folds.

Spread

- *Local*
  - *Superior*: Vallecula and base of tongue.
  - *Lateral*: Pyriform fossa.
  - *Anterior*: Growths of infrahyoid epiglottis and anterior ventricular band extend into preepiglottic space, and penetrate the thyroid cartilage.
  - *Inferior*: Glottic region.
- *Lymphatic:* Lymph node metastasis occurs early. It usually involves upper and middle jugular nodes. Bilateral metastases may be seen in epiglottic lesion.

Clinical Features

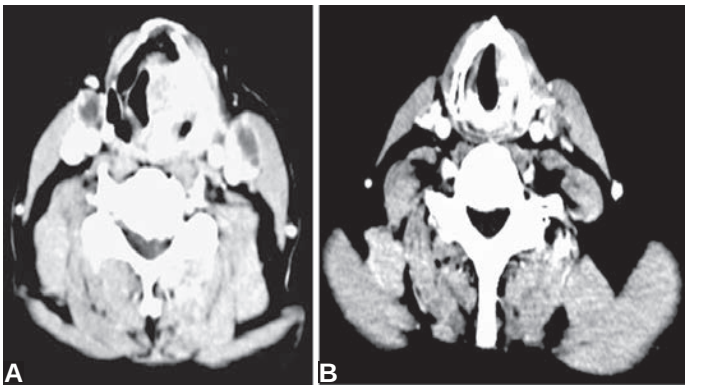
Patients report late because supraglottic growths (Figs 6A and B) remain silent for long time till they achieve enormous size and cause lymph node metastasis.

- Most common symptoms:
  - Throat pain
  - Dysphagia
  - Referred ear pain
  - Mass of lymph nodes in the neck
- Late symptoms:
  - Hoarseness of voice
  - Weight loss
  - Airway obstruction—stridor
  - Halitosis
- *Lesion*: Either exophytic (suprahyoid epiglottis) or ulcerative (infrahyoid epiglottis).

Treatment

In addition to the factors of TNM staging, age of the patient and status of lung functions should be considered before planning the treatment for supraglottic cancer.

- *Stage I*: $T_1\ N_0\ M_0$: Radiation or excision with CO_2 laser.
  - *Suprahyoid epiglottic tumors*: Endoscopic excision with electrocautery or CO_2 laser.
  - *Infrahyoid epiglottic tumors*: Endoscopic laser partial laryngectomy.
- *Stage II*: $T_2\ N_0\ M_0$
  - *Good lung function*: Supraglottic laryngectomy (when vocal cord is mobile) with or without neck dissection. Vallecula and base of tongue up to the circumvallate papilla can be removed. Patient must be able to tolerate



Figs 6A and B: CT scan of neck axial cuts. Mass in left pyriform sinus and AE fold extending to left larynx
Source: Courtesy Dr Swati Shah, Consultant, Radiodiagnosis, Apollo Hospital, Ahmedabad

mild-to-moderate aspiration that may last for the rest of life. Other surgical procedures include subtotal laryngectomy with cricohyoidopexy and supracricoid partial laryngectomy.
  - *Poor Lung Function:* Radiotherapy.
- *T_3 and T_4*: Total laryngectomy with neck dissection and postoperative radiotherapy.

SUBGLOTTIC CANCER

Subglottic region lies below the glottic area, and extends up to the lower border of cricoid cartilage. Subglottic cancer is the least common laryngeal cancer.

Spread

- *Local*: One side growth of subglottis spreads around the anterior wall, and to the opposite side.
  - *Inferior*: Trachea.
  - *Upward*: Vocal cords are involved late so the hoarseness is a late symptom.
  - *Anterior*: Cricothyroid membrane, thyroid gland and ribbon muscles of neck.
- *Lymphatic*: Prelaryngeal (Delphian node), pretracheal, paratracheal and lower jugular nodes.

Clinical Features

These patients report late when the growth is significantly big and extensive.

- *Stridor*: The most common symptom is airway obstruction.
- *Hoarseness of voice* due to infiltration of thyroarytenoid muscle or recurrent laryngeal nerve at the cricoarytenoid joint.
- *Lesion*: Raised submucosal nodule usually involving the anterior half.

Treatment

- *T_1 and T_2*: Radiotherapy.
- *T_3 and T_4*: Total laryngectomy and postoperative radiation, which include superior mediastinum.

VERRUCOUS CARCINOMA

- It is a slowly growing and locally aggressive well-differentiated squamous cell carcinoma. It accounts for 1–2% of laryngeal carcinomas.
- HPV 16 and 18 are associated with this lesion.
- Larynx is the second most common site after oral cavity.
- It presents as an exophytic and fungating carpet like lesion.
- Significant inflammation surrounds the lesion, and lead to enlarged inflammatory neck nodes.
- Biopsies are often repeated before satisfactory diagnosis.

Histology

Hyperkeratotic squamous epithelium, minimal cytologic atypia, papillomatosis and club-shaped rete ridges.

Treatment

Treatment is excision of the lesion, not radiotherapy. Failure to complete excision, results in local recurrence. After radiotherapy, they may undergo transformation to undifferentiated carcinoma.

ORGAN PRESERVATION THERAPY

The current trend in the management of cancer of larynx is developing towards chemotherapy+radiotherapy for preserving the larynx. Studies have shown that there is no difference in survival rate between the cases of total laryngectomy and combined chemotherapy+radiotherapy.

The patients who are candidates for total laryngectomy are given a course of chemotherapy. Those who have complete response (some include even partial response) are given complete radiotherapy. Those who have no response (some include even partial response) are treated with surgery, usually followed by radiotherapy. No difference in survival was found in these two groups; those who went for radiotherapy or surgery after the course of chemotherapy.

- *Chemotherapy Protocols*: They include:
  - Three-cycle regimen of cisplatinum (100 mg/m^2) and fluorouracil (5-FU) continuous infusion for 5 days at 15 or 21 days interval—most commonly used.
  - Three-cycle regimen of cisplatinum, bleomycin and 5-FU.

PHOTODYNAMIC THERAPY

There is preferential uptake of photosensitizer photofrin (dihematoporphyrin ether or DHE), which is given intravenously, by the malignant cells. Superficial cancers of larynx have been treated with photodynamic therapy (PDT), which has got US FDA approval for treating obstructing esophageal and endobronchial tumors, and minimally invasive endobronchial non-small cell carcinoma.

Light activation of photo concentrated DHE results in mitochondrial damage and apoptosis in malignant cells. Erythrocyte leakage and endothelial damage of vessels cause ischemic necrosis of tumor tissue.

The main side effect of PDT is generalized skin photosensitization.

POST-LARYNGECTOMY VOCAL REHABILITATION

Patient loses laryngeal speech after laryngectomy. Following methods may facilitate patient's communication. The current trend is for tracheoesophageal (TE) fistula speech. The artificial larynx is the least preferred ones. Good esophageal speech compares well with TE speech.

Esophageal Speech

Patient learns how to swallow air, and hold it in the upper esophagus. Then patient slowly ejects the air from upper esophagus, and speaks 6–10 words which are even though rough, are loud and understandable. Some patients learn to store air in stomach, which greatly improves duration of speech.

Artificial Larynx

Artificial larynx is useful when patients fail to learn esophageal speech. They are least socially acceptable because they draw

unwanted attention to the speaker. Monotone sound production is another limitation of artificial larynx.

- *Electrolarynx*: This is transistorized, battery operated, and portable device. It produces low-pitched sound which is further modulated into speech by the tongue, lips, teeth and palate. The vibrating disc of artificial larynx is held against the soft tissues of the neck or oral cavity.
- *Transoral Pneumatic Device*: The expired air from the tracheostome vibrates the diaphragm of this pneumatic device. These rubber diaphragm vibrations are carried by a plastic tube into the back of the oral cavity where modulators convert the sound into speech. Now, they are rarely used.

Tracheoesophageal Fistula Speech

The air is carried from trachea to esophagus/hypopharynx through a tracheoesophageal fistula or an artificial prosthesis. The vibrating column of air in the pharynx is modulated into speech. The lungs are used as bellows and produce greater air pressure across pharyngoesophageal (PE) segment than the patients who use esophageal speech. The main disadvantage is aspiration of food into the trachea. The current prosthesis shunts the air from trachea to esophagus, and their inbuilt one-way valves prevent problems of aspiration.

Tracheoesophageal puncture (TEP) may be either primary (at the time of laryngectomy) or secondary (2–4 weeks after surgery or postoperative radiotherapy).

- *Prosthesis*: There are two types of prostheses non-indwelling and indwelling.
  - *Non-indwelling devices:* The non-indwelling prosthesis is inserted in about 1–2 weeks after the TEP. Patients remove them daily for cleaning and maintenance. They include:
    a. Panje Voice Button (Hood Laboratories, USA)
    b. Blom-Singer Duckhill Prosthesis (InHealth Technologies, USA)
  - *Indwelling prostheses:* They are placed at the time of TEP, and remain in TEP tract. In case of problems they are replaced by the therapist or surgeon. They include:
    a. Groningen Voice Button (Hood Laboratories, USA)
    b. Provox and Provox 2 (Atos Medical AB, Horby, Sweden)
    c. Blom-Singer Indwelling Voice Prosthesis (InHealth Technologies, USA)

Clinical Highlights

1. **Demarquay's sign:** Absence of elevation of the larynx during swallowing.
2. **Carcinoma larynx:** In addition to genetic susceptibility, the risk factors of carcinoma larynx are smoking, alcohol, gastroesophageal reflux disease (GERD), and exposure to wood dust, asbestos and volatile chemicals, nitrogen mustard and previous ionizing radiation.
   a. **Carcinoma glottis:** The most common and earliest symptom is hoarseness of voice. In comparison to supraglottis, nasopharynx and subglottic cancers, carcinoma glottis is the most radiosensitive tumor.
   b. **Supraglottic cancer:** It is the most aggressive of laryngeal cancers and has highest incidence of cervical nodal metastases. The most common site in supraglottic region is epiglottis.
   c. **Synchronous second primary**: In cases of carcinoma larynx the most common site of second primary (synchronous second primary) is bronchus.
   d. **Distant metastases:** The highest incidence of distant metastases in laryngeal cancer is seen in lungs.
   e. **Treatment**: The first line of treatment of carcinoma larynx in Stage I (T_1, N_0, M_0) is radiotherapy**,** whereas in Stage III it is laryngectomy followed by postoperative radiotherapy.
   f. **Verrucous carcinoma of larynx:** The treatment of choice is surgery.
3. **Ashai technique:** It is a method of vocal rehabilitation in patients with laryngectomy.
4. **Esophageal speech:** The dynamic component of phonation in esophageal speech in a case of laryngectomy lies at pharyngoesophageal segment.
5. **Blom Singer prosthesis:** In laryngectomy patients, it is used to divert tracheal air into esophagus for voice production.

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51 Management of Impaired Airway

Struggle, struggle. Struggle still say I. Be not afraid. Have fire and spread all over. Work, work. Be the servant while leading. Be unselfish, and never listen to one friend in private accusing another. Have infinite patience, and success is yours.

—Swami Vivekananda

Points of Focus

- **INTRODUCTION**

TRACHEOSTOMY/TRACHEOTOMY

- ✓ Functions
- ✓ Indications
- ✓ Timings
- ✓ Level and Site: High, Mid and Low Tracheostomy
- ✓ Anesthesia, Position
- ✓ Steps of Operation
- ✓ Tracheostomy /Tracheotomy in Infants and Children
- ✓ Postoperative Care
- ✓ Decannulation in Infants and Young Child
- ✓ Causes of Failed Decannulation
- ✓ Complications

- **CRICOTHYROTOMY (LARYNGOTOMY OR CONIOTOMY)**
  - ✓ Indications
  - ✓ Contraindications
  - ✓ Technique
- **PERCUTANEOUS DILATIONAL TRACHEOSTOMY (PDA)**
  - ✓ Preoperative Criteria
  - ✓ Contraindications
  - ✓ Advantages and Disadvantages
  - ✓ Technique

CONGENITAL LESIONS OF LARYNX

- **LARYNGOMALACIA**
  - ✓ Awake Flexible Laryngoscopy
- **CONGENITAL VOCAL CORD PARALYSIS**
- **CONGENITAL SUBGLOTTIC STENOSIS**
- **LARYNGEAL WEB/ATRESIA**
- **SUBGLOTTIC HEMANGIOMAS**
- **LARYNGOESOPHAGEAL CLEFT**

FOREIGN BODIES OF AIR PASSAGES

- ✓ Predisposing Factors
- ✓ Nature of Foreign Bodies
- ✓ Clinical Features
- ✓ Differential Diagnoses
- ✓ Diagnosis
- ✓ Management-Heimlich Maneuver

LARYNGO TRACHEAL TRAUMA

- ✓ Modes of Injuries
- ✓ Pathology
- ✓ Clinical Features
- ✓ Diagnosis
- ✓ Associated Injuries
- ✓ Complications
- ✓ Treatments

- **CLINICAL HIGHLIGHTS**

INTRODUCTION

A patient with impaired airway demands prompt attention from doctors. Failure to achieve a patent airway in cases of impaired airway, inevitably results in hypoxic brain injury and death.

The various means of airway control in the anesthetized or unconscious patients are discussed in chapter Anesthesia.

TRACHEOSTOMY/TRACHEOTOMY

In tracheostomy, an opening in the anterior wall of trachea is created and converted into a stoma on the skin surface. The terms tracheostomy and tracheotomy have been used interchangeably. The tracheotomy means opening the trachea, which is a step of tracheostomy operation.

Tracheostomy facilitates suctioning, feeding, mobility, early return of speech, and decrease work of breathing.

Functions

- *Obstruction:* Bypass the obstruction in the upper airway from pharynx to larynx.
- *Ventilation:* Improves alveolar ventilation in cases of respiratory insufficiency in the following ways:
  - *Dead space*: Decrease the dead space by 30–50% (normal dead space is 150 ml).
  - *Resistance*: Reduces the resistance to airflow.
- *Protection:* Cuffed tracheostomy tube protects the tracheobronchial tree against aspiration of secretions and blood.
  - *Secretions*: Pharyngeal secretions in cases of bulbar paralysis and coma.
  - *Blood*: Bleeding from pharynx, larynx and maxillofacial injuries.
  - *Packing*: Allows packing of pharynx and larynx to control bleeding.
- *Suction:* Suction clearance of tracheobronchial secretions avoids the need for repeated bronchoscopy or intubation.
- Intermittent positive pressure respiration (IPPR).
- *Anesthesia:* Administration of anesthesia in cases of laryngopharyngeal growths and marked trismus of oral submucous fibrosis.

If IPPR is expected to prolong beyond 72 hours, tracheostomy is preferred over endotracheal intubation.

Indications

The indications can be categorized in three groups: upper respiratory obstruction, retained secretions, and respiratory insufficiency. Today, prolonged intubation usually with mechanical ventilation is the most common indication for tracheostomy; formerly, it was upper respiratory obstruction. Other indications include easier management of secretions and an adjuvant procedure during chest surgery, and major head and neck operations.

- *Upper Respiratory Tract Obstruction*. See Stridor in chapter Laryngeal Symptoms and Examination.
- Retained secretions
  - *Inability to cough*: Coma due to head injuries, cerebrovascular strokes, accidents, narcotic poisoning.
  - Paralysis of respiratory muscles due to spinal injuries, polio, Guillain-Barre syndrome and myasthenia gravis.
  - Spasm of respiratory muscles due to tetanus, eclampsia, strychnine poisoning.
  - *Painful cough*: Chest injuries, multiple rib fractures, pneumonia.
  - *Aspiration of secretions*: Bulbar polio, polyneuritis, bilateral laryngeal paralysis.
- *Respiratory insufficiency:* Emphysema, chronic bronchitis, bronchiectasis and atelectasis.

Timings

- *Emergent Tracheostomy (slash trach)*: It is required when intubation and laryngotomy (cricothyrotomy) are not feasible, and emergency airway distress is accompanied with impending death. The complication rate of emergency tracheostomy is as high as 21%. This emergent situation is an ideal indication for cricothyrotomy.

  The technique is more or less similar to elective tracheostomy. After the midline vertical cervical skin incision, a transverse incision is made along the lower border of cricoid cartilage in pretracheal fascia. The thyroid isthmus is dissected down (even cut with knife or diathermy) to expose upper three tracheal rings. Vertical tracheal incision involves second and third rings. Trachea is opened with a hemostat, and the tube inserted.
- *Urgent (awake) Tracheostomy*: This is done in the operation theater under local anesthesia with minimal sedation. The patient has respiratory distress, and needs immediate surgical intervention.
- *Elective Tracheostomy*: This tranquil, orderly and routine tracheostomy is a planned surgery. It is performed where all operative surgical facilities such as endotracheal intubation, local and general anesthesia, are available.
  - *Therapeutic*: It relieves not only the respiratory obstruction but also allows removing tracheobronchial secretions, and provides assisted ventilation.
  - *Prophylactic*: It is performed to prevent anticipated respiratory obstruction and aspiration of blood and secretions. The indications include extensive surgery of tongue, floor of mouth, mandibular resection or laryngofissure.
- *Temporary Tracheostomy*: Elective tracheostomy is usually temporary, and is closed when causative disease is cured.
- *Permanent Tracheostomy*: It is indicated in cases of bilateral abductor paralysis and laryngeal stenosis. In cases of laryngectomy and laryngopharyngectomy, a tracheostome is created where lower tracheal stump is stitched to the surface skin.

Tracheotomy: It is the general opinion that the best time of doing tracheostomy is when you first think that patient needs tracheotomy. No one would like to end up in doing a "slash" tracheotomy later!

Level and Site

On the bases of the site, tracheostomy has been divided into three groups: high, mid and low.

- *High Tracheostomy*: It is done at the level of first tracheal ring above the level of thyroid isthmus, which lies at the level of third and fourth tracheal rings. The high tracheostomy is generally avoided because of the postoperative risk of perichondritis of the cricoid cartilage and subglottic stenosis. In cases of carcinoma larynx with stridor when total laryngectomy would be done, high tracheostomy is indicated.
- *Mid Tracheostomy*: It is done through the second and third tracheal rings, and needs either division of the thyroid isthmus or its retraction upwards.
- *Low Tracheostomy*: It is done below the level of isthmus where trachea becomes deep, and lies close to large vessels. The tracheostomy tubes tend to impinge on suprasternal notch.

Anesthesia

General anesthesia with endotracheal intubation is preferred. It is particularly important in infants and children. No anesthesia is needed in unconscious patients and dire emergency conditions. In cases of malignancy larynx, it is usually done under local anesthesia (1–2% lignocaine with epinephrine infiltration).

Position

Patient lies in supine position. The neck is extended with a pillow under the shoulders to bring the trachea anteriorly.

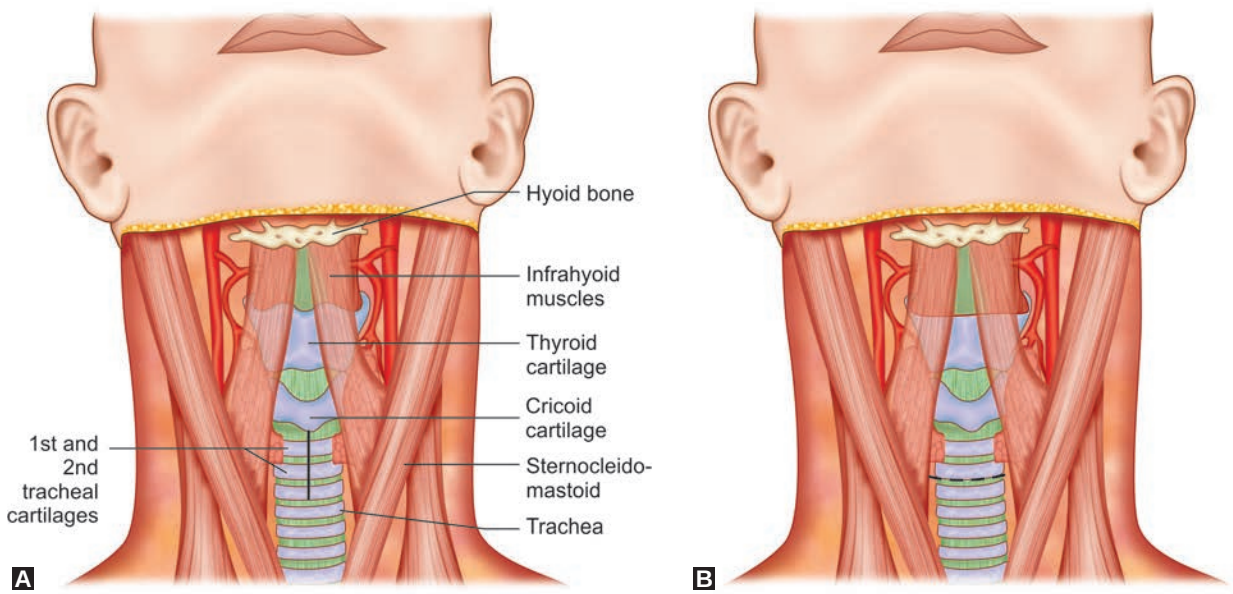
Steps of Operation (Figs 1 and 2)

- *Skin Incision*: A vertical midline cervical incision that extends from cricoid cartilage to just above the sternal notch is the most frequently used skin incision. A transverse incision—5 cm above the sternal notch—has the advantage of a cosmetically better scar.
- *Strap Muscles:* Strap muscles are first separated in the midline and then retracted laterally.
- *Thyroid Isthmus:* The thyroid isthmus is either displaced upwards or divided between the clamps.
- *Trachea:* After injecting few drops of 4% lignocaine into the trachea that suppresses the cough, trachea is incised with a vertical incision in the region anywhere between second to fourth tracheal rings. The incision is then converted into a circular opening. Trachea may be fixed with a hook before the incision. The first tracheal ring is never damaged as it may result in perichondritis of cricoid cartilage and stenosis.
- *Tracheal Hook (blunt and sharp):* Blunt tracheal hook retracts the thyroid isthmus, and exposes the trachea. It may also be used for retracting the strap muscles. When making incision in the tracheal wall, sharp tracheal hook is applied to the lower border of cricoid cartilage to stabilize the trachea.
- *Tracheal Dilator* keeps the cut tracheal-edges open so that tracheostomy tube can be easily introduced. Its tip is blunt (see chapter Instruments). The blades spread out on approximating its rings. A curved artery forceps can also serve the purpose.
- *Tracheostomy Tube:* An appropriate size of tracheostomy tube is inserted and secured by tapes (Fig. 3). *See* chapter 'Instruments' for different types and sizes of tracheostomy tubes.

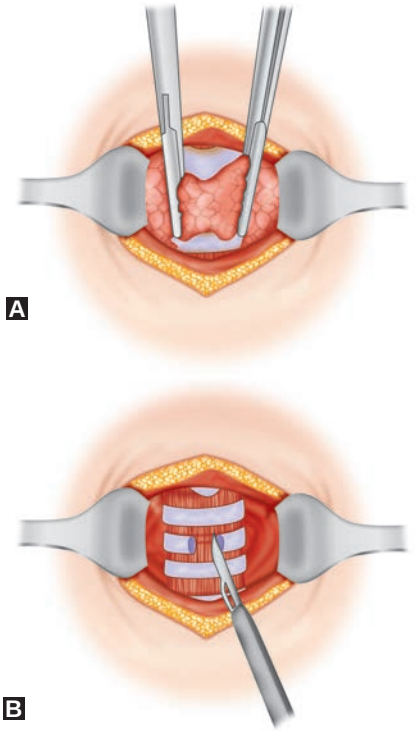
Tracheostomy in Infants and Children

Great care is needed to prevent avoidable complications, which are not uncommon while doing tracheostomy in infants and young children. The following facts and precautions must be kept in mind:

- Stay strictly in midline. Position the child without neck or head deviation. Fix the larynx by putting a finger on either side of larynx. Put a vertical incision through skin, and down to the trachea.
- Trachea is soft and compressible in infants and children. The surgeon tends to displace trachea, and go deep or lateral to it; possibly injuring recurrent laryngeal nerve and common carotid artery. A bronchoscope or an endotracheal intubation facilitates identification of trachea. Therefore, tracheostomy in infants and children is done under general endotracheal anesthesia.
- Too much extension of neck pulls thoracic structures into the neck. The vulnerable structures are pleura, innominate artery and thymus. The risk of making tracheostomy opening near suprasternal notch is high.
- The silk sutures, placed in the trachea on either side of midline, helps in proper incision of trachea. Trachea is only incised. The excision of a circular piece of anterior tracheal wall is never done.
- Tracheal lumen is narrow, and too much insertion of knife in trachea can easily injure posterior tracheal wall and esophagus causing tracheoesophageal fistula.
- Infolding of anterior tracheal wall while inserting the tracheostomy tube is avoided.
- Tracheostomy tube should be of proper diameter, length and curvature. A long tube can impinge on the carina, or go into right bronchus. Lower end of high curvature tube impinges on anterior tracheal wall, and its upper part



Figs 1A and B: Incisions for tracheostomy. (A) Surface landmarks for the midline skin vertical incision for tracheostomy; (B) Horizontal skin incision for cricothyrotomy



Figs 2A and B: Thyroid isthmus and tracheostomy. (A) Thyroid isthmus may be retracted superiorly or inferiorly, or transected and suture ligated; (B) Anterior section of a tracheal ring can be removed or a round window spanning two tracheal rings can be created

Fig. 3: Tracheostomy tube inserted and secured by tapes

compresses the tracheal rings or cricoid. Soft silastic and portex tubes are preferred over metallic tubes, which cause more traumas. If a proper tracheostomy tube is not available, a pediatric endotracheal tube can be used.

- Postoperative X-ray chest and neck ascertains the position of tracheostomy tube.

Postoperative Care

- Watch for bleeding and displacement, and blocking of tube.
- Paper pad and a pencil for patient's communication as these patients cannot speak.
- Regular suction (hourly or half-hourly) depending on the amount of secretion for their removal.
- Use sterile catheters with a Y-connector that breaks suction force and avoids suction injuries to tracheal mucosa. Apply suction only when the catheter is taken out.
- Proper humidification that prevents crusting is achieved by using humidifier, steam tent, ultrasonic nebulizer, or keeping a boiling kettle in the room.
- Crusting: Few drops of normal or hypotonic saline or Ringer's lactate are instilled into the trachea every 2–3 hours to loosen crusts. Instillation of acetylcysteine (mucolytic agent) solution liquefies tenacious secretions, and loosens the crusts.
- Tracheostomy tube: Inner cannula is removed, and cleaned regularly for the first 3 days to prevent respiratory distress. Outer tube is changed daily after 3–4 days of tracheostomy when a track is formed that facilitates easy tube placement.
- Periodical deflation of cuffed tube prevents pressure necrosis and dilatation of trachea.

Decannulation

Prolonged use of tracheostomy tube can cause tracheobronchial infections, tracheal ulceration, granulations, stenosis and unsightly scars. Therefore, decannulation (removal of tube) should be considered once the causative condition is under control.

Method: The tracheostomy tube is occluded, and the patient closely watched for respiratory distress. If there is no distress for 24 hours, the tube is removed, and the wound is taped. Wound healing usually takes place within a week. Rarely a secondary closure is required.

In children, decannulation is done using progressively smaller size tubes.

Decannulation in Infant and Young Child

As the decannulation in infant and young child carries more risk, the following care-listing should be observed:

- Decannulate the child in the operation theater where services of re-intubation, anesthetist, headlight, laryngoscope, proper sized endotracheal tubes, and a tracheostomy tray are kept ready.
- After decannulation, the child is watched for several hours for respiratory distress, tachycardia and cyanosis. Blood gas determinations may be required.

Causes of Failed Decannulation

When decannulation fails in spite of all the measures, the following causes must be looked for. The patient requires endoscopic examination of the larynx, trachea and bronchi using telescopes or a flexible endoscope.

- Persistence of the cause for which tracheostomy was required.
- Granulations around the stoma and in trachea, where tip of the tracheostomy tube impinge.
- Tracheal edema.
- Subglottic stenosis.
- Incurving of tracheal wall at the site of tracheostome.
- Tracheomalacia.
- Psychological dependence on tracheostomy.

- Inability to tolerate the resistance of the upper airways (physiological dependence).

Complications

Complications of tracheostomy (Box 1) are traditionally categorized into three groups: immediate (intraoperative), intermediate (few hours to days), and late (weeks and months). Pediatric, head injury, obese, burn and debilitated patients are more prone to complications.

Hemorrhage and tube displacement are the major causes of tracheostomy deaths. The most common complications include hemorrhage (3.7%), tube obstruction (2.7%), and tube displacement (1.5%).

The common sites of hemorrhage are anterior jugular veins and thyroid isthmus. Apnea due to sudden washing out of CO_2, which was acting as a respiratory stimulus, may occur in a patient who had prolonged respiratory obstruction. Treatment includes administration of 5% CO_2 in oxygen (carbogen) or assisted ventilation.

Pneumothorax occurs due to injury to apical pleura. Vocal cord palsy occurs due to injury to recurrent laryngeal nerves. Tracheoesophageal fistula is the result of injury to posterior wall of trachea and esophagus.

Obstruction of tube can occur due to blood clot, partial displacement, and impingement on posterior tracheal wall.

Box 1: Complications of tracheostomy

1. Immediate
   a. Hemorrhage
   b. Apnea
   c. Pneumothorax
   d. Vocal cord palsy
   e. Tracheoesophageal fistula
   f. Aspiration of blood
   g. False passage
2. Intermediate
   a. Bleeding: Reactionary and secondary
   b. Displacement of tube
   c. Obstruction of tube
   d. Subcutaneous emphysema
   e. Pneumomediastinum and pneumothorax
   f. Crusting in trachea
   g. Tracheitis and stomal cellulitis
   h. Tracheobronchitis
   i. Severe infections: mediastinitis, clavicular osteomyelitis, and necrotizing fasciitis
   j. Atelectasis and lung abscess
   k. Local wound infection and granulations
3. Late (with prolonged use of tube):
   a. Hemorrhage due to granulation tissue and innominate artery blowout
   b. Laryngeal stenosis
   c. Tracheal stenosis
   d. Tracheoesophageal fistula
   e. Difficult decannulation
   f. Persistent tracheocutaneous fistula
   g. Keloid or unsightly scar
   h. Corrosion of tracheostomy tube.

Innominate artery blows out due to erosion of its posterior walls. It carries 90% mortality rate. Direct digital pressure on the anterior wall of stoma tract is found effective. Laryngeal stenosis occurs due to cricoid cartilage perichondritis. Tracheal stenosis is the result of tracheal ulceration and infection. Tracheoesophageal fistula may occur due to cuffed tube and tip of tracheostomy tube. Problems of decannulation are more in infants and children.

CRICOTHYROTOMY (LARYNGOTOMY OR CONIOTOMY)

In this emergent procedure, an opening is made for airway through the cricothyroid membrane.

Indications

They include severe bleeding (maxillofacial injuries) foreign bodies, emeses, clenched teeth, repeated failed intubation, cervical spine injuries, burns and smoke inhalation.

Contraindications

They include infants and children, and inflammation and malignancy of larynx and surrounding area.

Technique

The neck is extended. Vertical midline cervical incision is made between the lower border of thyroid cartilage and upper border of the cricoid ring. Then, cricothyroid membrane is cut with a transverse incision, and subglottic larynx is entered. A small tracheostomy tube or any hollow tube like thing can be inserted to maintain airway. It is followed by tracheostomy, which is done at the earliest to avoid complications of cricothyrotomy such as perichondritis, subglottic edema and laryngeal stenosis.

PERCUTANEOUS DILATIONAL TRACHEOSTOMY

In percutaneous dilational tracheostomy (PDT), tracheostomy tube is introduced through the pretracheal skin and soft tissue without direct surgical visualization of trachea (Fig. 4).

Preoperative Criteria

- This minimally invasive, bedside procedure is performed only on intubated adult patients with long neck, which are admitted in the intensive care unit.
- Ability of the patient to hyperextend the neck.
- Easy reintubation in case of accidental extubation.

Contraindications

- *Absolute*
  - Need for emergency airway access
- *Relative*
  - Children younger than 12 years
  - History of difficult intubation
  - Anatomical
    - Cervical spine lesions causing limited extension of neck
    - Abnormality of trachea and larynx
    - Short and thick neck
  - Local tracheostomy site problems
    - Visible pulsating vessels

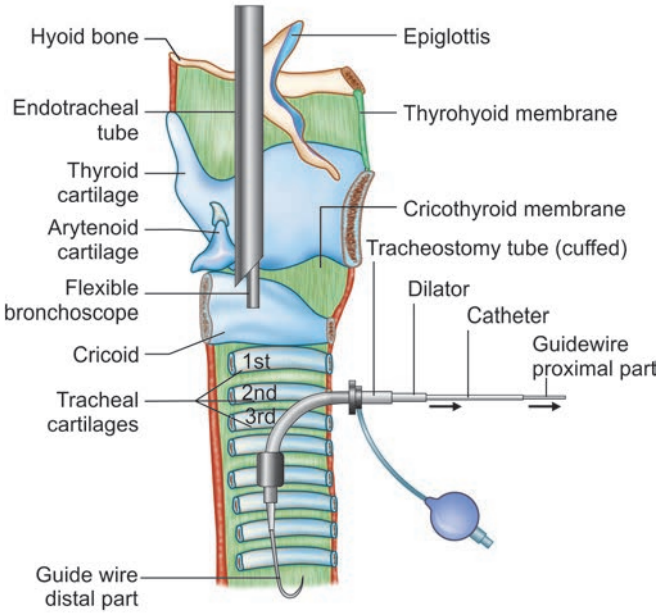


Fig. 4: Percutaneous dilational tracheostomy. After making skin incision and clearing pretracheal tissue, a guidewire along its sheath is introduced below the second tracheal ring. Guidewire, guide catheter, and dilator are advanced together after removing the guidewire sheath. Tracheostomy tube is loaded and advanced onto the dilator into the trachea. The procedure is done under bronchoscopic vision under general anesthesia

  - Active infection
  - Goiter
- Hematological
  - Platelet count: less than 40,000/mm^3
  - Bleeding time: greater than 10 minutes
  - Prothrombin time or partial thromboplastin time: greater than 1.5 times of control
- Positive end-expiratory pressure: greater than 15 cm H_2O

Advantages

- Relatively easy to learn.
- The time required is considerably shorter.
- No need of operation theater and anesthesiologists team.
- Saves fees of operation theater and anesthesiologists team.
- No need to shift the patient from ICU.

Disadvantages

The PDT system set is expensive.

Technique

There are different systems and approaches to perform a percutaneous tracheostomy; the details of which are beyond the scope of this book.

CONGENITAL LESIONS OF LARYNX

Although the congenital laryngeal anomalies present with stridor, they can also cause dysphagia, aspiration, failure to thrive and dysphonia. Laryngoscopy (direct or flexible) usually diagnoses the problem while laryngotracheobronchial endoscopy will detect other synchronous airway lesions.

Aspiration is more common with tracheoesophageal fistula, whereas hoarseness is more common with vocal cord palsy.

LARYNGOMALACIA

This most common congenital laryngeal abnormality is characterized by excessive flaccidity of supraglottic larynx which gets sucked in during inspiration.

Clinical Features

The newborn develops intermittent, low-pitched inspiratory stridor within the first 2 weeks of life. The severity of stridor increases on crying but reduces in prone position. The stridor usually starts resolving after 9 months, and disappears by 2 years of age.

The cry is normal. Feeding difficulty, failure to thrive, and cyanosis occur infrequently. Children with laryngomalacia have high prevalence of gastroesophageal reflux disease (50–100%) and second synchronous airway lesions (17%).

Awake Flexible Laryngoscopy:

- Anterior prolapse of arytenoids mucosa.
- Short aryepiglottic folds.
- Posterior collapse of epiglottis.
- Elongated epiglottis is curled upon itself (omega-shaped). Aryepiglottic folds are floppy, and arytenoids are prominent.

Treatment

About 10% patients need surgical intervention, which includes supraglottoplasty (aryepiglottoplasty). Tracheostomy may be needed.

Laryngomalacia: It is the most common cause of stridor in childhood and usually occurs within the first 2 weeks of life.

- *Pathology*: It has multiple anatomic abnormalities, which include exaggerated omega-shaped epiglottis and short and inward collapse of aryepiglottic folds. Gastroesophageal reflux is probably secondary to increased negative intrathoracic pressure.

CONGENITAL VOCAL CORD PARALYSIS

Vocal fold (cord) palsy is the second common cause of neonatal stridor after laryngomalacia. See chapter Neurological Disorders of Larynx.

CONGENITAL SUBGLOTTIC STENOSIS

It consists of abnormal thickening of cricoid cartilage or fibrous tissue below the level of the vocal cords. The subglottic diameter becomes less than 3.5 mm in full-term neonate (normal 4.5–5.5 mm) and 3 mm in premature neonate (normal 3.5 mm).

Clinical Feature

They depend upon the grade of stenosis. Upper respiratory infection (URI) causes increase in dyspnea and stridor. Cry is normal.

Direct Laryngoscopy and Bronchoscopy

The stenosis is confirmed and then grading is done.

1. Grade I: less than 50% obstruction

2. Grade II: 51–70% obstruction
3. Grade III: 71–99% obstruction
4. Grade IV: No detectable lumen

Treatment

Many children, even though need observation, improve as the larynx grows. Grade-II or Grade-III/IV patients need tracheostomy. Decannulation may be tried when cricoid grows to sufficient size. Some patients may need laryngotracheal reconstruction to expand cricoid ring.

LARYNGEAL WEB/ATRESIA

Laryngeal webs are rare. They can be of two types—anterior and posterior. Most severe type of laryngeal web is total atresia of larynx. The incomplete recanalization of larynx results in the web which is seen between the vocal cords, and has a concave posterior margin.

Clinical Features

The child presents with congenital airway obstruction, weak cry or aphonia. All patients need genetic screening and cardiovascular evaluation especially of aortic arch.

Treatment

- Tracheostomy: It is often required.
- Thin web needs cutting either with a knife or CO_2 laser.
- Thick web needs excision via laryngofissure, silicon keel placement, and subsequent dilatations.

SUBGLOTTIC HEMANGIOMAS

Subglottic hemangiomas are benign vascular malformations and consist of endothelial hyperplasia.

Clinical Features

The infant remains asymptomatic for 3–6 months after birth. When hemangioma increases in size, the child presents with inspiratory or biphasic stridor which progressively increases in severity. Crying increases airway obstruction due to venous filling. The child has a normal cry. Some infants present with nonresolving croup. About 50% of affected children have associated cutaneous hemangioma. Some children have associated mediastinal hemangioma.

- Direct Laryngoscopy with Telescope or Microscope: Reddish-blue mass below the vocal cords can be seen. Biopsy may be associated with hemorrhage.
- *CT or MRI:* Assist in diagnosis.

Treatment

- Tracheostomy and observation: Many hemangioma regresses spontaneously.
- Steroid:
  - Dexamethasome 1 mg/kg/day for 1 week and then prednisolone 3 mg/kg in divided doses for one year.
  - Intralesional steroids and short-term intubation.
- CO_2 and KTP lasers: For small lesions.
- Laryngofissure: Surgical resection of circumferential or bilateral subglottic hemangiomas.

Congenital vascular lesions: They grow a bit but then usually start regressing and do not need any treatment. Complete involution may take many years.

LARYNGOESOPHAGEAL CLEFT

The failure of the fusion of cricoid lamina results in laryngoesophageal cleft.

Clinical features: Coughing, choking and cyanosis occur during feeding. The child gets repeated aspiration pneumonia.

Cleft and craniofacial anomalies: The care of these patients needs many medical and surgical subspecialties and requires a team management.

- *Initial management*: The children need a secure airway and adequate nutrition.

FOREIGN BODIES OF AIR PASSAGES

Depending on the size and nature, a foreign body (FB) may lodge in the larynx, trachea, or bronchi. A large FB, which cannot pass through glottis, gets lodged in the supraglottic region. A small FB easily passes down into the trachea or bronchi. Sharp pointed foreign bodies (such as pins, needles, or fish bones) may stick in either larynx or tracheobronchial tree.

Predisposing Factors

- *Age:* Children less than 5 years of age constitute more than 50% of the patients. They have a tendency to put the things in their mouth, which can go in either the food or the air passages.
- *Unconsciousness:* Unconscious adults (coma, deep sleep, alcoholic intoxication, or anesthesia) can inhale food, saliva, liquids, denture, and blood.
- *Disturbed swallowing:* Coughing, laughing, talking, crying, or tapping on back during swallowing can result in inhalation of food into air passages.
- *CN IX and CN X lesions:* Paralysis of pharynx and larynx can lead to aspiration problems.

Nature of Foreign Bodies

- *Nonirritating:* Plastic, glass or metallic items may remain symptomless for a long time.
- *Irritating:* Vegetables (peanuts, beans, seeds) can cause diffuse reactive congestion and edema of the tracheobronchial mucosa (vegetable bronchitis). They may swell up, and cause airway obstruction and pulmonary suppuration.

Clinical Features

Children may either hide or forget the incidence of foreign body (FB) inhalation. It is the suspicion of clinician that helps in diagnosis. The patients having air passage FB can pass through three stages:

1. *Inhalation:* This is the initial period of choking, gagging and wheezing that lasts for a short time. Small FB may be coughed out. If it is large, patient may get cyanosis and die.
2. *Latent:* During this symptom-free interval mucosa adapts to the presence of FB. It varies with the size and nature of the FB. Non-vegetable FB may go unnoticed for a long period of time.

3. *Manifestation:* Depending on the site, size, shape and nature of the FB, the obstruction, inflammation and trauma to the airway can give rise to following features:
   - Laryngeal foreign body: A large FB can kill the person. A relatively small FB can present with throat discomfort, hoarseness of voice, croupy cough, aphonia, inspiratory stridor, dyspnea, and hemoptysis.
   - Tracheal foreign body: A sharp FB produces coughing and hemoptysis. A small seed may move up and down the trachea, and produce audible slap, tracheal flutter, and palpatory thud. Biphasic stridor or asthmatoid wheeze is best heard when child's mouth is open.
   - Bronchial foreign body: Right bronchus is the most common site for FBs because it is wider and more in line with the tracheal lumen. A FB may either totally obstruct a lobar/segmental bronchus (atelectasis), or produce a check valve obstruction (obstructive emphysema) which allows only ingress of air (Fig. 5).
     - Atelectasis: Complete stop valve obstruction results in atelectasis of lung or its segment. Poor air entry is found on auscultation of the lower lobe on the back of the thorax.
     - Obstructive emphysema: Bronchial edema proximal to the FB and bronchial dilatation during inspiration produce a sort of check valve obstruction that allows entry of air with only inspiration.
     - Shifting foreign body: Small FB may move from one bronchus to other bronchus or to trachea, and produce poor air entry in any lung.
     - Acute laryngotracheobronchitis: Infection and chemical reaction of vegetable FB can produce the picture of acute laryngotracheobronchitis.
     - Spontaneous pneumothorax: Emphysematous bulla may rupture.
     - Pulmonary infections: A retained FB can give rise to pneumonitis, bronchiectasis or lung abscess.

In cases of chronic cough and recurrent pulmonary lesions such pneumonia, atelectasis, emphysema, pneumothorax, lung abscess and bronchiectasis of the same side and region FB must be ruled out.

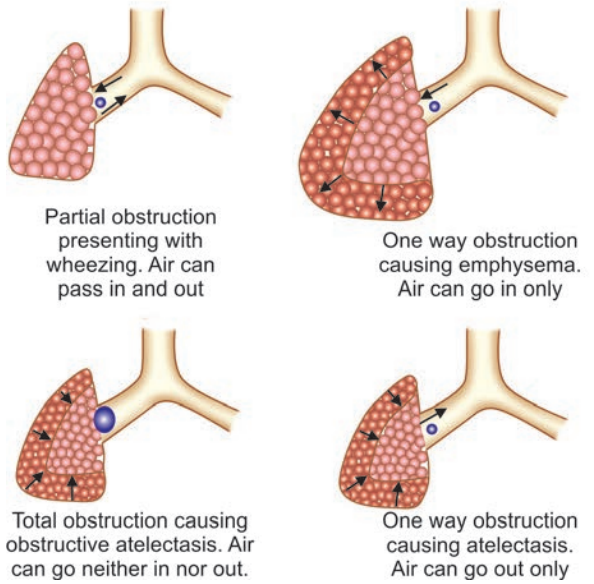


Fig. 5: Foreign body bronchus resulting in four types of bronchial obstruction

Differential Diagnoses

Laryngotracheobronchitis (see chapter Infections of Larynx).

Diagnosis

The detailed history, physical examination, and radiographs confirm the diagnosis.

- *X-ray soft tissue neck postero-anterior (PA) and lateral views in extended position:* Radio-opaque and sometimes, radiolucent foreign bodies can be seen in the larynx and trachea.
- *X-ray chest PA and lateral views especially at the end of inspiration and expiration:* They show atelectasis and obstructive emphysema, and provide indirect evidence of radiolucent FB.
- *Fluoroscopy/videofluoroscopy during inspiration and expiration*.
- *Bronchograms:* It delineates radiolucent foreign bodies and bronchiectasis.
- *CT scan:* HRCT/Virtual bronchoscopy
- *Laryngoscopy and bronchoscopy:* They are both diagnostic as well as therapeutic.

Jackson's dictum: In suspected cases of FB of air passages, bronchoscopy must be done as failure to do bronchoscopy is more disastrous than the complications of bronchoscopy.

Management

- *Medical:* Antibiotics, steroids and oxygen are administered immediately, and continued after the bronchoscopy.
- *Laryngeal foreign body*
  - Cricothyrotomy or emergency tracheostomy: These are done when Heimlich's maneuver fails.
  - Direct laryngoscopy or laryngofissure: FB is removed by direct laryngoscopy or by laryngofissure.
  - Heimlich's maneuver (Fig. 6):
    - Indication: A large bolus of food completely obstructing the larynx with total aphonia, and patient dying of asphyxia (impending death).
    - Contraindication: Partially obstructing FB because maneuver can cause total obstruction.

Fig. 6: Heimlich's maneuver. Standing behind the standing patient, sudden upward and backward thrusts just below the epigastric region squeeze the lungs' air, and may dislodge laryngeal foreign body

  - Method: First pounds on the back, and turns the patient upside down. Then, follows Heimlich maneuver. Standing behind the patient, and placing arms around patient's lower chest give four abdominal thrusts. The residual pulmonary air can dislodge the laryngeal FB.
- Tracheal and bronchial foreign bodies
  - Bronchoscopy under general anesthesia: Rigid bronchoscopy is the standard procedure for FB removal. Emergency removal is indicated only when the airway obstruction is present or FB is vegetable. The following methods are used for the removal of tracheobronchial FB:
    - Rigid bronchoscopy with or without a telescope.
    - Bronchoscopy with C- arm fluoroscopy.
    - Snares and basket or Fogarty's catheter balloon for rounded FB.
    - Bronchoscopy through the tracheostome.
    - Flexible fiberoptic bronchoscopy in selected cases.
    - Thoracotomy and bronchotomy for peripheral foreign bodies.
    - Lobectomy or pneumonectomy may be needed in cases of old unnoticed FBs.

Respiratory distress: If a child with respiratory distress is becoming quiet, it indicates that the child is about to have respiratory collapse.

LARYNGOTRACHEAL TRAUMA

Modes of Injuries

- Automobile accidents: Most common.
- Blow or kick on the neck.
- Neck hitting a stretched wire or cable.
- Strangulation.
- Penetrating injuries: Sharp objects or gun shot.

Pathology

Depending on the mode and severity of injury, the following lesions can occur:

- *Bruises:* External cervical bruises.
- *Tear and laceration of mucosa:* Tears in laryngeal or pharyngeal mucosa results in subcutaneous emphysema.
- *Fractures:* Compound (external or internal) or comminuted fractures of the laryngeal framework are common after 40 years of age because of calcification of the laryngeal framework. Children laryngeal cartilages are more resilient, and usually escape injury.
  - Hyoid bone
  - Thyroid cartilage (vertical or transverse): Fracture of upper part of thyroid cartilage may cause avulsion of epiglottis and false cords. Fractures of lower part of thyroid cartilage can disrupt true vocal cords
  - Cricoid cartilage
  - Upper tracheal cartilages
- *Hematoma and edema* of supraglottic or subglottic region.
- Dislocation of joints
  - Cricoarytenoid joints: The arytenoid cartilage can be displaced anteriorly, dislocated or avulsed
  - Cricothyroid joint: Dislocation of cricothyroid joint can cause paralysis of recurrent laryngeal nerve which lies just behind the joint
- *Laryngotracheal separation:* Trachea may get separated from the cricoid cartilage, and go into upper mediastinum injuring the recurrent laryngeal nerve

Clinical Features

Clinical features depend on the type and severity of the injury. They are following:

- Stridor.
- Hoarseness of voice or aphonia.
- Pain and difficulty in swallowing.
- Aspiration of food, blood and secretions.
- Local laryngeal pain may be marked on speaking or swallowing.
- Hemoptysis due to tears in laryngeal or tracheal mucosa.
- Cervical bruises or abrasions.
- Tenderness in the laryngeal area.
- Subcutaneous emphysema: It may increase on coughing and is due to mucosal tears.
- Flattening of thyroid prominence and anterior cervical contour.
- Gap and crepitus between fractured fragments of thyroid and cricoid cartilages or hyoid bone.
- Cricoid cartilage may get separated from larynx or trachea.

Diagnosis

- *Laryngoscopy:* When the patient's condition allows indirect laryngoscopy examination can reveal edema, hematoma, mucosal lacerations, displacement of epiglottis, fragments of cartilage, asymmetry of larynx. Fiberoptic laryngoscopy (rigid or flexible) provides easier visualization. Direct laryngoscopy is relatively contraindicated as it may precipitate respiratory distress.
- *X-ray soft tissue neck lateral view:* It may show subcutaneous emphysema, mucosal swelling, fracture/displacement of epiglottis, thyroid and cricoid cartilages, hyoid bone, or change in the air column.
- *CT scan:* It reveals injuries of laryngeal cartilages in a better way. Three-dimensional reconstructions are very useful.

Associated Injuries

It is essential to examine the patient to rule out other common injuries such as injury to head, cervical spine, chest, abdomen and extremities. X-ray chest may show pneumothorax. Gastrografin swallow will reveal esophageal tears.

Complications

- Laryngeal stenosis: Supraglottic, glottic or subglottic
- Perichondritis
- Laryngeal abscess
- Vocal cord paralysis

Treatment

- Watch out for respiratory distress.
- Voice rest.
- Humidification of inspired air.

- *Steroids*: Steroids resolve edema and hematoma, and prevent scarring and stenosis.
- *Antibiotics*: They prevent perichondritis and cartilage necrosis.
- *Management of impaired airway*: In cases of laryngeal trauma, tracheostomy is preferred over endotracheal intubation which may be difficult and hazardous.
- *Open reduction*: It should be done in 3–5 days after injury, and definitely before 10 days.
  - Wire and titanium miniplate: For fractures of hyoid bone, thyroid or cricoid cartilage.
  - Absorbable sutures: Mucosal lacerations are repaired with vicryl.
  - Removal of loose fragments of cartilage and avulsed epiglottis and arytenoids.
  - Anchoring: Arytenoid and epiglottis are repositioned in their normal position.
  - End to end anastomosis: In cases of laryngotracheal separation.
  - Internal splint of laryngeal structures: A laryngeal stent or silicone tube may have to be left for 2–6 weeks. A silastic keel prevents webbing of anterior commissure.

Clinical Highlights

1. **Respiratory dead *space*:** It is approximately 150 ml. Tracheostomy cuts down the dead space by 30–50%.
2. **Congenital stridor with a hoarse cry:** Some of the causes are laryngeal web, laryngeal paralysis and congenital laryngeal cyst.
3. **Laryngomalacia:** It is the commonest cause of "inspiratory stridor" shortly after the birth. Laryngomalacia is the most common congenital anomaly of the larynx. Flexible fiberoptic laryngoscopy is the best way to diagnose it. It is the most common cause of congenital laryngeal stridor. Stridor is relieved when child is put in prone position and disappears spontaneously as the child grows. Epiglottis appears omega shaped.
4. **Congenital subglottic *stenosis*:** In case of premature neonate with subglottic stenosis, tip of 3 mm diameter bronchoscope cannot be passed through subglottis. In a full term newborn with subglottic stenosis tip of 4 mm diameter bronchoscope cannot be passed.
5. **Laryngeal *web*:** Glottis is the most common site.
6. **Foreign body *bronchus*:** The features include history of coughing, choking and gagging. FB lodges more often in the right bronchus and can be expelled spontaneously with coughing. X-ray chest may show either hyper inflated (emphysema) lung on one side or unilateral atelectasis (collapse) of lung. Best management for inhaled FB is bronchoscopy removal.
7. **Laryngeal *trauma*:** The chief danger of laryngeal trauma is respiratory distress stridor.

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52 Cervical Symptoms and Examination

One should not trifle with a thing, though it may be very insignificant. If you respect a thing, the thing also respects you. Even a broomstick should be treated with respect. One should perform even an insignificant work with respect.

—Holy Mother Sarada Devi

Points of Focus

NECK

- **HISTORY**
  - ✓ Common Neck Masses occurrence by Location
  - ✓ Age Specific Swellings
- **PHYSICAL EXAMINATION**
  - ✓ Examination of Lymph Nodes of Neck
  - ✓ Clinical Features of Common Neck Lesions
- **DIAGNOSTIC TESTS**
  - ✓ Unknown Neck Mass (Unknown Primary Lesions)

THYROID GLAND

- **HISTORY**
- **EXAMINATION**
  - ✓ General Appearance
  - ✓ Local Examination
  - ✓ Toxic Manifestations
- **INVESTIGATIONS**
  - ✓ Levels of T_3, T_4 and TSH in Thyroid Disorders
- **CLINICAL HIGHLIGHTS**

NECK

HISTORY

The age of the patient (child, young adult or older adult) and location of the neck mass (midline anterior neck, anterior triangle and posterior triangle) are particularly important in differentiation of congenital/developmental, inflammatory and neoplastic neck masses (Table 1). Although the congenital/developmental swellings occur in children (< 15 years of age) and young adults (16–40 years of age), they are less common than inflammatory masses in these age groups. In contrast, neoplasia should be considered first in older adults (> 40 years of age). The specific history aspects and physical findings limit the number of diagnostic tests and can avoid unnecessary investigations.

Age specific swellings

- Since birth: Branchial fistula
- Newborn: Sternomastoid tumor (torticollis or wryneck)
- Young children: Branchial cyst and cystic hygroma
- Elderly people: Secondary metastatic neck nodes
- Adults: Inflammatory swellings

PHYSICAL EXAMINATION

Thorough physical examination gives an idea about the derivation of mass—inflammatory (Fig. 1), congenital or neoplastic; vascular, salivary, thyroid or nodal. The key features of physical examination include:

- Site of the swelling according to anatomic lymphatic drainage (Fig. 2), and developmental areas
- Size and extent of the mass and number of nodes
- Relationship (fixation and displacement) to surrounding structures
- Consistency of the mass
- Presence of pulsation, thrills and bruits
- Distinct odor of wet keratin and necrotic tumor
- Transillumination
- Radiotherapy effects (Fig. 3)
- Depth of the swelling: Cutaneous (Fig. 4), subcutaneous, deep or superficial to muscle
- Complete head and neck evaluation including palpation and indirect laryngoscopy or flexible nasopharyngolaryngoscopy

TABLE 1 Common neck masses occurrence by location and age

Groups	*Midline/Anterior neck*	*Anterior triangle of neck*	*Posterior triangle of neck*
Congenital/Developmental	Common (number two cause) in children (up to 15 years) and young adults (16–40 years)		
	Thyroglossal duct cyst, dermoid, laryngocele	Branchial cyst, thymic cyst, sialadenopathy (parotid, submandibular)	Cavernous lymphangioma (cystic hygroma)
Inflammatory	Most common and number one cause in children and young adults		
• Sialadenitis		Parotid, submandibular	
• Acute lymphadenitis	Bacterial: *Streptococcus pyogenes*, *Staphylococcus aureus*, mixed oral bacteria including anaerobes Viral: Epstein-Barr (infectious mononucleosis); HIV Protozoans: *Toxoplasma gondii*		
• Granulomatous lymphadenitis		Atypical tuberculosis (children), actinomycosis (adult; submandibular and upper jugular), cat-scratch disease (children, preauricular and submandibular)	Tuberculosis (adults): posterior triangle supraclavicular (scrofula)
Primary neoplastic	Thyroid; lymphoma; lipoma	Thyroid; lymphoma; salivary; lipoma; vascular: glomus, carotid body tumor	Lymphoma, Lipoma
Metastatic	Most common in old adults (> 40 years)		
• Primary unknown		Unknown primary lesion	
• Primary known: Secondary metastatic lymph nodes		Submandibular: oral cavity, nasal-sinus, face; Upper jugular: oral cavity, oropharynx, larynx; Lower jugular: hypopharynx, thyroid	Jugular; Occipital: nasopharynx, scalp; Supraclavicular: Infraclavicular primary

Fig. 1: Cold abscess neck pointing at two sites: posterior triangle and suprasternal region. See Figure 7 for ultrasonography neck findings

Fig. 2: Metastatic neck nodes level III right side of neck

for upper aerodigestive tract especially for the primary sites of lymphatic drainage to the location of neck mass

Examination of Lymph Nodes of Neck

Examination of neck nodes [Figs 5 and 6(A and B)] needs a systematic approach. It is an important part of otorhinolaryngology head and neck evaluation.

Palpation of cervical lymph nodes is carried out while standing behind the seated patient. Examiner uses both hands simultaneously, with fingers semiflexed and adducted, and thumbs in partial opposition. Slight flexion of neck achieves relaxation of muscles. Internal jugular chain (upper, middle and lower) groups lie deep to sternocleidomastoid muscle, which needs to be retracted. The nodes are explored systematically: the submental triangle, the submandibular glands and triangles (thumbs over buccinators); the retromandibular depressions (thumbs probing over parotids), the upper attachment of sternocleidomastoid, and the occipital attachment of trapezius. Palpation then continues along the vertical chains of cervical nodes which have superficial (associated with the external jugular and anterior jugular veins, the superficial cervical and anterior cervical groups) and deep nodes (upper, middle and

Fig. 3: Neck after 4 months of radiotherapy in a case of carcinoma, right tonsil, soft palate and base of tongue

Fig. 4: Fasciitis. Right side neck's firm well-defined swelling involving superficial fascia

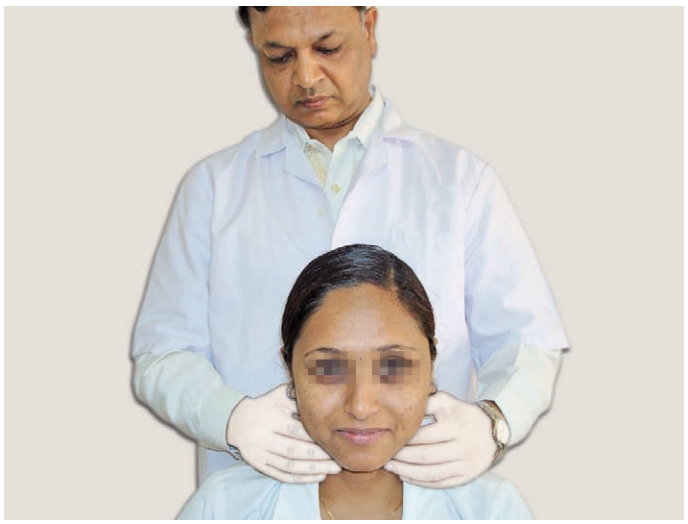
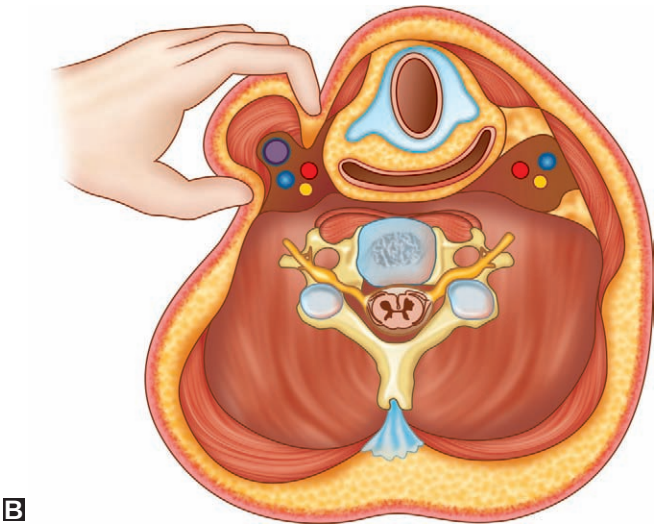


Fig. 5: Palpation of cervical lymph nodes

lower deep cervical jugular nodes, alongside and embedded in areolar tissue near the carotid sheath but particularly those aspects surrounding the internal jugular vein) and then the juxtavisceral chain (prelaryngeal, pretracheal and paratracheal).



Figs 6A and B: Method of palpating deep cervical lymph nodes. The thumb and index finger grasping the sternocleidomastoid muscle palpate the deep jugular lymph nodes of region II, III, and IV

During the inspection and palpation of these nodes, one must note their location, number, discrete/matted, size, consistency, tenderness and fixity. Inflammatory nodes are tender. Fixity can be either to overlying skin or deeper structures. Mobility of nodes is checked in both vertical and horizontal planes. Metastatic nodes are usually hard but metastatic melanoma and lymphoma nodes are soft.

Clinical Features of Common Neck Lesions

- *Common submandibular swellings:* Lymphadenopathy, submandibular salivary gland swelling, deep or plunging ranula, extension of growth from mandible.
- *Common swellings of lower part of posterior triangle:* Lymph nodes enlargement, cystic hygroma and cervical rib.
- *Swellings moving with deglutition:* Goiter, thyroglossal cyst, laryngeal swellings, subhyoid bursitis, and prelaryngeal and pretracheal tubercular and metastatic lymph nodes attached to larynx and trachea.
- *Branchial cyst:* This cystic swelling is partially covered by the anterior margin of upper third of sternocleidomastoid. It is seen in children and young adults.

- *Carotid body tumor:* Firm pulsatile swelling at the level of upper border of thyroid cartilage under the anterior margin of sternocleidomastoid.
- *Aneurysm:* Pulsatile swelling in line of carotid arteries.
- *Sternomastoid tumor:* An oval firm swelling in the line of sternomastoid muscle of a newborn baby with torticollis (head is bent to one side whilst the chin points to the other side).
- *Dermoid cyst:* This cystic swelling is common in the middle of lower part of neck.
- *Lymphosarcoma:* Subcutaneous dilated veins are characteristic features.
- *Secondary metastatic nodes:* Hard, painless, and progressively enlarging swelling at the levels of cervical lymph nodes (Fig. 2) in elderly people which may involve skin, and get fixed to deeper structures.
- *Tuberculous sinus or ulcer:* Transverse linear scar is characteristic.
- *Osteomyelitis sinus of mandible:* This single sinus lies slightly below the inferior margin of mandible.
- *Actinomycosis:* Sulfur-like grains in the pus coming out from multiple sinuses over an indurated mass at the upper part of neck.
- *Branchial fistula:* A pin-point opening present since birth at the lower third of anterior border of sternocleidomastoid muscle.
- *Cold abscess in posterior triangle:* It can be due to caries of cervical spine. In these cases, neck movements (flexion, extension, rotation and lateral flexion) are limited. While testing cervical movements, deaths have been reported from the dislocation of atlantoaxial joint and dens pressing the medulla.
- *Thyroglossal cyst and fistula*: This cervical midline cyst is present between hyoid and thyroid gland, and moves with deglutition. It moves upward with the protrusion of tongue. Cyst may get secondarily infected, and form abscess which may be either incised surgically or burst spontaneously, and results in thyroglossal fistula. The opening of fistula is indrawn, and frequently overlaid by a crescent of skin.

Thyroglossal cyst: This midline neck mass is recognized by its movement with swallowing. An ultrasound neck will evaluate for ectopic thyroid tissue.
Carotid triangle swellings: The most common causes of swellings in this region are paragangliomas, schwannomas, meningiomas and nodal metastases.

DIAGNOSTIC TESTS

In suspected cases of inflammatory adenopathy, a trial of antibiotic therapy and observation (not > 15 days) is well accepted as a clinical test. For a firm diagnosis, the following tests are helpful:

- *Ultrasonography* (Fig. 7): In the presence of bruit or thrill in vascular masses, ultrasonography (USG) can differentiate degenerative problems (aneurysm) from neoplastic conditions (glomus and carotid body tumors). When both A and B modes of scans are used, USG also facilitates differentiation between solid (lymph nodes, neurogenic tumors and ectopic thyroid tissue) and cystic masses (branchial and thyroglossal cysts).
- *Radionuclide scanning:* This scan shows whether the mass has functioning or nonfunctioning tissue. It is helpful in thyroid disorders, and in localizing the mass inside or outside of the salivary gland.

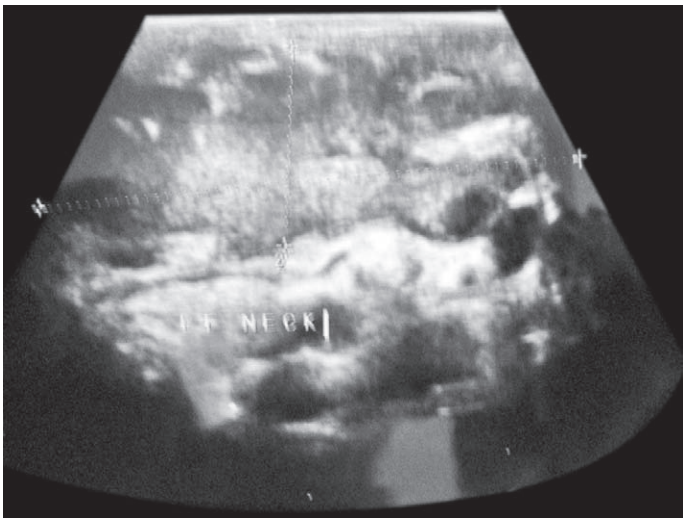


Fig. 7: Ultrasonography neck findings of patient of Figure 1. Cold abscess neck pointing at two sites: posterior triangle and suprasternal region

  - Positron emission tomography: Positron emission tomography (PET) indicates functional activity of a mass.
    - A more radio-intense mass has greater metabolic activity, and is usually neoplastic.
    - It differentiates recurrent cancer from postirradiation changes.
    - Identifies distant metastasis.
    - Diagnoses equivocal adenopathy found in CT/MRI.
    - It helps in determining primary site (in and beyond head and neck region) for metastatic neck carcinoma of unknown origin.
    - It can show concomitant distant metastasis.
    - Disadvantages include high expense, resolution limitations and high false-positive rate due to background salivary activity.
- *CT and MRI imaging*: Currently, these are the most helpful tests, and differentiate solid from cystic masses; locate a mass within a glandular structure or lymph node, and differentiate vascular masses. They enhance the mucosal disease, and show anatomic relationship.

T2W images of MRI can show submucosal neoplastic disease in nasopharynx and base of tongue in cases of unknown primary.

- They help in determining the extent of disease and planning surgery.
- *Fine needle aspiration cytology and open biopsy:* A 25-gauge needle biopsy is gold standard in diagnosis of a neck mass. In most cases, clinical judgment and needle biopsy establishes the diagnosis. Fine needle aspiration cytology (FNAC) differentiates a cystic lesion from inflammatory tissue. If needle biopsy is inconclusive or negative and suspicion is high, open biopsy should be performed. Biopsy evaluation with culture can show a specific cause.
  - Indications: The main indications for biopsy particularly in children are the following:
    - Progressively increasing size of nodes
    - Single asymmetric nodal mass
    - Persistent nodal mass in absence of features of infection
    - Infectious disorders not responding to antibiotics

Unknown Neck Mass (Unknown Primary Lesions)

Its management include complete ENT, head and neck examination, FNAC, histopathology, flow cytometry and polymerase chain reaction; imaging include X-ray chest, USG abdomen, T2W MRI and PET scanning, panendoscopy and guided biopsy, open excisional biopsy, and frozen section diagnosis.

THYROID GLAND

HISTORY

- *Age and sex*: Thyroid swellings (goiter) are more common in females (Fig. 8). Hashimoto's disease is a disease of middle-aged women. Physiological goiter occurs frequently at puberty in girls. Deficiency of iodine in water and food is the cause of endemic goiter. Thyroid adenoma of middle-aged patients is usually toxic. Papillary carcinoma is common in young girls whereas follicular carcinoma occurs in middle-aged women. The anaplastic carcinoma occurs in old age. The patients of primary toxic goiter are usually young.
- *Onset, duration and progress*: Inflammatory diseases have short history. Thyrotoxicosis may appear in women having stress and strain. Sudden increase in the size of previous goiter is an ominous sign. Simple goiters grow very slowly, and may remain of the same size for years at a stretch. Multinodular goiter and solitary nodules grow extremely slow. The papillary and follicular carcinomas also grow slowly for years before metastasizing. Anaplastic carcinoma grows fast.
- *Primary thyrotoxicosis*: These patients may or may not have goiter, and the brunt of attack falls on nervous system. The symptoms include loss of weight, staring or protruding eyes, preference for cold, excessive sweating, excitability, irritability, insomnia, tremors and muscle weakness. Enquire the patient regarding palpitation and exhaustion on strain.
- *Secondary thyrotoxicosis*: In this condition, longstanding thyroid nodule and simple or multinodular goiter develop toxic features. In this disease, the brunt of attack falls on cardiovascular systems, and patients do not have protruding eyes and tremors. The cardiovascular symptoms include palpitation, ectopic beats, chest pain and dyspnea on exertion. Other features are cardiac arrhythmias, congestive cardiac failure and swelling of ankles.
- *Myxedema (hypothyroidism)*: The symptoms include weight gain, intolerance to cold weather, dry skin, puffiness of face, dull expression, loss of hair and two-third of eye brows, muscle fatigue, lethargy and mild hoarseness of voice.
- *Pressure effects*: Enquire about dyspnea (pressure on trachea), dysphagia (pressure on esophagus) and hoarseness of voice (pressure on recurrent laryngeal nerve).
- *Drugs*: Enquire about the medicines taken before the goiter or for the goiter.
- *Personal history*: Cabbage, kale and rape are goitrogens. Sea fish has low iodine contents.

EXAMINATION

General Appearance

Look for the features of hyperthyroidism such as exophthalmos (staring), excitability, trembling, nervousness, sweating and wasting. Sleeping pulse rate must be recorded. Hypothyroidism causes weight gain, bradycardia, dry and rough skin and depression.

- *Abdominal examination*: In women with Hashimoto's disease, liver and spleen should be examined for cirrhosis.

Local Examination

Inspection (Figs 9 and 10) and palpation of goiter are similar to examination of the swelling (See Chapter: History and Examination) and neck, and include site, shape, size, extent, surface, pulsation, overlying skin, consistency, mobility and relation with neighboring structures, etc. The thyroid swellings are palpated both from the front and behind (Figs 11 and 12). Thyroid swellings move on swallowing.

- *Uniform enlargement*: It takes the shape of thyroid gland (Fig. 8).
- *Nodule*: It may be single or multiple, and may occur either in isthmus or lobe.
- *Mobility*: Movements with deglutition are greatly limited in cases of chronic thyroiditis inflammation and malignancy. Extent of mobility should be checked in both horizontal and vertical dimensions.

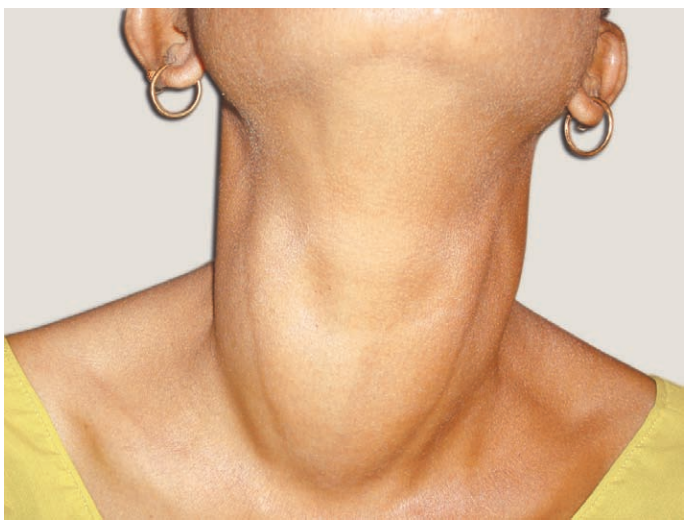


Fig. 8: Goiter. Generalized swelling of whole gland maintaining the shape of thyroid gland

Fig. 9: In normal position, goiter is not visible

Fig. 10: Pizzillo's technique of examining goiter. Pizzillo's method reveals the thyroid swelling better

Fig. 11: Thyroid palpation from behind. Fingers palpate lobes of the gland, and thumbs are placed on occipital region to keep the neck flexed

Fig. 12: Lahey's method. One hand pushes the gland to make the lobe prominent, and the other hand palpates the prominent lobe

- *Retrosternal extension* (Figs 13 and 14): It puts pressure on the great veins at the thoracic inlet and results in dilatation of subcutaneous veins over the upper part of chest. In these cases, it is not possible to get the lower limit of goiter on deglutition. Raising of both arms and touching both the ears result into facial congestion, cyanosis and distress.
- *Consistency*: Note whether the consistency is variable or uniform. Clinically, it is difficult to know whether the nodule is solid or cystic because the cystic fluid may be under great tension. The cellular swelling may have no tension, and feels soft. Ultrasonography neck examination easily differentiates between solid and cystic swellings. Stony hard consistency suggests malignancy and Riedel's thyroiditis.
- *Pulsation, bruit and murmurs* (Fig. 15): It is usually present in toxic goiter, and is due to high vascularity.
- *Surrounding structures*
  - Trachea: It can be displaced and compressed from both the sides (scabbard trachea).
    - Kocher's test: Slight pressure on lobes produces stridor in cases of scabbard trachea.
  - Esophagus: Displacement or compression of esophagus will cause dysphagia.
  - Recurrent laryngeal nerve: Its involvement leads to vocal cord palsy and hoarseness of voice, and suggests malignancy.

Fig. 13: Goiter; swelling is more on left side. Note the obliteration of suprasternal space that raises suspicion of retrosternal extension

Fig. 14: Palpation of the lower end of goiter to rule out retrosternal extension. Patient is first requested to swallow her saliva and then surgeon pushes his fingers in suprasternal region to feel the lower extent of thyroid mass

Fig. 15: Auscultation of thyroid

- Carotid artery: Benign large goiter just pushes the carotid sheath, and pulsation can be felt. Malignant goiter can engulf the carotid sheath and then carotid pulsations are not detected.
- Cervical sympathetic trunk: Its involvement gives rise to Horner's syndrome, and suggests malignancy.
  - Horner's syndrome: It consists of partial ptosis, enophthalmos, constricted pupil and anhidrosis of face.

- *Measurement of neck circumference*: Measurement of neck circumference at the most prominent part of goiter helps in determining the progress and results of treatment.
- *Metastasis*: In addition to the draining neck lymph nodes, surgeon should look for distant metastases in skull and pathological fractures of long bones.

Toxic Manifestations

The cardinal signs of primary toxic goiter include exophthalmos, enlargement of thyroid gland, tachycardia and tremors. In the secondary toxic goiter (toxicity occurring in preexisting longstanding adenoma or nodular goiter), exophthalmos and tremors are usually absent. The brunt of disease falls on cardiovascular system. Pulse becomes irregular in rate and rhythm. Atrial fibrillation and heart failure may supervene later on.

- *Exophthalmos*: It refers to protrusion of eyeball (Fig. 16). The following signs should be looked for:
  - Von Graefe's sign: The upper eyelid lags behind when patient looks downward.
  - Stellwag's sign: It consists of upper eyelid retraction and infrequent winking.
  - Joffroy's sign: There is no forehead wrinkling when patient looks towards ceiling keeping the head downward. In cases of facial palsy, it is unilateral.
  - Moebius sign of convergence: Convergence of eyes becomes difficult.
  - Dalrymple's sign: The upper sclera becomes visible when patient has exophthalmos.
  - Advanced features: They include chemosis and ophthalmoplegia.
- *Tachycardia*: The sleeping pulse rate rises, and may vary from 90–180. Pulse becomes water hammer type.
- *Tremors*: Fine tremors of stretched fingers of stretched out arms are seen (Fig. 17). In protruding tongue, fibrillar twitching can be seen (Fig. 18). In advanced cases, whole body may shake and tremble.

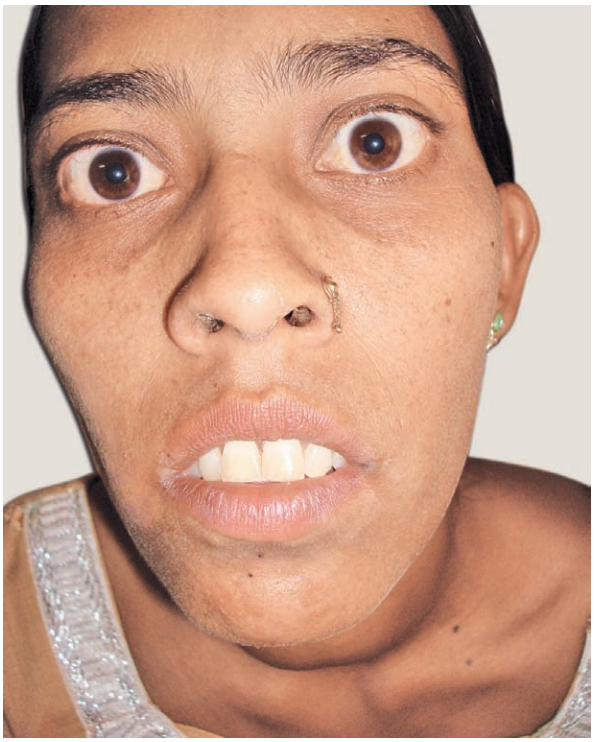


Fig. 16: Exophthalmus in case of toxic goiter

- *Thyroid bruit*: It is present in Grave's primary thyrotoxicosis, and indicates increased vascularity. It is better present on the upper pole.

INVESTIGATIONS

There is a long list of thyroid function tests but the essential tests done routinely are mentioned here.

- *T_3, T_4,* and TSH: The levels triiodothyronine (T_3), L-thyroxine (T_4) and TSH are reliable indicators of the functioning of thyroid gland (Table 2). If the TSH levels are normal, T_3 and T_4 estimations are usually not required.
  - Free T_3 and T_4: T_3 and T_4 are bound to serum proteins. The small amount biologically active hormones remain unbound and free, and have metabolic activity. T_3 is

Fig. 17: Fine tremors of stretched fingers of stretched out arms

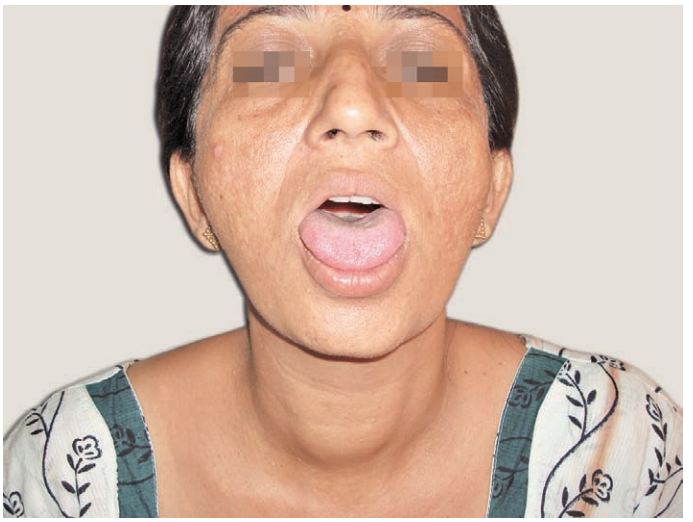


Fig. 18: See fibrillar twitching of the tongue

TABLE 2 Levels of T_3, T_4 and thyroid stimulating hormone in thyroid disorders

Thyroid functions	*Free T_3 (3.5–7.5 µmol l^{-1})*	*Free T_4 (10–30 nmol l^{-1})*	*TSH (0.3–3.3 mU l^{-1})*
Euthyroid	Normal	Normal	Normal
Thyrotoxicosis	High	High	Low/ Undetectable
Myxedema	Low	Low	High
Developing hypothyroidism	Lower normal range	Lower normal range	High
T_3 toxicity	High	Normal	Low/ Undetectable

quick acting (few hours), and is a more important physiological hormone. T_4 acts more slowly (4–14 days). The free T3 and T4 are preferred, and the assay of total T_3 and T_4 are now becoming obsolete.

- Pituitary-thyroid axis: The secretion of TSH from pituitary depends upon the levels of T_3 and T_4, and is modified in a negative feedback mechanism. Thyrotrophin-releasing hormone from hypothalamus also regulates the TSH secretion from pituitary gland.
- *Thyroid autoantibodies*: The levels of antibodies against thyroid peroxidase (thyroid microsomal antigen) and thyroglobulin are significantly high in many cases of autoimmune thyroiditis. In the follow-up cases of thyroid carcinomas, surgeon should not forget that anti-thyroglobulin antibody affects the levels of thyroglobulin levels.
- *X-ray chest and thoracic inlet*: It shows the retrosternal extension of goiter, tracheal deviation and compression and pulmonary metastasis.
- *Ultrasound neck*: It easily differentiates cystic nodules from the solid tumors. Ultrasound guided FNAC offer better results. Incidental thyroid swellings are frequently noticed during USG neck, and may not be clinically relevant.
- *CT, MRI and PET*: They are not suggested in routine study of thyroid swelling. Their indications include malignancy, retrosternal extension and recurrent thyroid swellings.
- *Isotope scanning*: It is indicated only in thyrotoxicosis to localize the area of overactivity in thyroid gland (single nodule or multinodular goiter) that has important implications for therapy. Whole-body scanning is indicated in operated patients of thyroid carcinoma to demonstrate metastases. Technetium (^{99m}Tc) is cheaper than radiolabelled iodine (^{123}I).
- *Fine-needle aspiration cytology*: This simple, quick, economical and OPD procedure is the investigation of choice for discrete thyroid nodules. US-guided FNAC achieve more accurate sample material. The FNAC results may be nondiagnostic, non-neoplastic, follicular, suspicious of malignancy or malignant.
- *Serum calcitonin and carcinoembryonic antigen*: They are screening tests for medullary carcinoma.

The essential thyroid function tests, which are done in routine, include T_3, T_4 and TSH, thyroid antibodies, USG neck and FNAC.

Clinical Highlights

1. **Thyroglossal cyst:** This midline neck mass is recognized by its movement with swallowing. An ultrasound neck will evaluate for ectopic thyroid tissue.
2. **Carotid triangle swellings:** The most common causes of swellings in this region are paragangliomas, schwannomas, meningiomas, and nodal metastases.
3. **Thyroid function tests:** The essential tests, which are done in routine, include T_3, T_4 and TSH, thyroid antibodies, USG neck and FNAC.

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53 Neck Nodes, Masses and Thyroid

If you project hatred and jealousy, they will rebound on you with compound interest. No power can avert them; when once you have put them in motion, you will have to bear them.

—Swami Vivekananda

Points of Focus

NECK NODES AND MASSES
- ✓ Unknown Primary Lesions
- ✓ Thyroid Neoplasms
- ✓ Lymphoma
- ✓ Salivary Neoplasms
- ✓ Carotid Body Tumors and Glomus Tumors
- ✓ Schwannomas or Neurilemmomas
- ✓ Lipomas
- ✓ Branchial Cysts
- ✓ Branchial Sinus or Fistula
- ✓ Thyroglossal Cyst
- ✓ Cervical Lymphangioma (Cystic Hygroma)
- ✓ Dermoid Cyst
- ✓ Acute Cervical Lymphadenitis
- ✓ Tuberculous (TB) Cervical Lymphadenitis (Scrofula)
- ✓ Nontuberculous Mycobacteria Cervical Lymphadenitis
- ✓ Sternomastoid Tumor
- ✓ Cervical Rib

THYROID NEOPLASMS
- ✓ Predisposing (Risk) Factors
- ✓ Thyroid Nodule
- ✓ Classification of Thyroid Neoplasms
- ✓ Adenoma
- ✓ Clinical Features of Malignant Tumors
- ✓ Papillary Adenocarcinoma
- ✓ Follicular Adenocarcinoma
- ✓ Medullary Thyroid Carcinoma
- ✓ Lymphoma and Anaplastic Carcinoma
- ✓ Investigations
- ✓ Staging
- ✓ Treatment
- ✓ Prognostic Factors

CLINICAL HIGHLIGHTS

NECK NODES AND MASSES

The age of the patient and location of the neck mass are particularly important in differentiation of congenital/developmental, inflammatory and neoplastic neck masses. In children and young adults, congenital/developmental swellings are more common than inflammatory masses. In contrast, neoplasia should be considered first in older adults. The specific history aspects and physical findings limit the number of diagnostic tests, and can avoid unnecessary investigations. Thorough physical examination gives an idea about the derivation of mass—inflammatory, congenital or neoplastic; vascular, salivary, thyroid or nodal. Diagnostic tests usually include ultrasonography (USG), positron emission tomography, CT and MRI imaging, fine needle aspiration cytology (FNAC) and open biopsy. See the table of common neck masses in Chapter "Cervical Symptoms and Examination.

Management of Unknown Neck Mass (Unknown Primary Lesions)

Any unknown asymptomatic asymmetric neck mass in the lymph nodes' region in adults which cannot be diagnosed with history, physical and routine investigations must be considered metastatic until proven otherwise (Fig. 1).

Fig. 1: Unknown primary. Left-side neck swelling involving lymph nodes level of I, II, III and IV. History of gradually increasing painless swelling for 3 months. Swelling was bony hard and fixed. Routine clinical ENT examination could not reveal any primary site

Fig. 2: Carcinoma right tonsil. Right-side metastatic nodes of level I, II and III. Note the mark of local traditional burn treatment

- *Complete ENT, head and neck examination*: When the enlarged nodes are present in the supraclavicular area or in the lower third of the neck, one must search for the complete digestive tract, tracheobronchial tree, breast, thyroid and the genitourinary tract.
- *Fine needle aspiration cytology*: In a patient who has a neck mass and in whom thorough ENT head and neck examination is negative, FNAC should be performed.
  - Histopathology: It is helpful in differentiating lymphoma from carcinoma.
  - Flow cytometry (for lymphoma diagnosis): It is a method of measuring fluorescence from stained cells that are in suspension and flowing through a narrow aperture. One or two lasers activate the dyes. It is used to measure cell size, number, viability and nucleic acid content.
  - Polymerase chain reaction: Polymerase chain reaction identifies Epstein-Barr virus for diagnosis of primary nasopharyngeal carcinoma.
  - After FNAC, which can be positive, equivocal or negative in a high-risk patient (tobacco/alcohol user), following diagnostic tests are performed.
- *Imaging*: X-ray chest and USG abdomen are frequently ordered, but are usually of not much help except in cases of supraclavicular nodal mass. The T2W MRI delineates submucosal disease. Positron emission tomography scanning is controversial in detecting unknown primary for head and neck tumors.
- *Panendoscopy and guided biopsy*: If FNAC remains elusive, panendoscopy of aerodigestive tract is done after appropriate scans are obtained. An obvious lesion must be biopsied. If no obvious lesion is found then depending upon the region of lymph nodes involved, deep submucosal biopsies should be performed from the following silent primary tumor sites:
  - Nodes high in the neck or posterior triangle: Fossa of Rosenmüller in nasopharynx.
  - Jugulodigastric nodes:
    - Tonsil: Tonsillectomy is done as the tumor arises deep in the crypts of tonsil (Fig. 2).
    - Base of tongue: Deep to the folds of lingual lymphoid tissue.
    - Supraglottic larynx: Less common.
  - Lower jugular: Pyriform sinus.
- *Open excisional biopsy*: It is performed when FNAC shows carcinoma, but no primary site is found.
- *Frozen section diagnosis*: If this facility is available, depending upon the reports, the following decisions can be taken.
  - Radical neck dissection: In cases of squamous cell carcinoma, melanoma and upper neck adenocarcinoma.
    - Postoperative radiotherapy: Though still controversial, postoperative radiotherapy of nasopharynx, ipsilateral tonsil, base of tongue, and contralateral side of neck is frequently advocated in cases of squamous cell carcinoma.
      Disadvantages:
      - Infraclavicular sites not covered
      - Compromise treatment of late appearing mucosal carcinoma
      - Induce later mucosal carcinoma
      - Increased morbidity: Xerostomia, dysphagia, dental caries
      - Improvement in cure rates not well proved
    - Adenocarcinoma: Most cases (86%) of adenocarcinoma have other distant metastases, and the cure rate is not affected by treatment of neck.
  - Closure: In cases of supraclavicular adenocarcinoma and lymphoma, staging before radiation and/or chemotherapy is important.

Silent primary sites and metastatic neck nodes: In patients with metastatic neck nodes when the primary lesion is not found upon physical examination including flexible/rigid endoscopy, biopsies from four silent primary sites should be taken. These four silent sites are nasopharynx, tongue base valleculae, pyriform sinus and tonsils (tonsillectomy).

Thyroid Neoplasms

These constitute a leading cause of anterior-compartment neck masses in all age groups.

- *Age*: Among children, boys are more affected than girls, and there is increased incidence of malignant disease. Adults have a greater incidence of benign lesions with a female predominance (Fig. 3).
- *Investigations*: Ultrasound, thyroid scans and thyroid function tests are usually done.
- *FNAC*: Currently, initial FNAC of all thyroid mass skipping both USG and thyroid scans, has become a standard practice.
- *Treatment*: It varies:
  - Cystic thyroid mass is aspirated. Recurrence needs surgery.
  - Solid thyroid lesions are managed according to their activity on thyroid scan.
    - Functioning nodules are managed by suppression.
    - Nonfunctioning cold nodules are explored, and managed on the basis of histology and extent of disease. Approximately 25% of solitary cold nodules are found cystic and another 25% prove to be cancerous. Neck node metastasis is very common in thyroid malignancy

(For details see other sections of this chapter).

Lymphoma

The five most frequent cancers in Indian males (in descending order) are mouth/oropharynx, trachea/bronchus/lungs, lymphomas/multiple myeloma, esophagus and leukemia. In women, these are (in descending order) breast, cervix, ovary, mouth/oropharynx, esophagus and lymphomas/multiple myeloma (ICMR, 2004).

Lymphoma, Hodgkin's disease and lymphosarcomas occur in all age groups, but are common in children and young adults. They account for up to 55% of all pediatric cancers.

- *Clinical features*: Features associated with lymphoma may include fever and hypertrophy of spleen, liver or Waldeyer's ring.
  - Eighty percent children with Hodgkin's disease have at least one neck mass whereas in lymphosarcomas, 40% children have this feature.

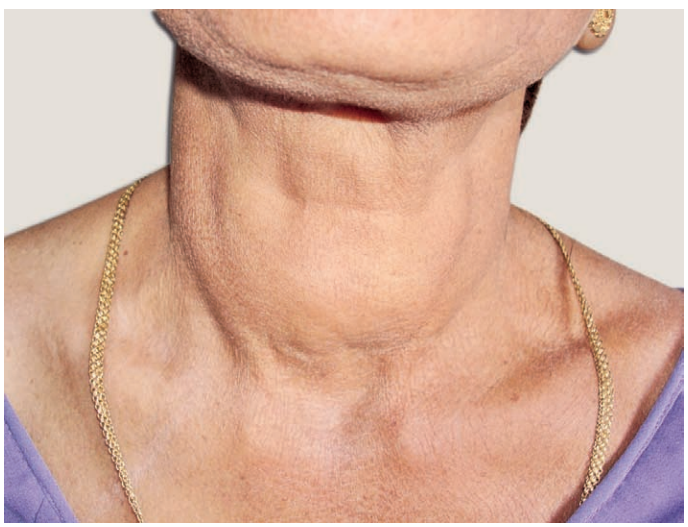

Fig. 3: Recurrent multinodular goiter with hypothyroidism. Note the previous scar of subtotal thyroidectomy which was done 21 years ago

  - Lymph nodes are usually discrete, rubbery and nontender, and have progressive enlargement. Usually, there are no other local head and neck symptoms.
- *FNAC*: Needle biopsy is taken from a suspicious mass (single, dominant, supraclavicular, asymmetric), and studied by flow cytometry.
- *Open biopsy*: It is done for complete histocytopathologic examination in uncooperative children, and when FNAC and flow cytometry results are equivocal or negative.
  - Biopsy from the site of abnormality of Waldeyer's ring, if any, is necessary for staging at the time of neck node biopsy.

Evaluation of gastrointestinal or central nervous system is required in cases of extranodal lymphomas.

Salivary Neoplasms

- *Clinical features*: Enlarging solid mass lies in front of and below the ear, at the angle of mandible (parotid) or in the submandibular triangle (submandibular gland).
  - Benign tumors are usually asymptomatic.
  - Malignant tumors present with pain, rapid growth, facial nerve palsy (parotid) or skin fixation.
- *Imaging*: Sialography, nuclear scans and CT scans just indicate whether the mass is salivary in origin. They are not preferred by many surgeons.
- *Open biopsy* (superficial parotidectomy or complete submandibular gland removal): This is the procedure of preference.

  For details see the Chapter "Disorders of Salivary Gland."

Carotid Body Tumors and Glomus Tumors

- *Clinical features*: They are:
  - Site: Upper anterior triangle around the carotid bifurcation.
  - Nature: Pulsatile and compressible, which often rapidly refill on release of pressure.
    - Both bruit and thrill are present.
  - Mobility: Carotid body tumor can be moved from side-to-side, but not up or down.
  - Tonsil: In glomus vagus tumor, the tonsil may pulsate and be pushed medially.
- *Diagnosis*: Angiography.
- *Treatment*: Small tumor in a young patient is resected.
  - Arterial embolization aids in clearance of large tumors with less blood loss.
  - Radiotherapy is preferred in elderly patients or extensive tumors in high-risk patients.

Schwannomas or Neurilemmomas

- *Clinical features*: These neurogenic tumors are solid.
  - They occur in parapharyngeal space, and displace the tonsil medially.
  - Vagus nerve tumor can cause hoarseness of voice.
  - Sympathetic chain tumors may present with Horner's syndrome.
- *Management*: Surgical exploration and excision are done after a thorough search for an unknown primary tumor.

Lipomas

- *Clinical features*: It is present as an ill-defined, asymptomatic, soft mass anywhere in the neck of patients who are older than 35 years of age.
- *CT scan*: The soft tissue mass shows fat-air density.
- *Treatment*: Surgical excision for the cosmetic purposes.

Branchial Cysts

Most commonly, they occur in late childhood or early adulthood.

- *Clinical features*: Initially, they appear as painful tender swelling with fever after an episode of upper respiratory tract infection (URI). They usually persist as soft, doughy, variably-sized masses even after the course of antibiotics.
- *Location*: Anterior triangle of neck
  - Second branchial cleft cyst: Deep to and along the anterior border of sternocleidomastoid muscle. Their remnant tract may course between carotid branches anterior to glossopharyngeal and hypoglossal nerves, and enter into oropharynx.
  - First branchial cleft cyst (less common): Along the inferior border of mandible, at the angle of mandible or just below the ear lobule. Their remnant tract may course towards the external auditory canal.
- *Ultrasonography*: It reveals cystic lesion.
- *FNAC*: Milky, mucoid or brownish fluid, which usually show cholesterol crystals.
- *Treatment*: Surgical excision of the cyst along with its tract after controlling the local infection with a course of antibiotics.

Branchial Sinus or Fistula

Second branchial cleft fistula is more common than third branchial cleft one. The tract ascends just deep-to-deep cervical fascia along the carotid artery.

- *Course of tracts*: They are:
  - Second branchial cleft sinus: The tract passes between the second arch structures (external carotid artery, stylohyoid muscle and posterior belly of digastric) and third arch structures (internal carotid artery). This tract remains superficial to hypoglossal nerve. It perforates pharyngeal wall, and ends in tonsillar fossa.
  - Third branchial cleft sinus: The tract passes deeper to both external and internal carotid arteries, but superficial to vagus and hypoglossal nerves. Its internal opening lies in piriform fossa.
- *Clinical features*: Both sinuses second, as well as third present with an external opening along the anterior border of sternocleidomastoid muscle.
- *Treatment:* Complete excision of the tract.

Thyroglossal Cyst

- *Clinical features*: Initially, it appears as painful tender swelling with fever after an episode of URI. It persists as soft, doughy, variably sized mass after a course of antibiotic.
  - Location: Midline of anterior neck (Fig. 4)
  - Pathognomonic sign: Vertical movement with swallowing and protrusion of tongue

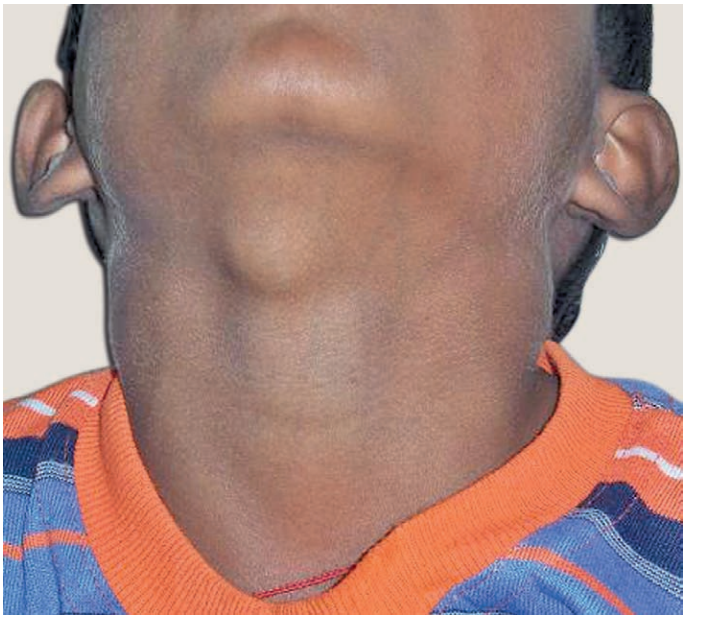

Fig. 4: Thyroglossal cyst. Soft and doughy swelling in the midline of anterior neck in a child

- *USG*: It differentiates the cyst from lymph node, dermoid cyst and thyroid tissue.
- *Radionuclide scan*: It is indicated if the cyst is present in the base of tongue to differentiate from undescended lingual thyroid.
- *Treatment*: Surgical excision of the cyst along with its tract and midportion of the hyoid (Sistrunk operation). The specimen should be sent for histopathological examination to rule out otherwise rare concomitant neoplastic disease.

Cervical Lymphangioma (Cystic Hygroma)

This cavernous lymphangioma is said to be the result of incomplete development and obstruction of normal lymphatic system. Majority of them are present at birth, and becomes evident within the 1st year of life. The extent is usually much more than what is apparent.

- *Most common site*: Posterior triangle.
- *Nature*: The swelling is fluctuant, diffuse, soft, spongy, and has indiscrete margins.
- *Diagnostic feature*: Transillumination.
- *Treatment*: The easily accessible mass is excised if it is affecting vital functions. Mutilating procedures are not performed.
  - Sclerotherapy: Due to the high-risk of recurrence and complications, it may be tried in extensive lesions.

Dermoid Cyst

This slow-growing painless cystic swelling is most commonly seen in children and young adults.

- *Most common sites*: Midline and anterior neck.
- *Differential diagnosis*: Epidermal or sebaceous cyst (Fig. 5) lies superficially in the skin whereas dermoid cyst lies deep in cervical fascia, and skin moves freely over it.
- *Treatment*: Complete surgical excision.

Acute Cervical Lymphadenitis

Acute, unilateral, pyogenic cervical lymphadenitis is mainly the disease of young children (1–5 years of age). For identifying the reason of reactive lymphadenopathy, review the source of drainage to that particular lymph node area (Fig. 6). In suspected cases of inflammatory adenopathy, a trial of antibiotic therapy and observation (not > 15 days) is well accepted as a clinical test.

- *Causative organisms*: *Streptococcus pyogenes* (Group A *Streptococcus*) or *Staphylococcus aureus*.
- *Focus of infection:* Tonsillitis or a dental abscess.
- *Clinical feature:* The child often has fever (37.8–39.6° C), anorexia and general malaise with an elevated leukocyte count.
  - Lymph nodes: They are enlarged and tender.
- *Treatment*: Antibiotic therapy is often curative.
  - Abscess formation in lymph nodes, parapharyngeal or retropharyngeal space requires surgical drainage.

Tuberculous Cervical Lymphadenitis (Scrofula)

Tuberculosis (TB) is a common cause of cervical lymphadenopathy. It is the most common presentation of extrapulmonary TB.

- *Organisms*: Both bovine and human TB.
- *Route of entry*: Ipsilateral tonsil.
- *Clinical features:* Patients who are often children and young adults are usually asymptomatic and have no evidence of active TB (Fig. 7). They present with:
  - Lymphadenitis: Unilateral, painless, firm, erythematous swelling in the posterior triangle. Matting together of a substantial number of lymph nodes is common, and tuberculous process is usually limited to clinically affected group of lymph nodes.
  - Cold abscess: The caseated node can liquefy and break down, and result in the formation of cold abscess (Figs 8 and 9).
  - Collar stud abscess: The pus can erode the deep cervical fascia, and through its point of erosion flows into the space beneath the superficial fascia.

Fig. 5: Sebaceous cyst

Fig. 7: Tuberculous cervical lymphadenitis. Unilateral, painless, firm swelling in left submandibular region of a young adult female mimicking submandibular gland swelling. Excisional biopsy showed evidence of tuberculosis

Fig. 6: Acute lymphadenitis. Submandibular region red tender swelling. Note the healing infected-lesions on the upper lip

Fig. 8: Tuberculous cervical lymphadenitis with cold abscess. Unilateral, painless, soft and erythematous swelling behind the left angle of mandible in a young adult female. Pus was aspirated

Fig. 9: Tuberculous cervical lymphadenitis with cold abscess. Right supraclavicular painless, soft and erythematous swelling in a 3 year child

- Discharging sinus: The overlying skin may become reddened, and lead to discharging sinus.
- *Tuberculin skin test*: It is almost always positive (> 10 mm induration).
- *X-ray chest posteroanterior view*: Children and immunosuppressed adults usually have concurrent pulmonary TB.
- *FNAC*: It generally shows granulomata but acid-fast bacilli are usually seen only in immunosuppressed patients.
- *Culture*: Culture of the specimen is important in differentiating tuberculous lymphadenitis from that caused by other mycobacteria or fungi.
- *Treatment*: Antituberculous chemotherapy (depending on the local susceptibility and culture, and sensitivity report) and complete excision without drains to avoid fistulization.
  - Unresolving abscess: It may need excision along with its surrounding fibrous capsule and relevant lymph nodes.
  - Active pulmonary TB: In these cases tubercular cervical lymph nodes are not removed.

Nontuberculous Mycobacteria Cervical Lymphadenitis

- *Common nontuberculous mycobacteria in neck*: *Mycobacterium avium* complex (*M. avium* and *M. intracellulare*), *M. fortuitum, M. kansasii, M. scrofulaceum*.
- *Clinical features*: Patients who are mostly children of less than 6 years of age usually present with unilateral, painless, and firm swelling.
- *Common sites*: Submandibular triangle (Fig. 10), upper jugular nodes or preauricular nodes.
- *Tuberculin test*: It is usually positive (10–15 mm), but a strongly positive test (> 20 mm) usually indicates TB.
- *Treatment*: Complete surgical excision is often curative.
  - Antituberculous chemotherapy: Although it should be given while culture report is pending or when a definite diagnosis is not made, it does not have any role in nontuberculous mycobacteria in neck.
  - Antimicrobial chemotherapy directed against nontuberculous mycobacteria even though tried, it has not been proved as beneficial.

Fig. 10: Submandibular lymphadenopathy in a child

Sternomastoid Tumor

Birth trauma results in fibrosis and later shortening of sternocleidomastoid muscle in newborns.

- *Clinical features*
  - Torticollis: Face is turned to opposite side and head is laterally flexed on ipsilateral shoulder.
  - Neck mass: Palpation of mass in sternocleidomastoid muscle.
- *Treatment*
  - Conservative: Active and passive neck movements and positions in early stages.
  - Surgery: Division of sternocleidomastoid muscle is done in persistent cases.

Cervical Rib

An extra rib occasionally arises from the seventh cervical vertebra, and attaches to first rib. It results in compression of subclavian artery and brachial plexus, which passes between anterior and middle scalene muscles over the first rib. Arterial compression may result in aneurysm and mural thrombus formation, and emboli to the distal arterial system of upper limb.

- *Clinical features*
  - Neck mass: Bony hard lump in supraclavicular region.
  - Compression of lower part of brachial plexus: Tingling and numbness along the upper side of forearm and hand.
  - Compression of subclavian artery: Cold and numb hand and intermittent claudication of upper limb.
- *Treatment*
  - Asymptomatic cases: No treatment is required.
  - Symptomatic cases: The cervical rib is excised by supraclavicular or transaxillary approach.

THYROID NEOPLASMS

Colloid and adenomatous goiters of multiple nodules of varying size and consistency are most common. Carcinoma of thyroid gland is uncommon. Adenoma is the most common benign thyroid neoplasm. In the endemic areas of goiter, mortality rate from thyroid cancer is about 10 times more. Although there

is no age bar, majority of follicular, medullary and anaplastic carcinoma patients are elderly. In adolescent and young adults, thyroid carcinoma is mainly of papillary type.

Predisposing (Risk) Factors

- *Iodine deficiency and elevated thyroid-stimulating hormone (TSH) levels*: Relationship between iodine deficiency, endemic goiter and thyroid cancer (especially follicular type) is well-known. Dietary deficiency of iodine affects T_3 and T_4 production. Prolonged TSH stimulation causes abnormal thyroid gland to under go malignant change.
  - Prevention: Dietary iodine supplementation reduces the incidence of thyroid cancer.
- *Solitary thyroid nodule*: The incidence of malignant change in solitary nodule is 10–20%.
- *Ionizing radiation*: This well recognized etiological factor usually causes papillary carcinomas.
- *Genetic predisposition*: Hyperthyroidism, goiter and thyroid cancer do occur in members of the same family. Genetic-based medullary carcinoma is found in several variants of multiple endocrine neoplasia (MEN) syndromes.
  - Multiple endocrine syndromes: In MEN IIA, which is an autosomal dominant condition, medullary thyroid carcinoma occurs in association with pheochromocytoma and hyperparathyroidism. If affected family member has genetic mutation on the RET proto-oncogene, prophylactic total thyroidectomy should be considered after excluding pheochromocytoma.
- *Autoimmunity*: Thyroid lymphoma frequently occurs in cases of autoimmune lymphocytic thyroiditis (Hashimoto's disease).

Thyroid Nodule

Box 1 shows the classification of thyroid neoplasms. Colloid and adenomatous goiters of multiple nodules of varying size and consistency are most common. The nodules are demarcated but not encapsulated. Many have gelatinous consistency with areas of degeneration or calcification.

- *Risk factors*: The malignancy risk factors of a solitary nodule (Fig. 11) include the following:
  - Age: Person under 14 or over 65 years of age
  - Sex: Male
  - Enlarging in spite of suppressive doses of thyroxine
  - Past history of:
    - Ionizing radiation
    - Thyroid cancer
  - Family history of thyroid cancer

Consistency of different types of thyroid nodules:
- *Cysts:* Cystic filled with fluid
- *Colloid nodules:* Doughy
- *Papillary and medullary cancers:* Hard rubber
- *Anaplastic cancer:* Hard, fixed and craggy
- *Lymphomas:* Diffuse swelling

Adenoma

Adenoma is the most common benign thyroid neoplasm. In contrast to nodule, it is encapsulated. They are not premalignant, and rarely become toxic. Capsular or vascular invasion indicates malignancy.

Box 1: Classification of thyroid neoplasms

- Benign—adenoma (most common)
  - Follicular adenoma
  - Microfollicular
  - Hurthle cell
  - Embryonal
- Malignant
  - Primary
    - ◊ Follicular epithelium
      - Papillary adenocarcinoma (60–80%)
      - Follicular adenocarcinoma (10–20%)
      - Anaplastic carcinoma (5–10%)
    - ◊ Parafollicular cells
      - Medullary carcinoma (5%)
    - ◊ Lymphoid cells
      - Lymphoma
  - Secondary
    - ◊ Metastatic from
      - Kidney, lung, colon and breast
    - ◊ Direct spread local infiltration from
      - Carcinoma of larynx and postcricoid

Fig. 11: Left lobe thyroid nodule in euthyroid patient

- *Clinical features*: It usually presents as a solitary nodule, or a dominant nodule in multinodular goiter in middle-aged females.

Thyroid nodules: Six to seven percent of adult females and 1–2% of adult males will show a thyroid nodule on thorough clinical examination.

The clinical features of malignant tumors include:
- Solitary nodule
- Rapidly enlarging preexisting goiter of long duration
- Hard and fixed thyroid
- Enlarged neck nodes
- Neck pain and referred otalgia
- Distant metastases
- Pressure features
  - Trachea: Stridor
  - Esophagus: Dysphagia
  - Recurrent laryngeal nerve: Hoarseness of voice and vocal cord palsy

Papillary Adenocarcinoma

It accounts for 60–80% of all thyroid malignancies.

- *Age*: It is most common in fifth decade of life. It is the only thyroid cancer of children.
- *Clinical features*: It presents as firm noncapsulated thyroid nodule. It is usually (80%) multicentric, and involves both thyroid lobes.
- *Lymphatic metastases*: There is high incidence of enlarged level III–VII neck nodes. Primary tumor may remain impalpable.
- *Distant metastases*: Pulmonary metastasis is more common than bony metastasis.
- *Classification:* On the bases of size and extension, they are categorized into the following groups:
  - Minimal (Microcarcinoma) less than 1.0 cm: They are often seen as incidental findings at USG neck. Their incidence far exceeds that of papillary carcinoma. Conservative approach is justified in management.
  - Intrathyroidal greater than 1.0 cm.
  - Extrathyroidal: Extend outside thyroid capsule or with lymph node metastases.
- *Prognosis*: Ten year survival from intrathyroid cancer is over 90% whereas from extrathyroidal, it falls to 60%. In older age group, it is more aggressive, and may invade larynx and trachea.

Follicular Adenocarcinoma

It accounts for only 10–20% of all thyroid malignancies.

- *Age*: It is most common in the sixth decade of life and seldom seen under 30 years of age.
- *Clinical features*: It usually presents as a new solitary thyroid nodule. It may present as malignant change in a thyroid swelling of many years duration. Some patients may present with features of secondary metastases.
- *Metastases*: In contrast to papillary carcinoma, bony and pulmonary metastases are more common than lymphatic metastasis.
- *Classification*: In contrast to papillary carcinoma, it has well defined capsule. Depending upon the breach of capsule, patients are divided into two groups.
- *FNAC*: It will not be able to differentiate between follicular adenoma and carcinoma. Lobectomy is required to know whether a follicular neoplasm is benign or malignant.

If God ask you to choose a cancer to suffer from, which one would you like to have?[1]

Medullary Thyroid Carcinoma

It accounts for only 5% of all thyroid malignancies.

- *Clinical features*: As a part of MEN, it is frequently multifocal and bilateral. Sporadic cases are often unifocal.
- *Tumor marker*: The levels of serum calcitonin are valuable.

Lymphoma and Anaplastic Carcinoma

They are uncommon tumors. It presents as a rapidly increasing swelling of neck in an elderly woman.

- *Lymphoma*: It usually occurs in cases of Hashimoto's autoimmune lymphocytic thyroiditis (Fig. 12).
- *Anaplastic cancers*: It can appear in long-standing goiter patients as a rapidly growing tumor associated with referred otalgia and hoarseness of voice. This aggressive tumor is highly metastatic, and invades larynx, pharynx and esophagus.
  - Prognosis: Treatment is not effective. Patient usually dies within 1 year of presentation. It has very poor prognosis.
- *Investigations*:
  - Thyroid function tests: T_3, T_4, TSH.
  - Thyroid antibodies.
  - FNAC and open biopsy (or tru-cut): Appropriate immunocytochemistry is essential to distinguish these two tumors because the treatment and prognosis of these cancers are different.
- CT scan: Neck and chest for anaplastic, and whole body for lymphoma.
- Lactate dehydrogenase, erythrocyte sedimentation rate, bone marrow and trephine: For thyroid lymphoma.

Investigations

- *T_3, T_4 and TSH*: Overt thyroid dysfunctions (hyper and hypothyroidism) with goiter do not require further investigation. They are not neoplastic (Fig. 13).
- *Thyroid antibodies*: They are not mandatory but help in interpretation of serum thyroglobulin levels after surgery.
- *FNAC*: This safe, quick, cheap and effective OPD procedure is mandatory. It requires experienced cytopathologist.
- *X-ray chest*: It can show tracheal deviation, mediastinal extension or lymph nodes enlargement and pulmonary metastasis.
- *X-ray neck and thoracic inlet*: In addition to the tracheal deviation and compression, other findings may include:
  - Benign lesion: Rim or eggshell calcification
  - Multinodular goiter: Heavy irregular calcification
  - Papillary carcinoma: Finely stippled area due to calcification in Psammoma bodies

Fig. 12: Hashimoto thyroiditis. Right-side painful thyroid swelling of 5 days duration in 45-year-old lady

[1] Papillary adenocarcinoma of thyroid

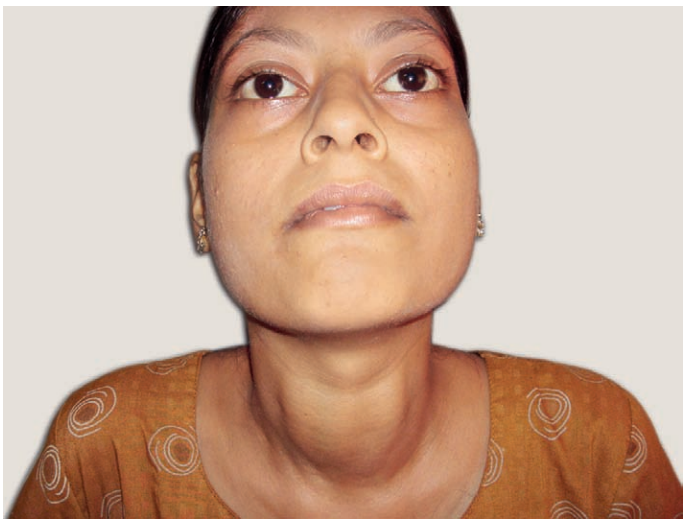

Fig. 13: Toxic goiter

- Medullary carcinoma: Bilateral calcification at the upper lateral portion of gland

- *Ultrasound*: This easy, quick and cheap method of assessment helps in the following ways:
  - Measure tumor size
  - Diagnose multinodular goiter
  - Exclude contralateral lesion
  - Distinguish purely cystic (up to 1 mm diameter) nodules from complex cysts with solid (up to 3 mm diameter) component
  - Demonstration of Psammoma bodies
  - Disadvantage: It cannot reliably distinguish benign from malignant lesions.
- *Scintigraphy*: It differentiates between cold (nonfunctional) and hot (functioning) nodule of more than 5 mm diameter.
  - Technetium-99m:
    - Solitary cold nodule: It may be adenoma, carcinoma, cyst or dominant nodule in non-palpable multinodular goiter. The chances of malignancy of cold nodule are 20%.
    - Hot nodules: They are unlikely to be malignant.
  - Metaiodobenzylguanidine: It is useful in suspected cases of MEN as it is taken up by pheochromocytoma and medullary thyroid carcinoma.
  - ^{67}Ga citrate (gallium scanning): It is useful in detecting lymphoma in long-standing cases of Hashimoto's disease.
  - ^{131}I: It is used after surgery to ablate residual thyroid tissue. It also searches and treats metastases.
- *Computed tomography*:
  - Neck and thorax: It shows involvement of larynx, trachea, pharynx, esophagus, major vessels and retrosternal extent, and pulmonary metastasis and nodal deposits in neck and mediastinum.
  - Abdominal: It is done for lymphoma staging, and in suspected cases of pheochromocytoma.
- *Magnetic resonance imaging*: It has good soft tissue contrast. MR angiography assesses vessel involvement.
- *Serum calcium and calcitonin:* They are indicated in medullary carcinoma.
- *Excision biopsy*: Small-lateral tumors need lobectomy with removal of isthmus and a small portion of opposite lobe. The small isthmus nodule needs removal of central portion of gland with 1 cm safe margin.

Staging

Table 1 shows the tumor nodal metastasis (TNM) classification and staging of thyroid carcinoma. Patients with differentiated papillary and follicular carcinoma under 45 years of age do not have stage III and IV, whereas histological diagnosis of anaplastic carcinoma puts the patient directly in stage IV.

Treatment

Treatment of thyroid carcinoma requires multidisciplinary approach and includes:

- Surgery for the primary tumor and lymphatic metastasis (neck dissection). Thyroid surgeries include:
  - Lobectomy or hemithyroidectomy: Complete removal of one thyroid lobe and isthmus

Treatment of thyroid carcinoma requires multidisciplinary approach and includes:

- Surgery for the primary tumor and lymphatic metastasis (neck dissection). Thyroid surgeries include:

  - Lobectomy or hemithyroidectomy: Complete removal of one thyroid lobe and isthmus
  - Subtotal thyroidectomy: Bilateral removal of more than one half of thyroid gland plus isthmus
  - Near-total thyroidectomy: Complete removal of one lobe, isthmus and removal of greater than 90% of opposite lobe
  - Total thyroidectomy: Removal of both sides lobes, and central isthmus
  - Completion thyroidectomy: Conversion of lesser operation into a near-total or total thyroidectomy
- Radioactive Iodine for ablating residual thyroid tissue and metastases
- External beam radiotherapy after surgery, and in cases of lymphoma
- Thyroxin therapy to suppress the TSH levels
- Chemotherapy in high-grade and more-advanced lymphoma

- *Papillary and follicular adenocarcinoma*: In contrast to squamous cell carcinoma of head and neck region, distant metastasis in this differentiated thyroid carcinoma is not a death sentence. About 50% patients receiving radioiodine for pulmonary metastasis live for 10–15 years. Osseous metastasis has poor prognosis. Hematogenous spread is more common with follicular than papillary carcinoma.
  - T_{1-2} N_0 M_0 in females under 45 years: Tumor in lobe: Lobectomy/isthmusectomy with 1 cm margin; TSH suppression with thyroxin and thyroglobulin surveillance.
  - T_2 N_0 M_0 in males or greater than 45 years: Total thyroidectomy, immediate postoperative radioiodine ablation of residual normal thyroid tissue, TSH suppression with thyroxine (usually 200 mcg daily), and thyroglobulin surveillance for life.
  - T_3 or T_{1-4} N_1 M_{0-1}: Total thyroidectomy, selective neck dissection (nodal surgery depending upon the extent of lymphatic involvement), immediate postoperative radioiodine ablation of residual normal thyroid tissue, TSH suppression with thyroxine, and thyroglobulin surveillance for life.

TABLE 1 Tumor nodal metastasis classification and staging of thyroid carcinoma

Primary tumor (T)	
T_x	Unable to assess primary tumor
T_0	No evidence of primary tumor
T_1	Intrathyroid tumor up to 1 cm greatest dimension
T_2	Intrathyroid tumor more than 1 cm, and less than 4 cm in greatest dimension
T_3	Intrathyroid tumor more than 4 cm in greatest dimension
T_4	Tumor extending beyond thyroid capsule
Regional lymphadenopathy (N) (cervical and upper mediastinal nodes)	
N_x	Unable to assess regional lymph nodes
N_0	No evidence of regional metastasis
N_{1a}	Metastasis in ipsilateral cervical lymph node
N_{1b}	Metastasis in bilateral, contralateral, or midline cervical or mediastinal lymph nodes
Distant metastases (M)	
M_x	Unable to assess for distant metastases
M_0	No distant metastases
M_1	Distant metastases
TNM staging (under 45 years) for papillary and follicular carcinoma*	
Stage I	Any T, any N, and M_0
Stage II	Any T, any N, and M_1
TNM staging (over 45 years) for papillary, follicular and medullary carcinoma	
Stage I	T_1 N_0 M_0
Stage II	T_{2-3} N_0 M_0
Stage III	T_4 N_0 M_0; T_{1-4} N_1 M_0
Stage IV	T_{1-4} N_{0-1} M_1
TNM staging for anaplastic carcinoma	
Stage IV	Any T, any N and any M

***** Patients under 45 years of age do not have stage III and IV.

  - Clinically N_0 neck: Level VI should be always dissected, and levels II–V and VII must be palpated during surgery.
  - Clinically N_1 neck (levels II–V): Selective neck dissection involving these levels; modified radical, radical or extended radical neck dissection depending on the size and extent of disease.
  - Postoperative external beam radiotherapy: It is indicated when operative clearance is doubtful, or in cases of extensive nodal involvement.
- *Medullary carcinoma*: It needs total thyroidectomy with modified radical or radical neck dissection, which may need to be extended into superior mediastinum.
  - Postoperative external beam radiotherapy: It is indicated when operative clearance is doubtful, or in cases of extensive nodal involvement with extracapsular extension.
  - 131I-MIBG: It may be used in cases of recurrence or metastasis.
- *Lymphoma*: Radiotherapy is the main treatment for this lesion.
  - Chemotherapy: It is indicated in high-grade histology and more advanced disease.
- *Anaplastic carcinoma*: Unfortunately, no treatment is effective in this thyroid cancer. Radiotherapy achieves regression but early recurrence usually occurs. Patient with stridor needs isthmus split tracheostomy.

Prognostic factors of thyroid carcinoma are the following:
- *Age*: Patients under 45 years of age do better.
- *Sex*: Females do better.
- *TNM staging*: Higher the stage, poorer is the prognosis.
- *Histology*: Tall cell variant of papillary carcinoma and marked invasion of follicular carcinoma have poorer prognosis. Anaplastic carcinoma has very poor prognosis.
- *Treatment*: Delay treatment, extensive surgeries and inadequate facilities offer poorer prognosis.
- *Surgeon*: Experienced surgeon can offer best treatment.

Clinical Highlights

1. **Carotid body tumor:** Embryologically it is believed to originate from neural-crest cell.
2. **Lyre sign:** Splaying apart of internal and external carotid arteries by carotid body tumor and seen on carotid angiography.
3. **Second arch branchial fistula:** Its external opening lies along the anterior border of sternocleidomastoid muscle. The fistulous tract passes deep to digastric muscle between the internal and external carotid arteries.
4. **Extranodal lymphoma:** It is a common manifestation of non-Hodgkin's lymphoma.
5. **Thyroglossal cyst:** Arises from remnants of thyroglossal duct.
6. **Biopsy:** Thyroid lymphoma and anaplastic carcinoma need open biopsy before the beginning of therapy.
7. **Papillary and follicular carcinoma of thyroid:** Low-risk patients are females under 45 years and high-risk patients are all males, and over 45 years females. Low-risk tumors include less than 1 cm size papillary and minimally invasive follicular carcinoma. High-risk tumors include more than 1 cm size papillary and follicular carcinomas. Multifocality, local or distant spread are also risk factors.

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54 Deep Neck Infections

Make your nerves strong. What you need is muscles of iron and nerves of steel. You have wept long enough. No more weeping, but stand on your feet and be men. Have faith in yourself. You all have infinite power in you. Rouse that up; arise, arise—apply yourselves heart and soul, gird up your loins.

—Swami Vivekananda

Points of Focus

- **PERTINENT ANATOMY**
  - ✓ Peritonsillar Space
  - ✓ Parapharyngeal Space
  - ✓ Retropharyngeal Space
  - ✓ Danger Space
  - ✓ Prevertebral Space
  - ✓ Submandibular Space
  - ✓ Space of Body of Mandible
  - ✓ Masticator Space
  - ✓ Submandibular Space
  - ✓ Masticator Space
- **SOURCES OF INFECTIONS**
- **MICROBIOLOGY**
- **CLINICAL FEATURES**
- **INVESTIGATIONS**
- **TREATMENT**
  - ✓ Antibiotic Therapy
  - ✓ Surgical Drainage
- **PERITONSILLAR INFECTIONS**
- **PARAPHARYNGEAL SPACE ABSCESS**
- **ACUTE RETROPHARYNGEAL ABSCESS**
- **PREVERTEBRAL SPACE ABSCESS**
- **LUDWIG'S ANGINA**
- **ABSCESS OF SPACE OF BODY OF MANDIBLE**
- **MASTICATOR SPACE ABSCESS**
- **TRISMUS**
- **CLINICAL HIGHLIGHTS**

Deep space neck infections are potentially lethal infections and need immediate appropriate treatment. These spaces are situated between the three layers of deep cervical fascia which have been described in chapter "Anatomy of Neck." On the basis of hyoid bone, deep neck spaces can be divided into several groups (Box 1).

PERTINENT ANATOMY

Figure 1 shows the sites of various deep neck space abscesses.

Peritonsillar Space

It is situated between the capsule of tonsil and the superior constrictor muscle. Medial wall of the parapharyngeal space is the lateral wall of the peritonsillar space. It is formed by the superior constrictor muscle.

Parapharyngeal Space or Pharyngomaxillary Space or Lateral Pharyngeal Space

This pyramid-shaped space has its base towards the skull base while its apex towards the hyoid bone.

- *Boundaries*
  - Medial: Buccopharyngeal fascia, which covers the superior constrictor muscles. Medial wall of the parapharyngeal space is the lateral wall of the peritonsillar space.
  - Posterior: Prevertebral fascia, which covers prevertebral muscles and transverse processes of the cervical vertebrae.

Box 1: Classification of deep neck spaces

- Suprahyoid
  - Face
    - ◊ Buccal
    - ◊ Canine
    - ◊ Space of body of mandible
    - ◊ Masticator
      - Masseteric
      - Temporal
      - Pterygoid
    - ◊ Temporal
    - ◊ Parotid
  - Neck
    - ◊ Peritonsillar
    - ◊ Parapharyngeal
    - ◊ Submandibular
      - Sublingual space (superior)
      - Submaxillary space (inferior)
- Infrahyoid
  - Visceral
- Entire length of neck
  - Vascular: Carotid sheath
  - Prevertebral
  - Danger space or alar space
  - Retropharyngeal

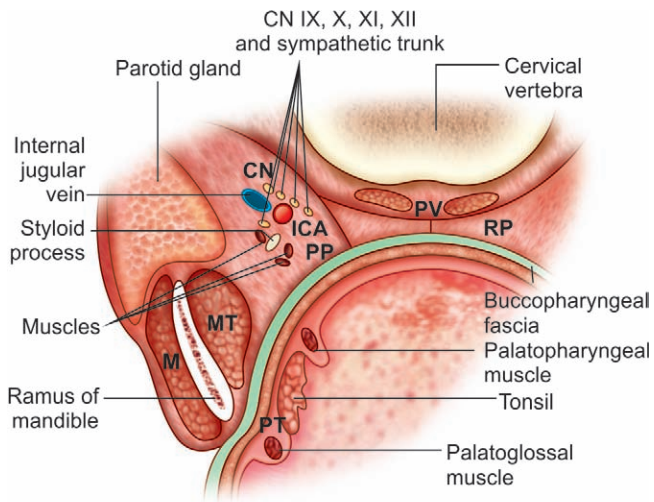


Fig. 1: Deep neck spaces for abscesses
Abbreviations: CN, Cranial nerves IX, X, XI and XII and sympathetic trunk; IC, Internal carotid artery; IJV, Internal jugular vein; M, Masseter muscle; MT, Medial pterygoid muscle; PP, Parapharyngeal space; PT, Peritonsillar space; PV, Prevertebral space; RP, Retropharyngeal space

- Lateral: Medial pterygoid muscle and mandible in anterior part, and the deep surface of the parotid gland in the posterior part.

- *Compartments*
  Styloid process and the structures attached to it divide the parapharyngeal space into anterior and posterior compartments.
  - Anterior compartment: It is medially related to the tonsillar fossa, and laterally to the medial pterygoid muscle.
  - Posterior compartment: It is related medially to the posterior part of lateral pharyngeal wall, and laterally to the parotid gland. The contents of this compartment are the following structures:
    - Internal carotid artery
    - Internal jugular vein
    - Cranial nerves 9th, 10th, 11th and 12th
    - Sympathetic trunk
    - Upper deep cervical nodes
- *Communications*: Parapharyngeal space communicates with the following spaces:
  - Retropharyngeal
  - Submandibular
  - Parotid
  - Carotid
  - Visceral

Retropharyngeal Space

- *Boundaries and Extent*:
  - The retropharyngeal space lies between the buccopharyngeal fascia covering the pharyngeal constrictor muscles, and the prevertebral fascia covering the vertebrae and prevertebral muscles.
  - It extends from the skull base to the bifurcation of trachea in mediastinum.
- *Compartments*:
  - A midline fibrous raphe divides this space into two lateral compartments (spaces of Gillette); one on each side.
  - An abscess of retropharyngeal space causes unilateral bulge.
- *Contents*:
  - The space contains retropharyngeal nodes, which usually disappear at 3–4 years of age.
- Communications:
  - The retropharyngeal space communicates with the parapharyngeal space.
  - Retropharyngeal space infection may pass down into the mediastinum behind the esophagus.

Danger Space

It lies just posterior to retropharyngeal space in between the alar fascia (anteriorly) and prevertebral fascia (posteriorly). These are the two layers of preverterbral layer of deep cervical fascia. It contains only loose connective tissue and extends from skull base to mediastinum. So the infection of this space can cause mediastinitis. During the surgical drainage, both the dangerous and retropharyngeal spaces are treated as one unit. A dissecting finger is used to disrupt the partition between these two spaces.

Danger space: It lies between the alar fascia and the prevertebral space. Infection of this space can lead to mediastinitis and death if not properly treated.

Prevertebral Space

- It lies between the vertebral bodies and the prevertebral fascia and extends from the skull base above to down up to the coccyx.
- Infection usually comes from the caries spine.
- An abscess in the prevertebral space produces a midline bulge.

Submandibular Space (Fig. 2)

- *Boundaries*:
  - The submandibular space lies between the mucous membranes of floor of the mouth and tongue, and the superficial layer of deep cervical fascia.
  - It extends between the hyoid bone and mandible.
- *Compartments*
  Mylohyoid muscle divides this space into two compartments. These compartments communicate with each other around the posterior border of the mylohyoid muscle.
  - Sublingual compartment: It lies above the mylohyoid.
  - Submaxillary and submental compartment: It lies below the mylohyoid.

Space of Body of Mandible

- *Boundaries*: It lies between the investing layer of deep cervical fascia and body of the mandible. Investing layer of deep cervical fascia envelops the lower part of body of mandible and attaches to its periosteum. The space is limited anteriorly by the submental muscles and posteriorly by the masseter (external surface) and medial pterygoid muscle (lingual surface).
- *Communications*: Infection can spread posteriorly to masticator space and medially to submandibular space.

Masticator Space

- *Boundaries and Extent*
  - Superior: Base of skull
  - Inferior: Lower border of mandible
  - Lateral: Superficial layer of deep cervical fascia making parotid capsule
  - Medial: Muscles of mastication (masseter, medial and lateral pterygoids, and insertion of temporalis) and mandible.

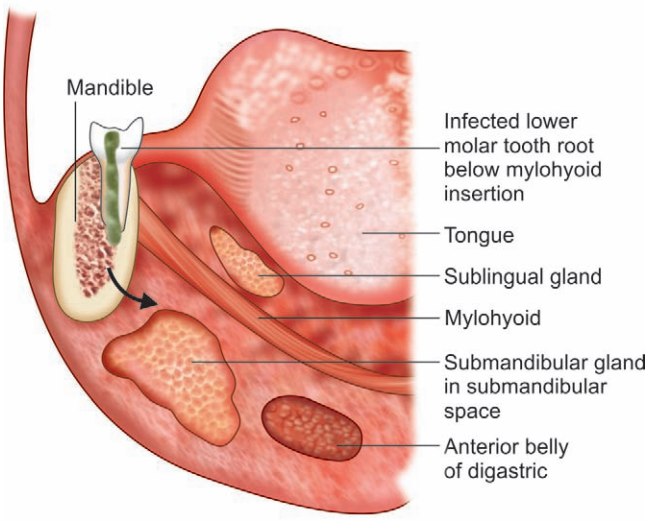


Fig. 2: Submandibular space. coronal section showing pathway for pus

SOURCES OF INFECTIONS

No apparent source is found in good number of cases. The sources, which have been encountered, include:

- Tonsillitis is the most common cause of pediatric deep neck infections
- Poor dental hygiene is the most common cause in adults
- Injection drug abuse
- Trauma
- Iatrogenic: Surgical trauma
- Esophageal perforation
- Complications of mastoiditis: Bezold's abscess and petrositis
- Thyroiditis
- Pot's cervical spine
- Laryngocele
- Infected branchial and thyroglossal cysts.

MICROBIOLOGY

They are often polymicrobial. The commonly encountered species are the following:

- Anaerobics comprise 90% of the bacteria in gingival crevice.
  - Fusobacteria species
  - Pigmented bacteroides species
  - Peptostreptococci species
- Facultative anaerobic streptococci
- Group A β-hemolytic streptococci pyogenes
- Aerobic Gram-negative bacilli: In injection drug abusers and seriously ill indoor patients
- Staphylococci.

CLINICAL FEATURES

The clinical features of deep neck infections include fever, chills, decreased appetite, sore throat, painful swelling, referred otalgia, trismus, dysphagia and respiratory distress.

INVESTIGATIONS

It is often difficult to differentiate between true abscess and cellulitis or adenitis clinically.

- CBC and serum electrolytes
- *Cultures*: Throat swab, blood, sputum, needle aspirations of possible abscess site.
- *Panorex oral view*: For dental infections.
- *X-ray soft tissue neck lateral view*: For assessing retropharyngeal and pretracheal spaces. Normal thickness of prevertebral soft tissue is 7 mm at axis (second cervical vertebra) and 14 mm (children) or 22 mm (adults) at sixth cervical vertebra.
- *X-ray chest*: For assessing mediastinal silhouette for any widening.
- *Ultrasonography*: It is an easy, economical, safe and widely available technology for seeing deep space neck abscesses.

It is used to guide needle aspiration and posttreatment surveillance.

- *CT and MRI*: CT or MRI is often indicated. CT is usually the examination of choice though the MRI provides better resolution for soft tissues and blood vessels. They can be invaluable in defining:
  - Boundaries of infection
    - Mass with air-fluid interface, cystic or multiloculated appearance
    - Edema
    - Contrast ring enhancement of tissue surrounding the mass
  - Involvement of great vessels and internal jugular vein thrombosis
  - Tracheal compression
  - Mediastinal spread.

The principles of managing deep neck abscesses are the following:

- *Airway*: Most patients need only humidified oxygen. Some patients need a secure airway in the form of intubation. In some patients with respiratory distress, trismus and soft tissue swelling make the intubation difficult and they need tracheostomy or cricothyrotomy.
- Ultrasound or CT-guided aspiration
- Fluid resuscitation
- Antibiotic therapy
- Surgical drainage.

TREATMENT

Antibiotic Therapy

It is not necessary to give antibiotics directed against all the cultured organisms because they are not equally pathogenic. Many fastidious organisms are difficult to culture. Usually, antibiotics which are effective against streptococci, anaerobes and β-lactamase-producing bacteria are used, and they include the following:

- Ampicillin-sulbactam or cefoxitin or ceftriaxone.
- Combination of clindamycin plus ciprofloxacin (to cover *Eikenella corrodens*, a Gram-negative facultative anaerobe and *Pseudomonas aeruginosa*) for penicillin allergic patients.
- Combination of trimethoprim-sulfamethoxazole plus metronidazole.
- Vancomycin is added if resistant *Staphylococci* are suspected.
- Ticarcillin-clavulanate or piperacillin-tazobactam or imipenem-cilastatin for compromised patients or suspected *Pseudomonas aeruginosa*.

Surgical Drainage

The principles of surgical drainage include:

- Wide exposure and blunt dissection.
- Identify the carotid sheath early to avoid damage.
- Pus or tissue specimen should be sent for:
  - Gram, acid-fast and fungal stains.
  - Culture of aerobic, anaerobic, and acid-fast bacteria and fungi
  - Tissue should be sent for pathologic evaluation.
- Complete drainage of abscess including breaking of loculations.
- Debride all devitalized tissue.

PERITONSILLAR INFECTIONS

Peritonsillar abscess (Quinsy) is a collection of pus in the peritonsillar space which lies between tonsil capsule and the superior constrictor muscle covered by the buccopharyngeal fascia.

Source of Infection

Tonsillitis is usually the source of infection. The crypta magna gets infected and sealed off. An intratonsillar abscess develops which subsequently bursts through the tonsillar capsule into the peritonsillar space. The *peritonsillitis* sets up, and results in an abscess. It is a mixed infection of *Streptococcus pyogenes*, *Staphylococcus aureus,* and anaerobic organisms.

Clinical Features

Clinical features include:

- *Age*: Most patients are adults. Children are rarely affected. Acute tonsillitis is more common in children.
- *Side*: Usually, it is unilateral.
- *Local features*: They are–
  - Pain: Unilateral severe throat pain.
  - Ipsilateral referred otalgia: Cranial nerve (CN) IX supplies tonsil, as well as ear.
  - Odynophagia may lead to drooling of saliva from the angle of mouth and dehydration.
  - Hot potato voice: Muffled and thick speech.
  - Halitosis: Foul breath due to oral sepsis and poor hygiene.
  - Trismus: Due to spasm of pterygoid muscles which are near to superior constrictor muscle.
- *General:* The general features due to septicemia, which is usually present in all deep space neck infections include:
  - Fever (up to 104° F) with chills and rigors
  - General malaise, body pain and headache
  - Nausea and constipation.
- *Physical findings*: They show
  - Anterior pillar and soft palate: Congestion and swelling anterior and superior to the tonsil (Fig. 3).
  - Tonsil: Enlarged but gets buried in and hidden behind the edematous pillar and soft palate. Tonsillar exudate may be seen.
  - Uvula: Swollen, edematous and pushed to the opposite side.
  - Cervical lymphadenopathy of jugulodigastric lymph nodes. Swelling is tender and painful.
  - Torticollis: Neck is tilted towards the side of the abscess.

Treatment

A. Medical

The conservative measures which are taken in case of all the deep-space neck infections, which may possibly cure the patient, include:

- Hospitalization.
- Intravenous fluids to combat dehydration.
- Suitable antibiotics covering both aerobic and anaerobic organisms.

- Analgesics and antipyretics (paracetamol): Aspirin can increase the chances of bleeding and should not be used.
- Oral hygiene: It is maintained by hydrogen peroxide or saline mouth washes.

B. Surgical
- Incision and drainage of abscess is indicated in cases of frank abscess formation.
  Method: With the help of a guarded knife, a small stab incision is made at the point of maximum bulge above the upper pole of tonsil, or the junction of anterior pillar and base of uvula (Fig. 4). The site is usually touched by phenol (carbolic acid) prior to incision. A sinus- or artery-forcep is inserted to open and drain the abscess. It may need to be repeated the following day to drain any reaccumulation.
  Peritonsillar (Quinsy) abscess forceps can also be used for drainage of peritonsillar abscess. For further details of these forceps, see chapter "Instruments."

Fig. 3: Peritonsillar abscess right side. Swelling and congestion of right anterior tonsillar pillar and soft palate. Tonsil gets buried in and hidden behind the edematous pillar and soft palate

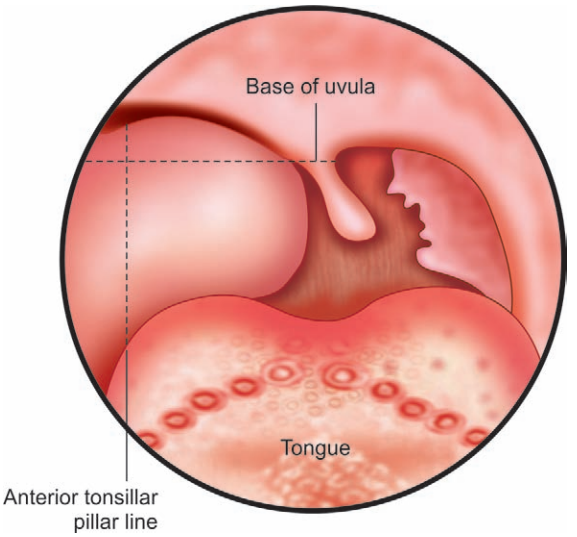


Fig. 4: Peritonsillar abscess. Site of incision and drainage lies just lateral to the junction of vertical anterior faucial pillar line and horizontal base of uvula line

- Interval tonsillectomy: Tonsillectomy is done 4–6 weeks after an attack of quinsy.
- Abscess or hot tonsillectomy: Abscess tonsillectomy cuts down the cost remarkably. It avoids risk and morbidity of second anesthesia and surgery and automatically drains the abscess in the same sitting. The risk of rupture of abscess during anesthesia and operative bleeding should be kept in mind.

Complications

Though rare in this era of antibiotics, they include:
- Parapharyngeal abscess may result in edema of larynx, jugular vein thrombosis and spontaneous carotid artery or jugular vein bleeding.
- Airway obstruction may need tracheostomy.
- Septicemia may lead to endocarditis, nephritis or brain abscess.
- Aspiration of pus (spontaneous rupture of abscess) may cause pneumonitis or lung abscess.

PARAPHARYNGEAL SPACE ABSCESS OR PHARYNGOMAXILLARY ABSCESS OR LATERAL PHARYNGEAL SPACE ABSCESS

Sources of Infections

Infection of parapharyngeal abscess can occur from:
- *Oropharynx*: Bursting of peritonsillar abscess, pharyngitis, tonsillitis and adenoiditis
- *Dental*: Infections of usually lower last molars.
- *Suppurative otitis media complications*: Bezold's abscess and petrositis.
- *Extensions*: Infections of parotid, retropharyngeal and submaxillary spaces.
- *Injuries*: Penetrating injuries of neck.
- *Iatrogenic*: Injection local anesthetic for tonsillectomy or mandibular nerve block.

Clinical Features

- *Common features*: The patients with parapharyngeal abscess usually present with
  - Fever
  - Odynophagia
  - Sore throat
  - Torticollis (due to spasm of prevertebral muscles)
  - Toxemia.

  Other features depend upon the compartment involved. They include the following:
- *Anterior compartment*:
  - Prolapse of tonsil and tonsillar fossa
  - Trismus due to spasm of medial pterygoid muscle
  - Swelling behind the angle of jaw
  - Odynophagia.
- *Posterior compartment*:
  - Pharyngeal bulging behind the posterior pillar.
  - Cranial nerve palsies: CN 9, 10, 11 and 12 palsies will present with dysphagia and hoarseness of voice, and ipsilateral nasal regurgitation and ipsilateral palsies of palate, larynx and tongue.

- Horner's syndrome due to the involvement of sympathetic chain. The syndrome consists of ipsilateral:
  - Anhidrosis
  - Partial ptosis
  - Enophthalmos
  - Constricted pupil
- Swelling in parotid region.

Complications

- Airway obstruction due to edema of larynx.
- Thrombophlebitis of jugular vein.
- Retropharyngeal abscess.
- Infection of mediastinum and carotid sheath.
- Carotid artery bleeding.

Treatment

- *Medical*: Intravenous antibiotics to combat infection.
- *Surgical drainage under general anesthesia*: Preoperative tracheostomy is required in cases of marked trismus or airway obstruction. A horizontal incision is made 2–3 cm below the angle of mandible. Abscess is approached and drained with blunt dissection along the inner surface of medial pterygoid muscle towards styloid process. A drain is usually inserted. Transoral drainage has the danger of injuring great vessels, and is avoided.

ACUTE RETROPHARYNGEAL ABSCESS

Most patients are children below 3 years of age.

Sources of Infection

Suppuration of retropharyngeal lymph nodes occur secondary to:

- Infection in the adenoids, nasopharynx, posterior nasal sinuses or nasal cavity.
- Penetrating injury of posterior pharyngeal wall or cervical esophagus.
- Petrositis due to acute mastoiditis.

Clinical Features

- Dysphagia and airway obstruction (stridor) are prominent symptoms because abscess obstructs the air and food passages.
- Croupy cough may be present.
- Torticollis: Stiff neck and extended head.
- Unilateral bulging in posterior pharyngeal wall on one side of the midline.

Diagnosis

- *X-ray soft tissue neck lateral view*: Shows widening of prevertebral space and presence of gas.
- *CT scan*: Figure 5 shows CT scan neck axial section of retropharyngeal abscess.

Treatment

- *Medical*: Systemic intravenous antibiotics.
- *Incision and drainage of abscess*: Rupture of abscess may occur during intubation. Child is kept in supine and head-low position. A vertical incision is made in the most fluctuant area of the abscess on the lateral part of the posterior pharyngeal wall. Suction must be ready and handy to prevent aspiration of pus.
- *Tracheostomy*: In cases of a large abscess causing airway obstruction and laryngeal edema.

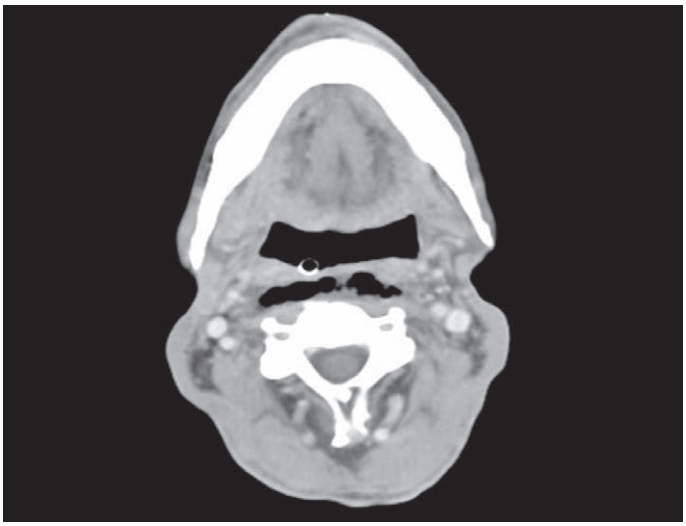


Fig. 5: CT scan neck axial section. Retropharyngeal abscess
Courtesy: Dr Swati Shah, Professor, Radio-diagnosis, GCR Medical College, Ahmedabad.

CHRONIC RETROPHARYNGEAL ABSCESS OR PREVERTEBRAL SPACE ABSCESS

It is common in adults and rare in children below 3 years of age.

Sources of Infection

The tubercular infection of prevertebral space can occur due to:

- *Caries of cervical spine (Pot's spine):* It presents swelling in the midline of posterior pharyngeal wall because the infection is behind the prevertebral fascia.
- *Tuberculosis of retropharyngeal lymph nodes:* It occurs due to tuberculosis of the deep cervical nodes. It presents swelling on the side of the midline as the infection is in the retropharyngeal space.

Clinical Features

- Mild discomfort in throat and dysphagia.
- Posterior pharyngeal wall: Fluctuant swelling centrally or on one side of midline.
- Cervical tuberculous lymph nodes.

Diagnosis

- *X-ray cervical spines:* Shows caries of the cervical spine.

Treatment

- *Medical*: Full course of antitubercular therapy
- *Incision and drainage of abscess*: A vertical incision is made along the anterior border (for low abscess) or posterior border (for high abscess) of sternocleidomastoid muscle.

LUDWIG'S ANGINA

It is the infection of the submandibular space.

Bacteriology

- Infections involve both aerobes and anaerobes.
- The most common causative microorganisms are α *Hemolytic streptococci*, *staphylococci* and bacteroides groups.
- Rarely *Haemophilus influenzae*, *Escherichia coli* and *Pseudomonas* are noted.

Sources of Infection

- *Dental infections* account for 80% of the cases. Roots of lower premolars lie above the mylohyoid line and cause sublingual space infection whereas roots of lower molar teeth extend below the mylohyoid line and cause submaxillary space infection.
- *Others*: Submandibular sialadenitis, injuries of oral mucosa and fractures of the mandible.

Clinical Features

- Presenting complaints are marked difficulty in swallowing, odynophagia and varying degrees of trismus.
- There occurs swelling in the floor of mouth. The tongue is pushed up and back if the infection happens to be involving the sublingual space.
- Once the infection spreads to the submaxillary and submental spaces, the submandibular regions become swollen and tender (Fig. 6), and feel woody hard. There is marked cellulitis of these areas. The frank abscess is uncommon.
- In advanced cases, airway is threatened. Tongue is progressively pushed upwards and backwards. Laryngeal edema may ensue.

Treatment

- Systemic antibiotics and incision and drainage are the main lines of management. Drainage material must be submitted for culture and sensitivity.
- *Incision and drainage*:
  - Intraoral approach is used when abscess is localized to sublingual space.
  - External approach is needed if swelling involves submaxillary space.
    - Method: A transverse incision extending between angles of mandible, is made. The vertical opening of midline musculature of tongue is made with the help of hemostat. Usually, it is serous fluid and not frank pus. It provides significant relief to the patient.
- Tracheostomy must be considered if the airway is compromised.

Fig. 6: Ludwig's angina. Submental and submandibular tender swelling in a 20-year old lady

Complications

- *Spread of infection*: The infection can spread to parapharyngeal and retropharyngeal spaces and then to the *mediastinum*.
- *Airway obstruction*: Airway is compromised not only due to *laryngeal edema* but also due to swelling and pushing back of the tongue.
- *Septicemia* and *aspiration pneumonia* need immediate attention.

ABSCESS OF SPACE OF BODY OF MANDIBLE

- *Sources of infection*: Infection (such as periapical abscess) of bicuspids and first and second molars of lower jaw.
- *Clinical features*: Patient presents with a painful tender swelling on the facial or/and lingual surface of the lower part of the body of mandible (Figs 7A and B). Redness of surrounding gingiva may be seen in some patients. The affected tooth feels long and becomes tender. Tenderness can be elicited in buccal or/and lingual sulcus where the abscess gets localized.
- *Treatment*: It consists of antibiotics and surgical drainage. Depending upon where the abscess is pointing, horizontal intraoral incision is made on either buccal or lingual surface. In some cases, it may also be drained externally by horizontal incision below and parallel to inferior border of the mandible. The incisions should be carried through the periosteum. The offending tooth needs either removal or root canal treatment.
- *Complications*: Infection can spread posteriorly (common) to masticator space and medially (uncommon) submandibular space. Osteomyelitis is also common complication.

MASTICATOR SPACE ABSCESS

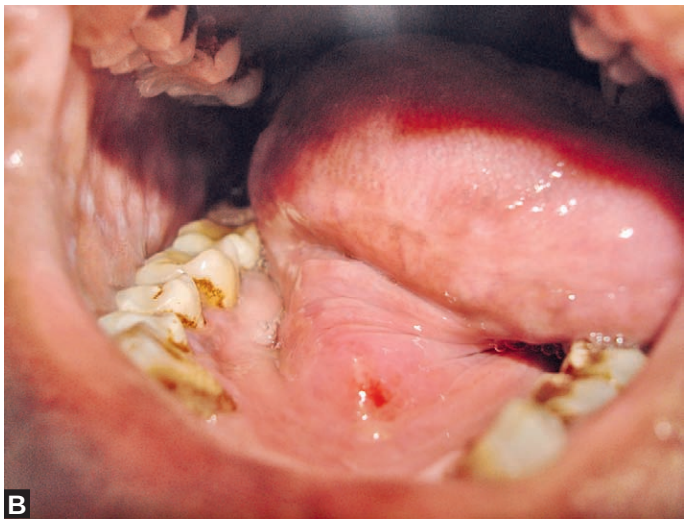
In comparison to parapharyngeal abscess patient is not acutely ill.

Sources of Infections

- Pericoronitis or impacted third molars
- Posterior spread of infection of body of mandible space

Clinical Features

- *Severe painful swelling* (Fig. 8): It extends over ramus of mandible and obliterates subangular depression.
  - Fluctuation is usually absent.
- *Marked trismus* (Fig. 9): It results from irritation of masseter and medical pterygoid muscle.
- Dysphagia.
- Induration of posterior sublingual tissue making the tongue depression difficult.



Figs 7A and B: (A) Abscess of space of mandible. Swelling of lower part of the body of mandible. Note mental region is not involved. Swelling in the submandibular region indicates that infection is spreading to the submandibular space; (B) Abscess of space of mandible. Swelling on the lingual surface of the lower part of the body of mandible. Swelling in the sublingual region indicates that infection is spreading to the submandibular space

Fig. 8: Masticator space abscess. Tender red swelling extending over ramus of mandible and obliterating subangular depression

Fig. 9: Trismus in patient with masticator space abscess (patient of Fig. 8)

Treatment

- Parenteral antibiotics for 1 week
- *Incision and drainage*:
  - External incision: In infections not resolving with antibiotics, an external incision is made below and behind the angle of mandible. Blunt dissection is carried up to the subperiosteal abscess.
  - Intraoral incision: When abscess points lingually, vertical incision along the anterior border of ramus is made. Blunt dissection is carried on both medial and lateral to ramus of mandible.

Complications

- Buccinator space infection – more common
- Osteomyelitis of mandible
- Temporal space infection
- Parotid space infection – occasional on superficial rupture
- Parapharyngeal space infection – rare.

TRISMUS

(L. fr. G. trismos, a creaking, rasping; Synonym: Lockjaw)
Persistent contraction of masseter muscle leads to inability to open the mouth. In the past, tetanus was a common cause but in India now, perhaps the most common cause is oral submucous fibrosis.

Causes

They can be divided into three groups: acute painless, acute painful, and chronic.

1. *Acute Painless Trismus*
   a. Tetanus: Spasms of neck and abdominal muscles. Convulsions occur due to the painful tonic muscular contractions.
   b. Tetany: It occurs due to calcium deficiency and hypoparathyroidism. It is characterized by muscle twitches, cramps and carpopedal spasm, severe laryngospasm and seizures.
   c. Strychnine poisoning: It stimulates all parts of the central nervous system. The muscles are relaxed in between the convulsions.

2. *Acute Painful Trismus*
   a. Peritonsillar abscess: Trismus occurs due to the spasm of pterygoid muscles. Drooling of saliva (due to odynophagia), fever, congestion and fullness in peritonsillar region will confirm the diagnosis.
   b. Parapharyngeal abscess of anterior compartment: Torticollis (due to spasm of prevertebral muscles), prolapse of tonsil and tonsillar fossa, trismus (due to spasm of medial pterygoid muscle) swelling behind the angle of jaw, and odynophagia will confirm the diagnosis.
   c. Ludwig's angina: Painful swelling in the floor of mouth and submandibular region can be seen.
   d. Alveolar infection especially of last molar region.
   e Acute parotitis: Mumps and bacterial parotitis, and abscess.
   f. Acute temporomandibular arthritis.
   g Condylar fracture of mandible: History of trauma and painful swelling. OPG will confirm the diagnosis.
   h. Acute otitis externa especially furuncle.

3. *Chronic Trismus*
   a. Oral submucous fibrosis: Whitish fibrous bands in buccal and palatal region and history of chewing *Paan, Sopari* and Tobacco will confirm the diagnosis.
   b. Ankylosis of temporomandibular joint.
   c. Malignancy of buccal mucosa, tonsil, retromolar trigone, pterygopalatine fossa, maxillary sinus, and parotid: These malignant tumors are visible on examination.

Sequela

- Poor oral hygiene.
- Difficulty in enjoying sumptuous meals.
- Difficulty in examination of oral cavity pharynx, and larynx.
- Endotracheal intubation: In cases of emergency and anesthesia, endotracheal intubation becomes troublesome, and patient may need tracheostomy.

Clinical Highlights

1. **CT with contrast enhancement:** It is the most valuable investigation in deep neck infections.
2. **Peritonsillar abscess (Quinsy):** There occurs collection of pus in the peritonsillar space, which lies medial to superior constrictor muscle of pharynx. Trismus is due to spasm of medial pterygoid muscle.
3. **Parapharyngeal abscess:** The parapharyngeal abscess collects medial to medial pterygoid muscle. The otogenic parapharyngeal abscess is caused by petrositis. In other instances it occurs due to caries/extraction of molar tooth. Patient develops trismus, fever and swelling that pushes the tonsil medially and spreads laterally posterior to sternocleidomastoid.
4. **Ludwig's angina:** The spaces involved in Ludwig's angina are sublingual, submental and submandibular.

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55 Middle Ear and Mastoid Surgeries

To succeed, you must have tremendous perseverance, tremendous will. 'I will drink the ocean,' says the persevering soul, 'at my will, mountains will crumble up'. Have that sort of energy, that sort of will; work hard, and you will reach the goal.

—Swami Vivekananda

Points of Focus

- **MYRINGOTOMY AND TYMPANOSTOMY TUBES**
  - ✓ Indications for Myringotomy
  - ✓ Indications for Grommet
  - ✓ Preoperative Evaluation
  - ✓ Anesthesia
  - ✓ Types of Tympanostomy Tubes
  - ✓ Technique
  - ✓ Postoperative Care
  - ✓ Complications
  - ✓ Indications for Removal of Retained Tympanostomy Tube
- **MASTOIDECTOMY**
  - ✓ Types
  - ✓ Surgical Approaches: Endaural, Postaural, Endomeatal
  - ✓ Anesthesia
  - ✓ Position of Patient
- **CORTICAL MASTOIDECTOMY**
  - ✓ Indications
  - ✓ Instruments
  - ✓ Steps of Operation
  - ✓ Postoperative Care
  - ✓ Complications
- **RADICAL MASTOIDECTOMY**
  - ✓ Indications
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- **TYMPANOPLASTY**
  - ✓ Types of Tympanoplasty
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MYRINGOTOMY AND TYMPANOSTOMY TUBES (GROMMET)

Myringotomy refers to an incision of the tympanic membrane to drain middle ear fluid, which may be suppurative or nonsuppurative (Fig. 1).

Ventilation tube (grommet) is inserted through myringotomy incision for draining middle ear fluid as well as for providing aeration in case of malfunctioning Eustachian tube.

Myringotomy/grommet insertion are not done in suspected cases of intratympanic glomus tumor because these procedures can cause profuse bleeding in such patients.

Indications for Myringotomy

- Indications in acute otitis media (AOM)
  - Bulging eardrum
  - Acute excruciating pain

- Unresponsive to antibiotics: Incomplete resolution with opaque drum and persistent conductive deafness
- Complications: Facial paralysis, labyrinthitis or meningitis with bulging tympanic membrane.

- Otitis media with effusion (OME).
- Aero-otitis media to drain middle ear fluid and "unlock" the Eustachian tube.
- Atelectatic ear: Grommet is often inserted for long-term aeration.

Short-term tympanostomy tubes (grommet) provide longer lasting drainage of middle ear effusion than myringotomy.

Indications for Grommet

- Recurrent AOM.
- Chronic secretory or otitis media with effusion (COME).
- Barotitis media.
- Tympanic membrane abnormalities
  - Severe retraction
  - Atelectasis
  - Retraction pocket with impending
    - Ossicular chain erosion
    - Cholesteatoma.
- Eustachian tube dysfunction.
- Cleft palate with long-standing OME and hearing loss
- AOM with mastoiditis, facial palsy and intracranial complications.
- Hyperbaric oxygen therapy: For the prevention of middle ear complications such as severe ear pain, hemotympanum and OME.

The common indications for grommet are recurrent AOM and chronic OME.

Preoperative Evaluation

- History and physical examination
- Audiometry
- Tympanometry

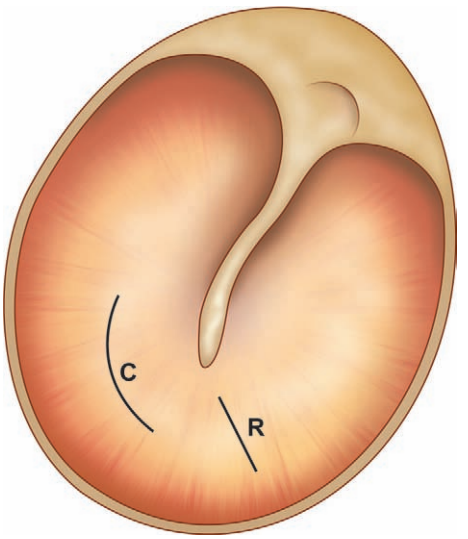


Fig. 1: Myringotomy incisions. Circumferential (C) for acute suppurative otitis media (ASOM) and radial (R) for otitis media with effusion (OME)

Anesthesia

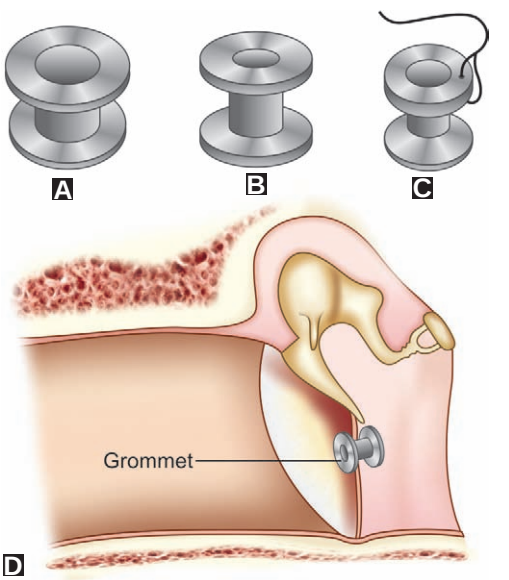
- *General anesthesia*
  - In children and uncooperative adults
  - Acutely inflamed tympanic membrane.
- *Local anesthesia*: In cooperative adults.
- *Topical anesthesia*
  - Phenol
  - Topical tetracaine-base powder suspended in alcohol
  - Lidocaine delivered with iontophoresis
  - Eutectic mixture of lidocaine and prilocaine.

Types of Tympanostomy Tubes (Figs 2A to D)

1. Titanium tympanostomy tubes
2. Silveroxide tympanostomy tubes
3. Small-flanged grommet tubes such as Donaldson tube.
4. Short-flanged T-tube of soft silicone.

Technique

- *Requirement*: The procedure is always done under operating microscope. Clean wax and debris from external auditory canal (EAC).
- *Site, size and shape of incision*: Perform myringotomy in anterior portion preferably anteroinferior quadrant of tympanic membrane using a sharp myringotome. Myringotomy may be done by a fine knife used in ear microsurgery. Either radial or circumferential incision are employed.
  - Acute otitis media: A circumferential incision in the posteroinferior quadrant of pars tensa avoids injury to incudostapedial joint.
  - Otitis media with effusion: A small radial incision may be used in the posteroinferior or anteroinferior quadrant of pars tensa.
  - Grommet: The size of incision should be just enough to admit the grommet.
- *Cautions*: Following precautions should be taken:



Figs 2A to D: Ventilation tubes. (A) Tuebingen ventilation tube; (B) Sheehy teflon ventilation tube; (C) Shepard teflon ventilation tube with wire; and (D) Grommet in position for middle ear ventilation

- Thick tympanic membrane: A deep incision that cuts through entire thickness of tympanic membrane is required.
- Acute otitis media: In cases of acute inflammation distinction between the tympanic membrane and posterior meatal wall is lost. Avoid putting incision in the posterior meatal wall.

Postoperative Care

- Topical antibiotic ear drops in presence of mucoid effusion reduce otorrhea.
- Swimming: Surprisingly, studies found no difference in rates of otorrhea in patients who swam or not with or without earplugs.
- Hearing tests.
- Follow-up: Twice year otoscopy/microscopy to assess
  - Status of tympanostomy tube
  - Tympanic membrane for perforation, retraction pocket, atelectasis and cholesteatoma.

Complications

Complications are uncommon in experienced hands. The potential morbidities are following:

- *Otorrhea*: Most common.
- *Myringosclerosis*: It is of little functional importance.
- *Trauma*: External auditory canal lacerations and injury to ossicles such as incudostapedial joint or stapes
  - If jugular bulb is high and floor of the middle ear dehiscent, injury to jugular bulb can cause profuse bleeding.
- Tympanic membrane:
  - Perforation
  - Atrophy
  - Retraction
  - Atelectasis
  - Cholesteatoma.
- Grommet:
  - Migration or loss of tube into middle ear
  - Early extrusion
  - Plugging of tube.
- Anesthetic complications.

Indications for Removal of Retained Tympanostomy Tube

- Chronic otorrhea.
- Granuloma formation.
- Blockage of tube.
- Migration of tube into middle ear.

MASTOIDECTOMY

Mastoidectomy is an operation in which mastoid antrum is opened and air cells are removed. This operation can be done either alone or in association with tympanoplasty, which consists of eradication of middle ear disease and reconstruction of the hearing mechanism. The various types of mastoidectomies are enumerated in Box 1.

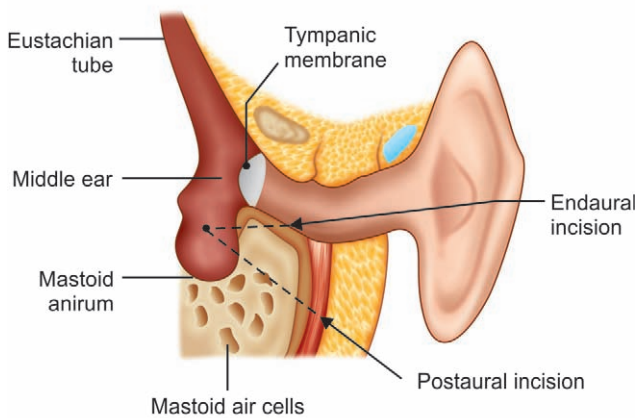
Surgical Approaches

Depending upon the surgeon and nature and extent of the lesion middle ear and mastoid operations can be done with two approaches endaural and postaural (Figs 3 and 4A to C).

1. *Postaural (or Wilde's) incision* (Fig. 4A): This postaural incision begins at the highest attachment of the pinna and ends at the mastoid tip. The incision may lie either 1 cm behind or in the retroauricular groove. In infants and children up to 2 years of age, the postaural incision must be slanting posteriorly to avoid lower part of the mastoid. In these children the mastoid process is not developed and stylomastoid foramen from where facial nerve emerges is located more superficially. The slanting incision avoids cutting facial nerve which is superficial.
   - Indications: The postaural incision is preferred by many for:
     - Cortical mastoidectomy
     - Modified radical and radical mastoidectomy
     - Tympanoplasty
     - Decompression of facial nerve
     - Endolymphatic sac operation.
2. *Endaural approach* (Fig. 4B): Lempert's incision employed for the endaural approach has two parts.
   a. Lempert I: This semicircular horizontal incision is made at the bony cartilaginous junction in the posterior meatal wall. The incision extends from 12 O'clock to 6 O'clock position.
   b. Lempert II: This curvilinear vertical incision begins from the 1st incision at 12 O'clock and passes upwards between tragus and the crus of helix (incisura terminalis) without cutting the aural cartilage.

Box 1: Types of mastoidectomies

1. *Simple mastoidectomy* (cortical mastoidectomy or Schwartz operation)
2. *Canal wall-up procedures* (intact posterior meatal wall or closed procedures)
   a. With facial recess approach
   b. Without facial recess approach
3. *Canal wall-down procedures* (open procedures)
   a. Radical mastoidectomy
   b. Modified radical mastoidectomy
   c. Bondy procedure
4. *Tympanoplasty* (reconstructive surgery)
   a. During the primary surgery
   b. Second stage



• Surgical approach to mastoid antrum schematic horizontal section trough external Auditory meatus by middle ear cleft

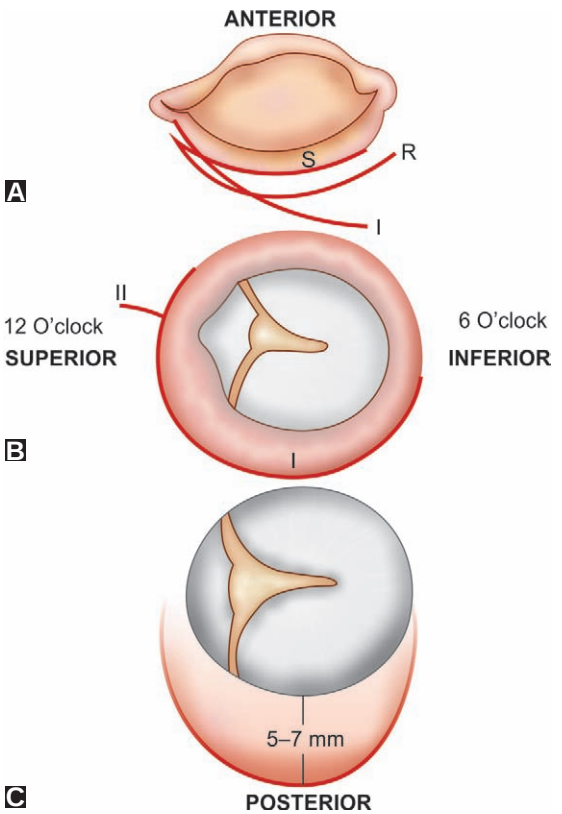
Fig. 3: Postaural and endaural approaches to mastoid antrum

- Indications: Both mastoid and external canal surgeries can be done. It is preferred for the following conditions:
  - Osteoma and exostosis of EAC
  - Large tympanic membrane perforation
  - Atticoantrotomy: Atticoantral cholesteatoma
  - Modified radical mastoidectomy: Cholesteatoma of attic, antrum and mastoid.

3. *Endomeatal or transcanal approach* (Fig. 4C): Rosen's stapedectomy incision is an example of endomeatal incision. A posterior tympanomeatal flap is raised to enter into the middle ear. Posterior meatal wall skin is raised and annulus dislocated from the sulcus. The posterosuperior overhang of bony meatus if hiding the view of stapes is removed. Rosen's incision consists of two parts:
   a. I incision: A small vertical incision begins at 12 O'clock position near the annulus.
   b. II incision: A curvilinear incision 5–7 mm away from the annulus begins at 6 O'clock position and meets the 1st incision in the posterosuperior region of deep bony canals.
      - Indications: It offers nice view of the middle ear and ossicles. The indications include:
        - Exploratory tympanotomy
        - Stapedectomy
        - Inlay myringoplasty
        - Ossicular reconstruction.

Anesthesia

The surgery is usually done under general anesthesia. Some surgeons prefer local anesthesia in selected cases.



Figs 4A to C: Surgical approaches. (A) Postaural incisions (S, Sulcus; R, Retrosulcus; and I, Infants); (B) Endaural incision (I, Lempert I posterior meatal wall semicircular incision 6 to 12 O'clock; and II, Lempert II upward curvilinear 12 O'clock incision in the incisura terminalis of cartilaginous EAC); and (C) Endomeatal incision (Rosen)

Position of Patient

Patient lies in supine position with head turned to the side so that operation ear is on upper side.

CORTICAL MASTOIDECTOMY

(Synonyms: Simple or Complete Mastoidectomy or Schwartz Operation)

Cortical mastoidectomy refers to complete exenteration of all accessible mastoid air cells without removing the posterior meatal wall. Middle ear structures are not disturbed.

Indications

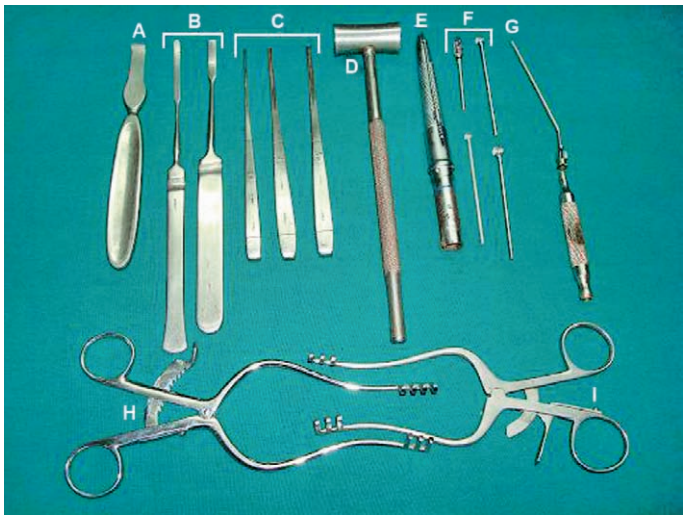
- Acute coalescent mastoiditis with or without subperiosteal abscess.
- Incompletely resolved AOM with reservoir sign.
- Masked mastoiditis.
- Primary step to perform:
  - Cochlear implantation.
  - Endolymphatic sac surgery.
  - Decompression of facial nerve.
  - Labyrinthectomy.
  - Cerebrospinal fluid (CSF) otorrhea.
  - Access to cerebellopontine angle (acoustic neuroma), skull base, and petrous apex (translabyrinthine or retrolabyrinthine procedures).

Instruments (Figs 5A to I and 6A to O)

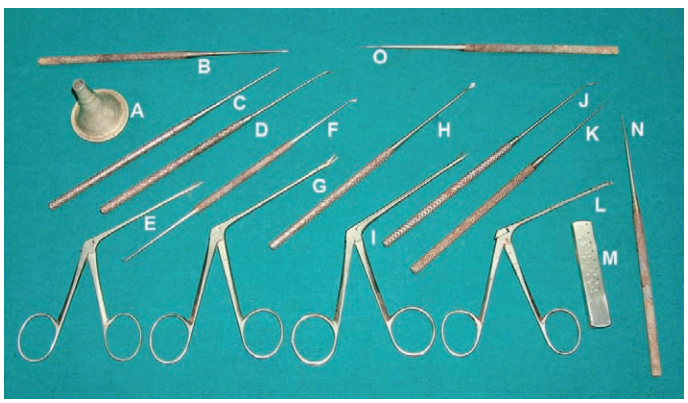
See also Chapter Instruments.

Steps of Operation

1. *Incision*: A postaural incision cuts through soft tissues and reaches up to the periosteum without cutting the temporalis muscle. Mastoid retractors (Jansen's or Mollison's) retract soft tissues after the postaural incision and elevation of flaps. The pressure on the edges of the incision provides hemostasis. Lempert's endaural retractor is used in cases of endaural approach. Lempert's endaural speculum can be used to spread open the meatus when giving local injection or making an endaural incision.
2. *Exposure of Mastoid*: Periosteum is cut in the line of postaural incision. A horizontal incision is made along the lower border of temporalis muscle. A branch of superficial temporal artery is usually encountered and may need ligation or cauterization. Periosteum is elevated from the lateral surface of mastoid and posterosuperior region of bony meatus. Farabeuf's periosteal elevator may be used for elevation of periosteum from the mastoid cortex. Tendinous fibers of sternocleidomastoid muscle need sharp cutting near the mastoid tip. A self-retaining mastoid retractor is applied for the exposure.
3. *Identification of mastoid antrum*: Mastoid cortex is drilled out to enter into the mastoid antrum. In adults mastoid antrum lies 12–15 mm deep to the suprameatal triangle (Macewen's triangle, Fig. 7). Mastoid gouges may be used to remove mastoid bone. Horizontal semicircular canal, aditus ad antrum and short process of incus are identified. Mastoid suction tips helps aspirating blood, discharge, irrigation water and bone dust and pieces.
4. *Removal of mastoid air cells*: Lempert's mastoid curette (scoop) removes bony septa and granulations in mastoid antrum



Figs 5A to I: Mastoidectomy instruments. (A) Farabeuf's cutting edge curved raspatory; (B) Elevator and raspatory (curved with straight tip); (C) Gauges; (D) Hammer; (E) Straight slip joint hand piece for burrs; (F) Burrs (G) Suction cannula and adaptor with finger cut-off; (H) Mastoid retractor with 3×4 prongs; and (I) Mastoid retractor 3×3 prongs



Figs 6A to O: Ear microsurgery instruments. Micro ear forceps are with tubular shafts. Their opening and closing of the jaw are by means of thumb ring-handle. (A) Ear speculum for adults, black finish in- and outside to eliminate reflection; (B) Straight needle perforator (for footplate); (C) Round cutting knife angled 90° for incising skin of external auditory canal; (D) Sickle knife slightly curved; (E) Teflon piston holding straight alligator forceps; (F) Double ended round cutting flap knife (blade in line with shaft) and angled (45°) round cutting knife; (G) Straight sharp micro scissors with tubular shaft; (H) Ear curette; (I) Straight alligator forceps; (J) Buttoned ear hook; (K) Sharp point curved needle (pick); (L) Upward cutting malleus nipper; (M) Teflon piston measuring rod; (N) Sharp point straight needle (pick); and (O) Measuring rod (caliper) marker pins at 3.5, 4 and 4.5 mm

and air cells. The electrical drill and burs are replacing the use of gouges and curettes. All the accessible mastoid air cells are removed. Zygomatic cells in the root of zygoma and retrosinus cells behind the sinus between sinus plate and cortex are also removed. The bony plate of tegmen tympani above, sinus plate behind and posterior meatal wall in front though exposed are not removed (Figs 8 and 9). Try to identify following structures in the newly created mastoid cavity:

a. *Sinodural angle*: It lies between the tegmen antri (middle cranial fossa) and the sigmoid sinus.
b. *Solid angle*: This solid bone angle, which lies medial to antrum, is formed by the three semicircular canals.

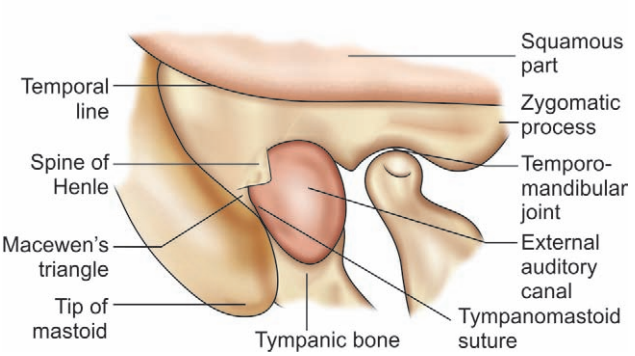


Fig. 7: Macewen's triangle. Surface landmark for mastoid antrum

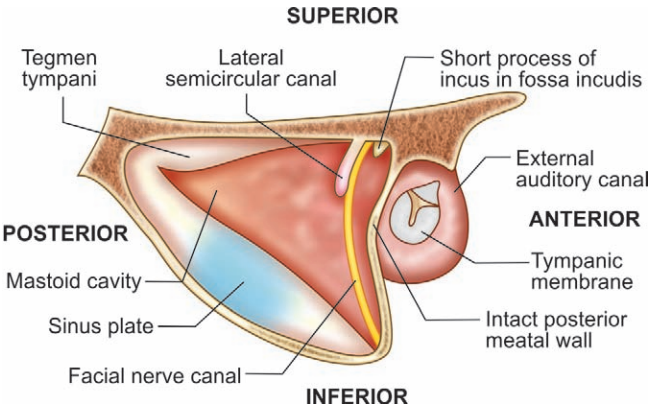


Fig. 8: Simple cortical mastoidectomy with intact posterior meatal wall

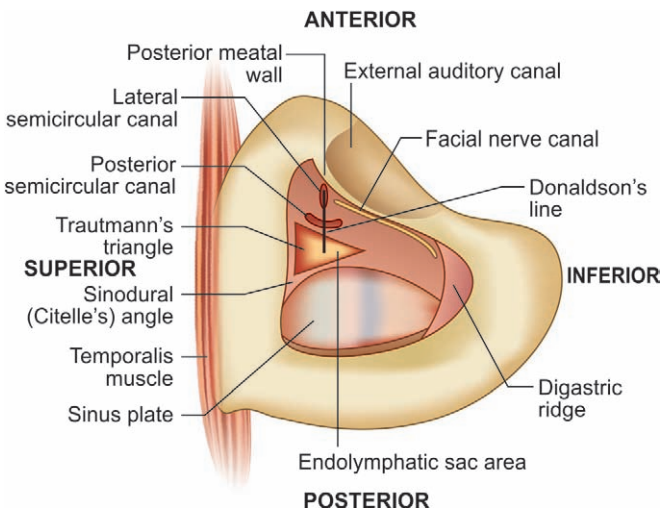


Fig. 9: Cortical mastoidectomy cavity—landmarks and structures seen

c. *Trautmann's triangle*: This bony plate of posterior surface of petrous bone lies behind the mastoid antrum. It is bounded by following structures:
   - Sigmoid sinus
   - Bony labyrinth
   - Superior petrosal sinus.
d. *Donaldson's line*: This line passes through the lateral semicircular canal bisecting the posterior semicircular canal. The endolymphatic sac that appears as thickening of the posterior cranial fossa dura (double layered) is situated inferior to Donaldson's line.

5. *Removal of mastoid tip*: Removal of the lateral wall of mastoid tip will expose muscle fibers of posterior belly of digastric. Edges of the mastoid cavity should be beveled (saucerization of edges) so that soft tissue sits in and obliterates the cavity.
6. *Closure of wound*: After cleaning the bone dust from the wound and mastoid cavity (thorough irrigation with saline), the wound is closed in two layers. If there is infection or excessive bleeding, a rubber drain may be kept at the lower end of incision for 1 to 2 days. The EAC is packed with ribbon gauze impregnated in antimicrobial agent. It avoids stenosis of ear canal. Mastoid bandage is applied.

Postoperative Care

- *Antibiotics*: They are started before the surgery and continued postoperatively for at least 1 week. Perioperative swab for culture from the mastoid helps in identifying the microorganisms and selecting the antibiotic.
- *Mastoid drain*: It is usually removed in 1–2 days.
- *Removal of stitches*: They are removed usually on the 6th day.

Complications

- Perioperative injuries to:
  - Facial nerve
  - Horizontal semicircular canal: Patient develops vertigo and jerky nystagmus
  - Sigmoid sinus (profuse bleeding)
  - Dura mater of middle cranial fossa-CSF leak
  - Dislocation of incus-conductive hearing loss
- Postoperative
  - Wound infection and wound break-down.

RADICAL MASTOIDECTOMY

Radical mastoidectomy eradicates disease from the middle ear and mastoid and exteriorizes mastoid, middle ear, attic and antrum into the external ear by removing the posterior meatal wall (Fig. 10). The structures removed include cholesteatoma, granulations and remnants of tympanic membrane, malleus, incus (not the stapes) chorda tympani and mucoperiosteal lining. The opening of Eustachian tube is closed with a piece of muscle or cartilage.

There is no reconstruction of hearing system. As the posterior meatal wall is removed, the entire area of middle ear, attic, antrum and mastoid is converted into a single cavity. The basic aim is to exteriorize the diseased area for inspection and cleaning. This radical surgery is infrequently required these days.

Indications

- CSOM with intracranial complications.
- Cholesteatoma invading Eustachian tube, round window niche, perilabyrinthine or hypotympanic cells
- Revision surgery to eradicate chronic inflammatory disease or remnant cholesteatoma
- Disease in petrous apex
- Glomus tumor
- Carcinoma middle ear: If en bloc removal of temporal bone is not feasible then radical mastoidectomy followed by radiotherapy may be considered.

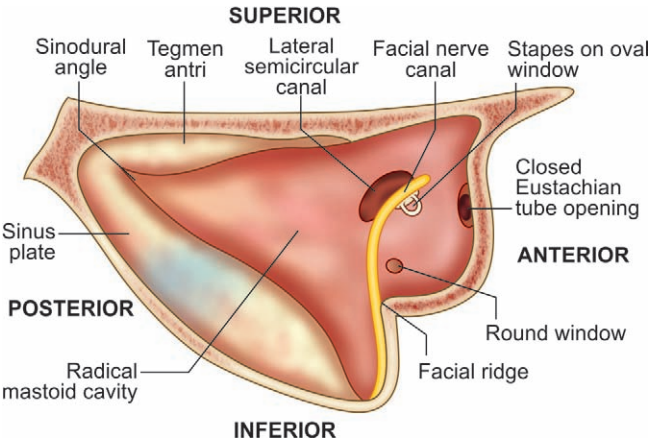


Fig. 10: Radical mastoidectomy cavity after removing the posterior meatal wall, exteriorizing the area of mastoid, middle ear, attic and antrum, removing malleus and incus and obliterating the Eustachian tube

Steps of Operation (*See* Steps of Cortical Mastoidectomy)

1. *Incision*: Postaural or endaural.
2. *Exposure of mastoid area:* The periosteum from the lateral mastoid surface is elevated. This extends from the root of zygoma to the area behind the suprameatal triangle and from suprameatal crest (continuation of inferior temporal line) to the lower part of mastoid tip. The wound is retracted.
3. *Exposure of attic and antrum*: The cortical mastoid bone is drilled out from the suprameatal triangle, spine of Henle, root of zygoma and upper part of superior meatal wall till the exposure of attic and antrum. The tegmen antri and lateral semicircular canal are identified.
4. *Removal of "bridge" and facial buttresses*: Deeper part of superior bony meatal wall bridging over the notch of Rivinus is taken off carefully without damaging the deeper middle ear structures. The removal of anterior buttress (anterior spine of the notch) and posterior buttress (posterior spine of the notch), which form the lateral attic wall, will expose the regions of attic, aditus ad antrum, facial canal and ossicles. The diseased incus and the malleus are removed.
5. *Lowering of facial ridge*: The deeper part of posterior meatal wall lying over the vertical part of facial nerve called facial ridge is removed as much as possible within the safety of facial nerve. It makes mastoid cavity easily inspected and cleaned during the postoperative care.
6. *Debridement of middle ear*: Remnants of tympanic membrane, annulus and sulcus tympanicus, middle ear mucoperiosteum, cholesteatoma, polyp and granulation tissue, malleus and incus are removed step by step. Stapes is left intact. Eustachian tube mucosa is curetted and plugged with tensor tympani muscle or piece of cartilage.
7. *Cavity inspection and irrigation*: Ensure complete exteriorization of the mastoid antrum and cavity and middle ear including attic into the EAC. Bony overhangs are removed and cavity is smoothened with polishing burr. Saucerization of the cortical edges of mastoid must be done. Irrigate the cavity and wound with saline or ringer to remove blood and bone dust.

8. *Meatoplasty*: A concha based flap from posterior and superior meatal wall is raised and turned into the mastoid cavity. It covers the area of the facial ridge and facilitates epithelialization of the mastoid cavity. In this operation, which is invariably combined with canal wall-down procedures, a crescent of conchal cartilage is excised to widen the meatus. This meatoplasty facilitates access to cavity for inspection and cleaning during the postoperative care.

   Meatoplasty is also done as an isolated procedure in cases of sagging auricle, which is seen in older people. The obstruction of the ear canal causes hearing loss and retention of wax.
9. *Mastoid obliteration*: If the ultimate mastoid cavity becomes very large, it may be obliterated with temporalis muscle or other musculofascial tissues. Due care must be taken to remove any vestige of disease (cholesteatoma), which may be buried underneath.
10. *Closure of wound*: The mastoid cavity is packed with ribbon gauze, impregnated with an antibiotic/antiseptic solution. The wound is closed in two layers. Mastoid bandage is applied.

Postoperative Care

- *Antibiotic*: It is given for 7–14 days.
- *The bandage and packing*: They are removed as per the liking of the surgeon from 1–7 days. Skin stitches are removed on 6th or 7th day. Some prefer changing of the pack at weekly intervals and others leave the cavity unpacked with regular suction and cleaning till the epithelialization of the cavity.
  - Look for any signs of perichondritis or infection.
- *Cavity care:* The epithelialization of cavity takes 2–3 months. The cavity is checked every 4–6 months in the first year and then annually. The debris is removed. Granulation tissue which delays the healing is either removed or chemically cauterized.

Complications

- *Perioperative injury* to:
  - Facial nerve resulting in facial paralysis.
  - Dura or sigmoid sinus.
  - Stapes dislocation can result in labyrinthitis sensorineural hearing loss (SNHL) and vertigo.
  - Severe conductive hearing loss due to removal of the ossicles and tympanic membrane.
- *Perichondritis*
- *Cavity problems*:
  - Nonhealing of cavity.
  - Regular after care.

MODIFIED RADICAL MASTOIDECTOMY

In this modification of radical mastoidectomy, hearing mechanism and middle ear cleft structures are preserved as far as possible. Only irreversibly damaged tissues are removed. Preservation and conservation of middle ear structures help in reconstruction of the hearing mechanism.

Modified radical mastoidectomy eradicates disease from the attic and mastoid and exteriorizes both into the EAC by removal of the posterior meatal and lateral attic walls (Fig. 11). The remnant of tympanic membrane, functioning ossicles and the reversible mucosa and function of the Eustachian tube, which are necessary to reconstruct hearing mechanism, are preserved.

Indications

- Atticoantral cholesteatoma.
- Chronic otitis media with limited disease.

Operation

The operative steps, postoperative care and complications are similar to radical mastoidectomy and the differences are following:

1. *Removal of disease and preservation of healthy tissue*: Cholesteatoma, granulations and unhealthy mucosa are removed. Incus and head of malleus are preserved. They are removed only if cholesteatoma engulfs them or extends medial to them. Healthy pars tensa and middle ear structures are left undisturbed.
2. *Tympanoplasty*: Reconstruction of tympanic membrane and ossicular chain depending upon the extent of damage can be done (mastoidectomy with tympanoplasty operation) at the same sitting or in second stage.

Bondy Procedure

The attic is exteriorized by removing portions of the adjacent superior and/or posterior canal wall. The uninvolved middle ear is not entered. The cholesteatoma matrix on the lateral surface of malleus head and incus body is maintained in place as a lining for the created cavity.

TYMPANOPLASTY

These surgeries are done under operating microscope with the help of microsurgical instruments and biocompatible implant materials.

- *Tympanoplasty*: The tympanoplasty operation consists of both eradication of middle ear disease and reconstruction of hearing mechanism including tympanic membrane and ossicles. It may be done with or without mastoidectomy.
- *Myringoplasty*: The limited repair of tympanic membrane is called myringoplasty.
- *Ossiculoplasty*: The limited reconstruction of ossicular chain is called ossiculoplasty.

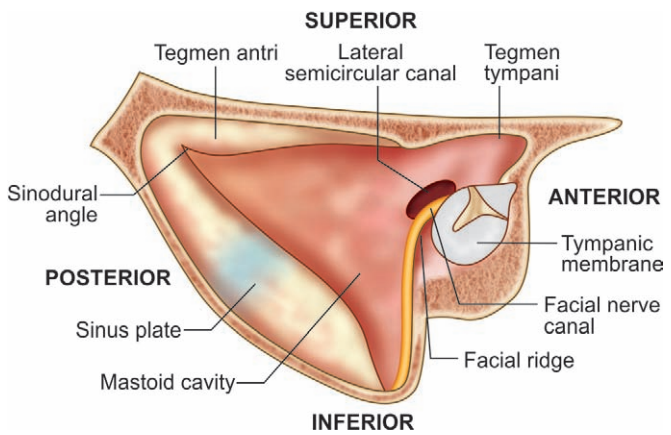


Fig. 11: Modified radical mastoidectomy cavity after removing the posterior meatal wall and exteriorizing the diseased area of attic and mastoid antrum without removing the normal ossicles and tympanic membrane

Types of Tympanoplasty (Wullstein) (Figs 12A to E)

Wullstein described five types of tympanoplasty (Table 1). Several modifications in the Wullstein classification have been reported in the literature, which mainly pertain to the types of ossicular reconstruction.

- *Cavum minor and tympanoplasty IV*: In Type IV tympanoplasty graft (Fig. 13) is placed between the oval and round windows to create an air pocket around the round window. This narrow middle ear space, which is called cavum minor, is a mucosa-lined space that extends from the Eustachian tube to the round window. Sound waves in this Type IV tympanoplasty directly hit on the footplate while the round window has been shielded by the cavum minor.
- *Tympanoplasty V:* In Type V tympanoplasty a window is created on horizontal semicircular canal that is covered with a graft.

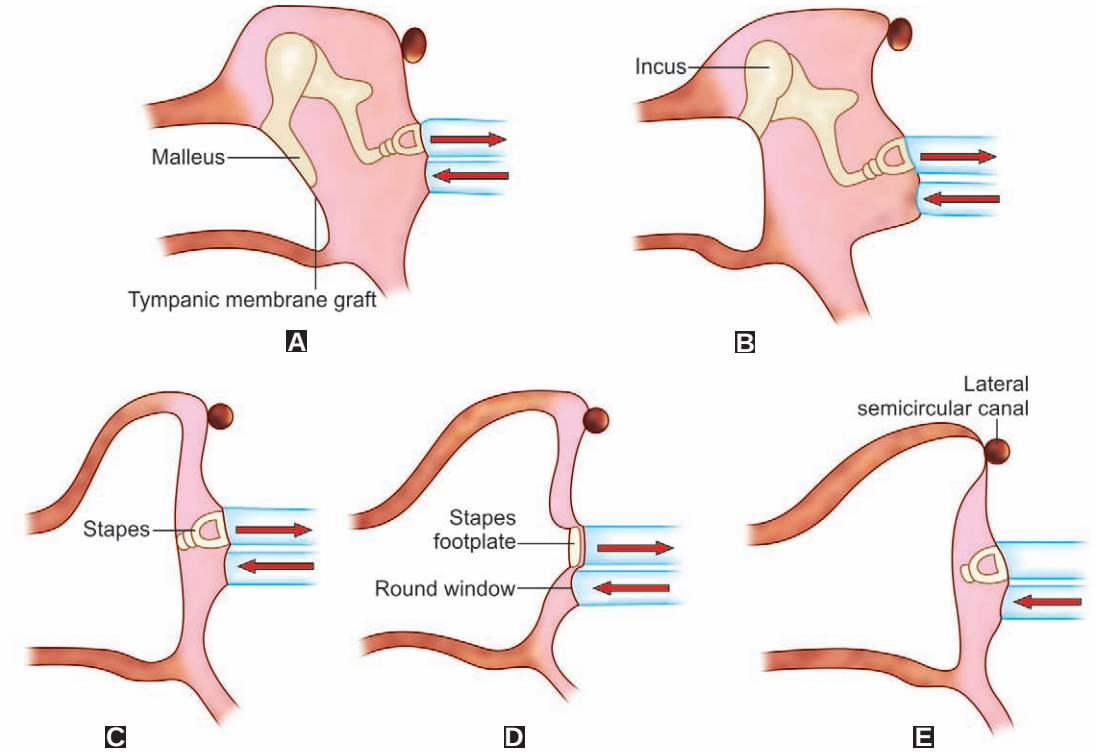
Myringoplasty

The simple closure of tympanic membrane perforation is called myringoplasty.

- *Graft materials*: The most commonly used graft materials are temporalis fascia and tragal perichondrium. The other graft material includes fascia lata, cartilage and homograft (dura, vein, cadaver tympanic membrane).
- *Techniques*: There are following two techniques underlay (inlay) and overlay (onlay).
  - Underlay technique: In this technique, graft is placed medial to the tympanic annulus. The underlay technique requires opening of the middle ear (tympanotomy), which provide an opportunity to examine the ossicles and other middle ear structures.
  - Overlay technique (Figs 14A to C): In this technique, graft is placed lateral to fibrous layer of the tympanic membrane. It requires careful removal of squamous epithelium from the lateral surface of remnant tympanic membrane.
  - Inlay technique: Graft is placed in between the fibrous and mucosal layers of tympanic membrane.

Ossiculoplasty

- *Indications*: Ossicular reconstruction is required in cases of destruction of ossicular chain mostly caused by chronic suppurative otitis media (CSOM)
  - Destruction of ossicles
    - Necrosis of long process of incus: It is the most common lesion.



Figs 12A to E: Types of tympanoplasty. (A) Tympanic membrane graft touches malleus (Type I); (B) Incus (Type II); (C) Stapes superstructure (Type III); (D) Mobile stapes footplate (Type IV); and (E) Lateral semicircular duct (Type V) stapes fixed

TABLE 1 Types of tympanoplasty (Wullstein)

Types	*Defect*	*Position or contact of graft*
Type I (myringoplasty)	Tympanic membrane perforation	Tympanic membrane or malleus
Type II	Tympanic membrane perforation with erosion of malleus	Incus or remnant of malleus
Type III Myringostapediopexy or columella tympanoplasty	Malleus and incus absent	Stapes superstructure
Type IV	Stapes superstructure absent	Stapes footplate
Type V Fenestration operation	Stapes footplate fixed	Fenestra in horizontal semicircular canal

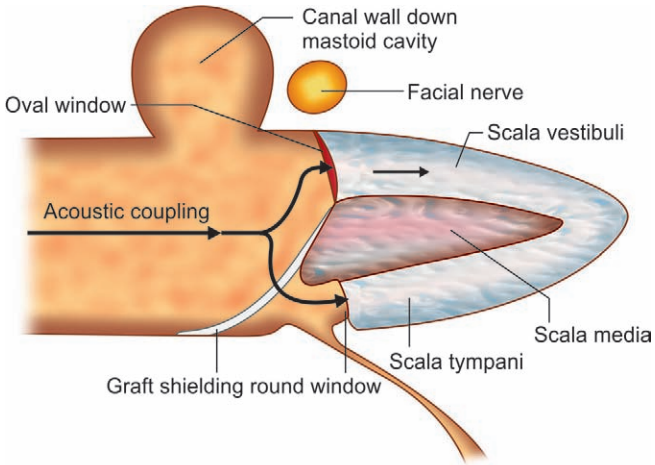
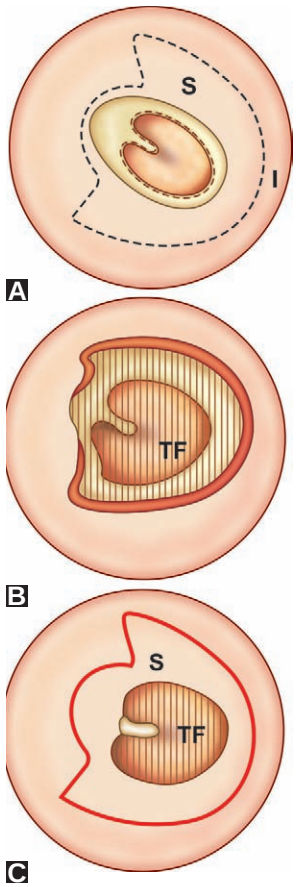


Fig. 13: Tympanoplasty IV. There is only acoustic coupling and no ossicular coupling. Graft acoustically protects round window while sound directly impinges stapes footplate

- Loss of stapes superstructure: It leaves behind a mobile footplate and malleus.
- Destruction of malleus, incus and the stapes superstructure: They leave behind only the mobile footplate. It is common in cholesteatoma.

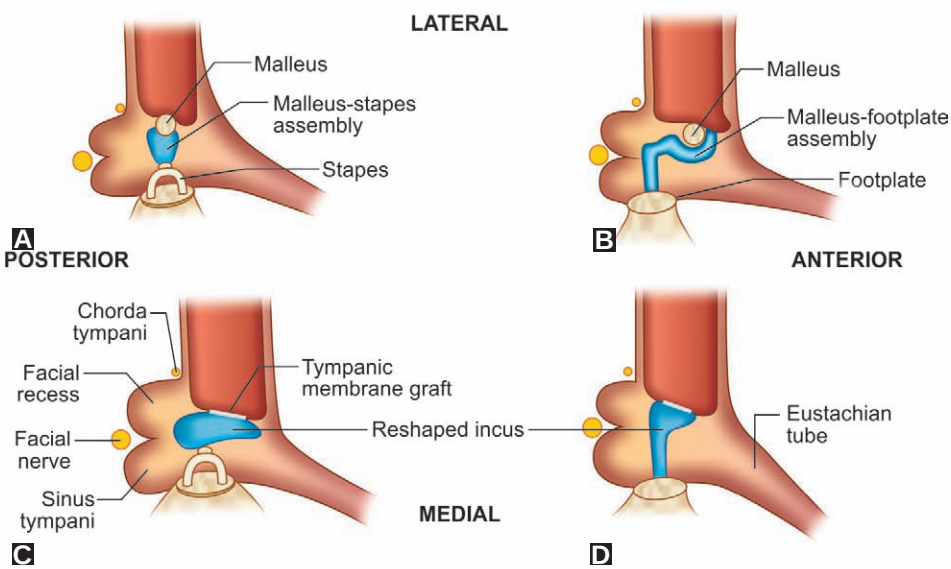
- Fixation of ossicles
  - Ankylosis of stapes footplate (otosclerosis and tympanosclerosis): The correction of ankylosis of stapes consists of removal of the superstructures stapes and its replacement by prosthesis.
  - Fixation of the head of malleus in the attic (tympanosclerosis and congenital): The attic fixation of malleus head needs removal of the head of malleus and entire incus. Then a contact is established between handle of malleus and the stapes.

- *Graft materials*: They are:
  - Autografts (Figs 15A to D): The most commonly employed graft materials are autograft ossicles (incus transposition and sculptured ossicles) and tragal cartilage.
  - Homograft: Ossicles and membrane.
  - Prosthetic implants: They are made of ceramic (hydroxy appetite) and teflon.



Figs 14A to C: Overlay myringoplasty. (A) Incision (I) to raise the meatal skin (S) and epithelium of tympanic membrane; (B) Placement of temporalis fascia (TF) graft; and (C) Replacement of meatal skin and epithelium of tympanic membrane

- Total ossicular replacement prosthesis: A total ossicular replacement prosthesis (TORP) bridges the gap between tympanic me mbrane and stapes footplate.
- Partial ossicular replacement prosthesis: A partial ossicular replacement prosthesis (PORP) provides a direct contact between tympanic membrane and stapes head.



Figs 15A to D: Ossicular reconstruction. (A) Malleus-stapes assembly; (B) Malleus-footplate assembly; (C) Reshaped incus between stapes head and tympanic membrane graft; and (D) Reshaped incus between stapes footplate and tympanic membrane graft

Clinical Highlights

1. **Operative microscope:** The objective piece of microscope commonly used for ear microsurgery is 200–300 mm (250 mm).
2. **Mastoid surgery:** Aim of mastoid surgery in unsafe chronic suppurative otitis media (CSOM) is rendering the ear safe. Techniques, which are used to control bleeding from bone during mastoid surgery, include bone wax, bipolar cautery over the bleeding area and diamond drill. Cutting drill over the bleeding area will not control bleeding.
3. **Radical mastoidectomy:** Eustachian tube is obliterated surgically in radical mastoidectomy. Closure of Eustachian tube prevents infection of middle ear from the nasopharynx.
4. **Grafts:** Autograft tympanic membrane acts as a scaffold. Denatured homograft tympanic membrane has no antigenic property.

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56 Operations of Nose and Paranasal Sinuses

It is fear alone that is death. You have to go beyond all fear. So from this day, be fearless.
It is the level-headed man, the calm man, of good judgment and cool nerves, of great sympathy and love, who does good work and so does good to himself.

—Swami Vivekananda

Points of Focus

SINUS OPERATIONS

- **PREOPERATIVE ASSESSMENT**
- **DIAGNOSTIC NASAL ENDOSCOPY**
  - ✓ Indications
  - ✓ Instruments
  - ✓ Techniques
- **ENDOSCOPIC SINUS SURGERY**
  - ✓ Indications, Contraindications
  - ✓ Anesthesia
  - ✓ Preparations
  - ✓ Position of Patient
  - ✓ Techniques
  - ✓ Postoperative Care
  - ✓ Complications
- **ANTRAL PUNCTURE**
  - ✓ Indications, Contraindications
  - ✓ Anesthesia and Position
  - ✓ Instruments
  - ✓ Techniques
  - ✓ Complications
- **INFERIOR MEATAL ANTROSTOMY**
- **CALDWELL-LUC OPERATION**
  - ✓ Indications, Contraindication
  - ✓ Anesthesia, Position
  - ✓ Instruments
  - ✓ Techniques
  - ✓ Postoperative Care
  - ✓ Complications

SURGERY OF NASAL SEPTUM

- ✓ Indications, Contraindications
- ✓ Anesthesia
- ✓ Techniques
- ✓ Instruments

- **SUBMUCOUS RESECTION OF NASAL SEPTUM**
- **SEPTOPLASTY**
- **POSTOPERATIVE CARE**
- **COMPLICATIONS**
- **CLINICAL HIGHLIGHTS**

SINUS OPERATIONS

The current trend for treating the sinonasal diseases is functional endoscopic sinus surgery (FESS), which will be dealt in detail but it is important to have knowledge of external approaches, which used to be the mainstay of treatment in recent past. FESS is considered for patients with refractory rhinosinusitis or its complications. The success rate of FESS is good in patients with chronic or recurrent rhinosinusitis while it's low in cases with recurrent polyps, severe allergies, past external procedures and immunocompromised state.

PREOPERATIVE ASSESSMENT

A thorough history and physical examination is of paramount importance and must include following points:

1. *History*: Chief complaints and associated history of allergy, asthma, aspirin sensitivity, polyps and facial pain, congestion, hyposmia, nasal obstruction and purulent discharge.

- Past medical and surgical history.

2. *Examination*: Complete head, neck and ocular examination and review of systems.
3. *Nasal Endoscopy Examination*: Look for septum deviation, character of mucosa and polyps. Detailed nasal endoscopy examination is described in other section of this chapter.
   a. *First pass*: Examine floor, nasolacrimal duct and nasopharynx on first pass.
   b. *Second pass*: Examine middle meatus, sphenoethmoidal recess on second pass at a 30° angle from floor. Notice anatomical variations especially of osteomeatal complex such as paradoxical, lateralized middle turbinate or concha bullosa, deviated nasal septum or spurs, medialized uncinate process, hypertrophy of ethmoidal bulla, polyps, purulent discharge and big agger nasi cells.
   c. *Third pass*: Examine frontal recess during the final pass.
4. *Preoperative Antibiotics and Steroids*: They help in infection and inflammation, especially in cases with polyps, chronic rhinosinusitis and reactive airways.
5. *Computerized Tomography of Nose and Paranasal Sinuses*: CT scan limited study coronal cuts in bone window is prerequisite for endoscopic sinus surgery (ESS).
   a. *Method*: Detailed sinus CT evaluation includes axial and coronal views, with 2–5 mm intervals, printed in both soft tissue and bony windows.
   b. *Prerequisite*: To evaluate the anatomy and the pattern of rhinosinusitis, CT is obtained 2–4 weeks after medical treatment.
   c. *Examination*: Look for the following findings:
      - Normal anatomy: Notice the presence of agger nasi cells, frontal cells, infraorbital cells, attachment of uncinate process, course of internal carotid artery and optic nerve.
      - Pathology: Retained fluid, thickened mucosa, polyps, anatomical abnormalities (septal deviations), and evidence of expansion, erosion, or dehiscence, calcifications and hyperintense signals (fungus and tumors).
      - Anatomic variations: The anatomic variations, which can predispose to complications during ESS, include lamina papyracea medial to maxillary ostium, dehiscence of lamina papyracea, internal carotid artery or optic nerve, hypoplasia of maxillary sinus, low or sloping fovea, sphenoethmoidal cells and sphenoid sinus septa attached to carotid.

Incidental sinus opacification is seen in 25–30% of sinus scans.

6. *Magnetic Resonance Imaging*: Magnetic resonance imaging (MRI) is indicated in cases of bony dehiscence or erosion, intracranial invasion, and encephalocele, and in differentiating tumors from fluid within sinuses.

DIAGNOSTIC NASAL ENDOSCOPY (SINUSCOPY) (FIG. 1)

The brighter illumination, magnification, and angled view of endoscopes (sinuscopes) facilitate examination of all the cleft and crevices of nose and nasopharynx.

Fig. 1: Sinuscopy examination with CCTV attachment. Right hand holds the suction cannula

Indications

Sinuscopy facilitates in diagnosing the nose, paranasal sinuses (PNS) and nasopharynx diseases:

- Locate the site of nose bleed.
- Biopsy from nose, PNS and nasopharynx lesions.
- Assess the response to treatment of nose, PNS and nasopharynx diseases.

Instruments

- 4 mm 0° and 30° sinuscopes.
- 2.7 mm 0° and 30° sinuscopes in cases of children and narrow nasal cavity.
- Freer's elevator.
- Suction cannula.
- Biopsy forceps.
- Antifog solution/savlon to avoid fogging.

Techniques

- *Anesthesia*: Mix 4% xylocaine in equal quantity of oxymetazoline hydrochloride. Instill these drops into the nose and then pack nasal cavity with packing impregnated with this solution.
- *Position of Patient*: Patient is kept in either sitting or supine position (Fig. 1).
- *Method*: For thorough and complete examination, the scope is passed through the standard three paths. The examination is conducted while inserting and withdrawing the scope.
  - *First pass (0° sinuscope)*: It examines the nasal vestibule, nasal cavity in general, septum, inferior meatus and nasopharynx.
    - The scope is passed up to the nasopharynx through the inferior meatus. Slight pressure over the lacrimal sac may show the opening of nasolacrimal duct in the inferior meatus.
    - Note the septal deviation or spur, nature of nasal discharge, and color of mucous membrane, opening of the eustachian tube, walls of nasopharynx, and upper surface of soft palate.
  - *Second pass (0° sinuscope)*: It examines the posterior part of middle turbinate, sphenoethmoidal recess, superior

meatus, superior turbinate and openings of the sphenoid sinus (in the posterior wall of sphenoethmoidal recess between the nasal septum and superior turbinate) and posterior ethmoid sinuses (in the superior meatus).
  - The scope is passed medial to the posterior part of middle turbinate and progresses up to the sphenoethmoidal recess between the nasal septum and superior turbinate.
- *Third pass (30° sinuscope)*: It examines the osteomeatal complex in the middle meatus such as uncinate process, bulla ethmoidalis, hiatus semilunaris, sinus of turbinate (space lateral to middle turbinate), basal lamina, and the frontal recess.
  - The sinuscope is passed into the middle meatus usually from the anterior aspect but in some cases it is entered from behind where it is wider and the structures are examined from behind forward. If needed middle turbinate can be gently retracted medially with the help of Freer elevator.
- *Complication*: Bleeding can occur due to improper manipulation of instruments and is usually controlled by the application of vasoconstrictor pledgets.

ENDOSCOPIC SINUS SURGERY

The endoscope has revolutionized the diagnosis and treatment of diseases of nose and paranasal sinuses. Now the first line of surgical treatment of rhinosinusitis is FESS. FESS has demonstrated success rates of 76–98%.

The philosophy of FESS is minimal (functional) surgery with mucosal preservation to achieve physiological drainage and ventilation of sinuses and healing. FESS is targeted to diseased sinuses and the normal sinuses are left alone. Aggressive removal of mucosa is avoided as it leads to postoperative healing problems.

Indications

- Recurrent and chronic rhinosinusitis, which do not respond to medical therapy (Fig. 2).
- Nasal polyps both ethmoidal and antrochoanal.
- Foreign body.
- Septoplasty.
- Dacryocystorhinostomy.
- Epistaxis especially uncontrolled posterior bleeding and ligation of sphenopalatine artery.
- Headache and facial pains: Due to nasal septal deviation and concha bullosa.
- Complications of rhinosinusitis such as orbital abscess.
- Cerebrospinal fluid rhinorrhea: Traumatic and iatrogenic. Patch material includes mucosal grafts, fascia grafts and synthetic products.
- Fungal mycetoma: CT shows heterogeneous and microcalcifications.
- Juvenile nasopharyngeal angiofibroma.
- Tumors of nose and paranasal sinuses such as inverted papillomas.
- Failed previous surgeries such as external maxillary, ethmoidal and frontal procedures.
- Mucoceles (frontoethmoid and sphenoid)–marsupialization.

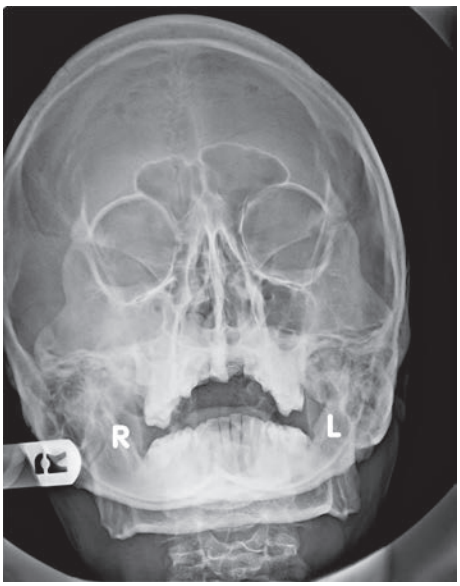


Fig. 2: X-ray PNS PA view Water's position showing right maxillary sinus haziness
Source: Dr Jayesh Patel, Consultant Radiologist, Anand, Gujarat

- Encephalocele.
- Pituitary tumors.
- Optic nerve decompression.
- Orbital decompression in Graves' disease.
- Choanal atresia.

Contraindications

The contraindications include following conditions, which are better tackled by the external approaches:
- Intracranial complications.
- Orbital cellulitis with visual field defects.
- Osteomyelitis.
- Aggressive fungal infections such as mucormycosis.

Anesthesia

- *Local Anesthesia with Sedation*: Endoscopic sinus surgery in adults is usually done under local anesthesia and sedation. It improves safety, as manipulations of orbital perisoteum and dura are painful. The standby anesthesiologist monitors the vital parameters such as blood pressure, pulse, respiration, temperature and oxygen saturation.
- *General Anesthesia*: It is preferred in pediatric patients, anxious adults, in anticipated long cases and computer-assisted navigation systems.

Preparations

Topical decongestants and anesthetics are administered in nose before the patient comes to operation theater (OT). Local injection with 1% lignocaine with 1:100,000 epinephrine is infiltrated to nasal septum and dorsum, inferior and middle turbinates (infraorbital block) canine fossa, and greater palatine foramen. A small Foley catheter No. 8 or expandable sponges in nasopharynx prevents blood pooling in oropharynx.

Position of Patient

Patient is placed in supine position. A slight reverse Trendelenburg position with patient rotation towards surgeon helps in reducing blood loss and makes surgeon comfortable.

Techniques

The endoscopes and microsurgical instruments provide better precision in the removal of tissue and avoid unnecessary stripping of mucosa. ESS instruments are described in chapter Instruments. There are two techniques of ESS: Messerklinger (anterior to posterior) and Wigand (posterior to anterior).

A. *Messerklinger Technique*: It consists of anterior-to-posterior approach. The extent of surgery can be tailored according to the extent of disease. It includes following steps:
   1. *Removal of uncinate process and exposure of infundibulum*: Uncinectomy and infundibulotomy are done with the help of either back-biting forceps or sickle knife and Blakesley forceps.
   2. *Identification and widening of maxillary sinus ostium*: Maxillary ostium is situated in the posterior part of infundibulum and becomes visible after the uncinectomy. It is enlarged anteriorly and posteriorly.
   3. *Anterior ethmoidectomy*: Removal of ethmoidal bulla (bullectomy) is performed with curette or Blakesley forceps.
   4. *Frontal sinusotomy*: Exposure and cleaning of frontal sinus ostium is done in the event of frontal sinus disease. Many surgeons prefer to do this step in last. The position of frontal sinus opening varies depending on the insertion of uncinate process. It is situated lateral to anterior attachment of middle turbinate, medial to lamina papyracea, anterior to anterior ethmoidal artery and posterior to agger nasi cells.
   5. *Identification of roof of ethmoid*: Remove the remaining anterior ethmoidal cells and identify the middle turbinate basal lamella.
   6. *Posterior ethmoidectomy (removal of posterior ethmoidal cells)*: This thin basal lamella separates the anterior ethmoidal cells from the posterior ethmoid cells. It is penetrated in the lower and medial part and the diseased posterior ethmoid cells are removed.

      The presence of posterior ethmoid Onodi cells, which extends into sphenoid bone lateral and superior to sphenoid sinus, places the optic nerve at risk.
   7. *Sphenoid sinusotomy*: Opening of the anterior wall of the diseased sphenoid sinus is done in last. The inspissated secretions and pus is aspirated. The sinus can be entered either by directly enlarging the opening of the sphenoid sinus or through the created anterior and inferior ethmoid cavity.
   8. *Packing*: Small Merocel packing in middle meatus keeps the middle turbinate medial and prevents adhesions.

B. *Wigand Technique*: It involves posterior-to-anterior approach and include following steps:
   1. Partial resection of middle turbinate.
   2. Opening of posterior ethmoidal cells.
   3. Removal of anterior wall of sphenoid sinus.
   4. Identification of skull base within sphenoid sinus.
   5. Removal of anterior ethmoids.

C. *Advanced Techniques*:
   a. *Powered instruments*: Powered instrument such as soft tissue shaver (microdebrider) helps in not only reducing bleeding but is also excellent in removing polyps and soft tissue masses. Bone cutting drills are used during the surgery of frontal sinus and lacrimal sac.
   b. *Image-guided navigation*: The computer-aided ESS is of great assistance in revision surgeries and when operating near the optic nerve and base of skull and anatomical landmarks are disturbed by the pathology. Anatomy can be distorted due to previous surgery, mucocele and extensive polyposis and intracranial and orbital extensions.

Postoperative Care

- *Watch for swelling*: Elevation of head and local ice to nose reduce swelling.
- Monitoring of visual and mental status.
- *Watch for subcutaneous emphysema*: Small fracture of lamina papyracea can cause subcutaneous emphysema, which can increase due to positive pressure ventilation, coughing, vomiting, and blowing of nose.
- *Antibiotics*: Intraoperative as well as postoperative for 7–10 days.
- *Steroids*: Reduces mucosal edema and manage allergy.
- Analgesics relieve the pain.
- *Other agents*: Allergy management, antifungal agents, and leukotriene inhibitors; and irrigations are administrated as per the need of the case.
- *Removal of nasal packing*: It is removed at the time of discharge that is usually 24 hours after the surgery.
- *Topical saline and decongestants*: Saline nasal spray and a short course of nasal decongestant after the removal of nasal packing.
  - Nasal saline irrigations remove blood clots, crusts and secretions.
- *Removal of stenting*: Plan for the stent removal if that is used.
- Avoid strenuous activity and nose blowing and medicines that increase risk of bleeding.
- *First postoperative visit*: It varies from patient to patient and is usually after 3–6 days. Some patients need frequent cleaning while others may need none. Debridement of old blood and crusts promotes healing and restores mucociliary function. Fixed clots and crusts are not removed as they cause damage to mucosa and bleeding. Middle turbinate should not get lateralized.

Complications

Complications of ESS are usually divided into two categories—minor and major (Table 1).

The most common minor complication of FESS is the adhesions and major complications are bleeding, blindness and intracranial injury.

- *Subcutaneous Emphysema*: Small fracture of lamina papyracea can cause subcutaneous emphysema, which can increase due to positive pressure ventilation, coughing, vomiting and blowing of nose.
- *Bleeding*: The common arteries, which can be injured and cause major bleeding, are posterior septal artery below sphenoid sinus and arteries from the internal maxillary artery into middle turbinate. Injury to carotid artery needs immediate angiography for balloon occlusion.
- *Orbital Hematoma*: Rapidly expanding orbital hematoma occurs due to the injury to anterior or posterior ethmoidal artery.

TABLE 1 Minor and major complications of endoscopic sinus surgery

Minor	*Major*	
	Orbital	**Intracranial**
Minor bleeding	Orbital hematoma	Fracture of skull base
Hyposmia	Extraocular muscle injury:	Intracranial bleeding
Synechia	Diplopia	CSF rhinorrhea, stroke
Headache	Injury optic nerve: Blindness	Meningitis, brain injury
Periorbital ecchymosis	Decreased visual acuity	Carotid injury, death
Periorbital emphysema	Nasolacrimal duct or sac injury	Pneumoencephalus
Dental pain	Lamina papyracea injury	**Others:** Anosmia, bleeding
Facial pain		

- *Clinical features*: It leads to elevated intraorbital pressure and blindness.
- *Treatment*: It needs immediate control of bleeding (cautery or clipping) and reduction of intraorbital pressure (orbital decompression or lateral canthotomy and cantholysis). The treatment also includes immediate removal of the nasal pack, administration of steroids and ophthalmologist consultation.

- *Cerebrospinal Fluid Leak*: The overlay and the underlay graft materials used in cases of cerebrospinal fluid leak (CSF) due to skull base injury include nasal mucosa, fascia (temporalis), fat, muscle, acellular dermal graft, and bone or cartilage. Fibrin glue adds support and assists healing.

ANTRAL PUNCTURE OR PROOF PUNCTURE

In antral puncture (AP), medial wall of maxillary sinus is punctured in the region of inferior meatus for antral irrigation (lavage).

Indications

- *Diagnostic*: For collecting the specimen of the antral contents for cytological (early malignancy) and identification of microorganisms (staining, culture and sensitivity).
  - Thin amber-colored fluid with cholesterol crystals indicates antral cyst.
  - Blobs of mucopus indicate hyperplastic sinusitis.
  - In cases of suppuration, foul smelling pus mixes with irrigating saline.
- *Therapeutic*: For washing out the pus in chronic and subacute maxillary sinusitis (Fig. 2).

Contraindications

- Acute maxillary sinusitis: In this condition AP can lead to osteomyelitis.
- Under-developed maxilla with thick bony wall
- Fracture Maxilla
- Children (<3 years)

Anesthesia and Position

- *Local Anesthesia and Sitting Position in Adults*: A pack of 4% lignocaine with adrenaline in inferior meatus is kept for 10–15 minutes. Middle meatus is decongested, which help in opening the maxillary ostium and easy return of fluid.
- *General Anesthesia and Tonsillectomy Position*: They are used in children and anxious uncooperative adults.

Instruments

Lichtwitz trocar and cannula (see chapter *Instruments*) is used for proof puncture (antral lavage). It perforates the lateral wall of inferior meatus. This area is easily accessible and safe region.

Technique

The medial wall of maxillary antrum is punctured through the lateral wall of inferior meatus with Lichwitz trocar and cannula (Fig. 3). A point 1.5–2.0 cm posterior to anterior end of inferior turbinate and near the attachment of concha is selected for puncture because the bone is very thin here and is easily pierced. The direction of the trocar and cannula is towards the ipsilateral ear. After piercing the nasoantral wall, trocar is removed. The cannula is advanced gradually till it reaches the posterior antral wall and then is withdrawn a little. The antrum is then irrigated with 37°C and 20 ml normal saline (Fig. 4) or Hagginson's syringe (Fig. 5) till the return is clear. The cannula is removed and a pack is kept in the inferior meatus if bleeding is present.

Complications

- *Swelling of Cheek*: It can occur if cannula goes into soft tissues over the anterolateral wall of the maxilla. Puncture of the posterior antral wall will result in the swelling of posterior part of cheek.
- *Orbital Injury and Cellulitis*: They occur when trocar and cannula perforates the floor of orbit, which is the roof of maxillary antrum.
- *Bleeding*: It may be brisk in some cases.
- *Air Embolism*: Though rare it may prove fatal. Avoid air insufflation into the antrum after irrigation.

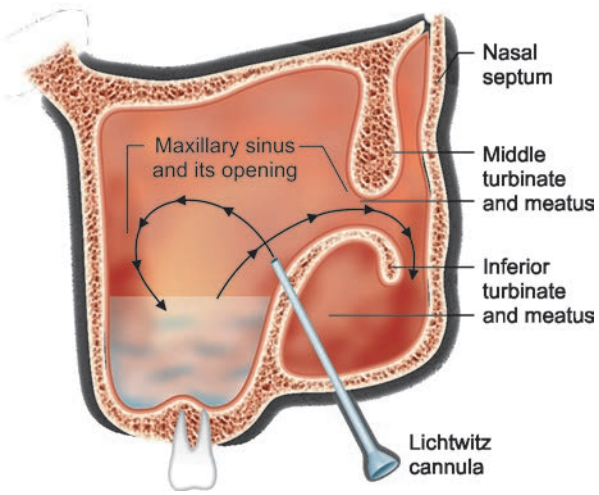


Fig. 3: Antrum puncture. The medial wall of maxillary antrum is punctured through the lateral wall of inferior meatus with Lichwitz trocar and cannula

Fig. 4: Proof puncture (antral lavage). Antrum is irrigated with 20 ml syringe

Fig. 5: Antrum irrigation with Hagginson's syringe

INFERIOR MEATAL ANTROSTOMY

In this procedure, an opening is made in the medial wall of maxillary antrum through the lateral wall of inferior meatus.

- *Indication*: Refractory cases of chronic purulent maxillary sinusitis.
- *Contraindications*: Polypoidal hypertrophy, osteitis and suspected malignancy.
- *Anesthesia and Position*: It can be done under local or general anesthesia. Patient is placed in same position as in submucous resection (SMR) operation.
- *Technique*: After fracturing the inferior turbinate medially and superiorly with a large periosteal elevator, the lateral wall of inferior meatus is perforated with a curved hemostat. This perforation is further enlarged to 1.5–2 cm diameter close to the floor of nose with Kerrison's bone forceps, Luc's or side-biting ring forceps. The sinus content pus/debris is aspirated by suction.
  - Instruments are described in chapter Instruments.
- *Complications*: They are occasional and include bleeding and injury to nasolacrimal duct. Packing into the sinus and nose is done in cases of severe bleeding.

CALDWELL-LUC OPERATION

Caldwell-Luc operation, which was described by George Caldwell in 1893 and Henry Luc in 1897, consists of two antrostomies: canine fossa (through sublabial approach) and inferior meatus (intranasally). It allows for both dependent drainage and irrigation. It has a long history in the treatment of paranasal sinus diseases.

This operation has held on its own for 100 years. Until the last quarter of the 20th century, this operation was the mainstay of treatment for chronic sinusitis. The postoperative factors coupled with advances in CT scan and endoscopes, culminated in the development of FESS. Though some infrequent indications still remain, FESS has now replaced the Caldwell-Luc operation as the treatment of choice in sinusitis.

Indications

The main indication is chronic maxillary sinusitis. Other indications include:

- Complications of acute maxillary sinusitis.
- Removal of foreign bodies or root of tooth.
- Dental cyst.
- Oroantral fistula.
- Biopsy in suspected cases of neoplasm.
- Recurrent antrochoanal polyp.
- Fracture of maxilla or blow out fractures of the orbit.
- Horgan's transantral ethmoidectomy.
- Pterygomaxillary space surgery such as ligation of maxillary artery through pterygopalatine fossa approach and vidian neurectomy.
- In combination with endoscopic approach: Orbital decompression and removal of inverting papillomas.

Contraindication

It is contraindicated in children below 17 years of age.

Anesthesia

General anesthesia with cuffed endotracheal intubation is preferred but can even be done under local anesthesia.

Position

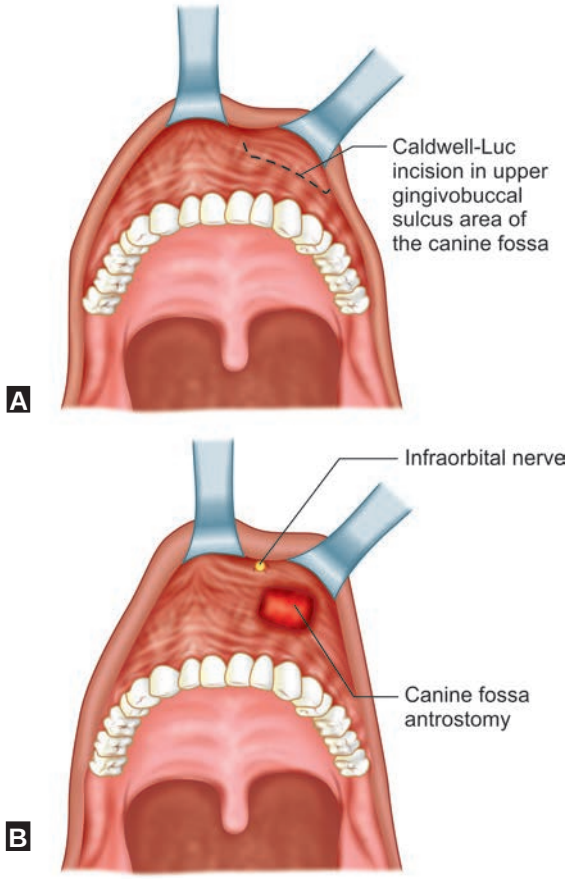
Patient is placed in supine position and face is turned slightly to the opposite side. Head end of the table is raised.

Instruments

See the chapter on Instruments.

Techniques (Figs 6A and B)

- *Preparation*: Local 1% or 2% lignocaine with epinephrine (1:100000) is infiltrated in gingivobuccal and gingivo labial sulcus. Nose is packed with cotton pledgets soaked in topical decongestant.
- *Incision*: Lip retractor retracts the upper lip. Cheek retractor can also be used for making a horizontal incision with its ends upward in the gingivolabial sulcus below the canine fossa. The incision cuts through mucous membrane and periosteum. In children this incision is made above the level of secondary dentition, which can be seen in plain radiograph.
- *Mucoperisoteal Flap*: Periosteum elevator is used for



Figs 6A and B: Caldwell-Luc operation. (A) Incision; (B) Canine fossa antrostomy

elevating the periosteum and soft tissues. The mucoperiosteal flap is raised superiorly up to the level infraorbital nerve, which should not be stretched and damaged.

- *Canine Fossa Antrostomy*: Using 4 mm osteotome and hammer or a drill, a window is made in the antrum through the canine fossa. Killian's nasal gouge (bayonet-shaped) can be used for opening the maxillary antrum. The window is enlarged as necessary using Kerrison's punch or Citelli sphenoid punch.
- *Removal of Disease*: The maxillary sinus pathology and diseased mucosa (cyst, benign tumor, and foreign body) can be removed with elevators, curettes and forceps. Luc's forceps can be used to remove polyps, growth and bone pieces.
- *Intranasal Antrostomy*: A curved hemostat is pushed into the antrum below the inferior turbinate at least 1 cm behind the anterior end of middle turbinate (to avoid damage to nasolacrimal duct). This opening is widened to about 1.5 cm in diameter with Kerrison's and side biting forceps.
- *Endoscopic Examination*: The maxillary sinus can be examined through both the antrostomies with the help of endoscopes.
- *Packing*: Ribbon gauze packing which is impregnated with liquid paraffin or Furacin (0.2% w/w nitrofurazone) ointment is prepared. One end of the packing is brought out from the nasoantral window into the nose and the rest is packed in maxillary antrum. Packing is also done in the nose. Dressing forceps (Tilley's, Wilde's or Hartmann's) can be used for packing. The packing takes care of bleeding.
- *Closure*: Sublabial incision is closed with absorbable suture.

Postoperative Care

- *Local Ice Packs*: They prevent edema, hematoma and discomfort.
- *Pack Removal*: Sinus and nose packing is usually removed in 24–48 hours.
- *Antibiotic*: Postoperative antibiotic is given for 1 week.
- *Instruction to Patient*: Patient is instructed to avoid blowing of nose for 2 weeks because it can cause surgical emphysema.

Complications

- Bleeding is controlled with packing.
- Anesthesia of the cheek may last for few weeks or months and occurs due to stretching or injury to infraorbital nerve. Gentle sublabial retraction prevents this complication.
- Anesthesia of teeth (devitalized teeth).
- Facial asymmetry.
- Recurrent sinusitis and polyposis.
- Dacryocystitis due to injury to nasolacrimal duct.
- Sublabial oroantral fistula.
- Osteomyelitis of maxilla.

SURGERY OF NASAL SEPTUM

If needed the trimming of enlarged turbinates, nasal valves repair and correction of external nose deformity are done together with the correction of deviated nasal septum.

Indications

- Deviated nasal septum (DNS) causing nasal obstruction, recurrent headaches, rhinosinusitis and otitis media.
- Recurrent epistaxis from spur and convex side of DNS.
- Septorhinoplasty for external nasal deformities.
- Hypophysectomy: Trans-septal trans-sphenoidal approach.
- Vidian neurectomy: Trans-septal approach.
- Septal cartilage graft: It is obtained for rhinoplasty and repair of CSF leak.

Contraindications

- *Children*: A conservative surgery (septoplasty) can be considered.
- *Acute URI*: Acute episode of nasal, sinus or respiratory infection.
- *Medical:* Bleeding diathesis and uncontrolled diabetes or hypertension.

Anesthesia

- Local anesthesia with sedation: It is preferred in adults. Topical and local xylocaine with epinephrine provides both analgesia and decongestion. Infiltration of nasal septum with 1% xylocaine and 1:100,000 epinephrine in subperichondrial and subperiosteal planes is done with 27-guage needle. The injection begins at caudal end of septum and then goes posteriorly and includes both sides of septum and floor around maxillary crest.
  - In cardiac patients, oxymetazoline is preferred over epinephrine. Maximum dose of xylocaine is 4–7 mg/kg.

- General anesthesia with endotracheal intubation is used for children and too much anxious adults.
- Position: Patient is placed in reclining position and head end of the table is raised.

Techniques

Septoplasty is replacing submucous resection of nasal septum. ESS provides better visualization and facilitates mucoperichondrial elevation. The CO_2 laser can be used in some selective cases of septal spurs.

Instruments

See the chapter on Instruments.

SUBMUCOUS RESECTION OF NASAL SEPTUM

- *Mucoperichondrial and Periosteal Flap*: It is elevated in the plane beneath the perichondrium and periosteum. Long bladed nasal speculums (Killian's or St Clair Thomson's) keep mucoperiosteal flaps away.
- *Incision of the Cartilage*: Cartilage is incised posterior to the first incision without cutting the opposite side of mucoperichondrium. An elevator is passed through the cartilage incision and mucoperichondrial and periosteal flap is raised from the opposite side of nasal septum. Freer's double ended elevator can be used for the elevation of mucoperichondrium/mucoperiosteum.
- *Removal of Cartilage and Bone*: The long bladed nasal speculum (Killian's or St Clair Thomson's) is introduced and flaps are separated from septal skeleton. The cartilage and bone that lie between the two flaps are removed with the help of Ballenger swivel knife and Luc's forceps. Bony spur or ridge is removed with the help of gouge (4 mm unguarded osteotome) and hammer. Preservation of 1 cm strip of cartilage along the dorsal and caudal border of the septum (L-strut) prevents collapse of the dorsum of nose and retraction of columella (Fig. 7). Killian's nasal gouge (bayonet-shaped) can be used for removal of septal spurs or bony crests and ridges in nasal septum. It may also be used for removing the maxillary crest of the deviated nasal bony septum. Double action bone nibbling forceps can also be used for removing the bony part of the deviated nasal septum.
- *Closing*: After achieving hemostasis, one or two stitches may be applied in mucoperichondrial incision.
- *Packing*: Ribbon gauze, which is smeared with furacin ointment or liquid paraffin, is packed in each side of nose. Dressing forceps (Tilley's, Wilde's or Hartmann's) can be used for nasal packing. It prevents collection of blood between the flaps.

SEPTOPLASTY (FIGS 8 TO FIGS 10)

Septoplasty is a conservative septal surgery and retains maximum possible septal framework. Mucoperichondrial and mucoperiosteal flap is usually raised only on concave side.

- *Incision*: A 2–3 mm curvilinear incision is made above the caudal end of septal cartilage on the concave side. A transfixion or hemitransfixion incision is employed in cases of caudal dislocation.
- *Flaps and Tunnels*: Mucoperichondrial flap is raised only on the concave side and creates a superior tunnel.

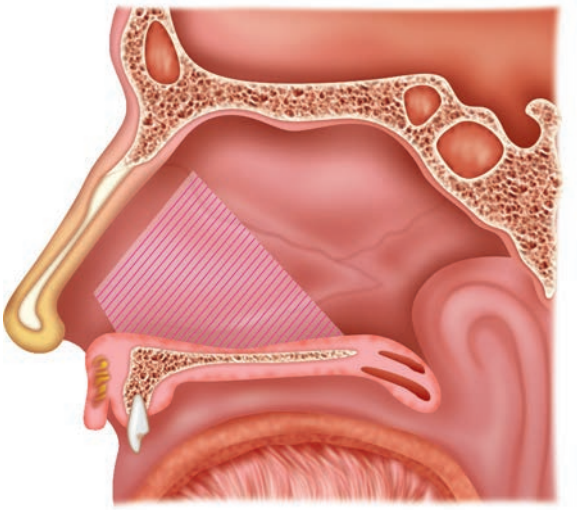
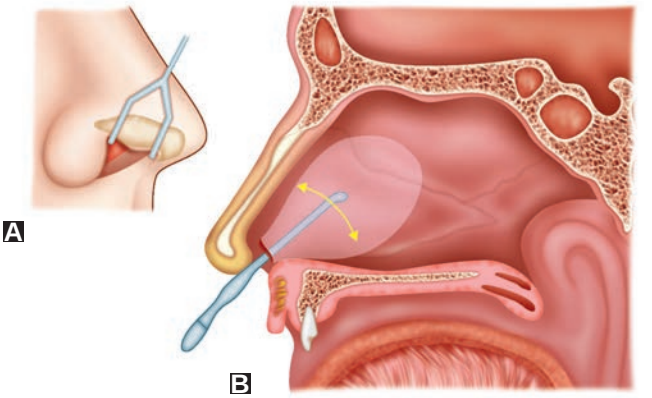


Fig. 7: Submucosal resection (SMR) of large central septal segment preserving L-strut (1 cm dorsal and 1 cm caudal septal segment)



Figs 8A and B: Septoplasty. (A) Incision (Killian and hemitransfixion); (B) Plane of dissection (Cottle elevator creating subperichondrial pocket)

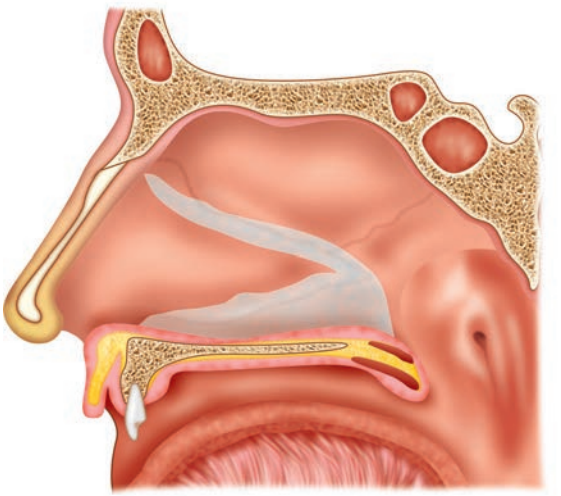
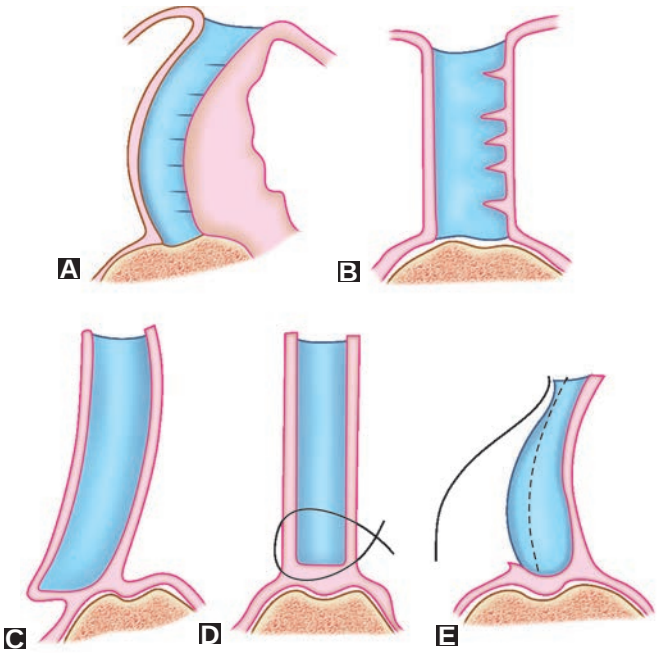


Fig. 9: Septoplasty. Separation of septum from the perpendicular plate of ethmoid and trimming of the inferior cartilaginous portion displaced from the maxillary crest

Mucoperiosteal flap is elevated on both the sides of maxillary crest and creates two inferior tunnels. The superior and inferior tunnels on concave side are joined after cutting the fibrous tissue with sharp knife.



Figs 10A to E: Septoplasty techniques. (A and B) Scoring of the cartilage on concave side removes the interlocked cartilage stresses; (C and D) Excising the dislocated lower end of septal cartilage and replacing on the anterior nasal spine or in the groove of nasal crest of maxilla; (E) Shaving the convexity of septal cartilage

- *Septal Cartilage***:** It is separated from the vomer and ethmoid plate.
- *Maxillary Crest*: If needed, it is fractured to realign the septal cartilage.
- *Bony Septum*: It is corrected with removal of the deformed parts.
- *Deformed Septal Cartilage*: It can be corrected by scoring, cross hatching, morcelizing, shaving or wedge excision.
- *Other Options*:
  - *Septorhinoplasty*: Some cases need separation of septal cartilage from upper lateral cartilages, implantation of cartilage either in the columella or dorsum of nose.
  - *Endoscopic sinus surgery*: Sinuscope provides better visualization and facilitates mucoperichondrial elevation.
  - *Laser surgery*: The CO_2 laser can be used in some selective cases of septal spurs, where 2–3 mm height mucoperichondrium and cartilaginous spur are completely excised. The opposite side mucoperichondrium is left intact.
- *Closure:* Trans-septal sutures keep the mucoperichondrial flaps together and prevent hematoma. Nose is packed.

POSTOPERATIVE CARE

- Patient is kept in semi-sitting position that prevents oozing of blood and swelling.
- A soft diet in the first two days minimizes active mastication and prevent bleeding.
- Analgesics control pain.
- Antibiotic for 5–7 days.
- Nasal packs: They are removed usually after 24 hours. Subsequently regular regimen of saline water flushing, gentle suctioning and instillation of antibiotic ointment is started.
- Decongestant nasal drops and steam inhalations for 5 days.
- Sutures are removed on 5th or 6th day.
- Patient is instructed to avoid strenuous exercise, trauma to the nose, blowing and frequent picking of nose for about 3 weeks.

COMPLICATIONS

In addition to the complications of anesthesia following complications can occur after septal surgery:

- *Hemorrhage*: Repacking of nose is required in severe reactionary bleeding. The other two types of hemorrhage include primary (during surgery) and secondary (5–7 days after surgery due to infection).
- *Septal Hematoma*: It needs immediate evacuation followed by intranasal packing on both sides of septum with equal pressure.
- *Septal Abscess*: The septal abscess usually occurs due to infection of septal hematoma. It needs immediate incision and drainage.
- *Perforation*: It occurs when both sides of septal mucosa are perforated at the same level.
- *Saddle Nose Deformity and Tip Ptosis*: Depression of dorsum of nose in supratip area and tip ptosis occurs when too much of septal cartilage is removed along the dorsal border.
- *Columella Retraction*: It can occur when caudal strip of nasal cartilage is removed.
- *Failure*: Persistence of deviation is usually the result of inadequate surgery, which needs revision operation.
- *Flapping of Nasal Septum*: In this condition two mucoperichondrial flaps move with respiration to the right or left. It happens when too much septal framework is removed.
- *CSF Leak*: Never manipulate perpendicular plate of ethmoid before incising it as it can damage cribriform plate and cause CSF leak.
- *Toxic Shock Syndrome*: This staphylococcal (sometimes streptococcal) infection is characterized by nausea, vomiting, purulent nasal secretions, hypotension and rash.
  - *Treatment*: It consists of removal of packing (which may be the cause), proper hydration of patient, maintenance of blood pressure and administration of proper antibiotics.
- *Synechia*: Injuries of mucosal fold and turbinates at the same level can lead to formation of adhesions. Asymptomatic synechia do not need any treatment. They need excision if they cause nasal obstruction. The two raw surfaces are kept apart for 2 weeks with polyethylene or silastic sheet.

Clinical Highlights

1. **External ethmoidectomy:** The anterior ethmoidal artery is located at frontoethmoidal suture line. It lies 24 mm posterior to the anterior lacrimal crest.
2. **Maxillary sinusitis:** Most common predisposing factor linked to maxillary sinusitis is mucosal swelling in ethmoid infundibulum.
3. **Caldwell Luc operation:** The entry into the maxillary sinus is made through the canine fossa (canine fossa antrostomy).
4. **SMR:** The complete removal of septal cartilage results in supratip depression of cartilaginous nasal dorsum.

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57 Adenotonsillectomy

Each one thinks his method is best. Very good! But remember, it may be good for you. One food which is very indigestible to one is very digestible to another. Because it is good for you, do not jump to the conclusion that your method is everybody's method.

—Swami Vivekananda

Points of Focus

- **PREOPERATIVE ASSESSMENT**
- **INDICATIONS OF TONSILLECTOMY**
  - ✓ Part of Other Operations
- **INDICATIONS OF ADENOIDECTOMY**
- **CONTRAINDICATIONS**
- **PREOPERATIVE MEASURES**
- **STEPS OF ADENOIDECTOMY**
- **STEPS OF TONSILLECTOMY**
- **POSTOPERATIVE CARE**
- **COMPLICATIONS**
  - ✓ Immediate
  - ✓ Delayed
  - ✓ Rare
- **CLINICAL HIGHLIGHTS**

PREOPERATIVE ASSESSMENT

1. Family history of coagulation disorders.
2. History of easy bruising.
3. Coagulation screening: Prothrombin time, partial thromboplastin time, bleeding time and platelet count.
4. If warranted, evaluation of coagulation studies.
5. In cases of adenotonsillar hypertrophy associated with severe airway obstruction (severe snoring, obstructive sleep apnea, (OSA) cor pulmonale)
   a. Polysomnography or sleep sonography
   b. X-ray chest
   c. ECG
   d. Cardiology consultation
6. Further testing for any present systemic medical conditions such as history of bronchospasm.
7. Screening for sickle cell disease.
8. Beta-human chorionic gonadotropin to rule out pregnancy.
9. Angiography to rule out medially placed carotid artery in velocardiofacial syndrome.

INDICATIONS FOR TONSILLECTOMY

Recurrent infections and upper airway obstruction are the most common indications for Tonsillectomy.

Absolute

- Chronic or recurrent tonsillitis
  - Seven episodes in 1 year, or
  - Five episodes per year for 2 years, or
  - Three episodes per year for 3 years, or
  - Two weeks or more of lost school or work in a year.
- Peritonsillar abscess: In children after 4–6 weeks; in adults after second attack.
- Tonsillitis causing febrile convulsions.
- Cardiac valvular disease associated with recurrent streptococcal tonsillitis.
- Hypertrophy of tonsils causing
  - Excessive snoring or sleep disturbances
  - Obstructive sleep apnea
  - Cor pulmonale

- Dysphagia
- Interfere with speech.

- Suspicion of malignancy–Asymmetric tonsillar hypertrophy.

A unilateral tonsil enlargement in children can be due to lymphoma while in adults epidermoid carcinoma may be the cause. An excisional biopsy is indicated in these conditions.

Relative

- Diphtheria carriers, who do not respond to antibiotics.
- Streptococcal carriers, who may be the source of infection to others.
- Chronic tonsillitis with bad taste or halitosis.
- Recurrent sore throats or upper respiratory infections. (URI)
- Recurrent streptococcal tonsillitis in a patient with valvular heart disease.
- Tender cervical adenitis.
- Difficulty in eating.
- Tonsillolithiasis.
- Orofacial and dental abnormalities.
- Failure to thrive.
- Enuresis.
- Obstructive tonsils in infectious mononucleosis not responding to medical therapy.

Part of Other Operations

- Palatopharyngoplasty for OSA syndrome.
- Glossopharyngeal neurectomy. CN IX is severed in the bed of tonsil.
- Removal of styloid process.

INDICATIONS FOR ADENOIDECTOMY

Adenoidectomy may be done alone or in combination with tonsillectomy. Adenoids are usually removed first and the nasopharynx is packed. Adenoidectomy prevents recurrence of dental abnormalities after orthodontic treatment.

- Adenoid hypertrophy causing
  - Excessive snoring
  - Mouth breathing
  - OSA syndrome or sleep disturbances
  - Cor pulmonale
  - Failure to thrive
  - Enuresis
  - Dysphagia
  - Speech abnormalities such as rhinolalia clausa
  - Craniofacial growth abnormalities.
- Purulent adenoiditis
- Recurrent rhinosinusitis
- Middle ear infections
  - Chronic secretory otitis media
  - Chronic recurrent acute otitis media
  - Benign CSOM with recurrent ear discharge.
- Dental malocclusion

The common indications of adenoidectomy include nasal obstruction due to adenoidal hyperplasia, recurrent otitis media and otitis media with effusion in children.

CONTRAINDICATIONS

- Anemia: Hemoglobin less than 10 g%.
- Acute upper respiratory tract infection or acute tonsillitis.
- Overt or submucous cleft palate: Conservative adenoidectomy. Leaving the lower portion of adenoidal pad intact prevents velopharyngeal insufficiency.
- Bleeding disorders such as leukemia, purpura, aplastic anemia, hemophilia.
- Epidemic of polio.
- Uncontrolled systemic disease such as diabetes, cardiac disease, hypertension or asthma.
- During the period of menses in females.

Children under 3 years are at more surgical risks.

SURGICAL TECHNIQUES

- *Guillotine method*: It may be done in mobile tonsil without any fibrosis. Fibrosis of the tonsillar bed occurs due to repeated infections. This technique has been abandoned. Currently, it is not used but gave good results in the expert hands.
- *Dissection and snare method* (with sharp instrumentation): More operative bleeding but less postoperative pain.
- *Electrocautery* (Bovie and bipolar): Less operative bleeding but more postoperative pain
- *Laser*: CO_2 or KTP laser surgery.
- *Cryosurgery*: Tonsil is frozen with two applications of cryoprobe for 3–4 minutes each time and then allowed to thaw. Tonsils undergo necrosis and then fall off and leave a granulating surface. Bleeding is less because of the thrombosis of vessels.

Newer Technologies

The newer methods are being employed to perform tonsillectomy and include:

- *Harmonic scalpel*: Ultrasonic technology cuts and coagulates tissues. Tonsillectomies have been done but cannot be used for adenoidectomy. Each blade used for operation increases the cost.
- *Coblation*: This technology utilizes the radio- frequency bipolar electrical current. It claims effective dissection with less postoperative pain from thermal injury. It has been used for completion of tonsillectomy, adenoidectomy (for small adenoid pads and not for large obstructive adenoids), and intracapsular tonsillectomy (tonsillotomy in which tonsil is debulked). Tonsillotomy leaves behind small amount of tonsil tissue covering the constrictor muscle.
- *Powered instrumentation*: Microdebrider shaver allows precise, rapid and safe removal of tissue. It can be used for adenoidectomy and tonsillotomy (intracapsular tonsillectomy).

PREOPERATIVE MEASURES

- No solid food by mouth for 8 hours; clear liquids may be allowed for 3 hours before surgery.
- Sedation with midazolam hydrochloride (0.5–1 mg/kg) 30 minutes before surgery.
- Intravenous antibiotic such as ampicillin 20 mg/kg up to 1 g.
- Intraoperative dexamethasone (0.5 mg/kg) especially in OSA and children younger than 3 years.

ANESTHESIA

The operation is usually done under general anesthesia with endotracheal intubation. In cooperative adults, it may be done under local anesthesia.

POSITION

Rose's position (Fig. 1): Supine position with extended head (Place a pillow under the shoulders). A rubber ring under the head stabilizes head and prevents its hyperextension.

SURGICAL INSTRUMENTS

See the chapter *Instruments*.

OPERATIVE STEPS

When both the tonsils and adenoids are removed in one sitting, adenoids are removed first. Depending upon the indication, either tonsils or adenoids may be removed alone.

Adenoidectomy

- *Opening of mouth*: Boyle-Davis mouth gag is used for opening the mouth and retracting the tongue anteriorly and inferiorly (Fig. 2). Instruments are discussed in chapter Instrument. The built in tongue depressor along with the closed mouth gag is inserted in the mouth after depressing the lower jaw. The mouth gag is opened gradually. It is suspended from Draffin's bipods and extends the neck and head. The two pods are assembled together as per the height at which the tongue blade of the Boyle-Davis mouth gag is suspended. The lower ends of the pods are placed in one of the several depressions of the Magauran's plate.
- *Laryngopharyngeal packing*: Put a throat pack, which prevents blood and secretions entering esophagus and aspiration of laryngeal clot and leakage of air, oxygen and anesthetic agent.
- *Examination*: Nasopharynx and adenoids are examined after retracting the soft palate with curved end of the tongue depressor. Tonsil dissector and anterior pillar retractor can also be used in retracting the soft palate and uvula.
  - Digital palpation: It helps in the following:
    - Assesses adenoids size
    - Pushes the lateral adenoid masses towards the midline
    - Assesses evidence of submucous cleft palate and abnormal vessels.
- *Adenoid curette*: St. Clair Thomson's curette with guard shaves off the adenoid mass and holds it and prevents from slipping. Introduce a proper size of adenoid curette with guard into nasopharynx and feel the posterior border of nasal septum. Press the adenoid curette backward and engage the adenoids. The LaForce adenotome may also be used instead of adenoid curette. Shave the adenoids with gentle sweeping movement of adenoid curette.
  - Caution: Flexion of head at this stage will avoid injury to the odontoid process.
    - Remove the lateral adenoid masses with smaller curettes.
    - Remove the remaining small tags of adenoids with punch forceps, smaller plain adenoid curette, Luc forceps or conchotome.

Fig. 1: Rose position. The patient lies supine. The head is extended by putting a sandbag beneath the shoulders

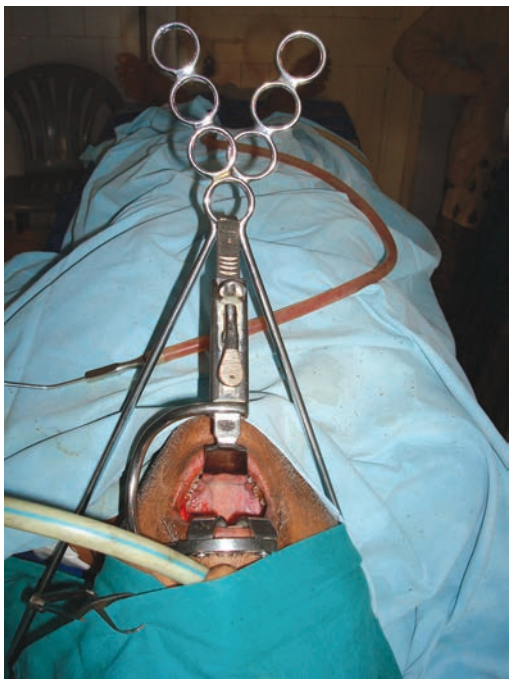


Fig. 2: Boyle-Davis gag in position

- *Hemostasis*: Achieve the hemostasis with packing. Persistent bleeders may be electrocoagulated under direct vision. Very occasionally, postnasal pack is required for refractory cases of bleeding.

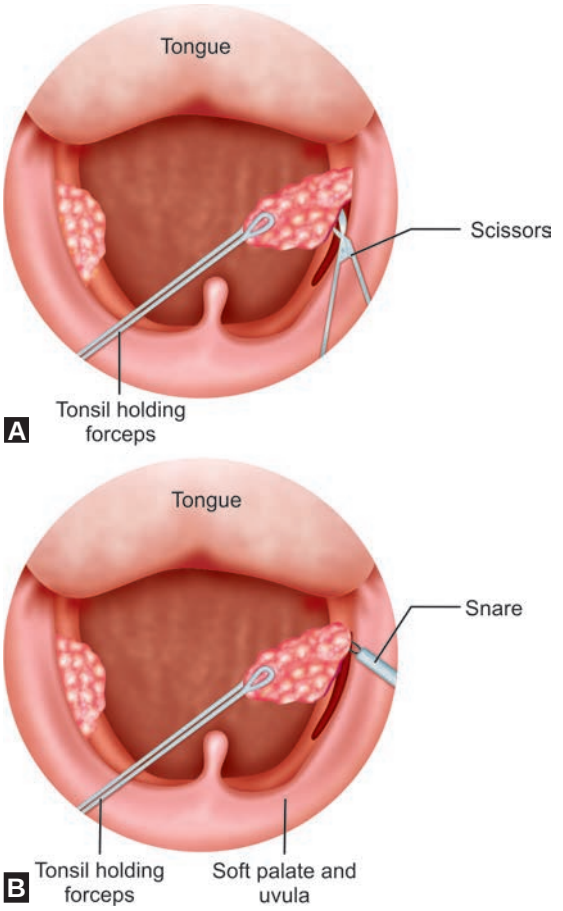
Adenoidectomy: It should be modified or avoided in cases of submucous cleft palate.

Tonsillectomy (Dissection and Snare Method)

- *Opening of mouth*: Boyle-Davis mouth gag is introduced and held in position with Draffin's bipods. See Adenoidectomy section.
- *Laryngopharyngeal packing*: Put a throat pack, which prevents blood and secretions entering esophagus and aspiration of laryngeal clot and leakage of air, oxygen and anesthetic agent.
- *Tonsil holding*: Tonsil is grasped and pulled medially with tonsil holding forceps (Allis clamp, tenaculum, or Denis

Browne) (Figs 3A and B). Luc forceps may also be used for this purpose.

- *Incision*: The mucous membrane where it reflects from the tonsil to anterior pillar is incised either with sharp instrument or electrocautery. Tonsil dissection forceps with teeth (Waugh's) can also be used for putting incision in the mucous membrane and dissection of tonsil. The incision is extended along the upper pole between the tonsil and posterior pillar. Electrocautery must not touch metal instruments such as mouth gag and Yankauer suction. Yankauer's suction tube is used for suction.
- *Dissection*: With the help of a blunt curved scissor or tonsil dissector (Thompson dissector or a Fischer knife) separate the tonsil capsule from the bed of tonsil. Begin the dissection from the upper pole. Tonsil dissector and anterior pillar retractor dissect the tonsil and retract the anterior pillar to inspect the fossa for any bleeding. Retraction of the upper pole medially and towards tongue facilitates the dissection. It is continued until lower pole is reached. Tonsil scissors can also be used for sharp dissection of the tonsils and cutting the ligatures.
  - *Caution*: Preserve the mucosa and muscle of the posterior pillar to prevent postoperative nasopharyngeal stenosis.
- *Snare*: Eve's tonsillar snare is threaded over the lower pole tonsil pedicle, which is amputated with snare and tonsil removed. When firmly closed, the snare crushes and cuts the pedicle and minimizes the bleeding.



Figs 3A and B: Tonsillectomy. (A) A blunt curved scissor is dissecting tonsil from the peritonsillar tissue while tonsil is grasped with tonsil holding forceps and pulled medially; (B) Wire loop of tonsillar snare threaded over the tonsil onto its pedicle and cut

- *Packing*: A gauze sponge placed in tonsillar fossa for few minutes obtains pressure hemostasis.
- *Other side*: Procedure is repeated on the other side.
- *Removal of packs*: Remove the packs from tonsillar fossa and nasopharynx.
- *Hemostasis*: Bleeding is usually controlled using electrocautery on a low wattage. Large bleeders may be clamped and ligated with the help of tonsillar artery forceps. Straight and curved tonsil artery forceps (such as Negus artery forceps) are used to catch the bleeding point and ligating the bleeder. Negus Knot tyer helps in tying the ligature knot up to the tip of curved artery forceps that holds the vessel. Tonsil needle is used for sewing the tonsillar pillars together for controlling the bleeding, when it is not controlled by ligation and cauterization of bleeding points. In these cases if gauze is kept in the tonsillar fossa that must be removed within 48 hours.
  - *Caution*: Suture ligatures with needle may inadvertently ligate external maxillary and lingual arteries or damage internal carotid artery.
- *Irrigation and cleaning*: Irrigate nasopharynx, oral and nasal cavity thoroughly and evacuate secretions and blood clots from laryngopharynx. Stomach may also be suctioned.

POSTOPERATIVE CARE

- *Immediate general care:* This includes:
  - Position: Patient is kept in tonsillar position till full recovery from anesthesia
  - Monitor vital signs such as pulse, respiration and blood pressure
  - Watch for bleeding from the nose and mouth, vomiting, swelling (surgical emphysema), pain, and respiratory distress
  - OSA and cor pulmonale: In these cases pulse oximetry and planned mechanical ventilation is important for persistent hypercapnia until Pco_2 levels return to normal.
- *Diet*: Plenty of cold fluids such as cold milk or ice cream. Sucking of ice cubes helps in relieving pain. The child may take soft diet on the second day such as custard, jelly, boiled eggs, porridge or slice of bread soaked in milk. Solid foods subsequently are allowed as tolerated.
- *Oral hygiene*: Saline gargles 3–4 times a day. Mouth should be washed with plain water after every feed.
- *Analgesics*: Paracetamol with or without codeine.
- *Antibiotics*: For a week.
- *Instruction*: Report immediately if there is any bright red colored bleeding from nose or mouth.

Inpatient Versus Outpatient Procedures

Most patients can be discharged after an observation period of 4–6 hours. Patients are instructed to report immediately if there is any bright red bleeding from nose or mouth. Following patients need overnight observation:

- History of OSA
- Younger than 3 years
- Presence of other medical problems
- History of abnormal bleeding disorder or abnormal coagulation profile
- Craniofacial abnormalities such as Down's syndrome
- Tonsillectomy for peritonsillar abscess.

COMPLICATIONS

The complications are mainly related to pain, bleeding, respiratory obstruction, and pulmonary edema and may even result in death. Complications are divided here into three groups: immediate, delayed and rare.

Immediate

- *Primary hemorrhage*: Bleeding during the operation is usually controlled by pressure, ligation or electrocoagulation. Certain analgesic such as ketorolac should be avoided. Application of tannic acid, bismuth subgallate or hemostatic agents may be helpful. Coagulopathy must be ruled out. Residual remnants of adenoid tissue should be completely removed.
- *Reactionary hemorrhage*: Bleeding after the recovery from anesthesia on the day of surgery is usually controlled by removing the clot, applying pressure or vasoconstrictor. Clot may prevent the clipping action of the superior constrictor muscle on the vessels. Immediate postoperative bleeding from nose and mouth or vomiting of dark colored blood and rising pulse rate indicate bleeding from the operative site. Topical decongestant nasal drops may help in controlling nasopharyngeal bleeding. In cases of refractory bleeding, patient is taken back to operation room and ligation or electrocoagulation of the bleeding vessels is done under general anesthesia. Postnasal pack may be needed in children with adenoidectomy.
- *Injury*: Oral cavity and oropharyngeal structures such as tonsillar pillars, uvula, soft palate, tongue, superior constrictor muscle, or teeth can be injured during tonsillectomy. Eustachian tube, pharyngeal musculature and vertebrae injuries during adenoidectomy can be prevented by avoiding hyperextension of neck and undue pressure of curette.
- *Aspiration and foreign bodies*: Such as blood, tissue of tonsil or adenoids or tooth.
- *Pulmonary edema*: It can occur in cases of OSA and cor pulmonale.
- *Edema of tongue, nasopharynx and palate*: Need replacement of nasal trumpet and intravenous steroids.
- *Edema*: Face or eyelids.
- *Surgical emphysema*: Due to superior constrictor muscle injury.

Delayed

- *Secondary hemorrhage*: Bleeding seen between 5th–10th postoperative days is the result of sepsis and premature separation of the membrane.
  - Clinical features: The common presentation is blood-stained sputum but bleeding may be profuse.
  - Management: If bleeding is not controlled after removal of clot and topical application of dilute adrenaline, hydrogen peroxide and with pressure, then patient is taken to operation room. Under general anesthesia, bleeding vessel is electrocoagulated or ligated. Approximation of pillars with mattress sutures or external carotid ligation may be required in rare cases. Transfusion of blood or plasma may be needed. Systemic antibiotics control the infection.
- *Infection*: Infection may cause parapharyngeal abscess or otitis media.
- *Pulmonary complications*: Aspiration of blood, mucus or tissue fragments may lead to atelectasis or lung abscess.
- *Scarring*: Soft palate and pillars.
- *Tonsillar remnants*: The remaining tonsil tags or tissue may cause repeated infection.
- *Hypertrophy of lingual tonsil*: It is compensatory to loss of palatine tonsils.
- *Velopharyngeal insufficiency*: Hypernasality is a complication of adenoidectomy. It is usually managed by speech pathologist. But refractory cases may need reconstructive surgery (pharyngeal flap, sphincteroplasty or posterior pharyngeal wall augmentation).
- *Nasopharyngeal stenosis*: It occurs due to scarring after excessive damage to nasopharyngeal mucosa (roof, posterior and lateral walls) and resection of posterior tonsillar pillar. These children are difficult to manage.
- *Recurrence*: Remaining adenoids and tonsil tissue may grow again. If plica triangularis near the lower pole of tonsil is not removed along with tonsil, it may get hypertrophied.
- *Atlantoaxial subluxation (Grisel's syndrome)*: It leads to stiff neck and spasm of sternocleidomastoid and deep cervical muscles. Treatment includes intravenous antibiotics and cervical traction.

Rare Complications

- Pulmonary abscess and mediastinal emphysema.
- Horner's syndrome, optic neuritis, paralysis of glossopharyngeal and recurrent laryngeal nerve.
- Meningitis and brain abscess.
- Submandibular salivary fistula in tonsillar fossa.
- Chronic postoperative pain and Eagle's syndrome (calcification of stylohyoid ligament).
- Fires and endotracheal tube explosions.
- Retropharyngeal abscess after adenoidectomy.
- Sloughing and severe hemorrhage after accidental suture ligature of lingual and maxillary arteries and pseudoaneurysm and excessive bleeding due to internal carotid artery.
- Injury to aberrant vessels such as carotid artery arising in fossa of Rosenmuller.

Clinical Highlights

1. **Contraindication of tonsillectomy:** Children with acute upper respiratory infections [URI (cough, cold, and fever)] should wait for 3 weeks and get treatment for URI.
2. **Post-tonsillectomy earache:** This referred otalgia occurs through the glossopharyngeal nerve.
3. **Indications of adenoidectomy:** The common indications of adenoidectomy include nasal obstruction due to adenoidal hyperplasia, recurrent otitis media and otitis media with effusion in children.
4. **Atlantoaxial dislocation:** This complication can occur in children due to adenotonsillectomy, nasopharyngeal infection, and esophagoscopy.

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58 Endoscopies

You fail only when you do not strive sufficiently to manifest infinite power. As soon as a man loses faith, death comes. He who is overcautious about himself falls into dangers at every step; he who is afraid of losing honor respect, gets only disgrace; he who is always afraid of loss always loses.

—Swami Vivekananda

Points of Focus

DIRECT LARYNGOSCOPY (DLS)/ MICROLARYNGOSCOPY (MLS)

- **INDICATIONS, CONTRAINDICATIONS**
- **ANESTHESIA, POSITION**
- **PROCEDURES**
- **POSTOPERATIVE CARE**
- **COMPLICATIONS**
- **FLEXIBLE NASOPHARYNGOLARYNGOSCOPY**

BRONCHOSCOPY

- **INDICATIONS: DIAGNOSTIC, THERAPEUTIC**
- **RIGID BRONCHOSCOPY**
  - ✓ Advantages
  - ✓ Anesthesia, Oxygenation and Ventilation, Position
  - ✓ Techniques, Precautions
  - ✓ Postoperative Care
  - ✓ Complications
- **FLEXIBLE FIBEROPTIC BRONCHOSCOPY (FOB)**
  - ✓ Types of FOB
  - ✓ Advantages

ESOPHAGOSCOPY

- **INDICATIONS, CONTRAINDICATIONS**
- **RIGID ESOPHAGOSCOPY**
  - ✓ Advantages, Disadvantages
  - ✓ Anesthesia, Position
  - ✓ Techniques
  - ✓ Postoperative Care
  - ✓ Complications
- **FLEXIBLE ESOPHAGOSCOPY**
  - ✓ Advantages, Disadvantages
  - ✓ Common Indications
  - ✓ Endoscope and Instruments
  - ✓ Techniques
- **CLINICAL HIGHLIGHTS**

DIRECT LARYNGOSCOPY/MICROLARYNGOSCOPY

Laryngoscope used for direct laryngoscopy (DLS) may have either single or twin light carrier which is connected to a cold light source through a flexible cable. There are several models of laryngoscopes, which are described in chapter of instruments. The shape and size of viewing and distal ends and body vary with the types of laryngoscopes. The size of laryngoscope is selected as per the age of the patient. The larynx, hypopharynx and oropharynx are visualized directly with the help of laryngoscope.

- Laryngoscope of Microlaryngoscopy (MLS) is self-retaining laryngoscope. It can be fixed on the chest by a chest piece so that hands of the surgeon remain free for the surgery. MLS is performed under the magnification of an operating microscope (Fig. 1). Microlaryngeal surgery needs special laryngoscopes, forceps, scissors, dissectors and knives

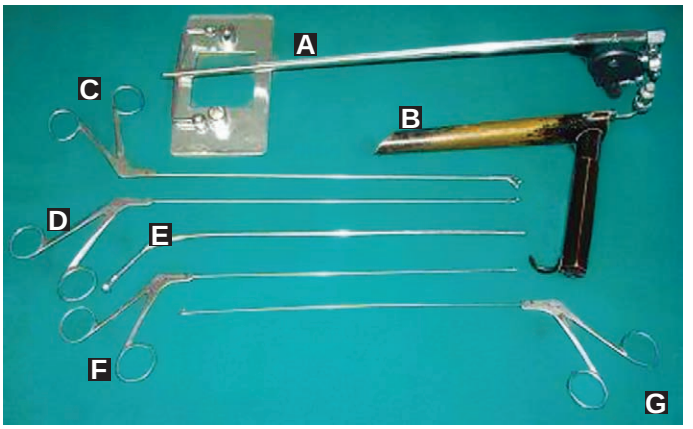
(Figs 2A to G). Other types of laryngoscopy have been described in chapter Laryngeal Symptoms and Examination of section Larynx, Trachea and Bronchus.

INDICATIONS

Direct laryngoscopy/microlaryngoscopy can be performed for both diagnostic as well as therapeutic purposes.

- *Diagnostic*
  - *Infants and young children*
  - *Anatomy*: Strong gag reflex or overhanging epiglottis
  - *Symptoms*: Hoarseness, dyspnea, stridor and dysphagia
  - *Examination of hidden areas*: Following areas cannot be adequately seen during mirror laryngoscopy:
  - Oropharynx: Base of tongue, valleculae
    - Hypopharynx: Lower part of pyriform fossa
    - Larynx: Infrahyoid epiglottis, anterior commissure, ventricles and subglottic region.
  - *Biopsy*: Tumors of base of tongue, vallecula, laryngopharynx and larynx.
- Therapeutic
  - *Benign swellings*: Removal of papilloma, fibroma, vocal nodule, polyp or cyst
  - *Malignant lesions*: Early carcinoma of larynx and laryngopharynx
  - *Foreign bodies*: Larynx and laryngopharynx
  - *Strictures*: Dilatation of laryngeal strictures.

Fig. 1: Microlaryngoscopy procedure



Figs 2A to G: Microlaryngoscopy instruments. (A) Laryngoscope holder and chest support; (B) Anterior commissure operating laryngoscope; (C) Laryngeal forceps, upward-angled large size cupped jaws; (D) Laryngeal forceps, side-angled medium size cupped jaws; (E) Suction tube; (F) Laryngeal forceps, upward-angled fine cupped jaws; and (G) Laryngeal forceps, up-cutting angled scissor

CONTRAINDICATIONS

- Lesions of cervical spines
- Stridor (usually need prior tracheostomy)
- Recent coronary occlusion
- Cardiac decompensation.

ANESTHESIA

- Usually done under general anesthesia
- Infants and young children do not need any anesthesia for diagnostic DLS.

POSITION

Patient lies in supine position. Head is elevated 10–15 cm by placing a pillow under the occipital region or by raising head flap of operation table. Neck is flexed on thorax and head is extended on atlanto-occipital joint (barking dog position).

PROCEDURES

- *Protection of teeth and lips*: Examine the patient for neck stability, loose teeth and dentures. Eyes are protected with a shield and the patient is draped. A gauze piece protects the upper teeth against trauma.
- *Lubrication*: Lubricate laryngoscope with liquid paraffin or xylocaine jelly.
- *Holding of scope*: Left hand holds the handle of laryngoscope. Right hand retracts the lips and guides the introduction of laryngoscope and handle suction and forceps.
- *Introduction of scope*: Laryngoscope is introduced usually right side of the tongue and is then moved to the midline. It brings the epiglottis in view.
- *Lifting of epiglottis*: The lifting of epiglottis forward (without levering laryngoscope on the upper teeth or jaw) provides view of the interior larynx.
- *Interior of larynx*: The tip of anterior commissure laryngoscope is advanced further between the vestibular folds (to examine the ventricles and anterior commissure) and the vocal cords (to examine the subglottic region).
- *Structures examined*: The structures, which are examined serially, include tongue base, valleculae, epiglottis, pyriform sinuses, aryepiglottic folds, arytenoids, postcricoid region, false cords, anterior and posterior commissure, ventricles, vocal cords, subglottic region and mobility of vocal cords and arytenoids.
- *Telescope*: Angled telescopes facilitate examination of the undersurface of vocal cords and subglottic region.

Practical tips for DLS/MLS

- Protects the lips and teeth. Lips are easily pinched between the laryngoscope and teeth or gingiva. Upper teeth are easily broken by levering the laryngoscope on them.
- Position the head. Extend the neck. Head rests on the occiput (sniffing position). This position provides the best access to anterior areas.
- Exclude contraindications for extending the neck: Fusion, instability, Down syndrome.

Do not biopsy both sides of the vocal folds (such as nodules) close to the anterior commissure to prevent formation of web

POSTOPERATIVE CARE

- *Position*: Patient is kept in coma position, which prevents aspiration of blood and secretions.
- *Observation*: Watch for any spitting of blood, and respiratory distress and cyanosis. They can be due to laryngeal spasm, laryngeal edema, or aspiration of blood.
- *Patient's follow up instructions*: See Box 1.

COMPLICATIONS

- Injury to lips, tongue and teeth
- Bleeding
- Laryngeal spasm and edema.

FLEXIBLE NASOPHARYNGOLARYNGOSCOPY

See chapter Laryngeal Symptoms and Examination in the section Larynx, Trachea and Bronchus.

- Flexible laryngoscopy: If you have difficulty in seeing into nasopharynx and oropharynx request the patient to breath through nose that will clear the soft tissue obstruction.
- Valsalva maneuver: By requesting the patient to perform this maneuver we can better visualize the pyriform sinuses because that will stent open them.

BRONCHOSCOPY

Bronchoscopy is of two types: rigid bronchoscopy and flexible fiberoptic bronchoscopy.

Box 1: Postoperative instructions to professional voice users after microlaryngeal surgery

- First 1–2 weeks: Complete voice rest
- Next 1–2 weeks: Graduated voice use. Beginning from 5 minutes on 1st day, phonation time is doubled each day until the full conversational speech
- Next 2–3 months: Relearn how to sing from the basic level and avoid maladaptive behaviors
- After 3 months: Return to professional activity after the permission from your doctor
- Observe general laryngeal hygiene and have proper hydration
- Take guaifenesin and perioperative antireflux medications
- Take professional help from speech language pathologists and vocal pedagogue
- Practice easy onset phonation
- Avoid vocal abusive behavior
- Use appropriate pitch and vocal intensity.

Laryngeal Hygiene

- Drink increased amounts of water (at least eight glasses, i.e. 64 ounces per day) so that urine becomes lightly colored. Professional voice users should further increase the amount of water if they are traveling by air, are ill, or have demanding or excessive performance schedule
- Avoid use of caffeine because of its diuretic nature
- Avoid dairy products as they increase the viscosity of secretions and hinder the healthy smooth vibratory function of vocal cords
- Avoid tobacco as it contributes to poor laryngeal hygiene
- Guaifenesin may thin the secretions.

INDICATIONS FOR BRONCHOSCOPY

- *Diagnostic*
  - *Symptoms*
    - Dyspnea, which may be associated with wheezing or stridor
    - Hemoptysis
    - Unexplained chronic cough
    - Hoarseness of voice
    - Fever or chest pain suggestive of pulmonary infections
  - *X-ray chest findings*
    - Atelectasis: Segment, lobe or lung
    - Opacity: Segment or lobe
    - Obstructive emphysema
    - Hilar or mediastinal shadows
    - Pleural effusions.
  - *Vocal cord paralysis*
  - *Collection of bronchial secretions*: Culture and sensitivity, acid-fast bacilli, fungus, or malignant cells.
- Therapeutic
  - Foreign body removal.
  - Suction clearance of secretions, blood clots or inspissated mucus plugs: Head injuries, chest trauma, thoracic or abdominal surgery, or coma.
  - Assessment or placement of endotracheal tubes or double lumen tubes.
  - Guided percutaneous tracheostomy.
  - Thermal ablation and removal of tumors.
  - Debridement of benign stenosis.
  - Balloon dilation.
  - Placement of airway stent.

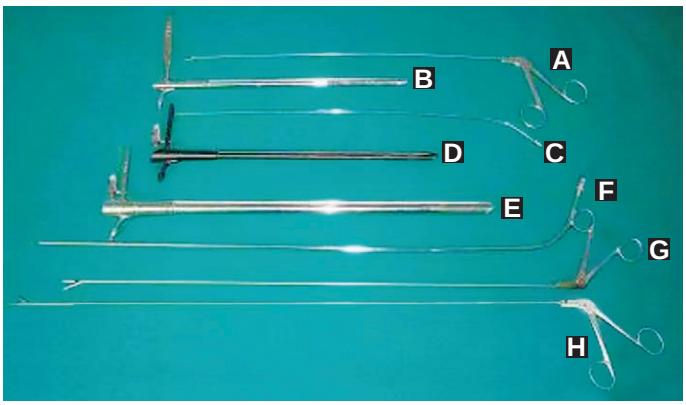
RIGID BRONCHOSCOPY

The instruments for rigid bronchoscopy are shown in Figs 3A to H. For the details of these instruments, see chapter Instruments. Size of bronchoscope will vary with the age of the patient.

Advantages

Large instruments can be passed through the larger lumen of rigid bronchoscope. It is advantageous in the following conditions:

- Removal of foreign bodies.



Figs 3A to H: Bronchoscopy (rigid) instruments. (A) Peanut foreign body grasping forceps, both jaws movable; (B, D and E) Different size bronchoscopes Jackson pattern; (C and F) Different size suction tubes; (G) Cup-shaped oval straight biopsy forceps; and (H) Peanut foreign body grasping forceps

- Suctioning in profuse hemoptysis.
- Placements of noncompressible silastic airway stent.
- Coring out tumors.
- Tamponading to a bleeding source.

Anesthesia

Usually done under general anesthesia but can be done with topical surface anesthesia and conscious sedation.

Oxygenation and Ventilation

- Intermittent apneic to spontaneous/assisted.
- Tidal volume through closed system or open system of side port Venturi jet ventilation.

Position

Patient lies in supine position. Neck is flexed on thorax and head is extended on atlanto-occipital joint.

Techniques

- *Lubrication of scope*: Lubricate proper size bronchoscope with liquid paraffin or xylocaine jelly.
- *Protection of teeth and lips*: Examine the patient for neck stability, loose teeth and dentures. Eyes are protected with a shield and the patient is draped. A gauze piece or teeth guard protects the upper teeth against injury. Left hand thumb retracts the upper lip and teeth while index finger lifts lower teeth.
- *Holding of scope*: The shaft of bronchoscope is held in right hand like a pen. Left hand thumb retracts the upper lip and teeth while index finger lifts lower teeth and guides the introduction of bronchoscope with bevel up.
- *Introduction of scope*: Bronchoscope is directed perpendicularly until the uvula is passed. It is introduced usually on the right side of the tongue and is then moved to the midline. It brings the epiglottis in view.
- *Larynx*: Tip of epiglottis is identified while lifting the base of the tongue. Glottis is exposed when epiglottis is lifted forward. Bronchoscope is introduced either directly or after exposing the glottis with the help of a spatular type laryngoscope especially in infants, young children and short neck or thick tongue patients.
  - Rotate the bronchoscope 90° clockwise to bring its beveled tip in the axis of glottis and enter into the trachea. Rotate back the scope into its original position.
- *Tracheobronchial tree*: Gradually advance the scope and examine the entire tracheobronchial tree. Head and neck are flexed to the left while examining the right bronchial tree and to the right for left side bronchial tree. In this way, axis of bronchoscope corresponds with trachea and bronchi. Examine openings of all the segmental bronchi.
- *Telescope*: Straight and angled telescopes provide magnification and facilitate detailed examination.
- *Biopsy*: Take biopsy of the lesion of suspicious area.
- *Collection of secretions*: Collect secretions for exfoliative cytology, or bacteriologic examination.

Precautions for Rigid Bronchoscopy

- Select proper size of bronchoscope as per the patient's age.
- No force is applied against closed glottis.
- Avoid repeated removal and introduction of bronchoscope as far as possible. Prolonged procedure (> 20 minutes) may cause postoperative subglottic edema in infants and children.
- Maintain intravenous line and administer injections of antibiotic and steroid especially in infants and children.

Postoperative Care

- *Position*: Patient is kept in coma position, which prevents aspiration of blood or secretions.
  - Patient is kept in humid atmosphere.
- *Observation*: Watch for any spitting of blood, respiratory distress (inspiratory stridor, suprasternal retraction) and cyanosis. They can be due to laryngeal spasm, laryngeal edema, or aspiration of blood. These patients may need tracheostomy.

Complications

- Injury to teeth and lips.
- Transient fever especially after bronchoalveolar lavage.
- Bleeding and hemoptysis in cases of inflamed or malignant tissue.
- Hypoxia and cardiac arrest.
- Laryngeal spasm or edema.

FLEXIBLE FIBEROPTIC BRONCHOSCOPY

Flexible fiberoptic bronchoscope (FOB) is replacing rigid bronchoscope but its utility is limited in children because of the problems of ventilation.

Types of Fiberoptic Bronchoscope

1. *Conventional*: Image is viewed at proximal end by the eyepiece or connected to a video display.
2. *Videoscopes*: High-resolution true-color rendition of the image captured by the true color-chip charged couple device cameras embedded in distal tip of scopes. The recording is possible with digital still images or video segments.
3. *Outer diameter*: It varies from 2.2 (fiber view only and no operating channel)–6.4 mm. An average FOB has an outer diameter of 5.0–5.5 mm. Ultra thin FOB can traverse 12–16 generations of airways.
4. *Length*: It varies from 40–60 cm.
5. *Diameter of operating channel*: It varies from 1.2–3.2 mm. An average FOB has 2.0–2.2 mm operating channel, which admits cytology brushes, biopsy forceps and aspiration needles. Larger diameter channels are used for larger laser fibers, electrocautery, cryotherapy probes and expandable balloons.

Advantages

- Magnification and better illumination
- Better documentation (Figs 4 to 6)
- Smaller size allows examination of subsegmental bronchi

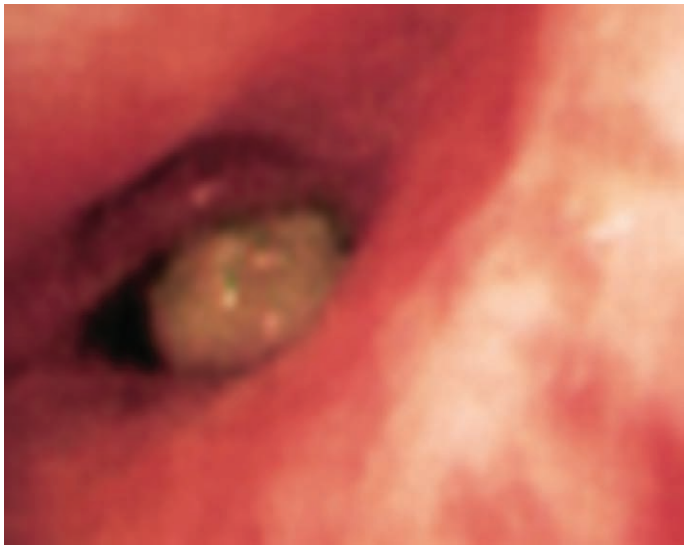


Fig. 4: Mucoid impaction with surrounding edema
Source: Dr Rajiv Paliwal, Professor, Chest Medicine, Pramukh Swami Medical College, Karamsad, Anand, Gujarat

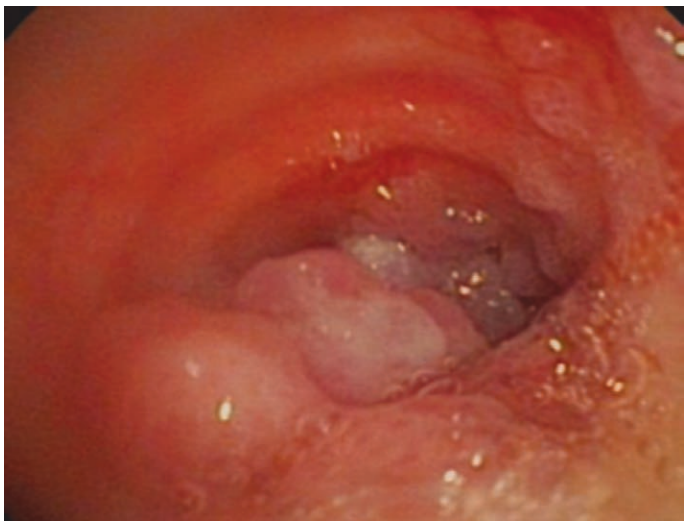


Fig. 5: Endobronchial mass lesion
Source: Dr Rajiv Paliwal, Professor, Chest Medicine, Pramukh Swami Medical College, Karamsad, Anand, Gujarat

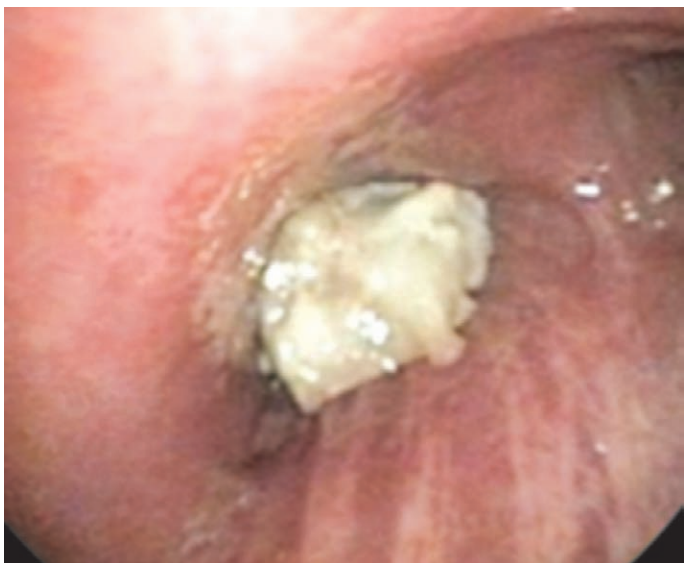


Fig. 6: Irregular mass occluding bronchus
Source: Dr Rajiv Paliwal, Professor, Chest Medicine, Pramukh Swami Medical College, Karamsad, Anand, Gujarat

- Easy to do even under topical anesthesia
- It can be performed in cases of neck or jaw abnormalities or critical illness (bedside examination)
- The suction/biopsy channel helps in removing secretions, inspissated mucous plugs or small foreign bodies
- Flexible bronchoscope can be passed through endotracheal tube.

Bronchoscopy biopsy of the right upper lobe carina/spur: It is the most dangerous site for biopsy because of the underlying right pulmonary artery.

ESOPHAGOSCOPY

Esophagoscopy is of two types: Rigid esophagoscopy and flexible fiberoptic esophagoscopy.

INDICATIONS

- *Diagnostic*
  - *Symptoms*: Dysphagia, odynophagia, aphagia, sensation of a lump or "sticking" in throat, retrosternal burning, hematemesis and persistent regurgitation.
  - *Signs*: Vocal cord palsy.
  - *Investigation findings*
    - Radiological evidence of extrinsic or intrinsic esophageal disorders
    - Abnormal esophageal manometry
    - Abnormal esophageal pH recording
  - *Diseases*: Malignancy esophagus, cardiac achalasia, strictures, infectious esophagitis, diverticulum, reflux esophagitis, hiatus hernia, esophageal varices, caustic ingestion, secondary neck node with unknown primary, surveillance for second primary, penetrating trauma to thorax to rule out esophageal injury.
- *Therapeutic*
  - Foreign body
  - Impacted food
  - Dilatation of esophageal strictures, stenosis or cardiac achalasia
  - Removal of benign neoplasms such as fibroma, papilloma, and cysts
  - Insertion of Soutar's or Mousseau Barbin tube in palliative treatment of esophageal carcinoma
  - Tracheoesophageal puncture after total laryngectomy
  - Treatment of diverticulum or varices.

CONTRAINDICATIONS OF ESOPHAGOSCOPY

- *Absolute*:
  - Coagulopathy.
  - Perforation of esophagus: Spontaneous, traumatic or iatrogenic.
- *Relative contraindications*: Advanced heart, liver or kidney disease.
- *Contraindications of rigid esophagoscopy*: In most of the following conditions, new generations of flexible gastroscopes can be used successfully.
  - Severe trismus: It does not allow passage for esophagoscope. Small size flexible gastroscopes can be passed through nose.

- Cervical spine lesions: Cervical trauma, spondylosis, Pot's spine, osteophytes, or kyphosis.
- Receding mandible.
- Aneurysm of aorta: May rupture and cause fatal hemorrhage.

RIGID ESOPHAGOSCOPY

Instruments for esophagoscopy (Figs 7A to G) are described in chapter Instruments. The size of the esophagoscope is selected as per the age of the patient. Handle at the proximal end of esophagoscope indicates the direction of the bevel at the distal end.

A rigid bronchoscope may be used for performing esophagoscopy but the rigid esophagoscope cannot be used for bronchoscopy.

Advantages

- More amenable to therapeutic indications especially removal of foreign bodies.
- Better visualization of proximal one-third of esophagus.

Disadvantages

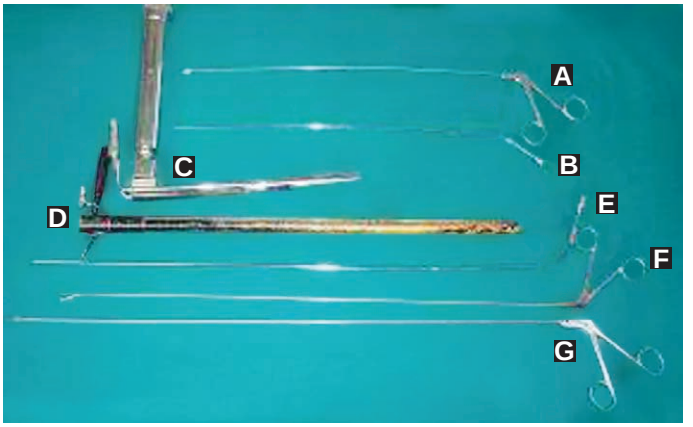
- General anesthesia.
- More cost and morbidity to patient.
- More complications such as dental trauma and esophageal perforation.
- Concomitant examination of stomach and intestine not possible.
- Not amenable to cases of trismus or cervical spine degenerative diseases.

Anesthesia

It is usually done under general anesthesia with endotracheal intubation.

Position

- Patient is placed in supine position. Head is elevated by 10–15 cm. Neck is flexed on chest. Head is extended at atlanto-occipital joint. This position brings the axes of mouth, pharynx and esophagus in a straight line and facilitates passage of esophagoscope. Shoulders are at the edge of operation table and head rests on a special headrest or hold by an assistant.
- A right-handed surgeon sits on the left of the long axis of the patient.
- The assistant, instrument table, light source, suction and video are on right side of surgeon.



Figs 7A to G: Instruments of rigid esophagoscopy. (A) Peanut foreign body grasping forceps, both jaws movable; (B) Suction tube; (C) Cervical esophageal speculum (Laryngopharyngoscope), Negus pattern; (D) Esophagoscope, Jackson pattern; (E) Suction tube; (F) Cup-shaped oval straight biopsy forceps; and (G) Peanut foreign body grasping forceps

Techniques (Fig. 8)

- *Protection of teeth and lips*: Examine the patient for neck stability and loose teeth or dentures. Eyes are protected with a shield and the patient is draped. A gauze piece or teeth guard protects the upper teeth against injury. Left hand thumb retracts the upper lip and teeth while index finger lifts lower teeth.
- *Lubrication of scope*: Lubricate proper size esophagoscope with liquid paraffin or xylocaine jelly.
- *Holding of scope*: Esophagoscope is held by its proximal end in right hand and introduced into right side of mouth lateral to the tongue and advanced towards the middle of base of tongue.
- *Laryngopharynx*: Esophagoscope is further advanced gently by the left thumb and index finger. Identify epiglottis, endotracheal tube and arytenoids.
- *Cricopharyngeal sphincter (upper esophageal sphincter)*: Keep the tip of esophagoscope in midline and behind the larynx. Lift the scope with the help of left thumb and open the hypopharynx. Slow, gentle and sustained pressure of the scope tip on the cricopharyngeal sphincter opens it. Then the tip of scope is guided into the esophagus. A fine bougie or an additional dose of muscle relaxant may be used if needed. Advance the scope constantly seeing the esophageal lumen.

Application of too much force on upper esophageal sphincter is the most common cause of cervical esophageal perforation.

- *Aortic arch and left bronchus*: Indentations of aortic arch (aortic pulsation seen and felt) and left bronchus lie about 25 cm from the incisors. During this time, head of the patient is slightly lowered, which brings the esophageal lumen in the line of the scope.

Fig. 8: Procedure of esophagoscopy

- *Cardiac end*: Head and shoulders are kept below the level of the table. The head, which is slightly higher than the shoulders, is moved slightly to the right. The esophagoscope now points to the left anterior superior iliac spine. Cardia has redder and more velvety or rugose mucosa.
- *Withdrawing*: Inspect the esophageal wall again while withdrawing the esophagoscope.

Postoperative Care

- *Features of esophageal perforation*: Watch for the features of esophageal perforation such as pain in the interscapular region, surgical emphysema of neck and chest, and abrupt rise of temperature.
- *Diet*: Sips of plain water followed by usual diet may be given in an uneventful esophagoscopy.

Complications

- *Injury* to lips, teeth, and pharynx.
- *Perforation of esophagus*: This is the most dreaded complication and usually occurs near cricopharyngeal sphincter (Killian's dehiscence). Surgical emphysema develops within an hour. It may be complicated with an abscess in retropharyngeal space or mediastinum. The features of thoracic esophageal perforation include–
  - Pain in the interscapular region
  - Surgical emphysema
  - Abrupt rise of temperature.
- *Compression of trachea*: It may occur especially in children when esophagoscope is pressed on posterior tracheal wall. It causes obstruction to respiration and cyanosis and needs immediate withdrawal of esophagoscope.

FLEXIBLE ESOPHAGOSCOPY

Advantages

- An outdoor procedure.
- No general anesthesia. It is done under local anesthesia with or without intravenous sedation.
- Less morbidity.
- It can be done in abnormalities of spine or jaw.
- Gastroscope allows examination of stomach and duodenum.
- Good illumination and magnification.
- Accurate diagnosis of the mucosal diseases.

Disadvantages

- Narrow channel limits the size of instruments and removal of certain foreign bodies.
- Foreign body cannot be retracted into the endoscope (like rigid esophagoscope) so more chances of injuring esophagus.
- Laryngopharynx and proximal one-third esophagus (less distensible with insufflations) may not be examined adequately.

Common Indications

- Precision biopsies
- Removal of small foreign bodies or benign tumors
- Dilatation of webs or strictures
- Injection of sclerosing agents in bleeding varices.

Endoscope and Instruments (Fig. 9)

Esophagoscope is available in wide range of diameter smallest being 5.0 mm. There are separate channels for optics, suctioning (secretions), insufflations, and instruments (for biopsy, foreign bodies, sclerotherapy and laser ablation). Set up also includes light source, camera and video processing unit.

Techniques

- The patient is usually in left lateral position or in supine and gentle extension of neck with a shoulder roll.
- Lubricated scope (with xylocaine jelly) is introduced into the mouth through a plastic mouth block (Fig. 10) and advanced into the pharynx, postcricoid region and esophagus (Fig. 11).
- The esophagoscope can be deflected in any direction and secretions can be aspirated.
- Air or water insufflation opens the lumen of esophagus and the endoscope is advanced further.

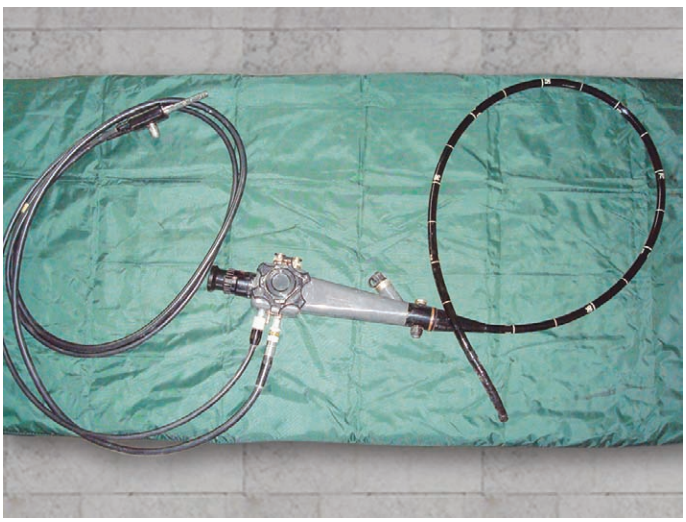


Fig. 9: Flexible esophagoscope

Fig. 10: Flexible esophagoscopy. Patient's position and plastic mouth block in mouth

Fig. 11: Flexible esophagoscopy. Note the biopsy forceps in the hand of assistant

Clinical Highlights

1. **Tracheobronchial tree and larynx:** Rate of topical absorption is highest.
2. **Objective piece of operative microscope:** 400 mm focal length of objective piece of operative microscope is usually used for microlaryngoscopy.
3. **Flexible bronchoscopy:** It offers visions of segmental bronchi and the upper lobe bronchi, which are beyond the reach of rigid bronchoscopes.
4. **Rigid bronchoscopy:** It is superior in taking biopsy and culture specimens, removing foreign bodies and in surgical intervention such as dilatation.
5. **Esophagoscopy:** Application of too much force on upper esophageal sphincter (UES) is the most common cause of cervical esophageal perforation.
6. **Esophageal perforation:**
   a. **Features:** Fever after esophagoscopy.
   b. **Diagnosis:** Swallow study confirms diagnosis.
   c. **Treatment:** Early intervention to repair is most desirable. Drain the perforation to prevent complications.

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59 Instruments

Misery is truly a gift of God. I believe it is a symbol of His compassion. It is idle to expect that dangers and difficulties will not come. They are bound to come. But, for a devotee of God they will pass away from under feet like water.

—Holy Mother Sarada Devi

Points of Focus

- **INTRODUCTION**
- **OPD INSTRUMENTS**
  - ✓ Dressing Forceps, Ear Speculum, Otoscope, Tuning Fork, Barany Noise Box, Aural Syringe, Blunt Probe, Laryngeal Mirror, Thudicum Nasal Speculum, Lack's L-shaped Tongue Depressor, Postnasal Mirror, Jobson-Horn Probe with Ring Curette, Siegel's Pneumatic Speculum, Eustachian Catheter and Politzer's Bag.
- **MASTOID AND EAR-MICROSURGERY**
  - ✓ Endaural Speculum, Myringotome, Mastoid Self-Retaining Retractors, Lempert's Endaural Retractor, Mastoid Suction Tips, Mastoid Gouges, Mastoid Curette (Scoop), Farabeuf's Periosteal Elevator, Macewen's Curette and Cell Seeker, Crocodile (Alligator) Scissors and Forceps, Microsurgery Instruments (Zoellner)
- **ANTRUM PUNCTURE**
  - ✓ Lichtwitz Trocar and Cannula, Higginson's Syringe
- **INFERIOR MEATAL ANTROSTOMY**
  - ✓ Tilley's Harpoon, Tilley' Antral Burr, Rose' Sinus Douching Cannula
- **NASAL FRACTURE REDUCTION FORCEPS**
  - ✓ Walsham's Forceps, Asch's Septum Forceps
- **NASAL SEPTAL AND SINUS SURGERY**
  - ✓ Endoscopic Sinus Surgery Instruments: Hopkins Telescope/Sinuscope, Bloberley Forceps, Punches, Bone Forceps, Chisels, Heymann Turbinectomy Scissors, Elevators, Nasal Speculum, Mattel, Nasal Gouges, Periosteum Elevator, Luc's Forceps, Ballenger's Swivel Knife
- **MOUTH GAGS AND RETRACTORS**
  - ✓ Jenning's Mouth Gag, Doyen's Mouth Gag, Cheek Retractor, Hajek Lip Retractor
- **ADENOTONSILLECTOMY**
  - ✓ Boyle-Davis Mouth Gag, Tonsil Holding Forceps, Tonsil Knife, Tonsil Dissector and Anterior Pillar Retractor, Tonsil Scissors, Negus Artery Forceps, Tonsil Needle, Yankauer's Suction Tube, Eve's Tonsil Snare, Conchotome, Draffin's Bipod, St. Clair Thomson's Adenoid Curette With Guard, Tonsil Dissection Forceps With Teeth (Waugh's), Tonsil Artery Forceps, Negus Knot Tyer
- **INCISION AND DRAINAGE OF QUINSY**
  - ✓ Peritonsillar (Quinsy) Abscess Forceps
- **ENDOSCOPES**
  - ✓ Comparison between Two Types of Illumination (Jackson and Negus)
  - ✓ Laryngoscopes: Direct Laryngoscope, Laryngoscopes for Microlaryngoscopy and Surgery:
  - ✓ Bronchoscope
  - ✓ Esophagoscope
  - ✓ Endoscope and Instruments
- **TRACHEOSTOMY**
  - ✓ Tracheal Dilator, Tracheal Hook (Blunt and Sharp)
  - ✓ Types of Tracheostomy Tubes
  - ✓ Fuller's Bivalved Tracheostomy Tube
  - ✓ Chevalier Jackson's Tracheostomy Tube
  - ✓ Nonmetallic Tracheostomy Tubes: Cuffed tracheostomy Tube, Cuffed Suction Aid Tracheostomy Tube
  - ✓ Size of Tracheostomy Tube
- **AIRWAY DEVICES**
  - ✓ Nasopharyngeal Airway, Oropharyngeal Airway, Laryngeal Mask Airway (LMA), Endotracheal Tubes

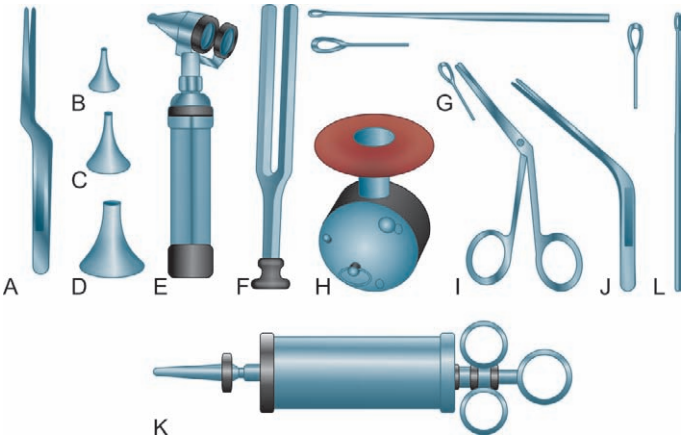
INTRODUCTION

There is vast number of ENT instruments used for diagnostic, therapeutic and surgical purposes. It is beyond the scope of this book to discuss each and every instrument. The figures show quite good number of instruments but the description covers only frequently asked instruments. For the further details regarding the method of use and indications, the reader should refer to the related chapters such as History and Examination and section of Operations.

OPD INSTRUMENTS

Figures 1A to L and 2A to J show OPD instruments. For the related details, *see* chapters of "Symptoms and Examination" of respective sections.

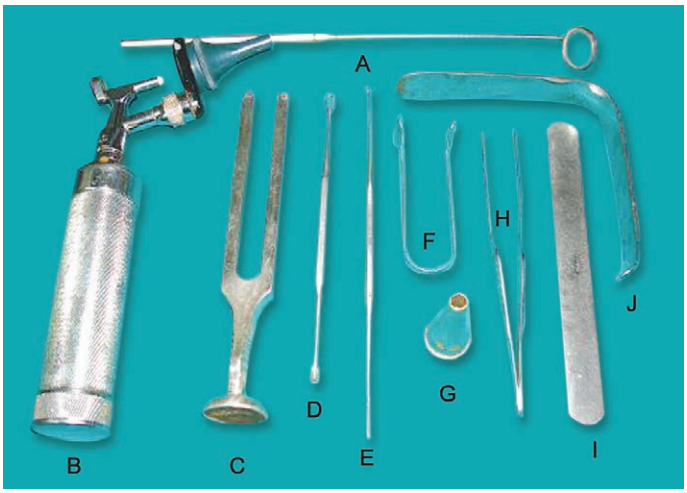
- *Dressing forceps* (Figs 1 A, I and J): Tilley's dressing forceps has a box joint. Its bayonet-shaped or bent at an obtuse angle prevents the hand of the surgeon from obstructing the line of vision. Hartmann's dressing forceps is similar to Tilley's forceps and has a screw joint and the serrated and grooved jaws. Wilde's dressing forceps acts on spring action.
  - Uses: They are used for nasal packing, ear dressing and removal of foreign bodies.
- *Ear speculum* (Figs 1B to D): This cone-shaped speculum has tapered end that is inserted into the cartilaginous portion of the external auditory canal (EAC) after retracting the pinna. The black or dull finished speculums are used in operations and prevent reflection of light. Various sizes and shapes of the ear speculums are available, which suit different sizes of the ear canal. The use of the largest ear speculum that can easily enter the canal is safe and provides better view.
  - Use: It is used for examination and operations of the EAC, tympanic membrane and middle ear.
- *Otoscope* (Fig. 1E): It has its own illumination, and the source of light that is housed in its handle. It also provides magnification. Some of the otoscopes have Siegel's speculum.
  - Use: It helps in examining the EAC and tympanic membrane. It is especially useful in examining the ears and nose of infants and bedridden patients.
- *Tuning fork* (Fig. 1F): For details of tuning fork tests, see chapter "Hearing Evaluation".
  - Use: They are used for Rinne, Weber and other hearing tests.
- *Barany noise box* (Fig. 1H): For further details, see chapter "Hearing Evaluation".
  - Use: It is used for masking purposes during the tuning fork hearing tests.
- *Aural syringe* (Fig. 1K): This metal syringe consists of a cylinder with a well-fitting piston and a nozzle. For details of ear syringing, see chapter "Diseases of External Ear".
  - Use: It is used for ear syringing to remove EAC wax and foreign bodies.
- *Blunt probe:*
  - Use: It is used for palpation of polyp, growths and swellings in the ear canal as well as nasal cavity.
- *Laryngeal mirror* (Fig. 2A): The shaft of the laryngeal mirror is straight. It is available in various sizes from 6–30 mm diameter. See chapter "Laryngeal Symptoms and Examination" in section larynx, trachea and bronchus.
  - Use: It is used for the indirect examination of oropharynx, laryngopharynx and larynx.



Figs 1A to L: OPD ear instruments. (A) Jansen dressing forceps bayonet shaped; (B) Plastic ear speculum; (C) Hartmann ear speculum; (D) Boucheron ear speculum; (E) Heine Otoscope with plastic ear specula; (F) Tuning fork; (G) Billeau ear loop; (H) Barany noise box with soft rubber; (I) Lucae ear dressing forceps; (J) Troeltsch dressing forceps angular; (K) Ear syringe; (L) Weber-Loch ear curette
Source: Karl Storz, Germany

- *Thudicum nasal speculum* (Fig. 2F): Thudicum nasal speculum consists of U-shaped metal spring with two blades at its ends. The size of the nasal speculum should be chosen according to the age of patient and size of the nose. A Thudicum or Vienna type of nasal speculum is held in the left hand, assists in widening the vestibule. For the method of examination, see chapter "Nasal Symptoms and Examination" in section "Nose and Paranasal Sinuses".
  - Uses: They are used for examination and during surgery of the nose.
- *Lack's L-shaped tongue depressor* (Fig. 2J): One blade of Lack's tongue depressor is slightly bent at the end. The bent end is used for holding the depressor and supports the little finger of the examiner. The other blade depresses the tongue and is used like a lever to depress anterior two-third of the tongue with the fulcrum over the lower teeth.
  - Caution: Touching of the posterior one-third of the tongue usually leads to the gag reflex and not tolerated by the patient.
  - Uses: It is used for examining the oral cavity and the pharynx. In addition to the depressing of tongue, it can also be used for:
    - Squeezing the tonsil
    - Retraction of cheek
    - Test for gag reflex
    - To check nasal air blast
    - Spatula test for suspected case of tetanus
    - Posterior rhinoscopy examination
    - Checking out for loose teeth
    - Intraoral surgical procedure
    - Checking postnasal bleeding.
- *Postnasal mirror*: Mirror is smaller than laryngeal mirror and the shaft is bayonet-shaped.
  - Use: It is used for examining the nasopharynx and posterior part of nasal cavity. See chapter "Nasal Symptoms and Examination" in section "Nose and Paranasal Sinuses".

The posterior rhinoscopy mirror is smaller and its shaft is bayonet-shaped, while the shaft of the laryngeal mirror is straight.



Figs 2A to J: Some of the routinely used OPD ear, nose, and throat instruments. (A) Laryngeal mirror; (B) Otoscope; (C) Tuning fork; (D) Ear vectis and curette; (E) Jobson Horne probe with round serrated end applicator and curette end; (F) Thudicum nasal speculum; (G) Ear speculum; (H) Bayonet-shaped ear dressing forceps; (I) Straight tongue depressor; (J) L-shaped tongue depressor

- *Jobson-Horne probe with ring curette* (Figs 2E and 3): It has two ends: round serrated end applicator and curette end.
  - Use: One end of the Jobson-Horne's probe is used for applying cotton to clean the ear of discharge. The other end (ring curette) is used to remove the wax and foreign body.
- *Siegel's pneumatic speculum*: It is (Figs 4A and B) fitted with a convex lens and is attached to a rubber bulb through plastic tubing. The rubber bulb assists in alternately increasing and decreasing pressure in the EAC. Its convex lens provides magnification.
  - Uses:
    - Mobility of tympanic membrane: It is tested with Siegel's speculum.
    - Fistula test: It is for labyrinthine fistula (see chapter Evaluation of Vertigo).
    - Aural toilet: Suction of middle ear secretions in cases of acute and chronic suppurative otitis media.

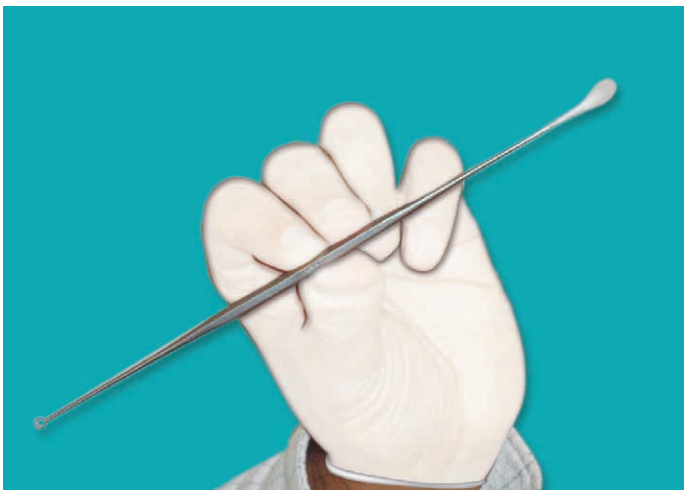
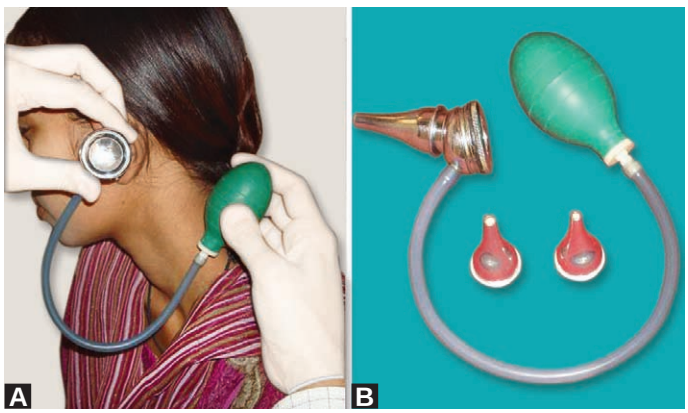


Fig. 3: Jobson Horne's probe. Applying cotton to clean the ear of discharge while the other end (ring curette) is used to remove the wax and foreign body

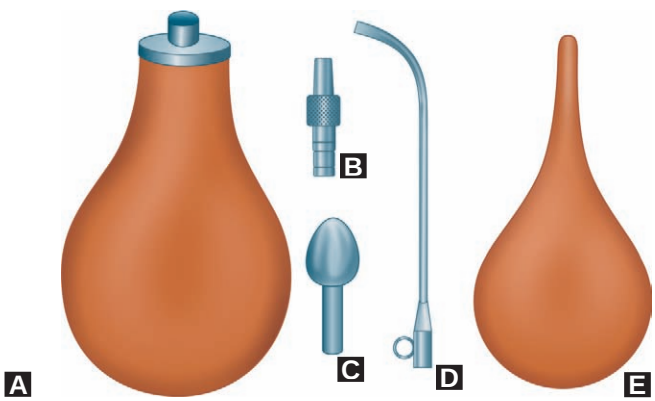
    - Topical ear medicines: Pushing of medicines through the central perforation of tympanic membrane.
- *Eustachian catheter and Politzer's bag* (Figs 5A to E): Eustachian catheter is a 12–15 cm metal cannula. It has a ring at its base that indicates the direction of its curved tip. It looks similar to the antral washing cannula, in which opening is not at the tip but a little proximal to it. An olive-shaped tip of the Politzer's bag is introduced into the nose.
  - Uses:
    - Politzer test and Eustachian tube catheterization for testing the functioning of Eustachian tube. For further details, see chapter "Disorders of Eustachian Tube".
    - To inflate the middle ear
    - To instill medicines into middle ear
    - To remove foreign body from the nose
    - For suctioning the secretions and discharge

MASTOID AND EAR MICROSURGERY

Figures 6A to E show speculum, retractors and suction cannula. Figures 7A to H and 8A to G show bone cutting instruments.



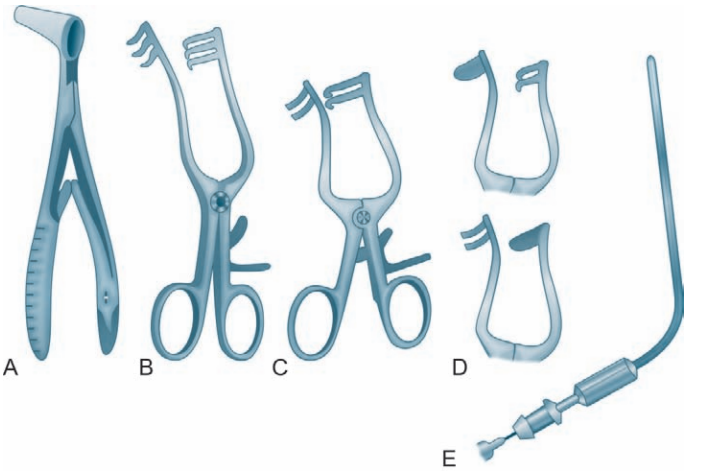
Figs 4A and B: Siegel's examination. Mobility of tympanic membrane is tested by alternately increasing and decreasing pressure in the external auditory canal. (A) Method of Siegel's examination; (B) Siegel's pneumatic speculum fitted with a convex lens and attached rubber bulb through plastic tubing



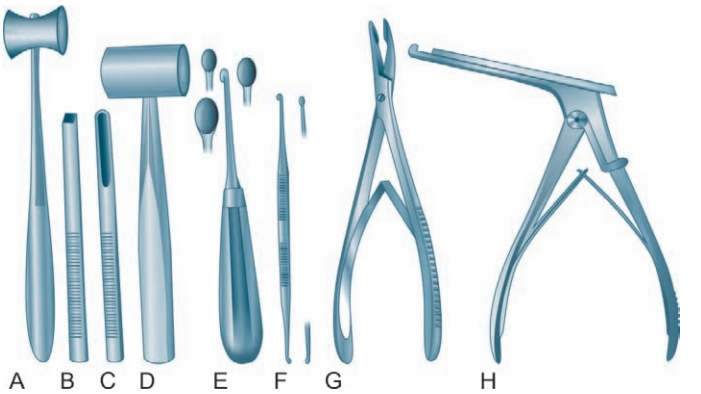
Figs 5A to E: Eustachian tube instruments. (A) Politzer air bag 8 ounce capacity; (B) Metal connector Eustachian catheter to Politzer air bag; (C) Nasal tip for use with Politzer air bag; (D) Hartmann Eustachian catheter; (E) Soft rubber air bag (2.5 ounce capacity) for use with Eustachian catheter
Source: Karl Storz, Germany

Figures 9A to I show elevators and suction cannula. Figures 10 A to F show ear snare and micro forceps. For details of the operations and more instruments, see chapter "Middle Ear and Mastoid Surgeries".

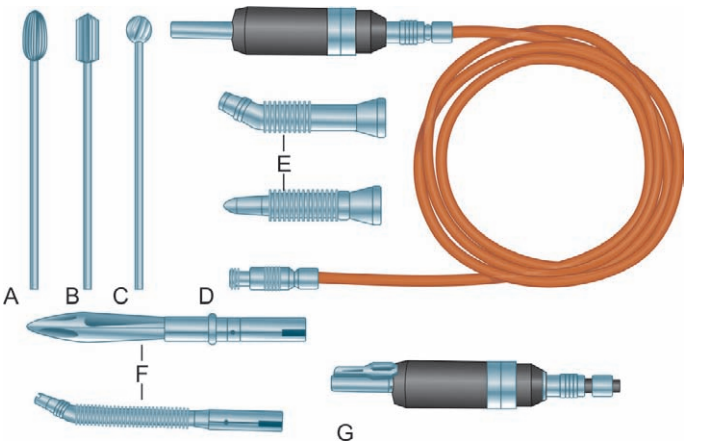
- *Endaural speculum* (Fig. 6A): There are many types of endaural speculum including Lempert's. This curved speculum is similar to Vienna model.
  - Use: It spreads open the meatus and is used when giving local injection or making an endaural incision.
- *Myringotome*:
  - Use: For puncturing tympanic membrane and placing grommets.
- *Mastoid self-retaining retractors* (Figs 6B to D): There are many types of mastoid retractors including Mollison's and Jansen's. The catch prevents its closure and the blades hold apart the edges of the incision.
  - Uses:
    - Mastoidectomy: They retract soft tissues after incision and elevation of flaps. The pressure on the edges of the incision provides hemostasis.
    - Other operations: This self-retaining retractor may be used in other surgeries, such as laryngofissure, craniotomy, burr-holing and external ethmoidectomy.
- *Lempert's endaural retractor*: It has three blades. The two lateral blades retract the flaps. The third central blade with holes retracts the temporalis muscle superiorly. The central blade with its hole is fixed to the body of the retractor.
  - Use: It is specifically used in the endaural approach mastoidectomy.
- *Mastoid suction tips* (Figs 6E and 9C to F): This bent cannula has an obtuse angle and a hole that can be used to regulate the force of suction. The hole can be closed by placing a finger tip on it. Fine suction tips are used for microsurgery.
  - Use: Suction of blood, secretions, irrigation water and bone dust.
- *Mastoid gouges* (Figs 7C and 15D): They are of various sizes and have a concave edge and rounded margins. The electrical drill and burrs (Figs 8 A to G) are replacing the use of gouges.
  - Uses:
    - Mastoidectomy: They are used to remove bone in mastoid surgery.
    - Caldwell-Luc operation: It is also used for canine fossa antrostomy for opening the maxillary antrum.
    - Exostosis: Excision of exostosis in the EAC.
- *Mastoid curette (scoop)* (Figs 7E and F): There are many types of mastoid curette including Lempert's.
  - Use: It removes bony septa and granulations in mastoid surgery.
- *Farabeuf's periosteal elevator* (Fig. 9H)
  - Use: It is used for elevation of periosteum from the mastoid cortex in mastoidectomy.
- *Macewen's curette and cell seeker*: It has two ends which have different functions.
  - Use: One end (seeker) explores the air cells and mastoid antrum. Other end curettes the intervening septa and granulations in mastoid surgery.
- *Crocodile (alligator) scissors and forceps* (Figs 10A to F): These thin and delicate forceps have crocodile or alligator jaws type tips and are used for microsurgery of the ear.



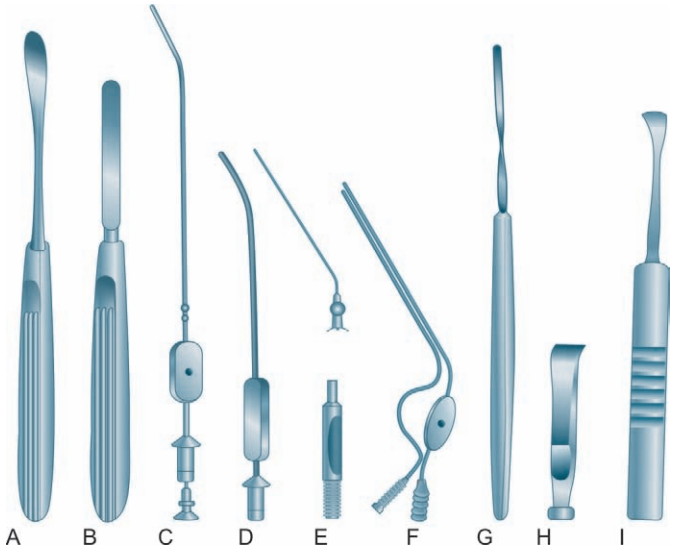
Figs 6A to E: Ear surgery instruments retractors. (A) Hartmann speculum; (B) Wullstein retractor; (C) Plester retractor 2 x 2 prongs; (D) Plester retractors with biprong blade and solid blade for left and right side; (E) Ferguson suction tube with finger cut-off and stylet
Source: Karl Storz, Germany



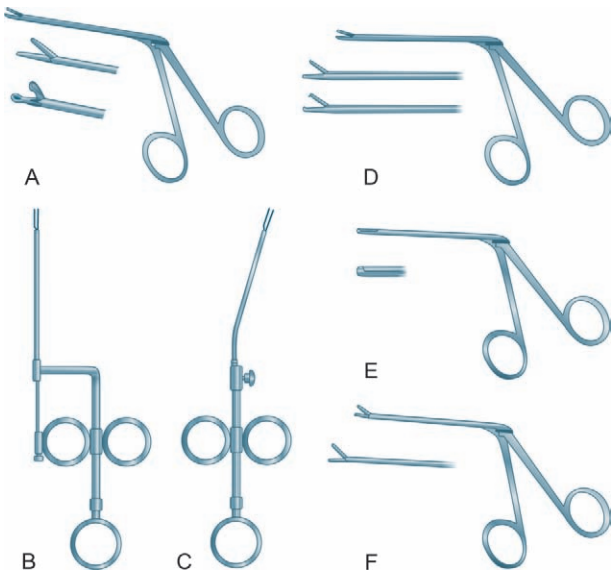
Figs 7A to H: Ear surgery bone cutting instruments. (A) Cottle mallet; (B) Trautmann mastoid chisel; (C) Trautmann mastoid gauge; (D) Lucae mallet; (E) Spratt mastoid curette; (F) House curette; (G) Beyer Rongeur light curved jaws; (H) Kerrison Rongeur
Source: Karl Storz, Germany



Figs 8A to G: Ear surgery instruments, electric motor and burrs. (A) Oval burr; (B) Oblong burr; (C) Standard burr; (D) Bone engine with hand held micromotor; (E) Intra hand piece straight and angled; (F) Slip joint hand piece straight and angled; (G) Bone engine with micromotor
Source: Karl Storz, Germany



Figs 9A to I: Ear surgery instruments, elevators and suctions. (A) Jansen elevator; (B) Plester elevator; (C) Plester suction tube with finger cut-off and stylet; (D) Zoellner suction tube with finger cut-off; (E) Fisch adaptor for suction cannula with finger cut-off luer cone; (F) Fisch suction irrigator; (G) Lempert elevator; (H) Farabeuf's elevator; (I) Fisch elevator
Source: Karl Storz, Germany

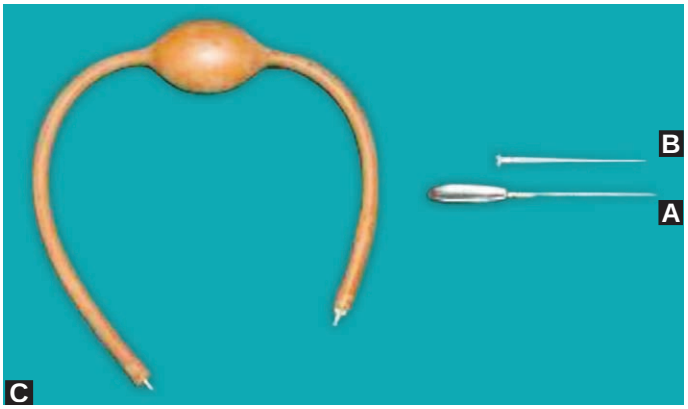


Figs 10A to F: Ear micro-forceps and snares. (A) Hartmann ear forceps serrated and cupped jaws; (B) Wilde ear snare bayonet type; (C) Krause ear snare; (D) Wullstein very fine ear forceps serrated and cupped jaws; (E) House-Dieter very fine malleus nipper; (F) Bellucci delicate scissors
Source: Karl Storz, Germany

- *Microsurgery instruments (Zoellner)*: This set of ear microsurgery instruments are bent at a right angle and have a handle to hold them between the thumb and fingers. The black or dull finish prevents the glare of light under the microscope.

ANTRUM PUNCTURE

Figures 11A to C show the instruments used for antrum puncture. For the detailed procedure and indications, see chapter "Operations of Nose and Paranasal Sinuses".



Figs 11A to C: Antrum puncture instruments. (A) Lichtwitz trocar; (B) Cannula; (C) Higginson's syringe

- *Lichtwitz trocar and cannula*: It perforates the lateral wall of inferior meatus. This area is easily accessible and safe region. After piercing the nasoantral wall, trocar is removed.
  - Use: It is used for proof puncture (antral lavage).
- *Higginson's syringe*:
  - Use: It is used in irrigating the maxillary antrum with normal saline.

INFERIOR MEATAL ANTROSTOMY

- *Tilley's harpoon*
  - Use: It is used for intranasal antrostomy in the inferior meatus.
  - Advantage: It removes the bony chips when it is withdrawn. So the bony chips do not fall in the sinus cavity.
- Tilley's antral burr
  - Use: It enlarges and smoothens the hole made by harpoon in intranasal inferior meatal antrostomy.
- *Rose's sinus douching cannula*: The hook outside indicates the direction of the tip.
  - Use: It is used for the irrigation of maxillary sinus through the nasoantral inferior meatus window after intranasal antrostomy or Caldwell-Luc operation.

NASAL FRACTURE REDUCTION FORCEPS (FIGS 12A AND B)

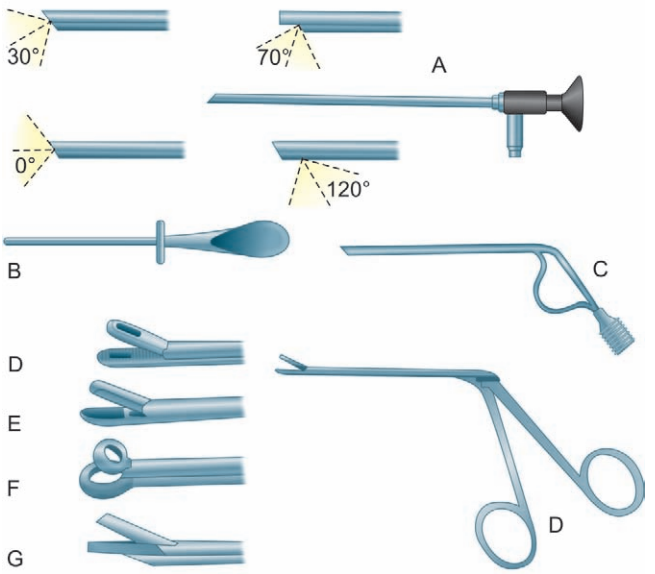
For other related details, see chapter "Maxillofacial Trauma".

- *Walsham's forceps* (Fig. 12A): Rubber tubing may be used to cover one blade to protect the skin of the external nose.
  - Use: They are used for the disimpaction and reduction of the fractures of nasal bones.
- *Asch's septum forceps* (Fig. 12B): The forceps are bent at an obtuse angle. When they are closed, there remains a gap between the blades that prevent the crushing of the nasal septum.
  - Use: They are used for reducing fractures of nasal septum. It lifts the nasal septum forwards.

NASAL SEPTAL AND SINUS SURGERY

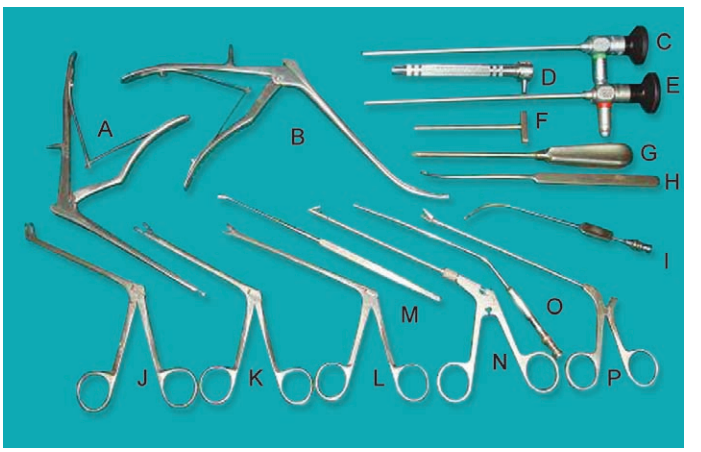
Figures 13A to G and 14A to P shows instruments of endoscopic sinus surgery. Figures 15A to O and 16A to J show instruments of nasal and septal surgeries. See chapter "Operations of Nose and Paranasal Sinuses" for the related details.

Figs 12A and B: (A) Walsham forceps; (B) Asche's forceps



Figs 13A to G: Instruments of endoscopic sinus surgery. (A) Hopkins telescopes 4 mm with fiber optic light transmission (0°, 30°, 70° and 120° Sinuscopes); (B) Trocar and cannula for sinuscopy; (C) Suction irrigation tube with channel for inflow and outflow; (D) Blakesley ethmoid forceps; (E) Takahashi ethmoid forceps; (F) Hartmann nasal cutting forceps; (G) Struycken narrow blade nasal cutting forceps for removal of turbinates
Source: Karl Storz, Germany

- *Endoscopic sinus surgery instruments*:
  - *Hopkins telescope/Sinuscope* (Figs 13A and 14 C and E): Hopkins Telescopes with fiber optic light transmission are available in various sizes and angles (0°, 30°, 70°, and 120° Sinuscopes).
    - Uses: They are used for examination of nasal cavity and nasopharynx and for endoscopic sinus surgery.
  - *Blakesley* (Figs 13D and 14J and K) *and Takahashi* (Fig. 13E) *ethmoid forceps*: Working length of straight Takahashi forceps is 10 cm. Blakesley forceps are cutting and are of different sizes and various angles, such as straight and upward curved (30°, 45°, 70° and 90°). Blakesley forceps have oval cup-shaped jaws.
    - Uses: They are used for removal of polyps and uncinate process and widening the middle meatus antrostomy. They can be used during septoplasty and endoscopic adenoidectomy.
  - *Punches* (Figs 14 A, B and L): There are many types of punches. Kerrison bone punches may be straight upwards cutting (Fig. 14A) or curved upwards cutting (Fig. 14B). Hajek's punch forceps are also similar. Sphenoid punch (Citelli) is upward biting or downward biting punches. This is a strong and heavy instrument.
    - Uses:
      - Caldwell-Luc operation: It is used for enlarging the opening of canine fossa antrostomy in Caldwell-Luc operation.
      - Sphenoid sinus and pituitary fossa surgery: It is also used in the surgeries of sphenoid sinus and pituitary fossa.
      - Endoscopic dacryocystorhinostomy: It is used to remove the medial wall of lacrimal groove.
      - Transpalatal approach: It can also be used in transpalatal operations.
- *Bone forceps* (Figs 15A and B): Double action bone nibbling forceps are bent at an obtuse angle and have four joints with double lever systems that allow the blades to close and open to a limited extent in a narrow deep cavity. They have strong grasp.
  - Use: They are used for removing the bony part such as deviated nasal septum.
- *Chisels* (Figs 15D, E and J): In comparison to an elevator they have head.
  - Uses: They are used in various nasal and paranasal sinus surgeries.
- *Heymann turbinectomy scissors* (Fig 15F): It has a bend in the center that offers better field of vision. Its narrow stout blades have blunt tips.
  - Use: Removal of inferior turbinate.
- *Elevators* (Figs 15G, I and 16E): Freer, Cottle and Killian have two dissecting faces (ends).
  - Septal surgery: They are used for the elevation of mucoperichondrium/mucoperiosteum in nasal septum

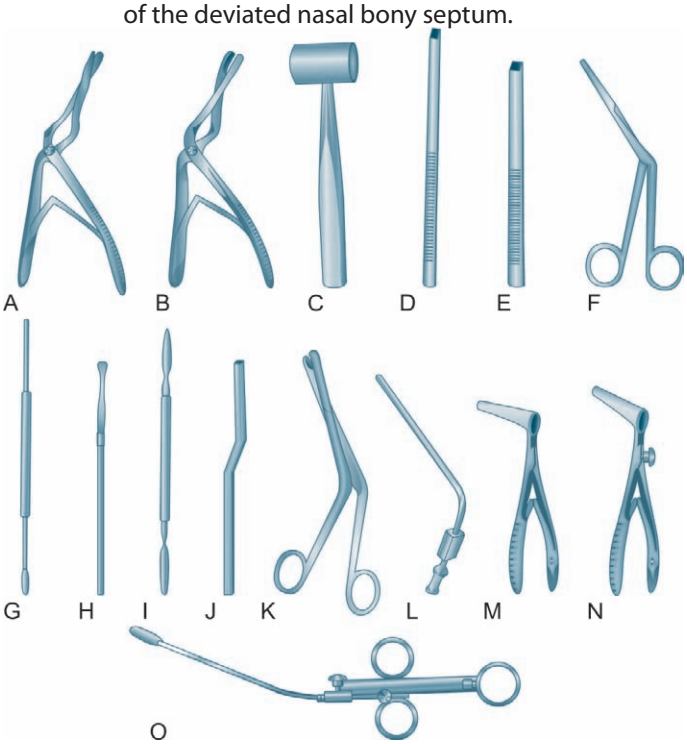


Figs 14A to P: Instruments of endoscopic sinus surgery. (A) Kerrison bone punches satraight upwards cutting; (B) Kerrison bone punches curved upwards cutting; (C) Sinuscope 4 mm 0°; (D) Sheath for the sinuscopes; (E) Sinuscope 4 mm 30°; (F and G) Sinuscopy trocar and cannula; (H) Antrum curved curette; (I) Antrum cannula with blunt end; (J) Blakesley forceps, straight, oval cup-shaped jaws; (K) Blakesley forceps, upward curved, oval cup-shaped jaws; (L) Antrum punch through cutting; (M) Sickle knife; (N) Reverse through cutting forceps; (O) Suction tube with finger cut-off; (P) Straight scissor upper part of jaw movable

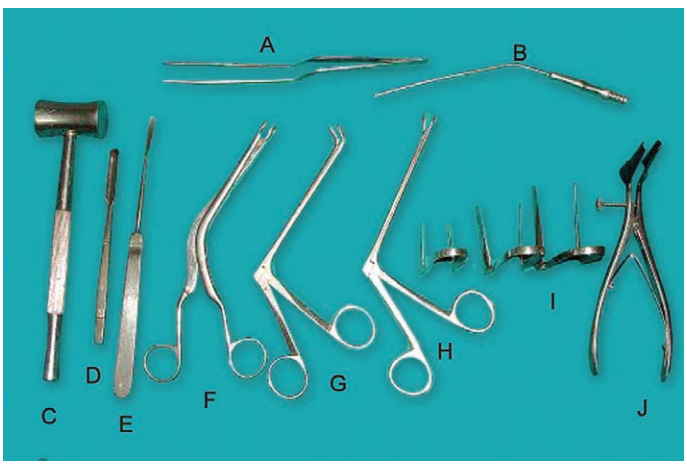
operation. The sharper spade like end begins the dissection in submucoperichondrial plane. Its flat and dull end elevates the flap in an atraumatic way especially in nasal septal surgery.
  - Endoscopic sinus surgery: For manipulating uncinate process, middle meatus and bulla ethmoidalis.
  - Tympanoplasty: For separating temporalis fascia from temporalis muscle.
- *Nasal speculum* (Figs 15M, N and 16I and J): There are many types of nasal speculum including St. Clair Thomson's and Killian's. Its long blades are concave from inside. They keep mucoperiosteal flaps away.
  - Uses: It is used in septal surgery such as submucosal resection (SMR) and septoplasty.

7. *Mallet or Hammer* (Figs 15C and 16C): It consists of handle, shaft and a head.
   - Uses: To give gentle blows on the gouge for removing spur and doing osteotomy in rhinoplasty.

- *Nasal gouges* (Fig. 16D): There are many types of nasal gouges. Killian's bayonet-shaped gouge has rounded and concave, or "V" shaped edge for a better grip on the septal bone.
  - *Uses*:
    - Nasal septum surgery: It is used for removal of septal spurs or bony crests and ridges in nasal septum operation. It is used for removing the maxillary crest of the deviated nasal bony septum.
    - Caldwell-Luc operation: For canine fossa antrostomy and opening the maxillary antrum.
- *Periosteum elevator* (Fig. 16E): There are various shapes and sizes of periosteum elevators. In comparison to a chisel, it does not have a head. A rest is provided for a finger and the edge is blunt.
  - Uses: It is used for elevating the periosteum and soft tissues in Caldwell-Luc operation and mastoidectomy operation.
- *Luc's forceps* (Fig. 16F): These have a screw joint. The blades are fenestrated and cup-shaped with sharp edges. The tissue caught by the blades may bulge out through the fenestra for a better grip.
  - Uses:
    - *Nasal surgery*: They are used in nasal surgery to remove polyps, growth and bone/cartilage during Caldwell-Luc operation, septal surgery, and polypectomy (to grasp and avulse polyp).
    - *Biopsy*: Punch biopsy from the nose, oral cavity and pharynx.
    - *Tonsillectomy*: They may be used in tonsillectomy for holding the tonsils.
    - *Adenoidectomy*: They may be used in adenoidectomy for removing the tags of adenoids after the use of adenoid curette.
- *Ballenger's swivel knife*: The blade of this knife (available in different sizes) revolves automatically and changes direction while cutting the cartilage anteroposteriorly, downwards and posteroanteriorly. The blade is fitted by swivel joints to the handle so that the blade can rotate through 360° in the joints.
  - Method: It is introduced in a cut of the nasal septal cartilage and is pushed backwards, then downwards and finally forwards to remove a quadrangular piece of the septal cartilage.
  - Use: It is used in removal of septal cartilage in SMR operation.



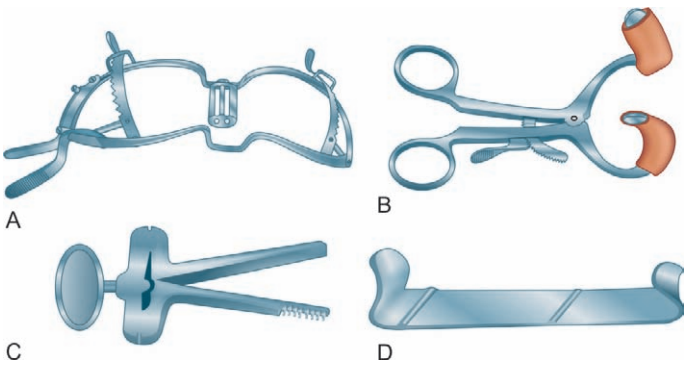
Figs 15A to O: Instruments for nasal surgery. (A) Jansen Middleton septum bone forceps with through cutting blades; (B) Jansen septum bone forceps; (C) Cottle mallet; (D) Cottle chisel with depth markings; (E) Masing chisel double guarded; (F) Heymann nasal scissors medium size; (G) Freer elevator double ended; (H) Freer septum chisel straight end; (I) Killian elevator double ended; (J) Killian claus septum chisel with V-shaped cutting edge; (K) Luc septum bone forceps; (L) Ferguson suction tube with finger cut-off and stylet; (M) Killian-Struycken nasal speculum; (N) Cottle nasal speculum with set screw; (O) Krause nasal snare
Source: Karl Storz, Germany)



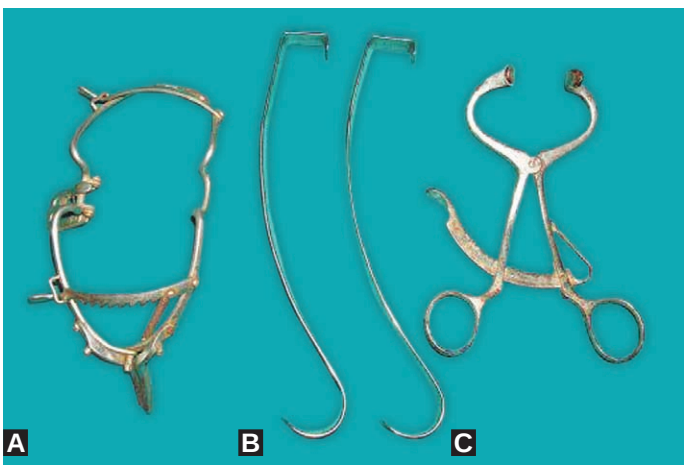
Figs 16A to J: Instruments for septal surgery. (A) Nasal dressing forceps bayonet-shaped; (B) Suction cannula; (C) Mallet with both sides plain faces; (D) Septum gouge, straight, round cutting edge, without notch; (E) Periosteum elevator; (F) Luc's forceps with fenestrated jaws; (G) Takahashi ethmoidal forceps upward curved with oval cup-shaped; (H) Takahashi ethmoidal forceps straight oval cup-shaped; (I) Three sizes of nasal speculum; (J) Septum speculum with joint and set screw

MOUTH GAGS AND RETRACTORS (FIGS 17A TO D AND 18A TO C)

- *Jennings mouth gag* (Figs 17A and 18A): The use is similar to the Doyen's mouth gag.
  - Method: It is applied in the centre of the mouth. The blades of this mouth gag fit over the alveolar margins and not over the teeth.
  - Advantages: Unlike Doyen's gag, it can be fitted to edentulous patients and does not damage the teeth.
- *Doyen's mouth gag* (Figs 17B and 18C): It keeps the mouth open for intraoral tongue surgery.
  - Method: It is applied on one side of the mouth on molar teeth. Its curved blades fit over the teeth. The lower jaw is depressed by the surgeon and the closed mouth gag is introduced in between the jaws and is gradually opened. It remains open by a ratchet.
  - Uses:
    - Operations of the oral cavity, oropharynx and nasopharynx, such as tongue-tie release and removal of large uvula.
    - Oral toilet in an unconscious patient for keeping the airway free.
    - Improving the movements of the temporomandibular joint ankylosis with trismus.
    - Prevent tongue bite in epileptics.



Figs 17A to D: Mouth gags. (A) Whitehead mouth gag with tongue depressor; (B) Doyen-Jansen mouth gag with tongue depressor; (C) Heister mouth gag; (D) Roux cheek retractor
Source: Aesculap®, Germany



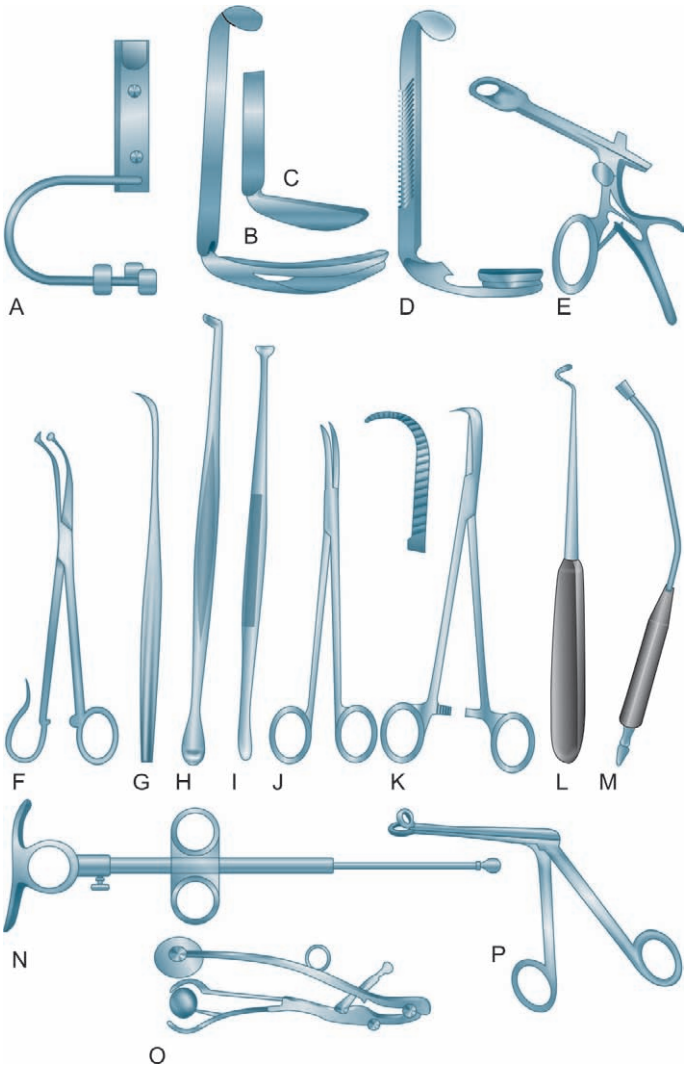
Figs 18A to C: Different types of mouth retractors. (A) Whitehead mouth gag with tongue depressor; (B) Hajek's retractors; (C) Doyen's pattern mouth gag

  - Disadvantages:
    - Chances of teeth damage.
    - Not used in edentulous patients.
- *Cheek retractor* (Fig. 17D): It has a handle and a blade which is molded to fit snugly into the angle of the mouth. It retracts the cheek and exposes the teeth, gums and jaws.
  - Uses: It is used for making an incision for Caldwell-Luc operation and for interdental wiring for fractures of the maxilla and mandible.
- *Hajek' lip (cheek) retractor* (Fig. 18B): It is S-shaped instrument has two different types of ends. One end is bent at right angle and is used for retracting the upper lip. The other end is bent in a curved manner and is used by the assistant or the surgeon for supporting his thumb against the head of the patient. This end can be used for cheek retraction.
  - Uses: It retracts the upper lip during Caldwell-Luc operation and maxillectomy operations.

ADENOTONSILLECTOMY (FIGS 19A TO P AND 21A TO K)

For the related details of these instruments, see chapter "Adenotonsillectomy".

- *Boyle-Davis mouth gag* (Figs 19 A to D and 20B): The mouth gag remains open because of the ratchet. Tongue blades of various sizes are available. It opens the mouth and retracts the tongue anteriorly and inferiorly. Other types of mouth gags are also described in the previous section of this chapter.
  - Uses: It is used for the surgeries of the oral cavity, oropharynx and nasopharynx such as
    - Tonsillectomy
    - Adenoidectomy
    - Snoring surgeries like uvulopalatopharyngoplasty
    - Palatal surgeries
    - Repair of cleft palate
    - Excision of angiofibroma
    - Removal of antrochoanal polyp
    - Craniovertebral anomalies
  - Method: The built in tongue depressor along with the closed mouth gag is inserted in the mouth after depressing the lower jaw. The mouth gag is opened gradually. It is suspended from Draffin's bipods (Fig. 20A).
- *Tonsil holding forceps (Denis Browne's)*: They resemble Luc forceps, but their fenestrated jaws are not sharp. The upper jaw sits within the larger lower jaw. There are many other types of tonsil holding forceps (Figs 19F and 20E and F).
  - Use: These are meant for holding the tonsil during tonsillar dissection. Luc forceps may also be used for this purpose.
- *Tonsil knife* (Fig. 19G): This sickle-shaped knife is made of stainless steel.
  - Use: It is used for making the first mucosal incision of tonsillectomy at the upper pole of tonsil.
- *Tonsil dissector and anterior pillar retractor* (Fig. 19I):
  - Uses:
    - Tonsillectomy: Its blunt end is used to dissect the tonsil. Its C-shaped bent end is used to retract the anterior tonsillar pillar and helps in inspecting the

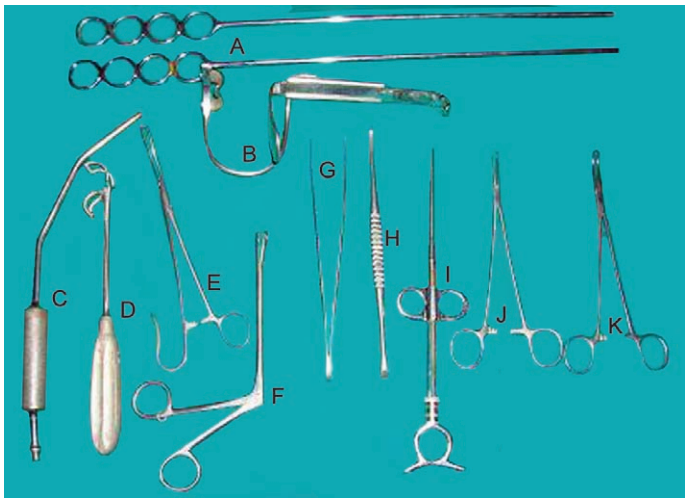


Figs 19A to P: Instruments of adenotonsillectomy. (A) Frame of Boyle Davis gag with interchangeable inserts suitable for many tongue depressors; (B) Russel Davis tongue blade with a groove for holding endotracheal tube, and protecting it from moving, kinking, or closure by teeth; (C) McIvor tongue blade; (D) Gast tongue blade with groove for endotracheal tube and lateral slot; (E) Sluder-Ballenger tonsillotome with blade and handle; (F) White tonsil holding forceps; (G) Abraham tonsil knife; (H) Henke Tonsil dissector; (I) Hurd tonsil dissector and pillar retractor; (J) Tonsil scissor; (K) Negus tonsil artery forceps; (L) Hurd ligature needle; (M) Yankauer suction tube; (N) Eves tonsil snare; (O) Corwin tonsil hemostats with spring clamp and sliding collar for holding gauze pad in position; (P) Hartmann tonsil punch forceps
Source: Aesculap, Germany

fossa for any bleeding. The retraction of anterior pillar also provides better visualization of tonsil especially fibrosed tonsil before surgery so that tonsil holding forceps can be applied properly.

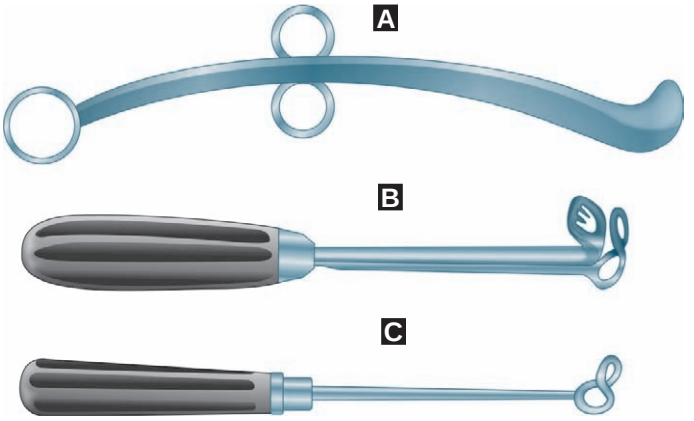
- Nasopharynx: It can be used in retracting the soft palate and uvula during adenoidectomy and for examining the nasopharynx.

- *Tonsil scissors* (Fig. 19J): These are long scissors with slightly bent blunt tips.
  - Use: They are used for sharp dissection of the tonsils and cutting the ligatures. They can also be used for tongue tie release and uvulectomy for long uvula.



Figs 20A to K: Instruments of adenotonsillectomy. (A) Draffin suspension apparatus (two bipods); (B) Boyle Davis mouth gag with tongue depressor; (C) Yankauer's tonsil suction tube; (D) St. Clair Thomson's adenoid curette with guard; (E) Peacock Vulsellum tonsil holding forceps; (F) Tonsil holding forceps; (G) Waugh's tenaculum tonsil dissection forceps with teeth; (H) Tonsil dissector and retractor; (I) Eves tonsil snare; (J) Straight tonsil artery forceps; (K) Curved tonsil artery forceps

- *Negus artery forceps* (Fig. 19K): Its tip is sharply curved.
  - Use: It is used as replacement forceps to ligate the bleeding point.
- *Tonsil needle* (Fig. 19L): It consists of needle and handle. The sharp curved needle is at a right angle to the long handle.
  - Use: It is used for sewing the tonsillar pillars together for controlling the bleeding, which is not controlled by ligation and cauterization of bleeding points.
- *Yankauer's suction tube* (Figs 19M and 20 C): This long suction cannula consists of large handle and covered tip that prevents damage to mucosa.
  - Uses: It is used for suction in tonsillectomy and other oral, oropharyngeal and nasopharyngeal operations.
- *Eve's tonsil snare* (Figs 19N and 20I):
  - Use: When the tonsil has been dissected till the lower pole, snare is passed round the tonsil to engage the pedicle. When firmly closed, it crushes and cuts the pedicle and minimizes the bleeding.
  - Method of application: The index and the middle fingers are passed into the two rings on the outer tube of the snare, and the thumb is introduced in the ring of the central movable slide. The loop of the fully opened snare wire is threaded on the tonsil holding forceps. The snare wire loop is pushed down up to the lower pole of the tonsil. On closing by pushing the slide into the tube with the thumb, the snare wire loop is withdrawn into the tube of the snare and the tonsil is excised.
- *Conchotome* (Fig. 19P): This punch biopsy forceps open like the jaws of a crocodile and is bent at an obtuse angle. The upper smaller ring sits inside the larger lower ring.
  - Uses: It is used for punching out pieces of the turbinates. It is also used for removing the tags of adenoids and taking punch biopsy.
- *Draffin's bipod* (Fig. 20A): There are two pods and each has four rings in a row. The two pods are assembled together as



Figs 21A to C: Instruments of adenoidectomy. (A) LaForce adenotome; (B) St. Clair-Thompson adenoid curette with catcher; (C) Beckmann adenoid curette
Source: Aesculap®, Germany

per the height at which the tongue blade of the Boyle-Davis mouth gag is suspended. The lower ends of the pods are placed in one of the several depressions of the Magauran's plate.
- Use: They are used along with Boyle-Davis mouth gag.

- *St. Clair Thomson's adenoid curette* (Figs 20D and 21): There are two varieties: (1) with cage (guard); and (2) plane without cage (guard). The curette shaves off the adenoid mass while the guard holds the adenoid tissue and prevents from slipping. The sharp transverse blade cuts adenoids and holds them in a cage with the help of the fangs in the cage. Remaining tags of adenoids may be removed by a smaller plain adenoid curette, Luc forceps or conchotome.
  - Use: For adenoidectomy it is held in dagger holding fashion. The plane curette is used to remove adenoid tissue around the Eustachian tube.
  - Caution: The neck flexion avoids subluxation of atlanto-occipital joint. Bring the adenoid tissue from lateral wall to midline by right index finger. Keep the curette with cage in midline.
- *Tonsil dissection forceps with teeth (Waugh's)* (Fig. 20G)
  - Use: It is used for putting incision in the mucous membrane and dissection of tonsil.
- *Tonsil artery forceps (straight and curved)* (Figs 20J and K):
  - Use: The straight forceps is used to catch the bleeding point and is replaced by curved forceps before tying a ligature.
- *Negus Knot tyer*: It has a blunt forked end and slips the ligature knot beyond the curved tip of the artery forceps.
  - Use: It helps in tying the ligature knot up to the tip of curved artery forceps that holds the vessel.

INCISION AND DRAINAGE OF QUINSY

- *Peritonsillar (Quinsy) abscess forceps* (Fig. 22): These bayonet-shaped quinsy forceps have a sharp trocar point with a shoulder that prevents deep entry. See chapter "Deep Neck Infections" for other related details.
  - Use: It is used for drainage of peritonsillar abscess.
  - Method: The sharp trocar tip of closed forceps is inserted into the abscess and then forceps are opened like a sinus forceps to drain the pus.

ENDOSCOPES

There are two types of illumination in the rigid hollow tube scopes, Jackson and Negus. The illumination is distal in Jackson type and proximal in Negus type (Table 1).

Laryngoscopes

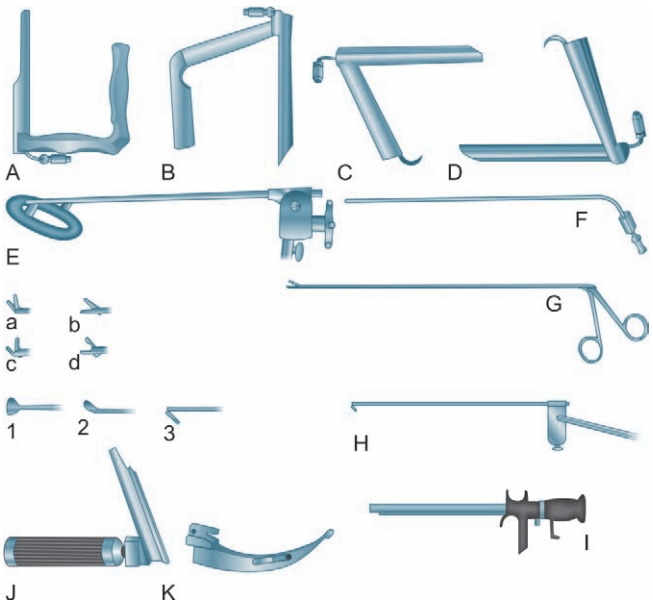
Laryngoscope used for direct laryngoscopy has light carrier which is connected to a cold light source through a flexible cable. There are several models of laryngoscopes (Figs 23A to K). The size of laryngoscope is selected as per the age of the patient. The larynx and hypopharynx are visualized directly with the help of laryngoscope. For further related details, see chapter Endoscopy.

- *Direct laryngoscope (Jackson)* (Fig. 23A): This U-shaped laryngoscope consists of a metal tube with illumination at the distal end (through which laryngoscopy is performed) and a right-angled handle.
  - Uses: They are used for direct laryngoscopy. Some laryngoscopes have a detachable posterior blade for inserting a bronchoscope or an esophagoscope.
- *Laryngoscopes for microlaryngoscopy and surgery*: These self-retaining laryngoscopes (Figs 23B to D) are fixed on the chest by a chest piece (Fig. 23E) so that hands of the surgeon remain free for the surgery, which is performed under the magnification of an operating microscope. Microlaryngeal surgery needs special laryngoscopes, forceps, scissors, dissectors and knives (Figs 23F to H). The shape and size of viewing and distal ends and body vary with the types of laryngoscopes, such as Holinger hourglass, Jako-Cherry, Bouchayer and Dedo.
  - Advantages: Magnification provides precision and ideal for Laser surgery. Both the hands of surgeon remain free for surgical manipulations.

TABLE 1 Comparison between two types of illumination (Jackson and Negus) in traditional hollow tube rigid scopes

	Jackson pattern	*Negus pattern*
Location of illumination	Distal	Proximal
Brightness	Less bright	More bright
Number of illuminants	One	Two
Visibility of forceps tip	Good	Poor
Chances of secretions covering illuminant	Frequent	Occasional
Width of scope	Narrow	Broader
Vision	Relatively poor	Better
Introduction	Easy	Relatively difficult

Fig. 22: Peritonsillar abscess forceps
Source: Aesculap®



Figs 23A to K: Laryngoscopic instruments. (A) Jackson laryngoscope; (B) Holinger anterior commissure laryngoscope; (C) Kleinsasser operating laryngoscope; (D) Stange hour glass operating laryngoscope; (E) Ricker Kleinsasser laryngoscope holder and chest support for adult and children; (F) Suction tube; (G) Kleinsasser laryngeal forceps: angled scissor (a), straight alligator (b), angled (c) straight (d) cupped jaws; (H) Handle for use with Kleinsasser laryngeal instruments: straight (1) curved (2) laryngeal knife (3) blunt laryngeal hook (3); (I) Hopkins lateral Telelaryngo-pharyngoscope 90^0; (J) Magill folding laryngoscope; (K) McIntosh folding laryngoscope
Source: Karl Storz, Germany

Bronchoscope

Figures 24A to L show the instruments used for bronchoscopy and esophagoscopy. Bronchoscope (Fig. 24A) consists of a hollow metal tube. Jackson type has distal illumination. There are openings (vents) at the distal part of the tube for the aeration of the side bronchi. The size of bronchoscope should be selected as per the age of the patient (Table 2). A bronchoscope may be used for performing esophagoscopy but the esophagoscope cannot be used for bronchoscopy. For further related details of bronchoscopy, see chapter "Endoscopy".

Esophagoscope

Figures 24A to L show the instruments used for bronchoscopy and esophagoscopy. Esophagoscope (Fig. 24K) appears similar to the bronchoscope but does not have vents. It is used for diagnostic and therapeutic esophagoscopy. The size of the esophagoscope is selected as per the age of the patient. Handle at the proximal end of esophagoscope indicates the direction of the bevel at the distal end. A bronchoscope may be used for performing esophagoscopy but the esophagoscope cannot be used for bronchoscopy. For further related details of esophagoscopy, see chapter "Endoscopy".

Endoscope and Instruments (Figs 24A to L)

- There are variety of shapes, diameter and lengths of endoscopes. Smallest being 3.0 mm inner diameter and 20 mm length.
- The thick and smooth distal tip of esophagoscope allows easy insertion and decreases chances of injuring mucosa.
- One or two smaller side channels are for suction catheters or light carriers.
- The large central channel allows passage for variety of instruments.
  - Hopkins road for better illumination and visualization (Figs 24C and D)
  - Foreign body removal forceps [Figs 24E (a, b, e and f)]
  - Biopsy forceps [Figs 24E (b and c)]
  - Foreign body basket [Fig. 24E (d)]
  - Suction catheters (Fig. 24F)
  - Scissors
  - Staplers
  - Dilators
  - Electrocautery
  - Laser cables
  - Syringes
- Oval open esophagoscope: For examination and removal of foreign bodies.
- Round open esophagoscope: For tumors and strictures.

TRACHEOSTOMY

- *Tracheal dilator* (Fig. 25, TD): The blades spread out on approximating its rings. Its tip is blunt. A curved artery forceps can also serve its purpose.
  - Use: It keeps the cut tracheal edges open so that tracheostomy tube can be easily introduced.
- *Tracheal hook (blunt and sharp)*
  - Blunt: It retracts the thyroid isthmus and exposes the trachea. It may also be used for retracting the strap muscles.
  - Sharp: It retracts the lower border of cricoid cartilage and thus stabilizes the trachea when making incision in the anterior wall of trachea.

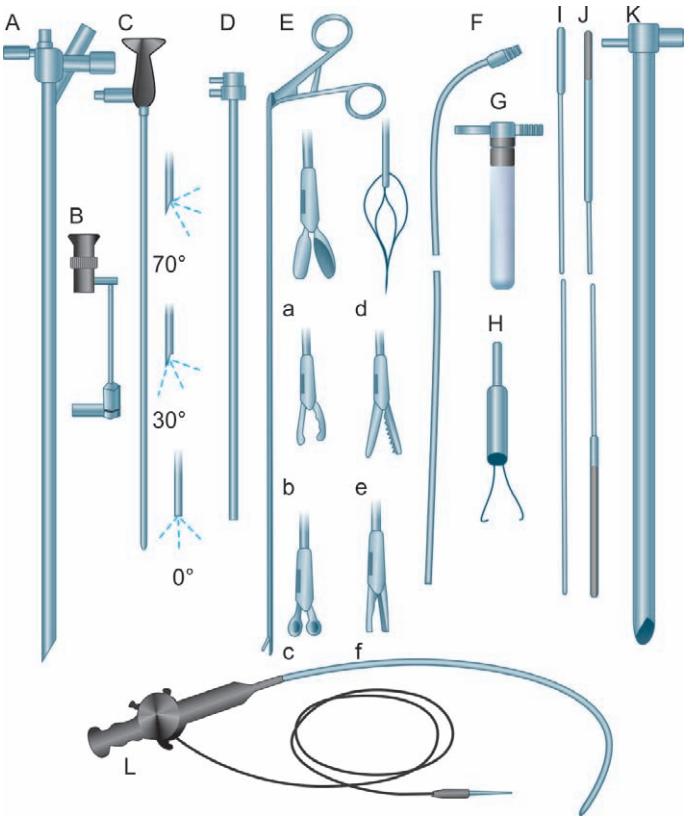
Types of Tracheostomy Tubes (Figs 25 and 26)

There are various types of tubes. See Box 1 for the classification of tracheostomy tubes.
They can be grouped on the bases of cuff, fenestra, length, number of lumen and the material. Some of the commonly used tubes will be described in this section.

- *Fenestra at the upper curvature of the tube*: It helps in speech production or in weaning from tracheostomy.
- *Extra length tracheostomy tubes*: They are used in cases of thick or swollen pretracheal tissue, a growth and stenosis in trachea, postcricoid growth pressing trachea.

TABLE 2 Size of bronchoscope and the age of patient

Age group	*Size of bronchoscope lumen (mm)*
Preterm neonates	2.5–3.0
1–2 years	3.5–4
3–9 years	4.5–5
9–14 years	6



Figs 24A to L: Bronchoesophagoscopy instruments. (A) Bronchoscope tube; (B) Adjustable magnifier; (C) Hopkins telescope: 0°, 30°, and 70°; (D) Rubber telescope guide; (E) Forceps: peanut grasping (a), universal biopsy and grasping (b), circular cup biopsy (c), foreign body basket (d), alligator grasping (e), and rotating sharp pointed (f); (F) Rigid suction tube; (G) Specimen collector may be attached directly to suction tubes; (H) Sponge holder for sterile smear cytology; (I) Cotton carrier; (J) Bougie; (K) Oval esophagoscope tube; (L) Broncho fiberscope with instrument channel
Source: Karl Storz, Germany

- *Adjustable flange long tube*: The flange can be adjusted.
- *Single lumen (cannula) tube*: It has only one cannula.
- *Double lumen (cannula) tube*: It has inner cannula and outer cannula.
- *Metallic tubes*: They are formed from the alloy of silver, copper and phosphorus.
- *Polyvinyl chloride (PVC) tubes*: They are disposable (single use) and thermolabile. They adjust to tracheal lumen.
- *Silicone tubes*: The advantages include less crusting and no adherence of bacteria and secretions.
- *Siliconized PVC tubes*: They offer advantages of both PVC and silicon tubes.
- *Silastic tubes*: They are soft and non-irritating and minimize the crusting.
- *Rubber tubes*: They are most economical.
- *Armored tubes*: The plastic tubes are reinforced by a spiral or rings of stainless steel. They do not get kinked.

Fuller's Bivalved Tracheostomy Tube

It has two tubes outer and inner (Figs 25, F1 and F2). The inner tube is slightly longer than outer tube. The two blades (bivalve) of the outer tube when pressed together can be easily inserted into the tracheostomy opening. There is no need of a pilot and a tracheal dilator. The central hole in the inner tube provides a chance to breathe when tube is blocked by a finger, cork or secretions. The tip of the compressed outer tube is not so blunt and may cause injury.

Box 1: Classification of tracheostomy tubes

- On the basis of cuff
  - Uncuffed
  - Cuffed tubes
    - Single cuff tube
    - Double cuff tube
    - Low pressure cuff tube
- On the basis of fenestra at the upper curvature of the tube
  - Tubes without fenestra
  - Single fenestrated tube
  - Multiple fenestrated tube
- On the basis of length of the tube
  - Standard length
  - Extra length tracheostomy tube
    - Adjustable flange long tube
- On the basis of number of lumens (cannula)
  - Single lumen (cannula) tube – Nonmetallic
  - Double lumen (cannula) tube – Jackson and Fuller
- Suction-aided tracheostomy tubes – Metallic
- On the basis of the material
  - Metallic
    - Jackson
    - Fuller
  - Nonmetallic
    - Polyvinyl chloride (PVC)
    - Silicone
    - Siliconized PVC
    - Silastic
    - Rubber tube
  - Mixed
    - Armored tubes

Chevalier Jackson's Tracheostomy Tube

Traditionally, it is made of silver or German silver which is nonirritating. It has three parts: outer tube, inner tube and an obturator (Figs 25, J1 to J3).

1. *Outer tube*: Outer tube fits into the tracheostomy opening. When the inner tube is removed, the outer tube continues to serve the purpose. Shield is attached to the proximal end of outer tube and is fixed to the neck by a tape threaded through the holes of the shield. Lock fitted on the shield fixes the inner tube inside the outer tube.
2. *Inner tube*: The inner tube is slightly longer than outer tube. It can be fixed to the shield of the outer tube by a lock. The longer inner tube does not allow the blocking of outer tube and can be cleaned and replaced regularly.
3. *Obturator (pilot)*: The obturator is passed in the outer tube and helps in the introduction of tube into the trachea. Its blunt rounded end is inserted into the outer tube after removing the inner tube. It is used for inserting the tracheostomy tube into the trachea. When tube is inserted in its position, pilot is withdrawn and inner tube is inserted back into the outer tube.

Metallic tubes cannot be used during radiotherapy and MRI. Metal tubes become radioactive by irradiation. Patient cannot speak with patent tube in position. The speaking valve can be used with metallic tubes.

Nonmetallic Tracheostomy Tubes (Fig. 26)

These tubes are usually larger than the metallic tubes and are available with or without pressure cuff. These tubes are useful for patients undergoing radiation therapy (metal tubes become radioactive by irradiation) and MRI.

- *Cuffed tracheostomy tube*: The external bulb in cuffed tubes tells the status of inside cuff.
  - Indications: Cuff tracheostomy tubes are required when patients need anesthesia or Intermittent Positive Pressure Respiration. It is used in unconscious patient or when patient is on a respirator.
  - Advantages: The inflated cuff prevents not only aspiration of secretions into the trachea but also prevents air-leak and keeps the position in tube.
  - Precautions: Cuff should be deflated every 2 hours for 5 minutes to prevent pressure damage to the trachea.

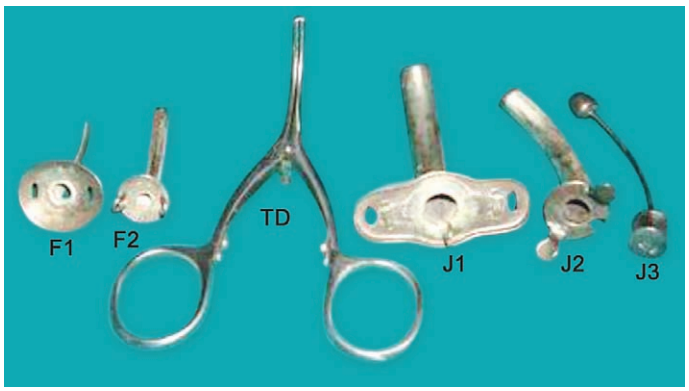


Fig. 25: Metallic tracheostomy tubes and tracheal dilator (TD). Fuller's bivalved tracheostomy tube has two tubes outer two blades (bivalve) tube (F1) and inner fenestrated tube (F2). Jackson's tracheostomy tube has three parts: outer tube (J1), inner tube (J2) and an obturator (J3)

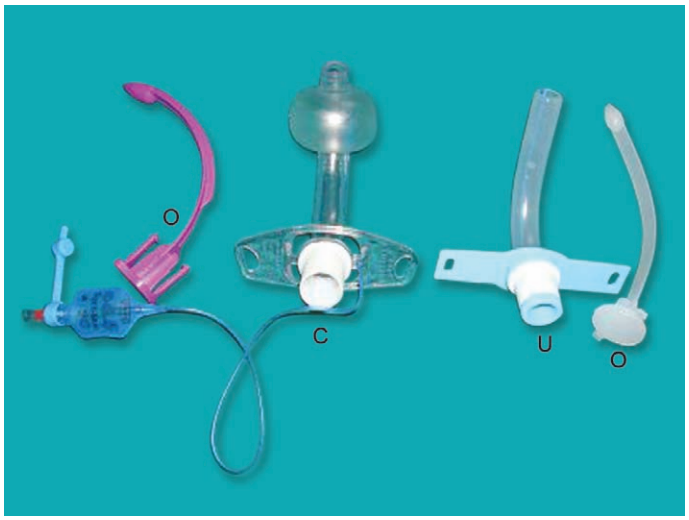


Fig. 26: Nonmetallic tracheostomy tubes. Cuffed (C) and Uncuffed (U) tubes with their obturator (O)

In the current two cuffs tube, inflation of the cuff can be alternated. The current low pressure cuff tubes may not damage the trachea and avoids tracheal stenosis.

- *Cuffed suction aid tracheostomy tube*: The inbuilt suction channel reaches above the cuff and helps in sucking out pharyngeal secretion collected above the cuff.

Suction is done before deflating the cuff. It avoids the aspiration of accumulated pharyngeal secretions into the trachea.

Nonmetallic tracheostomy tubes are useful for patients undergoing radiation therapy and MRI. They are less traumatic to trachea.

Size of Tracheostomy Tube

Tracheostomy tubes are of different sizes or numbers. Larger the number, greater is the inner diameter. In adults, tubes of inner diameter varying between 6 mm and 9 mm or 10 mm are used. Sometimes, size of tube (Jackson's or Negus) is expressed in French gauge (FG), which is about three times the size of internal diameter. It means 8 mm internal diameter will have approximately 24 FG.

FG = outer diameter x π (π = 3.14 or approx 3)

- *Children*: An appropriate size of tracheostomy tube should be selected as per the age of the patient (Table 3).

AIRWAY DEVICES (FIGS 27A TO H)

Nasopharyngeal Airway (Trumpet) (Fig. 27A)

It is inserted into the hypopharynx through the nose and provides better patent airway from nose to laryngopharynx. In conscious patients it is better tolerated than oropharyngeal airway.

Oropharyngeal Airway (Figs 27B to D)

This may be plastic or metallic. It brings the base tongue forward and widens the oropharynx. The ventilation can be provided with face mask or Ambu bag.

Laryngeal Mask Airway (Fig. 27E)

Laryngeal mask airway consists of a tube and a distal end triangular laryngeal mask which fits over the laryngeal inlet in supraglottic region. The weight of patient decides the size of mask.

- *Uses:* It is very useful in cases of unsuccessful intubation when standard mask ventilation is inadequate.
- *Method*: The deflated cuff of mask is positioned over the larynx and then inflated. It is useful when face mask is ineffective and intubation is difficult.
- *Disadvantages*: Laryngeal mask airway does not prevent aspiration of gastric secretions. It cannot be used in patients with trismus.
- *Contraindication*: It is contraindicated in cases of laryngeal obstruction.

Endotracheal Tubes (Figs 27G and H)

The endotracheal intubation is a quick method of establishing airway. For the detail procedure, see chapter "Anesthesia".

- *Size of endotracheal tube in children*: Roughly the size of endotracheal tube is equal to the size of the child's little

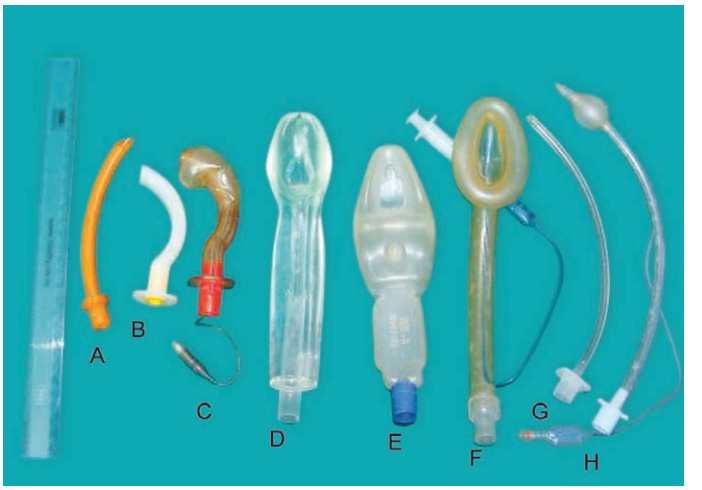
TABLE 3 Size of tracheostomy tube and the age of patient

Age group	*Tracheostomy tube size lumen (mm)*
Preterm neonates	2.5–3.0
1–2 years	3.5–4
3–6 years	4.5–5
6–12 years	5.5–6
12–14 years	7
Adults	8–9

finger but it can be calculated with the following formula:

- Size (number) of tube = (Age/4) + 4. It indicates internal diameter in mm.
- Length of the tube in mm = Size (number) of tube x 3.
- Example: If the patient is 2-year-old, then the size (number) of the tube = (2/4) +4 = 4.5, internal diameter 4.5 mm, and length of tube = 4.5 x 3 = 13.5 mm.

Specialized endotracheal tubes (RAE tubes) have a sharp bend to angle the tube so that surgeon's field of vision is not obstructed during oral cavity and oropharyngeal surgeries.



Figs 27A to H: Airway devices. (A) Nasopharyngeal airway; (B) Guedel oropharyngeal airway; (C) Cuffed oropharyngeal airway; (D) Sanjivani airway management oropharyngeal airway; (E) Laryngeal mask airway; (F) I-Gel Laryngeal airway; (G) Endotracheal uncuffed tube; (H) Endotracheal cuffed tube
Source: Dr AS Solanki, Consultant Anesthesiologist, Anand, Gujarat

60 Diagnostic Imaging

Adjustment or adaptation is the outcome of the self pitted against external forces tending to suppress it. He who adjusts himself best lives the longest. Even if I do not preach this, society is changing, it must change.

—Swami Vivekananda

Points of Focus

- **CONVENTIONAL RADIOLOGY**
  - ✓ Temporal Bone: Law's View, Schullar's View, Stenver's View, Towne's View, Transorbital View, Submentovertical View
  - ✓ Nose and Paranasal Sinuses: Water's View, Caldwell View, Lateral View, Right and Left Oblique Views, Lateral and Occlusal Views of Nasal Bones
  - ✓ Neck, Larynx and Pharynx: Lateral View and Antero-posterior Views of Neck, Soft Tissue Lateral View Nasopharynx, Submandibular Salivary Gland
- **ORTHOPANTOMOGRAM**
- **ULTRASOUND**
- **COMPUTERIZED TOMOGRAPHY**
- **MAGNETIC RESONANCE IMAGING**
  - ✓ Principle and Machines
  - ✓ Imaging Protocols
- **RADIONUCLIDE IMAGING**
- **INTERVENTIONAL RADIOLOGY**
- **APPLICATIONS OF CT, MRI AND US**
  - ✓ Comparison of CT and MRI
- **CT ANATOMY OF EAR, NOSE, THROAT, HEAD AND NECK**
- **CLINICAL HIGHLIGHTS**

The increased availability of new imaging modalities are replacing the conventional radiology and making the accurate diagnostic process easier. It is the key to good otolaryngology practice. ENT surgeon must be able to recognize the most appropriate test to fit the clinical context. Though the simple and least expensive imaging [conventional X-rays and ultrasound (US)] should be preferred but in complex problems, more expensive scans [Computerized tomography/Magnetic resonance imaging (CT/MRI)] are more cost-effective as they lead to early and confident diagnosis and management. If frequent imaging studies are expected, then nonradiation-dependent modalities (MRI and US) must be preferred as they avoid the adverse effects of the radiation (especially in children and young) such as cancer and genetic defects.

CONVENTIONAL RADIOLOGY

X-rays are absorbed differently by soft tissue, bone, gas and fat. X-ray of chest is requested commonly for preanesthetic assessment to exclude tuberculosis, pulmonary metastases and bronchogenic carcinoma.

Temporal Bone (Figs 1 and 2)

- *Areas of translucency (semitransparent)*: The causes include cholesteatoma, mastoidectomy, malignancy, eosinophil granuloma, tuberculosis of mastoid, multiple myeloma, glomus and large antral and periantral mastoid air cells.
- *Law's view* (lateral view of mastoid): Sagittal plane of the skull is parallel to the film; X-ray beam is projected 15° cephalocaudal. Key areas of the mastoid (attic, aditus and antrum) are not seen well.
  - Structures seen: External auditory canal (EAC) (superimposed on internal auditory canal (IAC)), mastoid air cells, tegmen, lateral sinus plate and temporomandibular joint.
- *Schullar's view*: This frequently used view is similar to Law's view but X-ray beam is projected 30° cephalocaudal and prevents superimposition of two sides of mastoid bones.

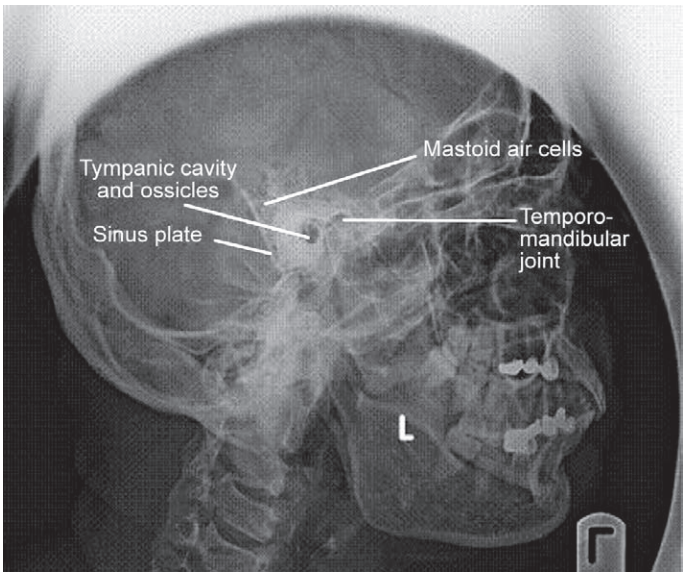


Fig. 1: X-ray mastoid left–Normal
Source: Dr Jayesh Patel, Consultant Radiologist, Anand, Gujarat

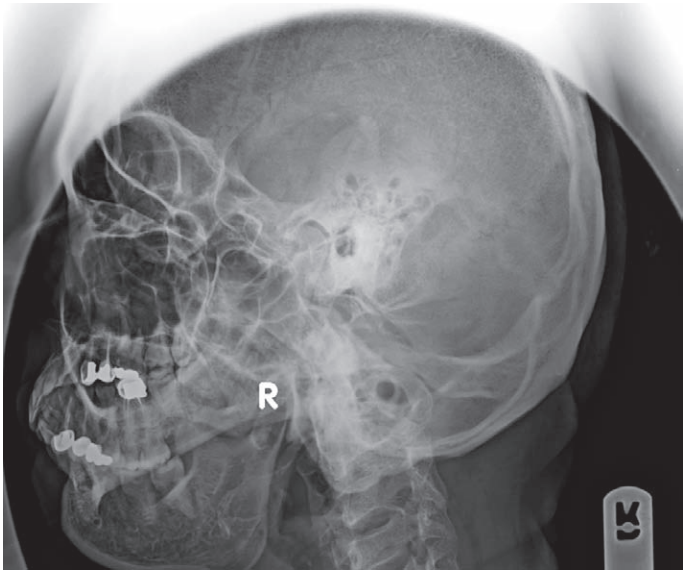


Fig. 2: X-ray mastoid right–Mastoiditis
Source: Dr Jayesh Patel, Consultant Radiologist, Anand, Gujarat

- Structures seen: EAC superimposed on IAC, mastoid air cells, tegmen, lateral sinus plate, condyle of mandible, sinodural angle and atticoantral region (key areas for cholesteatoma and its erosion).
- Clinical applications: Extent of pneumatization, sclerotic mastoid, destruction of intercellular septa (mastoiditis), location of sinus plate (position of sigmoid sinus) and tegmen (roof of middle ear and floor of middle cranial fossa), cholesteatoma and longitudinal fracture of petrous pyramid.

- *Stenver's view*: Long axis of the petrous bone lies parallel to the film.
  - Structures seen: Entire petrous pyramid, arcuate eminence, internal auditory meatus, labyrinth with its vestibule, cochlea and mastoid antrum.
- *Towne's view*: This is an anteroposterior view of skull with 30° tilt from above and in front. It shows both petrous pyramids, which can be compared.
  - Structures seen: Both side temporal bones, arcuate eminence and superior semicircular canal, mastoid antrum, IAC, tympanic cavity, cochlea and EAC.
  - Clinical applications: Acoustic neuroma and apical petrositis.
- *Transorbital view*: This is an anteroposterior view of skull. Orbitomeatal line is at right angles to the film. X-ray beam passes through the orbit.
  - Structures seen: IAC, cochlea, labyrinth and both petrous pyramids projected through the orbits.
  - Clinical applications: Acoustic neuroma and petrous pyramid.
- *Submentovertical view*: Vertex remains near the film and X-ray beam is projected from the submental area.
  - Structures seen: EAC, mastoid cells, middle ear, eustachian tube, IAC, foramen ovale and spinosum and carotid canal, and sphenoid, posterior ethmoid and maxillary sinuses (seen best in that order), zygoma, zygomatic arches, mandible along with coronoid and condyle processes.

Nose and Paranasal Sinuses

The most common view taken for paranasal sinus is Water's view and second one is Caldwell view.

- *Common clinical findings*: They include
  - Slight haziness in sinuses due to mucosal thickening
  - Presence of cyst (missing tooth in dentigerous cyst)
  - An opacity with horizontal level due to fluid in chronic sinusitis
  - Opacity due to sinusitis, antrochoanal polyp (maxillary sinus) and malignancy (opacity beyond the sinus limit due to erosion).
  - Dense opacity with regular outline (osteoma).
  - Multilocular swelling expanding the bone (osteoclastoma).
- *Water's view* (occipitomental view): Nose and chin touch the film and X-ray beam is projected from occipital side. Open mouth view shows sphenoid sinus. Petrous bones are projected below the maxillary sinuses. Fractures of right and left nasal bones and their lateral displacement can be seen.
  - Structures seen (Fig. 3): Maxillary (seen best), frontal and sphenoid sinuses, zygoma, zygomatic arch, nasal bones, frontal process of maxilla, superior orbital fissure and infratemporal fossa.
- *Caldwell view* (occipitofrontal view): Nose and forehead touch the film and X-ray beam is projected 15–20° caudally. Frontal and ethmoidal sinuses are seen well in this view.
  - Structures seen: Frontal, ethmoid and maxillary sinuses, frontal process of zygoma, zygomatic process of frontal bone, superior margins of orbits, lamina papyracea, superior orbital fissures and foramen rotundum.
- *Lateral view*
  - Structures seen (Fig. 4): Anterior and posterior extents of sphenoid, frontal and maxillary sinuses, sella turcica, ethmoid sinuses, alveolar process, condyle and neck of mandible.
  - *Right and left oblique views*: For seeing the posterior ethmoid sinuses and the optic foramen.
  - *Lateral views of nasal bones*: To see fracture line,

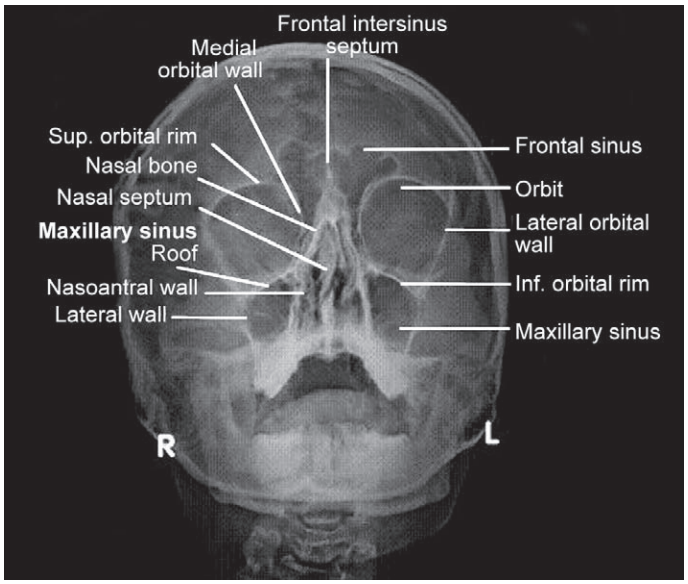


Fig. 3: X-ray PNS Water's view–Normal
Source: Dr Jayesh Patel, Consultant Radiologist, Anand, Gujarat

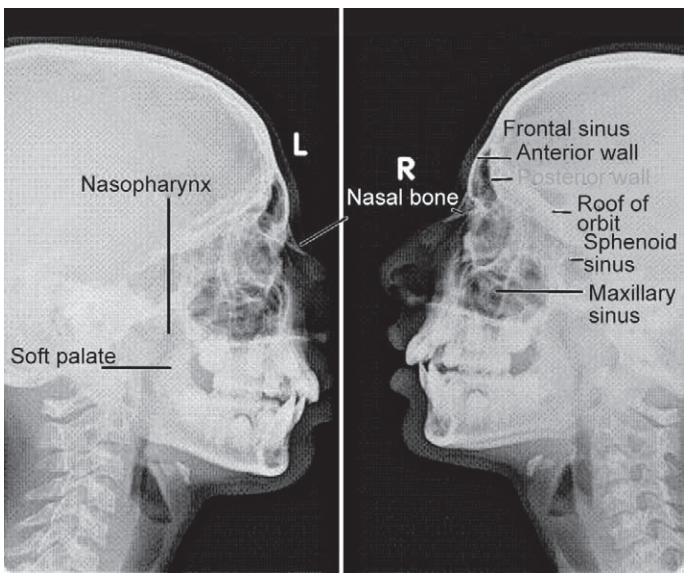


Fig. 4: X-ray nasal bones lateral view left and right showing fracture of septal cartilage
Source: Dr Jayesh Patel, Consultant Radiologist, Anand, Gujarat

depression or elevation of the fractures segment. Lower part of nasal bones, which is thin, fracture more frequently. Groove for ethmoidal nerve and vessels may look like fracture line (Fig. 4).

- *Occlusal view of nasal bone*: Film is held between the teeth and X-ray beam is projected perpendicular to the film. It shows fracture line and lateral displacement of the nasal pyramid clearly.

Neck, Larynx and Pharynx

- *Lateral view of neck (Fig. 5)*
  - Structure seen: Outline of base of tongue, vallecula, hyoid bone, epiglottis and aryepiglottic folds, arytenoids, false and true cords with ventricle in between them, thyroid and cricoid cartilages, subglottic space and trachea, prevertebral soft tissues, cervical spines and pretracheal soft tissues and thyroid.

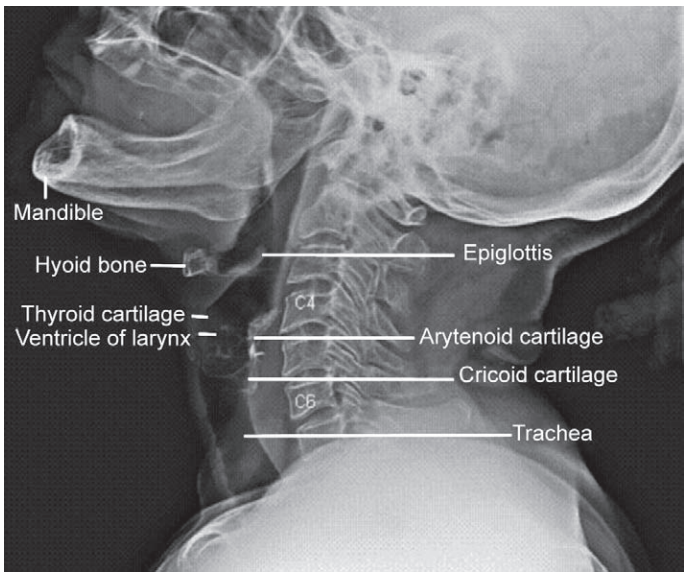


Fig. 5: X-ray soft tissue neck and nasopharynx lateral view
Source: Dr Jayesh Patel, Consultant Radiologist, Anand, Gujarat

  - Clinical applications:
    - Radio-opaque foreign bodies of larynx, pharynx and upper esophagus
    - Acute epiglottitis
    - Retropharyngeal abscess: Retropharyngeal widening with fluid level and straightening of cervical spines with or without foreign body and/or tuberculosis of cervical vertebrae
    - Position of tracheostomy tube, T-tube and laryngeal stent
    - Laryngeal stenosis
    - Fractures of larynx and hyoid bone and their displacement
    - Compression of trachea by thyroid or retropharyngeal masses
    - Caries of cervical spine associated with retropharyngeal abscess
    - Osteophytes in cervical vertebrae and injuries of spine.
- *Anteroposterior view of neck*: It helps in differentiating between a foreign body of larynx and esophagus (lateral view is also needed). It shows compression or displacement of trachea by lateral neck masses such as thyroid swellings.
- *Soft tissue lateral view nasopharynx*: For soft tissue masses in the nasopharynx, soft palate, roof and posterior wall of nasopharynx.
  - Clinical applications:
    - Adenoids
    - Angiofibroma: Soft tissue density arising from posterosuperior wall of nasopharynx and interrupting the airway.
    - Antrochoanal polyp: Soft tissue density with a column of air between the mass and posterior wall of nasopharynx.
    - Foreign body nose and tumor.
    - Choanal atresia: Interruption of air column from nose to nasopharynx.
- *Submandibular salivary gland*: Radio-opaque calculus can be seen.

Barium Swallow

For further detail, *See* "Barium Esophagography" in Chapter: "Pharyngeal Symptoms and Examination".

ORTHOPANTOMOGRAM

A pantomograph is a panoramic radiograph machine. It permits visualization of entire maxillary and mandibular dentition, alveolar arches and contiguous structures on a single extraoral film (Fig. 6).

ULTRASOUND

Ultrasound (US) requires an experienced operator. It is inexpensive, quick, reliable, noninvasive and effective investigation.

A transducer placed on skin produces high-frequency sound waves (3 MHz and 7 MHz), which are reflected back by body tissues. The echoes (echogenicity) generated are picked up back by the same transducer, which converts them into an image that is displayed in real-time on a monitor. Superficial lesions need high-frequency probe, whereas deeper lesions require lower-frequency probes. US allows differentiation between solid and cystic masses and assessment of margins and texture of neck swellings (Box 1 and Figs 7 and 8).

Ultrasound beam does not readily penetrate bone, cartilage and gas. It is attenuated as it passes through the tissues. These limitations make examination of fatty necks and deep structures (such as deep parotid lobe and parapharyngeal space) and staging of many head and neck cancers more difficult.

The probes, which can be combined with fiberoptic endoscopes, are available to visualize the esophagus wall.

- *Doppler ultrasound*: It measures blood flow of vessels.
- *Color Doppler*: Flowing blood appears either red or blue, which depends upon the blood direction, towards or away from the transducer.
- *Power Doppler*: It can demonstrate tissue perfusion.

COMPUTERIZED TOMOGRAPHY

Computerized tomography (CT) is now becoming readily available and fast replacing the need of plain X-rays (Figs 9 and 10). CT is quite commonly used to diagnose refractory cases of rhinosinusitis and stage head and neck malignancies.

- *Attenuation values* (Table 1): The attenuation values of tissues (gray-scale image) are related to water (0 HU). Tissue densities varies from + 1000 (bone) to - 1000 (air). Fat is about - 80 to - 100 HU, while calcium and bone are in the range of 100–400

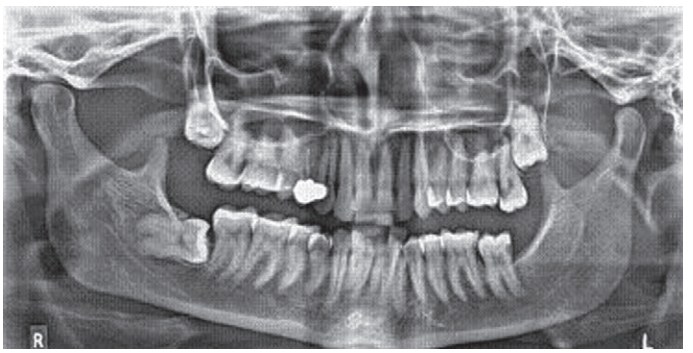


Fig. 6: Orthopantomogram: Impacted molars
Source: Dr Jaydeep Doshi, Consultant Radiologist, Anand, Gujarat

Box 1: Echogenicity of various tissues

- Fat: Moderate degree of internal echoes
- Skeletal muscle: Less echogenic than fat
- Solid mass: Well-defined margins with variable echogenicity but less echogenic than fat
- Cyst: Few internal echoes but strong echogenic back wall
- Calcium and bone: Strongly echogenic

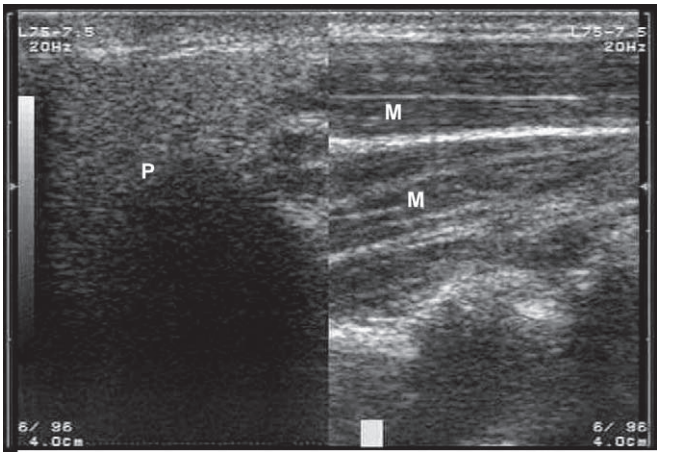


Fig. 7: USG neck showing echogenicity of parotid gland (P) and muscle (M)
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

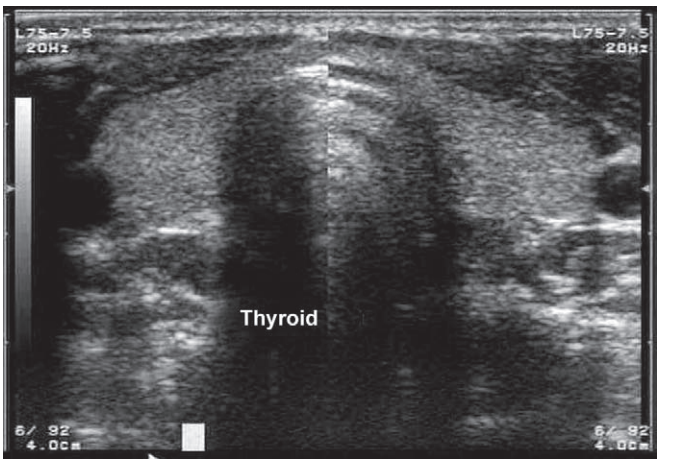


Fig. 8: USG neck showing echogenicity of thyroid gland
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

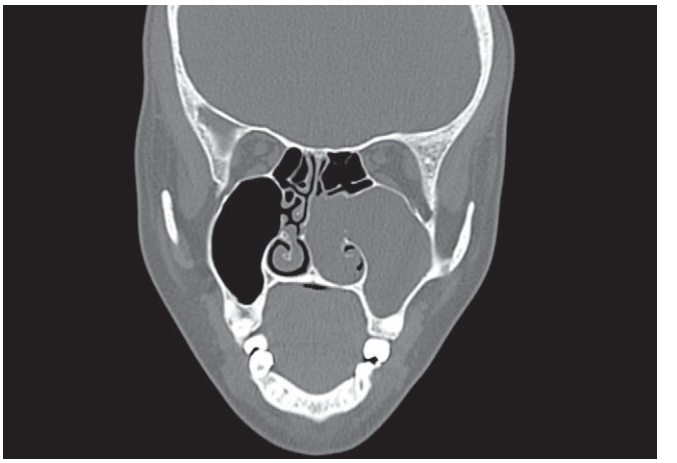


Fig. 9: CT scan plain coronal section nose and paranasal sinuses showing antrochoanal polyp
Source: Dr Amit Goyal, Shillong

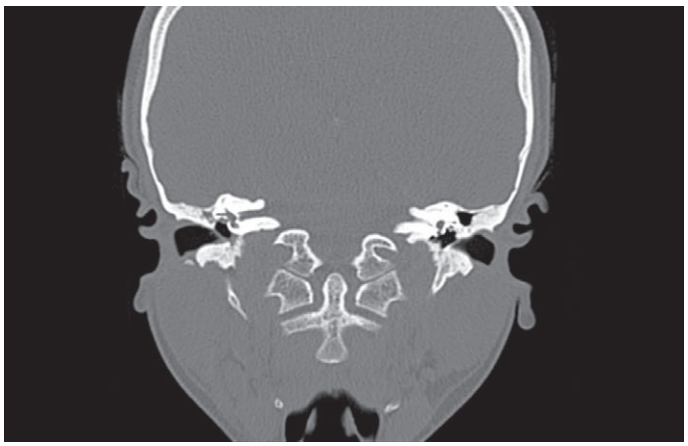


Fig. 10: HRCT temporal bones coronal sections. Contracted aditus of right side
Source: Dr Amit Goyal, Shillong

TABLE 1 Density levels of different types of structures

Average density in HU	*Structure*
250–1,000	Compact bone
130	Spongy bone
80	Clotted blood
70	Thyroid gland
65	Liver
55	Blood
45	Spleen/Muscle/Lymphoma
40	Pancreas
30	Kidney
25	Exudate/Effusion
18	Transudate
17	Suprarenal gland
00	Water
-15	Fat/Connective tissue
-90	Fat
-700	Lungs
-1,000	Air

HU. Most fluids are in the range of 0–30 HU. The density levels of all soft tissue organs except lungs lie within a narrow range between 10 HU and 90 HU.

Due to the high degree of over-lap between the densities (gray tones), it is difficult to differentiate between lymph nodes, muscles and vessels.

- *Window width/level*: By varying the range and centering of densities (window width/level) at a viewing console, an image appropriate to area of interest can be selected. So, CT scans can be displayed on soft tissue and bone settings (Box 2). Bone diseases and traumas are best visualized with bone detail algorithm.
- *Spiral CT*: Helical or spiral CT scans a volume of tissue and provides better quality images than the conventional CT. It covers more than 300 cm tissue during a single breathhold of 30 seconds.

Box 2: CT image display

- *Window width:* Range of attenuation values (HU), which make up the gray scale for a given image
- *Window level:* Center HU value (midpoint of densities) for the given window width (chosen for display)
- *Soft tissue windows:* Width: 250–400 HU; Level: 40–70 HU (equal to density of muscles). Display wider range of densities of calcification, intravenous contrast, muscle and fat
- *Bone windows:* Width: 2,000–4,000 HU; Level: 0– + 400 HU; Wide range of densities from cortical bone (+ 1,000 HU) to gas (- 1,000 HU). For paranasal sinuses and temporal bone
- *Brain windows:* Width: 80 HU; Level: + 40 HU (density of brain tissue); Density > 80 HU appear white while < 0 HU black. Intermediate densities are spread out evenly along the gray scale

- *Contrast CT*: Intravenous contrast agents allow identification of rim enhancement in pathological lymph nodes and increase the definition of primary tumors.
- *CT angiography*: The facilities, which enhance characteristics of tissues in both arterial and venous phases of imaging, are used in CT angiography.
- *Processing of volumetric data*: The volumetric data can be processed to produce
  - Multiplanar images: Sagittal and coronal
  - Three-dimensional (3D) images
  - Virtual endoscopy: Such as laryngoscopy, bronchoscopy and sinuscopy.

MAGNETIC RESONANCE IMAGING

High contrast resolution and multiplanar imaging capability make MRI ideal for intracranial, spinal and musculoskeletal imaging.

Principle

The protons in the nuclei of hydrogen behave like small spinning bar magnets and align with a strong external magnetic field. A radio frequency pulse knocks protons out of alignment, which release small amount of energy while gradually returning to their original position. This energy is detected by sensitive coils, which are placed around patient. Hydrogen atoms are abundantly present in body water and MRI can differentiate the water content of various tissues.

Machines

Clinical MRI machines operate at magnetic field strengths of between 0.5 and 2.0 tesla (T). One tesla (T) is equal to 10,000 Gauss (G). Higher the field strength of the magnet, higher the signal-to-noise ratio and higher is the quality of MRI. The open access magnets allow interventional procedures.

Imaging Protocols

The strength of signal depends on proton density and relaxation time. T1 is the time taken by proton to return to its original field and T2 is the time proton takes to de-phase.

Numerous imaging protocols are available but the commonly used pulse sequences are T1-weighted (T1W), T2-weighted (T2W), gadolinium-enhanced T1W, spin (proton) density, fat-suppressed and gradient echo imaging (Table 2). Fat suppression methods are T2W, short-tau inversion recovery (STIR), spectral presaturation inversion recovery (SPIR), and chemical shift selective presaturation (fat saturation).

- *T1W*: Because of high soft tissue discrimination, T1W images show exquisite anatomical details (Fig. 11).
- *T2W*: The pathological lesions increase T2 de-phase times, which produce higher signal than surrounding normal tissue in T2W images (Fig. 12).

  The combination of T1W and T2W images is good for characterizing fluid containing structures, solid components and hemorrhage.
- *Gadolinium-enhanced T1W*: Intravenous gadolinium (used in T1W) reduces T1 relaxation time and enhances lesions, which appear as high signal intensity areas (improved delineation of tumor margins relative to the lower signal of muscle, bone, vessel and globe). Gadolinium enhancement is optimally used with specific fat suppression techniques.
- *Short-tau inversion recovery*: The STIR sequence suppresses high signal intensity from fat (that turn fat black) and fluid containing structures remain high signal intensity. In STIR, decreased signal-to-noise ratio degrades the image.
- *Magnetic resonance angiography*: It uses specific sequences and demonstrates flowing blood.

RADIONUCLIDE IMAGING

Intravenously administered radiopharmaceuticals (radionuclides tagged to substances) such as technetium-99m (^{99m}Tc)-pertechnetate concentrate selectively in certain tissues and emit gamma radiation detected by a gamma camera. It provides two-dimensional display of physiological and functional changes in tissue (thyroid, parathyroid, salivary glands, brain and bone metastases). Single photon emission computed tomography (SPECT) and positron emission tomography (PET), which require special and dedicated set up, are not widely available.

- *Technetium-99m (^{99m}Tc)-pertechnetate scan*: In salivary glands, this scan is pathognomonic of Warthin's tumor and oncocytoma (Fig. 13).
- *I-123 scan*: It is used for thyroid uptake study to know whether nodule is hot or cold.
- Positron emission tomography: It is more sensitive and provides physiologic (blood flow and ischemia) and biochemical data (metabolism of DNA, glucose, amino acid). In the F-18 fluorodeoxyglucose (FDG) PET scan, a safe low dose of short-lived positron-emitting isotope of a glucose compound FDG is injected intravenously. Patient is asked not to take sugars or carbohydrates for the entire day prior to the scan. Patient observes complete fast (except water) after midnight for the morning PET scan.
- *Single photon emission computed tomography*: It provides 3D images.

INTERVENTIONAL RADIOLOGY

Ultrasound and CT guidance can be employed to choose shortest route from skin to the lesion. US, which is quick and flexible, allows needle path in real time without radiation hazard. Some of the many indications and techniques include:

1. Percutaneous biopsy
2. Drainage of abscess and fluid collection
3. Percutaneous gastrostomy
4. High local dose of chemotherapy to feeding vessels of tumor

TABLE 2 Differences between the images of various pulse sequences of MRI

	T1W	*T2W*	*Contrast T1W*	*STIR*
Fat	Bright or white	Low to int. signal	White	Complete black
CSF and vitreous	Black	Bright white	Black	White
Nasal mucosa	Low signal	White	White	White
Air, rapid blood flow and bone	Dark or black	Black	Black	Black
Muscles	Low to int. signal	Low to int. signal	Low to int. signal	Low to int. signal
Bone marrow	Bright	Low to int. signal	Bright	Black
Fluid secretions	Low signal	High signal	High signal	High signal
Viscus desiccated	Int. to high signal	Low signal	Low signal	Low signal
Mass	Low to int. signal	Higher signal	Higher signal	Higher signal
Best	Normal anatomy	Pathology lesions		
Slices	Few			6–8
Repetition time (TR)	Short (500–700 ms)	Long (2,000–4,000 ms)		
Echo time (TE)	Short (15–40 ms)	Long (50–90 ms)		
Signal-to-noise ratio	High			Low
Imaging time	Short			Long
Motion artifacts	Less			More

Int.: Intermediate; STIR: Short-tau inversion recovery

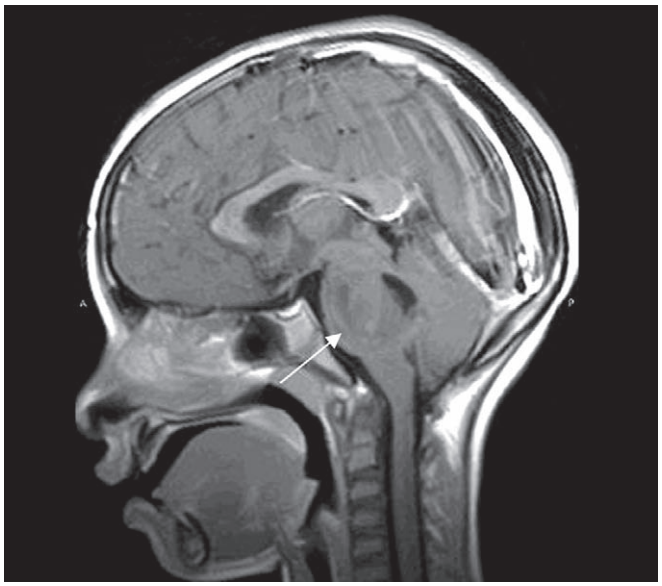


Fig. 11: MRI head sagittal section T1-weighted
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

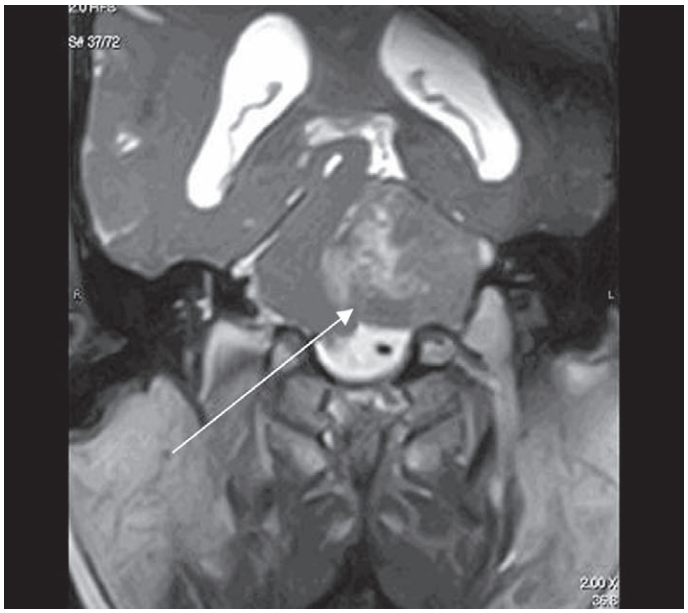


Fig. 12: MRI head axial section T2-weighted
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

5. Angioplasty and vascular stenting
6. Therapeutic embolization.

APPLICATIONS OF CT, MRI AND US

Though CT and MRI have their own advantages, disadvantages, characteristics (Table 3) and definite indications, yet in certain situations they prove to be not competing, but rather complementing to each other. Usually CT and MRI are performed before biopsy or resection of lesion.

CT

- Extension of mucosal tumors of suprahyoid neck (Fig. 14) and metastatic neck lymph nodes (ring enhancement)
- Postoperative neck
- Salivary gland tumors and metastatic neck lymph nodes
- Computed tomography sialography

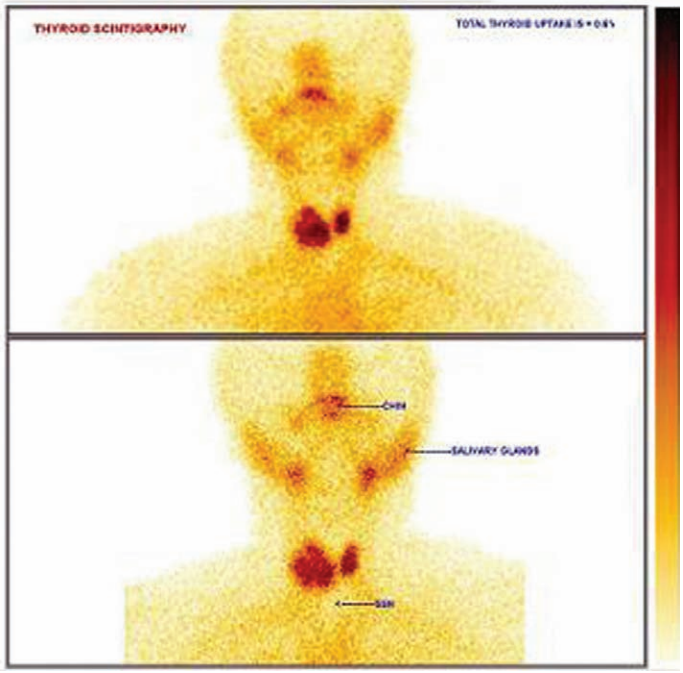


Fig. 13: Thyroid scan showing salivary glands

- Cervical lymphadenopathy
- Trauma, inflammation and cancer of larynx and laryngopharynx with metastatic neck nodes
- Large or fixed thyroid tumors invading and compressing larynx, laryngopharynx, trachea and mediastinum
- Paranasal sinuses prior to endoscopic sinus surgery, severe nasal polyposis, tumors
- Facial trauma (Fig. 15).
- Temporal bone and skull base tumors, semicircular canal fistulas, cochlear implants.

MRI

- Tumors of nasopharynx, oropharynx, oral cavity and tongue (Fig. 16)
- Extracapsular spread of tumor from nodes
- Perineural spread and extension beyond gland of salivary gland tumors
- Tumors of nose and paranasal sinuses: Distinguish between tumor and obstructed sinus secretions (hydrated fluid, viscous, desiccated); perineural spread to anterior cranial fossa, orbit, parapharyngeal space and pterygopalatine fossa and cribriform plate extension
- Lesions of IAC, facial nerve canal, and jugular foramen; acoustic schwannoma
- Skull base tumors.

CT and MRI temporal bone: Thin sectioned HRCT is ideal for looking temporal bone abnormalities or fractures, but MRI is gold standard for acoustic neuroma.

US

- Differentiating cystic from solid masses
- Metastatic lymph nodes of lower two-third of internal jugular vein, spinal accessory nerve, submental and submandibular triangles (more sensitive than CT)
- Tumor invasion of carotid vessels and internal jugular vein

TABLE 3 Comparison of advantages, disadvantages and characteristics of CT and MRI

	CT	*MRI*
Hazards ionizing radiations	Present	Absent
Spatial resolution	Good	Poor
Speed of acquisition	Fast with spiral CT	Slow
Dental amalgam artifacts	More	Less
Beam hardening artifacts from bony surfaces	Present	Absent
Motion artifacts: swallowing and respiration	Less chances	More chances
Eyelid mascara artifact	Absent	Present
Multiple pulse sequences	Absent	Present
Soft tissue/bone settings	Present	Absent
Plane of direct scanning	Axial, + coronal in skull	Multiplanar (axial, coronal, sagittal)
Ideal for	Bone lesions	Soft tissue lesions
Blood vessels	Seen with contrast	Seen without contrast
Availability	More	Less
Cost	Less	More
Claustrophobia	Less	More (less in open systems)
Direct imaging bone marrow	Poor	Good
Cortical and trabecular bone	Good delineation	Poor delineation
In metallic implants/foreign body	Can be done	Contraindicated
Contrast agent	Ionizing/nonionizing	Gadolinium
Appearance of bone	White	Black
Appearance of fat	Black	White

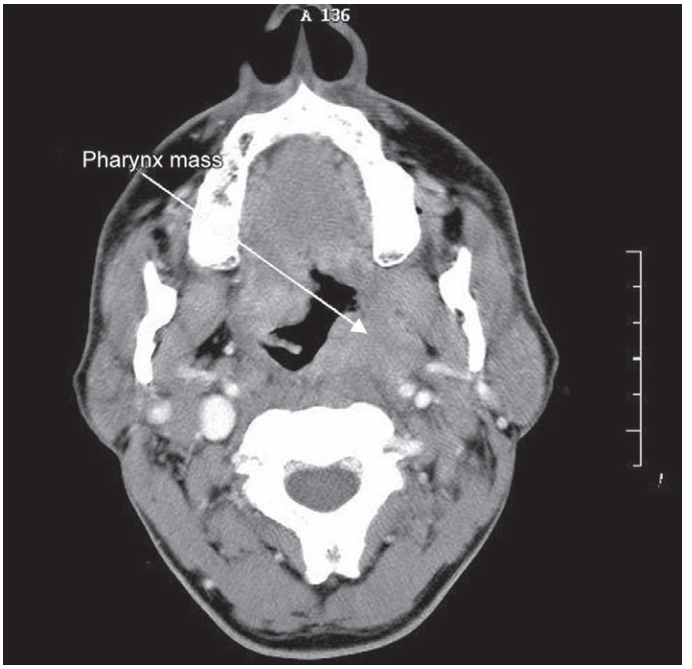


Fig. 14: CT scan neck axial section showing pharyngeal mass
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

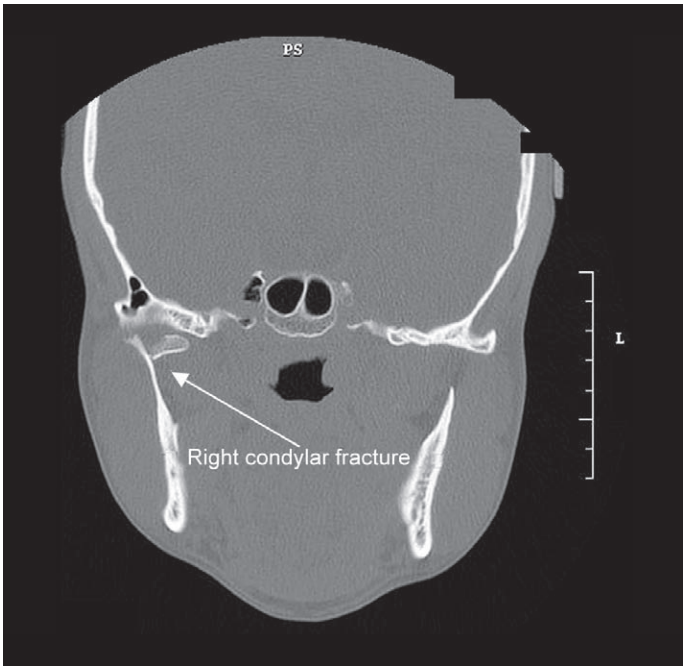


Fig. 15: CT scan head coronal section showing condylar fracture of mandible
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

(loss of echogenic fascial plane between vessel wall and tumor) better than CT/MRI
- Tumors of parotid and submandibular salivary glands
- Salivary duct stones even less than 2 mm
- Detection and drainage of salivary gland abscess under US guidance
- Masses of thyroid and parathyroid glands and US guided fine needle biopsy
- Neck lymphoma.

CT ANATOMY OF EAR, NOSE, THROAT, HEAD AND NECK

- Ear and temporal bone (Figs 17 to 20)
- Nose and paranasal sinuses (Figs 21 to 32)
- Oral cavity and oropharynx (Figs 33 and 34)
- Laryngopharynx and neck (Figs 35 to 37)
- Larynx and neck (Figs 38 and 39)

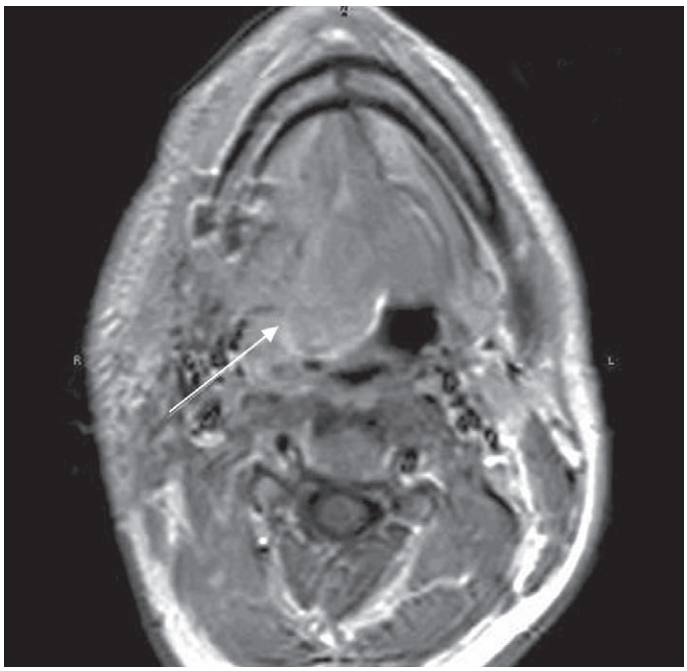


Fig. 16: MRI neck showing tongue lesion
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

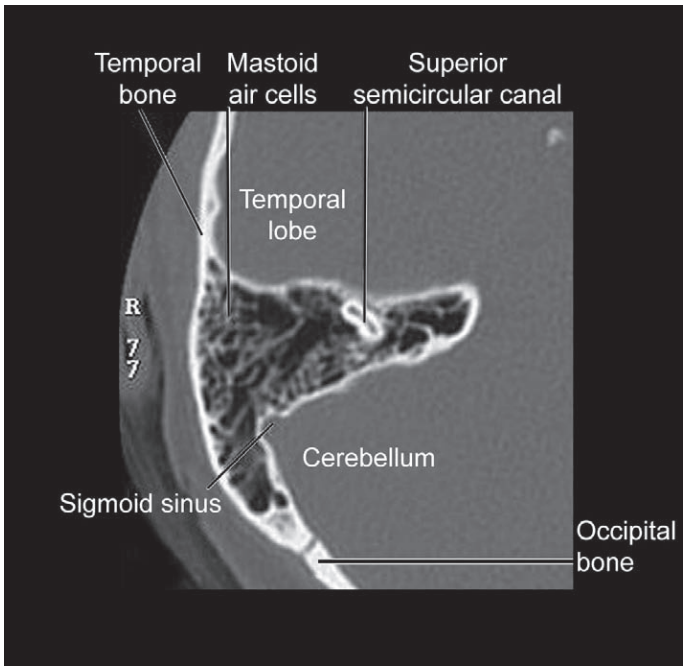


Fig. 17: CT scan axial section. Pneumatization of temporal bone

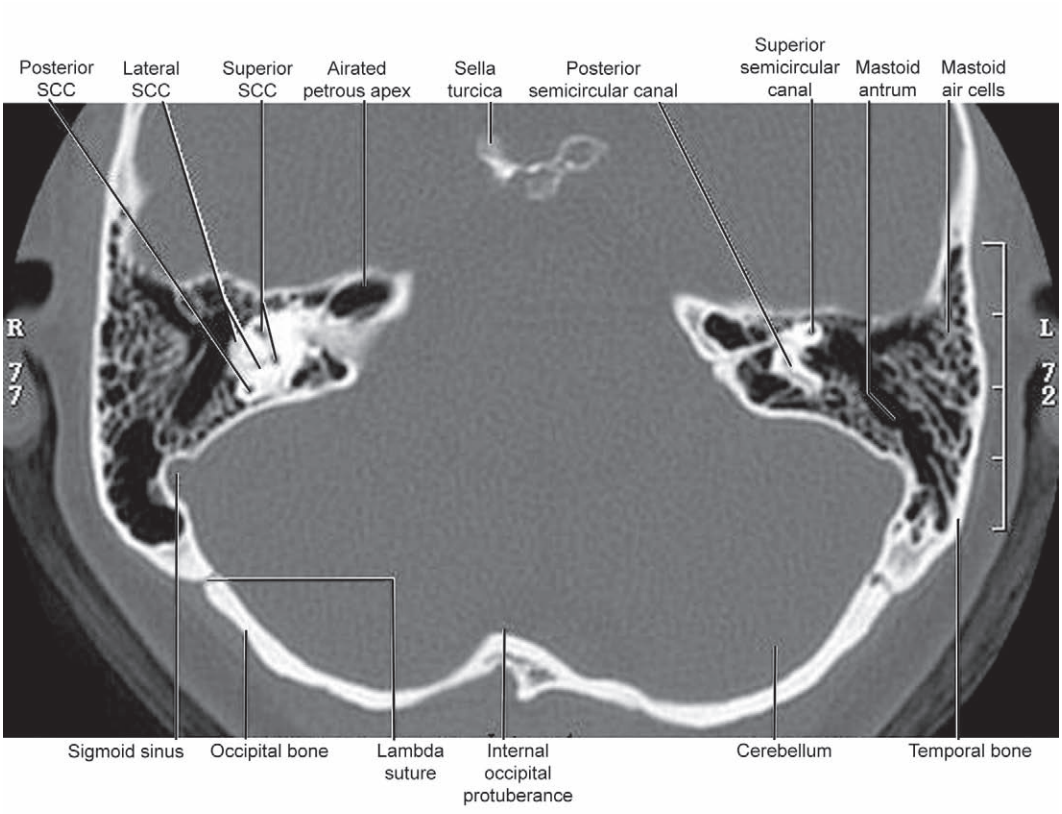


Fig. 18: HRCT temporal bone axial section showing vestibular anatomy

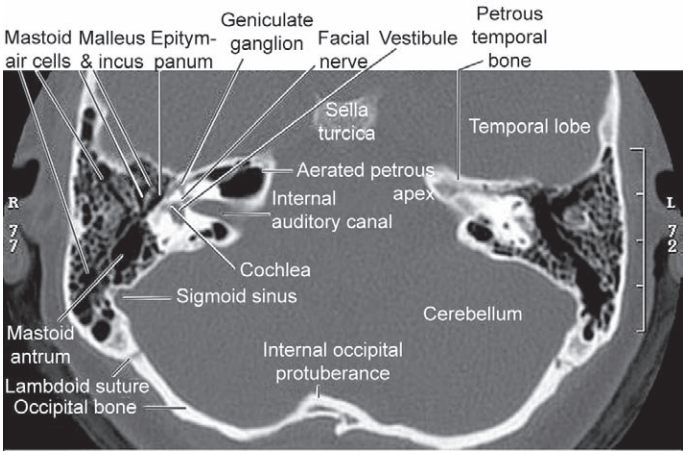


Fig. 19: HRCT temporal bone axial section showing internal auditory canal

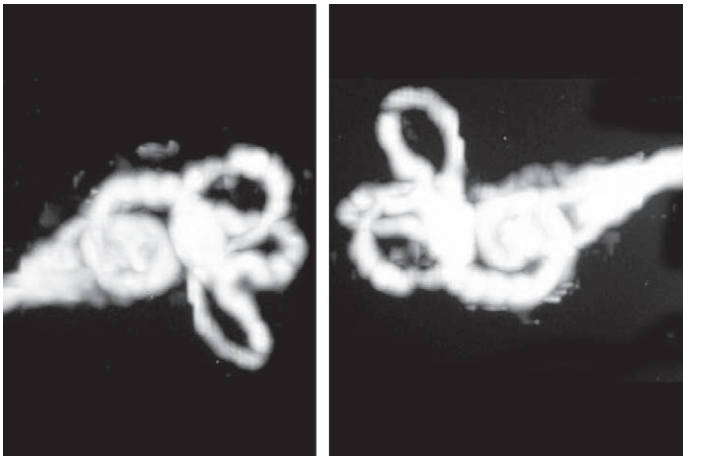


Fig. 20: MRI temporal bone showing membranous labyrinths of both the sides
Source: Dr Amit Goyal, Shillong

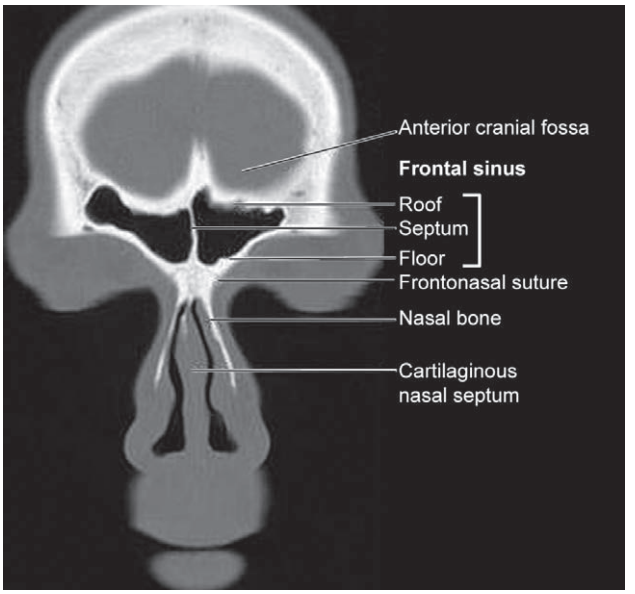


Fig. 21: CT PNS coronal section showing frontal sinuses

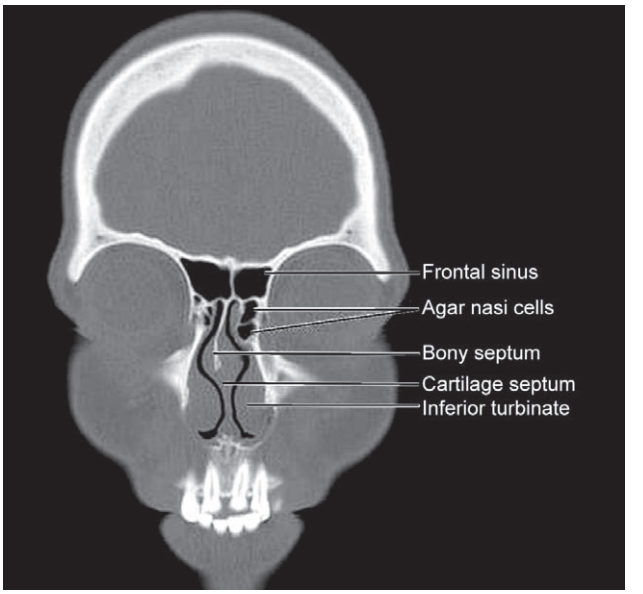


Fig. 22: CT PNS coronal section showing agger nasi cells

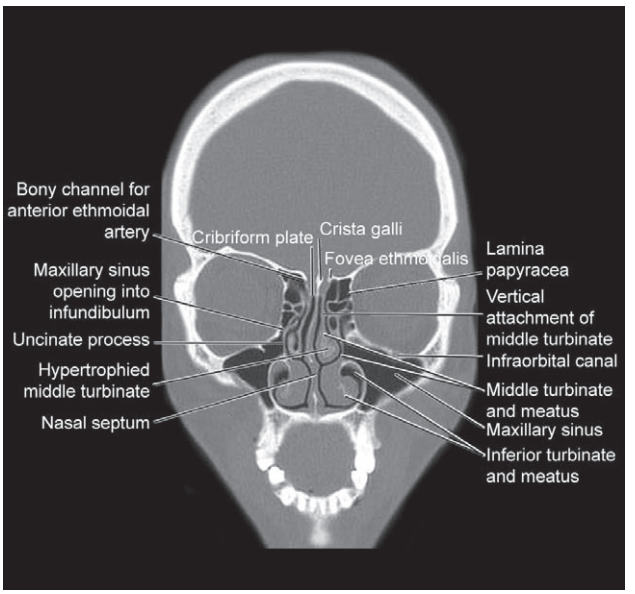


Fig. 23: CT PNS coronal section showing ethmoidal and maxillary sinuses

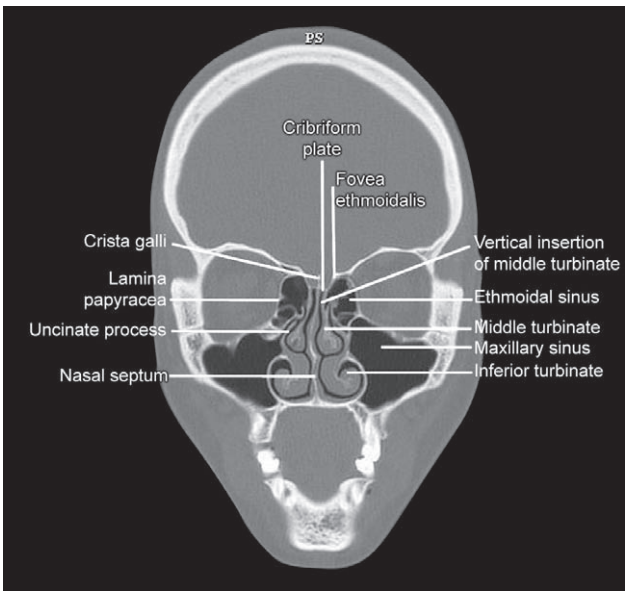


Fig. 24: CT scan nose and paranasal sinuses coronal section

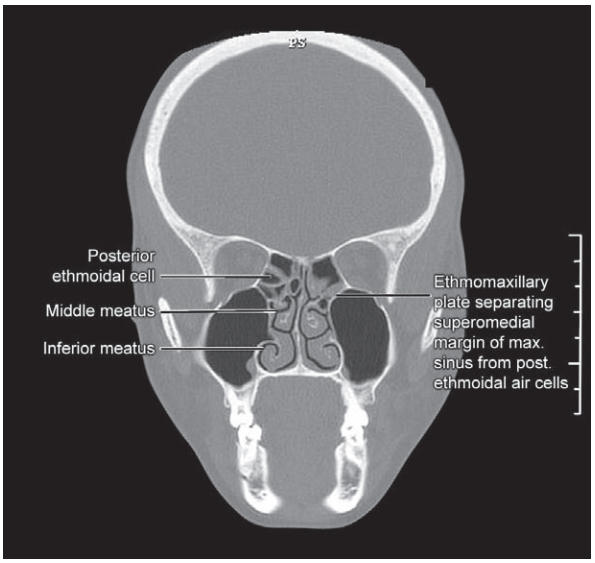


Fig. 25: CT scan nose and paranasal sinuses coronal section

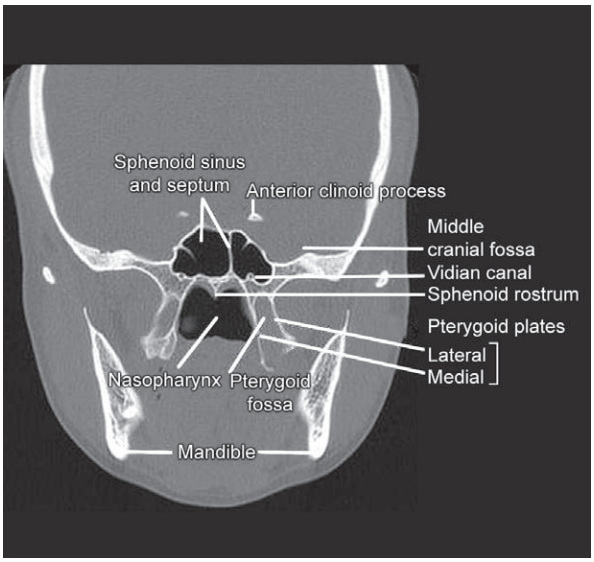


Fig. 26: CT PNS coronal section showing sphenoidal sinuses

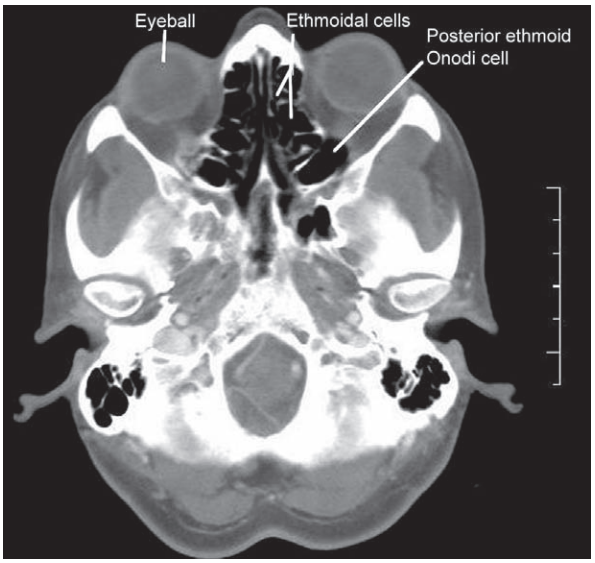


Fig. 27: CT scan head axial sections showing nose and paranasal sinuses

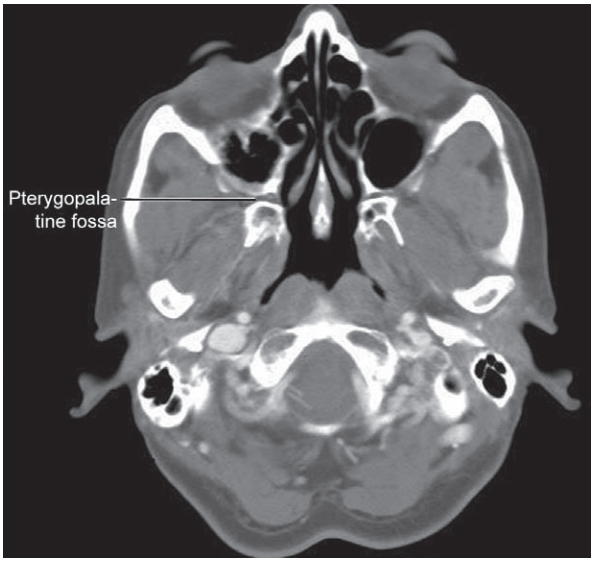


Fig. 28: CT scan head axial sections showing nose and paranasal sinuses

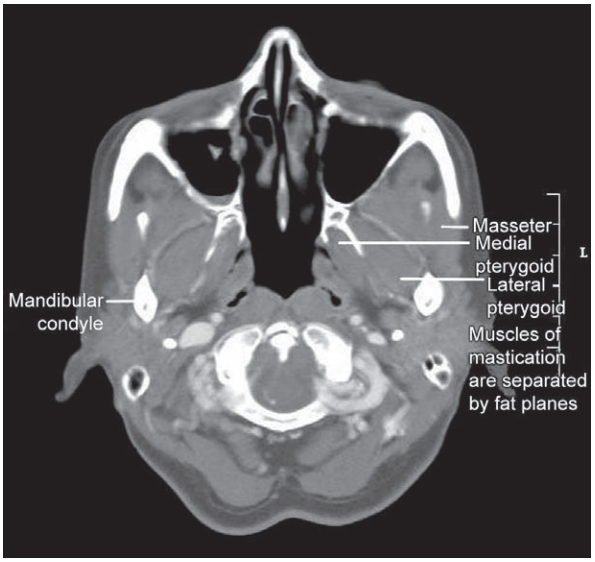


Fig. 29: CT scan head axial sections showing nose and paranasal sinuses

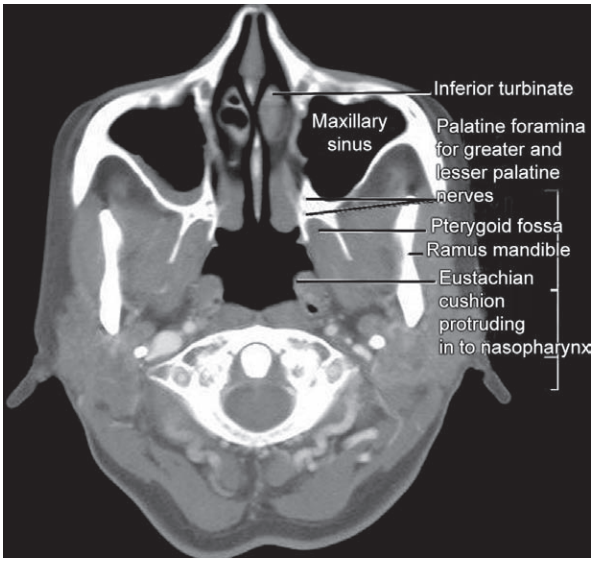


Fig. 30: CT scan head axial sections showing nose and paranasal sinuses

Fig. 31: CT scan nose and paranasal sinuses axial section

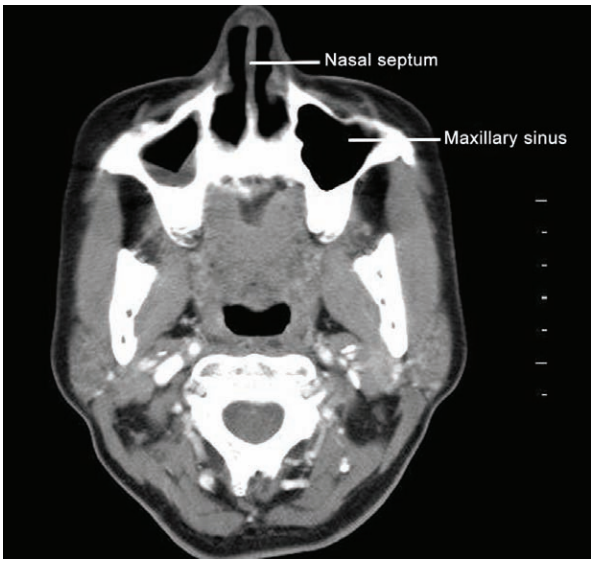


Fig. 32: CT scan nose and paranasal sinuses axial section

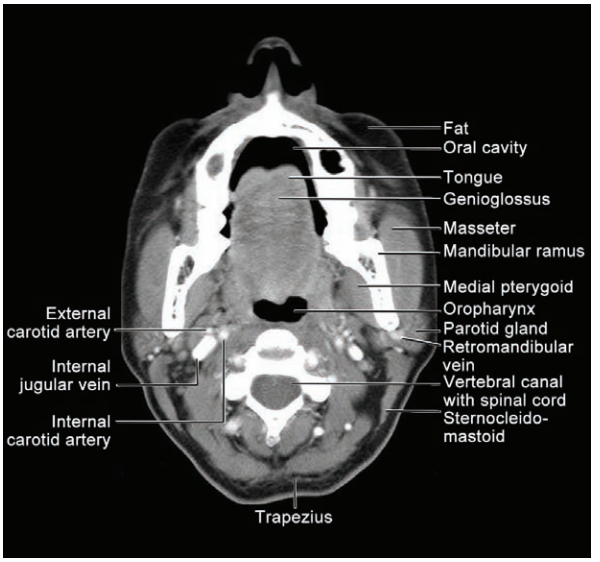


Fig. 33: CT scan oral cavity and oropharynx

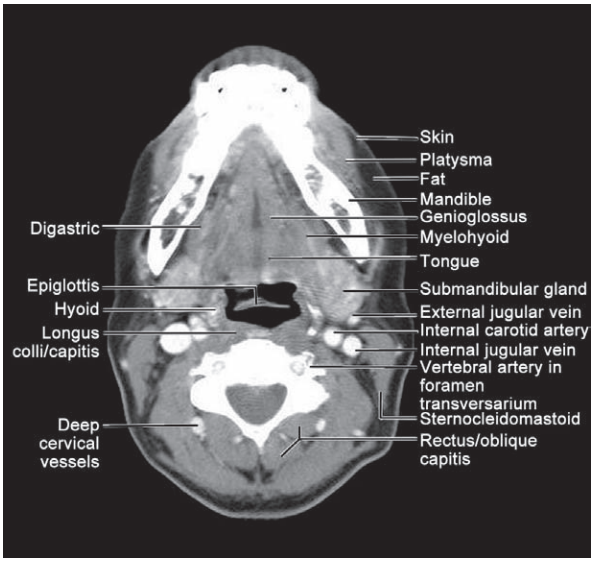


Fig. 34: CT scan oropharynx
Source: Dr Ritesh Prajapati, Consultant Radiologist, Anand, Gujarat

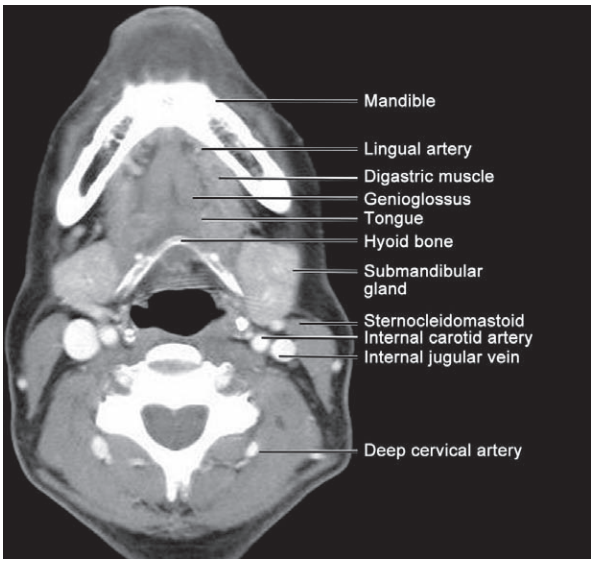


Fig. 35: CT scan laryngopharynx

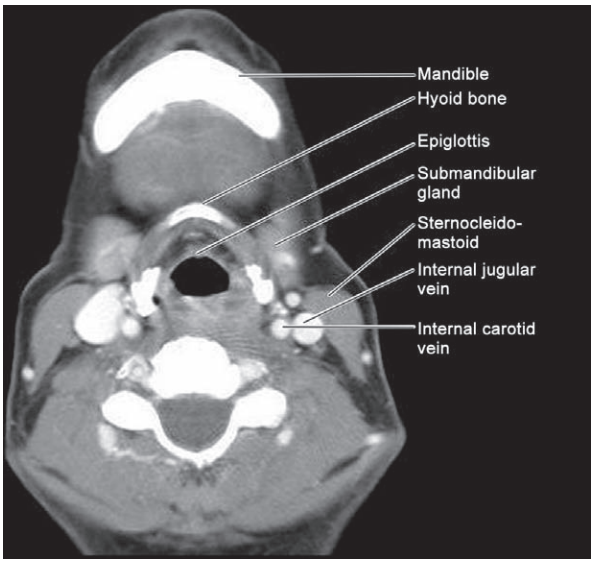


Fig. 36: CT scan laryngopharynx

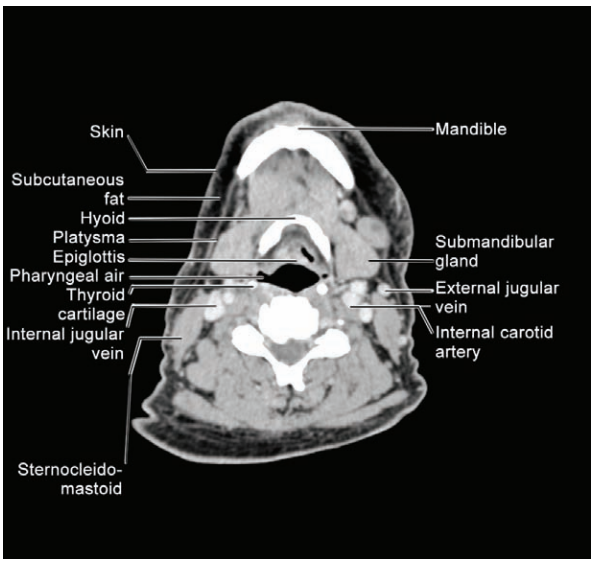


Fig. 37: Axial CT scan at hyoid bone level show necrotic nodes at level IA on left side

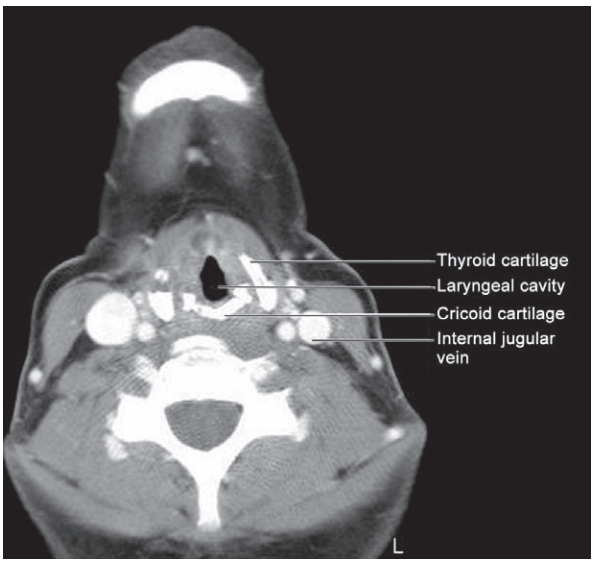


Fig. 38: CT neck with contrast showing larynx and other cervical structures

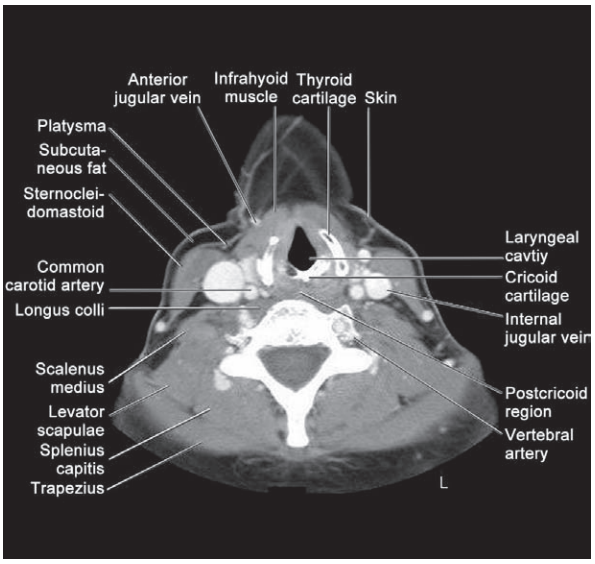


Fig. 39: CT neck with contrast showing larynx and other cervical structures

Clinical Highlights

1. **High resolution CT scan**: This is best to assess temporal bone fractures.
2. **Ultrasound**: It is better than CT in differentiating solid from cystic lesions.
3. **MRI, Doppler and ultrasound**: These imaging technologies do not use ionizing radiation.
4. **MRI contraindication**: It should not be used in the third trimester of pregnancy.
5. **Acoustic neuroma**: MRI is superior to CT in diagnosis of acoustic neuroma.

FURTHER READING

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3. Bradoo RA, Nerurkar NK, Joshi AR, et al. Virtual bronchoscopy: Our early experience. Indian J Otolaryngol Head Neck Surg. 2005;57;142-4.
4. Gupta SC, Singh M, Jain A, et al. A comparative study of radiological and antroscopic findings in the lesions of maxillary sinus. Indian J Otolaryngol Head Neck Surg. 2004;56;09-13.
5. Joshi AA, Shah KD, Bradoo R. Radiological correlation between the anterior ethmoidal artery and the supraorbital ethmoid cell. Indian J Otolaryngol Head Neck Surg. 2010;62:299-303.

61 Radiotherapy and Chemotherapy

Education is not the amount of information that is put into your brain and runs riot there, undigested all your life. We must have life-building, man-making, character-making, assimilation of ideas. If you have assimilated five ideas and made them your life and character, you have more education than any man who has got by heart a whole library.

—Swami Vivekananda

Points of Focus

RADIOTHERAPY

- **BASIC PHYSICS**
  - ✓ Electromagnetic Radiations
  - ✓ Ionizing Radiations
  - ✓ Effects of High-Energy Photons in Tissues
  - ✓ Radiation Units
  - ✓ Sources of Radiation
  - ✓ Depth-Dose
- **RADIOBIOLOGY**
  - ✓ Oxygen Enhancement Ratio (OER)
- **THERAPEUTIC WINDOW**
- **MODES OF RADIOTHERAPY**
  - ✓ External Beam Therapy (Teletherapy)
  - ✓ Brachytherapy
  - ✓ Unsealed Radionuclide Therapy
- **COMBINED MODALITY TREATMENT**
  - ✓ Preoperative Radiation
  - ✓ Postoperative Radiation
  - ✓ Intraoperative Radiotherapy (IORT)
  - ✓ Radiotherapy and Chemotherapy
- **PLANNING OF RADIOTHERAPY**
  - ✓ Factors Affecting Response to Radiotherapy
  - ✓ Fractionation of Radiotherapy
  - ✓ Radiotherapy Fractionation (FX) Scheme
  - ✓ Intensity Modulated Radiation Therapy (IMRT)
- **COMPLICATIONS OF RADIOTHERAPY**
  - ✓ Early Complications
  - ✓ Late Complications
  - ✓ Patient Care during Radiotherapy

CHEMOTHERAPY

- ✓ Principles
- ✓ Single Agent Doses and Toxicity
- ✓ Criteria for Response
- ✓ Prognostic Factors
- ✓ Performance Status
- ✓ Work-up Before Chemotherapy
- ✓ Toxicity of Anticancer Drugs
- **PALLIATIVE CHEMOTHERAPY**
- **COMBINED MODALITY THERAPY**
  - ✓ Induction Chemotherapy
  - ✓ Chemoradiation
  - ✓ Adjuvant or Posterior Chemotherapy
- **ORGAN PRESERVATION**
- **INTRA-ARTERIAL CHEMOTHERAPY**
- **PREVENTION OF CANCER**
  - ✓ Lifestyle Modifications
  - ✓ Chemoprevention
- **CLINICAL HIGHLIGHTS**

RADIOTHERAPY

Radiation oncology has emerged as a separate medical specialty. Radiotherapy forms an integral part in the management of head and neck malignancies. It may be used either alone or in combination with surgery and/or chemotherapy. It is used alone (curative radiotherapy) in early cancers (such as glottic cancer) to preserve the function of the organ. As an adjuvant to surgery or chemotherapy, it can increase the survival rate in more advanced lesions. Palliative radiotherapy in advanced lesions, when total control of disease is not expected, helps in controlling local symptoms of pain, bleeding or obstruction to air and food channels. It has also been found useful in the treatment of benign vascular lesions (angiofibroma and glomus tumor) and to control excessive scar formation in cases of keloids.

BASIC PHYSICS

A wave can be of various types. It is a periodic disturbance in a medium or space. The chief characteristics of a wave are its speed of propagation, its frequency, its wavelength and its amplitude (Box 1).

Electromagnetic Radiations

The energy of electromagnetic radiations (a stream of photons) can be regarded as waves propagated through space (need no supporting medium). They travel in a vacuum with the speed of light. Photon travels at speed of light. It is a particle with zero rest mass consisting of a quantum (minimum amount by which certain properties, such as energy or angular momentum of a system can change) of electromagnetic radiation. Electromagnetic spectrum is the range of wavelengths over which electromagnetic radiation extends. The longest waves are radio waves (Table 1) while the shortest are cosmic rays. Other waves in the descending order of their wavelengths are: infrared waves, narrow band of visible light, ultraviolet waves, X-rays and gamma rays. Gamma rays produced from a ^{60}Co correspond to a wavelength of 10^{-12} m and 1.3 million electron volts (MeV). Energies of 3–5 electron volts (eV) (correspond to less than 10^{-6} m) are required to break chemical bonds.

Box 1: Different types and characteristics of waves

- *Traveling wave* (*Progressive wave*): Energy is transferred from one place to another by the vibrations. The profile moves through the medium at the speed of wave.
- *Electromagnetic waves*: A type of transverse wave (such as water), where electric and magnetic fields vary in a periodic way at right angles to each other and to the direction of propagation.
- *Sound waves*: A type of longitudinal waves, where the air is alternately compressed and rarefied by displacements in the direction of propagation.
- *Speed of propagation*: Distance covered by the wave in unit time.
- *Frequency*: Number of complete disturbances (cycles) in unit time expressed in hertz.
- *Wavelength*: Distance in meters between successive points of equal phase in a wave.
- *Amplitude*: Maximum differences of the disturbed quantity from its mean value.

TABLE 1 Range of wavelength (in meters) in electromagnetic spectrum

Radio waves	10^{5}–10^{-3} m
Infrared waves	10^{-3}–10^{-6} m
Visible light	4–7 x 10^{-7} m
Ultraviolet	10^{-7}–10^{-9} m
X-rays	10^{-9}–10^{-11} m
Gamma rays	10^{-11}–10^{-14} m
Cosmic rays	Less than 10^{-14} m

Ionizing Radiations

The sufficiently high energy of ionizing radiations results in ionization in the medium through which it passes. In biological tissue, its effect can be very serious, as a consequence of the ejection of an electron from a water molecule and the oxidizing or reducing effects of the resulting highly reactive species. The ionizing radiations are enumerated in Box 2.

The principal processes by which high-energy electrons are produced in the body tissue by the high-energy photons (X-rays and gamma rays for radiotherapy) are shown in Table 2.

Photon beams include both X-rays and gamma rays. They are the most common forms of radiation. X-rays are produced when high-energy electrons bombard a metallic target. Gamma rays are emitted by radioactive sources, e.g. cobalt-60.

Electron beam is a type of particle radiation. It is the second most common forms of radiotherapy, which have rapid dose build up and sharp dose fall off with very little scatter. Electron beams are produced by linear accelerator, betatron and

Box 2: Ionizing radiations

- Stream of high energy particles
  - Electrons
  - Protons
  - α-particles
- Short wavelength electromagnetic radiations (photon beams)
  - Ultraviolet rays
  - X-rays
  - Gamma rays

TABLE 2 Effects of high-energy photons in tissues

Photoelectric effect	Photon, 10–50 kilo electron volts (keV) used in diagnostic radiology, excites a tightly bound inner shell electron and is completely annihilated
Compton effect	Photon, 500 keV–10 MeV used in radiotherapy, excites loosely bound outer electron and emerges at a lower energy (long wavelength)
Pair production	High-energy photon is annihilated in strong electromagnetic field of an atomic nucleus and produces an electron-positron pair

microtron. They boost up the radiation dose to the target area and avoid radiation to adjoining vital structures such as spinal cord. Other particle radiations, which are still under investigation, include fast neutrons, α-particles, protons or pions.

Radiation Units

They are expressed in terms of energy that is deposited in a unit of material.

- *Rad*: The conventional dose unit rad (radiation absorbed dose) is equivalent to 100 ergs deposited per gram of material.
- *Gray*: The recent dose unit gray (Gy) agreed by international commission is equivalent to 1 joule deposited per kilogram of material.
- *Rad and gray*: One hundred rads is equal to 1 Gy, which has 100 centigrays (cGy). Therefore, 1 rad is equal to 1 cGy.

Sources of Radiation

Higher the energy of radiations, deeper do they penetrate. Earlier X-ray machines, which could be used for superficial tumors of the skin or lip, produced energy only in kilovolts (kV). Newer machines produce radiations of high energy in million volts (MV), which have greater penetrating power. They can be used for deep seated tumors and spare untoward effects on the skin and bone. The common sources, which are used for radiotherapy, include following:

- *Kilovoltage machines*: They are becoming obsolete and consist of superficial (50–150 kV) and ortho (200–400 kV) voltage X-ray machines that produce X-rays of 50–400 kV.
- *Cobalt-60 machine*: This is the most commonly used source for radiotherapy. It uses radioactive cobalt-60 that produces gamma rays of 1.17 and 1.33 MeV. The radioactive cobalt-60 has its natural decay time. It needs replacement after every 5 years.
- *Linear accelerator*: These megavoltage machines can produce photon and electron beams of 4–25 MV depending on whether an intervening metallic target is used in machine or not. They work on electricity. Other examples of this category of machines include betatron and microtron.
- *Radioactive material*: Radium-226 used in the form of needles has been replaced by safer radionuclides, which include Cesium-137 pellets, Iridium-192 wire, Gold-198 seeds and Iodine-125.

Depth-Dose

The energy deposited by a radiation beam is characterized by its linear energy transfer (LET) spectrum. The deposited dose of photon beams (X-rays and gamma rays) is low at the surface and more in depth. After reaching to a maximum, then it decreases with depth because of attenuation. The skin and subcutaneous tissues are spared and delivery of radiation is more to deeper tumor.

Electron beam penetrates a given distance and then fall off rapidly. There occur slight amount of skin sparing with 6 MeV beam. These beams are used for cancers of skin and mucosa of oral cavity.

- D_{max}: It refers to the distance of the dose maximum from the surface, which varies with energy and source axis distance (SAD):
  - 1.2 cm for 4 MV (80 cm SAD)
  - 1.3 cm for 6 MV (100 cm SAD)
  - 3.0 cm for 15 MV (100 cm SAD)

RADIOBIOLOGY

Radiotherapy kills the cancer cells (in mitosis), which divide more frequently but repair less efficiently. There are primarily two mechanisms (direct and indirect) by which radiations act on biological cells.

1. *Direct mechanism*: In the direct mechanism of action, key targets of radiations are DNA, nuclear membrane and mitochondria, which result in cell killing.
2. *Indirect mechanism*: Indirect mechanism of action affects molecules in cell cytoplasm, which then result in sequence of complex chemical reactions.

Though radiations usually employ both the mechanisms, low-LET radiations primarily affect cell through indirect mechanism while high-LET kills cells through direct mechanism.

Oxygen Enhancement Ratio

A synergy has been reported between ionizing radiations and oxygenation, and hyperthermia. Indirect mechanism of radiation action on molecules of cytoplasm results in free radicals, which contain unpaired electrons and are highly reactive. Oxygen facilitates diffusion of these free radicals to DNA and other target regions where they produce damage. Regions of necrosis surrounded by hypoxic cells are present in large tumors, which make them relatively radioresistant. High-LET, which primarily kills cells directly, avoids this problem. Generally, oxygen enhancement rate decreases with increasing LET.

THERAPEUTIC WINDOW

There are some intrinsic differences in the properties of tumor and normal cells. A therapeutic dose of radiation in a radiosensitive tumor has wide "therapeutic window," which results in 95% chances of tumor control and 5% chances of normal tissue complications. In contrast radioresistant tumors have narrow "therapeutic window," which means a therapeutic dose of radiation has 95% chances of tumor control with very high chances of normal tissue damage. The following techniques exploit this therapeutic window concept:

- Three-dimensional treatment planning and delivery
- Brachytherapy
- Intraoperative radiotherapy
- Use of high-LET and charged particle radiations
- Altered fractionation schedule
- Use of radiosensitization and radioprotective schedule.

MODES OF RADIOTHERAPY

- *External beam therapy (teletherapy)*: This mode of radiotherapy uses photon and electron beams. They project to the target area through the skin from a distance. The main advantages of megavoltage radiotherapy are its better precision, skin sparing, diminished bone absorption and increased dose to deep tumor.
- *Brachytherapy*: Brachytherapy uses radioactive material, which is applied in the form of mould, interstitial implant

and intracavitary implant. They are placed in close contact with the tumor.

- Interstitial implants: The interstitial implants are in the form of needles (^{226}Ra and ^{137}Cs), wires, ribbons or seeds. They are inserted into the tumor.
- Shorter half-life: Some of the radioactive materials (^{198}Au and ^{125}I) have shorter half-life and are permanently left in the tissues.
- Intracavitary implants: The radioactive material is placed in a hollow cavity next to the tumor for either minutes, hours or days. Examples include nasopharynx and maxillary antrum.
- Recent developments: The following recent developments is again making brachytherapy popular, which was abandoned due to the hazards of radiation to the physician, nurses and other personnel:
  - After-loading techniques (^{192}Ir)
  - Safer radionuclides
  - Computerized dosimetry

  Currently, high-dose rate remote after-loading devices are available, which push ^{192}Ir via a set of interstitial catheters. Computers control dwelling time throughout the implant.

- *Unsealed radionuclide therapy*: Radioactive isotopes are given either orally or intravenously. They are concentrated by metabolic pathways in malignant tissue, which receive large radiation doses. The normal tissues are relatively spared. The best example is the radioactive iodine treatment of differentiated thyroid carcinoma of follicular cell origin.

COMBINED MODALITY TREATMENT

- Curative radiotherapy for small cancerous lesions not only cures the cancer but also has the advantage of preserving the function of the organ. Surgery does give good results in these early cases but functions are significantly affected. The curative dose of radiotherapy in head and neck cancer ranges from 65–75 Gy (6,500–7,500 rads/cGy).
- Palliative radiotherapy is given in advanced inoperable tumors, or patients with distant metastases, or poor general condition (poor nutrition and diseases of heart, lung, liver and kidney).
- Combination therapy combines radiotherapy with surgery and chemotherapy. Radiotherapy can be given before, after and even during the surgery to achieve better control of disease. Each modality has its own advantages and disadvantages; however postoperative radiotherapy is the most common and well established.

Preoperative Radiation

- *Advantages*
  - It reduces the tumor bulk and can convert inoperable tumor into operable one.
  - Response to radiation better because of the better oxygenation of tissues, which is affected in operated cases.
  - Radiation blocks the lymphatics and the chances of dissemination of tumor cells during surgery are less.
  - It decreases the chances of microscopic spread beyond palpable tumor mass or occult metastasis to lymph nodes.
- *Disadvantages*
  - The central part of large tumor responds poorly to radiation because of poor oxygenation.
  - Radiation induced reduction in the vitality of tissues increases the chances of postoperative complications, which include delayed healing, flap necrosis, fistulae formation and carotid blowouts.

Postoperative Radiation

- *Advantages*
  - It is more effective as the remaining tumor mass after the radical surgery is limited.
  - As the extent of disease is well defined after the surgery, radiotherapy can be better planned for the suspected areas of residual disease.
  - Surgical resection is easier and postoperative complications are lesser in comparison to the cases of preoperative radiotherapy.
- *Disadvantages*: Blood supply of the tissues is affected, which results in relative hypoxic cells that respond poorly to radiation.
- *Indications*: Postoperative radiotherapy is usually indicated in following conditions:
  - When the margins of growth are reported positive or very close.
  - In the presence of invasion of bone or cartilage.
  - Extracapsular invasion of lymph nodes.
  - Neck nodes are multiple or the size of a node is greater than 3 cm.

Intraoperative Radiotherapy

A single large fraction of radiation is given to the exposed tumor bed at the time of operation. This technique employs either orthovoltage X-rays or megavoltage electrons. During the surgery at the time of radiotherapy, critical structures are moved outside the field of radiation.

Radiotherapy and Chemotherapy

Chemotherapy is used either before, during or after radiotherapy.

- *Induction chemotherapy* (before radiotherapy) reduces the bulk of tumor and may help in organ preservation. The vascularity of the lesion is maintained and enhanced.
- *Concomitant chemotherapy* acts as a radiosensitizer and improves the effect of radiation on the tumor cells. Methotrexate and bleomycin are good radiosensitizers.
- *Adjuvant chemotherapy* is used after radiation. It helps in controlling distant metastases (see section "Chemotherapy").

PLANNING OF RADIOTHERAPY

Shells are prepared for accurate treatment of tumor and sparing adjacent normal and critical structures. The treatment simulator consists of diagnostic X-ray machine with image intensification facilities. It allows distance adjustment to make rectangles of any size. The introduction of blocks shapes the radiation beam to shield normal critical structures.

The histology nature, site, size and extent of tumor (T) and its draining lymph nodes (N) must be known. The primary tumor and its draining lymph nodes are included in the radiation field. Extent of tumor can be found by clinical examination (palpation under anesthesia) and radiological studies.

Factors Affecting Response to Radiotherapy

- Smaller the tumor, better is the response.
- Tumors of lymphoid tissues are very radiosensitive.
- Anaplastic tumors and embryonal tumors are also radiosensitive.
- The differentiated squamous cell carcinomas do not respond well.
- Adenocarcinomas, sarcomas and bone tumors have low sensitivity.

Fractionation of Radiotherapy

- *Curative*: The curative dose of 65–70 Gy (6,500–7,500 rad/cGy) is usually delivered in fractions of 2 Gy (200 rads/cGy) daily for 5 days in a week (Monday to Friday) till the completion of desired curative dose.
- *Preoperative*: Preoperative radiation, which is not used frequently, consists of 50 Gy (5,000 rads/cGy) delivered in fractions of 2 Gy (200 rads/cGy) daily for 5 days in a week for 5 weeks to the primary tumor and neck nodes. It is followed by surgery in 4–5 weeks. Surgery is not delayed beyond 6 weeks as it causes regrowth of the tumor cells.
- *Postoperative*: Postoperative radiation dose is 50–60 Gy (5,000–6,000 rads/cGy) and is started 4–6 weeks after surgery. Postoperative radiation should not be delayed. It can result in a regrowth of tumor cells.
- *Fractionation schedules*: They vary from center to center. Some of them are given in Table 3. Conventional fractionations use 2 Gy per day for 5 days a week. To avoid the prolong treatment time, multiple daily fractions with a minimum 6 hours interval are given in hyperfractionation and accelerated fractionation schemes.
  - Hypofractionation consists of smaller number of fractions, which are larger than 2 Gy.
  - In hyperfractionation, larger numbers of fractions, which are smaller than 2 Gy, are employed.
  - Accelerated fractions shorten the overall time.
  - Split courses provide gap, which allows acute reactions to settle.

Intensity Modulated Radiation Therapy

The aim of any radiotherapy is to deliver entire dose to the tumor and none to normal tissue. The intensity modulated radiation therapy (IMRT) is current step in that direction. In the past, customized cerrobend and multileaf collimators used to shape the beam, which allowed the field to be shaped to cover area of interest. In three-dimensional conformal radiotherapy, isodose curves conform closely to tumor shape but close normal tissues are not spared.

Intensity modulated radiation therapy delivers the dose as a gradient across the area of interest. The main advantage of IMRT is sharper gradient between tumor tissue and normal tissue.

- *Techniques*: There are three techniques of IMRT:
  1. Dynamic IMRT: It is similar to tomotherapy unit, which circle around patient.
  2. Stepwise delivery: It is similar to three-dimensional conformal radiotherapy.
  3. Step and soot IMRT: It is the hybrid of former two where multiple beam arrangements are used but leaves are changed in real time.

TABLE 3 Radiotherapy fractionation (fx) scheme

Hyperfractionation	81.6 Gy, 1.2 Gy/fx, 2 times a day, 68 fx, 5 days a week
Accelerated intense	54 Gy, 1.5 Gy/fx, 3 times a day, 36 fx, 7 days a week
Accelerated split course	67.2 Gy, 1.6 Gy/fx, 2 times a day, 42 fx, 5 days a week

COMPLICATIONS OF RADIOTHERAPY

Complications of radiotherapy depend on the site of radiation, dose delivered and its daily fractions. They can be early and late. Higher the total dose and larger the daily fraction, more are the complications. Acute radiation side effects are caused by changes in tissues, which are composed of rapidly proliferating cells. The delayed effects are caused by changes in tissues, which have slowly proliferating cells.

Early Complications

- Radiation sickness (anorexia and nausea).
- Mucositis (ulcers in mouth and pharynx).
- Dryness of mucous membranes (xerostomia).
- Skin reactions are erythema, dry and wet desquamation of skin (Fig. 1).
- Pharyngeal edema.
- Laryngeal edema (stridor).
- Fungal infections: Candida.
- Hematopoietic suppression.

Late Complications

- Permanent xerostomia (Dryness of mouth)
- Atrophy of skin and subcutaneous fibrosis
- Teeth decay

Fig. 1 Skin reactions after postoperative radiotherapy

- Osteoradionecrosis
- Trismus due to fibrosis of temporomandibular joint and muscles
- Transverse myelitis
- Eye: Retinopathy and cataract
- Endocrinal deficit: Thyroid and pituitary
- Malignancy: Thyroid cancer and orbital osteosarcoma.

Patient Care during Radiotherapy

- *Nutrition*: Diet should be rich in protein, vitamins and iron. Nasogastric tube feeding is started if needed. Blood transfusion is given if patient has severe anemia.
- *Teeth care*: Dental evaluation and extraction if needed are of paramount importance and prevent osteoradionecrosis of mandible. Wound of extraction must heal before the beginning of radiotherapy. Xerostomia facilitates caries of teeth.
- *Skin care*: Skin reactions were common with old superficial and orthovoltage X-ray machines. The modern megavoltage therapy machine has a skin sparing effect. Electron beam therapy to skin can also result in skin reactions. Advises, which are given to patients during radiotherapy, include:
  - Keep the skin dry and avoid soap, water and wet shaving.
  - Avoid sunlight.
  - Avoid abrasive dressing and clothing and adhesive plaster for dressings, which peels off the desquamated skin.
  - Skin should be covered with soft cloth, which should provide free aeration to the skin.
  - For moist desquamated skin, use antibiotic ointment. Topical steroid creams relieve itching and pain.
- *Mouth care*: Dryness of mouth (xerostomia) and ulcerations (mucositis) are common and interfere with feeding. Acute radiation mucositis generally persists for 8–12 weeks after radiation. Irradiated patients are prone to develop candida infection, which can affect mucosa of oral cavity and pharynx.
  - Patient is asked to avoid alcohol, tobacco and spicy food.
  - Irritating mouthwashes contain alcohol and should be avoided.
  - Milk of magnesia neutralizes the acid pH and prevents caries of teeth and soothes and protects inflamed mucosa.
  - Xylocaine viscous relieves pain and discomfort of mucositis and facilitates food taking.
  - Oral candida is usually treated by topical application of nystatin and clotrimazole. Systemic antifungal therapy is needed for the mycelia, which may lie protected beneath organic debris.

CHEMOTHERAPY

Chemotherapy can be used either alone or in combination with surgery and radiotherapy. The drugs, which are found effective in head and neck squamous cell cancers are, methotrexate, cisplatin, bleomycin and 5-fluorouracil. Adriamycin has been used for adenoid cystic carcinoma and dacabazine for melanomas. Lymphomas, both Hodgkin and non-Hodgkin types, which have multifocal origin and widespread involvement, are also treated by chemotherapy.

Principles

Though the single drug as well as combinations of various drugs have been employed to treat head and neck malignancies, it has been observed that combination of two or more drugs provide relatively better overall response. The drugs, which are found effective in head and neck malignancies, and their side effects and doses, are listed in Table 4.

- *Phases of drug evaluation*: The evaluation of drugs occurs in the following three phases:
  1. Phase I: Determination of toxic effects and establishment of highest safe dose.
  2. Phase II: Determination of activity (response rate) in patients with same tumor type.
  3. Phase III: Randomized comparisons (response duration and progression-free survival) of two or more treatment options.

Criteria for Response

The efficacy of chemotherapy is investigated through clinical trials, which use standard definitions that allow objective reporting of results (Table 5).

Prognostic Factors

The survival time may be prolonged in patients who achieve a complete response. Lower response rates are observed in T_4 and N_3 stage cancer. The overall survival time correlates with performance status, T and N stage, primary site and nodal extracapsular extension of tumor. After analyzing many chemotherapy trials for patients with recurrent disease, following poor prognostic factors have been identified:

- Low performance status (Box 3).
- Poor nutrition.
- Large tumor burden.
- Extensive previous radiotherapy and surgery.
- Tumor progression during or shortly after surgery or radiotherapy.

Biologic factors

The following biologic factors, which may be used to predict response and survival outcome, are under investigations:

- DNA content
- Immunologic status
- Circulating immune complexes.

Work-up Before Chemotherapy

Patients who are candidates for cancer chemotherapy should be subjected to following investigations:

- *History and clinical examination*: Exclude kidney, heart and lung disease.
- *CBC*: Hemoglobin, total and differential WBC count and platelet count. Many drugs are myelosuppressive.
- *Urine examination.*
- *Biochemistry*: Blood urea nitrogen and creatinine and liver function tests. Methotrexate and cisplatin are nephrotoxic.
- *Radiology*: X-ray chest, CT scan/MRI (to know extent of disease) and ultrasound abdomen for liver/spleen.

TABLE 4 Single agent doses and toxicity

Drug and doses	*Acute toxicity*	*Delayed toxicity*
• **Alkylating Agents**		
Cyclophosphamide 100–400 mg/m^2/d; 1–1.5 g/m^2 IV every 3–4 weeks	Nausea and vomiting with higher doses	Neutropenia, alopecia, hemorrhagic cystitis
Ifosfamide 1.5–2.5 g/m^2 twice weekly		Myelosuppression, alopecia, confusion; mesna prevents cystitis
Cisplatin 50–100 mg/m^2 IV 3 weekly	Severe nausea and vomiting	Nephrotoxicity, ototoxicity, neuropathy
Carboplatin 360 mg/m^2 IV, 4 weekly	Severe nausea and vomiting	Myelosuppression, prolonged anemia
• **Antimetabolites**		
Methotrexate 2.5–5 mg/d orally; 20–25 mg IM twice weekly; 500–1,000 mg/m^2 IV every 2–3 weeks	None Citrovorum factor (leucovorin) rescue for doses over 100 mg/m^2	Mucositis (oral and GIT ulcers), myelosuppression, acute renal failure, rash, hepatotoxic
5-Fluorouracil 15 mg/kg/d IV for 3–5 days every 3 weeks	None	Nausea, mucositis, diarrhea, myelosuppression and dacryocystitis
• **Antibiotics**		
Bleomycin Up to 15 units/m^2 IM/IV/SC twice weekly	Allergic reactions, fever and hypotension	Fever, pulmonary fibrosis, dermatitis (skin rash), mucositis
Adriamycin 60–90 mg/m^2 IV monthly	Vesicant, nausea, vomiting, diarrhea	Cardiotoxicity, mucositis, myelosuppression, alopecia
Mitomycin 10–20 mg/m^2, 6–8 weekly	Severe vesicant; nausea	Bone marrow depression, hemolytic-uremic syndrome
• **Vinca Alkaloids**		
Vincristine 1.5 mg/m^2/week (Max 2 mg)	Severe vesicant	Areflexia, neurotoxicity, alopecia, myelosuppression, paralytic ileus, syndrome of inappropriate antidiuretic hormone secretion constipation
Vinblastin 0.1–0.2 mg/kg (6 mg/m^2)IV	Mild nausea and vomiting; severe vesicant	
• **Taxanes**		
Paclitaxel (Taxol) 135 mg/m^2 IV infusion 24 hours every 3 weeks	Mild nausea vomiting; hypersensitivity reaction (premedicate with diphenhydramine and dexamethasone)	Myelosuppression, neuropathy
Docetaxel (Taxotere) 60–100 mg/m^2 IV 3 weekly		Edema (fluid retention), neutropenia, neuropathy
• **Supporting Agents**		
Leucovorin 10 mg/m^2, 6 hourly IV/oral	None	Enhances toxic effects of fluorouracil
Mesna 20% of ifosfamide dosage	Nausea, vomiting, diarrhea	None

New agents under investigation: Topoisomerase-I inhibitors, gemcitabine (pyrimidine antimetabolite), vinorelbine (semisynthetic vinca alkaloid), trimetrexate, edatrexate, and piritrexim.

TABLE 5 Criteria for response

Complete response	No evidence of tumor for at least 4 weeks
Partial response	50% tumor regression for at least 4 weeks
Minor response	Less than 50% tumor regression
Stable disease	No appreciable tumor regression
Progressive disease	25% increase in tumor size or appearance of new lesions

Box 3: Performance status Eastern Cooperative Oncology Group, Southwestern Oncology Group Zubrid scales

0. Fully active; carry on all the housework and office work without restriction
1. Restricted activity; carry on light house work and office work
2. No work activities but capable of all self-care; ambulatory for more than 50% of waking hours
3. Capable of only limited self-care; ambulatory for less than 50% of waking hours
4. Completely bed ridden; unable to do any self-care
5. Dead

- *Pulmonary function tests*: Bleomycin causes interstitial pulmonary fibrosis.
- *ECG*: Adriamycin has cardiotoxicity.
- *Audiogram*: Cisplatin causes high frequency hearing loss.
- *Nutritional status.*

Toxicity of Anticancer Drugs

Anticancer drugs act on not only rapidly dividing cancer cells but also normally dividing cells of hair follicles, gastrointestinal mucosa and bone marrow. They can cause alopecia, stomatitis, nausea, vomiting, diarrhea, anemia, leukopenia and thrombocytopenia (Table 4). Some drugs have selective action on certain systems, such as kidney (methotrexate and cisplatin), nerves (vincristine and cisplatin), heart (adriamycin) and bladder (cyclophosphamide).

Indications for chemotherapy include:

- To make radiotherapy more effective for local primary treatment.
- Combined with radiotherapy for organ-preservation therapy especially of laryngeal cancers.
- Induction or neoadjuvant therapy facilitates less extensive local surgery.
- To control symptoms in metastatic disease.

PALLIATIVE CHEMOTHERAPY

Palliative chemotherapy includes use of cytotoxic drugs either singly or in combination in advanced, recurrent or metastatic disease just to relieve the symptoms. It is not well proved that they prolong life. The response duration to commonly used single agents tends to be 2–4 months. The complete response was observed in less than 5%. Some of the combination chemotherapy regimens, which have been used and improve the response rates, are the following:

- Cisplatin and 5-fluorouracil
- Cisplatin and methotrexate
- Carboplatin and 5-fluorouracil
- Cisplatin and bleomycin
- Cisplatin, methotrexate and bleomycin
- Cisplatin, vinblastine and bleomycin
- Cisplatin, methotrexate, vinblastine and bleomycin.

COMBINED MODALITY THERAPY

The chemotherapy may be used either before, during or after other modalities of treatment, such as surgery and radiotherapy. The three general approaches, which have been undertaken for the sake of improving primary treatment program by using combined modality therapy, are: induction, chemoradiation and adjuvant therapy.

Induction or Anterior Chemotherapy (Neoadjuvant Chemotherapy)

When chemotherapy is used before surgery or radiation, it is called induction or anterior chemotherapy. It helps in reducing tumor burden and micrometastases, which can occur during surgery or before radiation. This neoadjuvant chemotherapy has better compliance and tolerance in downstaging patients. It eliminates problems of poor vascularity (after surgery and radiotherapy) and helps in preservation of organ function.

Response to anterior chemotherapy correlates with response to subsequent radiotherapy. It does not correlates with surgical and radiotherapy complication rates. Some of the most important randomized trials of neoadjuvant chemotherapy before surgery or radiotherapy are listed in Table 6.

Encouraging data emerge from neoadjuvant chemotherapy trials. The overall response rates range from 60–90% (after 3 cycles). Survival time is improved in cases with a complete response (Table 5), which ranges from 20–50%. The critical prognostic factors are: TN stage and type of chemotherapy. A significant decrease in distant metastases has been reported. It has the potential to improve quality of life with organ preservation.

Chemoradiation

- In cases of unresectable tumors, concurrent radiotherapy and chemotherapy, which may result in additive or synergistic enhancement, is used to improve local and regional control.
- It usually enhances acute radiation-induced toxicity (mainly mucosal), which can result in dose reduction and interruption in radiation.
- The survival benefits are not well established.
- The single agents, which have been combined with radiation to improve cellular cytotoxicity are: methotrexate, hydroxyurea, bleomycin, 5-fluorouracil, mitomycin and cisplatin.
- Multiple agents' trials of concomitant or alternating combination chemotherapy and radiation are listed in Table 7.

Adjuvant or Posterior Chemotherapy

Chemotherapy used after surgery or radiation is aimed to cure micrometastases and decrease distant metastases. It is also called posterior chemotherapy. There appears to be no role of maintenance chemotherapy in low-risk cases, although high-risk cases (extracapsular extension in cervical lymph nodes, carcinoma *in situ*, close surgical margins) may benefit.

- *Advantages*: The potential advantages are:
  - Surgery is not delayed.

TABLE 6 Trials of induction chemotherapy

Regimen	*Sites*
PF (most common)	OC, OP, HP, NP, L
PB with/without M	OC, L, HP, OP
CF	OC, OP, HP, L
BEP	NP
PBVM	OC, OP, HP, L
BVcBP	OC, OP
PFBM	OC, OP, HP, L
Key: B, Bleomycin; C, Carboplatin; E, Epirubicin; F, 5-fluorouracil; M, Methotrexate; P, Cisplatinum; V, Vinblastine; Vc, Vincristine;	*Key*: HP, Hypopharynx; L, Larynx; NP, Nasopharynx; OC, Oral cavity; OP, Oropharynx

TABLE 7 Multiple agents and radiotherapy

Concurrent MMC + F/RT	Sequential VBM → RT
Alternating PF/RT	Concurrent VBM with/without F/RT
Concurrent PF/RT	Sequential VBM with/without F → RT
Sequential PF → RT	Concurrent PFL/RT
Alternating VBM/RT	

Key: B, Bleomycin; F, 5-fluorouracil; L, Leucovorin; M, Methotrexate; MMC, Mitomycin-C; P, Cisplatin; Pc, Carboplatin; RT, Radiotherapy; V, Vinblastine

- No blurring of tumor margins as it occurs in neoadjuvant chemotherapy.

ORGAN PRESERVATION

The radical surgery of stage III and IV tumors results in loss of speech, loss of swallowing function, or disfigurement. The neoadjuvant chemotherapy followed by radiotherapy has been used to preserve functions for tumors of hypopharynx, larynx and oropharynx. The patients who achieve either partial or complete response with neoadjuvant chemotherapy go for radiotherapy, while nonresponders go for radical surgery. No significant survival difference is reported in either group.

Organ preservation therapy: Chemotherapy may play an important role in organ preservation. It avoids extensive surgery and is the main treatment for metastatic head and neck cancers. Radiotherapy becomes more effective as well as more toxic when combined with chemotherapy.

INTRA-ARTERIAL CHEMOTHERAPY

Intra-arterial chemotherapy, a type of regional drug delivery, which is not yet viewed as a standard of care, can increase tumor drug exposure (maximum cancer cell killing) and decrease systemic drug toxicity.

PREVENTION OF CANCER

Lifestyle Modifications

The lifestyle factors, which accounts for a majority of avoidable cancer, include:

- Chewing of *paan, sopari* and tobacco.
- Smoking and alcohol.
- Occupational risks.
- Obesity, parity and length of lactation (breast cancer).
- High intake of fat and specific fatty acids (breast, colon, prostate and lung cancer).
- Exposure to ultraviolet light (use of sunscreens protects skin from sunlight).
- Protective role of exercise, folate and calcium supplementation, fruit and vegetable intake and dietary fiber (colorectal cancer and adenomas).

Chemoprevention

Chemoprevention is an important area of research. It consists of administration of drugs, which inhibit carcinogenesis or reverse a premalignant condition.

- *Indications*: Following high-risk patients have been identified for chemoprevention:
  - Past history of cancer (for preventing second cancer).
  - Premalignant lesions, such as leukoplakia.
  - Risks due to family history, lifestyle and occupation.
- *Pharmacological agents*: The following agents have been tried in different designs (induction, maintenance and adjuvant) in head and necks malignancies:
  - Retinoids (synthetic and natural analogs of vitamin A): Retinoids have both acute toxicities (dryness of conjunctiva and oral mucous membrane, cheilitis, skin desquamation, hypertriglyceridemia, bone tenderness, arthralgia and myalgia) as well as chronic toxicities (teratogenic, hepatotoxic and bone remodeling).
    - Vitamin A (retinol)
    - β-all-transretinoic acid (retinoid)
    - 13-cis retinoic acid (isotretinoin)
    - Etretinate (aromatic ethyl ester derivative).
  - Carotenoids: β-carotene is a major source of vitamin A in diet. Carotenoids results in yellowing of skin.
  - α-tocopherol (vitamin E).
  - Calcium and selenium.
  - N-acetyl cysteine: N-acetyl cysteine is relatively nontoxic.
  - Epidermal growth factor receptor.
  - Farnesyl transferase.
  - Aspirin and cyclooxygenase-2 inhibitors.
  - Other: Other diet and pharmacological agents, which are under investigations include: soy isoflavones, folic acid, dietary fat and fish oils, raloxifene, sulindac and polyphenols in green tea.
- *Further information about ongoing trials*: It can be obtained from USA's Chemoprevention Branch of National Cancer Institute (301-496-8563).

Clinical Highlights

1. **Radiotherapy**: Sensitivity to radiation is more on the well oxygenated peripheral cells of carcinoma than the hypoxic central cells of carcinoma.
2. **Lhermitte's sign**: This rare sign is seen after radiation of cervical spine. On flexing the neck, patient develops electric current like sensation in arms, dorsal spine and both legs.
3. **Radioactive iodine**: It cannot be used in the ablation of medullary carcinoma of the thyroid.
4. **Adjuvant chemotherapy**: The chemotherapy used after surgery or radiotherapy for cancer is called as adjuvant chemotherapy.
5. **Bleomycin**: This antineoplastic drug causes pulmonary toxicity.

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62 Anesthesia

Be not a traitor to your thoughts. Be sincere; act according to your thoughts; and you shall surely succeed. Pray with a sincere and simple heart, and your prayer will be heard.

—Sri Ramakrishna Dev

Points of Focus

- **INTRODUCTION**
  - ✓ Preoperative Medications
- **GENERAL ANESTHESIA**
  - ✓ Inhaled Anesthetics
  - ✓ Intravenous Anesthetics
  - ✓ Skeletal Muscle Relaxants
  - ✓ Hemostasis and Blood Pressure Control
  - ✓ Monitoring During Anesthesia
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- **IMMEDIATE AIRWAY MANAGEMENT**
  - ✓ Endotracheal Intubation
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- **LOCAL ANESTHESIA**
  - ✓ Two Groups of Local Anesthetics
  - ✓ Types
- **CLINICAL HIGHLIGHTS**

INTRODUCTION

Advances in anesthesia have been driven by the advances in the surgical practices. The anesthetist prime duty is not only to maintain patient's safety but also to optimize operative conditions. An anesthetist care begins before surgery and extends into the postoperative period until the patient is in surgical ward or intensive care unit. The anesthetist should assess the patient preoperatively and discuss general medical condition and specific anesthetic risk factors with the patient and relatives.

Preoperative Medications

Following medications may be considered before otolaryngology operations:

- *Sedatives and tranquilizers*: Benzodiazepines (BZs) and barbiturates.
- *Analgesics*: Opioids.
- *Anticholinergics*: They prevent increased salivation and vagal tone.
- *Antiemetics*: Droperidol and ondansetron control postoperative nausea and vomiting.
- *Gastric emptying*: If needed it can be done with metoclopramide or intraoperative gastric suction.

GENERAL ANESTHESIA

General anesthetics are now safer and more controllable. General anesthesia is usually induced intravenously and maintained by halothane, enflurane, desflurane or sevoflurane. Propofol, which is being used as induction as well as maintenance anesthetic, has been replacing thiopentone as the intravenous induction agent. The nonpungent sevoflurane can also be used as an inhalational induction especially in children and needle phobic adults.

Analgesic agents at the time of induction reduce cardiovascular response to endotracheal intubation. Oxygen-enriched air mixtures are becoming more popular than nitrous oxide especially for longer surgeries. The newer agents, such as desflurane and sevoflurane, have more rapid action and recovery with fewer side effects.

Nitrous oxide is not commonly used during tympanoplasty and if used should be stopped 15 minutes or more before putting the tympanic membrane graft. Nitrous oxide is 34 times more soluble than nitrogen and rapidly enters in air filled middle ear cavity and results in increased middle ear pressure.

- *Total intravenous anesthesia*: It comprises propofol, alfentanil (short-acting opioid analgesic), neuromuscular blockade and pulmonary ventilation with air and oxygen. It is preferred during airway laryngeal surgery and endoscopy.
- *Upper respiratory infection*: A child with acute upper respiratory infection (URI) with fever and productive cough should not undergo general anesthesia as the chances of following complications are relatively more: bronchospasm, laryngospasm, desaturation, pulmonary aspiration, negative pressure pulmonary edema and aggravation in respiratory infection. The children with mild URI in the absence of fever and productive cough have minimal intraoperative and postoperative difficulties.

Inhaled Anesthetics

They include nitrous oxide and halogenated hydrocarbons and have varying potency in proportion to their lipid solubility (Table 1). They interact with neuronal membrane lipids and lead to inhibition of ion flux. Nitrous oxide (laughing gas) has euphoric and behavioral disinhibitory effects.

- *Minimal alveolar concentration value*: It is the minimal alveolar anesthetic concentration, (percentage of inspired air) at which 50% of patients do not respond to surgical stimulus.Minimal alveolar anesthetic concentration value is a measure of anesthetic potency. These values are additive. They are lower in elderly and in presence of opioid analgesic, local and intravenous anesthetics, and sedative and hypnotics.

Intravenous Anesthetics

- *Thiopental*: This rapid onset and short-acting barbiturate is used mainly for induction of anesthesia. It depresses respiratory and cardiac function. It is metabolized in liver. It does not increase cerebral blood flow.
- *Propofol*: It is particularly used in outpatient surgery and has antiemetic effect. Propofol has very rapid onset and recovery. It is used for both induction and maintenance.
- *Ketamine*: This fast and short acting anesthetic causes "dissociative anesthesia" that consists of amnesia, catatonia[1] and analgesia. It causes cardiovascular stimulation. Emergence reactions (vivid dreams and hallucinations) can be partly taken care of by BZs.
- *Midazolam*: This intravenous benzodiazepine (BZ) is used for conscious sedation and causes anterograde amnesia. It depresses respiratory function that is reversed by flumazenil (BZ receptor antagonist).
- *Fentanyl*: This short-acting strong opioid analgesic is used in anesthesia. Chest wall rigidity can occur with intravenous use. Oral and patch formulations are also available.

Neuromuscular Blockade (Skeletal Muscle Relaxants)

The neuromuscular blockade provides relaxation of muscle to facilitate surgery and mechanical positive pressure ventilation. These agents are not hypnotic and are used as an adjunct to general anesthesia. They interact with nicotinic receptors at the skeletal muscle end plate.

- *Depolarizing muscle relaxant (suxamethonium or succinylcholine)*: This rapid and short acting agent is excellent for brief duration during endotracheal intubation. Though rare it may cause prolonged muscular relaxation in patients with plasma pseudocholinesterase deficiency.
  - Side effects: It may cause hyperkalemia and is implicated in malignant hyperthermia (MH). It frequently causes postoperative diffuse muscle pains.
- *Competitive neuromuscular blocking agents (nondepolarizing)*: These long acting muscle relaxants are less dependent on hepatorenal function for metabolism and excretion. They need careful timing of doses so that their action is over by the end of surgery. A peripheral nerve stimulator can check the adequate depth of blockade

TABLE 1 Characteristic features of inhaled anesthetics

Inhaled anesthetic	*Heart rate*	*MAC values*	*Blood gas ratio**	*Characteristic features*
Nitrous oxide	Minimal effect	> 100%	0.5	Rapid onset and recovery, lowest potency, used in combinations
Halothane	Decrease	0.8%	2.3	Hepatitis, malignant hyperthermia, sensitizes heart to catecholamines, cardiac arrhythmias
Isoflurane	Increase	1.3%	1.4	Vasodilation, bronchial secretions, spasms
Enflurane	Decrease	1.7%	1.9	Tonic/clonic muscle spasms
Desflurane	Increase	7.2%	0.4	Most rapid onset, rapid recovery, airway irritation and coughing
Sevoflurane	Decrease	2.5%	0.7	Rapid onset and recovery

* Higher the solubility ratio slower the onset of action and recovery time
Key: MAC, minimal alveolar anesthetic concentration

[1](G. Katatonos, stretching down, depressed) A syndrome of psychomotor disturbances characterized by periods of physical rigidity, negativism (stubbornly resist or do opposite of what one is requested to do) or stupor.

during the surgery and confirm adequate recovery at the time of extubation.

- Mode of action: They bind competitively to one of the alpha subunits to prevent depolarization (receptor antagonists). They can be reversed by acetylcholinesterase inhibitors (e.g. neostigmine, pyridostigmine).
- Specific drugs: They include curare and its modern successor atracurium, cisatracurium, vencuronium, and rocuronium.
  - *d-Tubocurarine*: It blocks ganglia of autonomic nervous system and releases histamine. Its side effects include fall in blood pressure, bronchial secretions, bronchospasm and MH. It is not used now.
  - *Pancuronium*: More rapid onset and recovery. It is vagolytic and sympathomimetic and raises blood pressure.
  - *Atracurium*: This rapid onset and recovery agent is safer in hepatic and renal dysfunction as it is spontaneously inactivated.
  - *Mivacurium*: This very short acting agent is metabolized by plasma pseudocholinesterase and releases histamine.

Hemostasis and Blood Pressure Control

A 20–30% reduction of preoperative mean arterial blood pressure in otherwise healthy patient reduces not only the total blood loss but also facilitate surgery by providing the relatively bloodless operative field. The deep levels of anesthesia and slightly reduced arterial carbon dioxide (CO_2) tension also prevent excessive bleeding. If surgeon needs, induced controlled hypotension can be achieved with hypotensive drugs.

Monitoring During Anesthesia

The following parameters are monitored during the anesthesia:

- *Oxygen*:
  - Inspiratory oxygen concentration.
  - Pulse oxymetry for oxygen saturation.
- *Carbon dioxide*: CO_2 tension.
- *Blood pressure*: Careful control of blood pressure and blood loss.
- *Electrocardiogram:* Cardioscope for continuous electrocardiogram.
- *Temperature*: Vasodilation, cold infusions, loss of body heat and fluid evaporation from open wounds can cause hypothermia, which can be hazardous especially in children and elderly patients. Hypothermia and shivering increase the oxygen consumption and vascular resistance and predispose elderly patients to myocardial infarction.
- *Muscle relaxation*: It allows proper ventilation and surgical exposure.
- *Unconsciousness and pain control*: They minimize patient's distress.
- *Major surgery*: Hourly urine volume and invasive direct monitoring of circulation.

Recovery

Patient's respiration must be observed for any inadequacies, which may result from the following factors:

- Airway obstruction.
- Sedation from drugs and anesthetics.
- Hypoxia.
- Hypercarbia may be due to mechanical overventilation.
- Effect of muscle relaxants.

Day Care (Ambulatory) Surgery

Preoperative assessment should assess the medical, surgical, as well as social needs of the patient. Anesthetics that have rapid and short action and recovery, such as propofol, sevoflurane and desflurane, are preferred. Drugs having prolonged central depression including preanesthetic agents are avoided. Preferably these patients should be managed with laryngeal or face mask. Postoperative analgesics should be strong enough to cope with the pain.

Laser Surgery

The following anesthesia measures are taken to prevent laser related fires when surgery is in close proximity to endotracheal tube (ET):

- *Nitrous oxide*: It is avoided as it supports combustion.
- *Oxygen concentration*: Lowest possible inspired oxygen concentration guided by pulse oxymeter.
- *Endotracheal tubes*: If possible, perform laser surgery without ET or via a ventilating bronchoscope. Or use ET that resist laser ignition.
  - Cuff of ET is filled with saline. If laser accidentally strikes the cuff it will puncture and saline may extinguish a potential fire and irrigate tracheal tissue.
  - Methylene blue dye in the saline alerts the ENT surgeon to a punctured cuff. The blue dye will be seen leaking in the surgical field.
- *Management of ET ignition*: The smoke of burning tube contains toxic chemicals.
  - Remove the ET immediately.
  - Irrigate the burning tracheal tissue with normal saline.
  - Immediate reintubation: Trachea rapidly becomes edematous because of chemical and thermal injuries.

Endoscopy

During endoscopy airway is shared between ENT surgeon and anesthesiologist. The oxygenation and ventilation must be adequately maintained. The following measures facilitate better outcome:

- In patients with intubation ET is positioned to left side so that ENT surgeon has relatively better access to right side.
- Oxygenation can be maintained with ventilating bronchoscope or laryngoscope.
- Topical anesthesia aids to the outcome.
- Anticholinergics reduce the secretions.

Head and Neck Oncology

The following special considerations are worth remembering:

- *Airway difficulty*: The muscles of airway lose tone after the administration of muscle relaxant and tumor may unexpectedly increase the airway obstruction. So awake tracheostomy or awake fiberoptic intubation should be considered.
- *Alcohol abuse*: Many cancer patients have alcoholic hepatic

damage. The coagulopathies can cause extensive airway bleeding.
- *Smoking*: Chronic smokers have increased secretions and chronic obstructive pulmonary disease.
- *Radiation*: Preoperative radiotherapy can cause epiglottic fibrosis and laryngeal edema.

Thyroid Surgery

The following considerations are unique for thyroid surgery:
- *Hyperthyroidism*: These patients should be rendered euthyroid with antithyroid agents, which must be given even in the morning of surgery.
- *Tracheomalacia*: A large goiter can cause tracheomalacia and potential airway difficulties.
- *Depth of anesthesia*: The maintenance of appropriate depth of anesthesia and careful use of perioperative medications prevent exaggerated sympathetic response.
- *Muscle relaxant*: *Vecuronium* is preferred as it provides cardiovascular stability.

Complications

- *Intraoperative*:
  - Tachycardia: Hypoxia is the most common cause of tachycardia.
    - Cardiac causes: They include cardiac arrhythmia, ventricular tachycardia and supraventricular tachycardia.
    - Secondary causes: They include hypercarbia, pain, fever, sepsis, MH and hyperthyroidism.
    - Drugs: Atropine, glycopyrrolate, catecholamines, isoflurane, pancuronium and cocaine.
  - Malignant hyperthermia.
- *Postoperative:*
  - Respiratory acidosis: Residual anesthetics and muscle relaxants blunt the response to rising $PaCO_2$.
  - Metabolic acidosis: It occurs due to inadequate volume replacement after bleeding and third-space losses.
  - Respiratory alkalosis: It occurs due to pain or anxiety.
  - Dilutional thrombocytopenia: Blood transfusion when approaches the patient's entire blood volume can cause this coagulopathy because the original platelets are either lost or diluted. Stored red blood cells have almost no active platelets.

Dilutional thrombocytopenia is the most common coagulopathy that is seen after massive blood transfusion.

Malignant Hyperthermia (MH)

This autosomal dominant inherited anesthesia complication has 5–20% mortality. It seems that multiple genes play a role. The incidence in adults is about 1 in 40,000 anesthetics and 1 in 10,000–15,000 in children. It is exceedingly rare in children under 3 years of age.
- *Risk factors*: This rare condition occurs in patients with Duchenne's muscular dystrophy, family history of MH, and past history of anesthetic reaction.
- *Pathophysiology*: When patients with inherited membrane defect in skeletal muscle is exposed to potent inhalation anesthetics and succinylcholine, calcium is released at an enhanced rate from sarcoplasmic reticulum. The excess calcium causes hypermetabolic state which decreases the amount of available ATP. The actin-myosin bridge cannot detach and muscle remains contracted and rigid.
- *Clinical features*: Malignant hyperthermia has highly variable clinical presentation. The hypermetabolic skeletal muscles result in tachycardia, trismus, muscle rigidity, hyperventilation, cyanosis, sweating, fluctuating blood pressure and increased temperature.
  - Masseter muscle rigidity (MMR): It may be an early sign of MH in children. Malignant hyperthermia may develop within 10–20 minutes. If MMR lasts for more than 2–3 minutes halothane and succinylcholine must be stopped.
- *Laboratory findings*: Hypercarbia, hyperkalemia, increased creatinine kinase (often >20,000 IU), respiratory and metabolic acidosis, hypoxia and myoglobinuria.
- *Treatment*: Operation is stopped as soon as possible. Stop all triggering anesthesia medicines immediately. Team should be prepared for the management of ventricular tachycardia. The following medical treatment must be started immediately:
  - Hundred percent oxygen and hyperventilation.
  - Hyperthermia: Manage with cooling blankets.
  - Dantrolene 2.5 mg/kg: It blocks calcium release from sarcoplasmic reticulum and thus decrease muscle contractility and extreme rigidity.
  - Sodium bicarbonate 1m Eq/kg to begin with.
  - Glucose 0.5 gm/kg.
  - Insulin 0.15 U/kg.
  - IV fluids.
  - Diuretics like furosemide mannitol as per the need.

IMMEDIATE AIRWAY MANAGEMENT

About 30% surgical anesthesia deaths are caused by impaired airway, which is usually due to inadequate ventilation, unrecognized esophageal intubation and unanticipated difficult tracheal intubation. The methods to maintain airway during general anesthesia include jaw thrust, Guedel or laryngeal mask airway or an ET.

Endotracheal Intubation

Endotracheal intubation is a quick method of establishing airway. The larynx is visualized with a laryngoscope and an ET or a bronchoscope is inserted into the trachea. There is no need of anesthesia in emergency cases. Later on an orderly tracheostomy can be performed if needed. This avoids complications of an emergency tracheostomy, which are relatively frequent. Endotracheal tubes and their size selection are described in chapter "Instruments."

- In cases of difficult intubation, first pass a stylet and then rail road the ET or take the help of flexible bronchoscope.
- In cases of dental wiring and jaws shutting, nasal endotracheal intubation is done.

Other Procedures for Immediate Airway Management

- *Jaw thrust* (*Fig. 1*): In unconscious patients, extension of neck and lifting of mandible anteriorly displace the tongue base anteriorly away from the posterior pharyngeal wall and widens and straightens the airway. Neck extension is not indicated in cases of cervical spinal injuries. The ventilation can be provided with face mask or Ambu bag.
- *Oropharyngeal, nasopharyngeal and laryngeal mask airways*: They are described in chapter "Instruments."
- *Transtracheal jet ventilation* (*Fig. 2*): In this invasive method, an intravenous catheter (12 or 14-gauge) is inserted into the subglottic region through the cricothyroid membrane. The direction of the catheter is kept towards the trachea and intraluminal position is confirmed by the aspiration with a syringe. After withdrawing the needle jet ventilation can be started through the catheter.
  - *Caution*: Expiration of air must be insured to avoid pulmonary barotrauma with pneumothorax, pneumomediastinum and surgical emphysema.

Fig. 1: Jaw thrust

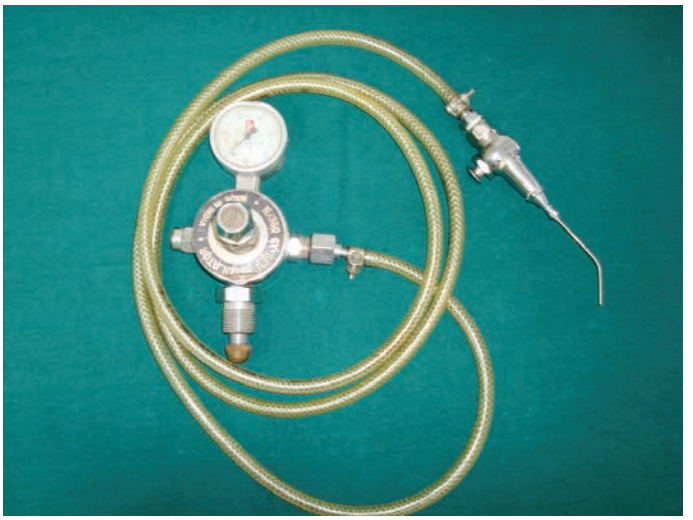


Fig. 2: Jet ventilation

Assessment for Difficult Intubation (Figs 3A and B)

Four distinct classes for difficult intubation are described on the bases of oral cavity view:

I. Uvula, faucial pillars, soft palate and hard palate visible.
II. Faucial pillars, soft palate and hard palate visible.
III. Soft palate and hard palate visible.
IV. Only hard palate visible.

In anticipated cases of difficult intubation (class III and IV), awake nonsurgical intubation techniques or awake surgical techniques must be employed.

Complications of Endotracheal Intubation

Tracheostomy must be done at the earliest when need for prolonged artificial airway is anticipated. The rate of complications of endotracheal intubation, which include following, significantly increases beyond the 10th day:

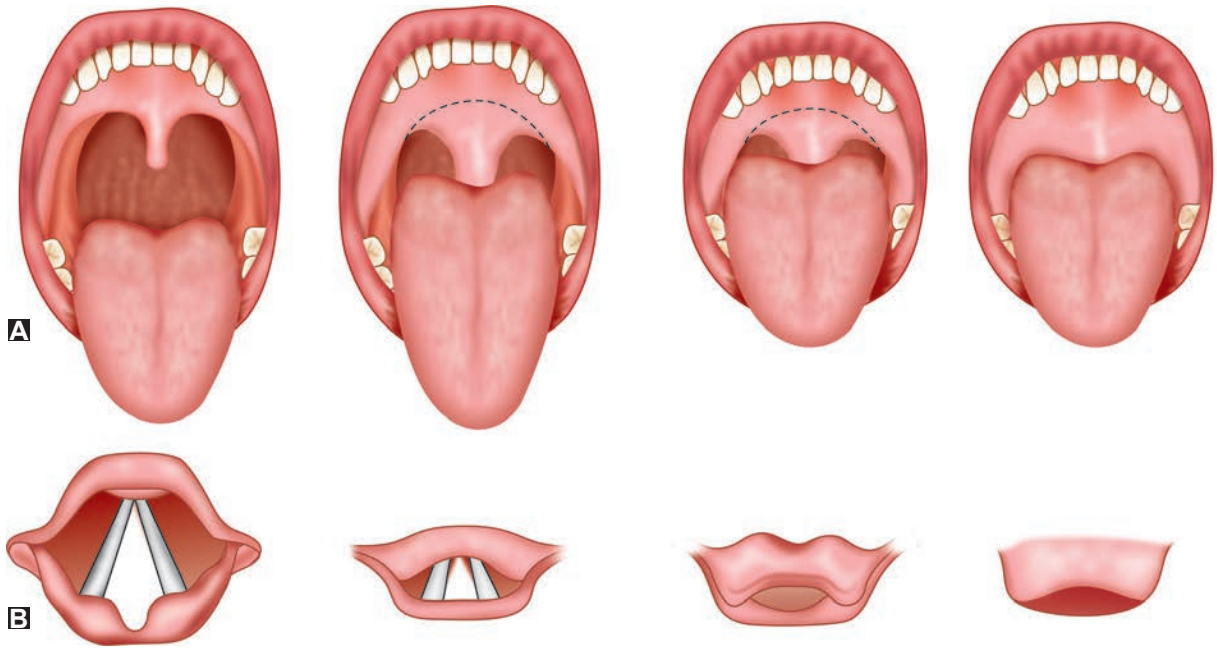
- Laryngeal injury
- Glottic and subglottic stenosis
- Infectious complications
- Tracheomalacia
- Tracheal stenosis

Tracheostomy facilitates suctioning, feeding, mobility, early return of speech and easy breathing.

LOCAL ANESTHESIA

For local anesthesia (LA) it is not necessary to keep the patient nil by mouth preoperatively. If the LA fails general anesthesia may be necessary. Compensation for an inadequate LA by heavy sedation carries risk of airway obstruction and aspiration of vomitus.

- *Prerequisites*: The patient's willingness and cooperation are important for LA. Resuscitation measures and oxygen supply should be available.
- *Contraindications*: They include allergy and local infection.
- *Side effects*: They may be local or systemic:
  - Local: Infection and hematoma.
  - Systemic: Recent agents, such as ropivacaine and levobupivacaine, are said to have better cardiovascular safety profiles. Accidental intravascular injection or overdoses can result in:
    - Neurotoxicity: Dizziness, nystagmus, depressed conscious level, convulsions.
    - Cardiovascular: Depresses cardiovascular parameters except cocaine that increase heart rate and blood pressure. Cardiac arrest (especially with bupivacaine) may be heralded by circumoral paresthesia and lightheadedness.
    - Methemoglobinemia (prilocaine).
    - Allergy: It is via para-aminobenzoic acid in esters.
- *Adrenaline*: Local anesthetics if given with adrenaline (1:200,000–1:125,000) has the following benefits:
    - Quick onset of action.
    - Prolong duration of action.
    - Slow absorption in circulation.
  - Contraindications: Hypertension, monoamine oxidase inhibitors, tricyclic antidepressants and end-arterial locations.



Figs 3A and B: Difficult airway. (A) Four classes of Mallampati oral view; (B) Four grades of Cormack and Lehand laryngoscopic view

Epinephrine (adrenaline) 1:200,000 contains 5 mcg/ml. Halothane may cause ventricular ectopy if patient has received adrenaline 2 µg/kg.

Local Anesthetics

They block sodium channels and the cell cannot achieve threshold potential. The nerve conduction is blocked as a propagated action potential fails to develop. They are categorized in two groups: ester and amide (Table 2).

1. *Cocaine*: This ester agent, which is used topically on mucosal surfaces, is almost exclusively used in otorhinolaryngology. It has a unique advantage of vasoconstriction. So, it provides topical anesthesia and vasoconstriction, which offer relatively blood-less operative field. In association with adrenaline or halothane, cocaine can sensitize myocardium to arrhythmias.

Cocaine was the first known local anesthetic agent which is still very popular especially in nasal surgeries.

2. *Lignocaine (lidocaine)*: Lignocaine 1% (10 mg/ml) is effective for most sensory blocks. It is a vasodilator and often prepared with epinephrine. About 50 ml 1% lignocaine with adrenaline can be safely used in a 70 kg person. Its early onset and short duration of action is good for sensory blocks. It is available in 2% xylocaine with and without adrenaline, 4% xylocaine, xylocaine viscus, xylocaine jelly and xylocaine ointment.
3. *Bupivacaine*: It has longer duration of action than lignocaine. Bupivacaine 0.25% is effective for sensory block. It is more cardiotoxic and is never given intravenously.
4. *Ropivacaine and Levobupivacaine*: These recent agents have better cardiovascular safety profile. They offer better sensory block than motor block.
5. *Prilocaine*: Methemoglobinemia causes blue-brown skin color.

Pain sensation is usually lost before the paralysis of motor activity by local anesthetics. Small, myelinated nerves (A-delta pain fibers) are blocked before small unmyelinated (C fibers) or large nerves.

Types

- *Topical anesthesia*: Topical anesthetic agents (e.g. lignocaine) are used on the skin and mucosa (such as nasal, oral, pharyngeal, laryngeal and tracheobronchial).

TABLE 2 Two groups of local anesthetics with their maximum doses

*Amide**	*Max dose*	*Ester***	*Max dose*
Lidocaine	3 mg/kg or 7 mg/kg with adrenaline	Cocaine	2–4 mg/kg
Bupivacaine	2 mg/kg	Tetracaine	1.4 mg/kg
Mepivacaine	4 mg/kg	Procaine	6 mg/kg
Etidocaine	4 mg/kg	Chloroprocaine	11 mg/kg
Prilocaine	400 mg		
Ropivacaine	225 mg		
Levobupivacaine	150 mg		

**Metabolism in plasma by pseudocholinesterase; May cause allergic reactions because they are metabolized to para-aminobenzoic acid; One i in names.

*Metabolism in liver; Allergy reactions rare; two i's in names.

- Amethocaine: It is well absorbed by mucosa.
- Cocaine: It has vasoconstrictive property.
- Lignocaine and prilocaine eutectic mixture on skin: It is used frequently in children before venepuncture.

- *Local infiltration*: Local anesthetic agent may be infiltrated into and around the wound and surgical site. It is contraindicated near infection site because it can spread the infection and acidity produced by infection blocks the action of local anesthetic agent and makes it ineffective. It is also contraindicated in patients with clotting disorders as it may cause hemorrhage.
- *Regional anesthesia*: It includes blockade of major nerve trunks, spinal and epidural anesthesia.
- *Adjunct to general anesthesia*: It provides pain relief during the postoperative period.

Peak anesthetic effect of topical lignocaine and cocaine occurs within 2–5 minutes and lasts for 30–45 minutes.

Clinical Highlights

1. **Down syndrome**: Children with Down syndrome have macroglossia and atlantoaxial instability. Preoperative airway assessment is important.
2. **Cocaine**: It is a catecholamine re-uptake inhibitor that causes vasoconstriction.
3. **Order of neuron sensitivity of LA**: Type B and C > Type A delta > Type A beta and gamma > Type A alpha.

FURTHER READING

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63 Laser Surgery and Cryosurgery

Each work has to pass through these stages—ridicule, opposition and then acceptance. Each man who thinks ahead of his time is sure to be misunderstood.

—Swami Vivekananda

Points of Focus

LASER
- **RELATED PHYSICS**
  - ✓ Properties of Radiant Laser Energy
- **CONTROL OF LASER**
  - ✓ Transverse Electromagnetic Mode
- **TISSUE EFFECT**
- **LASER IN OTOLARYNGOLOGY**
  - ✓ Properties of Commonly Used Lasers
  - ✓ Argon Laser
  - ✓ Potassium-Titanyl-Phosphate–532 Laser
  - ✓ Neodymium:Yttrium-Aluminium-Garnet Laser
  - ✓ Carbon Dioxide Laser
  - ✓ Complications and Safety
- **PHOTODYNAMIC THERAPY**
  - ✓ Principle
  - ✓ Indications
  - ✓ Side Effects

RADIOFREQUENCY SURGERY
- ✓ Indications
- ✓ Material and Method

CRYOSURGERY
- ✓ Principle
- ✓ Tissue Effect
- ✓ Technique
- ✓ Indications
- ✓ Advantages
- ✓ Disadvantages

HYPERBARIC OXYGEN THERAPY
- ✓ Mode of Action
- ✓ Material and Method
- ✓ Indications

- **CLINICAL HIGHLIGHTS**

LASER

LASER is an acronym for **L**ight **A**mplification by **S**timulated **E**mission of **R**adiation. Laser light is the brightest monochromatic (one wavelength) light. In addition to diagnostic medicine and surgery, the laser is used in research laboratories, communications, surveying, manufacturing, lecture pointers, printers, CD players and engraving. Bar code scanners are used in supermarkets and shops.

RELATED PHYSICS

- *Spontaneous emission of radiation*: In a stable atom, there are equal number of protons and electrons. Electrons revolve around the nucleus in one of several discrete orbits. The orbits closure to nucleus, have lower energy levels than the larger shells, which are away from nucleus. The interaction of electron with photon (called absorption), which is a quantum of light, makes the atom excited. During excitation, an electron of low energy level can go into higher energy orbit. But within a very short time (10–8 sec) the electron spontaneously drops back to its lower level and gives up energy difference. During this process, atom emits extra energy as photon of light, which is called as spontaneous emission of radiation.
- *Stimulated emission of radiation*: If a photon of correct energy hits an excited atom, it results in emission of two identical photons, which have same frequency and energy

and travel in same direction. This stimulated emission of radiation, which was described by Einstein, is the basic fundamental principle of laser science.

- *Radiant laser energy*: The stimulated radiation is amplified with the help of two mirrors in an optical resonating chamber, which is filled with an active medium, such as Ar, Nd:YAG or CO_2. An electric current excites this active medium, which can consist of molecules, atoms, ions semiconductors or even free electrons in an accelerator. Mirrors reflect the photons back and forth. One of the two mirrors is partially transmissive, which emits some of the radiant energy as laser.

Properties of Radiant Laser Energy

The radiant laser energy is a type of electromagnetic radiations. It has following qualities that distinguish it from disorganized light of a bulb:

- Monochromatic, i.e. same wavelength (single color)
- Collimated (unidirectional)
- Coherent: Both temporally (waves of light oscillating in a phase) and spatially (photons are equal and parallel)
- Extremely intense.

CONTROL OF LASER

The variables of lasers, which can be controlled, are power (watts), spot size (millimeters) and exposure time (seconds).

- Irradiance (W/cm^2): It considers surface area of focal spot. It is more useful measure than power, which may be kept constant. Irradiance varies directly with power and inversely with spot size. The laser lens setting (focal length) and working distance combinations decide the size of focal spot. Larger the focal spot (unfocussed, away from focal plane), lower the irradiance. Smallest the focal spot (focused in focal plane), highest the irradiance, which results in precise cutting and vaporization.
- Depth of focus: The beam waist presents over a range of distances called depth of focus.
- Fluence (J/cm^2): It is a measure of the total amount of laser energy per unit area. It varies directly with exposure time (seconds) of laser beam to a unit area. Working in pulsed mode or in continuous mode can change fluence.

Transverse Electromagnetic Mode

Transverse electromagnetic mode (TEM) determines the shape of laser spot. It refers to the distribution of radiant energy of laser beam across the focal spot. The different modes of TEM are:

- TEM_{00}: Laser spot is circular on cross-section. The power density is greatest at the center and progressively diminishes peripherally (Gaussian distribution).
- TEM_{01} and TEM_{11}: Beams cannot be focused to a small spot and have complex distribution of energy. It results in predictable tissue vaporization.

TISSUE EFFECT

The tissue deals with incident laser energy in four ways: reflects, absorbs, transmits and scatters. The energy, which is reflected from or transmitted through the tissue, will not have any effect on the tissue. Energy that is absorbed results in surgical interaction with tissue and varies with laser's wavelength.

- *Wavelength*: Shorter the wavelength, more is the scattering, which spreads the energy and limits the penetration depth.
- *Levels of heating and tissue changes*: The primary form of interaction of absorbed laser with tissue is heating, the level of which decides the following changes in the tissue:
  - 60–65°C: Protein denaturation and blanching of tissue
  - 100°C: Vaporization of intracellular water, vacuole formation, craters and tissue shrinkage
  - Several 100°C: Carbonization, disintegration, smoke, destruction and gas generation.
- *CO_2 laser*: In the center of the wound is an area of tissue vaporization. It which is surrounded by an area of thermal necrosis (small vessels, nerves and lymphatics are sealed) which is further surrounded by an area of thermal conductivity and repair. The short laser pulse minimizes the lateral thermal damage.

LASER IN OTOLARYNGOLOGY

The lasers beams are used to vaporize, cut and coagulate the tissue. The clinical applications depend on their wavelength and special absorptive powers of the target tissues. The laser can be ultraviolet, which result in heating and photodissociation of chemical bonds, but most commonly used lasers emit either visible [Argon (Ar) and Potassium-titanyl-phosphate-532 (KTP-532) lasers] or infrared light [carbon dioxide (CO_2) laser].

- Properties of commonly used lasers and ENT applications: They are given in Table 1.
- *Most commonly used*: They are CO_2, neodymium:yttrium-aluminium-garnet (Nd:YAG), KTP-532 and Ar.
- *Other*: Other lasers used in otolaryngology are Ar tunable dye laser and flash lamp pumped dye laser.
- *Under investigations*: The lasers under investigations include erbium:YAG (Er:YAG) and holmium:YAG (Ho:YAG) lasers.

Argon Laser

Argon laser passes through clear fluid and is absorbed by hemoglobin and pigmented tissues.

- Indications:
  - *Vascular lesions*: Photocoagulation of portwine stain, hemangioma and telangiectasia.
  - *Retinal lesions*: It passes through the clear aqueous tissues (cornea, lens and vitreous).
  - *Ear microsurgery*: Its uses in ear microsurgery are lysis of middle ear adhesions, spot welding or tympanoplasty grafts.
    - Stapedotomy: A drop of blood is kept on stapes footplate before its use in stapedotomy.

Potassium-Titanyl-Phosphate-532 Laser (KTR-532)

Potassium-titanyl-phosphate laser has wavelength of 532 nm (blue-green) and comparable with Ar laser. It falls in visible spectrum and is selectively absorbed by pigment and more

TABLE 1 Properties of commonly used lasers and their ENT applications

	Ar laser	*Nd:YAG laser*	*CO_2 laser*
Electromagnetic range	Visible	Invisible infrared	Invisible infrared
Color	Blue-green	Colorless	Red light of helium-neon
Wavelength	0.488 and 0.514 μm	1.064 μm	10.6 μm
Extinction length*	80 m	40 m	0.03 mm
Transmitted through	Clear aqueous tissue	Clear liquids	
Absorption by	Hemoglobin, pigmented tissue	Darkly pigmented tissue, charred debris	Water, tissue with high water content
Scattering	Less	More	Negligible
Clinical applications	Portwine stains, hemangiomas, telangiectasis, stapedotomy	Obstructing lesions of trachea, bronchus, esophagus; vascular, lymphatic lesions	Extremely versatile use in ear, nose and throat lesions
Precision	Good	Less	Good

Ar, Argon; Nd:YAG, Neodymium:Yttrium-Aluminium-Garnet; CO_2, Carbon Dioxide
*Extinction length: The thickness of water necessary to absorb 90% of the incident laser energy

strongly by hemoglobin. Hand-held probe facilitates its use in endoscopic sinus surgery and microlaryngeal surgery. The optical fiber delivery can be manipulated through rigid bronchoscope.

- Indications: It is first choice in the following conditions:
  - *Ear*: Stapedotomy
  - *Nose*: Polyps, concha bullosa, epistaxis, turbinate hypertrophy and telangiectasia
  - *Oral cavity*: Verrucous and T_1 carcinoma, leukoplakia, erythroplakia, early tongue cancer T_1, lymphangioma
  - *Oropharynx*: Recurrent tonsillitis and hypertrophy, uvulopalatopharyngoplasty in obstructive sleep apnea, T_1 and T_2 carcinoma
  - *Larynx*: Laryngocele, cyst, granulomas, stenosis (glottic, posterior and subglottic), bilateral vocal cord paralysis, recurrent respiratory papillomas, suprahyoid supraglottic T_1 carcinoma and obstructing carcinoma
  - *Skin*: Pigmented dermal lesions.

Neodymium:Yttrium-Aluminium-Garnet Laser (Nd:YAG)

Neodymium:yttrium-aluminium-garnet laser can be transmitted by flexible endoscopes and has effective coagulative properties. It controls the bleeding well. The flexible fiberoptic delivery system allows its use with flexible endoscope.

It is excellent for tissue coagulation, but the precision is poor as the tissue damage is widespread and depth of tissue penetration is less predictable. It can be used in combination with CO_2 laser.

- *Indications*: It is advantageously used for following lesions as control of bleeding (dangerous in bronchoscopy) is more secure.
  - Obstructing malignant tumor of trachea, bronchus and esophagus
  - Vascular lesions: Hereditary hemorrhagic telangiectasia of nose
  - Lymphatic disorders: Lymphangioma.

Carbon Dioxide Laser

Carbon dioxide laser requires aiming beam of helium neon laser. It is the most commonly used laser in ENT surgery. It is transmitted through an articulating arm and can be used free-hand for microscopic surgery, attached to microscope and adapted to rigid bronchoscope. Its main limitation is that it cannot pass through the flexible endoscopes.

It is effective not only in vaporizing tissues, but it also provides bloodless field. Surgery can be performed in cases of hypertension, bleeding dyscrasias and coagulopathies. The other advantages are precision surgery and less postoperative edema and pain.

- *Advantages*
  - Negligible scattering and reflection
  - Absorption independent of color
  - Minimal thermal effect on adjacent tissue
- *Indications*
  - *Nose*: Papillomas, rhinophyma, telangiectasis, nasal polyps, choanal atresia and turbinate hypertrophy.
  - *Oral cavity*: Leukoplakia, erythroplakia, small superficial cancers and debulking of large, recurrent or inoperable tumors
  - *Oropharynx*: Recurrent tonsillitis and hypertrophy, tonsillar and pharyngeal tumors, tongue T_1 and limited T_2 cancer
  - *Larynx*: Papilloma, web, stenosis (glottic, subglottic, and posterior), capillary hemangiomas, vocal nodule, Reinke's edema, leukoplakia of cord, polypoid degeneration of cord, arytenoidectomy, T_1 midcordal carcinoma without anterior commissure involvement, suprahyoid supraglottic T_1 cancer, laryngocele, cysts and granulomas
  - *Trachea and bronchi*: Recurrent papillomatosis, tracheal stenosis, granulation tissue and bronchial adenoma, debulking of obstructive malignant lesions of trachea or bronchi
  - *Plastic surgery*: Benign and malignant tumors of skin, vaporization of nevi and tattoos
  - *Ear*: Stapedotomy and acoustic neuroma.

Complications and Safety

The laser is a potentially dangerous instrument. The utmost caution is required to prevent accidents, which can injure not only patient but also health care personnel present in operation room.

- *Education of staff*: The operating surgeon and anesthesiologist must have proper experience and training. Nursing and operation theater personnel should be conversant with safety measures while operating laser.
- *Protection of eye*: Protective eyeglasses with side protectors, which are specific for the wavelength of each laser (blue-green glasses with optical density of 6 for Nd:YAG laser; orange yellow glasses for Ar, KTP or dye lasers), must be worn by the patient, surgeon, anesthetist, assistants, nurses and all other personnel present in operating room. They prevent accidental burns to cornea, retina and lens (lenticular opacities). Patient's eyes are protected by a double layer of saline moistened eye pads.
- *Protection of skin*: All exposed parts of the patient not in surgical field, which include skin, mucous membranes and teeth, are protected by saline soaked towels, pads or sponges that are moistened periodically.
- *Evacuation of smoke*: Two separate suctions, one for the blood and mucous and the other for smoke and steam (produced by laser vaporization of tissues) are used.
- *Anesthetic gases and equipment*: The endotracheal tube fire is the dreaded complication. Only noninflammable gases (such as halothane or enflurane) are used. During the CO_2 laser, red rubber or silicone tube is wrapped by reflective metallic foil. Cuff of endotracheal tube is inflated with saline water, which may be colored by methylene blue that helps in warning about the leak of cuff. Tubes are further protected with saline soaked cotton. The colorless or white polyvinyl or silicone tube that does not have any black or dark marking or a lead lined marking along the side, is safest with the use of Nd:YAG laser.

PHOTODYNAMIC THERAPY

Principle

Photodynamic therapy (PDT) is an upcoming modality. It is based on the principle that photosensitizing agent is taken up preferentially by the malignant cells, which are then exposed to specific wavelength of laser (such as Ar tunable dye laser with a wavelength of 630 nm). Laser activates the photosensitizing agent and thus destroys the cancer cells. There is preferential uptake of photosensitizer photofrin (dihematoporphyrin ether or DHE) (given intravenously) by the malignant cells.

Light activation of photoconcentrated DHE results in mitochondrial damage and apoptosis in malignant cells. Erythrocyte leakage and endothelial damage of vessels cause ischemic necrosis of tumor tissue.

Indications

- Photodynamic therapy is helpful in treating cancer of skin, larynx, nasopharynx, aerodigestive tract and endobronchial region.
- It has also been used in cases of recurrences after radiation or surgery.
- Superficial cancers of larynx have been treated with PDT. It has got US FDA approval for treating obstructing esophageal and endobronchial tumors and minimally invasive endobronchial non-small cell carcinoma.

Side Effects

The main side effect of PDT is generalized skin photosensitization. Patient should use sun-protective clothing to avoid exposure to sunlight.

RADIOFREQUENCY SURGERY

Radiofrequency (RF) surgery reduces the volume of tissues. This minimally invasive surgery can be done as an OPD procedure.

Indications

Radiofrequency cuts and coagulates tissues with minimal lateral tissue damage. It can be used in the following disorders:

- Nasal obstruction:
  - Reduction of hypertrophied inferior turbinates.
- Sleep apnea and sleep-disordered breathing including primary snoring, upper airway resistance and obstructive sleep apnea (OSA). See Chapter on "Sleep Apnea and Sleep-Disordered Breathing".
  - Reduction of redundant soft palate and uvulopalatoplasty
  - Reduction of fullness in base of tongue.
- Lingual thyroid
- Tonsillotomy
- Microlaryngeal surgery to remove granuloma, papilloma and cyst
- Myringotomy
- Rhinophyma
- Cosmetic: Removal of skin lesions.

Material and Method

The machine generates electromagnetic waves of very high frequency (350 kHz to 4 MHz). Usually 460 kHz RF is delivered through the probe, which is inserted into the tissue and causes ionic agitation. Heating of the tissue causes protein coagulation and tissue necrosis. There is no charring. The scar formation occurs in 3 weeks and reduces the size of tissue. The parameters, which can be controlled by the device, include:

- Power in watts
- Temperature in degrees of celsius
- Resistance in ohms
- Treatment time in seconds
- Energy in joules (watts × seconds).

CRYOSURGERY

Principle

At –30°C and below, rapid freezing of tissues and slow thawing results in the destruction of tissue. This principle is used in cryosurgery, which has been used to treat benign, premalignant and malignant lesions.

The freezing agents are used either by an open method

(liquid nitrogen spray or CO_2 snow) or through a closed system cryoprobe, which is based on Joule-Thomson effect (rapid expansion of compressed gas through a small hole produces cooling). The freezing agents employed in closed systems probes are: liquid nitrogen, nitrous oxide or CO_2. The probes are available in different sizes and designs and produce a tip temperature of –70°C. The thermocouples of probes can be inserted into the tissue to monitor the temperature.

Tissue Effect

The cell death occurs through following mechanisms:

- Dehydration occurs due to crystallization of intracellular and extracellular water with consequent rise in electrolytes concentration. The pH changes as the buffering substances crystallites out. Urea and dissolved gases increase to reach toxic concentrations, which cause cell death.
- Denaturation of cell membrane lipoproteins makes cell membrane permeable to cations. Thawing of cells, which become full of cations, results in lysis of cells.
- Thermal shock arrests the cellular respiration.
- Vascular stasis of both arterial and venous blood results in ischemic infarction. Cryosurgery is useful in the treatment of vascular lesions (hemangioma, angiofibroma and glomus tumors) because thrombosis of capillaries results in less bleeding.
- Autoantibodies specific to the frozen tumor tissues may provide tissue specific immunity to subsequent recurrence.

Technique

1. *Anesthesia*: Cryosurgery can be done under either local anesthesia or mild sedation or even without anesthesia because tissue freezing itself causes numbness.
2. *Freezing*: The cryoprobe is applied into or upon the tissues (which is insulated and include a margin of normal tissue) for 3–8 minutes. It results in rapid freezing.
3. *Thawing*: Then the frozen tissue is allowed to thaw slowly.
4. *Repetitions*: The procedure may be repeated as required once or twice to achieve the best result.
5. *Thermocouple*: If available, a thermocouple will ensure freezing at an adequate depth.
6. *Healing*: The area is allowed to heal by secondary intention. The necrotic slough usually falls off in 3–6 weeks. If needed, repeat the cycles of cryotherapy.

Indications

The increasing availability and popularity of laser is fast declining the indications of cryosurgery. Its lower cost still makes it an option in developing countries.

- *Benign vascular tumors*: Hemangiomas of skin, oral cavity and oropharynx, angiofibroma and glomus tumor
- *Premalignant lesions*: Leukoplakia of cheek, tongue, floor of mouth and solar keratosis (precancerous condition of skin). The scarring is less and quality of regenerated epithelium is better in comparison to diathermy.
- *Malignant lesions*: Intraepithelial carcinoma (Bowen's disease) and basal cell carcinoma of skin. Palliation of advanced cancers, recurrent and residual tumors. Debulking of tumor facilitates deglutition and respiration. It reduces bleeding and relieves pain. Cryotherapy does not cause necrosis of bone and cartilage, which may underlie the lesion. Recurrent tumors and ill-defined lesions are not good cases for cryotherapy.
- *Nose*: Reduction of turbinates improves the airway. In allergic rhinitis, it controls sneezing and rhinorrhea.
- *Tonsils*: Cryodestruction of tonsils can be done in poor risk patients.

Advantages

- No need of general anesthesia. Therefore, it is useful in poor risk patients.
- The patients with bleeding disorders or coagulopathies can be managed.
- Palliation in multiple and recurrent cancers where second course of radiotherapy cannot be used.
- Minimal after effects, such as discomfort and pain.
- Minimal scarring. Therefore, it can be used at sites well-known for keloid development, such as sternal area.
- Cryosurgery can be done as an OPD procedure.
- Lower cost in comparison to laser.

Disadvantages

- Excisional biopsy and histopathological assessment of tumor margins are not possible.
- Depth of freezing is unpredictable.
- Causes skin depigmentation and loss of hair (destruction of hair follicles).

HYPERBARIC OXYGEN THERAPY

Mode of Action

- Increased partial pressure of oxygen (up to 460% increase in pO_2) in perilymph and endolymph supplies oxygen to the inner hair cells. The pO_2 level remains 60% above the normal after 1 hour of termination of hyperbaric oxygen (HBO_2) therapy.
- Reduction of hematocrit and decrease in blood viscosity improves diffusion of oxygen to ischemic cells.

Material and Method

Patient inhales 100% oxygen for 1 hour at atmospheric pressure 2.4 in a HBO_2 chamber. In addition to 1 hour oxygen inhalation under pressure, the compression and decompression time of the chamber is 10–15 minutes respectively. Usually a total of 10–20 such sittings (six per week) are given.

Indications

Hyperbaric oxygen has been used with success in following disorders:

1. Sudden idiopathic sensorineural hearing loss and tinnitus: The results are better if therapy is started earlier. In various studies, improvements have been reported in 30–80% of the patients.
2. Acoustic trauma
3. Noise-induced hearing loss
4. Malignant otitis externa
5. Mucormycosis of paranasal sinuses
6. Skin flaps with compromised blood supply.

Clinical Highlights

1. **Argon laser**: It is useful for middle ear surgery.
2. **Cryosurgery**: To cause cell death, temperature should at least reach –30°C. In cryosurgery, liquid nitrogen is applied at –30°C. The cryoprobe is kept for 3–8 minutes so that area is frozen rapidly reaching a temperature of about –70°C.

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Appendix

Today the human body is, tomorrow it is not; even the shortest span of life is beset with pain and misery. Just surrender yourself to God; you will then feel His grace. Do not be afraid; the Master is behind you, and I am, too as your Mother. Whenever you are in distress, say to yourself 'I have a mother'.

—Holy Mother Sarada Devi

☞ **Top 101 Clinical Secrets** ☞ **Problem Oriented Cases** ☞ **Miscellaneous Key Points**

TOP 101 CLINICAL SECRETS

1. **Outer hair cells:** They serve an amplifying role in cochlea and are more sensitive to noise exposure and ototoxicity. They are responsible for generating the cochlear microphonic and otoacoustic emissions. Otoacoustic emissions are measured to test hearing in infants and others difficult to test patients.
2. **Danger space:** It lies between the alar fascia and the prevertebral space. Infection of this space can lead to mediastinitis and death if not properly treated.
3. **Drugs:** Physicians are supposed to know the indications, interactions and side effects of drugs they prescribe. They must update themselves regarding the newer drugs which can have advantages and disadvantages over existing drugs.
4. **Penicillin allergy and cephalosporins:** Cephalosporins can be given safely to patients with a history of rashes with penicillin but not in patients with a history of anaphylaxis.
5. **HIV/AIDS:** Biopsy of new oral lesions must be done early to rule out malignancy.
6. **ENT examination:** Keep a specific order with clinical examination (and even within each area) such as ear, nose, mouth, pharynx and larynx, face and scalp, neck, and neurologic exam. It ensures complete examination and avoids missing anything.
7. **Temporomandibular disorder:** Most patients respond to nonsurgical therapy that includes rest, and medical, physical and splint therapies.
8. **Surgical treatment of temporomandibular joint disorder:** It is indicated only for internal derangement of the temporomandibular joint (TMJ). It is not suggested for musculoskeletal disorders.
9. **Pneumatic otoscopy:** It is the gold standard for the diagnosis of otitis media.
10. **Tinnitus:** It is a symptom and not a disease. It can be so disabling that the patient can have suicidal tendencies.
11. **Subjective tinnitus:** It is most common type of tinnitus and is heard only by the patient.
12. **Objective tinnitus:** It can be heard by both patient as well as examiner.
13. **Pulsatile tinnitus:** In cases of pulsatile tinnitus always first rule out the paraganglioma (glomus tympanicum or jugulare).
14. **Tuning fork tests:** They should confirm the results of audiometry. Never trust the audiogram alone when surgical intervention is considered. The inconsistency must be resolved with the audiologist.
15. **Conductive hearing loss:** The most common causes are ear wax, otitis media, otomycosis, and otosclerosis.
16. **Otosclerosis:** Fifty percent of patients give positive family history. Patient has conductive hearing loss with normal tympanic membrane and impaired acoustic reflexes. The patients with negative Rinne [Bone conduction (BC) > Air

conduction (AC)] are candidate for stapedectomy, which provides very gratifying results.

17. **Reversible causes of sensorineural hearing loss:** They must be ruled out even in patients with the most common causes of hearing loss.Such as presbyacusis and noise-induced hearing losses
18. **Deaf and dumb:** Deaf persons are not intellectually dumb. Psychological tests usually overlook the intellectual capacity of deaf people as they consist of culturally biased material.
19. **Mild hearing loss in children:** Address even mild hearing loss early to prevent speech delay in children.
20. **MRI and cochlear implants:** MRI is contraindicated in patients with conventional cochlear implants. There is risk of implant movement.
21. **Live insect in external ear:** Never remove live insect from external ear. First either drown or anesthetize the insect. Live angry insect can further damage the canal.
22. **Otitis externa treatment:** First line treatment for otitis externa is topical antibiotics. Systemic antibiotics are given only for severe infections.
23. **Eustachian tube dysfunction:** It is the key to the pathogenesis of otitis media.
24. **Persistent unilateral otitis media in an adult:** Nasopharyngoscopy must be done to rule out neoplastic lesion of the nasopharynx.
25. **Congenital cholesteatoma:** A whitish mass behind a child's intact tympanic membrane with hearing loss indicates congenital cholesteatoma.
26. **Cholesteatoma:** It is collection of squamous epithelium and keratin debris in the middle ear. It which erodes bone and can cause life-threatening complications. Patient presents with purulent and putrid ear discharge and conductive hearing loss. Otoscopy may show a "trail sign" of debris along the posterosuperior canal wall to the marginal perforation. The usual treatment is modified radical mastoidectomy.
27. **Gradenigo's triad:** It consists of otorrhea, abducens palsy, and retrobulbar pain. It is the classic presentation of petrous apicitis.
28. **Tullio's phenomenon or Hennebert's sign:** They are present in patients with labyrinthine fistula.
29. **Dix-Hallpike examination:** It is the most important clinical test for dizzy patients because 25% of all dizzy patients have benign paroxysmal positioning vertigo.
30. **Downbeating nystagmus:** A pure downbeating nystagmus during Dix-Hallpike testing suggests Chiari malformation or other posterior fossa lesions. MRI must be done.
31. **Peripheral and central vertigo:** Patients with central dizziness complain few symptoms but have many findings (sensory/motor deficits). Patients with peripheral vertigo have many symptoms (severe whirling vertigo with or without otological symptoms) but few findings.
32. **Multisensory imbalance:** It is suggested when dizziness appears on walking but is relieved by pushing a grocery store cart.
33. **Vertigo spells in children:** Migraine is not uncommon in children. It must be considered when vertigo spells are not associated with ear malformations or middle ear infections.
34. **Facial nerve:** The regeneration and degree of return to normal is dependent on the degree of initial injury (neuropraxia vs. neurotmesis). The most important factor in history is whether the palsy develops slowly over days or immediately at the time of the injury.
35. **Clinical features of temporal bone fracture:** Clinical features include hearing loss, dizziness, facial weakness, ear bleeding, hemotympanum, raccoon eyes, and/or bruising over the mastoid cortex (Battle's sign).
36. **Radiological investigation for temporal bone fracture:** The best radiologic examination is a fine cut, axial and coronal, temporal bone high-resolution computed tomography (HRCT) scan.
37. **Indications for surgical exploration of the facial nerve in temporal bone fracture:** They include:
    a. Immediate onset of complete facial paralysis
    b. Delayed onset of complete facial paralysis associated with–
        i. Radiologic evidence of a fracture through the fallopian canal of facial nerve.
        ii. Poor prognostic testing with electroneuronography (EnoG) or electromyography (EMG).
38. **Anosmia:** The three most common causes are sinonasal disease, post-upper respiratory tract infection and trauma (injury to olfactory nerves at cribriform plate or brain injury).
39. **Posterior epistaxis:** One to two percent of patients die within 1 year.
40. **Antibiotic resistance patterns in acute bacterial rhinosinusitis:** Penicillin resistant Streptococcus pneumoniae (25–40%), beta lactamase producing Haemophilus influenzae (30–40%) and beta lactamase producing Moraxella catarrhalis (92%).
41. **Chronic hyperplastic rhinosinusitis:** Eosinophilic infiltration is the hallmark in most of the patients and about 50% of patients have asthma.
42. **Autoeczematization ("id" reaction):** This is a cutaneous response to remote infections.
43. **Allergic rhinitis:** Tree, grass, and weed pollens in their seasons may cause symptoms of allergic rhinitis in atopic individuals (hereditary predisposed). Immunotherapy attenuates allergic response by hyposensitization to the allergens.
44. **Septal hematoma:** Before reducing the nasal bone fractures, a septal hematoma must be ruled out because failure to drain it may result in a septal abscess, septal perforation and saddle nose deformity.
45. **Polychondritis:** Seventy to eighty percent patients have involvement of the nasal septum.
46. **Life-threatening injuries:** Most of them can be identified during the primary survey.
47. **Cervical spine immobilization in trauma patients:** It is vital to maintain cervical spine immobilization while managing the airway.
48. **Blood loss:** Young healthy patients can lose up to 30% of blood volume with minimal symptoms.
49. **Intubation:** It is required if the patient has Glasgow Coma Scale score of 8 or less.
50. **Facial trauma:** Thin cut (1–3 mm) facial CT with coronal reformatting is the ideal imaging study.
51. **Healing:** Primary closure of a facial defect is best if possible. Other options include healing by secondary intention, local flaps, skin grafts and regional or free flaps.
52. **Mandibular fractures:** Open reduction and internal fixation (ORIF) with bone plates and lag screws is the most accepted treatment. It provides stable rigidity and early patient function.
53. **Condylar fractures:** Severe fractures are problematic even with the open reduction and internal fixation.
54. **Orbital blowout fractures:** The simple forced duction testing detects extraocular muscle entrapment.

55. **Foreign body nose in children:** A unilateral foul smelling purulent nasal discharge in a child is mostly due to forgotten foreign body nose.
56. **Inverted papilloma:** The excision should be aggressive because this aggressive nasal benign tumor has frequent coexisting carcinoma.
57. **Jaw swellings:** The panoramic OPG (Panorex) X-ray is indispensable for the diagnosis.
58. **Oral and pharyngeal examination:** It also includes palpation. It is important because certain lesions such as submucosal mass/nodule of carcinoma tongue can be felt and not seen.
59. **Taste and flavor:** They are usually confused. The taste includes only the ability to sense sweet, salty, bitter, and sour tastes. Flavor includes both taste and smell (80%). Patients with taste problems may be having flavor and a smell disorder.
60. **Sense of taste:** It is highly redundant due to its innervation. It is nearly impossible to lose all sense of taste.
61. **Oral thrush in adults:** The common risk factors are corticosteroid and broad-spectrum antibiotics, pregnancy, diabetes mellitus, nutritional deficiency and human immunodeficiency virus.
62. **Viral sialadenitis:** Mumps is the most common parotid viral infection. Less common viral infections are cytomegalovirus, coxsackie and Epstein-Barr viruses.
63. **Bacterial sialadenitis:** It is usually caused by coagulase positive Staphylococcus aureus. S. pneumoniae, Escherichia coli, H. influenzae and oral anaerobe.
64. **Sjögren's syndrome:** A positive anti-nuclear antibodies (ANA), rheumatoid factor (RF), Sjögren's syndrome (SS)-a, SS-b and an elevated erythrocyte sedimentation rate (ESR) are indicative of Sjögren's syndrome. Biopsy from lip confirms the diagnosis and shows atrophy of minor salivary glands with an abundance of lymphocytes and histiocytes.
65. **Malignancy of salivary glands:** Their presentation may be similar to benign tumors and can lead to delay in diagnosis.
66. **Facial nerve in parotid:** The most common landmark for identifying facial nerve during parotid surgery is the tympanomastoid suture.
67. **Group A beta-hemolytic streptococci:** It is the most common cause of acute tonsillitis and can also result in rheumatic fever and post-streptococcal glomerulonephritis.
68. **Infectious mononucleosis:** Amoxicillin or ampicillin can cause a salmon-colored rash.
69. **Adenoidectomy:** It should be modified or avoided in cases of submucous cleft palate.
70. **Juvenile nasopharyngeal angiofibroma:** In an adolescent male, profuse recurrent episodes of nosebleed suggests juvenile nasopharyngeal angiofibroma until proven otherwise.
71. **Esophageal foreign body in children:** The most common site is just distal to the upper esophageal sphincter.
72. **Caustic ingestion:** Oral burns may not correlate with severity of esophageal lesions. Alkaline agents penetrate deeper tissue layers.
73. **Management of caustic ingestion:** Watch for sign of airway obstruction because airway control must be the prime concern. Watch also for the features of mediastinitis (such as tachycardia, chest pain, fever, sepsis) and peritonitis.
74. **Esophagoscopy in caustic ingestion:** It is controversial. It may lead to further injury. Early esophagoscopy may help in diagnosis and placements of feeding tube. It is terminated if a significant burn is seen. It is contraindicated after 12 hours.
75. **Symptoms of laryngopharyngeal reflux:** They include in descending order of frequency hoarseness, cough, globus pharyngeus, throat clearing and difficulty in swallowing. Gastrointestinal symptoms are absent in more than 50% of patients with gastroesophageal reflux diseases (GERD).
76. **Laryngoscopy in laryngopharyngeal reflux:** It shows posterior laryngitis, pachydermia, Reinke's edema, and granulomas.
77. **Complications of laryngopharyngeal reflux:** Gastroesophageal reflux disease may be a risk factor for subglottic stenosis and laryngeal cancer.
78. **Squamous cell carcinoma of esophagus:** Prior head and neck cancer increases the risk of this cancer eightfold.
79. **Dissemination of carcinoma of esophagus:** There occurs rapid dissemination of mucosal cancer because esophagus does not have serosal layer.
80. **Flexible laryngoscopy:** If you have difficulty in seeing into nasopharynx and oropharynx request the patient to breath through nose that will clear the soft tissue obstruction.
81. **Valsalva maneuver during flexible laryngoscopy:** It can better show the pyriform sinuses because it would stent open them.
82. **Reflux laryngitis:** In cases of chronic hoarseness and cough, or globus, first rule out neoplasm and then consider reflux laryngitis.
83. **Laryngeal cancer:** The most common malignancy of the larynx is squamous cell carcinoma.
84. **Tracheotomy:** It is the general opinion that the best time of doing tracheostomy is when you first think that patient needs tracheotomy. No one would like to end up in doing a "slash" tracheotomy later!
85. **Airway:** Before aggressive ventilation, always confirm the placement of an endotracheal or tracheotomy tube by observing:
    a. Humidification in the tube upon expiration,
    b. Affirm CO_2 return
    c. Auscultation for bilateral breath sounds to rule out main-stem bronchi placement.
86. **Respiratory distress:** If a child with respiratory distress is becoming quiet, it indicates that s/he is about to have respiratory collapse.
87. **Congenital laryngeal lesions:** Aspiration is more common with tracheoesophageal fistula, whereas hoarseness is more common with vocal cord palsy.
88. **Laryngomalacia:** It is the most common cause of stridor in childhood and usually occurs within the first 2 weeks of life.
89. **Features of laryngomalacia:** It has multiple anatomic abnormalities and includes exaggerated omega-shaped epiglottis and short and inward collapse of aryepiglottic folds. Gastroesophageal reflux is probably secondary to increased negative intrathoracic pressure.
90. **Congenital vascular lesions:** They grow a bit but then usually start regressing and do not need any treatment. Complete involution may take many years.
91. **Carotid triangle:** The most common causes of swellings in this region are paragangliomas, schwannomas, meningiomas, and nodal metastases.
92. **Silent primary sites and metastatic neck nodes:** In patients with metastatic neck nodes when the primary lesion is not found upon physical examination including flexible/rigid endoscopy, biopsies from four silent primary sites should be taken. These four silent sites are nasopharynx, tongue base valleculae, pyriform sinus and tonsils (tonsillectomy).
93. **Thyroid nodules:** Six to seven percent of adult females and 1–2% of adult males will show a thyroid nodule on thorough clinical examination.

94. **Thyroglossal cyst:** This midline neck mass is recognized by its movement with swallowing. An ultrasound neck will evaluate for ectopic thyroid tissue.
95. **Skin cancers:** Surgical excision is the treatment of choice.
96. **Paragangliomas:** These vascular tumors are strongly enhancing on CT. They contain small regions of single void on most MR pulse sequences.
97. **Bronchoscopy biopsy of the right upper lobe carina/spur:** It is the most dangerous site for biopsy because of the underlying right pulmonary artery.
98. **Esophagoscopy:** Application of too much force on upper esophageal sphincter (UES) is the most common cause of cervical esophageal perforation.
99. **Esophageal perforation:** Fever after esophagoscopy heralds perforation. Swallow study confirms diagnosis. Early surgical repair is important. Draining the perforation may prevent complications. Antibiotics can mask the clinical features.
100. **CT and MRI temporal bone:** Thin sectioned HRCT is ideal for looking temporal bone abnormalities or fractures, but MRI is gold standard for acoustic neuroma.
101. **Organ preservation therapy:** Chemotherapy may play an important role in organ preservation. It avoids extensive surgery and is the main treatment for metastatic head and neck cancers. Radiotherapy becomes more effective as well as more toxic when combined with chemotherapy.

PROBLEM-ORIENTED CASES

What is your spot diagnosis in the following clinical cases? See the footnotes (bottom of the page) for the answers.

1. A 30-year old female has been suffering from bilateral hearing loss for 5 years. She does not give any past history of ear discharge. She says that her hearing loss increased during her pregnancy.[1]
2. A patient comes to you with ear bleeding, pain, tinnitus and progressive deafness. On examination, you see a red swelling behind the intact tympanic membrane which blanches on pressure with pneumatic speculum.[2]
3. A 10-year old child develops torticollis, a tender swelling behind the angle of mandible and fever. He has been suffering from chronic foul smelling ear discharge for 5 years. Ear examination shows purulent putrid discharge with granulations in the ear canal.[3]
4. A child with an acute suppurative otitis media was being treated with antibiotic ear drops, oral antibiotics and analgesics. Two weeks later, the child develops a swelling over the mastoid, pain in the ear, fever, and pulsatile ear discharge.[4]
5. A pregnant lady in her third trimester starts hearing her own sounds. On otoscopy you could see movements of tympanic membrane that are synchronous with respiration. The tympanic membrane movements are exaggerated when patient breaths only through the ipsilateral side of nose.[5]
6. An adult lady develops hearing loss during pregnancy. Her mother had similar problem. Hearing loss is bilateral and slowly progressive. She feels that her hearing improves in noisy background. Pure tone audiometry shows conductive hearing loss. Bone conduction dips at 2,000 Hz.[6]
7. An elderly man, who has been suffering from chronic ear discharge for many years, develops facial palsy and pain in the ear which is worse at night. Otoscopy shows a friable ear polyp, which has a tendency to bleed.[7]
8. A 20-year-old female from Uttar Pradesh is having nasal obstruction and crusting of nose. Your examination of patient reveals an infiltrating lesion that involves nasal vestibule and upper lip with broadening of nasal dorsum.[8]
9. A biopsy taken from a granulomatous lesion of nose shows Mikulicz's cells and eosinophilic structures in the cytoplasm of the plasma cells.[9]
10. A 10-year boy presents with chronic sinusitis and multiple nasal polyps, recurrent chest infections and malabsorption syndrome. Sweat chloride test confirms the diagnosis.[10]
11. A 4-year-child presents with chronic right side serosanguinous (purulent blood tinged) nasal discharge, which is not getting cured with medical treatment.[11]
12. A 3-year-child presents with a unilateral single nasal polyp, which does not bleed on touch. Which condition would you like to rule out or confirm?[12]
13. An elderly diabetic patient develops left sided orbital cellulitis. CT scan shows evidence of left sided maxillary sinusitis. Gram stained smear of orbital exudates reveals irregularly branching septate hyphae.[13]
14. A 15-year old boy presents with right side nasal obstruction and episodes of profuse bleeding. On examination you see marked anemia and fullness of right cheek.[14]
15. A 45-year old man presents with right side facial pain in temporoparietal and the lower jaw area. On examination you

[1] Otosclerosis
[2] Glomus jugulare
[3] Bezold's abscess
[4] Coalescent mastoiditis
[5] Patulous Eustachian tube
[6] Otosclerosis
[7] Carcinoma of middle ear
[8] Rhinoscleroma
[9] Rhinoscleroma
[10] Cystic fibrosis
[11] Foreign body nose
[12] Intranasal meningocele
[13] Aspergillus maxillary sinusitis
[14] Juvenile nasopharyngeal angiofibroma
[15] Trotter's syndrome

find right side conductive hearing loss and immobile soft palate.[15]

16. A 5-year old boy presents with recurrent upper respiratory tract infections, mouth breathing, and nasal obstruction. His parents observed that the child's hearing has deteriorated.[16]
17. A neonate female child presents with a unilateral parotid swelling with bluish overlying skin. The swelling increases when child is crying. There is no other abnormality.[17]
18. HIV infected patient develops painless nontender parotid gland swelling. What is the most common cause?[18]
19. A 50-year old man had plenty of drinks followed by a heavy dinner. He developed severe vomiting and chest pain and collapsed. Hydropneumothaorx was seen on X-ray chest posteroanterior (PA) view.[19]
20. A 20-year old boy presents with throat pain and easy fatigable voice. During indirect laryngoscopy both vocal cords approximate well but leave a triangular gap in the interarytenoid region.[20]
21. A 40-year old patient presents with frequent episodes of transient but severe unbearable throat pain that radiates to the ear and posterior part of tongue and is aggravated on swallowing. ENT head and neck examination does not show any positive findings.[21]
22. A 35-year old patient develops trismus, fever, and swelling that is pushing the tonsils medially and spreading laterally posterior to sternocleidomastoid. He had extraction of third molar caries tooth few days back.[22]
23. A middle aged chronic smoker male patient has been suffering from hoarseness of voice for 2 years. On indirect laryngoscopy, reddish areas of mucosal irregularity are seen on both vocal cords. What will be your line of management?[23]
24. A 3-year old boy presents with hoarse voice and slight respiratory distress. The child is having multiple laryngeal papillomas, which also involve glottis. Which is the best line of treatment?[24]
25. A 4 year old child presents in emergency with mild respiratory distress. Laryngoscopy reveals multiple juvenile papillomatosis of the larynx. What will be your first line of treatment when the immediate facility for microlaryngoscopy surgery is not available?[25]
26. A 30-year old patient comes back to hospital with gradually increasing respiratory distress in 4 days. The patient was on mechanical ventilation and had orotracheal intubation for 2 weeks. The patient was fine after removing the endotracheal tube and was discharged. It seems that the patient has developed tracheal stenosis. What would be your next line of treatment?[26]
27. A 10-year old child after an episode of diphtheria develops hoarseness of voice. Laryngoscopy reveals right vocal cord palsy. In addition to the medical treatment what else would you like to do for the paralyzed vocal cord?[27]
28. A patient of carcinoma larynx involving anterior commissure and vocal cord has perichondritis of thyroid cartilage. What will be your line of treatment?[28]
29. An adult male patient of 35 years of age has been diagnosed as squamous cell carcinoma of left lateral border of anterior two-third tongue. The size of local ulceroinfiltrative lesion is 6 cm. The size of swelling of level 3 left cervical lymphadenopathy is 4 cm. General and systemic examination do not show any evidence of distant metastasis. Do TNM classification and staging of his cancer disease.[29]
30. What will be the staging of a squamous cell carcinoma if the TNM classification has T_1, N_1 and M_0?[30]
31. An adult patient who is a chronic smoker has squamous cell carcinoma of larynx. The local laryngeal lesion is involving unilateral vocal and false cords. Vocal cords are mobile. There are no neck nodes. Do the TNM classification and staging according to American Joint Committee on Cancer (AJCC) classification.[31]
32. A child aged 2 years has severe bilateral sensorineural hearing loss. Hearing aids were used but with no benefit. What for will you further evaluate the child?[32]
33. A 50-year-old patient presents with 3 cm indurated infiltrative ulcerative lesion of right buccal mucosa. Cervical palpation reveals multiple nodes on the right side in the submandibular region. None of the nodes is more than 3 cm. Biopsy confirms the diagnosis of squamous cell carcinoma. Stage the disease and tell the best line of treatment.[33]
34. A 50-year-old man who is smoker and alcoholic develops gradually progressive lymphadenopathy in the upper cervical region. On thorough ENT examination, no primary site of cancer was seen. What will be your next diagnostic step?[34]
35. A child was scheduled for tonsillectomy. On the day of surgery he comes with upper respiratory infection (URI, cough, cold, and fever). Will you proceed for surgery?[35]
36. A tracheostomy patient with Portex tracheostomy tube in position develops severe respiratory distress due to the blockage of the tube. What immediate action will you take?[36]

[16] Adenoid hypertrophy and otitis media with effusion
[17] Hemangioma
[18] Lymphoepithelial cysts
[19] Boerhaave's syndrome
[20] Phonasthenia
[21] Glossopharyngeal neuralgia
[22] Parapharyngeal abscess
[23] Microlaryngoscopy and biopsy
[24] Microlaryngoscopy and excision
[25] Tracheostomy
[26] Intravenous steroids
[27] Wait and watch for the spontaneous recovery
[28] Laryngectomy and postoperative radiotherapy
[29] Stage IV: T_3 N_2 M_0
[30] Stage III
[31] Stage II: T_2 N_0 M_0
[32] Cochlear implant
[33] Stage III: $T_2N_1M_0$; Surgical excision of growth with supra-omohyoid neck dissection and post operative radiotherapy
[34] Fine needle aspiration cytology (FNAC)
[35] No. Wait for 3 weeks and treat the URI if needed with antibiotics.
[36] Immediate removal of Portex tracheostomy tube

(*Hint*: Portex tracheostomy tube does not have separate inner cannula).

37. During microlaryngoscopy CO_2 laser surgery an ignition of the endotracheal tube is seen. What will be your immediate step?[37]
38. An unknown microorganism from a very ill patient is isolated. The microorganism has all the features of aerobic filamentous fungus except that the ribosomes are prokaryotic. Which antibiotic would most likely be successful?[38]
39. A 5-year old child is afebrile but weak and exhausted from a week of paroxysmal coughing with inspiratory whoops, frequently associated with vomiting. The parents admit that childhood vaccinations were not given. Blood samples reveals lymphocytosis of 44,000/mm^3. What is your most likely diagnosis?[39]
40. An 85-year old male patient is admitted in the hospital with an infection. He has creatinine clearance of 25 ml/min. Which antibiotic is safe and would not need dose adjustment?[40]
41. A 6-year old child presents with sore-throat and fever of 2 days duration. Microbiology of throat swab reveals small, translucent beta-hemolytic colonies sensitive to *in-vitro* bacitracin. Past history revealed severe allergic reaction to amoxicillin. Infection is not that serious and does not need parenteral antibiotic. Which antibiotic you think would be safe and effective?[41]
42. An immunocompromised patient with severe aspergillosis is hospitalized. He is suffering from invasive fungal sinusitis. He has been administered nonsteroidal anti-inflammatory drugs (NSAIDs), antihistamines, and adrenal corticoids. Now which parenteral antifungal agent you would like to start?[42]
43. A pregnant lady comes to you with community-acquired pneumonia. The usual pathogens responsible include pneumococci, Gram-negative rods, and atypicals (Mycoplasma pneumoniae, *Chlamydophila pneumoniae*). Which one antibiotic would be safe and effective?[43]

MISCELLANEOUS KEY POINTS

The following important points for quick revision are prepared for answering frequently asked MCQs, short answer questions and oral/viva questions.

1. **Abducent:** CN VI has the longest intracranial course. Dorello's canal transmits abducent nerve.
2. **Accessory nerve:** It innervates sternocleidomastoid and trapezius. On the lesion of CN IX, patient cannot turn head to the opposite side and has difficulty raising ipsilateral arm above the head to comb hair.
3. **Adenohypophysis:** This anterior lobe of pituitary gland is derived from an outgrowth of oral ectoderm called Rathke's pouch.
4. **Apical lung tumor:** It can manifest into Horner syndrome because of the compression of the structures that pass through the scalene interval and cross the first rib. Preganglionic sympathetic axons leave the spinal cord in the T1 ventral ramus and synapse in the superior cervical ganglion and innervate the face, scalp, and orbit.
5. **Arnold-Chiari malformation:** Cerebellar tonsils project into the vertebral canal through the foramen magnum.
6. **Asthmatic bronchitis:** Asthmatic patients, frequently, have allergy to aspirin (acetyl salicylic acid).
7. **Ataxic gait:** The most common cause is chronic alcohol abuse, which preferentially affects Purkinje cells of anterior vermis of cerebellum that control proximal musculature.
8. **Brook's tumor:** It is of basal cell origin.
9. **Caroticotympanic artery:** It is a branch of internal carotid artery, which anastomoses with branches of external carotid system in the middle ear.
10. **Cervical spinal cord lesion:** It might affect the descending hypothalamic fibers, which control all preganglionic sympathetic neurons and result in Horner syndrome.
11. **CHARGE syndrome:** It consists of **C**oloboma, **H**eart defects, choanal **A**tresia, **R**etarded growth, **G**enital hypoplasia and **E**ar anomalies.
12. **Chordoma:** It arises from the remnants of notochord. Characteristic histological features include physaliferous cells (foamy cells with compressed nuclei).
13. **Cleft and craniofacial anomalies:** The care of these patients needs many medical and surgical subspecialties and requires a team management.
    a. Initial management: The children need a secure airway and adequate nutrition.
14. **Cleft lip:** Maxillary prominence fails to fuse with intermaxillary segment.
15. **Cleft palate:** Eustachian tube dysfunction can cause persistent otitis media with effusion and recurrent acute otitis media.
16. **CN V1:** The ophthalmic division of trigeminal nerve carries sensory fibers from cornea (sensory limb of corneal reflex), bridge of nose, forehead, and scalp (anterior to mid coronal plane). The lesion of CN V1 will result in ipsilateral loss of corneal reflex and forehead sensations. On stimulation of opposite side of cornea with a wisp of cotton, both eyes blink.
17. **Contraindications for biopsy:** They include nasopharyngeal angiofibroma, glomus tumor of the middle ear, carotid body tumor of the neck and benign parapharyngeal tumors.

[37] Remove endotracheal tube and reestablish the airway
[38] Tetracycline
[39] Whooping cough (pertussis)
[40] Erythromycin
[41] Azithromycin
[42] Amphotericin B
[43] Azithromycin

18. **Costen's syndrome:** This disorder of temporomandibular joint, which is due to defective bite, is characterized by otalgia, feeling of blocked ear, tinnitus and sometimes vertigo. The ear pain radiates to frontal, parietal and occipital region.
19. **Dalrymple's sign:** This is the retraction of upper eyelid showing sclera above the cornea and is present in Grave's disease hyperthyroidism.
20. **DiGeorge sequence:** The absence of thymus (few T cells in the paracortex of lymph nodes) and parathyroid glands (hypocalcemia) results from the improper development of the third and fourth pharyngeal pouches. Young girls get repeated episodes of viral and fungal infections.
21. **Dysmetria:** It is an inability to stop a movement at the proper place.
22. **Epidural (extradural) hematoma:** The middle meningeal artery that enters the cranium through the foramen spinosum is usually lacerated in lateral skull trauma. The initial examination is normal but later in the day patient becomes lethargic and drowsy. The hematoma displaces the temporal lobe and results into contralateral Babinski sign and brisker patellar tendon reflex.
23. **Extraocular muscles:** The oblique muscles are tested by requesting the patient to adduct the eye first, then look up (for testing inferior oblique) or down (for testing superior oblique). The superior and inferior rectus muscles are tested by asking the patient to abduct the eye first then look up and down respectively.
24. **Facial plastic surgery:** Good results depend on two factors: (1) reasonable patient expectations, and (2) proper surgical capability.
    a. *Rhinoplasty*: The success depends mainly on the esthetic liking of the patient.
    b. *Blepharoplasty*: Preoperative basic ophthalmologic examination is must including visual acuity, visual fields and dry eyes.
    c. *Rhytidectomy*: It can correct wrinkling in only the lower two-third of the face and the neckline.
25. **First pharyngeal arch:** Mandible, malleus, and incus are derived from first pharyngeal arch.
26. **Glossopharyngeal nerve:** It supplies to oropharynx and stylopharyngeus muscle. It also carries preganglionic parasympathetic secretomotor axons and traverses the jugular foramen. Its tympanic branch synapses in the otic ganglion through the lesser petrosal nerve (arising from the tympanic plexus, which is formed by tympanic branch of CN IX), and provide innervation to the parotid gland.
27. **Glossopharyngeal neuralgia:** The clinical features include paroxysmal attacks of pain that radiates in the area of tongue, tonsil and ear. The pain is precipitated by swallowing, talking and even laughing and responds to carbamazepine.
28. **Hereditary lipid proteinosis:** It most commonly involves larynx.
29. **Horizontal gaze:** The frontal eye field lesion would result in an inability to look towards contralateral side with both eyes. The medial longitudinal fasciculus (MLF) lesion would result in an inability to adduct the ipsilateral eye on the attempted gaze to opposite side.
30. **Horner's syndrome:** It consists of partial ptosis, miosis (constriction of pupil), anhidrosis and apparent enophthalmos. It occurs due to an ipsilateral lesion of cervical sympathetic chain or its central pathways.
31. **Hypothyroidism:** The ENT symptoms include hoarseness of voice, nasal stuffiness, vertigo and hearing loss.
32. **Internuclear ophthalmoplegia:** Medial longitudinal fasciculus lesions present with weakness of adduction during gaze and nystagmus of the abducting or normal eye. Patient can look straight ahead with both eyes. Convergence remains intact as the oculomotor nerve is normal.
33. **Jugular foramen syndrome:** It is caused by a tumor, which compresses contents that are passing through the jugular foramen. It consists of paralysis of CN IX, X and XI, which pass through the jugular foramen along with the internal jugular vein. It can be seen in patients of malignancy nasopharynx, glomus jugular, large acoustic neuroma or thrombophlebitis of jugular bulb. The syndrome consists of following features:
    a. Reduction in the parotid gland secretion is the autonomic deficit.
    b. Axonal lesions result in retrograde chromatolysis (changes in the neuronal cell bodies) in the nucleus ambiguus.
    c. Loss of gag reflex
    d. Uvula deviated towards the opposite side of the lesion.
34. **Kinesin:** Defects in the production of kinesin (ATPase motor protein) might result in an inability of axons to transport substances in the anterograde direction.
35. **"Locked in" syndrome:** It occurs in elderly people due to the occlusion of basilar artery in the caudal pons. It has following lesions and features:
    a. Bilateral abducens and horizontal gaze centers: Patient can only move eyes vertically and is able to blink bilaterally. Patient can read and understand speech and remains aware of surroundings. Patient responds by blinking as s/he has quadriplegia.
    b. Bilateral corticobulbar fibers to nucleus ambiguus and hypoglossal nuclei: Patient becomes unable to speak and swallow.
    c. Bilateral corticospinal tracts: Patient becomes tetraplegic (quadriplegia).
36. **Marcus Gunn pupil:** When light is put on the blind eye, the pupils of both eyes remain dilated. But when the light is put on the normal eye, pupils of both eyes constrict. It happens due to the consensual pupillary light reflex because the efferent pathway on diseased side is normal. This interruption of afferent papillary pathway occurs in retrobulbar neuritis or optic nerve lesions (such as injury to the optic nerve during endoscopic sinus surgery).
37. **Mesencephalic nucleus of CN V:** Lesion results in a loss of touch sensations in the face. Facial sensations of pain and temperature remain intact and there is no jaw weakness.
38. **Montogomery tube:** This silicon T-tube is used in ENT procedure.
39. **Muller's muscles:** These extraocular smooth muscles of eyelids are supplied by sympathetic fibers. Its paralysis in Horner syndrome causes partial ptosis.
40. **Nasogastric feeding:** About 2,000 kcal can be given without any untoward side effect by nasogastric feeding.
41. **Neurohypophysis:** Pituicytes, a form of glial cells, are found in neurohypophysis, which is the posterior lobe of pituitary gland. They are present along with axons of supraoptic and paraventricular neurons, which are found in the hypothalamus.
42. **Nucleus solitarius in medulla:** Most visceral sensation other than pain such as taste sensations, chemoreceptor and baroreceptor information from carotid sinus and carotid body go to this nucleus through facial, glossopharyngeal and vagus nerves. Lesions of carotid body and carotid sinus result in problems in self regulating blood pressure. The nuclei found throuhout the length of the medulla are solitary nucleus and spinal nucleus of CN V.
43. Oligodendrocytes are derived from neuroectoderm.
44. **Palatal muscles:** All the muscles of palate are supplied by the vagal-accessory nerve except the tensor palati, which is innervated by mandibular division of trigeminal nerve.

45. **Parafollicular cells of thyroid:** These cells are neural crest cells that migrate into 4th pharyngeal pouch and present adjacent to thyroid follicles.
46. **Peutz-Jegher's syndrome:** The benign intestinal polyps are associated with perinasal, perioral and buccal mucosal pigmentation.
47. **Psammoma bodies:** They are seen in the papillary carcinoma of thyroid gland.
48. **Schaumann's bodies:** These are seen in sarcoid granuloma.
49. **Schwann cells:** They are derived from neural crest.
50. Skin resurfacing
    a. *Phenol*: This chemical agent can be cardiotoxic, hepatotoxic and nephrotoxic. Proper steps must be taken to avoid these toxicities.
    b. *Croton oil*: It allows deeper penetration of the phenol. This is an epidermolytic agent in the Baker Gordon phenol formula.
    c. *Carbon dioxide laser*: A yellow chamois color indicates level of the reticular dermis.
    d. *Dermabrasion infection*: The most common organisms resulting in post laser, chemical peel, or dermabrasion infection are streptococci, staphylococci, herpes simplex virus and candidal species.
51. **Sluder's neuralgia:** This neuralgia of sphenopalatine ganglion presents with neuralgic pain in lower half of face, nasal congestion, rhinorrhea and increased lacrimation.
52. **Styloid process:** All the three muscles arising form the styloid process have different nerve supplies: stylohyoid by facial nerve; styloglossus by hypoglossal nerve; and stylopharyngeus by glossopharyngeal nerve.
53. **Submucous cleft palate:** The clinical features include bifid uvula, notch in posterior border of hard palate, and deficient palatal muscles.
54. **Third aortic arch:** It gives rise to the common carotid arteries.
55. **Tongue:** All the intrinsic and extrinsic muscles of tongue are supplied by the hypoglossal nerve except the palatoglossal, which is innervated by vagal-accessory nerve.
56. **Thyroglossal duct:** The pyramidal lobe of the thyroid gland, which may be seen in an infant as a small lump on the anterior aspect of thyroid cartilage near the midline, is a remnant of thyroglossal duct. The thyroid gland develops from thyroglossal duct, which descends in the midline from the apex (foramen cecum) of the sulcus terminalis in the tongue.
57. **Trigeminal nerve:** CN V is the largest cranial nerve. Muscles supplied by trigeminal nerve include muscles of mastication (masseter, temporalis, and medial and lateral pterygoids), mylohyoid, anterior belly of digastric, tensor tympani, tensor veli palatini.
58. **Trigeminal neuralgia:** Carbamazepine is the drug of choice.

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